Dark Matter Detection in Solids

Volume I

Foundations

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Plan of the three volumes

The book is in three volumes. Volume I, *Foundations* (Chaps. 1 to 8), teaches the quantum mechanics, solid-state physics and response theory that the dark-matter event rates rest on, and ends with the master rate formula and the response framework built on it (Chap. 8). Volume II, *Signals and Detection* (Chaps. 9 to 22), derives the rate of each excitation channel and follows the deposited energy to the measured signal and to an exclusion limit. Volume III, *Computations* (Chaps. 23 to 33), gives the first-principles methods and the specification of each calculation behind the numbers of Volumes I and II.

Chapter numbers and appendix letters run through the three volumes, so a reference names its target without the volume: Sect. 4.4 is in Volume I and Sect. 10.2 in Volume II. Appendices A and B (conventions and notation) are printed in every volume, Appendices C and D (mathematical tools and long derivations) in Volume I, and Appendices E and F (long derivations and an index of literature formulas) in Volume II.

Conventions and units

We use natural units, $\hbar = c = k_{\mathrm{B}} = 1$, unless stated otherwise, and quote physical values in conventional units with SI prefixes.

A widely used formula from the literature appears in a gray box called a Limit Check, which plots the formula against a fuller result. A Limit Check has three parts: the formula, a plot, and one sentence. The formula is written in the book's notation, and a one-line citation names the equation of the source in which the formula was first given. The fuller result describes the same quantity as the formula and is either a measurement or a calculation in the book, for a real crystal or for a model that keeps the physics the formula drops. Most boxes plot the ratio between the formula and the fuller result, so that the formula is the horizontal line at one; some boxes plot the formula and the fuller result as two curves, when the shapes of the curves matter more than their ratio. The horizontal axis carries the *controlling parameter*: along this axis the relative difference between the formula and the fuller result runs from negligible at one end to order one at the other. The *leading correction* is the first term of this relative difference in the controlling parameter, and it is drawn as a third curve, usually dotted or dashed. A gray band marks the values of the controlling parameter at which the formula agrees with the fuller result, to a tolerance stated in the caption, usually 10 %, and the caption prints the edge of the band. In Limit Check 1.1 the free-recoil formula $\omega = q^2/2M$ is compared with the energy deposited in a nucleus bound in an Einstein potential, as a function of the recoil energy in units of the phonon energy. The leading correction in that box is the rms width $(E_R\omega_0)^{1/2}$ of the deposited energy. The sentence under the plot says what breaks down outside the band. The running text next to the box derives the formula from the book's general result.

Inside the gray band the formula is established to the stated tolerance, with the same confidence as the fuller result. Where the fuller result is a model, the edge of the band is an estimate of where the formula fails, not a bound. Where nothing is shaded, the formula fails over the whole plotted range, and the box gives the size of the failure in place of a band edge. Some boxes compare two published forms of one result, or test a sum rule; these boxes are read in the same way, with the exact relation as the fuller result. Appendix F lists the most used of the Limit Check formulas with their source equations and the sizes of their leading corrections.

Chapter 1

Light dark matter

Most of the matter in the universe has been detected only through its gravity. It shapes galaxies and the cosmic microwave background and outweighs the atoms we know five to one, yet decades of searches for a heavy particle that recoils off a nucleus have found nothing. Those null results have shifted attention to lighter particles, below a GeV, which carry too little momentum to move a nucleus measurably but enough energy to lift an electron across a band gap or to set a crystal lattice vibrating. Detecting these lighter particles turns a dark-matter search into a problem in condensed-matter physics: the target is no longer a collection of free nuclei but a solid, with all of its electronic and vibrational structure. What a detector can see is set by the rate at which it registers dark-matter events. This rate factorizes into a dark-matter part, which contains the halo and the coupling, and the response of the solid.

1.1 The master equation of direct detection

A dark-matter detector counts events. Each one is a small burst of electron–hole pairs, phonons, quasiparticles or photons, released when a particle from the Galactic halo gives up an energy ω and a momentum $\boldsymbol{q}$ to the target. Theory has to predict the number of events that a kilogram of target collects per year, the *event rate* $\mathcal{R}$, and the distribution of these events in ω. Take dark matter χ of mass m_χ that scatters on the electrons of the target, through an interaction that couples to the electron density and not to the electron spins. The event rate then has the form

$$\mathcal{R} = \frac{1}{\rho_T} \underbrace{\frac{\rho_\chi}{m_\chi} \int \mathrm{d}^3 v \, f_\chi(\boldsymbol{v})}_{\text{halo}} \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \underbrace{|\tilde{V}(\boldsymbol{q})|^2}_{\text{coupling}} \int \mathrm{d}\omega \, \underbrace{S(\boldsymbol{q},\omega)}_{\text{target}} \underbrace{\delta\!\left(\omega - \boldsymbol{q}\cdot\boldsymbol{v} + \frac{q^2}{2m_\chi}\right)}_{\text{kinematics}} . \tag{1.1}$$

We will derive it in the chapters that follow: Chap. 2 obtains it from Fermi's golden rule (Eq. (2.67)), and once the directions of $\boldsymbol{q}$ are averaged it reduces to Eq. (1.46), the form used for limits [1].

The integrand is the product of the halo, the coupling and the response of the target, joined by a delta function. The halo enters through ρ_χ/m_χ, the number density of dark-matter particles near the Earth, with ρ_χ their mass density, and through $f_\chi(\boldsymbol{v})$, the distribution of their velocities $\boldsymbol{v}$ in the laboratory, normalized to one. The coupling is $\tilde{V}(\boldsymbol{q})$, the Fourier transform of the potential between a dark-matter particle and an electron, and $|\tilde{V}(\boldsymbol{q})|^2$ weights each momentum transfer by the strength of the dark-matter–electron interaction at that momentum. The response is the *dynamic structure factor* $S(\boldsymbol{q},\omega)$ of the electron density. The dynamic structure factor measures how readily the solid absorbs a momentum $\boldsymbol{q}$ together

with an energy ω. At zero temperature it is a sum over the excited states of the solid that lie an energy ω above its ground state. Each term is the squared matrix element of the electron density at wavevector $\boldsymbol{q}$ between the ground state and the excited state. A factor 2π comes from the golden rule, and a division by the volume of the target makes S independent of the size of the target (Eq. (2.67); Sect. 4.2). The delta function is energy conservation. A particle whose momentum changes by $\boldsymbol{q}$ loses the kinetic energy $\boldsymbol{q} \cdot \boldsymbol{v} - q^2/2m_\chi$, and the target must take up exactly that amount. Finally, $1/\rho_T$, with ρ_T the mass density of the target, turns a rate per unit volume into a rate per unit target mass, the unit in which experiments quote event rates.

Each factor of Eq. (1.1) belongs to a different body of physics. The halo is astrophysics, the coupling is particle physics, and the response is condensed-matter physics; only the kinematics of the delta function connects them, and it involves only $\boldsymbol{q}$, $\boldsymbol{v}$, ω and m_χ. As a result, a new dark-matter model changes $|\tilde{V}|^2$ and leaves S alone, while a new target changes S and leaves the coupling alone. A response computed once for silicon serves every model that couples to the electron density, and once S is tabulated, a limit can be recast for another model by weighting S with the new coupling, without computing the response of the solid again. The response involves of order 10^{23} interacting electrons per cubic centimeter in a periodic lattice, and computing it is most of what this book does.

The halo needs no new physics. Its local density and velocity distribution are adopted from astronomy by convention: $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$ and a Maxwellian with speeds near 10^{-3} in units of the speed of light, seen from a laboratory that moves through the halo (Sect. 1.2). Since the delta function involves no property of the target, much of what a detector can reach follows before any target is specified (Sect. 1.3). Kinematics fixes the smallest speed v_min that can deposit a given ω at a given q, and it fixes the lightest mass that can get over a detector threshold. Once the velocity integral is done, the halo enters only through $\eta(v_\mathrm{min})$, the mean inverse speed above v_min. The coupling is a choice of model. Its simplest form is the exchange of a *mediator*, a particle that carries the interaction between the dark matter and the electron. The coupling then condenses into $|\tilde{V}(q)|^2 = \pi\bar{\sigma}_e F_\mathrm{DM}(q)^2/\mu_{\chi e}^2$ (Eq. (1.33)), with the reduced mass $\mu_{\chi e}$ of the dark matter and the electron, a reference cross-section $\bar{\sigma}_e$ that sets the strength, and a form factor $F_\mathrm{DM}(q)$ that carries the momentum dependence (Sect. 1.4). An experiment then quotes its result as a limit on $\bar{\sigma}_e$ at each mass. The response, however, is a many-body problem. It needs the electronic states of a crystal, the linear response of its electrons, the vibrations of its lattice and the electron–phonon coupling (Chaps. 3 to 7). The response also needs a framework that holds for any coupling (Chap. 8) and a separate treatment of each excitation channel. Even an exact S does not finish the calculation, because a detector does not record ω. It records the charges, phonons, quasiparticles or photons into which the deposited energy relaxes, and this relaxation is the last link between the event rate and a measured count (Sect. 1.5).

The product in Eq. (1.1) is the simplest case, one coupling to one density, and the book generalizes it in several directions without losing the separation of halo, coupling and response. Dark matter can couple to more than the electron density, to nuclei, spins or currents as well, and the single S then becomes a matrix of response functions, with one entry for each pair of couplings. The off-diagonal entries of this matrix make the contributions of different couplings interfere (Chap. 8). The electrons of the target also rearrange to screen the dark-matter potential. For a mediator that mixes with the photon, the screened response that takes the place of S is the *loss function*, the imaginary part of the inverse dielectric function (Sect. 10.2). A bosonic dark-matter particle can instead be absorbed. It deposits its whole mass m_ϕ at almost zero momentum, and the absorption rate is then set by the optical conductivity of the target at $\omega = m_\phi$ instead of by a scattering response (Sect. 1.4). Other processes need more than one response at once. In the Migdal effect a recoiling nucleus shakes off an electron,

and the Migdal rate needs the joint response of electrons and lattice (Chap. 12). At an indirect gap the highest filled and the lowest empty electron states differ in crystal momentum (Sect. 1.3). An electron excited across an indirect gap needs a phonon to supply this difference in crystal momentum, and the excitation is then a second-order process (Chap. 13). Finally, no computed response is exact. Each S in the book names the approximation behind it and comes with an estimate of its error, and Chap. 21 carries those errors into the exclusion limits.

1.1.1 Gravitational evidence and the mass range

All of the evidence for dark matter is gravitational. Stars and gas in spiral galaxies orbit at a circular speed v_c that stays flat far outside the luminous disk, where the visible mass alone would make v_c fall as $r^{-1/2}$ with the galactocentric radius r. A flat rotation curve needs an enclosed mass that grows linearly with r where almost no light is emitted [2, 3]. The anisotropies of the cosmic microwave background (CMB) fix the total amount of dark matter, because dark matter does not take part in the acoustic oscillations of the photon–baryon plasma before recombination, and its potential wells shape the relative heights of the acoustic peaks [4].

None of this gravitational evidence identifies the dark-matter particle. The observations measure a mass density, and for detection they supply two numbers. The local density ρ_χ comes from the motion of stars near the Sun and from the rotation curve of the Milky Way [5, Sect. 3.1]; Sect. 1.2 gives the measured range and the value we use. The cosmological abundance is quoted as $\Omega_c h^2$, where Ω_c is the mean density of cold dark matter as a fraction of the critical density (the density of a spatially flat universe), and h is the present Hubble rate in units of $100\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The CMB anisotropies, with CMB lensing, give [4, Table 2 and Eq. (23)]

$$\Omega_c h^2 = 0.1200 \pm 0.0012,$$

usually quoted as 0.120 ± 0.001. The baryon density from the same fit, 0.02237 ± 0.00015 in the same units, is smaller than $\Omega_c h^2$ by a factor of 5.4.

Gravity cannot fix the mass m_χ of the dark-matter particle: by the equivalence principle, a test body falls the same way whatever its mass, and every observation above measures the mass density $n_\chi m_\chi$ alone, with n_χ the number density. The mass enters only where the particle picture itself fails. At the light end, a boson with $m_\chi \approx 10^{-22}\,\mathrm{eV}$ has a de Broglie wavelength comparable to the size of a dwarf galaxy, and lighter fields cannot cluster on that scale. Fermions need m_χ above a few hundred eV so that halo phase-space densities stay below the Pauli limit, the largest density that the exclusion principle allows; the argument is related to the Tremaine–Gunn bound [6, 7]. A particle once in equilibrium with the plasma must be heavier than roughly a keV, or it moves fast enough to erase observed small-scale structure [7, Sect. I A]. At the heavy end, fundamental particles reach the Planck mass, and compact objects such as primordial black holes extend to tens of solar masses [7, Sect. I C]. The allowed range spans about 90 decades (Fig. 1.1).

Within this range, the window between a keV and a GeV has been tested least. Dark matter bound to the Galaxy moves at $v \approx 10^{-3}$, and its kinetic energy $\frac{1}{2} m_\chi v^2$ is 0.5 keV for 1 GeV, 0.5 eV for 1 MeV and 0.5 meV for 1 keV. A 1 GeV particle can deposit at most about 0.1 keV in a xenon nucleus (Sect. 1.3). A deposit of 0.1 keV is below the thresholds of ton-scale xenon detectors, which read the scintillation and ionization of the recoiling nucleus: Ref. [1, Sect. I] places these thresholds at the few-hundred-eV scale, and the thresholds of analyses that require both signals lie nearer a keV. Solids, however, have excitations across the range of these kinetic energies: typical electronic band gaps of an eV and optical phonons of tens of meV [1, Table I], and acoustic phonons and superconducting gaps at the meV scale and below

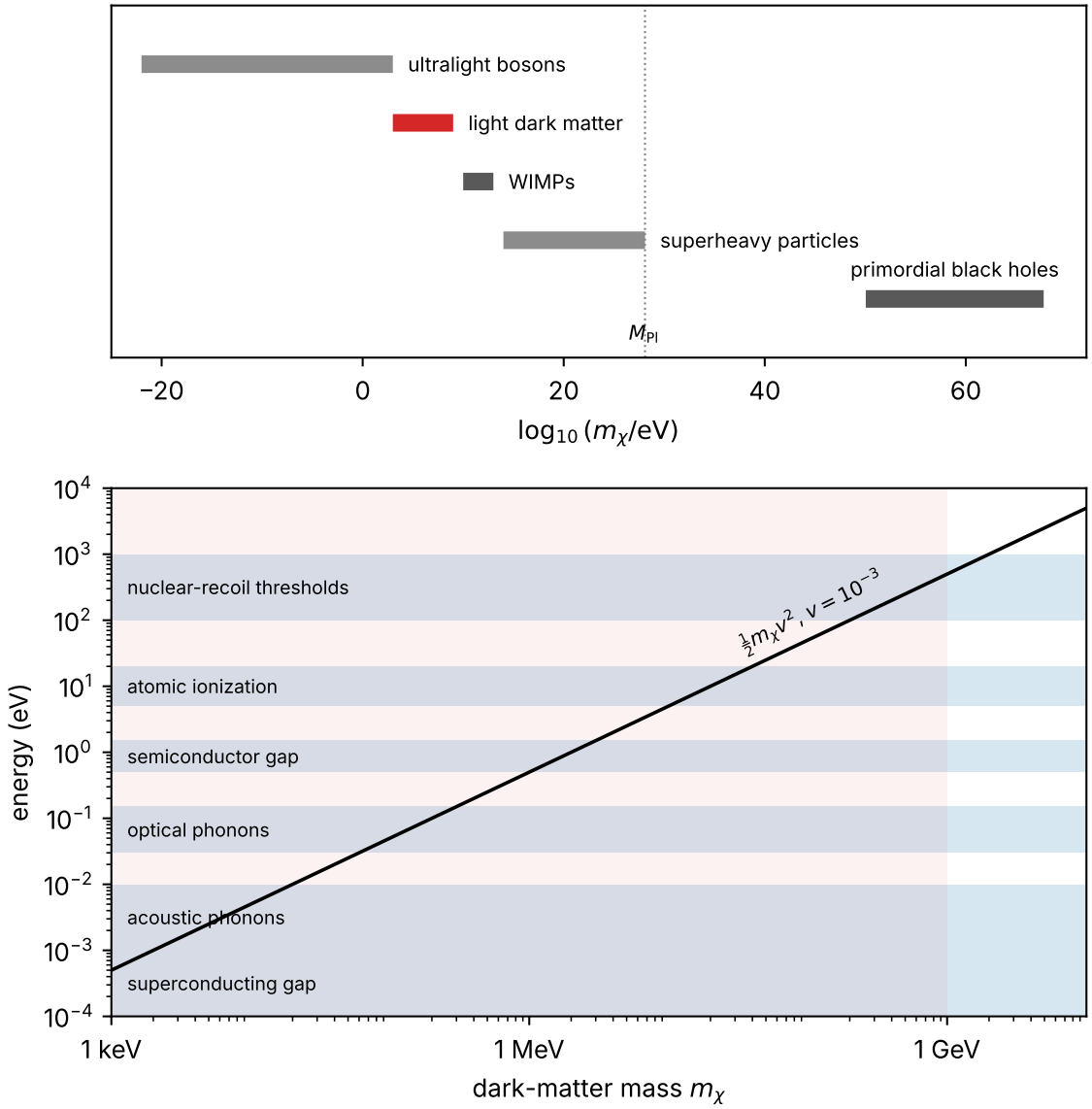


Figure 1.1: Top: masses of dark-matter candidates, from ultralight bosonic fields to primordial black holes; the dotted line marks the Planck mass. Ranges follow Ref. [7, Sect. I]. Bottom: the keV–GeV window (shaded red). The solid line is the kinetic energy $\frac{1}{2}m_\chi v^2$ of dark matter moving at $v = 10^{-3}$. The blue bands are the energy scales of excitations in solids and the thresholds of nuclear-recoil detectors [1].

(Sects. 1.3 and 1.5). Particles in this window, between a keV and a GeV, are what the field calls *light dark matter*, or sub-GeV dark matter.

1.2 The local halo

Every rate in this book is an average over the dark matter that passes through the detector. That average is fixed by the local mass density, which sets how many particles there are, and by the velocity distribution in the laboratory, which sets how fast they move and from which direction they come.

1.2.1 Density and velocity distribution

Measurements of the local dark-matter density, from the motion of stars near the Sun and from the rotation curve of the Galaxy, give values between 0.2 and $0.6\,\mathrm{GeV\,cm^{-3}}$ [8, Sect. 3.1.1]. Mass models with different halo shapes and data sets disagree by more than their statistical errors [5, Sect. 3.1]. For this reason direct-detection results are quoted for a fixed reference value; we use the recommended one [8, Table 1],

$$\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}} = 2.31\times10^{-6}\,\mathrm{eV}^4, \tag{1.2}$$

where the value in natural units follows from multiplying by $(\hbar c)^3$, with $\hbar c = 1.973\times10^{-5}\,\mathrm{eV\,cm}$ (Appendix A.1). Since the density enters every rate as an overall factor, a different value rescales a limit without changing its shape [8, Sect. 3.1.1]. The number density of particles of mass m_χ is $n_\chi = \rho_\chi/m_\chi$: three particles per cubic centimeter at 100 MeV, three thousand at 100 keV. For comparison, $0.3\,\mathrm{GeV\,cm^{-3}}$ is the mass of one hydrogen atom in roughly $3\,\mathrm{cm}^3$, while a crystal holds about 10^{23} atoms in each cubic centimeter.

The velocity distribution is modeled first in the rest frame of the Galaxy, in which the halo as a whole does not rotate. We write $\boldsymbol{v}_\mathrm{gal}$ for a dark-matter velocity in that frame and use Cartesian galactic axes: the first points from the Sun to the Galactic center, the second along the direction of Galactic rotation, and the third to the north Galactic pole [5, Sect. 5]. The *standard halo model* (SHM) takes the halo to be a singular isothermal sphere, a self-gravitating gas with density $\rho_\mathrm{halo}(r) \propto r^{-2}$ at galactocentric radius r. This density profile encloses a mass that grows linearly with r and so gives the flat rotation curve of Sect. 1.1 [5, Sect. 2.1]. The halo particles have an isotropic Maxwellian distribution,

$$f_\mathrm{gal}(\boldsymbol{v}_\mathrm{gal}) = \frac{1}{N_\mathrm{esc}}\,\mathrm{e}^{-v_\mathrm{gal}^2/v_0^2}\,\Theta(v_\mathrm{esc}-v_\mathrm{gal}), \tag{1.3}$$

normalized to $\int \mathrm{d}^3v_\mathrm{gal}\, f_\mathrm{gal} = 1$, with $v_\mathrm{gal} = |\boldsymbol{v}_\mathrm{gal}|$ and $\Theta(x)$ the Heaviside step function [8, Eq. (10)]. The distribution is specified by the speed parameter v_0 and the escape speed v_esc, which together fix the normalization N_esc.

The speed parameter v_0 is the local circular speed. Each Cartesian component of $\boldsymbol{v}_\mathrm{gal}$ in Eq. (1.3) has variance $v_0^2/2$, and for $\rho_\mathrm{halo} \propto r^{-2}$ the Jeans equation, which balances the gradient of the velocity pressure $\rho_\mathrm{halo}v_0^2/2$ against gravity, gives $v_0 = v_\mathrm{c}$ [5, Eq. (11)]. We take v_0 to be the speed of the local standard of rest, which moves on the circular orbit at the Sun's radius: $v_0 = 238\,\mathrm{km\,s^{-1}}$ (Table 1.1).

The escape speed v_esc truncates the distribution. A particle faster than the local escape speed is not bound to the Galaxy and is assumed absent. Since the isothermal sphere itself has an infinite escape speed, the cut at v_esc is imposed by hand, and the sharp step is an idealization [5, Sect. 2.1]. The value $v_\mathrm{esc} = 544\,\mathrm{km\,s^{-1}}$ comes from the fastest stars in the Radial Velocity Experiment (RAVE) survey; later surveys give medians from 485 to $580\,\mathrm{km\,s^{-1}}$ [8, Sect. 3.1.2 and Table 2].

The normalization N_esc follows in spherical coordinates [9, Eq. (15)]. Since the integrand depends only on v_gal, the angles contribute 4π, and

$$N_\mathrm{esc} = 4\pi\int_0^{v_\mathrm{esc}} \mathrm{d}y\, y^2\mathrm{e}^{-y^2/v_0^2} = \pi^{3/2}v_0^3\left[\mathrm{erf}\left(\frac{v_\mathrm{esc}}{v_0}\right) - \frac{2}{\sqrt{\pi}}\,\frac{v_\mathrm{esc}}{v_0}\,\mathrm{e}^{-v_\mathrm{esc}^2/v_0^2}\right], \tag{1.4}$$

Table 1.1: Standard-halo-model parameters adopted in this book, with their sources. Vectors are given in galactic axes (toward the Galactic center, along Galactic rotation, toward the north Galactic pole). The last two rows follow from the others. The reference Earth speed is the adopted magnitude of the solar velocity $(0, 238, 0) + (11.1, 12.2, 7.3)\,\mathrm{km\,s^{-1}}$. The largest laboratory speed is the sum of the escape speed and the reference Earth speed.

Symbol	Quantity	Value	Source
ρ_χ	local dark-matter density	$0.3\,\mathrm{GeV\,cm^{-3}}$	[8, Table 1]
v_0	circular speed at the Sun's radius	$238\,\mathrm{km\,s^{-1}}$	[8, Table 1]
v_{esc}	Galactic escape speed	$544\,\mathrm{km\,s^{-1}}$	[8, Table 1]
–	solar peculiar velocity	$(11.1, 12.2, 7.3)\,\mathrm{km\,s^{-1}}$	[8, Table 1]
–	mean orbital speed of the Earth	$29.8\,\mathrm{km\,s^{-1}}$	[8, Eq. (13)]
$\boldsymbol{v}_\oplus(t)$	Earth velocity relative to the Sun	Eq. (1.12)	[8, Eq. (11)]
$v_{\mathrm{E}}^{\mathrm{ref}}$	reference Earth speed (adopted $\lvert\boldsymbol{v}_\odot\rvert$)	$250.5\,\mathrm{km\,s^{-1}}$	Eq. (1.5)
v_{max}	largest laboratory speed $v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}}$	$794.5\,\mathrm{km\,s^{-1}}$	Eq. (1.9)

with the error function $\mathrm{erf}(x) = (2/\sqrt{\pi})\int_0^x \mathrm{e}^{-y^2}\mathrm{d}y$ (Appendix C.8). The second equality integrates $y^2\mathrm{e}^{-y^2/v_0^2}$ once by parts; Appendix D.1 writes the step out. Without the cut, N_{esc} would be the Gaussian integral $\pi^{3/2}v_0^3$; the bracket is the fraction of this Gaussian integral that comes from speeds below v_{esc}.

The Sun moves through this distribution. Its velocity in the galactic frame is the circular velocity of the local standard of rest plus the Sun's peculiar motion relative to the local standard of rest [8, Table 1 and Eq. (14)],

$$\boldsymbol{v}_\odot = (0, 238, 0) + (11.1, 12.2, 7.3)\ \mathrm{km\,s^{-1}} = (11.1, 250.2, 7.3)\ \mathrm{km\,s^{-1}}, \tag{1.5}$$

in the galactic axes above. The magnitude is $|\boldsymbol{v}_\odot| = (11.1^2 + 250.2^2 + 7.3^2)^{1/2}\,\mathrm{km\,s^{-1}} = 250.55\,\mathrm{km\,s^{-1}}$, which we round to the reference value $250.5\,\mathrm{km\,s^{-1}}$. Table 1.1 collects the parameters.

1.2.2 The laboratory frame

A detector shares the motion of the Earth. Its velocity in the galactic frame is

$$\boldsymbol{v}_{\mathrm{E}}(t) = \boldsymbol{v}_\odot + \boldsymbol{v}_\oplus(t), \tag{1.6}$$

where $\boldsymbol{v}_\oplus(t)$ is the Earth's orbital velocity relative to the Sun. Since galactic speeds are below 10^{-2} in natural units, the change of frame is Galilean, with relativistic corrections of relative size $v^2 \approx 10^{-6}$. A particle with laboratory velocity $\boldsymbol{v}$ has galactic velocity

$$\boldsymbol{v}_{\mathrm{gal}} = \boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t). \tag{1.7}$$

The velocity shift of Eq. (1.7) has unit Jacobian, $\mathrm{d}^3v_{\mathrm{gal}} = \mathrm{d}^3v$, and a velocity cell holds the same number of particles in both frames. The laboratory distribution is the galactic one of Eq. (1.3) [8, Eq. (10)] evaluated at the boosted velocity, $f_\chi(\boldsymbol{v}) = f_{\mathrm{gal}}(\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t))$:

$$f_\chi(\boldsymbol{v}, t) = \frac{1}{N_{\mathrm{esc}}}\,\mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}}(t)|^2/v_0^2}\,\Theta\big(v_{\mathrm{esc}} - |\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t)|\big). \tag{1.8}$$

We write $f_\chi(\boldsymbol{v})$ when the time is not needed, and $v = |\boldsymbol{v}|$ for the laboratory speed. The normalization is unchanged, $\int \mathrm{d}^3v\, f_\chi = 1$, and we write angle brackets for an average over the

distribution, $\langle\cdots\rangle = \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,(\cdots)$. In the laboratory the distribution is no longer centered on zero: the most probable velocity is $-\boldsymbol{v}_\mathrm{E}$. Dark matter arrives as a wind from the direction in which the Earth moves. Because f_gal is even in $\boldsymbol{v}_\mathrm{gal}$, its mean velocity vanishes, and by Eq. (1.7) the mean laboratory velocity is exactly $\langle\boldsymbol{v}\rangle = -\boldsymbol{v}_\mathrm{E}(t)$, cut or no cut.

The spread about this mean is the galactic one: $\langle v_\mathrm{gal}^2\rangle^{1/2} = (3/2)^{1/2} v_0 = 291\,\mathrm{km\,s^{-1}}$ for the Gaussian and $284\,\mathrm{km\,s^{-1}}$ with the cut. The spread is larger than the magnitude of the drift $\langle\boldsymbol{v}\rangle = -\boldsymbol{v}_\mathrm{E}(t)$. A numerical integration of Eq. (1.8) at $v_\mathrm{E} = v_\mathrm{E}^\mathrm{ref}$ shows that 94 % of the particles move with a positive component along $-\boldsymbol{v}_\mathrm{E}$, and 59 % within 45° of it; the mean cosine of the angle to $-\boldsymbol{v}_\mathrm{E}$ is 0.66.

The fastest particle in the laboratory is one that moves at v_esc in the galactic frame, opposite to $\boldsymbol{v}_\mathrm{E}$; its laboratory speed is $v_\mathrm{esc} + |\boldsymbol{v}_\mathrm{E}(t)|$. We quote thresholds and kinematic edges at a fixed reference speed, the rounded solar speed $v_\mathrm{E}^\mathrm{ref} = 250.5\,\mathrm{km\,s^{-1}}$ (Eq. (1.5); Appendix A.8). The *maximum speed* is then

$$v_\mathrm{max} = v_\mathrm{esc} + v_\mathrm{E}^\mathrm{ref} = 794.5\,\mathrm{km\,s^{-1}} = 2.650 \times 10^{-3}. \tag{1.9}$$

Over a year $v_\mathrm{esc} + |\boldsymbol{v}_\mathrm{E}(t)|$ ranges from 781 to $810\,\mathrm{km\,s^{-1}}$ (see the annual modulation below). With the reference value, thresholds stay time-independent, while integrals over f_χ use the full $\boldsymbol{v}_\mathrm{E}(t)$, averaged over the exposure.

Many rates depend only on the speed. The *speed distribution* $F_\chi(v)$, normalized to $\int_0^\infty \mathrm{d}v\, F_\chi = 1$, is v^2 times the integral of f_χ over directions. Take the polar axis along $\boldsymbol{v}_\mathrm{E}$, write ϑ for the angle between $\boldsymbol{v}$ and $\boldsymbol{v}_\mathrm{E}$, and $v_\mathrm{E} = |\boldsymbol{v}_\mathrm{E}|$. Then $|\boldsymbol{v} + \boldsymbol{v}_\mathrm{E}|^2 = v^2 + v_\mathrm{E}^2 + 2vv_\mathrm{E}\cos\vartheta$, and the azimuth contributes 2π:

$$F_\chi(v) = \frac{2\pi v^2}{N_\mathrm{esc}}\, \mathrm{e}^{-(v^2+v_\mathrm{E}^2)/v_0^2} \int_{-1}^{\cos\vartheta_\mathrm{cut}} \mathrm{d}(\cos\vartheta)\, \mathrm{e}^{-2vv_\mathrm{E}\cos\vartheta/v_0^2}.$$

The upper limit $\cos\vartheta_\mathrm{cut}$ is where $|\boldsymbol{v} + \boldsymbol{v}_\mathrm{E}| = v_\mathrm{esc}$, or 1 if no direction of $\boldsymbol{v}$ reaches the cut. For $v < v_\mathrm{esc} - v_\mathrm{E}$ no direction reaches the cut and $\cos\vartheta_\mathrm{cut} = 1$. For $v_\mathrm{esc} - v_\mathrm{E} < v < v_\mathrm{esc} + v_\mathrm{E}$ the cut removes the directions close to $\boldsymbol{v}_\mathrm{E}$, and at $\cos\vartheta_\mathrm{cut}$ the exponent equals $-v_\mathrm{esc}^2/v_0^2$. Doing the elementary integral, we find

$$F_\chi(v) = \frac{\pi v_0^2\, v}{v_\mathrm{E} N_\mathrm{esc}} \times \begin{cases} \mathrm{e}^{-(v-v_\mathrm{E})^2/v_0^2} - \mathrm{e}^{-(v+v_\mathrm{E})^2/v_0^2}, & v < v_\mathrm{esc} - v_\mathrm{E}, \\ \mathrm{e}^{-(v-v_\mathrm{E})^2/v_0^2} - \mathrm{e}^{-v_\mathrm{esc}^2/v_0^2}, & v_\mathrm{esc} - v_\mathrm{E} < v < v_\mathrm{esc} + v_\mathrm{E} \end{cases}, \tag{1.10}$$

and $F_\chi = 0$ above $v_\mathrm{esc} + v_\mathrm{E}$. The two branches join continuously at $v = v_\mathrm{esc} - v_\mathrm{E}$, and the second vanishes at $v_\mathrm{esc} + v_\mathrm{E}$. Eq. (1.10) assumes $v_\mathrm{E} < v_\mathrm{esc}$, and v_esc exceeds v_E by a factor of two. Fig. 1.2 shows F_χ on the days of largest and smallest Earth speed. In the galactic frame the speed distribution $4\pi v_\mathrm{gal}^2 f_\mathrm{gal}$ peaks at v_0; in the laboratory the drift moves the peak to $335\,\mathrm{km\,s^{-1}}$ (at $v_\mathrm{E} = v_\mathrm{E}^\mathrm{ref}$) and stretches the tail to $v_\mathrm{esc} + v_\mathrm{E}$.

The tail of F_χ matters for light dark matter, which must often use the fastest particles to deposit enough energy (Sect. 1.3). We integrate Eq. (1.10) numerically at $v_\mathrm{E} = v_\mathrm{E}^\mathrm{ref}$. The fraction of particles faster than 400, 500, 600 and $700\,\mathrm{km\,s^{-1}}$ is 36 %, 15 %, 4.2 % and 0.58 %; above $750\,\mathrm{km\,s^{-1}}$ it is 0.10 %. A threshold that needs $v \gtrsim 700\,\mathrm{km\,s^{-1}}$ sees a rate suppressed by more than two orders of magnitude, and this rate depends on v_esc and on the exact shape of the cut, which are the most uncertain parts of the model.

Scattering rates contain the speed through $1/v$: integrating the energy-conservation delta function of a collision over the direction of $\boldsymbol{v}$ leaves a factor $1/v$ (Sect. 1.3). In the SHM the mean inverse speed $\langle 1/v\rangle$ has a closed form.

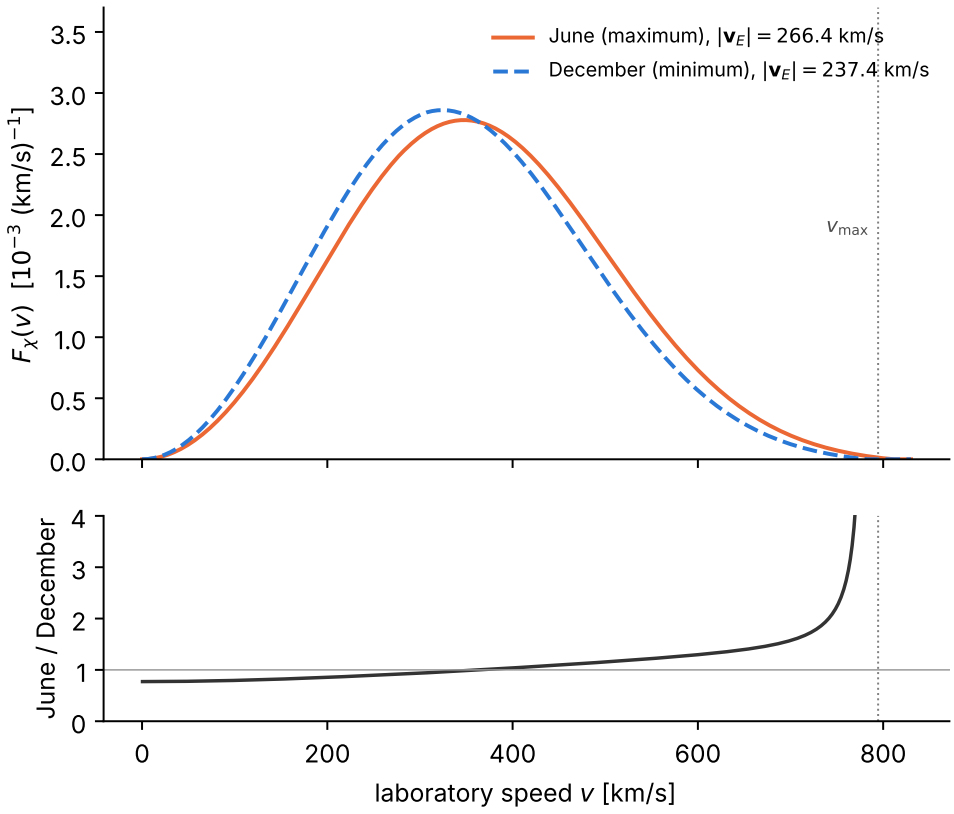


Figure 1.2: Top: laboratory-frame speed distribution $F_\chi(v)$ of the standard halo model with $v_0 = 238\,\mathrm{km\,s^{-1}}$ and $v_{\mathrm{esc}} = 544\,\mathrm{km\,s^{-1}}$, in June (solid, Earth speed at its annual maximum $266.4\,\mathrm{km\,s^{-1}}$) and December (dashed, $237.4\,\mathrm{km\,s^{-1}}$), the days of largest and smallest Earth speed for the Earth velocity of Eq. (1.12) [8, Eq. (11)]. Bottom: ratio of the June to the December distribution. The dotted line marks $v_{\mathrm{max}} = 794.5\,\mathrm{km\,s^{-1}}$; in June the distribution extends to $810\,\mathrm{km\,s^{-1}}$, in December to $781\,\mathrm{km\,s^{-1}}$.

Worked example 1.1: Normalization and mean inverse speed

With $v_{\rm esc}/v_0 = 544/238 = 2.286$, the error function is $\mathrm{erf}(2.286) = 0.99877$ and the second term in the bracket of Eq. (1.4) is $(2/\sqrt{\pi})(2.286)\,\mathrm{e}^{-5.224} = 2.579 \times 0.005383 = 0.01388$. The bracket is 0.98489: the cut removes 1.5 % of the Gaussian integral. With $\pi^{3/2} v_0^3 = 5.5683 \times (238)^3\,(\mathrm{km/s})^3 = 7.507 \times 10^7\,(\mathrm{km/s})^3$,

$$N_{\rm esc} = 0.98489 \times 7.507 \times 10^7\,(\mathrm{km/s})^3 = 7.393 \times 10^7\,(\mathrm{km/s})^3.$$

The mean inverse speed is $\langle 1/v \rangle = \int_0^\infty \mathrm{d}v\, F_\chi(v)/v$. Dividing Eq. (1.10) by v leaves Gaussians in $v \mp v_{\rm E}$ and a constant. After the substitutions $y = v - v_{\rm E}$ in the first exponential and $y = v + v_{\rm E}$ in the second, the two Gaussians cancel on $v_{\rm E} < y < v_{\rm esc}$ and leave $\int_{-v_{\rm E}}^{v_{\rm E}} \mathrm{e}^{-y^2/v_0^2}\mathrm{d}y = \sqrt{\pi}\, v_0\, \mathrm{erf}(v_{\rm E}/v_0)$; the constant term, integrated over an interval of length $2v_{\rm E}$, contributes $-2v_{\rm E}\mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ (Appendix D.1). Hence

$$\left\langle \frac{1}{v} \right\rangle = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left[\sqrt{\pi}\, v_0\, \mathrm{erf}\left(\frac{v_{\rm E}}{v_0} \right) - 2 v_{\rm E}\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right]. \tag{1.11}$$

At $v_{\rm E} = v_{\rm E}^{\rm ref} = 250.5\,\mathrm{km\,s^{-1}}$, $v_{\rm E}/v_0 = 1.0525$ and $\mathrm{erf}(1.0525) = 0.86338$. The bracket is $\sqrt{\pi} \times 238 \times 0.86338 - 2 \times 250.5 \times 0.005383 = 364.21 - 2.70 = 361.51\,\mathrm{km\,s^{-1}}$, and the prefactor is $\pi(238)^2/(250.5 \times 7.393 \times 10^7) = 9.608 \times 10^{-6}\,(\mathrm{km/s})^{-2}$. The result is

$$\left\langle \frac{1}{v} \right\rangle = 3.474 \times 10^{-3}\,(\mathrm{km/s})^{-1} = \frac{1}{287.9\,\mathrm{km\,s^{-1}}}, \quad \text{or } 1041 \text{ in natural units.}$$

Without the cut, Eq. (1.11) reduces to $\mathrm{erf}(v_{\rm E}/v_0)/v_{\rm E} = 1/(290.1\,\mathrm{km\,s^{-1}})$; the cut removes fast particles and raises $\langle 1/v \rangle$ by 0.8 %. In June and December the same formula yields $1/(298.3\,\mathrm{km\,s^{-1}})$ and $1/(279.8\,\mathrm{km\,s^{-1}})$, a 6 % swing.

The mean inverse speed is the value at zero of the function $\eta(v_{\rm min}) = \langle \Theta(v - v_{\rm min})/v \rangle$ that Sect. 1.3 introduces, with $v_{\rm min}$ the minimum velocity of Eq. (1.22); Appendix D.1 gives η for every $v_{\rm min}$.

1.2.3 Annual modulation

The Earth's orbital velocity, in the galactic axes of Eq. (1.5), is [8, Eq. (11)]

$$\boldsymbol{v}_\oplus(t) = 29.8\,\mathrm{km\,s^{-1}} \left[\begin{pmatrix} 0.9941 \\ 0.1088 \\ 0.0042 \end{pmatrix} \cos \omega_{\rm yr} t + \begin{pmatrix} -0.0504 \\ 0.4946 \\ -0.8677 \end{pmatrix} \sin \omega_{\rm yr} t \right], \tag{1.12}$$

with $\omega_{\rm yr} = 0.0172\,\mathrm{d^{-1}}$ the orbital angular frequency and t counted in days from 22 March. The two column vectors span the plane of the orbit and are orthogonal unit vectors to an accuracy of 10^{-4}, and $29.8\,\mathrm{km\,s^{-1}}$ is the mean orbital speed [8, Eq. (13)]. We treat the orbit as a circle. The eccentricity of the orbit shifts the date of the largest Earth speed by 1.2 days and changes the modulation amplitude by 0.4 % [10, Table I]; both effects are too small to matter here.

Since the orbital speed is small next to the solar speed, $29.8/250.5 = 0.12$, we can expand the Earth's speed to first order in $\boldsymbol{v}_\oplus$:

$$|\boldsymbol{v}_{\rm E}(t)| = \left(|\boldsymbol{v}_\odot|^2 + 2\boldsymbol{v}_\odot \cdot \boldsymbol{v}_\oplus + |\boldsymbol{v}_\oplus|^2 \right)^{1/2} \approx v_{\rm E}^{\rm ref} + \hat{\boldsymbol{v}}_\odot \cdot \boldsymbol{v}_\oplus(t),$$

with $\hat{\boldsymbol{v}}_\odot = \boldsymbol{v}_\odot/|\boldsymbol{v}_\odot| = (0.0443, 0.9986, 0.0291)$ from Eq. (1.5). Projected onto $\hat{\boldsymbol{v}}_\odot$, the two vectors of Eq. (1.12) have components 0.1528 and 0.4664, and

$$|\boldsymbol{v}_\mathrm{E}(t)| \approx v_\mathrm{E}^\mathrm{ref} + 14.6\,\mathrm{km\,s^{-1}} \cos\bigl[\omega_\mathrm{yr}(t - t_\mathrm{peak})\bigr], \tag{1.13}$$

with amplitude $29.8 \times (0.1528^2 + 0.4664^2)^{1/2} = 29.8 \times 0.4908 = 14.6\,\mathrm{km\,s^{-1}}$ and phase $\omega_\mathrm{yr} t_\mathrm{peak} = \arctan(0.4664/0.1528) = 71.9^\circ = 1.254\,\mathrm{rad}$, which is $t_\mathrm{peak} = 72.9$ days after 22 March, that is, 2 June. Only half of the orbital speed modulates the Earth's speed because the Sun's velocity makes an angle of $\arccos 0.4908 = 60.6^\circ$ with the plane of the orbit. The second-order term, $(|\boldsymbol{v}_\oplus|^2 - (\hat{\boldsymbol{v}}_\odot \cdot \boldsymbol{v}_\oplus)^2)/(2v_\mathrm{E}^\mathrm{ref})$, adds $1.3\,\mathrm{km\,s^{-1}}$ at both extremes and $1.6\,\mathrm{km\,s^{-1}}$ on average over the year. The exact speed from Eqs. (1.6) and (1.12) is largest on 2 June, $266.4\,\mathrm{km\,s^{-1}}$, smallest on 2 December, $237.4\,\mathrm{km\,s^{-1}}$, and averages $252.1\,\mathrm{km\,s^{-1}}$ over the year.

A rate follows the seasonal change of F_χ, and the sign of its modulation depends on which speeds contribute to the rate. In June the distribution is shifted to higher speeds: it holds more particles above $364\,\mathrm{km\,s^{-1}}$, where the curves of Fig. 1.2 cross, and fewer below. A rate that needs fast particles, such as one near a kinematic threshold, peaks in June. A rate that weights all speeds by $1/v$ peaks in December, since $\langle 1/v \rangle$ is 6 % larger in December than in June (Worked example 1.1). For the integral $\eta(v_\mathrm{min})$ of Sect. 1.3, computed with the formulas of Appendix D.1, the phase of the modulation reverses at $v_\mathrm{min} = 211\,\mathrm{km\,s^{-1}}$. Above this speed the modulation grows from a few percent to +21 % and −16 % about the reference value at $v_\mathrm{min} = 600\,\mathrm{km\,s^{-1}}$, and to +42 % and −28 % at $700\,\mathrm{km\,s^{-1}}$. Averaged over the year, $\langle 1/v \rangle$ lies 0.3 % below its value at $v_\mathrm{E}^\mathrm{ref}$ and agrees to 10^{-4} with its value at the annual-mean speed. In the tail the difference between the yearly average and the reference value grows: at $v_\mathrm{min} = 600$ and $700\,\mathrm{km\,s^{-1}}$ the yearly average of η exceeds its value at $v_\mathrm{E}^\mathrm{ref}$ by 2.6 % and 5.6 %, because η there is a convex function of v_E.

The size of the modulation follows from the closed forms by differentiation. For a rate that depends on time only through v_E, Eq. (1.13) makes the relative modulation about the yearly mean

$$(14.6\,\mathrm{km\,s^{-1}})\,\frac{\partial \ln(\mathrm{rate})}{\partial v_\mathrm{E}}\,\cos\bigl[\omega_\mathrm{yr}(t - t_\mathrm{peak})\bigr],$$

with the derivative taken at $v_\mathrm{E}^\mathrm{ref}$. For a rate proportional to $\langle 1/v \rangle$, write Eq. (1.11) as $\pi v_0^2 \mathcal{I}(v_\mathrm{E})/(v_\mathrm{E} N_\mathrm{esc})$ with $\mathcal{I}(v_\mathrm{E})$ the bracket. Since $\mathrm{d\,erf}(x)/\mathrm{d}x = (2/\sqrt{\pi})\mathrm{e}^{-x^2}$, $\mathrm{d}\mathcal{I}/\mathrm{d}v_\mathrm{E} = 2\mathrm{e}^{-v_\mathrm{E}^2/v_0^2} - 2\mathrm{e}^{-v_\mathrm{esc}^2/v_0^2} = 0.6498$, and

$$\frac{\partial \ln\langle 1/v \rangle}{\partial v_\mathrm{E}} = -\frac{1}{v_\mathrm{E}} + \frac{1}{\mathcal{I}}\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}v_\mathrm{E}} = -\frac{1}{250.5} + \frac{0.6498}{361.51} = -2.195 \times 10^{-3}\,(\mathrm{km/s})^{-1}.$$

The amplitude is $14.6 \times (-2.195 \times 10^{-3}) = -3.2\,\%$: the mean inverse speed is largest in December (Worked example 1.1).

Annual modulation is a signature because each of its features is fixed by the halo and the orbit, not by the detector. The period is one year, the phase is set by the June maximum and December minimum, the phase flips at a definite v_min, and the amplitude grows toward the kinematic edge. A background would have to reproduce all four of these features. The amplitude of the modulation is also more sensitive to the shape of the speed distribution than the time-averaged rate is, and an observed modulation would test the halo model itself [5, Sect. 6.2].

1.2.4 Daily modulation

The Earth also rotates, and a crystal fixed in the laboratory turns with the Earth relative to $\boldsymbol{v}_\mathrm{E}$. The rotation speed is $2\pi \times 6378\,\mathrm{km}/86\,164\,\mathrm{s} = 0.47\,\mathrm{km\,s^{-1}}$ at the equator and less by the

cosine of the latitude elsewhere; it is 3 % of the annual amplitude of Eq. (1.13) and leaves the speed distribution practically unchanged. The rotation does change the direction, in crystal axes, from which the wind arrives.

The rotation is simplest in equatorial coordinates. Their Cartesian axes are fixed relative to the distant stars: the third points along the Earth's rotation axis to the north celestial pole, the first to the vernal equinox in the celestial equator, and the second completes a right-handed set. A direction on the sky is given by its *declination*, the angle above the celestial equator, and its *right ascension*, the angle measured eastward in the equatorial plane from the vernal equinox. Right ascension is often quoted in hours ($24\,\mathrm{h} = 360°$). In equatorial Cartesian axes a direction has the polar angle 90° minus its declination and the azimuth equal to its right ascension. A laboratory turns about the third axis once per sidereal day, the rotation period relative to the stars. The *meridian* of the laboratory is the half-plane that is bounded by the rotation axis and contains the local vertical. The *local sidereal time* of the laboratory is the right ascension of its meridian, and the *hour angle* of a direction is the local sidereal time minus the right ascension of that direction.

The rotation from equatorial to galactic Cartesian axes is fixed by three angles: the equatorial coordinates of the north Galactic pole, right ascension 192.85948° and declination 27.12825°, and the galactic longitude of the north celestial pole, 122.932° [10, App. B, Eq. (B3)]. The rotation matrix built from these angles takes a unit vector with equatorial components $(x_{\mathrm{eq}}, y_{\mathrm{eq}}, z_{\mathrm{eq}})$ to its galactic components,

$$\begin{pmatrix} x_{\mathrm{gal}} \\ y_{\mathrm{gal}} \\ z_{\mathrm{gal}} \end{pmatrix} = \begin{pmatrix} -0.0549 & -0.8734 & -0.4838 \\ 0.4941 & -0.4448 & 0.7470 \\ -0.8677 & -0.1981 & 0.4560 \end{pmatrix} \begin{pmatrix} x_{\mathrm{eq}} \\ y_{\mathrm{eq}} \\ z_{\mathrm{eq}} \end{pmatrix} . \tag{1.14}$$

The rows are the galactic axes in equatorial components; the first, the direction of the Galactic center, has right ascension 266.4° and declination −28.9°. The third column is the north celestial pole in galactic axes. Its scalar product with $\hat{\boldsymbol{v}}_\odot$ is 0.7378: the solar velocity makes the angle

$$\theta_{\mathrm{E}} = \arccos 0.7378 = 42.5°, \tag{1.15}$$

with the rotation axis. The inverse rotation, given by the transpose of the matrix in Eq. (1.14), takes $\hat{\boldsymbol{v}}_\odot$ to the equatorial components $(0.4657, -0.4887, 0.7378)$, that is, right ascension $\arctan(-0.4887/0.4657) + 360° = 313.6° = 20.9\,\mathrm{h}$ and declination $47.5° = 90° - \theta_{\mathrm{E}}$, in the constellation Cygnus. The orbital motion moves this direction during the year, to declination 42.8° in June and 52.6° in December, so the angle to the rotation axis varies between 47.2° and 37.4°. We use $\hat{\boldsymbol{v}}_\odot$ below; a prediction at the level of a few degrees computes the direction of $\boldsymbol{v}_{\mathrm{E}}(t)$ from Eqs. (1.6) and (1.12) on each day, and so combines the annual and daily modulations.

Let $\hat{\boldsymbol{n}}$ be an axis fixed in the laboratory, for example a crystal axis, at polar angle $\theta_{\hat{n}}$ from the north celestial pole. As the Earth turns, $\hat{\boldsymbol{n}}$ keeps its polar angle and its right ascension advances at the sidereal rate $\omega_{\mathrm{sid}} = 2\pi/(23\,\mathrm{h}\,56\,\mathrm{min}\,4\,\mathrm{s})$. In equatorial Cartesian axes the two unit vectors are $\hat{\boldsymbol{v}}_\odot = (\sin\theta_{\mathrm{E}}\cos\alpha_{\mathrm{E}}, \sin\theta_{\mathrm{E}}\sin\alpha_{\mathrm{E}}, \cos\theta_{\mathrm{E}})$, with α_{E} its right ascension, and $\hat{\boldsymbol{n}}(t) = (\sin\theta_{\hat{n}}\cos\alpha_{\hat{n}}(t), \sin\theta_{\hat{n}}\sin\alpha_{\hat{n}}(t), \cos\theta_{\hat{n}})$, with $\alpha_{\hat{n}}(t) = \alpha_{\mathrm{E}} + \omega_{\mathrm{sid}}(t - t_{\mathrm{E}})$ and t_{E} the time at which both lie in one meridian plane. The scalar product of the two unit vectors fixes the angle $\theta_{\mathrm{lab}}(t)$ between $\boldsymbol{v}_{\mathrm{E}}$ and the laboratory axis,

$$\cos\theta_{\mathrm{lab}}(t) = \cos\theta_{\mathrm{E}}\cos\theta_{\hat{n}} + \sin\theta_{\mathrm{E}}\sin\theta_{\hat{n}}\cos\bigl[\omega_{\mathrm{sid}}(t - t_{\mathrm{E}})\bigr]. \tag{1.16}$$

Over one sidereal day θ_{lab} oscillates between $|\theta_{\mathrm{E}} - \theta_{\hat{n}}|$ and $\theta_{\mathrm{E}} + \theta_{\hat{n}}$ (for $\theta_{\mathrm{E}} + \theta_{\hat{n}} \le 180°$). An axis along the rotation axis, $\theta_{\hat{n}} = 0$, keeps the fixed angle θ_{E}; an axis in the equatorial plane swings between 47.5° and 132.5°. For the local vertical at latitude ℓ_{lat} north, $\theta_{\hat{n}} = 90° - \ell_{\mathrm{lat}}$

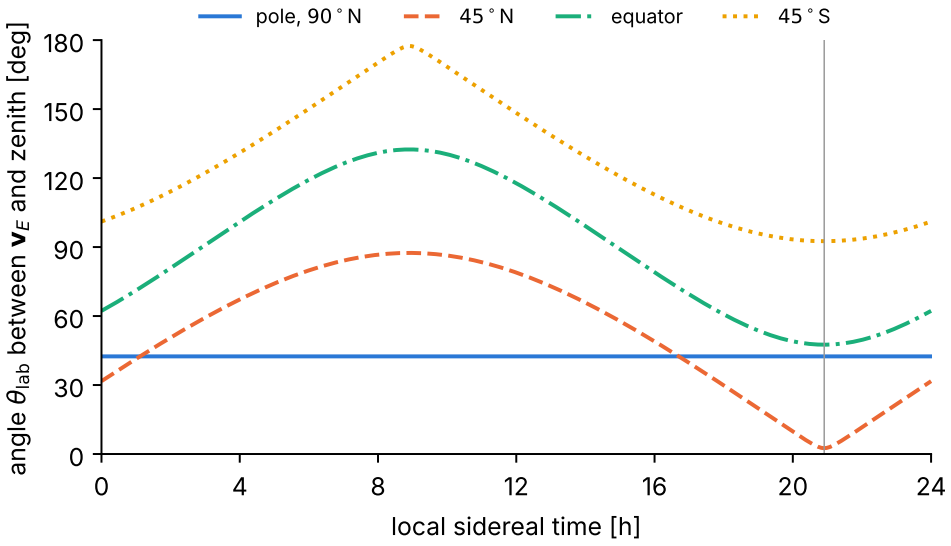


Figure 1.3: Angle $\theta_{\rm lab}$ between the Earth's galactic-frame velocity and the local vertical over one sidereal day, as a function of local sidereal time, at latitudes 90°N, 45°N, 0° and 45°S. The velocity direction is the direction of the solar velocity, right ascension 20.9 h and declination 47.5° (vertical line: its transit), from Ref. [8, Table 1] and the coordinate rotation of Ref. [10, Eq. (B3)]. The dark-matter wind arrives from this direction.

and $\alpha_{\hat{n}}$ is the local sidereal time. Fig. 1.3 plots Eq. (1.16) for the vertical at four latitudes. At 45°N the angle runs from 2.5°, when Cygnus is nearly overhead and the wind comes from above, to 87.5° twelve sidereal hours later.

A crystal is usually mounted with its axes along the local vertical and the local north and east directions. The full direction of $\hat{\boldsymbol{v}}_\odot$ in these axes follows from the same construction. The local vertical at latitude $\ell_{\rm lat}$ is $(\cos\ell_{\rm lat}\cos\alpha_{\rm z}, \cos\ell_{\rm lat}\sin\alpha_{\rm z}, \sin\ell_{\rm lat})$ in equatorial axes, with $\alpha_{\rm z}$ the local sidereal time. The local north is the derivative of this vector with respect to $\ell_{\rm lat}$, $(-\sin\ell_{\rm lat}\cos\alpha_{\rm z}, -\sin\ell_{\rm lat}\sin\alpha_{\rm z}, \cos\ell_{\rm lat})$. The local east is the unit vector $(-\sin\alpha_{\rm z}, \cos\alpha_{\rm z}, 0)$ in the direction of increasing $\alpha_{\rm z}$. With $\delta_{\rm E} = 90° - \theta_{\rm E}$ the declination of $\hat{\boldsymbol{v}}_\odot$ and $h_{\rm E} = \alpha_{\rm z} - \alpha_{\rm E}$ its hour angle, the scalar products of the local vertical, north and east vectors with $\hat{\boldsymbol{v}}_\odot$ are the local components of $\hat{\boldsymbol{v}}_\odot$

$$\begin{aligned} \hat{v}_{\rm east} &= -\cos\delta_{\rm E}\sin h_{\rm E}, \\ \hat{v}_{\rm north} &= \cos\ell_{\rm lat}\sin\delta_{\rm E} - \sin\ell_{\rm lat}\cos\delta_{\rm E}\cos h_{\rm E},, \\ \hat{v}_{\rm up} &= \sin\ell_{\rm lat}\sin\delta_{\rm E} + \cos\ell_{\rm lat}\cos\delta_{\rm E}\cos h_{\rm E} \end{aligned} \tag{1.17}$$

with $h_{\rm E} = \omega_{\rm sid}(t - t_{\rm E})$ advancing at the sidereal rate. The vertical component is Eq. (1.16) for $\theta_{\hat{n}} = 90° - \ell_{\rm lat}$. At 45°N and $h_{\rm E} = 0$ the (east, north, up) components are $(0, 0.04, 1.00)$, a nearly vertical direction; six sidereal hours later, $h_{\rm E} = 90°$, they are $(-0.67, 0.52, 0.52)$, toward the northwest at 31° above the horizon. The wind arrives from the direction $\hat{\boldsymbol{v}}_\odot$; the dark-matter velocities point along $-\hat{\boldsymbol{v}}_\odot$.

For an isotropic target, such as a liquid or a polycrystal, the rate depends on $\boldsymbol{v}_{\rm E}$ only through its magnitude, and the rotation changes nothing. A single crystal is anisotropic: its response to a momentum transfer depends on the direction of $\boldsymbol{q}$ relative to the crystal axes, and slow dark matter transfers momentum preferentially along the wind. As $\theta_{\rm lab}$ sweeps through its daily range, the wind samples different crystal directions and the rate varies with

the sidereal period.

The simplest anisotropy shows how Eq. (1.16) enters a rate. Take a uniaxial crystal, one in which a single axis $\hat{\boldsymbol{n}}$ (the optic axis) differs from the two directions perpendicular to it, and whose response is unchanged under $\hat{\boldsymbol{n}} \to -\hat{\boldsymbol{n}}$. Since the halo of Eq. (1.8) is symmetric under rotations about $\boldsymbol{v}_\mathrm{E}$, after the average over velocities the rate can depend on $\hat{\boldsymbol{n}}$ and $\boldsymbol{v}_\mathrm{E}$ only through the scalar $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{v}}_\mathrm{E} = \cos\theta_\mathrm{lab}$, and evenness in $\hat{\boldsymbol{n}}$ makes the rate a function of $\cos^2\theta_\mathrm{lab}$. To lowest order in the anisotropy the rate is a constant plus a constant times $\cos^2\theta_\mathrm{lab}(t)$; the crystal response fixes both constants. With the optic axis in the equatorial plane, $\theta_{\hat{n}} = 90°$ and Eq. (1.16) reduces to $\cos\theta_\mathrm{lab} = \sin\theta_\mathrm{E}\cos[\omega_\mathrm{sid}(t - t_\mathrm{E})]$; $\cos^2\theta_\mathrm{lab}$ then swings between 0 and $\sin^2\theta_\mathrm{E} = 0.456$ twice each sidereal day. With the optic axis along the Earth's axis the rate is constant.

Polar crystals, whose ions carry opposite charges, show sizable daily modulation of phonon rates when the lattice is strongly anisotropic, as in sapphire, while the cubic crystal GaAs shows almost none. The sidereal period is a distinctive signature: the sidereal day is about four minutes shorter than the solar day, and a background that follows the solar day, or does not vary at all, falls out of phase with the signal over many periods [9, Sect. III C].

1.2.5 Number flux

The number flux of dark matter is the number of particles that cross a unit area per unit time. Each particle is counted across an area perpendicular to its own velocity:

$$\Phi_\chi = n_\chi \langle v \rangle = \frac{\rho_\chi}{m_\chi} \langle v \rangle. \tag{1.18}$$

Multiplied by a velocity-independent cross-section, it is the rate per target particle. A target particle at rest presents a fixed cross-section to each incoming particle, and the particles with velocity in d^3v sweep a volume v per unit time per unit of cross-section; they hit the target particle at the rate $n_\chi f_\chi(\boldsymbol{v})\,\mathrm{d}^3v\,v$ times the cross-section. When the cross-section does not depend on v, summing over velocities replaces v by $\langle v \rangle$, and the rate is Φ_χ times the cross-section. When it does depend on v, as in most of this book, the average must be taken with the cross-section inside, and Φ_χ serves only as a measure of how much dark matter passes through. A detector face of fixed orientation intercepts a different number of particles, and this number depends on the angle between the face and the wind through a geometric factor that Eq. (1.18) avoids. At $v_\mathrm{E} = v_\mathrm{E}^\mathrm{ref}$ a numerical integration of $\langle v \rangle = \int_0^\infty \mathrm{d}v\, v F_\chi(v)$ with Eq. (1.10) gives $\langle v \rangle = 354.1\,\mathrm{km\,s^{-1}}$; without the cut it would be $358.2\,\mathrm{km\,s^{-1}}$.

Worked example 1.2: Number flux of 100 MeV dark matter

At $m_\chi = 100\,\mathrm{MeV}$ the number density is $n_\chi = (0.3\,\mathrm{GeV\,cm^{-3}})/(0.1\,\mathrm{GeV}) = 3\,\mathrm{cm^{-3}}$, so

$$\Phi_\chi = 3\,\mathrm{cm^{-3}} \times 3.541 \times 10^7\,\mathrm{cm\,s^{-1}} = 1.06 \times 10^8\,\mathrm{cm^{-2}\,s^{-1}}.$$

About 10^8 particles cross each square centimeter every second, 3.4×10^{15} in a year. A kilogram of silicon, of density $2.33\,\mathrm{g\,cm^{-3}}$, is a cube of volume $429\,\mathrm{cm^3}$ and side 7.5 cm, and this cube holds about 1300 dark-matter particles of 100 MeV at any instant. The flux scales as $1/m_\chi$: at 1 MeV it is $10^{10}\,\mathrm{cm^{-2}\,s^{-1}}$. These large numbers make feeble couplings testable; the event rate is small because the cross-section is small.

Eq. (1.8) leaves out the gravity of the Sun and of the Earth, which bends and accelerates the particles that approach them. By energy conservation, a particle with speed v far away arrives with speed $(v^2 + u_\mathrm{esc}^2)^{1/2}$, where u_esc is the local escape speed of the attracting body:

$42.1\,\mathrm{km\,s^{-1}}$ from the Sun at the Earth's orbit and $11.2\,\mathrm{km\,s^{-1}}$ from the Earth at its surface. For $v = 354\,\mathrm{km\,s^{-1}}$ the speed increases by $u_{\mathrm{esc}}^2/2v = 2.5\,\mathrm{km\,s^{-1}}$ and $0.18\,\mathrm{km\,s^{-1}}$, below 1 % and comparable to the effect of the orbital eccentricity; the resulting modulation from solar focusing is small [5, Sect. 5]. We neglect these speed shifts.

The standard halo model is a reference, not a measured distribution. Its density is uncertain by roughly a factor of two, its speed parameters by several percent, and stellar surveys reveal kinematic substructure that a single Maxwellian does not describe: debris of galaxies that merged with ours [8, Sect. 3.1]. Near a kinematic threshold the rate probes only the tail of F_χ, where these uncertainties matter most [5, Sect. 6]. The closed form of η in Appendix D.1 makes the sensitivity of η to the halo parameters explicit. Changing v_0 by $\pm 6\,\mathrm{km\,s^{-1}}$ changes $\langle 1/v\rangle$ by ∓ 1 %, but it changes η at $v_{\mathrm{min}} = 600\,\mathrm{km\,s^{-1}}$ by factors 1.11 and 0.89. Changing v_{esc} by $\pm 25\,\mathrm{km\,s^{-1}}$ leaves $\langle 1/v\rangle$ within 0.4 % and changes $\eta(600\,\mathrm{km\,s^{-1}})$ by factors 1.06 and 0.92, but it changes $\eta(760\,\mathrm{km\,s^{-1}})$ by factors 2.26 and 0.10. Close to v_{max}, the escape speed is the dominant uncertainty, larger than the annual modulation. Results in this book use the parameters of Table 1.1.

1.3 Kinematics of slow dark matter

A dark-matter particle that crosses a detector carries at most 3.5 eV of kinetic energy at 1 MeV and 3.5 meV at 1 keV, and about a seventh of this maximum at a typical speed (Sect. 1.1). Energy and momentum conservation decide, before any matrix element is computed, whether it can excite the target and which excitations it can reach. Sign conventions follow Appendix A.4.

1.3.1 Energy transfer and minimum velocity

We start from the elastic collision with a free particle at rest. A dark-matter particle of mass m_χ and laboratory velocity $\boldsymbol{v}$ has momentum $m_\chi \boldsymbol{v}$ and kinetic energy $\frac{1}{2} m_\chi v^2$; at speeds $v \lesssim 3 \times 10^{-3}$ the nonrelativistic forms are exact to a part in 10^5. It scatters on a nucleus of mass M at rest. We write $\boldsymbol{q}$ for the momentum transferred *from* the dark matter *to* the target; the dark matter leaves with $m_\chi \boldsymbol{v} - \boldsymbol{q}$ and the nucleus with $\boldsymbol{q}$. In a crystal we need four terms of lattice theory, which Sect. 3.1 develops. A *reciprocal-lattice vector* $\boldsymbol{G}$ is a wavevector whose plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{x}}$ takes the same value at every lattice site, and a crystal conserves momentum only up to a reciprocal-lattice vector. As a result, a state of an electron or phonon in a crystal is labeled by a *crystal momentum*, defined only up to a reciprocal-lattice vector. The *first Brillouin zone* is the set of wavevectors closer to the origin than to any nonzero $\boldsymbol{G}$, and *folding* a momentum means subtracting the $\boldsymbol{G}$ that brings it into that zone. We never fold $\boldsymbol{q}$: it is the physical momentum lost by the dark matter (the reason is given below Eq. (1.26)). Energy conservation reads

$$\tfrac{1}{2} m_\chi v^2 = \frac{(m_\chi \boldsymbol{v} - \boldsymbol{q})^2}{2 m_\chi} + \frac{q^2}{2M} \quad \Longrightarrow \quad \boldsymbol{q}\cdot\boldsymbol{v} = \frac{q^2}{2\mu_{\chi N}}, \tag{1.19}$$

with $q = |\boldsymbol{q}|$ and $\mu_{\chi N} = m_\chi M/(m_\chi + M)$ the dark-matter–nucleus reduced mass [7, Eqs. (75)–(77)]. We write $\mu_{\chi e}$ and $\mu_{\chi p}$ for the reduced masses with an electron and a proton, defined in the same way. Since $\boldsymbol{q}\cdot\boldsymbol{v} \le qv$, there is a maximum momentum transfer, $q \le 2\mu_{\chi N} v$, and the energy deposited in the nucleus, the *nuclear recoil energy* $E_R = q^2/2M$, obeys [7, Eq. (78)]

$$E_R \le \frac{2\mu_{\chi N}^2 v^2}{M}. \tag{1.20}$$

For $m_\chi \ll M$ the bound is $(4m_\chi/M) \times \frac{1}{2} m_\chi v^2$: a light particle bounces off a heavy nucleus like a ball off a wall and keeps almost all of its energy.

The free particle at rest is an idealization, and in a solid a poor one: electrons are bound, with typical momenta $\sim \alpha m_e$, and nuclei are tied to a lattice. The target can absorb a momentum $\boldsymbol{q}$ and an energy ω that are not related by a free dispersion $q^2/2M$, so we treat the dark-matter side and the target side separately. We write ω for the *energy deposited* in the target. In this book it is defined as the energy given to the target, so it can be negative at $T > 0$. All of the target's physics is then in which pairs $(\boldsymbol{q}, \omega)$ it can absorb, and how often. The dark-matter side alone fixes ω once $\boldsymbol{q}$ and $\boldsymbol{v}$ are given.

To keep the models of Sect. 1.4 in view we allow the scattering to change the dark-matter state, $\chi_1 \to \chi_2$, with a mass splitting $\Delta_\chi = m_{\chi_2} - m_{\chi_1}$. The energy the dark matter loses is its initial kinetic energy minus its final kinetic and extra rest energy:

$$\omega = \tfrac{1}{2} m_\chi v^2 - \frac{(m_\chi \boldsymbol{v} - \boldsymbol{q})^2}{2m_\chi} - \Delta_\chi .$$

We have neglected Δ_χ against m_χ in the final kinetic energy; the relative error is $|\Delta_\chi|/m_\chi$, negligible for splittings small compared with the mass. When the square is expanded, the terms $\frac{1}{2} m_\chi v^2$ cancel, and energy conservation becomes $\omega = \omega_{\boldsymbol{q}}$, with

Energy transfer and minimum velocity

$$\omega_{\boldsymbol{q}} = \boldsymbol{q} \cdot \boldsymbol{v} - \frac{q^2}{2m_\chi} - \Delta_\chi , \tag{1.21}$$

$$v_{\min}(q, \omega) = \left| \frac{\omega + \Delta_\chi}{q} + \frac{q}{2m_\chi} \right| . \tag{1.22}$$

A pair $(\boldsymbol{q}, \omega)$ is open to a particle of speed v if and only if $v \geq v_{\min}(q, \omega)$.

Eq. (1.21) is the form used for electron and phonon targets [1, Eq. (9), 7, Eq. (116), 11, Eq. (3.2)]. The splitting is endothermic for $\Delta_\chi > 0$, when part of the kinetic energy is spent on the heavier final state, and exothermic for $\Delta_\chi < 0$, when the dark matter releases $|\Delta_\chi|$. For elastic scattering $\Delta_\chi = 0$. A free nucleus at rest absorbs only $\omega = q^2/2M$, and Eq. (1.21) with $\Delta_\chi = 0$ then returns Eq. (1.19). Some papers take $\boldsymbol{q}$ as the momentum gained by the dark matter instead. The choice of sign matters for a crystal without inversion symmetry or with magnetic order, where $\boldsymbol{q}$ and $-\boldsymbol{q}$ are physically different (Appendix A.4).

Bound electrons show why the dark-matter side and the target side must be treated separately. Treat an outer electron as free but moving with a typical velocity $\sim \alpha$, about ten times faster than the dark matter. Its momentum is then $\sim \alpha m_e$, and a momentum transfer of that size is what changes its state. The typical momentum transfer is set by the electron, not by the dark matter: $q \approx \mu_{\chi e} \alpha$, which is αm_e for $m_\chi \gg m_e$ [7, Eqs. (117) and (118)]. By Eq. (1.21) the energy deposited is then $\sim qv \approx \alpha m_e v \approx 4\,\mathrm{eV}$ at $v = 10^{-3}$, independent of m_χ, and is not tied to q by any dispersion of the dark matter. For $m_\chi < m_e$ the estimate $q \approx m_\chi \alpha$ fails: with $\Delta_\chi = 0$, Eq. (1.21) would allow at most $qv - q^2/2m_\chi = m_\chi \alpha (v - \alpha/2)$, which is negative because $\alpha/2 = 3.6 \times 10^{-3}$ exceeds every halo speed. The momentum transfer is then bounded by roughly $m_\chi v$, and the dark matter can give up at most its kinetic energy [7, Eq. (119)]. A real target has a spectrum of states rather than a single electron velocity, and Eq. (1.21) holds for all of them: it constrains only the dark-matter side.

The minimum velocity follows from the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$. Let θ be this angle. Eq. (1.21) with $\omega = \omega_{\boldsymbol{q}}$ fixes

$$\cos\theta = \frac{1}{v} \left(\frac{\omega + \Delta_\chi}{q} + \frac{q}{2m_\chi} \right) ,$$

and a direction with this cosine exists only if the right side lies between -1 and 1. This requirement on the right side is the condition $v \geq v_{\min}$ of Eq. (1.22). The absolute value matters only when $\omega + \Delta_\chi < 0$, which happens in two cases: for exothermic dark matter, and for a target at temperature $T > 0$ that gives energy to the dark matter, $\omega < 0$ (Sect. 2.5 sets the thermal weights of transitions with $\omega < 0$). For $\omega + \Delta_\chi < 0$ the expression inside the absolute value can be negative, and reading a negative value as "every particle qualifies" would be wrong: at $\omega = 0$ and small q, Eq. (1.22) demands $v \geq |\Delta_\chi|/q$, which no halo particle meets once $q < |\Delta_\chi|/v_{\max}$. With the absolute value, $v_{\min} \geq 0$ always. At $\omega = -\Delta_\chi = |\Delta_\chi|$ it reduces to $q/2m_\chi$, which vanishes as $q \to 0$: an exothermic particle deposits its splitting at any speed, as in the exothermic models of Sect. 1.4.

Without any knowledge of the target, Eq. (1.21) fixes the largest energy a particle can deposit and the minimum velocity needed to deposit a given energy. First, at fixed v, Eq. (1.21) is largest for $\boldsymbol{q}$ parallel to $\boldsymbol{v}$ and $q = m_\chi v$, where

$$\omega \leq \tfrac{1}{2} m_\chi v^2 - \Delta_\chi. \tag{1.23}$$

For elastic scattering the dark matter can give up its entire kinetic energy, provided the target absorbs the momentum $m_\chi v$ at that energy [7, Eq. (119)]. A free target of mass M cannot, because part of the energy goes into its recoil: the most energy that can go into an internal excitation of a free target is $\frac{1}{2}\mu_{\chi T} v^2$, with $\mu_{\chi T}$ the reduced mass of the dark matter and the target [11, Eq. (3.4)]. A crystal absorbs momentum as a whole, its mass is macroscopic, and the full kinetic energy is available. Second, for elastic scattering at fixed $\omega > 0$, Eq. (1.22) as a function of q is smallest where its derivative $-\omega/q^2 + 1/2m_\chi$ vanishes, at $q = (2m_\chi\omega)^{1/2}$, with the value $v_{\min} = (2\omega/m_\chi)^{1/2}$. A particle of speed v can deposit ω at some q only if $\frac{1}{2} m_\chi v^2 \geq \omega$.

1.3.2 Threshold mass

A detector registers an event only if the deposited energy exceeds a *threshold energy* E_{th}, set by the target (the lowest excitation energy) or by the readout. No particle in the laboratory is faster than $v_{\max} = 794.5\,\text{km}\,\text{s}^{-1} = 2.650 \times 10^{-3}$ (Eq. (1.9)). By Eq. (1.23), an elastic scatter above threshold then requires $\frac{1}{2} m_\chi v_{\max}^2 \geq E_{\text{th}}$, that is $m_\chi \geq m_{\min}$ with the *threshold mass*

$$m_{\min} = \frac{2E_{\text{th}}}{v_{\max}^2}. \tag{1.24}$$

The bound is attained only if the target has an excitation of energy E_{th} at the momentum $q = m_{\min} v_{\max}$. If the lowest excitation energy of the target depends on q, the threshold condition becomes the requirement that $v_{\min}(q, \omega) \leq v_{\max}$ for some q at which the target can absorb $\omega \geq E_{\text{th}}$. Eq. (1.24) with the global minimum of the excitation energy is then a strict lower bound. Sect. 3.3 applies this requirement to silicon, whose lowest electronic excitations lie at a nonzero crystal momentum. For inelastic dark matter E_{th} is replaced by $E_{\text{th}} + \Delta_\chi$ when $E_{\text{th}} + \Delta_\chi$ is positive; if $E_{\text{th}} + \Delta_\chi \leq 0$, kinematics alone sets no lower mass. The two signs of Δ_χ change the spectrum in opposite ways. An endothermic splitting removes Δ_χ from every deposit and raises $m_{\min}$. An exothermic splitting adds $|\Delta_\chi|$: by Eq. (1.21) the deposits then lie between $|\Delta_\chi| - qv - q^2/2m_\chi$ and $|\Delta_\chi| + qv - q^2/2m_\chi$, a band of width $2qv$ about $|\Delta_\chi|$, and for small q the spectrum approaches a line at the splitting.

The scale of $m_{\min}$ follows from $v_{\max}^2 = 7.02 \times 10^{-6}$. A threshold of 1 eV needs $m_\chi \geq 0.28\,\text{MeV}$, and a threshold of 1 meV needs $m_\chi \geq 0.28\,\text{keV}$. Every factor of ten in threshold is a factor of ten in reach. The threshold mass depends on the halo through $v_{\max}^2$ alone. Shifting the escape speed by its uncertainty of $25\,\text{km}\,\text{s}^{-1}$ (Sect. 1.2) changes $v_{\max}$ by 3.1 % and $m_{\min}$

by 6.3 %. Over the year the largest speed moves between 781 and 810 km s^{-1} (Eq. (1.9)), and the threshold mass computed with that speed moves by −3.9 % in June and 3.4 % in December about its reference value.

Worked example 1.3: Threshold masses for an electronic and a phonon threshold

The band gap of silicon is the lowest energy that frees an electron from the occupied states, and it depends on temperature: it is 1.12 eV at 300 K [12, Table I] and 1.1692 eV near zero temperature [12, Eq. (11)], where detectors run. With the low-temperature gap, $E_g = 1.17\,\text{eV}$, as the threshold, Eq. (1.24) gives

$$m_{\text{min}} = \frac{2 \times 1.17\,\text{eV}}{(2.650 \times 10^{-3})^2} = \frac{2.34\,\text{eV}}{7.023 \times 10^{-6}} = 3.33 \times 10^5\,\text{eV} = 0.333\,\text{MeV}.$$

The optimal momentum transfer is $q = m_{\text{min}} v_{\text{max}} = 3.33 \times 10^5\,\text{eV} \times 2.650 \times 10^{-3} = 0.88\,\text{keV}$. Sect. 3.3 computes the gap. For a phonon threshold of 30 meV,

$$m_{\text{min}} = \frac{2 \times 0.030\,\text{eV}}{7.023 \times 10^{-6}} = 8.5 \times 10^3\,\text{eV} = 8.5\,\text{keV},$$

at $q = 8.5 \times 10^3\,\text{eV} \times 2.650 \times 10^{-3} = 23\,\text{eV}$. Lowering the threshold from the electronic to the phonon scale lowers the reach by a factor of $1.17/0.030 = 39$. Both threshold masses are limits set by kinematics alone: at $m_\chi = m_{\text{min}}$ only particles moving at exactly v_{max} in exactly the right direction contribute, and the rate vanishes (Sect. 1.3.5).

1.3.3 Momentum and energy scales

Eq. (1.21) defines, for each mass, the region of the (q, ω) plane that the halo can reach. With $\Delta_\chi = 0$ and the fastest particles moving along $\boldsymbol{q}$,

$$\omega \leq q v_{\text{max}} - \frac{q^2}{2 m_\chi}. \tag{1.25}$$

The boundary is a parabola that leaves the origin with slope v_{max}, peaks at $\omega = \frac{1}{2} m_\chi v_{\text{max}}^2$ at $q = m_\chi v_{\text{max}}$, and returns to zero at $q = 2 m_\chi v_{\text{max}}$, the maximum momentum transfer. Whatever the mass, every allowed point lies below the line $\omega = q v_{\text{max}}$, and depositing ω needs $q \geq \omega / v_{\text{max}}$ [1, Eq. (122)]; no allowed point lies beyond $q = 2 m_\chi v_{\text{max}}$. Fig. 1.4 draws the region for four masses, after Ref. [1, Fig. 1], and Table 1.2 lists the corresponding extremes. The energy deposited is small compared with the momentum transfer, $\omega \leq q v_{\text{max}} < 3 \times 10^{-3} q$. As a result, scattering of halo dark matter is always far from the light cone $\omega = q$, and for this reason the static potential of Sect. 1.4 suffices.

Fig. 1.4 answers at a glance which excitations each mass can reach. At 0.1 MeV the largest deposit, 0.35 eV, is below the silicon gap: only phonons, and other excitations below an eV, are open. At 1 MeV the gap is open in a narrow band of momentum transfers, from $q = 0.49\,\text{keV}$ to 4.8 keV, the two roots of Eq. (1.25) at $\omega = 1.17\,\text{eV}$. At 10 MeV the band widens to 0.45–53 keV, and deposits up to 35 eV are allowed. A free silicon nucleus, by contrast, receives 1.17 eV only for $m_\chi \gtrsim 47\,\text{MeV}$, where $2\mu_{\chi N}^2 v_{\text{max}}^2 / M$ reaches that value.

What these momentum transfers probe is decided by two momentum scales of the target, one for bound electrons and one for the lattice. The first is the *electron momentum scale*

$$q_0 = \alpha m_e = 3.73\,\text{keV}, \tag{1.26}$$

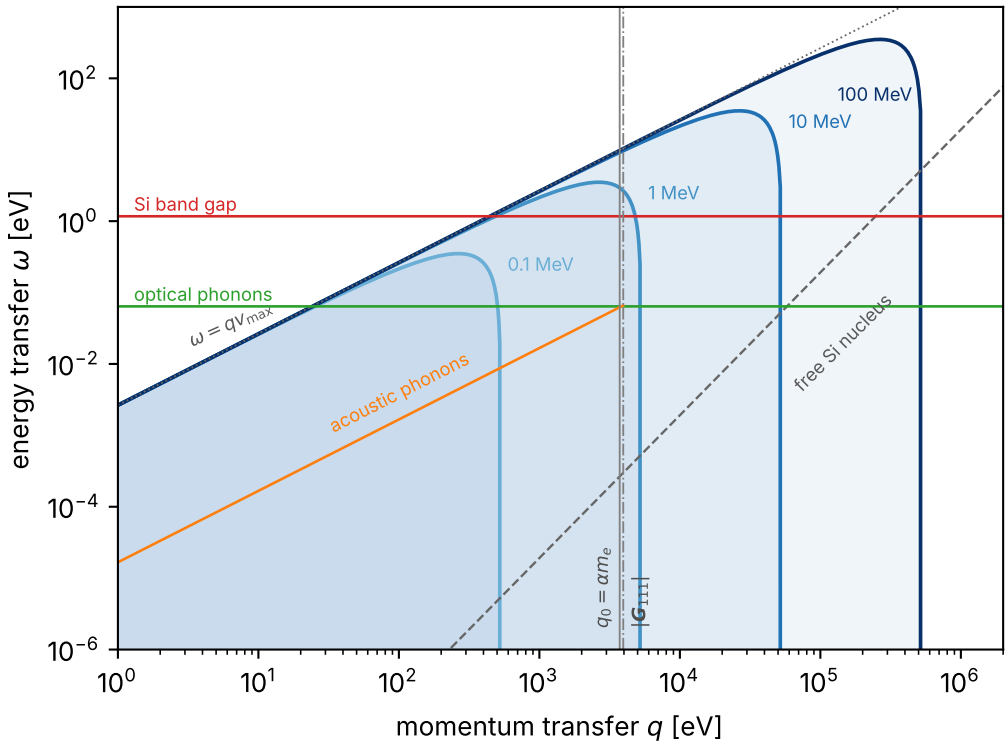


Figure 1.4: Momentum transfer q and energy transfer ω accessible to elastic dark matter of mass 0.1, 1, 10 and 100 MeV (blue curves, light to dark; shaded below), from $\omega \leq qv_{\max} - q^2/2m_\chi$ with $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. The dotted line $\omega = qv_{\max}$ bounds all masses. Overlaid for silicon are the low-temperature band gap, 1.17 eV (red), and the optical-phonon energy at the zone center, 64 meV (green). The orange line is an acoustic branch $\omega =$ sound speed $\times\, q$ for a sound speed of $5\,\mathrm{km\,s^{-1}}$, drawn up to $q = |\boldsymbol{G}_{111}|$, and the dashed curve is the recoil $q^2/2M$ of a free silicon nucleus, with $M = 26.16\,\mathrm{GeV}$ the mean atomic mass of natural silicon. Vertical lines: the electron momentum scale $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$ and the length $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a \approx 4.0\,\mathrm{keV}$ of the shortest reciprocal-lattice vector of silicon, whose lattice constant is $a = 5.431\,\text{Å}$. Scales from Refs. [1, Eqs. (50) and (121), Sect. IV A, 12, Eq. (11), 13, lattice parameter of silicon].

Table 1.2: Kinematic extremes for elastic dark matter at $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. Columns: the dark-matter mass; the largest momentum transfer to a target that absorbs any momentum; the largest energy transfer; the largest recoil energy of a free silicon nucleus ($M = 26.16\,\mathrm{GeV}$, the mean atomic mass of natural silicon). The $\frac{1}{2}m_\chi v_{\max}^2$ column is also the threshold energy for which $m_{\min} = m_\chi$.

m_χ	$2m_\chi v_{\max}$	$\frac{1}{2}m_\chi v_{\max}^2$	free Si nucleus
0.1 MeV	0.53 keV	0.35 eV	5.4 µeV
1 MeV	5.3 keV	3.5 eV	0.54 meV
10 MeV	53 keV	35 eV	54 meV
100 MeV	530 keV	351 eV	5.3 eV
1 GeV	5.3 MeV	3.5 keV	0.50 keV

the inverse Bohr radius [1, Eq. (121)]. A bound outer electron has a wavefunction of size $1/(\alpha m_e)$ and typical momenta $\sim q_0$ [7, Sect. V A]. A momentum transfer much smaller than q_0 barely changes the electron's state; a much larger one reaches only the high-momentum tail of its wavefunction. The second is the *interatomic scale*, the length of the shortest reciprocal-lattice vector. For a cubic crystal with lattice constant a the natural unit is $2\pi/a$. Silicon has $a = 5.431\,\text{Å}$ [13, lattice parameter of silicon], about ten Bohr radii [1, Sect. V], and $2\pi/a = 2.28\,\text{keV} = 0.61\,q_0$. Its lattice is face-centered cubic, and the shortest reciprocal-lattice vectors, along the cube diagonals, are longer by $\sqrt{3}$: $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\text{keV} \approx 1.06\,q_0$. A momentum transfer well below $|\boldsymbol{G}_{111}|$ has a wavelength longer than the interatomic spacing and sees the crystal as a whole; a larger one resolves the atoms. Sect. 3.1 makes the comparison with $|\boldsymbol{G}_{111}|$ precise with the reciprocal lattice.

The two scales are close to each other, and light dark matter straddles both. For an electronic excitation of a few eV, $q \geq \omega/v_{\text{max}}$ requires $q \gtrsim 1\,\text{keV}$, roughly $0.3\,q_0$; the silicon gap alone needs $q \geq 1.17\,\text{eV}/2.650 \times 10^{-3} = 0.44\,\text{keV} = 0.12\,q_0$ [1, Eq. (123)]. At the upper end, interpolation in Table 1.2 gives $2m_\chi v_{\text{max}} = q_0$ at $m_\chi = 0.70\,\text{MeV}$. Between about 1 MeV and 1 GeV, then, the momentum transfers match the momenta at which electrons in atoms and solids have most of their weight. Lighter dark matter cannot reach q_0 and probes the long-wavelength response of the crystal. Heavier dark matter can reach far above q_0, but momentum transfers of that size find little electronic weight.

The comparison with $|\boldsymbol{G}_{111}|$ explains why we never fold $\boldsymbol{q}$. Above about 0.75 MeV, $2m_\chi v_{\text{max}}$ exceeds $|\boldsymbol{G}_{111}|$. Because the crystal conserves momentum only up to a reciprocal-lattice vector, the same excitation can be reached by momentum transfers that differ by a reciprocal-lattice vector, and these have different magnitudes. The dark-matter side, Eqs. (1.21) and (1.22), depends on the magnitude of the physical momentum transfer $\boldsymbol{q}$, not on its image in the first Brillouin zone. Folding $\boldsymbol{q}$ would assign the wrong v_{min} and the wrong dark-matter form factor $F_{\text{DM}}(q)$ of Sect. 1.4. Sect. 3.1 writes $\boldsymbol{q}$ as a first-zone part plus a reciprocal-lattice vector and keeps both.

On the energy side, electronic excitations and phonons set the scales. *Electronic excitations* move an electron from an occupied to an empty state and typically cost an eV or more. In an insulator or semiconductor the lowest energy of an electronic excitation is the *band gap* E_{g} (Worked example 1.3; Sect. 3.3). *Phonons* are the quantized vibrations of the lattice, with energies from zero to roughly 100 meV [1, Sect. III A]; the optical phonon of silicon at the zone center has 64 meV, or 15.5 THz [14] (Sect. 5.3). Two kinds, optical and acoustic, matter here. Optical phonons, in which neighboring atoms move against each other, keep a finite energy as their momentum goes to zero. Acoustic phonons, long-wavelength sound waves, have energy equal to the sound speed times the momentum, with sound speeds of 3 to $10\,\text{km}\,\text{s}^{-1}$ [1, Eq. (50)], that is 1 to 3×10^{-5}.

The kinematics treats optical and acoustic phonons differently. A nearly flat optical phonon branch of energy ω can be excited at any q between ω/v_{max} and the upper root of Eq. (1.25), close to $2m_\chi v_{\text{max}}$. A flat optical branch can absorb nearly the full kinetic energy, and Eq. (1.24) applies; the 30 meV threshold of Worked example 1.3 needs a momentum transfer of just $q \approx 23\,\text{eV}$, below a hundredth of $|\boldsymbol{G}_{111}|$. An acoustic phonon of branch ν has $\omega = c_\nu q$, with c_ν its sound speed (Sect. 5.2). By Eq. (1.22) it needs $v_{\text{min}} = c_\nu + q/2m_\chi \leq v$, and this condition bounds $q \leq 2m_\chi(v - c_\nu)$ and $\omega \leq 2c_\nu m_\chi v$ to leading order in c_ν/v [1, Eq. (93)]. The acoustic branch takes at most a fraction $4c_\nu/v$, a few percent, of the kinetic energy. In Fig. 1.4 it is the orange line, which for the lightest mass leaves the parabola near its right end.

The band gap and the phonon energies are material properties, and later chapters use computed values for them. The band gap is computed from first principles with density-functional theory and then corrected for the systematic underestimate of gaps that the method

makes (Sects. 23.3 and 23.4); phonon energies come from density-functional perturbation theory (Sect. 25.2). Measured values serve as checks.

Nuclear recoils. Eq. (1.20) shows why scattering on free nuclei loses its reach for light dark matter. The free nucleus is the dashed parabola $\omega = q^2/2M$ of Fig. 1.4. It meets each dark-matter region only where $q^2/2M$ is small, because M is large; for $m_\chi \ll M$ the nucleus receives at most the fraction $4m_\chi/M$ of the kinetic energy, 1.5×10^{-2} at 100 MeV on silicon. The same parabola explains when a nucleus in a crystal is not free. A recoil energy comparable to the phonon energies, tens of meV, is reached at $m_\chi \approx 10\,\text{MeV}$ (Table 1.2). Below that mass the nucleus cannot be treated as free: the dark matter excites one or a few phonons of the crystal instead, with the energies set by the lattice, and Chap. 5 develops this phonon description.

For a free nucleus, Eq. (1.22) at $\omega = E_R = q^2/2M$ gives $v_\text{min} = q/2\mu_{\chi N} = (ME_R/2\mu_{\chi N}^2)^{1/2}$. The crossover to phonons is already visible for a single nucleus held in a harmonic well of frequency ω_0. At zero temperature a transfer q excites n quanta with the Poisson probability $P_n = \mathrm{e}^{-E_R/\omega_0}(E_R/\omega_0)^n/n!$, where $E_R = q^2/2M$ is the free recoil energy [1, Eqs. (90) and (91)]. The energy deposit $n\omega_0$ therefore has mean exactly E_R and rms spread $(E_R\omega_0)^{1/2}$. The free formula is the limit $E_R \gg \omega_0$, in which the spread is small compared with E_R.

Limit Check 1.1. Elastic nuclear recoil

A free nucleus of mass M at rest that receives momentum q absorbs the energy $\omega = E_R = q^2/2M$. A dark-matter particle of speed v gives it at most $E_R \le 2\mu_{\chi N}^2 v^2/M$, with $\mu_{\chi N}$ the dark-matter–nucleus reduced mass, and a recoil E_R needs the speed $v \ge v_\text{min} = (ME_R/2\mu_{\chi N}^2)^{1/2}$ [7, Eqs. (77) and (78)].

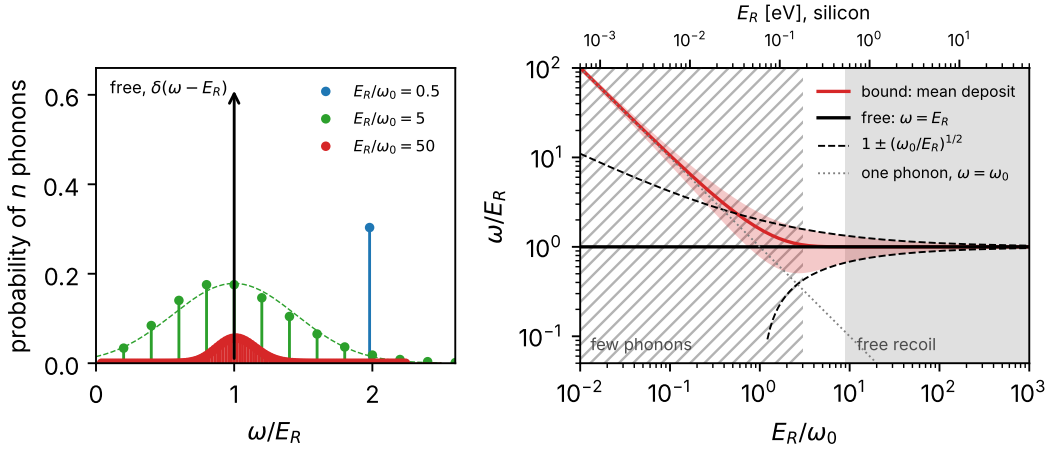


Figure 1.5: Free nuclear recoil against a nucleus bound in a harmonic (Einstein) potential of frequency $\omega_0 = 60\,\text{meV}$, within the range of the optical phonons of silicon at the zone boundary, 51 to 61 meV (64 meV at the zone center), at zero temperature [1, Sect. IV A, Eqs. (80), (90) and (91)]. Left: probability of exciting n phonons, a deposit $\omega = n\omega_0$, for $E_R/\omega_0 = 0.5$, 5 and 50 (dots; the term $n = 0$, which deposits nothing, is omitted), with Gaussians of rms width $(E_R\omega_0)^{1/2}$ (dashed) and the free response $\delta(\omega - E_R)$ (arrow). Right: mean deposit of the scatterings that excite the lattice (red line) and its rms spread (red band), against E_R/ω_0, with the free value $\omega = E_R$ (black), the estimate $1 \pm (\omega_0/E_R)^{1/2}$ (dashed) and one phonon, $\omega = \omega_0$ (dotted); the top axis gives E_R for silicon. Gray: $E_R \ge 9\omega_0 = 0.54\,\text{eV}$, where the rms width is below a third of E_R and

the free formula holds. Hatched: $E_R \leq 3\omega_0$, the single- and few-phonon regime.

Below a few ω_0 the deposit comes in whole phonons and spreads as widely as E_R itself; below ω_0 most scatterings leave the lattice unexcited, and the remaining scatterings deposit at least one phonon, ω_0, well above E_R.

Worked example 1.4: Largest recoil energy of a silicon nucleus

For $m_\chi = 1\,\mathrm{GeV}$ on silicon, $M = 26.16\,\mathrm{GeV}$ (the mean atomic mass of natural silicon) and $\mu_{\chi N} = (1 \times 26.16/27.16)\,\mathrm{GeV} = 0.963\,\mathrm{GeV}$. The maximum momentum transfer at v_max is $q = 2\mu_{\chi N} v_\mathrm{max} = 2 \times 0.963\,\mathrm{GeV} \times 2.650 \times 10^{-3} = 5.10\,\mathrm{MeV}$, and Eq. (1.20) gives

$$E_R = \frac{q^2}{2M} = \frac{(5.10\,\mathrm{MeV})^2}{2 \times 26\,160\,\mathrm{MeV}} = 5.0 \times 10^{-4}\,\mathrm{MeV} = 0.50\,\mathrm{keV}.$$

The kinetic energy at v_max is $\frac{1}{2} m_\chi v_\mathrm{max}^2 = 3.5\,\mathrm{keV}$, so the nucleus receives at most 14 % of it; the exact fraction is $4\mu_{\chi N}^2/(m_\chi M) = 0.142$. A typical particle, at $v \approx 300\,\mathrm{km\,s^{-1}}$ and with an average over angles, deposits far less. On xenon, $M = 122.3\,\mathrm{GeV}$, the same formula gives 0.11 keV, the value quoted in Sect. 1.1.

1.3.4 The speed integral

A rate sums the probability of each momentum transfer over the dark-matter velocities. Energy conservation enters it as a delta function $\delta(\omega - \omega_{\boldsymbol{q}})$, integrated against $f_\chi(\boldsymbol{v})$ (Sect. 2.2 derives the rate). The integral over $\boldsymbol{v}$ can be done once for all targets whose response depends on the magnitude q but not on the direction of $\boldsymbol{q}$, as for a liquid, an atom, or a crystal averaged over orientations. For these targets only the angle θ between $\boldsymbol{q}$ and $\boldsymbol{v}$ matters. The average of the delta function over this angle is

$$\frac{1}{2}\int_{-1}^{1} \mathrm{d}(\cos\theta)\, \delta\!\left(\omega + \Delta_\chi + \frac{q^2}{2m_\chi} - qv\cos\theta\right) = \frac{1}{2qv}\,\Theta(v - v_\mathrm{min}(q,\omega))\,.$$

The delta function fires once if its root $\cos\theta = (\omega + \Delta_\chi + q^2/2m_\chi)/qv$ lies in $[-1, 1]$, and this requirement is the condition $v \geq v_\mathrm{min}$. The argument of the delta function has the derivative $-qv$ with respect to $\cos\theta$, and the factor $1/qv$ is one over the modulus of this derivative (Appendix C.1). This factor is the origin of the factor $1/v$ promised in Sect. 1.2. Averaging over the halo, we find $\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\Theta(v - v_\mathrm{min})/(2qv) = \eta(v_\mathrm{min})/(2q)$, with

$$\eta(v_\mathrm{min}) \equiv \int \mathrm{d}^3v\, \frac{f_\chi(\boldsymbol{v})}{v}\,\Theta(v - v_\mathrm{min}) = \left\langle \frac{\Theta(v - v_\mathrm{min})}{v} \right\rangle. \tag{1.27}$$

The function $\eta(v_\mathrm{min})$ is the *mean inverse speed above* v_min. It carries all of the halo dependence of a rate for a target whose response depends only on q [15, below Eq. (5), 1, Eqs. (11) and (12), 11, Eqs. (A.12)–(A.15)]. Its value at zero is the mean inverse speed $\langle 1/v \rangle$ of Eq. (1.11), and it decreases monotonically to zero at the largest speed in the halo. Because $v_\mathrm{min} \geq 0$ by Eq. (1.22), η is never needed at negative arguments.

Eq. (1.27) fixes how the halo enters the integral over momentum transfers. For an integrand that depends only on q, $\mathrm{d}^3q/(2\pi)^3 = 4\pi q^2 \mathrm{d}q/(2\pi)^3$, and with the factor $\eta/2q$ the measure becomes [1, Eq. (11), 11, Eq. (A.14)]

$$\int \frac{\mathrm{d}^3q}{(2\pi)^3}\,\frac{\eta(v_\mathrm{min})}{2q}\,(\cdots) = \int \frac{q\,\mathrm{d}q}{4\pi^2}\,\eta(v_\mathrm{min}(q,\omega))\,(\cdots).$$

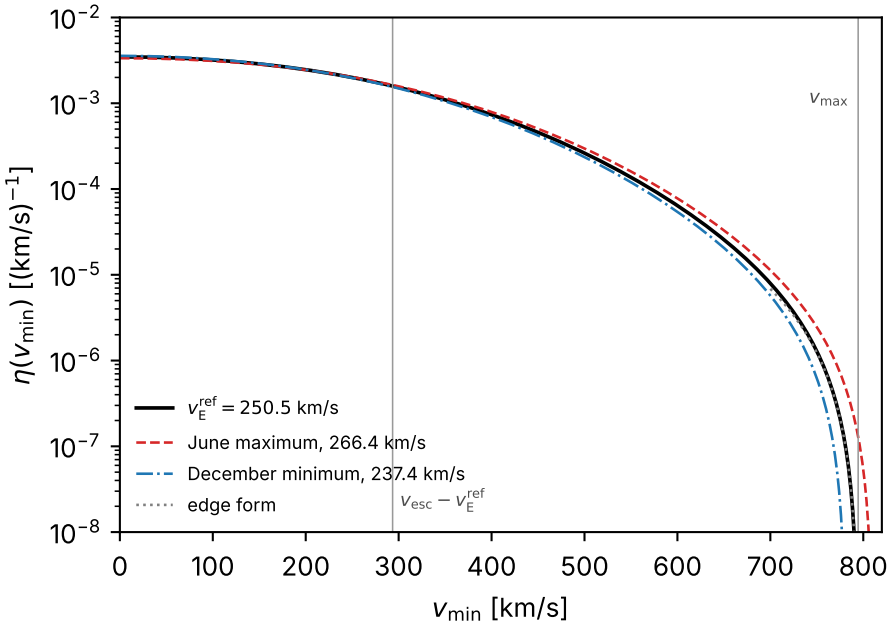


Figure 1.6: Mean inverse speed above $v_{\rm min}$, $\eta(v_{\rm min}) = \langle \Theta(v - v_{\rm min})/v \rangle$, in the standard halo model ($v_0 = 238\,\mathrm{km\,s^{-1}}$, $v_{\rm esc} = 544\,\mathrm{km\,s^{-1}}$). Solid: Earth speed $v_{\rm E}^{\rm ref} = 250.5\,\mathrm{km\,s^{-1}}$; dashed: June maximum ($266.4\,\mathrm{km\,s^{-1}}$); dash-dotted: December minimum ($237.4\,\mathrm{km\,s^{-1}}$). Vertical lines: $v_{\rm esc} - v_{\rm E}^{\rm ref} = 293.5\,\mathrm{km\,s^{-1}}$, where the closed form changes branch, and $v_{\rm max} = 794.5\,\mathrm{km\,s^{-1}}$, where η vanishes. The dotted curve above $700\,\mathrm{km\,s^{-1}}$ is the edge form of Eq. (1.28) with its first correction; it lies within 10 % of the exact curve above $720\,\mathrm{km\,s^{-1}}$, within 5 % above $745\,\mathrm{km\,s^{-1}}$, and merges with it toward $v_{\rm max}$. At $v_{\rm min} = 0$, $\eta = 1/(287.9\,\mathrm{km\,s^{-1}})$, or 1041 in natural units. Closed form from Appendix D.1; halo parameters from Ref. [8, Table 1].

Each momentum transfer enters with the weight $q\,\mathrm{d}q$, and η cuts the integral off wherever $v_{\rm min}(q, \omega) > v_{\rm max}$, at both small and large q. Sect. 1.5 assembles the full rate from this measure, the cross-section of Sect. 1.4 and the response of the target. For an anisotropic crystal at a fixed orientation the angular average does not factor out; the rate then depends on the direction of $\boldsymbol{v}_{\rm E}$ relative to the crystal axes, and this dependence is the daily modulation of Sect. 1.2.

In the standard halo model the integral is elementary. With $F_\chi(v)$ the speed distribution of Eq. (1.10), $\eta(v_{\rm min}) = \int_{v_{\rm min}}^{\infty} \mathrm{d}v\, F_\chi(v)/v$, and the factor v in F_χ cancels the $1/v$. Appendix D.1 integrates the Gaussians and finds three branches in closed form: Eq. (D.2) for $0 \le v_{\rm min} < v_{\rm esc} - v_{\rm E}$, Eq. (D.3) for $v_{\rm esc} - v_{\rm E} \le v_{\rm min} < v_{\rm esc} + v_{\rm E}$, and $\eta = 0$ above. The first two branches are differences of error functions (Appendix C.8) minus a constant from the cut, with $v_{\rm E} = |\boldsymbol{v}_{\rm E}(t)|$.

Fig. 1.6 shows η at the reference Earth speed and on the days of largest and smallest Earth speed. Up to $v_{\rm min} \approx 200\,\mathrm{km\,s^{-1}}$, η stays within a factor of 1.41 of $\langle 1/v \rangle = 1/(287.9\,\mathrm{km\,s^{-1}})$: almost every particle qualifies. Between 300 and $700\,\mathrm{km\,s^{-1}}$ it falls roughly as a Gaussian, $\mathrm{e}^{-(v_{\rm min} - v_{\rm E})^2/v_0^2}$, by a factor of 190, from $1.53 \times 10^{-3}\,(\mathrm{km/s})^{-1}$ to $7.95 \times 10^{-6}\,(\mathrm{km/s})^{-1}$. Above $v_{\rm esc} - v_{\rm E}^{\rm ref} = 293.5\,\mathrm{km\,s^{-1}}$ the second branch applies; every $v_{\rm min}$ near a kinematic threshold lies on it. In natural units, $\eta(0) = 1041$, $\eta(600\,\mathrm{km\,s^{-1}}) = 19.2$ and $\eta(700\,\mathrm{km\,s^{-1}}) = 2.38$.

Near the end point $v_{\rm esc} + v_{\rm E}$, the function η changes with the season mainly because the end point itself moves. Relative to the reference curve, η in June and December is 1.42 and 0.72 times as large at $v_{\rm min} = 700\,\mathrm{km\,s^{-1}}$, 2.2 and 0.37 times at $760\,\mathrm{km\,s^{-1}}$, and 4.5 and 0.009 times at $780\,\mathrm{km\,s^{-1}}$, where the December halo has almost no particles left.

1.3.5 The kinematic edge

The *kinematic edge* of a rate is the boundary beyond which no particle in the halo is fast enough. At fixed m_χ the spectrum ends at $\omega = \frac{1}{2} m_\chi v_{\max}^2 - \Delta_\chi$, by Eq. (1.23). At a fixed threshold the rate, as a function of mass, ends at $m_{\min}$. The edge in energy and the edge in mass are both reached as $v_{\min} \to v_{\max}$, and the shape of a rate near an edge is controlled by the shape of η there.

Close to $v_{\max}$, η vanishes as a power. Write $v_{\min} = v_{\rm esc} + v_{\rm E} - \delta$ with small $\delta > 0$. On the second branch, Eq. (D.3), the difference of error functions is an integral, $(\sqrt{\pi} v_0/2)[\mathrm{erf}(v_{\rm esc}/v_0) - \mathrm{erf}((v_{\rm esc} - \delta)/v_0)] = \int_{v_{\rm esc}-\delta}^{v_{\rm esc}} \mathrm{e}^{-y^2/v_0^2} \mathrm{d}y$, and the constant term is $\delta\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} = \int_{v_{\rm esc}-\delta}^{v_{\rm esc}} \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \mathrm{d}y$. With $s = v_{\rm esc} - y$, the integrand of their difference is

$$\mathrm{e}^{-v_{\rm esc}^2/v_0^2} \left[\mathrm{e}^{(2 v_{\rm esc} s - s^2)/v_0^2} - 1 \right] = \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \left[\frac{2 v_{\rm esc} s}{v_0^2} + \left(\frac{2 v_{\rm esc}^2}{v_0^4} - \frac{1}{v_0^2} \right) s^2 + O(s^3) \right].$$

Integrating over $0 < s < \delta$ and multiplying by the prefactor $\pi v_0^2/(v_{\rm E} N_{\rm esc})$ gives

$$\eta(v_{\min}) = \frac{\pi v_{\rm esc}\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2}}{v_{\rm E} N_{\rm esc}}\, \delta^2 \left[1 + \frac{(2 v_{\rm esc}^2 - v_0^2)\, \delta}{3 v_{\rm esc} v_0^2} + O(\delta^2) \right], \qquad \delta = v_{\rm esc} + v_{\rm E} - v_{\min}. \quad (1.28)$$

The first correction in the bracket of Eq. (1.28), linear in δ, has the velocity scale $3 v_{\rm esc} v_0^2/(2 v_{\rm esc}^2 - v_0^2) = 173\,\mathrm{km\,s^{-1}}$. At the reference Earth speed and $v_{\min} = 760$ and $780\,\mathrm{km\,s^{-1}}$ ($\delta = 34.5$ and $14.5\,\mathrm{km\,s^{-1}}$), the quadratic term alone captures 0.81 and 0.92 of the exact value, and with the first correction 0.98 and 1.00. The edge is quadratic because two things vanish together: the speed window $v_{\max} - v_{\min}$, and the density of particles inside it, which drops linearly to zero at the escape speed of the truncated Maxwellian. The prefactor $\mathrm{e}^{-v_{\rm esc}^2/v_0^2}/N_{\rm esc}$ is the height of the distribution at the cut. The prefactor and the quadratic power of δ are both fixed by the most uncertain parts of the halo model (Sect. 1.2); a smoother truncation would raise the power of δ.

The edge in mass follows. For a threshold $E_{\rm th}$ the smallest $v_{\min}$ available to a particle of mass m_χ is $(2E_{\rm th}/m_\chi)^{1/2}$ (below Eq. (1.23)). Its distance from $v_{\max}$ is

$$v_{\max} - \left(\frac{2E_{\rm th}}{m_\chi} \right)^{1/2} = v_{\max} \left[1 - \left(\frac{m_{\min}}{m_\chi} \right)^{1/2} \right] \simeq \frac{v_{\max}}{2}\, \frac{m_\chi - m_{\min}}{m_{\min}},$$

so a mass one percent above $m_{\min}$ uses only particles within $4\,\mathrm{km\,s^{-1}}$ of $v_{\max}$. Just above the edge the rate is the product of two factors: η, which is quadratic in the speed window $v_{\max} - v_{\min}$, and the phase space of the pairs (q, ω) that fit inside this window. The phase space shrinks as a further power of the window, and how fast it shrinks depends on how the target's response begins at threshold. For silicon the electronic response begins at a crystal momentum away from the zone center, and the question becomes which excitations the dark matter can reach at the edge (Sect. 3.3).

The recovery above the edge is slow. Take the silicon gap of Worked example 1.3 as threshold, $m_{\min} = 0.333\,\mathrm{MeV}$, and evaluate η at the most favorable speed $(2E_{\rm g}/m_\chi)^{1/2}$, an upper bound on what the halo offers at each mass. Relative to $\eta(0)$, this value of η is 6.0×10^{-5} at 0.35 MeV, 1.1×10^{-3} at 0.4 MeV, 0.76 % at 0.5 MeV, 12 % at 1 MeV and 84 % at 10 MeV. Between 0.35 MeV, 5 % above $m_{\min}$, and $2m_{\min}$ the halo supply grows by a factor of more than 500. Over this whole mass range the halo supply comes from particles faster than about $560\,\mathrm{km\,s^{-1}}$, the most favorable speed at $2m_{\min}$; these particles are 7 % of the halo. A limit in this mass range is a statement about the tail of the velocity distribution as much as about the target.

The position of the edge, by contrast, depends on the target and the halo only through $E_{\mathrm{th}}/v_{\mathrm{max}}^2$. The uncertainties of E_{th} and v_{max} enter as $\delta \ln m_{\mathrm{min}} = \delta \ln E_{\mathrm{th}} - 2\,\delta \ln v_{\mathrm{max}}$. A 5 % change in the gap moves m_{min} by 5 %.

1.4 Light dark-matter models

Kinematics (Sect. 1.3) decides which pairs $(\boldsymbol{q}, \omega)$ are open; the interaction with electrons and nuclei decides how often they occur.

For a particle in the keV–GeV window of Sect. 1.1, thermal freeze-out is the simplest explanation of the observed abundance, and it constrains how light dark matter interacts. Suppose χ was once in equilibrium with the hot plasma of the early universe. Once the temperature drops below m_χ, the equilibrium density falls exponentially, and annihilation stops when a particle no longer meets a partner within one expansion time. The number left over is then inversely proportional to the thermally averaged annihilation cross-section times relative speed, $\langle \sigma v \rangle$ [7, Eq. (32)]: the more efficiently dark matter annihilates, the less of it survives. The observed $\Omega_{\mathrm{c}} h^2$ requires $\langle \sigma v \rangle \approx 2 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$ to $3 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$ for masses above about 10 GeV. The commonly quoted benchmark is $3 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$. A numerical solution of the Boltzmann equation for the number density gives $2.2 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$ for a Majorana particle (one that is its own antiparticle), and the required value rises to $5.2 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$ near 0.3 GeV [16, abstract and Sect. II C].

The same requirement rules out light dark matter that annihilates only through heavy particles. If χ annihilates through the weak bosons, or through any mediator much heavier than itself, $\langle \sigma v \rangle$ scales as the square of the dark-matter mass divided by the fourth power of the mediator mass, as in Fermi's theory of the weak interaction [7, Eq. (44)]. Below a few GeV it then falls short of the thermal value, too many particles survive, and the universe is overclosed. Thermal light dark matter therefore needs a new, light mediator, with a mass at or below the GeV scale, to carry the interaction between χ and ordinary matter [17]. Alternatively, χ was never in equilibrium and was produced by *freeze-in* (slow leakage from the plasma through a feeble coupling) [18] or by another non-thermal history. A thermal and a non-thermal history both fix the coupling factor of Eq. (1.1).

1.4.1 Mediators and the Yukawa potential

We take the mediator to be either a massive vector field A'_μ or a real scalar field ϕ_V, with mass m_V and couplings y_χ to dark matter, y_e to electrons and y_N to nucleons. The vector couples to a current of each species and the scalar to a density ($\bar\psi\gamma^\mu\psi$ and $\bar\psi\psi$ in field-theory notation). For slow particles the current and the density both reduce, at leading order, to the number density [19, Eq. (7)], and we use only that limit. The interaction Lagrangian densities are then

$$\mathcal{L}_{\mathrm{int}} = -A'_\mu (y_\chi J^\mu_\chi + y_e J^\mu_e + y_N J^\mu_N) \quad \text{or} \quad \mathcal{L}_{\mathrm{int}} = -\phi_V (y_\chi n_\chi + y_e n_e + y_N n_N), \tag{1.29}$$

where J^μ_χ, J^μ_e and J^μ_N are number currents whose time components are the number densities n_χ, n_e and n_N. The spatial components carry a factor of the particle velocity; they lead to velocity-dependent interactions, mentioned at the end of this section, and do not enter the potential below.

The mediator can be treated as a classical field sourced by the particles. A dark-matter particle at rest at the origin is a point source $y_\chi \delta^3(\boldsymbol{x})$. For this static source the Euler–Lagrange equation of the vector field reduces to the static Klein–Gordon equation $(-\nabla^2 + m_V^2) A'^0 = y_\chi \delta^3(\boldsymbol{x})$ (Eq. (C.78)). Its solution is y_χ times the Green's function of Eq. (C.80),

$A'^0(\boldsymbol{x}) = y_\chi \, \mathrm{e}^{-m_V r}/(4\pi r)$ with $r = |\boldsymbol{x}|$. By Eq. (1.29), an electron at $\boldsymbol{x}$ has the potential energy $y_e A'^0(\boldsymbol{x})$. The dark-matter–electron potential is then

$$V(\boldsymbol{x}) = y_\chi y_e \, \frac{\mathrm{e}^{-m_V |\boldsymbol{x}|}}{4\pi |\boldsymbol{x}|}, \tag{1.30}$$

the Yukawa potential. It has range $1/m_V$ and becomes Coulomb-like as $m_V \to 0$. A scalar mediator leads to the same function with the opposite sign: vector exchange repels sources whose couplings have equal signs, and scalar exchange attracts them (Eq. (C.82), where the signs are derived from the field energy). With the Fourier convention of Appendix A.3 and Eq. (C.79), the transform is

Yukawa potential in momentum space

$$\tilde{V}(\boldsymbol{q}) = \int \mathrm{d}^3x \, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} \, V(\boldsymbol{x}) = \frac{y_\chi y_e}{q^2 + m_V^2}. \tag{1.31}$$

For a scalar mediator the sign is reversed. Since every rate in this book depends on $|\tilde{V}(\boldsymbol{q})|^2$, the sign does not enter.

Here $\boldsymbol{q}$ is the momentum the dark matter transfers to the target (Sect. 1.3). Because $V(\boldsymbol{x})$ depends only on the separation of the dark-matter particle and the electron, the amplitude for transferring $\boldsymbol{q}$ is proportional to $\tilde{V}(\boldsymbol{q})$ (derived in Sect. 2.2). Eq. (1.31) has two limits, which the field calls the heavy- and light-mediator limits. For a heavy mediator, with m_V above every relevant q, $\tilde{V}$ is a constant: a contact interaction. For a light mediator, with m_V below every relevant q, $\tilde{V} \propto 1/q^2$, as in Coulomb scattering. The potential is static because the energy deposited is small: a moving source adds $-\omega^2$ to $q^2 + m_V^2$ in the denominator. Since $\omega \leq qv$ in scattering and $v \lesssim 3 \times 10^{-3}$, the $-\omega^2$ correction stays below 10^{-5} relative to q^2 (Appendix C.9).

Which couplings a mediator has depends on the model. A vector coupled to electric charge, the dark photon below, couples to electrons and protons and not to neutrons. A vector coupled to baryon minus lepton number couples to electrons, protons and neutrons with equal magnitude. A scalar that mixes with the Higgs boson couples to each fermion in proportion to its mass, and this mass dependence makes y_N/y_e large [7, Sect. III C]. Accelerator, stellar and cosmological data constrain each mediator: they bound the products $y_\chi y_e$ and $y_\chi y_N$ that enter rates, and they bound the couplings to ordinary matter separately [7, Sect. III C]. We will not need these bounds, since the rates below depend on the models only through $\tilde{V}(\boldsymbol{q})$ and the absorption couplings.

1.4.2 Reference cross-section and form factor

Experiments compare models through the reference cross-section $\bar{\sigma}_e$ and the dark-matter form factor $F_{\mathrm{DM}}(q)$, both fixed by $\tilde{V}(\boldsymbol{q})$. The reference momentum is $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$ (Sect. 1.3), the inverse Bohr radius, typical of electrons bound in atoms and solids. We define [15, Eqs. (3) and (4), 19, Eqs. (11) and (12)]

$$\bar{\sigma}_e \equiv \frac{\mu_{\chi e}^2}{\pi} \, |\tilde{V}(q_0)|^2, \qquad F_{\mathrm{DM}}(q) \equiv \frac{\tilde{V}(q)}{\tilde{V}(q_0)}, \tag{1.32}$$

with $\mu_{\chi e}$ the dark-matter–electron reduced mass (Sect. 1.3). Equivalently,

$$|\tilde{V}(q)|^2 = \frac{\pi \bar{\sigma}_e}{\mu_{\chi e}^2} \, F_{\mathrm{DM}}(q)^2, \tag{1.33}$$

the form in which the interaction enters every electron-scattering rate in this book. For the Yukawa potential of Eq. (1.31),

$$\bar{\sigma}_e = \frac{\mu_{\chi e}^2\, y_\chi^2 y_e^2}{\pi\,(m_V^2 + \alpha^2 m_e^2)^2}, \qquad F_{\rm DM}(q) = \frac{m_V^2 + \alpha^2 m_e^2}{m_V^2 + q^2}, \tag{1.34}$$

as in Ref. [19, Eqs. (11) and (12)].

Which process does $\bar{\sigma}_e$ describe? For a potential whose transform is a constant $\tilde{V}$, the first Born approximation yields the free elastic cross-section $\mu_{\chi e}^2|\tilde{V}|^2/\pi$ (derived in Sect. 2.2). The reference cross-section $\bar{\sigma}_e$ is this free cross-section with $\tilde{V}$ frozen at q_0; for a heavy mediator it is the actual cross-section of dark matter on a free electron. For a light mediator, however, $\bar{\sigma}_e$ only normalizes $|\tilde{V}(q_0)|^2$: the free cross-section of a $1/q^2$ potential diverges at small angles, as Rutherford's does. In a target the divergence is cut off by the minimum momentum transfer needed to excite it and by screening. Since $\bar{\sigma}_e$ is defined in vacuum, with screening and every other in-medium effect assigned to the target response (Sect. 4.4), a quoted $\bar{\sigma}_e$ means the same thing for every target. Other papers use other reference points and normalizations (Appendix A.9).

Two limits of Eq. (1.34) cover most of the literature. A heavy mediator, with m_V above both q_0 and every momentum transfer that contributes, gives $F_{\rm DM} = 1$ and $\bar{\sigma}_e = \mu_{\chi e}^2 y_\chi^2 y_e^2/(\pi m_V^4)$. A light mediator, with m_V below q_0 and below every contributing momentum transfer, gives $F_{\rm DM} = (q_0/q)^2$ and $\bar{\sigma}_e = \mu_{\chi e}^2 y_\chi^2 y_e^2/(\pi q_0^4)$. To first order in the small ratio between m_V^2 and the squared momenta, $F_{\rm DM} \simeq 1 + (q_0^2 - q^2)/m_V^2$ for the heavy mediator and $F_{\rm DM} \simeq (q_0/q)^2\,(1 + m_V^2/q_0^2 - m_V^2/q^2)$ for the light one. The momentum transfers that contribute run from $\omega/v_{\rm max}$, $0.44\,\text{keV} = 0.12\,q_0$ for the silicon gap, up to $2m_\chi v_{\rm max}$ (Sect. 1.3).

Limit Check 1.2. Heavy and light mediators

For a mediator of mass m_V and a momentum transfer q, the form factor of Eq. (1.32) is $F_{\rm DM}(q) = 1$ when $m_V \gg q, q_0$ (heavy mediator) and $F_{\rm DM}(q) = (q_0/q)^2$ when $m_V \ll q, q_0$ (light mediator), with $q_0 = \alpha m_e$ [15, Eq. (15) and the display above it].

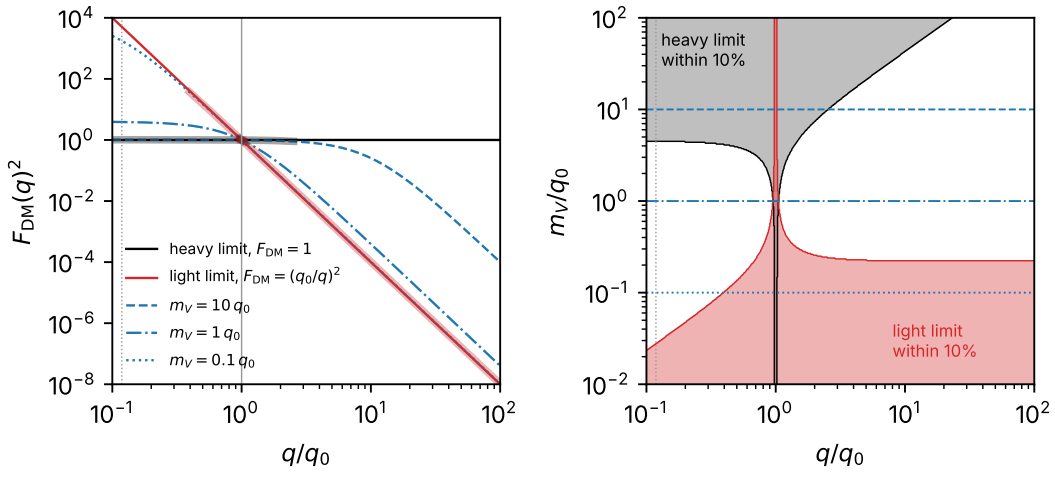


Figure 1.7: Squared dark-matter form factor $F_{\rm DM}(q)^2 = [(m_V^2 + q_0^2)/(m_V^2 + q^2)]^2$ [19, Eq. (12)], with $q_0 = \alpha m_e = 3.73\,\text{keV}$. Left: $F_{\rm DM}^2$ against q/q_0 for $m_V = 10\,q_0$ (dashed), q_0 (dash-dotted) and $0.1\,q_0$ (dotted), between the heavy limit $F_{\rm DM} = 1$ (black) and the light limit $(q_0/q)^4$ (red). A curve is overdrawn in gray or light red where it lies within 10 % of the heavy or light limit, $|F_{\rm DM}^2/\text{limit} - 1| \leq 0.1$. Right: the same two

regions in the plane of q/q_0 and m_V/q_0; the horizontal lines are the three mediator masses of the left panel. Every curve equals both limits at $q = q_0$ by construction, and this coincidence produces the narrow vertical strip. Dotted vertical line: $q = 0.12\,q_0$, the smallest momentum transfer that crosses the low-temperature silicon gap. Across $0.12\,q_0 \le q \le q_0$ the heavy limit holds for $m_V \ge 4.5\,q_0 = 17\,\mathrm{keV}$; for all $q \ge 0.12\,q_0$ the light limit holds for $m_V \le 0.028\,q_0 = 0.10\,\mathrm{keV}$.

For m_V within a factor of a few of the momentum transfers that the target samples, neither limit holds, and rates must use the full form of Eq. (1.34).

A light mediator weights the rate toward small momentum transfers, by $(q_0/q)^4$, and favors excitations near the kinematic threshold at small q. The target also screens most strongly at small q (Sect. 4.4).

1.4.3 The dark photon

The most studied vector mediator is the dark photon: the gauge boson of a new U(1) symmetry under which the dark matter carries charge y_χ and ordinary matter carries none. It reaches ordinary matter through kinetic mixing, a term $(\zeta/2)F_{\mu\nu}F'^{\mu\nu}$ that couples the photon field strength $F_{\mu\nu}$ to the dark-photon field strength $F'_{\mu\nu}$, with a dimensionless strength ζ [20]. The literature writes ϵ, κ or χ for ζ (Appendix B.2). Gauge invariance permits the term, and loops of heavy particles charged under both the new U(1) and electromagnetism generate it [20]. The mixing is removed by shifting the photon field, $A_\mu = \tilde{A}_\mu + \zeta A'_\mu$, and this shift leaves the mass term of A'_μ unchanged (Eqs. (C.83) and (C.84)).

Dark-photon couplings

To first order in ζ, the massive dark photon A'_μ couples to the dark-matter current with strength y_χ and to the electromagnetic current J^μ_{em} with strength ζe, through $\mathcal{L} \supset -\zeta e\, A'_\mu J^\mu_{\mathrm{em}} - y_\chi A'_\mu J^\mu_\chi$. Here $e > 0$ is the elementary charge, $e^2 = 4\pi\alpha$ (Appendix A.2), and J^μ_{em} counts each particle with its charge in units of e. In the notation of Eq. (1.29) the electron coupling is $y_e = -\zeta e$, since the electron charge is $-e$; protons have $+\zeta e$ and neutrons zero. The dark matter stays neutral under the photon.

Inserting $y_e^2 = 4\pi\alpha\zeta^2$ into Eq. (1.34), we find [7, Eq. (74), 15, display above Eq. (15)]

$$\bar{\sigma}_e = \frac{4\alpha\,\zeta^2 y_\chi^2\,\mu_{\chi e}^2}{(m_V^2 + \alpha^2 m_e^2)^2}. \tag{1.35}$$

Worked example 1.5: Reference cross-section of a heavy dark photon

We take the thermal-relic benchmark of Ref. [7, Sect. III D]: $m_\chi = 10\,\mathrm{MeV}$, $m_V = 30\,\mathrm{MeV}$, $y_\chi = 0.3$ and $\zeta = 10^{-4}$. The reduced mass is $\mu_{\chi e} = m_e m_\chi/(m_e + m_\chi) = (0.511 \times 10/10.511)\,\mathrm{MeV} = 0.486\,\mathrm{MeV}$, so $\mu_{\chi e}^2 = 0.236\,\mathrm{MeV}^2$. The maximum momentum transfer is $2m_\chi v_{\max} = 0.053\,\mathrm{MeV}$, smaller than m_V by a factor of 570: the mediator is heavy, $F_{\mathrm{DM}} = 1$, and $\alpha^2 m_e^2/m_V^2 = 1.5 \times 10^{-8}$ can be dropped in Eq. (1.35). With $4\alpha = 0.0292$,

$\zeta^2 = 10^{-8}$ and $y_\chi^2 = 0.09$,

$$\bar{\sigma}_e = \frac{4\alpha\,\zeta^2 y_\chi^2\,\mu_{\chi e}^2}{m_V^4} = \frac{0.0292 \times 10^{-8} \times 0.09 \times 0.236\,\mathrm{MeV}^2}{(30\,\mathrm{MeV})^4} = \frac{6.21 \times 10^{-12}\,\mathrm{MeV}^2}{8.10 \times 10^5\,\mathrm{MeV}^4} = 7.67 \times 10^{-18}\,\mathrm{MeV}^{-2}.$$

With $(\hbar c)^2 = 3.894 \times 10^{-22}\,\mathrm{MeV}^2\,\mathrm{cm}^2$ this is $\bar{\sigma}_e = 3.0 \times 10^{-39}\,\mathrm{cm}^2$. At fixed couplings $\bar{\sigma}_e$ scales as m_V^{-4}, and for $m_\chi \gg m_e$ it hardly depends on m_χ, because $\mu_{\chi e} \to m_e$.

Freeze-out and freeze-in, the two production histories above, lead to the heavy- and light-mediator limits of the form factor. Thermal freeze-out through a dark photon needs couplings large enough to deplete the dark matter, and couplings this large favor a heavy mediator, as in Worked example 1.5. Freeze-in, by contrast, needs couplings so feeble that the dark matter never reaches equilibrium; with an ultralight dark photon the rate of scattering on electrons is still observable, because the light-mediator limit enhances small momentum transfers. The freeze-in benchmark of Ref. [7, Sect. III D] has $m_\chi = 1\,\mathrm{MeV}$, $m_V = 10^{-12}\,\mathrm{eV}$, $y_\chi = 3 \times 10^{-6}$ and $\zeta = 10^{-6}$, and its form factor is $F_\mathrm{DM} = (q_0/q)^2$. The light limit of Eq. (1.35), with $\mu_{\chi e} = 0.338\,\mathrm{MeV}$, gives $\bar{\sigma}_e = 4\alpha\zeta^2 y_\chi^2 \mu_{\chi e}^2/q_0^4 = 6.1 \times 10^{-38}\,\mathrm{cm}^2$.

The field redefinition $A_\mu = \tilde{A}_\mu + \zeta A'_\mu$ holds in vacuum. In a medium the photon has a self-energy Π_em from the charges it polarizes, and the medium modifies the mixing. For absorption of a dark photon at rest, with $\omega = m_V$ and $q \to 0$, the effective mixing is $\zeta_\mathrm{eff}^2 = \zeta^2 m_V^4/\{[m_V^2 - \mathrm{Re}\,\Pi_\mathrm{em}(m_V)]^2 + [\mathrm{Im}\,\Pi_\mathrm{em}(m_V)]^2\}$ [21, Eqs. (5) and (6), with κ for ζ]. The effective mixing is suppressed where m_V lies below the scale of $|\Pi_\mathrm{em}|$ and approaches ζ well above that scale; in metals and semiconductors the scale of $|\Pi_\mathrm{em}|$ is set by the plasma frequency (Sect. 4.3). For scattering, the same physics is the screening of the dark photon's potential by the target. In the convention of this section, where $\tilde{V}(\boldsymbol{q})$ is the unscreened potential, the screening is a factor $1/|\varepsilon(\boldsymbol{q},\omega)|^2$ inside the target response [19, Eqs. (6) and (8)], and $\bar{\sigma}_e$ stays a vacuum quantity. The self-energy Π_em and the longitudinal dielectric function $\varepsilon(\boldsymbol{q},\omega)$ are material quantities: they are measured optically or computed from first principles, in the random-phase approximation with local fields or in time-dependent density-functional theory (Sect. 26.1).

1.4.4 Scattering on nuclei

Take a mediator that couples to nucleons with strength y_N, the same for protons and neutrons. It leads to a dark-matter–nucleon potential of the form of Eq. (1.31) with $y_e \to y_N$. For a heavy mediator the dark-matter–proton cross-section is $\sigma_p = \mu_{\chi p}^2 y_\chi^2 y_N^2/(\pi m_V^4)$ [7, Eq. (99)], the analogue of $\bar{\sigma}_e$, with $\mu_{\chi p}$ the dark-matter–proton reduced mass. A nucleus of mass number A and atomic number Z holds A nucleons within a radius of a few fm. When q is small compared with the inverse nuclear radius, the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ is the same for every nucleon, and the Fourier transform of the summed potential is A times the single-nucleon transform. The amplitudes of the A nucleons add coherently. For silicon the nuclear radius is $R \approx 3.47\,\mathrm{fm}$ (Sect. 2.2), and $1/R \approx 57\,\mathrm{MeV}$.

Replacing y_N by Ay_N and $\mu_{\chi p}$ by the dark-matter–nucleus reduced mass $\mu_{\chi N}$ in σ_p gives

$$\sigma_N = \sigma_p \left(\frac{\mu_{\chi N}}{\mu_{\chi p}}\right)^2 A^2, \tag{1.36}$$

the standard convention for spin-independent scattering [22, 7, Eq. (104)]. If protons and

neutrons couple differently, A becomes $Z + (A - Z)y_n/y_p$, with y_p and y_n the two couplings [7, Eq. (104)]. Since a dark photon couples only to charge, A^2 becomes Z^2 for it. For silicon, $A = 28$ and $Z = 14$, with the mean atomic mass of natural silicon, 26.16 GeV. Worked example 2.4 evaluates σ_N/σ_p for silicon.

At larger q the nuclear form factor $F_{\mathrm{Helm}}(q)$ of Sect. 2.2 multiplies the amplitude. For the light dark matter of this book the momentum transfer stays below $2m_\chi v_{\max} \approx 5\,\mathrm{MeV}$ even at $m_\chi = 1\,\mathrm{GeV}$, and the nucleus is always coherent. At q below the inverse size of an atom, the electrons screen the nuclear charge seen by a dark photon; this screening enters through the target response (Sect. 4.4). In a crystal, however, the nuclei are bound, and at low q the dark matter excites phonons instead of free recoils (Chap. 5).

1.4.5 Absorption of bosonic dark matter

Dark matter lighter than about a keV cannot be a thermal relic (Sect. 1.1), but bosons in this range can be produced non-thermally, for example by misalignment, and behave as a classical field oscillating at their mass [7, Sect. I B]. A bosonic dark-matter particle ϕ of mass m_ϕ can be absorbed: it disappears and deposits its whole energy in the target. Bound to the Galaxy, it moves at $v \approx 10^{-3}$ and carries energy $m_\phi(1 + v^2/2) \approx m_\phi$ and momentum $\boldsymbol{p}_\phi = m_\phi \boldsymbol{v}$. Absorption then deposits

$$\omega \approx m_\phi, \qquad \boldsymbol{q} = \boldsymbol{p}_\phi, \qquad |\boldsymbol{p}_\phi| \approx m_\phi v \approx 10^{-3}\,\omega. \tag{1.37}$$

The momentum is much smaller than the energy, whereas in scattering $\omega \leq qv$ (Sect. 1.3). Absorption probes the response of the target at $q \to 0$, the regime of optical absorption, at a single energy, $\omega = m_\phi$. Compared to scattering, absorption also reaches lower masses. A scattering particle can deposit at most its kinetic energy, and crossing a gap E_g needs $m_\chi \geq 2E_\mathrm{g}/v_{\max}^2$ (Sect. 1.3), about 0.3 MeV for $E_\mathrm{g} \approx 1\,\mathrm{eV}$; an absorbed boson deposits its rest energy and needs only $m_\phi \geq E_\mathrm{g}$, about 1 eV. Below the gap, absorption on phonons extends the reach to m_ϕ of tens of meV (Chap. 5).

The standard absorbed bosons are the dark photon, the axion-like particle and the scalar. Appendix C.9 derives the classical-field steps that the dark photon shares with scattering, Chap. 2 derives absorption rates from the golden rule, and Sect. 8.2.3 gives the general rate. A dark photon that is itself the dark matter, with $m_V = m_\phi$, couples to J^μ_em exactly as a photon does with e replaced by ζe. Its absorption rate is the photon absorption rate times ζ_eff^2. In the normalization of the event rate $\mathcal{R}$ of Sect. 1.5, per unit target mass, the dark-photon absorption rate is [21, Eq. (7)]

$$\mathcal{R} = \frac{1}{\rho_T}\,\frac{\rho_\chi}{m_\phi}\,\zeta_\mathrm{eff}^2\,\sigma_1(m_\phi), \tag{1.38}$$

with ρ_χ the local dark-matter density (Sect. 1.2), ρ_T the target mass density and σ_1 the real part of the optical conductivity (Sect. 4.4), measured or computed at $\omega = m_\phi$. The flux of dark matter does not enter, only its number density ρ_χ/m_ϕ: an absorbed field deposits energy at a rate set by its amplitude, whatever its speed.

An axion-like particle (ALP), a light pseudoscalar, couples to electrons through $\mathcal{L} \supset g_{aee}\,\phi\,\bar\psi \mathrm{i}\gamma^5\psi$ [23, Eq. (3)] and to photons through $\mathcal{L} \supset -\frac{1}{4} g_{a\gamma\gamma}\,\phi\,F_{\mu\nu}\tilde F^{\mu\nu} = g_{a\gamma\gamma}\,\phi\,\boldsymbol{E}\cdot\boldsymbol{B}$, with $\tilde F^{\mu\nu}$ the dual field strength [24, Eqs. (2) and (3)]; the subscript a names the ALP. At low energy the electron field ψ reduces to a two-component field $\hat\psi$ (the field operator of Sect. 2.3), and the electron coupling becomes [23, Eq. (18) and Table I]

$$\mathcal{L}_{aee} = -\frac{g_{aee}}{2m_e}\,(\nabla\phi)\cdot(\hat\psi^\dagger\boldsymbol{\sigma}\hat\psi) + \frac{\mathrm{i}g_{aee}}{4m_e^2}\,(\partial_t\phi)\,\hat\psi^\dagger\boldsymbol{\sigma}\cdot\overleftrightarrow{\nabla}\hat\psi, \tag{1.39}$$

Table 1.3: Interactions of light dark matter used in this book. Each row gives the coupling, the target quantity that sets the rate and the excitation that dominates in a semiconductor. The density response and conductivity are defined in Chap. 4, the spin and kinetic-energy responses in Chap. 8, and phonons in Chap. 5.

Interaction	Coupling	Target quantity	Dominant excitation
Scattering, heavy mediator	$\bar{\sigma}_e$, $F_{\rm DM} = 1$	electron density response at all q	electron–hole pairs
Scattering, light mediator	$\bar{\sigma}_e$, $F_{\rm DM} = (q_0/q)^2$	electron density response at small q, screened	electron–hole pairs near threshold
Scattering on nuclei	σ_p, σ_N	nuclear density response	phonons; nuclear recoils above the binding scale
Dark-photon absorption	ζ	optical conductivity at $\omega = m_\phi$	electron–hole pairs above the gap; optical phonons in polar crystals below it
ALP absorption	g_{aee}; $g_{a\gamma\gamma}$	electron spin-current response; field $\boldsymbol{B}_0$	electron–hole pairs; photons in $\boldsymbol{B}_0$
Scalar absorption	$g_{\phi ee}$	electron kinetic-energy response	electron–hole pairs

with $\boldsymbol{\sigma}$ the Pauli matrices and $\hat{\psi}^\dagger \overleftrightarrow{\nabla} \hat{\psi} \equiv \hat{\psi}^\dagger(\nabla\hat{\psi}) - (\nabla\hat{\psi}^\dagger)\hat{\psi}$. The first term couples $\nabla\phi$, of size $|\boldsymbol{p}_\phi|\phi$, to the electron spin density and is suppressed by $|\boldsymbol{p}_\phi|$. The second couples $\partial_t\phi$, of size $m_\phi\phi$, to the electron spin-current. Compared with the first, it trades the dark-matter velocity, typically 10^{-3}, for the electron velocity $|\nabla|/m_e$, typically $\sim\alpha$ in a solid; as a result it dominates absorption, although it is formally of higher order in $1/m_e$ [23, Sect. II]. When spin–orbit coupling is negligible, the ALP absorption rate follows from the optical conductivity, like the dark-photon absorption rate [23, Table I]. The photon coupling contributes to absorption in a solid only in a static magnetic field $\boldsymbol{B}_0$: there ϕ drives an electric field along $\boldsymbol{B}_0$ that oscillates at m_ϕ, and the ALP is absorbed like a dark photon polarized along $\boldsymbol{B}_0$.

A scalar couples through $\mathcal{L} \supset g_{\phi ee}\,\phi\,\bar{\psi}\psi$ [23, Eq. (3)], which at low energy becomes [23, Eq. (18) and Table I]

$$\mathcal{L}_{\phi ee} = g_{\phi ee}\,\phi\,\hat{\psi}^\dagger\hat{\psi} + \frac{g_{\phi ee}}{8m_e^2}\,\phi\,\hat{\psi}^\dagger\overleftrightarrow{\nabla}^2\hat{\psi}, \tag{1.40}$$

where a third term of Ref. [23, Eq. (18)], involving the photon field, is omitted because it does not contribute to absorption in an isotropic medium [23, Table I], and $\hat{\psi}^\dagger\overleftrightarrow{\nabla}^2\hat{\psi} \equiv \hat{\psi}^\dagger(\nabla^2\hat{\psi}) + (\nabla^2\hat{\psi}^\dagger)\hat{\psi} - 2(\nabla\hat{\psi}^\dagger)\cdot(\nabla\hat{\psi})$. The first term couples to the electron number density, whose matrix elements between different states vanish as $\boldsymbol{p}_\phi \to 0$ by charge conservation. The second term, a kinetic-energy density, dominates, and the scalar absorption rate is not fixed by the optical conductivity [23, Table I]. Table 1.3 collects the interactions of this section.

1.4.6 Inelastic dark matter and further interactions

The dark matter may have two nearly degenerate states, χ_1 and χ_2, split by $\Delta_\chi = m_{\chi_2} - m_{\chi_1}$. The two states arise, for example, when a small mass term splits one dark-matter particle into χ_1 and χ_2, and the mediator couples χ_1 to χ_2 but couples neither state to itself. Every scattering then changes the state [25]; in field-theory language, a Dirac fermion splits into two Majorana states. The potential and $\bar{\sigma}_e$ keep their form; only the kinematics changes, through the term $-\Delta_\chi$ in the energy transfer $\omega_{\boldsymbol{q}}$ of Eq. (1.21), with the endothermic and exothermic cases of Sect. 1.3.

The mediators of this section lead to the simplest non-relativistic interaction, a spin-independent potential. A general basis of non-relativistic interactions exists, built from $\boldsymbol{q}$, the

relative velocity and the spins of the dark matter and the target [26]; the Yukawa potential is its first, spin-independent operator multiplied by $F_{\mathrm{DM}}(q)$. Chap. 8 treats this basis and the target responses it requires.

1.5 Detector signals

A detector records none of three quantities directly: the dark-matter flux (Sect. 1.2), the energy and momentum that the dark matter can transfer (Sect. 1.3), and the strength of the dark-matter interaction (Sect. 1.4). It counts the electrons, phonons, quasiparticles or photons produced as the deposited energy relaxes. Fig. 1.8 shows the chain that the rest of the book follows.

1.5.1 Readout quanta

A deposit of a few eV or less does not stay in the excitation that the dark matter created. Hot electrons and holes lose energy to further electron–hole pairs and to phonons; quasiparticles in a superconductor emit phonons; phonons decay into lower-energy phonons. Detectors read the products at one of four stages: as charge, as heat carried by phonons, as quasiparticles of a superconductor, or as photons [1, Sect. VII]. Each channel counts a different quantum, and the energy per quantum sets how finely a deposit is resolved.

Charge. In a semiconductor an excitation across the band gap promotes an electron to the conduction band and leaves a vacancy, a *hole*, in the valence band. The electron and the hole together form an *electron–hole pair*. Both carriers are mobile, and an applied electric field drifts them to electrodes before they recombine [1, Sect. VII A]. The number of pairs produced by a deposit ω is the *ionization yield* N_{eh}. A primary carrier with energy well above the gap creates further pairs by impact ionization, knocking further electrons across the gap, and also emits phonons; on average, then, each pair costs more than E_{g}. For deposits well above 20 eV the mean yield is proportional to the deposit [12, abstract and Eq. (1)],

$$\langle N_{\mathrm{eh}} \rangle = \frac{\omega}{\varepsilon_{\mathrm{eh}}}, \tag{1.41}$$

with the mean energy per pair $\varepsilon_{\mathrm{eh}}$. Measurements in silicon give 3.62 eV to 3.66 eV at 300 K and 3.70 eV to 3.75 eV between 5 K and 123 K [12, Table I], roughly three times the gap. The energy in excess of E_{g} per pair goes into phonons. The fluctuations of N_{eh} are smaller than Poisson fluctuations: the variance is 0.1 to 0.15 times the mean [11, Eq. (5.4)]. A 100 eV deposit produces on average $100/3.7 = 27$ pairs with a spread of $(0.12 \times 27)^{1/2} = 1.8$, against $27^{1/2} = 5.2$ for a Poisson count. The pairs are not produced independently: the energy is fixed, and a cascade that makes one more pair has less energy left for the next.

Light dark matter, however, deposits only a few eV, where Eq. (1.41) fails. Below $2E_{\mathrm{g}}$ energy conservation allows one pair at most. Between $2E_{\mathrm{g}}$ and a few tens of eV the yield depends on how the primary carriers share the energy, and the silicon data cover 3 to 8 eV and then 50 eV upward, with no measurement between 8 and 50 eV [12, Sect. III]. A widely used step model assigns one pair for the first E_{g} and one more for each further $\varepsilon_{\mathrm{eh}}$ [11, Eq. (5.1)],

$$N_{\mathrm{eh}}(\omega) = 1 + \left\lfloor \frac{\omega - E_{\mathrm{g}}}{\varepsilon_{\mathrm{eh}}} \right\rfloor \qquad (\omega \geq E_{\mathrm{g}}), \tag{1.42}$$

where $\lfloor x \rfloor$ is the largest integer not above x. With the low-temperature gap $E_{\mathrm{g}} = 1.17$ eV of Worked example 1.3 and $\varepsilon_{\mathrm{eh}} = 3.6$ eV [11, Eq. (5.2)], a 5 eV deposit makes $1 + \lfloor 1.06 \rfloor = 2$ pairs and a 10 eV deposit $1 + \lfloor 2.45 \rfloor = 3$. Chap. 18 derives the yield from the carrier cascade.

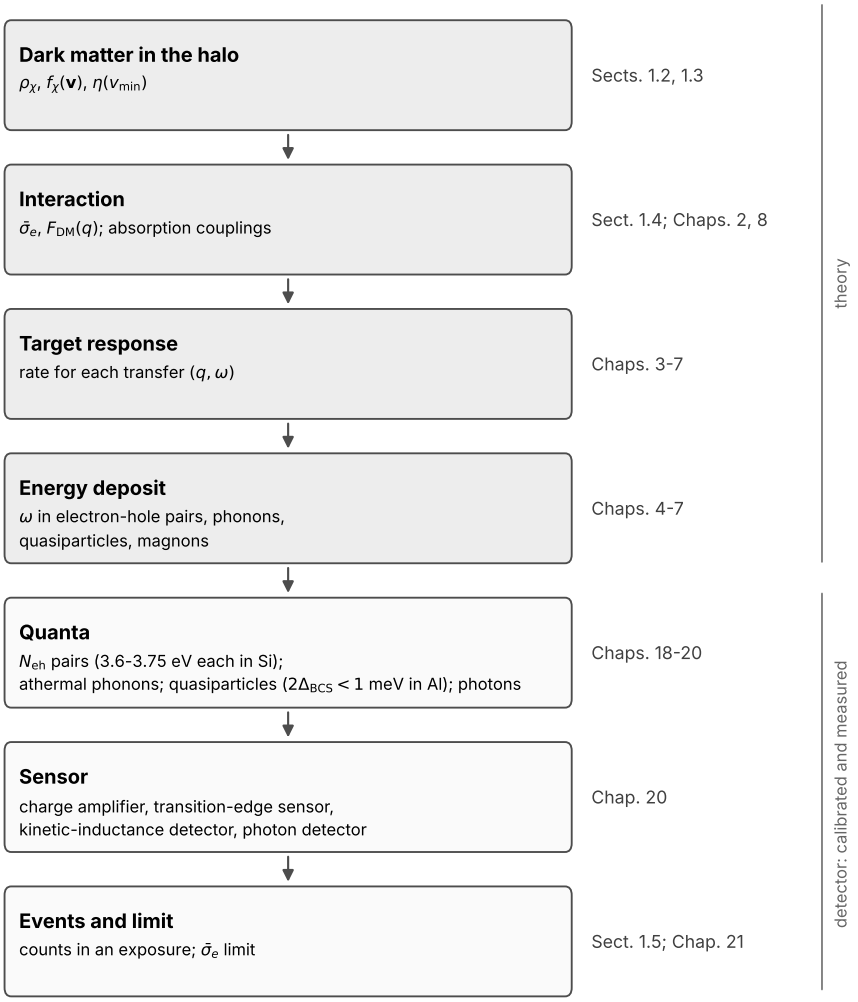


Figure 1.8: The signal chain from dark matter to sensor. The upper four steps are computed from theory: the halo (Sects. 1.2 and 1.3), the interaction (Sect. 1.4; Chaps. 2 and 8), the target response and the primary excitations (Chaps. 3 to 7). The lower three are fixed by calibration and measurement: the conversion of the deposit into countable quanta (Chaps. 18 and 20), the sensor (Chap. 20) and the statistical step from counts to a limit (Sect. 1.5; Chap. 21). Energies per quantum are from Refs. [12, Table I, 27, Sect. 2.2.1].

A detector that counts pairs states its threshold as a number of pairs, and Eq. (1.42) turns that number into an energy. The n-th pair appears at $\omega = E_\mathrm{g} + (n-1)\varepsilon_\mathrm{eh}$, and thresholds of one, two and three pairs then correspond to 1.17, 4.77 and 8.37 eV. Through $m_\mathrm{min} = 2E_\mathrm{th}/v_\mathrm{max}^2$ (Eq. (1.24)) with $v_\mathrm{max} = 2.650 \times 10^{-3}$, these threshold energies translate into threshold masses of 0.33, 1.36 and 2.38 MeV. A search can discard the single-pair bin, where the dark counts described below accumulate; it then gives up a factor of four in mass reach for a cleaner sample. Since the yield has not been measured in silicon between 8 and 50 eV, the energy assigned to a three-pair threshold, and with it m_min, carries a model uncertainty (Sect. 1.3.5).

Skipper charge-coupled devices and calorimetric detectors count single charges. A skipper charge-coupled device measures the charge in each pixel many times without destroying it; the noise falls with the square root of the number of measurements, below 0.1 electron per pixel [1, Sect. VII A]. A calorimetric detector instead drifts the carriers across a bias voltage V_bias. Each carrier then emits phonons in proportion to the work done on it (the Neganov–Trofimov–Luke effect). The phonon energy is [1, Eq. (195)]

$$E_\mathrm{ph} = E_\mathrm{prim} + N_\mathrm{eh}\, eV_\mathrm{bias}, \tag{1.43}$$

where E_prim is the part of the deposit that went directly into phonons, at most ω; the second term is usually much larger. At $V_\mathrm{bias} = 150$ V each pair adds 150 eV, and the spectrum shows peaks at integer N_eh [1, Sect. VII A]. In both devices the lowest bins are polluted by a dark rate (charge that leaks across the gap without any deposit), and single-pair events carry the largest background.

Phonons. Every excitation of a solid ends as phonons, the quantized lattice vibrations of Sect. 1.3. A calorimeter measures the heat carried by these phonons and with it the total deposit, whatever the primary excitation [1, Sect. VII B]. The phonons first created in silicon are optical phonons of about 64 meV; through several anharmonic decays they eventually become acoustic phonons of a few meV. At an operating temperature of 15 mK the thermal phonons carry about 1 µeV, and the signal phonons are far from equilibrium (*athermal*) and cross the crystal ballistically before they reach a sensor on the surface [1, Sect. VII B]. The signal phonons are too numerous and too soft to count one by one. The sensor integrates their energy, and its noise sets the threshold. A transition-edge sensor is a superconducting film held at the edge of its transition, where a small heat input produces a large change in resistance. Phonons are treated precisely in Sect. 5.3.

Quasiparticles. The Fermi level is the energy that separates occupied from empty states. In a superconductor the electrons near the Fermi level bind into Cooper pairs. Breaking a pair costs $2\Delta_\mathrm{BCS}$, where Δ_BCS is the superconducting gap, and creates two *quasiparticles*, the excitations of the paired state (made precise in Sect. 7.2). The Bardeen–Cooper–Schrieffer (BCS) theory relates the gap to the critical temperature, $\Delta_\mathrm{BCS} \approx 1.76\, k_\mathrm{B} T_\mathrm{c}$ [28, Sect. 3.3]. With $T_\mathrm{c} = 1.2$ K for bulk aluminum [29], $2\Delta_\mathrm{BCS} \approx 0.36$ meV. Thin films have higher critical temperatures [29], and dark-matter projections adopt the film value $2\Delta_\mathrm{BCS} = 0.6$ meV [27, Sect. 2.2.1, 23, Table II]. A deposit in a superconducting absorber cascades through quasiparticles and phonons until no phonon can break another pair; about 60 % of the energy ends as quasiparticles and the rest as phonons near $2\Delta_\mathrm{BCS}$ [27, Sect. 2.2.1]. Quasiparticles also convert phonons into a signal. Kinetic-inductance detectors are superconducting resonators whose frequency shifts when pairs are broken. In transition-edge sensors and in kinetic-inductance detectors, phonons from the crystal break Cooper pairs in a superconducting film, and the change in the film's resistance or impedance is read out [28, Sect. 3.3.1]. The pair-breaking

Table 1.4: Readout channels, the quantum each counts and its energy in silicon or aluminum, with the sensors that read them. The phonon and quasiparticle energies set the scale of the quanta; calorimetric sensors integrate many of them.

Channel	Quantum	Energy per quantum	Sensor
Charge	electron–hole pair	$\varepsilon_{\mathrm{eh}} = 3.6$ to 3.75 eV in Si (at least E_{g} for the first pair)	skipper charge-coupled device (CCD); phonon calorimeter with bias
Heat	athermal phonon	64 meV optical at creation, a few meV after decay, in Si	transition-edge sensor
Quasiparticles	broken Cooper pair	$2\Delta_{\mathrm{BCS}} \approx 0.36$ meV in bulk Al	transition-edge sensor; kinetic-inductance detector
Light	photon	a few eV (optical, near-ultraviolet); infrared in GaAs	photomultiplier; transition-edge sensor; Si photodetector

energy is the smallest quantum in Table 1.4. If a single broken pair in an aluminum absorber could be detected, Eq. (1.24) with the bulk value $E_{\mathrm{th}} = 0.36$ meV would place the threshold mass at 0.10 keV (0.17 keV for the film value), below the keV lower end of the window of Sect. 1.1. The thresholds of present sensors are set by noise, far above this quantum.

Photons. A scintillator returns part of the deposit as photons. In NaI and CsI, with band gaps of about 6 eV, these are optical or near-ultraviolet photons of a few eV; GaAs, with its gap of 1.52 eV, emits in the infrared [1, Sect. VII C]. The photons are counted by a photomultiplier, by a transition-edge sensor or by a silicon device, in which each photon above the silicon gap, 1.17 eV at low temperature (Worked example 1.3), creates electron–hole pairs.

Table 1.4 collects the four channels. In every case the detector sees a number of quanta, or their summed energy, after the cascade. The relation between that number and ω is an input to every limit, and for deposits of a few eV it is the most poorly known step of the chain.

1.5.2 Event rate

The event rate $\mathcal{R}$ of Eq. (1.1) is the expected number of events per unit time per unit target mass, quoted in $\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$. Its *spectrum* $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ is the rate per unit deposited energy, and $\mathcal{R}$ is the integral of the spectrum above the threshold E_{th} (Sect. 1.3). A detector of mass M_{det} that runs for a live time t_{exp} collects the *exposure* $M_{\mathrm{det}} t_{\mathrm{exp}}$, quoted in kg yr. The expected number of signal events is

$$N_{\mathrm{sig}} = M_{\mathrm{det}}\, t_{\mathrm{exp}}\, \mathcal{R}, \tag{1.44}$$

where $\mathcal{R}$ includes the probability that an event passes the selection. The absorption rate of Eq. (1.38) uses the same normalization per unit target mass.

A charge detector measures N_{eh} instead of ω. The rate of events with n pairs collects the spectrum over the deposits that Eq. (1.42) maps to n,

$$\mathcal{R}_n = \int_{E_{\mathrm{g}}+(n-1)\varepsilon_{\mathrm{eh}}}^{E_{\mathrm{g}}+n\varepsilon_{\mathrm{eh}}} \mathrm{d}\omega\, \frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega}, \tag{1.45}$$

and a search compares the counts in each bin with the expected background in that bin. Since for scattering through a heavy mediator in silicon the typical deposit is a few eV, the spectrum peaks in the two-pair bin [11, Fig. 7].

The schematic rate of Eq. (1.1) becomes concrete for a target whose response depends on the magnitude q of the momentum transfer but not on its direction, as for a liquid, an

atom or a crystal averaged over orientations. The velocity integral and the average over the direction of $\boldsymbol{q}$ then act on the delta function alone, and Sect. 1.3.4 turned them into the measure $q\,\mathrm{d}q\,\eta(v_{\min})/4\pi^2$. With the coupling of Eq. (1.33) and the halo of Sect. 1.2, the factors of Sect. 1.1 combine into [1, Eq. (11)]

$$\mathcal{R} = \frac{1}{\rho_T}\,\underbrace{\frac{\rho_\chi}{m_\chi}}_{\text{halo}} \int \frac{q\,\mathrm{d}q}{(2\pi)^2} \int \mathrm{d}\omega\; \underbrace{\eta(v_{\min}(q,\omega))}_{\text{halo, kinematics}}\; \underbrace{\frac{\pi\bar{\sigma}_e F_{\mathrm{DM}}(q)^2}{\mu_{\chi e}^2}}_{\text{coupling}}\; \underbrace{S(q,\omega)}_{\text{target}}. \tag{1.46}$$

Sect. 2.5 derives it term by term from the thermal golden rule, and Sect. 4.2 develops the properties of $S(q,\omega)$.

For a crystal, $S(q,\omega)$ is computed from the electronic states of the material. Density-functional theory supplies the bands and wavefunctions as Kohn–Sham states, the eigenstates of an effective one-electron problem rather than of the interacting electrons. The overlaps of the Kohn–Sham states at momentum transfer q, the crystal form factors, fix the transition probabilities; Sect. 26.2 describes how the crystal form factors are evaluated and converged. Most of the theory uncertainty of a limit originates in this step.

Define the response-weighted average of the squared form factor,

$$\langle F_{\mathrm{DM}}^2 \rangle = \frac{\int q\,\mathrm{d}q \int \mathrm{d}\omega\; \eta(v_{\min})\, S(q,\omega)\, F_{\mathrm{DM}}(q)^2}{\int q\,\mathrm{d}q \int \mathrm{d}\omega\; \eta(v_{\min})\, S(q,\omega)}, \tag{1.47}$$

which is 1 for a heavy mediator and $\langle (q_0/q)^4 \rangle$ for a light one. Then Eq. (1.46) reads

$$\mathcal{R} = \frac{\pi\bar{\sigma}_e\,\rho_\chi\,\langle F_{\mathrm{DM}}^2\rangle}{\rho_T\, m_\chi\, \mu_{\chi e}^2} \int \frac{q\,\mathrm{d}q}{(2\pi)^2} \int \mathrm{d}\omega\; \eta(v_{\min})\, S(q,\omega). \tag{1.48}$$

We see that the rate is linear in $\bar{\sigma}_e$ and in ρ_χ, and $\bar{\sigma}_e$ appears nowhere else. The mass enters through the number density, the reduced mass and the integration region of the last integral. The number density brings $1/m_\chi$. The reduced mass tends to m_e once $m_\chi \gg m_e$, and the factor $\mu_{\chi e}^2$ stops changing above a few MeV. The last integral depends on m_χ only through the momentum transfers allowed by $v_{\min}(q,\omega) \le v_{\max}$ (Eq. (1.25)): it vanishes for $m_\chi \le m_{\min}$ and stops growing once $2m_\chi v_{\max}$ exceeds the momenta at which $S(q,\omega)$ is appreciable. Well above threshold, then,

$$\mathcal{R} \propto \frac{\bar{\sigma}_e\,\rho_\chi\,\langle F_{\mathrm{DM}}^2\rangle}{m_\chi}. \tag{1.49}$$

This proportionality is a statement about scaling only; the absolute rate needs $S(q,\omega)$ and the full integral. As a check of Eq. (1.46), in natural units $\rho_\chi/(\rho_T m_\chi)$ has dimension eV^{-1} (an inverse mass), $q\,\mathrm{d}q\,\mathrm{d}\omega$ has eV^3, $\bar{\sigma}_e/\mu_{\chi e}^2$ has eV^{-4}, $S(q,\omega)$ has eV^2 with the normalization of Sect. 4.2, and η, an inverse speed, is dimensionless. The product is dimensionless, as a rate (eV) per unit mass (eV) must be.

1.5.3 Counting and the upper limit

Since dark-matter events are rare and independent, their number in a fixed exposure follows the Poisson distribution. To derive the Poisson distribution, divide the exposure into K equal parts, each so small that it holds one event with probability $p = N_{\mathrm{sig}}/K$ and two events with negligible probability. The probability of N_{obs} events is binomial,

$$\binom{K}{N_{\mathrm{obs}}} p^{N_{\mathrm{obs}}} (1-p)^{K-N_{\mathrm{obs}}} = \frac{K(K-1)\cdots(K-N_{\mathrm{obs}}+1)}{K^{N_{\mathrm{obs}}}}\, \frac{N_{\mathrm{sig}}^{N_{\mathrm{obs}}}}{N_{\mathrm{obs}}!} \left(1 - \frac{N_{\mathrm{sig}}}{K}\right)^{K-N_{\mathrm{obs}}}.$$

As $K \to \infty$ at fixed $N_{\rm obs}$, the first fraction tends to 1 and the last factor to $\mathrm{e}^{-N_{\rm sig}}$, and we are left with

$$P(N_{\rm obs} \,|\, N_{\rm sig}) = \frac{N_{\rm sig}^{N_{\rm obs}}}{N_{\rm obs}!}\, \mathrm{e}^{-N_{\rm sig}}. \tag{1.50}$$

Suppose a search sees no events and expects no background. A large $N_{\rm sig}$ would make this outcome improbable, since $P(0 \,|\, N_{\rm sig}) = \mathrm{e}^{-N_{\rm sig}}$. The upper limit at 90 % confidence, N_{90}, is the expected count at which the observed outcome, or one with fewer events, would occur in only 10 % of repeated experiments. Every larger $N_{\rm sig}$ would yield at least one event more than 90 % of the time, and is excluded at that confidence. The confidence level describes the procedure, not the parameter: if the experiment were repeated many times and the limit computed the same way each time, the interval from 0 to N_{90} would contain the true $N_{\rm sig}$ in at least 90 % of the repetitions. The confidence level is not the probability that $\bar{\sigma}_e$ lies below the limit; that probability would require a prior distribution for $\bar{\sigma}_e$. Setting $\mathrm{e}^{-N_{90}} = 0.1$ gives

$$N_{90} = \ln 10 = 2.303. \tag{1.51}$$

At 95 % confidence the same condition leads to $\ln 20 = 3.00$, the three events assumed in many projections [23, Sect. V]. If instead $N_{\rm obs}$ events are seen and all are kept as possible signal, the 90 % limit solves $\sum_{n=0}^{N_{\rm obs}} P(n \,|\, N_{90}) = 0.1$. A numerical solution yields $N_{90} = 3.89$ for one event and 5.32 for two. A limit that keeps every observed event as possible signal holds whatever the origin of these events.

Since $\mathcal{R}$ is proportional to $\bar{\sigma}_e$ (Eq. (1.48)), we write $\mathcal{R} = \bar{\sigma}_e\, r(m_\chi)$ with $r(m_\chi)$ the rate per unit cross-section. Setting $N_{\rm sig} = N_{90}$ in Eq. (1.44) gives the limit:

Zero-background limit

A search with exposure $M_{\rm det} t_{\rm exp}$ that observes no events and expects no background excludes at 90 % confidence

$$\bar{\sigma}_e > \bar{\sigma}_e^{\rm lim}(m_\chi) = \frac{N_{90}}{M_{\rm det}\, t_{\rm exp}\, r(m_\chi)}, \qquad N_{90} = 2.303. \tag{1.52}$$

The limit improves in proportion to the exposure, $\bar{\sigma}_e^{\rm lim} \propto 1/(M_{\rm det} t_{\rm exp})$.

Background changes the scaling. Suppose the background is known to give $N_{\rm bkg} = \mathcal{R}_{\rm bkg}\, M_{\rm det} t_{\rm exp}$ events, with $\mathcal{R}_{\rm bkg}$ the background rate per unit exposure, and cannot be removed event by event. The observed count then fluctuates by roughly $N_{\rm bkg}^{1/2}$, and a signal is excluded only when it exceeds a fixed multiple k of that fluctuation, $\bar{\sigma}_e\, r\, M_{\rm det} t_{\rm exp} \geq k\, (\mathcal{R}_{\rm bkg}\, M_{\rm det} t_{\rm exp})^{1/2}$. Solving for $\bar{\sigma}_e$, we find $\bar{\sigma}_e^{\rm lim} = k\, \mathcal{R}_{\rm bkg}^{1/2} / [r\, (M_{\rm det} t_{\rm exp})^{1/2}]$: the limit now improves only as the inverse square root of the exposure.

Counting alone cannot tell signal from background, but the annual modulation of Sect. 1.2 can. For momentum transfers that need $v_{\rm min} = 600\,\mathrm{km\,s^{-1}}$, η is 21 % above its reference value in June and 16 % below it in December. Take the modulation as a cosine with amplitude $A_{\rm mod} = 0.185$, the mean of the June and December deviations. Averaged over the half-year centered on June, a cosine of amplitude $A_{\rm mod}$ has mean $2A_{\rm mod}/\pi = 0.118$; with N events expected per half-year, the June half then exceeds the December half by $2 \times 0.118\, N = 0.236\, N$. The Poisson fluctuation of the difference of two counts of about N each is $(2N)^{1/2}$. A difference of two standard deviations requires $0.236\, N \geq 2(2N)^{1/2}$, that is $N \geq 144$ events per half-year. A modulation test needs a detected signal of a few hundred events, far beyond the few events that set a limit. Chap. 21 treats the statistics in full.

Worked example 1.6: Expected events and a limit for a toy rate

We take a silicon target, $m_\chi = 100\,\mathrm{MeV}$, a heavy mediator and $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$. For the toy rate we keep the prefactor of the silicon rate of Ref. [11, Eq. (3.13)]. That equation is written for the spectrum per unit logarithm of the deposit, $\mathrm{d}\mathcal{R}/\mathrm{d}\ln\omega$, and the reference describes its remaining integral as roughly unity for typical q and ω. We set that integral equal to one between $E_\mathrm{g} = 1.17\,\mathrm{eV}$ and $30\,\mathrm{eV}$, the range of the toy response of Fig. 1.9, and to zero elsewhere. Integrating over $\ln\omega$ then multiplies the prefactor by $\ln(30/1.17) = 3.24$, and the toy rate is

$$\mathcal{R}_\mathrm{toy} = \ln\!\left(\frac{30\,\mathrm{eV}}{E_\mathrm{g}}\right) \frac{\rho_\chi}{m_\chi}\,\frac{\bar\sigma_e}{M_\mathrm{cell}}\,\alpha\,\frac{m_e^2}{\mu_{\chi e}^2}.$$

Here $M_\mathrm{cell} = 52.33\,\mathrm{GeV}$ is the mass of the two-atom silicon unit cell [11, below Eq. (3.13)], so $1/M_\mathrm{cell}$ is the number of cells per unit mass. The factor $\alpha = q_0/m_e$ is the typical speed of a bound electron, αc in conventional units. The factors are: $\rho_\chi/m_\chi = 3\,\mathrm{cm^{-3}}$; $1\,\mathrm{kg} = 5.610\times10^{26}\,\mathrm{GeV}$, so a kilogram holds 1.072×10^{25} cells; $\alpha c = 2.188\times10^{8}\,\mathrm{cm\,s^{-1}}$; $\mu_{\chi e} = 0.5084\,\mathrm{MeV}$, so $m_e^2/\mu_{\chi e}^2 = 1.010$. Without the logarithm their product is $7.106\times10^{33}\,\mathrm{cm^{-2}\,s^{-1}\,kg^{-1}}$; with it, $\mathcal{R}_\mathrm{toy}/\bar\sigma_e = 2.305\times10^{34}\,\mathrm{cm^{-2}\,s^{-1}\,kg^{-1}}$. With $1\,\mathrm{yr} = 3.156\times10^{7}\,\mathrm{s}$ this is $r = 7.28\times10^{41}\,\mathrm{cm^{-2}\,kg^{-1}\,yr^{-1}}$. For an exposure of $1\,\mathrm{kg\,yr}$ and $\bar\sigma_e = 10^{-40}\,\mathrm{cm^2}$, Eq. (1.44) predicts $N_\mathrm{sig} = 73$ events. With no events observed, Eq. (1.52) gives

$$\bar\sigma_e^\mathrm{lim} = \frac{2.303}{(1\,\mathrm{kg\,yr})(7.28\times10^{41}\,\mathrm{cm^{-2}\,kg^{-1}\,yr^{-1}})} = 3.2\times10^{-42}\,\mathrm{cm^2},$$

and ten times the exposure brings it to $3.2\times10^{-43}\,\mathrm{cm^2}$. Since the toy has no material response and no yield model, these numbers show only the arithmetic and the scaling; a real silicon search must compute $S(q,\omega)$ and fold in Eq. (1.42) and the efficiency.

1.5.4 Exclusion plots

Results of light-dark-matter searches are shown as *exclusion limits*: the curve $\bar\sigma_e^\mathrm{lim}(m_\chi)$ of Eq. (1.52) in the plane of m_χ and $\bar\sigma_e$, drawn separately for a heavy and a light mediator. Every point above the curve would have produced more than N_{90} events and is excluded. By Eqs. (1.48) and (1.52),

$$\bar\sigma_e^\mathrm{lim}(m_\chi) \propto \frac{m_\chi\,\mu_{\chi e}^2}{\rho_\chi\,\langle F_\mathrm{DM}^2\rangle\,M_\mathrm{det}\,t_\mathrm{exp}} \left[\int \frac{q\,\mathrm{d}q}{(2\pi)^2}\int \mathrm{d}\omega\;\eta(v_\mathrm{min})\,S(q,\omega)\right]^{-1}, \tag{1.53}$$

and the shape of the curve follows from the mass dependence discussed after Eq. (1.48). Fig. 1.9 shows the shapes for a toy response.

The low-mass edge. Below $m_\mathrm{min} = 2E_\mathrm{th}/v_\mathrm{max}^2$ (Eq. (1.24)) no dark-matter particle in the halo can deposit E_th, the integral in Eq. (1.53) vanishes and the limit is infinite. Just above m_min only the fastest particles contribute, and η grows from zero quadratically in the distance from the escape speed (Eq. (1.28)): the curve falls steeply from a vertical wall.

The heavy-mediator slope. With $F_\mathrm{DM} = 1$ the integrand of Eq. (1.53) grows with q through the factor $q\,\mathrm{d}q$, and the maximum momentum transfer, $2m_\chi v_\mathrm{max}$, grows with m_χ. The maximum momentum transfer equals q_0 at $0.70\,\mathrm{MeV}$ (Sect. 1.3). The integral keeps growing until $2m_\chi v_\mathrm{max}$ passes the momenta of the valence electrons, a few q_0, at masses

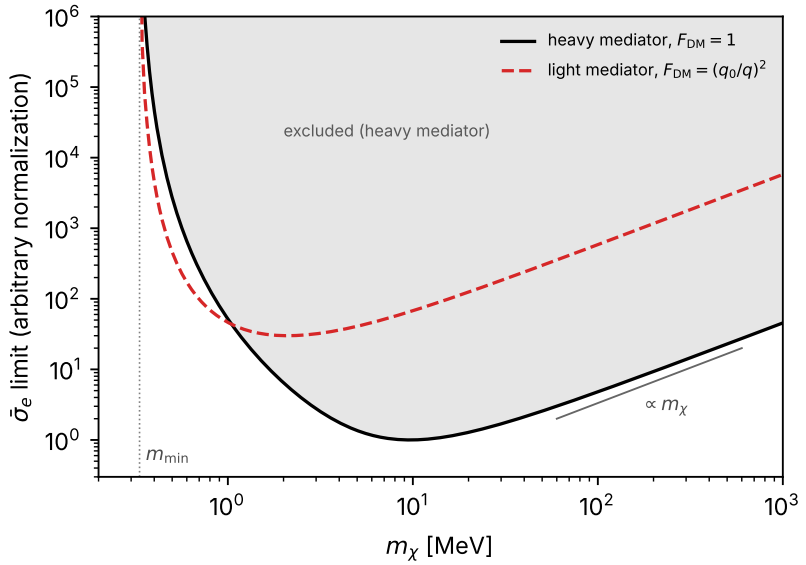


Figure 1.9: Schematic exclusion limits on $\bar{\sigma}_e$ for a silicon-like target. The threshold is the low-temperature gap $E_{\rm g} = 1.17\,\text{eV}$. The curves follow from Eq. (1.53) with the standard-halo $\eta(v_{\rm min})$ of Sect. 1.3 and a toy response: $S(q,\omega)$ flat between $E_{\rm g}$ and 30 eV, multiplied by the Gaussian cutoff $\exp[-q^2/2(2q_0)^2]$ with $q_0 = \alpha m_e = 3.73\,\text{keV}$. Each curve has an arbitrary normalization (the light-mediator curve is shifted up for legibility). Both curves rise without bound at the threshold mass $m_{\rm min} = 0.33\,\text{MeV}$ (dotted) and grow as m_χ at large mass. The heavy-mediator curve ($F_{\rm DM} = 1$) reaches its minimum at 9.7 MeV, where the allowed momenta cover the cutoff; the light-mediator curve ($F_{\rm DM} = (q_0/q)^2$), dominated by the smallest q, reaches its minimum at 2.1 MeV. The shaded region is excluded by the heavy-mediator search.

around 10 MeV; the curve falls over this whole range. Above about 10 MeV the integral is constant, $\mu_{\chi e} = m_e$, and only the factor m_χ remains: $\bar{\sigma}_e^{\lim} \propto m_\chi$, a line of slope one on logarithmic axes. Heavier dark matter is less numerous at fixed ρ_χ.

The light-mediator slope. With $F_{\rm DM} = (q_0/q)^2$ the weight $(q_0/q)^4\, q\, \mathrm{d}q$ favors the smallest momentum transfer that can deposit the threshold, $q \approx E_{\rm th}/v_{\max}$ (Sect. 1.3). That momentum transfer is reachable as soon as m_χ is a few times $m_{\rm min}$, and the light-mediator curve turns over close to the edge, growing as m_χ from a few MeV upward. Because the light-mediator rate comes from small q, the light-mediator limit is also the one most affected by screening (Sect. 4.4) and by the low-energy end of the ionization yield.

As a result, the two curves are not related by a constant factor, and a limit quoted for one mediator cannot be rescaled to the other without the response. In the toy of Fig. 1.9 the heavy-mediator limit at 1 MeV stands a factor of 49 above its minimum at 9.7 MeV, while the light-mediator limit is already within a factor of 1.5 of its minimum at 2.1 MeV. Real exclusion plots add curves from several experiments and the benchmark cross-sections of thermal and freeze-in models (Worked example 1.5).

1.5.5 Inputs of a limit

Every exclusion curve combines measured inputs from the detector with theory inputs, and Fig. 1.8 marks the boundary between the two kinds of input. The measured inputs belong to the detector: the exposure $M_{\rm det} t_{\rm exp}$, the threshold, the efficiency with which events pass the selection, the background and the observed count. Their errors are statistical or come from calibration, and experiments propagate them.

The theory inputs are the halo parameters, the particle-physics model and the response of the target. The halo parameters ρ_χ, v_0 and $v_{\rm esc}$ are fixed by convention to the values of Sect. 1.2 so that experiments can be compared (Appendix A.8); the limit scales as $1/\rho_\chi$, and $v_{\rm esc}$ moves the edge (Sect. 1.3.5). The particle physics enters only through $F_{\rm DM}$, which is a definition of the model rather than an approximation (Sect. 1.4). The target enters through $S(q,\omega)$ and through the conversion from ω to quanta, $N_{\rm eh}(\omega)$ for charge detectors. The response and the yield are both computed, not measured, in the eV range that matters for light dark matter. Their errors come from approximations, such as the treatment of exchange and correlation in the band structure, the neglect of screening or of phonons, or the step model of Eq. (1.42). A longer exposure does not reduce these errors, and their effect on a limit follows from Eq. (1.53). A fractional error in $S(q,\omega)$ at the momentum transfers that dominate the integral moves the limit by the same fraction. An error in the gap or in the threshold energy moves the edge, $\delta \ln m_{\rm min} = \delta \ln E_{\rm th}$ (Sect. 1.3.5), and near the edge it changes the limit by a much larger factor, because η varies steeply there. Together these errors make up the *theory uncertainty* of a limit.

Summary

The event rate is an integral over a product of three factors, the halo, the coupling $|\tilde{V}(\boldsymbol{q})|^2$ and the dynamic structure factor $S(\boldsymbol{q},\omega)$ of the target, tied together by energy conservation (Sect. 1.1). The particle physics and the condensed-matter physics therefore separate, and most of the book computes S. Gravity measures the density of dark matter but not its mass: a local density of a few tenths of a GeV cm^{-3} (Sect. 1.2) and a cosmological density $\Omega_{\rm c} h^2 = 0.120$ from the cosmic microwave background. The mass is open over some 90 decades, and the keV–GeV window is the range where the kinetic energy of halo dark matter, $\frac{1}{2} m_\chi v^2$ with $v \approx 10^{-3}$, matches the meV–eV excitations of solids. A thermal relic needs

$\langle\sigma v\rangle \approx 2\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$ to $3\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$, and below a few GeV this cross-section requires a light mediator; freeze-in is the main alternative (Sect. 1.4).

The local halo is described by the standard halo model (Sect. 1.2): a truncated Maxwell–Boltzmann distribution with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $v_0 = 238\,\mathrm{km\,s^{-1}}$ and $v_\mathrm{esc} = 544\,\mathrm{km\,s^{-1}}$, seen in the laboratory through the Earth's motion. The largest laboratory speed is $v_\mathrm{max} = 794.5\,\mathrm{km\,s^{-1}}$. The Earth's orbit modulates the rate annually, with a phase that reverses at $v_\mathrm{min} \approx 211\,\mathrm{km\,s^{-1}}$, and the Earth's rotation turns a crystal through the dark-matter wind once per sidereal day.

Kinematics (Sect. 1.3) relates the momentum q and energy ω given to the target through $\omega_{\boldsymbol{q}} = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi - \Delta_\chi$, and every scattering rate contains $\eta(v_\mathrm{min})$, the mean inverse speed above the minimum speed v_min. A threshold E_th is reachable only above $m_\mathrm{min} = 2E_\mathrm{th}/v_\mathrm{max}^2$, 0.33 MeV for the low-temperature silicon gap. Near m_min the rate vanishes quadratically with the tail of the halo. Electron transitions involve momentum transfers comparable to $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, close to the shortest reciprocal-lattice vector of silicon, and free nuclear recoils deposit little energy for sub-GeV dark matter.

A mediator of mass m_V produces the potential $\tilde{V}(q) = y_\chi y_e/(q^2 + m_V^2)$ (Sect. 1.4). Its strength is quoted as the vacuum reference cross-section $\bar{\sigma}_e$ and its momentum dependence as $F_\mathrm{DM}(q)$, equal to 1 for a heavy and $(q_0/q)^2$ for a light mediator. The dark photon is the standard example; bosonic dark matter can instead be absorbed, depositing $\omega \approx m_\phi$ at $q \approx 0$.

A detector counts electron–hole pairs (3.6 eV to 3.75 eV each in silicon), phonons, quasiparticles (about 0.36 meV per pair in bulk aluminum) or photons (Sect. 1.5). The rate is linear in $\bar{\sigma}_e$ and ρ_χ and falls as $1/m_\chi$ well above threshold. With no events and no background the 90 % upper limit is 2.3 expected events, and the cross-section limit improves in proportion to the exposure. Exclusion curves have a vertical edge at m_min and grow as m_χ at large mass. Their accuracy rests on the computed response of the target, which Chaps. 2 to 8 develop. The errors of that computation, the theory uncertainty, do not shrink with exposure, and the book estimates them with each approximation it makes.

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Chapter 2

Transition rates

About 10^8 dark-matter particles of 100 MeV cross each square centimeter of a detector every second (Worked example 1.2), each with at most 350 eV of kinetic energy (Sect. 1.3), and Eq. (1.46) turns them into an event rate that Chap. 1 quoted without derivation. Deriving the event rate is a question for quantum mechanics: how often does a weak interaction move a kilogram of silicon, with some 10^{26} electrons, from its ground state into one of its excited states? Scattering theory for two particles in vacuum answers only part of this question, because a solid is not a free electron. The excited states of a solid are states of many identical electrons, some of them can be reached only through an intermediate state, with a phonon completing the transition, and at any finite temperature the crystal can also hand energy back to the dark matter. Bosonic dark matter raises a related question, why its absorption rate depends on the local density and not on the flux. The answers need Fermi's golden rule and four extensions of it: the Born approximation, second quantization, second-order perturbation theory and the thermal average. With the golden rule and these extensions we derive the cross-sections and form factors behind the reference quantities of Sect. 1.4; Fig. 2.1 shows how each extension builds on the golden rule.

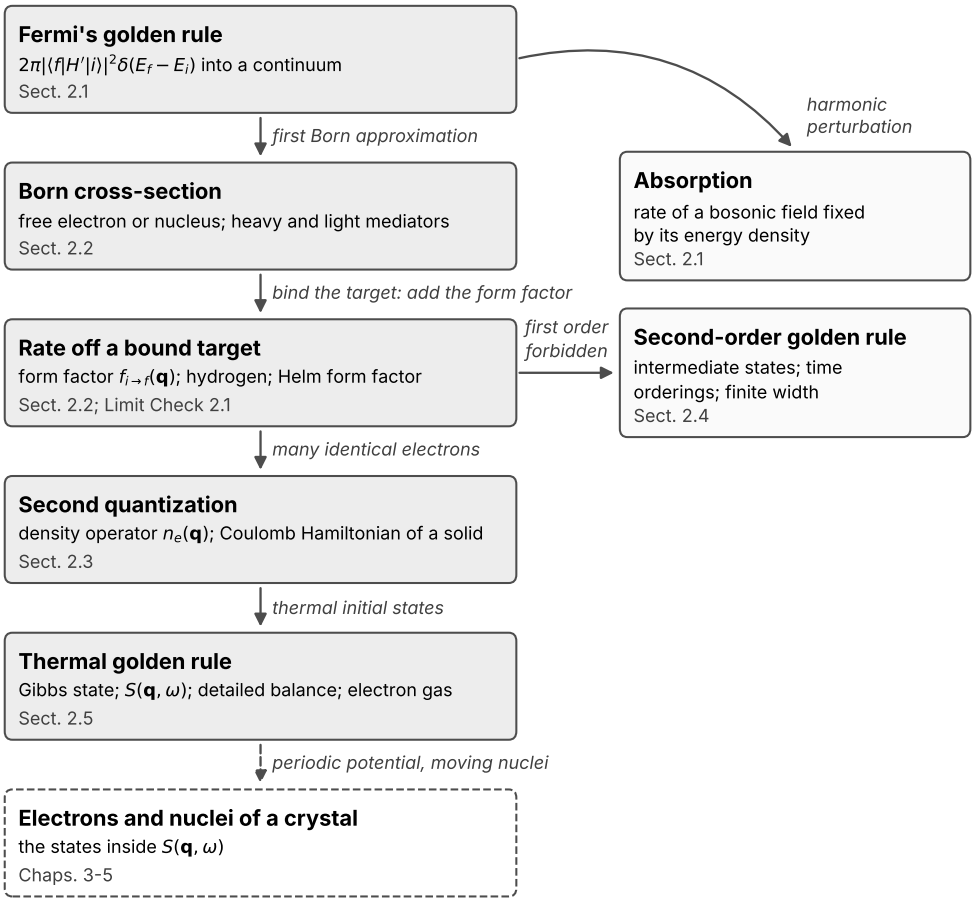


Figure 2.1: From Fermi's golden rule to the dynamic structure factor. Arrows name the step between levels: the first Born approximation gives the cross-section on a free electron or nucleus (Sects. 2.1 and 2.2); binding the target adds the transition form factor $f_{i\to f}(\boldsymbol{q})$, checked on the ionization of hydrogen (Sect. 2.2; Limit Check 2.1); identical electrons call for occupation numbers and the density operator $n_e(\boldsymbol{q})$ (Sect. 2.3); thermal initial states turn the rate into the dynamic structure factor $S(\boldsymbol{q}, \omega)$ with detailed balance (Sect. 2.5). Side branches: a harmonic perturbation gives the absorption rate of a bosonic field, set by its energy density (Sect. 2.1), and a transition forbidden at first order proceeds through an intermediate state (Sect. 2.4). The dashed box is the crystal whose states fill S (Chaps. 3 to 5).

2.1 Fermi's golden rule

In the toy rate of Worked example 1.6, dark matter of 100 MeV with $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$ produces 73 events in a kilogram of silicon in one year, and twice as many in two years: the count is a rate times an exposure (Eq. (1.44)). Quantum mechanics gives probabilities, not rates. A probability that grows linearly in time appears only when a weak perturbation couples an initial state to a continuum of final states, and only within a window of times. The derivation of Fermi's golden rule follows Refs. [1, Chap. 5] and [2, Chap. XIII].

2.1.1 First-order amplitude

The target and a distant dark-matter particle evolve under an unperturbed Hamiltonian H_0 with eigenstates $|n\rangle$ and energies E_n. At $t = 0$ a perturbation $H'(t)$ is switched on, and the system starts in the eigenstate $|i\rangle$. We expand the state in the unperturbed basis, with the free phases taken out:

$$|\psi(t)\rangle = \sum_n c_n(t)\, \mathrm{e}^{-\mathrm{i}E_n t}\, |n\rangle, \qquad c_n(0) = \delta_{ni}.$$

When this expansion is inserted into $\mathrm{i}\,\partial_t|\psi\rangle = (H_0 + H')|\psi\rangle$, the time derivative of each phase cancels the action of H_0. Projecting onto $\mathrm{e}^{-\mathrm{i}E_f t}|f\rangle$ leaves an exact set of equations for the amplitudes,

$$\mathrm{i}\,\dot{c}_f(t) = \sum_n \langle f|H'(t)|n\rangle\, \mathrm{e}^{\mathrm{i}(E_f - E_n)t}\, c_n(t). \tag{2.1}$$

To first order in H' we replace $c_n(t)$ on the right by its initial value δ_{ni} and integrate. For $f \neq i$,

$$c_f(t) = -\mathrm{i}\int_0^t \mathrm{d}t'\, \langle f|H'(t')|i\rangle\, \mathrm{e}^{\mathrm{i}(E_f - E_i)t'}, \tag{2.2}$$

and the transition probability is $P_{i\to f}(t) = |c_f(t)|^2$. The replacement is consistent only while the initial state is barely depleted, $\sum_{f\neq i} P_{i\to f}(t) \ll 1$. Eq. (2.2) is the first term of the Dyson series of the interaction picture; Sect. 2.4 takes the second.

2.1.2 Constant perturbation

Let H' be independent of time for $t > 0$. Dark-matter scattering has a time-independent perturbation: the interaction H_{int} does not depend on time, and energy passes between the dark matter and the target, whose kinetic and internal energies are both in E_i and E_f. With $\xi = E_f - E_i$ the integral in Eq. (2.2) is elementary, $c_f(t) = -\langle f|H'|i\rangle(\mathrm{e}^{\mathrm{i}\xi t} - 1)/\xi$. Since $|\mathrm{e}^{\mathrm{i}\xi t} - 1|^2 = 2 - 2\cos\xi t = 4\sin^2(\xi t/2)$,

$$P_{i\to f}(t) = 4\,|\langle f|H'|i\rangle|^2\, \frac{\sin^2(\xi t/2)}{\xi^2}. \tag{2.3}$$

All the time dependence is in the function $\sin^2(\xi t/2)/\xi^2$, drawn in Fig. 2.2(a). At $\xi = 0$ the function equals $t^2/4$, and its zeros are at $\xi = 2\pi k/t$ with k a nonzero integer, so the central peak has width $4\pi/t$. The substitution $u = \xi t/2$ gives the area under the function,

$$\int_{-\infty}^{\infty} \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2} = \frac{t}{2}\int_{-\infty}^{\infty} \mathrm{d}u\, \frac{\sin^2 u}{u^2} = \frac{\pi t}{2},$$

with $\int \mathrm{d}u\, \sin^2 u/u^2 = \pi$ (Appendix C.1). The peak height grows as t^2, the width shrinks as $1/t$, and the area grows as t; the central peak holds 90.3 % of the area. Integrated against

any function $w(\xi)$ that varies little over a range $2\pi/t$, $\sin^2(\xi t/2)/\xi^2$ returns $(\pi t/2)\,w(0)$, with a relative error $\sim 1/(t\,\Delta_{\mathrm{w}})$, where Δ_{w} is the range over which w varies. In the sense of distributions,

$$\frac{\sin^2(\xi t/2)}{\xi^2} \;\longrightarrow\; \frac{\pi t}{2}\,\delta(\xi) \qquad (t\to\infty), \tag{2.4}$$

which Appendix C.1 proves. Applied to Eq. (2.3), the limit Eq. (2.4) gives a probability that grows linearly in time, $P_{i\to f}(t)\to 2\pi t\,|\langle f|H'|i\rangle|^2\,\delta(E_f-E_i)$. The same limit can be taken after differentiating. The time derivative of Eq. (2.3) is $2|\langle f|H'|i\rangle|^2\sin(\xi t)/\xi$, a function of ξ whose area is π at every $t>0$ and whose width shrinks as $1/t$; it tends to $\pi\delta(\xi)$. Since the area stays fixed while the width shrinks, the probability per unit time stops depending on t; a probability per unit time independent of t is what we mean by a rate. The transition rate is

Fermi's golden rule

$$\Gamma_{i\to f} = 2\pi\,|\langle f|H'|i\rangle|^2\,\delta(E_f-E_i). \tag{2.5}$$

The rate is defined only after summation over a continuum of final states f, for times in the window of Eq. (2.7).

In natural units the rate is an energy: $|\langle f|H'|i\rangle|^2$ carries eV^2 and the delta function eV^{-1}. Dividing by $\hbar$ converts the rate to a frequency: a rate of 10 meV is $1.52\times10^{13}\,\mathrm{s}^{-1}$. The delta function is energy conservation, enforced to within about $2\pi/t$ after a time t. For a dark-matter particle it becomes $\delta(\omega-\omega_{\boldsymbol{q}})$ of Sect. 1.3 once the dark-matter part of E_f-E_i is written out (Sect. 2.2).

The sudden switch-on at $t=0$ adds a transient but does not change the rate. A perturbation that is turned on smoothly over a time τ_{s} replaces $\sin^2(\xi t/2)/\xi^2$ by a function whose tails fall off faster in ξ. The area under the new function still grows as $\pi t/2$ once $t\gg\tau_{\mathrm{s}}$, because the area counts only the time during which the perturbation acts at full strength.

2.1.3 Conditions for a rate

For a single discrete final state, however, Eq. (2.3) does not grow linearly. Off resonance it oscillates below $4|\langle f|H'|i\rangle|^2/\xi^2$; exactly on resonance it grows as t^2 until first order fails. A rate requires a continuum. Let the *density of final states* $\mathcal{D}_f(E)$ be such that $\mathcal{D}_f(E)\,\mathrm{d}E$ is the number of final states with energies in $[E,E+\mathrm{d}E]$, and let $\overline{|\langle f|H'|i\rangle|^2}(E)$ be the average of the squared matrix element over those states. The total transition probability is then

$$\sum_f P_{i\to f}(t) = \int \mathrm{d}E\,\mathcal{D}_f(E)\;\overline{|\langle f|H'|i\rangle|^2}(E)\;\frac{4\sin^2[(E-E_i)t/2]}{(E-E_i)^2}.$$

By Eq. (2.4) it becomes $\Gamma_i t$, with the total rate

$$\Gamma_i = 2\pi\;\overline{|\langle f|H'|i\rangle|^2}(E_i)\;\mathcal{D}_f(E_i). \tag{2.6}$$

The time t must lie in a window. Let E_{w} be the energy scale over which the product $\mathcal{D}_f\,\overline{|\langle f|H'|i\rangle|^2}$ changes appreciably: the width of the continuum, or of a structure within it. The delta-function limit needs the peak to be narrow on that scale, $t\gg 1/E_{\mathrm{w}}$. The sum over discrete levels becomes an integral only when the peak covers many levels. With a level spacing of $1/\mathcal{D}_f$, the peak covers many levels for $t\ll 2\pi\mathcal{D}_f$. First-order perturbation theory requires $\Gamma_i t\ll 1$. Together, the three conditions are

$$\frac{1}{E_{\mathrm{w}}} \ll t \ll \min\!\left(\frac{1}{\Gamma_i},\; 2\pi\mathcal{D}_f\right). \tag{2.7}$$

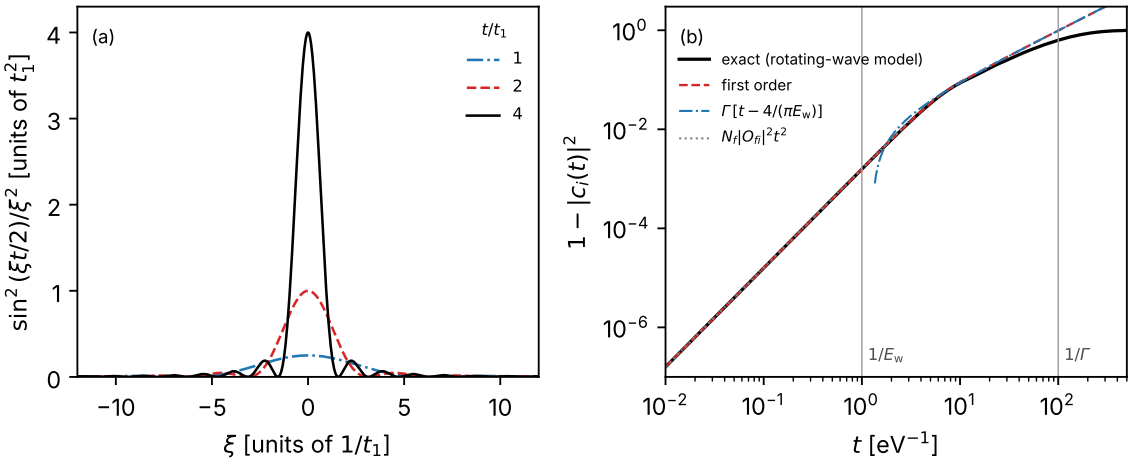


Figure 2.2: (a) The function $\sin^2(\xi t/2)/\xi^2$ of the first-order transition probability, Eq. (2.3), against the energy mismatch $\xi = E_f - E_i$, for times $t = t_1$, $2t_1$ and $4t_1$, where t_1 is an arbitrary unit of time ($\hbar = 1$). The peak height grows as t^2, the width as $1/t$, and the area, $\pi t/2$, as t. (b) Depletion $1 - |c_i(t)|^2$ of a discrete state coupled to a flat continuum by the absorption term of a harmonic perturbation, for the model of Worked example 2.1. The model has 2000 final states spread uniformly over a width $E_w = 1\,\text{eV}$ centered on resonance, all with the same matrix element, chosen so that the golden-rule rate is $\Gamma = 10\,\text{meV}$. Solid: exact solution of this model with the counter-rotating (emission) term dropped, by diagonalization of the 2001-level Hamiltonian in the frame rotating with the perturbation. Dashed: first-order probability, Eq. (2.9). Dash-dotted: its linear long-time form. Dotted: its quadratic short-time form. Vertical lines mark $t = 1/E_w$ (0.66 fs) and $t = 1/\Gamma$ (66 fs).

In dark-matter scattering, however, the continuum is always present, even for a target with discrete levels, such as an atom below its ionization threshold. The final state contains the scattered dark-matter particle, whose momentum ranges over a continuum, and $\mathcal{D}_f$ counts target levels and dark-matter momenta together. Energy conservation then selects, for each target level, a shell of outgoing momenta, and the delta function of Eq. (2.5) is integrated over that shell rather than over the target spectrum (Sect. 2.2).

The upper bound $2\pi\mathcal{D}_f$ grows with the volume of the system, because $\mathcal{D}_f$ does (Eq. (2.11)); the bound drops out when the infinite-volume limit is taken before the long-time limit, as we do throughout. The window then exists whenever $\Gamma_i \ll E_\mathrm{w}$: the transition must be slow compared with the inverse energy scale of the continuum.

At times beyond $1/\Gamma_i$ the initial state is depleted. The exact evolution then becomes exponential decay at the same rate Γ_i (Wigner–Weisskopf; Sect. 2.4), and Γ_i keeps its meaning beyond the first-order window. For a dark-matter particle crossing a detector the condition $\Gamma_i t \ll 1$ holds by a wide margin, because the probability of any scattering during the passage is the small number that makes the searches of Sect. 1.5 difficult.

The golden rule also fails when the first-order matrix element vanishes and when the perturbation shifts the energy of the initial state. If $\langle f|H'|i\rangle$ vanishes for every final state at energy E_i, by a selection rule or because no final state exists at that energy, Eq. (2.6) vanishes, and the leading rate comes from the second-order amplitude through intermediate states (Sect. 2.4). If the perturbation shifts the energy of $|i\rangle$ itself, by $\langle i|H'|i\rangle \neq 0$, the delta function must be evaluated at the shifted energy; the shift is of first order, while the rate is of second order in H'. For dark-matter scattering the diagonal element is the $\boldsymbol{q} = 0$ term of Eq. (2.12), $\tilde{V}(\mathbf{0})/\mathcal{V}$, which for a massive mediator vanishes as the volume grows, and the shift plays no role. For a massless mediator $\tilde{V}(\mathbf{0})$ itself diverges; in a neutral target the charges cancel this term, as Sect. 2.3 shows for the Coulomb interaction.

2.1.4 Harmonic perturbations

An absorbed bosonic dark-matter field oscillates in time at its mass (Sect. 1.4), and an applied field oscillates at its frequency. An oscillating field gives a harmonic perturbation,

$$H'(t) = \hat{O}\,\mathrm{e}^{-\mathrm{i}\omega t} + \hat{O}^\dagger\,\mathrm{e}^{\mathrm{i}\omega t}, \qquad \omega > 0,$$

where $\hat{O}$ is an operator on the target and the second term makes H' Hermitian. Eq. (2.2) now has two terms of the static form, with ξ shifted by $\mp\omega$:

$$c_f(t) = -\langle f|\hat{O}|i\rangle\,\frac{\mathrm{e}^{\mathrm{i}(\xi-\omega)t}-1}{\xi-\omega} - \langle f|\hat{O}^\dagger|i\rangle\,\frac{\mathrm{e}^{\mathrm{i}(\xi+\omega)t}-1}{\xi+\omega}.$$

The first term peaks at $\xi = \omega$, the second at $\xi = -\omega$, each with width of order $2\pi/t$. For $\omega t \gg 1$ the peaks do not overlap. The cross term in $|c_f|^2$ is then bounded in time instead of growing, and each squared term turns into a delta function by Eq. (2.4):

$$\Gamma_{i\to f} = 2\pi\,|\langle f|\hat{O}|i\rangle|^2\,\delta(E_f - E_i - \omega) + 2\pi\,|\langle f|\hat{O}^\dagger|i\rangle|^2\,\delta(E_f - E_i + \omega). \tag{2.8}$$

The first term is absorption: the target gains the energy ω. The second is emission: the target gives ω to the field, so the energy given to the target is $-\omega$. With the temporal Fourier convention of Appendix A.3, $f(\omega) = \int \mathrm{d}t\,\mathrm{e}^{+\mathrm{i}\omega t} f(t)$, the factor $\mathrm{e}^{-\mathrm{i}\omega t}$ is the component of H' at positive frequency, and positive frequency means energy given to the target. Since $|\langle f|\hat{O}^\dagger|i\rangle| = |\langle i|\hat{O}|f\rangle|$, emission from i to f and absorption from f to i have the same matrix element; Sect. 2.5 combines this equality of matrix elements with thermal populations into detailed balance.

For a classical field the emission term is stimulated emission, driven by the field itself. When the field is a quantized mode, the dark-matter or phonon state is part of $|i\rangle$ and $|f\rangle$, the perturbation is static, and Eq. (2.5) applies to the combined system. Absorption removes a quantum and emission adds one, with matrix elements proportional to $\sqrt{n}$ and $\sqrt{n+1}$ for a mode holding n quanta (Sects. 2.3 and 2.5). The extra 1 is spontaneous emission, which a classical field misses. At $\omega = 0$ the two peaks coincide, the cross term no longer averages out, and Eq. (2.8) reduces to Eq. (2.5) with $H' = \hat{O} + \hat{O}^\dagger$.

The harmonic form explains why the absorption rates of Sect. 1.4 depend on the local density of dark matter and not on its flux. Let a bosonic field of mass m_ϕ make up the local dark matter. The field oscillates as $\phi(t) = \phi_0 \cos(m_\phi t)$, with energy density $\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}m_\phi^2\phi^2 = \frac{1}{2}m_\phi^2\phi_0^2 = \rho_\chi$. Its spatial phase $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$ changes over a length $1/|\boldsymbol{p}_\phi| \approx 1/(m_\phi v)$, a thousand times the Compton wavelength $1/m_\phi$, and we neglect the spatial phase here. An interaction linear in the field, $H' = \phi(t)$ times a target operator, is harmonic with $\omega = m_\phi$, and $\hat{O}$ is $\phi_0/2$ times that operator. The absorption term of Eq. (2.8) is therefore proportional to $\phi_0^2 = 2\rho_\chi/m_\phi^2$: the rate is fixed by the energy density of the field, and the speed of the dark matter does not enter. Chap. 8 evaluates the target operators for each boson.

Worked example 2.1: A harmonic perturbation coupled to a flat continuum

A discrete state $|i\rangle$ is coupled by the absorption term $\hat{O}\,\mathrm{e}^{-\mathrm{i}\omega t}$ to N_f final states spread uniformly over an energy width E_w centered on the resonance $E_f = E_i + \omega$. The density of final states is $\mathcal{D}_f = N_f/E_\mathrm{w}$, and every final state has the same $|\langle f|\hat{O}|i\rangle|^2 \equiv |O_{fi}|^2$. With $\xi_f = E_f - E_i - \omega$, the first-order depletion is Eq. (2.3) with $H' \to \hat{O}$ and $\xi \to \xi_f$, summed over f. For $t \ll 2\pi\mathcal{D}_f$ the sum is an integral, and the substitution $u = \xi t/2$ gives

$$P(t) = 4|O_{fi}|^2\mathcal{D}_f \int_{-E_\mathrm{w}/2}^{E_\mathrm{w}/2} \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2} = 4|O_{fi}|^2\mathcal{D}_f\, t \int_0^{E_\mathrm{w}t/4} \mathrm{d}u\, \frac{\sin^2 u}{u^2}.$$

Integration by parts, with $\mathrm{d}(-\sin^2 u/u)/\mathrm{d}u = \sin^2 u/u^2 - \sin 2u/u$, gives $\int_0^U \mathrm{d}u\, \sin^2 u/u^2 = \mathrm{Si}(2U) - \sin^2 U/U$, where $\mathrm{Si}(z) = \int_0^z \mathrm{d}s\, \sin s/s$ is the sine integral. Hence

$$P(t) = 4|O_{fi}|^2\mathcal{D}_f\, t \left[\mathrm{Si}\left(\frac{E_\mathrm{w}t}{2}\right) - \frac{\sin^2(E_\mathrm{w}t/4)}{E_\mathrm{w}t/4}\right]. \tag{2.9}$$

At short times, $E_\mathrm{w}t \ll 1$, the expansions $\mathrm{Si}(z) = z - z^3/18 + \ldots$ and $\sin^2 U/U = U - U^3/3 + \ldots$ make the bracket $E_\mathrm{w}t/4$, so $P(t) = N_f|O_{fi}|^2t^2$. Every final state is populated, as if each were resonant: the energy is not yet resolved. At long times, $E_\mathrm{w}t \gg 1$, the asymptotic form $\mathrm{Si}(z) = \pi/2 - \cos z/z + O(z^{-2})$ and the identity $2\cos 2U + 4\sin^2 U = 2$ make the bracket $\pi/2 - 2/(E_\mathrm{w}t)$, so

$$P(t) = \Gamma\left(t - \frac{4}{\pi E_\mathrm{w}}\right), \qquad \Gamma = 2\pi|O_{fi}|^2\mathcal{D}_f,$$

up to terms that decay as $1/t$. We see that the probability grows linearly at the golden-rule rate Eq. (2.6), with a delay of $4/(\pi E_\mathrm{w})$, the time the system needs to resolve the energy. For numbers we take $E_\mathrm{w} = 1\,\mathrm{eV}$, $N_f = 2000$ (level spacing $0.5\,\mathrm{meV}$, $\mathcal{D}_f = 2000\,\mathrm{eV}^{-1}$) and $|O_{fi}| = 0.892\,\mathrm{meV}$, for which $\Gamma = 10\,\mathrm{meV}$. With $\hbar = 6.582\times10^{-16}\,\mathrm{eV\,s}$, the time scales are $1/E_\mathrm{w} = 0.66\,\mathrm{fs}$, the delay $4/(\pi E_\mathrm{w}) = 0.84\,\mathrm{fs}$, $1/\Gamma = 66\,\mathrm{fs}$ and $2\pi\mathcal{D}_f = 8.3\,\mathrm{ps}$, so the window Eq. (2.7) spans two orders of magnitude, from 0.66 fs to 66 fs. Fig. 2.2(b) compares Eq. (2.9) with the exact evolution of the same model, obtained by solving Eq. (2.1) with the emission term dropped. The substitution $\tilde{c}_f = c_f\,\mathrm{e}^{\mathrm{i}\xi_f t}$ (and $\tilde{c}_i = c_i$)

removes every time-dependent phase from Eq. (2.1), since the absorption term carries $\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}(E_f-E_i)t} = \mathrm{e}^{\mathrm{i}\xi_f t}$; the substitution is a change to a frame that rotates with the perturbation. In the rotating frame the coefficients are constant and the model is a time-independent Hamiltonian of 2001 levels, which we diagonalize numerically. First order exceeds the exact depletion by 10 % at $t = 21.9\,\mathrm{eV}^{-1}$, where $\Gamma t = 0.22$; at $t = 1/\Gamma$ it predicts 0.99 against the exact 0.63. First order therefore breaks down at $\Gamma t \approx 0.2$. At later times the exact depletion follows $1 - \mathrm{e}^{-\Gamma t}$, within 0.4 % from $t = 1/\Gamma$ on. The function $1 - \mathrm{e}^{-\Gamma t}$ is the exponential decay of Sect. 2.4 with the golden-rule rate as its decay constant.

2.1.5 Box normalization and final-state sums

With plane waves normalized in infinite volume, $\langle \boldsymbol{p}|\boldsymbol{p}'\rangle = (2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{p}')$, momentum conservation in a matrix element is a delta function, and Eq. (2.5) squares the delta function. The square $[\delta^3(\boldsymbol{P}_f - \boldsymbol{P}_i)]^2$ is undefined. A finite box makes every step well defined: momenta are discrete, momentum conservation is a Kronecker delta, whose square is itself, and the golden rule needs countable states to sum over. We therefore place the system in a cube of volume $\mathcal{V} = L^3$ with periodic boundary conditions. The plane waves are $\langle \boldsymbol{x}|\boldsymbol{p}\rangle = \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}/\sqrt{\mathcal{V}}$, with $\boldsymbol{p} = 2\pi\boldsymbol{n}/L$ and $\boldsymbol{n}$ a vector of integers, and they obey $\langle \boldsymbol{p}|\boldsymbol{p}'\rangle = \delta_{\boldsymbol{p}\boldsymbol{p}'}$: each state holds one particle in the volume $\mathcal{V}$. The script $\mathcal{V}$ always denotes this volume; an italic V is a potential and always carries its argument, $V(\boldsymbol{x})$, or a tilde, $\tilde{V}(\boldsymbol{q})$. Since each allowed momentum occupies a cell of volume $(2\pi)^3/\mathcal{V}$, for any function that varies little over $2\pi/L$ [3, Eqs. (A1) and (A2)]

$$\sum_{\boldsymbol{p}} w(\boldsymbol{p}) \;\to\; \mathcal{V}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\, w(\boldsymbol{p}), \qquad \delta_{\boldsymbol{p}\boldsymbol{p}'} \;\to\; \frac{(2\pi)^3}{\mathcal{V}}\,\delta^3(\boldsymbol{p}-\boldsymbol{p}'). \tag{2.10}$$

Appendix A.3 collects these conventions and Appendix C.2 the continuum limit of sums over momenta, including sums over the Brillouin zone. For one free particle of mass m with energy $E = p^2/2m$, the number of states in a shell is $\mathcal{V}\,4\pi p^2\,\mathrm{d}p/(2\pi)^3$, and with $\mathrm{d}E = p\,\mathrm{d}p/m$ we find

$$\mathcal{D}_f(E) = \frac{\mathcal{V}\,m\,p}{2\pi^2}, \qquad p = (2mE)^{1/2}. \tag{2.11}$$

The density of final states is proportional to $\mathcal{V}$. In a crystal the box is a block of N unit cells, and the same counting applies to crystal momenta (Sect. 3.1). Discrete quantum numbers, such as spin, are summed separately in the final state. When the initial state is not known exactly, the rate is averaged over it: over the two spin states of an unpolarized particle, and over a thermal ensemble of target states (Sect. 2.5).

The volume must drop out of every observable, and in a finite box we can follow it through every rate as a check. Consider a dark-matter particle and an electron, both box-normalized, interacting through a potential $V(\boldsymbol{x}_\chi - \boldsymbol{x}_e)$ of their separation (Sect. 1.4). Writing $\boldsymbol{r} = \boldsymbol{x}_\chi - \boldsymbol{x}_e$ and integrating over $\boldsymbol{x}_e$ at fixed $\boldsymbol{r}$ gives

$$\langle \boldsymbol{p}'_\chi, \boldsymbol{p}'_e|V(\boldsymbol{x}_\chi - \boldsymbol{x}_e)|\boldsymbol{p}_\chi, \boldsymbol{p}_e\rangle = \frac{1}{\mathcal{V}^2}\int \mathrm{d}^3x_e\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{P}_f-\boldsymbol{P}_i)\cdot\boldsymbol{x}_e}\int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,V(\boldsymbol{r}) = \frac{\tilde{V}(\boldsymbol{q})}{\mathcal{V}}\,\delta_{\boldsymbol{P}_f\boldsymbol{P}_i}, \tag{2.12}$$

with $\boldsymbol{q} = \boldsymbol{p}_\chi - \boldsymbol{p}'_\chi$ the momentum transferred to the electron, $\boldsymbol{P}_{i,f}$ the initial and final total momenta, and $\tilde{V}(\boldsymbol{q})$ the transform of Appendix A.3. Each plane wave brings $\mathcal{V}^{-1/2}$ and the integral over the position of the pair brings $\mathcal{V}$ times momentum conservation. In Eq. (2.5) the squared matrix element brings $1/\mathcal{V}^2$, and the sum over one final momentum brings $\mathcal{V}$,

since $\delta_{\boldsymbol{P}_f \boldsymbol{P}_i}$ fixes the other final momentum. We see that the rate is proportional to $1/\mathcal{V}$: it is the rate at which two particles placed in the same box find each other.

A cross-section removes the factor $1/\mathcal{V}$. Since the incident particle has density $1/\mathcal{V}$ and speed v relative to the target, the flux of incident particles is $v/\mathcal{V}$ (Eq. (1.18), with a single velocity), and the rate per target particle is that flux times the cross-section σ_T for scattering on one target particle. Hence

$$\sigma_T = \frac{\mathcal{V}}{v}\,\Gamma = \frac{\mathcal{V}}{v}\sum_f 2\pi\,|\langle f|H'|i\rangle|^2\,\delta(E_f - E_i), \tag{2.13}$$

which is independent of $\mathcal{V}$. The rate in a detector follows by reversing the argument. Dark matter with velocity in d^3v crosses a target particle with flux $n_\chi f_\chi(\boldsymbol{v})\,\mathrm{d}^3v\,v$ (Sect. 1.2). The number of target particles per unit mass is $1/m_T$, where m_T is the mass of one target particle. Integrating over velocities and multiplying by $1/m_T$ leads to an event rate per unit mass $(\rho_\chi/m_\chi m_T)\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\, v\,\sigma_T(v)$, the structure that Sect. 1.5 assembled for a target with a response.

Particle-physics papers use relativistic normalization, $\langle \boldsymbol{p}|\boldsymbol{p}'\rangle_{\mathrm{rel}} = 2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{p}')$ [4, Chap. 2]; we use it again only to compare with the literature (Sect. 2.4 and Appendix A.9). In the box, $(2\pi)^3\delta^3(\boldsymbol{0}) \to \mathcal{V}$ by Eq. (2.10), and $\langle \boldsymbol{p}|\boldsymbol{p}\rangle_{\mathrm{rel}} = 2E_{\boldsymbol{p}}\mathcal{V}$: the state holds $2E_{\boldsymbol{p}}$ particles per unit volume, and $|\boldsymbol{p}\rangle_{\mathrm{rel}} = (2E_{\boldsymbol{p}}\mathcal{V})^{1/2}|\boldsymbol{p}\rangle$. At first order the invariant amplitude $\mathcal{M}$ of a process $1+2\to 3+4$ is defined by $\langle f|H'|i\rangle_{\mathrm{rel}} = -\mathcal{M}\,(2\pi)^3\delta^3(\boldsymbol{P}_f - \boldsymbol{P}_i)$, the first-order term of $\langle f|\mathrm{i}\hat{T}|i\rangle = \mathrm{i}(2\pi)^4\delta^4(P_f - P_i)\,\mathcal{M}$ once the time integral has produced $2\pi\delta(E_f - E_i)$ [4, Chap. 4]. Dividing by the four normalization factors and using $(2\pi)^3\delta^3 \to \mathcal{V}\,\delta_{\boldsymbol{P}_f\boldsymbol{P}_i}$ gives the box matrix element $\langle f|H'|i\rangle = -\mathcal{M}\,\delta_{\boldsymbol{P}_f\boldsymbol{P}_i}/[\mathcal{V}\prod_{j=1}^4(2E_j)^{1/2}]$. Inserting the box matrix element into Eq. (2.13), and turning the sum over $\boldsymbol{p}_3$ into an integral, gives

$$\sigma_T = \frac{1}{2E_1\,2E_2\,v}\int \frac{\mathrm{d}^3p_3}{(2\pi)^3\,2E_3}\,\frac{2\pi}{2E_4}\,|\mathcal{M}|^2\,\delta(E_f - E_i),$$

the relativistic formula with its flux factor and two-body phase space [4, Chap. 4]. The relativistic normalization puts factors $2E$ into the states, and these factors reappear as $1/2E$ in the flux and the phase space. We use box normalization throughout, because a crystal is not Lorentz invariant and its states are naturally normalized to one per crystal volume. In the nonrelativistic limit $2E_j \to 2m_j$, and for elastic dark-matter–electron scattering comparison with Eq. (2.12) leads to $|\mathcal{M}|^2 = 16m_\chi^2 m_e^2|\tilde{V}(\boldsymbol{q})|^2$. The definition of $\bar{\sigma}_e$ in Eq. (1.32) then takes the form $\mu_{\chi e}^2|\mathcal{M}(q_0)|^2/(16\pi m_\chi^2 m_e^2)$ used in the particle-physics literature [5, Eq. (3)].

2.2 Born approximation and form factors

When the mediator is heavy, a dark-matter particle of 100 MeV scatters on a free electron with the cross-section $\mu_{\chi e}^2|\tilde{V}|^2/\pi$. Sect. 1.4 defined $\bar{\sigma}_e$ through this cross-section. The same golden-rule calculation yields the rate off an electron bound in an atom and the rate off a nucleus. For the bound electron, its wavefunction enters through a single overlap, the transition form factor; in the nucleus, the nucleons scatter in phase and multiply the cross-section by A^2. We derive the three results, for free electrons, bound electrons and nuclei, from Eqs. (2.5) and (2.12), to first order in the potential: the first Born approximation.

2.2.1 The Born cross-section

Take a dark-matter particle and an electron in a box, interacting through a potential $V(\boldsymbol{x}_\chi - \boldsymbol{x}_e)$ of their separation. Since the total momentum is conserved, it is simplest to work in the

center-of-mass frame, where the dark matter has momentum $\boldsymbol{p}$, the electron $-\boldsymbol{p}$, and the relative speed is $v = p/\mu_{\chi e}$. The kinetic energy of relative motion is $p^2/2\mu_{\chi e}$. In general the Hamiltonian splits as $P_{\rm tot}^2/2(m_\chi + m_e) + p_{\rm rel}^2/2\mu_{\chi e} + V(\boldsymbol{r})$, with $\boldsymbol{P}_{\rm tot}$ the total momentum, $\boldsymbol{p}_{\rm rel}$ the relative momentum and $\boldsymbol{r}$ the separation. The potential acts on the relative motion alone, which is the motion of a single particle of mass $\mu_{\chi e}$; the center of mass moves freely and drops out of the cross-section. A transition to $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$ has the box matrix element $\tilde{V}(\boldsymbol{q})/\mathscr{V}$ of Eq. (2.12), with $\boldsymbol{q}$ the momentum transferred to the electron. The potential does not act on spin, and the spins need no sum. Inserting this matrix element into the cross-section of Eq. (2.13) and summing over the one free final momentum,

$$\sigma_T = \frac{\mathscr{V}}{v} \sum_{\boldsymbol{p}'} 2\pi \, \frac{|\tilde{V}(\boldsymbol{q})|^2}{\mathscr{V}^2} \, \delta\!\left(\frac{p'^2}{2\mu_{\chi e}} - \frac{p^2}{2\mu_{\chi e}} \right) = \frac{1}{v} \int \frac{p'^2 \, \mathrm{d}p' \, \mathrm{d}\Omega_{\boldsymbol{q}}}{(2\pi)^3} \, 2\pi \, |\tilde{V}(\boldsymbol{q})|^2 \, \frac{\mu_{\chi e}}{p} \, \delta(p' - p),$$

where we used Eq. (2.10) for the sum and $\delta(p'^2/2\mu_{\chi e} - p^2/2\mu_{\chi e}) = (\mu_{\chi e}/p)\,\delta(p' - p)$, an instance of the rule $\delta(g(x)) = \delta(x - x_0)/|g'(x_0)|$ for a simple root x_0 (Appendix C.1). Here $\mathrm{d}\Omega_{\boldsymbol{q}}$ is the element of solid angle of the scattered momentum $\boldsymbol{p}'$ (the literature writes $\mathrm{d}\Omega$). The volume has dropped out, as it must. Doing the radial integral and using $p = \mu_{\chi e} v$,

Born cross-section

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}\Omega_{\boldsymbol{q}}} = \left(\frac{\mu_{\chi e}}{2\pi} \right)^2 |\tilde{V}(\boldsymbol{q})|^2, \qquad |\boldsymbol{q}| = 2p \sin(\theta_{\rm s}/2), \tag{2.14}$$

with $\theta_{\rm s}$ the scattering angle between $\boldsymbol{p}$ and $\boldsymbol{p}'$. For dark-matter–nucleus scattering $\mu_{\chi e}$ becomes $\mu_{\chi N}$.

Eq. (2.14) is the textbook first Born approximation [1, Chap. 6]: the scattering amplitude is $-(\mu_{\chi e}/2\pi)\tilde{V}(\boldsymbol{q})$, and its square is the differential cross-section. The minus sign makes the amplitude positive for an attractive potential, and the sign drops out of every rate. The momentum transfer $\boldsymbol{q} = \boldsymbol{p} - \boldsymbol{p}'$ is the same in the laboratory frame, because a Galilean boost shifts $\boldsymbol{p}$ and $\boldsymbol{p}'$ by the same amount (Sect. 1.2).

A quick check is Coulomb scattering. For $m_V \to 0$ and couplings $y_\chi y_e \to e^2 = 4\pi\alpha$, Eq. (1.31) becomes $\tilde{V} = 4\pi\alpha/q^2$, and Eq. (2.14) becomes $\mathrm{d}\sigma_T/\mathrm{d}\Omega_{\boldsymbol{q}} = 4\mu_{\chi e}^2\alpha^2/q^4 = \alpha^2/[4\mu_{\chi e}^2 v^4 \sin^4(\theta_{\rm s}/2)]$, Rutherford's formula. Its divergence at small angle, $\theta_{\rm s}^{-4}$, is the divergence of $\bar{\sigma}_e$ for a light mediator noted in Sect. 1.4.

The first Born approximation keeps one power of $V(\boldsymbol{x})$ in the amplitude. The next term, of second order (Sect. 2.4), is small when the potential barely distorts the incoming wave over the range of the potential. For a Yukawa potential of range $1/m_V$ and strength $y_\chi y_e$ (Eq. (1.30)) the second-order term is small, up to $O(1)$ factors, when $y_\chi y_e \, \mu_{\chi e}/(4\pi m_V) \ll 1$ at low energy and $y_\chi y_e/(4\pi v) \ll 1$ when the de Broglie wavelength is shorter than the range [1, Chap. 6]. The second condition is the familiar $\alpha/v \ll 1$ of Coulomb scattering. For the heavy dark photon of Worked example 1.5, $y_\chi y_e = y_\chi \zeta e = 9 \times 10^{-6}$, and $y_\chi y_e \, \mu_{\chi e}/(4\pi m_V)$ and $y_\chi y_e/(4\pi v)$ are both below 10^{-3} at $v = 10^{-3}$. Couplings of dark matter to ordinary matter are always this weak, because larger ones are excluded (Sect. 1.4), and first order is sufficient for our purposes. The exception is the self-interaction within the dark sector, which can be strong but does not enter a detector. Screening in a target is not a higher Born term either. It comes from the target's electrons rearranging in response to the potential, at first order in $V(\boldsymbol{x})$ but to all orders in the Coulomb interaction among themselves, and it belongs to the target response (Sect. 4.4).

The rate needs the distribution in q rather than in angle. From $q^2 = 2p^2(1 - \cos\theta_{\rm s})$, the solid angle is $\mathrm{d}\Omega_{\boldsymbol{q}} = 2\pi \, \mathrm{d}\cos\theta_{\rm s} = \pi \, \mathrm{d}q^2/p^2 = 2\pi q \, \mathrm{d}q/(\mu_{\chi e}^2 v^2)$ after the trivial azimuthal

integral. Eq. (2.14) then gives

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}q} = \frac{q\,|\tilde{V}(\boldsymbol{q})|^2}{2\pi v^2} = \frac{q\,\bar{\sigma}_e}{2\mu_{\chi e}^2 v^2}\,F_{\mathrm{DM}}(q)^2, \qquad 0 \le q \le 2\mu_{\chi e} v, \tag{2.15}$$

where the form in $\bar{\sigma}_e$ uses $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ (Eq. (1.33)). The upper limit is backscattering, $\theta_{\mathrm{s}} = \pi$. For a contact interaction, $F_{\mathrm{DM}} = 1$, the distribution rises linearly in q and integrates to

$$\sigma_T = \frac{|\tilde{V}|^2}{2\pi v^2}\,\frac{(2\mu_{\chi e} v)^2}{2} = \frac{\mu_{\chi e}^2 |\tilde{V}|^2}{\pi} = \bar{\sigma}_e, \tag{2.16}$$

independent of the speed. Eq. (2.16) is the statement behind the definition of $\bar{\sigma}_e$ in Eq. (1.32): for a heavy mediator, $\bar{\sigma}_e$ is the total cross-section of dark matter on a free electron. For a nucleus, $\mu_{\chi e} \to \mu_{\chi N}$ and $\bar{\sigma}_e \to \sigma_N$; we return to the nucleus at the end of the section.

2.2.2 Scattering off a bound electron

The momentum of an electron bound in an atom or a crystal is not conserved. The nucleus, or the lattice, absorbs whatever momentum the electron does not keep, and the electron jumps between states $|i\rangle$ and $|f\rangle$ of its own Hamiltonian instead of between plane waves. Treating the binding potential as fixed in space, we neglect the recoil of a nucleus thousands of times heavier than the electron. The dark matter is still a plane wave. The matrix element follows from the same step as Eq. (2.12): integrating over the dark-matter position at fixed electron position $\boldsymbol{r}$ (Eq. (A.8)) gives

$$\langle \boldsymbol{p}-\boldsymbol{q}, f|V(\boldsymbol{x}_\chi - \boldsymbol{r})|\boldsymbol{p}, i\rangle = \frac{\tilde{V}(\boldsymbol{q})}{\mathscr{V}}\,f_{i\to f}(\boldsymbol{q}), \qquad f_{i\to f}(\boldsymbol{q}) \equiv \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle = \int \mathrm{d}^3 r\;\psi_f^*(\boldsymbol{r})\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\psi_i(\boldsymbol{r}). \tag{2.17}$$

The *transition form factor* $f_{i\to f}(\boldsymbol{q})$ is the amplitude for the electron to absorb the momentum $\boldsymbol{q}$ and end up in $|f\rangle$. It is a property of the target alone; the dark matter enters only through $\tilde{V}(\boldsymbol{q})$. The sign convention of $f_{i\to f}$ follows the convention for $\tilde{V}$ (Appendix A.3). For free electrons, ψ_i and ψ_f are plane waves and $f_{i\to f}$ is a Kronecker delta that conserves momentum, so Eq. (2.12) is a special case of Eq. (2.17).

The dark matter loses the energy $p^2/2m_\chi - (\boldsymbol{p}-\boldsymbol{q})^2/2m_\chi = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi$, which is the energy deposit $\omega_{\boldsymbol{q}}$ of Eq. (1.21) (with $\Delta_\chi = 0$; an inelastic dark-matter particle adds $-\Delta_\chi$). Summed over the final dark-matter momentum by Eq. (2.10) and over the final electron state, the golden rule yields the rate Γ per dark-matter particle in the box. The box holds one dark-matter particle, a density $1/\mathscr{V}$. The volume-independent combination is $\mathscr{V}\Gamma$, which is $\sigma_T v$ by Eq. (2.13):

Rate off a bound target

$$\sigma_T v = \mathscr{V}\Gamma = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\,|\tilde{V}(\boldsymbol{q})|^2 \sum_f |f_{i\to f}(\boldsymbol{q})|^2\;2\pi\,\delta(E_f - E_i - \omega_{\boldsymbol{q}}). \tag{2.18}$$

The target enters only through $|f_{i\to f}(\boldsymbol{q})|^2$ and the excitation energies $E_f - E_i$; the dark matter only through $|\tilde{V}(\boldsymbol{q})|^2$ and $\omega_{\boldsymbol{q}}$.

With $|\tilde{V}|^2$ written through Eq. (1.33), Eq. (2.18) becomes [6, Eqs. (3.9) and (3.10)]

$$\sigma_T v = \frac{\bar{\sigma}_e}{\mu_{\chi e}^2} \int \frac{\mathrm{d}^3 q}{4\pi}\,F_{\mathrm{DM}}^2 \sum_f |f_{i\to f}|^2\,\delta(E_f - E_i - \omega_{\boldsymbol{q}}).$$

Depositing an energy ω needs $q \geq \omega/v$, since $\omega_{\boldsymbol{q}} \leq qv$, and this lower bound on q decides which momentum transfers Eq. (2.18) weights. A bound electron absorbs momentum transfers up to about its own momentum, $q_0 = \alpha m_e$ for an outer electron, and beyond q_0 the squared form factor $|f_{i\to f}|^2$ collapses. The lower bound ω/v reaches q_0 at $\omega \approx q_0 v \approx \alpha m_e v \approx 4\,\mathrm{eV}$ for $v \approx 10^{-3}$, independent of m_χ (Sect. 1.3). The crossing at about 4 eV makes eV-gap targets a good kinematic match to sub-GeV dark matter: the low-temperature silicon gap of 1.17 eV lies below this energy, and the momentum transfers that cross the gap, $q \gtrsim 0.44\,\mathrm{keV}$ at $v_{\max}$ (Sect. 1.3), are ones the valence electrons absorb readily. Ionizing hydrogen, $\omega \geq 13.6\,\mathrm{eV}$, needs $q \geq 5.1\,\mathrm{keV} = 1.4\,q_0$ even at the largest halo speed $v_{\max} = 2.65 \times 10^{-3}$, and a dark-matter mass of at least $2 \times 13.6\,\mathrm{eV}/v_{\max}^2 = 3.9\,\mathrm{MeV}$ (Eq. (1.24)). The form factor at the required momentum transfers is not yet small, as the hydrogen example shows, but the halo factor $\eta(v_{\min})$ is, because only the fastest dark matter can supply the energy.

Closure fixes the total size of the form factor. Summed over a complete set of final states, bound and free, and including $|i\rangle$ itself,

$$\sum_f |f_{i\to f}(\boldsymbol{q})|^2 = \langle i|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle = 1. \tag{2.19}$$

The electron absorbs the momentum $\boldsymbol{q}$ with total probability one; the form factor distributes that probability over final states. The term $f = i$ is the *elastic form factor* $f_{i\to i}(\boldsymbol{q}) = \int \mathrm{d}^3r\,|\psi_i(\boldsymbol{r})|^2\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, the Fourier transform of the electron's charge cloud. An electron bound in an orbital of radius r has momenta $\sim 1/r$. A momentum transfer much smaller than $1/r$ barely changes the state, so the elastic term takes almost all of the probability, $f_{i\to i} \to 1$. A momentum transfer much larger than $1/r$ must be absorbed by the high-momentum tail of the wavefunction, and every overlap with a bound final state falls off. The two limits of small and large momentum transfer form the physical picture behind $q_0 = \alpha m_e$ in Sect. 1.3, and the hydrogen atom checks the picture below.

An atom with Z electrons shows the same split between elastic and inelastic terms on a larger scale. The target operator is now $\sum_{j=1}^{Z}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$, and its elastic matrix element in the atomic ground state is the transform of the total electron density, a transform that tends to Z as $q \to 0$. Elastic scattering at small q is coherent, with a cross-section proportional to Z^2. Inelastic transitions, in which one electron changes its orbital, add incoherently, with total weight at most Z by closure applied to each electron. The nucleus has the same coherent and incoherent parts, with A nucleons; we treat it at the end of the section. In a crystal the coherent part is the Bragg-like elastic response at reciprocal-lattice vectors, which deposits no energy, and the incoherent part is the electronic excitation a detector sees (Chap. 4).

The rate in a detector averages Eq. (2.18) over the dark-matter velocities. For a target whose $\sum_f |f_{i\to f}(\boldsymbol{q})|^2$ at fixed energy depends only on $q = |\boldsymbol{q}|$, such as an atom with closed shells, only the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$ enters the delta function. The average of the delta function over that angle is $\Theta(v - v_{\min})/(2qv)$, and the halo average turns $\Theta(v - v_{\min})/v$ into $\eta(v_{\min})$, the mean inverse speed of the dark matter above $v_{\min}$ (Eq. (1.27)). Inserting $1 = \int \mathrm{d}\omega\,\delta(\omega - E_f + E_i)$ and writing $\mathrm{d}^3q = 4\pi q^2\,\mathrm{d}q$ for the remaining integral,

$$\frac{\mathrm{d}\langle\sigma_T v\rangle}{\mathrm{d}\omega} = \frac{1}{2\pi}\int q\,\mathrm{d}q\,|\tilde{V}(q)|^2\,\eta\big(v_{\min}(q,\omega)\big)\sum_f |f_{i\to f}(q)|^2\,\delta(\omega - E_f + E_i), \tag{2.20}$$

with $\langle\cdot\rangle$ the average over the halo. A potential that does not act on spin leaves the spin of the electron unchanged, and each occupied orbital contributes once for each of its two spin states. A full shell of angular momentum l therefore holds $2(2l+1)$ electrons. Its density is spherical, because $\sum_m |Y_{lm}|^2 = (2l+1)/4\pi$ for the spherical harmonics Y_{lm} (Appendix C.3), and so is its summed response to $\boldsymbol{q}$. A crystal is not spherical, and Eq. (2.18) must then be kept with its angular integral; Sect. 1.2 gave the resulting daily modulation.

For a gas of atoms whose response depends only on q, with number density n_T, the event rate per unit target mass is $(\rho_\chi/m_\chi)(n_T/\rho_T)\langle\sigma_T v\rangle$. Comparing with Eq. (1.46) identifies the target response of a gas of independent atoms: $S(q,\omega) = 2\pi\, n_T \sum_i \sum_f |f_{i\to f}(q)|^2\, \delta(\omega - E_f + E_i)$, summed over the occupied initial states i of each atom. For any target, $S(q,\omega)$ is the dynamic structure factor of the density operator $n_e(\boldsymbol{q}) = \sum_j e^{i\boldsymbol{q}\cdot\boldsymbol{r}_j}$ of Eq. (A.7) (Eqs. (2.67) and (A.30)). The single-electron matrix elements of $n_e(\boldsymbol{q})$ are the $f_{i\to f}$ of Eq. (2.17).

For several electrons the sum over i runs over occupied states and the final states must be empty; second quantization (Sect. 2.3) is the bookkeeping that handles these occupation constraints, and Sect. 4.2 recovers Eq. (2.20) as the independent-particle limit. A target at finite temperature starts in a mixture of states $|i\rangle$, and the rate is averaged over them with thermal weights (Sect. 2.5).

2.2.3 The hydrogen atom

The electron in the hydrogen ground state is the one bound electron whose form factor we can write in closed form. Its wavefunction is $\psi_{1s}(\boldsymbol{r}) = e^{-r/a_0}/(\pi a_0^3)^{1/2}$, with the Bohr radius $a_0 = 1/(\alpha m_e) = 0.5292\,\text{Å}$ (Appendix A.1). The reference momentum of Sect. 1.3 is the inverse of the Bohr radius, $q_0 = \alpha m_e = 1/a_0$, so $q_0 a_0 = 1$ exactly.

Worked example 2.2: The hydrogen 1s elastic form factor

The elastic form factor is the Fourier transform of $|\psi_{1s}|^2 = e^{-2r/a_0}/(\pi a_0^3)$. Since the density is spherical, the integral of $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ over the directions of $\boldsymbol{r}$ gives $4\pi\sin(qr)/(qr)$ (Appendix C.3), and

$$F_{1s}(q) \equiv f_{1s\to 1s}(\boldsymbol{q}) = \frac{4\pi}{\pi a_0^3}\int_0^\infty dr\, r^2\, e^{-2r/a_0}\,\frac{\sin qr}{qr}$$
$$= \frac{4}{a_0^3 q}\int_0^\infty dr\, r\, e^{-\lambda_{1s} r}\sin qr, \qquad \lambda_{1s} = \frac{2}{a_0}.$$

The remaining integral is the imaginary part of $\int_0^\infty dr\, r\, e^{-(\lambda_{1s}-iq)r} = 1/(\lambda_{1s}-iq)^2 = (\lambda_{1s}+iq)^2/(\lambda_{1s}^2+q^2)^2$, namely $2\lambda_{1s}q/(\lambda_{1s}^2+q^2)^2$. So $F_{1s}(q) = (4/a_0^3 q)\,(4q/a_0)/(4/a_0^2+q^2)^2$, or

$$F_{1s}(q) = \frac{16}{(4+q^2a_0^2)^2}. \tag{2.21}$$

At small q, $F_{1s} = (1+q^2a_0^2/4)^{-2} \simeq 1 - q^2a_0^2/2$. The expansion $1 - q^2a_0^2/2$ is the general small-q form $1 - q^2\langle r^2\rangle/6$ of an elastic form factor, with $\langle r^2\rangle = 3a_0^2$ for the 1s state. At large q, $F_{1s} \simeq 16/(qa_0)^4$: the elastic amplitude falls as the fourth power of the momentum transfer. The value at the reference momentum is $F_{1s}(q_0) = 16/25 = 0.64$: the electron stays in its ground state with probability $F_{1s}^2 = 0.41$ and, by Eq. (2.19), is excited or ionized with probability 0.59. The form factor falls to one half at $qa_0 = 1.29$ and to 0.25 at $qa_0 = 2$; its square falls to one half at $qa_0 = 0.87$. The 1s state is the most compact state of hydrogen, and q_0 is close to the momentum transfer at which the 1s electron stops responding as a whole and starts to be knocked out.

Where does the inelastic probability go? The lowest excitation, $1s \to 2p$ at 10.2 eV, also has a form factor in closed form. Take $\boldsymbol{q}$ along z. Only the $m = 0$ state $\psi_{2p} = (32\pi a_0^5)^{-1/2}\, z\, e^{-r/2a_0}$ couples, because the operator e^{iqz} does not change the angular momentum about z. The angular integral is $2\pi\int_{-1}^{1} du\, u\, e^{iqru} = 4\pi i\, j_1(qr)$, with $j_1(x) = (\sin x - x\cos x)/x^2$ the spherical Bessel function of order one (Appendix C.3), and the radial integral, done as in the elastic case with $\lambda_{2p} = 3/2a_0$, is $\int_0^\infty dr\, r^3 e^{-\lambda_{2p}r} j_1(qr) =$

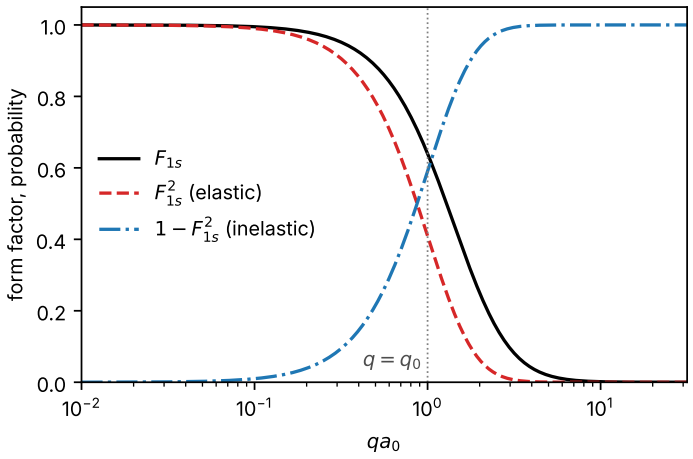


Figure 2.3: Hydrogen 1s elastic form factor $F_{1s}(q) = 16/(4 + q^2a_0^2)^2$ (solid); its square F_{1s}^2, which is the probability that the electron stays in the ground state after absorbing momentum q (dashed); and the probability $1 - F_{1s}^2$ that the electron is excited or ionized (dash-dotted), against qa_0, with a_0 the Bohr radius. The dotted line marks the reference momentum $q_0 = \alpha m_e = 3.73\,\text{keV}$, where $qa_0 = 1$ exactly and $F_{1s} = 0.64$. The inelastic probability follows from the closure relation $\sum_f |\langle f|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|1s\rangle|^2 = 1$.

$8\lambda_{2p}q/(\lambda_{2p}^2 + q^2)^3$. Collecting the normalizations,

$$|f_{1s\to 2p}(q)|^2 = \frac{72\,(qa_0)^2}{(9/4 + q^2a_0^2)^6}.$$

At small q it grows as $(2^{15}/3^{10})\,q^2a_0^2 = 0.555\,q^2a_0^2$, which is $q^2|\langle 2p|z|1s\rangle|^2$: a momentum transfer much smaller than $1/a_0$ acts on the atom as a weak uniform field, through its dipole. The squared form factor $|f_{1s\to 2p}(q)|^2$ peaks at $qa_0 = 0.67$ with the value 0.084 and is 0.061 at q_0. So at q_0 the $2p$ level takes about a tenth of the inelastic probability of 0.59, and the rest of the inelastic probability goes to higher levels and, mostly, to ionization.

Fig. 2.3 shows the crossover from elastic absorption to excitation and ionization. Below $q \approx q_0/3$ the atom absorbs the momentum transfer elastically: the electron and the nucleus move together, and the energy deposit is the recoil of the whole atom. Above $q \approx 3q_0$ almost every scattering excites or ionizes the atom. The elastic form factor also explains how a neutral atom screens the charge of its nucleus. A dark photon couples to the electron and to the proton with opposite signs (Sect. 1.4), and the elastic amplitude off the atom is proportional to $1 - F_{1s}(q)$: the proton, a point, contributes 1, and the electron cloud $-F_{1s}$. At $q \ll q_0$ the two contributions cancel and the dark matter sees a neutral atom; at q_0 the squared factor is $(1 - 0.64)^2 = 0.13$. The cancellation at small q is screening in its simplest form, and Sect. 4.4 treats screening in solids.

For ionization the final state is an unbound electron of momentum p_{ej} and kinetic energy $E_{\text{ej}} = p_{\text{ej}}^2/2m_e$, and the energy transfer is $\omega = E_B + E_{\text{ej}}$, with $E_B > 0$ the binding energy of the shell. Far from the atom the ejected electron is a spherical wave; near the atom, the wave is distorted by the ion. The sum over final states in Eq. (2.20) becomes an integral over p_{ej} and a sum over the angular quantum numbers of the outgoing wave, and the result is conventionally written as the *atomic ionization form factor* $|f_{\text{ion}}(p_{\text{ej}}, q)|^2$ of each shell [5,

Eq. (6)].

The outgoing waves are labeled by $p_{\rm ej}$ and by the angular quantum numbers l', m', and normalized to $\langle\tilde\psi_{p'l'm'}|\tilde\psi_{plm}\rangle = (2\pi)^3\delta_{l'l}\delta_{m'm}\,\delta(p'-p)/p^2$. The sum over final states is then $\sum_f = \sum_{l'm'}\int p_{\rm ej}^2\,{\rm d}p_{\rm ej}/(2\pi)^3$. With ${\rm d}E_{\rm ej} = p_{\rm ej}\,{\rm d}p_{\rm ej}/m_e$, $p_{\rm ej}^2\,{\rm d}p_{\rm ej} = (p_{\rm ej}^3/2)\,{\rm d}\ln E_{\rm ej}$. Define

$$|f_{\rm ion}(p_{\rm ej},q)|^2 = \frac{2p_{\rm ej}^3}{(2\pi)^3}\sum_{l'm'}\sum_i\left|\int {\rm d}^3r\;\tilde\psi^*_{p_{\rm ej}l'm'}(\boldsymbol{r})\,\psi_i(\boldsymbol{r})\,{\rm e}^{{\rm i}\boldsymbol{q}\cdot\boldsymbol{r}}\right|^2, \tag{2.22}$$

with i running over the occupied states of the shell, both spins included. At fixed E_B, ${\rm d}\omega = {\rm d}E_{\rm ej} = E_{\rm ej}\,{\rm d}\ln E_{\rm ej}$. In Eq. (2.20) the delta function $\delta(\omega - E_B - E_{\rm ej})$ then turns $(p_{\rm ej}^3/2)\,{\rm d}\ln E_{\rm ej}$ into $p_{\rm ej}^3/(2E_{\rm ej})$, and passing from ${\rm d}/{\rm d}\omega$ to ${\rm d}/{\rm d}\ln E_{\rm ej}$ multiplies by $E_{\rm ej}$ again. The sum over final states is therefore $|f_{\rm ion}|^2/4$ per unit $\ln E_{\rm ej}$. Inserting this sum and $|\tilde V|^2 = \pi\bar\sigma_e F_{\rm DM}^2/\mu_{\chi e}^2$ into Eq. (2.20) leaves the prefactor $(1/2\pi)(\pi\bar\sigma_e/\mu_{\chi e}^2)(1/4) = \bar\sigma_e/(8\mu_{\chi e}^2)$:

$$\frac{{\rm d}\langle\sigma v\rangle}{{\rm d}\ln E_{\rm ej}} = \frac{\bar\sigma_e}{8\mu_{\chi e}^2}\int q\,{\rm d}q\,|f_{\rm ion}(p_{\rm ej},q)|^2\,F_{\rm DM}(q)^2\,\eta(v_{\rm min}), \qquad v_{\rm min} = \frac{E_B+E_{\rm ej}}{q} + \frac{q}{2m_\chi}, \tag{2.23}$$

the ionization rate of Ref. [5, Eqs. (5) and (6)]. It rests on the first Born approximation in the dark-matter potential and treats one electron at a time in a fixed atomic potential, with no correlation and no recoil of the ion.

The exact outgoing waves are tedious to compute, and the standard shortcut replaces them by plane waves. The plane-wave shortcut also shows where the spin factor enters. Replace the outgoing wave by ${\rm e}^{{\rm i}\boldsymbol{p}_{\rm ej}\cdot\boldsymbol{r}}$. The overlap in Eq. (2.17) is then the momentum-space wavefunction $\tilde\psi_i(\boldsymbol{p}_{\rm ej}-\boldsymbol{q})$, with $\tilde\psi_i(\boldsymbol{k}) = \int {\rm d}^3r\,{\rm e}^{-{\rm i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi_i(\boldsymbol{r})$. For a full shell, $\sum_m|\tilde\psi_{nlm}(\boldsymbol{k})|^2 = (2l+1)|\tilde\psi_{nl}(k)|^2/4\pi$, where $\tilde\psi_{nl}(k)$ is the radial momentum wavefunction of the shell with principal and orbital quantum numbers n, l, normalized to $\int k^2\,{\rm d}k\,|\tilde\psi_{nl}|^2 = (2\pi)^3$. The two spin states double this sum. The sum over final states is $\int {\rm d}^3p_{\rm ej}/(2\pi)^3$. At fixed $p_{\rm ej}$ the integral over its direction relative to $\boldsymbol{q}$ becomes an integral over $k = |\boldsymbol{p}_{\rm ej}-\boldsymbol{q}|$, since $k^2 = p_{\rm ej}^2 + q^2 - 2p_{\rm ej}q\cos\theta'$ gives $2\pi\,{\rm d}\cos\theta' = 2\pi k\,{\rm d}k/(p_{\rm ej}q)$, with θ' the angle between $\boldsymbol{p}_{\rm ej}$ and $\boldsymbol{q}$ and k running between $|p_{\rm ej}-q|$ and $p_{\rm ej}+q$. Collecting the factors as for Eq. (2.22),

$$|f_{\rm ion}(p_{\rm ej},q)|^2_{\rm PW} = \frac{(2l+1)\,p_{\rm ej}^2}{4\pi^3 q}\int_{|p_{\rm ej}-q|}^{p_{\rm ej}+q} k\,{\rm d}k\,|\tilde\psi_{nl}(k)|^2, \tag{2.24}$$

which is the expression given in Ref. [5, text after Eq. (6)]; it counts all $2(2l+1)$ electrons of a full shell, and for the single electron of hydrogen 1s the rate is half of the full-shell rate. The result has a direct reading: the ejected electron leaves with the momentum it had in the atom plus the momentum transfer, and the form factor counts how much of the bound momentum distribution lies on the shell $|\boldsymbol{k}+\boldsymbol{q}| = p_{\rm ej}$.

The plane wave ignores the ion. The ion attracts a slow electron and raises its wavefunction near the nucleus, where the overlap with ψ_i is taken, and the enhancement grows as $E_{\rm ej}$ falls. For an electron of momentum p leaving a point charge $Z_{\rm eff}e$, the ratio of the exact to the free probability density at the origin is the Fermi factor familiar from beta decay,

$$F_{\rm Fermi}(p) = \frac{2\pi\xi_{\rm C}}{1-{\rm e}^{-2\pi\xi_{\rm C}}}, \qquad \xi_{\rm C} = \frac{Z_{\rm eff}\,\alpha m_e}{p}, \tag{2.25}$$

which grows as $1/p$ for a slow electron [5, Eq. (1)]. Multiplying Eq. (2.24) by $F_{\rm Fermi}(p_{\rm ej})$ with $Z_{\rm eff} = 1$ is the standard approximation. For the outer shells of xenon and argon, which dominate the rates of these atoms, Ref. [5, text after Eq. (6)] reports agreement with continuum states computed in the atomic potential to within $O(30\,\%)$ in the total rate. Hydrogen 1s is the one case in which the exact continuum states are known in closed form, and Fig. 2.4 uses it to test the approximation on a single compact shell. For hydrogen 1s the same recipe reaches only about half of the exact rate: the quality of the Fermi factor depends on the shell.

Limit Check 2.1. Atomic ionization rate

The ionization rate of a shell with binding energy E_B, per unit logarithm of the ejected electron's kinetic energy $E_{\rm ej} = p_{\rm ej}^2/2m_e$, is

$$\frac{\mathrm{d}\langle\sigma v\rangle}{\mathrm{d}\ln E_{\rm ej}} = \frac{\bar{\sigma}_e}{8\mu_{\chi e}^2}\int q\,\mathrm{d}q\,|f_{\rm ion}|^2\,F_{\rm DM}(q)^2\,\eta(v_{\rm min}),$$

$$|f_{\rm ion}|^2 \simeq \frac{2\pi\xi_{\rm C}}{1-\mathrm{e}^{-2\pi\xi_{\rm C}}}\,\frac{(2l+1)\,p_{\rm ej}^2}{4\pi^3 q}\int_{|p_{\rm ej}-q|}^{p_{\rm ej}+q} k\,\mathrm{d}k\,|\tilde{\psi}_{nl}(k)|^2,$$

with $\xi_{\rm C} = \alpha m_e/p_{\rm ej}$ (the Fermi factor of Eq. (2.25) for $Z_{\rm eff} = 1$), q the momentum transfer, $\tilde{\psi}_{nl}(k)$ the radial momentum wavefunction of the shell with principal and orbital quantum numbers n, l, and $v_{\rm min} = (E_B + E_{\rm ej})/q + q/2m_\chi$ [5, Eqs. (1), (5) and (6)].

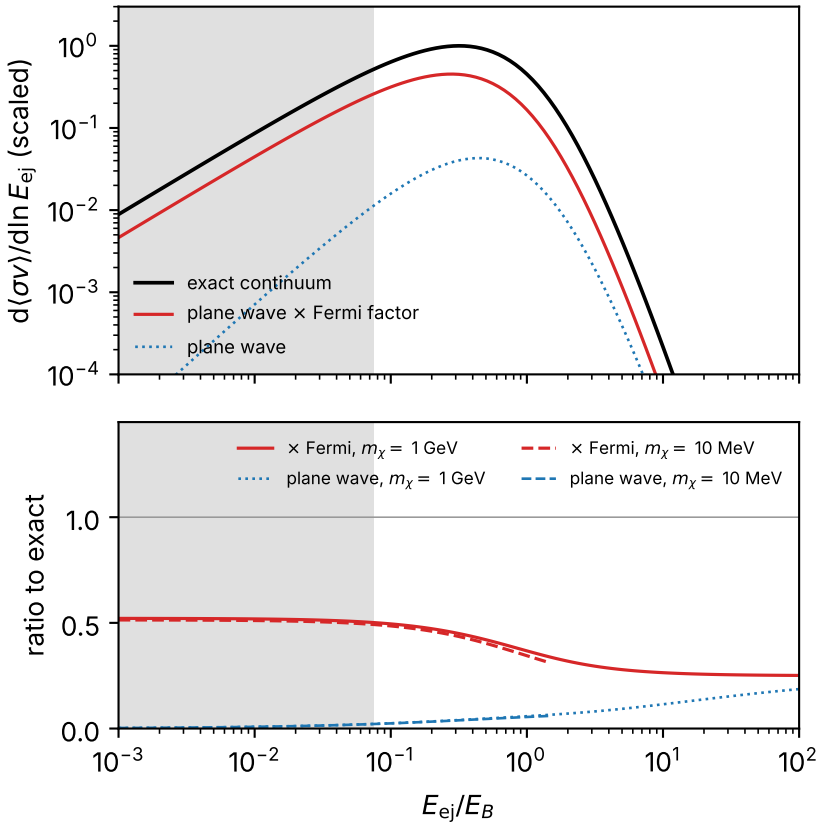


Figure 2.4: Ionization spectrum of hydrogen 1s (one electron, so half of the two-spin formula above; $E_B = 13.6\,\mathrm{eV}$) for a heavy mediator, $F_{\rm DM} = 1$, in the standard halo model with the parameters of Table 1.1, against $E_{\rm ej}/E_B$. Top, for $m_\chi = 1\,\mathrm{GeV}$, scaled to the peak of the exact result: exact Coulomb continuum states (black), plane wave times the Fermi factor with $Z_{\rm eff} = 1$ (red), plane wave alone (blue dotted). Bottom: the two approximations divided by the exact spectrum, for $m_\chi = 1\,\mathrm{GeV}$ and $10\,\mathrm{MeV}$. Gray: $E_{\rm ej} \leq 0.075\,E_B$, where the Fermi-corrected plane wave reaches about half of the exact spectrum. Integrated over $E_{\rm ej}$ at 1 GeV, the Fermi-corrected plane wave gives 0.46 of the exact rate and the plane wave alone 0.036. The exact form factor sums the overlaps with Coulomb partial waves of the hydrogen continuum.

Above the shaded region, where most of the rate lies, the halo forces momentum transfers $q \geq (E_B + E_{\rm ej})/v$ well above $p_{\rm ej}$. Momentum transfers well above $p_{\rm ej}$ probe the Coulomb wave at $r \sim 1/q$, where the Coulomb wave is not a rescaled plane wave, and the approximation falls to between a quarter and a third of the exact spectrum for $E_{\rm ej}$ of several E_B.

Hydrogen is the only atom with closed-form orbitals. For every other atom the initial orbitals ψ_i are not known exactly: they are computed self-consistently, each electron moving in the average field of the others, as Hartree–Fock orbitals, and the ionization form factors are integrals over them. Sect. 26.2 describes how atomic ionization form factors are computed and converged. In a crystal the same overlap is taken between Bloch states of the valence and conduction bands, summed over their crystal momenta, and the atomic $|f_{\rm ion}|^2$ becomes the crystal form factor [6, Eqs. (3.13) and (3.17)] of Chap. 10.

2.2.4 Free electrons

Eq. (2.18) contains free scattering as a special case, and checking this special case, now in the laboratory frame, shows how the mediator mass enters the free cross-section.

Worked example 2.3: Scattering off a free electron

Take an electron at rest, $|i\rangle = |\mathbf{0}\rangle$, with final states $|f\rangle = |\boldsymbol{k}\rangle$ in the box. By Eq. (2.17), $f_{i\to f}(\boldsymbol{q}) = \langle \boldsymbol{k}|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|\mathbf{0}\rangle = \delta_{\boldsymbol{k}\boldsymbol{q}}$: the electron takes exactly the momentum it is given. The sum over f in Eq. (2.18) keeps $\boldsymbol{k} = \boldsymbol{q}$, with $E_f - E_i = q^2/2m_e$, and with $\omega_{\boldsymbol{q}} = qv\cos\theta - q^2/2m_\chi$ the delta function is $\delta(q^2/2\mu_{\chi e} - qv\cos\theta)$, where θ is the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$ and $1/\mu_{\chi e} = 1/m_e + 1/m_\chi$. The root of the delta function is at $\cos\theta = q/(2\mu_{\chi e}v)$, inside $[-1, 1]$ for $q \leq 2\mu_{\chi e}v$, and the weight of the root is $1/(qv)$. With $\mathrm{d}^3q = 2\pi q^2\,\mathrm{d}q\,\mathrm{d}\cos\theta$,

$$\sigma_T v = \int_0^{2\mu_{\chi e}v} \frac{2\pi q^2\,\mathrm{d}q}{(2\pi)^3}\,|\tilde{V}(q)|^2\,\frac{2\pi}{qv} = \frac{1}{2\pi v}\int_0^{2\mu_{\chi e}v} q\,\mathrm{d}q\,|\tilde{V}(q)|^2.$$

Dividing by v gives $\mathrm{d}\sigma_T/\mathrm{d}q = q|\tilde{V}(q)|^2/(2\pi v^2)$, which is Eq. (2.15), obtained before in the center-of-mass frame.

For a mediator of mass m_V, Eq. (1.34) has $F_{\rm DM}^2 = (m_V^2 + q_0^2)^2/(m_V^2 + q^2)^2$, and the integral over q^2 from 0 to $q_{\rm max}^2 = (2\mu_{\chi e}v)^2$ is elementary, $\int_0^{q_{\rm max}^2} \mathrm{d}x\,(m_V^2 + x)^{-2} = q_{\rm max}^2/[m_V^2(m_V^2 + q_{\rm max}^2)]$. So

$$\sigma_T = \bar{\sigma}_e\,\frac{(m_V^2 + q_0^2)^2}{m_V^2\,(m_V^2 + q_{\rm max}^2)}.$$

For concreteness take $m_\chi = 100\,\mathrm{MeV}$, with $\mu_{\chi e} = 0.508\,\mathrm{MeV}$, and the largest halo speed, $v_{\rm max} = 2.65 \times 10^{-3}$ (Eq. (1.9)). Then $q_{\rm max} = 2.69\,\mathrm{keV} = 0.72\,q_0$. A free electron at rest cannot even absorb the reference momentum q_0 from this dark-matter particle: $\bar{\sigma}_e$ is defined at a momentum transfer that free kinematics does not reach. For a heavy mediator the definition at q_0 does not matter. With $m_V = 10\,q_0$ we find $\sigma_T = 1.015\,\bar{\sigma}_e$, and at $m_V = 30\,\mathrm{MeV}$ the relative difference between σ_T and $\bar{\sigma}_e$ is 2.3×10^{-8}. At $m_V = q_0$, $\sigma_T = 2.63\,\bar{\sigma}_e$, and at $m_V = 0.1\,q_0$, $\sigma_T = 192\,\bar{\sigma}_e$. As $m_V \to 0$, σ_T grows as $q_0^4/(m_V^2 q_{\rm max}^2)$, the small-angle divergence of Sect. 1.4.

The free electron at rest is a poor model of a bound one. A bound electron moves with a typical speed $\sim \alpha$, about ten times the dark-matter speed (Sect. 1.3), so it can absorb momentum transfers well above $2\mu_{\chi e}v$, and it needs an energy $E_f - E_i$ that free kinematics does not

supply. The transition form factor $f_{i\to f}$ contains both the absorption of large momentum transfers and the need for the energy $E_f - E_i$.

For a light mediator the bound target also removes the divergence of Worked example 2.3. In Eq. (2.20) the factor $F_{\rm DM}^2 = (q_0/q)^4$ multiplies $q\,\mathrm{d}q$, and the integrand grows as q^{-3} toward small q. The integral is cut off at $q = \omega/v_{\max}$, below which no dark-matter particle can deposit ω. In addition, the squared form factor of every inelastic transition vanishes at small q, as q^2 for a dipole-allowed one (Worked example 2.2). The rate is then dominated by the smallest q that kinematics allows, just above the threshold $q = \omega/v_{\max}$.

2.2.5 Nuclei and the Helm form factor

A nucleus of mass number A is a bound target of A nucleons, and Eq. (2.17) applies with one change: every nucleon couples. With a potential $V_N(\boldsymbol{x}_\chi - \boldsymbol{r}_j)$ between the dark matter and nucleon j, the same for protons and neutrons, the target operator is $\sum_{j=1}^{A} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$. Write $\boldsymbol{r}_j = \boldsymbol{R}_{\rm cm} + \boldsymbol{r}'_j$, with $\boldsymbol{R}_{\rm cm}$ the center of mass of the nucleus and $\boldsymbol{r}'_j$ the internal coordinates. The center of mass is a free particle and its plane waves give momentum conservation, as in Eq. (2.12). The internal state is the nuclear ground state $|0\rangle$, before and after. The lowest excited levels lie at 1.78 MeV in ^{28}Si, 1.27 MeV in ^{29}Si and 2.24 MeV in ^{30}Si, and between 0.56 MeV and 1.04 MeV in the even germanium isotopes. The isotope ^{73}Ge, 7.8 % of natural germanium, has levels at the keV scale: its first excited state is at 13.3 keV and its second at 66.7 keV [7]. Sub-GeV dark matter carries at most a few keV of kinetic energy (Sect. 1.3), below all of these levels. The internal matrix element is

$$\Big\langle 0\Big|\sum_{j=1}^{A} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'_j}\Big|0\Big\rangle = \int \mathrm{d}^3r\; \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, \boldsymbol{n}_N(\boldsymbol{r}) \equiv A\, F_N(q), \tag{2.26}$$

with $\boldsymbol{n}_N(\boldsymbol{r})$ the nucleon number density of the nucleus, normalized to $\int \mathrm{d}^3r\, \boldsymbol{n}_N = A$, and $F_N(q)$ the *nuclear form factor*, the elastic form factor of the nucleus with $F_N(0) = 1$. Eq. (2.26) is the coherent sum: at $q \to 0$ the phases of all nucleons agree and the amplitude is A times that of one nucleon. The two-body result Eq. (2.15), with $\mu_{\chi e} \to \mu_{\chi N}$ and $\tilde{V} \to A\tilde{V}_N F_N$, gives

$$\frac{\mathrm{d}\sigma_N}{\mathrm{d}q} = \frac{q\,A^2|\tilde{V}_N(q)|^2}{2\pi v^2}\,F_N(q)^2 = \frac{q\,\sigma_N}{2\mu_{\chi N}^2 v^2}\,F_N(q)^2, \qquad \sigma_N = \frac{\mu_{\chi N}^2 A^2 |\tilde{V}_N|^2}{\pi} = \sigma_p \left(\frac{\mu_{\chi N}}{\mu_{\chi p}}\right)^2 A^2, \tag{2.27}$$

for a heavy mediator, $\tilde{V}_N = y_\chi y_N/m_V^2$. The expression through σ_p uses the proton cross-section $\sigma_p = \mu_{\chi p}^2|\tilde{V}_N|^2/\pi$, which is Eq. (2.16) with $\mu_{\chi e} \to \mu_{\chi p}$. Eq. (2.27) derives Eq. (1.36), and σ_N is the cross-section at $q \to 0$; the form factor then cuts off the distribution at large q. Unequal proton and neutron couplings replace A by $Z + (A - Z)y_n/y_p$ (Sect. 1.4).

Experiments measure the recoil energy $E_R = q^2/2M$ rather than q, with M the nuclear mass. With $\mathrm{d}E_R = q\,\mathrm{d}q/M$, $\mathrm{d}\sigma_N/\mathrm{d}E_R = M\sigma_N F_N^2/(2\mu_{\chi N}^2 v^2)$. We multiply by the number of nuclei per unit mass, $1/M$, and by the dark-matter flux $(\rho_\chi/m_\chi)\, v\, f_\chi(\boldsymbol{v})\,\mathrm{d}^3v$. Integrating over velocities above $v_{\min} = (ME_R/2\mu_{\chi N}^2)^{1/2}$ (Sect. 1.3) then turns the factor v/v^2 into $\eta(v_{\min})$:

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}E_R} = \frac{\rho_\chi\,\sigma_N}{2m_\chi\mu_{\chi N}^2}\,F_N(q)^2\,\eta(v_{\min}), \qquad q = (2ME_R)^{1/2},$$

the standard spin-independent recoil spectrum, Ref. [3, Eq. (104)] with σ_N written through Eq. (1.36) and equal couplings to protons and neutrons.

The nucleon density is close to uniform inside the nucleus and falls off over a skin about 1 fm thick. Helm's model [8] writes the nucleon density as a uniform sphere of radius $R_{\rm nuc}$ smeared

by a Gaussian of width s_{skin}. The transform of a convolution is the product of the transforms. For the uniform sphere, the angular integral is $4\pi \sin(qr)/(qr)$ again (Appendix C.3), and $\int_0^R \mathrm{d}r\, r \sin qr = (\sin qR - qR\cos qR)/q^2$, so the normalized transform of the uniform sphere is $3(\sin x - x\cos x)/x^3 = 3j_1(x)/x$ with $x = qR_{\mathrm{nuc}}$ and j_1 the spherical Bessel function. The normalized Gaussian $\mathrm{e}^{-r^2/2s_{\mathrm{skin}}^2}/(2\pi s_{\mathrm{skin}}^2)^{3/2}$ transforms to $\mathrm{e}^{-q^2 s_{\mathrm{skin}}^2/2}$. Multiplying the two transforms gives [9, Eq. (17)]

$$F_{\mathrm{Helm}}(q) = \frac{3j_1(qR_{\mathrm{nuc}})}{qR_{\mathrm{nuc}}}\, \mathrm{e}^{-q^2 s_{\mathrm{skin}}^2/2}. \tag{2.28}$$

The standard parameters come from Lewin and Smith [10]: $R_{\mathrm{nuc}}^2 = c_{\mathrm{nuc}}^2 + \frac{7}{3}\pi^2 a_{\mathrm{nuc}}^2 - 5s_{\mathrm{skin}}^2$, with $c_{\mathrm{nuc}} = (1.23A^{1/3} - 0.60)\,\mathrm{fm}$, $a_{\mathrm{nuc}} = 0.52\,\mathrm{fm}$ and $s_{\mathrm{skin}} = 0.9\,\mathrm{fm}$ [9, Eqs. (11) and (12), quoting Lewin and Smith]. Here c_{nuc} and a_{nuc} are the half-density radius and surface diffuseness of a Woods–Saxon density, and the formula for R_{nuc} matches the Helm form to the Woods–Saxon density. At small q, $3j_1(x)/x \simeq 1 - x^2/10$ and $F_{\mathrm{Helm}} \simeq 1 - q^2(R_{\mathrm{nuc}}^2/10 + s_{\mathrm{skin}}^2/2)$, the form $1 - q^2\langle r^2\rangle/6$ with $\langle r^2\rangle = \frac{3}{5}R_{\mathrm{nuc}}^2 + 3s_{\mathrm{skin}}^2$. For silicon, $c_{\mathrm{nuc}} = 3.14\,\mathrm{fm}$ and $R_{\mathrm{nuc}} = 3.47\,\mathrm{fm}$; for germanium ($A = 73$), $c_{\mathrm{nuc}} = 4.54\,\mathrm{fm}$ and $R_{\mathrm{nuc}} = 4.77\,\mathrm{fm}$.

The Helm form is a model. Electron scattering measures the charge density of the nucleus, that is, the density of the protons. The form factor of Eq. (2.26) needs the density of all nucleons. The standard assumption is that the two densities are proportional, in which case the nucleon form factor equals the charge form factor [9, Eqs. (7) and (8)]; for the dark photon, which couples to charge, the charge form factor is exact. The physical form factor is then the transform of the measured charge density, and Ref. [9, Sect. V] compares the Helm form factor with that transform. For ^{28}Si the Lewin–Smith Helm form factor differs from the measured one negligibly at small q and by up to 5 % at the momentum transfer of a 100 keV recoil; for ^{70}Ge the difference is 2 % at 10 keV and 20 % at 100 keV. The Helm form is accurate at the momentum transfers of light dark matter.

Fig. 2.5 shows that light dark matter never resolves the nucleus. The silicon radius $R_{\mathrm{nuc}} = 3.47\,\mathrm{fm}$ corresponds to $1/R_{\mathrm{nuc}} = 57\,\mathrm{MeV}$, and germanium's 4.77 fm to 41 MeV. The form factor matters once qR_{nuc} approaches one, at tens of MeV; momentum transfers of tens of MeV need dark matter of several GeV.

Worked example 2.4: The A^2 enhancement for silicon

For ^{28}Si, $A^2 = 784$. With the mean atomic mass of natural silicon, 26.16 GeV, the reduced masses at $m_\chi = 100\,\mathrm{MeV}$ are $\mu_{\chi p} = 90.4\,\mathrm{MeV}$ and $\mu_{\chi N} = 99.6\,\mathrm{MeV}$, and Eq. (2.27) gives $\sigma_N/\sigma_p = 1.215 \times 784 = 953$; at 1 GeV, $\mu_{\chi p} = 484\,\mathrm{MeV}$, $\mu_{\chi N} = 963\,\mathrm{MeV}$, and $\sigma_N/\sigma_p = 3.96 \times 784 = 3100$. Coherence turns the A of 28 nucleons scattering independently into A^2, a gain of 28; the reduced-mass factor is a separate effect of the heavier recoiling target. Per kilogram of target the rate still grows as A, because A^2 in the cross-section outweighs the $1/A$ in the number of nuclei per unit mass. The form factor is close to one. At $m_\chi = 1\,\mathrm{GeV}$ the maximum momentum transfer is $q_{\mathrm{max}} = 2\mu_{\chi N} v_{\mathrm{max}} = 5.10\,\mathrm{MeV}$ (Worked example 1.4). The suppression is greatest at q_{max}, where $q_{\mathrm{max}} R_{\mathrm{nuc}} = 0.090$ and $F_{\mathrm{Helm}}^2 = 0.998$. Form factors shape the recoil spectra of weak-scale dark matter but are a 0.2 % correction at 1 GeV and far less for the sub-GeV dark matter of this book.

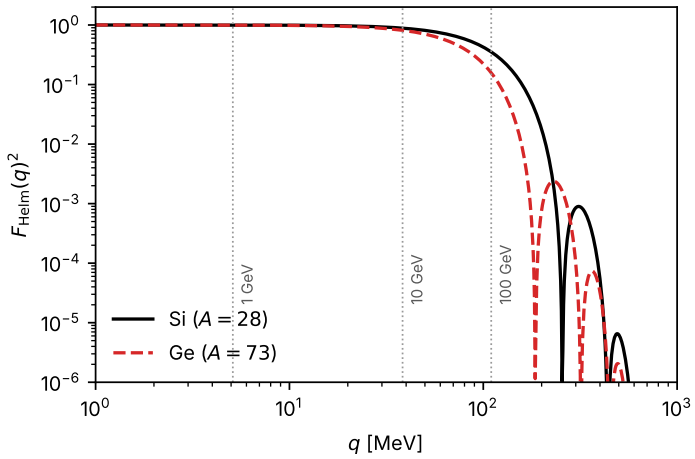


Figure 2.5: Squared Helm nuclear form factor $F_{\mathrm{Helm}}(q)^2 = [3j_1(qR_{\mathrm{nuc}})/(qR_{\mathrm{nuc}})]^2 \mathrm{e}^{-q^2 s_{\mathrm{skin}}^2}$ of silicon ($A = 28$, solid) and germanium ($A = 73$, the integer nearest the mean mass number 72.6 of natural germanium; dashed), against the momentum transfer q. Parameters of Lewin and Smith: $R_{\mathrm{nuc}}^2 = c_{\mathrm{nuc}}^2 + \frac{7}{3}\pi^2 a_{\mathrm{nuc}}^2 - 5s_{\mathrm{skin}}^2$, $c_{\mathrm{nuc}} = (1.23A^{1/3} - 0.60)\,\mathrm{fm}$, $a_{\mathrm{nuc}} = 0.52\,\mathrm{fm}$, $s_{\mathrm{skin}} = 0.9\,\mathrm{fm}$, which give $R_{\mathrm{nuc}} = 3.47\,\mathrm{fm}$ (Si) and $4.77\,\mathrm{fm}$ (Ge). Dotted lines: the maximum momentum transfer $2\mu_{\chi N} v_{\mathrm{max}}$ to a silicon nucleus for dark-matter masses of 1, 10 and 100 GeV, with $v_{\mathrm{max}} = 794.5\,\mathrm{km/s}$. The first zero is at $qR_{\mathrm{nuc}} = 4.49$, the first zero of j_1: $q = 256\,\mathrm{MeV}$ for Si and 186 MeV for Ge.

2.3 Second quantization

When dark matter lifts one electron of silicon across the gap, the other 10^{26} electrons stay where they were. Written as a wave function, the final state is still an antisymmetric function of all the electron coordinates, and so is the ground state. Written as a list of which single-particle orbitals are filled, the two states differ in two entries: one orbital emptied, one filled. Second quantization is the bookkeeping that works with lists of filled orbitals. Its operators create and destroy particles in orbitals and take care of the exchange signs automatically. Every operator we need takes a compact form in creation and annihilation operators, from the density $n_e(\boldsymbol{q})$ that dark matter couples to, to the Coulomb Hamiltonian of electrons and nuclei. Phonons and magnons later fit into the same language (Sects. 5.3 and 7.1). We follow the standard treatments of Refs. [11, Chap. 1], [12, Chap. 1] and [13, Chap. 1].

2.3.1 Identical particles

Two electrons cannot be told apart, and swapping their labels changes no measurement. Write $x_i = (\boldsymbol{r}_i, s_i)$ for the position and spin projection $s_i = \uparrow, \downarrow$ of particle i. The N-particle wave function $\Psi(x_1, \ldots, x_N)$ can then change under the exchange of two arguments only by a phase, and for the particles of this book that phase is $+1$ or -1. Electrons, and fermions generally, take -1: Ψ is antisymmetric under every exchange. Bosons, such as phonons and photons, take $+1$.

For particles that do not interact, the eigenstates are built from single-particle orbitals $\psi_\gamma(x) = \langle x|\gamma\rangle$, where γ labels a complete orthonormal set of one-particle states, spin included: plane waves $\gamma = (\boldsymbol{k}, s)$ in a box, atomic orbitals, or the Bloch states of Sect. 3.2. The product $\psi_{\gamma_1}(x_1)\cdots\psi_{\gamma_N}(x_N)$ is not antisymmetric. Summing the product over all permutations P of

the arguments, with the sign $(-1)^P$ of each, gives the antisymmetric Slater determinant

$$\Psi_{\gamma_1\ldots\gamma_N}(x_1,\ldots,x_N) = \frac{1}{\sqrt{N!}}\sum_P (-1)^P\,\psi_{\gamma_1}(x_{P1})\cdots\psi_{\gamma_N}(x_{PN}) = \frac{1}{\sqrt{N!}}\det\bigl[\psi_{\gamma_j}(x_i)\bigr]. \quad (2.29)$$

The Pauli principle follows: two equal labels give two equal columns, and the determinant vanishes. Swapping two labels swaps two columns and flips the sign. Orthonormal orbitals make Ψ normalized, and Slater determinants with different sets of labels are orthogonal. The determinants built from all N-element subsets of the labels form a basis of the antisymmetric N-particle states, interacting or not. A correlated ground state is a superposition of many of them.

A determinant is fixed, up to its sign, by which orbitals it contains. We therefore fix an order of the labels once and for all, and write the state as a list of occupation numbers,

$$|n_{\gamma_1}, n_{\gamma_2}, n_{\gamma_3}, \ldots\rangle, \qquad n_\gamma \in \{0,1\},$$

with the determinant's columns in the fixed order. For bosons the same construction uses the permanent (all signs +), any $n_\gamma \geq 0$ is allowed, and a normalization factor $(\prod_\gamma n_\gamma!)^{-1/2}$ is needed because repeated labels make some permutations identical. Fock space is the space spanned by the occupation-number states of every total number $N = 0, 1, 2, \ldots$. Its state with no particles is the vacuum $|0\rangle$. Fock space lets the particle number change. The operators below need a changing particle number even when the physical number of electrons is fixed: removing an electron and putting it back elsewhere passes through the space with one electron fewer.

2.3.2 Fermion operators

The creation operator $c^\dagger_\gamma$ adds an electron in orbital γ and puts it at the front of the list:

$$c^\dagger_\gamma\,|\gamma_1\ldots\gamma_N\rangle = |\gamma\,\gamma_1\ldots\gamma_N\rangle, \quad (2.30)$$

where $|\gamma_1\ldots\gamma_N\rangle$ is the determinant of Eq. (2.29) as a ket. Every determinant is then a string of creation operators acting on the vacuum, $|\gamma_1\ldots\gamma_N\rangle = c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}\cdots c^\dagger_{\gamma_N}|0\rangle$. Since swapping two labels flips the sign of the determinant, $c^\dagger_\gamma c^\dagger_{\gamma'} = -c^\dagger_{\gamma'}c^\dagger_\gamma$, and in particular $(c^\dagger_\gamma)^2 = 0$. The annihilation operator c_γ is the adjoint of $c^\dagger_\gamma$.

To see what c_γ does, act on an occupation-number state. Let $S_\gamma = \sum_{\gamma'<\gamma} n_{\gamma'}$ count the occupied orbitals that come before γ in the fixed order. To bring a new electron in γ from the front of the string to its own place, $c^\dagger_\gamma$ must pass S_γ creation operators. Since each pass costs a sign,

$$\begin{aligned} c^\dagger_\gamma\,|\ldots n_\gamma\ldots\rangle &= (-1)^{S_\gamma}\,(1-n_\gamma)\,|\ldots 1_\gamma\ldots\rangle,\\ c_\gamma\,|\ldots n_\gamma\ldots\rangle &= (-1)^{S_\gamma}\,n_\gamma\,|\ldots 0_\gamma\ldots\rangle. \end{aligned} \quad (2.31)$$

The second relation follows from the first: the only nonzero matrix element of $c^\dagger_\gamma$ takes 0_γ to 1_γ and equals $(-1)^{S_\gamma}$, and its complex conjugate is the matrix element of c_γ between the same states in the other direction. In words, c_γ empties a full orbital and kills an empty one, and neither c_γ nor $c^\dagger_\gamma$ can act twice on the same orbital.

The algebra follows in two lines. For $\gamma = \gamma'$, Eq. (2.31) leads to $c_\gamma c^\dagger_\gamma|\ldots n_\gamma\ldots\rangle = (1-n_\gamma)|\ldots n_\gamma\ldots\rangle$ and $c^\dagger_\gamma c_\gamma|\ldots n_\gamma\ldots\rangle = n_\gamma|\ldots n_\gamma\ldots\rangle$, and the two products add to one. For $\gamma \neq \gamma'$, the two orders of c_γ and $c^\dagger_{\gamma'}$ reach the same final state. In one of the two orders, however, the second operator sees the occupation of the orbital of the first operator changed by one, and the sign factors of the two orders differ by -1. With $\{A,B\} = AB + BA$,

$$\{c_\gamma, c^\dagger_{\gamma'}\} = \delta_{\gamma\gamma'}, \qquad \{c_\gamma, c_{\gamma'}\} = \{c^\dagger_\gamma, c^\dagger_{\gamma'}\} = 0. \quad (2.32)$$

These anticommutation relations contain the whole of Fermi statistics: the Pauli principle is $(c_\gamma^\dagger)^2 = 0$, and the exchange signs of Eq. (2.29) are the signs picked up by reordering operators.

The product $c_\gamma^\dagger c_\gamma$ counts the electrons in γ: its eigenvalue on $|\dots n_\gamma \dots\rangle$ is n_γ. The total number operator is $\hat{N}_e = \sum_\gamma c_\gamma^\dagger c_\gamma$, with eigenvalue N_e on every state with N_e electrons. A Hamiltonian of independent electrons with orbital energies E_γ is $\sum_\gamma E_\gamma\, c_\gamma^\dagger c_\gamma$: each filled orbital adds its energy. In operators, the excitation of the opening paragraph is $c_{\gamma_c}^\dagger c_{\gamma_v}$ acting on the ground state: it moves an electron from a filled orbital γ_v to an empty one γ_c, and its energy is $E_{\gamma_c} - E_{\gamma_v}$ in this approximation.

The commutator of a bilinear with a single operator does most of the work later. By Eq. (2.32),

$$[c_\gamma^\dagger c_{\gamma'},\, c_{\gamma''}^\dagger] = \delta_{\gamma'\gamma''}\, c_\gamma^\dagger, \qquad [c_\gamma^\dagger c_{\gamma'},\, c_{\gamma''}] = -\delta_{\gamma\gamma''}\, c_{\gamma'}, \tag{2.33}$$

because $c_\gamma^\dagger c_{\gamma'} c_{\gamma''}^\dagger = \delta_{\gamma'\gamma''} c_\gamma^\dagger - c_\gamma^\dagger c_{\gamma''}^\dagger c_{\gamma'}$ and the last term equals $-c_{\gamma''}^\dagger c_\gamma^\dagger c_{\gamma'}$ after one swap of creation operators. The sign problems of fermions cancel in pairs here: a bilinear behaves like an ordinary operator that moves one particle. As a result, every bilinear commutes with $\hat{N}_e$, and operators built from bilinears conserve the number of electrons. And for independent electrons, $H_0 = \sum_\gamma E_\gamma c_\gamma^\dagger c_\gamma$, the Heisenberg equation $\mathrm{i}\,\mathrm{d}c_\gamma^\dagger/\mathrm{d}t = [c_\gamma^\dagger, H_0] = -E_\gamma c_\gamma^\dagger$ integrates to $\mathrm{e}^{\mathrm{i}H_0 t} c_\gamma^\dagger \mathrm{e}^{-\mathrm{i}H_0 t} = \mathrm{e}^{\mathrm{i}E_\gamma t} c_\gamma^\dagger$. The bilinear $c_\gamma^\dagger c_{\gamma'}$, which moves an electron from γ' to γ, then oscillates as $\mathrm{e}^{\mathrm{i}(E_\gamma - E_{\gamma'})t}$, at the energy of the transition it causes. The energy denominators of Sect. 2.4 and the response functions of Sect. 4.1 arise from this oscillation.

2.3.3 Boson operators

Bosons work the same way with every sign turned positive. Generic boson operators d_γ, $d_\gamma^\dagger$ act on occupation numbers $n_\gamma = 0, 1, 2, \dots$ as

$$d_\gamma^\dagger\,|\dots n_\gamma \dots\rangle = \sqrt{n_\gamma + 1}\,|\dots n_\gamma + 1 \dots\rangle, \qquad d_\gamma\,|\dots n_\gamma \dots\rangle = \sqrt{n_\gamma}\,|\dots n_\gamma - 1 \dots\rangle, \tag{2.34}$$

with no sign factor, because bosons are symmetric under exchange. The square roots come from the normalization factor of the symmetrized states: $d_\gamma^\dagger$ builds the state with $n_\gamma + 1$ quanta from the state with n_γ, and the unnormalized result has norm $n_\gamma + 1$. Then $d_\gamma d_\gamma^\dagger - d_\gamma^\dagger d_\gamma$ gives $(n_\gamma + 1) - n_\gamma = 1$ on every state, and operators of different modes commute:

$$[d_\gamma, d_{\gamma'}^\dagger] = \delta_{\gamma\gamma'}, \qquad [d_\gamma, d_{\gamma'}] = [d_\gamma^\dagger, d_{\gamma'}^\dagger] = 0. \tag{2.35}$$

The number of quanta in mode γ is $d_\gamma^\dagger d_\gamma$.

The operators d_γ and $d_\gamma^\dagger$ are the ladder operators of the harmonic oscillator. An oscillator of frequency E_γ has $H = E_\gamma(d_\gamma^\dagger d_\gamma + \frac{1}{2})$, and its nth level reads as n quanta in a single mode. A set of uncoupled oscillators, such as the normal modes of a crystal, is a set of bosonic modes. As a result, the quantized lattice vibrations of Sect. 5.3 are particles, phonons, with operators $b_{\boldsymbol{Q}\nu}$. The magnon operators $a_{\boldsymbol{k}\nu}$ of Sect. 7.1 are bosons in the same sense. Eq. (2.34) is the origin of the matrix elements $\sqrt{n}$ for absorption and $\sqrt{n+1}$ for emission quoted in Sect. 2.1, and Sect. 2.5 puts thermal occupations into n.

Unlike electrons, these bosons are created and destroyed in physical processes, and the Hamiltonians that describe their creation and destruction are not built from bilinears $d^\dagger d$ alone. A term of the form $c_\gamma^\dagger c_{\gamma'}(b_{\boldsymbol{Q}\nu} + b_{-\boldsymbol{Q}\nu}^\dagger)$ describes an electron that emits or absorbs a phonon, with phonon wavevector $\boldsymbol{Q}$ and branch ν (Sect. 6.2): the electron bilinear conserves electron number and the boson operator changes the number of phonons by one. Absorption of bosonic dark matter fits the same pattern. If the dark-matter field is quantized, the oscillating amplitude ϕ_0 of Sect. 2.1 becomes an annihilation operator for the dark-matter mode. Between

states with n and $n-1$ quanta the annihilation operator has the matrix element $\sqrt{n}$, which leads to a rate proportional to the number of quanta in the mode, that is, to the density ρ_χ. The classical field of Sect. 2.1 is the limit of large n, where $\sqrt{n}$ and $\sqrt{n+1}$ no longer differ.

2.3.4 One-body operators and the density

The operator that dark matter couples to in electron scattering is $n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$ (Eq. (A.7)). It is a sum over electrons of the same single-particle operator, and so are the kinetic energy $\sum_i \boldsymbol{p}_i^2/2m_e$ and the potential energy $\sum_i U_1(\boldsymbol{r}_i)$ in an external potential U_1. A one-body operator is a sum $\hat{O}_1 = \sum_i \hat{o}_i$, where $\hat{o}_i$ is a fixed one-particle operator $\hat{o}$ acting on particle i. In Fock space it reads

One-body operators

$$\hat{O}_1 = \sum_{i=1}^{N} \hat{o}_i \;=\; \sum_{\gamma\gamma'} \langle\gamma|\hat{o}|\gamma'\rangle\, c_\gamma^\dagger c_{\gamma'}. \tag{2.36}$$

The right-hand side is the same for every N, holds for bosons with $c \to d$, and has the same form in every orthonormal single-particle basis.

Acting on a determinant, $\sum_i \hat{o}_i$ replaces each occupied orbital in turn by $\hat{o}|\gamma_j\rangle = \sum_\gamma |\gamma\rangle\langle\gamma|\hat{o}|\gamma_j\rangle$ and leaves the others unchanged; the bilinear $c_\gamma^\dagger c_{\gamma_j}$ makes exactly this replacement. Appendix D.2 writes out this argument, extends it to bosons, and proves the basis independence stated in the box.

Expressed through the field operator,

$$\hat{\psi}_s(\boldsymbol{r}) = \sum_\gamma \psi_\gamma(\boldsymbol{r}, s)\, c_\gamma, \qquad \{\hat{\psi}_s(\boldsymbol{r}), \hat{\psi}_{s'}^\dagger(\boldsymbol{r}')\} = \delta_{ss'}\, \delta^3(\boldsymbol{r}-\boldsymbol{r}'), \tag{2.37}$$

Eq. (2.36) takes a basis-free form. The anticommutator follows from Eq. (2.32) and the completeness of the orbitals, $\sum_\gamma \psi_\gamma(\boldsymbol{r}, s)\psi_\gamma^*(\boldsymbol{r}', s') = \delta_{ss'}\delta^3(\boldsymbol{r}-\boldsymbol{r}')$. We see that $\hat{\psi}_s^\dagger(\boldsymbol{r})$ creates an electron at the point $\boldsymbol{r}$ with spin s. The density of electrons at $\boldsymbol{r}$, which Appendix A.3 writes as $n_e(\boldsymbol{r}) = \sum_i \delta^3(\boldsymbol{r}-\boldsymbol{r}_i)$, becomes $\sum_s \hat{\psi}_s^\dagger(\boldsymbol{r})\hat{\psi}_s(\boldsymbol{r})$. For an operator that is local in position and spin-independent, such as $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ or $U_1(\boldsymbol{r})$, Eq. (2.36) becomes $\sum_s \int \mathrm{d}^3r\, \hat{\psi}_s^\dagger(\boldsymbol{r})\, o(\boldsymbol{r})\, \hat{\psi}_s(\boldsymbol{r})$; the kinetic energy puts $-\nabla^2/2m_e$ between the two field operators.

In a box with plane waves, $\psi_{\boldsymbol{k}s}(\boldsymbol{r}, s') = \delta_{ss'}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}/\sqrt{\mathcal{V}}$ (Eq. (2.10)), and the field operator is $\hat{\psi}_s(\boldsymbol{r}) = \mathcal{V}^{-1/2}\sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\, c_{\boldsymbol{k}s}$. The matrix element of the density is a Kronecker delta,

$$\langle \boldsymbol{k}'s'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\boldsymbol{k}s\rangle = \frac{\delta_{ss'}}{\mathcal{V}} \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{r}} = \delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}\, \delta_{ss'},$$

so

$$n_e(\boldsymbol{q}) = \sum_{\gamma\gamma'} \langle\gamma|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma'\rangle\, c_\gamma^\dagger c_{\gamma'} = \sum_{\boldsymbol{k}s} c_{\boldsymbol{k}+\boldsymbol{q},s}^\dagger\, c_{\boldsymbol{k}s}. \tag{2.38}$$

The density operator takes an electron from $\boldsymbol{k}$ and puts it at $\boldsymbol{k}+\boldsymbol{q}$, keeping its spin. The move from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$ is the momentum bookkeeping of a scattering event: the dark matter goes from $\boldsymbol{p}$ to $\boldsymbol{p}-\boldsymbol{q}$, and Eq. (A.8) shows that, between these two dark-matter states, the interaction of the dark matter with the electrons is $\tilde{V}(\boldsymbol{q})\, n_e(\boldsymbol{q})/\mathcal{V}$. We can therefore write every electronic scattering rate of this book as a matrix element of $n_e(\boldsymbol{q})$ between many-electron states, squared and summed (Sect. 4.2). Note that $n_e(\boldsymbol{q})^\dagger = n_e(-\boldsymbol{q})$, and that $n_e(\boldsymbol{0}) = \hat{N}_e$. In a crystal the same formula in the Bloch basis gives the density vertex of Sect. 3.2.

The plane-wave form also shows the Pauli principle at work in a scattering event. Acting on a state in which the electrons fill all plane waves inside some region of $\boldsymbol{k}$ space, the term

$c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s}c_{\boldsymbol{k}s}$ survives only if $\boldsymbol{k}$ is filled and $\boldsymbol{k}+\boldsymbol{q}$ is empty. For a momentum transfer q small compared with the size of the filled region, only electrons within about q of its surface can take the momentum, and the rest are blocked. Pauli blocking makes the electrons of a metal respond to slow dark matter as a thin shell rather than as the whole electron density, and Sects. 2.5 and 4.5 make the thin-shell response quantitative. At q larger than the filled region no electron is blocked, and every term contributes. In a semiconductor the same counting sets the electronic excitations, with the valence bands in place of the filled region and the conduction bands in place of the empty states.

For independent electrons the many-body matrix element of $n_e(\boldsymbol{q})$ reduces to a single-particle one. Let $|\Phi_0\rangle$ be a determinant with orbital γ_v filled and γ_c empty, and $c^\dagger_{\gamma_c}c_{\gamma_v}|\Phi_0\rangle$ the excited determinant, whose bra is $\langle\Phi_0|c^\dagger_{\gamma_v}c_{\gamma_c}$. In $\langle\Phi_0|c^\dagger_{\gamma_v}c_{\gamma_c}\,c^\dagger_{\gamma}c_{\gamma'}|\Phi_0\rangle$ the bilinear $c^\dagger_{\gamma}c_{\gamma'}$ must undo the excitation, and only $\gamma=\gamma_c$, $\gamma'=\gamma_v$ survives, with weight one. With Eq. (2.38),

$$\langle\Phi_0|c^\dagger_{\gamma_v}c_{\gamma_c}\;n_e(\boldsymbol{q})\;|\Phi_0\rangle = \langle\gamma_c|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma_v\rangle, \tag{2.39}$$

which is the transition form factor $f_{i\to f}(\boldsymbol{q})$ of Sect. 2.2 for the electron that moves, whatever the other electrons do. The other electrons enter only through which orbitals are blocked. Interactions spoil the single-particle reduction of Eq. (2.39), and Sects. 4.2 and 4.6 show what replaces it.

A one-body potential in second quantization follows from the inverse transform. With Eqs. (A.5) and (A.9), $U_1(\boldsymbol{r}) = \mathscr{V}^{-1}\sum_{\boldsymbol{q}}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}U_1(\boldsymbol{q})$, and $\sum_i U_1(\boldsymbol{r}_i) = \mathscr{V}^{-1}\sum_{\boldsymbol{q}}U_1(\boldsymbol{q})\,n_e(-\boldsymbol{q})$: a potential with a Fourier component at $\boldsymbol{q}$ moves electrons by $-\boldsymbol{q}$. For plane waves the kinetic energy is diagonal, $\sum_{\boldsymbol{k}s}(k^2/2m_e)\,c^\dagger_{\boldsymbol{k}s}c_{\boldsymbol{k}s}$. Table 2.1 collects these forms with the two-body operators derived below.

2.3.5 Change of basis

A calculation often starts in one basis and ends in another: plane waves for the Coulomb interaction, Bloch states for the band structure (Sect. 3.2), localized Wannier orbitals for interpolation. A new orthonormal basis $|\tilde\gamma\rangle = \sum_\gamma |\gamma\rangle\langle\gamma|\tilde\gamma\rangle$ comes with new operators, defined by the rule of Eq. (2.30). A creation operator adds a particle in a given orbital, so it expands like the orbital:

$$c^\dagger_{\tilde\gamma} = \sum_\gamma \langle\gamma|\tilde\gamma\rangle\, c^\dagger_\gamma, \qquad c_{\tilde\gamma} = \sum_\gamma \langle\tilde\gamma|\gamma\rangle\, c_\gamma. \tag{2.40}$$

The coefficients form a unitary matrix, so the anticommutators survive,

$$\{c_{\tilde\gamma}, c^\dagger_{\tilde\gamma'}\} = \sum_\gamma \langle\tilde\gamma|\gamma\rangle\langle\gamma|\tilde\gamma'\rangle = \delta_{\tilde\gamma\tilde\gamma'}.$$

Inserting Eq. (2.40) into Eq. (2.36) and using completeness twice leads to $\sum_{\tilde\gamma\tilde\gamma'}\langle\tilde\gamma|\hat o|\tilde\gamma'\rangle c^\dagger_{\tilde\gamma}c_{\tilde\gamma'}$: one-body operators look the same in every basis, with the matrix elements of the new one. The field operator of Eq. (2.37) is the special case in which the new basis is the position eigenbasis, $\langle\gamma|\boldsymbol{r}s\rangle = \psi^*_\gamma(\boldsymbol{r},s)$. In the basis that diagonalizes $\hat o$, the one-body operator reduces to $\sum_\gamma o_\gamma c^\dagger_\gamma c_\gamma$. Because a Slater determinant in the old basis becomes a sum of determinants in the new one (Appendix D.2), the occupation-number language, and not the wave function, is the economical one.

Take two orthonormal orbitals $|A\rangle$ and $|B\rangle$, one on each of two neighboring atoms, and the bonding and antibonding combinations $|\pm\rangle = (|A\rangle \pm |B\rangle)/\sqrt{2}$, a caricature of the covalent bond of silicon. Two electrons of opposite spin in the bonding orbital form one determinant, $c^\dagger_{+\uparrow}c^\dagger_{+\downarrow}|0\rangle$. By Eq. (2.40) the determinant is

$$c^\dagger_{+\uparrow}c^\dagger_{+\downarrow}|0\rangle = \tfrac{1}{2}(c^\dagger_{A\uparrow}c^\dagger_{A\downarrow} + c^\dagger_{B\uparrow}c^\dagger_{B\downarrow} + c^\dagger_{A\uparrow}c^\dagger_{B\downarrow} + c^\dagger_{B\uparrow}c^\dagger_{A\downarrow})|0\rangle,$$

Table 2.1: Operators used in this book in first quantization (sums over electrons i and nuclei I) and in second quantization, in a general orthonormal single-particle basis $\{|\gamma\rangle\}$ (spin included in γ) and in box-normalized plane waves $|\boldsymbol{k}s\rangle$ of a box of volume $\mathcal{V}$. Fourier transforms carry $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$; $v_{\mathrm{C}}(k) = e^2/k^2$ is the Heaviside–Lorentz Coulomb kernel, $e^2 = 4\pi\alpha$; Z_κ is the charge number of nuclear species κ and $n_\kappa(\boldsymbol{k}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}$ its density, with $\boldsymbol{X}_I$ the position of nucleus I. Two-body elements are $\langle\gamma_1\gamma_2|U_2|\gamma_3\gamma_4\rangle = \int \mathrm{d}x\,\mathrm{d}x'\,\psi^*_{\gamma_1}(x)\psi^*_{\gamma_2}(x')U_2(\boldsymbol{r}-\boldsymbol{r}')\psi_{\gamma_3}(x)\psi_{\gamma_4}(x')$, with $x = (\boldsymbol{r}, s)$ and $\int \mathrm{d}x = \sum_s \int \mathrm{d}^3r$. The $\boldsymbol{k} = 0$ Coulomb terms are removed by charge neutrality (Worked example 2.5).

First quantization	General basis	Plane waves
$N_e = \sum_i 1$	$\hat{N}_e = \sum_\gamma c^\dagger_\gamma c_\gamma$	$\sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s}$
$\sum_i \boldsymbol{p}_i^2/2m_e$	$\sum_{\gamma\gamma'}\langle\gamma\|\frac{\boldsymbol{p}^2}{2m_e}\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$	$\sum_{\boldsymbol{k}s}\frac{k^2}{2m_e} c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s}$
$\sum_i U_1(\boldsymbol{r}_i)$	$\sum_{\gamma\gamma'}\langle\gamma\|U_1\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$	$\frac{1}{\mathcal{V}}\sum_{\boldsymbol{q}} U_1(\boldsymbol{q})\, n_e(-\boldsymbol{q})$
$n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$	$\sum_{\gamma\gamma'}\langle\gamma\|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$	$\sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$
dark matter $\boldsymbol{p} \to \boldsymbol{p} - \boldsymbol{q}$: $\frac{1}{\mathcal{V}}\tilde{V}(\boldsymbol{q})\sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$	$\frac{1}{\mathcal{V}}\tilde{V}(\boldsymbol{q})\, n_e(\boldsymbol{q})$	$\frac{1}{\mathcal{V}}\tilde{V}(\boldsymbol{q})\sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$
$\frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i - \boldsymbol{r}_j)$	$\frac{1}{2}\sum\langle\gamma_1\gamma_2\|U_2\|\gamma_3\gamma_4\rangle$ $\times c^\dagger_{\gamma_1} c^\dagger_{\gamma_2} c_{\gamma_4} c_{\gamma_3}$	$\frac{1}{2\mathcal{V}}\sum_{\boldsymbol{k}\boldsymbol{p}\boldsymbol{p}'ss'} U_2(\boldsymbol{k})$ $\times c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'} c_{\boldsymbol{p}s}$
$\sum_{i<j}\dfrac{e^2}{4\pi\lvert\boldsymbol{r}_i - \boldsymbol{r}_j\rvert}$	$U_2 \to e^2/(4\pi\lvert\boldsymbol{r} - \boldsymbol{r}'\rvert)$	$\frac{1}{2\mathcal{V}}\sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k)\big[n_e(\boldsymbol{k})^\dagger n_e(\boldsymbol{k}) - \hat{N}_e\big]$
$-\sum_{iI}\dfrac{Z_{\kappa(I)}e^2}{4\pi\lvert\boldsymbol{r}_i - \boldsymbol{X}_I\rvert}$	$\sum_{\gamma\gamma'}\langle\gamma\|U_1\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$, $U_1(\boldsymbol{r}) = -\sum_I \dfrac{Z_{\kappa(I)}e^2}{4\pi\lvert\boldsymbol{r} - \boldsymbol{X}_I\rvert}$	$-\frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k)\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k})\, n_e(\boldsymbol{k})^\dagger$

four determinants in the atomic basis: two with both electrons on one atom, and two with one electron on each. The same state is simple in one basis and a superposition in the other, and the operators of Eqs. (2.36) and (2.42) handle both without any change of form.

2.3.6 Two-body operators and normal ordering

The Coulomb repulsion $\frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i-\boldsymbol{r}_j)$, with $U_2(\boldsymbol{r})=e^2/(4\pi r)$, is a sum over pairs, and the condition $i\neq j$ matters: an electron does not repel itself. To see what the Coulomb repulsion becomes, split the pair sum into all pairs minus the diagonal, $\sum_{i\neq j}=\sum_{i,j}-\sum_{i=j}$, and write each sum through the density $n_e(\boldsymbol{r})=\sum_i \delta^3(\boldsymbol{r}-\boldsymbol{r}_i)$:

$$\frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i-\boldsymbol{r}_j)=\frac{1}{2}\int \mathrm{d}^3r\,\mathrm{d}^3r'\,U_2(\boldsymbol{r}-\boldsymbol{r}')\,[n_e(\boldsymbol{r})n_e(\boldsymbol{r}')-\delta^3(\boldsymbol{r}-\boldsymbol{r}')\,n_e(\boldsymbol{r})].$$

Each density is now the bilinear $\sum_s \hat\psi_s^\dagger\hat\psi_s$. Moving the inner annihilation operator in $\hat\psi_s^\dagger(\boldsymbol{r})\hat\psi_s(\boldsymbol{r})\hat\psi_{s'}^\dagger(\boldsymbol{r}')\hat\psi_{s'}(\boldsymbol{r}')$ to the right with Eq. (2.37) produces a term $\delta_{ss'}\delta^3(\boldsymbol{r}-\boldsymbol{r}')\hat\psi_s^\dagger(\boldsymbol{r})\hat\psi_s(\boldsymbol{r})$, which cancels the self-interaction term exactly. For $\hat O_2=\frac{1}{2}\sum_{i\neq j}U_2(\boldsymbol{r}_i-\boldsymbol{r}_j)$, the remaining term is

$$\hat O_2=\frac{1}{2}\sum_{ss'}\int \mathrm{d}^3r\,\mathrm{d}^3r'\,U_2(\boldsymbol{r}-\boldsymbol{r}')\,\hat\psi_s^\dagger(\boldsymbol{r})\hat\psi_{s'}^\dagger(\boldsymbol{r}')\hat\psi_{s'}(\boldsymbol{r}')\hat\psi_s(\boldsymbol{r}). \tag{2.41}$$

All creation operators stand to the left of all annihilation operators. An operator product is called normal ordered, and written $:\cdots:$, when it has been rearranged into this order, with the sign of the permutation for fermions. Thus $\hat O_2=\frac{1}{2}\int\!\int U_2 :n_e(\boldsymbol{r})n_e(\boldsymbol{r}'):$, and normal ordering removes the self-interaction. Since a normal-ordered product of two annihilation operators vanishes on any state with fewer than two particles, a single electron feels no Coulomb energy of its own, as it should, and the vacuum has none either. Expanding the field operators in a basis gives the general form of Table 2.1,

$$\hat O_2=\frac{1}{2}\sum_{\gamma_1\gamma_2\gamma_3\gamma_4}\langle\gamma_1\gamma_2|U_2|\gamma_3\gamma_4\rangle\,c_{\gamma_1}^\dagger c_{\gamma_2}^\dagger c_{\gamma_4}c_{\gamma_3}. \tag{2.42}$$

The order of the last two indices matters: particle 2 is annihilated first and created last, so the two operators of each particle stand symmetrically about the middle. Reversing the last two indices flips the sign. Appendix D.2 proves Eq. (2.42) for any two-body operator, spin-dependent or not. The proof applies the one-body result twice, in the operator form of the split $\sum_{i,j}-\sum_{i=j}$ used above.

2.3.7 Coulomb interaction in momentum space

The Coulomb potential $e^2/(4\pi r)$ has the Fourier transform (Eq. (A.3))

$$v_{\mathrm{C}}(k)=\frac{e^2}{k^2}=\frac{4\pi\alpha}{k^2}\qquad\text{(Heaviside–Lorentz)}, \tag{2.43}$$

the $m_V\to 0$ limit of the Yukawa transform of Eq. (C.79). In Gaussian units, where the electron charge e_{G} obeys $e_{\mathrm{G}}^2=\alpha$, the same kernel reads $4\pi e_{\mathrm{G}}^2/k^2$, and in SI units $e_{\mathrm{SI}}^2/(\varepsilon_0^{\mathrm{SI}}k^2)$ (Eq. (A.4)).

In the box, the inverse transform is a sum, $e^2/(4\pi|\boldsymbol{r}|)=\mathscr{V}^{-1}\sum_{\boldsymbol{k}} v_{\mathrm{C}}(k)\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, and the pair sum factorizes into densities at once. Using the split $\sum_{i\neq j}=\sum_{i,j}-\sum_{i=j}$ again, with $\sum_{i,j}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)}=n_e(-\boldsymbol{k})\,n_e(\boldsymbol{k})$ and $\sum_{i=j}1=N_e$,

$$H_{ee}=\frac{1}{2}\sum_{i\neq j}\frac{e^2}{4\pi|\boldsymbol{r}_i-\boldsymbol{r}_j|}=\frac{1}{2\mathscr{V}}\sum_{\boldsymbol{k}} v_{\mathrm{C}}(k)\,[n_e(\boldsymbol{k})^\dagger n_e(\boldsymbol{k})-\hat N_e]. \tag{2.44}$$

The Coulomb energy of the electrons is a sum over Fourier components of the density fluctuation, each weighted by $v_{\mathrm{C}}(k)$, with the self-energy of each point charge subtracted. Dimensions check: v_{C} is in eV^{-2}, $1/\mathcal{V}$ in eV^3, and the densities are numbers. To get the plane-wave form, insert Eq. (2.38) with $n_e(\boldsymbol{k})^\dagger = \sum_{\boldsymbol{p}'s'} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'}$ and normal order. The anticommutator $c_{\boldsymbol{p}'s'} c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} = \delta_{\boldsymbol{p}',\boldsymbol{p}+\boldsymbol{k}}\delta_{ss'} - c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c_{\boldsymbol{p}'s'}$ gives a one-body piece equal to $\hat{N}_e$, which cancels the subtracted $\hat{N}_e$. In the two-body piece that remains, swapping the two creation operators removes the minus sign:

$$n_e(\boldsymbol{k})^\dagger n_e(\boldsymbol{k}) - \hat{N}_e = \sum_{\boldsymbol{p}\boldsymbol{p}'ss'} c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}\, c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}\, c_{\boldsymbol{p}'s'}\, c_{\boldsymbol{p}s}. \tag{2.45}$$

Eq. (2.45) is the matrix element of Eq. (2.42) for plane waves: two electrons with momenta $\boldsymbol{p}$ and $\boldsymbol{p}'$ exchange a momentum $\boldsymbol{k}$, each keeps its spin, and the amplitude is $v_{\mathrm{C}}(k)/\mathcal{V}$. Without normal ordering, the $\hat{N}_e$ term would carry $\mathcal{V}^{-1}\sum_{\boldsymbol{k}} v_{\mathrm{C}}(k) = \int \mathrm{d}^3k\,(2\pi)^{-3}e^2/k^2$, which diverges linearly at large k: the infinite Coulomb self-energy of a point charge, which has nothing to do with the solid.

The exchange sign shows up as soon as we take a matrix element. Let two electrons start in $c^\dagger_{\boldsymbol{p}s}c^\dagger_{\boldsymbol{p}'s'}|0\rangle$ and end in $c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}|0\rangle$, with $\boldsymbol{k}\neq 0$. The two annihilation operators of Eq. (2.45) can remove the initial pair in two ways, with signs $+1$ and -1. Each of the two removals combines with two matchings of the creation operators to the final pair, one at momentum exchange $\boldsymbol{k}$ and the other at $\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}$. The four terms add to $2[v_{\mathrm{C}}(k)-\delta_{ss'}v_{\mathrm{C}}(|\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}|)]/\mathcal{V}$, and the factor $\frac{1}{2}$ of Eq. (2.42) cancels the 2, which counts the two equivalent labelings of the pair. The result is

$$\langle \boldsymbol{p}+\boldsymbol{k}\,s,\ \boldsymbol{p}'-\boldsymbol{k}\,s'|H_{ee}|\boldsymbol{p}s,\ \boldsymbol{p}'s'\rangle = \frac{1}{\mathcal{V}}\Big[v_{\mathrm{C}}(k) - \delta_{ss'}\,v_{\mathrm{C}}(|\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}|)\Big]. \tag{2.46}$$

The first term is the direct process: electron $\boldsymbol{p}$ takes the momentum $\boldsymbol{k}$. The second is the exchange process, in which the electrons end up in the same final orbitals but each in the other's: electron $\boldsymbol{p}$ goes to $\boldsymbol{p}'-\boldsymbol{k}$, so the momentum it takes is $\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}$. Because the two processes lead to the same final state, their amplitudes add, and Fermi statistics attaches a minus sign to the exchange. Electrons with opposite spins, however, can be told apart by their spins, and for them the exchange term is absent. The exchange term lowers the energy of parallel spins, and it is the origin of the exchange energy and of the exchange coupling of magnetic order (Sect. 7.1).

The same counting gives the Coulomb energy of a single determinant, the starting point of the Hartree–Fock method. In $\langle\Phi_0|c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}c_{\gamma_4}c_{\gamma_3}|\Phi_0\rangle$ the annihilation operators must remove two occupied orbitals and the creation operators must put the same two back, either in the same places or swapped. For orthonormal orbitals Eq. (2.42) then reduces to

$$\langle\Phi_0|\hat{O}_2|\Phi_0\rangle = \frac{1}{2}\sum_{\gamma,\gamma'\ \text{occupied}} \Big[\langle\gamma\gamma'|U_2|\gamma\gamma'\rangle - \langle\gamma\gamma'|U_2|\gamma'\gamma\rangle\Big],$$

a direct (Hartree) term, the classical electrostatic energy of the charge distribution, minus an exchange (Fock) term. The diagonal $\gamma=\gamma'$ cancels between the direct and exchange terms: no orbital interacts with itself. Sect. 3.4 minimizes this energy and then improves on it.

2.3.8 Coulomb Hamiltonian of a solid

Eq. (2.44) has one more infinity, at $\boldsymbol{k}=0$, where $v_{\mathrm{C}}(0)$ diverges. The $\boldsymbol{k}=0$ term is the energy of a uniform charge spread over the whole box; this energy grows faster than the volume, and in a neutral system the positive charge cancels it. In jellium, the simplest model of a metal or

of a valence electron gas, a uniform positive background replaces the nuclei. Jellium (used again in Sects. 2.5 and 4.5) shows how the cancellation works, in Worked example 2.5 below.

How large is the Coulomb term? A rough answer comes from the mean spacing $d = \boldsymbol{n}_e^{-1/3}$ of the electrons. For the four valence electrons per atom of silicon, $\boldsymbol{n}_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ (Appendix A.2), and $d = 1.71\,\text{Å} = 3.2\,a_0$, with $a_0 = 1/(\alpha m_e)$ the Bohr radius (Sect. 2.2). The Coulomb energy of two neighbors, α/d, is then about 8.4 eV. The kinetic energy must respect the Pauli principle, and for a free gas at this density its mean is $\frac{3}{5}E_\mathrm{F} = 7.5\,\mathrm{eV}$ (Sect. 2.5), about the same. The field measures the ratio of the Coulomb to the kinetic energy by $r_s = (3/4\pi\boldsymbol{n}_e)^{1/3}/a_0$, the radius per electron in units of a_0: the Coulomb energy scales as $1/r_s$ and the kinetic energy as $1/r_s^2$, and their ratio grows as r_s. For silicon's valence density $r_s = 2.0$. The interaction is therefore not a small perturbation of free electrons. It is treated instead by screening (Sect. 4.4) and by self-consistent fields, and both start from the Coulomb Hamiltonian of Worked example 2.5.

Worked example 2.5: The Coulomb Hamiltonian of jellium and of a solid

Take N_e electrons in the box with a uniform positive background of number density $N_e/\mathcal{V}$ and charge $+e$ per unit, so the system is neutral. To give the $\boldsymbol{k} = 0$ terms a meaning, screen the Coulomb potential at a large length $1/k_\mathrm{reg}$, so $e^2/(4\pi r) \to e^2\mathrm{e}^{-k_\mathrm{reg}r}/(4\pi r)$ and $v_\mathrm{C}(k) \to e^2/(k^2 + k_\mathrm{reg}^2)$ by Eq. (C.79). We take $\mathcal{V} \to \infty$ at fixed $N_e/\mathcal{V}$ first, with $\mathcal{V}^{1/3} \gg 1/k_\mathrm{reg}$, and $k_\mathrm{reg} \to 0$ at the end. The integral of the screened potential over all space is $v_\mathrm{C}(0) = e^2/k_\mathrm{reg}^2$. The uniform background has a Fourier component only at $\boldsymbol{k} = 0$, where three terms meet:

- the background with itself, $+\frac{1}{2}(N_e/\mathcal{V})^2\,\mathcal{V}\,v_\mathrm{C}(0) = +N_e^2\,v_\mathrm{C}(0)/2\mathcal{V}$;
- the electrons with the background, $-(N_e/\mathcal{V})\sum_i \int \mathrm{d}^3r\, e^2\mathrm{e}^{-k_\mathrm{reg}|\boldsymbol{r}_i-\boldsymbol{r}|}/(4\pi|\boldsymbol{r}_i-\boldsymbol{r}|) = -N_e^2\,v_\mathrm{C}(0)/\mathcal{V}$;
- the $\boldsymbol{k} = 0$ term of Eq. (2.44), $+v_\mathrm{C}(0)\,(N_e^2 - N_e)/2\mathcal{V}$, since $n_e(\boldsymbol{0}) = \hat{N}_e = N_e$.

The terms in N_e^2, each of order $N_e \cdot (N_e/\mathcal{V})/k_\mathrm{reg}^2$ and so proportional to the volume, cancel exactly. The term linear in N_e remains, $-N_e\,v_\mathrm{C}(0)/2\mathcal{V} = -N_e e^2/(2Vk_\mathrm{reg}^2)$, and per electron it vanishes as $\mathcal{V} \to \infty$. Then $k_\mathrm{reg} \to 0$ restores $v_\mathrm{C}(k) = e^2/k^2$ at every $\boldsymbol{k} \neq 0$. With the kinetic energy of Table 2.1 and Eq. (2.45), the jellium Hamiltonian is [11, Chap. 2]

$$H_\mathrm{jel} = \sum_{\boldsymbol{k}s} \frac{k^2}{2m_e}\, c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s} + \frac{1}{2\mathcal{V}} \sum_{\boldsymbol{k}\neq 0} \sum_{\boldsymbol{p}\boldsymbol{p}' ss'} v_\mathrm{C}(k)\, c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}\, c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}\, c_{\boldsymbol{p}'s'}\, c_{\boldsymbol{p}s}. \tag{2.47}$$

The background has disappeared, apart from the instruction to drop $\boldsymbol{k} = 0$. The order of limits matters. With $k_\mathrm{reg} \to 0$ first, the screening length would exceed the box, and the three terms would each be infinite.

A solid replaces the background by point nuclei. Nucleus I sits at $\boldsymbol{X}_I$, belongs to a species $\kappa(I)$ with charge number Z_κ and mass M_κ (for silicon one species, $Z_\kappa = 14$; species are treated systematically in Sect. 5.2), and the species density is $n_\kappa(\boldsymbol{k}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}$. Every pair of point charges interacts through the product of their charge numbers, -1 for an electron and Z_κ for a nucleus, times $e^2/(4\pi r)$ at separation r. Splitting the pair sum as in Eq. (2.44) leads to

$$H_\mathrm{Coul} = \frac{1}{2\mathcal{V}} \sum_{\boldsymbol{k}} v_\mathrm{C}(k) \Big[n_Z(\boldsymbol{k})^\dagger n_Z(\boldsymbol{k}) - N_e - \sum_\kappa Z_\kappa^2 N_\kappa \Big], \quad n_Z(\boldsymbol{k}) = -n_e(\boldsymbol{k}) + \sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k}),$$

where N_κ counts the nuclei of species κ. Neutrality, $\sum_\kappa Z_\kappa N_\kappa = N_e$, now holds as an operator identity, $n_Z(\mathbf{0}) = 0$. The $\boldsymbol{k} = 0$ term reduces to the self-energy subtraction, $-v_{\mathrm{C}}(0)(N_e + \sum_\kappa Z_\kappa^2 N_\kappa)/2\mathcal{V}$, which vanishes per particle in the same limit as before. The nuclei are their own neutralizing background. Expanding $n_Z^\dagger n_Z$ leads to three pieces. The electron–electron piece is Eq. (2.45). The cross terms $-Z_\kappa[n_e(\boldsymbol{k})^\dagger n_\kappa(\boldsymbol{k}) + n_\kappa(\boldsymbol{k})^\dagger n_e(\boldsymbol{k})]$ are equal after $\boldsymbol{k} \to -\boldsymbol{k}$, since v_{C} is even and electron and nuclear coordinates commute, and add to twice the first cross term. The nuclear piece keeps its own self-energy subtraction.

Collecting the pieces, with the nuclei left in first quantization because their quantization belongs to the lattice (Sect. 5.3):

Coulomb Hamiltonian of electrons and point nuclei

$$H_C = \sum_{\boldsymbol{k}s} \frac{k^2}{2m_e} c_{\boldsymbol{k}s}^\dagger c_{\boldsymbol{k}s} - \sum_I \frac{\nabla_I^2}{2M_{\kappa(I)}} + \frac{1}{2\mathcal{V}} \sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k) \Big[\sum_{\boldsymbol{p}\boldsymbol{p}'ss'} c_{\boldsymbol{p}+\boldsymbol{k},s}^\dagger c_{\boldsymbol{p}'-\boldsymbol{k},s'}^\dagger c_{\boldsymbol{p}'s'} c_{\boldsymbol{p}s} - 2\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k}) \sum_{\boldsymbol{p}s} c_{\boldsymbol{p}-\boldsymbol{k},s}^\dagger c_{\boldsymbol{p}s} + \sum_{\kappa\kappa'} Z_\kappa Z_{\kappa'} \big(n_\kappa(\boldsymbol{k})^\dagger n_{\kappa'}(\boldsymbol{k}) - \delta_{\kappa\kappa'} N_\kappa\big) \Big]. \quad (2.48)$$

The three terms in the bracket are the electron–electron, electron–nucleus and nucleus–nucleus interactions; ∇_I acts on $\boldsymbol{X}_I$, and $\boldsymbol{k} = 0$ is removed by neutrality.

The powers of $\mathcal{V}$ in Eq. (2.48) are a useful check, in the spirit of Sect. 2.1. The energy of a large system must be extensive, proportional to $\mathcal{V}$ at fixed density. In the electron–electron term, take the expectation value in a determinant of plane waves. Only terms that put both electrons back where they started contribute. The direct one has $\boldsymbol{k} = 0$ and is gone with the background; the exchange one is the diagonal case of Eq. (2.46) with $\boldsymbol{k} = \boldsymbol{p}' - \boldsymbol{p}$ and leaves $-(1/2\mathcal{V}) \sum_{\boldsymbol{p}\neq\boldsymbol{p}'} v_{\mathrm{C}}(|\boldsymbol{p} - \boldsymbol{p}'|)$ for each spin, summed over occupied momenta. By Eq. (2.10) each sum over momenta brings a factor $\mathcal{V}$, and we see that the total is $\mathcal{V}^2/\mathcal{V} = \mathcal{V}$ times an integral over the occupied region: extensive, as it must be. Applied to the subtracted self-energy, $(1/2\mathcal{V}) \sum_{\boldsymbol{k}} v_{\mathrm{C}}(k) N_e$, the same count yields N_e times a divergent integral, and this divergence is the reason the self-energy is removed rather than kept. A stray power of $\mathcal{V}$ in a Coulomb expression nearly always means a misplaced $\boldsymbol{k} = 0$ term or a forgotten box normalization.

The electron–nucleus term is the one-body potential $\sum_i U_1(\boldsymbol{r}_i)$ of Table 2.1 with $U_1(\boldsymbol{k}) = -v_{\mathrm{C}}(k) \sum_\kappa Z_\kappa n_\kappa(\boldsymbol{k})$. As a check, consider the matrix element of U_1 for an electron going from $\boldsymbol{p}$ to $\boldsymbol{p} - \boldsymbol{k}$,

$$\langle \boldsymbol{p} - \boldsymbol{k} | U_1 | \boldsymbol{p} \rangle = -\frac{1}{\mathcal{V}} \sum_I Z_{\kappa(I)}\, v_{\mathrm{C}}(k)\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I},$$

which is what Eq. (A.8) gives for the potential $-\sum_I Z_{\kappa(I)} e^2/(4\pi|\boldsymbol{r} - \boldsymbol{X}_I|)$, with the nuclei in the role of the dark matter. When the nuclei are held at fixed positions, $n_\kappa(\boldsymbol{k})$ is a number, and for a perfect crystal it is nonzero only at reciprocal-lattice vectors. The components of $n_\kappa(\boldsymbol{k})$ at reciprocal-lattice vectors lead to the periodic potential of the band structure (Sects. 3.1 and 3.2).

Momentum conservation reads off Eq. (2.48) directly. In every electron–electron term the electrons gain $\boldsymbol{k}$ and lose $\boldsymbol{k}$, and the total electron momentum is conserved by the Coulomb repulsion. The electron–nucleus term moves electron momentum by $-\boldsymbol{k}$ and multiplies by $n_\kappa(\boldsymbol{k})$, which, for mobile nuclei, shifts the nuclear momenta by $+\boldsymbol{k}$: the total momentum of electrons and nuclei is conserved. With the nuclei clamped in a crystal, the electrons can gain

or lose only reciprocal-lattice vectors, and their momentum is conserved only modulo these vectors. Conservation modulo reciprocal-lattice vectors is the origin of crystal momentum (Sect. 3.1).

Eq. (2.48) is the starting point of everything the book computes about the target. It is nonrelativistic and treats the nuclei as point charges; spin–orbit coupling and the relativistic corrections of core electrons are added where they matter (Sect. 3.3). Sect. 5.1 separates the motion of the nuclei from that of the electrons, and the dark-matter coupling enters only through $n_e(\boldsymbol{q})$ for electrons, or $n_\kappa(\boldsymbol{q})$ for nuclei. No one, however, solves Eq. (2.48) for 10^{26} electrons as it stands. Density-functional theory replaces the electron–electron term by an effective one-body potential.

2.4 Second-order processes

A photon of 1.2 eV carries a momentum of about $6 \times 10^{-4}\,\text{Å}^{-1}$. In silicon, the crystal momentum (Sect. 1.3) of the lowest conduction states differs from the crystal momentum of the top of the valence band by a large fraction of $2\pi/a = 1.16\,\text{Å}^{-1}$ (Sect. 3.3). The photon has enough energy to cross the gap, 1.12 eV at 300 K, but too little crystal momentum, and the golden rule of Sect. 2.1 gives zero for the absorption of the photon. Silicon nevertheless absorbs these photons, because a phonon can supply the crystal momentum that the photon lacks. The absorption is a second-order process. The photon first takes the system from the initial state to an intermediate state, in a transition that conserves crystal momentum but not energy. Emitting or absorbing a phonon then takes the system from the intermediate state to a final state, and the whole process, from the initial to the final state, conserves both energy and crystal momentum. The intermediate state is never observed; it enters the absorption rate through an energy denominator, which is the difference between the energy of the initial state and the energy of the intermediate state. Our treatment of second-order processes follows Refs. [1, Chap. 5] and [14, Chaps. I and III].

2.4.1 Interaction picture and Dyson series

To go beyond first order we need the amplitude equations Eq. (2.1) in a form that can be iterated; the interaction picture provides this form. Define the state $|\psi_\text{I}(t)\rangle = \mathrm{e}^{\mathrm{i}H_0 t}|\psi(t)\rangle$, whose components are exactly the amplitudes, $c_n(t) = \langle n|\psi_\text{I}(t)\rangle$, and the perturbation $H'_\text{I}(t) = \mathrm{e}^{\mathrm{i}H_0 t} H'(t)\,\mathrm{e}^{-\mathrm{i}H_0 t}$. The Schrödinger equation becomes

$$\mathrm{i}\,\partial_t |\psi_\text{I}(t)\rangle = H'_\text{I}(t)\,|\psi_\text{I}(t)\rangle, \tag{2.49}$$

whose matrix elements in the basis $|n\rangle$ are Eq. (2.1). The free evolution is now in the operator: $\langle m|H'_\text{I}(t)|n\rangle = H'_{mn}(t)\,\mathrm{e}^{\mathrm{i}(E_m - E_n)t}$, with $H'_{mn}(t) \equiv \langle m|H'(t)|n\rangle$. For independent electrons this phase factor is the oscillation $\mathrm{e}^{\mathrm{i}(E_\gamma - E_{\gamma'})t}$ of $c^\dagger_\gamma c_{\gamma'}$ found in Sect. 2.3.

Integrating Eq. (2.49) from t_0 to t and inserting the result into its own right-hand side, again and again, leads to the evolution operator of the interaction picture as a power series in H', the Dyson series,

$$U_\text{I}(t,t_0) = 1 - \mathrm{i}\int_{t_0}^{t} \mathrm{d}t_1\, H'_\text{I}(t_1) + (-\mathrm{i})^2 \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2\, H'_\text{I}(t_1)\, H'_\text{I}(t_2) + \ldots, \tag{2.50}$$

with $|\psi_\text{I}(t)\rangle = U_\text{I}(t,t_0)|\psi_\text{I}(t_0)\rangle$. In each term the operators stand in order of decreasing time from left to right, which is why the series is also called the time-ordered exponential of $-\mathrm{i}\int H'_\text{I}$. The first-order term reproduces Eq. (2.2). Inserting a complete set of states $|m\rangle$ between the

two factors of the second-order term gives the amplitude for $f \neq i$,

$$c_f^{(2)}(t) = (-\mathrm{i})^2 \sum_m \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \; H'_{fm}(t_1)\, \mathrm{e}^{\mathrm{i}(E_f - E_m)t_1} \; H'_{mi}(t_2)\, \mathrm{e}^{\mathrm{i}(E_m - E_i)t_2}. \tag{2.51}$$

Read from right to left, Eq. (2.51) says that at t_2 the perturbation takes the system from i to m, that the system evolves freely in m until t_1, and that at t_1 the perturbation takes the system from m to f. Every state m contributes, $m = i$ and $m = f$ included, and the amplitudes of the different m add. Each further order adds one more factor of H' and one more time integral. For a constant perturbation acting for a time t, the terms of Eq. (2.50) grow as powers of $|H'|t$, the expansion parameter of Sect. 2.1. Once the time integrals are done in the long-time limit, the powers of t turn into powers of $|H'|$ divided by energy differences.

2.4.2 Second-order golden rule

For a constant perturbation the time integrals in Eq. (2.51) are elementary once we decide how H' is switched on. Sect. 2.1 showed that a rate does not depend on the switching. The most convenient choice is to turn H' on slowly in the remote past, $H'(t) = H'\,\mathrm{e}^{t/\tau_\mathrm{s}}$ with $t_0 \to -\infty$, and to let the switching time τ_s go to infinity at the end, at fixed t. The inner integral is then

$$\int_{-\infty}^{t_1} \mathrm{d}t_2\, \mathrm{e}^{[\mathrm{i}(E_m - E_i) + 1/\tau_\mathrm{s}]t_2} = \frac{\mathrm{i}\,\mathrm{e}^{[\mathrm{i}(E_m - E_i) + 1/\tau_\mathrm{s}]t_1}}{E_i - E_m + \mathrm{i}/\tau_\mathrm{s}}.$$

The free phase that the system accumulates in m has turned into the factor $1/(E_i - E_m)$, whose denominator $E_i - E_m$ is called the *energy denominator*, and the switching has added a small positive imaginary part to the energy denominator. The rest of the integrand of Eq. (2.51) has the form of the first-order integrand, with $\xi = E_f - E_i$ and the switching rate doubled. The amplitude to second order is

$$c_f(t) = -\mathrm{e}^{\mathrm{i}\xi t}\left[\frac{A_1\,\mathrm{e}^{t/\tau_\mathrm{s}}}{\xi - \mathrm{i}/\tau_\mathrm{s}} + \frac{A_2\,\mathrm{e}^{2t/\tau_\mathrm{s}}}{\xi - 2\mathrm{i}/\tau_\mathrm{s}}\right], \qquad A_1 = H'_{fi}, \quad A_2 = \sum_m \frac{H'_{fm}H'_{mi}}{E_i - E_m + \mathrm{i}0^+},$$

where $\mathrm{i}0^+$ stands for $\mathrm{i}/\tau_\mathrm{s}$ in the limit. The transition rate is the time derivative of $|c_f|^2$. The squared first-order term contributes $(2/\tau_\mathrm{s})\,\mathrm{e}^{2t/\tau_\mathrm{s}}|A_1|^2/(\xi^2 + 1/\tau_\mathrm{s}^2)$, a Lorentzian of area $2\pi|A_1|^2$ whose width vanishes with $1/\tau_\mathrm{s}$, and this Lorentzian tends to $2\pi|A_1|^2\delta(\xi)$; the squared second-order term does the same with $2/\tau_\mathrm{s}$ in place of $1/\tau_\mathrm{s}$. The cross term needs more care, since the first- and second-order terms of c_f carry different switching rates. The time derivative of the cross term is

$$2\,\mathrm{Re}\left[A_1 A_2^*\,\mathrm{e}^{3t/\tau_\mathrm{s}}\,\frac{3/\tau_\mathrm{s}}{(\xi - \mathrm{i}/\tau_\mathrm{s})(\xi + 2\mathrm{i}/\tau_\mathrm{s})}\right].$$

The fraction falls off as $1/\xi^2$, and its only pole in the upper half plane is at $\xi = \mathrm{i}/\tau_\mathrm{s}$. Closing the contour in the upper half plane gives the integral of the fraction over ξ, $2\pi\mathrm{i}\,(3/\tau_\mathrm{s})/(3\mathrm{i}/\tau_\mathrm{s}) = 2\pi$, independent of τ_s, while its width shrinks as $1/\tau_\mathrm{s}$. Its imaginary part is odd in ξ and drops out against any smooth function. The cross term therefore tends to $2\pi \cdot 2\,\mathrm{Re}(A_1 A_2^*)\,\delta(\xi)$. The three terms combine into a single squared modulus:

Second-order golden rule

$$\Gamma_{i\to f} = 2\pi\,|T_{fi}|^2\,\delta(E_f - E_i), \qquad T_{fi} = H'_{fi} + \sum_m \frac{H'_{fm}\,H'_{mi}}{E_i - E_m + \mathrm{i}0^+} + \ldots. \tag{2.52}$$

The sum runs over all intermediate states m; the rate is defined after summation over a continuum of final states, as in Eq. (2.5).

The quantity T_{fi} is the element of the transition matrix, or T-matrix. Continued to all orders, the Dyson series becomes a geometric series in the energy denominators, and the geometric series sums to the Lippmann–Schwinger equation [1, Chap. 6]

$$T_{fi} = H'_{fi} + \sum_m \frac{H'_{fm}\, T_{mi}}{E_i - E_m + \mathrm{i}0^+}. \tag{2.53}$$

Iterating the Lippmann–Schwinger equation once reproduces Eq. (2.52). In the language of Sect. 2.1, the S-matrix element is $\delta_{fi} - 2\pi\mathrm{i}\,\delta(E_f - E_i)\,T_{fi}$, and at every order T_{fi} equals the invariant amplitude $-\mathcal{M}$ divided by the normalization factors of the box matrix element.

Energy is conserved between the initial and the final state, through $\delta(E_f - E_i)$, but not in the intermediate state: E_m can be anything. The intermediate state is *virtual*. The phase $\mathrm{e}^{\mathrm{i}(E_m - E_i)t_2}$ in Eq. (2.51) oscillates with period $2\pi/|E_i - E_m|$, and contributions from times t_2 further than about $1/|E_i - E_m|$ before t_1 cancel. The system therefore spends a time of about $1/|E_i - E_m|$ in m. The further E_m is from E_i, the shorter this time and the smaller the amplitude. The physical content of the energy denominator is the energy–time relation. A state m with $E_m = E_i$ is *on shell*, any other one *off shell*, and an on-shell intermediate state makes the denominator vanish.

The second-order term is small compared to the first by a factor $\sim |H'_{fm}|/|E_i - E_m|$, the coupling divided by the energy difference between the initial and intermediate states. For dark-matter scattering this factor is the Born parameter of Sect. 2.2, below 10^{-3} for the couplings that are still allowed, and the second-order term of Eq. (2.52), with plane-wave intermediate states, is the second Born term. Second-order processes matter in this book for another reason: when first order is closed, an internal interaction of the target, such as the coupling of electrons to phonons, can open a second-order channel.

Classical fields fit the same formula. If H' contains harmonic terms $\hat{O}_j \mathrm{e}^{-\mathrm{i}\omega_j t}$, as in Sect. 2.1, each factor H'_{I} in Eq. (2.51) carries its own frequency. The inner time integral then produces the denominator $E_i + \omega_j - E_m$ for the field j that acts first. The outer time integral leads to $\delta(E_f - E_i - \omega_j - \omega_k)$ in the rate, with ω_k the frequency of the field that acts second. An emission term enters with $-\omega$ in place of ω. The result is the same as if each field were a quantized mode whose photon energy is part of E_i, E_m and E_f, and we use whichever language is clearer.

The second-order golden rule also inherits the time window of Eq. (2.7), with one more condition. The energy denominator replaces the free phase in m only after that phase has oscillated many times, and the derivation took $\tau_{\mathrm{s}} \to \infty$ to ensure many oscillations. As long as we watch for longer than $1/|E_i - E_m|$, about a femtosecond below a direct gap of a few eV, the denominator form holds, and every measurement lasts that long. Near a resonance this condition fails and is replaced by $t \gg 1/\Gamma_{\mathrm{w}}$, where Γ_{w} is the decay width of the real intermediate state (Eq. (2.57)).

2.4.3 Time orderings

The phonon-assisted absorption of the opening paragraph has the general structure of a second-order process in which a target absorbs one quantum and emits another. Let quantum 1, of energy ω_1, be absorbed, and quantum 2, of energy ω_2, be emitted, while the target goes from a state $|i_{\mathrm{T}}\rangle$ to a state $|f_{\mathrm{T}}\rangle$, with the subscript T for target. In the language of Sect. 2.3 the two quanta are boson modes with operators d_1 and d_2, and the perturbation is a sum of

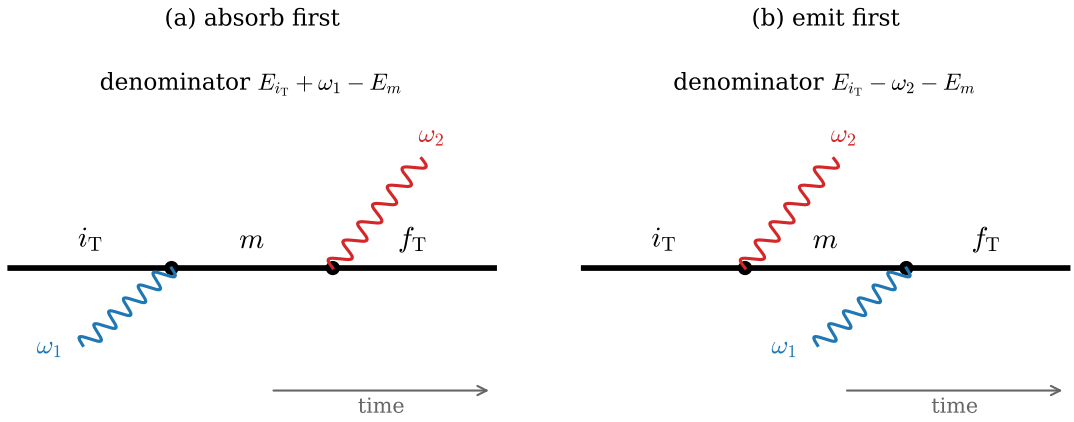


Figure 2.6: The two time orderings of a second-order process in which a target goes from state i_T to state f_T while absorbing a quantum of energy ω_1 and emitting one of energy ω_2. Time runs to the right; the thick line is the target and the wavy lines are the quanta (absorbed from below, emitted upward). (a) Absorb first: the intermediate state is the target in m with no quanta, of energy E_m. (b) Emit first: the intermediate state, of energy $E_m + \omega_1 + \omega_2$, is the target in m with both quanta present. Each panel gives the energy denominator of its ordering in Eq. (2.54): the initial energy $E_{i_T} + \omega_1$ minus the intermediate energy.

two terms that each change one occupation by one,

$$H' = H_1 + H_2, \qquad H_1 = \hat{O}_a\, d_1 + \hat{O}_a^\dagger\, d_1^\dagger, \qquad H_2 = \hat{O}_b\, d_2 + \hat{O}_b^\dagger\, d_2^\dagger,$$

with $\hat{O}_a$ and $\hat{O}_b$ operators on the target. The initial state $|i\rangle = |i_T;\, n_1 = 1, n_2 = 0\rangle$ has energy $E_i = E_{i_T} + \omega_1$, and the final state $|f\rangle = |f_T;\, 0, 1\rangle$ has $E_f = E_{f_T} + \omega_2$. Neither H_1 nor H_2 connects i and f, since each changes only one occupation, and $H'_{fi} = 0$. At second order there are two routes, drawn in Fig. 2.6. In the absorb-first ordering, H_1 acts first and removes quantum 1; the intermediate state $|m;\, 0, 0\rangle$ has energy E_m. In the emit-first ordering, H_2 acts first and creates quantum 2 while quantum 1 is still present; the intermediate state $|m;\, 1, 1\rangle$ has energy $E_m + \omega_1 + \omega_2$. Inserting both orderings into Eq. (2.52), with $d_1|1\rangle = |0\rangle$ and $d_2^\dagger|0\rangle = |1\rangle$,

$$T_{fi} = \sum_m \left[\frac{\langle f_T|\hat{O}_b^\dagger|m\rangle\langle m|\hat{O}_a|i_T\rangle}{E_{i_T} + \omega_1 - E_m + \mathrm{i}0^+} + \frac{\langle f_T|\hat{O}_a|m\rangle\langle m|\hat{O}_b^\dagger|i_T\rangle}{E_{i_T} - \omega_2 - E_m + \mathrm{i}0^+} \right], \tag{2.54}$$

and energy conservation, $E_{f_T} + \omega_2 = E_{i_T} + \omega_1$, holds between i and f but not in the intermediate states.

Both orderings connect the same states, and their amplitudes add before squaring and can interfere. If i_T is the ground state of the target, $E_m \geq E_{i_T}$ for every m, and the emit-first denominator $E_{i_T} - \omega_2 - E_m$ is at most $-\omega_2$: it never vanishes, and this ordering is always off shell. The absorb-first denominator vanishes when $\omega_1 = E_m - E_{i_T}$, that is, when quantum 1 alone could excite the target to a real state m. This condition is the resonance, and near the resonance the absorb-first ordering dominates. Far from every resonance the two orderings are of similar size, and dropping one changes the rate by a factor of a few or more, as Worked example 2.6 shows. In silicon, with quantum 1 a photon and quantum 2 a phonon, Eq. (2.54) is the bracket in Hall, Bardeen and Blatt's theory of phonon-assisted absorption [15], which also includes phonon absorption in place of emission. Ref. [16, Eq. (200)] writes the bracket in first-principles form. The absorb-first term, in which the photon acts first, has the denominator

of a direct, vertical transition at fixed crystal momentum, and the emit-first term, in which the phonon acts first, has the denominator of an electron scattered by the phonon before the photon arrives. Sect. 6.5 evaluates both terms. When the modes hold n_1 and n_2 quanta rather than one and zero, the matrix elements of d_1 and $d_2^\dagger$ bring factors $\sqrt{n_1}$ and $\sqrt{n_2+1}$, in both orderings alike; Sect. 2.5 averages these factors over thermal populations.

In a target with many electrons, i_T, m and f_T are the many-electron states of Sect. 2.3, and the denominators are differences of total energies. For independent electrons the denominators reduce to differences of orbital energies, because an operator that moves one electron from γ' to γ changes the total energy by $E_\gamma - E_{\gamma'}$, the frequency at which $c_\gamma^\dagger c_{\gamma'}$ oscillates in the interaction picture. If the photon moves an electron from an occupied valence orbital to an empty conduction orbital at the same crystal momentum, the absorb-first denominator is the photon energy minus the energy difference between the two orbitals, whatever the other electrons do. The Pauli principle changes the list of intermediate states. An intermediate state in which the moved electron would enter an occupied orbital does not exist. In place of this intermediate state there are orderings in which a different electron moves first, and the electron moved by the photon then fills the hole left by that other electron, with the anticommutation sign of Sect. 2.3 attached to the amplitude. The single-particle picture of one electron passing through an intermediate band is a convenient way of organizing these orderings, not a literal history, and Sect. 6.5 counts the orderings for silicon.

2.4.4 Resonance and finite width

At resonance the absorb-first term of Eq. (2.54) is infinite, and the rate with it. No physical rate diverges at resonance; the divergence comes from Eq. (2.51), which lets the system stay in m forever with unit probability, while a real excited state decays. Quantum 1 can take the target to m, and m then empties into f_T by emitting quantum 2, or into any other state coupled to m. The time dependence of a state that decays into a continuum is the Wigner–Weisskopf problem [17]. A state that empties at rate Γ_w survives for a time of about $1/\Gamma_\mathrm{w}$, so its energy is defined only to within about Γ_w, and a denominator that would vanish should instead saturate at a size of order Γ_w.

Take a discrete state $|i\rangle$ coupled by a constant H' to a continuum of states $|f\rangle$, with no other couplings, as in Sect. 2.1. Appendix D.3 solves the amplitude equations Eq. (2.1) for this case without assuming that c_i stays close to one. Under the condition that made the golden rule possible, $\Gamma_i \ll E_\mathrm{w}$, the memory of the continuum is short compared with the decay time. At times $t \gg 1/E_\mathrm{w}$ the amplitude of the initial state is then

$$c_i(t) = \mathrm{e}^{-(\Gamma_i/2 + \mathrm{i}\Delta E_i)\,t}, \qquad \Delta E_i = \mathrm{PV}\sum_f \frac{|H'_{fi}|^2}{E_i - E_f}, \tag{2.55}$$

where Γ_i is the golden-rule rate Eq. (2.6) and PV denotes the principal value of the sum, made precise in the next subsection. The probability $|c_i|^2 = \mathrm{e}^{-\Gamma_i t}$ decays exponentially at the golden-rule rate. This exponential decay justifies the statement of Sect. 2.1 that Γ_i keeps its meaning beyond the first-order window (Worked example 2.1). The energy of the state is shifted by ΔE_i, the second-order level shift of stationary perturbation theory, now with the principal value that a continuum requires; in Worked example 2.1 the shift vanishes because the continuum is symmetric about resonance. The populations of the final states at late times trace out the line,

$$|c_f(\infty)|^2 = \frac{|H'_{fi}|^2}{(E_f - E_i - \Delta E_i)^2 + \Gamma_i^2/4}, \tag{2.56}$$

a Lorentzian of full width Γ_i at half maximum, centered at the shifted energy. Summed over the continuum with density $\mathcal{D}_f$, the Lorentzian integrates to $\overline{|H'_{fi}|^2}\mathcal{D}_f \cdot 2\pi/\Gamma_i = 1$: the probability

that leaves the decaying state ends in the continuum, and the golden-rule rate fixes both the speed of the decay and the width of the line. The same shape describes the cross-section near a resonance in scattering, where it is called a Breit–Wigner line. Appendix D.3 also states the limits of the exponential law. The exponential law fails at times shorter than $1/E_{\mathrm{w}}$, where the decay starts quadratically as in Worked example 2.1, and again after many lifetimes, when the edges of the continuum leave a slow power-law tail; for $\Gamma_i \ll E_{\mathrm{w}}$ that tail is far below any measurable level.

We return to the intermediate state m, in which the system evolves between t_2 and t_1 in Eq. (2.51). Eq. (2.55) applied to m replaces the free phase of m, $\mathrm{e}^{-\mathrm{i}E_m(t_1-t_2)}$ by $\mathrm{e}^{-\mathrm{i}(\tilde{E}_m-\mathrm{i}\Gamma_{\mathrm{w}}/2)(t_1-t_2)}$, where $\tilde{E}_m = E_m + \Delta E_m$ is the shifted energy and Γ_{w} the total decay rate of m, summed over all of its decay channels. The inner time integral then converges without any switching and gives

$$\frac{1}{E_i - E_m + \mathrm{i}0^+} \;\longrightarrow\; \frac{1}{E_i - \tilde{E}_m + \mathrm{i}\Gamma_{\mathrm{w}}/2}. \tag{2.57}$$

The infinitesimal becomes a finite width: at resonance the amplitude carries the factor $-2\mathrm{i}/\Gamma_{\mathrm{w}}$, large but finite, in place of the divergent inverse energy denominator. In the language of the Dyson series, Eq. (2.57) keeps, from every order, the terms in which the system emits and reabsorbs while it stays in m. These terms grow near resonance, and summing them is consistent under the condition used for Eq. (2.55): Γ_{w} small compared with the energy over which the decay continuum of m changes. Off resonance, with $|E_i - \tilde{E}_m| \gg \Gamma_{\mathrm{w}}$, the width is a small correction and the shift is usually absorbed in the measured energy $\tilde{E}_m$. From now on we write E_m for the shifted energy. In a solid the width of an excited electron comes from the emission of phonons and from scattering with other electrons, and the width depends on how far above the band edge the electron lies; Sect. 6.3 computes the phonon part. For the off-resonant denominators that dominate below a direct gap, an infinitesimal $\mathrm{i}0^+$ is enough.

2.4.5 Principal value and finite part

An energy denominator regularized by a width, $1/(x + \mathrm{i}\Gamma_{\mathrm{w}}/2)$ with $x = E_i - E_m$, has a real and an imaginary part,

$$\frac{1}{x + \mathrm{i}\Gamma_{\mathrm{w}}/2} = \frac{x}{x^2 + \Gamma_{\mathrm{w}}^2/4} - \mathrm{i}\,\frac{\Gamma_{\mathrm{w}}/2}{x^2 + \Gamma_{\mathrm{w}}^2/4},$$

drawn in Fig. 2.7(a). The real part equals $1/x$ for $|x| \gg \Gamma_{\mathrm{w}}$ and turns over smoothly through zero inside $|x| < \Gamma_{\mathrm{w}}/2$, with extrema $\pm 1/\Gamma_{\mathrm{w}}$. The imaginary part is a Lorentzian of area π and depth $2/\Gamma_{\mathrm{w}}$. As $\Gamma_{\mathrm{w}} \to 0$ the Lorentzian becomes $\pi\delta(x)$. The real part, integrated against a smooth function $w(x)$, becomes $\int \mathrm{d}x\, x[w(x) - w(0)]/(x^2 + \Gamma_{\mathrm{w}}^2/4)$, since $x/(x^2 + \Gamma_{\mathrm{w}}^2/4)$ is odd, and the integral tends to $\int \mathrm{d}x\,[w(x) - w(0)]/x$. This limit is the principal value of $\int \mathrm{d}x\, w(x)/x$: the integral with a symmetric interval $(-x_{\mathrm{c}}, x_{\mathrm{c}})$ cut out around the pole, in the limit $x_{\mathrm{c}} \to 0$, where the divergences on the two sides of the pole cancel. Together these limits are the Sokhotski–Plemelj formula,

$$\frac{1}{x + \mathrm{i}0^+} = \mathrm{PV}\,\frac{1}{x} - \mathrm{i}\pi\,\delta(x), \tag{2.58}$$

which Appendix C.4 proves for the general case. Applied to a second-order amplitude, the Sokhotski–Plemelj formula splits the sum over intermediate states into two parts with different physics. The principal-value part collects the off-shell intermediate states and is real; it shifts energies, as ΔE_i does in Eq. (2.55), and produces the dispersive effects of a medium. The delta part picks out the on-shell intermediate states, with $E_m = E_i$, and is imaginary; it

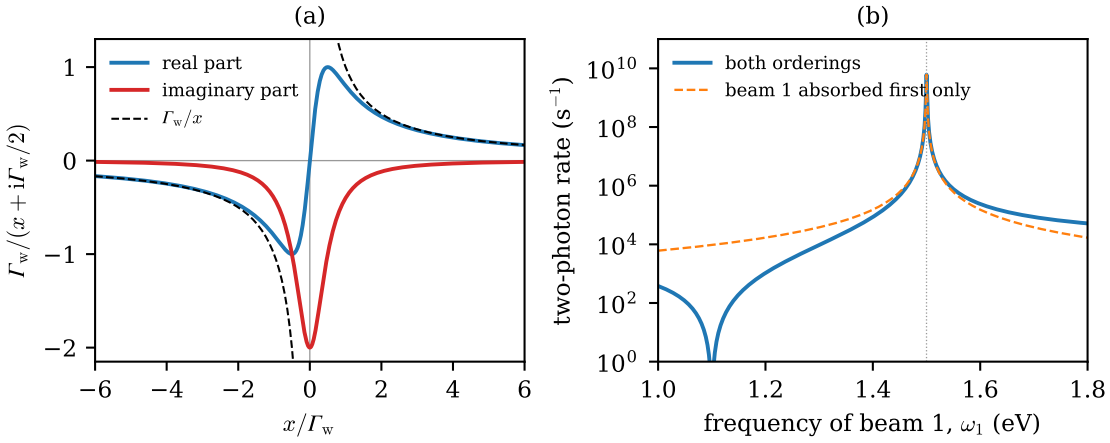


Figure 2.7: (a) Real and imaginary parts of an energy denominator regularized by a width, $1/(x+\mathrm{i}\Gamma_\mathrm{w}/2)$, multiplied by Γ_w and plotted against x/Γ_w, where x is the initial energy minus the intermediate energy. Dashed: Γ_w/x, to which the real part tends for $|x| \gg \Gamma_\mathrm{w}$. As $\Gamma_\mathrm{w} \to 0$ the real part becomes the principal value of $1/x$ and the imaginary part $-\pi\delta(x)$. (b) Rate of two-photon absorption in the three-level model of Worked example 2.6, against the frequency ω_1 of beam 1 at fixed $\omega_2 = 1.90\,\mathrm{eV}$, for the channel with final energy $\omega_1+\omega_2$. The model has a ground level i_T, an intermediate level m at 1.50 eV with total width $\Gamma_\mathrm{w} = 1\,\mathrm{meV}$, and a flat band of final states above 3.20 eV. Either beam drives $i_\mathrm{T} \to m$ with matrix element 0.1 meV, and beam 2 empties a real level m into the band at the rate 0.1 meV. Solid: both time orderings. Dashed: the ordering in which beam 1 is absorbed first, alone. The peak at $\omega_1 = 1.50\,\mathrm{eV}$ is the resonance with m; the zero at 1.10 eV is destructive interference between the two orderings.

produces the decay. Eq. (2.55) is the example: the self-coupling of i through the continuum, $\sum_f |H'_{fi}|^2/(E_i - E_f + \mathrm{i}0^+)$, is $\Delta E_i - \mathrm{i}\Gamma_i/2$ by Eq. (2.58). The same statement can be read off the T-matrix. The forward element at second order, $T_{ii} = \sum_f |H'_{fi}|^2/(E_i - E_f + \mathrm{i}0^+)$, has imaginary part $-\pi\sum_f |H'_{fi}|^2\delta(E_f - E_i) = -\Gamma_i/2$. The total decay rate of a state is $-2\,\mathrm{Im}\,T_{ii}$. This relation is the lowest-order form of the optical theorem, which follows in general from the unitarity of the evolution: the probability that leaves i is the probability that arrives in all the states f.

The principal-value part has a familiar instance in light scattering. Let quanta 1 and 2 be photons of the same frequency, so that the target absorbs one photon and emits another and returns to $f_\mathrm{T} = i_\mathrm{T}$. Eq. (2.54) is then the amplitude of elastic (Rayleigh) scattering, and below every resonance both orderings are off shell and real. Summed over intermediate states, the two orderings combine into $\sum_m 2(E_m - E_{i_\mathrm{T}})|\langle m|\hat{O}|i_\mathrm{T}\rangle|^2/[\omega^2 - (E_m - E_{i_\mathrm{T}})^2]$, for a Hermitian $\hat{O}$. For a dipole coupling, $\hat{O}$ is the field amplitude times the dipole operator along the field, and the sum is minus the square of the field amplitude times the dynamic polarizability of the target; the denominators of the polarizability are $(E_m - E_{i_\mathrm{T}})^2 - \omega^2$. The polarizability acquires an imaginary part once a width or a continuum puts the absorb-first denominator on shell, and this imaginary part describes absorption. The dielectric function of Sect. 4.4 has the same structure, with a real part made of off-shell transitions and an imaginary part made of on-shell ones.

The rate contains the squared modulus of the inverse energy denominator, and this squared modulus has a double pole at resonance. With $|E_i - E_m + \mathrm{i}\Gamma_\mathrm{w}/2|^{-2} = 1/(x^2 + \Gamma_\mathrm{w}^2/4)$, the limit $\Gamma_\mathrm{w} \to 0$ cannot be taken in Eq. (2.58): the square of $\mathrm{i}\pi\delta(x)$ has no meaning, as the square of

the momentum delta function had none in Sect. 2.1. The function $1/x^2$ has no principal value either. Its singularity is even, the divergences on the two sides add instead of cancelling, and $\int_{|x|>x_c} \mathrm{d}x/x^2 = 2/x_c$. The Hadamard finite part removes exactly this divergence:

$$\mathrm{fp}\!\int \mathrm{d}x\, \frac{w(x)}{x^2} = \lim_{x_c\to 0}\left[\int_{|x|>x_c} \mathrm{d}x\, \frac{w(x)}{x^2} - \frac{2w(0)}{x_c}\right] = \int_0^\infty \mathrm{d}x\, \frac{w(x)+w(-x)-2w(0)}{x^2}. \quad (2.59)$$

The integrand on the right is finite at $x = 0$, where it tends to $w''(0)$. With the finite part, the Lorentzian of the rate splits as follows. Write $\int \mathrm{d}x\, w(x)/(x^2+\Gamma_\mathrm{w}^2/4)$ as $w(0)\int \mathrm{d}x/(x^2+\Gamma_\mathrm{w}^2/4) = 2\pi w(0)/\Gamma_\mathrm{w}$ plus $\int \mathrm{d}x\,[w(x)-w(0)]/(x^2+\Gamma_\mathrm{w}^2/4)$. The remainder tends to the finite part as $\Gamma_\mathrm{w}\to 0$, since the integrand of the remainder is integrable at $x = 0$. The difference between the remainder and its limit is $-(\Gamma_\mathrm{w}^2/4)\int \mathrm{d}x\,[w(x)-w(0)]/[x^2(x^2+\Gamma_\mathrm{w}^2/4)]$, an integral dominated by $|x| \lesssim \Gamma_\mathrm{w}$. In that range the odd part of $w(x)-w(0)$ drops out, the even part is $w''(0)x^2/2$, and the difference is $-(\Gamma_\mathrm{w}^2/8)\,w''(0)\cdot 2\pi/\Gamma_\mathrm{w} = -\pi\Gamma_\mathrm{w}w''(0)/4$. In the sense of distributions,

$$\frac{1}{x^2+\Gamma_\mathrm{w}^2/4} = \frac{2\pi}{\Gamma_\mathrm{w}}\,\delta(x) + \mathrm{fp}\,\frac{1}{x^2} - \frac{\pi\Gamma_\mathrm{w}}{4}\,\delta''(x) + O(\Gamma_\mathrm{w}^2). \quad (2.60)$$

For a function w that varies on an energy scale Δ_w, the delta term is of size w/Γ_w, the finite part of size w/Δ_w, and the δ'' term a fraction $(\Gamma_\mathrm{w}/\Delta_\mathrm{w})^2$ of the delta term. Appendix C.4 collects these definitions.

The delta term and the finite part of Eq. (2.60) are the on-shell and the off-shell parts of a resonant second-order rate, and the on-shell part has a direct interpretation. Inserted in Eq. (2.52), the delta term of the absorb-first ordering is a product of two factors. The first factor is the first-order golden-rule rate $2\pi|\langle m|\hat{O}_a|i_\mathrm{T}\rangle|^2\delta(E_{i_\mathrm{T}}+\omega_1-E_m)$ at which quantum 1 creates a real state m; the second factor is $2\pi|\langle f_\mathrm{T}|\hat{O}_b^\dagger|m\rangle|^2\delta(E_f-E_i)/\Gamma_\mathrm{w}$. Summed over final states, the second factor is the fraction of the decays of m that end in f_T. The on-shell part is therefore a cascade of two first-order processes: the excitation of m, and the decay of m into f_T weighted by the branching ratio of that decay. The finite part is the genuinely second-order process, the virtual passage through m. The finite part is not positive: against a function w peaked at $x = 0$ with width Δ_w, Eq. (2.59) is about $-w(0)/\Delta_\mathrm{w}$, because $w(x)-w(0) < 0$ near the peak. Only the sum of the two parts is a rate. Care is needed in any calculation that counts the cascade separately, as two first-order events, and also includes a second-order rate: the delta part of the second-order rate is then counted twice. The resonant denominators of phonon-assisted absorption return in Sect. 6.5, where the electron in the intermediate state becomes real once the photon energy exceeds the direct gap.

2.4.6 Forbidden first-order transitions

Since each order costs a factor $|H'|/|E_i-E_m|$, a second-order process competes with a first-order one only when the first-order rate vanishes, by a selection rule or by kinematics.

If a symmetry of the target makes $\langle f|H'|i\rangle = 0$ for every final state at the right energy, the leading amplitude is the second-order sum, which involves different matrix elements. The classic case is parity. The electric-dipole operator is odd, and a single photon cannot connect two states of the same parity. Two photons can, through intermediate states of the opposite parity. The metastable $2s$ level of hydrogen decays by emitting two photons, and two-photon absorption spectroscopy reaches levels that one photon cannot. Spin, angular momentum and the symmetries of a crystal impose rules of the same kind.

Kinematics closes the first-order channel when no final state conserves both energy and momentum, as in the phonon-assisted absorption of the opening paragraph. A vertical transition at fixed crystal momentum needs an energy well above the 1.12 eV gap at 300 K,

because the conduction-band minimum and the valence-band maximum lie at different crystal momenta (Sect. 3.3). Below the energy of the lowest vertical transition the first-order rate is zero, and phonon-assisted absorption, with the amplitude Eq. (2.54), is the leading process. The phonon costs or supplies tens of meV of energy [18, Table I], but it can carry any crystal momentum in the zone. Absorbed dark matter, like the photon, carries too little crystal momentum to cross the indirect gap at first order. A boson of mass m_ϕ is absorbed with momentum $m_\phi v \approx 10^{-3} m_\phi$ (Sect. 2.1), a smaller fraction of its energy than for a photon. In an indirect-gap material, absorption of the boson at energies between the indirect and the direct gap is a second-order process (Chap. 8). Dark-matter scattering is different. The momentum transfer is typically $q \sim q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, which is $1.89\,\text{Å}^{-1}$, larger than $2\pi/a$, so scattering reaches the indirect transitions at first order. At small momentum transfer, however, scattering supplies too little crystal momentum, as a photon does, and scattering with the simultaneous emission of a phonon dominates [18, Sect. VI]. This process is the dark-matter analogue of phonon-assisted absorption. The Migdal effect is a process of the same structure, with a nuclear recoil in place of the phonon (Sect. 12.2).

In each case the size of the second-order rate relative to a first-order one is set by $|H_2|^2/(E_i - E_m)^2$, with H_2 the coupling that the second-order route adds and $E_i - E_m$ the off-shell distance of the intermediate state. Sect. 8.5 names this ratio for the electron–phonon coupling and uses it to organize the expansion.

When the first-order amplitude does not vanish, second order corrects the first-order amplitude and enters the rate through the cross term $2\,\mathrm{Re}(A_1 A_2^*)$, at relative order $|H'|/|E_i - E_m|$ rather than its square. Corrections of this kind are part of the target response when H_2 is internal to the target. They are already contained in an exact treatment of the target's states, such as the band structure with its phonon renormalization (Sect. 6.3), and are not added again as second-order rates. A second-order rate is a separate channel only when it leads to a final state that first order cannot reach, such as a state with one more phonon.

Worked example 2.6: Two-photon absorption in a three-level system

A target has a ground level i_T at zero energy, an intermediate level m at $E_m = 1.50\,\mathrm{eV}$, and a flat band of final states f_T above $3.20\,\mathrm{eV}$, with density of states $\mathcal{D}_{f_\mathrm{T}}$. Two light beams of frequencies ω_1 and ω_2 couple to the target through the same operator $\hat{O}$ (equal polarization and intensity); Sect. 2.1 showed that a classical beam acts like a quantized mode holding many photons, and we use the language of Eq. (2.54). Here, however, both photons are absorbed, so the ordering that absorbs ω_2 first has the denominator $E_{i_\mathrm{T}} + \omega_2 - E_m$, with $+\omega_2$ in place of the $-\omega_2$ of emission in Eq. (2.54). We fix $\omega_2 = 1.90\,\mathrm{eV}$ and tune ω_1 near $1.50\,\mathrm{eV}$. Neither photon alone reaches the band, and neither do two photons of beam 1 for $\omega_1 < 1.60\,\mathrm{eV}$: the first-order rate is kinematically zero. The pair $\omega_1 + \omega_2 \approx 3.40\,\mathrm{eV}$ reaches the band. Two photons of beam 2 reach the band too, at $3.80\,\mathrm{eV}$; their final state differs from the final state of the pair $\omega_1 + \omega_2$ and does not interfere, and we follow the channel with final energy $\omega_1 + \omega_2$. The model numbers are $|\langle m|\hat{O}|i_\mathrm{T}\rangle| = 0.1\,\mathrm{meV}$, a rate $\Gamma_{m\to f_\mathrm{T}} = 2\pi|\langle f_\mathrm{T}|\hat{O}|m\rangle|^2\,\mathcal{D}_{f_\mathrm{T}} = 0.1\,\mathrm{meV}$ at which beam 2 empties a real level m into the band (beam 1 cannot, since $E_m + \omega_1 < 3.20\,\mathrm{eV}$), and a total width $\Gamma_\mathrm{w} = 1\,\mathrm{meV}$ of m, which includes $\Gamma_{m\to f_\mathrm{T}}$ and all other decays.
Both orderings absorb one photon from each beam. Absorbing ω_1 first leaves the target in m with photon ω_2 still present, and the denominator is $\delta_1 = E_{i_\mathrm{T}} + \omega_1 - E_m = \omega_1 - 1.50\,\mathrm{eV}$; absorbing ω_2 first leads to $\delta_2 = \omega_2 - 1.50\,\mathrm{eV} = 0.40\,\mathrm{eV}$. With Eq. (2.57),

$$T_{fi} = \langle f_\mathrm{T}|\hat{O}|m\rangle\langle m|\hat{O}|i_\mathrm{T}\rangle\,\mathcal{A}(\omega_1), \qquad \mathcal{A}(\omega_1) = \frac{1}{\delta_1 + \mathrm{i}\Gamma_\mathrm{w}/2} + \frac{1}{\delta_2 + \mathrm{i}\Gamma_\mathrm{w}/2},$$

and summing Eq. (2.52) over the flat band gives $\Gamma_{i_T \to f_T} = \Gamma_{m \to f_T} |\langle m|\hat{O}|i_T\rangle|^2 |\mathcal{A}|^2$. Fig. 2.7(b) plots the rate against ω_1.
Off resonance, at $\omega_1 = 1.20\,\text{eV}$, $\delta_1 = -0.30\,\text{eV}$ and $\mathcal{A} = -3.33 + 2.50 = -0.833\,\text{eV}^{-1}$. The two orderings have denominators of opposite sign and interfere destructively. Beam 1 is below the resonance of m and beam 2 above it, and the virtual excitation of m follows the two beams with opposite phases, as a driven oscillator follows a force below and above its natural frequency. The rate is $6.9 \times 10^{-13}\,\text{eV}$, or $1.1 \times 10^3\,\text{s}^{-1}$; keeping the absorb-$\omega_1$-first ordering alone would overestimate the rate by a factor of 16. At $\omega_1 = 1.10\,\text{eV}$ the two denominators are equal and opposite, and the rate vanishes up to terms of relative order $(\Gamma_w/\delta_2)^2$.
On resonance, at $\omega_1 = 1.50\,\text{eV}$, the first denominator is $-2i/\Gamma_w$ and $|\mathcal{A}|^2 = 4.0 \times 10^6\,\text{eV}^{-2}$, so $\Gamma_{i_T \to f_T} = 4.0 \times 10^{-6}\,\text{eV} = 6.1 \times 10^9\,\text{s}^{-1}$, larger than the off-resonant rate by 5.8×10^6. The cascade picture reproduces this number. On resonance, beam 1 excites m at the first-order rate $2\pi|\langle m|\hat{O}|i_T\rangle|^2$ times the peak $2/(\pi\Gamma_w)$ of the Lorentzian of m, that is $4|\langle m|\hat{O}|i_T\rangle|^2/\Gamma_w = 4.0 \times 10^{-5}\,\text{eV}$; a fraction $\Gamma_{m \to f_T}/\Gamma_w = 0.10$ of the excited levels end in the band. The product of the excitation rate and this fraction agrees with the second-order rate to five parts in 10^6, the contribution of the off-resonant ordering. Perturbation theory holds, since the probability of finding the target in m on resonance is $|\langle m|\hat{O}|i_T\rangle|^2/(\Gamma_w/2)^2 = 0.04$.
A broadband beam 1 tests Eq. (2.60). Let its spectrum be a Gaussian of rms width 10 meV in δ_1, normalized to one; the measured rate is $|\mathcal{A}|^2$ averaged over it. Centered on resonance, the exact average is $2.410 \times 10^5\,\text{eV}^{-2}$. The delta term of Eq. (2.60) is $(2\pi/\Gamma_w)$ times the spectral density at resonance, $2.507 \times 10^5\,\text{eV}^{-2}$, an overestimate by 4.0 %. The finite part, $-1/(10\,\text{meV})^2 = -1.00 \times 10^4\,\text{eV}^{-2}$ for a centered Gaussian, is negative, as it must be for a spectrum peaked at the pole. Delta and finite part together are within 0.13 % of the exact value, and the difference between the exact average and the sum of the two terms matches the term $-\pi\Gamma_w w''(0)/4$. Centered 30 meV away from resonance, the delta term is $2785\,\text{eV}^{-2}$, the finite part $1795\,\text{eV}^{-2}$, now positive, the principal-value cross term between the two orderings $197\,\text{eV}^{-2}$, and the term of the off-resonant ordering $6\,\text{eV}^{-2}$. Their sum is within 0.6 % of the exact average, $4753\,\text{eV}^{-2}$, while the delta term alone misses it by 41 %. The cascade accounts for nearly the whole rate only when the spectrum is centered on the resonance and is broad compared with Γ_w. Even then the finite part corrects the cascade by about Γ_w divided by the spectral width. Away from resonance, the finite part and the cross term between the orderings are as large as the cascade itself.

2.5 Thermal states and detailed balance

At 300 K each zone-center optical phonon mode of silicon, at 64 meV, holds on average 0.09 quanta; at the 4 K of a liquid-helium cryostat it holds 10^{-81}. The rates of Sects. 2.1 and 2.2 assumed that the target is in an eigenstate of its Hamiltonian. A crystal at a finite temperature is instead in a statistical mixture of eigenstates, and an excited component of the mixture can give its energy to the dark matter. An energy transfer from the crystal to the dark matter is the negative ω that Sect. 1.3 allowed. At zero temperature the thermal occupation numbers of the electrons produce the filled Fermi sea of a free electron gas, whose Fermi momentum and density of states Chap. 4 uses throughout. We follow Refs. [19] and [11, Chap. 1] for the ensembles and Ref. [20, Chap. 2] for the free electron gas.

2.5.1 Gibbs state

A mixture that holds the eigenstate $|i\rangle$ with probability p_i is described by the density matrix

$$\hat{\varrho} = \sum_i p_i \, |i\rangle\langle i|, \qquad \langle \hat{A} \rangle = \mathrm{Tr}(\hat{\varrho}\, \hat{A}) = \sum_i p_i \, \langle i|\hat{A}|i\rangle, \tag{2.61}$$

with $\sum_i p_i = 1$; the thermal average $\langle \hat{A} \rangle$ is the quantum expectation value averaged over the mixture. A target in equilibrium with a bath at temperature T has the Gibbs weights

$$p_i = \frac{\mathrm{e}^{-\beta E_i}}{\mathcal{Z}}, \qquad \mathcal{Z} = \sum_i \mathrm{e}^{-\beta E_i} = \mathrm{Tr}\ \mathrm{e}^{-\beta H}, \qquad \beta = \frac{1}{T}, \tag{2.62}$$

so that $\hat{\varrho} = \mathrm{e}^{-\beta H}/\mathcal{Z}$, with $\mathcal{Z}$ the partition function. In natural units temperature is an energy, $1\,\mathrm{K} = 86.17\,\mu\mathrm{eV}$, and a detector at 10 mK has $T = 0.86\,\mu\mathrm{eV}$.

Since the Gibbs state is a function of H, it commutes with H, and its time evolution $\hat{\varrho}(t) = \mathrm{e}^{-\mathrm{i}Ht}\hat{\varrho}\,\mathrm{e}^{\mathrm{i}Ht}$ leaves it unchanged. The Gibbs state is a stationary state, and every thermal average is independent of time. A correlation $\langle \hat{A}(t)\hat{B}(t')\rangle$ of Heisenberg operators $\hat{A}(t) = \mathrm{e}^{\mathrm{i}Ht}\hat{A}\,\mathrm{e}^{-\mathrm{i}Ht}$ depends only on $t - t'$, so its Fourier transform in time is a function of one frequency (Sect. 4.1). Note that stationarity does not need the Gibbs form; any $\hat{\varrho}$ diagonal in the energy basis is stationary. Thermal equilibrium fixes the weights.

Particles can also be exchanged with a reservoir, or we may prefer to fix their mean number instead of the exact one. The grand-canonical state then replaces H by $H - \mu_{\mathrm{ch}}\hat{N}$, with $\hat{N}$ the particle-number operator and μ_{ch} the chemical potential:

$$\hat{\varrho} = \frac{\mathrm{e}^{-\beta(H-\mu_{\mathrm{ch}}\hat{N})}}{\mathcal{Z}}, \qquad p_i = \frac{\mathrm{e}^{-\beta(E_i-\mu_{\mathrm{ch}}N_i)}}{\mathcal{Z}}. \tag{2.63}$$

The value of μ_{ch} is fixed by $\langle \hat{N} \rangle = N$. A detector holds a fixed number of electrons, and the canonical state is the physical one. However, the two ensembles differ in a macroscopic body by fluctuations of relative size $N^{-1/2}$, and the grand-canonical state is much easier to use, since for independent particles it factorizes into one factor per single-particle state (Eq. (2.69) below).

2.5.2 Thermal golden rule

The probability of finding the system in $|f\rangle$ at time t is linear in the initial density matrix. With $U(t)$ the evolution operator,

$$\langle f|U(t)\,\hat{\varrho}\,U(t)^\dagger|f\rangle = \sum_i p_i \, |\langle f|U(t)|i\rangle|^2 = \sum_i p_i \, P_{i\to f}(t).$$

Different initial states do not interfere, because $\hat{\varrho}$ has no off-diagonal elements between them. Each $P_{i\to f}$ grows at its golden-rule rate (Eq. (2.5)), and the rate out of the thermal state is the weighted sum

$$\Gamma = \sum_i p_i \sum_f \Gamma_{i\to f}. \tag{2.64}$$

For dark-matter scattering the initial state is the product of the thermal target and a dark-matter plane wave of momentum $\boldsymbol{p}$, and the final state has the target in $|f\rangle$ and the dark matter at $\boldsymbol{p} - \boldsymbol{q}$. The interaction between these two dark-matter states is $\tilde{V}(\boldsymbol{q})\, n_e(\boldsymbol{q})/\mathcal{V}$ (Eq. (2.38)), and energy conservation, with the dark-matter energies written out, is $\delta(\omega_{\boldsymbol{q}} - E_f + E_i)$, where

$\omega_{\boldsymbol{q}}$ is the energy lost by the dark matter (Eq. (1.21)). Summing Eq. (2.64) over the final momenta,

$$\Gamma = \sum_{\boldsymbol{q}} \frac{2\pi}{\mathscr{V}^2} \, |\tilde{V}(\boldsymbol{q})|^2 \, \mathcal{S}_{n_e(\boldsymbol{q})}(\omega_{\boldsymbol{q}}),$$

where all the target physics is contained in the transition sum of an operator $\hat{O}$,

$$\mathcal{S}_O(\omega) = \sum_{i,f} p_i \, |\langle f|\hat{O}|i\rangle|^2 \, \delta(\omega - E_f + E_i). \tag{2.65}$$

The transition sum counts the transitions that $\hat{O}$ drives from the thermal state, weighted by their probability, at each energy ω given to the target. It is also the Fourier transform of a thermal correlation function. Insert a complete set of states between the operators of $\langle \hat{O}^\dagger(t)\hat{O}(0)\rangle$. Each term then oscillates as $\mathrm{e}^{\mathrm{i}(E_i - E_f)t}$, and the time integral turns it into a delta function:

$$\int_{-\infty}^{\infty} \mathrm{d}t \, \mathrm{e}^{\mathrm{i}\omega t} \, \langle \hat{O}^\dagger(t) \, \hat{O}(0)\rangle = 2\pi \, \mathcal{S}_O(\omega). \tag{2.66}$$

The rate measures how the fluctuations of $\hat{O}$ in the equilibrium state are correlated in time; Sect. 4.2 develops this viewpoint. With Eq. (2.10) for the sum over $\boldsymbol{q}$,

Thermal golden rule

$$\Gamma = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \, \underbrace{|\tilde{V}(\boldsymbol{q})|^2}_{\text{coupling}} \, \underbrace{S(\boldsymbol{q}, \omega_{\boldsymbol{q}})}_{\text{target}}, \qquad S(\boldsymbol{q}, \omega) = \frac{2\pi}{\mathscr{V}} \, \mathcal{S}_{n_e(\boldsymbol{q})}(\omega). \tag{2.67}$$

Γ is the scattering rate of one dark-matter particle of momentum $\boldsymbol{p}$ in a box of volume $\mathscr{V}$ filled by the target, which is in thermal equilibrium at temperature T.

The function $S(\boldsymbol{q}, \omega)$ is the dynamic structure factor of the electron density (Eq. (A.30)) [21, Eq. (4)]; that reference writes the density at $-\boldsymbol{k}$ where we write $n_e(\boldsymbol{q})$ (Appendix A.7); Sect. 4.2 develops S. The volume drops out as it should. The transition sum is extensive, since a target twice as large offers twice as many transitions, and S is intensive. The rate of one dark-matter particle, present with density $1/\mathscr{V}$ in a target of volume $\mathscr{V}$, is then independent of $\mathscr{V}$.

The rate assembled in Sect. 1.5 now follows. For a target whose response depends only on $q = |\boldsymbol{q}|$, we write $S(q, \omega_{\boldsymbol{q}}) = \int \mathrm{d}\omega \, S(q, \omega) \, \delta(\omega - \omega_{\boldsymbol{q}})$. The average of the delta function over the direction of $\boldsymbol{q}$ and over the halo is $\eta(v_{\min})/2q$ (Sect. 1.3; Eq. (2.20)), and the measure becomes $\int \mathrm{d}^3 q/(2\pi)^3 \, (2q)^{-1} = \int q \, \mathrm{d}q/(2\pi)^2$. We multiply by the number density ρ_χ/m_χ of dark matter, divide by the target mass density ρ_T and insert $|\tilde{V}|^2 = \pi \bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ (Eq. (1.33)), and we recover Eq. (1.46) term by term. Unlike the version of Sect. 1.5, the derivation holds at any temperature, with the thermal weights inside S, and the integral over ω in the rate runs over negative values too.

2.5.3 Detailed balance

Negative ω means that the target gives energy to the dark matter. The weight of transitions at negative ω is tied to the weight of the reverse transitions by relabeling the sum. Write the transition sum of $\hat{O}^\dagger$ at $-\omega$ and exchange the names of the summed states i and f:

$$\mathcal{S}_{O^\dagger}(-\omega) = \sum_{i,f} p_i \, |\langle f|\hat{O}^\dagger|i\rangle|^2 \, \delta(-\omega - E_f + E_i) = \sum_{i,f} p_f \, |\langle i|\hat{O}^\dagger|f\rangle|^2 \, \delta(\omega - E_f + E_i).$$

Since $|\langle i|\hat{O}^\dagger|f\rangle| = |\langle f|\hat{O}|i\rangle|$, the identity of Sect. 2.1, the matrix elements are the same as in $\mathcal{S}_O(\omega)$. The weights differ: the delta function sets $E_f = E_i + \omega$, where $p_f = p_i \, \mathrm{e}^{-\beta\omega}$. Hence

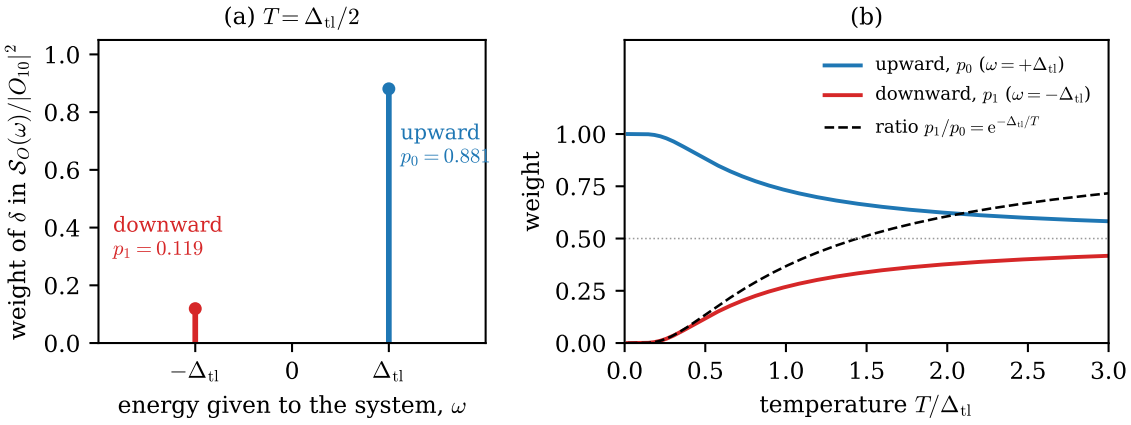


Figure 2.8: Upward and downward transitions of a two-level system with splitting Δ_{tl} in thermal equilibrium at temperature T, driven by a Hermitian operator $\hat{O}$ with matrix element O_{10}. (a) At $T = \Delta_{tl}/2$ the transition sum $\mathcal{S}_O(\omega)$ consists of two delta functions, drawn as stems with their weights in units of $|O_{10}|^2$. Upward, at energy $\omega = +\Delta_{tl}$ given to the system, with the ground-state population $p_0 = 0.881$; downward, at $\omega = -\Delta_{tl}$, with the excited-state population $p_1 = 0.119$. (b) The two weights against T/Δ_{tl}, and their ratio $p_1/p_0 = e^{-\Delta_{tl}/T}$ (detailed balance). The weights approach $\frac{1}{2}$ (dotted line) from either side as T grows and never cross.

Detailed balance

$$\mathcal{S}_{O^\dagger}(-\omega) = e^{-\beta\omega}\,\mathcal{S}_O(\omega). \tag{2.68}$$

For the electron density, $n_e(\boldsymbol{q})^\dagger = n_e(-\boldsymbol{q})$, and $S(-\boldsymbol{q}, -\omega) = e^{-\beta\omega} S(\boldsymbol{q}, \omega)$. For a target with inversion or time-reversal symmetry, $S(\boldsymbol{q}, -\omega) = e^{-\beta\omega} S(\boldsymbol{q}, \omega)$. The density form is the diagonal element of Eq. (A.33).

In the grand-canonical state the weights carry $-\mu_{ch}N$ as well, and the step $p_f = p_i e^{-\beta\omega}$ needs $N_f = N_i$. The condition $N_f = N_i$ holds for every operator that conserves particle number, such as $n_e(\boldsymbol{q})$; an operator that adds ΔN particles acquires the factor $e^{-\beta(\omega - \mu_{ch}\Delta N)}$ instead. The symmetric form uses $S(-\boldsymbol{q}, \omega) = S(\boldsymbol{q}, \omega)$, which inversion or time reversal guarantees (Appendix A.4); silicon has both.

Detailed balance holds for every target in equilibrium, whatever its interactions, because the argument used only the Gibbs weights. In a two-level system with splitting Δ_{tl}, driven by a Hermitian $\hat{O}$ with matrix element $O_{10} = \langle 1|\hat{O}|0\rangle$, the transition sum is

$$\mathcal{S}_O(\omega) = |O_{10}|^2 \left[p_0\,\delta(\omega - \Delta_{tl}) + p_1\,\delta(\omega + \Delta_{tl})\right], \qquad \frac{p_1}{p_0} = e^{-\beta\Delta_{tl}}.$$

The upward transition, at $\omega = +\Delta_{tl}$, starts from the ground state; the downward one, at $\omega = -\Delta_{tl}$, starts from the excited state and carries its population (Fig. 2.8). At $T = \Delta_{tl}/2$ the weights are 0.881 and 0.119. At $T = 0$ the downward weight vanishes and $\omega \geq 0$. As $T \to \infty$ both weights tend to $\frac{1}{2}$, but the upward weight never falls below the downward one: a thermal target always absorbs more often than it emits, by the factor $e^{\beta\omega}$.

The negative-ω part of the rate is suppressed, relative to the positive-ω part at the same $|\omega|$, by the Boltzmann factor $e^{-\beta|\omega|}$ of the excitation that a negative-ω transition removes. For a 1 meV acoustic phonon at 1 K, $\beta\omega = 11.6$ and the Boltzmann factor is 9×10^{-6}; Appendix A.4 gives the case of an electronic gap. Note that a negative-ω event deposits no energy in the

detector. The target loses energy to the dark matter, and a calorimeter or a charge readout registers nothing. Negative-ω transitions nevertheless matter in three places: in the total scattering rate, in the ω integrals of sum rules (Sect. 4.3), and in the factor $1-\mathrm{e}^{-\beta\omega}$ that relates S to the dissipative part of the response through the fluctuation–dissipation theorem (Eq. (A.32), Sect. 4.2).

2.5.4 Occupation of a mode

The thermal factors n and $n+1$ of a bosonic mode (Sect. 2.3) follow from the grand-canonical state of independent particles. For $H-\mu_{\mathrm{ch}}\hat{N}=\sum_\gamma(E_\gamma-\mu_{\mathrm{ch}})\,c_\gamma^\dagger c_\gamma$ the exponential in Eq. (2.63) is a product over single-particle states γ, and so is $\mathcal{Z}$. Each fermion factor has $n_\gamma=0$ or 1, and each boson factor $n_\gamma=0,1,2,\ldots$, a geometric series. The mean occupation of one state follows by weighting n_γ with its factor:

$$\langle c_\gamma^\dagger c_\gamma\rangle=\frac{\mathrm{e}^{-\beta(E_\gamma-\mu_{\mathrm{ch}})}}{1+\mathrm{e}^{-\beta(E_\gamma-\mu_{\mathrm{ch}})}}=n_{\mathrm{F}}(E_\gamma-\mu_{\mathrm{ch}}),\qquad \langle d_\gamma^\dagger d_\gamma\rangle=\frac{\sum_n n\,\mathrm{e}^{-\beta nE_\gamma}}{\sum_n \mathrm{e}^{-\beta nE_\gamma}}=n_{\mathrm{B}}(E_\gamma),\tag{2.69}$$

with the Fermi and Bose functions $n_{\mathrm{F}}(x)=(\mathrm{e}^{\beta x}+1)^{-1}$ and $n_{\mathrm{B}}(x)=(\mathrm{e}^{\beta x}-1)^{-1}$ [19]. A mode with $E_\gamma\ll T$ holds $n_{\mathrm{B}}\approx T/E_\gamma$ quanta, the classical equipartition value, and a mode with $E_\gamma\gg T$ holds $n_{\mathrm{B}}\approx\mathrm{e}^{-\beta E_\gamma}$. The boson form has no chemical potential. Phonons and magnons are created and destroyed freely, their number is not conserved, and the free energy is at its minimum when $\mu_{\mathrm{ch}}=0$. The reversed products follow from Eqs. (2.32) and (2.35): $\langle c_\gamma c_\gamma^\dagger\rangle=1-n_{\mathrm{F}}$, the probability that the state is free to receive an electron, and $\langle d_\gamma d_\gamma^\dagger\rangle=n_{\mathrm{B}}+1$. Absorbing a boson from a thermal mode has the weight n_{B} and emitting one has $n_{\mathrm{B}}+1$, and $(n_{\mathrm{B}}+1)/n_{\mathrm{B}}=\mathrm{e}^{\beta E_\gamma}$ is detailed balance for a single mode. Off-diagonal averages $\langle c_\gamma^\dagger c_{\gamma'}\rangle$ with $\gamma\neq\gamma'$ vanish, since $\hat{\varrho}$ conserves each occupation.

The fermion factors decide the transition sum of the density for independent electrons. By Eq. (2.36), $n_e(\boldsymbol{q})=\sum_{\gamma\gamma'}\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\,c_{\gamma'}^\dagger c_\gamma$. Acting on an occupation-number state $|i\rangle$, each term with $\gamma'\neq\gamma$ moves one electron and leads to its own final state, and the sum of $|\langle f|c_{\gamma'}^\dagger c_\gamma|i\rangle|^2$ over those final states is $\langle i|c_\gamma^\dagger c_{\gamma'}c_{\gamma'}^\dagger c_\gamma|i\rangle=n_\gamma(1-n_{\gamma'})$. The terms with $\gamma'=\gamma$ leave the state unchanged and contribute only at $\omega=0$. The Gibbs state of independent electrons is a product over orbitals, the average of $n_\gamma(1-n_{\gamma'})$ factorizes, and for $\omega\neq0$

$$\mathcal{S}_{n_e(\boldsymbol{q})}(\omega)=\sum_{\gamma\neq\gamma'}|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle|^2\,n_{\mathrm{F}}(E_\gamma-\mu_{\mathrm{ch}})\big[1-n_{\mathrm{F}}(E_{\gamma'}-\mu_{\mathrm{ch}})\big]\,\delta(\omega-E_{\gamma'}+E_\gamma).\tag{2.70}$$

An electron must be present in γ, with probability n_{F}, and γ' must be free to receive it, with probability $1-n_{\mathrm{F}}$. The factor $1-n_{\mathrm{F}}$ is Pauli blocking at finite temperature. The identity $n_{\mathrm{F}}(E')[1-n_{\mathrm{F}}(E)]=\mathrm{e}^{-\beta(E'-E)}\,n_{\mathrm{F}}(E)[1-n_{\mathrm{F}}(E')]$, with both energies measured from μ_{ch}, makes Eq. (2.68) visible term by term. In a semiconductor at $T\ll E_{\mathrm{g}}$ the occupation factors are one for a filled valence state γ and an empty conduction state γ' and zero otherwise, and Eq. (2.70) becomes a sum over valence-to-conduction transitions of the squared single-particle form factor of Eq. (2.39). Written in Bloch states, with the spin sum supplying the factor 2 of a spin-independent vertex, that sum is the crystal form factor.

Worked example 2.7: Scattering off a thermal harmonic oscillator

Let the target be a single mode, $H=\omega_0(d^\dagger d+\frac{1}{2})$, in equilibrium at temperature T, and let the dark matter couple to its displacement, $\hat{O}=\lambda_{\mathrm{osc}}(d+d^\dagger)$, with a coupling λ_{osc} that depends on $\boldsymbol{q}$. The single-mode target describes a phonon mode coupled to a nuclear recoil (Sect. 5.3). By Eq. (2.34), $\hat{O}$ connects $|n\rangle$ to $|n+1\rangle$ with the matrix element $\lambda_{\mathrm{osc}}\sqrt{n+1}$

and to $|n-1\rangle$ with $\lambda_{\rm osc}\sqrt{n}$. The transition sum Eq. (2.65) is therefore

$$\begin{aligned}\mathcal{S}_O(\omega) &= |\lambda_{\rm osc}|^2 \sum_n p_n \left[(n+1)\,\delta(\omega-\omega_0) + n\,\delta(\omega+\omega_0)\right] \\ &= |\lambda_{\rm osc}|^2 \left[(n_{\rm B}+1)\,\delta(\omega-\omega_0) + n_{\rm B}\,\delta(\omega+\omega_0)\right],\end{aligned}$$

with $n_{\rm B} = n_{\rm B}(\omega_0) = \sum_n p_n n$ by Eq. (2.69). Emitting a quantum takes energy from the dark matter and has the weight $n_{\rm B}+1$; absorbing a quantum gives energy to the dark matter and has the weight $n_{\rm B}$. At the same $\boldsymbol{q}$ the ratio of the two weights is

$$\frac{\mathcal{S}_O(+\omega_0)}{\mathcal{S}_O(-\omega_0)} = \frac{n_{\rm B}+1}{n_{\rm B}} = \mathrm{e}^{\beta\omega_0},$$

as Eq. (2.68) requires, since $\hat{O}$ is Hermitian. The ratio of the energy-loss and energy-gain scattering rates also contains the dark-matter kinematics: at fixed q the loss rate carries $\eta(v_{\rm min}(q,+\omega_0))$ and the gain rate $\eta(v_{\rm min}(q,-\omega_0))$ (Eq. (1.22)). The minimum velocity for energy gain, $|q/2m_\chi - \omega_0/q|$, vanishes at $q = (2m_\chi\omega_0)^{1/2}$, and gain is kinematically allowed for every dark-matter mass.

For numbers, take the 64 meV optical phonon of silicon at the zone center. At 300 K, $\beta\omega_0 = 2.48$, $n_{\rm B} = 0.092$, and the loss-to-gain ratio of the target factors is $\mathrm{e}^{2.48} = 11.9$. At 4 K, $\beta\omega_0 = 186$ and $n_{\rm B} = 2.3\times10^{-81}$. For a 1 meV acoustic phonon, $n_{\rm B} = 9.1\times10^{-6}$ at 1 K and 10^{-252} at 20 mK, but 1.27 at 20 K, where the ratio is 1.79. A cryogenic detector is in its ground state for every mode above a few tens of μeV, and the "+1" of spontaneous emission is the whole signal: the dark matter creates a phonon in an empty mode.

The total weight $\int \mathrm{d}\omega\,\mathcal{S}_O$ and the first moment $\int \mathrm{d}\omega\,\omega\,\mathcal{S}_O$ respond differently to temperature. The total weight is $\int \mathrm{d}\omega\,\mathcal{S}_O = |\lambda_{\rm osc}|^2(2n_{\rm B}+1)$, which grows with T because the thermal fluctuations of the displacement grow; at 300 K it is 1.18 times the value at $T=0$ for the optical phonon. The first moment is $\int \mathrm{d}\omega\,\omega\,\mathcal{S}_O = |\lambda_{\rm osc}|^2\omega_0[(n_{\rm B}+1)-n_{\rm B}] = |\lambda_{\rm osc}|^2\omega_0$, independent of T, because the thermal populations cancel between emission and absorption. The T-independent first moment is the oscillator's version of the f-sum rule (Eq. (A.38), Sect. 4.3), and the cancellation requires the negative-ω weight.

2.5.5 Free electron gas at zero temperature

Jellium with its Coulomb interaction dropped (Worked example 2.5) is a gas of free electrons, $H = \sum_{\boldsymbol{k}s}(k^2/2m_e)\,c^\dagger_{\boldsymbol{k}s}c_{\boldsymbol{k}s}$, and its occupations are $n_{\rm F}(k^2/2m_e - \mu_{\rm ch})$. As $T\to0$ the Fermi function becomes a step, $n_{\rm F}(x)\to\Theta(-x)$: every state below $\mu_{\rm ch}$ is filled and every state above it is empty. The ground state is the Fermi sea, $\prod_{k<k_{\rm F},\,s} c^\dagger_{\boldsymbol{k}s}|0\rangle$, the filled sphere of radius $k_{\rm F}$ in $\boldsymbol{k}$ space, and the chemical potential at $T=0$ is the Fermi energy $E_{\rm F} = k_{\rm F}^2/2m_e$. With two spin states and the volume $(2\pi)^3/\mathcal{V}$ per wavevector (Eq. (2.10)), $N_e = 2\mathcal{V}(4\pi k_{\rm F}^3/3)/(2\pi)^3 = Vk_{\rm F}^3/3\pi^2$. The density of states per spin and per unit volume at energy E is $\mathcal{D}_f(E)/\mathcal{V} = m_e k/2\pi^2$, from Eq. (2.11), and at the Fermi level it is $\mathcal{N}_{\rm F}$.

Free electron gas at $T=0$

$$k_{\rm F} = (3\pi^2\boldsymbol{n}_e)^{1/3}, \qquad E_{\rm F} = \frac{k_{\rm F}^2}{2m_e}, \qquad v_{\rm F} = \frac{k_{\rm F}}{m_e}, \qquad \mathcal{N}_{\rm F} = \frac{m_e k_{\rm F}}{2\pi^2}, \tag{2.71}$$

where $\boldsymbol{n}_e = N_e/\mathcal{V}$ is the mean electron density and $\mathcal{N}_{\rm F}$ counts one spin. As a check, $2\mathcal{N}_{\rm F}E_{\rm F} = \frac{3}{2}\boldsymbol{n}_e$.

Eq. (2.71) collects the results of Ref. [20, Chap. 2]. At the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ (Sect. 2.3), Eq. (2.71) yields $k_\mathrm{F} = 3.57\,\mathrm{keV} = 0.96\,q_0$, $E_\mathrm{F} = 12.5\,\mathrm{eV}$, $v_\mathrm{F} = 7.0 \times 10^{-3} = 2090\,\mathrm{km/s}$ and $\mathcal{N}_\mathrm{F} = 9.2 \times 10^7\,\mathrm{eV}^2$. Silicon is not a metal: its valence electrons fill bands separated from the empty ones by a gap (Sect. 3.3). The free electron gas sets the scales, however. Its Fermi momentum is the typical electron momentum $q_0 = \alpha m_e$ of Sect. 1.3, and its Fermi velocity is 2.6 times the maximum dark-matter speed $v_\mathrm{max} = 2.65\times10^{-3}$. Temperature hardly matters here: T/E_F is 2×10^{-3} at 300 K and 7×10^{-8} at 10 mK, and the step is sharp. In a real crystal the same occupations apply to the computed bands; for metals, codes replace the step by a smooth function of width much larger than T to converge the sums over the Brillouin zone (Sect. 23.3).

The Fermi sea quantifies the Pauli blocking of Sect. 2.3. The term $c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$ of $n_e(\boldsymbol{q})$ survives if $\boldsymbol{k}$ lies inside the sphere and $\boldsymbol{k}+\boldsymbol{q}$ outside it. The blocked electrons are those in the overlap of the sphere with its copy displaced by $\boldsymbol{q}$, a lens of volume $\pi(4k_\mathrm{F}+q)(2k_\mathrm{F}-q)^2/12$. The fraction of the sea that can take the momentum q is therefore

$$\frac{3q}{4k_\mathrm{F}} - \frac{q^3}{16k_\mathrm{F}^3} \quad (q < 2k_\mathrm{F}), \qquad 1 \quad (q \geq 2k_\mathrm{F}). \tag{2.72}$$

At small q it grows linearly: the active electrons form a shell of thickness $\sim q$ at the Fermi surface. At $q = 0.1\,q_0$, where a light mediator puts much of its weight, 7.8 % of the silicon-density sea is active. Energy conservation narrows the active fraction further, since the dark matter supplies at most qv_max and $v_\mathrm{max} < v_\mathrm{F}$. Combining Pauli blocking and energy conservation with the occupations of Eq. (2.69) is the Lindhard calculation of Sect. 4.5.

Summary

The rate at which dark matter excites a target follows from one formula, Fermi's golden rule, $\Gamma_{i\to f} = 2\pi|\langle f|H'|i\rangle|^2\delta(E_f - E_i)$ (Eq. (2.5)). It is a rate only after a sum over a continuum of final states and within a window of times, which in dark-matter scattering is always open because the outgoing dark matter supplies the continuum. In a periodic box every factor of the volume can be tracked, and a rate turns into a cross-section by dividing by the flux (Eq. (2.13)). A harmonic perturbation separates absorption from emission, and the absorption rate of a bosonic field is fixed by its energy density, not its flux.

The first Born approximation suffices, because every allowed coupling of dark matter to ordinary matter keeps the Born parameters below 10^{-3}. In this approximation the cross-section on a free electron is $\bar\sigma_e$ for a heavy mediator (Eq. (2.16)). On a bound target the potential factorizes from the transition form factor $f_{i\to f}(\boldsymbol{q}) = \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$ (Eq. (2.18)), whose squares sum to one over final states, and the halo enters only through $\eta(v_\mathrm{min})$. Coherent addition of the nucleons at small q is the origin of the A^2 of nuclear scattering, and the Helm form factor describes how the coherence is lost at large q.

Second quantization writes the many-electron states as occupation numbers, and the operator that dark matter couples to as $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$ (Eq. (2.38)). The Coulomb Hamiltonian of electrons and nuclei, with its $\boldsymbol{k} = 0$ terms removed by neutrality (Eq. (2.48)), is the starting point of Chaps. 3 and 5.

Where first order vanishes, by a selection rule or by kinematics, the rate comes from the second-order T-matrix (Eq. (2.52)), with virtual intermediate states and energy denominators. When one quantum is absorbed and another emitted, two time orderings connect the initial and final states, and their amplitudes interfere (Eq. (2.54)). A real intermediate state decays exponentially at its golden-rule rate (Wigner–Weisskopf, Eq. (2.55)), and its width turns a vanishing energy denominator into a nonzero one. A resonant second-order rate then splits

into a cascade of two first-order processes and a finite part (Eq. (2.60)), and counting the cascade separately as well would count it twice. Phonon-assisted absorption in an indirect-gap crystal is the case that returns in Sect. 6.5.

A target at temperature T is a Gibbs mixture, and its rate is the thermal average of the golden rule. For scattering, all the target physics is then contained in the dynamic structure factor $S(\boldsymbol{q},\omega)$ (Eq. (2.67)), which reproduces the rate formula of Sect. 1.5. Its negative-ω weight, which describes energy given to the dark matter, is fixed by detailed balance, $S(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q},\omega)$; it is suppressed by $\mathrm{e}^{-\beta|\omega|}$ and negligible in a cryogenic detector for every excitation above a few tens of μeV. At $T = 0$ the free electron gas fills a Fermi sphere with $k_\mathrm{F} \approx q_0$ at silicon's valence density, and only a shell of the sphere can absorb a small momentum transfer. The gas can take up any energy, while the electrons of silicon need at least 1.17 eV to be excited; Chap. 3 finds the crystal states responsible for the difference between silicon and the free electron gas. Chap. 4 builds on these results the response of an interacting target: $S(\boldsymbol{q},\omega)$, its sum rules, and screening.

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Chapter 3

Electrons in crystals

A plane wave with the typical momentum of a valence electron, $q_0 = 3.73\,\mathrm{keV}$ (Sect. 1.3), has a wavelength of 3.3 Å, not much longer than the 2.35 Å bond between two silicon atoms. Dark matter that scatters on electrons with momentum transfers of the size of q_0 resolves the arrangement of the atoms. Chap. 2 reduced every scattering rate to matrix elements of the density operator $n_e(\boldsymbol{q})$ between states of the target and collected these matrix elements in the dynamic structure factor, but it left open what the target states are. The free electron gas of Sect. 2.5 has the right density and the right momentum scale, $k_\mathrm{F} \approx q_0$, yet it can take up any energy, while the electrons of silicon cannot be excited by less than 1.17 eV. The difference between the electron gas and silicon comes from the periodic potential of the nuclei. This potential groups the electron states into bands separated by gaps, decides which momentum transfers connect which states, and shapes the wavefunctions on the scale that a few-keV momentum transfer resolves. The electronic structure of the crystal therefore fixes the threshold of an electron-recoil detector and the growth of the detector's rate above threshold. The computed bands of silicon and GaAs in this chapter are Kohn–Sham bands: density-functional theory turns the Coulomb Hamiltonian of Eq. (2.48) into a problem of independent electrons in an effective potential. Density-functional theory is accurate enough to compute the bands and wavefunctions of real materials, but not the band gap. By the end of the chapter, along the path of Fig. 3.1, we will have the states of an electron in silicon, the matrix elements of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between these states, and the band edges that set the threshold, with the band gap corrected.

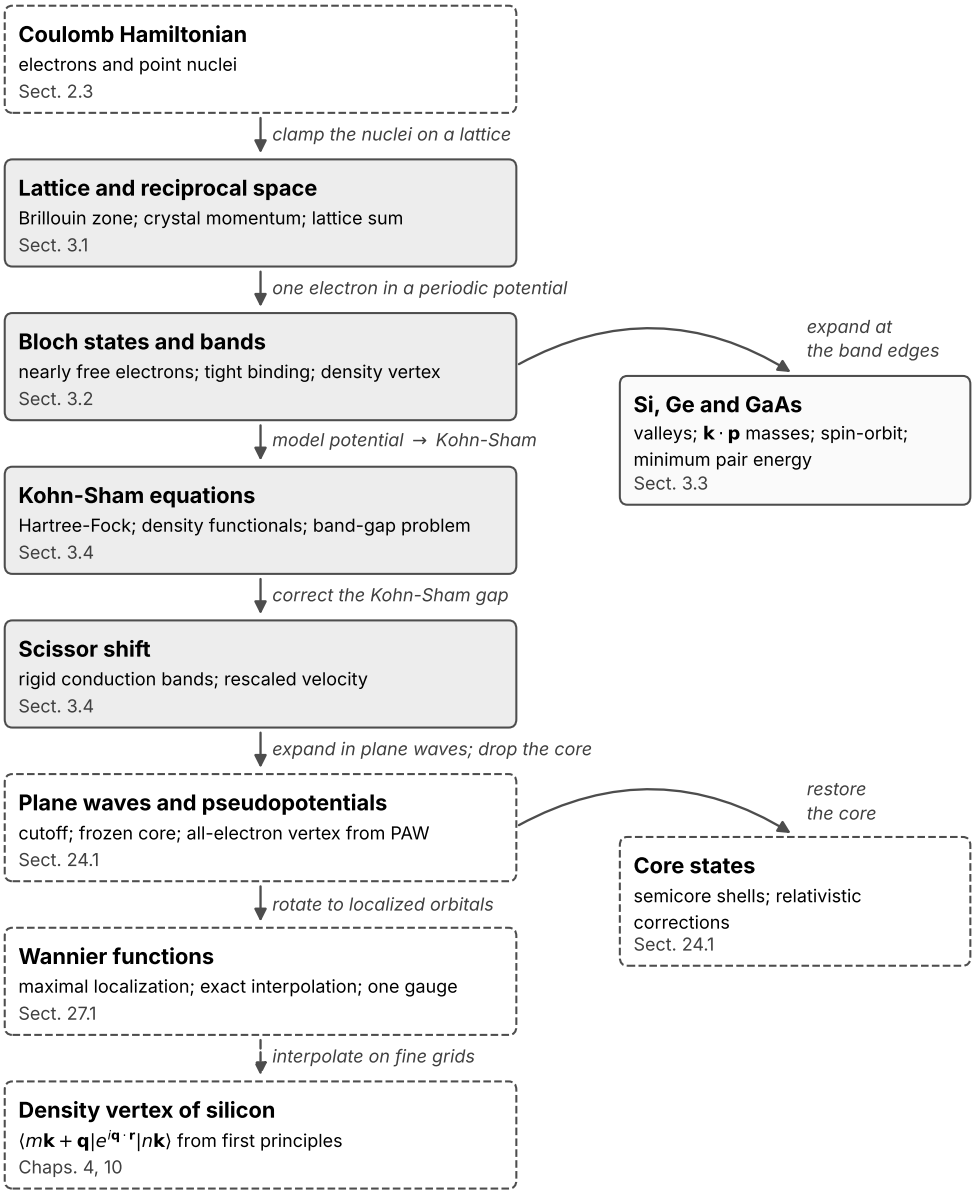


Figure 3.1: From the Coulomb Hamiltonian to the density vertex of silicon. Clamping the nuclei on a lattice gives reciprocal space and crystal momentum (Sect. 3.1); one electron in a periodic potential gives Bloch bands and the density vertex (Sect. 3.2); the band edges fix the valleys, masses and minimum pair energy of Si, Ge and GaAs (Sect. 3.3). The Kohn–Sham potential replaces the model potential, and the scissor shift corrects the Kohn–Sham gap (Sect. 3.4). Dashed boxes: the starting Hamiltonian (Sect. 2.3), the use of the vertex (Chaps. 4 and 10) and the practical calculation of bands and wavefunctions, with plane waves and pseudopotentials and with the core restored for large momentum transfers (Chap. 24), and Wannier interpolation (Chap. 27).

3.1 Lattices and reciprocal space

A kilogram of silicon holds 2.1×10^{25} atoms, and every one of them sits at a point of a lattice that repeats the same pair of atoms in every direction. Because of the repetition, a crystal conserves momentum only up to a reciprocal-lattice vector, as in the working definition of Sect. 1.3. The momenta that a crystal can tell apart fill a single cell of reciprocal space, the Brillouin zone. Both the conservation law and the zone enter every later rate. We follow Refs. [1, Chaps. 4–6 and 8] and [2, Chaps. 1–2].

Throughout Chaps. 3 and 4 the nuclei sit at fixed equilibrium positions: we work with a *static lattice*. The approximation rests on the ratio of the nuclear to the electron mass. A silicon nucleus is $M/m_e = 5.1 \times 10^4$ times heavier than an electron, and the electrons rearrange long before the nuclei have moved appreciably. Sect. 5.1 makes the argument precise with the Born–Oppenheimer expansion. The static lattice leaves out the lattice vibrations (phonons) and the electron–phonon coupling; Chaps. 5 and 6 add both back.

3.1.1 Bravais lattice and basis

A *Bravais lattice* is the set of points

$$\boldsymbol{R} = n_1\boldsymbol{a}_1 + n_2\boldsymbol{a}_2 + n_3\boldsymbol{a}_3, \qquad n_i \in \mathbb{Z}, \tag{3.1}$$

generated by three linearly independent *primitive vectors* $\boldsymbol{a}_i$. Every lattice vector $\boldsymbol{R}$ maps the lattice onto itself. The parallelepiped spanned by the $\boldsymbol{a}_i$ is a primitive cell: its translations by all $\boldsymbol{R}$ fill space without overlap, and it contains one lattice point. Its volume,

$$\Omega = \boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3), \tag{3.2}$$

is the *cell volume*, the only meaning of a bare Ω in this book (a solid angle always carries a subscript, as in $\mathrm{d}\Omega_q$). The choice of primitive vectors is not unique, but every primitive cell has the same volume, since each holds one lattice point.

A crystal places a group of atoms, the *basis*, at every lattice point. We label the atoms of the basis by $\kappa = 1, \ldots, n_{\mathrm{at}}$ and their positions within the cell by $\boldsymbol{\tau}_\kappa$, so that the nuclei sit at

$$\boldsymbol{X}_{\boldsymbol{R}\kappa} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa. \tag{3.3}$$

This is the κ that labels Z_κ and $n_\kappa(\boldsymbol{k})$ in Sect. 2.3; two sites of one element still carry different κ.

Silicon has the *diamond structure* drawn in Fig. 3.2. Its Bravais lattice is face-centered cubic (fcc): the corners and face centers of a cube of side a, the *lattice constant*. A symmetric choice of primitive vectors joins a corner to the three nearest face centers,

$$\boldsymbol{a}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{a}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{a}_3 = \frac{a}{2}(1,1,0), \tag{3.4}$$

and Eq. (3.2) yields $\Omega = a^3/4$: the cube contains four lattice points. The basis consists of two silicon atoms,

$$\boldsymbol{\tau}_1 = 0, \qquad \boldsymbol{\tau}_2 = \frac{a}{4}(1,1,1), \tag{3.5}$$

displaced along the cube diagonal by a quarter of its length. Each atom then has four nearest neighbors at the corners of a regular tetrahedron, at the distance $\sqrt{3}\,a/4$ and with bond angles of $109.47°$. The bonds to these neighbors are the four covalent bonds of a silicon atom, one for each valence electron, and Sect. 3.2 returns to the sp^3 orbitals that form the bonds. Note that the diamond structure is not a Bravais lattice: the environment of an atom at $\boldsymbol{\tau}_2$ is the

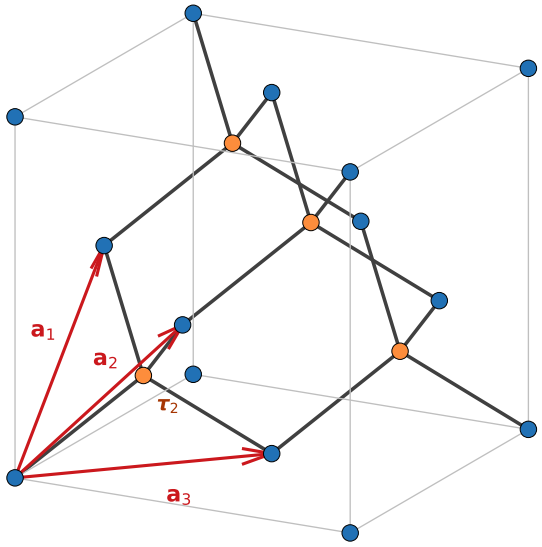


Figure 3.2: The diamond structure, drawn in a cube of side a, the lattice constant. Blue: the face-centered cubic lattice of sites $\boldsymbol{R}$ (cube corners and face centers). Orange: the second atom of the basis, at $\boldsymbol{R}+\boldsymbol{\tau}_2$ with $\boldsymbol{\tau}_2 = (a/4)(1,1,1)$; the four orange atoms inside the cube are shown. Dark lines join nearest neighbors, at a distance $\sqrt{3}\,a/4$, 2.352 Å in silicon ($a = 5.431$ Å); each atom is bonded to four others at the corners of a regular tetrahedron. Red arrows: the primitive vectors $\boldsymbol{a}_1 = (a/2)(0,1,1)$, $\boldsymbol{a}_2 = (a/2)(1,0,1)$, $\boldsymbol{a}_3 = (a/2)(1,1,0)$, which span a primitive cell of volume $a^3/4$ containing two atoms. In the zincblende structure (GaAs) the blue and orange sites hold different elements.

environment of an atom at $\boldsymbol{\tau}_1$ inverted through the bond center, and no lattice vector maps the site $\boldsymbol{\tau}_1$ onto the site $\boldsymbol{\tau}_2$.

The lattice constant of silicon is $a = 5.431\,\text{Å}$ [3, lattice parameter of silicon], or $2.752 \times 10^{-3}\,\text{eV}^{-1}$ in natural units. The cell volume is $\Omega = 40.05\,\text{Å}^3$, the nearest-neighbor distance is $2.352\,\text{Å}$, and the cube of side a holds eight atoms. These numbers fix the densities used in the earlier chapters. The atomic density is $8/a^3 = 4.99 \times 10^{22}\,\text{cm}^{-3}$, which with the atomic weight 28.0855 yields the mass density $2.33\,\text{g}\,\text{cm}^{-3}$ of Sect. 1.2. With four valence electrons per atom, the valence density $32/a^3 = 2.00 \times 10^{23}\,\text{cm}^{-3}$ is the density used for the free electron gas of Sect. 2.5. Germanium has the same structure with a lattice constant a few percent larger. Gallium arsenide has the *zincblende structure*, which has the lattice and the $\boldsymbol{\tau}_\kappa$ of silicon, but with gallium at $\boldsymbol{\tau}_1$ and arsenic at $\boldsymbol{\tau}_2$ [1, Chap. 4]. Since the two sites now differ, the inversion that exchanged them in silicon is no longer a symmetry. As a result GaAs is polar, a property that matters for its phonons (Sect. 5.4). Sect. 3.3 gives the band structures of all three materials.

A first-principles calculation, in density-functional theory (Sect. 3.4), either takes a from experiment or finds it by minimizing the computed total energy of the crystal with respect to a. For silicon in the local-density approximation the minimum lies at 5.401 Å, 0.5 % below the measured value [4, Sect. 10.2]. A rate is usually computed at the measured a. Since band energies depend on a, the choice between the measured and the computed a shifts the computed gap and is one of the reasons why computed and measured thresholds differ.

3.1.2 The reciprocal lattice

Every function with the periodicity of the lattice is built from plane waves $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ that take the same value at every lattice point, $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$ for all $\boldsymbol{R}$. The wavevectors $\boldsymbol{G}$ with this property are the *reciprocal-lattice vectors*. By Eq. (3.1) the condition holds for all $\boldsymbol{R}$ if and only if $\boldsymbol{G}\cdot\boldsymbol{a}_i$ is a multiple of 2π for each i. We therefore look for vectors $\boldsymbol{b}_j$ with

$$\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}. \tag{3.6}$$

They are

$$\boldsymbol{b}_1 = \frac{2\pi}{\Omega}\,\boldsymbol{a}_2 \times \boldsymbol{a}_3, \qquad \boldsymbol{b}_2 = \frac{2\pi}{\Omega}\,\boldsymbol{a}_3 \times \boldsymbol{a}_1, \qquad \boldsymbol{b}_3 = \frac{2\pi}{\Omega}\,\boldsymbol{a}_1 \times \boldsymbol{a}_2. \tag{3.7}$$

To check Eq. (3.6), note that $\boldsymbol{a}_1 \cdot \boldsymbol{b}_1 = 2\pi\,\boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3)/\Omega = 2\pi$ by Eq. (3.2), while $\boldsymbol{a}_2 \cdot \boldsymbol{b}_1$ and $\boldsymbol{a}_3 \cdot \boldsymbol{b}_1$ vanish because a triple product with a repeated vector is zero; the other components follow by cyclic permutation. Since the $\boldsymbol{b}_j$ are linearly independent, any wavevector can be written as $\boldsymbol{G} = \sum_j m_j \boldsymbol{b}_j$ with real m_j, and then $\boldsymbol{G}\cdot\boldsymbol{a}_i = 2\pi m_i$. The condition $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$ thus requires integer m_i, and the reciprocal-lattice vectors form a Bravais lattice of their own,

$$\boldsymbol{G} = m_1\boldsymbol{b}_1 + m_2\boldsymbol{b}_2 + m_3\boldsymbol{b}_3, \qquad m_i \in \mathbb{Z}, \tag{3.8}$$

with primitive vectors $\boldsymbol{b}_i$. In the book's convention (Appendix A.3), a function with the periodicity of the lattice is a sum $f(\boldsymbol{r}) = \sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ over these vectors, with $f_{\boldsymbol{G}} = \Omega^{-1}\int_\Omega \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f(\boldsymbol{r})$ (Eq. (A.11)). The plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ is constant on planes perpendicular to $\boldsymbol{G}$, and for the shortest $\boldsymbol{G}$ in a given direction the wavelength $2\pi/|\boldsymbol{G}|$ of the plane wave is the spacing of the lattice planes perpendicular to $\boldsymbol{G}$.

The volume of the cell spanned by the $\boldsymbol{b}_i$ follows from Eq. (3.7). With the identity $(\boldsymbol{u}\times\boldsymbol{v})\times(\boldsymbol{u}\times\boldsymbol{w}) = \boldsymbol{u}\,[\boldsymbol{u}\cdot(\boldsymbol{v}\times\boldsymbol{w})]$ for $\boldsymbol{u} = \boldsymbol{a}_1$,

$$\boldsymbol{b}_2 \times \boldsymbol{b}_3 = \frac{(2\pi)^2}{\Omega^2}\,(\boldsymbol{a}_3 \times \boldsymbol{a}_1) \times (\boldsymbol{a}_1 \times \boldsymbol{a}_2) = \frac{(2\pi)^2}{\Omega^2}\,\boldsymbol{a}_1\,[\boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3)] = \frac{(2\pi)^2}{\Omega}\,\boldsymbol{a}_1,$$

where reversing the first cross product brings $\boldsymbol{a}_1$ to the front of both factors at the cost of a sign, and reversing the cross product inside the resulting triple product restores the sign. Then Eq. (3.6) gives

$$\boldsymbol{b}_1 \cdot (\boldsymbol{b}_2 \times \boldsymbol{b}_3) = \frac{(2\pi)^3}{\Omega}. \tag{3.9}$$

A small cell in real space is a large one in reciprocal space, as the Fourier transform requires.

For the fcc vectors of Eq. (3.4), Eq. (3.7) yields

$$\boldsymbol{b}_1 = \frac{2\pi}{a}(-1,1,1), \qquad \boldsymbol{b}_2 = \frac{2\pi}{a}(1,-1,1), \qquad \boldsymbol{b}_3 = \frac{2\pi}{a}(1,1,-1), \tag{3.10}$$

as we can confirm directly from Eq. (3.6): for instance, $\boldsymbol{a}_1 \cdot \boldsymbol{b}_1 = \pi(0+1+1) = 2\pi$ and $\boldsymbol{a}_2 \cdot \boldsymbol{b}_1 = \pi(-1+0+1) = 0$. These are the primitive vectors of a body-centered cubic lattice. With $\boldsymbol{G} = (2\pi/a)(h,k,l)$, Eq. (3.8) reads $h = -m_1 + m_2 + m_3$ and the cyclic permutations of this relation, so that $h + k = 2m_3$, $k + l = 2m_1$ and $l + h = 2m_2$. The reciprocal-lattice vectors of the fcc lattice are the vectors $(2\pi/a)(h,k,l)$ with integers h,k,l that are all even or all odd, and every such triple occurs. The integers h,k,l are the Miller indices of the lattice planes perpendicular to $\boldsymbol{G}$, and we write $\boldsymbol{G}_{hkl}$. The shortest nonzero vectors are the eight $\boldsymbol{G}_{111}$ along the cube diagonals, with $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a$, followed by the six $\boldsymbol{G}_{200}$ along the cube axes, with $|\boldsymbol{G}_{200}| = 2 \cdot 2\pi/a$. In silicon $2\pi/a = 2.28\,\mathrm{keV} = 1.16\,\text{Å}^{-1}$, $|\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$ and $|\boldsymbol{G}_{200}| = 4.57\,\mathrm{keV}$, the scales already met in Sects. 1.3 and 2.4.

3.1.3 The Brillouin zone

Two wavevectors that differ by a reciprocal-lattice vector give the same phase at every lattice point, $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{R}} = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$, and a lattice cannot tell them apart. The wavevectors that a crystal can distinguish fill one primitive cell of the reciprocal lattice. The conventional choice of cell is the *first Brillouin zone*, the set of points closer to the origin than to any other reciprocal-lattice vector (Sect. 1.3). Around a point of the direct lattice, the set of points closer to that lattice point than to any other is the *Wigner–Seitz cell*, so the first Brillouin zone is the Wigner–Seitz cell of the reciprocal lattice. The zone boundary is made of pieces of the planes $\boldsymbol{k}\cdot\boldsymbol{G} = |\boldsymbol{G}|^2/2$ that bisect the vectors $\boldsymbol{G}$, and a point on the boundary is equidistant from the origin and from at least one $\boldsymbol{G}$. The zone is preferred to the parallelepiped spanned by the $\boldsymbol{b}_i$ because the zone has the full point symmetry of the lattice. The zone and the parallelepiped both tile reciprocal space under translations by $\boldsymbol{G}$, and the zone has the volume of Eq. (3.9), $(2\pi)^3/\Omega$. For fcc the zone volume is $4(2\pi/a)^3$.

For the fcc lattice the zone is the truncated octahedron of Fig. 3.3. The planes that bisect the eight $\boldsymbol{G}_{111}$ cut out an octahedron, $|k_x| + |k_y| + |k_z| \le \frac{3}{2}(2\pi/a)$, and the planes that bisect the six $\boldsymbol{G}_{200}$, $|k_x|, |k_y|, |k_z| \le 2\pi/a$, cut off its six corners; no $\boldsymbol{G}$ longer than $\boldsymbol{G}_{200}$ contributes a face. Points of high symmetry carry standard labels. The zone center is Γ. The centers of the square faces are the six $\mathrm{X} = (2\pi/a)(1,0,0)$, at a distance $2\pi/a$, and the centers of the hexagonal faces are the eight $\mathrm{L} = (2\pi/a)(\frac{1}{2},\frac{1}{2},\frac{1}{2})$, at $\frac{\sqrt{3}}{2}\,2\pi/a$, the closest boundary points to Γ. The midpoints of the edges shared by two hexagons are the points $\mathrm{K} = (2\pi/a)(\frac{3}{4},\frac{3}{4},0)$, and the corners are the points $\mathrm{W} = (2\pi/a)(1,\frac{1}{2},0)$, at $\frac{\sqrt{5}}{2}\,2\pi/a$, the farthest. The zone points are set upright, Γ, X, L, K, W, to keep them apart from the italic rates Γ of Chap. 2 and the loss function L of Sect. 4.4. A band structure is plotted along straight lines between these points, such as L–Γ–X, and the band extrema of silicon, germanium and GaAs lie on these lines (Sect. 3.3).

For silicon the zone is about 2.3 keV in radius: every momentum shorter than $|\Gamma\mathrm{L}| = 1.98\,\mathrm{keV}$ lies inside it, and every momentum longer than $|\Gamma\mathrm{W}| = 2.55\,\mathrm{keV}$ lies outside. A sphere of the same volume has the radius $(3/\pi)^{1/3}\,2\pi/a = 0.985\,(2\pi/a) = 2.25\,\mathrm{keV}$.

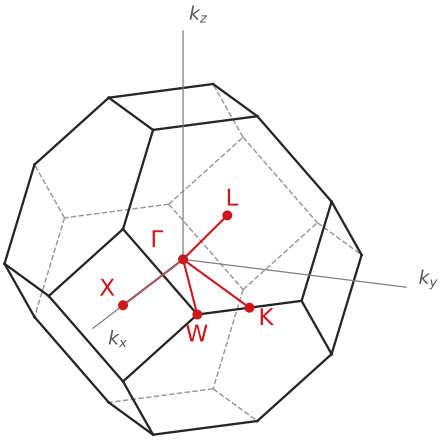


Figure 3.3: The first Brillouin zone of the face-centered cubic lattice, a truncated octahedron with eight hexagonal faces, which bisect the vectors $\boldsymbol{G}_{111} = (2\pi/a)(\pm1, \pm1, \pm1)$, and six square faces, which bisect $\boldsymbol{G}_{200} = (2\pi/a)(\pm2, 0, 0)$ and its permutations. Dashed edges are hidden. The high-symmetry points, in units of $2\pi/a$, are $\Gamma = (0,0,0)$, $\mathrm{X} = (1,0,0)$, $\mathrm{L} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $\mathrm{K} = (\frac{3}{4}, \frac{3}{4}, 0)$ and $\mathrm{W} = (1, \frac{1}{2}, 0)$. The distances of X, L, K and W from Γ are 1, 0.866, 1.061 and 1.118, or 2.28, 1.98, 2.42 and 2.55 keV for silicon ($a = 5.431\,\text{Å}$). The zone volume is $4(2\pi/a)^3 = (2\pi)^3/\Omega$.

3.1.4 Crystal momentum and the lattice sum

With the nuclei clamped at the sites of Eq. (3.3), the Hamiltonian of the electrons is Eq. (2.48) without the nuclear kinetic energy. The Hamiltonian of the electrons is unchanged when all electrons are moved by the same lattice vector $\boldsymbol{R}$, but not when they are moved by an arbitrary vector. Let $T_{\boldsymbol{R}}$ be the operator that moves all electrons by $\boldsymbol{R}$, $(T_{\boldsymbol{R}}\Psi)(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}) = \Psi(\boldsymbol{r}_1 + \boldsymbol{R}, \ldots, \boldsymbol{r}_{N_e} + \boldsymbol{R})$ for a wavefunction of N_e electrons (spins are not moved). It is unitary, it commutes with the Hamiltonian, and translations compose, $T_{\boldsymbol{R}}T_{\boldsymbol{R}'} = T_{\boldsymbol{R}+\boldsymbol{R}'}$. As a result, the eigenstates of the Hamiltonian can be chosen as eigenstates of all $T_{\boldsymbol{R}}$ at once, with eigenvalues $c(\boldsymbol{R})$ that are phases and obey $c(\boldsymbol{R}+\boldsymbol{R}') = c(\boldsymbol{R})c(\boldsymbol{R}')$. With $c(\boldsymbol{a}_i) = \mathrm{e}^{2\pi\mathrm{i}x_i}$ and real x_i, the composition rule requires $c(\boldsymbol{R}) = \mathrm{e}^{2\pi\mathrm{i}\sum_i n_i x_i}$, and with Eq. (3.6) the eigenvalue equation becomes

$$T_{\boldsymbol{R}}\Psi = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\,\Psi, \qquad \boldsymbol{k} = \sum_i x_i \boldsymbol{b}_i. \tag{3.11}$$

The wavevector $\boldsymbol{k}$ is the *crystal momentum* of the state. It is defined only up to a reciprocal-lattice vector, since $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{G}$ give the same eigenvalues, and we take it in the first Brillouin zone. Eq. (3.11) makes precise the working definition of Sect. 1.3. For a free particle every translation is a symmetry, and $\boldsymbol{k}$ is the momentum; in a crystal just the lattice translations survive, and momentum is conserved modulo $\boldsymbol{G}$, as Sect. 2.3 found from the Coulomb Hamiltonian. The literature also writes "quasi-momentum"; the two terms mean the same. For a single electron, Eq. (3.11) becomes Bloch's theorem, which Sect. 3.2 derives together with the form of the single-electron states.

A finite crystal needs boundary conditions, and the *Born–von Karman* conditions are the periodic boundary conditions of the box of Sect. 2.1, adapted to the lattice. The crystal is a

block of $N = N_1 N_2 N_3$ cells at the vectors $\boldsymbol{R} = \sum_i n_i \boldsymbol{a}_i$ with $0 \le n_i < N_i$, and a translation by $N_i \boldsymbol{a}_i$ acts as the identity. The volume of the block is

$$\mathcal{V} = N\Omega, \tag{3.12}$$

the normalization volume of Sect. 2.1. The condition $c(N_i \boldsymbol{a}_i) = \mathrm{e}^{\mathrm{i}N_i \boldsymbol{k}\cdot\boldsymbol{a}_i} = 1$ restricts the crystal momenta to

$$\boldsymbol{k} = \sum_i \frac{m_i}{N_i}\, \boldsymbol{b}_i, \qquad m_i \in \mathbb{Z}. \tag{3.13}$$

Adding N_i to m_i adds $\boldsymbol{b}_i$, a reciprocal-lattice vector. The distinct crystal momenta are those with $0 \le m_i < N_i$, and there are N of them, one per cell of the crystal. They form a uniform grid that fills one primitive cell of the reciprocal lattice, of volume $(2\pi)^3/\Omega$, and each occupies $(2\pi)^3/(N\Omega) = (2\pi)^3/\mathcal{V}$: the density of allowed wavevectors, $\mathcal{V}/(2\pi)^3$, is the density for plane waves in a box of the same volume. Shifting points by reciprocal-lattice vectors maps the grid onto the first Brillouin zone, which then holds the same N points.

The identity behind every crystal-momentum selection rule in this book is the lattice sum. For $\boldsymbol{k}$ on the grid of Eq. (3.13), $\boldsymbol{k}\cdot\boldsymbol{R} = 2\pi \sum_i m_i n_i / N_i$, so the sum of $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$ over the block factorizes into three geometric sums. Each geometric sum is

$$\sum_{n=0}^{N_i-1} \mathrm{e}^{2\pi \mathrm{i} m_i n/N_i} = \begin{cases} N_i, & m_i \text{ a multiple of } N_i, \\ \dfrac{1-\mathrm{e}^{2\pi\mathrm{i}m_i}}{1-\mathrm{e}^{2\pi\mathrm{i}m_i/N_i}} = 0, & \text{otherwise,} \end{cases}$$

by the geometric series. The product is nonzero only when all three m_i are multiples of the N_i, that is, when $\boldsymbol{k}$ is a reciprocal-lattice vector. It then equals N:

Lattice sum

$$\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N \sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}}. \tag{3.14}$$

The sum runs over the N cells of the Born–von Karman block and $\boldsymbol{k}$ over the grid of Eq. (3.13). The crystal momenta form N points in a zone of volume $(2\pi)^3/\Omega$, and $\sum_{\boldsymbol{k}\in\mathrm{BZ}} \to \mathcal{V} \int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3$.

The N phases add up only when they agree at every lattice site, and cancel otherwise. Appendix C.2 derives the infinite-crystal form, $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = [(2\pi)^3/\Omega] \sum_{\boldsymbol{G}} \delta^3(\boldsymbol{k}-\boldsymbol{G})$, and the error made when a calculation replaces the integral over the zone by a sum over a coarse grid.

The count of states fixes the filling of the bands. Each band of a crystal (Sect. 3.2) has one state per crystal momentum and spin, $2N$ states in all, two per cell. A primitive cell of silicon holds two atoms and eight valence electrons, which fill exactly four bands. A kilogram of silicon has $N = 1.07 \times 10^{25}$ cells, and as many crystal momenta in each band. The same count connects to the free electron gas of Sect. 2.5. Eight electrons per cell fill four zone volumes, and a Fermi sphere of four zone volumes has the radius $4^{1/3} \times 2.25\,\mathrm{keV} = 3.57\,\mathrm{keV}$, the k_F of silicon's valence density. Silicon's valence electrons are not a free gas, as Sect. 2.5 noted, and a gap opens at the zone boundary (Sect. 3.2). However, the comparison shows that in the free-electron picture the occupied states reach well beyond the first zone, and that the momentum scale of the valence electrons, $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, is 1.7 times the radius of a sphere with the volume of the zone.

3.1.5 Momentum transfer in a crystal

A dark-matter particle that scatters on the crystal transfers a momentum $\boldsymbol{q}$ through the density operator $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ (Sect. 2.3). Translating the electrons multiplies each term

by $e^{i\boldsymbol{q}\cdot\boldsymbol{R}}$, $T_{\boldsymbol{R}} n_e(\boldsymbol{q}) T_{\boldsymbol{R}}^{-1} = e^{i\boldsymbol{q}\cdot\boldsymbol{R}} n_e(\boldsymbol{q})$. If $|i\rangle$ and $|f\rangle$ have crystal momenta $\boldsymbol{k}_i$ and $\boldsymbol{k}_f$, the unitarity of $T_{\boldsymbol{R}}$ and Eq. (3.11) give

$$\langle f|n_e(\boldsymbol{q})|i\rangle = \langle T_{\boldsymbol{R}} f|\, T_{\boldsymbol{R}} n_e(\boldsymbol{q}) T_{\boldsymbol{R}}^{-1}\, |T_{\boldsymbol{R}} i\rangle = e^{i(\boldsymbol{k}_i+\boldsymbol{q}-\boldsymbol{k}_f)\cdot\boldsymbol{R}} \langle f|n_e(\boldsymbol{q})|i\rangle$$

for every $\boldsymbol{R}$. The matrix element vanishes unless the phase is one at every lattice site, that is, unless

$$\boldsymbol{k}_f = \boldsymbol{k}_i + \boldsymbol{q} - \boldsymbol{G} \tag{3.15}$$

for some reciprocal-lattice vector $\boldsymbol{G}$. The argument rests on symmetry alone, and it holds for the interacting many-electron states as well as for single electrons; Sect. 3.2 works out the single-electron matrix element.

Since only $\boldsymbol{q}$ modulo $\boldsymbol{G}$ enters Eq. (3.15), it is useful to split the momentum transfer into its part in the first zone and a reciprocal-lattice vector,

$$\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q, \qquad \boldsymbol{q}_r \in \mathrm{BZ}, \tag{3.16}$$

where $\boldsymbol{G}_q$ is the reciprocal-lattice vector closest to $\boldsymbol{q}$. The split is unique except on the zone boundary, a set of measure zero. Fig. 3.4 shows an example. The crystal momentum of the target changes by $\boldsymbol{q}_r$, up to a reciprocal-lattice vector, and the rest of the momentum transfer, $\boldsymbol{G}_q$, goes to the lattice as a whole. The static lattice takes up $\boldsymbol{G}_q$ with no recoil energy, since the lattice has infinite mass. A transition in which a nonzero $\boldsymbol{G}$ appears in Eq. (3.15) is an *Umklapp* process, from the German for "flipping over", and a transition with $\boldsymbol{G} = 0$ is a normal process. Whether a transition is an Umklapp process depends on $\boldsymbol{k}_i$ as well as on $\boldsymbol{q}$: even for $\boldsymbol{q}$ inside the zone, $\boldsymbol{k}_i + \boldsymbol{q}$ can leave it and be folded back. For $|\boldsymbol{q}|$ larger than the diameter of the zone, $2|\Gamma\mathrm{W}|$, $\boldsymbol{k}_i + \boldsymbol{q}$ always lies outside, and every transition is an Umklapp process.

The momentum transfer itself is never reduced to the first zone. The dark matter loses $\boldsymbol{q}$, not $\boldsymbol{q}_r$, and its kinematics (through $v_{\min}$) and the dark-matter form factor $F_{\mathrm{DM}}(q)$ (Sects. 1.3 and 1.4) depend on $|\boldsymbol{q}|$. The target does not treat $\boldsymbol{q}$ and $\boldsymbol{q}_r$ alike either, although $\boldsymbol{q}$ and $\boldsymbol{q}_r$ change the crystal momentum by the same amount. The plane wave $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ varies on the scale $2\pi/|\boldsymbol{q}|$ and resolves the electron density inside the cell on that scale. At large $|\boldsymbol{G}_q|$ the plane wave probes the region near the nuclei, where the core electrons and the steep part of the valence wavefunctions are. The response of the crystal is therefore a function of $\boldsymbol{q}_r$ and $\boldsymbol{G}_q$ separately, not of $\boldsymbol{q}_r$ alone, and the book keeps both (Appendix A.3).

Worked example 3.1: Momentum transfers against the zone of silicon

With $\hbar c = 1973.27\,\mathrm{eV\,\AA}$ and $a = 5.431\,\mathrm{\AA}$,

$$\frac{2\pi}{a} = \frac{2\pi \times 1973.27\,\mathrm{eV\,\AA}}{5.431\,\mathrm{\AA}} = 2283\,\mathrm{eV} = 2.28\,\mathrm{keV}.$$

The shortest reciprocal-lattice vectors are $|\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$ and $|\boldsymbol{G}_{200}| = 4.57\,\mathrm{keV}$. An electronic excitation across the low-temperature gap, $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$ (Sect. 3.3), needs $q \geq E_{\mathrm{g}}/v_{\max} = 1.17\,\mathrm{eV}/(2.650 \times 10^{-3}) = 0.44\,\mathrm{keV}$, or $0.19\,(2\pi/a)$, deep inside the zone. The largest momentum transfer that can still deposit E_{g} follows from Eq. (1.25), $q = m_\chi v_{\max}[1 + (1 - 2E_{\mathrm{g}}/m_\chi v_{\max}^2)^{1/2}]$. It is $4.81\,\mathrm{keV} = 2.11\,(2\pi/a)$ for $m_\chi = 1\,\mathrm{MeV}$, $52.6\,\mathrm{keV} = 23.0\,(2\pi/a)$ for 10 MeV and $530\,\mathrm{keV} = 232\,(2\pi/a)$ for 100 MeV. The smallest momentum transfer that can deposit E_{g} is 0.49 keV at 1 MeV and tends to $E_{\mathrm{g}}/v_{\max} = 0.44\,\mathrm{keV}$ for heavier particles, hardly depending on the mass; the largest grows in proportion to m_χ. Already at 1 MeV the momentum transfers that can deposit E_{g} extend past the eight $\boldsymbol{G}_{111}$ and the six $\boldsymbol{G}_{200}$, and the typical momentum of a valence electron, $q_0 = 3.73\,\mathrm{keV} =$

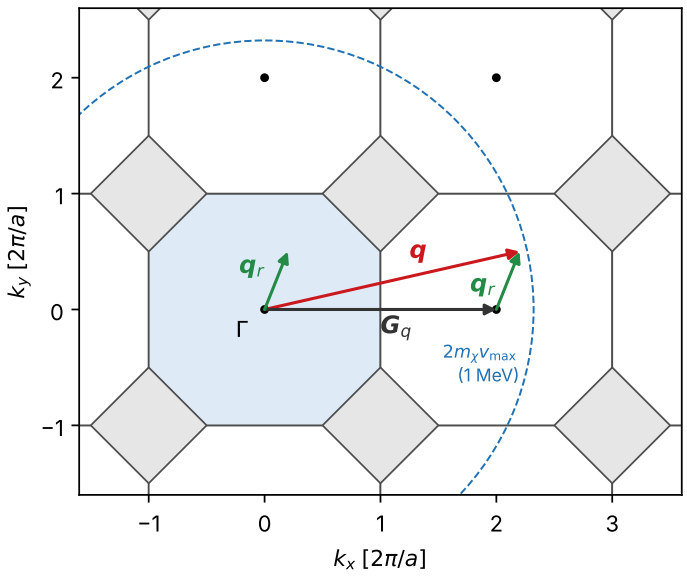


Figure 3.4: A momentum transfer reduced to the first Brillouin zone, in the plane $k_z = 0$ of the fcc reciprocal lattice, with momenta in units of $2\pi/a$ (2.28 keV for silicon). Dots: the reciprocal-lattice vectors in this plane, $(2\pi/a)(2m, 2n, 0)$. Octagons: the cuts of their Brillouin zones by the plane; the first zone is shaded blue. The grey squares belong to zones centered on reciprocal-lattice vectors outside the plane, $(2\pi/a)(h, k, \pm 1)$ with h, k odd. The momentum transfer $\boldsymbol{q} = (2\pi/a)(2.2, 0.5, 0)$, of length 5.15 keV, is the sum of the reciprocal-lattice vector $\boldsymbol{G}_q = (2\pi/a)(2, 0, 0)$ nearest to it and of $\boldsymbol{q}_r = (2\pi/a)(0.2, 0.5, 0)$, of length 1.23 keV, drawn both at $\boldsymbol{G}_q$ and at the origin. Dashed circle: the maximum momentum transfer $2m_\chi v_{\max} = 5.30$ keV of dark matter of mass 1 MeV, with $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$.

$1.63\,(2\pi/a)$, lies outside the zone. A ball of radius q contains about $(4\pi/3)q^3/[4(2\pi/a)^3]$ reciprocal-lattice vectors, one per zone volume. For q equal to the maximum momentum transfer $2m_\chi v_{\max}$, the number of reciprocal-lattice vectors is 1.3×10^4 at 10 MeV, which a direct count confirms, and 1.3×10^7 at 100 MeV. Above about a MeV, most of the kinematically allowed momentum transfers have $\boldsymbol{G}_q \neq 0$, and a calculation of the rate must carry the response out to reciprocal-lattice vectors whose number grows as m_χ^3.

3.1.6 The basis structure factor

The positions of the atoms within the cell enter every Fourier component of a lattice-periodic quantity. Consider a function built from one term per atom, $f(\boldsymbol{r}) = \sum_{\boldsymbol{R}}\sum_\kappa v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa)$, such as the potential of the clamped nuclei or a superposition of atomic charge densities. The Fourier components of f, Eq. (A.11), are

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\int_\Omega \mathrm{d}^3r\; \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} \sum_{\boldsymbol{R}}\sum_\kappa v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa) = \frac{1}{\Omega}\sum_\kappa \int \mathrm{d}^3r\; \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}\, v_\kappa(\boldsymbol{r}-\boldsymbol{\tau}_\kappa).$$

In the second step, the translate of the cell by $-\boldsymbol{R}$ was used for each term; since $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}}=1$, the phase is unchanged, and the cells together fill all of space. Shifting $\boldsymbol{r}$ by $\boldsymbol{\tau}_\kappa$ then gives

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}\,\tilde{v}_\kappa(\boldsymbol{G}), \qquad \tilde{v}_\kappa(\boldsymbol{G}) = \int \mathrm{d}^3x\; \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{x}}\, v_\kappa(\boldsymbol{x}). \tag{3.17}$$

Each atom contributes the transform of its own term at $\boldsymbol{G}$, with a phase set by its position in the cell. When all atoms of the basis are of one element, $\tilde{v}_\kappa = \tilde{v}$, the positions factor out into the *basis structure factor*

$$S^{\mathrm{bas}}(\boldsymbol{G}) = \sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}, \tag{3.18}$$

and $f_{\boldsymbol{G}} = S^{\mathrm{bas}}(\boldsymbol{G})\,\tilde{v}(\boldsymbol{G})/\Omega$. A zero of S^{bas} is an *extinction*: the Fourier component $f_{\boldsymbol{G}}$ vanishes at that reciprocal-lattice vector, whatever the atomic term, because the contributions of the atoms in the cell cancel.

The nuclei of Sect. 2.3 are the simplest case, with the density $n_\kappa(\boldsymbol{k}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}$ for site κ. With Eq. (3.14) the sum of these densities weighted by Z_κ becomes

$$\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k}) = N\sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}} \sum_\kappa Z_\kappa\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa},$$

nonzero only at reciprocal-lattice vectors, as Sect. 2.3 anticipated. The electron–nucleus potential of Eq. (2.48) then has the components of Eq. (3.17) with $\tilde{v}_\kappa(\boldsymbol{G}) = -Z_\kappa v_{\mathrm{C}}(|\boldsymbol{G}|)$. The same sum $\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k})$ describes dark matter that couples to the clamped nuclei: this dark matter can transfer only a reciprocal-lattice vector, elastically, with no energy deposited. The elastic transfer of a reciprocal-lattice vector is the coherent, Bragg-like part of the response mentioned in Sect. 2.2, and a static lattice has no other nuclear response. Energy reaches the nuclei once they are allowed to move (Chap. 5).

Worked example 3.2: The diamond structure factor and the absent (200) reflection

For $\boldsymbol{G} = (2\pi/a)(h,k,l)$ and the basis of Eq. (3.5), $\boldsymbol{G}\cdot\boldsymbol{\tau}_2 = (\pi/2)(h+k+l)$, and Eq. (3.18)

gives

$$S^{\rm bas}(\boldsymbol{G}_{hkl}) = 1 + \mathrm{e}^{\mathrm{i}\pi(h+k+l)/2}.$$

For h, k, l all odd, $h+k+l$ is odd and $S^{\rm bas} = 1 \pm \mathrm{i}$, with $|S^{\rm bas}|^2 = 2$; for (111), $h+k+l = 3$ and $S^{\rm bas} = 1 - \mathrm{i}$. For h, k, l all even, $h+k+l$ is even, and two cases arise. If $h+k+l$ is a multiple of four, as for (220) and (400), the two atoms add in phase, $S^{\rm bas} = 2$ and $|S^{\rm bas}|^2 = 4$. If $h+k+l = 4n+2$, as for (200) and (222), $S^{\rm bas} = 1 + \mathrm{e}^{\mathrm{i}\pi} = 0$. The plane wave at $\boldsymbol{G}_{200}$ has its crests on the atoms at $\boldsymbol{\tau}_1$ and its troughs on those at $\boldsymbol{\tau}_2$, a quarter cube diagonal away, and the contributions of the two atoms cancel.
The (200) and (222) reflections of silicon are therefore absent in X-ray diffraction from spherical atoms, although $\boldsymbol{G}_{200}$ and $\boldsymbol{G}_{222}$ are reciprocal-lattice vectors of its fcc lattice and $\boldsymbol{G}_{200}$ is the second-shortest. The same cancellation removes these components from any part of the crystal potential that is a sum of identical spherical atomic terms. The (222) component returns through the part of the potential that is not a sum of spherical atomic terms, such as the charge that piles up in the bonds, and is observed as a weak X-ray reflection. The (200) component does not. The diamond structure is mapped onto itself by *glides*, each a reflection in a plane parallel to a cube face followed by a translation by a quarter of a face diagonal within that plane. The glides force the (200) component to vanish for every function with the symmetry of the crystal (Appendix C.17). In zincblende GaAs the two sites hold different nuclei, $Z_\kappa = 31$ (gallium) and 33 (arsenic), and the nuclear factor $\sum_\kappa Z_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}$ equals $31 - 33 = -2$ at (200), against $31 + 33 = 64$ at (220). The (200) component is weak, smaller in square by a factor of $4/4096 \approx 10^{-3}$, but not zero, and distinguishes the zincblende structure from diamond in diffraction.

3.2 Bloch states and band structure

An electron in silicon moves in a potential that repeats itself after every lattice vector: the attraction of the clamped nuclei, the repulsion of the other electrons, and the exchange and correlation effects that come with the electron repulsion (Sect. 2.3). The full problem is a many-electron one. Most of the electronic input to a dark-matter rate, however, can be understood with each electron moving independently in an effective potential $U(\boldsymbol{r})$ with the periodicity of the lattice, $U(\boldsymbol{r} + \boldsymbol{R}) = U(\boldsymbol{r})$. Density-functional theory supplies the best effective potential and tells us what the effective potential leaves out. Here U is any lattice-periodic function, and its periodicity alone fixes the form of the eigenstates and the momentum transfers that connect them. Bands and gaps are understood in two opposite limits, nearly free electrons and tight binding. We then compute the two matrix elements that every electronic rate in this book contains: the matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, to which dark matter couples, and the matrix element of the velocity, to which light couples. We follow Refs. [1, Chaps. 8–10], [5, Chap. 3] and [6, Chap. 2].

3.2.1 Bloch's theorem

The one-electron Hamiltonian is

$$H_{\rm e} = \frac{\hat{\boldsymbol{p}}^2}{2m_e} + U(\boldsymbol{r}), \qquad U(\boldsymbol{r} + \boldsymbol{R}) = U(\boldsymbol{r}), \tag{3.19}$$

with $\hat{\boldsymbol{p}} = -\mathrm{i}\nabla$. We keep the electron spinless and count the two spin states with an explicit factor of 2; spin–orbit coupling, which mixes them, enters in Sect. 3.3. The translation operator of Sect. 3.1, acting now on one electron, is $(T_{\boldsymbol{R}}\psi)(\boldsymbol{r}) = \psi(\boldsymbol{r} + \boldsymbol{R})$. The kinetic energy

is unchanged by any translation, and U by a lattice translation, so every $T_{\boldsymbol{R}}$ commutes with H_{e}. The argument that led to Eq. (3.11) then applies to a single electron: the eigenstates of H_{e} can be chosen so that $\psi(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r})$ for every $\boldsymbol{R}$, with $\boldsymbol{k}$ in the first Brillouin zone and on the Born–von Karman grid of Eq. (3.13). Define $u(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi(\boldsymbol{r})$. It is periodic, since $u(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r}) = u(\boldsymbol{r})$. Each eigenstate is therefore a plane wave modulated by a function with the period of the lattice:

Bloch's theorem

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \int_{\Omega}\mathrm{d}^3r\,|u_{n\boldsymbol{k}}|^2 = 1. \tag{3.20}$$

Equivalently, $\psi_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{n\boldsymbol{k}}(\boldsymbol{r})$. The state is normalized to one over the crystal of volume $\mathcal{V} = N\Omega$, and its periodic part to one over a cell.

The two normalizations agree because $|\psi_{n\boldsymbol{k}}|^2 = |u_{n\boldsymbol{k}}|^2/N$ is periodic and the crystal holds N cells; these are the conventions of Appendix A.5. The label n, the *band index*, appears because there are many solutions at each $\boldsymbol{k}$. Inserting $\psi = N^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u$ into $H_{\mathrm{e}}\psi = \varepsilon\psi$ and using $\hat{\boldsymbol{p}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}(\hat{\boldsymbol{p}}+\boldsymbol{k})u$ leaves an equation for u alone,

$$H_{\boldsymbol{k}}\,u_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}}\,u_{n\boldsymbol{k}}, \qquad H_{\boldsymbol{k}} \equiv \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}H_{\mathrm{e}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = \frac{(\hat{\boldsymbol{p}}+\boldsymbol{k})^2}{2m_e} + U(\boldsymbol{r}), \tag{3.21}$$

to be solved in one cell with periodic boundary conditions. Like a particle in a box, $H_{\boldsymbol{k}}$ has a discrete spectrum at each $\boldsymbol{k}$, which we order as $\varepsilon_{1\boldsymbol{k}} \le \varepsilon_{2\boldsymbol{k}} \le \dots$. Since $\boldsymbol{k}$ enters $H_{\boldsymbol{k}}$ as a parameter, each *band energy* $\varepsilon_{n\boldsymbol{k}}$ varies continuously with $\boldsymbol{k}$ and sweeps out an energy band; the set of all energy bands, plotted along lines through the zone, is the *band structure*. The bands are periodic in reciprocal space: $\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$ is periodic and satisfies $H_{\boldsymbol{k}+\boldsymbol{G}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}H_{\boldsymbol{k}}u_{n\boldsymbol{k}}$, so $\varepsilon_{n,\boldsymbol{k}+\boldsymbol{G}} = \varepsilon_{n\boldsymbol{k}}$. We use the *periodic gauge* of Eq. (A.15), which makes the states themselves periodic in reciprocal space, $\psi_{n,\boldsymbol{k}+\boldsymbol{G}} = \psi_{n\boldsymbol{k}}$. A crystal momentum outside the first zone can then be used directly.

A Bloch state is not a state of definite momentum. Its periodic part is a Fourier series over reciprocal-lattice vectors, and we write it in the convention of plane-wave codes (Appendix A.3),

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}}\sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad \psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\mathcal{V}}}\sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}, \qquad \sum_{\boldsymbol{G}}|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2 = 1. \tag{3.22}$$

The coefficient $c_{n\boldsymbol{k}}(\boldsymbol{G})$ multiplies $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$, so it is the lattice Fourier component of Eq. (A.11) at $-\boldsymbol{G}$, times $\Omega^{1/2}$. The electron in state $n\boldsymbol{k}$ has the momenta $\boldsymbol{k}+\boldsymbol{G}$, with probabilities $|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2$. A smooth valence state is dominated by a few small $\boldsymbol{G}$; the part of the wavefunction near the nuclei, where it oscillates rapidly, puts weight at large $|\boldsymbol{G}|$, and a momentum transfer of many keV probes these large-$|\boldsymbol{G}|$ components.

The same expansion turns Eq. (3.21) into an algebraic equation and provides a second proof of Bloch's theorem. Write the potential as $U(\boldsymbol{r}) = \sum_{\boldsymbol{G}''} U_{\boldsymbol{G}''}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}''\cdot\boldsymbol{r}}$, with the components $U_{\boldsymbol{G}} = \Omega^{-1}\int_{\Omega}\mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}U(\boldsymbol{r})$ of Eq. (A.11). Acting on a plane wave, U produces $U(\boldsymbol{r})\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G}')\cdot\boldsymbol{r}} = \sum_{\boldsymbol{G}''}U_{\boldsymbol{G}''}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G}'-\boldsymbol{G}'')\cdot\boldsymbol{r}}$. Collecting the coefficient of $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}$ in $H_{\mathrm{e}}\psi_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}}\psi_{n\boldsymbol{k}}$ picks the term $\boldsymbol{G}'' = \boldsymbol{G}' - \boldsymbol{G}$, and

$$\Big[\frac{(\boldsymbol{k}+\boldsymbol{G})^2}{2m_e} - \varepsilon_{n\boldsymbol{k}}\Big]c_{n\boldsymbol{k}}(\boldsymbol{G}) + \sum_{\boldsymbol{G}'}U_{\boldsymbol{G}'-\boldsymbol{G}}\,c_{n\boldsymbol{k}}(\boldsymbol{G}') = 0. \tag{3.23}$$

A periodic potential couples a plane wave only to those that differ from it by a reciprocal-lattice vector. Bloch's theorem follows again: a state built from the plane waves at one $\boldsymbol{k}$ and at the

translates $\boldsymbol{k}+\boldsymbol{G}$ stays among these plane waves. The matrix $(\boldsymbol{k}+\boldsymbol{G})^2\delta_{\boldsymbol{G}\boldsymbol{G}'}/2m_e + U_{\boldsymbol{G}'-\boldsymbol{G}}$ is Hermitian, since a real U has $U_{-\boldsymbol{G}} = U^*_{\boldsymbol{G}}$. Its eigenvalues are the $\varepsilon_{n\boldsymbol{k}}$, and its eigenvectors the $c_{n\boldsymbol{k}}(\boldsymbol{G})$. A plane-wave code diagonalizes this matrix, truncated at a cutoff in $|\boldsymbol{k}+\boldsymbol{G}|$.

Bloch states with different $\boldsymbol{k}$ are orthogonal, because they belong to different eigenvalues of the unitary $T_{\boldsymbol{R}}$, and states with the same $\boldsymbol{k}$ and different n are orthogonal as eigenvectors of the Hermitian $H_{\boldsymbol{k}}$. Together the Bloch states form an orthonormal basis, and the change of basis of Eq. (2.40) turns the plane-wave operators of Sect. 2.3 into operators $c^\dagger_{n\boldsymbol{k}s}$ and $c_{n\boldsymbol{k}s}$ that create and destroy an electron of spin s in a Bloch state. For independent electrons the Hamiltonian is $\sum_{n\boldsymbol{k}s}\varepsilon_{n\boldsymbol{k}}\, c^\dagger_{n\boldsymbol{k}s}c_{n\boldsymbol{k}s}$, and in thermal equilibrium the *occupation* of a state is, by Eq. (2.69),

$$f_{n\boldsymbol{k}} = \langle c^\dagger_{n\boldsymbol{k}s}c_{n\boldsymbol{k}s}\rangle = n_{\mathrm{F}}(\varepsilon_{n\boldsymbol{k}} - \mu_{\mathrm{ch}}). \tag{3.24}$$

Each band holds $2N$ electrons, as Sect. 3.1 counted. In silicon the eight valence electrons per cell fill the four lowest bands, and a gap of about an electronvolt separates these bands from the next ones (Sect. 3.3). At any temperature at which a detector runs, $f_{n\boldsymbol{k}} = 1$ in these four bands and 0 above them, to an accuracy set by $\mathrm{e}^{-E_{\mathrm{g}}/2T}$, the factor that the intrinsic carrier density of Eq. (3.48) carries.

3.2.2 Nearly free electrons

Switch the potential off. The eigenstates of Eq. (3.23) are then single plane waves, $c_{n\boldsymbol{k}}(\boldsymbol{G}) = \delta_{\boldsymbol{G},\boldsymbol{G}_n}$ for some vector $\boldsymbol{G}_n$ that labels the band, with energies $\varepsilon^0_{\boldsymbol{k}+\boldsymbol{G}_n}$. Here $\varepsilon^0_{\boldsymbol{k}} = k^2/2m_e$ is the free-electron energy at wavevector $\boldsymbol{k}$, inside the zone or not. Nothing physical has changed: the free parabola has been cut into pieces and each piece shifted by a reciprocal-lattice vector into the first zone, so that at a given $\boldsymbol{k}$ the bands are the parabolas centered on all the $-\boldsymbol{G}$. The result is the *empty-lattice* band structure, drawn dashed in Fig. 3.5(a). Its bands cross wherever two plane waves have equal energy, $\varepsilon^0_{\boldsymbol{k}} = \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}$, that is, on the planes $\boldsymbol{k}\cdot\boldsymbol{G} = |\boldsymbol{G}|^2/2$ that bisect the reciprocal-lattice vectors. These are the Bragg planes whose pieces bound the Brillouin zone (Sect. 3.1).

Now let U be weak. Away from a crossing, the coupling $U_{\boldsymbol{G}'-\boldsymbol{G}}$ mixes plane waves whose energies differ by much more than $|U|$, and second-order perturbation theory shifts each level by $\sum_{\boldsymbol{G}\neq 0}|U_{\boldsymbol{G}}|^2/(\varepsilon^0_{\boldsymbol{k}} - \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}})$, a small correction. Near a crossing, however, two plane waves are degenerate and perturbation theory in this form fails. We keep just those two, $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{G})\cdot\boldsymbol{r}}$. Eq. (3.23) for the coefficients $c(0)$ and $c(-\boldsymbol{G})$ reads

$$\begin{pmatrix} \varepsilon^0_{\boldsymbol{k}} & U_{-\boldsymbol{G}} \\ U_{\boldsymbol{G}} & \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}} \end{pmatrix} \begin{pmatrix} c(0) \\ c(-\boldsymbol{G}) \end{pmatrix} = \varepsilon \begin{pmatrix} c(0) \\ c(-\boldsymbol{G}) \end{pmatrix}, \tag{3.25}$$

where the diagonal U_0, a constant shift, has been absorbed into the energy zero. The eigenvalues are

$$\varepsilon_\pm = \frac{\varepsilon^0_{\boldsymbol{k}} + \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}}{2} \pm \sqrt{\Big(\frac{\varepsilon^0_{\boldsymbol{k}} - \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}}{2}\Big)^2 + |U_{\boldsymbol{G}}|^2}. \tag{3.26}$$

Far from the Bragg plane the square root tends to $|\varepsilon^0_{\boldsymbol{k}} - \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}|/2$, and the two free parabolas are recovered. On the Bragg plane the two free energies are equal, and the levels are pushed apart symmetrically:

Nearly-free-electron gap

On the Bragg plane $\boldsymbol{k}\cdot\boldsymbol{G} = |\boldsymbol{G}|^2/2$, a weak periodic potential splits the two degenerate

plane waves into $\varepsilon_\pm = \varepsilon^0_{\boldsymbol{k}} \pm |U_{\boldsymbol{G}}|$. The gap,

$$\varepsilon_+ - \varepsilon_- = 2|U_{\boldsymbol{G}}|, \tag{3.27}$$

is set by the Fourier component of the potential at the reciprocal-lattice vector that the plane bisects. Eq. (3.27) holds to first order in U, as long as $|U_{\boldsymbol{G}}|$ is small compared with the energy separation to the next plane waves.

At the zone boundary the band structure therefore has a gap, and no state of the crystal has an energy inside it, whatever the direction of $\boldsymbol{k}$ along the plane. The sizes of the $U_{\boldsymbol{G}}$ and the shape of the zone decide whether the gaps on different faces of the zone overlap in energy and leave a gap for the whole crystal. In a nearly-free metal such as aluminium the gaps do not overlap, and a Fermi surface survives [1, Chap. 9]. Note that a component that vanishes by extinction (Worked example 3.2) opens no gap at first order. In silicon, the square faces of the zone bisect $\boldsymbol{G}_{200}$, and $U_{\boldsymbol{G}_{200}} = 0$ exactly, by the glide symmetry of the diamond structure (Worked example 3.2, Appendix C.17).

The states at the gap are standing waves. For a potential with inversion symmetry and real $U_{\boldsymbol{G}} = U_{-\boldsymbol{G}}$, the eigenvectors of Eq. (3.25) on the plane are $c(-\boldsymbol{G}) = \pm c(0)$, so that $\psi \propto \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}/2} \pm \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}/2}$ for $\boldsymbol{k} = \boldsymbol{G}/2$: a cosine and a sine of $\boldsymbol{G}\cdot\boldsymbol{r}/2$. The cosine and the sine have the same kinetic energy, but the cosine piles the electron density up at the points where $\boldsymbol{G}\cdot\boldsymbol{r}$ is a multiple of 2π, the lattice planes, and the sine piles it up halfway between them. If the potential is attractive at the planes, $U_{\boldsymbol{G}} < 0$, the cosine lies lower by $|U_{\boldsymbol{G}}|$ and the sine higher by the same amount. A running wave at the boundary would be Bragg-reflected into its partner at $\boldsymbol{k} - \boldsymbol{G}$; the two standing waves remain, and neither carries a current. The group velocity, the slope of the band (Eq. (3.41) below), vanishes on the Bragg plane in the direction of $\boldsymbol{G}$, and the bands meet the zone boundary with zero slope, as Fig. 3.5(a) shows.

Worked example 3.3: The one-dimensional gap

Take the one-dimensional potential $U(x) = 2U_{G_1}\cos(G_1 x)$, whose only components are $U_{\pm G_1} = U_{G_1}$, with $U_{G_1} = -0.5\,\mathrm{eV}$ and the period $a = 5.431\,\text{Å}$, so that $G_1 = 2\pi/a = 2283\,\mathrm{eV}$. The potential is attractive at the lattice sites, $U(0) = -1\,\mathrm{eV}$. The free-electron energy at the zone boundary, $k = G_1/2$, is

$$\varepsilon^0_{G_1/2} = \frac{(G_1/2)^2}{2m_e} = \frac{(1141.5\,\mathrm{eV})^2}{2\times 511.0\,\mathrm{keV}} = 1.275\,\mathrm{eV},$$

and Eq. (3.26) puts the band edges at $1.275\,\mathrm{eV} \mp 0.5\,\mathrm{eV}$, that is, at 0.775 and 1.775 eV, with the gap $2|U_{G_1}| = 1.000\,\mathrm{eV}$. The lower edge is the cosine, concentrated on the sites, and the upper edge the sine, concentrated between them. The full calculation of Fig. 3.5 confirms that the density $|\psi|^2$ of the lower state vanishes at $x = a/2$ and the density of the upper state vanishes at $x = 0$.

Diagonalizing Eq. (3.23) in 25 plane waves gives the edges 0.752 and 1.749 eV, and the gap 0.998 eV. The lower edge has moved down by 0.023 eV and the upper by 0.026 eV. The shifts of the two edges come from the next plane waves, $\mathrm{e}^{\mathrm{i}(G_1/2+G_1)x}$ and $\mathrm{e}^{\mathrm{i}(-G_1/2-G_1)x}$, which each of the two boundary states can reach with one factor of U_{G_1}. Their energy is $9\times 1.275\,\mathrm{eV} = 11.47\,\mathrm{eV}$, and second-order perturbation theory yields $U^2_{G_1}/(1.275\,\mathrm{eV} - 11.47\,\mathrm{eV}) = -0.025\,\mathrm{eV}$ for each edge, the average of the two computed shifts. The gap $2|U_{G_1}|$ is thus accurate to 0.2 %, although $|U_{G_1}|$ is 40 % of the kinetic energy at the boundary.

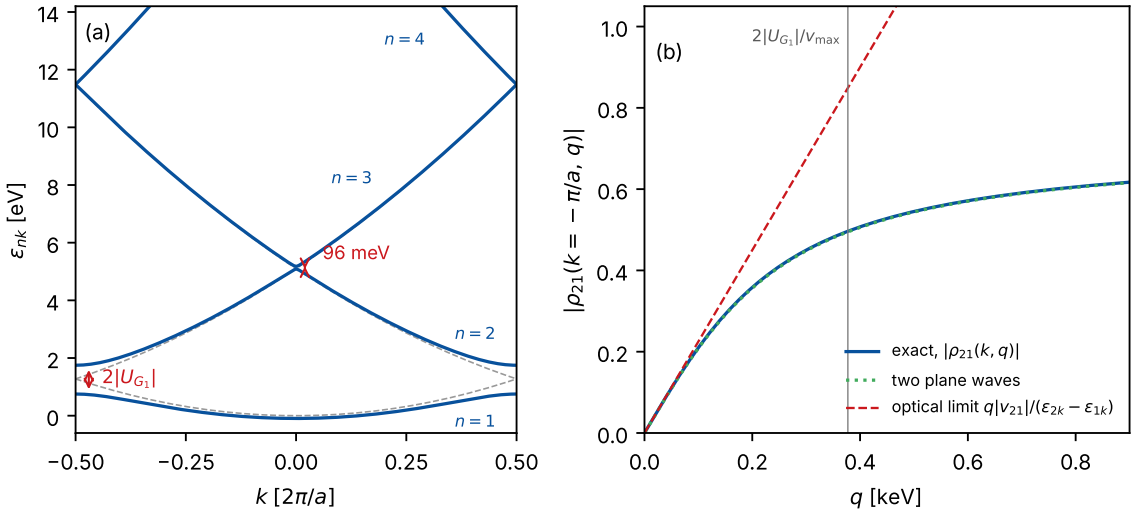


Figure 3.5: Nearly free electrons in one dimension, for the potential $U(x) = 2U_{G_1}\cos(G_1 x)$ with $G_1 = 2\pi/a$, $a = 5.431\,\text{Å}$ (the lattice constant of silicon, used only to set the scale) and $U_{G_1} = -0.5\,\text{eV}$. The bands are computed by diagonalizing Eq. (3.23) in 25 plane waves $e^{i(k+jG_1)x}$, $|j| \leq 12$. (a) The four lowest bands ε_{nk} (solid) and the free-electron parabolas $(k + jG_1)^2/2m_e$ folded into the first zone (dashed). The first gap, at the zone boundary $k = \pm\pi/a$, is 0.998 eV, close to $2|U_{G_1}| = 1\,\text{eV}$; the second gap, at $k = 0$, is 96 meV, second order in U_{G_1}, since the potential has no component at $2G_1$. (b) The density vertex $|\rho_{21}(k, q)|$ between the two lowest bands at $k = -\pi/a$, for momentum transfers $0 < q < 0.9\,\text{keV}$, which keep $k + q$ in the first zone. Solid: the full calculation. Dotted: the two-plane-wave model, indistinguishable from the full calculation at this scale. Dashed: the optical limit $q|v_{21}(k)|/(\varepsilon_{2k} - \varepsilon_{1k})$. The vertical line marks the smallest momentum transfer at which dark matter can deposit 1 eV, $2|U_{G_1}|/v_{\max} = 0.38\,\text{keV}$ for $v_{\max} = 2.650 \times 10^{-3}$.

The second gap opens at $k = 0$, where the plane waves $e^{\pm iG_1 x}$ are degenerate at $\varepsilon^0_{G_1} = 5.099\,\text{eV}$. The two plane waves differ by $2G_1$, and $U_{2G_1} = 0$, so they are not coupled directly but at second order, through the plane wave e^{i0x}, with the effective matrix element $U^2_{G_1}/(\varepsilon^0_{G_1} - \varepsilon^0_0) = 0.049\,\text{eV}$. The two-level splitting is twice the effective matrix element, $0.098\,\text{eV}$. The full calculation finds $0.096\,\text{eV}$. A gap at a Bragg plane is first order in the component of U at the $\boldsymbol{G}$ that the plane bisects, and higher order if that component vanishes. On the square faces of the silicon zone the gap is of higher order, because the extinction of $\boldsymbol{G}_{200}$ removes the first-order term.

Silicon is not in the regime where the nearly-free-electron estimates are quantitative. A three-dimensional gap requires the gaps on all faces of the zone to overlap in energy, and in silicon the band edges lie at different crystal momenta (Sect. 3.3). The potential of silicon is not weak either: the free-electron energy at L is $|\boldsymbol{G}_{111}|^2/8m_e = 3.82\,\text{eV}$, and the strong attraction of the ion cores mixes many plane waves into each state. Pseudopotentials remove this mixing. The nearly-free-electron mechanism carries over to silicon: gaps open where the free-electron energy surfaces cross the Bragg planes, and the Fourier components of the potential set the size of the gaps. The opposite limit, in which electrons hop between atomic orbitals, explains why the gap of silicon lies where it does.

3.2.3 Tight binding and the sp^3 bond

A silicon atom in the gas has two $3s$ and two $3p$ electrons in sharp atomic levels. Bring the atoms together into a crystal of N cells, and each atomic level of each atom in the cell broadens into a band of N states, because the electrons can now hop from atom to atom. When the hopping is weak compared with the spacing of the atomic levels, the natural starting point is the atomic orbitals, not plane waves. A calculation that starts from the atomic orbitals is the *tight-binding* method, or the linear combination of atomic orbitals (LCAO).

Let $\varphi^{\text{at}}_j(\boldsymbol{r})$ be atomic orbitals, with j labeling both the orbital and the site $\boldsymbol{\tau}_\kappa$ of the basis on which it is centered, and $\varphi^{\text{at}}_{j\boldsymbol{R}}(\boldsymbol{r}) = \varphi^{\text{at}}_j(\boldsymbol{r} - \boldsymbol{R})$ its copy in cell $\boldsymbol{R}$. No single orbital has the form Eq. (3.20), but the *Bloch sum*

$$|j\boldsymbol{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\, |\varphi^{\text{at}}_{j\boldsymbol{R}}\rangle, \tag{3.28}$$

does: translating it by a lattice vector $\boldsymbol{R}'$ relabels the sum and multiplies it by $e^{i\boldsymbol{k}\cdot\boldsymbol{R}'}$. Since H_e commutes with lattice translations, it does not connect Bloch sums of different $\boldsymbol{k}$, and we look for the states as $|\psi_{n\boldsymbol{k}}\rangle = \sum_j C_{nj}(\boldsymbol{k})\,|j\boldsymbol{k}\rangle$, with one coefficient per orbital in the cell. The matrix elements between Bloch sums reduce to sums over one lattice vector. Translation invariance gives $\langle\varphi^{\text{at}}_{j\boldsymbol{R}}|H_e|\varphi^{\text{at}}_{j'\boldsymbol{R}'}\rangle = \langle\varphi^{\text{at}}_{j\boldsymbol{0}}|H_e|\varphi^{\text{at}}_{j',\boldsymbol{R}'-\boldsymbol{R}}\rangle$. The double sum over $\boldsymbol{R}$ and $\boldsymbol{R}'$ in $\langle j\boldsymbol{k}|H_e|j'\boldsymbol{k}\rangle$ becomes N times a single sum over $\boldsymbol{R}'' = \boldsymbol{R}' - \boldsymbol{R}$:

$$H^{\text{TB}}_{jj'}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\, \langle\varphi^{\text{at}}_{j\boldsymbol{0}}|H_e|\varphi^{\text{at}}_{j'\boldsymbol{R}}\rangle, \qquad S^{\text{ov}}_{jj'}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\, \langle\varphi^{\text{at}}_{j\boldsymbol{0}}|\varphi^{\text{at}}_{j'\boldsymbol{R}}\rangle. \tag{3.29}$$

Orbitals on neighboring atoms overlap, so the Bloch sums are not orthonormal, and the *overlap matrix* $S^{\text{ov}}(\boldsymbol{k})$ is not the identity. Requiring the energy $\langle\psi|H_e|\psi\rangle/\langle\psi|\psi\rangle$ to be stationary with respect to the coefficients C^*_{nj} turns the Schrödinger equation into the generalized eigenvalue problem

$$\sum_{j'} H^{\text{TB}}_{jj'}(\boldsymbol{k})\, C_{nj'}(\boldsymbol{k}) = \varepsilon_{n\boldsymbol{k}} \sum_{j'} S^{\text{ov}}_{jj'}(\boldsymbol{k})\, C_{nj'}(\boldsymbol{k}). \tag{3.30}$$

There are as many bands as orbitals in the cell. The result is only as good as the basis, which in practice is a few orbitals per atom. In return, a handful of numbers fix the band structure: the on-site energies and the hopping and overlap integrals between neighbors.

The simplest case is one s orbital per site of a Bravais lattice, which yields one band, $n = 1$. The on-site element of the s orbital, $\langle\varphi^{\rm at}_{\mathbf{0}}|H_{\rm e}|\varphi^{\rm at}_{\mathbf{0}}\rangle = E_s$, is the atomic level, shifted slightly by the potential of the neighbors. For nearest neighbors we write $\langle\varphi^{\rm at}_{\mathbf{0}}|H_{\rm e}|\varphi^{\rm at}_{\boldsymbol{R}}\rangle = -t_{\rm hop}$, where the *hopping amplitude* $t_{\rm hop}$ is positive for s orbitals in an attractive potential. We neglect hopping beyond nearest neighbors and the overlap between different sites; neglecting the overlap sets $S^{\rm ov} = 1$. Eq. (3.30) then reduces to

$$\varepsilon_{1\boldsymbol{k}} = E_s - t_{\rm hop} \sum_{\boldsymbol{R}\in{\rm nn}} {\rm e}^{{\rm i}\boldsymbol{k}\cdot\boldsymbol{R}}. \tag{3.31}$$

The face-centered cubic lattice has 12 nearest neighbors, at $(a/2)(\pm1, \pm1, 0)$ and the permutations of these components. The phases of the four neighbors with a given pair of nonzero components sum to $4\cos(k_x a/2)\cos(k_y a/2)$, and

$$\varepsilon_{1\boldsymbol{k}} = E_s - 4t_{\rm hop}\Big[\cos\frac{k_x a}{2}\cos\frac{k_y a}{2} + \cos\frac{k_y a}{2}\cos\frac{k_z a}{2} + \cos\frac{k_z a}{2}\cos\frac{k_x a}{2}\Big]. \tag{3.32}$$

Fig. 3.6 plots this band along the lines between the high-symmetry points of Sect. 3.1. At Γ all phases are one and $\varepsilon = E_s - 12t_{\rm hop}$: the orbitals add in phase and the state is bonding between every pair of neighbors. At $\mathrm{L} = (2\pi/a)(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ every cosine is $\cos(\pi/2) = 0$, and $\varepsilon = E_s$. At $\mathrm{X} = (2\pi/a)(1, 0, 0)$ two of the three products are -1 and one is $+1$, and $\varepsilon = E_s + 4t_{\rm hop}$; the band stays at this maximum along the whole line from X to W. The bandwidth is $16t_{\rm hop}$, proportional to the number of neighbors and to the hopping between them. Near Γ, expanding each cosine to second order leads to $\varepsilon_{1\boldsymbol{k}} = E_s - 12t_{\rm hop} + t_{\rm hop}a^2k^2$, the dispersion of a free particle with the *effective mass* $1/(2t_{\rm hop}a^2)$. A narrow band is a heavy band: an electron that tunnels slowly between atoms responds to a force as if it were massive. Sect. 3.3 defines effective masses precisely and lists those of silicon.

Silicon needs more than one orbital per atom. A silicon atom has its $3s$ level several electronvolts below its $3p$ level, and in the crystal it uses both: it forms four bonds to the four neighbors of Eq. (3.5), at the corners of a tetrahedron. The orbitals that point along these bonds are the four sp^3 *hybrids*

$$\begin{aligned} h_1 &= \tfrac{1}{2}(s + p_x + p_y + p_z), & h_2 &= \tfrac{1}{2}(s + p_x - p_y - p_z), \\ h_3 &= \tfrac{1}{2}(s - p_x + p_y - p_z), & h_4 &= \tfrac{1}{2}(s - p_x - p_y + p_z). \end{aligned} \tag{3.33}$$

These are orthonormal combinations of one s and three p orbitals on the same atom. Each is directed along one of the cube diagonals $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$ and $(-1,-1,1)$ that join an atom at $\boldsymbol{\tau}_1$ to its neighbors. Each hybrid has the on-site energy $(E_s + 3E_p)/4$, a quarter of the way from E_p toward E_s. Forming the hybrids costs promotion energy, since an electron is moved from s to p character, and the energy gained in the bonds compensates the promotion energy [5, Chap. 3].

A large matrix element couples two hybrids that point at each other across a bond, one on each atom; we write it $-V_{\rm cov}$ with $V_{\rm cov} > 0$ (Harrison's covalent energy). Treated alone, the pair is the two-orbital problem of Sect. 2.3: a bonding combination $(h + h')/\sqrt{2}$ at $(E_s + 3E_p)/4 - V_{\rm cov}$ and an antibonding combination $(h - h')/\sqrt{2}$ at $(E_s + 3E_p)/4 + V_{\rm cov}$. A primitive cell holds two atoms and eight hybrids, which make four bonds, hence four bonding and four antibonding orbitals. The eight valence electrons of the cell exactly fill the four bonding orbitals, two per bond, with opposite spins. The smaller couplings, between hybrids on the same atom and between neighboring bonds, then broaden the four bonding levels

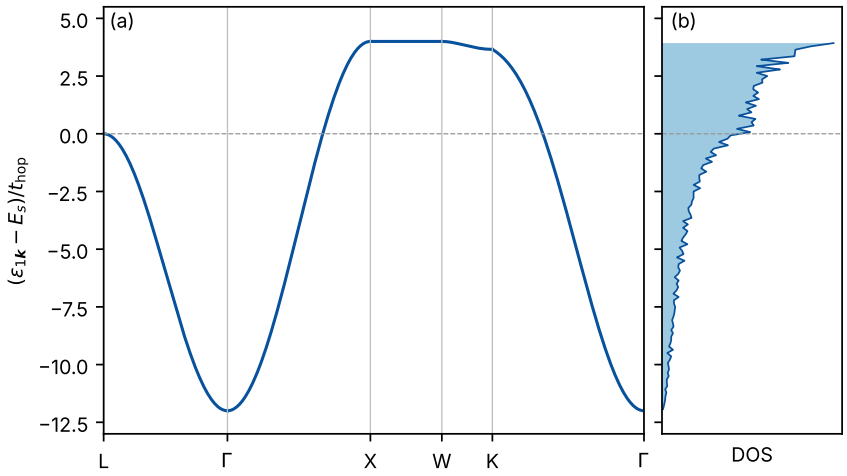


Figure 3.6: Tight-binding s band of the face-centered cubic lattice, Eq. (3.32): one orbital per site with on-site energy E_s, hopping $-t_{\mathrm{hop}}$ between the 12 nearest neighbors, no longer-range hopping, and the overlap between orbitals on different sites neglected. (a) $(\varepsilon_{1\boldsymbol{k}} - E_s)/t_{\mathrm{hop}}$ along the path L–Γ–X–W–K–Γ, with L $= (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, X $= (1, 0, 0)$, W $= (1, \frac{1}{2}, 0)$ and K $= (\frac{3}{4}, \frac{3}{4}, 0)$ in units of $2\pi/a$. The band runs from $-12t_{\mathrm{hop}}$ at Γ to $+4t_{\mathrm{hop}}$, which it reaches at X and keeps along the line X–W; its value at L is 0 and at K $3.66\,t_{\mathrm{hop}}$. (b) Density of states, in arbitrary units, from a uniform grid of 160^3 crystal momenta. Its peak at the top of the band comes from the flat line X–W.

into four valence bands and the four antibonding levels into four conduction bands. The gap of silicon is the remnant of the bonding–antibonding splitting $2V_{\mathrm{cov}}$, reduced by these broadenings, and it survives because the splitting is larger than the bands are wide [5, Chap. 3]. The bond picture accounts for the count of Sect. 3.1, four filled bands, without any reference to Bragg planes. It also explains where the valence charge lies: the bonding orbitals concentrate it between the atoms, in the bonds. The bond charge is not a sum of spherical atomic terms, and it gives the potential of silicon its (222) component. Germanium is bonded the same way, more weakly. In GaAs the two hybrids of a bond have different energies, the bond is partly ionic, and the gap is wider than the covalent splitting alone would give (Sect. 3.3).

The nearly-free and the tight-binding pictures are two limits of the same problem, Eq. (3.21), and both are approximations to it. Real valence states of silicon are well described by neither picture: they have the bonding character of the tight-binding picture, and the plane-wave content near the zone boundary that the nearly-free picture predicts. A first-principles calculation solves Eq. (3.21) in a plane-wave basis, and Wannier functions (Sect. 27.1) recover from the plane-wave solution a set of localized orbitals in which the band structure has the tight-binding form of Eq. (3.30), with orthonormal orbitals and hopping integrals computed rather than guessed.

3.2.4 The density vertex

A dark-matter particle couples to the electrons through $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$, and for independent electrons each transition moves one electron, with the single-particle matrix element $f_{i\to f}(\boldsymbol{q}) = \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$ of Eq. (2.17) (Eq. (2.39)). Between Bloch states this matrix element has a simple structure. Take $\psi_{n\boldsymbol{k}}$ as the initial state and $\psi_{m\boldsymbol{k}'}$ as the final one, with $\boldsymbol{k}$ and $\boldsymbol{k}'$ in the first zone. The momentum transfer lies on the same grid as the crystal momenta, since the dark-matter states are normalized in the same periodic box of volume $\mathcal{V}$ (Sect. 2.1). With

Eq. (3.20),

$$\langle\psi_{m\boldsymbol{k}'}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle = \frac{1}{N}\int_{\mathscr{V}} \mathrm{d}^3r\; \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{r}}\, u^*_{m\boldsymbol{k}'}(\boldsymbol{r})\, u_{n\boldsymbol{k}}(\boldsymbol{r}).$$

Split the crystal into cells, $\boldsymbol{r} = \boldsymbol{R} + \boldsymbol{x}$ with $\boldsymbol{x}$ in the cell at the origin. The product u^*u is the same in every cell, and the phase factorizes into $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{R}}$ times the same phase evaluated at $\boldsymbol{x}$. The sum over cells is the lattice sum of Eq. (3.14), which equals N if $\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}'$ is a reciprocal-lattice vector $\boldsymbol{G}_\mathrm{U}$ and vanishes otherwise:

$$\langle\psi_{m\boldsymbol{k}'}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle = \delta_{\boldsymbol{k}',\,\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}_\mathrm{U}} \int_\Omega \mathrm{d}^3x\; \mathrm{e}^{\mathrm{i}\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{x}}\, u^*_{m\boldsymbol{k}'}(\boldsymbol{x})\, u_{n\boldsymbol{k}}(\boldsymbol{x}).$$

The Kronecker delta is the selection rule Eq. (3.15), now for one electron. The vector $\boldsymbol{G}_\mathrm{U}$ folds $\boldsymbol{k}+\boldsymbol{q}$ back into the first zone. It depends on $\boldsymbol{k}$ as well as on $\boldsymbol{q}$, and it is not in general the vector $\boldsymbol{G}_q$ of Eq. (3.16), which is the reciprocal-lattice vector nearest to $\boldsymbol{q}$. With the periodic gauge, $u_{m,\boldsymbol{k}+\boldsymbol{q}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}} u_{m\boldsymbol{k}'}$ by Eq. (A.15), and the phase is absorbed into the final state:

Density vertex

For $\boldsymbol{k}$ in the first zone and any $\boldsymbol{q}$, $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ connects $\psi_{n\boldsymbol{k}}$ only to the states at crystal momentum $\boldsymbol{k}+\boldsymbol{q}$, folded back as $\boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}_\mathrm{U}$. The amplitude is the *density vertex*

$$\begin{aligned}\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \equiv \langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle &= \int_\Omega \mathrm{d}^3r\; u^*_{m,\boldsymbol{k}+\boldsymbol{q}}(\boldsymbol{r})\, u_{n\boldsymbol{k}}(\boldsymbol{r}) \\ &= \int_\Omega \mathrm{d}^3r\; u^*_{m\boldsymbol{k}'}(\boldsymbol{r})\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}\, u_{n\boldsymbol{k}}(\boldsymbol{r}),\end{aligned} \tag{3.34}$$

with $\rho_{nn}(\boldsymbol{k},0) = 1$. The density vertex is of order one and carries no factor of N. The Umklapp vector $\boldsymbol{G}_\mathrm{U}$ stays in the plane wave.

The normalization $\rho_{nn}(\boldsymbol{k},0) = \int_\Omega |u_{n\boldsymbol{k}}|^2 = 1$ follows from the normalization in Eq. (3.20). The plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}$ in Eq. (3.34) cannot be dropped. Without the plane wave the cell integral would be the overlap $\int_\Omega u^*_{m\boldsymbol{k}'}u_{n\boldsymbol{k}}$, a different number, and the vertex would depend on $\boldsymbol{q}$ only through the reduced part of $\boldsymbol{q}$. The crystal momentum changes by $\boldsymbol{q}$ modulo a reciprocal-lattice vector, but the whole $\boldsymbol{q}$ probes the wavefunction; for this reason Sect. 3.1 never folds $\boldsymbol{q}$.

The plane-wave form of the vertex shows what happens to the momentum. We insert Eq. (3.22) for both states in the cell integral of Eq. (3.34) that carries $\mathrm{e}^{\mathrm{i}\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}$. The cell integral of $\mathrm{e}^{\mathrm{i}(\boldsymbol{G}_\mathrm{U}+\boldsymbol{G}-\boldsymbol{G}')\cdot\boldsymbol{r}}$ is $\Omega\,\delta_{\boldsymbol{G}',\boldsymbol{G}+\boldsymbol{G}_\mathrm{U}}$, and

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}'}(\boldsymbol{G}+\boldsymbol{G}_\mathrm{U})\, c_{n\boldsymbol{k}}(\boldsymbol{G}), \qquad \boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}_\mathrm{U}. \tag{3.35}$$

Each plane-wave component of the initial state, with momentum $\boldsymbol{k}+\boldsymbol{G}$, is shifted by exactly $\boldsymbol{q}$ to $\boldsymbol{k}'+\boldsymbol{G}+\boldsymbol{G}_\mathrm{U}$, and the vertex is the overlap of the shifted state with the final one. Momentum is conserved component by component; the crystal only decides how the final momentum is labeled. In the empty lattice, where each state is one plane wave, Eq. (3.35) reduces to the momentum conservation of free electrons, and the distinction between normal and Umklapp processes is a matter of band labels. In a crystal the coefficients spread each state over many $\boldsymbol{G}$, and a large $|\boldsymbol{q}|$ picks out the overlap between the high-momentum tail of the initial state and the low-momentum part of the final one. In the notation of Ref. [7, Eqs. (3.14)–(3.16)], $c_{n\boldsymbol{k}}(\boldsymbol{G})$ is their $u_n(\boldsymbol{k}+\boldsymbol{G})$ and Eq. (3.35) is their $f_{[n\boldsymbol{k},m\boldsymbol{k}',\boldsymbol{G}_\mathrm{U}]}$.

The vertices out of one state exhaust the probability. The periodic functions $u_{m,\boldsymbol{k}+\boldsymbol{q}}$, $m = 1, 2, \ldots$, are the eigenfunctions of the Hermitian operator $H_{\boldsymbol{k}+\boldsymbol{q}}$ on one cell and form a

complete orthonormal set there. Expanding $u_{n\boldsymbol{k}}$ in them and using its normalization gives

$$\sum_m |\rho_{mn}(\boldsymbol{k},\boldsymbol{q})|^2 = \sum_m \Big| \int_\Omega u^*_{m,\boldsymbol{k}+\boldsymbol{q}}\, u_{n\boldsymbol{k}} \Big|^2 = \int_\Omega |u_{n\boldsymbol{k}}|^2 = 1 \tag{3.36}$$

for every $\boldsymbol{q}$. Eq. (3.36) is the closure relation of Eq. (2.19) for a Bloch electron: a momentum transfer redistributes the electron over the bands at $\boldsymbol{k}+\boldsymbol{q}$ without changing its total weight. At small q the weight stays in band n; at q beyond a few reciprocal-lattice vectors it spreads over many high bands. Worked example 3.4 illustrates the fraction that reaches the conduction bands just above the gap, which receive the transitions that a detector sees.

In second quantization, Eq. (2.38) in the Bloch basis becomes

$$n_e(\boldsymbol{q}) = \sum_s \sum_{\boldsymbol{k}\in\mathrm{BZ}} \sum_{mn} \rho_{mn}(\boldsymbol{k},\boldsymbol{q})\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{q},s}\, c_{n\boldsymbol{k}s}, \tag{3.37}$$

with $c_{m,\boldsymbol{k}+\boldsymbol{q},s} = c_{m\boldsymbol{k}'s}$ in the periodic gauge. In the Bloch basis the particle–hole matrix element of Eq. (2.39), which takes an electron from a valence state $v\boldsymbol{k}$ to a conduction state $c\boldsymbol{k}'$, is $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$. The independent-electron transition sum of Eq. (2.70) then becomes a sum of $|\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2$ over $\boldsymbol{k}$, the valence and conduction bands, and spin. That sum, weighted by energy conservation, is the crystal form factor of Ref. [7, Eq. (3.17)], which Sect. 10.1 derives in the book's normalization. The counting of Appendix A.3 explains why no factor of N appears in Eq. (3.34): one sum over $\boldsymbol{k}$ supplies a factor N, and with the prefactor $2\pi/\mathscr{V}$ of the structure factor this factor N gives the intensive $2\pi/\Omega$.

3.2.5 Velocity matrix elements

Light of energy ω has the momentum $q = \omega$, a few electronvolts for visible and ultraviolet photons, and 10^{-3} of the zone size. Optical absorption probes the density vertex at small q, where the vertex is controlled by a second matrix element, the matrix element of the velocity. The momentum and velocity matrix elements between Bloch states at the same $\boldsymbol{k}$ are

$$\begin{aligned} \boldsymbol{p}_{mn}(\boldsymbol{k}) &= \langle \psi_{m\boldsymbol{k}} | \hat{\boldsymbol{p}} | \psi_{n\boldsymbol{k}} \rangle = \int_\Omega \mathrm{d}^3 r\; u^*_{m\boldsymbol{k}}\, (\hat{\boldsymbol{p}} + \boldsymbol{k})\, u_{n\boldsymbol{k}}, \\ \boldsymbol{v}_{mn}(\boldsymbol{k}) &= \langle \psi_{m\boldsymbol{k}} | \boldsymbol{v} | \psi_{n\boldsymbol{k}} \rangle, \qquad \boldsymbol{v} = \mathrm{i}[H_\mathrm{e}, \boldsymbol{r}]. \end{aligned} \tag{3.38}$$

The velocity operator is the time derivative of the position in the Heisenberg picture. For the local potential of Eq. (3.19), $U(\boldsymbol{r})$ commutes with $\boldsymbol{r}$ and $\boldsymbol{v} = \hat{\boldsymbol{p}}/m_e$, so that $\boldsymbol{v}_{mn} = \boldsymbol{p}_{mn}/m_e$. In plane waves, $\boldsymbol{p}_{mn}(\boldsymbol{k}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}}(\boldsymbol{G})(\boldsymbol{k}+\boldsymbol{G})c_{n\boldsymbol{k}}(\boldsymbol{G})$. The matrix elements of $\boldsymbol{r}$ itself between extended states are not defined as integrals over the crystal, but for $m \neq n$ the commutator gives them a meaning. Taking the matrix element $\langle \psi_{m\boldsymbol{k}} | \cdot | \psi_{n\boldsymbol{k}} \rangle$ of $\boldsymbol{v} = \mathrm{i}[H_\mathrm{e}, \boldsymbol{r}]$ gives

$$\langle \psi_{m\boldsymbol{k}} | \boldsymbol{r} | \psi_{n\boldsymbol{k}} \rangle = \frac{-\mathrm{i}\, \boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}}, \qquad m \neq n, \tag{3.39}$$

which is the form in which interband position matrix elements enter first-principles codes.

The dependence of these matrix elements on $\boldsymbol{k}$ is easiest to handle through the operator $H_{\boldsymbol{k}}$ of Eq. (3.21). Differentiating the definition of $H_{\boldsymbol{k}}$ gives $\nabla_{\boldsymbol{k}}(\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} H_\mathrm{e} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}(-\mathrm{i}\boldsymbol{r} H_\mathrm{e} + \mathrm{i} H_\mathrm{e}\boldsymbol{r})\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, so

$$\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\, \boldsymbol{v}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}, \qquad \boldsymbol{v}_{mn}(\boldsymbol{k}) = \int_\Omega \mathrm{d}^3 r\; u^*_{m\boldsymbol{k}}\, (\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}})\, u_{n\boldsymbol{k}}. \tag{3.40}$$

For the local potential, $\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}} = (\hat{\boldsymbol{p}} + \boldsymbol{k})/m_e$; Eq. (3.40) holds for any Hamiltonian, nonlocal ones included. The group velocity and the $\boldsymbol{k}\cdot\boldsymbol{p}$ relation both follow by differentiating $H_{\boldsymbol{k}} u_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} u_{n\boldsymbol{k}}$ with respect to $\boldsymbol{k}$,

$$(\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}})\, u_{n\boldsymbol{k}} + H_{\boldsymbol{k}} \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} = (\nabla_{\boldsymbol{k}} \varepsilon_{n\boldsymbol{k}})\, u_{n\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}},$$

and projecting onto $u_{m\boldsymbol{k}}$. For $m = n$ the terms with $\nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}}$ cancel, because $\langle u_{n\boldsymbol{k}}| H_{\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} \langle u_{n\boldsymbol{k}}|$, and normalization leaves

Group velocity

$$\boldsymbol{v}_n^{\mathrm{g}}(\boldsymbol{k}) \equiv \nabla_{\boldsymbol{k}} \varepsilon_{n\boldsymbol{k}} = \boldsymbol{v}_{nn}(\boldsymbol{k}). \tag{3.41}$$

The slope of a band is the expectation value of the velocity in the Bloch state.

Eq. (3.41) is the Hellmann–Feynman theorem for the parameter $\boldsymbol{k}$. A wave packet built from states of band n near $\boldsymbol{k}$ moves with $\boldsymbol{v}_n^{\mathrm{g}}$, the *group velocity* [1, Chap. 8 and App. E]. For free electrons the group velocity is $\boldsymbol{k}/m_e$. On a Bragg plane its component along $\boldsymbol{G}$ vanishes, as the standing waves of the nearly-free picture require, and in the tight-binding band Eq. (3.32) it vanishes at Γ, X and L and is at most of order $t_{\mathrm{hop}} a$. For $m \neq n$, and $\varepsilon_{m\boldsymbol{k}} \neq \varepsilon_{n\boldsymbol{k}}$, the projection gives instead

$$\langle u_{m\boldsymbol{k}} | \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} \rangle = \frac{\boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}, \tag{3.42}$$

the first-order change of $u_{n\boldsymbol{k}}$ along band m when $\boldsymbol{k}$ moves, as $\boldsymbol{k} \cdot \boldsymbol{p}$ perturbation theory gives it [6, Chap. 2].

To obtain the optical limit of the vertex, expand the final state, $u_{m,\boldsymbol{k}+\boldsymbol{q}} = u_{m\boldsymbol{k}} + \boldsymbol{q} \cdot \nabla_{\boldsymbol{k}} u_{m\boldsymbol{k}} + O(q^2)$, in the overlap $\int_\Omega u^*_{m,\boldsymbol{k}+\boldsymbol{q}} u_{n\boldsymbol{k}}$ of Eq. (3.34). The zeroth-order term $\langle u_{m\boldsymbol{k}} | u_{n\boldsymbol{k}} \rangle$ vanishes for $m \neq n$, and the first-order term is $\boldsymbol{q} \cdot \langle \nabla_{\boldsymbol{k}} u_{m\boldsymbol{k}} | u_{n\boldsymbol{k}} \rangle = \boldsymbol{q} \cdot \langle u_{n\boldsymbol{k}} | \nabla_{\boldsymbol{k}} u_{m\boldsymbol{k}} \rangle^*$. Using Eq. (3.42) with n and m exchanged, and the hermiticity of the velocity, $\boldsymbol{v}^*_{nm} = \boldsymbol{v}_{mn}$, gives:

Optical limit of the density vertex

$$\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) = \frac{\boldsymbol{q} \cdot \boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}} + O(q^2), \qquad m \neq n. \tag{3.43}$$

An interband vertex vanishes linearly as $q \to 0$, with a slope set by the velocity matrix element and the transition energy. The intraband vertex tends to one, $\rho_{nn}(\boldsymbol{k}, \boldsymbol{q}) = 1 + O(q)$.

The same result follows from expanding the plane wave, $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \simeq 1 + \mathrm{i}\boldsymbol{q} \cdot \boldsymbol{r}$, between states at the same $\boldsymbol{k}$, and using Eq. (3.39); Eq. (A.17) takes this route. The expansion of $u_{m,\boldsymbol{k}+\boldsymbol{q}}$ assumes that the phases of the states are chosen smoothly in $\boldsymbol{k}$ between $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{q}$. Any other choice multiplies both sides of Eq. (3.43) by phases (Eq. (A.18)), and leaves unchanged $|\rho_{mn}|^2$, the quantity that a first-order rate contains. When band m is degenerate at $\boldsymbol{k}$, only the sum of $|\rho_{mn}|^2$ over the degenerate partners is defined (Appendix A.5).

The interband probability $|\rho_{mn}|^2$ therefore grows as q^2 at small q, like the dipole transition $1s \to 2p$ of hydrogen in Sect. 2.2. The intraband vertex does not vanish, but in a full band it cannot act: the state at $\boldsymbol{k} + \boldsymbol{q}$ in the same band is occupied, and the intraband transition is Pauli blocked. Intraband transitions matter in metals, at the Fermi surface (Sect. 4.5). For photons, Eq. (3.43) is all that is needed. The absorption spectrum of a semiconductor is a sum over the squared components of $\boldsymbol{v}_{cv}(\boldsymbol{k})$ along the polarization of the light (Sects. 4.4 and 4.6). The same matrix elements govern absorbed bosonic dark matter (Sect. 10.7), whose momentum is also far below the zone scale (Sect. 3.3).

The expansion of $u_{m,\boldsymbol{k}+\boldsymbol{q}}$ is useful while its first-order term is small, and by Eq. (3.42) the first-order term is small when

$$q \ll \frac{|\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}|}{|\boldsymbol{v}_{mn}(\boldsymbol{k})|}. \tag{3.44}$$

For a photon, with $q = \omega \simeq |\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}|$, the condition reads $|\boldsymbol{v}_{mn}| \ll 1$, the speed of light, and it always holds. For dark matter that deposits the same energy, the momentum transfer is at least $\omega/v_{\max}$ (Sect. 1.3), and the condition becomes $|\boldsymbol{v}_{mn}| \ll v_{\max} = 2.650 \times 10^{-3}$. Valence electrons, however, move with typical velocities $\sim \alpha$, and the Fermi velocity of silicon's valence density is 7.0×10^{-3} (Sect. 2.5). Parametrically, then, the optical limit ends at or below the momentum transfers at which dark matter can excite an electron across the gap. Eq. (3.43) is a check on a calculation of ρ_{mn} at small q, and a scattering rate needs the full vertex of Eq. (3.35).

In practice, the band energies $\varepsilon_{n\boldsymbol{k}}$ and the periodic parts $u_{n\boldsymbol{k}}$ of a real material come from density-functional theory. A code solves Eq. (3.23) for the Kohn–Sham potential in a basis of plane waves up to a cutoff and stores the coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ on a grid of crystal momenta. The vertex is then the sum Eq. (3.35), with the coefficients of the two states matched by the integer coordinates of their $\boldsymbol{G}$. For all $\boldsymbol{G}_{\mathrm{U}}$ at once the vertex is a fast Fourier transform of the product $u^*_{m\boldsymbol{k}'} u_{n\boldsymbol{k}}$, and the momentum matrix element is the corresponding sum over $\boldsymbol{k} + \boldsymbol{G}$. The pseudopotentials of these codes are nonlocal, and then $\boldsymbol{v} \neq \hat{\boldsymbol{p}}/m_e$: by Eq. (3.40), $\boldsymbol{v}_{mn}$ acquires the derivative of the nonlocal potential with respect to $\boldsymbol{k}$, a correction that Sect. 26.2 evaluates and that changes the computed absorption of diamond by 15 % to 20 % [8, Fig. 1].

Worked example 3.4: The vertex between the two lowest nearly-free bands

Return to the one-dimensional crystal of Worked example 3.3, and take the initial state at the left zone boundary, $k = -\pi/a = -G_1/2$, in the lowest band, with the final state in the second band. Near the boundary the two bands are made of the plane waves $\mathrm{e}^{\mathrm{i}kx}$ and $\mathrm{e}^{\mathrm{i}(k+G_1)x}$, and Eq. (3.25), with $\boldsymbol{G} \to -G_1$, mixes them. Its lower eigenvector is $(c(0), c(G_1)) = (\cos\theta_k, \sin\theta_k)$ and its upper one $(-\sin\theta_k, \cos\theta_k)$, with $\tan 2\theta_k = 2|U_{G_1}|/(\varepsilon^0_{k+G_1} - \varepsilon^0_k)$; for $U_{G_1} < 0$ both components of the lower state have the same sign. At the boundary the two free energies are equal and $\theta = \pi/4$.

Small q. For $0 < q < G_1$, $k + q$ stays in the first zone, $\boldsymbol{G}_{\mathrm{U}} = 0$, and Eq. (3.35) reads $\rho_{21}(k, q) = -\sin\theta_{k+q}\cos\theta_k + \cos\theta_{k+q}\sin\theta_k = \sin(\theta_k - \theta_{k+q})$. With $\varepsilon^0_{k+G_1} - \varepsilon^0_k = G_1(k + G_1/2)/m_e$, the angle falls from $\pi/4$ with the slope $\mathrm{d}\theta/\mathrm{d}k = -G_1/(4m_e|U_{G_1}|)$ at the boundary, and

$$|\rho_{21}(k, q)| \simeq \frac{qG_1}{4m_e|U_{G_1}|} = \frac{q}{448\,\mathrm{eV}}.$$

The optical limit agrees. In the two-wave model $|v_{21}| = G_1/2m_e = 2.23 \times 10^{-3}$, the velocity of a free electron at the zone boundary, and Eq. (3.43) gives $q|v_{21}|/2|U_{G_1}|$, the same expression. The full calculation of Fig. 3.5(b) has $|v_{21}| = 2.24 \times 10^{-3}$ and $\varepsilon_{2k} - \varepsilon_{1k} = 0.998\,\mathrm{eV}$. At $q = 10\,\mathrm{eV}$ the vertex is 0.0225 in the full calculation and in the optical limit; at 100 eV the full calculation gives 0.210 and the optical limit 0.225. At $q = 2|U_{G_1}|/v_{\max} = 377\,\mathrm{eV}$, the smallest momentum transfer with which dark matter can deposit the 1 eV gap, the full vertex is 0.496 and the optical limit 0.849, too large by a factor of 1.7, and $|\rho_{21}|^2$ by a factor of 2.9. This q is a reference scale, not an allowed event, since the vertex is evaluated at fixed k: at $q = 377\,\mathrm{eV}$ the final state from this k has moved up the second band, the transition costs 1.6 eV, and depositing 1.6 eV needs $q \geq 0.61\,\mathrm{keV}$. The vertex saturates instead of growing: as $k + q$ moves away from the boundary, the final state becomes a single plane wave, and $|\rho_{21}|$ tends to $\sin(\pi/4) = 0.71$ in the two-wave model. The scale on which the linear law fails, $4m_e|U_{G_1}|/G_1 = 448\,\mathrm{eV}$, is the $|\varepsilon_{2k} - \varepsilon_{1k}|/|v_{21}|$ of Eq. (3.44), and $|v_{21}| = 0.85\,v_{\max}$ puts the momentum transfers of dark matter right at this scale.

Umklapp. Now take $q = G_1 = 2.28\,\mathrm{keV}$. Then $k + q = +\pi/a$, which is folded back to $k' = -\pi/a$ with $\boldsymbol{G}_{\mathrm{U}} = G_1$: the final state has the same reduced crystal momentum

as the initial one, and the transition energy is the same 0.998 eV as at $q \to 0$. At the boundary the states are $(c(0), c(G_1)) = (1,1)/\sqrt{2}$ in band 1 and $(-1,1)/\sqrt{2}$ in band 2. In Eq. (3.35) only the term $\boldsymbol{G} = 0$ survives in the two-wave model, $c^*_{2k'}(G_1)\, c_{1k}(0) = \frac{1}{2}$, so $|\rho_{21}(k, G_1)| = \frac{1}{2}$; the full calculation finds 0.501. The same pair of states, with the same transition energy, has the vertex 0.02 at $q = 10$ eV and 0.50 at $q = G_1$. The difference between the two vertices comes from $\boldsymbol{G}_\mathrm{U}$ in the plane wave: the momentum transfer G_1 moves the component $\mathrm{e}^{\mathrm{i}kx}$ of the initial state onto the component $\mathrm{e}^{\mathrm{i}(k'+G_1)x}$ of the final one, and the two components overlap at order one. In the two-wave model band 2 receives a quarter of the probability and band 1 keeps another quarter. The remaining half, from the component $\mathrm{e}^{\mathrm{i}(k+G_1)x}$ shifted to $\mathrm{e}^{\mathrm{i}(k+2G_1)x}$, goes into bands 3 and 4, whose energies at the boundary are near $(3G_1/2)^2/2m_e = 11.5$ eV. The full calculation finds $|\rho_{m1}|^2 = 0.298$, 0.251, 0.226 and 0.225 for $m = 1$ to 4, and the closure relation Eq. (3.36) holds to 10^{-10}.

3.3 Silicon, germanium and gallium arsenide

The highest occupied electron state of silicon is at the center of the Brillouin zone, and the lowest empty one 85 % of the way to the square face of the zone, at a crystal momentum of 1.94 keV. A photon with the 1.17 eV needed to cross the gap carries a momentum a thousand times smaller than 1.94 keV and cannot connect the two states by itself. A dark-matter particle that scatters with a momentum transfer of a few keV can connect them. The bands just below and just above the gap answer the target's half of the kinematic question of Sect. 1.3: which momentum transfers reach which excitation energies. Table 3.1 collects the band edges, effective masses and spin–orbit splittings of silicon, germanium and GaAs, the book's running materials.

3.3.1 Band edges and valleys

At the temperatures at which detectors run, the four lowest bands of silicon are full and all higher bands are empty (Sect. 3.2). The full bands are the *valence bands* and the empty ones the *conduction bands*. We label them with the subscripts v and c, as in $\varepsilon_{v\boldsymbol{k}}$ and $\varepsilon_{c\boldsymbol{k}}$, and the two letters never appear on their own. The highest valence energy is the *valence-band maximum* (VBM), and the lowest conduction energy the *conduction-band minimum* (CBM). The band gap is the distance between the VBM and the CBM,

$$E_\mathrm{g} = \min_{c,\boldsymbol{k}} \varepsilon_{c\boldsymbol{k}} - \max_{v,\boldsymbol{k}} \varepsilon_{v\boldsymbol{k}}, \tag{3.45}$$

wherever in the zone the two extremes occur. When they occur at different crystal momenta the gap is *indirect*. A transition at fixed crystal momentum is called *vertical*, after its appearance in a band diagram, and the smallest vertical transition energy is the *minimum direct gap*,

$$E_\mathrm{dir} = \min_{v,c,\boldsymbol{k}} (\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}) \geq E_\mathrm{g}. \tag{3.46}$$

In a direct-gap crystal E_dir and E_g are equal.

Fig. 3.7(a) shows the bands of silicon along the lines of Sect. 3.1. The VBM lies at Γ and is threefold degenerate, sixfold with spin. The three VBM states are the bonding states of the sp^3 picture of Sect. 3.2 with pure p character: they transform like x, y and z under the rotations of the crystal, and an s orbital, unchanged by every rotation, cannot enter them. The CBM lies on the line from Γ to X, at the crystal momentum

$$\boldsymbol{k}_\mathrm{c} = 0.85\, \frac{2\pi}{a}\, (1,0,0) \tag{3.47}$$

and at the five equivalent points along $\pm x$, $\pm y$ and $\pm z$; the position was measured by electron–nuclear double resonance on donor electrons, which occupy these states [9]. The six minima are the *valleys* of the conduction band. An electron excited across the gap relaxes into one of the valleys, and near the valley bottom the electron behaves like a free particle with an anisotropic mass, which we derive below. The gap is $E_g = 1.17\,\text{eV}$ at low temperature, 1.1692 eV in the extrapolation to zero temperature [10, Eq. (11)], and 1.12 eV at 300 K [11, Table I]. It shrinks on warming because the lattice expands and the electron–phonon interaction renormalizes the band energies (Sect. 13.7). The vertical transitions of silicon cost much more than E_g. The smallest, $E_{dir} = 3.4\,\text{eV}$, joins the valence and conduction states at Γ [12, Table II], and vertical transitions near L start only slightly higher, at 3.45 eV [12, Table VI]. Silicon is thus indirect by a wide margin, $E_{dir} \approx 3E_g$.

Germanium has the same diamond structure, with $a = 5.658\,\text{Å}$ at room temperature [12, Sect. II A], and the same threefold VBM at Γ. Its CBM, however, lies at L, the center of a hexagonal face of the zone (Fig. 3.8(a)). The eight L points come in pairs that differ by a vector $\boldsymbol{G}_{111}$ and are the same crystal momentum, so germanium has four valleys, each made of two half-ellipsoids on opposite faces of the zone. The gap is 0.744 eV at low temperature [12, Table VI] and 0.66 eV at 300 K [11, Table I]. The conduction state at Γ is only 0.15 eV above the L minimum, $E_{dir} = 0.89\,\text{eV}$ [12, Table II]: germanium is indirect by a narrow margin, and a calculation can easily get the order of the two states wrong.

GaAs has the zincblende structure, with $a = 5.653\,\text{Å}$ at room temperature [16], and both band edges at Γ (Fig. 3.8(b)). Its gap is direct, 1.52 eV at low temperature and 1.42 eV at 300 K [11, Table I], and the conduction band has a single, isotropic valley. Valleys at L and X follow at 1.815 and 1.981 eV, 0.30 and 0.46 eV above the minimum [16]. Gallium and arsenic flank germanium in the periodic table, yet the gap of GaAs is twice the gap of germanium. The larger gap of GaAs comes from the partly ionic bond of Sect. 3.2: the two hybrids of a bond have different energies, and their energy difference adds to the covalent splitting.

The bands in Fig. 3.7 and Fig. 3.8(b) come from density-functional theory, which yields the potential from the electron density. A second route is the empirical pseudopotential, which we use for the germanium bands of Fig. 3.8(a) and, below, as a model to test the velocity sum rules. It solves the central equation Eq. (3.23) with a potential that has nonzero Fourier components $U_{\boldsymbol{G}}$ on just the three shortest shells of reciprocal-lattice vectors, and fits these three numbers per element to measured optical transitions [17]. The success of a fit with three numbers per element shows how much of the band structure the lattice fixes and how little of the potential the band structure needs. The potential is fitted rather than computed, and the fit does not constrain the wavefunctions at large momentum. Because the empirical pseudopotential is local, it is a convenient test bed for the sum rules. The band parameters of Table 3.1 are measured. A first-principles calculation is checked against these numbers: the measured gaps, the valley positions, the masses from cyclotron resonance, and the electronic dielectric constant of silicon, $\varepsilon_\infty = 11.7$, the square of the long-wavelength refractive index of silicon, 3.42 [18] (Sect. 4.4). For the gaps the comparison needs one more step, a *quasiparticle correction* that turns the Kohn–Sham eigenvalues into the energies to add or remove an electron (Sect. 23.4). The PBE functional of Fig. 3.7 is *semilocal*: its exchange–correlation energy at each point depends on the density and its gradient there and nowhere else. It places the CBM of silicon at $k_c = 0.843\,(2\pi/a)$ and the longitudinal valley mass, $0.95\,m_e$, within 4 % of experiment, but it puts the gap at 0.56 eV, half the measured gap.

A gap of an electronvolt is enormous compared with the temperature of a detector, and the number of electrons excited across it by heat alone is a useful check of what "empty" means. Far from the chemical potential μ_{ch} the occupation Eq. (3.24) of a conduction state is the Boltzmann factor $e^{-(\varepsilon_{c\boldsymbol{k}} - \mu_{ch})/T}$. For a parabolic band of mass m, summing the occupation over crystal momenta gives the integral $\int d^3k\,(2\pi)^{-3} e^{-k^2/2mT} = (mT/2\pi)^{3/2}$. With a factor

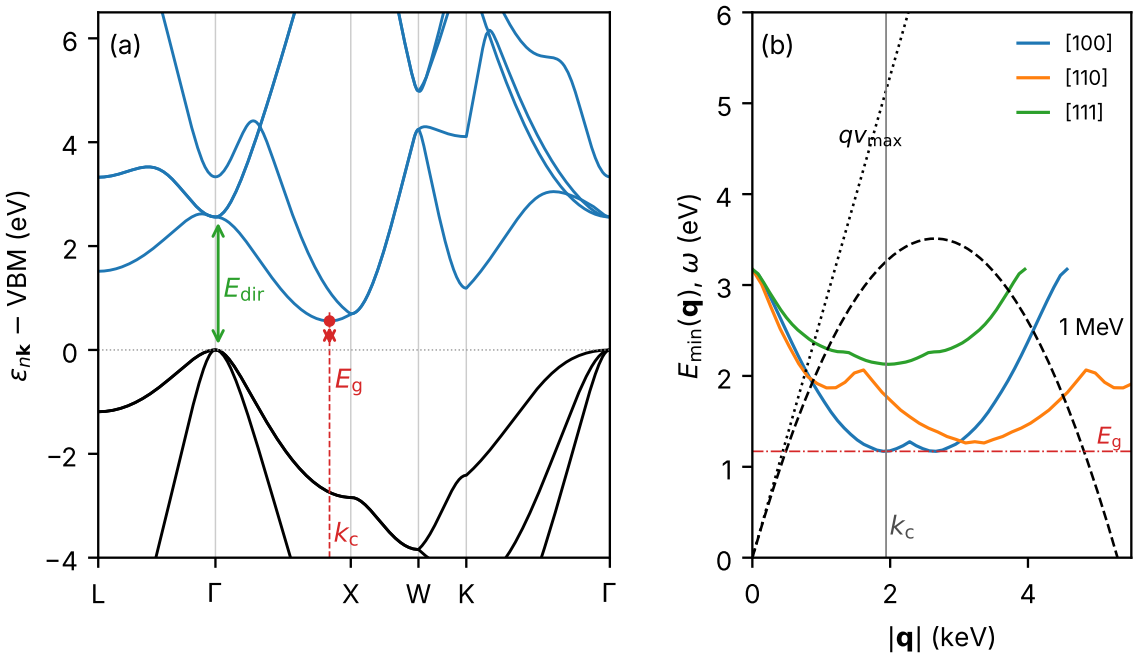


Figure 3.7: Silicon. (a) Band structure along L–Γ–X–W–K–Γ (points of Sect. 3.1), valence bands black and conduction bands blue, energies from the valence-band maximum. The bands are the Kohn–Sham bands of the plane-wave calculation of Sect. 23.3 at the measured lattice constant: generalized-gradient functional of Perdew, Burke and Ernzerhof (PBE), SG15 norm-conserving pseudopotential, cutoff 60 Ry, 12^3 grid for the density, $a = 5.431$ Å, no spin–orbit coupling [13–15]. The computed bands put the conduction-band minimum at $k_{\mathrm{c}} = 0.843\,(2\pi/a)$ (measured 0.85) and the indirect gap E_{g} at 0.557 eV (measured 1.17 eV), the gap problem of density-functional theory. The vertical gap at Γ, marked E_{dir}, is 2.56 eV (measured 3.4 eV) and is the smallest vertical gap in the zone; the vertical gap at L is 2.70 eV (measured 3.45 eV). (b) Minimum pair energy $E_{\mathrm{min}}(\boldsymbol{q})$, Eq. (3.57), for $\boldsymbol{q}$ along [100], [110] and [111], from the same calculation on a grid of 48^3 crystal momenta, with the conduction bands shifted rigidly up by 0.61 eV (a scissor, Sect. 3.4) so that the indirect gap equals the measured 1.17 eV (dash-dotted line); at $\boldsymbol{q} = 0$ the shifted bands give $E_{\mathrm{min}} = 3.17$ eV. Dashed: the largest energy that dark matter of 1 MeV can deposit, $qv_{\mathrm{max}} - q^2/2m_\chi$; dotted: the bound qv_{max} for any mass, with $v_{\mathrm{max}} = 2.650 \times 10^{-3}$. The vertical line marks $|\boldsymbol{k}_{\mathrm{c}}| = 1.94$ keV. Along [100], E_{min} returns to its value at $\boldsymbol{q} = 0$ at $|\boldsymbol{G}_{200}| = 4.57$ keV.

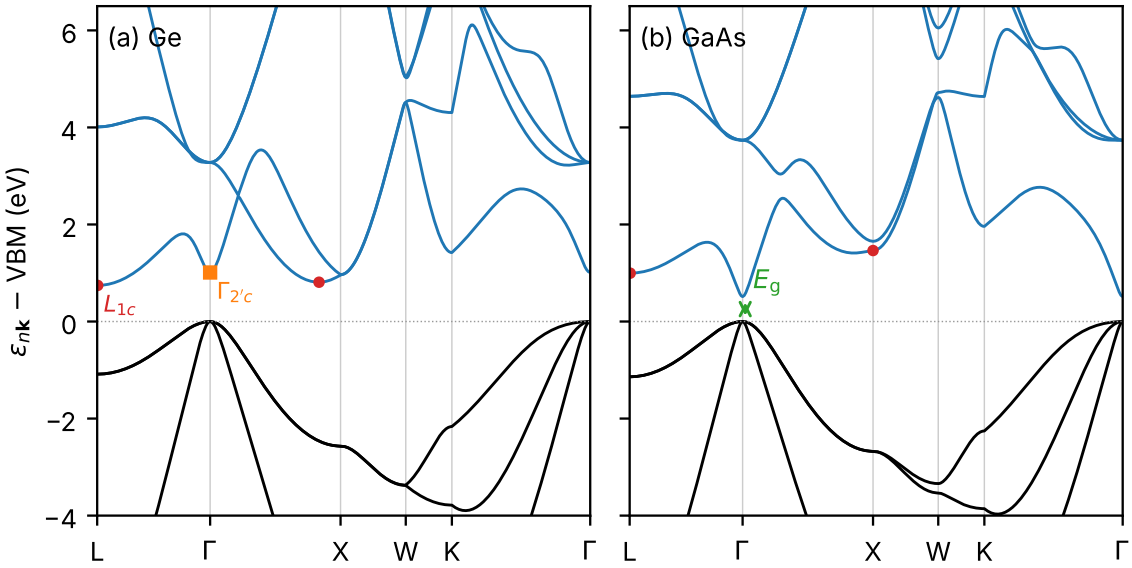


Figure 3.8: Band structures of (a) germanium and (b) GaAs along the path of Fig. 3.7(a). Energies from the valence-band maximum; red dots mark the lowest conduction states away from Γ. (a) Local empirical pseudopotential of Cohen and Bergstresser [17] at $a = 5.66$ Å, the lattice constant of their fit, without spin–orbit coupling, with the conduction bands shifted down by 0.21 eV so that the L minimum lies at the measured 0.744 eV. The crystal is indirect, with the s-like state $\Gamma_{2'c}$ (square) at 1.01 eV (measured 0.89 eV) and the Δ minimum, at $0.83\,(2\pi/a)$ along Γ–X, at 0.81 eV. Plain PBE with spin–orbit coupling at the measured lattice constant closes the germanium gap, with Γ_{7c} 0.05 eV below the valence maximum. (b) PBE plane-wave calculation of Fig. 3.7 (SG15 pseudopotentials with the Ga $3d$ shell in the valence, 60 Ry, 12^3 grid, no spin–orbit coupling) at the room-temperature lattice constant $a = 5.653$ Å [13–15]. The gap is direct, as measured, but PBE underestimates it: 0.51 eV against the measured 1.52 eV at low temperature, with the L and X states at 1.00 and 1.46 eV (measured 1.815 and 1.981 eV).

Table 3.1: Band parameters of silicon, germanium and GaAs. Lattice constant a at room temperature: silicon [3, lattice parameter of silicon], germanium [12, Sect. II A], GaAs [16]. Gaps at low temperature: silicon [10, Eq. (11)], germanium [12, Table VI], GaAs [11, Table I]; at 300 K [11, Table I]. Minimum direct gaps E_{dir} and spin–orbit splittings Δ_{SO} of silicon and germanium [12, Table II]. Conduction valleys: position and number; masses along ($m_\parallel^*$) and across ($m_\perp^*$) the valley axis, silicon [19] as listed in Ref. [12, Table VII], germanium [12, Table VII]. Heavy- and light-hole masses m_{hh}, m_{lh} along [100] and [111] from Eq. (3.56) with the Luttinger parameters $(\gamma_1^{\mathrm{Lu}}, \gamma_2^{\mathrm{Lu}}, \gamma_3^{\mathrm{Lu}}) = (4.285, 0.339, 1.446)$ for silicon and $(13.25, 4.20, 5.56)$ for germanium [12, Table VII], and $(6.98, 2.06, 2.93)$ for GaAs. All GaAs entries not credited otherwise are from Ref. [16]. Masses in units of m_e.

	Si	Ge	GaAs
a (Å)	5.431	5.658	5.653
E_{g}, low T / 300 K (eV)	1.17 / 1.12	0.744 / 0.66	1.52 / 1.42
gap	indirect	indirect	direct
E_{dir} (eV)	3.4	0.89	1.52
valleys	0.85 X, 6	L, 4	Γ, 1
$m_\parallel^*$ / $m_\perp^*$	0.916 / 0.191	1.59 / 0.0815	0.067 (isotropic)
Δ_{SO} (eV)	0.044	0.296	0.341
m_{hh} [100] / [111]	0.277 / 0.718	0.206 / 0.469	0.350 / 0.893
m_{lh} [100] / [111]	0.201 / 0.139	0.046 / 0.041	0.090 / 0.078

of 2 for spin, the sum yields the electron density $2(m_{\mathrm{dc}}T/2\pi)^{3/2}$ times the Boltzmann factor of the CBM. The empty valence states, the holes, give the same expression with the valence mass and the factor $\mathrm{e}^{-(\mu_{\mathrm{ch}}-\varepsilon_{\mathrm{VBM}})/T}$. In a pure crystal the two densities are equal, the chemical potential drops out of their product, and the *intrinsic carrier density* is [1, Chap. 28]

$$n_{\mathrm{i}} = 2\left(\frac{T}{2\pi}\right)^{3/2} (m_{\mathrm{dc}} m_{\mathrm{dv}})^{3/4}\, \mathrm{e}^{-E_{\mathrm{g}}/2T}. \tag{3.48}$$

Here m_{dc} and m_{dv} are the density-of-states masses of the conduction and valence bands: for silicon $m_{\mathrm{dc}} = 6^{2/3}(m_\parallel^* m_\perp^{*2})^{1/3} = 1.06\, m_e$, which counts the six valleys, and $m_{\mathrm{dv}} = (m_{\mathrm{hh}}^{3/2} + m_{\mathrm{lh}}^{3/2})^{2/3} = 0.50\, m_e$, which adds the heavy and light holes with their spherically averaged masses $m_e/(\gamma_1^{\mathrm{Lu}} \mp 2\bar\gamma^{\mathrm{Lu}})$, $\bar\gamma^{\mathrm{Lu}} = (2\gamma_2^{\mathrm{Lu}} + 3\gamma_3^{\mathrm{Lu}})/5$, from the Luttinger parameters of Eq. (3.56) below. At 300 K, with $E_{\mathrm{g}} = 1.12$ eV, Eq. (3.48) yields $n_{\mathrm{i}} \approx 6 \times 10^9\,\mathrm{cm}^{-3}$ for parabolic bands. With the low-temperature gap it falls to $10^{-20}\,\mathrm{cm}^{-3}$ at 77 K and to $10^{-720}\,\mathrm{cm}^{-3}$ at 4 K. A kilogram of silicon has a volume of $429\,\mathrm{cm}^3$, so a cold crystal holds no thermally excited carriers, and the dark counts of a charge detector (Sect. 1.5) come from other sources.

3.3.2 Effective masses from $k \cdot p$ theory

Near the bottom of a valley the band is a quadratic function of the distance $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_0$ from the minimum $\boldsymbol{k}_0$. Its curvature decides how an electron there moves, and how many states lie within an energy ω of the edge. The curvature follows from the states at $\boldsymbol{k}_0$ alone. With Eq. (3.21), the Bloch Hamiltonian a little away from $\boldsymbol{k}_0$ is

$$H_{\boldsymbol{k}_0+\delta\boldsymbol{k}} = \frac{(\hat{\boldsymbol{p}} + \boldsymbol{k}_0 + \delta\boldsymbol{k})^2}{2m_e} + U(\boldsymbol{r}) = H_{\boldsymbol{k}_0} + \delta\boldsymbol{k}\cdot\frac{\hat{\boldsymbol{p}} + \boldsymbol{k}_0}{m_e} + \frac{|\delta\boldsymbol{k}|^2}{2m_e}.$$

The operator multiplying $\delta\boldsymbol{k}$ is $\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}}$ at $\boldsymbol{k}_0$, whose matrix elements between the periodic parts are the velocity matrix elements $\boldsymbol{v}_{mn}(\boldsymbol{k}_0)$ of Eq. (3.40). Treating the two $\delta\boldsymbol{k}$-dependent

terms as a perturbation of $H_{\boldsymbol{k}_0}$, whose eigenstates are the $u_{m\boldsymbol{k}_0}$, and going to second order in $|\delta\boldsymbol{k}|$ for a nondegenerate band n gives

$$\varepsilon_{n,\boldsymbol{k}_0+\delta\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}_0} + \delta\boldsymbol{k}\cdot\boldsymbol{v}_{nn} + \frac{|\delta\boldsymbol{k}|^2}{2m_e} + \sum_{m\neq n}\frac{|\delta\boldsymbol{k}\cdot\boldsymbol{v}_{mn}|^2}{\varepsilon_{n\boldsymbol{k}_0}-\varepsilon_{m\boldsymbol{k}_0}} + O(|\delta\boldsymbol{k}|^3), \tag{3.49}$$

with all velocities at $\boldsymbol{k}_0$. The linear term is the group velocity Eq. (3.41), which vanishes at an extremum. Writing $|\delta\boldsymbol{k}\cdot\boldsymbol{v}_{mn}|^2 = \sum_{\alpha\beta}\delta k_\alpha\,\delta k_\beta\,v^\alpha_{nm}v^\beta_{mn}$, with $v^\alpha_{nm} = (v^\alpha_{mn})^*$, and comparing with $\varepsilon_{n\boldsymbol{k}_0} + \frac{1}{2}\sum_{\alpha\beta}\delta k_\alpha (m^{*-1})_{\alpha\beta}\delta k_\beta$ gives:

Effective mass from $k\cdot p$ theory

At a nondegenerate extremum $\boldsymbol{k}_0$ of band n, the inverse effective-mass tensor is

$$(m^{*-1})_{\alpha\beta} \equiv \frac{\partial^2\varepsilon_{n\boldsymbol{k}}}{\partial k_\alpha\,\partial k_\beta} = \frac{\delta_{\alpha\beta}}{m_e} + 2\sum_{m\neq n}\frac{\mathrm{Re}\big[v^\alpha_{nm}(\boldsymbol{k}_0)\,v^\beta_{mn}(\boldsymbol{k}_0)\big]}{\varepsilon_{n\boldsymbol{k}_0}-\varepsilon_{m\boldsymbol{k}_0}}, \tag{3.50}$$

and near the extremum $\varepsilon_{n,\boldsymbol{k}_0+\delta\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}_0} + \frac{1}{2}\,\delta\boldsymbol{k}\cdot m^{*-1}\delta\boldsymbol{k}$.

The same result follows from differentiating the group velocity Eq. (3.41) once more and using Eq. (3.42); $\boldsymbol{k}\cdot\boldsymbol{p}$ theory is usually presented this way [6, Chap. 2]. For a nonlocal potential the term $\delta_{\alpha\beta}/m_e$ becomes the expectation value of $\partial^2 H_{\boldsymbol{k}}/\partial k_\alpha\partial k_\beta$, while the sum keeps its form with the velocities of Eq. (3.40).

Eq. (3.50) reads like level repulsion. Every other band m pushes band n away from itself as $\boldsymbol{k}$ moves, and pushes harder the closer it lies and the larger the velocity matrix element between the two. Bands below a conduction minimum push the conduction band up on both sides of the minimum and make the electron light; bands above a valence maximum push the valence band down and make the hole light. A small gap therefore means a light mass, provided the velocity matrix element across the gap is sizable.

Silicon provides a check. The sum rule is exact only for a local potential, so we test it in the empirical pseudopotential of Cohen and Bergstresser described above. The model has form factors of -0.21, $+0.04$ and $+0.08$ Ry on the shells $|\boldsymbol{G}|^2 = 3$, 8 and 11 $(2\pi/a)^2$, atoms at $\pm(a/8)(1,1,1)$, $a = 5.43$ Å, the 113 plane waves with $|\boldsymbol{G}|^2 \le 20\,(2\pi/a)^2$, and no spin–orbit coupling [17]. Its conduction minimum, at $k_\mathrm{c} = 0.853\,(2\pi/a)$, has the curvature masses $0.872\,m_e$ along the valley axis and $0.184\,m_e$ across it, from finite differences of the band energies. The sum Eq. (3.50) over all 113 bands of the model reproduces both curvature masses to three digits, as it must in a complete basis. The PBE bands of Fig. 3.7 give $0.95\,m_e$ and $0.193\,m_e$, and the measured values are $0.916\,m_e$ and $0.191\,m_e$ [19]. In the model, the light transverse mass comes mostly from the doubly degenerate valence level 3.73 eV below the minimum. Its two states together contribute 4.53 of the total $m_e/m^*_\perp = 5.43$, with a summed squared velocity matrix element across the valley axis of $(4.1\times10^{-3})^2$.

By the symmetry of the valley, the mass tensor of silicon is diagonal in axes along and across the line Γ–X on which the minimum lies, with two equal transverse components. The same holds for the L valleys of germanium, with axes along ⟨111⟩. With energies measured from the VBM, each valley is an ellipsoid of revolution,

$$\varepsilon_{c,\boldsymbol{k}_\mathrm{c}+\delta\boldsymbol{k}} = E_\mathrm{g} + \frac{(\delta k_\parallel)^2}{2m^*_\parallel} + \frac{(\delta k_\perp)^2}{2m^*_\perp}, \tag{3.51}$$

where $\delta k_\parallel$ is the component of $\delta\boldsymbol{k}$ along the valley axis and $\delta k_\perp$ the one across it. The ellipsoids of silicon are elongated by $(m^*_\parallel/m^*_\perp)^{1/2} = 2.2$ along their axis, and the ellipsoids of

germanium, with a mass ratio of 20, are needles of aspect ratio 4.4. The mass of GaAs at Γ is isotropic.

GaAs shows the level repulsion at work. Its conduction minimum at Γ is an s-like state, 1.52 eV above the three p-like valence states, while the next conduction states at Γ lie about 3 eV higher still. The sum in Eq. (3.50) is then dominated by the valence states. By symmetry, only the valence state that transforms like x, denoted $|X\rangle$, has a nonzero matrix element of v^x with the conduction state. The square of this matrix element defines the *Kane energy*

$$E_P = 2m_e\,|v^x_{cX}|^2.$$

Keeping only the valence terms of Eq. (3.50), of which the $|X\rangle$ term alone survives, gives

$$\frac{m_e}{m^*} \simeq 1 + \frac{E_P}{E_\mathrm{g}}. \tag{3.52}$$

In the empirical pseudopotential of Cohen and Bergstresser, whose velocity is $\boldsymbol{p}/m_e$ exactly, $E_P = 18.32\,\mathrm{eV}$ and the gap is 1.417 eV, so that Eq. (3.52) predicts $m^* = 0.0718\,m_e$; the curvature of the computed band is $0.0721\,m_e$. The measured mass is $0.067\,m_e$ [16]; with the measured gap and the spin–orbit splitting of the valence states, to which we turn next, the same argument reproduces the measured mass. Since E_P varies little, between 16 and 23 eV for the phosphides, arsenides and antimonides of boron, aluminium, gallium and indium [20, Table VIII], Eq. (3.52) makes the mass at a direct gap roughly proportional to the gap. As a result, a calculation that underestimates the gap also underestimates the mass: the PBE bands of Fig. 3.8(b), with a gap of 0.51 eV, have a Γ-valley mass of $0.030\,m_e$.

The Kane energy also fixes the interband velocity that enters the optical limit of Sect. 3.2. With $E_P = 22.9\,\mathrm{eV}$ [20, Table VIII], $|v^x_{cX}| = (E_P/2m_e)^{1/2} = 4.7\times10^{-3}$, or $1.8\,v_\mathrm{max}$. By Eq. (3.44), dark matter excites GaAs across its gap at momentum transfers beyond the reach of the optical limit, as in the one-dimensional model of Worked example 3.4.

3.3.3 Spin–orbit coupling

An electron moving through the electric field of a nucleus sees, in its rest frame, a magnetic field proportional to its orbital angular momentum $\boldsymbol{L}$, and its spin magnetic moment couples to that field. This spin–orbit coupling produces the fine structure of hydrogen, and it is the largest effect of spin on the bands near the gap. So far (Sect. 3.2) we have treated these bands as spinless. With the Thomas precession included, the interaction for an electron of potential energy $U_\mathrm{at}(r)$ in a central field is [21, Chap. 5]

$$H_\mathrm{so} = \frac{1}{2m_e^2}\,\frac{1}{r}\,\frac{\mathrm{d}U_\mathrm{at}}{\mathrm{d}r}\,\boldsymbol{L}\cdot\boldsymbol{S}. \tag{3.53}$$

In a level of orbital angular momentum l, $\boldsymbol{L}\cdot\boldsymbol{S} = \frac{1}{2}[j(j+1) - l(l+1) - \frac{3}{4}]$ equals $l/2$ for $j = l + \frac{1}{2}$ and $-(l+1)/2$ for $j = l - \frac{1}{2}$. With the radial average $\xi_\mathrm{SO} = \langle(2m_e^2)^{-1}r^{-1}\mathrm{d}U_\mathrm{at}/\mathrm{d}r\rangle$, a p level ($l = 1$) splits into a fourfold $j = \frac{3}{2}$ level at $+\xi_\mathrm{SO}/2$ and a twofold $j = \frac{1}{2}$ level at $-\xi_\mathrm{SO}$, a splitting of $\frac{3}{2}\xi_\mathrm{SO}$; the weighted mean, $4\times\frac{1}{2} - 2\times1 = 0$, stays where it was.

In a heavier atom the valence electron spends part of its time inside the core, where it sees the nearly unscreened nuclear charge and $\mathrm{d}U_\mathrm{at}/\mathrm{d}r$ is large. The splitting then grows steeply with the atomic number. The most direct measurement of the splitting uses an ion with a single p electron outside closed shells, since the ground term of the ion is a ^2P doublet split by exactly $\frac{3}{2}\xi_\mathrm{SO}$. The $3s^23p$ ion Si^+ has a splitting of $287.24\,\mathrm{cm}^{-1}$, or 35.6 meV. For the $4s^24p$ ion Ge^+ the splitting is 219 meV, six times more [22].

In the crystal the valence-band maximum inherits the atomic splitting. The three VBM states at Γ transform like x, y and z and are built from p orbitals on the atoms of the cell.

Within these three states, the orbital angular momentum of each atom acts on the p orbitals as it acts on an atomic p level, turning p_x into p_y and so on. The spin–orbit operator of the crystal, $\sum_\kappa \xi_{\mathrm{SO},\kappa} \boldsymbol{L}_\kappa \cdot \boldsymbol{S}$ summed over the atoms, reduces to $\xi_{\mathrm{eff}} \boldsymbol{L} \cdot \boldsymbol{S}$, with $\xi_{\mathrm{eff}} = \sum_\kappa w_\kappa \xi_{\mathrm{SO},\kappa}$ and w_κ the p weight of the VBM states on atom κ. Cross terms between orbitals on different atoms drop out, because $\mathrm{d}U_{\mathrm{at}}/\mathrm{d}r$ is large only near each nucleus, where the orbitals of that atom alone have weight. The six states, three orbital states times two spin states, then split like an atomic p level. A fourfold $j = \frac{3}{2}$ level forms the new VBM, and a twofold $j = \frac{1}{2}$ level lies below it by the *spin–orbit splitting*

$$\Delta_{\mathrm{SO}} = \tfrac{3}{2}\, \xi_{\mathrm{eff}} = \tfrac{3}{2} \sum_\kappa w_\kappa\, \xi_{\mathrm{SO},\kappa}. \tag{3.54}$$

In an elemental crystal with w_κ summing to nearly one, Δ_{SO} is close to the atomic splitting. The measured values agree with the atomic splittings to about 30 %: 0.044 eV in silicon against the ionic 0.036 eV, and 0.296 eV in germanium against 0.219 eV [12, Table II], with a growth from silicon to germanium by a factor of 6.7 in the crystal and 6.2 in the ions. A calculation of the crystal that includes spin–orbit coupling closes the remaining difference between the crystal and the ionic splittings; the quasiparticle calculation of Ref. [12, Table II] finds 0.050 and 0.312 eV. In GaAs the valence states have most of their weight on the arsenic atom, the anion, whose hybrid is the lower one in the ionic bond of Sect. 3.2. The splitting of GaAs, 0.341 eV [16], lies close to the value for the $4s^2 4p$ ion As^{2+}, 0.365 eV, and far from the value for the gallium atom, 0.102 eV [22].

Spin–orbit coupling lowers the gap. A calculation without it places the VBM at the mean of the $j = \frac{3}{2}$ and $j = \frac{1}{2}$ levels, while the true VBM is the $j = \frac{3}{2}$ level, $\Delta_{\mathrm{SO}}/3$ higher. The spin–orbit correction to the gap is 15 meV in silicon, about 1 % of the gap; for this reason silicon is often computed without spin–orbit coupling. In germanium it is 0.10 eV, 13 % of the gap, and cannot be omitted.

The splitting also refines the Kane estimate Eq. (3.52). The conduction state of GaAs is coupled by v^x to the orbital state $|X\rangle$ with either spin, and the spin–orbit levels share that state between them. For $l = 1$ the projector onto $j = \frac{3}{2}$ is $\frac{2}{3}(1 + \boldsymbol{L} \cdot \boldsymbol{S})$, since $\boldsymbol{L} \cdot \boldsymbol{S}$ is $\frac{1}{2}$ there and -1 on $j = \frac{1}{2}$. A real orbital such as p_x has $\langle \boldsymbol{L} \rangle = 0$, so $|X\uparrow\rangle$ has weight $\frac{2}{3}$ in the $j = \frac{3}{2}$ level and $\frac{1}{3}$ in the $j = \frac{1}{2}$ level. The valence terms of Eq. (3.50) then carry the energy denominators E_{g} and $E_{\mathrm{g}} + \Delta_{\mathrm{SO}}$ with these weights,

$$\frac{m_e}{m^*} \simeq 1 + \frac{E_P}{3} \left(\frac{2}{E_{\mathrm{g}}} + \frac{1}{E_{\mathrm{g}} + \Delta_{\mathrm{SO}}} \right), \tag{3.55}$$

the result of Kane [23]. With $E_{\mathrm{g}} = 1.52\,\mathrm{eV}$, $\Delta_{\mathrm{SO}} = 0.341\,\mathrm{eV}$ and $E_P = 22.9\,\mathrm{eV}$, it yields $m^* = 0.066\, m_e$, against the measured $0.067\, m_e$. The same formula with the gap of a semilocal functional, about 0.5 eV, yields $0.025\, m_e$, and with the 1.77 eV of a calculation that overestimates the gap it yields $0.076\, m_e$, close to the $0.077\, m_e$ that this calculation finds (Sect. 23.4).

At the VBM itself Eq. (3.50) does not apply, because the states are degenerate and the perturbation mixes them among themselves. Second-order perturbation theory within the fourfold $j = \frac{3}{2}$ level yields a 4×4 matrix in $\delta\boldsymbol{k}$, and cubic symmetry leaves three independent constants in it, the Luttinger parameters γ_1^{Lu}, γ_2^{Lu} and γ_3^{Lu}. The eigenvalues of the 4×4 matrix, measured downward from the VBM, are [24][6, Chap. 2]

$$\begin{aligned} \varepsilon_{\mathrm{hh,lh}}(\delta\boldsymbol{k}) = -\frac{1}{2m_e} \Big[\gamma_1^{\mathrm{Lu}} |\delta\boldsymbol{k}|^2 \mp 2 \Big\{ (\gamma_2^{\mathrm{Lu}})^2 |\delta\boldsymbol{k}|^4 + 3\big((\gamma_3^{\mathrm{Lu}})^2 - (\gamma_2^{\mathrm{Lu}})^2\big) \\ \times \big((\delta k_x)^2 (\delta k_y)^2 + (\delta k_y)^2 (\delta k_z)^2 + (\delta k_z)^2 (\delta k_x)^2 \big) \Big\}^{1/2} \Big], \end{aligned} \tag{3.56}$$

two doubly degenerate branches that touch at Γ: the *heavy holes* (upper sign) and the *light holes*. Along [100] the cross terms vanish and the masses are $m_e/(\gamma_1^{\mathrm{Lu}} \mp 2\gamma_2^{\mathrm{Lu}})$; along [111] the square root is $\gamma_3^{\mathrm{Lu}}|\delta\boldsymbol{k}|^2$ and the masses are $m_e/(\gamma_1^{\mathrm{Lu}} \mp 2\gamma_3^{\mathrm{Lu}})$. The $j = \frac{1}{2}$ level forms the *split-off* band, Δ_{SO} lower. The constant-energy surfaces of the heavy holes are strongly warped spheres: in silicon the heavy-hole mass is $0.28\,m_e$ along [100] and $0.72\,m_e$ along [111] (Table 3.1). Removing an electron near the VBM leaves behind a hole with this direction-dependent mass, and the hole's share of the momentum of a pair matters in the next subsection.

3.3.4 The minimum pair energy

A momentum transfer $\boldsymbol{q}$ moves an electron from $v\boldsymbol{k}$ to $c, \boldsymbol{k}+\boldsymbol{q}$ (Eq. (3.34)), and the energy it deposits is $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$. The smallest deposited energy at fixed $\boldsymbol{q}$ is the *minimum pair energy*,

Minimum pair energy

$$E_{\min}(\boldsymbol{q}) = \min_{v,c,\boldsymbol{k}} \left(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}\right). \tag{3.57}$$

No electron–hole pair with momentum transfer $\boldsymbol{q}$ costs less. $E_{\min}$ is periodic, $E_{\min}(\boldsymbol{q} + \boldsymbol{G}) = E_{\min}(\boldsymbol{q})$; it equals E_{dir} at $\boldsymbol{q} = 0$, and its smallest value over all $\boldsymbol{q}$ is E_{g}.

The periodicity of $E_{\min}$ follows from the periodicity of the bands, $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}} = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}}$, and the value at $\boldsymbol{q} = 0$ is the definition Eq. (3.46). The minimum over $\boldsymbol{q}$ is reached when $\boldsymbol{k}$ is at the VBM and $\boldsymbol{k}+\boldsymbol{q}$ at the CBM: in silicon at $\boldsymbol{q} = \boldsymbol{k}_{\mathrm{c}}$, at its five images and at all vectors that differ from these by a $\boldsymbol{G}$. $E_{\min}$ depends only on the reduced momentum $\boldsymbol{q}_r$ of Eq. (3.16), although the strength of the transition, $|\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})|^2$, does not (Sect. 3.2).

Near a valley, the shape of $E_{\min}$ follows from the masses. Take $\boldsymbol{q}$ along the axis of a silicon valley, at a distance $\Delta q = q - k_{\mathrm{c}}$ from its minimum, and a hole of mass m_{h} in the same direction. The pair energy $E_{\mathrm{g}} + (k+\Delta q)^2/2m_{\parallel}^* + k^2/2m_{\mathrm{h}}$ is smallest at $k = -m_{\mathrm{h}}\Delta q/(m_{\parallel}^* + m_{\mathrm{h}})$, where

$$E_{\min}(\boldsymbol{q}) \simeq E_{\mathrm{g}} + \frac{(q - k_{\mathrm{c}})^2}{2(m_{\parallel}^* + m_{\mathrm{h}})}. \tag{3.58}$$

The electron and the hole share the momentum that the valley does not absorb, in proportion to their masses, and the pair recoils like a single particle of mass $m_{\parallel}^* + m_{\mathrm{h}}$. In silicon, with $m_{\mathrm{h}} = 0.28\,m_e$ for heavy holes along [100], the pair mass $m_{\parallel}^* + m_{\mathrm{h}}$ is $1.2\,m_e$, and $E_{\min}$ rises by 0.1 eV within 0.35 keV of the minimum. Far from the valley the parabola fails, and Fig. 3.7(b) shows $E_{\min}$ computed from the PBE bands, with the gap set to the measured one, along three directions. Along [100] $E_{\min}$ falls from the vertical gap at $\boldsymbol{q} = 0$ to E_{g} at $|\boldsymbol{q}| = k_{\mathrm{c}}$, stays near E_{g} up to $|\boldsymbol{G}_{200}| - k_{\mathrm{c}} = 2.63$ keV, where an Umklapp process reaches the opposite valley at $-\boldsymbol{k}_{\mathrm{c}}$, and returns to the vertical gap at $|\boldsymbol{G}_{200}| = 4.57$ keV. Along [110] and [111] it stays higher, since those directions pass the valleys at a distance: its lowest values are 1.26 eV near 3.1 keV along [110] and 2.13 eV along [111].

Elastic dark matter of mass m_χ deposits at most $qv_{\max} - q^2/2m_\chi$ at momentum transfer q (Eq. (1.25)). An electron–hole pair at $\boldsymbol{q}$ is therefore kinematically allowed only if

$$E_{\min}(\boldsymbol{q}) \le qv_{\max} - \frac{q^2}{2m_\chi} \tag{3.59}$$

for some $\boldsymbol{q}$. Eq. (3.59) is the condition of Sect. 1.3 with the target's lowest excitation energy at each q in place of a single threshold. The threshold mass $m_{\min} = 2E_{\mathrm{g}}/v_{\max}^2$ of Eq. (1.24) assumes that the target can absorb E_{g} at the momentum transfer $m_\chi v_{\max}$ where the reach peaks. Silicon cannot. At that momentum transfer $E_{\min}$ is well above E_{g}, and the true

threshold of the indirect gap lies above $m_{\min}$. Worked example 3.5 works out the numbers for silicon.

A photon carries a negligible momentum. Its momentum equals its energy, $q = \omega$, which is 6×10^{-4} of k_c for $\omega = E_g$, and on the scale of the zone every photon absorption is vertical. Without a phonon, a photon needs at least $E_{\min}(\boldsymbol{q} \approx 0) = E_{\text{dir}}$ to create a pair. Between E_g and E_{dir} silicon absorbs light only with the help of a phonon, which supplies the crystal momentum $\boldsymbol{k}_c$ at a cost or gain of tens of meV (Sect. 2.4); Sect. 6.5 derives the rate of this *phonon-assisted* absorption. Absorbed bosonic dark matter, with momentum $m_\phi v \approx 10^{-3} m_\phi$, carries too little crystal momentum, like the photon, to cross the indirect gap at first order. The need for a phonon in scattering depends on the dark-matter mass. Above a few hundred keV in mass, scattering dark matter reaches k_c with energy to spare, and it excites the indirect gap of silicon directly, at first order. Just above $m_{\min}$, however, it delivers enough energy only at momentum transfers well below k_c, and there a phonon can again supply the difference between $\boldsymbol{k}_c$ and the momentum transfer. In a window of masses, between the threshold mass of the phonon-assisted process and the threshold mass of the zero-phonon process, silicon detects dark matter only through phonon-assisted scattering.

Worked example 3.5: The momentum to reach the silicon conduction-band minimum

With $2\pi/a = 2283\,\text{eV}$ (Worked example 3.1), the CBM of silicon lies at $k_c = 0.85 \times 2283\,\text{eV} = 1.94\,\text{keV}$, or $0.98\,\text{Å}^{-1}$, along each of the six directions $\langle 100 \rangle$. A transition from the VBM to the CBM needs a momentum transfer of this size, up to a reciprocal-lattice vector.

Dark matter of 1 MeV deposits $\omega = E_g = 1.17\,\text{eV}$ for momentum transfers between the two roots of $q v_{\max} - q^2/2m_\chi = E_g$,

$$q = m_\chi v_{\max} \pm \sqrt{(m_\chi v_{\max})^2 - 2 m_\chi E_g} = 2650\,\text{eV} \pm 2164\,\text{eV},$$

that is, from 0.49 keV to 4.81 keV, as in Sect. 1.3. The momentum k_c lies well inside this window. At $q = k_c$ a particle of 1 MeV can deposit up to

$$k_c v_{\max} - \frac{k_c^2}{2m_\chi} = 1940\,\text{eV} \times 2.650 \times 10^{-3} - \frac{(1940\,\text{eV})^2}{2 \times 1\,\text{MeV}} = 5.14\,\text{eV} - 1.88\,\text{eV} = 3.26\,\text{eV},$$

almost three times the gap. Its largest momentum transfer, $2m_\chi v_{\max} = 5.30\,\text{keV}$, also covers the images $\boldsymbol{k}_c + \boldsymbol{G}$ nearest to the origin. For a particle of 1 MeV, the indirect gap of silicon is open at first order.

Lighter particles have less room. The lightest mass that deposits E_g at exactly $q = k_c$ solves $k_c v_{\max} - k_c^2/2m_\chi = E_g$:

$$m_\chi = \frac{k_c^2}{2(k_c v_{\max} - E_g)} = \frac{(1940\,\text{eV})^2}{2\,(5.14\,\text{eV} - 1.17\,\text{eV})} = 0.474\,\text{MeV}.$$

The kinematic bound of Worked example 1.3 at the low-temperature gap is $m_{\min} = 0.333\,\text{MeV}$, attained at $q = m_{\min} v_{\max} = 0.88\,\text{keV} = 0.39\,(2\pi/a)$. That momentum transfer reaches less than halfway to the valley, and there $E_{\min}$ is about 1.9 eV in Fig. 3.7(b), not 1.17 eV.

A particle of 0.333 MeV cannot create a pair without a phonon, whatever the details of the bands: it deposits 1.17 eV only at $q = 0.88\,\text{keV}$, and that momentum transfer does not connect the VBM to any valley, so that $E_{\min} > E_g$ there. Slightly below 0.474 MeV,

particles can still reach the valley at a momentum transfer a little smaller than k_c, where E_{min} is a little higher, so the threshold mass of the indirect gap lies between 0.333 and 0.474 MeV; Sect. 13.1 locates it.

Worked example 3.6: Threshold masses for the direct gaps of silicon and germanium

Worked example 1.3 evaluated $m_{min} = 2E_{th}/v_{max}^2$ for the indirect gap of silicon. With $v_{max}^2 = 7.023 \times 10^{-6}$, the other gaps of Table 3.1 lead to

$$\text{Si, direct:} \quad m_{min} = \frac{2 \times 3.4\,\text{eV}}{7.023 \times 10^{-6}} = 0.968\,\text{MeV}, \quad q = m_{min} v_{max} = 2.57\,\text{keV},$$

$$\text{Ge, direct:} \quad m_{min} = \frac{2 \times 0.89\,\text{eV}}{7.023 \times 10^{-6}} = 0.253\,\text{MeV}, \quad q = 0.67\,\text{keV},$$

$$\text{Ge, indirect:} \quad m_{min} = \frac{2 \times 0.744\,\text{eV}}{7.023 \times 10^{-6}} = 0.212\,\text{MeV}, \quad q = 0.56\,\text{keV}.$$

The direct gap of silicon would triple the threshold mass, but dark-matter scattering also makes nonvertical transitions: at $q = 2.57\,\text{keV}$, a particle of 0.968 MeV reaches pairs well below 3.4 eV in every direction of Fig. 3.7(b). The direct gap matters for photons and for absorbed dark matter, not for the threshold of scattering. In germanium the two gaps differ by 0.15 eV, and the bound of the indirect gap, 0.212 MeV, is 36 % below the bound for silicon. As in silicon, however, the bound assumes that the target absorbs the gap at $q = 0.56\,\text{keV}$, while the L valleys lie at $|\Gamma\text{L}| = \frac{\sqrt{3}}{2}\,(2\pi/a) = 1.90\,\text{keV}$, almost exactly the k_c of silicon. Depositing 0.744 eV at $q = 1.90\,\text{keV}$ requires $m_\chi \geq (1898\,\text{eV})^2/[2\,(5.03\,\text{eV} - 0.744\,\text{eV})] = 0.420\,\text{MeV}$, the counterpart of the 0.474 MeV of silicon. As in silicon, lighter particles reach the valleys at a smaller momentum transfer, where E_{min} is a little higher, so the threshold mass of the indirect gap lies between 0.212 and 0.420 MeV; Sect. 13.1 locates it by minimizing over the momentum transfer and the crystal momentum of the hole. The direct gap does not lower the threshold mass. The bound of 0.253 MeV needs a pair at Γ that absorbs $q = 0.67\,\text{keV}$ without kinetic energy. At that mass the particle can deposit only the 0.89 eV of the gap itself. The pair, a light electron with a heavy hole and about half an electron mass in total, needs a further 0.8 eV to carry that momentum. The route through Γ therefore opens only at masses far above the threshold of the L valleys, at a few MeV for parabolic bands.

The band edges, masses and splittings of Table 3.1 are all a threshold calculation needs from the band structure, and every one of them is measured. A rate needs more: the full bands and the wavefunctions behind $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$ at every $\boldsymbol{k}$ and up to large $\boldsymbol{q}$, which no table provides. Computing the full bands and wavefunctions requires a potential $U(\boldsymbol{r})$ that is derived rather than fitted, the Kohn–Sham potential of Sect. 3.4.

3.4 Many electrons and density-functional theory

The many-electron problem of Eq. (2.48) can be reduced to one electron in an effective potential, first by the best single Slater determinant and then by the Kohn–Sham construction. The Kohn–Sham eigenvalues are not the energies to add or remove an electron. The standard repair of the gap shifts the conduction bands rigidly, and this shift changes the energies and some of the matrix elements of a rate.

3.4.1 Hartree–Fock theory

The simplest many-electron state with the right antisymmetry is a single Slater determinant $|\Phi_0\rangle$, Eq. (2.29), built from N_e orthonormal orbitals $\psi_\gamma(x)$, with $x = (\boldsymbol{r}, s)$ the position and spin. Its Coulomb energy is a direct (Hartree) term minus an exchange (Fock) term (Sect. 2.3). By the variational principle the energy of any state is at least the exact ground-state energy E_0, so the best determinant is the one of lowest energy. The orbitals are varied under the constraint that they stay orthonormal. The minimum condition is the set of *Hartree–Fock equations* (Sect. 23.1):

$$\Big[-\frac{\nabla^2}{2m_e} + U_1(\boldsymbol{r}) + v_{\mathrm{H}}(\boldsymbol{r})\Big]\psi_\gamma(x) + \int \mathrm{d}x'\, \Sigma_{\mathrm{x}}(x,x')\,\psi_\gamma(x') = E_\gamma\,\psi_\gamma(x). \tag{3.60}$$

Here U_1 is the attraction of the clamped nuclei, $v_{\mathrm{H}}(\boldsymbol{r}) = \int \mathrm{d}^3r'\, e^2 \boldsymbol{n}(\boldsymbol{r}')/4\pi|\boldsymbol{r}-\boldsymbol{r}'|$ is the electrostatic potential of the electron density $\boldsymbol{n}(\boldsymbol{r})$, and the exchange term

$$\Sigma_{\mathrm{x}}(x,x') = -\frac{e^2}{4\pi|\boldsymbol{r}-\boldsymbol{r}'|}\sum_{\gamma'\,\mathrm{occ}} \psi_{\gamma'}(x)\psi^*_{\gamma'}(x'),$$

couples each electron only to occupied orbitals of the same spin. The equations are nonlinear, since the potentials are built from the occupied solutions, and the equations are iterated until the orbitals that go in are the ones that come out. The converged solution is the *self-consistent field*. The ionization form factors of Sect. 2.2 are built from self-consistent Hartree–Fock orbitals of the atoms.

The Hartree potential treats the other electrons as a smooth charge cloud. On its own it would also let each electron repel its own charge. The exchange term removes that self-repulsion exactly. Around every electron it digs a hole in the density of the electrons of the same spin, the exchange hole, which integrates to one missing electron. The exchange term is, however, an integral operator, not a function of $\boldsymbol{r}$. It acts on ψ_γ through the values of ψ_γ everywhere, and it is a nonlocal one-electron operator, with the consequences for the velocity and the f-sum rule that Sect. 4.3 works out.

The eigenvalues E_γ have a precise meaning, given by *Koopmans' theorem* [25]. If the other orbitals are held fixed, $-E_\gamma$ is the energy needed to remove an electron from the occupied orbital γ, and the eigenvalue E_γ of an empty orbital is the change in energy when an electron is added to it (Eq. (23.4)). In a crystal the frozen-orbital hypothesis is exact: a Bloch state is spread over all N cells, and the energy gained by relaxing the other orbitals vanishes as $N \to \infty$ (Sect. 23.1). Hartree–Fock bands are therefore the removal and addition energies of the Hartree–Fock state, and their gap is its fundamental gap.

A single determinant has no correlation, and the electron gas shows how much correlation matters. In the electron gas the Hartree–Fock bands can be computed in closed form, Eq. (23.5). At the valence density of silicon the exchange energy is $-6.22\,\mathrm{eV}$ per electron, comparable to the kinetic energy of $7.5\,\mathrm{eV}$ (Eq. (23.6)). Exchange widens the occupied band from the free $12.5\,\mathrm{eV}$ to $20.8\,\mathrm{eV}$, and it makes the group velocity diverge logarithmically at the Fermi surface. Real metals show neither the wide band nor the divergent velocity, and both features come from the long range of the bare exchange, $v_{\mathrm{C}}(k) \propto 1/k^2$, which in a real solid the other electrons screen. In a semiconductor the same unscreened exchange makes the Hartree–Fock gap far too large. An added electron is not allowed to push the others away, and a removed one does not let them close in, so both the addition and the removal energies come out too high in magnitude.

3.4.2 Density-functional theory

The electron density of silicon is a function of three variables, while its ground-state wavefunction depends on the $3N_e$ coordinates of all the electrons. Hohenberg and Kohn showed that the density is enough [26]. Their first theorem states that the ground-state density $n(\boldsymbol{r})$ determines the external potential U_1 up to a constant. With U_1 known up to a constant, n fixes the Hamiltonian and every property of the system, excited states included. Their second theorem makes the density the variable of a variational principle. In the constrained-search form of Levy [27], define the universal functional

$$\mathcal{F}[n] = \min_{\Psi \to n} \langle \Psi | H_{\text{kin}} + H_{ee} | \Psi \rangle,$$

the lowest kinetic plus Coulomb energy of any state of N_e electrons with density n. It does not depend on the nuclei and is the same for silicon, germanium and a single atom. The ground-state energy in the potential U_1 is then the minimum over densities,

$$E_0 = \min_{n} \Big(\mathcal{F}[n] + \int \mathrm{d}^3 r \, U_1(\boldsymbol{r}) \, n(\boldsymbol{r}) \Big), \tag{3.61}$$

reached at the ground-state density. Sect. 23.2 proves both theorems. Neither theorem says how to find $\mathcal{F}$.

Kohn and Sham made Eq. (3.61) usable by computing the largest and hardest piece of $\mathcal{F}$ exactly [28]. That piece is the kinetic energy. For independent electrons with the density n it is $K_{\text{s}}[n]$, the lowest kinetic energy of a single determinant with that density (written T_{s} in the literature). Subtracting $K_{\text{s}}[n]$ and the Hartree energy $E_{\text{H}}[n]$, the classical electrostatic energy of the density, from $\mathcal{F}$ leaves a remainder,

$$\mathcal{F}[n] = K_{\text{s}}[n] + E_{\text{H}}[n] + E_{\text{xc}}[n],$$

which defines the *exchange–correlation energy* E_{xc}. E_{xc} contains three pieces: the exchange energy, which removes the self-repulsion in E_{H} as in Hartree–Fock; the correlation energy that a determinant misses; and the part of the true kinetic energy that independent electrons do not have. E_{xc} is the one part of $\mathcal{F}$ that must be approximated. Minimizing over the orbitals of the determinant leads to one-electron equations.

Kohn–Sham equations

The ground-state density is $n(\boldsymbol{r}) = \sum_s \sum_{\gamma\,\text{occ}} |\psi_\gamma(\boldsymbol{r}, s)|^2$, summed over the N_e lowest solutions of

$$\begin{gathered} \Big[-\frac{\nabla^2}{2m_e} + V_{\text{KS}}(\boldsymbol{r}) \Big] \psi_\gamma(x) = E_\gamma \, \psi_\gamma(x), \\ V_{\text{KS}}(\boldsymbol{r}) = U_1(\boldsymbol{r}) + v_{\text{H}}(\boldsymbol{r}) + v_{\text{xc}}(\boldsymbol{r}), \qquad v_{\text{xc}}(\boldsymbol{r}) = \frac{\delta E_{\text{xc}}}{\delta n(\boldsymbol{r})}. \end{gathered} \tag{3.62}$$

With the exact E_{xc}, the density and the ground-state energy are exact. In a crystal $\gamma \to n\boldsymbol{k}$ and $E_\gamma \to \varepsilon_{n\boldsymbol{k}}$.

Eq. (3.62) has the form of the Hartree–Fock equations Eq. (3.60), with one difference: the Kohn–Sham potential is a local function of $\boldsymbol{r}$, where the Fock term was an integral operator. The Kohn–Sham potential is the potential $U(\boldsymbol{r})$ of Eq. (3.19). It is lattice-periodic, because the density is, and every result of Sects. 3.2 and 3.3 applies to its Bloch states. Unlike the fitted potential of Sect. 3.3, the Kohn–Sham potential is computed, and the density that enters it is the density of the bonds, not a sum of spherical atoms. Through v_{H} and v_{xc}, the charge

piled up between neighboring atoms gives V_{KS} a (222) component, which the basis structure factor removes from the nuclear part; the (200) component vanishes for any potential with the symmetry of the crystal (Worked example 3.2, Appendix C.17).

The ground-state energy, unlike the eigenvalues, has a physical meaning, and it is given by Eq. (23.8). Minimizing it with respect to the lattice constant, or the positions of the atoms, yields the equilibrium structure, and the second derivatives of the ground-state energy yield the force constants of the phonons (Sect. 5.2). In the local-density approximation, introduced next, the computed lattice constant of silicon is 5.401 Å, 0.5 % below the measured one [4, Sect. 10.2].

Exchange–correlation functionals. The *local-density approximation* (LDA) assumes that each small volume of the solid contributes the exchange–correlation energy of a uniform electron gas of the local density,

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int \mathrm{d}^3 r \, n(\boldsymbol{r}) \, \varepsilon_{\mathrm{xc}}(n(\boldsymbol{r})), \tag{3.63}$$

with $\varepsilon_{\mathrm{xc}}(n)$ the energy per electron of the gas [28]. The exchange part of $\varepsilon_{\mathrm{xc}}$ is the Hartree–Fock result Eq. (23.6), $\varepsilon_{\mathrm{x}} = -3e^2 k_{\mathrm{F}}/16\pi^2$ with the local $k_{\mathrm{F}} = (3\pi^2 n)^{1/3}$, and the exchange potential that follows is $v_{\mathrm{x}}^{\mathrm{LDA}} = \frac{4}{3}\varepsilon_{\mathrm{x}}$, the Hartree–Fock exchange of an electron at the Fermi surface of the gas: −8.29 eV at the valence density of silicon. The correlation part comes from quantum Monte Carlo energies of the gas [29], fitted as a function of the density [30]. The approximation is exact for a uniform density and good when the density changes little over the local $1/k_{\mathrm{F}}$. Valence densities do not satisfy this condition, yet LDA works far better than the condition suggests, because the exchange–correlation hole it implies obeys the same sum rule as the exact one. The *generalized-gradient approximations* (GGA) add a dependence on $|\nabla n|$; PBE [31] is the one used most in solids. *Hybrid functionals* replace part of the local exchange by the Fock term of Eq. (3.60) built from the Kohn–Sham orbitals, and the Fock term makes the potential nonlocal again; the hybrid functional of Heyd, Scuseria and Ernzerhof (HSE) [32] takes one quarter of the exchange from Fock, and only at short range. Each step from LDA to GGA to hybrids costs more, the hybrid step by one to three orders of magnitude (Sect. 23.3).

In practice, a Kohn–Sham calculation of silicon expands the periodic parts $u_{n\boldsymbol{k}}$ in plane waves, replaces the nuclei and core electrons by pseudopotentials (Sect. 24.1), solves Eq. (3.62) on a grid of crystal momenta, and iterates the density until it no longer changes. The calculation of Worked example 23.1, with the PBE functional and a norm-conserving silicon pseudopotential of the SG15 library [15], takes seconds. At the measured lattice constant it places the conduction-band minimum at $k_{\mathrm{c}} = 0.843\,(2\pi/a)$ against the measured 0.85, with a longitudinal valley mass $0.95\,m_e$ against $0.92\,m_e$ and a gap of 0.557 eV, half the measured value (Fig. 3.7). Minimizing the total energy puts the lattice constant at 5.477 Å, 0.85 % above the measured one, and at this lattice constant the gap is 0.602 eV. The electronic dielectric constant that the calculation gives is 13.04 against the measured 11.7 (Sect. 3.3). The positions and curvatures of the bands are trustworthy; the gap is not, and the screening inherits part of the gap error.

3.4.3 The band-gap problem

A dark-matter threshold is an energy difference between many-electron states. The gap that a detector sees is the energy to take an electron out of the crystal at the top of the valence band and put it back at the bottom of the conduction band. The electron is put back far

enough away that it does not interact with the hole it left. With E_{N_e} the ground-state energy of N_e electrons, it is the *fundamental gap*

$$E_{\mathrm{g}} = (E_{N_e+1} - E_{N_e}) - (E_{N_e} - E_{N_e-1}) = I - A, \tag{3.64}$$

the ionization energy I minus the electron affinity A. More generally, the *quasiparticle energy* $E_{\mathrm{qp}}(n\boldsymbol{k})$ is the energy to add an electron in an empty state $n\boldsymbol{k}$, or minus the energy to remove one from an occupied state; Sect. 4.7 defines it precisely as a pole of the one-electron Green's function. The bands that photoemission and inverse photoemission measure are quasiparticle bands, and the threshold of Sect. 3.3 is the difference of two of them.

The Kohn–Sham eigenvalues $\varepsilon_{n\boldsymbol{k}}$ are not quasiparticle energies. They are the Lagrange multipliers of a construction designed to reproduce the density, and only one of them has an exact meaning. By Janak's theorem [33], $\varepsilon_{n\boldsymbol{k}}$ is the derivative of the total energy with respect to the occupation of the state, and since the exact energy is a straight line between integer electron numbers [34], the highest occupied eigenvalue of the exact functional equals $-I$ (Sect. 23.2). The lowest empty eigenvalue would equal $-A$ only if the potential were the same on both sides of N_e. It is not. Since the exact energy is a straight line between integers, its slope, the chemical potential, jumps at N_e from $-I$ to $-A$, by the fundamental gap. The slope of the Kohn–Sham kinetic energy K_{s} jumps only by the Kohn–Sham gap, because an added electron enters the lowest empty orbital. The Hartree energy is a smooth function of the density, so the rest of the jump in the chemical potential must be in E_{xc}. When the electron number crosses N_e, the exact exchange–correlation potential jumps by a constant Δ_{xc}, the *derivative discontinuity*, and every eigenvalue jumps with it. Even with the exact functional

$$E_{\mathrm{g}} = E_{\mathrm{g}}^{\mathrm{KS}} + \Delta_{\mathrm{xc}}, \tag{3.65}$$

with $E_{\mathrm{g}}^{\mathrm{KS}}$ the gap between the Kohn–Sham eigenvalues [35, 36]. LDA and GGA are smooth functions of the density and have no discontinuity. For silicon, most of the difference between the LDA gap and the measured gap is the Δ_{xc} that LDA lacks, not an error of the LDA potential [37]. The *band-gap problem* follows: the Kohn–Sham gap is too small for a reason that a better semilocal functional cannot remove.

LDA puts the silicon gap at 0.52 eV, PBE at 0.56 to 0.60 eV, against the measured 1.17 eV (Table 23.3 and Worked example 23.1). The gap error is similar in other semiconductors. GaAs comes out at 0.43 eV in LDA against 1.52 eV. Germanium, with a measured gap of 0.744 eV, has no gap in LDA (Table 23.1), and in PBE with spin–orbit coupling its s-like conduction state at Γ falls below the valence-band maximum, so that germanium becomes a semimetal (Fig. 3.8). Fig. 3.9(a) shows LDA bands of silicon. Their minimum gap lies along Γ–X at the measured position, while quasiparticle energies and measured ones lie higher across the zone. The G_0W_0 squares, from a separate calculation, lie 0.66 to 0.70 eV above the solid LDA bands at the conduction states, and the dashed bands, shifted by one constant, pass close to these squares. To within about 0.1 eV, the error of the LDA conduction bands of silicon is a rigid shift.

Quasiparticle gaps are larger than Kohn–Sham gaps and smaller than Hartree–Fock ones because of what each theory lets the other electrons do. In Hartree–Fock, an electron in an occupied state is bound by the exchange with the other electrons of its spin, while an electron in an empty state has no exchange with itself. The difference in exchange between occupied and empty states is of order the exchange energy, several eV, and it opens the gap. But the other electrons are frozen. A real added electron repels the electrons around it and polarizes the crystal, and the polarization lowers its energy; a real hole attracts the surrounding electrons, with the same effect on the energy to create the hole. Both polarization effects narrow the gap. They come from screening, which at long range cuts the exchange

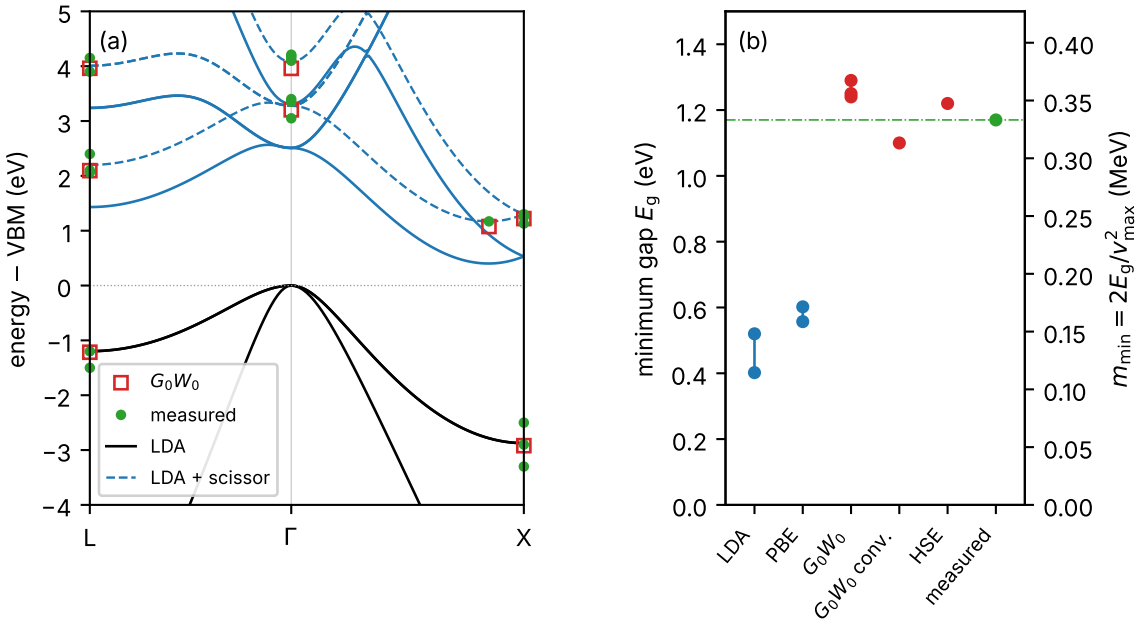


Figure 3.9: The gap of silicon in density-functional theory, in quasiparticle calculations and in experiment. (a) Solid: Kohn–Sham bands in the local-density approximation along L–Γ–X, valence black and conduction blue, from the VBM; Slater exchange and Perdew–Zunger correlation [30], in the plane-wave calculation of Sect. 23.3 (SG15 norm-conserving pseudopotential, 60 Ry, 12^3 grid, $a = 5.431$ Å, no spin–orbit coupling) [13–15]. Their minimum gap is 0.40 eV at $0.848\,(2\pi/a)$; it changes by less than 1 meV at 80 Ry or on a 16^3 grid, and the conduction states at Γ, X and L lie at 2.51, 0.53 and 1.43 eV. The pseudopotential was generated with PBE; a consistent LDA potential would raise the gap to about 0.50 eV, close to the tabulated LDA value of 0.52 eV (Table 23.3). Dashed: the same conduction bands shifted up by $\Delta_{scis} = 0.77$ eV, so that the minimum gap equals the measured 1.17 eV. Squares: one-shot G_0W_0 quasiparticle energies, and circles: measured energies (all values listed there), both as given in Ref. [12, Tables II and VI], at Γ, L, X and the conduction-band minimum. The G_0W_0 calculation is a different one from the LDA bands shown and includes spin–orbit coupling, whose splittings, below 0.05 eV, are not resolved. (b) Minimum gap from LDA [38, Table I] and from the LDA bands of panel (a), from PBE at the measured lattice constant (Fig. 3.7) and at the PBE lattice constant (Worked example 23.1), from G_0W_0 calculations started from LDA [39–41] as collected in Ref. [38, Table I] and converged [42, Sect. IV C], from the hybrid functional HSE [11, Table I], and measured at low temperature [10, Eq. (11)]. Right axis: the threshold mass $m_{min} = 2E_g/v^2_{max}$ of Eq. (1.24), $v_{max} = 2.650 \times 10^{-3}$.

interaction down by $1/\varepsilon_\infty$, a factor of 12 in silicon. The result is screened exchange together with a Coulomb-hole term. Screened exchange is the exchange of Hartree–Fock with the bare Coulomb interaction replaced, at long range, by the interaction screened by ε_∞; the Coulomb-hole term is the energy of the polarization cloud that the particle carries along. A local Kohn–Sham potential is at the other extreme. It acts in the same way on occupied and empty states, and does not contain the discontinuity in the cost of adding an electron. The quasiparticle gap lies between the Hartree–Fock and the Kohn–Sham limits, and in a strongly screening semiconductor it lies closer to the Kohn–Sham gap. The *GW* approximation of Sect. 4.8 makes this picture quantitative: its self-energy is the exchange with the dynamically screened interaction W^{scr}. Started from LDA and converged, it gives the silicon gap at 1.10 eV [42, Sect. IV C]. Hybrid functionals use the same idea crudely, with a fixed fraction of the Fock exchange; HSE gives 1.22 eV (Fig. 3.9(b)).

The gap error reaches every quantity of a dark-matter rate that depends on energy differences across the gap. The threshold mass $m_{\min} = 2E_{\mathrm{g}}/v_{\max}^2$ is proportional to the gap, and the minimum pair energy $E_{\min}(\boldsymbol{q})$ of Eq. (3.57) moves with the conduction bands. Effective masses at a direct gap scale with the gap through Eq. (3.52), as the Kane estimate of Sect. 3.3 shows for GaAs. The silicon valleys are less sensitive, since their mass comes from distant bands, and the PBE longitudinal mass is only 4 % too heavy. The screening is affected as well. Every transition contributes to $\varepsilon_\infty - 1$ more strongly the smaller its energy, and a gap that is too small makes the screening too strong. The in-medium screening of Chap. 4 suppresses a rate by $|\varepsilon|^{-2}$, and with a gap that is too small the screening suppresses the rate too much. Where the gap is measured, it can be imposed; where it is not, the quasiparticle and hybrid-functional values in Fig. 3.9(b) spread from 1.10 to 1.29 eV in silicon, against the measured 1.17 eV, and this spread is a theory uncertainty of the threshold.

3.4.4 The scissor shift

Since the LDA or PBE conduction bands of silicon are nearly right in shape, the cheapest repair of the gap is to move them. The *scissor shift* raises every conduction band by the same amount Δ_{scis} and leaves the valence bands and all the states as they are:

$$H^{\mathrm{s}}_{\boldsymbol{k}} = H_{\boldsymbol{k}} + \Delta_{\mathrm{scis}}\, P_{\mathrm{c}}(\boldsymbol{k}), \qquad P_{\mathrm{c}}(\boldsymbol{k}) = \sum_c |u_{c\boldsymbol{k}}\rangle\langle u_{c\boldsymbol{k}}|, \tag{3.66}$$

with $H_{\boldsymbol{k}}$ the Bloch Hamiltonian Eq. (3.21) of the Kohn–Sham potential and the sum over all conduction bands. The eigenstates of $H^{\mathrm{s}}_{\boldsymbol{k}}$ are the eigenstates of $H_{\boldsymbol{k}}$, with eigenvalues $\varepsilon_{v\boldsymbol{k}}$ and $\varepsilon_{c\boldsymbol{k}} + \Delta_{\mathrm{scis}}$. The shift is chosen to bring the computed gap to a measured or a quasiparticle one: for the PBE calculation of Worked example 23.1, $\Delta_{\mathrm{scis}} = 1.170 - 0.602 = 0.568$ eV at the PBE lattice constant, and 0.613 eV at the measured one. Against full quasiparticle calculations, a rigid shift is good to about 0.1 eV across the bands in silicon, and to 0.2 and 0.4 eV in GaAs and diamond [40][38, Sect. V B].

Under the shift every interband energy grows by Δ_{scis}, and so does $E_{\min}(\boldsymbol{q})$ at every $\boldsymbol{q}$. The density vertex $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ of Eq. (3.34) is an overlap of states, and the states do not change. The velocity matrix elements do change. P_{c} is a nonlocal operator, and the velocity, $\mathrm{i}[H,\boldsymbol{r}]$, picks up a commutator with it. By Eq. (3.40), the velocity matrix elements of H^{s} are the matrix elements of $\nabla_{\boldsymbol{k}} H^{\mathrm{s}}_{\boldsymbol{k}}$. This operator contains the derivative of the projector,

$$\nabla_{\boldsymbol{k}} H^{\mathrm{s}}_{\boldsymbol{k}} = \nabla_{\boldsymbol{k}} H_{\boldsymbol{k}} + \Delta_{\mathrm{scis}} \sum_{c'} \Big(|\nabla_{\boldsymbol{k}} u_{c'\boldsymbol{k}}\rangle\langle u_{c'\boldsymbol{k}}| + |u_{c'\boldsymbol{k}}\rangle\langle \nabla_{\boldsymbol{k}} u_{c'\boldsymbol{k}}| \Big).$$

Between a conduction state c and a valence state v at the same $\boldsymbol{k}$, the first term in the bracket contributes $\sum_{c'} \langle u_c|\nabla_{\boldsymbol{k}} u_{c'}\rangle\langle u_{c'}|u_v\rangle = 0$, since conduction and valence states are orthogonal.

The second contributes $\langle \nabla_{\boldsymbol{k}} u_c | u_v \rangle$, the term $c' = c$ of $\sum_{c'} \langle u_c | u_{c'} \rangle \langle \nabla_{\boldsymbol{k}} u_{c'} | u_v \rangle$. Differentiating $\langle u_c | u_v \rangle = 0$ gives $\langle \nabla_{\boldsymbol{k}} u_c | u_v \rangle = -\langle u_c | \nabla_{\boldsymbol{k}} u_v \rangle$, and Eq. (3.42) gives $\langle u_c | \nabla_{\boldsymbol{k}} u_v \rangle = \boldsymbol{v}_{cv}/(\varepsilon_{v\boldsymbol{k}} - \varepsilon_{c\boldsymbol{k}})$. Altogether,

$$\boldsymbol{v}^{\mathrm{s}}_{cv}(\boldsymbol{k}) = \boldsymbol{v}_{cv}(\boldsymbol{k}) - \Delta_{\mathrm{scis}} \frac{\boldsymbol{v}_{cv}(\boldsymbol{k})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{c\boldsymbol{k}}} = \boldsymbol{v}_{cv}(\boldsymbol{k}) \Big(1 + \frac{\Delta_{\mathrm{scis}}}{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}} \Big).$$

Between two conduction states the two terms of the bracket give

$$\Delta_{\mathrm{scis}} \big(\langle u_c | \nabla_{\boldsymbol{k}} u_{c'} \rangle + \langle \nabla_{\boldsymbol{k}} u_c | u_{c'} \rangle \big) = \Delta_{\mathrm{scis}} \nabla_{\boldsymbol{k}} \langle u_c | u_{c'} \rangle = 0,$$

and between two valence states the projector does not act, so the velocity matrix elements within the conduction bands and within the valence bands do not change.

Scissor rescaling of the velocity

Under the scissor shift Eq. (3.66), energies and states transform as

$$\begin{aligned} \varepsilon_{c\boldsymbol{k}} &\to \varepsilon_{c\boldsymbol{k}} + \Delta_{\mathrm{scis}}, \qquad \rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \to \rho_{mn}(\boldsymbol{k},\boldsymbol{q}), \\ \boldsymbol{v}_{cv}(\boldsymbol{k}) &\to \boldsymbol{v}_{cv}(\boldsymbol{k}) \, \frac{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} + \Delta_{\mathrm{scis}}}{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}}, \end{aligned} \tag{3.67}$$

while the velocity matrix elements within the valence bands and within the conduction bands, the group velocities included, are unchanged.

The interband velocity grows in proportion to the transition energy, and the optical limit Eq. (3.43) shows why it must. The vertex $\rho_{cv} = \boldsymbol{q} \cdot \boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ is fixed because the states are fixed, so a larger denominator needs a larger numerator. The same follows from Eq. (3.39): the position matrix element between the unchanged states stays the same, and the velocity is i times the energy difference times the position matrix element. Sect. 23.4 extends the rescaling to quasiparticle corrections that differ from state to state. At the direct gap of silicon at Γ, 2.55 eV in the PBE calculation at its own lattice constant, the shift of 0.568 eV multiplies the velocity by $3.12/2.55 = 1.22$ and its square by 1.50. An absorption rate built from the shifted energies but the unshifted velocities comes out too small by the factor 1.50.

A scissor keeps the order of the Kohn–Sham bands, and where a functional gets that order wrong, no rigid shift repairs the order (Sect. 23.3). PBE gets the order wrong in germanium, where the conduction state at Γ falls below the L valleys. A scissor also leaves the states unchanged while the energies of all transitions rise. As a result the f-sum rule of Sect. 4.3, a sum of transition energies times $|\rho_{cv}|^2$, exceeds its exact value by $\Delta_{\mathrm{scis}} \sum_c |\rho_{cv}|^2$, the excess that a nonlocal operator in the Hamiltonian produces (Sect. 23.4). And a scissor to a measured gap imports everything the measurement contains, including the renormalization of the gap by the zero-point motion of the lattice (Sect. 13.7), which must then not be added a second time.

Worked example 3.7: The silicon threshold mass from computed gaps

The threshold mass of Eq. (1.24) is $m_{\mathrm{min}} = 2E_{\mathrm{g}}/v^2_{\mathrm{max}}$, with $v^2_{\mathrm{max}} = 7.023 \times 10^{-6}$. For the gaps of Fig. 3.9(b),

LDA, 0.52 eV:	$m_{\mathrm{min}} = \dfrac{2 \times 0.52\,\mathrm{eV}}{7.023 \times 10^{-6}} = 0.148\,\mathrm{MeV}$,
PBE, 0.602 eV:	$m_{\mathrm{min}} = 0.171\,\mathrm{MeV}$,
G_0W_0, 1.10 to 1.29 eV:	$m_{\mathrm{min}} = 0.313$ to $0.367\,\mathrm{MeV}$,
measured, 1.17 eV:	$m_{\mathrm{min}} = 0.333\,\mathrm{MeV}$.

Taken at face value, an LDA band structure lowers the threshold mass by a factor $1.17/0.52 = 2.25$, to less than half its true value. A projection computed with uncorrected bands would claim reach between 0.148 and 0.333 MeV, where a silicon detector has none. The bound of Worked example 3.5, which requires the particle to deposit the gap at the conduction-band minimum $k_c = 1.94\,\mathrm{keV}$, responds much less to the gap error. With the LDA gap,

$$m_\chi = \frac{k_c^2}{2(k_c v_{\max} - E_g)} = \frac{(1940\,\mathrm{eV})^2}{2\,(5.14\,\mathrm{eV} - 0.52\,\mathrm{eV})} = 0.407\,\mathrm{MeV},$$

against 0.474 MeV with the measured gap. Near k_c the momentum needed to reach the valley, not the energy, limits light particles, and the error of the gap is diluted by the kinetic term $k_c v_{\max} = 5.14\,\mathrm{eV}$. Sect. 13.1 locates the true threshold between the bound $m_{\min}$ and the bound at k_c; both bounds, however, move down when the gap is too small. Once the gap is fixed by a scissor to the measured value, the remaining uncertainty, a few meV, comes from the measurement and its temperature. Without a measurement, the spread of 0.1 eV among quasiparticle methods translates through $\delta \ln m_{\min} = \delta \ln E_g$ (Sect. 1.3.5) into $0.1/1.17 = 8.5\,\%$ in $m_{\min}$, more than the 6 % that the uncertainty of the escape speed contributes. Just above the edge, where the halo supply grows by orders of magnitude over a factor of two in mass, a shift of 8.5 % in $m_{\min}$ changes the expected rate by large factors.

The Kohn–Sham bands, corrected for the gap, are the input of every electronic rate. They come from a calculation with pseudopotentials. The states of a pseudopotential calculation are smooth where the true states oscillate near the nuclei, and the density vertex at the momentum transfers that dark matter reaches depends on exactly those oscillations. Chap. 24 restores them.

Summary

With the nuclei clamped at their equilibrium sites, a crystal is a Bravais lattice with a basis (Sect. 3.1). Silicon is a face-centered cubic lattice with two atoms per cell, $a = 5.431\,\text{Å}$ and $\Omega = 40.05\,\text{Å}^3$. Its reciprocal lattice is body-centered cubic, and the shortest reciprocal-lattice vector, $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\mathrm{keV}$, is close to the typical momentum q_0 of electron recoils. Translation symmetry assigns every state a crystal momentum, defined modulo $\boldsymbol{G}$. The lattice sum $\sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}} = N \sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}}$ gives the selection rule for crystal momentum: dark matter that transfers $\boldsymbol{q}$ changes the crystal momentum of the target by $\boldsymbol{q}$ up to a reciprocal-lattice vector, which the lattice absorbs without recoil energy. The momentum transfer itself is never folded into the zone, since the halo integral needs $|\boldsymbol{q}|$.

For independent electrons in a periodic potential the states are Bloch states, $\psi_{n\boldsymbol{k}} = N^{-1/2} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{n\boldsymbol{k}}$, with band energies $\varepsilon_{n\boldsymbol{k}}$ (Sect. 3.2). Nearly free electrons and tight binding are the two limits; tight binding, with sp^3 bonds, accounts for the four filled bands of silicon and for its bond charge. Every electronic rate in the book contains the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$, an overlap of periodic parts that keeps the Umklapp vector in its plane wave and obeys $\sum_m |\rho_{mn}|^2 = 1$. At small q it reduces to a velocity matrix element, but the momentum transfers of light dark matter lie at or beyond the edge of that optical limit, and the full vertex is needed.

Silicon has an indirect gap of 1.17 eV at low temperature, between the valence-band maximum at Γ and six conduction valleys at $0.85\,(2\pi/a)$ along the cube axes, with valley masses $0.916\,m_e$ and $0.191\,m_e$; its smallest vertical gap is 3.4 eV (Sect. 3.3). Germanium is

indirect with its minima at L, GaAs direct. Effective masses follow from $\boldsymbol{k} \cdot \boldsymbol{p}$ theory, and spin–orbit coupling splits the valence-band maximum by an amount inherited from the atoms. The minimum pair energy $E_{\min}(\boldsymbol{q})$ combines bands and kinematics into the condition for a zero-phonon transition. In silicon $E_{\min}(\boldsymbol{q})$ shows that dark matter just above the kinematic threshold mass of 0.333 MeV cannot reach the conduction minimum without a phonon, so phonon-assisted transitions open a window of masses.

The effective potential of a crystal comes from the many-electron problem (Sect. 3.4). Hartree–Fock exchange is unscreened and opens the gap far too wide; the Kohn–Sham equations of density-functional theory reproduce the ground-state density and energy, and with local or semilocal functionals give good lattice constants, band shapes and masses. Their eigenvalues are not quasiparticle energies, however, and the Kohn–Sham gap of silicon is about half the measured gap: this is the band-gap problem. The standard repair, a scissor shift of the conduction bands, leaves the density vertex unchanged and rescales interband velocities by the ratio of corrected to Kohn–Sham transition energies. Chap. 4 adds the response of the other electrons to the perturbation itself, the screening.

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Chapter 4

Linear response

A dark-matter particle that scatters in silicon meets 2.00×10^{23} valence electrons per cubic centimeter (Appendix A.2), and they do not respond independently. They repel one another, so when the particle's potential pushes on some of them the others move to screen the potential, and at long wavelengths all of them can oscillate together, near 16.4 eV (Sect. 4.4). The thermal golden rule of Sect. 2.5 includes this screening and this collective oscillation, since it puts the whole target into one function, the dynamic structure factor $S(\boldsymbol{q}, \omega)$, built from exact many-body states. However, nobody can write down the exact states of the 10^{26} interacting electrons in a kilogram of silicon, and every computed S of a real crystal is an approximation. Linear response tells us which properties of S hold exactly, so that they can test an approximation, and how the approximations are organized. It describes how a target in equilibrium reacts to a weak external field, and the correlation functions that measure that reaction are the ones the golden rule needs. Causality then ties the real part of a response to its imaginary part, and commutators fix its frequency moments whatever the interactions. The long-range Coulomb force can be separated once and for all as screening by a dielectric function, the same function that modifies a dark photon inside the medium (Sect. 1.4). We start from the response to a weak field and end with the hierarchy of approximations from which the computed response of silicon is built, each a limit of the next (Fig. 4.1). The last step of the hierarchy is the Bethe–Salpeter equation of excitons, which needs quasiparticle energies and a screened interaction; the Green's functions and diagrams that supply these two inputs come in the sections just before it.

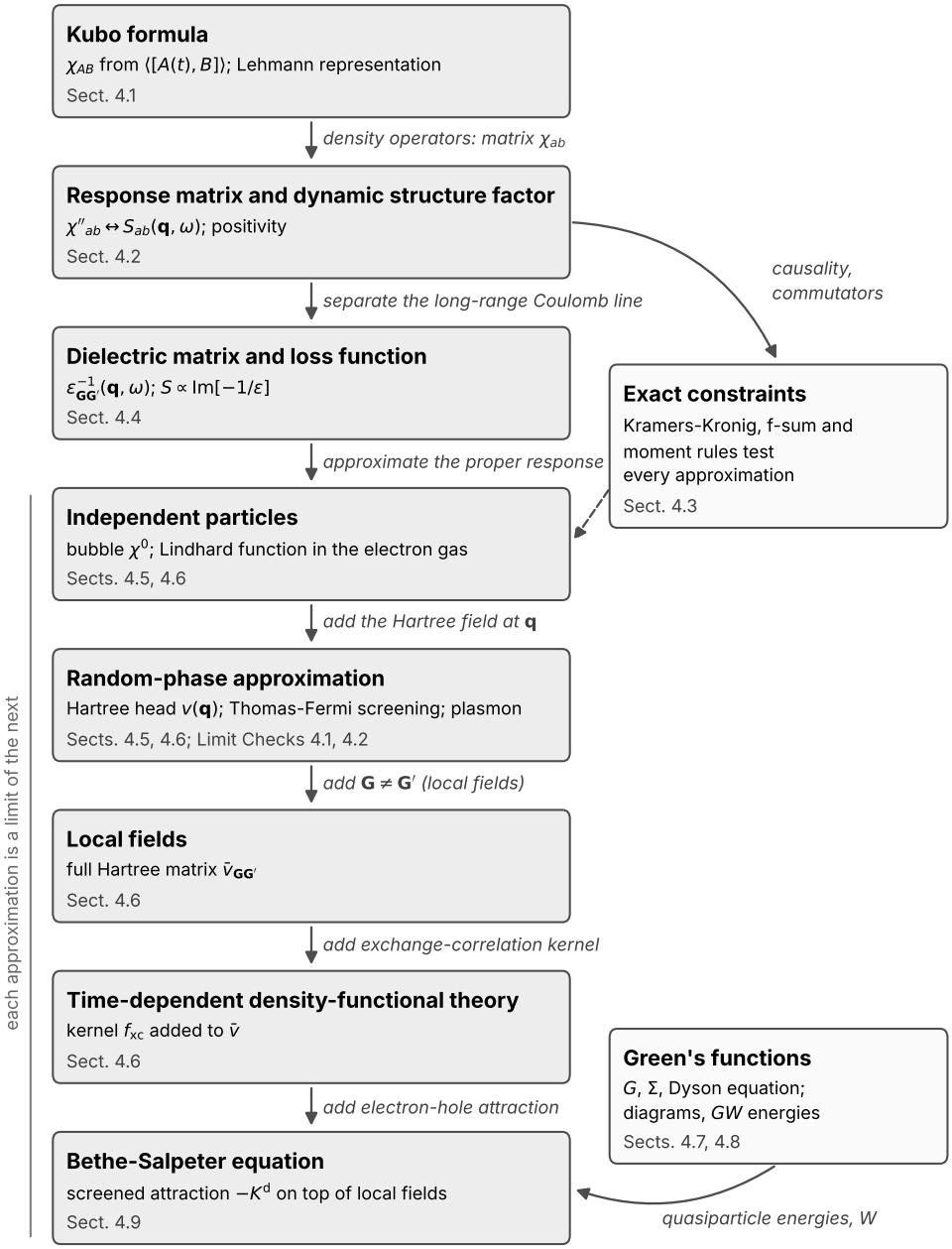


Figure 4.1: The exact backbone and the hierarchy of approximations of the density response. The Kubo formula (Sect. 4.1) leads to the response matrix and the dynamic structure factor (Sect. 4.2); the exact constraints on both, the Kramers–Kronig relations and sum rules (Sect. 4.3), hold in every approximation; separating the long-range Coulomb line gives the dielectric matrix and loss function (Sect. 4.4). The approximations to the proper response add, in turn, the Hartree field at $\boldsymbol{q}$ (random-phase approximation, Sect. 4.5, with Thomas–Fermi screening and the plasmon, Limit Checks 4.1 and 4.2), the local fields $\boldsymbol{G} \neq \boldsymbol{G}'$ and an exchange–correlation kernel (Sect. 4.6), and the screened electron–hole attraction (Sect. 4.9), with quasiparticle energies and the screened interaction W from the Green's functions and diagrams of Sects. 4.7 and 4.8.

4.1 The Kubo formula

Apply to a target crystal a weak potential that varies in space as a plane wave of wavevector $\boldsymbol{q}$ and in time at frequency ω, and the electron density of the target develops a ripple with the same wavevector and frequency. The amplitude of the ripple is proportional to the potential, and in general the ripple lags in phase. The ratio of ripple to potential is a response function. It is a property of the target alone, and it is fixed by correlations that the target has in equilibrium, before any potential is applied. The potential can be an applied electric field, a passing charge or the dark matter itself. A dark-matter particle that goes from one plane-wave state to another acts on the electrons through the density operator $n_e(\boldsymbol{q})$ (Eq. (2.38)), the same operator through which a plane-wave potential acts. The treatment follows Refs. [1, Chap. 3] and [2, Chap. 6].

4.1.1 Response to a weak field

The target has the Hamiltonian H, with all of its interactions, eigenstates $|i\rangle$ and energies E_i. Before the field acts, the target is in the Gibbs state $\hat{\varrho} = \mathrm{e}^{-\beta H}/\mathcal{Z}$ of Eq. (2.62). The field couples to a target operator O_b,

$$H_{\mathrm{tot}}(t) = H + O_b\,\phi_{\mathrm{ext}}(t), \qquad \phi_{\mathrm{ext}}(t) \to 0 \ \text{ as } t \to -\infty, \tag{4.1}$$

where the external potential $\phi_{\mathrm{ext}}(t)$ is a number, not an operator: the source of the field is outside the target and is not changed by the target. For a dimensionless O_b, such as a density, ϕ_{ext} is an energy.

A real potential energy $\phi_{\mathrm{ext}}(t)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \text{c.c.}$, acting on every electron, has exactly this form. Summed over electrons with Eq. (A.7), it is

$$\sum_j \left[\phi_{\mathrm{ext}}(t)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j} + \text{c.c.}\right] = n_e(\boldsymbol{q})\,\phi_{\mathrm{ext}}(t) + n_e(\boldsymbol{q})^\dagger\,\phi_{\mathrm{ext}}(t)^*,$$

two terms of the form Eq. (4.1), with $O_b = n_e(\boldsymbol{q})$ and $O_b = n_e(\boldsymbol{q})^\dagger$. To first order the responses to the two terms add, and we treat one at a time. The dark matter supplies one of the two terms. Between its plane-wave states $\boldsymbol{p}$ and $\boldsymbol{p}-\boldsymbol{q}$ the interaction is $\tilde{V}(\boldsymbol{q})\,n_e(\boldsymbol{q})/\mathcal{V}$ (Eq. (A.8)): the term with $O_b = n_e(\boldsymbol{q})$, which gives the target the momentum $\boldsymbol{q}$.

The thermal state is a mixture, and each of its components obeys the Schrödinger equation with $H_{\mathrm{tot}}(t)$. The density matrix therefore evolves as $\mathrm{i}\,\partial_t\hat{\varrho}(t) = [H_{\mathrm{tot}}(t), \hat{\varrho}(t)]$. As in Sect. 2.4, we take the known part of the evolution out by going to the interaction picture, with one difference: the part we take out is the evolution under the full Hamiltonian H of the target, interactions included, not under a free H_0. With $\hat{\varrho}_{\mathrm{I}}(t) = \mathrm{e}^{\mathrm{i}Ht}\hat{\varrho}(t)\,\mathrm{e}^{-\mathrm{i}Ht}$,

$$\mathrm{i}\,\partial_t\hat{\varrho}_{\mathrm{I}}(t) = \left[O_b(t)\,\phi_{\mathrm{ext}}(t),\ \hat{\varrho}_{\mathrm{I}}(t)\right], \qquad O_b(t) = \mathrm{e}^{\mathrm{i}Ht}O_b\,\mathrm{e}^{-\mathrm{i}Ht},$$

the counterpart of Eq. (2.49) for a density matrix. The Heisenberg operator $O_b(t)$ evolves under the target's own Hamiltonian, without the field. In the remote past $\hat{\varrho}_{\mathrm{I}}$ is the Gibbs state, which commutes with H and is unchanged by the transformation. Integrating once from $-\infty$ and keeping the term linear in ϕ_{ext} gives the first-order term of the Dyson series Eq. (2.50),

$$\hat{\varrho}_{\mathrm{I}}(t) = \hat{\varrho} - \mathrm{i}\int_{-\infty}^{t} \mathrm{d}t'\,\left[O_b(t'),\,\hat{\varrho}\right]\phi_{\mathrm{ext}}(t') + O(\phi_{\mathrm{ext}}^2).$$

The average of an operator A at time t is $\mathrm{Tr}[\hat{\varrho}(t)A] = \mathrm{Tr}[\hat{\varrho}_{\mathrm{I}}(t)A(t)]$, with $A(t)$ the Heisenberg operator. The zeroth-order term is the equilibrium average. In the first-order term the cyclic

property of the trace moves $\hat{\varrho}$ to the front, $\mathrm{Tr}([B,\hat{\varrho}]A) = \mathrm{Tr}(\hat{\varrho}[A,B])$, and the change of the average is

$$\delta\langle A\rangle(t) = -\mathrm{i}\int_{-\infty}^{t} \mathrm{d}t'\,\langle[A(t),\,O_b(t')]\rangle\,\phi_{\mathrm{ext}}(t'), \tag{4.2}$$

with the average taken in the Gibbs state. The field at time t' disturbs the target, the disturbance is carried to time t by the target's own dynamics, and the commutator measures how much the disturbance shows up in A. Times $t' > t$ do not contribute. Since the Gibbs state is stationary, the commutator depends on $t - t'$ alone (Sect. 2.5).

Which average responds to O_b? Suppose the target is invariant under translations, as the electron gas is. Then $O_b = n_e(\boldsymbol{q})$ adds the momentum $\boldsymbol{q}$ to every state it acts on, and the equilibrium average of $[A(t), O_b(t')]$ vanishes unless A removes the same momentum. An operator that removes $\boldsymbol{q}$ is the conjugate of an operator at $\boldsymbol{q}$, such as $n_e(\boldsymbol{q})^\dagger = n_e(-\boldsymbol{q})$. For the same reason the term with $n_e(\boldsymbol{q})^\dagger$ in the real potential above cannot change $\langle n_e(\boldsymbol{q})^\dagger\rangle$: the field operator and the measured operator both remove $\boldsymbol{q}$, and the average of their commutator would need the target to lose $2\boldsymbol{q}$. In a crystal, momentum is conserved only up to a reciprocal-lattice vector $\boldsymbol{G}$ (Sect. 3.1). The field at $\boldsymbol{q}$ then also changes the averages at $\boldsymbol{q}+\boldsymbol{G}$, which are the local fields of Sect. 4.4, and the two terms of a real potential mix only when $2\boldsymbol{q}$ is a reciprocal-lattice vector. Taking $A = O_a(\boldsymbol{q})^\dagger$ in Eq. (4.2) for a second target operator $O_a(\boldsymbol{q})$, with the Heisenberg operator $O_a(\boldsymbol{q},t)$ defined as $O_b(t)$, we arrive at the Kubo formula.

Kubo formula

$$\begin{aligned}\delta\langle O_a(\boldsymbol{q})^\dagger\rangle(t) &= \mathscr{V}\int_{-\infty}^{\infty}\mathrm{d}t'\,R_{ab}(\boldsymbol{q},t-t')\,\phi_{\mathrm{ext}}(t'),\\ R_{ab}(\boldsymbol{q},t) &= -\frac{\mathrm{i}}{\mathscr{V}}\,\Theta(t)\,\langle[O_a(\boldsymbol{q},t)^\dagger,\,O_b(\boldsymbol{q},0)]\rangle.\end{aligned} \tag{4.3}$$

The change of $\langle O_a(\boldsymbol{q})^\dagger\rangle$, to first order in a field coupled as $O_b(\boldsymbol{q})\,\phi_{\mathrm{ext}}(t)$, is fixed by the commutator of the two operators in the unperturbed equilibrium state of the target.

The step function $\Theta(t)$ imposes the restriction $t' < t$, and R_{ab} is called the retarded response function. The equilibrium state enters through the average alone: the formula holds for any H, with any interactions, at any temperature. The factor $1/\mathscr{V}$ makes R_{ab} intensive. For density operators the commutator is a sum over all the electrons and grows with the size of the target, and so does the induced $\delta\langle n_e(\boldsymbol{q})^\dagger\rangle$; divided by $\mathscr{V}$, neither grows. In a box, the inverse of Eq. (A.5) writes the electron density as $\mathscr{V}^{-1}\sum_{\boldsymbol{q}'}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}'\cdot\boldsymbol{r}}n_e(\boldsymbol{q}')$, and the induced change of the density in position space is the ripple

$$\frac{1}{\mathscr{V}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\delta\langle n_e(\boldsymbol{q})^\dagger\rangle(t) + \text{c.c.} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\int\mathrm{d}t'\,R_{ee}(\boldsymbol{q},t-t')\,\phi_{\mathrm{ext}}(t') + \text{c.c.},$$

where R_{ee} is the density response, with $O_a = O_b = n_e(\boldsymbol{q})$, in the labeling of Appendix A.7. The response function is the induced density per unit applied potential energy, so $R_{ee}(\boldsymbol{q},\omega)$ has the dimension $\mathrm{eV}^3/\mathrm{eV} = \mathrm{eV}^2$; the function of time, $R_{ee}(\boldsymbol{q},t)$, has one power of energy more, eV^3.

Kubo derived Eq. (4.2) in this generality [3], and the result carries his name. Many texts write the response function as χ_{AB} and call it a susceptibility; others write C^R_{AB}. They usually omit the factor $1/\mathscr{V}$ and write the responding operator A in place of our $O_a^\dagger$. A text that couples the field with a minus sign, $-O_b\,\phi_{\mathrm{ext}}$, has a response function of the opposite sign. In this book χ is the dark-matter particle and response functions are always R; Appendix B.2 lists the correspondence.

4.1.2 Frequency dependence

A field that oscillates at one frequency produces a response at the same frequency, with an amplitude set by the transform of R_{ab}. With the time convention of Appendix A.3, the convolution in Eq. (4.3) becomes a product,

$$\delta\langle O_a(\boldsymbol{q})^\dagger\rangle(\omega) = \mathscr{V} R_{ab}(\boldsymbol{q},\omega)\,\phi_{\rm ext}(\omega), \qquad R_{ab}(\boldsymbol{q},\omega) = -\frac{\mathrm{i}}{\mathscr{V}}\int_0^\infty \mathrm{d}t\; \mathrm{e}^{\mathrm{i}\omega t}\,\langle [O_a(\boldsymbol{q},t)^\dagger,\, O_b(\boldsymbol{q},0)]\rangle, \tag{4.4}$$

which is the definition Eq. (A.29). For a target with a discrete spectrum the commutator oscillates forever, and the integral needs a prescription to converge. Switching the field on slowly supplies one: $\phi_{\rm ext}(t) = \phi_0\,\mathrm{e}^{-\mathrm{i}\omega t+\eta_{\rm b} t}$ with a small rate $\eta_{\rm b} > 0$. Substituting $s = t - t'$ in Eq. (4.3),

$$\delta\langle O_a(\boldsymbol{q})^\dagger\rangle(t) = \mathscr{V}\phi_{\rm ext}(t)\int_0^\infty \mathrm{d}s\; \mathrm{e}^{\mathrm{i}(\omega+\mathrm{i}\eta_{\rm b})s}\,R_{ab}(\boldsymbol{q},s).$$

The response follows the field, and its amplitude is the transform of R_{ab} at the complex frequency $\omega + \mathrm{i}\eta_{\rm b}$. The physical response function is the limit $\eta_{\rm b} \to 0^+$, and ω in Eq. (4.4) always stands for $\omega + \mathrm{i}0^+$. This slow turn-on is the adiabatic switching of Sect. 2.4, which put $+\mathrm{i}0^+$ into the energy denominators there.

For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, the response at $-\boldsymbol{q}$ is the complex conjugate of the response at $\boldsymbol{q}$. Since $\langle X\rangle^* = \langle X^\dagger\rangle$ and $[A,B]^\dagger = -[A^\dagger, B^\dagger]$, the complex conjugate of $R_{ab}(\boldsymbol{q},t)$ is $-(\mathrm{i}/\mathscr{V})\,\Theta(t)\,\langle [O_a(\boldsymbol{q},t), O_b(\boldsymbol{q})^\dagger]\rangle$. For densities and other operators with this property, the complex conjugate is the response at $-\boldsymbol{q}$:

$$R_{ab}(\boldsymbol{q},t)^* = R_{ab}(-\boldsymbol{q},t), \qquad R_{ab}(\boldsymbol{q},\omega)^* = R_{ab}(-\boldsymbol{q},-\omega) \quad (\omega \text{ real}). \tag{4.5}$$

Eq. (4.5) is what makes the ripple produced by a real potential a real function of $\boldsymbol{r}$ and t. In a target with inversion or time-reversal symmetry, where $R_{ee}(-\boldsymbol{q},\omega) = R_{ee}(\boldsymbol{q},\omega)$, the real part of the density response is even in ω and its imaginary part odd.

4.1.3 Lehmann representation

In the eigenstates of H every Heisenberg operator oscillates at the energies of the transitions it causes, $\langle i|O(t)|f\rangle = \mathrm{e}^{\mathrm{i}(E_i-E_f)t}\langle i|O|f\rangle$, and the commutator in Eq. (4.4) becomes a sum of oscillating exponentials. Writing out the two products in the commutator, with the Gibbs weights $p_i = \mathrm{e}^{-\beta E_i}/\mathscr{Z}$, and inserting a complete set of states $|f\rangle$ between the operators,

$$\langle [O_a(\boldsymbol{q},t)^\dagger, O_b(\boldsymbol{q})]\rangle = \sum_{i,f} p_i\Big[\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\,\mathrm{e}^{-\mathrm{i}(E_f-E_i)t} - \langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle\,\mathrm{e}^{+\mathrm{i}(E_f-E_i)t}\Big],$$

where the operators without a time argument are taken at $t = 0$ and we drop the argument $\boldsymbol{q}$. The time integral of each oscillation, with $\omega + \mathrm{i}0^+$, is elementary, $-\mathrm{i}\int_0^\infty \mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega\mp(E_f-E_i)+\mathrm{i}0^+)t} = 1/[\omega \mp (E_f - E_i) + \mathrm{i}0^+]$.

Lehmann representation

$$R_{ab}(\boldsymbol{q},\omega) = \frac{1}{\mathscr{V}}\sum_{i,f} p_i\left[\frac{\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\omega - (E_f - E_i) + \mathrm{i}0^+} - \frac{\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle}{\omega + (E_f - E_i) + \mathrm{i}0^+}\right]. \tag{4.6}$$

The response is a sum of simple poles at the transition energies of the target, each displaced by $-\mathrm{i}0^+$ below the real axis and weighted by the thermal population of its initial state and a product of matrix elements.

Eq. (4.6) is Eq. (A.31). The term with the pole at $\omega = E_f - E_i$ is what the optics literature calls the resonant term, and the one with the pole at $\omega = -(E_f - E_i)$ the antiresonant term. For a target in its ground state, $E_f - E_i \geq 0$, and the resonant term has its poles at positive ω. The interactions of the target, however strong, enter through the exact energies and matrix elements. No approximation has been made beyond first order in ϕ_{ext}.

The poles split the response into two parts with different meanings. By the Sokhotski–Plemelj formula Eq. (2.58) (Appendix C.4), each denominator is a principal value minus $i\pi$ times a delta function. The principal value collects transitions that the field drives off resonance. They are never completed, they shift the target's energy, and they make the target polarizable, as the off-shell intermediate states of Sect. 2.4 did. The delta functions collect the transitions that conserve energy, and they make up the imaginary part. For a diagonal element, $a = b$, the weights are squared matrix elements, $|\langle f|O_a|i\rangle|^2$ in the resonant term and $|\langle f|O_a^\dagger|i\rangle|^2$ in the antiresonant one. The imaginary part is then a difference of two transition sums Eq. (2.65):

$$\operatorname{Im} R_{aa}(\boldsymbol{q},\omega) = -\frac{\pi}{\mathcal{V}}\Big[\mathcal{S}_{O_a}(\omega) - \mathcal{S}_{O_a^\dagger}(-\omega)\Big]. \tag{4.7}$$

The transition sum $\mathcal{S}_{O_a}(\omega)$ counts the transitions in which the target takes the energy ω; $\mathcal{S}_{O_a^\dagger}(-\omega)$ counts those in which it gives up ω. At $T = 0$ the target starts in its ground state, nothing can give up energy, and $\operatorname{Im} R_{aa} < 0$ for $\omega > 0$. At finite temperature both sums contribute, and detailed balance Eq. (2.68) makes $\mathcal{S}_{O_a}(\omega)$ larger than $\mathcal{S}_{O_a^\dagger}(-\omega)$ by the factor $\mathrm{e}^{\beta\omega}$. Sect. 4.2 turns this detailed-balance relation into the fluctuation–dissipation theorem, which ties $\operatorname{Im} R_{aa}$ to the dynamic structure factor.

Absorbed power. The sign of $\operatorname{Im} R$ is the statement that a target in equilibrium absorbs energy from a field. Take a Hermitian O_b, so that $R_{bb}(t)$ is real (the average of the commutator of two Hermitian operators is imaginary), and a real field $\phi_{\text{ext}}(t) = \phi_0 \cos\omega t$ with $\omega > 0$. The field does work on the target at the rate $\langle \partial H_{\text{tot}}/\partial t\rangle = \langle O_b\rangle(t)\, \dot\phi_{\text{ext}}(t)$. The field has components $\frac{1}{2}\phi_0$ at the frequencies $\pm\omega$, and by Eq. (4.4) the induced $\delta\langle O_b\rangle$ has components $\frac{1}{2}\phi_0 \mathcal{V} R_{bb}(\pm\omega)$, with $R_{bb}(-\omega) = R_{bb}(\omega)^*$ since $R_{bb}(t)$ is real. In the time average of the product the cross terms between $+\omega$ and $-\omega$ alone survive, and the mean absorbed power is

$$P = -\frac{\omega\,\phi_0^2}{2}\,\mathcal{V}\ \operatorname{Im} R_{bb}(\omega) = \frac{\pi}{2}\,\omega\,\phi_0^2\,\Big[\mathcal{S}_{O_b}(\omega) - \mathcal{S}_{O_b}(-\omega)\Big], \tag{4.8}$$

where the expression in transition sums follows from Eq. (4.7). The power P is ω times the difference between the absorption rate and the emission rate of the golden rule for a harmonic perturbation, Eq. (2.8) with $\hat O = \frac{1}{2}\phi_0 O_b$, averaged over the thermal state as in Sect. 2.5. The imaginary part of a diagonal response is therefore its dissipative part. The same formula describes a bosonic dark-matter field that couples to O_b. With $\omega = m_\phi$, the amplitude ϕ_0 is then the field amplitude of Sect. 2.1, $\phi_0^2 = 2\rho_\chi/m_\phi^2$, times the coupling constant of the field to O_b, which we set to one here. Then P/m_ϕ is the rate at which the field is absorbed in the volume $\mathcal{V}$, and absorption rates are therefore proportional to $-\operatorname{Im} R$ (Chap. 8).

Static limit. At $\omega \to 0$ the Lehmann form measures how much the target deforms in a static field. For a Hermitian O_b the resonant and antiresonant terms of a pair i, f have the same weight, so pairs with $E_f = E_i$ cancel, and the pairs with $E_f \neq E_i$ add:

$$\mathcal{V} R_{bb}(0) = -2\sum_{i,f} p_i\, \frac{|\langle f|O_b|i\rangle|^2}{E_f - E_i} \qquad (E_f \neq E_i).$$

At $T = 0$ every term is negative. Compare second-order perturbation theory for a static $H_{\text{ext}} = O_b\phi_0$: the ground-state energy is shifted by $-\phi_0^2 \sum_f |\langle f|O_b|0\rangle|^2/(E_f - E_0)$, which is

$\frac{1}{2}\mathcal{V}R_{bb}(0)\,\phi_0^2$. A static response is negative because the target always lowers its energy by yielding to a static field; for the density, electrons leave the regions where the potential energy is high. At finite temperature the limit $\omega \to 0$ describes a target that is deformed by a slowly applied field while isolated from any bath, Kubo's isolated response. The isothermal response, in which the target stays in equilibrium with a bath, has an extra term proportional to β from matrix elements of O_b between states of equal energy; the isolated and the isothermal response agree when all these matrix elements vanish [3].

Worked example 4.1: The driven harmonic oscillator

Take one mode of frequency ω_0, $H = \omega_0 d^\dagger d$, and the operator $O_a = O_b = \lambda_{\rm osc}(d + d^\dagger)$ of Worked example 2.7, with $\lambda_{\rm osc}$ real. A single oscillator has no volume of its own, and the combination $\mathcal{V}R$ is its response, independent of the box. The Heisenberg operators follow from the commutator $[d, H] = \omega_0 d$, which integrates, as for the fermion operators of Sect. 2.3, to $d(t) = d\,\mathrm{e}^{-\mathrm{i}\omega_0 t}$. The commutator in Eq. (4.3) is then a number,

$$[O(t), O] = \lambda_{\rm osc}^2([d, d^\dagger]\,\mathrm{e}^{-\mathrm{i}\omega_0 t} + [d^\dagger, d]\,\mathrm{e}^{\mathrm{i}\omega_0 t}) = -2\mathrm{i}\lambda_{\rm osc}^2 \sin\omega_0 t,$$

the same in every state, and $\mathcal{V}R(t) = -2\lambda_{\rm osc}^2\,\Theta(t)\sin\omega_0 t$. With $\int_0^\infty \mathrm{d}t\,\mathrm{e}^{\mathrm{i}zt}\sin\omega_0 t = \omega_0/(\omega_0^2 - z^2)$ for $\mathrm{Im}\,z > 0$, the transform of $\mathcal{V}R(t)$ is

$$\mathcal{V}R(\omega) = \frac{2\lambda_{\rm osc}^2\omega_0}{(\omega + \mathrm{i}0^+)^2 - \omega_0^2} = \lambda_{\rm osc}^2\left[\frac{1}{\omega - \omega_0 + \mathrm{i}0^+} - \frac{1}{\omega + \omega_0 + \mathrm{i}0^+}\right]. \tag{4.9}$$

The Lehmann form gives the same. From the level n, O reaches $n+1$ with $|\langle n+1|O|n\rangle|^2 = \lambda_{\rm osc}^2(n+1)$ and $n-1$ with $\lambda_{\rm osc}^2 n$, and averaged with the thermal weights of Eq. (2.69) the resonant term of Eq. (4.6) is $\lambda_{\rm osc}^2[(n_{\rm B}(\omega_0)+1)/(\omega - \omega_0) + n_{\rm B}(\omega_0)/(\omega + \omega_0)]$, with $n_{\rm B}(\omega_0)$ the Bose occupation of the mode. The antiresonant term has the same weights with the denominators exchanged. The resonant term minus the antiresonant term is Eq. (4.9), and $n_{\rm B}(\omega_0)$ cancels. The oscillator absorbs and emits more often at higher temperature, by the factors $n_{\rm B}(\omega_0)+1$ and $n_{\rm B}(\omega_0)$ of Worked example 2.7, but the net response to a field is independent of temperature.

The reason is that the oscillator is a linear system, and the quantum response is the classical one. For a particle of mass $M_{\rm osc}$ in a harmonic well, the displacement is $x = (2M_{\rm osc}\omega_0)^{-1/2}(d + d^\dagger)$, so $O = x$ has $\lambda_{\rm osc}^2 = 1/(2M_{\rm osc}\omega_0)$. The field $H_{\rm ext} = x\,\phi_{\rm ext}$ is a force $-\phi_{\rm ext}$, and Newton's equation $M_{\rm osc}(\ddot{x} + \omega_0^2 x) = -\phi_{\rm ext}$, with $x \propto \mathrm{e}^{-\mathrm{i}\omega t}$, gives $x(\omega) = \phi_{\rm ext}(\omega)/[M_{\rm osc}(\omega^2 - \omega_0^2)]$, which is Eq. (4.9) with this $\lambda_{\rm osc}$. The static limit is the stretched spring, $x = -\phi_{\rm ext}/(M_{\rm osc}\omega_0^2)$, and $\mathcal{V}R(0) = -2\lambda_{\rm osc}^2/\omega_0$; the energy $\frac{1}{2}\mathcal{V}R(0)\phi_0^2 = -\lambda_{\rm osc}^2\phi_0^2/\omega_0$ is the second-order shift of the ground state by $\lambda_{\rm osc}\phi_0(d + d^\dagger)$, as the static limit above requires.

A real oscillator also loses energy, to other modes or to electrons. With a damping term $M_{\rm osc}\Gamma_{\rm w}\dot{x}$ added to Newton's equation, where $\Gamma_{\rm w}$ is the rate at which the energy of the free oscillation decays, the response becomes

$$\mathcal{V}R(\omega) = \frac{2\lambda_{\rm osc}^2\omega_0}{\omega^2 + \mathrm{i}\Gamma_{\rm w}\omega - \omega_0^2}, \qquad -\,\mathrm{Im}\,\mathcal{V}R(\omega) = \frac{2\lambda_{\rm osc}^2\omega_0\,\Gamma_{\rm w}\omega}{(\omega^2 - \omega_0^2)^2 + \Gamma_{\rm w}^2\omega^2}.$$

The infinitesimal 0^+ has become the finite width $\Gamma_{\rm w}$, as in Eq. (2.57). The absorptive part $-\,\mathrm{Im}\,\mathcal{V}R$ is a resonance of height $2\lambda_{\rm osc}^2/\Gamma_{\rm w}$ at $\omega = \omega_0$, ten times $\lambda_{\rm osc}^2/\omega_0$ for $\Gamma_{\rm w} = 0.2\,\omega_0$, and it is positive at every $\omega > 0$ (Fig. 4.2b). The real part is negative below the resonance, where the oscillator follows the force in phase, and positive above the resonance, where the oscillator moves against the force. At large ω the real part falls as $2\lambda_{\rm osc}^2\omega_0/\omega^2$.

4.1.4 Causality and analyticity

Causality means that a response cannot precede its cause. In Eq. (4.3) causality appears as the step function $\Theta(t)$, and in frequency it becomes a statement about complex ω. Define the transform at a complex frequency z,

$$R_{ab}(\boldsymbol{q}, z) = \int_0^\infty \mathrm{d}t\ \mathrm{e}^{\mathrm{i}zt}\, R_{ab}(\boldsymbol{q}, t). \tag{4.10}$$

For $\operatorname{Im} z > 0$ the factor $\mathrm{e}^{\mathrm{i}zt} = \mathrm{e}^{\mathrm{i}t \operatorname{Re} z}\mathrm{e}^{-t \operatorname{Im} z}$ decays, while $R_{ab}(\boldsymbol{q}, t)$ stays bounded: in a stable target all energies are real, and the commutator average oscillates without growing. The integral then converges, it may be differentiated under the integral sign with respect to z, and $R_{ab}(\boldsymbol{q}, z)$ is analytic everywhere in the upper half plane. The argument used causality, which restricts the integral to $t > 0$, and stability, which keeps $R_{ab}(\boldsymbol{q}, t)$ from growing. The physical response is the boundary value on the real axis approached from above, $R_{ab}(\boldsymbol{q}, \omega + \mathrm{i}0^+)$, which is the prescription that the adiabatic switching gave. For $\operatorname{Im} z < 0$ the integral diverges, and the response there is defined, if at all, by analytic continuation. Appendix C.5 states these results in general.

The Lehmann form shows the analyticity pole by pole. With $\omega + \mathrm{i}0^+$ replaced by z, Eq. (4.6) is a sum of terms $1/[z \mp (E_f - E_i)]$, whose poles lie on the real axis, and the physical frequency passes just above them (Fig. 4.2a). In a finite target the poles are isolated, and the imaginary part on the real axis is a comb of delta functions. In a macroscopic target the transition energies crowd together, with spacings that shrink as the volume grows. If the limit $\mathcal{V} \to \infty$ is taken first, at a fixed small η_b, the sums over f become integrals, and $R_{ab}(\boldsymbol{q}, \omega + \mathrm{i}\eta_\mathrm{b})$ tends to a smooth function as $\eta_\mathrm{b} \to 0^+$. The poles have merged into a branch cut along the parts of the real axis where the spectrum is continuous. The opposite order of limits returns the comb. Taking $\mathcal{V} \to \infty$ before $\eta_\mathrm{b} \to 0^+$ is the time window of the golden rule (Sect. 2.1) once more. The switching rate η_b must be small compared with the scale on which the response varies, and large compared with the level spacing, which a macroscopic target makes arbitrarily small.

Across the cut, the continuation of R_{ab} into the lower half plane can have poles at complex frequencies. They are resonances, long-lived collective or single-particle excitations. A pole at z_p contributes $\mathrm{e}^{-\mathrm{i}z_\mathrm{p}t}$ to $R_{ab}(\boldsymbol{q}, t)$, which decays only if $\operatorname{Im} z_\mathrm{p} < 0$. The damped oscillator of Worked example 4.1 shows them in closed form. Its response has poles where $\omega^2 + \mathrm{i}\Gamma_\mathrm{w}\omega - \omega_0^2 = 0$,

$$\omega = -\frac{\mathrm{i}\Gamma_\mathrm{w}}{2} \pm \Big(\omega_0^2 - \frac{\Gamma_\mathrm{w}^2}{4}\Big)^{1/2}.$$

For $\Gamma_\mathrm{w} < 2\omega_0$ they lie on the circle $|\omega| = \omega_0$ in the lower half plane, at a depth $\Gamma_\mathrm{w}/2$ that is the decay rate of the amplitude. At critical damping, $\Gamma_\mathrm{w} = 2\omega_0$, they meet at $-\mathrm{i}\omega_0$, and for larger Γ_w they move apart along the negative imaginary axis. For no positive Γ_w do they cross into the upper half plane, where a pole would describe a response that grows exponentially after the field is gone. The plasmon (Sect. 4.5) and the quasiparticles of Sect. 4.7 are also resonances, poles of the continued response in the lower half plane.

How fast R_{ab} falls at large $|z|$ follows from the Lehmann form too. Expanding $1/[z \mp (E_f - E_i)] = 1/z \pm (E_f - E_i)/z^2 + \dots$ in Eq. (4.6) shows that the coefficient of $1/z$ is the equal-time commutator. In the coefficient of $1/z^2$ the resonant and antiresonant terms combine into a double commutator:

$$R_{ab}(\boldsymbol{q}, z) = \frac{\langle [O_a^\dagger, O_b] \rangle}{\mathcal{V} z} + \frac{\langle [O_a^\dagger, [H, O_b]] \rangle}{\mathcal{V} z^2} + O(z^{-3}). \tag{4.11}$$

The coefficient of $1/z^2$ is checked by writing out $\langle i|O_a^\dagger H O_b - O_a^\dagger O_b H - H O_b O_a^\dagger + O_b H O_a^\dagger|i\rangle$ in eigenstates, which gives $\sum_f (E_f - E_i)\big[\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle + \langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle\big]$. Density operators

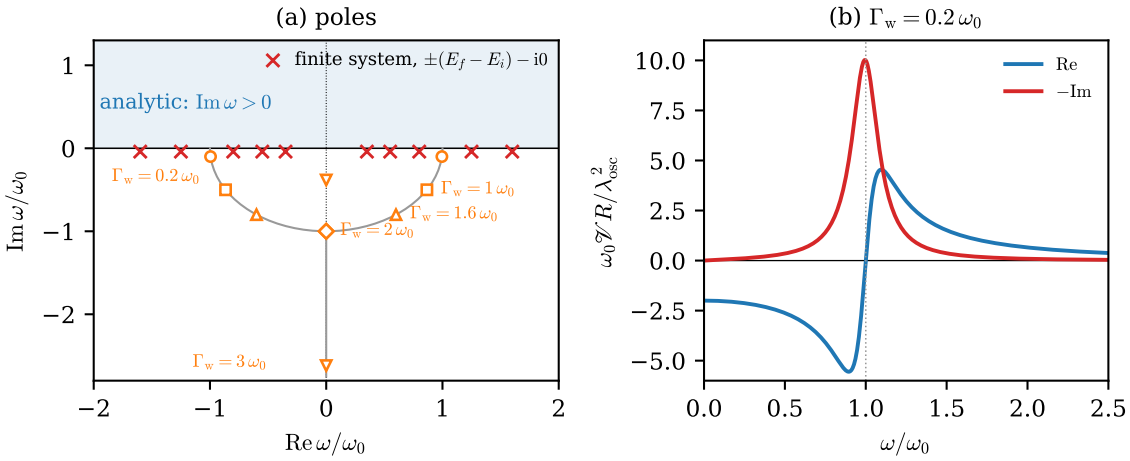


Figure 4.2: Poles of the retarded response function in the complex frequency plane ω, in units of an oscillator frequency ω_0. (a) The response is analytic in the upper half plane (shaded). A finite target has poles at its transition energies $\pm(E_f - E_i)$, displaced by an infinitesimal $-\mathrm{i}0$ below the real axis (crosses, at illustrative transition energies). A harmonic oscillator with energy damping rate Γ_w has poles at $-\mathrm{i}\Gamma_\mathrm{w}/2 \pm (\omega_0^2 - \Gamma_\mathrm{w}^2/4)^{1/2}$ (open symbols, for the values of Γ_w marked). They lie on the circle $|\omega| = \omega_0$ below critical damping, $\Gamma_\mathrm{w} = 2\omega_0$, and on the negative imaginary axis above it; for $\Gamma_\mathrm{w} = 3\omega_0$ the upper of the two is at $-0.38\,\mathrm{i}\omega_0$. (b) Real part and minus the imaginary part of $\mathscr{V} R$ for the damped oscillator with $\Gamma_\mathrm{w} = 0.2\,\omega_0$ and coupling $\lambda_\mathrm{osc}(d + d^\dagger)$, on the real axis. The absorptive part peaks at $2\lambda_\mathrm{osc}^2/\Gamma_\mathrm{w}$ at $\omega = \omega_0$; the real part is negative below the resonance and positive above it, and tends to $-2\lambda_\mathrm{osc}^2/\omega_0$ at $\omega = 0$.

at different wavevectors commute, since both are functions of the electron positions alone, and the density response therefore falls as $1/z^2$. When the potential acting on the electrons is local, only the kinetic energy contributes to the coefficient of $1/z^2$, and the result, the f-sum rule of Sect. 4.3, depends only on the electron density. For the oscillator, $[O,[H,O]] = 2\lambda_{\mathrm{osc}}^2\omega_0$, the coefficient found in Worked example 4.1. Analyticity in the upper half plane together with this decay leads to the Kramers–Kronig relations, which reconstruct the real part of R_{ab} on the real axis from its imaginary part (Sect. 4.3).

Worked example 4.2: The two-level system

Take the two-level system of Sect. 2.5: states $|0\rangle$ and $|1\rangle$ with splitting Δ_{tl}, Gibbs weights p_0 and p_1, and a Hermitian operator $O_a = O_b = O$ with only the off-diagonal element $O_{10} = \langle 1|O|0\rangle$. The Lehmann form Eq. (4.6) has four entries. In the resonant term, $i = 0$ contributes $p_0|O_{10}|^2/(\omega - \Delta_{\mathrm{tl}})$ and $i = 1$ contributes $p_1|O_{10}|^2/(\omega + \Delta_{\mathrm{tl}})$. In the antiresonant term the same weights appear with the poles exchanged and a minus sign. Altogether,

$$\mathscr{V}R(\omega) = (p_0 - p_1)\,|O_{10}|^2 \left[\frac{1}{\omega - \Delta_{\mathrm{tl}} + \mathrm{i}0^+} - \frac{1}{\omega + \Delta_{\mathrm{tl}} + \mathrm{i}0^+}\right] = \frac{2\Delta_{\mathrm{tl}}\,|O_{10}|^2 \tanh(\beta\Delta_{\mathrm{tl}}/2)}{(\omega + \mathrm{i}0^+)^2 - \Delta_{\mathrm{tl}}^2}, \tag{4.12}$$

with $p_0 - p_1 = \tanh(\beta\Delta_{\mathrm{tl}}/2)$ from Eq. (2.62). Eq. (4.12) has the form of the oscillator response, Eq. (4.9), with $\omega_0 \to \Delta_{\mathrm{tl}}$ and $\lambda_{\mathrm{osc}}^2 \to (p_0 - p_1)|O_{10}|^2$, but now the temperature matters. At $T = \Delta_{\mathrm{tl}}/2$ the weights are 0.881 and 0.119 (Fig. 2.8), the response is reduced by the factor 0.762, and $\mathscr{V}R(0) = -1.523\,|O_{10}|^2/\Delta_{\mathrm{tl}}$. At $T \gg \Delta_{\mathrm{tl}}$, $\tanh(\beta\Delta_{\mathrm{tl}}/2) \to \Delta_{\mathrm{tl}}/2T$ and $\mathscr{V}R(0) \to -|O_{10}|^2/T$. For a spin $\frac{1}{2}$ in a magnetic field, with Δ_{tl} the Zeeman splitting and O a transverse component of its magnetic moment, the high-temperature limit is the Curie law of a paramagnet: the static susceptibility falls as $1/T$.

The temperature dependence comes from the weight $p_0 - p_1$. The field drives the upward transition from $|0\rangle$ and stimulated emission from $|1\rangle$ with the same matrix element, so the excited population cancels part of the absorption, and the absorbed power Eq. (4.8) is proportional to $p_0 - p_1$. Since $p_1 < p_0$ at every temperature (Sect. 2.5), a two-level system in equilibrium always absorbs, and becomes transparent only as $T \to \infty$. The oscillator response does not depend on temperature because the matrix elements of the oscillator grow up the ladder: the extra absorption from level n, weighted by $n + 1$, exactly compensates the emission, weighted by n, and $n_{\mathrm{B}}(\omega_0)$ therefore dropped out of Eq. (4.9). The tail coefficient of Eq. (4.11) is $\langle[O,[H,O]]\rangle = 2\Delta_{\mathrm{tl}}(p_0 - p_1)|O_{10}|^2$, temperature dependent here and constant for the oscillator.

The two examples also mark where linear response stops. For the oscillator, the commutator $[O(t), O(t')]$ is a number, every nested commutator of higher order vanishes, and so does every higher-order term of the Dyson series, so Eq. (4.9) holds at any field strength. The two-level system saturates. On resonance the field moves population between $|0\rangle$ and $|1\rangle$, and Eq. (4.12) holds only while that transfer is small: the drive $|O_{10}\phi_0|$ must be small compared with the width of the transition, the weak-drive condition of Appendix D.3. A stronger field makes the populations oscillate at the Rabi frequency, a response that is not linear in the field.

4.2 Correlations and the dynamic structure factor

A dark-matter particle in silicon scatters at a rate fixed by the dynamic structure factor $S(\boldsymbol{q},\omega)$ of Eq. (2.67), which counts the transitions the target makes. When a weak potential acts on the target from outside, through the same density operator (Sect. 4.1), the density ripples by an amount fixed by the response function R_{ee} of Eq. (4.3). The structure factor and the response function are built from the same matrix elements and the same Gibbs weights, and at each $\boldsymbol{q}$ and ω either one fixes the other. This relation is the fluctuation–dissipation theorem, and it lets every method that computes a response also compute a rate. The treatment follows Refs. [1, Chaps. 3 and 4] and [4, Chap. 2].

4.2.1 The response matrix

A dark photon couples to the charge of the electrons and of the nuclei, with opposite signs, and other mediators couple to spin or to the nucleon number (Sect. 1.4). In general the interaction between the dark-matter plane waves $\boldsymbol{p}$ and $\boldsymbol{p}-\boldsymbol{q}$ is a sum $\mathscr{V}^{-1}\sum_b \tilde{V}_b(\boldsymbol{q})\,O_b(\boldsymbol{q})$ over target operators $O_b(\boldsymbol{q})$, each with its own potential $\tilde{V}_b(\boldsymbol{q})$; the density coupling of Eq. (A.8) is the case of a single term, $O_e = n_e(\boldsymbol{q})$ with $\tilde{V}_e = \tilde{V}$. The golden-rule amplitude to reach $|f\rangle$ from $|i\rangle$ is then $\mathscr{V}^{-1}\sum_b \tilde{V}_b\langle f|O_b|i\rangle$. Its square has cross terms $\tilde{V}_a^*\tilde{V}_b\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$, and the target enters through the matrix of their thermal sums.

Response matrix

$$\begin{aligned} S_{ab}(\boldsymbol{q},\omega) &= \frac{2\pi}{\mathscr{V}}\sum_{i,f} p_i\,\langle i|O_a(\boldsymbol{q})^\dagger|f\rangle\langle f|O_b(\boldsymbol{q})|i\rangle\,\delta(\omega - E_f + E_i) \\ &= \frac{1}{\mathscr{V}}\int_{-\infty}^{\infty}\mathrm{d}t\;\mathrm{e}^{\mathrm{i}\omega t}\,\big\langle O_a(\boldsymbol{q},t)^\dagger\,O_b(\boldsymbol{q},0)\big\rangle. \end{aligned} \tag{4.13}$$

Its diagonal elements are the transition sums of Eq. (2.65), $S_{aa} = (2\pi/\mathscr{V})\,\mathcal{S}_{O_a}$, and the dynamic structure factor is the density element, $S \equiv S_{ee}$.

The correlation-function form follows as Eq. (2.66) did. With a complete set of states inserted between the two operators, each term oscillates as $\mathrm{e}^{-\mathrm{i}(E_f-E_i)t}$, and the time integral turns the oscillation into $2\pi\,\delta(\omega - E_f + E_i)$. Eq. (4.13) is the definition Eq. (A.30), and its density element is $S(\boldsymbol{q},\omega)$ of Eq. (2.67), Ref. [5, Eq. (4)], with the same normalization. Exchanging a and b complex-conjugates each term, so $S_{ab}^* = S_{ba}$ and the matrix is Hermitian. Like R_{ab}, S_{ab} is intensive, and for density operators it has the dimension eV^2. Detailed balance carries over with the indices exchanged. For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, relabeling $i \leftrightarrow f$ as in Eq. (2.68) gives $S_{ab}(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega}S_{ba}(\boldsymbol{q},\omega)$, which is Eq. (A.33).

The sum over final dark-matter momenta is the one of Sect. 2.5, term by term, and the rate per dark-matter particle is

$$\Gamma = \int\frac{\mathrm{d}^3q}{(2\pi)^3}\sum_{a,b}\tilde{V}_a(\boldsymbol{q})^*\,\tilde{V}_b(\boldsymbol{q})\,S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}}), \tag{4.14}$$

with $\omega_{\boldsymbol{q}}$ the energy deposited at $\boldsymbol{q}$ (Sect. 2.5). For a single operator the rate reduces to $\int \mathrm{d}^3q\,(2\pi)^{-3}|\tilde{V}_a|^2 S_{aa}$, and for the density to Eq. (2.67). The off-diagonal elements carry the interference between two couplings, and Chap. 8 builds the general rate on Eq. (4.14).

4.2.2 Correlations in space and time

At a typical momentum transfer $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$ the dark matter probes the electron density on a length $2\pi/q_0 = 3.32\,\text{Å}$, and at an energy of $10\,\mathrm{eV}$ over a time $1/\omega = 0.066\,\mathrm{fs}$.

The structure factor can be read as a statement about the density on these scales. Van Hove introduced this reading for neutron scattering [6]: the rate measures how the density at one place and time is correlated with the density at another. To make this reading explicit, write the density in position space, $\hat{n}_e(\boldsymbol{r}) = \sum_j \delta^3(\boldsymbol{r} - \boldsymbol{r}_j)$, the operator whose transform Eq. (A.5) is $n_e(\boldsymbol{q})$, and define the Van Hove correlation function

$$\mathcal{G}_{\mathrm{VH}}(\boldsymbol{r}, t) = \frac{1}{N_e} \int \mathrm{d}^3 r' \, \langle \hat{n}_e(\boldsymbol{r}', 0) \, \hat{n}_e(\boldsymbol{r}' + \boldsymbol{r}, t) \rangle, \tag{4.15}$$

with N_e the number of electrons in the volume $\mathcal{V}$ and the Heisenberg operators of Sect. 4.1. In the correlation form Eq. (4.13), stationarity moves the time from one operator to the other, $\langle n_e(\boldsymbol{q}, t)^\dagger n_e(\boldsymbol{q}, 0) \rangle = \langle n_e(\boldsymbol{q}, 0)^\dagger n_e(\boldsymbol{q}, -t) \rangle$. Writing both operators as integrals over position, $n_e(\boldsymbol{q})^\dagger = \int \mathrm{d}^3 r' \, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'} \hat{n}_e(\boldsymbol{r}')$, and writing the position in $n_e(\boldsymbol{q}, -t)$ as $\boldsymbol{r}' + \boldsymbol{r}$,

$$\begin{aligned} \langle n_e(\boldsymbol{q}, 0)^\dagger n_e(\boldsymbol{q}, -t) \rangle &= \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \int \mathrm{d}^3 r' \, \langle \hat{n}_e(\boldsymbol{r}', 0) \, \hat{n}_e(\boldsymbol{r}' + \boldsymbol{r}, -t) \rangle \\ &= N_e \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \, \mathcal{G}_{\mathrm{VH}}(\boldsymbol{r}, -t). \end{aligned}$$

Dividing by $\mathcal{V}$, integrating over t with $\mathrm{e}^{\mathrm{i}\omega t}$ and reversing the sign of t,

$$S(\boldsymbol{q}, \omega) = n_e \int_{-\infty}^{\infty} \mathrm{d}t \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}(\boldsymbol{q}\cdot\boldsymbol{r} - \omega t)} \, \mathcal{G}_{\mathrm{VH}}(\boldsymbol{r}, t), \tag{4.16}$$

with $n_e = N_e/\mathcal{V}$. The dynamic structure factor is the space–time Fourier transform of the Van Hove function, per unit volume. Inserted into Eq. (2.67), Eq. (4.16) gives the Born rate of any target from its density correlations.

For classical, commuting densities, $\mathcal{G}_{\mathrm{VH}}$ of Eq. (4.15) becomes $N_e^{-1} \sum_{j,l} \langle \delta^3(\boldsymbol{r} + \boldsymbol{r}_j(0) - \boldsymbol{r}_l(t)) \rangle$: given an electron at the origin at $t = 0$, it is the density of electrons at $\boldsymbol{r}$ at time t. The terms with $j = l$ follow the electron itself. They integrate to one and spread out as the electron moves away. The terms with $j \neq l$ describe all the other electrons, and they tend to n_e at large r. At $t = 0$ the function is $\delta^3(\boldsymbol{r}) + n_e \, g_{\mathrm{pair}}(\boldsymbol{r})$, with $g_{\mathrm{pair}}(\boldsymbol{r})$ the pair distribution, the probability of finding a second electron at $\boldsymbol{r}$ relative to the first, normalized to one at large separation. Quantum mechanically, however, the densities at different times do not commute, and $\mathcal{G}_{\mathrm{VH}}$ is complex for $t \neq 0$. Its imaginary part is what makes $S(\boldsymbol{q}, \omega)$ and $S(-\boldsymbol{q}, -\omega)$ differ by the factor $\mathrm{e}^{-\beta\omega}$ of detailed balance, and it drops out when $|\omega| \ll T$.

A single free electron shows how the time dependence encodes the recoil. Take it at rest, in the plane wave $\boldsymbol{p} = 0$. Then $n_e(\boldsymbol{q}, t) = \mathrm{e}^{\mathrm{i}Ht} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \mathrm{e}^{-\mathrm{i}Ht}$ acting on $|\boldsymbol{0}\rangle$ gives $\mathrm{e}^{\mathrm{i}q^2 t/2m_e} |\boldsymbol{q}\rangle$, and the correlation $\langle n_e(\boldsymbol{q}, t)^\dagger n_e(\boldsymbol{q}, 0) \rangle = \mathrm{e}^{-\mathrm{i}q^2 t/2m_e}$ oscillates at the recoil energy forever. Its time transform is $2\pi \, \delta(\omega - q^2/2m_e)$: the electron takes the momentum $\boldsymbol{q}$ and exactly the free recoil energy, as in Worked example 2.3. In position space the oscillating correlation is the self term of $\mathcal{G}_{\mathrm{VH}}$, spreading as the struck electron moves off. A bound or interacting electron cannot keep oscillating at one frequency, its correlation decays, and the delta function broadens into the spectrum of the target.

Integrating Eq. (4.16) over ω sets $t = 0$, since $\int \mathrm{d}\omega \, \mathrm{e}^{-\mathrm{i}\omega t} = 2\pi \, \delta(t)$. The result is the static structure factor,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega \; S(\boldsymbol{q}, \omega) = 2\pi \, n_e \, S^{\mathrm{st}}(\boldsymbol{q}), \qquad S^{\mathrm{st}}(\boldsymbol{q}) = \frac{1}{N_e} \langle n_e(\boldsymbol{q})^\dagger n_e(\boldsymbol{q}) \rangle = \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \, \mathcal{G}_{\mathrm{VH}}(\boldsymbol{r}, 0), \tag{4.17}$$

where the middle form follows from summing Eq. (4.13) over f at fixed i. S^{st} is the total weight, per electron, of the transitions in which the target takes up the momentum $\boldsymbol{q}$ at any energy. It is not a probability and can exceed one, as it does at a Bragg vector. The static structure factor

is the many-electron version of the closure relation Eq. (2.19), and it tells us how the electrons share a momentum transfer. With $\mathcal{G}_{\mathrm{VH}}(\boldsymbol{r},0)=\delta^3(\boldsymbol{r})+n_e g_{\mathrm{pair}}(\boldsymbol{r})$, the constant part of g_{pair} transforms into a term $(2\pi)^3 n_e\,\delta^3(\boldsymbol{q})$ alone, and at $\boldsymbol{q}\neq 0$ $S^{\mathrm{st}}=1+n_e\int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}[g_{\mathrm{pair}}(\boldsymbol{r})-1]$. At large q the phase averages the pair term away and $S^{\mathrm{st}}\to 1$: the dark matter scatters incoherently, on one electron at a time. At small q, however, the pair term matters. In a uniform gas it cancels most of the self term, because each electron digs a hole in the density of the others, and long-wavelength density fluctuations are suppressed (Worked example 4.3). In a crystal $\langle n_e(\boldsymbol{q})\rangle$ itself is nonzero at $\boldsymbol{q}=\boldsymbol{G}$, the elastic Bragg term of Sect. 2.2, which in the ground state contributes $(2\pi/\mathcal{V})|\langle n_e(\boldsymbol{q})\rangle|^2\,\delta(\omega)$ to S at a fixed lattice and deposits no energy.

4.2.3 The fluctuation–dissipation theorem

The response function measures dissipation: by Eq. (4.8), $-\operatorname{Im} R_{bb}$ is the power a field coupled to O_b loses to the target. The structure factor measures fluctuations, the correlations of O_b in the undisturbed equilibrium state. The response function and the structure factor were already close in Sect. 4.1. The imaginary part of a diagonal response is a difference of two transition sums, $\operatorname{Im} R_{aa}=-(\pi/\mathcal{V})[\mathcal{S}_{O_a}(\omega)-\mathcal{S}_{O_a^\dagger}(-\omega)]$ (Eq. (4.7)), and detailed balance Eq. (2.68) turns the emission sum $\mathcal{S}_{O_a^\dagger}(-\omega)$ into $\mathrm{e}^{-\beta\omega}\mathcal{S}_{O_a}(\omega)$. Hence $\operatorname{Im} R_{aa}=-(\pi/\mathcal{V})(1-\mathrm{e}^{-\beta\omega})\,\mathcal{S}_{O_a}(\omega)$, and with $S_{aa}=(2\pi/\mathcal{V})\mathcal{S}_{O_a}$ the diagonal relation is $(1-\mathrm{e}^{-\beta\omega})S_{aa}=-2\operatorname{Im} R_{aa}$.

The off-diagonal elements follow from the Lehmann form. Write $c_{if}=\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$ and $\omega_{fi}=E_f-E_i$, so that Eq. (4.6) reads $R_{ab}=\mathcal{V}^{-1}\sum_{i,f}p_i\,[c_{if}/(\omega-\omega_{fi}+\mathrm{i}0^+)-\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle/(\omega+\omega_{fi}+\mathrm{i}0^+)]$. For real ω the complex conjugate of R_{ba} has the same numerators, since $(\langle i|O_b^\dagger|f\rangle\langle f|O_a|i\rangle)^*=c_{if}$, and the same denominators with $-\mathrm{i}0^+$. In the difference $R_{ab}-R_{ba}^*$ the principal values cancel, and each pair of denominators leaves $1/(x+\mathrm{i}0^+)-1/(x-\mathrm{i}0^+)=-2\pi\mathrm{i}\,\delta(x)$ (Eq. (2.58)). The resonant terms give $-\mathrm{i}S_{ab}$ directly. In the antiresonant terms, exchanging the names of i and f turns the weight p_i into $p_f=p_i\,\mathrm{e}^{-\beta\omega}$ on the support of the delta function, as in the proof of detailed balance, and they give $+\mathrm{i}\,\mathrm{e}^{-\beta\omega}S_{ab}$.

Fluctuation–dissipation theorem

$$\begin{aligned}(1-\mathrm{e}^{-\beta\omega})\,S_{ab}(\boldsymbol{q},\omega)&=\mathrm{i}\big[R_{ab}(\boldsymbol{q},\omega)-R_{ba}(\boldsymbol{q},\omega)^*\big],\\ (1-\mathrm{e}^{-\beta\omega})\,S_{aa}(\boldsymbol{q},\omega)&=-2\,\operatorname{Im} R_{aa}(\boldsymbol{q},\omega).\end{aligned}\tag{4.18}$$

The spectrum of equilibrium fluctuations of the target fixes its dissipative response, and conversely, for any H and any temperature.

Eq. (4.18) is Eq. (A.32), and the diagonal density element is Ref. [5, Eq. (3)], whose χ is our R_{ee} (Appendix A.7). The relation also runs the other way, and fixes all of R, not only its imaginary part. Writing the antiresonant terms of the Lehmann form with i and f exchanged, $R_{ab}=\mathcal{V}^{-1}\sum_{i,f}(p_i-p_f)\,c_{if}/(\omega-\omega_{fi}+\mathrm{i}0^+)$, and since $(p_i-p_f)=p_i(1-\mathrm{e}^{-\beta\omega_{fi}})$,

$$R_{ab}(\boldsymbol{q},\omega)=\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega'}{2\pi}\,\frac{(1-\mathrm{e}^{-\beta\omega'})\,S_{ab}(\boldsymbol{q},\omega')}{\omega-\omega'+\mathrm{i}0^+}.\tag{4.19}$$

Given S_{ab} at all frequencies, Eq. (4.19) reconstructs the response, real part included; Sect. 4.3 turns it into the Kramers–Kronig relations.

The factor $1-\mathrm{e}^{-\beta\omega}$ is one at $T=0$ and $\beta\omega$ for $|\omega|\ll T$. At $T=0$ the theorem reads $S_{aa}=-2\operatorname{Im} R_{aa}$ for $\omega>0$, and both sides vanish for $\omega<0$. For an electronic excitation in a detector the zero-temperature form is all we need: across the silicon gap of 1.12 eV at 300 K, $\mathrm{e}^{-\beta\omega}=2\times10^{-19}$. For $|\omega|\ll T$, $S_{aa}=-2T\operatorname{Im} R_{aa}/\omega$ grows with temperature. This

high-temperature relation is the classical form of the theorem, the one behind the thermal noise of a resistor, and it applies to the low-lying phonon and magnon excitations of Chaps. 5 and 7.

The thermal oscillator shows the two sides in closed form. Its response, Eq. (4.9), does not depend on temperature, and its imaginary part is $\mathcal{V}\,\mathrm{Im}\,R = -\pi\lambda_{\mathrm{osc}}^2[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$. Its structure factor, from Worked example 2.7, is $\mathcal{V}S = 2\pi\lambda_{\mathrm{osc}}^2[(n_{\mathrm{B}}(\omega_0)+1)\,\delta(\omega-\omega_0)+n_{\mathrm{B}}(\omega_0)\,\delta(\omega+\omega_0)]$ and does depend on temperature. The theorem connects them through $1/(1-\mathrm{e}^{-\beta\omega_0}) = n_{\mathrm{B}}(\omega_0)+1$ at $\omega=\omega_0$ and $1/(1-\mathrm{e}^{\beta\omega_0}) = -n_{\mathrm{B}}(\omega_0)$ at $\omega=-\omega_0$. The thermal populations appear in S alone. A temperature-independent response is compatible with a rate that grows with temperature, because the target's fluctuations grow while its net absorption does not.

For dark matter, Eq. (4.18) converts a response into a rate. The rate needs S, while most of the theory of solids, and every optical and electron-energy-loss measurement, delivers a response. Sect. 4.4 shows that the density response of the electrons is the inverse dielectric function, and the theorem then writes S as the energy-loss function times $2q^2/e^2$ [5, Eq. (6)], so that the rate includes screening. For silicon, S is computed in two ways. At the level of independent electrons, it is a sum over transitions between Kohn–Sham bands weighted by the density vertex $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ (Eq. (A.37); Sect. 26.2). With screening and local fields, a first-principles calculation computes R_{ee} in the random-phase approximation or with a kernel of time-dependent density-functional theory, and the theorem converts it (Sect. 4.6).

4.2.4 Positivity

A rate cannot be negative, and for one operator the structure factor is manifestly a sum of positive terms. With several operators the off-diagonal elements are complex and of either sign, and positivity becomes a statement about the matrix. Take any coefficients x_b and the combined operator $O_x = \sum_b x_b O_b$. Its transition sum is

$$\sum_{a,b} x_a^*\, S_{ab}(\boldsymbol{q},\omega)\, x_b = \frac{2\pi}{\mathcal{V}} \sum_{i,f} p_i \left|\sum_b x_b \langle f|O_b|i\rangle\right|^2 \delta(\omega - E_f + E_i) \geq 0,$$

since $\sum_a x_a^* \langle i|O_a^\dagger|f\rangle$ is the complex conjugate of $\sum_b x_b\langle f|O_b|i\rangle$.

Positivity of the response matrix

At every $\boldsymbol{q}$ and ω, $S_{ab}(\boldsymbol{q},\omega)$ is a Hermitian positive semidefinite matrix: $\sum_{ab} x_a^* S_{ab} x_b \geq 0$ for all x_b. In particular $S_{aa} \geq 0$ and

$$|S_{ab}(\boldsymbol{q},\omega)|^2 \leq S_{aa}(\boldsymbol{q},\omega)\, S_{bb}(\boldsymbol{q},\omega). \tag{4.20}$$

The inequality is the Cauchy–Schwarz inequality for a positive semidefinite matrix: every 2×2 principal submatrix has a nonnegative determinant (Eq. (C.53); Appendix C.7 gives the equivalent conditions). The rate Eq. (4.14) is nonnegative at every $\boldsymbol{q}$ for any set of couplings, since its integrand has the form $x^\dagger S x$ with $x_b = \tilde{V}_b$. The interference term between two couplings is bounded in magnitude by twice the geometric mean of their separate rates, and Chap. 8 uses this inequality to bound mixed electron–nuclear signals. Through Eq. (4.18), the matrix $\mathrm{i}(R_{ab} - R_{ba}^*)$ is positive semidefinite for $\omega > 0$, at any temperature, since there $1-\mathrm{e}^{-\beta\omega} > 0$. A target in equilibrium therefore absorbs power from any weak field at positive frequency. Eq. (4.8) showed this absorption for a single Hermitian operator; positivity extends the result to operators that are not Hermitian, such as $n_e(\boldsymbol{q})$, and to any combination of them. On the diagonal the condition reads $\mathrm{Im}\,R_{aa} \leq 0$ for $\omega > 0$, at any temperature. The

real part of R obeys no positivity constraint, and neither does the static response, which can have either sign off the diagonal. In practice positivity is a check: a computed S_{ab} with a negative eigenvalue at some $(\boldsymbol{q},\omega)$ contains an error.

4.2.5 Independent particles

When the electrons do not interact, the many-body states are occupation-number states of single-particle orbitals γ, and the structure factor is a sum over moves of one electron. The transition sum was worked out in Eq. (2.70), and multiplying by $2\pi/\mathcal{V}$,

$$S^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathcal{V}} \sum_{\gamma\neq\gamma'} \left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2 n_{\mathrm{F}}(E_\gamma-\mu_{\mathrm{ch}})\left[1-n_{\mathrm{F}}(E_{\gamma'}-\mu_{\mathrm{ch}})\right]\delta(\omega-E_{\gamma'}+E_\gamma), \quad (4.21)$$

with the spin included in γ. The elastic term, the many-body diagonal $f=i$ of Eq. (4.13), adds to S^0; it contains cross terms between different occupied orbitals, and it is not part of the single-particle sum, whose terms with $\gamma'=\gamma$ carry $n(1-n)=0$ at $T=0$. We write S^0 for the independent-particle structure factor; through Eq. (4.18) it is the imaginary part of the independent-particle response R^0 of Sect. 4.5. Each term of Eq. (4.21) is a single-particle transition form factor $f_{\gamma\to\gamma'}(\boldsymbol{q})$ of Eq. (2.17), squared, with the Pauli factors of the many-electron state. For a real target S^0 is the starting point. Built from Kohn–Sham orbitals it is the independent-particle structure factor of Eq. (A.37), and it misses the screening and the collective oscillation that the Coulomb interaction produces (Sects. 4.4 and 4.5). The free electron gas and a gas of atoms show what S^0 contains.

Worked example 4.3: The independent-particle structure factor of the electron gas

For free electrons at $T=0$ the orbitals are plane waves, $|\langle\boldsymbol{k}'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\boldsymbol{k}\rangle| = \delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$, and the occupations are those of the Fermi sea of Eq. (2.71). A transition moves an electron from $\boldsymbol{k}$ inside the sphere to $\boldsymbol{k}+\boldsymbol{q}$ outside it, leaving a hole behind, and deposits $(\boldsymbol{k}+\boldsymbol{q})^2/2m_e - k^2/2m_e = \boldsymbol{k}\cdot\boldsymbol{q}/m_e + q^2/2m_e$. With two spins and $\sum_{\boldsymbol{k}} \to \mathcal{V}\int \mathrm{d}^3k/(2\pi)^3$,

$$S^0(q,\omega) = 4\pi \int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\Theta(k_{\mathrm{F}}-k)\,\Theta(|\boldsymbol{k}+\boldsymbol{q}|-k_{\mathrm{F}})\,\delta\!\left(\omega - \frac{kqu}{m_e} - \frac{q^2}{2m_e}\right),$$

with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. The delta function fixes $u=u_0=(m_e\omega/q - q/2)/k$ with the weight $m_e/(kq)$. The angle must exist, $|u_0|\le 1$, which is $k \ge |m_e\omega/q - q/2|$. The final state must be empty, $k^2+2kqu_0+q^2 = k^2+2m_e\omega > k_{\mathrm{F}}^2$. With $\mathrm{d}^3k = 2\pi k^2\,\mathrm{d}k\,\mathrm{d}u$, the integral over k runs from the larger of the two lower limits, $k_{\min}$, to k_{F}:

$$S^0(q,\omega) = \frac{m_e}{\pi q}\int_{k_{\min}}^{k_{\mathrm{F}}} k\,\mathrm{d}k = \frac{m_e}{2\pi q}\left(k_{\mathrm{F}}^2 - k_{\min}^2\right), \qquad k_{\min} = \max\!\left(\left|\frac{m_e\omega}{q}-\frac{q}{2}\right|,\ \left(k_{\mathrm{F}}^2-2m_e\omega\right)^{1/2}\right), \quad (4.22)$$

for $k_{\min} < k_{\mathrm{F}}$ and zero otherwise, with the square root set to zero when $2m_e\omega > k_{\mathrm{F}}^2$. Through Eq. (4.18), Eq. (4.22) fixes the imaginary part of the independent-particle response R^0 of the gas, the Lindhard function of Sect. 4.5 [7].

The weight is confined to the particle–hole continuum (Fig. 4.3a). Its upper edge, $\omega_+ = qv_{\mathrm{F}} + q^2/2m_e$, is an electron at the top of the sea kicked along $\boldsymbol{q}$. Its lower edge is zero for $q < 2k_{\mathrm{F}}$, since an electron at the Fermi surface can be moved to an empty state just outside with almost no energy deposited. For $q > 2k_{\mathrm{F}}$ even an electron moving against $\boldsymbol{q}$ at v_{F} ends up outside the sea, and the edge rises to $\omega_- = q^2/2m_e - qv_{\mathrm{F}}$. Inside,

Eq. (4.22) has a linear and a parabolic form. Below the curve $\omega = qv_\mathrm{F} - q^2/2m_e$, which exists for $q < 2k_\mathrm{F}$, the Pauli condition decides, $k_\mathrm{min}^2 = k_\mathrm{F}^2 - 2m_e\omega$, and

$$S^0(q,\omega) = \frac{m_e^2\,\omega}{\pi q}.$$

The weight grows linearly with ω, because only electrons within a shell of thickness $\sim m_e\omega/k_\mathrm{F}$ below the Fermi surface can be promoted. Above the curve the angle decides, and $S^0 = (m_e/2\pi q)[k_\mathrm{F}^2 - (m_e\omega/q - q/2)^2]$, a parabola in ω. The two forms meet at $S^0 = m_e(2k_\mathrm{F} - q)/2\pi$; at $q = k_\mathrm{F}/2$ this value is $0.239\,m_e k_\mathrm{F}$, at $\omega = 0.75\,E_\mathrm{F}$. For $q \gg k_\mathrm{F}$ only the parabola is left. It is centered on the free recoil $\omega = q^2/2m_e$ of Worked example 2.3, with the half-width qv_F of the Doppler shift from the electron's motion in the sea: the parabola is the result of the impulse approximation, in which the struck electron recoils as if free (Fig. 4.3b).

The total weight and the first moment check the result. Integrating the sum form over ω removes the delta function, and what is left counts the electrons that can take the momentum q, $\int \mathrm{d}\omega\, S^0 = 2\pi n_e \times (3q/4k_\mathrm{F} - q^3/16k_\mathrm{F}^3)$ for $q < 2k_\mathrm{F}$ and $2\pi n_e$ above. By Eq. (4.17) the static structure factor of the free gas is the unblocked fraction Eq. (2.72): Pauli exclusion is the hole around each electron that suppresses the long-wavelength fluctuations, $S^\mathrm{st} \to 3q/4k_\mathrm{F}$. The first moment weights each transition with its energy $\Delta_{\boldsymbol{k}} = (\boldsymbol{k}+\boldsymbol{q})^2/2m_e - k^2/2m_e$. With the occupations $n_{\boldsymbol{k}} = \Theta(k_\mathrm{F} - k)$, the sum $\sum_{\boldsymbol{k}s} n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}})\Delta_{\boldsymbol{k}}$ splits in two. The part with $n_{\boldsymbol{k}}n_{\boldsymbol{k}+\boldsymbol{q}}$ changes sign under $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$, which exchanges the two occupations and reverses $\Delta_{\boldsymbol{k}}$, and it vanishes. In the part with $n_{\boldsymbol{k}}$ alone, $\Delta_{\boldsymbol{k}} = \boldsymbol{k}\cdot\boldsymbol{q}/m_e + q^2/2m_e$, the term in $\boldsymbol{k}\cdot\boldsymbol{q}$ averages to zero over the sphere, and $N_e q^2/2m_e$ is left. Multiplying by $2\pi/\mathcal{V}$ gives $\int \mathrm{d}\omega\,\omega\, S^0 = \pi n_e q^2/m_e$ at every q, the f-sum rule Eq. (A.38) that Sect. 4.3 derives for any target.

For dark matter at the silicon valence density, $k_\mathrm{F} = 3.57\,\mathrm{keV}$, $E_\mathrm{F} = 12.5\,\mathrm{eV}$ and $v_\mathrm{F} = 7.0\times10^{-3}$ (Sect. 2.5), and the energy deposit is at most qv_max with $v_\mathrm{max} = 2.65\times10^{-3} = 0.379\,v_\mathrm{F}$. The dark matter samples only the corner of the continuum where $\omega \le qv_\mathrm{max}$. For $q < 2k_\mathrm{F}(1 - v_\mathrm{max}/v_\mathrm{F}) = 1.24\,k_\mathrm{F} = 4.43\,\mathrm{keV}$ this corner lies in the linear region, and on its upper edge $S^0 = m_e^2 v_\mathrm{max}/\pi = 2.2\times10^8\,\mathrm{eV}^2$, or $0.121\,m_e k_\mathrm{F}$, independent of q. A slow probe excites only electrons near the Fermi surface. A real semiconductor differs in the corner that matters most: its gap removes all weight below 1.17 eV, and the Coulomb interaction moves much of the weight at small q into the plasmon (Sect. 4.5).

A gas of atoms is the other independent-particle target, and the target of Sect. 2.2. The many-body definition produces its structure factor directly.

Worked example 4.4: Hydrogen from the response matrix

Take N_at hydrogen atoms at fixed positions $\boldsymbol{R}_\ell$, far enough apart that their wavefunctions do not overlap, in a box of volume $\mathcal{V}$, with number density $n_T = N_\mathrm{at}/\mathcal{V}$. At any temperature far below the 10.2 eV of the lowest excitation, every atom is in its 1s state. The density operator is a sum over atoms, $n_e(\boldsymbol{q}) = \sum_\ell \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_\ell}$, with $\boldsymbol{r}_\ell$ the position of the electron of atom ℓ. Acting on the ground state it changes one atom at a time. A final state in which atom ℓ is in the level f and all others are in 1s is reached by the term of atom ℓ alone, with the matrix element $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\ell} f_{1s\to f}(\boldsymbol{q})$, since $\boldsymbol{r}_\ell = \boldsymbol{R}_\ell + \boldsymbol{r}$ with $\boldsymbol{r}$ measured from the nucleus. The phase drops out of the square, every atom contributes the same,

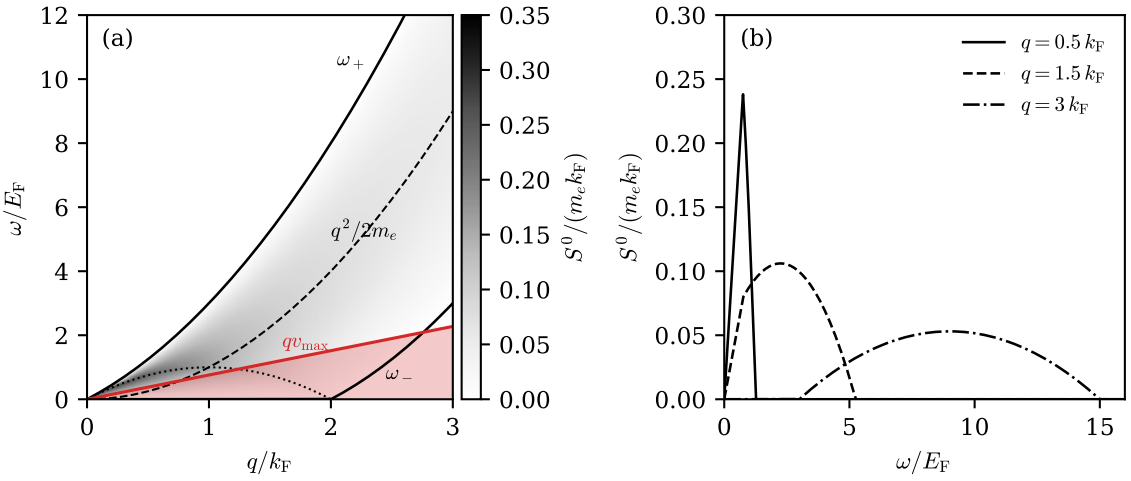


Figure 4.3: Dynamic structure factor $S^0(q,\omega)$ of the non-interacting electron gas at zero temperature, in units of $m_e k_F$, against the momentum transfer q in units of the Fermi wavevector k_F and the energy ω in units of the Fermi energy $E_F = k_F^2/2m_e$. (a) Gray scale: S^0 inside the particle–hole continuum, bounded above by $\omega_+ = qv_F + q^2/2m_e$ and below by $\omega_- = q^2/2m_e - qv_F$ for $q > 2k_F$ (solid). Below the dotted curve $\omega = qv_F - q^2/2m_e$, $S^0 = m_e^2\omega/\pi q$ grows linearly in ω; above it S^0 is a parabola in ω centered on the free recoil $q^2/2m_e$ (dashed). Red: the energies $\omega \le qv_{\max}$ that dark matter with the maximum halo speed $v_{\max} = 2.65 \times 10^{-3}$ can deposit, for the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ ($k_F = 3.57\,\mathrm{keV}$, $E_F = 12.5\,\mathrm{eV}$, $v_{\max} = 0.379\,v_F$). (b) S^0 against ω at $q = 0.5$, 1.5 and $3\,k_F$. At $q = 0.5\,k_F$ the triangle peaks at $0.239\,m_e k_F$ at $\omega = 0.75\,E_F$; at $q = 3\,k_F$ the parabola spans $\omega_- = 3\,E_F$ to $\omega_+ = 15\,E_F$. Every curve integrates to $2\pi n_e$ times the fraction of the Fermi sea that can take the momentum q, and its first moment is $\pi n_e q^2/m_e$.

and Eq. (4.13) becomes

$$S(\boldsymbol{q},\omega) = 2\pi\, n_T \sum_{f\neq 1s} |f_{1s\to f}(\boldsymbol{q})|^2\, \delta(\omega - E_f + E_{1s}) + \frac{2\pi}{\mathcal{V}}\, F_{1s}(q)^2 \left|\sum_\ell \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\ell}\right|^2 \delta(\omega), \quad (4.23)$$

with F_{1s} the elastic form factor Eq. (2.21). The inelastic term is the structure factor that Sect. 2.2 identified by comparing rates, and inserted into Eq. (2.67) it gives n_T times the $\sigma_T v$ of one atom, Eq. (2.18). Each atom scatters on its own, and the rate counts atoms; the angle and halo averages of Sect. 2.2 then produce the spectrum Eq. (2.20). For an atom with several electrons, Eq. (4.21) supplies what Sect. 2.2 added by hand: the initial orbitals must be occupied and the final ones empty.

In the elastic term the atoms act together. At a fixed $\boldsymbol{q}$ its weight depends on the arrangement of the nuclei. For atoms at random positions the cross terms of $|\sum_\ell \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\ell}|^2$ average to zero at $q \neq 0$ and the sum is N_{at}. In a crystal the phases add at the reciprocal-lattice vectors instead. With fixed nuclei the elastic term deposits nothing; free atoms would take up the recoil of the whole atom.

With the numbers of Worked example 2.2, at $q = q_0$ the integral of the inelastic term over ω is $2\pi n_T(1 - F_{1s}^2) = 2\pi n_T \times 0.59$, of which the $2p$ level takes 0.061, about a tenth. The elastic term of the random gas carries $2\pi n_T \times 0.41$, and by Eq. (4.17) the inelastic and elastic terms add to $S^{\mathrm{st}}(q_0) = 1$. A gas of independent atoms scatters incoherently at every q, and the form factor decides only how much of each scattering excites the atom.

4.3 Causality and sum rules

No calculation has the exact excited states of silicon, yet one number about its dynamic structure factor is known exactly at every momentum transfer: the energy-weighted integral $\int \mathrm{d}\omega\, \omega\, S(\boldsymbol{q},\omega)$ equals $\pi n_e q^2/m_e$, fixed by the electron density alone. Another exact statement ties the real part of every response function to its imaginary part. The energy-weighted integral follows from a commutator, the link between real and imaginary parts from causality, and neither depends on the interactions, the temperature or the crystal structure. The treatment follows Refs. [1, Sects. 3.2–3.4] and [4, Chaps. 2 and 3].

4.3.1 Kramers–Kronig relations

The spectral form Eq. (4.19) already reconstructs all of R from S, and for a diagonal element it becomes a relation between the real and imaginary parts of R itself. By the fluctuation–dissipation theorem Eq. (4.18), the numerator $(1 - \mathrm{e}^{-\beta\omega'})S_{aa}(\boldsymbol{q},\omega')$ is $-2\,\mathrm{Im}\, R_{aa}(\boldsymbol{q},\omega')$, a real function of ω', and

$$R_{aa}(\boldsymbol{q},\omega) = -\frac{1}{\pi}\int_{-\infty}^{\infty} \mathrm{d}\omega'\, \frac{\mathrm{Im}\, R_{aa}(\boldsymbol{q},\omega')}{\omega - \omega' + \mathrm{i}0^+}.$$

The Sokhotski–Plemelj formula Eq. (2.58) splits the denominator into a principal value and $-\mathrm{i}\pi\,\delta(\omega - \omega')$. The delta function returns $\mathrm{Im}\, R_{aa}(\omega)$ on the right, as it must, and the principal value is the real part.

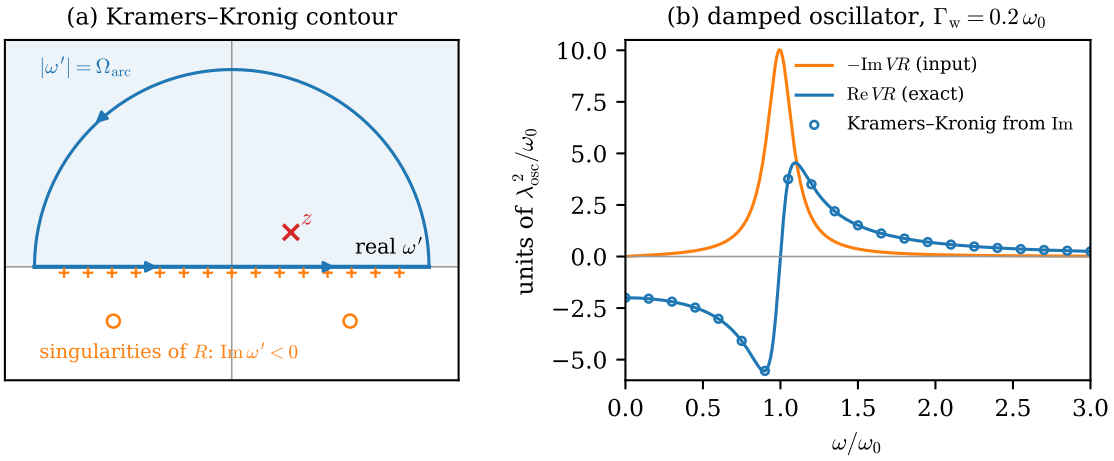


Figure 4.4: The Kramers–Kronig relations. (a) The contour in the complex frequency plane ω': the real axis, closed by a semicircle of radius $\Omega_{\mathrm{arc}} \to \infty$ in the upper half plane (shaded), where the retarded response R is analytic. The only singularity enclosed is the pole of $1/(\omega' - z)$ at a point z of the upper half plane (cross). The singularities of R, the poles at transition energies that merge into a branch cut and the resonance poles of its continuation (schematic), all lie below the axis. Letting z approach the real axis yields the relations. (b) Test on the damped oscillator with energy damping rate $\Gamma_{\mathrm{w}} = 0.2\,\omega_0$ and coupling $\lambda_{\mathrm{osc}}(d + d^\dagger)$. The real part of $\mathcal{V}R$ computed from $-\mathrm{Im}\,\mathcal{V}R$ alone, with the folded relation for an even real part and an odd imaginary part (circles), agrees with the exact real part (line) to better than $10^{-11}\,\lambda^2_{\mathrm{osc}}/\omega_0$. At $\omega = 0$ both give $-2\lambda^2_{\mathrm{osc}}/\omega_0$.

Kramers–Kronig relations

For a response function analytic in the upper half plane and falling off at least as $1/|\omega|$,

$$\begin{aligned} \mathrm{Re}\,R_{aa}(\boldsymbol{q},\omega) &= \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Im}\,R_{aa}(\boldsymbol{q},\omega')}{\omega' - \omega}, \\ \mathrm{Im}\,R_{aa}(\boldsymbol{q},\omega) &= -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Re}\,R_{aa}(\boldsymbol{q},\omega')}{\omega' - \omega}. \end{aligned} \tag{4.24}$$

The dissipative part of a response, at all frequencies, fixes its reactive part at each frequency, and conversely.

The relations Eq. (4.24) are Eq. (C.36) [8, 9]. The route through Eq. (4.19) yields the relation for the real part from the Lehmann form. The pair of relations needs less than the Lehmann form, just the two properties of Sect. 4.1: R is analytic in the upper half plane, and it falls off at large $|z|$, as $1/z^2$ for a density by Eq. (4.11). Appendix C.5 derives them from these two properties alone, by closing the integral of $R(\omega')/(\omega' - z)$ over a large semicircle in the upper half plane (Fig. 4.4a). Since that argument refers to no eigenstates, the relations hold for a measured response, for a model such as the damped oscillator of Worked example 4.1, and for any approximation that respects causality. Fig. 4.4b rebuilds the real part of the oscillator response from its imaginary part alone. For an off-diagonal element the same argument applies to the analytic function R_{ab}, with the combination $\mathrm{i}(R_{ab} - R^*_{ba})$ of Eq. (4.18) in the role of the absorptive part.

Most responses have a symmetry that folds the integral onto positive frequencies. For the

density response of a target with inversion or time-reversal symmetry, the real part is even in ω and the imaginary part odd (Sect. 4.1). Splitting the integral at zero and combining $1/(\omega' - \omega) + 1/(\omega' + \omega) = 2\omega'/(\omega'^2 - \omega^2)$,

$$\operatorname{Re} R_{ee}(\boldsymbol{q}, \omega) = \frac{2}{\pi} \operatorname{PV} \int_0^\infty \mathrm{d}\omega' \, \frac{\omega' \operatorname{Im} R_{ee}(\boldsymbol{q}, \omega')}{\omega'^2 - \omega^2}, \tag{4.25}$$

which is Eq. (C.37). At $\omega = 0$, with $-2 \operatorname{Im} R_{ee} = (1 - \mathrm{e}^{-\beta\omega}) S$,

$$R_{ee}(\boldsymbol{q}, 0) = -\frac{1}{\pi} \int_0^\infty \mathrm{d}\omega \, \frac{(1 - \mathrm{e}^{-\beta\omega}) \, S(\boldsymbol{q}, \omega)}{\omega}. \tag{4.26}$$

The static response is an inverse moment of the structure factor, and it is negative because $S \geq 0$. At $T = 0$, inserting Eq. (4.13) returns the static limit of Sect. 4.1, $\mathscr{V} R_{ee}(0) = -2 \sum_f |\langle f | n_e(\boldsymbol{q}) | 0 \rangle|^2 / (E_f - E_0)$. Transitions at low energy dominate, since they are weighted by $1/\omega$: a target with a large gap is hard to deform. At large ω, on the other hand, the denominator of Eq. (4.25) becomes $-\omega^2$, and $\operatorname{Re} R_{ee} \to -(2/\pi\omega^2) \int_0^\infty \mathrm{d}\omega' \, \omega' \operatorname{Im} R_{ee}$. The tail Eq. (4.11) fixes the same coefficient by a double commutator. Equating the two coefficients fixes the energy-weighted integral of the absorptive part by an equal-time average, and the resulting relation is the first of the sum rules.

4.3.2 Moments from commutators

The k-th frequency moment of S_{ab} weights each transition $i \to f$ with ω_{fi}^k, where $\omega_{fi} = E_f - E_i$. The same power appears in a nested commutator. For any operator X, $\langle f | [H, X] | i \rangle = \omega_{fi} \langle f | X | i \rangle$, and k commutators with H give ω_{fi}^k. Take operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, as for every density, and write $[H, O_b]_k$ for O_b commuted k times with H. In the average of $[O_a^\dagger, [H, O_b]_k]$ over any state diagonal in the energy basis, with weights p_i, we insert a complete set $|f\rangle$ between the two operators. The term $O_a^\dagger [H, O_b]_k$ contributes $\sum_{i,f} p_i \, \omega_{fi}^k \langle i | O_a^\dagger | f \rangle \langle f | O_b | i \rangle$, the k-th moment of $\mathscr{V} S_{ab}(\boldsymbol{q}, \omega)/2\pi$ by Eq. (4.13). The term $[H, O_b]_k O_a^\dagger$ contributes $\sum_{i,f} p_i \, \omega_{if}^k \langle i | O_b | f \rangle \langle f | O_a^\dagger | i \rangle$. Since $O_b = O_b(-\boldsymbol{q})^\dagger$ and $O_a^\dagger = O_a(-\boldsymbol{q})$, this second contribution is $(-1)^k$ times the k-th moment of $\mathscr{V} S_{ba}(-\boldsymbol{q}, \omega)/2\pi$. Altogether,

$$\frac{1}{\mathscr{V}} \langle [O_a(\boldsymbol{q})^\dagger, [H, O_b(\boldsymbol{q})]_k] \rangle = \int_{-\infty}^\infty \frac{\mathrm{d}\omega}{2\pi} \, \omega^k \, [S_{ab}(\boldsymbol{q}, \omega) - (-1)^k S_{ba}(-\boldsymbol{q}, \omega)]. \tag{4.27}$$

Since the derivation used only the stationarity of the state, Eq. (4.27) holds in the Gibbs state at any temperature, in any eigenstate, and with any interactions. The integral runs over negative ω too, where a thermal target gives up energy. For $k = 0$ the left side is an equal-time commutator, and for densities, which commute with one another, it vanishes. For densities the first nonvanishing case is $k = 1$.

First-moment identity

For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$ and any stationary state,

$$\mathscr{M}_{ab}^1(\boldsymbol{q}) \equiv \frac{1}{\mathscr{V}} \langle [O_a(\boldsymbol{q})^\dagger, [H, O_b(\boldsymbol{q})]] \rangle = \int_{-\infty}^\infty \frac{\mathrm{d}\omega}{2\pi} \, \omega \, [S_{ab}(\boldsymbol{q}, \omega) + S_{ba}(-\boldsymbol{q}, \omega)]. \tag{4.28}$$

The energy-weighted spectrum, summed over $\pm\boldsymbol{q}$, is an equal-time double commutator.

The first-moment matrix $\mathscr{M}_{ab}^1$ is the coefficient of $1/z^2$ in the tail Eq. (4.11), now read as a statement about S. On the diagonal, for a target with $S_{aa}(-\boldsymbol{q}, \omega) = S_{aa}(\boldsymbol{q}, \omega)$, the identity

reads $\int \mathrm{d}\omega\, \omega\, S_{aa} = \pi \mathscr{M}^1_{aa}$. The identity is useful because a double commutator is often far simpler than the states it sums over. The oscillator of Worked example 4.1 is the smallest example: $[O,[H,O]] = 2\lambda^2_{\mathrm{osc}}\omega_0$ is a number, and the first moment of its structure factor, $2\pi\lambda^2_{\mathrm{osc}}\omega_0$, does not depend on temperature, the result found from the populations in Worked example 2.7. Chap. 8 builds the higher moments into matrices for several operators.

4.3.3 The f-sum rule

For the electron density the double commutator is a number. The Hamiltonian of electrons and nuclei of Sect. 2.3 is the kinetic energy of the particles plus a potential energy that depends on their positions alone, and so do any external static potentials. The density $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ commutes with every position and with every nuclear momentum. Only the electron kinetic energy is left in $[H, n_e(\boldsymbol{q})]$. For one electron, $[\boldsymbol{p}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}] = \boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and $[\boldsymbol{p}^2, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}] = \boldsymbol{p}\cdot\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\boldsymbol{q}\cdot\boldsymbol{p}$. Summed over electrons,

$$[H, n_e(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q}), \qquad \boldsymbol{j}_e(\boldsymbol{q}) = \sum_j \frac{1}{2}\Big\{\frac{\boldsymbol{p}_j}{m_e}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\Big\}, \tag{4.29}$$

exactly, with $\{A,B\} = AB + BA$. Eq. (4.29) is the continuity equation in operator form. In the Heisenberg picture $\partial_t n_e(\boldsymbol{q},t) = \mathrm{i}[H, n_e(\boldsymbol{q},t)] = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},t)$, the transform with $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ of $\partial_t \hat{n}_e(\boldsymbol{r}) + \nabla\cdot\boldsymbol{j}_e(\boldsymbol{r}) = 0$ for the density operator of Sect. 4.2 and the current density in position space: density at $\boldsymbol{q}$ changes only because electrons carry it away. The symmetrized $\boldsymbol{j}_e(\boldsymbol{q})$ is what the literature calls the paramagnetic current, the current of the electrons when no vector potential acts.

The outer commutator, with $n_e(\boldsymbol{q})^\dagger$, is a number. Conjugating by the plane wave shifts the momentum, $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{p}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \boldsymbol{p} + \boldsymbol{q}$, so $[\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}, \boldsymbol{p}] = \boldsymbol{q}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and each electron contributes q^2/m_e to $[n_e(\boldsymbol{q})^\dagger, \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})]$ (Appendix D.4). The commutator is $N_e q^2/m_e$ in every state, and $\mathscr{M}^1_{ee} = n_e q^2/m_e$.

f-sum rule

For electrons that move in local potentials, at any temperature and with any interactions,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega\, \omega\, S(\boldsymbol{q},\omega) = \frac{\pi\, n_e\, q^2}{m_e} \qquad \big(S(-\boldsymbol{q},\omega) = S(\boldsymbol{q},\omega)\big), \tag{4.30}$$

and in general the same for the average of S over $\boldsymbol{q}$ and $-\boldsymbol{q}$. The energy-weighted spectrum of density fluctuations counts the electrons.

Eq. (4.30) is Eq. (A.38) [1, Sect. 3.4]. The number n_e counts every electron, core electrons included, and the nuclei may move. The normalization is π over the whole real line. Many-body texts write the rule per particle and over positive frequencies, and the conversion is one line. At $T = 0$, S vanishes for $\omega < 0$, and dividing by $2\pi n_e$ leads to $\int_0^\infty \mathrm{d}\omega\, \omega\, S/(2\pi n_e) = q^2/2m_e$: per electron, and without the factor 2π of Eq. (4.13), the first moment is the free recoil energy [4, Chap. 2]. At $T > 0$, detailed balance $S(\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q},\omega)$ folds the negative frequencies onto the positive ones, $\int_0^\infty \mathrm{d}\omega\, \omega\, (1 - \mathrm{e}^{-\beta\omega})\, S = \pi n_e q^2/m_e$. By the fluctuation–dissipation theorem the integrand is $-2\omega\, \mathrm{Im}\, R_{ee}$, and the rule fixes the coefficient of the large-ω limit of Eq. (4.25).

The rule has a direct reading at large q. Divided by the total weight $2\pi n_e S^{\mathrm{st}}(\boldsymbol{q})$ of Eq. (4.17), it gives the mean energy deposited at $T = 0$, $q^2/[2m_e S^{\mathrm{st}}(\boldsymbol{q})]$. Where $S^{\mathrm{st}} \to 1$ and the dark matter scatters on one electron at a time, the mean energy deposited tends to the free recoil energy, however strongly the electron is bound. Binding moves the weight

around, broadening it as in the impulse approximation of Worked example 4.3, but binding cannot shift the center of the weight. At small q, where S^{st} is suppressed, the mean energy lies far above $q^2/2m_e$, and in the interacting electron gas the plasmon carries all of the sum (Sect. 4.5).

The right side of Eq. (4.30) defines the frequency scale of the electrons. With $e^2 = 4\pi\alpha$,

$$\frac{\pi \boldsymbol{n}_e q^2}{m_e} = \frac{\pi q^2}{e^2}\,\omega_p^2, \qquad \omega_p^2 = \frac{e^2 \boldsymbol{n}_e}{m_e}, \tag{4.31}$$

where ω_p is the plasma frequency, the frequency at which the electrons, treated as free, would oscillate together against a uniform positive background (Sect. 4.5). For the valence density of silicon, $\boldsymbol{n}_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, $\omega_p = 16.6\,\mathrm{eV}$; for all 14 electrons per atom, $7.00 \times 10^{23}\,\mathrm{cm}^{-3}$, it is 31.1 eV. Sect. 4.4 converts Eq. (4.30) into the form $\int_0^\infty \mathrm{d}\omega\,\omega\,L = (\pi/2)\,\omega_p^2$ for the loss function, the form that a computed or measured spectrum is usually tested against.

4.3.4 Nonlocal potentials and the f-sum rule

The derivation of Eq. (4.30) needed one hypothesis: every potential energy in H is a function of positions. The Hamiltonian of electrons and nuclei satisfies the hypothesis, but the effective Hamiltonians with which solids are computed often do not. A pseudopotential (Sect. 24.1) replaces the nucleus and the core electrons by a local potential and a nonlocal part V_{NL}, the sum of projections Eq. (24.9), which acts on a valence wavefunction through its values over a whole core region rather than at one point. The Fock exchange operator is nonlocal in the same sense. A nonlocal operator does not commute with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, the commutator $[H, n_e(\boldsymbol{q})]$ acquires $[V_{\mathrm{NL}}, n_e(\boldsymbol{q})]$ besides $\boldsymbol{q}\cdot\boldsymbol{j}_e$, and

$$\mathscr{M}^1_{ee}(\boldsymbol{q}) = \frac{\boldsymbol{n}_e q^2}{m_e} + \frac{1}{\mathscr{V}}\langle [n_e(\boldsymbol{q})^\dagger, [V_{\mathrm{NL}}, n_e(\boldsymbol{q})]]\rangle, \tag{4.32}$$

with $\boldsymbol{n}_e$ now the density of the valence electrons that the calculation keeps. The exact rule still holds, since it belongs to the full Hamiltonian. The pseudo-Hamiltonian is built to reproduce the valence states outside the cores, and its commutators need not agree with those of the full Hamiltonian. A pseudo-wavefunction is smooth where the true valence state oscillates to stay orthogonal to the core states, and V_{NL} carries part of the effect of that oscillation. The nonlocal term of Eq. (4.32) is a sum of projections and costs almost nothing to evaluate. Sect. 24.2 writes it for one electron, Eq. (24.13), and finds that in a representative silicon calculation this term and the truncated band sum together put the computed sum 12 to 19 percent below the local value at small q. A sum-rule test of a pseudopotential calculation therefore compares with Eq. (4.32), not with Eq. (4.30). At $q \to 0$ the same algebra turns the momentum into the velocity $\mathrm{i}[H, \boldsymbol{r}] = \boldsymbol{p}/m_e + \mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$, the form that Ismail-Beigi, Chang and Louie derived for the coupling of a nonlocal potential to a long-wavelength field [10].

Relativistic electrons break the hypothesis in a different way. To order $1/m_e^2$, an electron in a one-body potential $U_1(\boldsymbol{r})$ has three corrections, derived in Appendix C.15: the mass–velocity term $-\boldsymbol{p}^4/8m_e^3$, the Darwin term $\nabla^2 U_1/8m_e^2$ and the spin–orbit term $\boldsymbol{\sigma}\cdot(\nabla U_1 \times \boldsymbol{p})/4m_e^2$. The Darwin and spin–orbit terms drop out of the double commutator, and the mass–velocity term does not (Appendix D.4). In an atom, or in a crystal of cubic symmetry,

$$\mathscr{M}^1_{ee}(\boldsymbol{q}) = \frac{\boldsymbol{n}_e q^2}{m_e}\Big[1 - \frac{5}{3}\,\frac{\bar{K}}{m_e} - \frac{q^2}{4m_e^2}\Big], \tag{4.33}$$

where $\bar{K}$ is the mean kinetic energy per electron. A moving electron is heavier, and the f-sum, which counts electrons weighted by $1/m_e$, is reduced by the relative order $\bar{K}/m_e$.

For silicon the virial theorem makes the total kinetic energy of the atom minus its total energy, 7.86 keV from the Hartree–Fock value of −288.854 hartree [11], so $\bar{K} = 561\,\mathrm{eV}$ and $(5/3)\bar{K}/m_e = 1.8 \times 10^{-3}$, or 0.026 electrons per atom. The two 1s electrons dominate: each has a kinetic energy near $Z^2\alpha^2 m_e/2 = 2.67\,\mathrm{keV}$, which reduces its own share of the sum by 8.7×10^{-3}. The term in q^2 is 2.7×10^{-3} at the 53 keV that a 10 MeV particle can transfer at most. The two-body Breit terms, which depend on the momenta of pairs of electrons, contribute at the same relative order, and we do not estimate them. All of these corrections lie below the accuracy of any calculation of S for dark matter. They do mean, however, that an all-electron relativistic calculation with positive-energy states alone should not reproduce Eq. (4.30) to better than a few parts in 10^3. The expansion in $1/m_e$ describes positive-energy states alone. The full Dirac Hamiltonian is linear in the momentum, its double commutator with the density vanishes, and the f-sum is then carried by the creation of electron–positron pairs, at energies near $2m_e$.

The nuclei obey the same algebra as the electrons. With the density $n_\kappa(\boldsymbol{q}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ of the nuclei of species κ, their mass M_κ and number density $\boldsymbol{n}_\kappa$ (Sect. 2.3), the kinetic energy of the nuclei gives $[H, n_\kappa(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_\kappa(\boldsymbol{q})$, with a current $\boldsymbol{j}_\kappa$ built like $\boldsymbol{j}_e$ from nuclear momenta and M_κ. A current of one species contains only that species' momenta, and it commutes with the density of every other species. The mixed double commutators therefore vanish.

Species-resolved f-sum rule

For electrons and nuclei in local potentials, with $s, s' \in \{e, \kappa\}$ and m_s equal to m_e or M_κ,

$$\mathscr{M}^1_{ss'}(\boldsymbol{q}) = \delta_{ss'}\,\frac{q^2\,\boldsymbol{n}_s}{m_s}. \tag{4.34}$$

Each species has its own f-sum, fixed by its density and mass, and the energy-weighted electron–nucleus interference vanishes.

The nuclear rule, $\int \mathrm{d}\omega\,\omega\,S_{\kappa\kappa} = \pi\boldsymbol{n}_\kappa q^2/M_\kappa$, makes the mean energy deposited on a nucleus equal to its free recoil energy $q^2/2M_\kappa$ wherever S^{st} of the nuclei is one, the incoherent limit of the lattice response of Chaps. 5 and 6. The mixed rule has a less obvious consequence. For a target with inversion symmetry, $S_{\kappa e}(-\boldsymbol{q},\omega) = S_{\kappa e}(\boldsymbol{q},\omega)$, which Hermiticity makes $S_{e\kappa}(\boldsymbol{q},\omega)^*$, and Eq. (4.28) with Eq. (4.34) reads $\int \mathrm{d}\omega\,\omega\,\mathrm{Re}\,S_{e\kappa}(\boldsymbol{q},\omega) = 0$. At $T = 0$ the integral runs over $\omega > 0$ alone, and the real part of the electron–nucleus interference must change sign: the electrons follow a recoiling nucleus in phase at some energies and against it at others. A computed interference spectrum of one sign, for the Migdal effect for instance (Chap. 8), is wrong. Finally, for the total charge density $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ of Sect. 2.3, the diagonal terms add: $\mathscr{M}^1_{ZZ} = q^2[\boldsymbol{n}_e/m_e + \sum_\kappa Z_\kappa^2\boldsymbol{n}_\kappa/M_\kappa]$. For silicon, with $Z_\kappa = 14$ and all 14 electrons per atom, the nuclear share is $14\,m_e/M_\kappa = 2.73 \times 10^{-4}$. A probe of the total charge, such as a dark photon, sees an f-sum set by the electrons to that accuracy.

4.3.5 Oscillator strengths

For a single atom the f-sum rule becomes a statement about its spectral lines. Take an atom with Z electrons in its ground state $|0\rangle$. Its double commutator is Zq^2/m_e, and inserting the excited states $|n\rangle$ as in Eq. (4.27) turns it into $\sum_n (E_n - E_0)\big[|\langle n|n_e(\boldsymbol{q})|0\rangle|^2 + |\langle n|n_e(-\boldsymbol{q})|0\rangle|^2\big]$. For a ground state with inversion symmetry the two terms are equal, and

$$\sum_n (E_n - E_0)\,\big|\langle n|n_e(\boldsymbol{q})|0\rangle\big|^2 = \frac{Zq^2}{2m_e}, \tag{4.35}$$

at every $\boldsymbol{q}$. For the gas of atoms of Worked example 4.4, whose structure factor is $2\pi n_T$ times the sum of these transition weights, Eq. (4.35) is Eq. (4.30) with $n_e = Z n_T$. As $q \to 0$, the matrix element to an excited state starts at first order in q, $\langle n|n_e(\boldsymbol{q})|0\rangle = \mathrm{i}\boldsymbol{q}\cdot\langle n|\sum_j \boldsymbol{r}_j|0\rangle$, since the zeroth order vanishes by orthogonality. The dipole matrix elements that remain define the oscillator strengths,

$$f_{n0}^{\mathrm{osc}} = 2m_e\,(E_n - E_0)\left|\langle n|\sum_j \hat{\boldsymbol{q}}\cdot\boldsymbol{r}_j|0\rangle\right|^2, \tag{4.36}$$

and Eq. (4.35) divided by $q^2/2m_e$ becomes the Thomas–Reiche–Kuhn rule [12–14],

$$\sum_n f_{n0}^{\mathrm{osc}} = Z, \tag{4.37}$$

where the sum includes the continuum, as an integral of the oscillator-strength density $\mathrm{d}f^{\mathrm{osc}}/\mathrm{d}\omega$ over the energy $\omega = E_n - E_0$ of the absorbed quantum. In the classical picture an atom was a set of electrons bound by springs, and f_{n0}^{osc} counted how many of them oscillated at the frequency of line n. The Thomas–Reiche–Kuhn rule says that the lines of an atom share exactly Z of these classical electrons. The rule is used as a check. A table of computed or measured oscillator strengths, or of photoabsorption cross-sections, must add up to Z, and a set of ionization form factors (Eq. (2.22)) must satisfy Eq. (4.35) at each q, a test of the normalization of the continuum.

Worked example 4.5: The Thomas–Reiche–Kuhn sum of hydrogen

For hydrogen, $Z = 1$, and the oscillator strengths from 1s are known in closed form [15, Sect. 61]. The dipole operator reaches only p states. For the bound levels,

$$f_{np,1s}^{\mathrm{osc}} = \frac{2^8\, n^5\,(n-1)^{2n-4}}{3\,(n+1)^{2n+4}},$$

which is 0.4162 for 2p, and 0.0791, 0.0290 and 0.0139 for 3p to 5p. For large n, $[(n-1)/(n+1)]^{2n} \to \mathrm{e}^{-4}$ and $f_{np,1s}^{\mathrm{osc}} \to (2^8/3)\,\mathrm{e}^{-4}/n^3$. The levels crowd toward the binding energy $E_B = \alpha^2 m_e/2 = 13.6\,\mathrm{eV}$ of Sect. 2.2 with spacing $2E_B/n^3$, and their strength per unit energy tends to $(2^7/3)\,\mathrm{e}^{-4}/E_B = 0.782/E_B$. The continuum starts at the same value. With the Coulomb parameter of Sect. 2.2 for the ejected electron, $\xi_{\mathrm{C}} = \alpha m_e/p = [E_B/(\omega - E_B)]^{1/2}$ for $\omega > E_B$, the oscillator-strength density of the continuum is [15, Sect. 61]

$$\frac{\mathrm{d}f^{\mathrm{osc}}}{\mathrm{d}\omega} = \frac{2^7}{3E_B}\Big(\frac{E_B}{\omega}\Big)^4 \frac{\exp(-4\xi_{\mathrm{C}}\,\mathrm{arccot}\,\xi_{\mathrm{C}})}{1-\mathrm{e}^{-2\pi\xi_{\mathrm{C}}}},$$

which tends to $0.782/E_B$ at threshold, where $\xi_{\mathrm{C}} \to \infty$ and $\xi_{\mathrm{C}}\,\mathrm{arccot}\,\xi_{\mathrm{C}} \to 1$.
Adding up (Fig. 4.5a), the bound levels carry 0.565, of which 2p alone carries 0.416 and the first three 0.524. The continuum carries 0.435: 0.316 within E_B of the threshold, and only 0.0042 above $10E_B$, since at large ω, where $\xi_{\mathrm{C}} \to 0$, the density falls as $\omega^{-7/2}$. The total is 1.000, as Eq. (4.37) requires. A calculation that keeps the bound states alone misses the 43 percent carried by the continuum, the ionization channel of Sect. 2.2.
The finite-q rule Eq. (4.35) tests the same states at a dark-matter momentum transfer. At $q_0 = \alpha m_e$ its right side is exactly $q_0^2/2m_e = \alpha^2 m_e/2 = E_B$. The 2p level has $|f_{1s\to 2p}(q_0)|^2 = 0.061$ (Worked example 2.2) at the energy $\frac{3}{4}E_B$, and contributes $0.046\,E_B$, under 5 percent of the sum, although it takes a tenth of the inelastic weight at that q (Worked example 4.4). The rest of the energy-weighted sum lies in the continuum, at energies $\sim E_B$ and above.

Worked example 4.6: The f-sum of the free electron gas

Worked example 4.3 obtained the first moment of S^0 from the sum over occupied states. The closed form Eq. (4.22) must reproduce this first moment, and integrating the closed form shows where the weight lies. For $q > 2k_\mathrm{F}$, S^0 is the parabola between ω_- and ω_+. We change variable to $x = m_e\omega/q - q/2$, which runs from $-k_\mathrm{F}$ to k_F. Then $\omega = (q/m_e)(x + q/2)$, $\mathrm{d}\omega = (q/m_e)\,\mathrm{d}x$ and $S^0 = (m_e/2\pi q)(k_\mathrm{F}^2 - x^2)$, so that

$$\int \mathrm{d}\omega\, \omega\, S^0 = \frac{q}{2\pi m_e} \int_{-k_\mathrm{F}}^{k_\mathrm{F}} \mathrm{d}x \left(x + \frac{q}{2}\right)(k_\mathrm{F}^2 - x^2) = \frac{q}{2\pi m_e}\, \frac{q}{2}\, \frac{4k_\mathrm{F}^3}{3} = \frac{q^2 k_\mathrm{F}^3}{3\pi m_e}.$$

The term odd in x, the Doppler shift from the motion of the struck electron, integrates to zero, and with $n_e = k_\mathrm{F}^3/3\pi^2$ the result is $\pi n_e q^2/m_e$. For $q < 2k_\mathrm{F}$ the linear region, $0 < \omega < \omega_1 = (q/m_e)(k_\mathrm{F} - q/2)$, contributes $\int \mathrm{d}\omega\, \omega\, m_e^2\omega/\pi q = m_e^2\omega_1^3/3\pi q = q^2(k_\mathrm{F} - q/2)^3/3\pi m_e$. The linear region thus carries the fraction $(1 - q/2k_\mathrm{F})^3$ of the sum, $27/64 = 0.42$ at $q = k_\mathrm{F}/2$, and the parabola carries the remaining fraction. As $q \to 0$ the linear region carries the whole sum, and the mean energy $(q^2/2m_e)/S^\mathrm{st}$, with $S^\mathrm{st} \to 3q/4k_\mathrm{F}$, is $\frac{2}{3}qv_\mathrm{F}$, which vanishes. A gapless target can place its f-sum at vanishing energies, but only because its long-wavelength fluctuations are suppressed by the Pauli hole of Sect. 4.2.

Take the valence electrons of an insulator at $T = 0$, with the nuclei clamped and the potentials local, and their share of the f-sum, collected below the core edges (Worked example 4.7). Their inelastic S vanishes below the gap E_g, so $E_\mathrm{g} \int \mathrm{d}\omega\, S \le \int \mathrm{d}\omega\, \omega\, S$. By Eqs. (4.17) and (4.30), the inelastic part of the static structure factor then obeys, to the accuracy with which the valence electrons carry their own f-sum (about 2 percent in silicon, Worked example 4.7),

$$S^\mathrm{st}(\boldsymbol{q}) \le \frac{q^2}{2m_e E_\mathrm{g}}.$$

At the valence density of silicon, $k_\mathrm{F} = 3.57\,\mathrm{keV}$, and with $E_\mathrm{g} = 1.17\,\mathrm{eV}$ this bound lies below the free-gas value $3q/4k_\mathrm{F}$ for $q < 3m_e E_\mathrm{g}/2k_\mathrm{F} = 0.25\,\mathrm{keV}$; at $q = 0.1\,\mathrm{keV}$ the bound is 0.40 of the free-gas value. The long-wavelength density fluctuations of an insulator vanish as q^2, not as q, and the f-sum rule derives this q^2 suppression from the gap.

Worked example 4.7: The f-sum of silicon

A silicon atom has 14 electrons, four in the valence shell and ten in the core, two 1s and eight 2s and 2p. With the full, local Hamiltonian the f-sum counts all of them, and at $q \to 0$ the Thomas–Reiche–Kuhn rule Eq. (4.37), applied per atom of the crystal, gives 14. The four valence electrons absorb from the gap upward; the core electrons cannot absorb below their binding energies, which put the L edge near 100 eV and the K edge at 1.84 keV.

Henke, Gullikson and Davis tabulated the photoabsorption of silicon from 50 eV to 30 keV. Their current distribution extends down to 30 eV, and we integrate it from there. The tables express the photoabsorption through the imaginary part $f_2(\omega)$ of the atomic scattering factor, which in the convention of the tables is $f_2 = (\pi/2)\,\omega\,\mathrm{d}f^\mathrm{osc}/\mathrm{d}\omega$ per atom [16]. The two edges appear in the tables as steps of f_2, at 99.8 eV and 1.839 keV. Integrating $2f_2/\pi\omega$ over the regions the two edges bound gives 0.35 between 30 eV and the L edge, 8.46 between the L and K edges, and 1.61 above the K edge. Of the 1.61 above the K edge, 0.007 comes from extending the last decade of the tables beyond 30 keV as a power law. Above 30 eV the tables therefore hold 10.42 of the 14. The rule then requires

3.58 below 30 eV (3.55 with the mass–velocity reduction of Eq. (4.33)), and 3.93 below the L edge, against the four valence electrons (Fig. 4.5b).
The sum saturates slowly, with the same pattern in each region. The valence part is complete to 2 percent below the L edge, at six times ω_p. The region between the edges collects 97 percent of its strength by 1 keV, ten times the L-edge energy, and the region above the K edge collects 84 percent by 5 keV and 96 percent by 10 keV. Each shell gathers its strength over energies from its edge to several times the edge energy, with the slow $\omega^{-5/2}$ approach of the remaining strength that hydrogen showed.
The regions are not the shells. The region above the K edge holds 1.61, not 2. The 1s electrons cannot be promoted into the full 2p shell, a transition that carries 0.42 of the sum in hydrogen. In a picture of independent electrons, the rule holds for each electron only when transitions into occupied orbitals are counted too, upward with positive and downward with negative strength. In the atom both the upward and the downward transition into an occupied orbital are blocked by the Pauli principle, and since their strengths cancel, the total stays 14 while strength is transferred from the inner shells to the outer ones. The excess of the region between the edges over 8 is the receiving side of this transfer, together with the overlap of tails: the L absorption continues above the K edge, and the valence absorption continues above the L edge.
For dark matter the split matters at finite q. At $q \to 0$ the core part of the sum lies above 100 eV, far from the valence part. At larger q the valence weight moves toward the free recoil energy $q^2/2m_e$, which is 98 eV at $q = 10$ keV, where the dark matter can deposit at most $qv_{\max} = 26.5$ eV. When a valence-only calculation is tested against its f-sum at momentum transfers of this size, it must follow the valence spectrum to energies where the core shells it omits absorb as well. The sum rule constrains the total over all energies, not the low-energy corner that the dark matter samples.

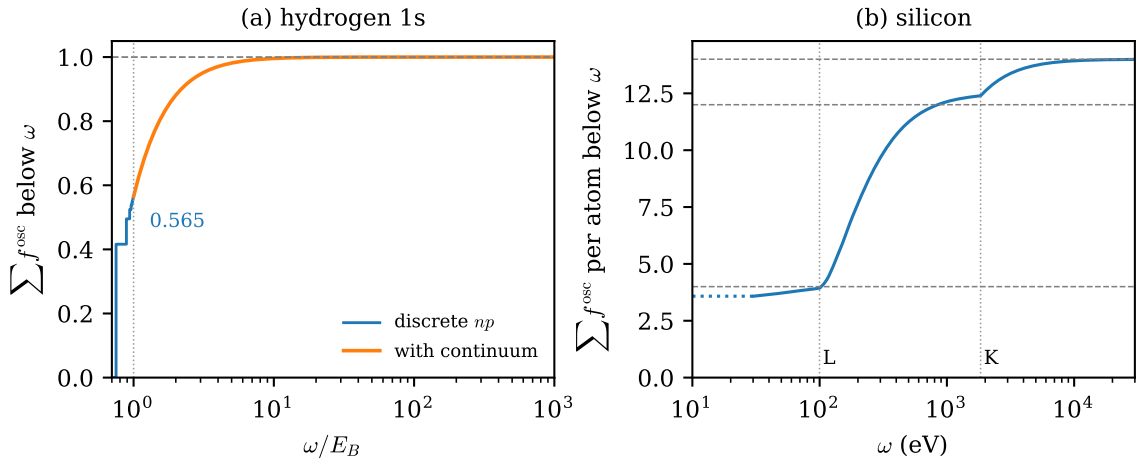


Figure 4.5: The Thomas–Reiche–Kuhn sum accumulated with energy. (a) Hydrogen in its ground state: oscillator strength of all transitions below the absorbed energy ω, in units of the binding energy $E_B = 13.6\,\text{eV}$, from the closed forms of Bethe and Salpeter. The bound 1s–np lines (steps, starting with 2p at $\frac{3}{4}E_B$) carry 0.565 and the continuum 0.435; the total is one electron (dashed). (b) Silicon, per atom: oscillator strength below ω from the photoabsorption tables of Henke and coworkers above 30 eV, integrated as $\mathrm{d}f^{\rm osc}/\mathrm{d}\omega = 2f_2/\pi\omega$, and offset so that the total, with the region above 30 keV extrapolated as a power law, is 14. The offset, 3.58, is the strength below 30 eV that the rule requires (dotted). Dashed lines mark 4 valence electrons, 4 + 8 and 14; dotted vertical lines the L and K edges at 99.8 eV and 1.839 keV.

4.4 Screening and the dielectric function

A dark-matter particle that transfers the typical momentum $q_0 = \alpha m_e = 3.73\,\text{keV}$ to silicon does not meet the bare valence electrons. Its potential pushes the electrons it reaches, the displaced charge produces a potential of its own, and the electrons respond to the sum of the two potentials. A crude estimate (Worked example 4.8) reduces the potential at q_0 by a factor of about two, and the rate, which goes as the square of the potential, by about five. Full calculations for silicon and germanium find that screening changes the rate by a factor of a few [5]. The response function R_{ee} of Sect. 4.1 contains this screening, but it mixes two kinds of physics: the long-range Coulomb field of the displaced charge, which is classical and universal, and everything else, which depends on the material. Separating the two kinds defines the dielectric function. We will see that the structure factor that sets the rate is, up to a factor q^2, the imaginary part of the inverse dielectric function, the loss function, which is measured in optical and electron-energy-loss experiments. The treatment follows Refs. [1, Chaps. 3 and 5] and [17].

4.4.1 Proper response and the dielectric function

Apply to the electrons an external potential energy that varies as $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, with amplitude $\phi_{\rm ext}(\boldsymbol{q},\omega)$ at frequency ω, and keep the nuclei clamped. By the Kubo formula Eq. (4.3), the electron density develops a ripple with amplitude $\delta n_e = R_{ee}(\boldsymbol{q},\omega)\,\phi_{\rm ext}$, where δn_e is $\delta\langle n_e(\boldsymbol{q})^\dagger\rangle/\mathcal{V}$, the coefficient of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in the induced density (Sect. 4.1). The ripple is a charge, and by Coulomb's law it acts on every electron with the potential energy $v_{\rm C}(q)\,\delta n_e$, with the kernel Eq. (2.43). Each electron then feels the total potential energy

$$\phi_{\rm tot}(\boldsymbol{q},\omega) = \phi_{\rm ext}(\boldsymbol{q},\omega) + v_{\rm C}(q)\,\delta n_e(\boldsymbol{q},\omega), \tag{4.38}$$

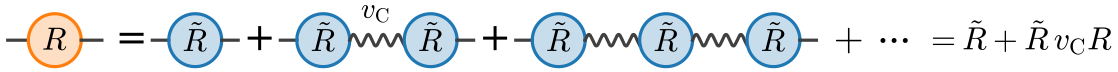


Figure 4.6: Schematic of the relation between the density response R and its Coulomb-proper part $\tilde{R}$. Each shaded blob is the proper response of the electrons, which includes every interaction effect except the long-range Coulomb field at the momentum transfer $\boldsymbol{q}$. Each wavy line is the Coulomb kernel $v_C(q) = e^2/q^2$, which carries the potential of the charge induced in one blob to the next. Summed, the chain is the geometric series $R = \tilde{R} + \tilde{R}v_C\tilde{R} + \cdots$, whose closed form is $R = \tilde{R} + \tilde{R}v_C R$. The drawing is a schematic, not a set of Feynman rules.

the external part plus the Hartree potential of the induced density. The induced density is a response to ϕ_{tot} as much as to ϕ_{ext}, and the response to the total potential is the one that excludes the long-range field. We define it as the Coulomb-proper (or irreducible) response $\tilde{R}_{ee}$,

$$\delta n_e = \tilde{R}_{ee}(\boldsymbol{q}, \omega)\, \phi_{\text{tot}}(\boldsymbol{q}, \omega). \tag{4.39}$$

It contains every effect of the interactions except the classical Hartree field at the wavevector $\boldsymbol{q}$: exchange, correlation, the band structure, the excitations of the target. The full response R_{ee} is called the improper response.

Inserting Eq. (4.38) into Eq. (4.39) and comparing with $\delta n_e = R_{ee}\phi_{\text{ext}}$, for any ϕ_{ext},

$$R_{ee} = \tilde{R}_{ee} + \tilde{R}_{ee}\, v_C\, R_{ee}, \qquad R_{ee} = \frac{\tilde{R}_{ee}}{1 - v_C\tilde{R}_{ee}}. \tag{4.40}$$

Eq. (4.40) is a Dyson equation of the form Eq. (C.57), with $\tilde{R}_{ee}$ as the bare object and v_C as the kernel. Iterating it gives the geometric series

$$R_{ee} = \tilde{R}_{ee} + \tilde{R}_{ee}v_C\tilde{R}_{ee} + \tilde{R}_{ee}v_C\tilde{R}_{ee}v_C\tilde{R}_{ee} + \cdots,$$

in which the external potential polarizes the electrons, the polarization creates a Coulomb potential, that potential polarizes them again, and so on. Fig. 4.6 draws the series with the proper parts as blobs joined by Coulomb lines. The name "proper" says that $\tilde{R}_{ee}$ is the part of the response that cannot be cut into two pieces joined by a single Coulomb line at the momentum $\boldsymbol{q}$; Sect. 4.8 reads the figure as a sum of Feynman diagrams. The closed form holds beyond the radius of convergence of the series, and its poles, at $1 - v_C\tilde{R}_{ee} = 0$, are oscillations that the electrons sustain without an external field.

The ratio of the external to the total potential is the dielectric function. From Eqs. (4.38) and (4.39), $\phi_{\text{ext}} = (1 - v_C\tilde{R}_{ee})\,\phi_{\text{tot}}$ and $\phi_{\text{tot}} = (1 + v_C R_{ee})\,\phi_{\text{ext}}$. Hence

$$\varepsilon(\boldsymbol{q}, \omega) \equiv \frac{\phi_{\text{ext}}}{\phi_{\text{tot}}} = 1 - v_C(q)\,\tilde{R}_{ee}(\boldsymbol{q}, \omega), \qquad \varepsilon^{-1}(\boldsymbol{q}, \omega) = 1 + v_C(q)\,R_{ee}(\boldsymbol{q}, \omega), \tag{4.41}$$

and the two forms are each other's inverse by Eq. (4.40). The two forms are the relations Eqs. (A.34) and (A.35) stated in Appendix A.7, and the second is Ref. [5, Eq. (5)] in our units. We write $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$. In Heaviside–Lorentz units the external charge is the source of $\boldsymbol{D}$ and the total charge the source of $\boldsymbol{E}$. For a longitudinal field the potential energies are proportional to the potentials, so ε is the ratio D/E of the Fourier components, the dielectric function of classical electromagnetism, now for every $\boldsymbol{q}$ and ω. It is the longitudinal dielectric function, because a potential produces a field along $\boldsymbol{q}$.

The inverse function $\varepsilon^{-1} - 1 = v_C R_{ee}$ is a retarded response, the induced potential per unit external one, and it has every property Sects. 4.1 and 4.3 proved for R_{ee}: analyticity in

the upper half plane, the Kramers–Kronig relations, and $\operatorname{Im}\varepsilon^{-1} \le 0$ for $\omega > 0$. The function $\varepsilon - 1$ obeys the Kramers–Kronig relations only because a stable target has no zeros of ε in the upper half plane (Eq. (C.39)). The static limit of $\varepsilon^{-1}v_{\mathrm{C}}$ is the statically screened Coulomb interaction: an external point charge, or a second electron held at rest, acts through $v_{\mathrm{C}}(q)/\varepsilon(\boldsymbol{q},0)$ rather than $v_{\mathrm{C}}(q)$ (Sect. 4.8 uses it in the electron self-energy). At small q, v_{C} grows as $1/q^2$, and screening matters most there.

In an insulator the static dielectric function at long wavelength is finite. With the lattice clamped, its value $\varepsilon_\infty = \varepsilon(\boldsymbol{q} \to 0, \omega \to 0)$ is the electronic, or high-frequency, dielectric constant, so called because it is measured at frequencies above those of the lattice vibrations and below the gap; the static constant ε_0 with the lattice included is the subject of Sect. 5.4. For silicon $\varepsilon_\infty = 11.7$ [18]: a static charge in the crystal is screened to a twelfth of its bare strength at distances of many cells. Where ε_1 crosses zero and ε_2 is small, the external potential produces a large total one, and the electrons oscillate collectively. This collective oscillation is the plasmon, a pole of Eq. (4.40) near the real axis, which appears as a peak of $\operatorname{Im}(-\varepsilon^{-1})$; Sect. 4.5 makes it precise.

4.4.2 Local fields and the dielectric matrix

In a crystal a potential at $\boldsymbol{q}$ also induces density at $\boldsymbol{q}+\boldsymbol{G}$ for every reciprocal-lattice vector $\boldsymbol{G}$ (Sect. 4.1): the electrons pile up near the atoms and in the bonds, on the scale of the cell. The density and the potentials are then vectors indexed by $\boldsymbol{G}$, at a wavevector $\boldsymbol{q}$ in the first zone, and the response is a matrix,

$$R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = R_{ab}(\boldsymbol{q},\omega)\big|_{O_a = n_e(\boldsymbol{q}+\boldsymbol{G}),\ O_b = n_e(\boldsymbol{q}+\boldsymbol{G}')}, \tag{4.42}$$

the density response at $\boldsymbol{q}+\boldsymbol{G}$ to a potential at $\boldsymbol{q}+\boldsymbol{G}'$, from the Kubo formula of Sect. 4.1. Each Fourier component of the induced density produces its own Hartree potential, $v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta n_{e,\boldsymbol{G}}$, and Eqs. (4.38) and (4.39) become matrix equations. The same steps lead to $R = \tilde{R} + \tilde{R}\,v_{\mathrm{C}}\,R$, with v_{C} the diagonal matrix of $v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)$, and to the dielectric matrix

$$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\tilde{R}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega), \qquad \left(\varepsilon^{-1}\right)_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} + v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,R_{\boldsymbol{G}\boldsymbol{G}'}, \tag{4.43}$$

for any proper response $\tilde{R}$, exact or approximate. That the two matrices are inverses of each other takes one line of matrix algebra (Appendix D.5). Eq. (4.43) is Eq. (A.34), the unsymmetrized counterpart of the symmetrized longitudinal dielectric matrix of Ref. [5, Eqs. (A5)–(A7)], which is built with $v_{\mathrm{C}}^{1/2}\tilde{R}v_{\mathrm{C}}^{1/2}$ and has the same diagonal elements of the inverse (Appendix D.5).

An experiment with a long-wavelength field sees the average of the total field over a cell. The external field has only the component $\boldsymbol{G}' = 0$, and the cell average of the total field is its component $\boldsymbol{G} = 0$ (Eq. (A.11)), which is $(\varepsilon^{-1})_{\boldsymbol{0}\boldsymbol{0}}$ times the external one. The macroscopic dielectric function is therefore

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega) = \frac{1}{\left(\varepsilon^{-1}\right)_{\boldsymbol{0}\boldsymbol{0}}(\boldsymbol{q},\omega)}, \tag{4.44}$$

the inverse of the head of the inverse matrix, not the head $\varepsilon_{\boldsymbol{0}\boldsymbol{0}}$ itself [19, 20]. The head $\varepsilon_{\boldsymbol{0}\boldsymbol{0}}$ answers a different question: the external potential needed at $\boldsymbol{G} = 0$ to produce a total potential that has only a $\boldsymbol{G} = 0$ component. No macroscopic external field produces such a total potential in a crystal. The function $\varepsilon_{\mathrm{mac}}$ and the head $\varepsilon_{\boldsymbol{0}\boldsymbol{0}}$ differ because the macroscopic field drives density at all $\boldsymbol{G}$, and the Hartree potentials of the components at nonzero $\boldsymbol{G}$ feed

back on the $\boldsymbol{G} = 0$ component. Inverting the matrix in blocks (Eq. (C.55)) gives the difference between ε_{mac} and $\varepsilon_{\boldsymbol{00}}$ explicitly,

$$\varepsilon_{\text{mac}} = \varepsilon_{\boldsymbol{00}} - \sum_{\boldsymbol{G},\boldsymbol{G}'\neq 0} \varepsilon_{\boldsymbol{0G}} \left[(\varepsilon_{\neq 0})^{-1}\right]_{\boldsymbol{GG}'} \varepsilon_{\boldsymbol{G}'\boldsymbol{0}},$$

with $\varepsilon_{\neq 0}$ the block with $\boldsymbol{G}, \boldsymbol{G}' \neq 0$. The sum is the local-field effect: the difference between the response to the average field and the response to the field that the electrons actually feel inside the cell. Appendix D.5 shows that it stays finite as $\boldsymbol{q} \to 0$ although single elements of the matrix diverge there, and writes ε_{mac} in the equivalent form $1 - v_{\text{C}}(q)\bar{R}_{\boldsymbol{00}}$, with $\bar{R}$ the response computed with the Coulomb kernel at $\boldsymbol{G} = 0$ removed [17]. In a homogeneous system the off-diagonal elements vanish and $\varepsilon_{\text{mac}} = \varepsilon_{\boldsymbol{00}} = \varepsilon$; how large the local-field effects of silicon are in the random-phase approximation is the subject of Sect. 4.6. The macroscopic function is the one measured optically, and ε_∞ is its static limit at $\boldsymbol{q} \to 0$. For a momentum transfer beyond the first zone, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ as in Eq. (3.16), the same reasoning applies at the row and column of $\boldsymbol{G}_q$, which is the unreduced indexing of Eq. (A.12): $(\varepsilon^{-1})_{00}(\boldsymbol{q}, \omega)$ means $(\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r, \omega)$.

4.4.3 Structure factor and loss function

The rate needs S, and the fluctuation–dissipation theorem Eq. (4.18) ties S to $\operatorname{Im} R_{ee}$, which Eq. (4.41) ties in turn to the inverse dielectric function. Taking the imaginary part of $\varepsilon^{-1} = 1 + v_{\text{C}} R_{ee}$, with v_{C} real, leads to $\operatorname{Im} \varepsilon^{-1} = v_{\text{C}} \operatorname{Im} R_{ee}$, and with $S = -2 \operatorname{Im} R_{ee}/(1 - \mathrm{e}^{-\beta\omega})$,

Structure factor and loss function

$$S(\boldsymbol{q}, \omega) = \frac{2q^2}{e^2} \frac{L(\boldsymbol{q}, \omega)}{1 - \mathrm{e}^{-\beta\omega}} = \frac{q^2}{2\pi\alpha} \frac{L(\boldsymbol{q}, \omega)}{1 - \mathrm{e}^{-\beta\omega}}, \qquad L(\boldsymbol{q}, \omega) = \operatorname{Im}\left[-(\varepsilon^{-1})_{00}(\boldsymbol{q}, \omega)\right], \quad (4.45)$$

for the density response of the electrons with the nuclei clamped. The loss function L is nonnegative for $\omega > 0$, and the rate of any density coupling is fixed by it.

This is Eq. (A.36), Ref. [5, Eq. (6)], and at $T = 0$ Ref. [21, Eqs. (S.7) and (S.24)]. In a crystal the derivation goes through element by element. By Eq. (4.42) the diagonal element of $R_{\boldsymbol{GG}'}(\boldsymbol{q}_r, \omega)$ at $\boldsymbol{G} = \boldsymbol{G}' = \boldsymbol{G}_q$ is the density response $R_{ee}(\boldsymbol{q}, \omega)$ at the unreduced $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$, and Eq. (4.43) relates it to the diagonal element of ε^{-1} at the same place. The loss function is this diagonal element of ε^{-1}, local fields included, and the relation holds at each vector $\boldsymbol{q}$ with no assumption of isotropy. The name comes from the energy loss of fast electrons in thin films, and the energy-loss spectrum of these electrons measures L directly; Eq. (4.54) below makes the connection.

The relation needs weak single scattering, a coupling to the density, and a probe that sees the electrons alone, and each of these fails somewhere. The probe acts once and weakly, so that the golden rule and linear response apply: the Born approximation of Sect. 2.2. It couples to the electrons through their density alone, through a potential that depends on their positions, and not through spin or current [21]; spin couplings need the spin response of Chap. 7, and velocity-dependent couplings the current response of Chap. 8. Finally, it couples to the electrons only, or the nuclei are clamped.

The last condition separates two kinds of probe. Three probes act on the total charge, $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ of Sect. 2.3: a test charge, a photon, and a dark photon, whose coupling to matter is the photon's scaled by the mixing ζ (Sect. 1.4). The dielectric function measured with these probes is $\varepsilon^{-1} = 1 + v_{\text{C}} R_{ZZ}$, with the response of the total charge, and its loss

function L fixes the total-charge structure factor exactly, $S_{ZZ} = (2q^2/e^2)\, L/(1 - \mathrm{e}^{-\beta\omega})$. A mediator that couples to electrons but not to nucleons sees $S = S_{ee}$ instead, and the relation Eq. (4.45) holds exactly with the electron-channel function $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$ and its loss L_e (Appendix A.7). With the nuclei clamped, n_κ contributes only static Bragg terms, $R_{ZZ} = R_{ee}$, and $L_e = L$. With moving nuclei L_e and L differ at phonon energies, where the nuclear charge oscillates and the electrons partly follow it (Chap. 5). Optical and electron-energy-loss experiments measure the total-charge L. They yield the dark-photon rate exactly at every energy [21], and the rate of an electron-only coupling at energies well above the phonon band, where the electronic excitations lie. For a mediator that couples to electrons and nuclei with weights other than those of the total charge, the interference between the electron and nuclear densities is treated in Chap. 8.

For a scalar dielectric function, with the local fields neglected or in a homogeneous system, $L = \varepsilon_2/|\varepsilon|^2$. Since $\varepsilon_2 = -v_{\mathrm{C}} \operatorname{Im} \tilde{R}_{ee}$, the absorptive part of the proper response, Eq. (4.45) becomes $S = S^{\mathrm{prop}}/|\varepsilon|^2$, where $S^{\mathrm{prop}} = (2q^2/e^2)\, \varepsilon_2/(1 - \mathrm{e}^{-\beta\omega})$ is the structure factor built from $\tilde{R}_{ee}$ (Eq. (A.36)). The rate is the rate of the proper response, screened by $1/|\varepsilon|^2$. In the random-phase approximation the proper response is the independent-particle one, $S^{\mathrm{prop}} = S^0$ of Eq. (4.21), and earlier calculations of the dark-matter–electron rate left out the factor $1/|\varepsilon|^2$ [5].

Fig. 4.7 shows what the factor does in silicon at $q \to 0$. Absorption, ε_2, is strong between the gap and about 10 eV, with peaks at 3.4 and 4.2 eV from interband transitions. In this range, however, $|\varepsilon|$ is large, and the loss function is small: at 3 eV, $\varepsilon_2 = 2.4$ but $L = 3.1 \times 10^{-3}$. The real part ε_1 is negative from 4.2 eV and crosses zero again at 16.3 eV, where ε_2 has fallen below one. The loss function peaks near this zero crossing, at 16.4 eV, with a height of 4.5 and a full width of 3.7 eV. This peak is the plasmon of the valence electrons; how its position relates to the free-electron ω_p is the subject of Sect. 4.5. A dark-matter particle can deposit at most $qv_{\max}$, a few eV at typical momentum transfers, and at these energies the target absorbs strongly and screens strongly at once.

4.4.4 Sum rules for the loss function

The sum rules of Sect. 4.3 carry over. For the f-sum we start from its form at any temperature, $\int_0^\infty \mathrm{d}\omega\, \omega(1 - \mathrm{e}^{-\beta\omega}) S = \pi \boldsymbol{n}_e q^2/m_e$, which folds the negative frequencies onto the positive ones with detailed balance. By Eq. (4.45) the factor $(1 - \mathrm{e}^{-\beta\omega})S$ is $(2q^2/e^2)L$, the temperature drops out, and with $\omega_p^2 = e^2 \boldsymbol{n}_e/m_e$ from Eq. (4.31),

$$\int_0^\infty \mathrm{d}\omega\; \omega\, L(\boldsymbol{q}, \omega) = \frac{\pi}{2}\, \omega_p^2, \tag{4.46}$$

at every $\boldsymbol{q}$ and temperature, with L for the electron channel, or the total one at clamped nuclei. Eq. (4.46) is Eq. (A.39) and Ref. [21, Eq. (S.10)]. With moving nuclei the total-charge L collects in addition the nuclear share of the charge f-sum of Sect. 4.3, 2.73×10^{-4} of the total for silicon.

The absorptive part obeys the same rule. The macroscopic ε_2 is $-v_{\mathrm{C}}(q) \operatorname{Im} \bar{R}_{\mathbf{00}}$ (Appendix D.5), and $\bar{R}$ is the density response of the electrons with the Coulomb kernel at the single wavevector $\boldsymbol{q}$ removed from the Hamiltonian. Removing one Fourier component out of a continuum of them does not change the equilibrium state in the thermodynamic limit (Sect. 4.8.3), and the double commutator of Sect. 4.3 does not involve the interaction. The response $\bar{R}$ therefore has the same f-sum as R, and

$$\int_0^\infty \mathrm{d}\omega\; \omega\, \varepsilon_2(\boldsymbol{q}, \omega) = \frac{\pi}{2}\, \omega_p^2, \tag{4.47}$$

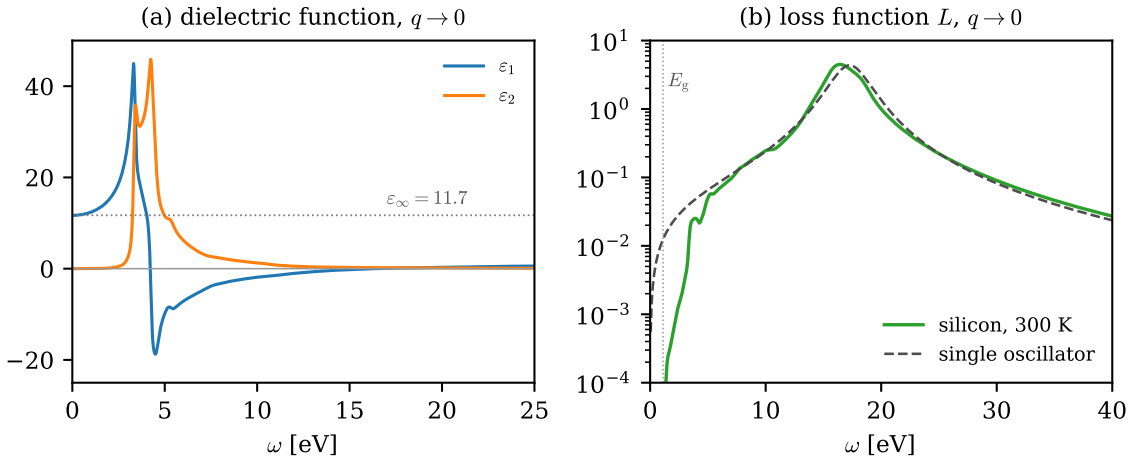


Figure 4.7: Dielectric function and loss function of crystalline silicon at $q \to 0$ and 300 K, from the optical constants of Franta and coworkers, a Kramers–Kronig-consistent dispersion model fitted to ellipsometric and spectrophotometric measurements and tabulated from 4 meV to 40 eV [18]. The data are optical, that is transverse; at $q \to 0$ the transverse and longitudinal functions coincide (Eq. (4.52)), and $L = \varepsilon_2/|\varepsilon|^2$. (a) ε_1 and ε_2; at low energy $\varepsilon_1 = \varepsilon_\infty = 11.7$ (dotted). The interband peaks of ε_2 lie at 3.4 and 4.2 eV, and ε_1 crosses zero upward at 16.3 eV. (b) The loss function on a logarithmic scale (solid), with its peak of 4.5 at 16.4 eV. Dashed: the single damped oscillator of Worked example 4.9 at $q = 0$, which has the same f-sum and the same ε_∞, and a damping equal to the measured full width, 3.66 eV. Dotted: the room-temperature gap, 1.12 eV, to match the data. Below 40 eV the tabulated spectrum collects 96 percent of the f-sum of the four valence electrons in $\int \omega \varepsilon_2 \, \mathrm{d}\omega$ and 92 percent in $\int \omega L \, \mathrm{d}\omega$.

which Appendix C.5 also obtains from the Kramers–Kronig relation for $\varepsilon - 1$ at large frequency. The two functions ε_2 and L distribute the same weight differently: ε_2 puts it at the transition energies, and $L = \varepsilon_2/|\varepsilon|^2$ moves it up to where $|\varepsilon|$ is small, toward the plasmon. In the tabulated silicon data of Fig. 4.7, the spectrum below 40 eV holds 96 percent of $(\pi/2)\omega_p^2 = 433\,\mathrm{eV}^2$ for the four valence electrons in $\int \omega\varepsilon_2$, and 92 percent in $\int \omega L$, which has more of its weight at high energy. The remaining weight lies above the table, where the valence absorption overlaps the core absorption (Worked example 4.7).

The inverse moments follow from the Kramers–Kronig relations at $\omega = 0$ (Eqs. (C.38) and (C.39)). For an insulator at $\boldsymbol{q} \to 0$,

$$\frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{L(\omega)}{\omega} = 1 - \frac{1}{\varepsilon_\infty}, \qquad \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{\varepsilon_2(\omega)}{\omega} = \varepsilon_\infty - 1. \tag{4.48}$$

In a metal static screening is perfect at long wavelength, $\varepsilon^{-1} \to 0$, and the first integral becomes $\pi/2$ (Appendix C.5). Weighted by $1/\omega$, these integrals converge much faster than the f-sum: the tabulated data of Fig. 4.7 return 99.8 percent of $\varepsilon_\infty - 1$ and 97.8 percent of $1 - 1/\varepsilon_\infty$ below 40 eV.

Together, the f-sum and the inverse moment already say where the weight of L lies at small q. Their ratio defines a mean excitation energy, $\bar{E}^2 = \int \omega L / \int (L/\omega)$, and

$$\bar{E}^2 = \frac{\omega_p^2}{1 - 1/\varepsilon_\infty}. \tag{4.49}$$

With $\omega_p = 16.6\,\mathrm{eV}$ for the valence electrons and $\varepsilon_\infty = 11.7$, $\bar{E} = 16.6/0.956 = 17.4\,\mathrm{eV}$. If the loss spectrum were a single line, it would lie at $\bar{E}$, more than fifteen times above the gap of 1.17 eV. By Eq. (4.45), the density fluctuations S at $T = 0$ have the same mean excitation energy: at long wavelengths the valence electrons respond collectively, at energies $\sim \omega_p$, and not at the gap, a point Sect. 4.5 makes precise.

4.4.5 Optical conductivity and transverse response

Light, and a dark photon absorbed at rest, act through a transverse field rather than a potential, and the experiment measures a current. The induced current density $\boldsymbol{J}$ is linear in the total field $\boldsymbol{E}$, $\boldsymbol{J} = \sigma_{\mathrm{cond}}\boldsymbol{E}$, which defines the optical conductivity $\sigma_{\mathrm{cond}}(\omega) = \sigma_1 + \mathrm{i}\sigma_2$. With fields varying as $\mathrm{e}^{-\mathrm{i}\omega t}$ (Appendix A.3), the Ampère–Maxwell law in Heaviside–Lorentz units is $\nabla \times \boldsymbol{B} = \boldsymbol{J} + \partial_t \boldsymbol{E}$. Written as $\nabla \times \boldsymbol{B} = \partial_t \boldsymbol{D}$ with $\boldsymbol{D} = \varepsilon\boldsymbol{E}$, the law requires $-\mathrm{i}\omega\varepsilon = \sigma_{\mathrm{cond}} - \mathrm{i}\omega$, so

$$\varepsilon = 1 + \frac{\mathrm{i}\,\sigma_{\mathrm{cond}}}{\omega}, \qquad \sigma_1 = \omega\,\varepsilon_2, \qquad \sigma_2 = \omega\,(1 - \varepsilon_1). \tag{4.50}$$

Texts in Gaussian units write $\varepsilon = 1 + 4\pi\mathrm{i}\sigma_{\mathrm{cond}}/\omega$ (Appendix A.2). The real part σ_1 is the dissipative one, and by Eq. (4.47) it obeys $\int_0^\infty \mathrm{d}\omega\, \sigma_1 = (\pi/2)\omega_p^2$, the f-sum rule in its optical form. Written with the loss function, $\sigma_1 = \omega|\varepsilon|^2 L$ [21]. For the absorption of a dark photon, the in-medium photon self-energy of Sect. 1.4 is $\Pi_{\mathrm{em}} = \omega^2(1 - \varepsilon^{\mathrm{T}}) = -\mathrm{i}\omega\sigma_{\mathrm{cond}}$: the wave equation $\omega^2\varepsilon^{\mathrm{T}}\boldsymbol{A} = k^2\boldsymbol{A}$ for a transverse vector potential in the medium is $(\omega^2 - k^2 - \Pi_{\mathrm{em}})\boldsymbol{A} = 0$. The imaginary part of Π_{em}, $-\omega\sigma_1$, sets the absorption rate of Eq. (1.37).

The function that enters here is the transverse dielectric function ε^{T}, the response to a field perpendicular to $\boldsymbol{q}$, while Eq. (4.41) defines the longitudinal one. The transverse and longitudinal functions come from one tensor. A vector potential $\boldsymbol{A}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in the gauge with no scalar potential, $\boldsymbol{E} = -\partial_t\boldsymbol{A} = \mathrm{i}\omega\boldsymbol{A}$, enters the kinetic energy of each electron as $(\boldsymbol{p} + e\boldsymbol{A})^2/2m_e$. To first order it couples to the paramagnetic current $\boldsymbol{j}_{\mathrm{e}}(\boldsymbol{q})$ of Eq. (4.29)

through $e\boldsymbol{A}\cdot\boldsymbol{j}_e(\boldsymbol{q})$, and it changes the velocity of every electron from $\boldsymbol{p}/m_e$ to $(\boldsymbol{p}+e\boldsymbol{A})/m_e$. The induced particle current per volume therefore has two parts. The diamagnetic part is the density times the added velocity, $n_e\, e\boldsymbol{A}/m_e$. The paramagnetic part is the Kubo response Eq. (4.3) of $\boldsymbol{j}_e$ to the perturbation $O_b\phi_{\text{ext}} = j_e^\beta(\boldsymbol{q})\, eA^\beta$, which is $\tilde{R}_{jj}^{\alpha\beta}\, eA^\beta$, with $\tilde{R}_{jj}^{\alpha\beta}$ the proper current–current response, proper because $\boldsymbol{E}$ is the total field. Together,

$$\frac{\delta\langle j_e^\alpha(\boldsymbol{q})^\dagger\rangle}{\mathcal{V}} + n_e\frac{eA^\alpha}{m_e} = e\Big[\frac{n_e}{m_e}\delta^{\alpha\beta} + \tilde{R}_{jj}^{\alpha\beta}(\boldsymbol{q},\omega)\Big]A^\beta.$$

Multiplying by the electron charge $-e$ and inserting $\boldsymbol{A} = -\mathrm{i}\boldsymbol{E}/\omega$,

$$\varepsilon^{\alpha\beta}(\boldsymbol{q},\omega) = \delta^{\alpha\beta} - \frac{e^2}{\omega^2}\Big[\frac{n_e}{m_e}\delta^{\alpha\beta} + \tilde{R}_{jj}^{\alpha\beta}(\boldsymbol{q},\omega)\Big].$$

For free electrons $\tilde{R}_{jj}$ vanishes at $\boldsymbol{q}=0$, since the total momentum is conserved, and $\varepsilon = 1-\omega_p^2/\omega^2$. The transverse function is $\varepsilon^{\mathrm{T}} = \hat{\boldsymbol{e}}\cdot\varepsilon\cdot\hat{\boldsymbol{e}}$ for a unit vector $\hat{\boldsymbol{e}}\perp\boldsymbol{q}$, and the longitudinal one is $\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}}$.

That the longitudinal projection is the ε of Eq. (4.41) follows from the continuity equation Eq. (4.29), $\partial_t n_e(\boldsymbol{q},t) = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},t)$. In the Kubo integral for $R_{ee}(\boldsymbol{q},\omega)$ we integrate by parts in t twice, using $\omega+\mathrm{i}0^+$ so that the boundary terms at large t vanish. The first boundary term contains the equal-time commutator of two densities, which is zero. The first integration replaces $n_e(\boldsymbol{q},t)^\dagger$ by $-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},t)^\dagger$; the second, after moving the time to the other operator by stationarity, replaces $n_e(\boldsymbol{q},0)$ by $\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},0)$ and leaves the boundary term $\langle[\boldsymbol{q}\cdot\boldsymbol{j}_e^\dagger, n_e]\rangle/\mathcal{V} = n_e q^2/m_e$, the double commutator of the f-sum rule. The result is a Ward identity, the statement of charge conservation for the response functions,

$$\omega^2 R_{ee}(\boldsymbol{q},\omega) = \frac{n_e q^2}{m_e} + q_\alpha q_\beta\, R_{jj}^{\alpha\beta}(\boldsymbol{q},\omega), \tag{4.51}$$

exactly, and the same identity holds for the proper functions, by the argument that gave Eq. (4.47) [1, Chap. 3]. Inserting the Ward identity into the tensor gives $\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}} = 1-(e^2/q^2)\tilde{R}_{ee} = \varepsilon$. As $\boldsymbol{q}\to 0$, the proper tensor $\tilde{R}_{jj}^{\alpha\beta}$ tends to a limit that does not depend on the direction of $\boldsymbol{q}$, since the proper response has no long-range Coulomb kernel and so no singularity at $\boldsymbol{q}=0$. Both projections are then taken of the same tensor $\varepsilon^{\alpha\beta}(\omega)$. In a cubic crystal that tensor is a multiple of the unit matrix (Appendix C.3), and

$$\varepsilon^{\mathrm{T}}(\boldsymbol{q},\omega)\to\varepsilon(\boldsymbol{q},\omega)\qquad(\boldsymbol{q}\to 0). \tag{4.52}$$

In a crystal of lower symmetry the longitudinal function depends on the direction $\hat{\boldsymbol{q}}$ even at $\boldsymbol{q}\to 0$, but it is still the projection of the optical tensor. Optical data at $q\to 0$ therefore give the longitudinal loss function, as in Fig. 4.7, and the absorption of a dark photon and the scattering of dark matter at small q are governed by the same ε_2. At finite q the two functions separate; for dark-matter scattering, which needs momentum transfers of keV, the longitudinal function is the one that matters.

4.4.6 Thomas–Fermi screening

The simplest proper response is that of a gas of electrons in a static potential that varies slowly in space. Each region of the gas is then in local equilibrium, with the potential energy $\phi_{\text{tot}}(\boldsymbol{r})$ shifting all its levels. The electrochemical potential must be the same everywhere, so the local chemical potential is $\mu_{\text{ch}} - \phi_{\text{tot}}(\boldsymbol{r})$, and the density changes by $\delta n_e = -(\partial n_e/\partial\mu_{\text{ch}})\,\phi_{\text{tot}}$.

For the free gas, $\partial n_e/\partial\mu_{\rm ch} = 2\mathcal{N}_{\rm F}$ at $T = 0$, with $\mathcal{N}_{\rm F}$ per spin from Eq. (2.71). The proper response is $\tilde{R}_{ee}(q, 0) = -2\mathcal{N}_{\rm F}$, and Eq. (4.41) gives

$$\varepsilon_{\rm TF}(q) = 1 + \frac{k_{\rm TF}^2}{q^2}, \qquad k_{\rm TF}^2 = 2e^2\mathcal{N}_{\rm F} = \frac{4\alpha\, m_e k_{\rm F}}{\pi} = \frac{3\,\omega_p^2}{v_{\rm F}^2}, \tag{4.53}$$

where the last two forms use Eqs. (2.71) and (4.31). The statically screened interaction is

$$\frac{v_{\rm C}(q)}{\varepsilon_{\rm TF}(q)} = \frac{e^2}{q^2 + k_{\rm TF}^2},$$

the transform of a Yukawa potential $\alpha\, \mathrm{e}^{-k_{\rm TF} r}/r$, the same form as a massive mediator (Eq. (1.30)) with mass $k_{\rm TF}$. An impurity charge in a metal is screened within a distance $1/k_{\rm TF}$. Eq. (4.53) is the Thomas–Fermi theory of screening, after the semiclassical model of the atom by Thomas and Fermi. Its assumptions are a static potential and a slow variation, $q \ll k_{\rm F}$, so that the local-equilibrium picture holds; Sect. 4.5 recovers Eq. (4.53) as the static long-wavelength limit of the Lindhard function and gives the corrections.

Worked example 4.8: The Thomas–Fermi length of the silicon valence electrons

Treat the four valence electrons per atom of silicon as a free gas, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, with $k_{\rm F} = 3.57\,\mathrm{keV}$ and $\mathcal{N}_{\rm F} = 9.2 \times 10^7\,\mathrm{eV}^2$ (Sect. 2.5). By Eq. (4.53), $k_{\rm TF}^2 = 4\alpha m_e k_{\rm F}/\pi = 1.69 \times 10^7\,\mathrm{eV}^2$, or $k_{\rm TF} = 4.12\,\mathrm{keV}$, and a screening length $1/k_{\rm TF} = 0.48\,\text{Å}$, a fifth of the bond length. At the typical momentum transfer $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, $\varepsilon_{\rm TF} = 1 + (4.12/3.73)^2 = 2.22$, and the squared potential, which sets the rate, is reduced by $|\varepsilon_{\rm TF}|^2 = 4.9$.

The estimate is crude because $k_{\rm TF}$ is not small against $k_{\rm F}$ and because silicon is an insulator. With $k_{\rm TF} = 1.15\,k_{\rm F}$, the screening happens at the wavevectors where the condition $q \ll k_{\rm F}$ fails, and the Lindhard function of Sect. 4.5 screens less than Eq. (4.53) at $q \sim k_{\rm F}$. In an insulator, moreover, the electrons cannot follow a potential that varies over many cells, and the static screening of an insulator at long wavelength saturates at $\varepsilon_\infty = 11.7$, while the gas formula diverges. The two descriptions meet where $1 + k_{\rm TF}^2/q^2 = \varepsilon_\infty$, at $q = k_{\rm TF}/(\varepsilon_\infty - 1)^{1/2} = 1.26\,\mathrm{keV}$, a length $1/q$ of 1.6 Å. As a rough picture, then, silicon screens like a metal at the momentum transfers of a few keV that dark matter typically deposits, and like an insulator with a dielectric constant of 11.7 below about 1 keV. Both statements concern the static response; the rate needs ε at the energy deposited, and the calculations of Chap. 10 therefore use the computed loss function, not Eq. (4.53).

4.4.7 Dielectric stopping

A charged particle crossing a solid slows down by exciting it, and the loss function fixes how fast. Take a particle of charge $Z_1 e$, heavy enough that its recoil can be neglected, moving with velocity $\boldsymbol{v}$. Its potential energy with an electron is $-Z_1 e^2/(4\pi r)$, so $\tilde{V}(\boldsymbol{q}) = -Z_1 v_{\rm C}(q)$, and a momentum transfer $\boldsymbol{q}$ deposits $\omega = \boldsymbol{q}\cdot\boldsymbol{v}$, the energy change $\boldsymbol{q}\cdot\boldsymbol{v} - q^2/2M_1$ of a particle of mass $M_1 \to \infty$. Weighting each momentum transfer in the rate Eq. (4.14) by the energy it deposits gives the energy lost per unit time, and dividing by v the energy lost per unit path. At $T = 0$, with Eq. (4.45) and $|\tilde{V}|^2 = Z_1^2 e^4/q^4$,

$$-\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{1}{v}\int \frac{\mathrm{d}^3 q}{(2\pi)^3}\,\frac{Z_1^2 e^4}{q^4}\,\frac{2q^2}{e^2}\,\omega\, L(\boldsymbol{q},\omega)\Big|_{\omega=\boldsymbol{q}\cdot\boldsymbol{v}}.$$

For an isotropic L we use polar coordinates about $\boldsymbol{v}$, $\mathrm{d}^3q = 2\pi q^2 \mathrm{d}q\,\mathrm{d}u$ with u the cosine of the angle, and change variables from u to $\omega = quv$, $\mathrm{d}u = \mathrm{d}\omega/(qv)$. Only $\omega > 0$ contributes at $T = 0$, and

$$-\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{Z_1^2 e^2}{2\pi^2 v^2}\int_0^\infty \frac{\mathrm{d}q}{q}\int_0^{qv} \mathrm{d}\omega\;\omega\,L(q,\omega) = \frac{2Z_1^2\alpha}{\pi v^2}\int_0^\infty \frac{\mathrm{d}q}{q}\int_0^{qv} \mathrm{d}\omega\;\omega\,L(q,\omega). \tag{4.54}$$

Eq. (4.54) is Lindhard's dielectric formula for the stopping power [7], in Heaviside–Lorentz units. The fast and the slow limits show what it contains. For a fast particle, qv reaches beyond the whole loss spectrum at all but the smallest q, and the inner integral is the f-sum $(\pi/2)\omega_p^2$. The stopping power is $Z_1^2\alpha\omega_p^2/v^2$ times a logarithm of the range of q that contributes: the Bethe formula, set by the electron density alone. For a slow particle, qv stays below the main peak, and the inner integral samples only the low-energy side of L, which the sum rules do not fix. A dark-matter particle samples the same low-energy side; it is a slow probe in exactly this sense.

Worked example 4.9: A single damped oscillator

The simplest loss function consistent with the f-sum and with ε_∞ puts all the weight in one damped oscillator. Write

$$\varepsilon^{-1}(q,\omega) = 1 + \frac{\omega_p^2}{\omega^2 - \omega_{\mathrm{res}}^2 + \mathrm{i}\Gamma_{\mathrm{w}}\omega}, \qquad \omega_{\mathrm{res}}(q)^2 = \bar{E}^2 + \Big(\frac{q^2}{2m_e}\Big)^2,$$

with a damping rate Γ_{w}. The resonance frequency moves from $\bar{E}$ at small q to the free recoil energy $q^2/2m_e$ at large q; this dispersion is our modeling choice, made so that the large-q limit has the free-recoil center that Sect. 4.3 requires. The function $\varepsilon^{-1} - 1$ has the form of the damped-oscillator response of Worked example 4.1, with ω_p^2 for $2\lambda_{\mathrm{osc}}^2\omega_0$ and ω_{res} for ω_0. It is causal, and its imaginary part is

$$L(q,\omega) = \frac{\omega_p^2\,\Gamma_{\mathrm{w}}\,\omega}{(\omega^2 - \omega_{\mathrm{res}}^2)^2 + \Gamma_{\mathrm{w}}^2\omega^2}.$$

Integrating, $\int_0^\infty \omega L\,\mathrm{d}\omega = (\pi/2)\omega_p^2$ for any Γ_{w} and q, the f-sum Eq. (4.46), and $\int_0^\infty (L/\omega)\,\mathrm{d}\omega = (\pi/2)\omega_p^2/\omega_{\mathrm{res}}^2$. At $q \to 0$ the inverse moment Eq. (4.48) then requires $1 - 1/\varepsilon_\infty = \omega_p^2/\bar{E}^2$, which is Eq. (4.49): the single oscillator that satisfies both sum rules has its loss peak at the mean excitation energy, $\bar{E} = 17.4\,\mathrm{eV}$ for silicon. Inverting, ε itself has its resonance lower, at $(\bar{E}^2 - \omega_p^2)^{1/2} = \bar{E}/\varepsilon_\infty^{1/2} = 5.08\,\mathrm{eV}$, and $\varepsilon(0,0) = 11.7$. We take $\Gamma_{\mathrm{w}} = 3.66\,\mathrm{eV}$, the measured full width of the silicon loss peak, which sets the peak height $\omega_p^2/(\Gamma_{\mathrm{w}}\bar{E}) = 4.34$ against the measured 4.5 (Fig. 4.7b).
By Eq. (4.45), the structure factor at $T = 0$ is

$$S(q,\omega) = \frac{q^2}{2\pi\alpha}\,\frac{\omega_p^2\,\Gamma_{\mathrm{w}}\,\omega}{(\omega^2 - \omega_{\mathrm{res}}^2)^2 + \Gamma_{\mathrm{w}}^2\omega^2},$$

and its first moment is $(q^2/2\pi\alpha)(\pi/2)\omega_p^2 = \pi n_e q^2/m_e$, the f-sum Eq. (4.30). As $\Gamma_{\mathrm{w}} \to 0$, the weight collapses onto $\omega = \omega_{\mathrm{res}}$ with $\int \mathrm{d}\omega\, S = \pi n_e q^2/(m_e\omega_{\mathrm{res}})$, and by Eq. (4.17) $S^{\mathrm{st}} = q^2/(2m_e\omega_{\mathrm{res}})$: the density fluctuations vanish as q^2 at small q, below the bound $q^2/(2m_e E_{\mathrm{g}})$ of Sect. 4.3 since $\omega_{\mathrm{res}} > E_{\mathrm{g}}$.
For the stopping power of a slow particle, take $qv \ll \omega_{\mathrm{res}}$ at every q. Since ω_{res}/q is smallest, $(\bar{E}/m_e)^{1/2} = 5.8 \times 10^{-3}$, at $q = (2m_e\bar{E})^{1/2} = 4.2\,\mathrm{keV}$, this condition means $v \ll 5.8 \times 10^{-3}$. Then $L \simeq \omega_p^2\Gamma_{\mathrm{w}}\omega/\omega_{\mathrm{res}}^4$ on the whole range of the inner integral of

Eq. (4.54), which becomes $\omega_p^2 \Gamma_\mathrm{w} q^3 v^3/(3\omega_\mathrm{res}^4)$. With $q = (2m_e\bar{E})^{1/2}x$, the remaining integral is $\int_0^\infty q^2\mathrm{d}q/\omega_\mathrm{res}^4 = (2m_e\bar{E})^{3/2}\bar{E}^{-4}\int_0^\infty x^2\mathrm{d}x/(1+x^4)^2$, and the last integral is $\pi/(8\sqrt{2})$. Altogether

$$-\frac{\mathrm{d}E}{\mathrm{d}x} \simeq \frac{Z_1^2\,\alpha\,\Gamma_\mathrm{w}\,\omega_p^2\,m_e^{3/2}}{6\,\bar{E}^{5/2}}\,v.$$

The slow particle feels a friction proportional to its speed, with a coefficient set by the damping. For silicon's numbers and $Z_1 = 1$ the stopping power is 0.18 eV/Å at $v = 10^{-3}$; the full integral Eq. (4.54) gives 0.186 eV/Å, 3 percent more, and 22 percent more at $v_\mathrm{max} = 2.65 \times 10^{-3}$, where the condition $v \ll 5.8 \times 10^{-3}$ is only marginal.

The coefficient tests only the tail of the model. The energies that matter are up to $qv \approx 4\,\mathrm{eV}$ at the dominant $q \approx 4\,\mathrm{keV}$, and there the oscillator and silicon disagree badly: at 3 eV the model has $L = 0.035$, eleven times the measured value, and it has weight below the gap, where silicon has none (Fig. 4.7b). Both loss functions satisfy the f-sum and give the same ε_∞. The sum rules fix the main peak, while slow particles, and dark matter, sample the low-energy side of L, which only a calculation or a measurement of the actual material can supply.

4.5 The homogeneous electron gas

Spread the four valence electrons of each silicon atom uniformly over the crystal, replace the ions by a uniform positive background, and ask where the loss function of this electron gas peaks at long wavelength. The answer is the plasma frequency, $\omega_p = 16.6\,\mathrm{eV}$, and the measured loss peak of silicon in Fig. 4.7 lies at 16.4 eV. A model with no lattice, no gap and no bonds places the main feature of the spectrum within 1.2 percent of the measured peak. The reason is that at long wavelength and high frequency the valence electrons respond collectively, and the details of the crystal matter less than the Coulomb force between them. The electron gas is also the one interacting system in which the independent-particle response and its screened version can be written in closed form, and the gas therefore serves as the reference model for valence electrons, plasmons and screening.

The model is jellium (Worked example 2.5) at the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$. Its density parameter, the radius per electron in units of the Bohr radius, is $r_s = 2.0$ (Sect. 2.3), and the free-gas scales of Eq. (2.71) are $k_\mathrm{F} = 3.57\,\mathrm{keV}$, $E_\mathrm{F} = 12.5\,\mathrm{eV}$ and $v_\mathrm{F} = 7.0 \times 10^{-3}$. In these units the plasma frequency is $\omega_p = 1.33\,E_\mathrm{F}$ and the Thomas–Fermi wavevector of Eq. (4.53) is $k_\mathrm{TF} = 1.15\,k_\mathrm{F}$. Both ratios grow with r_s, since $k_\mathrm{TF}^2/k_\mathrm{F}^2 = 4\alpha m_e/(\pi k_\mathrm{F}) = 0.663\,r_s$ with $k_\mathrm{F} = (9\pi/4)^{1/3}/(r_s a_0)$: the denser the gas, the weaker the Coulomb force compared with the kinetic energy.

4.5.1 The independent-particle bubble

Without the Coulomb interaction the electrons of jellium are free, and the Lehmann representation Eq. (4.6) of their density response is a sum over the moves of one electron. Take an occupation-number state $|i\rangle$ of plane waves, with the spin included in the label $\boldsymbol{k}$. The operator $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}} c_{\boldsymbol{k}}$ of Eq. (2.38) moves one electron from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$, and the move is possible only if $\boldsymbol{k}$ is occupied and $\boldsymbol{k}+\boldsymbol{q}$ empty. In the resonant term of Eq. (4.6) the final state $|f\rangle$ is this move, with weight $n_{\boldsymbol{k}}(1-n_{\boldsymbol{k}+\boldsymbol{q}})$ and energy $E_f - E_i = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}}$. In the antiresonant term $n_e(\boldsymbol{q})^\dagger$ makes the reverse move, from $\boldsymbol{k}+\boldsymbol{q}$ to $\boldsymbol{k}$, with weight $n_{\boldsymbol{k}+\boldsymbol{q}}(1-n_{\boldsymbol{k}})$ and energy $\xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}$. The two terms then share the denominator $\omega + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+$, and

their numerators combine to $n_{\boldsymbol{k}}(1-n_{\boldsymbol{k}+\boldsymbol{q}})-n_{\boldsymbol{k}+\boldsymbol{q}}(1-n_{\boldsymbol{k}})=n_{\boldsymbol{k}}-n_{\boldsymbol{k}+\boldsymbol{q}}$. The Pauli factors have cancelled: a move into an occupied state appears in both terms with opposite signs. The thermal average of $n_{\boldsymbol{k}}$ over the Gibbs state is $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, and

$$R^0(\boldsymbol{q},\omega)=\frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}}\frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}})-n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\omega+\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}}+\mathrm{i}0^+}, \tag{4.55}$$

the independent-particle bubble. Eq. (4.55) is the Matsubara bubble Eq. (4.93) continued to real frequencies, obtained here without imaginary time. Its imaginary part counts the transitions of Eq. (2.70), and by the fluctuation–dissipation theorem it is $-\frac{1}{2}S^0$ at $T=0$, with the structure factor of Worked example 4.3.

At $T=0$ the sum is a set of elementary integrals over the Fermi sphere, which Appendix D.6 carries out for any complex frequency z in the upper half plane. The result is the Lindhard function [7].

Lindhard function

$$R^0(q,\omega)=-2\mathcal{N}_{\mathrm{F}}\left\{\frac{1}{2}+\frac{k_{\mathrm{F}}}{4q}\Big[(1-\nu_+^2)\,\ell(\nu_+)-(1-\nu_-^2)\,\ell(\nu_-)\Big]\right\}, \tag{4.56}$$

with $\nu_\pm=(\omega+\mathrm{i}0^+)/qv_{\mathrm{F}}\pm q/2k_{\mathrm{F}}$, $\ell(w)=\ln[(w+1)/(w-1)]$ on the principal branch and $\mathcal{N}_{\mathrm{F}}=m_e k_{\mathrm{F}}/2\pi^2$ per spin. It is Eq. (D.21) at $z=\omega+\mathrm{i}0^+$, and it holds with $\omega+\mathrm{i}0^+$ replaced by any z with $\operatorname{Im} z>0$.

Apart from the prefactor $2\mathcal{N}_{\mathrm{F}}$, the response depends on two numbers only, the momentum in units of the diameter of the Fermi sphere, $y=q/2k_{\mathrm{F}}$, and the frequency in units of the largest Doppler shift, ω/qv_{F}. On the real axis the logarithms take the branch Eq. (C.73), $\ell(\nu+\mathrm{i}0^+)=\ln|(\nu+1)/(\nu-1)|-\mathrm{i}\pi\,\Theta(1-|\nu|)$. The real part, Eq. (D.22), keeps the form of Eq. (4.56) with the logarithms of the moduli. The imaginary part is a sum of steps, and it is nonzero only between the edges

$$\omega_+=qv_{\mathrm{F}}+\frac{q^2}{2m_e},\qquad \omega_-=\max\Big(0,\ \frac{q^2}{2m_e}-qv_{\mathrm{F}}\Big),$$

the particle–hole continuum of Fig. 4.3. Inside the continuum, Eq. (D.23) is $-\frac{1}{2}$ times the linear and parabolic forms of S^0 found by counting transitions in Worked example 4.3, as the fluctuation–dissipation theorem requires. The real part, which counts the same transitions off resonance through their principal values, is the new information.

The static limit and the high-frequency tail of the real part are needed below. At $\omega=0$ (Appendix D.6),

$$R^0(q,0)=-2\mathcal{N}_{\mathrm{F}}\,F_{\mathrm{L}}\Big(\frac{q}{2k_{\mathrm{F}}}\Big),\qquad F_{\mathrm{L}}(y)=\frac{1}{2}+\frac{1-y^2}{4y}\ln\Big|\frac{1+y}{1-y}\Big|, \tag{4.57}$$

with the function F_{L} of Eq. (C.72), which falls from one at $q=0$ to $\frac{1}{2}$ at $q=2k_{\mathrm{F}}$ and decays as $4k_{\mathrm{F}}^2/(3q^2)$ at large q. At high frequency it is simplest to start again from the sum. Let $\Delta_{\boldsymbol{k}}=q^2/2m_e+kqu/m_e$ be the energy of the move, with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$, and let $\langle\cdot\rangle$ be the average over the Fermi sphere. Relabeling $\boldsymbol{k}\to-\boldsymbol{k}-\boldsymbol{q}$ in the term with $n_{\boldsymbol{k}+\boldsymbol{q}}$, as Appendix D.6 does, puts Eq. (4.55) at $T=0$ in the form $R^0=n_e\langle 2\Delta_{\boldsymbol{k}}/(z^2-\Delta_{\boldsymbol{k}}^2)\rangle$. For $|z|>\omega_+$, the largest $|\Delta_{\boldsymbol{k}}|$, the geometric series converges,

$$R^0(q,z)=\frac{2n_e}{z^2}\sum_{j=0}^{\infty}\frac{\langle\Delta_{\boldsymbol{k}}^{2j+1}\rangle}{z^{2j}}=\frac{n_e q^2}{m_e z^2}\left[1+\frac{\frac{3}{5}q^2v_{\mathrm{F}}^2+(q^2/2m_e)^2}{z^2}+\frac{\frac{3}{7}q^4v_{\mathrm{F}}^4+O(q^6)}{z^4}+O(z^{-6})\right]. \tag{4.58}$$

The averages are elementary. Odd powers of u vanish, and $\langle k^2u^2\rangle = \frac{1}{5}k_{\mathrm{F}}^2$ and $\langle k^4u^4\rangle = \frac{3}{35}k_{\mathrm{F}}^4$ from $\langle k^{2n}\rangle = 3k_{\mathrm{F}}^{2n}/(2n+3)$ and $\langle u^{2n}\rangle = 1/(2n+1)$. Hence $\langle\Delta\rangle = q^2/2m_e$, $\langle\Delta^3\rangle = (q^2/2m_e)[(q^2/2m_e)^2 + \frac{3}{5}q^2v_{\mathrm{F}}^2]$ and $\langle\Delta^5\rangle = (q^2/2m_e)[\frac{3}{7}q^4v_{\mathrm{F}}^4 + O(q^6)]$. The leading term is the free-particle tail whose coefficient is the f-sum. The coefficients are the odd moments of the energies of the moves, the frequency moments of S^0 of Sect. 4.3.

The crystal bubble of Sect. 4.6 is computed on a grid of $\boldsymbol{k}$ points, over a finite number of bands and with a broadening in place of $\mathrm{i}0^+$. The closed form Eq. (4.56) has no grid, band cutoff or broadening, and the gas is the standard test of the codes that compute the crystal bubble.

4.5.2 The random-phase approximation

The Coulomb interaction enters through the proper response of Sect. 4.4. The random-phase approximation replaces the proper response by the independent-particle bubble,

$$\tilde{R}_{ee} \simeq R^0, \qquad \varepsilon_{\mathrm{Lin}}(q,\omega) = 1 - v_{\mathrm{C}}(q)\,R^0(q,\omega), \qquad R_{ee} = \frac{R^0}{\varepsilon_{\mathrm{Lin}}}, \tag{4.59}$$

by Eqs. (4.40) and (4.41), and $\varepsilon_{\mathrm{Lin}}$ with R^0 of Eq. (4.56) is the Lindhard dielectric function. In this approximation each electron responds as a free particle to the total potential, external plus Hartree, and to nothing else. The approximation drops the hole that exchange and correlation dig around each electron, a hole that makes the potential an electron feels differ from the average Hartree potential; Sect. 4.6 puts the hole back as an exchange–correlation kernel. The name goes back to Bohm and Pines, who derived the collective oscillation from the equation of motion of $n_e(\boldsymbol{q})$ by dropping the terms that couple it to density fluctuations at other wavevectors, on the argument that their phases are random and average out [22]. The approximation is exact to leading order at high density, $r_s \ll 1$, where the kinetic energy dominates and the correlation hole matters least. At $r_s = 2$ it is not exact, but it gets the long-wavelength physics right, since at small q the Hartree term $v_{\mathrm{C}}R^0 \propto 1/q^2$ dominates every correction.

In the static limit the random-phase approximation screens a slowly varying potential exactly as Thomas–Fermi theory does, and it corrects that theory at shorter wavelengths. By Eq. (4.57) and $k_{\mathrm{TF}}^2 = 2e^2\mathcal{N}_{\mathrm{F}}$,

$$\varepsilon_{\mathrm{Lin}}(q,0) = 1 + \frac{k_{\mathrm{TF}}^2}{q^2}\,F_{\mathrm{L}}\Big(\frac{q}{2k_{\mathrm{F}}}\Big),$$

and $F_{\mathrm{L}} = 1 - q^2/12k_{\mathrm{F}}^2 + O(q^4)$ recovers Eq. (4.53) at $q \ll k_{\mathrm{F}}$. As the wavelength shrinks to the spacing of the electrons, the gas screens less than the local-equilibrium picture of Sect. 4.4 assumes, because a potential that varies on the scale $1/k_{\mathrm{F}}$ cannot be followed by a shift of the local Fermi level. At the silicon valence density and the typical momentum transfer $q_0 = 3.73\,\mathrm{keV}$, $\varepsilon_{\mathrm{Lin}}(q_0,0) = 2.10$ against $\varepsilon_{\mathrm{TF}} = 2.22$, and the screening factor $1/|\varepsilon|^2$ of the static potential is $1/4.41$ rather than $1/4.9$ (Worked example 4.8). At $q = 2k_{\mathrm{F}}$, the momentum that spans the Fermi sphere, F_{L} has a logarithmic kink (Appendix C.8). In position space it turns the screening cloud around an impurity into a damped oscillation at the wavevector $2k_{\mathrm{F}}$, the Friedel oscillation; in the lattice dynamics of a metal the same kink is the Kohn anomaly.

Limit Check 4.1. Thomas–Fermi screening

A static potential that varies slowly in an electron gas is screened by the dielectric function $\varepsilon_{\mathrm{TF}} = 1 + k_{\mathrm{TF}}^2/q^2$, with $k_{\mathrm{TF}}^2 = 3\omega_p^2/v_{\mathrm{F}}^2$ [21, Eq. (S.14)].

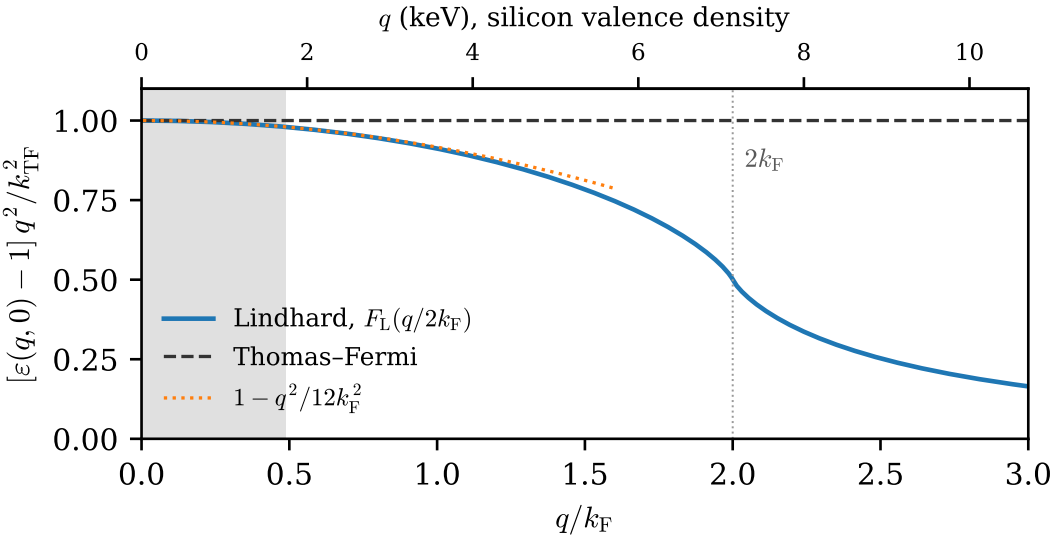


Figure 4.8: Static screening of the electron gas in the random-phase approximation against Thomas–Fermi theory. The induced part of the dielectric function, $[\varepsilon(q,0)-1]\,q^2/k_{\mathrm{TF}}^2$, is the Lindhard function $F_{\mathrm{L}}(q/2k_{\mathrm{F}})$ (solid) and one for Thomas–Fermi (dashed); dotted, the leading correction $1-q^2/12k_{\mathrm{F}}^2$. The curve depends on q/k_{F} alone; the top axis gives q for the silicon valence density, $k_{\mathrm{F}} = 3.57\,\mathrm{keV}$. Gray: $q < 0.49\,k_{\mathrm{F}}$, where the two agree within 2 percent; they differ by 10 percent at $q = 1.06\,k_{\mathrm{F}}$. The kink at $2k_{\mathrm{F}}$ is the edge of the Fermi sphere.

At $q \gtrsim k_{\mathrm{F}}$ the local-equilibrium picture fails and the gas screens less; beyond $2k_{\mathrm{F}}$ the induced term falls as $1/q^4$ instead of $1/q^2$, and screening dies out.

Static screening at long wavelength is perfect. As $q \to 0$, $\varepsilon_{\mathrm{Lin}}(q,0) \to \infty$, and an external charge is completely neutralized by the charge it induces: the induced density at $q \to 0$ is $R_{ee}\phi_{\mathrm{ext}} = -(1-1/\varepsilon)\,\phi_{\mathrm{ext}}/v_{\mathrm{C}}$, the opposite of the external charge. Through the inverse-moment relation Eq. (C.38) perfect screening becomes a sum rule for the loss function, $(2/\pi)\int_0^\infty \mathrm{d}\omega\, L(q,\omega)/\omega = 1 - 1/\varepsilon(q,0) \to 1$, the perfect-screening sum rule of a metal. At the silicon valence density and $q = 0.1\,k_{\mathrm{F}}$ the right side is 0.9925. In an insulator the same integral is $1 - 1/\varepsilon_\infty$ (Eq. (4.48)), 0.915 for silicon; the gas, with no gap, screens like a metal.

Fig. 4.10a shows the loss function $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{Lin}})$ of the gas at $r_s = 2$ over the (q,ω) plane. Inside the particle–hole continuum, $\varepsilon_2 = -v_{\mathrm{C}}\,\mathrm{Im}\,R^0$ and $L = \varepsilon_2/|\varepsilon_{\mathrm{Lin}}|^2$. At $q \gg k_{\mathrm{F}}$, where $v_{\mathrm{C}}R^0$ is small, $\varepsilon_{\mathrm{Lin}} \to 1$, and the loss function is the free-recoil parabola of Worked example 4.3, centered on $q^2/2m_e$: at short wavelength the electrons scatter one at a time, and screening is irrelevant. At $q \lesssim k_{\mathrm{F}}$ the factor $1/|\varepsilon_{\mathrm{Lin}}|^2$ suppresses the continuum, and the weight moves out of the continuum into a line above it. Since the f-sum rule fixes the total weight at every q, the weight that screening removes from the pairs goes into the line.

4.5.3 The plasmon

The line is the plasma oscillation, and its frequency follows from a classical argument. Displace all the electrons of a slab of jellium by a small distance s along x, against the fixed positive background. The two faces of the slab acquire surface charges $\pm e n_e s$ per unit area, and between them the field, in Heaviside–Lorentz units, is $E = e n_e s$. Each electron feels the restoring force $-eE$, and $m_e \ddot{s} = -e^2 n_e s$: the electrons oscillate together at $\omega_p = (e^2 n_e/m_e)^{1/2}$,

the plasma frequency of Eq. (4.31), whatever the size of the slab. No electron moves relative to its neighbors, and no kinetic energy of the Fermi sea enters. At finite wavelength the kinetic energy of the Fermi sea does enter, and the random-phase approximation includes it.

A pole of the response with nothing to drive it is a zero of the dielectric function, by Eq. (4.40). Above the continuum, $\omega > \omega_+$, $\operatorname{Im} R^0 = 0$ and $\varepsilon_{\rm Lin}$ is real, so a zero there is a pole of $1/\varepsilon_{\rm Lin}$ on the real axis, a mode that does not decay. Inserting Eq. (4.58) into Eq. (4.59), with $v_{\rm C}\boldsymbol{n}_e q^2/m_e = \omega_p^2$,

$$\varepsilon_{\rm Lin}(q,\omega) = 1 - \frac{\omega_p^2}{\omega^2}\left[1 + \frac{A}{\omega^2} + \frac{B}{\omega^4} + \dots\right], \qquad A = \tfrac{3}{5}q^2 v_{\rm F}^2 + \Big(\frac{q^2}{2m_e}\Big)^2, \qquad B = \tfrac{3}{7}q^4 v_{\rm F}^4 + O(q^6).$$

Setting $\varepsilon_{\rm Lin}$ to zero gives $\omega^2 = \omega_p^2 + A\omega_p^2/\omega^2 + B\omega_p^2/\omega^4 + \dots$, which we solve by iteration. At zeroth order $\omega = \omega_p$. Substituting $\omega^2 = \omega_p^2 + A$ on the right, $A\omega_p^2/\omega^2 = A - A^2/\omega_p^2 + O(q^6)$ and $B\omega_p^2/\omega^4 = B/\omega_p^2 + O(q^6)$, so

Plasmon dispersion in the random-phase approximation

$$\omega_{\rm pl}(q)^2 = \omega_p^2 + \frac{3}{5}v_{\rm F}^2 q^2 + \Big(\frac{1}{4m_e^2} + \frac{12}{175}\,\frac{v_{\rm F}^4}{\omega_p^2}\Big)q^4 + O(q^6), \tag{4.60}$$

where $\frac{12}{175} = \frac{3}{7} - \frac{9}{25}$ collects B and $-A^2$. The plasmon of the electron gas starts at ω_p and rises quadratically with q.

The q^2 term is the pressure of the Fermi sea. A classical plasma at temperature T has $3T/m_e$ in place of $\frac{3}{5}v_{\rm F}^2$. In the degenerate gas the electrons already move at speeds up to $v_{\rm F}$ at $T = 0$, and $\frac{3}{5}v_{\rm F}^2$ is three times their mean square velocity along $\boldsymbol{q}$, $\langle k^2u^2\rangle/m_e^2 = \frac{1}{5}v_{\rm F}^2$. Written as $\omega_{\rm pl} \simeq \omega_p + \alpha_{\rm pl}q^2/m_e$, the dispersion coefficient is $\alpha_{\rm pl} = \frac{3}{5}E_{\rm F}/\omega_p$, 0.45 at the silicon valence density. The expansion converges only outside the continuum, $\omega_{\rm pl} > \omega_+$, and Fig. 4.9 compares it with the exact zero of $\varepsilon_{\rm Lin}$.

Limit Check 4.2. Plasmon dispersion

In the random-phase approximation the plasmon of an electron gas disperses as $\omega_{\rm pl}^2 \simeq \omega_p^2 + \frac{3}{5}q^2v_{\rm F}^2$ and exists up to a cutoff $q_{\rm c} \simeq \omega_p/v_{\rm F}$ [23, Eq. (171) and the text following it].

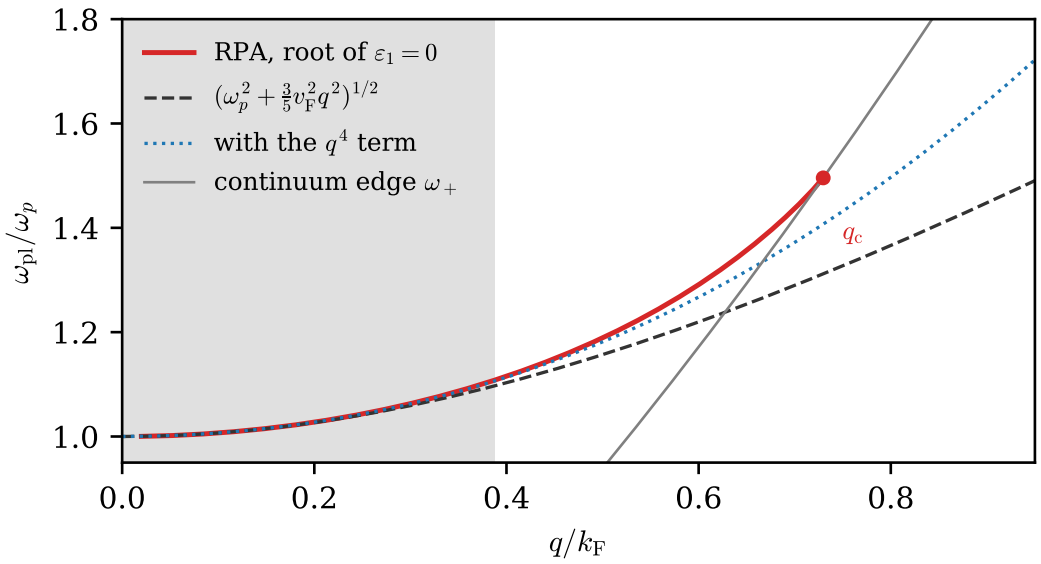


Figure 4.9: Plasmon dispersion of the electron gas at $r_s = 2$, the silicon valence density, in the random-phase approximation. Solid: the zero of $\mathrm{Re}\,\varepsilon_{\mathrm{Lin}}$ above the continuum, computed from Eq. (4.56), ending at $q_c = 0.730\,k_F$ where it meets the continuum edge $\omega_+ = qv_F + q^2/2m_e$ (thin line). Dashed: the quadratic formula; dotted: with the q^4 term of Eq. (4.60). Gray: $q < 0.39\,k_F$, where the quadratic formula is within one percent; there the q^4 term reduces the error to below 1.6×10^{-3}, and to below 3×10^{-4} for $q < 0.30\,k_F$. The estimate $q_c \simeq \omega_p/v_F = 0.666\,k_F$ is 9 percent low.

Near q_c the series in q^2 fails, since the plasmon approaches the pair energies that set its radius of convergence; beyond q_c there is no undamped plasmon.

The weight of the plasmon follows from the slope of $\varepsilon_{\mathrm{Lin}}$ at its zero. Near ω_{pl}, $\varepsilon_{\mathrm{Lin}} \simeq (\partial\varepsilon_1/\partial\omega)(\omega - \omega_{\mathrm{pl}} + \mathrm{i}0^+)$, with the $\mathrm{i}0^+$ carried from ω, and the Sokhotski–Plemelj formula gives

$$L(q,\omega) = \frac{\pi}{\partial\varepsilon_1/\partial\omega|_{\omega_{\mathrm{pl}}}}\,\delta(\omega - \omega_{\mathrm{pl}}(q)) \;\to\; \frac{\pi\omega_p}{2}\,\delta(\omega - \omega_p) \qquad (q \to 0), \tag{4.61}$$

since $\varepsilon_1 \to 1 - \omega_p^2/\omega^2$ with slope $2/\omega_p$ at ω_p. At long wavelength the line alone satisfies both sum rules. Its first moment is $(\pi/2)\omega_p^2$, the whole f-sum Eq. (4.46), and its inverse moment is $\pi/2$, the whole perfect-screening rule. The continuum, whose weight vanishes as $q \to 0$, has nothing left. By Eq. (4.45), the structure factor at $T = 0$ is then $S = (\pi q^2\omega_p/e^2)\,\delta(\omega - \omega_p)$. Its integral over ω is $2\pi n_e S^{\mathrm{st}}$ by Eq. (4.17), and with $e^2 n_e = m_e\omega_p^2$

$$S^{\mathrm{st}}(q) \to \frac{q^2}{2m_e\omega_p} \qquad (q \to 0).$$

The mean energy $q^2/(2m_e S^{\mathrm{st}})$ of Sect. 4.3 is ω_p at small q, and the density fluctuations of the interacting gas vanish as q^2, much faster than the $3q/4k_F$ of the free gas. At $q = 0.1\,k_F$ the interacting and free values are 7.5×10^{-3} and 7.5×10^{-2}. The Coulomb force suppresses long-wavelength fluctuations of the charge, since any long-wavelength charge fluctuation costs the energy of a plasma oscillation. At finite q the plasmon shares the f-sum with the continuum. Integrating the gridded loss function of Fig. 4.10 and adding the weight of Eq. (4.61) returns $(\pi/2)\omega_p^2$ to 10^{-5} at $q = 0.1\,k_F$ and $0.5\,k_F$; the plasmon carries $1 - 4 \times 10^{-5}$ of it at $0.1\,k_F$ and 0.957 at $0.5\,k_F$.

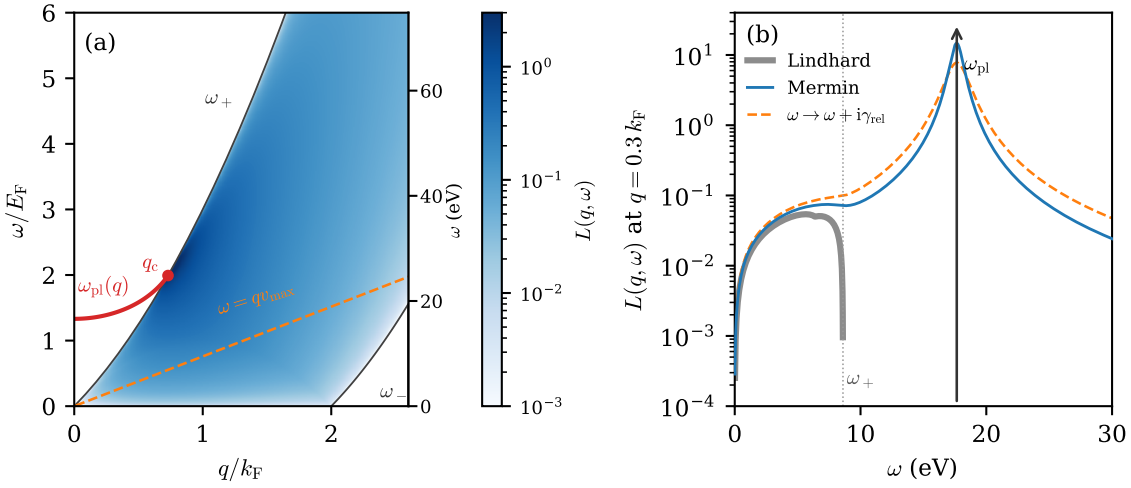


Figure 4.10: Loss function of the electron gas at $r_s = 2$, the valence density of silicon ($k_F = 3.57\,\text{keV}$, $E_F = 12.5\,\text{eV}$, $\omega_p = 16.6\,\text{eV}$), in the random-phase approximation at $T = 0$. (a) $L(q,\omega) = \text{Im}[-1/\varepsilon_{\text{Lin}}]$, logarithmic color scale, inside the particle–hole continuum bounded by $\omega_\pm$ (thin lines); the right axis gives ω in eV. Red: the undamped plasmon $\omega_{\text{pl}}(q)$, a delta function in L, drawn as a line up to $q_c = 0.730\,k_F$ (dot), where it enters the continuum and is Landau damped. Dashed: the largest energy a dark-matter particle can deposit at momentum q, $\omega = qv_{\text{max}}$ with $v_{\text{max}} = 0.379\,v_F$; it stays below the plasmon at every q. (b) L at $q = 0.3\,k_F$ on a logarithmic scale. Thick gray: Lindhard, with the continuum ending at ω_+ and the plasmon as a delta function (arrow, at 17.65 eV). Solid: the Mermin function with $\gamma_{\text{rel}} = 1\,\text{eV}$, an illustrative relaxation rate. Dashed: the naive substitution $\omega \to \omega + i\gamma_{\text{rel}}$ in R^0 with the same rate. Its plasmon peak is twice as wide, and its tail between the continuum and the plasmon is up to 1.7 times higher.

The plasmon cannot decay as long as no electron–hole pair has its energy and momentum, that is, as long as $\omega_{\text{pl}}(q) > \omega_+(q)$. The dispersion rises as q^2 from ω_p, while the continuum edge rises linearly from zero, and the two meet at the momentum q_c defined by $\varepsilon_{\text{Lin}}(q_c, \omega_+(q_c)) = 0$. A rough estimate sets $\omega_+ \simeq qv_F$ equal to ω_p, $q_c \simeq \omega_p/v_F$. At $r_s = 2$ the exact root is $q_c = 0.730\,k_F = 2.60\,\text{keV}$, at $\omega_{\text{pl}} = 24.8\,\text{eV}$; the estimate is $0.666\,k_F$. Beyond q_c, $\varepsilon_2 \neq 0$ where ε_1 vanishes, the zero of ε_{Lin} continued below the real axis moves into the lower half plane, and the plasmon becomes a resonance of the kind described in Sect. 4.1, with a width that grows quickly with q. Its energy goes into single electron–hole pairs, a decay called Landau damping, after the analogous damping of waves in a classical plasma by particles that move with the wave.

In Fig. 4.10a the plasmon line enters the continuum at q_c, where the loss function of the continuum is largest. From there a broad Landau-damped ridge continues toward the free-recoil parabola. Dark matter samples the low-energy corner of the continuum, where the electrons near the Fermi surface absorb the energy and the factor $1/|\varepsilon_{\text{Lin}}|^2$ screens the absorption. At small q and $\omega \ll qv_F$ the corner has a simple form. There $\varepsilon_2 = (e^2/q^2)\, m_e^2\omega/(2\pi q)$ from the linear form of $\text{Im}\, R^0$, and $\varepsilon_1 \simeq k_{\text{TF}}^2/q^2 \gg \varepsilon_2$, so

$$L(q,\omega) \simeq \frac{\varepsilon_2}{\varepsilon_1^2} = \frac{e^2 m_e^2}{2\pi k_{\text{TF}}^4}\, q\,\omega \qquad (q \ll k_F,\ \omega \ll qv_F),$$

linear in both q and ω, and suppressed by the fourth power of the screening wavevector. At the typical momentum transfer q_0 and the largest energy $q_0 v_{\text{max}} = 9.88\,\text{eV}$, the full Lindhard

function yields $|\varepsilon_{\mathrm{Lin}}|^2 = 3.96$ and $L = 0.183$: dynamic screening at the energy deposited is slightly weaker than the static estimate of Worked example 4.8.

4.5.4 Collisions and the Mermin function

In a real metal, or a doped semiconductor, the electrons scatter off impurities and phonons at some rate γ_{rel}, and the sharp edges of the Lindhard function and its undamped plasmon are smeared on that scale. The obvious way to include the collisions is to give the frequency an imaginary part, $\omega \to \omega + \mathrm{i}\gamma_{\mathrm{rel}}$, in R^0. This substitution is equivalent to letting the electrons relax toward the uniform equilibrium state at the rate γ_{rel}. It has an unphysical consequence: the excess density at a crest of the disturbance decays at that rate even when no current flows to carry it away. Electrons are destroyed at the crests and created in the troughs. Collisions with impurities and phonons conserve the number of electrons, and they relax the shape of the distribution at each point while leaving the number of electrons there unchanged. Mermin [24] therefore let the electrons relax toward a local equilibrium that has, at each instant, the same density as the disturbed gas. Appendix D.7 solves the equation of motion of the one-electron density matrix with this relaxation term and finds the response $\delta n_e/\phi = R^{\mathrm{Mer}}$, which in the random-phase approximation gives the Mermin dielectric function.

Mermin dielectric function

$$\varepsilon_{\mathrm{Mer}}(q,\omega) = 1 + \frac{(1 + \mathrm{i}\gamma_{\mathrm{rel}}/\omega)\,[\varepsilon_{\mathrm{Lin}}(q,\omega+\mathrm{i}\gamma_{\mathrm{rel}}) - 1]}{1 + (\mathrm{i}\gamma_{\mathrm{rel}}/\omega)\,[\varepsilon_{\mathrm{Lin}}(q,\omega+\mathrm{i}\gamma_{\mathrm{rel}}) - 1]/[\varepsilon_{\mathrm{Lin}}(q,0) - 1]}, \tag{4.62}$$

with $\varepsilon_{\mathrm{Lin}}$ at the complex frequency $\omega + \mathrm{i}\gamma_{\mathrm{rel}}$ from Eq. (4.56) and its static value from Eq. (4.57). It conserves the number of electrons at every q and ω, and it reduces to $\varepsilon_{\mathrm{Lin}}$ as $\gamma_{\mathrm{rel}} \to 0$.

This is Eq. (D.26), Mermin's form [24], and Refs. [5, Eq. (19)] and [23, Eq. (168)], where the rate is written Γ and Γ_p. Analyticity of $R^0(q, z + \mathrm{i}\gamma_{\mathrm{rel}})$ in the upper half plane does not by itself exclude a zero of the denominator there. Appendix D.7 shows that relaxation toward local equilibrium has no growing free mode, so the Mermin response has no pole in the upper half plane and is causal.

In the continuity equation Eq. (D.27), the naive substitution adds a source term, $-\mathrm{i}\gamma_{\mathrm{rel}}$ times the density, and Mermin's reference state makes this term vanish identically. A static potential and the long-wavelength limit tell the two prescriptions apart, as Appendix D.7 works out; the f-sum rule does not. For a static potential, $\omega \to 0$ at fixed γ_{rel}, $\varepsilon_{\mathrm{Mer}} \to \varepsilon_{\mathrm{Lin}}(q,0)$: a static potential leaves the electrons in equilibrium in that potential, with or without collisions, so the static screening, perfect screening included, is that of the collisionless gas. The naive substitution yields $\varepsilon_{\mathrm{Lin}}(q, \mathrm{i}\gamma_{\mathrm{rel}})$ instead; at $q = 0.3\,k_{\mathrm{F}}$ and $\gamma_{\mathrm{rel}} = 1\,\mathrm{eV}$ in the silicon-density gas the naive value is 12.8 against the correct 15.7. At long wavelength, $q \to 0$ at fixed ω, the Mermin function becomes the Drude dielectric function,

$$\varepsilon_{\mathrm{Mer}} \to 1 - \frac{\omega_p^2}{\omega(\omega + \mathrm{i}\gamma_{\mathrm{rel}})} = 1 + \frac{\mathrm{i}\sigma_{\mathrm{cond}}}{\omega}, \qquad \sigma_{\mathrm{cond}}(\omega) = \frac{\omega_p^2}{\gamma_{\mathrm{rel}} - \mathrm{i}\omega},$$

by Eq. (4.50), with the dc conductivity $\omega_p^2/\gamma_{\mathrm{rel}}$ of a metal whose electrons are scattered at the rate γ_{rel}. The naive substitution leads to $1 - \omega_p^2/(\omega + \mathrm{i}\gamma_{\mathrm{rel}})^2$, which tends to $1 + \omega_p^2/\gamma_{\mathrm{rel}}^2$ as $\omega \to 0$: no dc current and a finite static dielectric constant, the response of an insulator. Both functions are analytic in the upper half plane and both fall off as ω_p^2/ω^2, so both obey Eq. (4.46) by the argument of Appendix C.5.

At $q \to 0$ the Drude form puts the whole loss function in one Lorentzian line [23, Eq. (172)],

$$L(0,\omega) = \frac{\omega_p^2\,\gamma_{\mathrm{rel}}\,\omega}{(\omega^2-\omega_p^2)^2+\gamma_{\mathrm{rel}}^2\omega^2}.$$

This Lorentzian is the damped oscillator of Worked example 4.9 with $\omega_{\mathrm{res}} = \omega_p$, no gap, and γ_{rel} in the role of Γ_{w}. The plasmon now has the full width γ_{rel}, and below it the loss function has a tail, $L \simeq \gamma_{\mathrm{rel}}\omega/\omega_p^2$ for $\omega \ll \omega_p$, linear in ω down to zero frequency. The naive substitution puts the pole at $\pm\omega_p - \mathrm{i}\gamma_{\mathrm{rel}}$, with twice the width and twice the tail, $2\gamma_{\mathrm{rel}}\omega/\omega_p^2$. Fig. 4.10b compares the three functions at $q = 0.3\,k_{\mathrm{F}}$. For a dark-matter particle, which deposits energies far below the plasmon, the tail is the part that matters. The tail is weight that the Lindhard function does not have at small q, and whether a real material has it is a question about the material, not about the gas [21, 23].

In practice the Mermin function is used as a fitting form. The plasmon of silicon has a width of a few eV, but it does not come from collisions: the plasmon decays into interband electron–hole pairs, as Worked example 4.10 shows. A rate γ_{rel} fitted to that width is a parameter of the fit, not a collision rate. The Mermin oscillator method of Ref. [5, Eqs. (20) and (21)] writes the measured optical loss function as a sum of Mermin terms,

$$L(q,\omega) = \sum_i A_i(q)\ \mathrm{Im}\big[-1/\varepsilon_{\mathrm{Mer}}(q,\omega;\omega_{p,i},\gamma_i)\big]\,\Theta(\omega-\omega_{\mathrm{edge},i}).$$

Each term has its own plasma frequency $\omega_{p,i}$, rate γ_i and onset $\omega_{\mathrm{edge},i}$. The parameters are fitted at $q = 0$, and the Mermin function carries each term to finite q. The amplitudes $A_i(q)$ are rescaled at each q so that every term keeps its share of the f-sum. The step functions remove the weight below the gap that a single gas would put there. The method is a way to extrapolate optical data to the momentum transfers of dark-matter scattering while keeping the sum rules; the physics of the extrapolation is that of the gas.

Worked example 4.10: The silicon valence electrons as an electron gas

Treat the four valence electrons per atom of silicon as jellium at $n_e = 2.00\times10^{23}\,\mathrm{cm}^{-3}$, $r_s = 2.0$. The plasma frequency is $\omega_p = 16.6\,\mathrm{eV}$ (Eq. (4.31)), and by Eq. (4.60) the plasmon disperses as $\omega_{\mathrm{pl}} \simeq \omega_p + 0.45\,q^2/m_e$, up to $q_{\mathrm{c}} = 2.60\,\mathrm{keV}$ and $\omega_{\mathrm{pl}} = 24.8\,\mathrm{eV}$. The measured loss function at $q \to 0$ (Fig. 4.7) peaks at 16.4 eV, and ε_1 crosses zero at 16.3 eV; the gas places the zero 1.9 percent higher.

Position. Near 16 eV the valence electrons of silicon respond almost as if they were free. In the tabulated data, 87 percent of their f-sum weight $\int \omega\varepsilon_2\,\mathrm{d}\omega$ lies below 16.3 eV, at a root-mean-square energy of 6.5 eV, in the interband transitions of Fig. 4.7a. A transition at an energy ω_i well below ω contributes to ε_1 as an oscillator driven above its resonance, $-f_i/(\omega^2-\omega_i^2) \simeq -(f_i/\omega^2)(1+\omega_i^2/\omega^2)$, which is the free-electron term with a correction of order $(\omega_i/\omega)^2 \approx 0.16$. The slope of the data confirms the picture: at the zero, $\partial\varepsilon_1/\partial\omega = 0.128\,\mathrm{eV}^{-1}$, against $2\omega_p^2/\omega^3 = 0.127\,\mathrm{eV}^{-1}$ for the gas. The correction from binding has the sign that pushes the zero up; if all the weight were in one oscillator at 5.08 eV, the resonance of the single-oscillator model, the zero would move to $\bar{E} = 17.4\,\mathrm{eV}$ (Worked example 4.9). The remaining weight, 9.4 percent between 16.3 eV and 40 eV and the rest above 40 eV, lies above the plasmon and contributes positively to ε_1, which pulls the zero down. The two shifts nearly cancel.

Width. In the gas at $q \to 0$ the plasmon cannot decay, since the continuum shrinks to $\omega < qv_{\mathrm{F}} \to 0$. In silicon, vertical interband transitions exist at all energies up to well above 16 eV, and at the zero of ε_1 the data have $\varepsilon_2 = 0.224$. Near the zero $L = \varepsilon_2/(\varepsilon_1^2+\varepsilon_2^2)$

with $\varepsilon_1 \simeq (\partial\varepsilon_1/\partial\omega)(\omega - \omega_{\rm pl})$, a Lorentzian of full width $2\varepsilon_2/(\partial\varepsilon_1/\partial\omega) = 3.5\,\text{eV}$, against the measured 3.66 eV. The plasmon of silicon decays into interband electron–hole pairs: the crystal potential supplies the momentum that, in the gas, only $q > q_{\rm c}$ could provide. A Mermin function with $\gamma_{\rm rel} = 3.66\,\text{eV}$ reproduces the line: at $q \to 0$ its peak height is $\omega_p/\gamma_{\rm rel} = 4.54$, against the measured 4.5. Its tail, however, is $L = 0.042$ at 3 eV, 14 times the measured 3.1×10^{-3}, and it continues below the gap, where silicon absorbs nothing, just as the single oscillator of Worked example 4.9 did.
Dark matter. The plasmon is out of reach. A halo particle deposits at most $qv_{\rm max}$, and with $v_{\rm max} = 0.379\,v_{\rm F}$ the line $\omega = qv_{\rm max}$ reaches $\omega_p = 16.6\,\text{eV}$ only at $q = \omega_p/v_{\rm max} = 6.26\,\text{keV} = 1.75\,k_{\rm F}$, far beyond $q_{\rm c}$, where the gas has no plasmon left (Fig. 4.10a). The plasmon, which carries most of the weight at small q, is kinematically inaccessible [23]. The particle samples the low-energy side of the loss function, at a few eV, where the gas and the crystal differ most. The gas has a continuum down to $\omega = 0$, screened by $|\varepsilon_{\rm Lin}|^2 \approx 4$ at q_0, while silicon has a gap, interband peaks, and an optical loss function of 3.1×10^{-3} at 3 eV. The electron gas is a good model of the peak that holds most of the f-sum and a poor one of the corner that sets the rate. Its use in the dark-matter literature is accordingly limited to energies well above the gap, between about 5 and 15 eV, where it describes the computed spectrum of silicon and germanium fairly well [23].

4.6 Local fields and exchange–correlation kernels

The absorption spectrum of silicon has two peaks, at 3.42 and 4.24 eV (Fig. 4.7), and its static screening saturates at $\varepsilon_\infty = 11.7$. A calculation that lets each electron respond on its own, with band energies fitted to experiment, puts the upper peak in nearly the right place, turns the lower one into a shoulder, and finds a dielectric constant within a few percent of 11.7. Adding the microscopic fields that the induced charge produces inside each cell lowers both peaks and lowers ε_∞ by about a tenth. Adding the simplest exchange–correlation kernel undoes about half of this reduction of ε_∞ and barely changes the shape of the spectrum. Only the attraction between the excited electron and its hole, the subject of Sect. 4.9, restores the lower peak. Each of these steps is a different approximation to the proper response $\tilde{R}$ of Sect. 4.4. For dark matter the differences between these approximations matter where the rate is largest for light masses: in the first few eV above the gap, where the loss function is small and the screening strong.

4.6.1 The crystal bubble

Take the Kohn–Sham states of Sect. 3.4, with band energies $\varepsilon_{n\boldsymbol{k}}$ and occupations $f_{n\boldsymbol{k}}$ (Eq. (3.24)), as a system of independent electrons, and compute its density response matrix $R_{\boldsymbol{G}\boldsymbol{G}'}$ of Eq. (4.42) from the Lehmann representation Eq. (4.6). The steps are those of the electron gas in Sect. 4.5, with Bloch states in place of plane waves. In the Bloch basis the density operator at $\boldsymbol{q} + \boldsymbol{G}'$ is Eq. (3.37) with $\boldsymbol{q} \to \boldsymbol{q} + \boldsymbol{G}'$,

$$n_e(\boldsymbol{q} + \boldsymbol{G}') = \sum_s \sum_{\boldsymbol{k}} \sum_{mn} \rho_{mn}(\boldsymbol{k}, \boldsymbol{q} + \boldsymbol{G}')\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{q},s}\, c_{n\boldsymbol{k}s},$$

with $\boldsymbol{q}$ in the first zone. The reciprocal vector does not change the crystal momentum of the final state, $\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}' \equiv \boldsymbol{k}+\boldsymbol{q}$; it only changes the plane wave that probes the two states, through the density vertex ρ_{mn} of Eq. (3.34). Acting on an occupation-number state $|i\rangle$ of Bloch states, the operator moves one electron from $n\boldsymbol{k}$ to $m, \boldsymbol{k}+\boldsymbol{q}$. In the resonant term of Eq. (4.6) the

final state is this move, with weight $f_{n\boldsymbol{k}}(1-f_{m,\boldsymbol{k}+\boldsymbol{q}})$, energy $\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}}-\varepsilon_{n\boldsymbol{k}}$ and matrix elements $\langle f|n_e(\boldsymbol{q}+\boldsymbol{G}')|i\rangle=\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')$ and $\langle i|n_e(\boldsymbol{q}+\boldsymbol{G})^\dagger|f\rangle=\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*$. In the antiresonant term the final state has the reverse move, from $m,\boldsymbol{k}+\boldsymbol{q}$ to $n\boldsymbol{k}$, with weight $f_{m,\boldsymbol{k}+\boldsymbol{q}}(1-f_{n\boldsymbol{k}})$; its two matrix elements are the same pair, and its denominator $\omega+(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}})+\mathrm{i}0^+$ is the denominator of the resonant term. As in the gas, the Pauli factors cancel in the sum of the two numerators, $f_{n\boldsymbol{k}}(1-f_{m,\boldsymbol{k}+\boldsymbol{q}})-f_{m,\boldsymbol{k}+\boldsymbol{q}}(1-f_{n\boldsymbol{k}})=f_{n\boldsymbol{k}}-f_{m,\boldsymbol{k}+\boldsymbol{q}}$, and the sum over the two spins gives the degeneracy factor $g_{\mathrm{deg}}=2$.

Independent-particle bubble of a crystal

$$R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega)=\frac{g_{\mathrm{deg}}}{\mathscr{V}}\sum_{\boldsymbol{k}}\sum_{n,m}(f_{n\boldsymbol{k}}-f_{m,\boldsymbol{k}+\boldsymbol{q}})\,\frac{\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*\,\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')}{\omega+\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}}+\mathrm{i}0^+},\qquad(4.63)$$

with $\mathscr{V}=N\Omega$, the sum over the N crystal momenta of the first zone, and all pairs of bands. It is the density response at $\boldsymbol{q}+\boldsymbol{G}$ to a potential at $\boldsymbol{q}+\boldsymbol{G}'$ of electrons that move independently in the Kohn–Sham potential.

Eq. (4.63) is the result of Adler and Wiser [19, 20], and it is Eq. (26.1), written there with a finite broadening. For one band of plane waves, $\rho_{mn}\to 1$ at $\boldsymbol{G}=\boldsymbol{G}'=0$ and the matrix reduces to the bubble Eq. (4.55) of the gas. In a crystal the off-diagonal elements do not vanish: a potential at $\boldsymbol{q}$ moves electrons into states whose overlap with the initial ones is modulated on the scale of the cell, and the induced density has components at every $\boldsymbol{q}+\boldsymbol{G}$.

At small q the vertex at $\boldsymbol{G}=0$ between different bands is the optical one, Eq. (3.43), $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})=\boldsymbol{q}\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})/(\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}})$, while the vertices at $\boldsymbol{G}\neq 0$ tend to the finite $\rho_{cv}(\boldsymbol{k},\boldsymbol{G})$. In an insulator at $T=0$ only valence-to-conduction pairs contribute, with $f_{v\boldsymbol{k}}=1$ and $f_{c\boldsymbol{k}}=0$. Write $\Delta_{cv\boldsymbol{k}}=\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}}$. The pairs $(n,m)=(v,c)$ and (c,v) of Eq. (4.63) give the head

$$R^0_{\boldsymbol{0}\boldsymbol{0}}(\boldsymbol{q}\to 0,\omega)=\frac{g_{\mathrm{deg}}q^2}{\mathscr{V}}\sum_{\boldsymbol{k}}\sum_{v,c}\frac{|\hat{\boldsymbol{q}}\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})|^2}{\Delta^2_{cv\boldsymbol{k}}}\left[\frac{1}{\omega-\Delta_{cv\boldsymbol{k}}+\mathrm{i}0^+}-\frac{1}{\omega+\Delta_{cv\boldsymbol{k}}+\mathrm{i}0^+}\right],\qquad(4.64)$$

of order q^2, as Appendix D.5 requires of every proper response of an insulator. Its imaginary part, multiplied by $-v_{\mathrm{C}}(q)=-e^2/q^2$, is the absorptive part of the head of the dielectric matrix at positive frequency,

$$\varepsilon_2^{\mathrm{IP}}(\omega)=\frac{\pi g_{\mathrm{deg}}e^2}{\mathscr{V}}\sum_{\boldsymbol{k}}\sum_{v,c}\frac{|\hat{\boldsymbol{q}}\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})|^2}{\Delta^2_{cv\boldsymbol{k}}}\,\delta(\omega-\Delta_{cv\boldsymbol{k}}),\qquad(4.65)$$

where the label IP marks independent particles. It counts the vertical transitions at energy ω, each weighted by its squared velocity matrix element along the field: the count is the golden-rule absorption of light, through $\sigma_1=\omega\varepsilon_2$ of Eq. (4.50). The real part at $\omega=0$ is

$$\varepsilon_1^{\mathrm{IP}}(0)=1+\frac{2g_{\mathrm{deg}}e^2}{\mathscr{V}}\sum_{\boldsymbol{k}}\sum_{v,c}\frac{|\hat{\boldsymbol{q}}\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})|^2}{\Delta^3_{cv\boldsymbol{k}}},\qquad(4.66)$$

since the bracket of Eq. (4.64) is $-2/\Delta_{cv\boldsymbol{k}}$ at $\omega=0$. The third power of the transition energy makes the static screening sensitive to the smallest gaps. A Kohn–Sham gap that is too small therefore makes ε_∞ too large, as the PBE value 13.04 against the measured 11.7 shows. The scissor of Eq. (3.66) leaves ρ_{cv}, and so the ratio $\boldsymbol{v}_{cv}/\Delta_{cv\boldsymbol{k}}$, unchanged (Eq. (3.67)). It then shifts $\varepsilon_2^{\mathrm{IP}}$ rigidly to higher energy and lowers Eq. (4.66), whose terms go as $|\rho_{cv}/q|^2/\Delta_{cv\boldsymbol{k}}$.

In practice Eq. (4.63) is summed on a grid of $\boldsymbol{k}$ points, over the occupied bands and a finite number of empty ones, with a broadening in place of $\mathrm{i}0^+$ that must be larger than the

spacing of the transition energies on the grid. The head and the wings, the elements with one index zero, are computed analytically in $\hat{\boldsymbol{q}}$ from velocity matrix elements, and the matrix is cut off at a finite number of reciprocal vectors.

4.6.2 The random-phase approximation in a crystal

The random-phase approximation of Sect. 4.5 replaces the proper response by the bubble, now as a matrix,

$$\tilde{R}_{\boldsymbol{G}\boldsymbol{G}'} \simeq R^0_{\boldsymbol{G}\boldsymbol{G}'}, \qquad \varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, R^0_{\boldsymbol{G}\boldsymbol{G}'}, \tag{4.67}$$

with Eq. (4.43). Each electron responds as an independent particle to the total potential, external plus Hartree, at every $\boldsymbol{q}+\boldsymbol{G}$. The macroscopic function then comes at two levels. Keeping only the head gives $\varepsilon_{\mathrm{mac}} \simeq \varepsilon_{\boldsymbol{00}} = 1 - v_{\mathrm{C}}(q) R^0_{\boldsymbol{00}}$, the random-phase approximation without local fields. Inverting the whole matrix gives $\varepsilon_{\mathrm{mac}} = 1/(\varepsilon^{-1})_{\boldsymbol{00}}$, the random-phase approximation with local fields, which by Eq. (D.20) is

$$\varepsilon_{\mathrm{mac}} = 1 - v_{\mathrm{C}}(q)\, \bar{R}_{\boldsymbol{00}}, \qquad \bar{R} = R^0 + R^0\, \bar{v}\, \bar{R}, \tag{4.68}$$

with $\bar{v}$ the Coulomb matrix without its element at $\boldsymbol{G}=0$. In this form the singular head of v_{C} never appears in the inversion, and the limit $\boldsymbol{q}\to 0$ is taken in R^0 alone (Appendix D.5).

Since v_{C} is real, the imaginary part of $\varepsilon_{\boldsymbol{00}}$ is Eq. (4.65) exactly: without local fields the random-phase approximation has the absorption spectrum of independent particles. Screening enters through the real part, and so through the loss function $L = \varepsilon_2/|\varepsilon|^2$ of Sect. 4.4, in which screening suppresses ε_2 by $|\varepsilon|^2$ wherever ε_1 is large. Going from independent particles to the random-phase approximation without local fields changes the rate, through $1/|\varepsilon|^2$, and leaves the optical spectrum unchanged. The local fields are the first correction that changes ε_2.

The local fields have a classical analog. A capacitor filled with layers of different dielectric constants, stacked along the field, has a uniform D and a field $E = D/\varepsilon(x)$ that varies from layer to layer. The capacitance measures the average field, $\langle E\rangle = D\langle 1/\varepsilon\rangle$, so the effective constant is $1/\langle 1/\varepsilon\rangle$, the harmonic mean, which is never larger than the arithmetic mean $\langle\varepsilon\rangle$. The macroscopic function Eq. (4.44) has exactly this structure: it is the inverse of the average of the inverse, while the head $\varepsilon_{\boldsymbol{00}}$ is the average itself. In silicon the polarizable charge lies in the bonds. A uniform field pushes it along the bonds, the displaced charge forms dipoles on the scale of the cell, and their microscopic Hartree field, at $\boldsymbol{G}\neq 0$, points against the applied field in the regions where the polarizable electrons are. The electrons feel less than the average field, and the crystal screens less than the head alone predicts. The macroscopic function and the head would agree if the induced density were uniform, as in the gas, and they disagree most where the response is concentrated in a small part of the cell.

Fig. 4.11 shows the consequence for silicon in a model calculation with the empirical pseudopotential of Cohen and Bergstresser (Sect. 3.3), whose form factors were fitted to measured transition energies; no scissor is needed. The independent-particle spectrum, panel (a), has its upper peak at 4.26 eV, close to the measured 4.24 eV, with a height of 41 against the measured 46. At 3.42 eV, where the measured spectrum has its lower peak of height 36, the model has only 15, a shoulder. The local fields lower the upper peak to 34 and move it to 4.32 eV; at 3.42 eV they leave 13. Spectral weight moves up, as the f-sum rule with local fields (Appendix D.5) requires if the low-energy peaks lose some, and the agreement with experiment gets worse. The same happens in first-principles calculations on top of quasiparticle bands [25]. Since the static constant is the inverse moment $1 + (2/\pi)\int \varepsilon_2/\omega$ (Eq. (4.48)), weight moved up lowers it, and the local fields lower ε_∞ (Worked example 4.11).

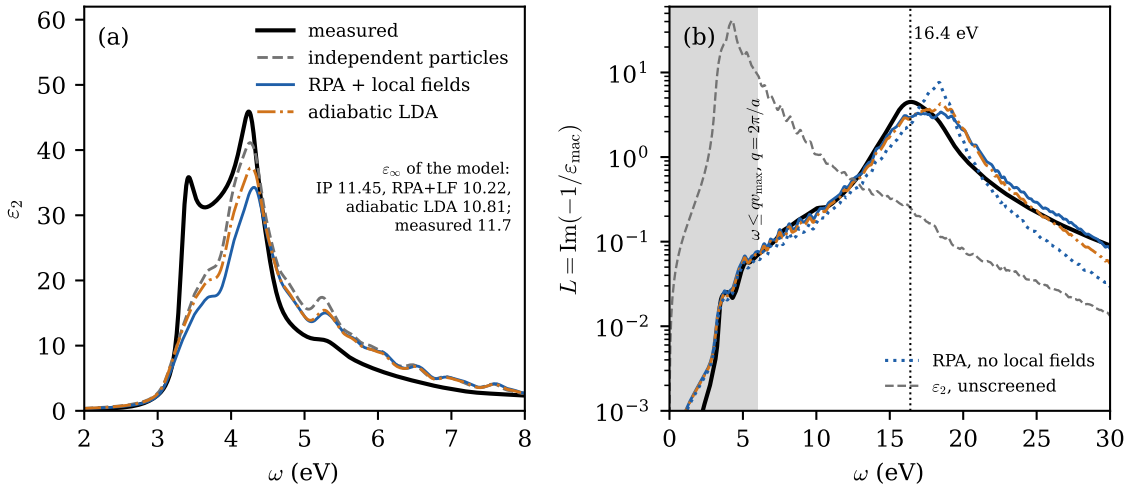


Figure 4.11: Three many-body approximations to the dielectric function of crystalline silicon at $\boldsymbol{q} \to 0$, in a model calculation. Band energies and Bloch states come from the local empirical pseudopotential of Cohen and Bergstresser [26], fitted to measured transition energies (no scissor shift). The sums run over a shifted 20^3 grid of $\boldsymbol{k}$ points over the full zone, with 4 valence and 26 conduction bands, a local-field matrix of the 59 shortest reciprocal vectors, and a Lorentzian broadening of 0.10 eV. The exchange–correlation kernel is the adiabatic local-density one, evaluated on the model's valence density with the Perdew–Zunger correlation energy [27]. Black: measured at 300 K [18]; panel (a) lists the static limits of the three approximations against the measured $\varepsilon_\infty = 11.7$ (Sect. 4.4). (a) Absorptive part ε_2: independent particles (dashed gray), which is also the random-phase approximation without local fields; the random-phase approximation with local fields (solid blue); adiabatic LDA (dash-dotted orange). The strength of the measured lower peak at 3.42 eV comes from the electron–hole attraction (Sect. 4.9), and all three underestimate it. (b) Loss function on a logarithmic scale: the random-phase approximation without local fields (dotted blue), with local fields (solid blue), and adiabatic LDA (dash-dotted orange). Dashed gray: the unscreened ε_2 of independent particles, which a rate without screening would use in place of L. The model's loss peak lies about 2 eV above the measured 16.4 eV (dotted line), because its high conduction bands are not fitted. Shaded: energies up to $qv_{\max} = 6.05$ eV at a momentum transfer of one reciprocal-lattice unit, $q = 2\pi/a$, the range that dark matter of any mass can reach there.

Panel (b) shows the loss function that sets the dark-matter rate. Up to about 6 eV the loss function lies two to three orders of magnitude below the unscreened ε_2; this suppression is the factor $1/|\varepsilon|^2$ that the head of the dielectric matrix already contains. On this scale the local fields are a correction of tens of percent. Between 3 and 12 eV they raise L by 10 to 35 percent, because they lower $|\varepsilon|$, and they cut the height of the plasmon peak from 7.6 to 3.4. A dark-matter rate integrates the loss function over $\boldsymbol{q}$ and over ω up to $qv_{\max}$, and for silicon and germanium the difference between rates computed with and without local fields is small [5]. Local fields matter more at momentum transfers of a few reciprocal vectors, where the off-diagonal elements of R^0 are as large as the diagonal ones, and at energies where semicore shells, localized near the nuclei, contribute.

Local fields have a second reading, which Sect. 4.9 uses. In the space of electron–hole pairs, the microscopic Hartree potential $\bar{v}$ is the bare Coulomb interaction between a pair and another pair at the same total momentum, with the long-range term at $\boldsymbol{G} = 0$ removed. This interaction $\bar{v}$ is the exchange kernel K^{x} of the Bethe–Salpeter equation. The random-phase approximation with local fields is that equation with the electron–hole attraction switched off, and it cannot bind an exciton.

4.6.3 Time-dependent density-functional theory

The bubble Eq. (4.63) is built from Kohn–Sham states, and until now the choice of Kohn–Sham states was a convenience: the states of some independent-particle model were needed, and these were at hand. Time-dependent density-functional theory makes the choice exact. Runge and Gross proved that the time-dependent density of an interacting system, evolving from its ground state, determines the time-dependent external potential up to a function of time alone [28]. It follows, as in the static theory, that a system of independent electrons in a suitable time-dependent Kohn–Sham potential has the same density at all times. That potential is the external one plus the Hartree potential plus an exchange–correlation potential $v_{\mathrm{xc}}(\boldsymbol{r}, t)$, which now depends on the density at all earlier times.

Linear response follows in three lines. A weak external potential energy ϕ_{ext} changes the Kohn–Sham potential by

$$\delta V_{\mathrm{KS}}(\boldsymbol{r}, t) = \phi_{\mathrm{ext}}(\boldsymbol{r}, t) + \int \mathrm{d}^3 r'\, v_{\mathrm{C}}(\boldsymbol{r} - \boldsymbol{r}')\, \delta n(\boldsymbol{r}', t) + \int \mathrm{d}^3 r' \int \mathrm{d}t'\, f_{\mathrm{xc}}(\boldsymbol{r}, \boldsymbol{r}', t - t')\, \delta n(\boldsymbol{r}', t'),$$

where the exchange–correlation kernel

$$f_{\mathrm{xc}}(\boldsymbol{r}, \boldsymbol{r}', t - t') = \left. \frac{\delta v_{\mathrm{xc}}(\boldsymbol{r}, t)}{\delta n(\boldsymbol{r}', t')} \right|_{n_{\mathrm{gs}}}, \tag{4.69}$$

is the functional derivative taken at the ground-state density n_{gs}. Since the Kohn–Sham electrons are independent, their density responds to their own potential through the bubble: $\delta n = R^0\, \delta V_{\mathrm{KS}}$, with R^0 of Eq. (4.63). And δn is, by construction, the density change of the interacting system, $\delta n = R\, \phi_{\mathrm{ext}}$. In reciprocal space, with $f_{\mathrm{Hxc}} = v_{\mathrm{C}} + f_{\mathrm{xc}}$ the matrix $(f_{\mathrm{Hxc}})_{\boldsymbol{G}\boldsymbol{G}'} = v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}|)\delta_{\boldsymbol{G}\boldsymbol{G}'} + (f_{\mathrm{xc}})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega)$, the two statements combine to $R\phi_{\mathrm{ext}} = R^0(\phi_{\mathrm{ext}} + f_{\mathrm{Hxc}} R \phi_{\mathrm{ext}})$ for every ϕ_{ext}.

Dyson equation of time-dependent density-functional theory

$$R = R^0 + R^0 f_{\mathrm{Hxc}} R, \qquad \tilde{R} = R^0 + R^0 f_{\mathrm{xc}} \tilde{R}, \tag{4.70}$$

as matrices in the reciprocal-lattice vectors at fixed $(\boldsymbol{q}, \omega)$. With the exact kernel, R is the exact density response, although R^0 is built from Kohn–Sham states. The random-phase approximation is $f_{\mathrm{xc}} = 0$.

The proper response follows from the first form by splitting off the Coulomb kernel, since $R = \tilde{R} + \tilde{R} v_\mathrm{C} R$ (Eq. (4.40)) has the solution $\tilde{R} = (1 - R^0 f_\mathrm{xc})^{-1} R^0$. This expression is Eq. (26.2), and Petersilka, Gossmann and Gross used it to compute excitation energies as the poles of R [29]. For the macroscopic function, the kernel of Eq. (4.68) becomes $\bar{v} + f_\mathrm{xc}$, with the long-range Coulomb term still removed.

The exact kernel, however, is unknown. The Kohn–Sham gap is smaller than the true one (Sect. 3.4), and the poles of R^0 lie at the wrong energies; the kernel must move them. It must also bind the electron and the hole into excitons, which takes an attraction of long range. Both tasks, moving the poles and binding the exciton, are hard to achieve with a function of the density at one point.

The most widely used kernel is the *adiabatic local-density approximation* (ALDA). It takes the exchange–correlation potential of Eq. (3.63) at the instantaneous local density, so that the kernel is local in space and instantaneous in time [30],

$$f_\mathrm{xc}^\mathrm{ALDA}(\boldsymbol{r}, \boldsymbol{r}', t - t') = \delta(\boldsymbol{r} - \boldsymbol{r}')\, \delta(t - t')\, \frac{\mathrm{d}^2 \left[n\, \varepsilon_\mathrm{xc}(n) \right]}{\mathrm{d}n^2} \bigg|_{n_\mathrm{gs}(\boldsymbol{r})} . \tag{4.71}$$

It is negative, since exchange and correlation make the potential more attractive where the density grows. In reciprocal space it is a function of $\boldsymbol{G} - \boldsymbol{G}'$ alone and does not depend on $\boldsymbol{q}$ or ω.

These properties fix what the kernel misses. Being instantaneous, it has no frequency dependence and cannot open the gap: the absorption edge stays at the Kohn–Sham gap, and a scissor or quasiparticle energies are still needed. Because the kernel is local, its matrix is finite as $\boldsymbol{q} \to 0$. In the optical limit the head element enters $\tilde{R}_{\boldsymbol{00}}$ multiplied by $R^0_{\boldsymbol{00}} = O(q^2)$ and drops out, and the wings drop out with one power of q. The kernel acts only through the body of the matrix, at $\boldsymbol{G}, \boldsymbol{G}' \neq 0$, alongside $\bar{v}$, which it partly cancels. An exciton, however, needs the attraction between an electron and a hole at large separation, which in an insulator behaves as $-\alpha_\mathrm{xc}/|\boldsymbol{q} + \boldsymbol{G}|^2$ in reciprocal space [31]; a local kernel has no long-range term. In Fig. 4.11 the adiabatic LDA spectrum lies between the two random-phase ones and close to the one with local fields, with the upper peak at a height of 37, and its static constant lies between theirs (Worked example 4.11). Its lower peak is as weak as theirs. The same holds generally for semiconductors: for absorption the adiabatic LDA is little better than the random-phase approximation with local fields, and for energy-loss spectra it does well because the random-phase approximation already does [17, 32].

4.6.4 Local-field factor of the electron gas

In the homogeneous gas every matrix is diagonal, and Eq. (4.70) gives $\tilde{R} = R^0/(1 - f_\mathrm{xc} R^0)$ with the Lindhard function R^0 of Eq. (4.56). The many-body literature on the gas writes the same proper response with a static *local-field factor* $G_\mathrm{LF}(q)$. It is defined by

$$\varepsilon(q, \omega) = 1 - \frac{v_\mathrm{C}(q)\, R^0(q, \omega)}{1 + v_\mathrm{C}(q)\, G_\mathrm{LF}(q)\, R^0(q, \omega)} .$$

An electron feels the Hartree potential of the induced density reduced by the factor $1 - G_\mathrm{LF}$, because the exchange–correlation hole keeps the other electrons away from it. The name local-field factor comes from the local fields of a crystal: in both cases the field an electron actually feels differs from the average one. Comparing with $\varepsilon = 1 - v_\mathrm{C} R^0/(1 - f_\mathrm{xc} R^0)$,

$$f_\mathrm{xc}(q) = -v_\mathrm{C}(q)\, G_\mathrm{LF}(q) = -\frac{e^2}{q^2}\, G_\mathrm{LF}(q). \tag{4.72}$$

A static local-field factor is thus a static kernel, and every model of G_{LF} in the literature on the gas [1, Chap. 5] is a kernel of time-dependent density-functional theory for the gas.

The long-wavelength limit is fixed by thermodynamics. A static potential of long wavelength shifts the local chemical potential, as in the Thomas–Fermi argument of Sect. 4.4, and the proper response is $\tilde{R}(q \to 0, 0) = -\partial n_e / \partial \mu_{\mathrm{ch}}$, with μ_{ch} the chemical potential without the Hartree term. With $R^0(q \to 0, 0) = -2\mathcal{N}_{\mathrm{F}}$, the inverse of $\tilde{R} = R^0/(1 - f_{\mathrm{xc}} R^0)$ is

$$\frac{\partial \mu_{\mathrm{ch}}}{\partial n_e} = \frac{1}{2\mathcal{N}_{\mathrm{F}}} + f_{\mathrm{xc}}(q \to 0), \tag{4.73}$$

which is the compressibility sum rule: the kernel at long wavelength is the second derivative of the exchange–correlation energy density, $\mathrm{d}^2(n\varepsilon_{\mathrm{xc}})/\mathrm{d}n^2$. The adiabatic LDA satisfies it by construction. For exchange alone, $v_{\mathrm{x}} = \frac{4}{3}\varepsilon_{\mathrm{x}} \propto n^{1/3}$ with $\varepsilon_{\mathrm{x}} = -3e^2 k_{\mathrm{F}}/16\pi^2$, so $f_{\mathrm{x}} = \mathrm{d}v_{\mathrm{x}}/\mathrm{d}n = v_{\mathrm{x}}/3n$, and with $n = k_{\mathrm{F}}^3/3\pi^2$,

$$f_{\mathrm{x}} = -\frac{e^2}{4k_{\mathrm{F}}^2}, \qquad G_{\mathrm{LF}}(q) = \frac{q^2}{4k_{\mathrm{F}}^2} \quad (q \ll k_{\mathrm{F}}).$$

Correlation adds to the exchange kernel. At $r_s = 2$, the Perdew–Zunger correlation energy [27] raises the coefficient from $\frac{1}{4}$ to 0.267, and Eq. (4.73) with $k_{\mathrm{TF}}^2/k_{\mathrm{F}}^2 = 1.33$ (Sect. 4.5) says that the gas is 1.55 times more compressible than free electrons of the same density. An electron surrounded by its exchange–correlation hole repels the others less, the gas yields more easily to a slow potential, and it screens more strongly than the random-phase approximation says.

The adiabatic LDA fails at short wavelengths, where dark matter scatters. Its $G_{\mathrm{LF}} = 0.267\, q^2/k_{\mathrm{F}}^2$ grows without bound, while the factor of the real gas follows the q^2 law only for $q \ll k_{\mathrm{F}}$; beyond $q \sim k_{\mathrm{F}}$ it stops growing and stays of order one [1, Chap. 5]. At the typical momentum transfer $q_0 = 3.73\,\mathrm{keV}$, which is $1.04\,k_{\mathrm{F}}$ at the silicon valence density, the local kernel is at the edge of its range of validity. There its factor is 0.29, and it raises the static Lindhard value $\varepsilon(q_0, 0) = 2.10$ of Sect. 4.5 to 2.62. The correction to the screening at q_0 is thus of order a quarter. Its sign is reliable, its size at $q \sim k_{\mathrm{F}}$ only to the extent that the local kernel is.

4.6.5 Many-body spread as an error measure

Independent particles, the random-phase approximation with the head alone, the same with local fields, and an exchange–correlation kernel on top form a hierarchy, which the Bethe–Salpeter equation of Sect. 4.9 completes. The hierarchy is not an expansion in a small number: each level adds a physical effect whose size depends on the material, the energy and the momentum, so no level bounds the error of the one below. The hierarchy does measure how much a quantity depends on the many-body treatment. In silicon the static constant changes by about a tenth across the approximations of Fig. 4.11 (Worked example 4.11), and the screening factor $1/\varepsilon_\infty^2$ at small q by about a quarter. Near the gap, where excitons move weight, the spread is larger, and only the Bethe–Salpeter equation captures the shape of the spectrum there. Well above the gap, at the energies that hold most of the f-sum, the approximations agree to within the accuracy of the bands. This many-body spread, computed with the same bands, $\boldsymbol{k}$ grid and cutoffs in every approximation, is the many-body part of the theory uncertainty of a loss function. For heavier dark matter, whose rate is dominated by energies of several eV, it is small. For light dark matter, which deposits at most $\frac{1}{2} m_\chi v_{\mathrm{max}}^2$ (Eq. (1.23)) and so samples the first eV above the gap, it is the leading part.

Worked example 4.11: The local-field reduction of the dielectric constant of silicon

The calculation of Fig. 4.11 yields the static limit of each approximation at $\boldsymbol{q} \to 0$ from the same states, from Eq. (4.66) for independent particles and from Eqs. (4.68) and (4.70) at $\omega = 0$ for the others. Independent particles, and equally the random-phase approximation without local fields, give $\varepsilon_\infty = 11.45$, within 2 percent of the measured 11.7. The small difference is a property of the model, whose form factors were fitted to measured transition energies. With the local fields of the 59 shortest reciprocal vectors, $\varepsilon_\infty = 10.22$, unchanged to three digits with 113 vectors; the local fields lower the dielectric constant by 11 percent. The adiabatic LDA kernel raises it again to 10.81, 6 percent above the random-phase value with local fields, so that it cancels about half of the local-field reduction. All three are converged in the $\boldsymbol{k}$ grid to better than 0.1 percent.

The PBE value 13.04 of Sect. 3.4 is a static linear-response calculation with Kohn–Sham bands, local fields and the adiabatic PBE kernel: it is the third of our approximations, with Kohn–Sham energies in place of fitted ones. It lies above the measured value by 11 percent, and the reason is Eq. (4.66). The PBE gap is about half the measured one, and the static sum weights small transition energies by their inverse cube. The same representative PBE setting as Sect. 23.3, at the measured lattice constant on a 20^3 grid, gives the other two approximations as well: 13.61 for independent particles and 12.20 with local fields, against 13.04 with the PBE kernel. The first-principles values follow the trend of the model values 11.45, 10.22 and 10.81: the local fields lower ε_∞ by 10 percent, and the kernel restores part of this reduction. Independent particles with fitted energies happen to land close to experiment, but the adiabatic kernel with fitted energies does not: at 10.81 it is 8 percent below 11.7. The adiabatic kernel lacks the electron–hole attraction, which lowers the energies of the transitions near the gap and so raises ε_∞ (Sect. 4.9).

For a dark-matter rate at momentum transfers small enough that $L \simeq \varepsilon_2/\varepsilon_\infty^2$, the local-field reduction of ε_∞ alone changes the screening factor by $(11.45/10.22)^2 \simeq 1.25$. The kernel undoes part of this change, and the electron–hole attraction more than makes up the rest of the change.

4.7 Green's functions

The Bethe–Salpeter equation, which completes the hierarchy of Sect. 4.6, needs quasiparticle energies and a screened interaction, and both are properties of one added electron. Put one extra electron into the conduction band of silicon. The other electrons move away from it and screen its charge, the lattice relaxes around it, and if it lies high enough in the band it can lose energy by creating an electron–hole pair or by emitting a phonon. The electron propagates dressed by its surroundings, with an energy that differs from its Kohn–Sham band energy and, in general, with a finite lifetime. The density response R_{ee} of Sect. 4.1 does not follow this dressed electron, since the density operator moves an electron from one state to another and never changes their number. The function that follows the dressed electron is the single-particle Green's function G: the amplitude that an electron added to the target at one time is still found in the same state at a later time. Its poles are the energies of the dressed electrons, the quasiparticles of the target, and the distance of each pole below the real axis is a decay rate. The treatment follows Refs. [2], [33, Chaps. 3 and 7] and [34, Chaps. 2 and 3].

4.7.1 The one-electron Green's function

The Green's function changes the number of electrons, so we work in the grand-canonical state Eq. (2.63), with $\hat{N}_e$ the electron number operator of Sect. 2.3. Every Heisenberg operator in this section evolves with $H_\mu = H - \mu_{\rm ch}\hat{N}_e$, so that $c(t) = \mathrm{e}^{\mathrm{i}H_\mu t} c\, \mathrm{e}^{-\mathrm{i}H_\mu t}$, and E_i denotes an eigenvalue of H_μ. Energies are then measured from the chemical potential: adding an electron to the N-electron state $|i\rangle$ to reach the $(N+1)$-electron state $|f\rangle$ costs $E_f - E_i$ in eigenvalues of H_μ, which is the change of the total energy minus $\mu_{\rm ch}$. Let $c^\dagger_{\boldsymbol{k}}$ create an electron in a single-particle state labeled $\boldsymbol{k}$, with the spin and, in a crystal, the band index included in the label. The retarded Green's function is

$$\mathsf{G}^{\rm R}(\boldsymbol{k},t) = -\mathrm{i}\,\Theta(t)\,\langle\{c_{\boldsymbol{k}}(t),\, c^\dagger_{\boldsymbol{k}}(0)\}\rangle, \tag{4.74}$$

with the anticommutator $\{A,B\} = AB + BA$ and the average in the grand-canonical state. It has the form of the response function Eq. (4.3), with an anticommutator in place of the commutator. The term $\langle c_{\boldsymbol{k}}(t) c^\dagger_{\boldsymbol{k}}\rangle$ is the overlap of the state with an electron added in $\boldsymbol{k}$ at time 0 with the same state at time t: the amplitude that the added electron is still there. The term $\langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}(t)\rangle$ is the same amplitude for a hole, an electron removed from $\boldsymbol{k}$. The anticommutator makes $\mathsf{G}^{\rm R}(\boldsymbol{k},0^+) = -\mathrm{i}$ whatever the state, since $\{c_{\boldsymbol{k}}, c^\dagger_{\boldsymbol{k}}\} = 1$ (Eq. (2.32)).

For independent electrons the amplitude is a pure phase. With $H_\mu = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}$, where $\xi_{\boldsymbol{k}}$ is the energy of the orbital $\boldsymbol{k}$ of Sect. 2.3 measured from $\mu_{\rm ch}$, the Heisenberg operator is $c_{\boldsymbol{k}}(t) = c_{\boldsymbol{k}}\, \mathrm{e}^{-\mathrm{i}\xi_{\boldsymbol{k}} t}$ (Sect. 2.3). The anticommutator in Eq. (4.74) is then the number $\mathrm{e}^{-\mathrm{i}\xi_{\boldsymbol{k}} t}$, and with the transform of Appendix A.3 and $\omega \to \omega + \mathrm{i}0^+$, as for every retarded function (Sect. 4.1),

$$\mathsf{G}^{\rm R}_0(\boldsymbol{k},t) = -\mathrm{i}\,\Theta(t)\,\mathrm{e}^{-\mathrm{i}\xi_{\boldsymbol{k}} t}, \qquad \mathsf{G}^{\rm R}_0(\boldsymbol{k},\omega) = \frac{1}{\omega - \xi_{\boldsymbol{k}} + \mathrm{i}0^+}. \tag{4.75}$$

The subscript 0 marks the function without interactions. It has one pole, at the energy of the orbital, and it is independent of temperature and of which orbitals are occupied.

In an interacting target the same steps as in Sect. 4.1 lead to the eigenstates. Inserting a complete set of states between the operators, $\langle c_{\boldsymbol{k}}(t) c^\dagger_{\boldsymbol{k}}\rangle = \sum_{i,f} p_i\, |\langle f|c^\dagger_{\boldsymbol{k}}|i\rangle|^2\, \mathrm{e}^{-\mathrm{i}(E_f - E_i)t}$ and $\langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}(t)\rangle = \sum_{i,f} p_i\, |\langle f|c_{\boldsymbol{k}}|i\rangle|^2\, \mathrm{e}^{+\mathrm{i}(E_f - E_i)t}$, and the time integrals are those of Eq. (4.6). Since the two terms are added, not subtracted, both enter with a plus sign:

Lehmann representation of the Green's function

$$\begin{aligned}\mathsf{G}^{\rm R}(\boldsymbol{k},\omega) &= \sum_{i,f} p_i \left[\frac{|\langle f|c^\dagger_{\boldsymbol{k}}|i\rangle|^2}{\omega - (E_f - E_i) + \mathrm{i}0^+} + \frac{|\langle f|c_{\boldsymbol{k}}|i\rangle|^2}{\omega + (E_f - E_i) + \mathrm{i}0^+}\right] \\ &= \int \mathrm{d}\omega'\, \frac{\mathsf{A}(\boldsymbol{k},\omega')}{\omega - \omega' + \mathrm{i}0^+},\end{aligned} \tag{4.76}$$

with the spectral function

$$\begin{aligned}\mathsf{A}(\boldsymbol{k},\omega) &= -\frac{1}{\pi}\,\mathrm{Im}\,\mathsf{G}^{\rm R}(\boldsymbol{k},\omega) \\ &= \sum_{i,f} p_i \Big[|\langle f|c^\dagger_{\boldsymbol{k}}|i\rangle|^2\, \delta(\omega - (E_f - E_i)) + |\langle f|c_{\boldsymbol{k}}|i\rangle|^2\, \delta(\omega + (E_f - E_i))\Big].\end{aligned} \tag{4.77}$$

The poles are at the energies needed to add an electron in $\boldsymbol{k}$, and at minus the energies released by removing one, each weighted by the probability that the added electron or hole is found in an exact eigenstate.

Eq. (4.76) is the representation of Lehmann [35], and it shows that A is positive, normalized, and tied to the occupations. It is positive since every weight in Eq. (4.77) is. Its integral over ω is $\sum_i p_i \langle i|c_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger + c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}|i\rangle = 1$, which makes $\mathsf{A}(\boldsymbol{k},\omega)$ a probability density in energy: the probability that an electron or hole put into $\boldsymbol{k}$ has the energy ω. And the removal part alone counts the electrons already in $\boldsymbol{k}$. To see that it does, relabel $i \leftrightarrow f$ in the removal term; it becomes $\sum_{i,f} p_f\, |\langle f|c_{\boldsymbol{k}}^\dagger|i\rangle|^2\, \delta(\omega - (E_f - E_i))$, and on the energy shell $p_f = p_i\, \mathrm{e}^{-\beta\omega}$, the detailed-balance step of Eq. (2.68). The removal part is thus $\mathrm{e}^{-\beta\omega}$ times the addition part, the addition and removal parts add up to A, and the removal part is $\mathsf{A}\,\mathrm{e}^{-\beta\omega}/(1+\mathrm{e}^{-\beta\omega}) = \mathsf{A}\, n_{\mathrm{F}}(\omega)$. The integral of the removal part is $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}\rangle$:

$$\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}\rangle = \int \mathrm{d}\omega\, \mathsf{A}(\boldsymbol{k},\omega)\, n_{\mathrm{F}}(\omega). \tag{4.78}$$

For independent electrons $\mathsf{A}_0(\boldsymbol{k},\omega) = \delta(\omega - \xi_{\boldsymbol{k}})$ and Eq. (4.78) is the Fermi function of Eq. (2.69). With interactions the occupation of a plane-wave state is no longer $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, even at $T = 0$, because A has weight at energies other than $\xi_{\boldsymbol{k}}$. Note that we normalize A to one. Many texts, Refs. [2, 34] among them, write $-2\,\mathrm{Im}\,\mathsf{G}^{\mathrm{R}}$ for the spectral function, which integrates to 2π.

The spectral function is close to a measured quantity. In photoemission a photon removes an electron from the crystal, and the rate at a given energy and momentum of the outgoing electron is, in the simplest approximation, proportional to the removal part $\mathsf{A}\, n_{\mathrm{F}}$; inverse photoemission measures the addition part. The analytic structure is that of Sect. 4.1. By Eq. (4.76), $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},z) = \int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k},\omega')/(z-\omega')$ is analytic in the upper half plane, and the advanced function $\mathsf{G}^{\mathrm{A}}(\boldsymbol{k},\omega) = \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega)^*$, nonzero only for $t < 0$, is its boundary value from below.

4.7.2 The time-ordered function

At $T = 0$ the perturbation theory of Sect. 4.8 produces a third function. The Dyson series Eq. (2.50) is a time-ordered exponential, and when it is used to expand a ground-state amplitude the operators come out in time order. The time-ordered Green's function is

$$\mathsf{G}(\boldsymbol{k},t) = -\mathrm{i}\,\langle 0|\mathrm{T}\, c_{\boldsymbol{k}}(t)\, c_{\boldsymbol{k}}^\dagger(0)|0\rangle = -\mathrm{i}\,\Theta(t)\,\langle 0|c_{\boldsymbol{k}}(t)\, c_{\boldsymbol{k}}^\dagger|0\rangle + \mathrm{i}\,\Theta(-t)\,\langle 0|c_{\boldsymbol{k}}^\dagger\, c_{\boldsymbol{k}}(t)|0\rangle, \tag{4.79}$$

where $|0\rangle$ is the ground state of H_μ and the time-ordering symbol T puts the operator with the later time on the left, with a minus sign when two fermion operators are exchanged. For $t > 0$ it propagates an added electron forward in time, and for $t < 0$ a hole. The integral over $t > 0$ is that of the retarded function. The integral over $t < 0$ needs a factor $\mathrm{e}^{+0^+ t}$ to converge, $\mathrm{i}\int_{-\infty}^0 \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega + E_f - E_0)t + 0^+ t} = 1/(\omega + E_f - E_0 - \mathrm{i}0^+)$, so

$$\mathsf{G}(\boldsymbol{k},\omega) = \sum_f \left[\frac{|\langle f|c_{\boldsymbol{k}}^\dagger|0\rangle|^2}{\omega - (E_f - E_0) + \mathrm{i}0^+} + \frac{|\langle f|c_{\boldsymbol{k}}|0\rangle|^2}{\omega + (E_f - E_0) - \mathrm{i}0^+} \right]. \tag{4.80}$$

The removal poles now lie above the real axis. At $T = 0$ every $E_f - E_0 \geq 0$, because $|0\rangle$ is the ground state of H_μ in all particle-number sectors. The addition poles therefore lie at $\omega > 0$ and the removal poles at $\omega < 0$. As a result, the time-ordered function equals G^{R} for $\omega > 0$ and G^{A} for $\omega < 0$, and both carry the same spectral function. In this section a G without superscript and with a real time or frequency is the time-ordered function. With an imaginary argument it is the Matsubara function introduced below, and with a complex argument z it is the function $\int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k},\omega')/(z-\omega')$, which unites them all.

4.7.3 Self-energy and the Dyson equation

How does an interaction change G^{R}? Split $H_\mu = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}} + H_{\mathrm{int}}$, where H_{int} holds the Coulomb interaction between the electrons, their coupling to phonons, or any other term that is not a one-body energy. The Heisenberg operator obeys $\mathrm{i}\,\partial_t c_{\boldsymbol{k}}(t) = [c_{\boldsymbol{k}}, H_\mu](t) = \xi_{\boldsymbol{k}} c_{\boldsymbol{k}}(t) + [c_{\boldsymbol{k}}, H_{\mathrm{int}}](t)$, by Eq. (2.33). When Eq. (4.74) is differentiated, the derivative of the step function gives $\delta(t)\langle\{c_{\boldsymbol{k}}, c^\dagger_{\boldsymbol{k}}\}\rangle = \delta(t)$, and

$$\mathrm{i}\,\partial_t \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t) = \delta(t) + \xi_{\boldsymbol{k}}\, \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t) + \mathsf{F}^{\mathrm{R}}(\boldsymbol{k},t), \qquad \mathsf{F}^{\mathrm{R}}(\boldsymbol{k},t) = -\mathrm{i}\,\Theta(t)\,\langle\{[c_{\boldsymbol{k}}, H_{\mathrm{int}}](t),\, c^\dagger_{\boldsymbol{k}}\}\rangle.$$

The function F^{R} is a retarded correlation of the same kind, built from the operator $[c_{\boldsymbol{k}}, H_{\mathrm{int}}]$, which removes an electron from $\boldsymbol{k}$ and scatters the others. With $\mathrm{e}^{\mathrm{i}\omega t}$, the derivative $\mathrm{i}\,\partial_t$ becomes ω, and $(\omega - \xi_{\boldsymbol{k}})\,\mathsf{G}^{\mathrm{R}} = 1 + \mathsf{F}^{\mathrm{R}}$. We define the self-energy $\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)$ as the ratio $\mathsf{F}^{\mathrm{R}}/\mathsf{G}^{\mathrm{R}}$. The ratio exists because $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},z)$ has no zeros in the upper half plane: there $\operatorname{Im}\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},z) = -\operatorname{Im} z \int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k},\omega')/|z-\omega'|^2 < 0$ by the positivity of A. Multiplying by $\mathsf{G}^{\mathrm{R}}_0$ of Eq. (4.75) gives the Dyson equation.

Dyson equation

$$\begin{aligned} \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega) &= \mathsf{G}^{\mathrm{R}}_0(\boldsymbol{k},\omega) + \mathsf{G}^{\mathrm{R}}_0(\boldsymbol{k},\omega)\,\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)\,\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega), \\ \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega) &= \frac{1}{\omega - \xi_{\boldsymbol{k}} - \Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)}. \end{aligned} \tag{4.81}$$

All the effects of the interaction on the propagation of one electron are in the self-energy Σ^{R}, a complex function of energy.

In a crystal, with $\boldsymbol{k}$ the crystal momentum and n the band, the same steps make G^{R} and Σ^{R} matrices in the band indices, and Eq. (4.81) a matrix equation of the type Eq. (C.57). In a translation-invariant target they are diagonal in $\boldsymbol{k}$, as we assume here.

So far Eq. (4.81) is a definition. Its content appears when it is iterated: $\mathsf{G}^{\mathrm{R}} = \mathsf{G}^{\mathrm{R}}_0 + \mathsf{G}^{\mathrm{R}}_0\Sigma^{\mathrm{R}}\mathsf{G}^{\mathrm{R}}_0 + \mathsf{G}^{\mathrm{R}}_0\Sigma^{\mathrm{R}}\mathsf{G}^{\mathrm{R}}_0\Sigma^{\mathrm{R}}\mathsf{G}^{\mathrm{R}}_0 + \cdots$. The electron propagates freely, is scattered out of $\boldsymbol{k}$ and back by the interaction, propagates freely again, and so on, and Σ^{R} is the amplitude of one such excursion. Sect. 4.8 computes Σ with diagrams and shows that it is the sum of the excursions that cannot be cut into two by removing a single free propagator. A self-energy known only to low order then yields a Green's function to all orders, and no finite order of the series has the shifted, broadened pole that the closed form has (Appendix C.7). The width of the intermediate state of Sect. 2.4 arose in the same way: summing the repeated emission and reabsorption turned $\mathrm{i}0^+$ into $\mathrm{i}\Gamma_{\mathrm{w}}/2$ in Eq. (2.57).

The sign of the width is fixed before any calculation. On the real axis, $\Sigma^{\mathrm{R}} = \omega - \xi_{\boldsymbol{k}} - 1/\mathsf{G}^{\mathrm{R}}$ has the imaginary part $\operatorname{Im}\mathsf{G}^{\mathrm{R}}/|\mathsf{G}^{\mathrm{R}}|^2$, so

$$\operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega) = -\frac{\pi\,\mathsf{A}(\boldsymbol{k},\omega)}{\left|\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega)\right|^2} \le 0. \tag{4.82}$$

The large-frequency limit is fixed as well. At large $|\omega|$, the expansion $1/(\omega-\omega') = \omega^{-1} + \omega'\omega^{-2} + \cdots$ in Eq. (4.76) gives $\mathsf{G}^{\mathrm{R}} = \omega^{-1} + m_1\omega^{-2} + O(\omega^{-3})$, with $m_1 = \int \mathrm{d}\omega\,\omega\,\mathsf{A} = \langle\{[c_{\boldsymbol{k}}, H_\mu], c^\dagger_{\boldsymbol{k}}\}\rangle$ by the same steps as Eq. (4.11). Then $1/\mathsf{G}^{\mathrm{R}} = \omega - m_1 + O(\omega^{-1})$, and $\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega) \to m_1 - \xi_{\boldsymbol{k}} = \langle\{[c_{\boldsymbol{k}}, H_{\mathrm{int}}], c^\dagger_{\boldsymbol{k}}\}\rangle$. The self-energy tends to a static potential, linear in H_{int} and evaluated in the exact state. For the Coulomb interaction it is the mean-field (Hartree and exchange) potential, built with the exact occupations. Everything that depends on frequency, the retardation of the screening and the decay into other excitations, is in $\Sigma^{\mathrm{R}} - \Sigma^{\mathrm{R}}(\boldsymbol{k},\infty)$, which is analytic in the upper half plane and falls off at large $|\omega|$, and as a result obeys the Kramers–Kronig relations of Appendix C.5.

4.7.4 Quasiparticles

Suppose the self-energy is small and varies slowly with ω. The Green's function Eq. (4.81) is then large where its real denominator vanishes, at the quasiparticle energy $E_{\mathrm{qp}}(\boldsymbol{k})$ defined by

$$E_{\mathrm{qp}}(\boldsymbol{k}) - \xi_{\boldsymbol{k}} - \operatorname{Re}\Sigma^{\mathrm{R}}(\boldsymbol{k}, E_{\mathrm{qp}}(\boldsymbol{k})) = 0.$$

Near this energy we expand the real part to first order, $\omega - \xi_{\boldsymbol{k}} - \operatorname{Re}\Sigma^{\mathrm{R}} \simeq (\omega - E_{\mathrm{qp}})/z_{\mathrm{qp}}$, and keep the imaginary part at its value at E_{qp}.

Quasiparticle

$$\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega) \simeq \frac{z_{\mathrm{qp}}(\boldsymbol{k})}{\omega - E_{\mathrm{qp}}(\boldsymbol{k}) + \mathrm{i}\Gamma_{\mathrm{qp}}(\boldsymbol{k})/2},$$
$$z_{\mathrm{qp}} = \Big[1 - \partial_\omega \operatorname{Re}\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)\big|_{E_{\mathrm{qp}}}\Big]^{-1}, \qquad \Gamma_{\mathrm{qp}} = -2 z_{\mathrm{qp}} \operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k}, E_{\mathrm{qp}}). \tag{4.83}$$

The real part of the self-energy shifts the energy, its slope sets the residue z_{qp}, and its imaginary part sets the width Γ_{qp}.

In time, the pole at $E_{\mathrm{qp}} - \mathrm{i}\Gamma_{\mathrm{qp}}/2$, in the lower half plane of the continued function, gives $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t) \simeq -\mathrm{i} z_{\mathrm{qp}}\, \mathrm{e}^{-\mathrm{i}E_{\mathrm{qp}}t - \Gamma_{\mathrm{qp}}t/2}$: the added electron keeps the fraction z_{qp} of its amplitude in a state that oscillates at a definite energy and decays at the rate Γ_{qp}. The pole is a resonance, as in Sect. 4.1, and $1/\Gamma_{\mathrm{qp}}$ is the quasiparticle lifetime. In the spectral function the pole is a Lorentzian of weight z_{qp} and full width Γ_{qp}. In an interacting system z_{qp} is typically below one, and the rest of the sum rule, $1 - z_{\mathrm{qp}}$, is spread over a broad incoherent background: states in which the added electron has created electron–hole pairs, plasmons or phonons. Since the background is broad, its contribution to $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t)$ dephases within a time of order the inverse of its spread, and at later times the quasiparticle term is all that is left. The expansion requires Γ_{qp} to be small compared with the energy over which Σ^{R} changes, the condition of the Wigner–Weisskopf result of Sect. 2.4. The quasiparticle is a useful concept only if it also lives long compared with its own period, $\Gamma_{\mathrm{qp}} \ll |E_{\mathrm{qp}}|$.

Whether the condition $\Gamma_{\mathrm{qp}} \ll |E_{\mathrm{qp}}|$ holds is a matter of phase space, and an estimate shows why quasiparticles are sharp near the Fermi level of a metal. An electron added in a state of energy $\xi_{\boldsymbol{k}} > 0$ above μ_{ch} can decay by scattering off an electron of the Fermi sea, which leaves a hole below μ_{ch} and two electrons above it. At $T = 0$ the Pauli principle forces all three energies into a window of width $\xi_{\boldsymbol{k}}$: the energy ω' lost by the added electron lies between 0 and $\xi_{\boldsymbol{k}}$, and the partner must start between $-\omega'$ and 0. The allowed region in these two energy variables is a triangle of area $\xi_{\boldsymbol{k}}^2/2$, so $\Gamma_{\mathrm{qp}} \propto \xi_{\boldsymbol{k}}^2$ and $\Gamma_{\mathrm{qp}}/\xi_{\boldsymbol{k}} \to 0$ as $\xi_{\boldsymbol{k}} \to 0$. Luttinger showed that $\operatorname{Im}\Sigma^{\mathrm{R}}$ vanishes as $(\omega - \mu_{\mathrm{ch}})^2$ at every order of perturbation theory [36]. The vanishing width at the Fermi level is the basis of Landau's picture of a metal as a gas of weakly interacting quasiparticles. In a semiconductor the window is set by the gap instead. An electron at the bottom of the conduction band of silicon is in the lowest state with one extra electron, it has nothing to decay into, and its width is zero. Higher in the band, the electron–electron part of $\operatorname{Im}\Sigma^{\mathrm{R}}$ stays zero until the electron has enough energy to create an electron–hole pair, at least E_{g} above the band edge. Below that energy the width comes from phonon emission alone, which Sect. 6.3 computes from the same Eq. (4.83).

The quasiparticle energies of silicon are material quantities, and the dark-matter rate needs them: the band edges set the threshold for electron excitation. They are computed by starting from the Kohn–Sham Hamiltonian in place of the free one. Its eigenvalues $\varepsilon_{n\boldsymbol{k}}$ already contain the exchange–correlation potential v_{xc}, so $\Sigma - v_{\mathrm{xc}}$ takes the place of Σ in the Dyson equation. With the approximation that the Kohn–Sham states diagonalize Σ, the

quasiparticle condition expanded to first order about $\varepsilon_{n\boldsymbol{k}}$ gives

$$E_{\mathrm{qp}}(n\boldsymbol{k}) \simeq \varepsilon_{n\boldsymbol{k}} + z_{\mathrm{qp}}\ \mathrm{Re}\langle\psi_{n\boldsymbol{k}}|\Sigma(\varepsilon_{n\boldsymbol{k}}) - v_{\mathrm{xc}}|\psi_{n\boldsymbol{k}}\rangle, \tag{4.84}$$

the form in which first-principles quasiparticle corrections are evaluated [37]. A PBE calculation of silicon at its own lattice constant puts the indirect gap at 0.602 eV (Sect. 3.4), against the measured 1.17 eV at low temperature [38, Eq. (11)] (1.12 eV at 300 K); the self-energy in the *GW* approximation of Sect. 4.8 supplies most of the difference. Sect. 23.4 describes the calculation and its convergence.

Worked example 4.12: A level coupled to a flat band

The simplest self-energy that can be computed exactly is that of a localized orbital d, of energy E_{lev}, hybridized with a band of orbitals $\boldsymbol{k}$ of energies $\xi_{\boldsymbol{k}}$: an impurity level in a metal or an atomic orbital near a surface. The electrons do not interact with each other, and the model is Anderson's without his on-site repulsion [39]:

$$H_\mu = E_{\mathrm{lev}}\, d^\dagger d + \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}}\, c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} + \sum_{\boldsymbol{k}} (V_{\boldsymbol{k}}\, c_{\boldsymbol{k}}^\dagger d + V_{\boldsymbol{k}}^*\, d^\dagger c_{\boldsymbol{k}}),$$

with energies measured from μ_{ch} and d, $d^\dagger$ the fermion operators of the level. The commutators $[d, H_\mu] = E_{\mathrm{lev}} d + \sum_{\boldsymbol{k}} V_{\boldsymbol{k}}^* c_{\boldsymbol{k}}$ and $[c_{\boldsymbol{k}}, H_\mu] = \xi_{\boldsymbol{k}} c_{\boldsymbol{k}} + V_{\boldsymbol{k}} d$ follow from Eq. (2.33). Let $\mathsf{G}^{\mathrm{R}}_{dd}$ be defined by Eq. (4.74) with d for $c_{\boldsymbol{k}}$. Its equation of motion contains the mixed function $\mathsf{G}^{\mathrm{R}}_{\boldsymbol{k}d}(t) = -\mathrm{i}\Theta(t)\langle\{c_{\boldsymbol{k}}(t), d^\dagger\}\rangle$, whose own equation closes, since $\{c_{\boldsymbol{k}}, d^\dagger\} = 0$:

$$(\omega - E_{\mathrm{lev}})\, \mathsf{G}^{\mathrm{R}}_{dd} = 1 + \sum_{\boldsymbol{k}} V_{\boldsymbol{k}}^*\, \mathsf{G}^{\mathrm{R}}_{\boldsymbol{k}d}, \qquad (\omega - \xi_{\boldsymbol{k}})\, \mathsf{G}^{\mathrm{R}}_{\boldsymbol{k}d} = V_{\boldsymbol{k}}\, \mathsf{G}^{\mathrm{R}}_{dd}.$$

Eliminating $\mathsf{G}^{\mathrm{R}}_{\boldsymbol{k}d}$ gives Eq. (4.81) with

$$\Sigma^{\mathrm{R}}_d(\omega) = \sum_{\boldsymbol{k}} \frac{|V_{\boldsymbol{k}}|^2}{\omega - \xi_{\boldsymbol{k}} + \mathrm{i}0^+}, \tag{4.85}$$

exact at every order in $V_{\boldsymbol{k}}$, and the same at every temperature and filling of the band: the anticommutators are numbers, and the averages never enter. The electron leaves d for the band and returns, and Eq. (4.85) is the amplitude of one such excursion.

Eq. (4.85) is the level-shift function of the Wigner–Weisskopf problem, Eq. (D.10), at $z = \omega - E_{\mathrm{lev}} + \mathrm{i}0^+$, with $|V_{\boldsymbol{k}}|^2$ in place of $|H'_{fi}|^2$. Take the flat band of Worked example 2.1: states spread uniformly over a width $E_{\mathrm{w}} = 1$ eV centered on E_{lev}, all with the same $|V_{\boldsymbol{k}}|^2$, chosen so that the golden-rule rate $\Gamma = 2\pi|V_{\boldsymbol{k}}|^2\mathcal{D}_f$ is 10 meV. Appendix D.3 found $\Sigma^{\mathrm{R}}_d(E_{\mathrm{lev}}) = -\mathrm{i}\Gamma/2$ with no shift, and a slope $\partial_\omega\, \mathrm{Re}\,\Sigma^{\mathrm{R}}_d = 2\Gamma/(\pi E_{\mathrm{w}}) = 0.0064$ at E_{lev}. By Eq. (4.83), $E_{\mathrm{qp}} = E_{\mathrm{lev}}$, $z_{\mathrm{qp}} = 1/(1 - 0.0064) = 1.0064$ and $\Gamma_{\mathrm{qp}} = z_{\mathrm{qp}}\Gamma$. The spectral function is a Lorentzian of full width Γ with peak height $2/(\pi\Gamma) = 63.7\,\mathrm{eV}^{-1}$, and the exact A_{dd} follows it to within 1.3 percent out to 10Γ from the peak (Fig. 4.12). This Lorentzian is the line of Eq. (2.56), now as the energy distribution of an electron put into d, and its width is $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}}_d$.

The residue exceeds one because the exact A_{dd} vanishes outside the band, where $\mathrm{Im}\,\Sigma^{\mathrm{R}}_d = 0$, while a Lorentzian of width Γ puts the weight $2\Gamma/(\pi E_{\mathrm{w}}) = 0.0064$ beyond the band edges. The pole carries that weight back to the center, and the sum rule $\int \mathsf{A}_{dd} = 1$ is satisfied by the exact function inside the band. The quasiparticle formulas also reproduce the corrections of Appendix D.3: there the decay rate was Γ divided by one minus the slope of the level-shift function, and the residue was the inverse of the same factor, which are Γ_{qp} and z_{qp}. In the state with the band empty, $\mathsf{G}^{\mathrm{R}}_{dd}(t) = -\mathrm{i}\Theta(t)\langle 0|d\,\mathrm{e}^{-\mathrm{i}H_\mu t}d^\dagger|0\rangle$

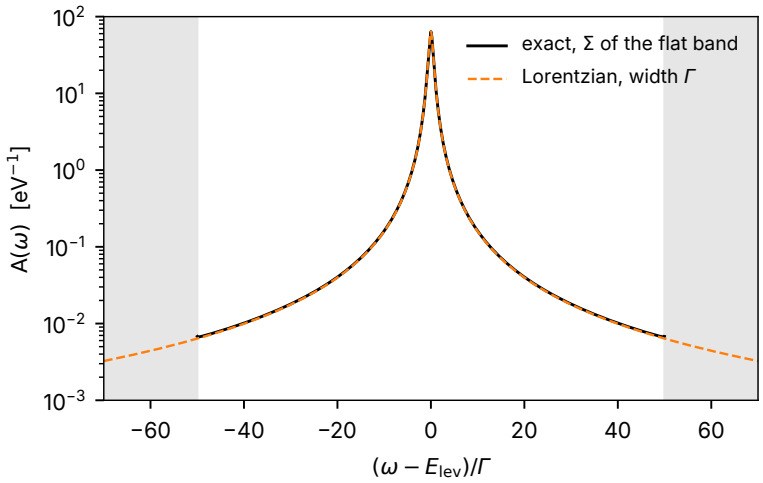


Figure 4.12: Spectral function of a level of energy E_{lev} coupled to a flat band of width $E_{\mathrm{w}} = 1\,\mathrm{eV}$ centered on E_{lev}, with golden-rule rate $\Gamma = 10\,\mathrm{meV}$ (the model of Worked example 4.12), on a logarithmic scale against $(\omega - E_{\mathrm{lev}})/\Gamma$. Solid: exact result, $\mathsf{A}_{dd} = -\pi^{-1}\,\mathrm{Im}[\omega - E_{\mathrm{lev}} - \Sigma_d^{\mathrm{R}}(\omega)]^{-1}$ with the self-energy of the band in the limit of a continuum, $\Sigma_d^{\mathrm{R}} = (\Gamma/2\pi)\ln[(\omega - E_{\mathrm{lev}} + E_{\mathrm{w}}/2)/(\omega - E_{\mathrm{lev}} - E_{\mathrm{w}}/2)]$ at $\omega + \mathrm{i}0^+$. Dashed: Lorentzian of full width Γ and unit weight. Shaded: outside the band, where the exact function vanishes (its two bound states lie exponentially close to the band edges and carry negligible weight) and the Lorentzian has the weight $2\Gamma/(\pi E_{\mathrm{w}}) = 0.0064$. Inside the band the exact function exceeds the Lorentzian by 1.0 percent at $|\omega - E_{\mathrm{lev}}| = \Gamma$ and by 1.3 percent at 10Γ, because its quasiparticle residue is 1.0064.

is $-\mathrm{i}$ times the amplitude that an electron put into d is still there, the $c_i(t)\,\mathrm{e}^{-\mathrm{i}E_{\mathrm{lev}}t}$ of Eq. (2.55), and $|\mathsf{G}_{dd}^{\mathrm{R}}(t)|^2$ is the survival probability. The Green's function contains the Wigner–Weisskopf result, with the condition $\Gamma \ll E_{\mathrm{w}}$ appearing as the slow variation of Σ_d^{R}. Temperature enters only the occupation: at $T = 20\,\mathrm{meV}$ and $E_{\mathrm{lev}} = -30\,\mathrm{meV}$, Eq. (4.78) gives $\langle d^\dagger d\rangle = 0.787$, against $n_{\mathrm{F}}(E_{\mathrm{lev}}) = 0.818$ for the isolated level.

4.7.5 Imaginary time

At finite temperature two exponentials of the Hamiltonian appear in every average: $\mathrm{e}^{-\mathrm{i}H_\mu t}$ moves the operators in time, and $\mathrm{e}^{-\beta H_\mu}$ weights the states. The weight $\mathrm{e}^{-\beta H_\mu}$ is the evolution $\mathrm{e}^{-\mathrm{i}H_\mu t}$ at the imaginary time $t = -\mathrm{i}\beta$. Matsubara used this coincidence to treat the two exponentials on the same footing [40]. Define operators in imaginary time τ by

$$O(\tau) = \mathrm{e}^{\tau H_\mu}\, O\, \mathrm{e}^{-\tau H_\mu}, \qquad 0 \le \tau \le \beta,$$

which is the Heisenberg operator $O(t)$ at $t = -\mathrm{i}\tau$. Note that $c^\dagger(\tau)$ is not the adjoint of $c(\tau)$. The imaginary-time, or Matsubara, Green's function is

$$\mathsf{G}(\boldsymbol{k}, \tau) = -\big\langle \mathrm{T}_\tau\, c_{\boldsymbol{k}}(\tau)\, c_{\boldsymbol{k}}^\dagger(0)\big\rangle, \tag{4.86}$$

where T_τ puts the operator with the larger τ on the left, with a minus sign for an exchange of fermion operators. The Matsubara function is useful because a perturbation expansion of $\mathrm{e}^{-\beta H_\mu}$ and of $\mathrm{e}^{\pm\tau H_\mu}$ in H_{int} is one Dyson series along the imaginary-time axis, with the

same structure as Eq. (2.50), and Wick's theorem then reduces every term to products of free Green's functions (Sect. 4.8). The real-time time-ordered function of Eq. (4.79), however, has no expansion of this single-exponential form at $T > 0$ (Appendix D.9), and the Matsubara function therefore replaces it.

The cyclic property of the trace makes Eq. (4.86) antiperiodic, $\mathsf{G}(\boldsymbol{k}, \tau - \beta) = -\mathsf{G}(\boldsymbol{k}, \tau)$ for $0 < \tau < \beta$ (Appendix D.8). A function on $-\beta < \tau < \beta$ with this property is a Fourier series in the odd multiples of πT, the fermionic Matsubara frequencies $\omega_j^{\mathrm{F}} = (2j+1)\pi T$ with integer j:

$$\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = \int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}\tau}\, \mathsf{G}(\boldsymbol{k}, \tau), \qquad \mathsf{G}(\boldsymbol{k}, \tau) = T \sum_j \mathrm{e}^{-\mathrm{i}\omega_j^{\mathrm{F}}\tau}\, \mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}). \tag{4.87}$$

A correlation of two operators that each change the electron number by an even amount, such as two densities, is periodic instead, and its series runs over the bosonic frequencies $\omega_j^{\mathrm{B}} = 2j\pi T$. As $T \to 0$ both sets become dense on the imaginary axis.

The Lehmann form follows as before. For $0 < \tau < \beta$ only the ordering $c_{\boldsymbol{k}}(\tau) c_{\boldsymbol{k}}^\dagger$ appears, and inserting states leads to $\mathsf{G}(\boldsymbol{k}, \tau) = -\sum_{i,f} p_i |\langle f | c_{\boldsymbol{k}}^\dagger | i \rangle|^2\, \mathrm{e}^{-(E_f - E_i)\tau}$. The integral in Eq. (4.87) is elementary, and $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}\beta} = -1$:

$$\int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{[\mathrm{i}\omega_j^{\mathrm{F}} - (E_f - E_i)]\tau} = -\frac{1 + \mathrm{e}^{-\beta(E_f - E_i)}}{\mathrm{i}\omega_j^{\mathrm{F}} - (E_f - E_i)}.$$

Since $p_i\, \mathrm{e}^{-\beta(E_f - E_i)} = p_f$, the weight becomes $(p_i + p_f) |\langle f | c_{\boldsymbol{k}}^\dagger | i \rangle|^2$, and the relabeling that led to Eq. (4.78) shows that this weight is the weight of the spectral function at $\omega = E_f - E_i$.

Matsubara Green's function and analytic continuation

$$\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = \int \mathrm{d}\omega'\, \frac{\mathsf{A}(\boldsymbol{k}, \omega')}{\mathrm{i}\omega_j^{\mathrm{F}} - \omega'}, \qquad \mathsf{G}^{\mathrm{R}}(\boldsymbol{k}, \omega) = \mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}} \to \omega + \mathrm{i}0^+). \tag{4.88}$$

The Matsubara and retarded functions are values of one analytic function, $\int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k}, \omega')/(z - \omega')$, at the points $z = \mathrm{i}\omega_j^{\mathrm{F}}$ and just above the real axis.

In practice a Matsubara function computed in closed form is turned into the retarded one by the substitution $\mathrm{i}\omega_j^{\mathrm{F}} \to \omega + \mathrm{i}0^+$, made after every factor $\mathrm{e}^{\mathrm{i}\beta\omega_j^{\mathrm{F}}}$ has been replaced by -1. The continuation is unique, because the function falls off at large $|z|$ (Appendix C.5) [41]. The Dyson equation carries over with the same self-energy evaluated at $z = \mathrm{i}\omega_j^{\mathrm{F}}$, $\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = [\mathrm{i}\omega_j^{\mathrm{F}} - \xi_{\boldsymbol{k}} - \Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}})]^{-1}$, and for free electrons $\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = 1/(\mathrm{i}\omega_j^{\mathrm{F}} - \xi_{\boldsymbol{k}})$.

Diagrams return their results as sums over Matsubara frequencies, and these are done by contour integration. The occupation is the simplest case. At $\tau = 0^-$ the time ordering puts $c_{\boldsymbol{k}}^\dagger$ on the left, $\mathsf{G}(\boldsymbol{k}, 0^-) = \langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} \rangle$, and by Eq. (4.87)

$$\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} \rangle = T \sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}} 0^+}\, \mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}). \tag{4.89}$$

The Fermi function $n_{\mathrm{F}}(z)$ has poles at the points $z = \mathrm{i}\omega_j^{\mathrm{F}}$, each with residue $-T$, and the sum is a contour integral of $n_{\mathrm{F}}(z)\, \mathsf{G}(\boldsymbol{k}, z)$ around small circles that enclose them (Fig. 4.13a). Opened up, the circles become two lines that run along the real axis, above and below the cut of $\mathsf{G}(\boldsymbol{k}, z)$ (Fig. 4.13b). The jump of G across the cut is $-2\pi\mathrm{i}\mathsf{A}$, by Eq. (4.76), and the sum becomes $\int \mathrm{d}\omega\, \mathsf{A}(\boldsymbol{k}, \omega)\, n_{\mathrm{F}}(\omega)$, the occupation Eq. (4.78). Appendix D.8 writes out the steps and the sum of two propagators that appears in every bubble diagram.

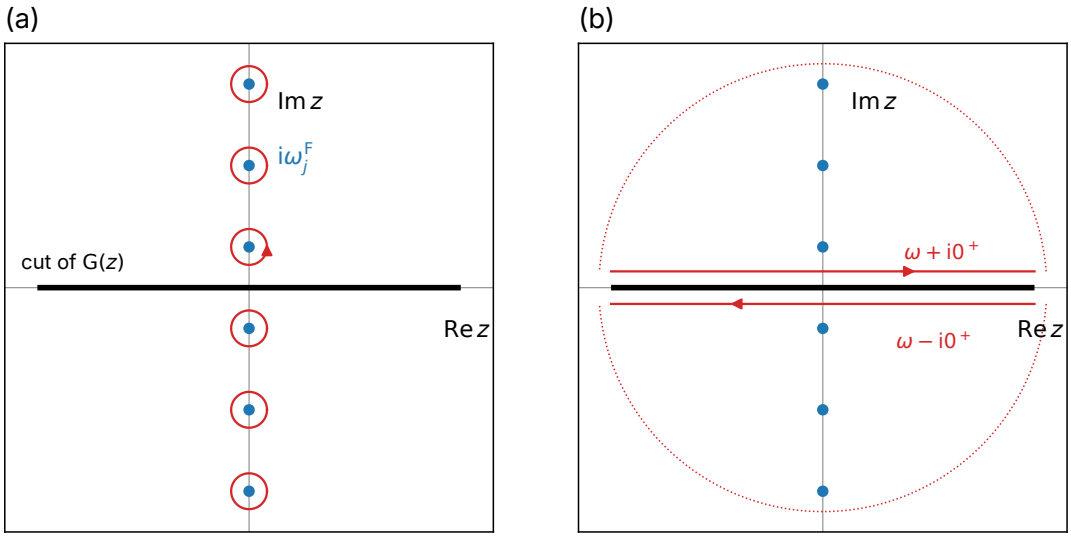


Figure 4.13: Frequency sums by contour integration in the complex frequency plane z. (a) The fermionic Matsubara frequencies $\mathrm{i}\omega_j^{\mathrm{F}} = (2j+1)\pi\mathrm{i}T$ (dots) are the poles of the Fermi function $n_{\mathrm{F}}(z)$, each with residue $-T$. A sum $T\sum_j g(\mathrm{i}\omega_j^{\mathrm{F}})$ is minus the integral of $n_{\mathrm{F}}(z)\,g(z)/2\pi\mathrm{i}$ around small circles that enclose them counterclockwise. For $g = \mathsf{G}(\boldsymbol{k}, z)$ the only other singularity is the cut on the real axis (thick line), where the spectral function is nonzero. (b) The circles are merged into the boundaries of the upper and lower half planes: a line just above the real axis, at $\omega + \mathrm{i}0^+$, run from left to right, and a line just below, at $\omega - \mathrm{i}0^+$, run from right to left. The large arcs (dotted) contribute nothing when $n_{\mathrm{F}}(z)\,g(z)$ falls off faster than $1/|z|$ in every direction. The two lines see the retarded and advanced functions, and their difference, $-2\pi\mathrm{i}\mathsf{A}$, turns the sum into an integral over real frequencies.

4.7.6 The density response in imaginary time

The density response obeys the same rules. For two operators O_a, O_b that conserve the electron number, define

$$R_{ab}(\boldsymbol{q},\tau) = -\frac{1}{\mathscr{V}}\langle \mathrm{T}_\tau\, O_a(\boldsymbol{q},\tau)^\dagger\, O_b(\boldsymbol{q},0)\rangle, \tag{4.90}$$

with $O_a(\boldsymbol{q},\tau)^\dagger$ meaning $\mathrm{e}^{\tau H_\mu} O_a(\boldsymbol{q})^\dagger \mathrm{e}^{-\tau H_\mu}$ and no sign on exchange, since the operators are bosonic. The function is periodic in τ with period β, and its coefficients are taken at $\mathrm{i}\omega_j^{\mathrm{B}}$. For $0<\tau<\beta$ it is $-\mathscr{V}^{-1}\sum_{i,f} p_i \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\, \mathrm{e}^{-(E_f-E_i)\tau}$. Integrated with $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{B}}\tau}$, now with $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{B}}\beta}=+1$, each term gives $[\mathrm{e}^{-\beta(E_f-E_i)}-1]/[\mathrm{i}\omega_j^{\mathrm{B}}-(E_f-E_i)]$ if the denominator is nonzero, and β if it is zero, which happens only for $j=0$ and $E_f=E_i$. With $p_i\,\mathrm{e}^{-\beta(E_f-E_i)}=p_f$,

$$R_{ab}(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{i,f}^{E_f\neq E_i} \frac{(p_i-p_f)\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\mathrm{i}\omega_j^{\mathrm{B}}-(E_f-E_i)} - \delta_{j0}\,\frac{\beta}{\mathscr{V}}\sum_{i,f}^{E_f=E_i} p_i\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle. \tag{4.91}$$

In the retarded Lehmann form Eq. (4.6), relabeling $i\leftrightarrow f$ in the antiresonant term turns its weight into $-p_f\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$ and its denominator into $\omega-(E_f-E_i)+\mathrm{i}0^+$, and the two terms combine into (p_i-p_f) times one pole. The result is the sum in Eq. (4.91) with $\mathrm{i}\omega_j^{\mathrm{B}}\to\omega+\mathrm{i}0^+$.

Response function from imaginary time

$$R_{ab}(\boldsymbol{q},\omega) = R_{ab}(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}\to\omega+\mathrm{i}0^+), \qquad j>0. \tag{4.92}$$

The retarded response function of Sect. 4.1 is the analytic continuation of the imaginary-time correlation function Eq. (4.90) from the positive bosonic Matsubara frequencies.

The term with δ_{j0} is not part of the continuation, which uses the points $j>0$ alone, and it has a physical meaning. It comes from matrix elements between states of equal energy, and it is the extra term proportional to β that distinguishes the isothermal static response from the limit $\omega\to 0$ of $R_{ab}(\boldsymbol{q},\omega)$, the isolated response of Sect. 4.1 [3]. The value at $\omega_0^{\mathrm{B}}=0$ is the isothermal one, since it is a derivative of an equilibrium average in which the field has been present long enough for the target to stay in equilibrium with the bath.

For independent electrons the imaginary-time route writes the response as a sum over one fermionic frequency of a product of two free Green's functions. With $O_a=O_b=n_e(\boldsymbol{q})$ from Eq. (2.38) and occupations $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, Appendix D.8 evaluates Eq. (4.90) directly and shows that it equals

$$R^0(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}} T\sum_{j'} \mathsf{G}_0(\boldsymbol{k},\mathrm{i}\omega_{j'}^{\mathrm{F}})\,\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q},\mathrm{i}\omega_{j'}^{\mathrm{F}}+\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}} \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}})-n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{B}}+\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}}}, \tag{4.93}$$

with the sum over $\boldsymbol{k}$ including spin. Continued to $\omega+\mathrm{i}0^+$, it is the independent-electron response whose imaginary part is the transition sum Eq. (2.70), and Sect. 4.5 evaluates it for the electron gas. The product of two propagators, one carrying the electron from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$ and one carrying the hole back, is the bubble diagram of Sect. 4.8. With interactions the same continuation holds term by term, and the diagrams of the next section compute Σ and R as Matsubara sums, which Eqs. (4.88) and (4.92) then carry to the real frequencies at which a dark-matter particle deposits its energy.

4.8 Feynman diagrams

An electron injected 5 eV above the conduction-band edge of silicon does not stay there for long. It knocks electrons across the gap and emits phonons, and its quasiparticle peak acquires the width $\Gamma_{\text{qp}} = -2z_{\text{qp}} \operatorname{Im} \Sigma^{\text{R}}$ of Eq. (4.83). To get a number we need Σ, and the self-energy of an interacting system can only be computed as a series in the interaction. The density response beyond the independent-electron bubble of Sect. 4.5 also has to be computed as such a series. Feynman diagrams are the bookkeeping for these series: each term of the expansion is a picture, and a small set of rules turns the picture into an integral. The treatment follows Refs. [2], [33, Chaps. 3 and 7] and [34, Chaps. 2 and 3].

4.8.1 Wick's theorem and the rules

Split $H_\mu = H_0 + H_{\text{int}}$, with $H_0 = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}$ the independent electrons of Sect. 4.7. In imaginary time the Boltzmann factor and the Heisenberg operators can both be written with the ordered exponential $U_{\text{I}}(\beta, 0) = \mathrm{T}_\tau \exp[-\int_0^\beta \mathrm{d}\tau\, H_{\text{int,I}}(\tau)]$, where the subscript I marks operators that evolve with H_0 alone. The Matsubara Green's function Eq. (4.86) becomes the ratio Eq. (D.32),

$$\mathsf{G}(\boldsymbol{k}, \tau) = -\frac{\langle \mathrm{T}_\tau\, U_{\text{I}}(\beta, 0)\, c_{\boldsymbol{k},\text{I}}(\tau)\, c^\dagger_{\boldsymbol{k},\text{I}}(0) \rangle_0}{\langle U_{\text{I}}(\beta, 0) \rangle_0},$$

with $\langle \cdots \rangle_0$ the average in the free grand-canonical state. In the expansion of the exponential, the term of order n in H_{int} is an integral over n imaginary times of a free average of a time-ordered string of creation and annihilation operators. Nothing has been approximated so far.

Wick's theorem evaluates these averages [42]. In the free state, the time-ordered average of a string of fermion operators is a sum over all ways of grouping the operators into pairs. Each grouping contributes the product of its pair averages, with the sign of the permutation that brings the members of every pair next to each other. Only pairs of one creation and one annihilation operator of the same mode survive. Their time-ordered averages, the contractions, are free Green's functions:

$$\langle \mathrm{T}_\tau\, c_{\boldsymbol{k},\text{I}}(\tau)\, c^\dagger_{\boldsymbol{k},\text{I}}(\tau') \rangle_0 = -\mathsf{G}_0(\boldsymbol{k}, \tau - \tau'). \tag{4.94}$$

For bosons the same holds without signs, and a contraction of the field $d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}}$ with itself is the free boson propagator. The theorem is the quantum version of the statement that every moment of a Gaussian distribution is a sum over pairings (Appendix C.6): the free grand-canonical state is the "Gaussian" state of the many-body problem. Appendix D.9 proves the theorem with Gaudin's trace argument [43], which uses only the anticommutators and the cyclic property of the trace. At $T = 0$ the same theorem holds for averages in the Fermi sea $|\Phi_0\rangle$, the limit $\beta \to \infty$ of the free state, with the contraction $\mathrm{i}\mathsf{G}_0(\boldsymbol{k}, t - t')$ of the time-ordered function Eq. (4.79). The interacting ground state is reached from $|\Phi_0\rangle$ by switching on the interaction slowly, as long as no level crossing intervenes (the theorem of Gell-Mann and Low [44]), and the real-time expansion Eq. (D.33) has the same structure.

Each pairing is drawn as a picture. A contraction Eq. (4.94) is a solid line with an arrow, from the point where the electron is created to the point where it is annihilated. A factor H_{int} is a vertex, or for the Coulomb interaction a dashed line whose two ends are the two density operators of the pair interaction. Some pairings fall into a piece attached to the external points and a vacuum piece that is not; the denominator cancels the vacuum pieces, by the linked-cluster theorem, and only connected diagrams remain. Relabeling the n identical interaction factors, and exchanging the two ends of each line, produces $n!\, 2^n$ pairings with

the same value, which cancel the $1/n!$ of the exponential and the $\frac{1}{2}$ of the pair interaction. Collecting the signs and the factors of T and $\mathcal{V}$, Appendix D.9 arrives at the following rules.

Feynman rules in imaginary time

The Green's function $\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}})$ is the sum of all connected, topologically distinct diagrams with two external points, each evaluated with the following factors.

1. Each solid line carrying $(\boldsymbol{p}, \mathrm{i}\omega^{\mathrm{F}})$ is $\mathsf{G}_0(\boldsymbol{p}, \mathrm{i}\omega^{\mathrm{F}}) = 1/(\mathrm{i}\omega^{\mathrm{F}} - \xi_{\boldsymbol{p}})$; spin is conserved along it.
2. Each Coulomb line carrying $(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{B}})$ is $-v_{\mathrm{C}}(k)/\mathcal{V}$. Each boson line is $-|g(\boldsymbol{Q})|^2 \mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}})/\mathcal{V}$, with $\mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}) = 2\omega_{\boldsymbol{Q}}/[(\mathrm{i}\omega^{\mathrm{B}})^2 - \omega_{\boldsymbol{Q}}^2]$ for a boson of energy $\omega_{\boldsymbol{Q}}$ coupled through $H_{\mathrm{int}} = \mathcal{V}^{-1/2} \sum g(\boldsymbol{Q})\, c^\dagger_{\boldsymbol{k}+\boldsymbol{Q}} c_{\boldsymbol{k}} (d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}})$.
3. Momentum and frequency are conserved at every vertex. Each independent internal momentum is summed, and each independent internal frequency carries $T\sum_j$.
4. Each closed fermion loop gives -1 and a sum over its spin.
5. A solid line that begins and ends on the same interaction carries $\mathrm{e}^{\mathrm{i}\omega^{\mathrm{F}} 0^+}$.

The response function Eq. (4.90) of two densities is $-1/\mathcal{V}$ times the sum of connected diagrams with two density vertices, at which $\boldsymbol{q}$ and $\mathrm{i}\omega_j^{\mathrm{B}}$ enter and leave, evaluated with the same rules. At $T = 0$ the rules for the time-ordered functions are the same with $\mathsf{G}_0(\boldsymbol{p}, \omega) = 1/(\omega - \xi_{\boldsymbol{p}} + \mathrm{i}0^+ \operatorname{sgn} \xi_{\boldsymbol{p}})$, each Coulomb line $+\mathrm{i}v_{\mathrm{C}}(k)/\mathcal{V}$, and each internal frequency $\int \mathrm{d}\omega/2\pi$, and the time-ordered response $-(\mathrm{i}/\mathcal{V})\langle \mathrm{T}\, O_a(\boldsymbol{q}, t)^\dagger O_b(\boldsymbol{q}, 0)\rangle$ carries $+\mathrm{i}/\mathcal{V}$ in place of $-1/\mathcal{V}$.

The sign of a loop is the one rule that has no counterpart in the old-fashioned perturbation theory of Sect. 2.4. It comes from the last contraction around the loop, which pairs the first creation operator with the last annihilation operator and costs an odd number of exchanges. The rule for the response function needs one line of its own. The definition Eq. (4.90) carries $-1/\mathcal{V}$. A response diagram has $2n + 2$ propagators at order n, an even number, so the -1 of each contraction drops out. The signs left over are those of the expansion, $(-1)^n$, which the rule puts into the lines, and of the loops. Appendix D.9 checks the rules for G at first order against the Hartree–Fock self-energy. The next subsection checks the rule for R against Eq. (4.93) and against the screening of Sect. 4.4.

In the frequency representation every line is a simple pole, every frequency sum is done by the residue formula Eq. (C.40), and the temperature enters only through these sums: G_0 and D_0 do not depend on temperature, and the occupation factors n_{F} and n_{B} appear as residues of $n_{\mathrm{F}}(z)$ and $n_{\mathrm{B}}(z)$.

4.8.2 The bubble and the random-phase chain

The density response of independent electrons is the simplest diagram there is: one fermion loop through the two density vertices (Fig. 4.14a). An electron is created in $\boldsymbol{k}+\boldsymbol{q}$ and a hole in $\boldsymbol{k}$ at one vertex, and both are annihilated at the other. By the rules, the loop contributes -1 and a spin sum, the two lines give $\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega_{j'}^{\mathrm{F}})\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega_{j'}^{\mathrm{F}} + \mathrm{i}\omega_j^{\mathrm{B}})$, and the internal momentum

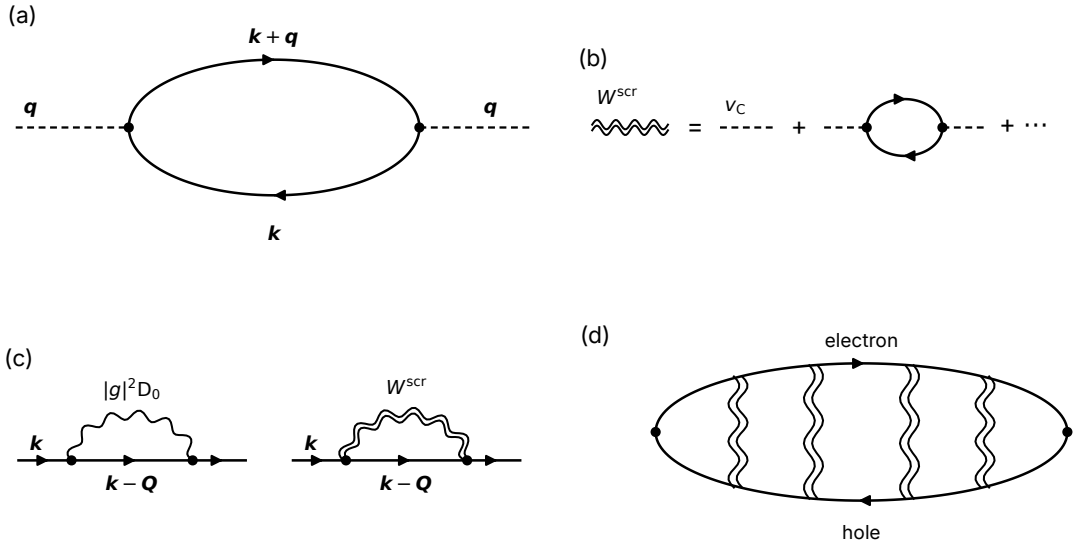


Figure 4.14: The diagrams of Sect. 4.8. Solid lines with arrows are free electron propagators G_0; an arrow against the direction of the momentum label is a hole. Dashed lines are the bare Coulomb interaction v_C, except the short dashed stubs in (a), which mark the external momentum $\boldsymbol{q}$ entering and leaving at the density vertices; single wavy lines a boson, double wavy lines the screened interaction W^scr. Dots are vertices. (a) The bubble: a density vertex creates an electron in $\boldsymbol{k}+\boldsymbol{q}$ and a hole in $\boldsymbol{k}$, and a second vertex annihilates them; it is the independent-electron response R^0. (b) The screened interaction as the random-phase chain of bubbles joined by Coulomb lines, $W^\mathrm{scr} = v_\mathrm{C} + v_\mathrm{C}R^0v_\mathrm{C} + \cdots$. (c) The self-energy of an electron that emits and reabsorbs a boson, with the line $|g|^2\mathsf{D}_0$, and the GW self-energy, the same diagram with the screened line. (d) The ladder: the electron and the hole of the bubble interact through screened interaction lines. Summed together with the chain of exchange lines, it becomes the Bethe–Salpeter equation of Sect. 4.9.

and frequency are summed. With the $-1/\mathscr{V}$ of the response,

$$R^0(\boldsymbol{q}, \mathrm{i}\omega_j^\mathrm{B}) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}s} T \sum_{j'} \mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega_{j'}^\mathrm{F})\, \mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega_{j'}^\mathrm{F} + \mathrm{i}\omega_j^\mathrm{B}), \tag{4.95}$$

which is the first form of Eq. (4.93). The frequency sum is Eq. (D.31), and it produces the Pauli difference $[n_\mathrm{F}(\xi_{\boldsymbol{k}}) - n_\mathrm{F}(\xi_{\boldsymbol{k}+\boldsymbol{q}})]/(\mathrm{i}\omega_j^\mathrm{B} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})$. Continued to $\omega + \mathrm{i}0^+$ by Eq. (4.92), the bubble is the independent-electron response Eq. (4.55), whose evaluation for the electron gas is the Lindhard function of Sect. 4.5. In a crystal the plane waves become Bloch states $\psi_{n\boldsymbol{k}}$, the density vertex $n_e(\boldsymbol{q}+\boldsymbol{G})$ acquires the matrix element $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{G})$ between the two states it connects, and the same diagram is the crystal bubble Eq. (4.63) (Eqs. (D.40) and (D.41)). The rule gives the right sign: the static bubble is negative, as a response must be.

At first order in v_C the bubble is corrected by three kinds of diagram, which differ in where the Coulomb line attaches. It can join two bubbles, the line carrying the momentum $\boldsymbol{q}$ from one to the other. It can run across the bubble, from the electron line to the hole line. Or it can begin and end on one propagator, which is a first-order self-energy inserted into that line. The

first kind has two loops, and by the rules its value is $-(1/\mathcal{V})(-1)^2(-v_{\mathrm{C}}(q)/\mathcal{V})\,(\mathcal{V}R^0)(\mathcal{V}R^0) = R^0 v_{\mathrm{C}}(q) R^0$, with a plus sign, where each loop without its sign is $\sum_{\boldsymbol{k}\boldsymbol{s}} T \sum_{j'} \mathsf{G}_0\mathsf{G}_0 = \mathcal{V}R^0$. The second kind has the same propagators and the same line but only one loop, and the opposite sign. Repeating the first kind gives the chain $R^0 + R^0 v_{\mathrm{C}} R^0 + R^0 v_{\mathrm{C}} R^0 v_{\mathrm{C}} R^0 + \cdots$, a geometric series with the sum

$$R = \frac{R^0}{1 - v_{\mathrm{C}}(q) R^0}. \tag{4.96}$$

The chain sum Eq. (4.96) is the random-phase approximation Eq. (4.59), now as a definite selection of diagrams: the chain of bubbles, and nothing inside them. Its sign is the one Sect. 4.4 found from the classical argument, $\varepsilon = 1 - v_{\mathrm{C}} R^0 > 1$ in the static limit. A diagram for R either can or cannot be cut into two by cutting a single Coulomb line at the momentum $\boldsymbol{q}$. Call the sum of those that cannot the proper part $\tilde{R}$. Every diagram that can is a chain of proper pieces joined by such lines, and the sum of the chains is $R = \tilde{R} + \tilde{R} v_{\mathrm{C}} R$, the Dyson equation Eq. (4.40). The blobs of Fig. 4.6 are sums of Feynman diagrams, and the random-phase approximation keeps the bubble alone in each blob.

In a crystal the Coulomb line between two bubbles carries $\boldsymbol{q} + \boldsymbol{G}$ for every reciprocal-lattice vector $\boldsymbol{G}$, and the chain becomes the matrix equation of Sect. 4.6. The line with $\boldsymbol{G} = 0$ carries the macroscopic field. The lines with $\boldsymbol{G} \neq 0$ are the local fields, and in the language of transitions they form the exchange kernel K^{x} of Sect. 4.6, the Coulomb interaction without its $\boldsymbol{G} = 0$ term. The second kind of first-order diagram, the line across the bubble, is the attraction between the electron and the hole that the chain leaves out. Repeated, with each interaction line screened by its own chain of bubbles, it forms the ladder of Fig. 4.14d. The sum of ladders and exchange chains is the Bethe–Salpeter equation, the equation for an electron–hole pair that propagates while the two particles attract each other, and its bound states are the excitons of Sect. 4.9. Appendix D.10 derives it from these diagrams, with the direct kernel K^{d} built from the screened interaction line.

4.8.3 Reducible lines and the thermodynamic limit

The f-sum rule for ε_2 in Sect. 4.4 rested on one claim. Deleting the Coulomb interaction at the single wavevector $\boldsymbol{q}$ from the Hamiltonian leaves the equilibrium state unchanged in the thermodynamic limit, while it turns the full response into its proper part. Diagrams make both halves precise. Let $\bar{H}$ be H with the Coulomb terms at $\pm\boldsymbol{q}$ removed, $\boldsymbol{q} \neq 0$, and compare the diagrams of the two Hamiltonians.

Take first any equilibrium average: the Green's function at some $\boldsymbol{k}$, an occupation, or the free energy per volume. In the diagrams of such an average, a Coulomb line has its momentum fixed by conservation only if cutting it separates the diagram into two pieces. The piece without external points then exchanges no momentum with the rest of the diagram, and the line carries $\boldsymbol{k} = 0$, which is absent in a neutral system. Every other line carries a loop momentum, summed as $\mathcal{V}^{-1}\sum_{\boldsymbol{k}}$ with $v_{\mathrm{C}}(k)$ finite at $\boldsymbol{k} = \pm\boldsymbol{q}$. Deleting the two terms $\boldsymbol{k} = \pm\boldsymbol{q}$ from such a sum changes it by a relative amount of order $1/N$, with N the number of electrons, and the change vanishes in the thermodynamic limit. The equilibrium state of $\bar{H}$ is therefore that of H, at every order.

Now take the response at $\boldsymbol{q}$. Here a line can carry exactly $\boldsymbol{q}$, and conservation forces it to when, and only when, the line is a bridge: cutting it splits the diagram into a piece attached to one density vertex and a piece attached to the other. The bridge lines are the lines of the chain. In $\bar{H}$ they are gone, while the lines at $\pm\boldsymbol{q}$ inside loops lose a negligible term as before.

Removing one Fourier component

In the thermodynamic limit, the Hamiltonian $\bar{H}$ without the Coulomb terms at $\pm\boldsymbol{q}$ has the same equilibrium state as H, and its density response at $\boldsymbol{q}$ is the proper part of the response of H,

$$\bar{R}(\boldsymbol{q},\omega) = \tilde{R}(\boldsymbol{q},\omega). \tag{4.97}$$

In a crystal the same argument with the $\boldsymbol{G} = 0$ term alone removed identifies $\bar{R}$ of Appendix D.5 with the response that is proper with respect to the macroscopic line only.

Because $\bar{R}$ is the response function of a Hamiltonian in its own equilibrium state, every exact property proved in Sects. 4.1 to 4.3 holds for it: the Lehmann form, positivity of the absorption and the f-sum rule with the same density $\boldsymbol{n}_e$. Sect. 4.4 used these properties for ε_2 and for the Ward identity of the proper current response. The argument is diagrammatic, and it holds at every order of perturbation theory for $\boldsymbol{q} \neq 0$. At $\boldsymbol{q} \to 0$ the deleted terms are at the edge of the sum and the limit is taken after the thermodynamic one, which is the order in which the optical limit of Appendix D.5 is defined.

4.8.4 An electron coupled to a boson

An electron in a crystal shakes the lattice as it moves: it emits a phonon, propagates in a different state, and absorbs the phonon again. The lowest self-energy diagram of this process is the Fock diagram with a boson line (Fig. 4.14c), second order in the coupling g. It shows every feature of finite-temperature perturbation theory in a few lines, and Sect. 6.3 builds the electron–phonon self-energy of silicon on it. By the rules, with the boson line carrying $(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}_{j'})$ and the electron line $(\boldsymbol{k}-\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{F}}_{j} - \mathrm{i}\omega^{\mathrm{B}}_{j'})$, and no loop,

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_{j}) = -\frac{1}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 \, T \sum_{j'} \mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}_{j'}) \, \mathsf{G}_0(\boldsymbol{k}-\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{F}}_{j} - \mathrm{i}\omega^{\mathrm{B}}_{j'}).$$

The sum runs over a bosonic frequency, and by Eq. (C.40) it is minus the sum of the residues of $n_{\mathrm{B}}(z)h(z)$ at the poles of

$$h(z) = \Big[\frac{1}{z-\omega_{\boldsymbol{Q}}} - \frac{1}{z+\omega_{\boldsymbol{Q}}}\Big] \frac{1}{\mathrm{i}\omega^{\mathrm{F}}_{j} - z - \xi'}, \qquad \xi' = \xi_{\boldsymbol{k}-\boldsymbol{Q}},$$

which falls off as $1/|z|^3$, so no convergence factor is needed. The pole at $z = \omega_{\boldsymbol{Q}}$ has the residue $n_{\mathrm{B}}(\omega_{\boldsymbol{Q}})/(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi' - \omega_{\boldsymbol{Q}})$. The pole at $z = -\omega_{\boldsymbol{Q}}$ has the residue $-n_{\mathrm{B}}(-\omega_{\boldsymbol{Q}})/(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi' + \omega_{\boldsymbol{Q}})$, and $n_{\mathrm{B}}(-x) = -1 - n_{\mathrm{B}}(x)$ turns its numerator into $1 + n_{\mathrm{B}}$. The pole at $z = \mathrm{i}\omega^{\mathrm{F}}_{j} - \xi'$, where the electron line is singular, has the residue $-n_{\mathrm{B}}(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi')\,\mathsf{D}_0(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi')$. Before anything is continued, the Bose function at a fermionic frequency is simplified with $\mathrm{e}^{\mathrm{i}\beta\omega^{\mathrm{F}}_{j}} = -1$: $n_{\mathrm{B}}(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi') = 1/(-\mathrm{e}^{-\beta\xi'} - 1) = -[1 - n_{\mathrm{F}}(\xi')]$. This residue is then $[1 - n_{\mathrm{F}}(\xi')][1/(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi' - \omega_{\boldsymbol{Q}}) - 1/(\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi' + \omega_{\boldsymbol{Q}})]$. Adding the three residues, changing the sign, and multiplying by $-|g|^2/\mathscr{V}$ gives the self-energy.

Second-order self-energy with a boson

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_{j}) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 \left[\frac{n_{\mathrm{B}}(\omega_{\boldsymbol{Q}}) + 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{Q}})}{\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi_{\boldsymbol{k}-\boldsymbol{Q}} - \omega_{\boldsymbol{Q}}} + \frac{n_{\mathrm{B}}(\omega_{\boldsymbol{Q}}) + n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{Q}})}{\mathrm{i}\omega^{\mathrm{F}}_{j} - \xi_{\boldsymbol{k}-\boldsymbol{Q}} + \omega_{\boldsymbol{Q}}} \right]. \tag{4.98}$$

The retarded self-energy is the same expression with $\mathrm{i}\omega_j^{\mathrm{F}} \to \omega + \mathrm{i}0^+$. The term with $-\omega_{\boldsymbol{Q}}$ in the denominator is the emission of a boson, the term with $+\omega_{\boldsymbol{Q}}$ its absorption.

The thermal factors have a direct reading. In the emission term, $n_{\mathrm{B}}+1$ is the stimulated plus spontaneous emission of Worked example 2.7, and $-n_{\mathrm{F}}$ removes the intermediate states that are already occupied: the Pauli principle acts on the virtual state as well as on a real one. In the absorption term, n_{B} counts the thermal bosons available, and $+n_{\mathrm{F}}$ adds the process in which an electron already in $\boldsymbol{k}-\boldsymbol{Q}$ makes the transition instead. Eq. (4.103) below shows that the two factors are sums of a forward and a reverse rate. For an electron in an empty band at $T=0$, $n_{\mathrm{B}}=n_{\mathrm{F}}=0$, and the retarded self-energy is

$$\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega) = \frac{1}{\mathcal{V}}\sum_{\boldsymbol{Q}} \frac{|g(\boldsymbol{Q})|^2}{\omega - \xi_{\boldsymbol{k}-\boldsymbol{Q}} - \omega_{\boldsymbol{Q}} + \mathrm{i}0^+},$$

the second-order sum of Eq. (2.52) over intermediate states with one electron in $\boldsymbol{k}-\boldsymbol{Q}$ and one boson, with ω in place of the initial energy. It has the form of the self-energy Eq. (4.85) of the level coupled to a band, and the matrix element $g/\sqrt{\mathcal{V}}$ of H_{int} plays the role of $V_{\boldsymbol{k}}$.

Worked example 4.13: An electron coupled to an Einstein boson

Take the electron in an empty conduction band of a semiconductor, with μ_{ch} near midgap, 0.5 eV below the band bottom E_{c}. Then $n_{\mathrm{F}} \le 4\times10^{-9}$ in the band at 300 K, and the n_{F} terms of Eq. (4.98) drop out. Let the boson be dispersionless, an Einstein mode with $\omega_{\boldsymbol{Q}} = \omega_{\mathrm{Ein}} = 60$ meV, a typical silicon phonon energy (Sect. 1.3), and let the coupling g be constant. The sum over $\boldsymbol{Q}$ then runs over all final electron states $\boldsymbol{k}-\boldsymbol{Q}$ with the same weight, and the self-energy depends on ω alone. For the band we take the flat continuum of Worked example 2.1: $\mathcal{D}_f$ states per unit energy, spread uniformly over a width $E_{\mathrm{w}} = 1$ eV above E_{c}. The golden-rule rate for emitting a boson from deep inside the band at $T=0$ is $\Gamma_{\mathrm{em}} = 2\pi(|g|^2/\mathcal{V})\mathcal{D}_f$, and we choose the coupling so that $\Gamma_{\mathrm{em}} = 20$ meV. With $\sum_{\boldsymbol{Q}} \to \mathcal{D}_f \int \mathrm{d}\epsilon$ and the integral $\int_{E_{\mathrm{c}}}^{E_{\mathrm{c}}+E_{\mathrm{w}}} \mathrm{d}\epsilon/(x-\epsilon+\mathrm{i}0^+) = \ell_{\mathrm{b}}(x)$,

$$\Sigma^{\mathrm{R}}(\omega) = \frac{\Gamma_{\mathrm{em}}}{2\pi}\Big[(n_{\mathrm{B}}+1)\,\ell_{\mathrm{b}}(\omega-\omega_{\mathrm{Ein}}) + n_{\mathrm{B}}\,\ell_{\mathrm{b}}(\omega+\omega_{\mathrm{Ein}})\Big], \qquad \ell_{\mathrm{b}}(x) = \ln\frac{x-E_{\mathrm{c}}+\mathrm{i}0^+}{x-E_{\mathrm{c}}-E_{\mathrm{w}}+\mathrm{i}0^+},$$

with $n_{\mathrm{B}} = n_{\mathrm{B}}(\omega_{\mathrm{Ein}})$. The logarithm has the imaginary part $-\pi$ where its argument lies in the band and zero outside, and each term contributes a width only where its final electron state exists (Fig. 4.15).

At $T=0$ an electron at the band bottom, $\omega = E_{\mathrm{c}}$, cannot emit: the final state would lie ω_{Ein} below the band. Its width is zero, as Sect. 4.7 argued for the conduction-band minimum. The virtual emission still shifts its energy, by $\mathrm{Re}\,\Sigma^{\mathrm{R}}(E_{\mathrm{c}}) = (\Gamma_{\mathrm{em}}/2\pi)\ln[\omega_{\mathrm{Ein}}/(E_{\mathrm{w}}+\omega_{\mathrm{Ein}})] = -9.14$ meV, and the slope $\partial_\omega \mathrm{Re}\,\Sigma^{\mathrm{R}} = -0.0500$ sets $z_{\mathrm{qp}} = 0.952$. The electron at the band bottom carries a cloud of virtual bosons, which lowers its energy and takes 5 percent of its weight into the incoherent background. Solving $E_{\mathrm{qp}} = E_{\mathrm{c}} + \mathrm{Re}\,\Sigma^{\mathrm{R}}(E_{\mathrm{qp}})$ instead of evaluating Σ at E_{c} moves the energy by -8.73 meV. An electron 0.3 eV up in the band emits freely, and $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}} = \Gamma_{\mathrm{em}}$ is the golden-rule rate.

At 300 K, $T = 25.85$ meV and $n_{\mathrm{B}} = 0.109$. The electron at the band bottom can now absorb a thermal boson, and its width is $\Gamma_{\mathrm{qp}} = z_{\mathrm{qp}}\Gamma_{\mathrm{em}} n_{\mathrm{B}} = 2.08$ meV, with $z_{\mathrm{qp}} = 0.953$: a lifetime of 0.32 ps. The shift grows to -11.09 meV, which is the zero-temperature shift times $2n_{\mathrm{B}}+1$ to within 0.4 percent, since emission and absorption contribute nearly equal logarithms when $\omega_{\mathrm{Ein}} \ll E_{\mathrm{w}}$. Inside the band the width rises to $\Gamma_{\mathrm{em}}(2n_{\mathrm{B}}+1) = 24.4$ meV. These two features, a shift that scales as $2n_{\mathrm{B}}+1$ and a width at the band edge set by

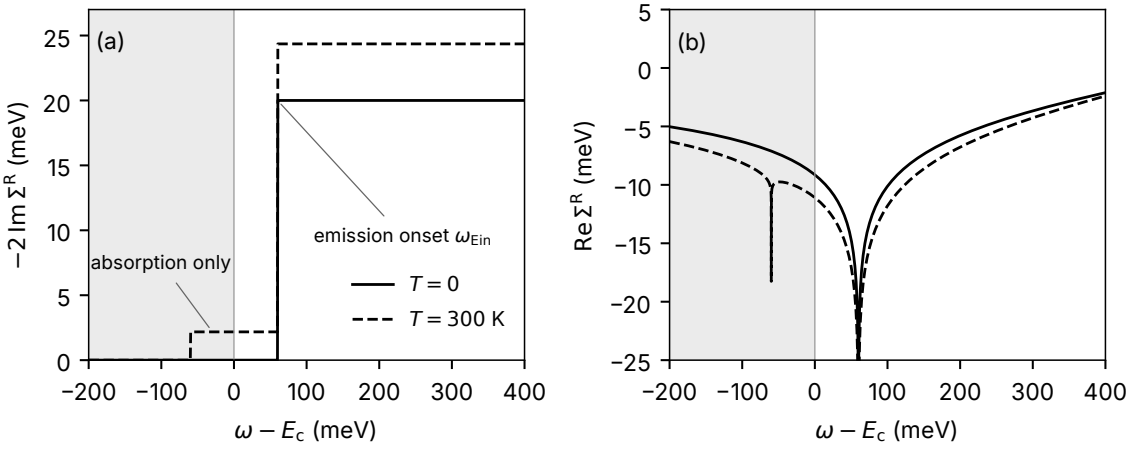


Figure 4.15: Self-energy of an electron coupled to a dispersionless boson of energy $\omega_{\mathrm{Ein}} = 60\,\mathrm{meV}$ (Worked example 4.13), against the energy ω measured from the bottom E_{c} of an empty band with a flat density of states over $E_{\mathrm{w}} = 1\,\mathrm{eV}$, for a constant coupling with the golden-rule emission rate $\Gamma_{\mathrm{em}} = 20\,\mathrm{meV}$; solid at $T = 0$, dashed at 300 K. The model and its parameters are illustrative. (a) The width $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}}$: zero below the emission threshold $E_{\mathrm{c}} + \omega_{\mathrm{Ein}}$ at $T = 0$; at 300 K, $\Gamma_{\mathrm{em}} n_{\mathrm{B}} = 2.2\,\mathrm{meV}$ from absorption above $E_{\mathrm{c}} - \omega_{\mathrm{Ein}}$, and $\Gamma_{\mathrm{em}}(2n_{\mathrm{B}} + 1)$ where both processes are allowed. (b) The shift $\mathrm{Re}\,\Sigma^{\mathrm{R}}$, $-9.14\,\mathrm{meV}$ at E_{c} for $T = 0$ and $-11.09\,\mathrm{meV}$ at 300 K, with logarithmic singularities at the thresholds caused by the sharp edge of the flat band. Shaded: below the band bottom.

absorption alone, are the temperature dependence of band energies that Sect. 6.3 computes for silicon. The logarithmic dip of $\mathrm{Re}\,\Sigma^{\mathrm{R}}$ at the emission threshold, $\omega = E_{\mathrm{c}} + \omega_{\mathrm{Ein}}$, comes from the step of the flat density of states at E_{c}; with the square-root density of states of a three-dimensional band edge it becomes a finite cusp.

4.8.5 The screened interaction and the GW self-energy

The Coulomb version of the same diagram, with a bare line, is the exchange self-energy Eq. (D.36), $\Sigma_{\mathrm{x}}(\boldsymbol{k}) = -\mathscr{V}^{-1}\sum_{\boldsymbol{p}} v_{\mathrm{C}}(|\boldsymbol{k} - \boldsymbol{p}|)\, n_{\mathrm{F}}(\xi_{\boldsymbol{p}})$, the Fock term. It is static, it has no imaginary part, and in a metal it makes the group velocity of the electron gas infinite at the Fermi surface through a logarithm (Sect. 23.1). It misses screening: while the electron propagates, the other electrons rearrange around it and weaken the interaction at long distances. The rules say how to put screening in. Every chain of bubbles that can be inserted into the Coulomb line is a self-energy diagram, and summing the chains replaces the line, exactly as in the random-phase chain:

$$W^{\mathrm{scr}}(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}}) = v_{\mathrm{C}}(q) + v_{\mathrm{C}}(q)\, R(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}})\, v_{\mathrm{C}}(q) = \varepsilon^{-1}(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}})\, v_{\mathrm{C}}(q). \tag{4.99}$$

The last step is Eq. (4.41). With the bubble chain alone, R is the random-phase response Eq. (4.96) and ε the Lindhard function of Sect. 4.5 or its crystal form; with the proper blobs it is exact. The screened interaction is the Coulomb potential of a test charge seen through the dielectric response of the target, the same $\varepsilon^{-1} v_{\mathrm{C}}$ that Sect. 4.4 found for an external charge, but now at every frequency. In a crystal it is the matrix $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'} = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'} v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}'|)$, and the steps below hold element by element.

Replacing the bare line of the exchange diagram by W^{scr}, and the free propagator by the full one, is the GW approximation of Hedin [45]. In imaginary time,

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = -\frac{1}{\mathscr{V}} \sum_{\boldsymbol{q}} T \sum_{j'} W^{\mathrm{scr}}(\boldsymbol{q}, \mathrm{i}\omega_{j'}^{\mathrm{B}})\, \mathsf{G}(\boldsymbol{k}-\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{F}} - \mathrm{i}\omega_{j'}^{\mathrm{B}}), \tag{4.100}$$

with the convergence factor of rule 5 on the static part v_{C}, which reproduces Σ_{x}. At $T = 0$ the real-time rules give the same diagram as

$$\Sigma(\boldsymbol{k}, \omega) = \frac{\mathrm{i}}{\mathscr{V}} \sum_{\boldsymbol{q}} \int \frac{\mathrm{d}\omega'}{2\pi}\, \mathsf{G}(\boldsymbol{k}-\boldsymbol{q}, \omega-\omega')\, W^{\mathrm{scr}}(\boldsymbol{q}, \omega'), \tag{4.101}$$

which in space and time is a product, $\Sigma(1,2) = \mathrm{i}\mathsf{G}(1,2)W^{\mathrm{scr}}(1,2)$, the form usually written $\Sigma = \mathrm{i}GW$ [46, Eqs. (60) and (61)]. Note that G and W^{scr} here are time-ordered functions: by the convention of Sect. 4.7, a function without superscript and with a real argument is time-ordered, and $W^{\mathrm{scr}}(\boldsymbol{q}, \omega')$ equals the retarded $\varepsilon^{-1}v_{\mathrm{C}}$ of Sect. 4.4 for $\omega' > 0$ and the advanced one for $\omega' < 0$. Lifetimes need the retarded self-energy, which the Matsubara form Eq. (4.100) delivers by continuation. The approximation keeps the screening to all orders and drops the vertex corrections, the diagrams in which a further line crosses the interaction line of Fig. 4.14c; Hedin's exact equations contain them in a vertex function, which the GW approximation sets to one [46, Eqs. (43)–(46)]. With the Hartree term, which is the electrostatic potential of the density, GW is the self-energy behind the quasiparticle gap of silicon in Sects. 3.4 and 4.7.

The lifetime follows from the same frequency sum as for a boson. The difference $W^{\mathrm{scr}} - v_{\mathrm{C}} = v_{\mathrm{C}}^2 R$ is a response function, and like every response it has the spectral representation Eq. (4.19), $W^{\mathrm{scr}}(\boldsymbol{q}, z) - v_{\mathrm{C}}(q) = \int \mathrm{d}\omega'\, B(\boldsymbol{q}, \omega')/(z-\omega')$, with the weight $B = -\operatorname{Im} W^{\mathrm{scr,R}}/\pi = v_{\mathrm{C}}(q) L(\boldsymbol{q}, \omega')/\pi$, where L is the loss function Eq. (4.45). The spectral form makes the screened line a superposition of bosons: the free boson line has $B = |g|^2[\delta(\omega' - \omega_{\boldsymbol{Q}}) - \delta(\omega' + \omega_{\boldsymbol{Q}})]$, and the screened Coulomb line is a continuum of them, the electron–hole pairs and plasmons of the target, weighted by the loss function. The residue sum that led to Eq. (4.98) then leads, with the free propagator in Eq. (4.100),

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = \Sigma_{\mathrm{x}}(\boldsymbol{k}) + \frac{1}{\mathscr{V}} \sum_{\boldsymbol{q}} \int \mathrm{d}\omega'\, B(\boldsymbol{q}, \omega')\, \frac{n_{\mathrm{B}}(\omega') + 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{F}} - \xi_{\boldsymbol{k}-\boldsymbol{q}} - \omega'},$$

and the imaginary part of its continuation, $\operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k}, \omega) = -\mathscr{V}^{-1} \sum_{\boldsymbol{q}} v_{\mathrm{C}}(q) L(\boldsymbol{q}, \omega')[n_{\mathrm{B}}(\omega') + 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}})]$ with $\omega' = \omega - \xi_{\boldsymbol{k}-\boldsymbol{q}}$. On the shell $\omega = \xi_{\boldsymbol{k}}$, with $z_{\mathrm{qp}} \simeq 1$, the width is $\Gamma_{\mathrm{qp}} = -2 \operatorname{Im}\Sigma^{\mathrm{R}}$. The structure-factor relation $2L/v_{\mathrm{C}} = (1 - \mathrm{e}^{-\beta\omega'})S$ of Eq. (4.45), and the identity $(1 - \mathrm{e}^{-\beta\omega'})(n_{\mathrm{B}}(\omega') + 1) = 1$, turn the width into a rate.

Electron–electron lifetime in the *GW* approximation

$$\Gamma_{\mathrm{qp}}(\boldsymbol{k}) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\, v_{\mathrm{C}}(q)^2\, S(\boldsymbol{q}, \omega') \Big[1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}}) + \mathrm{e}^{-\beta\omega'} n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}})\Big], \qquad \omega' = \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}-\boldsymbol{q}}. \tag{4.102}$$

The inverse lifetime of an electron is the rate Eq. (4.14) at which it would scatter off the target as a projectile with the potential $\tilde{V} = v_{\mathrm{C}}$, with the Pauli factor of its final state, plus the rate of the reverse process.

An injected electron is thus a dark-matter particle with a Coulomb coupling. It loses energy to exactly the excitations that the loss function describes, through the same structure

factor, and Eq. (4.102) is a stopping-power calculation of the kind in Eq. (4.54), with a quantum recoil and a Pauli-blocked final state. The reverse term, $\mathrm{e}^{-\beta\omega'} n_\mathrm{F}$, is the rate at which electrons in $\boldsymbol{k}-\boldsymbol{q}$ absorb energy from the target and refill $\boldsymbol{k}$, and it is negligible for an electron far above μ_ch. Near the Fermi level of a metal, $1-n_\mathrm{F}$ and the linear low-frequency loss function of Sect. 4.5 give the $\xi_{\boldsymbol{k}}^2$ of Sect. 4.7. In silicon, $S(\boldsymbol{q},\omega')$ vanishes for ω' below the gap, and an electron cannot decay by creating a pair until it lies at least E_g above the band edge. Higher up, the loss function peaks at the plasmon near 16.4 eV (Sect. 4.4), and an electron with tens of eV loses its energy mostly by plasmon emission. The exchange of the projectile with the electrons it excites, a vertex correction, is absent from Eq. (4.102).

In practice the *GW* quasiparticle energies and lifetimes of silicon are computed in the basis of Kohn–Sham states. The independent-electron response $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ of Sect. 4.6 yields $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}$ in the random-phase approximation, and Eq. (4.99) gives the matrix $W^\mathrm{scr}_{\boldsymbol{G}\boldsymbol{G}'}$. The matrix elements $\langle\psi_{n\boldsymbol{k}}|\Sigma(\omega)|\psi_{n\boldsymbol{k}}\rangle$ of Eq. (4.101), built from the same plane-wave matrix elements, enter the linearized quasiparticle equation Eq. (4.84). The frequency integral is the expensive step: a plasmon-pole model, which puts each element of $\operatorname{Im}\varepsilon^{-1}$ into a single delta function, is accurate for the energies; however, it produces no lifetimes, which need $\operatorname{Im}\varepsilon^{-1}$ resolved down to the gap.

4.8.6 The cutting rule

The imaginary parts found so far have all been golden-rule rates: the transition sums in Eq. (4.93), the emission and absorption rates of Worked example 4.13, the scattering rate Eq. (4.102). The link between imaginary parts and golden-rule rates is general, and it has a graphical form. After the frequency sums, a diagram at real frequency is a sum of terms, each a product of occupation factors divided by $\omega - E_\mathrm{cs} + \mathrm{i}0^+$, where E_cs is the total energy of a set of lines that together separate the two external points. The imaginary part of the diagram replaces each such denominator by $-\pi\delta(\omega - E_\mathrm{cs})$, by the Sokhotski–Plemelj formula Eq. (2.58). Drawing a line through the diagram across these propagators, the cut, puts them on shell. The pieces on either side of the cut are the amplitudes to reach the cut state and to return from it, and the imaginary part is π times their product summed over the cut states: a golden-rule rate. Cutkosky formulated this rule for relativistic amplitudes [47]. At finite temperature the occupation factors combine into rates in two ways, and Weldon gave the general form [48]; we show it for the two diagrams that the book uses.

For the boson self-energy Eq. (4.98), the cut runs through the electron line and the boson line. Write each thermal factor on its shell as a sum of products of occupations, $n_\mathrm{B}+1-n_\mathrm{F} = (1+n_\mathrm{B})(1-n_\mathrm{F}) + n_\mathrm{B}n_\mathrm{F}$ and $n_\mathrm{B}+n_\mathrm{F} = n_\mathrm{B}(1-n_\mathrm{F}) + (1+n_\mathrm{B})n_\mathrm{F}$. With $\xi' = \xi_{\boldsymbol{k}-\boldsymbol{Q}}$ and $n_\mathrm{F} = n_\mathrm{F}(\xi')$,

$$
\begin{aligned}
-2\operatorname{Im}\Sigma^\mathrm{R}(\boldsymbol{k},\omega) &= \Gamma_\mathrm{out}(\boldsymbol{k},\omega) + \Gamma_\mathrm{in}(\boldsymbol{k},\omega),\\
\Gamma_\mathrm{out} &= \frac{2\pi}{\mathscr{V}}\sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 (1-n_\mathrm{F})\Big[(1+n_\mathrm{B})\,\delta(\omega-\xi'-\omega_{\boldsymbol{Q}}) + n_\mathrm{B}\,\delta(\omega-\xi'+\omega_{\boldsymbol{Q}})\Big],\\
\Gamma_\mathrm{in} &= \frac{2\pi}{\mathscr{V}}\sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2\, n_\mathrm{F}\Big[n_\mathrm{B}\,\delta(\omega-\xi'-\omega_{\boldsymbol{Q}}) + (1+n_\mathrm{B})\,\delta(\omega-\xi'+\omega_{\boldsymbol{Q}})\Big].
\end{aligned}
\tag{4.103}
$$

Γ_out is the golden-rule rate Eq. (2.5) at which an electron in $\boldsymbol{k}$ with energy ω scatters out, by emission or absorption, into an empty state. Γ_in is the rate at which electrons scatter into an empty $\boldsymbol{k}$ from occupied states. On the shell, $n_\mathrm{B}/(1+n_\mathrm{B}) = \mathrm{e}^{-\beta\omega_{\boldsymbol{Q}}}$ and $n_\mathrm{F}/(1-n_\mathrm{F}) = \mathrm{e}^{-\beta\xi'}$, so that $\Gamma_\mathrm{in} = \mathrm{e}^{-\beta\omega}\Gamma_\mathrm{out}$, the fermion counterpart of detailed balance. The retarded Green's function is built from an anticommutator, and its width counts the decay of an added electron

and the filling of an added hole together. The density response is built from a commutator, and the corresponding combination is a difference.

Cutting rule

The imaginary part of the retarded continuation of a diagram is the sum over its cuts, each cut line on shell. For a fermion self-energy, $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}} = \Gamma_{\mathrm{out}}(1+\mathrm{e}^{-\beta\omega})$ with Γ_{out} the golden-rule rate out of the state; for a density response, $-2\,\mathrm{Im}\,R = (1-\mathrm{e}^{-\beta\omega})S$ with S the golden-rule sum into the cut states.

Worked example 4.14: The cut of the bubble

The bubble has a single cut, through its two propagators (Fig. 4.16), and its cut state is one electron–hole pair. With orbitals γ in place of plane waves, the density vertex between γ and γ' is $\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle$, and the frequency sum Eq. (D.31) gives

$$R^0(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\gamma\gamma'}\left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2 \frac{n_\gamma - n_{\gamma'}}{\mathrm{i}\omega_j^{\mathrm{B}} + E_\gamma - E_{\gamma'}},$$

with $n_\gamma = n_{\mathrm{F}}(E_\gamma - \mu_{\mathrm{ch}})$; for plane waves the matrix element is $\delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$ and the sum reduces to Eq. (4.93). The denominator is $\omega - E_{\mathrm{cs}}$ with $E_{\mathrm{cs}} = E_{\gamma'} - E_\gamma$, the energy of the pair. Continuing to $\omega + \mathrm{i}0^+$ and cutting,

$$-2\,\mathrm{Im}\,R^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}}\sum_{\gamma\gamma'}\left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2 (n_\gamma - n_{\gamma'})\,\delta(\omega - E_{\gamma'} + E_\gamma).$$

The Pauli difference is a difference of two rates. On the shell, $n_{\gamma'}(1-n_\gamma) = \mathrm{e}^{-\beta\omega}n_\gamma(1-n_{\gamma'})$, and $n_\gamma - n_{\gamma'} = n_\gamma(1-n_{\gamma'}) - n_{\gamma'}(1-n_\gamma) = (1-\mathrm{e}^{-\beta\omega})\,n_\gamma(1-n_{\gamma'})$: an electron moves from occupied γ to empty γ', minus the reverse move. The factor $1-\mathrm{e}^{-\beta\omega}$ is that of the fluctuation–dissipation theorem Eq. (4.18), and what it multiplies is

$$S^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}}\sum_{\gamma\gamma'}\left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2 n_\gamma(1-n_{\gamma'})\,\delta(\omega - E_{\gamma'} + E_\gamma),$$

the golden-rule sum Eq. (4.21) over single electron–hole pairs, term by term; for the electron gas at $T = 0$ it is the closed form Eq. (4.22). The terms with $\gamma' = \gamma$ have $n_\gamma - n_{\gamma'} = 0$ and drop out of the bubble; they are the elastic term of Eq. (4.21), which carries no dissipation.

Whatever diagram is drawn, its imaginary part counts real final states with positive weights, and every approximation that is a selection of diagrams, with the cuts it allows, has a golden-rule reading. The random-phase chain cuts through electron–hole pairs and plasmons. The ladder, which Sect. 4.9 sums, cuts through pairs whose electron and hole are bound into excitons.

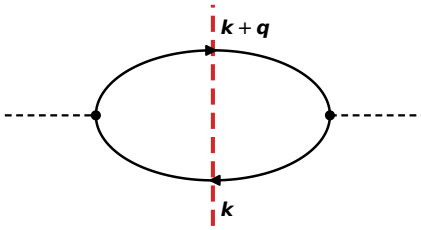


Figure 4.16: The cut of the bubble. The dashed stubs mark the external momentum $\boldsymbol{q}$ at the two density vertices. The cut (dashed vertical line) passes through the electron line $\boldsymbol{k}+\boldsymbol{q}$ and the hole line $\boldsymbol{k}$ and puts both on shell, $\omega = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}}$; each half of the diagram is the amplitude for the density operator to create the pair.

4.9 Excitons and the Bethe–Salpeter equation

The measured absorption spectrum of silicon has a peak of height 36 at 3.42 eV, and every approximation of Sect. 4.6 leaves only a shoulder of 13 to 15 in its place (Fig. 4.11). The bubble, the local fields and the adiabatic kernel all treat the electron lifted across the gap and the hole it leaves behind as independent. The electron and the hole carry opposite charges and attract, and the other electrons screen their attraction without cancelling it. In GaAs the attraction binds them into a hydrogen-like state 4.2 meV below the gap, which shows up as a sharp line at the absorption edge [49], and in silicon the bound pair lies 14.7 meV below the indirect gap [50]. Such correlated electron–hole pairs, bound or not, are excitons. For dark matter, excitons change the response exactly where light particles probe the target: in the first eV above the gap, where the loss function is smallest and the attraction moves spectral weight downward and adds lines below the continuum.

4.9.1 The electron–hole Hamiltonian

In the bubble of Sect. 4.8 one density vertex creates a pair, an electron in $(c, \boldsymbol{k}+\boldsymbol{q})$ and a hole in $(v, \boldsymbol{k})$, and the other vertex destroys it; this pair is a transition, labeled $cv\boldsymbol{k}$ at fixed $\boldsymbol{q}$. Between the two vertices the pair propagates freely, and in an insulator at $T = 0$ each transition is an eigenstate of the problem with energy $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$. The ladder of Fig. 4.14(d) lets the electron and the hole interact any number of times on the way, and Appendix D.10 sums it with the rules of Appendix D.9. The two first-order insertions of Sect. 4.8 become two kernels, and with a statically screened interaction line the sum closes as a matrix equation in the space of transitions, which is an eigenvalue problem.

Bethe–Salpeter equation in the transition basis

At momentum transfer $\boldsymbol{q}$, the exciton energies $E_\lambda(\boldsymbol{q})$ and envelopes $A^\lambda_{cv\boldsymbol{k}}(\boldsymbol{q})$ solve

$$\sum_{c'v'\boldsymbol{k}'} H^{\mathrm{exc}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'}\, A^\lambda_{c'v'\boldsymbol{k}'} = E_\lambda\, A^\lambda_{cv\boldsymbol{k}},$$
$$H^{\mathrm{exc}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = (\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})\delta_{cc'}\delta_{vv'}\delta_{\boldsymbol{k}\boldsymbol{k}'} + g_{\mathrm{deg}} K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} - K^{\mathrm{d}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'}, \tag{4.104}$$

normalized to $\sum_{cv\boldsymbol{k}} |A^\lambda_{cv\boldsymbol{k}}|^2 = 1$, with $g_{\mathrm{deg}} = 2$ for the spin singlets that the density creates. The exchange and direct kernels are

$$K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{G}\neq 0} v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})\, \rho_{c'v'}(\boldsymbol{k}',\boldsymbol{q}+\boldsymbol{G})^*,$$
$$K^{\mathrm{d}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{G}\boldsymbol{G}'} \rho_{cc'}(\boldsymbol{k}'+\boldsymbol{q},\boldsymbol{p}+\boldsymbol{G})\, W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p},\omega=0)\, \rho_{vv'}(\boldsymbol{k}',\boldsymbol{p}+\boldsymbol{G}')^*, \quad \boldsymbol{p} = \boldsymbol{k}-\boldsymbol{k}', \tag{4.105}$$

with the density vertices of Eq. (3.34), the screened interaction $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'} = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'} v_{\mathrm{C}}(|\boldsymbol{p}+\boldsymbol{G}'|)$ of Eq. (4.99), and quasiparticle band energies.

Eq. (4.104) is Eq. (26.4) at finite $\boldsymbol{q}$. The exchange kernel is the one of Sects. 4.6 and 4.8, the Coulomb interaction without its $\boldsymbol{G} = 0$ term. For a momentum transfer beyond the first zone, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ in the unreduced indexing of Sect. 4.4, the kernels are written with $\boldsymbol{q}_r$ in place of $\boldsymbol{q}$, and the omitted term of K^{x} is then the one at $\boldsymbol{G} = \boldsymbol{G}_q$.

The diagonal of H^{exc} is the energy of the free pair. It must be a quasiparticle energy (Sect. 4.7), because the self-energy insertions of the ladder are summed into the propagators, and with Kohn–Sham bands a scissor (Eq. (3.66)) takes the place of the quasiparticle energies. The direct kernel is the Coulomb interaction between the electron and the hole, screened by all the other electrons. The electron scatters from $(c, \boldsymbol{k}+\boldsymbol{q})$ to $(c', \boldsymbol{k}'+\boldsymbol{q})$ and the hole from $(v, \boldsymbol{k})$ to $(v', \boldsymbol{k}')$, so the pair keeps its total momentum $\boldsymbol{q}$ and its constituents exchange $\boldsymbol{p}$. The diagonal element of K^{d} is the Coulomb energy of two positive densities, $|\psi_{c,\boldsymbol{k}+\boldsymbol{q}}|^2$ and $|\psi_{v\boldsymbol{k}}|^2$. Since that energy is positive, $-K^{\mathrm{d}}$ lowers the pair energy: it is the attraction. At large separation the screened interaction of two charges in an insulator is $e^2/(4\pi\varepsilon_\infty r)$, and this long-range tail binds the pairs of the next subsection. The exchange kernel describes something else. The pair annihilates at one point and a pair is re-created at another, and the Coulomb line between them is the microscopic Hartree potential of the induced density. The exchange kernel is the local-field kernel of Sect. 4.6, positive on the diagonal and repulsive. The long-range term $\boldsymbol{G} = 0$ is left out because it is the macroscopic field, which the definition of $\varepsilon_{\mathrm{mac}}$ already treats on its own (Eq. (D.49)); added back, it yields the full density response (Appendix D.10). Exchange is also what distinguishes the spin states. The density vertex couples to both spins alike, so K^{x} couples a transition of either spin to one of either spin and appears with g_{deg} in the singlet, while the triplet feels only $-K^{\mathrm{d}}$ and is not excited by a density coupling.

Behind Eq. (4.104) stand a kernel without the change of the screening, a static screened interaction and the Tamm–Dancoff approximation, and Appendix D.10 makes each one explicit. The kernel is the change of the Hartree potential plus the GW self-energy of Sect. 4.8 when the Green's function changes, with the change of W^{scr} itself neglected [51, 52]. The screened interaction is static: W^{scr} is taken at zero frequency, which is good when the binding energies and the spread of transition energies that matter are small against the frequencies at which ε^{-1} changes, set in silicon by the plasma frequency of 16.6 eV. The static interaction misses the retarded response of the other electrons to the moving pair. And the coupling between the creation and the annihilation of a pair is dropped, the Tamm–Dancoff approximation, whose error on E_λ is of second order in the kernel divided by twice the gap. With these choices H^{exc} is Hermitian (Appendix D.10), and its eigenvectors are orthonormal.

Without kernels each transition is its own eigenstate and the response is the bubble of Eq. (4.63). With K^{d} switched off, $H^{\mathrm{exc}} = \mathbf{D} + g_{\mathrm{deg}}\mathbf{X}\mathbf{X}^\dagger$, where $\mathbf{D}$ is the diagonal matrix of transition energies and the column of $\mathbf{X}$ for each $\boldsymbol{G} \neq 0$ holds the vertices $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{G})$ times $[v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)/\mathscr{V}]^{1/2}$. Projected on the vertices, the resolvent $(\omega - H^{\mathrm{exc}})^{-1}$ becomes, by the

Woodbury identity Eq. (C.58),

$$\mathbf{X}^\dagger(\omega - \mathbf{D} - g_{\mathrm{deg}}\mathbf{X}\mathbf{X}^\dagger)^{-1}\mathbf{X} = \mathbf{M}(\mathbf{1} - g_{\mathrm{deg}}\mathbf{M})^{-1}, \qquad \mathbf{M} = \mathbf{X}^\dagger(\omega - \mathbf{D})^{-1}\mathbf{X}.$$

Here $g_{\mathrm{deg}}\mathbf{M}$ is the resonant part of R^0 of Eq. (4.63) between the vertices at $\boldsymbol{G} \neq 0$, sandwiched between factors $\bar{v}^{1/2}$, and $g_{\mathrm{deg}}\mathbf{M}(\mathbf{1} - g_{\mathrm{deg}}\mathbf{M})^{-1}$, with the same factors, is the geometric series of $\bar{R} = R^0 + R^0\bar{v}\bar{R}$ on the block $\boldsymbol{G}, \boldsymbol{G}' \neq 0$; the head $\bar{R}_{\mathbf{00}}$ follows from the same equation with the row $\boldsymbol{G} = 0$ of R^0, which is Eq. (4.68). The random-phase approximation with local fields is thus the Bethe–Salpeter equation without the attraction, as Sect. 4.6 anticipated.

In practice the equation is set up from the same states as the random-phase response, with the screened interaction $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p})$ computed beforehand for every difference $\boldsymbol{p}$ of two grid points. The dimension of the equation is the number of transitions, $N_v N_c N_k$, and the $\boldsymbol{k}$ grid sets the cost: the kernel has $(N_v N_c N_k)^2$ elements, and a converged spectrum of silicon needs several thousand points in the zone. The spectrum then comes from an iterative (Lanczos–Haydock) solver, which needs products of H^{exc} with a vector and nothing else, and a full diagonalization is needed when individual envelopes are wanted.

4.9.2 Wannier–Mott excitons

When the attraction is weak, the pair it binds is large, and a large pair samples just the bands near their extrema and the screening at long wavelengths. Eq. (4.104) then reduces to the hydrogen atom, a result of Wannier and of Mott [53, 54]. Take a direct gap at $\boldsymbol{k} = 0$, as in GaAs, with one isotropic conduction band of mass m^*, one valence band of hole mass m_{h}, and $\boldsymbol{q} \to 0$. Near the extrema the transition energy is

$$\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} = E_{\mathrm{g}} + \frac{k^2}{2m^*} + \frac{k^2}{2m_{\mathrm{h}}} = E_{\mathrm{g}} + \frac{k^2}{2\mu_x}, \qquad \frac{1}{\mu_x} = \frac{1}{m^*} + \frac{1}{m_{\mathrm{h}}},$$

with μ_x the reduced mass of the exciton. Suppose the envelope $A_{\boldsymbol{k}}$ is confined to $|\boldsymbol{k}|$ small against the zone; we check the assumption at the end. Then $\boldsymbol{p} = \boldsymbol{k} - \boldsymbol{k}'$ is small too, and in the direct kernel only $\boldsymbol{G} = \boldsymbol{G}' = 0$ survives: the terms with $\boldsymbol{G} \neq 0$ carry vertices between nearly equal states at momentum transfer $|\boldsymbol{p} + \boldsymbol{G}| \approx |\boldsymbol{G}|$, which are small, and a Coulomb factor that is finite instead of singular. The two vertices at $\boldsymbol{G} = 0$ are overlaps of the periodic parts of the same band at nearby $\boldsymbol{k}$, $\rho_{cc}(\boldsymbol{k}, \boldsymbol{p}) \simeq \rho_{vv}(\boldsymbol{k}, \boldsymbol{p}) \simeq 1$ (Eq. (3.34)), and the screened interaction at small $\boldsymbol{p}$ is the bare one divided by ε_∞. The direct kernel becomes

$$K^{\mathrm{d}}_{\boldsymbol{k}\boldsymbol{k}'} \simeq \frac{1}{\mathscr{V}} \frac{e^2}{\varepsilon_\infty |\boldsymbol{k} - \boldsymbol{k}'|^2}.$$

The exchange kernel, without its $\boldsymbol{G} = 0$ term, is however short-ranged in the relative coordinate: it acts only when the electron and the hole are in the same cell. For a pair spread over many cells its effect is smaller than that of K^{d} by the probability of finding the electron and the hole in one cell, and we drop it. The exchange kernel splits the singlet from the triplet by a small amount, the electron–hole exchange splitting. Eq. (4.104) now reads

$$\Big(E_{\mathrm{g}} + \frac{k^2}{2\mu_x}\Big) A_{\boldsymbol{k}} - \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}'} \frac{e^2}{\varepsilon_\infty |\boldsymbol{k} - \boldsymbol{k}'|^2} A_{\boldsymbol{k}'} = E\, A_{\boldsymbol{k}}.$$

Define the envelope function $F(\boldsymbol{r}) = \mathscr{V}^{-1/2} \sum_{\boldsymbol{k}} A_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, the Fourier transform of the envelope in the relative coordinate $\boldsymbol{r}$ of the electron and the hole; by Parseval, $\int \mathrm{d}^3 r\, |F|^2 = \sum_{\boldsymbol{k}} |A_{\boldsymbol{k}}|^2 = 1$. Multiplying by $\mathscr{V}^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and summing over $\boldsymbol{k}$ turns k^2 into $-\nabla^2$, and the kernel, a convolution

in $\boldsymbol{k}$, into a product in $\boldsymbol{r}$ with the transform of $e^2/(\varepsilon_\infty k^2)$, which is $e^2/(4\pi\varepsilon_\infty r)$ by Eq. (A.3). The result is a Schrödinger equation for the relative motion,

$$\Big[-\frac{\nabla^2}{2\mu_x} - \frac{\alpha}{\varepsilon_\infty r}\Big] F(\boldsymbol{r}) = (E - E_\mathrm{g})\, F(\boldsymbol{r}), \tag{4.106}$$

the hydrogen atom with $m_e \to \mu_x$ and $\alpha \to \alpha/\varepsilon_\infty$. Its bound states follow from those of hydrogen by the same substitution.

Wannier–Mott excitons

Below the gap the pair has the levels $E_n = E_\mathrm{g} - E_x^b/n^2$, $n = 1, 2, \ldots$, with the binding energy and radius

$$E_x^b = \frac{\mu_x \alpha^2}{2\varepsilon_\infty^2} = \frac{\mu_x}{m_e}\,\frac{13.61\,\mathrm{eV}}{\varepsilon_\infty^2}, \qquad a_x = \frac{\varepsilon_\infty}{\alpha\mu_x} = \varepsilon_\infty\,\frac{m_e}{\mu_x}\,a_0, \tag{4.107}$$

and hydrogenic envelopes; the ground state is $F_{1s}(\boldsymbol{r}) = \mathrm{e}^{-r/a_x}/(\pi a_x^3)^{1/2}$, and $|F_{ns}(0)|^2 = 1/(\pi a_x^3 n^3)$. Above E_g the unbound solutions form the continuum of scattering states.

The approximation holds when the exciton is large compared with the lattice constant, $a_x \gg a$. Then the envelope occupies a region of size $1/a_x$ around the band extremum, where the bands are parabolic and the vertices close to one, and the electron and the hole are far enough apart for the macroscopic ε_∞ to screen them. Static screening needs a second condition: the internal motion of the pair, at frequencies $\sim E_x^b$, must be slow against the response that screens it. For the electrons, which respond at energies of the gap and above, this condition holds whenever $E_x^b \ll E_\mathrm{g}$. The lattice of a polar crystal also screens, with the ions responding at the optical-phonon energies. If E_x^b is small against these, the ions follow the pair, and the static constant ε_0 with the lattice included (Sect. 5.4) takes the place of ε_∞; if it is large, the ions cannot follow and ε_∞ is right. When E_x^b is comparable with the optical-phonon energies, the interaction is not of the form $1/r$.

In silicon the pair is indirect. Its lowest state has the electron in one of the conduction valleys near $\boldsymbol{k}_\mathrm{c}$ and the hole at the valence maximum at Γ, and its total crystal momentum is $\boldsymbol{k}_\mathrm{c}$ (Sect. 3.3). The relative motion is still governed by Eq. (4.106), now with the anisotropic valley mass and the degenerate, warped valence band. A spherical estimate brackets the binding energy. The kinetic energy of an s state averaged over directions involves $(1/3)(1/m_\parallel^* + 2/m_\perp^*)$, an electron mass of $0.26\,m_e$ from the masses of Table 3.1; with the heavy-hole mass between its values along [100] and [111], 0.28 and $0.72\,m_e$, Eq. (4.107) with $\varepsilon_\infty = 11.7$ gives 13 to 19 meV. The measured binding is 14.7 meV [50]. Read through Eq. (4.107), it corresponds to $\mu_x = 0.148\,m_e$ and to a radius $a_x = 4.2$ nm, nearly eight lattice constants, and the large-exciton picture holds. In crystal momentum the envelope has a width $1/a_x = 0.024\,\text{Å}^{-1}$, about one percent of the zone, $|\boldsymbol{G}_{111}| = 2.0\,\text{Å}^{-1}$. A calculation of this state with Eq. (4.104) on a uniform grid would need hundreds of points per direction, and bound excitons near the gap are therefore computed with grids refined around the band extrema, or in the effective-mass limit itself. Since silicon is not polar, $\varepsilon_0 = \varepsilon_\infty$ and the question of the lattice does not arise.

Worked example 4.15: The Wannier–Mott exciton of GaAs

The conduction band of GaAs at Γ has $m^* = 0.067\,m_e$, and the valence maximum is the fourfold level of heavy and light holes (Table 3.1). For a hydrogenic estimate the hole needs one mass. The two hole energies of Eq. (3.56) at the same $\boldsymbol{\kappa}$ differ by the sign of the square root, so their mean is $\gamma_1^\mathrm{Lu}\kappa^2/2m_e$ in every direction: averaged over the two

branches, the hole moves with the inverse mass $\gamma_1^{\mathrm{Lu}}/m_e$. With $\gamma_1^{\mathrm{Lu}} = 6.98$,

$$\frac{m_e}{\mu_x} = \frac{1}{0.067} + 6.98 = 21.9, \qquad \mu_x = 0.0457\, m_e.$$

With the measured electronic constant $\varepsilon_\infty = 10.88$ [55], Eq. (4.107) gives

$$E_x^b = 0.0457 \times \frac{13.61\,\mathrm{eV}}{10.88^2} = 5.25\,\mathrm{meV}, \qquad a_x = 10.88 \times \frac{0.529\,\text{Å}}{0.0457} = 12.6\,\mathrm{nm},$$

22 lattice constants, which keeps the envelope well inside the parabolic region. The measured binding energy is 4.2 meV [49], and the estimate with ε_∞ overbinds by a quarter. The reason is the lattice. The optical phonons of GaAs lie near 35 meV [56, Fig. 2], eight times the binding energy, so the ions follow the slow relative motion of the pair and add their own screening. With the static constant $\varepsilon_0 = 12.9$ [56, Table I], the binding falls to $0.0457 \times 13.61\,\mathrm{eV}/12.9^2 = 3.74\,\mathrm{meV}$ and the radius grows to 14.9 nm. Most of the remaining difference from the measurement, 0.5 meV, comes from the part of the hole kinetic energy that the average over the two branches throws away. That part mixes heavy and light holes, has no first-order effect on the 1s state by symmetry, and lowers its energy in second order [57]. Either way the binding is a few meV, small against the gap of 1.52 eV, and the exciton is a line just below the absorption edge. Sect. 5.4 derives ε_0 and the lattice contribution to the screening.

At finite momentum transfer the pair moves as a whole. Take the transitions from $(v, \boldsymbol{k})$ to $(c, \boldsymbol{k}+\boldsymbol{q})$ with a conduction minimum at $\boldsymbol{k}_\mathrm{c}$ (zero for GaAs) and $\boldsymbol{Q} = \boldsymbol{q} - \boldsymbol{k}_\mathrm{c}$, small. The transition energy is $E_\mathrm{g} + (\boldsymbol{k}+\boldsymbol{Q})^2/2m^* + k^2/2m_\mathrm{h}$. With $\boldsymbol{k} = \boldsymbol{\kappa} - m_\mathrm{h}\boldsymbol{Q}/(m^* + m_\mathrm{h})$, the terms linear in $\boldsymbol{\kappa}$ cancel and the transition energy becomes $E_\mathrm{g} + Q^2/[2(m^* + m_\mathrm{h})] + \kappa^2/2\mu_x$. Since the direct kernel depends only on $\boldsymbol{k} - \boldsymbol{k}' = \boldsymbol{\kappa} - \boldsymbol{\kappa}'$, the relative motion obeys Eq. (4.106) unchanged, and

$$E_n(\boldsymbol{q}) = E_\mathrm{g} - \frac{E_x^b}{n^2} + \frac{(\boldsymbol{q} - \boldsymbol{k}_\mathrm{c})^2}{2(m^* + m_\mathrm{h})}. \tag{4.108}$$

The exciton has the center-of-mass dispersion of a particle with the total mass of the pair, the same mass that governs the pair threshold $E_\mathrm{min}(\boldsymbol{q})$ near a valley in Eq. (3.58). The bound levels follow the threshold, E_x^b/n^2 below it. In silicon the lowest exciton lies at $\boldsymbol{q} \simeq \boldsymbol{k}_\mathrm{c}$, 14.7 meV below the indirect gap, at a crystal momentum that a photon cannot supply. Light reaches it only with a phonon (Sect. 6.5), while dark matter that transfers $\boldsymbol{q} \approx \boldsymbol{k}_\mathrm{c}$ can create it without a phonon.

4.9.3 Excitonic absorption and scattering

The eigenpairs of Eq. (4.104) give the macroscopic dielectric function directly. In the resonant block the resolvent of the pair is $(\omega - H^\mathrm{exc} + \mathrm{i}0^+)^{-1} = \sum_\lambda A^\lambda A^{\lambda\dagger}/(\omega - E_\lambda + \mathrm{i}0^+)$, and projecting it on the density vertices at $\boldsymbol{q}$, as in Eq. (D.47), gives the response $\bar{R}_{\boldsymbol{00}}$ that is proper with respect to the long-range Coulomb term. With Eq. (4.68), $\varepsilon_\mathrm{mac} = 1 - v_\mathrm{C}(q)\bar{R}_{\boldsymbol{00}}$,

$$\varepsilon_\mathrm{mac}(\boldsymbol{q}, \omega) = 1 - \frac{g_\mathrm{deg} e^2}{V q^2} \sum_\lambda \Big| \sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\, \rho_{cv}(\boldsymbol{k}, \boldsymbol{q})^* \Big|^2 \left[\frac{1}{\omega - E_\lambda + \mathrm{i}0^+} - \frac{1}{\omega + E_\lambda + \mathrm{i}0^+} \right], \tag{4.109}$$

where the antiresonant term, with its poles at $-E_\lambda$, is the mirror image that the Tamm–Dancoff approximation keeps. The imaginary part of ε_mac at $\omega > 0$ and its static limit

are

$$\begin{aligned}\varepsilon_2(\boldsymbol{q},\omega) &= \frac{\pi g_{\mathrm{deg}} e^2}{V q^2} \sum_\lambda \Big| \sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^* \Big|^2 \delta(\omega - E_\lambda), \\ \varepsilon_{\mathrm{mac}}(\boldsymbol{q},0) &= 1 + \frac{2 g_{\mathrm{deg}} e^2}{V q^2} \sum_\lambda \frac{\left| \sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}} \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^* \right|^2}{E_\lambda}.\end{aligned} \tag{4.110}$$

At $\boldsymbol{q} \to 0$ the vertex is the optical one, $\rho_{cv} = \boldsymbol{q} \cdot \boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ (Eq. (3.43)), the factor q^2 cancels, and the static limit is the constant ε_∞. Without kernels each λ is a single transition, and the two reduce to Eqs. (4.65) and (4.66).

The new object is the exciton oscillator strength, the squared modulus of a coherent sum. Independent transitions contribute $|\rho_{cv}|^2$ each, whatever their phases. An exciton, however, contributes the square of the sum of the amplitudes of the transitions it is made of, weighted by A^λ. When the attraction lowers a state, the transitions in it add in phase, the low-lying excitons collect the oscillator strength of many transitions, and the weight moves to lower energy. Through the static limit in Eq. (4.110), with E_λ in the denominator, the downward move of weight also raises ε_∞. The f-sum rule of Sect. 4.3 fixes the total weight of the exact response, so the weight gained at low energies is lost higher up.

For a Wannier–Mott exciton the coherent sum has a closed form. The vertex varies little across the region of size $1/a_x$ that the envelope occupies, and with $\sum_{\boldsymbol{k}} A^\lambda_{\boldsymbol{k}} = \mathscr{V}^{1/2} F_\lambda(0)$, from the definition of F,

$$\Big| \sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^* \Big|^2 \simeq \mathscr{V}\, |F_\lambda(0)|^2\, |\rho_{cv}(\boldsymbol{k}_0,\boldsymbol{q})|^2, \tag{4.111}$$

with $\boldsymbol{k}_0$ the valence extremum. Only s states, whose envelope does not vanish where the electron and the hole coincide, are excited. Their strengths fall as $|F_{ns}(0)|^2 = 1/(\pi a_x^3 n^3)$, and since the levels crowd toward the gap as n^{-2}, the weight per unit energy just below E_{g} joins smoothly onto that of the continuum above it. The continuum itself is enhanced over the independent-particle one near the edge, since the attraction raises the probability of finding the electron at the hole. Both results are Elliott's [58]. Compared with independent particles, whose absorption at a direct edge rises from zero as $(\omega - E_{\mathrm{g}})^{1/2}$, the excitonic edge starts with a series of lines and a finite step.

Fig. 4.17 shows what the attraction does to silicon at $\boldsymbol{q} \to 0$, in a model calculation with the empirical pseudopotential states of Sect. 3.3. The bands were fitted to the measured positions of the absorption peaks, and those positions already contain the downward shift that the electron–hole attraction produces. Adding the attraction on top of the fitted bands counts that shift a second time, and every peak of the model lands too low. The figure therefore tests what the attraction does to the shape and the weight of the spectrum, not where the peaks lie. All three spectra come from one Hamiltonian, Eq. (4.104), on the same transitions: without kernels, with $g_{\mathrm{deg}} K^{\mathrm{x}}$ alone, and with $g_{\mathrm{deg}} K^{\mathrm{x}} - K^{\mathrm{d}}$. The screened interaction in K^{d} is the model's own random-phase one with local fields. On this grid the diagonal elements of K^{d} average 81 meV, a measure of the attraction between an electron and a hole spread over the crystal. The average depends on the grid, since the term at $\boldsymbol{p} \to 0$ grows with the volume of the zone per $\boldsymbol{k}$ point. The full random-phase result with local fields, computed from Eq. (4.68) on the same states without the Tamm–Dancoff approximation, differs from the spectrum with K^{x} alone by 3.4 percent of the peak, which is the Tamm–Dancoff error here.

The attraction restores the lower peak. At 3.42 eV independent particles give 15 and local fields 13; with the direct kernel the spectrum has a peak of height 34 at 3.36 eV, against the measured 36 at 3.42 eV. This peak, however, is not a bound state. The vertical transitions of silicon start near 3.1 eV, and the strength comes from many transitions along a region of

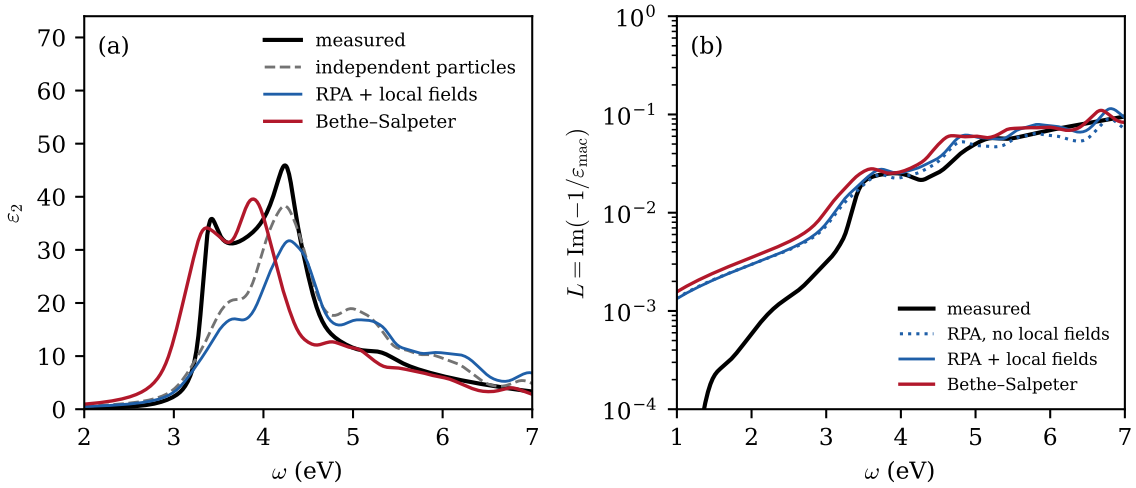


Figure 4.17: The electron–hole attraction in the optical spectrum of crystalline silicon at $\boldsymbol{q} \to 0$, in a model calculation. Band energies and states come from the local empirical pseudopotential of Cohen and Bergstresser [26], fitted to measured transition energies, as in Fig. 4.11. All curves come from the transition-basis Hamiltonian Eq. (4.104) in the Tamm–Dancoff approximation. The basis has 4 valence and 3 conduction bands on a shifted 12^3 grid of $\boldsymbol{k}$ points over the full zone (20,736 transitions); both kernels use the 59 shortest reciprocal vectors, and the spectra carry a Lorentzian broadening of 0.15 eV. The direct kernel uses the diagonal of the model's static inverse dielectric matrix in the random-phase approximation with local fields, computed with 26 conduction bands; spectra by the Lanczos–Haydock method. Black: measured at 300 K [18]. (a) ε_2: independent particles (dashed gray), the random-phase approximation with local fields, that is, exchange kernel only (blue), and the Bethe–Salpeter equation with both kernels (red). The attraction turns the shoulder near 3.4 eV into a peak; the whole spectrum shifts down because the fitted bands already contain part of the excitonic shift. (b) Loss function L on a logarithmic scale for the same three cases (dotted blue: no local fields, which has the independent-particle ε_2). Below about 3 eV the model curves are the Lorentzian tails of transitions that start at 3.13 eV on this grid, and lie above the measured L, which has no such tails.

the zone where conduction and valence bands run nearly parallel, and the attraction mixes these transitions coherently. The upper peak moves down to 3.88 eV; the 0.36 eV by which it falls below the measured 4.24 eV is the double counting described above. First-principles calculations avoid it by starting from quasiparticle energies of a GW calculation, which contain no excitonic shift. The Bethe–Salpeter spectrum of silicon computed that way by Albrecht and coworkers places both main peaks at the measured energies with the measured ratio of their heights, and overestimates their absolute heights slightly [25]. In the model, the static constant rises by 17 percent over the random-phase value with local fields in the same basis of bands. Part of that rise is again the double counting, since ε_∞ weights each exciton by $1/E_\lambda$, and the 17 percent rise is an upper estimate of what the attraction adds to the local-field value of Worked example 4.11.

For dark matter the loss function in panel (b) is the relevant quantity. At 3 eV the attraction raises L by 36 percent over the random-phase value with local fields; from 4 to 6 eV the two agree within 3.5 percent. The effect is concentrated in the first eV of the absorption, exactly where a slow particle deposits its energy.

At finite $\boldsymbol{q}$ the dark-matter rate needs the structure factor rather than ε_2. At $T = 0$ and $\omega > 0$, Eq. (4.45) gives $S = (2q^2/e^2)L$, with $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{mac}}) = \varepsilon_2/|\varepsilon_{\mathrm{mac}}|^2$, and Eq. (4.110) then yields the exciton contribution

$$S(\boldsymbol{q},\omega) = \frac{2\pi g_{\mathrm{deg}}}{\mathcal{V}} \sum_{\lambda} \frac{\left|\sum_{cv\boldsymbol{k}} A^{\lambda}_{cv\boldsymbol{k}}(\boldsymbol{q})\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\right|^2}{|\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)|^2}\, \delta(\omega - E_\lambda(\boldsymbol{q})). \tag{4.112}$$

Each exciton is a line at $E_\lambda(\boldsymbol{q})$ whose weight is its oscillator strength, screened by the factor $1/|\varepsilon_{\mathrm{mac}}|^2$ of Sect. 4.4. The exact statement is $S = (2q^2/e^2)\varepsilon_2/|\varepsilon_{\mathrm{mac}}|^2$, and the line form holds when the line's own contribution to $\varepsilon_{\mathrm{mac}}$ near its energy is small against the background, whose value then goes into the denominator. A line strong enough to change $\varepsilon_{\mathrm{mac}}$ near its own energy is moved up by the long-range exchange energy, since the pole of L lies at the zero of $\varepsilon_{\mathrm{mac}}$. This upward shift is the electronic analog of the splitting between the longitudinal optical (LO) and transverse optical (TO) phonons of a polar crystal, in which the macroscopic field of a longitudinal lattice vibration raises its frequency above that of the transverse one (Sect. 5.4). With the $\boldsymbol{G} = 0$ term kept in the kernel, the eigenvalues are these shifted energies and the weights are not divided by $|\varepsilon_{\mathrm{mac}}|^2$ (Eq. (D.48)). At the momentum transfers of a few keV that dark matter typically deposits, the Coulomb factor e^2/q^2 is small and the shift is negligible. Without kernels, Eq. (4.112) is the independent-particle structure factor Eq. (4.21) screened by $1/|\varepsilon_{\mathrm{mac}}|^2$, the random-phase result for the electrons.

For a Wannier–Mott exciton, Eq. (4.111) and Eq. (4.108) make the lines explicit,

$$S(\boldsymbol{q},\omega) = 2\pi g_{\mathrm{deg}}\, \frac{|\rho_{cv}(\boldsymbol{k}_0,\boldsymbol{q})|^2}{|\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)|^2} \sum_n \frac{1}{\pi a_x^3 n^3}\, \delta(\omega - E_n(\boldsymbol{q})),$$

a weight per unit volume set by the probability $|F_{ns}(0)|^2$ of finding the electron at the hole, times the squared vertex between the two band extrema. In silicon the lowest line follows the indirect threshold of Sect. 3.3 at $\boldsymbol{q} \approx \boldsymbol{k}_{\mathrm{c}}$ and lies 14.7 meV below it. As a shift of the threshold the 14.7 meV is 1.3 percent of the gap, small for the threshold mass of Sect. 3.3. As a redistribution of weight, however, it matters more: the edge of the zero-phonon response starts with a line, and the continuum above it is enhanced. Light dark matter samples exactly this region, and Sect. 10.5 computes the excitonic loss function at finite $\boldsymbol{q}$.

4.9.4 Levels of approximation

Each approximation contains the previous one as a limit: $K^{\mathrm{d}} \to 0$ gives local fields, $\bar{v} \to 0$ the random-phase head, and the head alone at fixed energies the independent particles. The

Bethe–Salpeter equation in this form leaves out the dynamical screening of the interaction, the coupling of pair creation and annihilation, and every process with a phonon, which Chap. 6 adds.

In silicon the approximations of Figs. 4.11 and 4.17 differ in characteristic ways. The static constant falls with the local fields and rises part of the way back with the adiabatic kernel (Worked example 4.11), and in the model the attraction raises it by 17 percent above the local-field value of 10.22, to about 11.9. The screening factor $1/\varepsilon_\infty^2$ at small q thus varies by up to a third across the hierarchy. At the lower absorption peak the spread is a factor of more than two, 13 to 34, and the loss function at 3 eV changes by a third between the adiabatic kernel and the Bethe–Salpeter equation alone. Above about 4 eV, which holds most of the f-sum, the approximations agree within the accuracy of the bands. As in Sect. 4.6, the spread is an estimate, not a bound. Light dark matter samples the first eV above the gap; for it the attraction is the leading correction, and the Bethe–Salpeter result is the reference against which the cheaper approximations are judged.

Summary

A weak field that couples to an operator O_b of a target in equilibrium changes the average of any other operator O_a through the retarded commutator, the Kubo formula $R_{ab}(\boldsymbol{q},t) = -(\mathrm{i}/\mathcal{V})\Theta(t)\langle[O_a(\boldsymbol{q},t)^\dagger, O_b(\boldsymbol{q},0)]\rangle$ (Sect. 4.1). The formula holds for any Hamiltonian and at any temperature, and causality makes R_{ab} analytic in the upper half of the frequency plane.

The rate of a dark-matter particle that couples through several operators is an integral over $\boldsymbol{q}$ of $\sum_{ab}\tilde{V}_a^*\tilde{V}_b S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}})$ (Sect. 4.2). The fluctuation–dissipation theorem, $(1-\mathrm{e}^{-\beta\omega})S = -2\,\mathrm{Im}\,R$ on the diagonal, ties the rate to the response, and S_{ab} is a positive semidefinite matrix at every $(\boldsymbol{q},\omega)$, so that every rate is positive and every interference is bounded.

The Kramers–Kronig relations rebuild the real part of a response from the imaginary one, and the moments of S are equal-time commutators (Sect. 4.3). The first moment gives the f-sum rule, $\int \omega S$ fixed by the density and the mass whatever the interactions, which defines the plasma frequency, 16.6 eV for the valence electrons of silicon.

The long-range Coulomb interaction is separated by the dielectric function (Sect. 4.4), and the fluctuation–dissipation theorem writes the structure factor as the loss function, $S = (2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$ with $L = \mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$, so that measured optical and energy-loss spectra enter the rate directly. Silicon has $\varepsilon_\infty = 11.7$ and a loss peak of height 4.5 at 16.4 eV. The f-sum and inverse-moment rules fix the main loss peak, but not the low-energy side of L that slow particles, and dark matter, sample.

The homogeneous electron gas is the one system whose response is known in closed form (Sect. 4.5): the Lindhard function, screened in the random-phase approximation. Its plasmon meets the pair continuum at $q_\mathrm{c} = 2.60$ keV, while a dark-matter particle reaches the plasmon energy only at 6.26 keV, so the plasmon is kinematically out of reach.

In a crystal the bubble becomes the Adler–Wiser matrix of Bloch states (Sect. 4.6). The full matrix adds local fields, and time-dependent density-functional theory adds an exchange–correlation kernel. In silicon the local fields lower ε_∞ by about a tenth, and none of these approximations gives the lower absorption peak.

Green's functions describe one added electron or hole (Sect. 4.7). Their poles near the real axis are quasiparticles with energy E_qp, residue z_qp and width Γ_qp, and the quasiparticle correction to Kohn–Sham energies repairs the gap, 0.602 eV with PBE at its own lattice constant against the measured 1.17 eV in silicon. Feynman diagrams organize the perturbation series (Sect. 4.8): the random-phase approximation is the chain of bubbles, and the screened interaction $W^\mathrm{scr} = \varepsilon^{-1}v_\mathrm{C}$ replaces the bare one in the GW self-energy.

Summing the ladder diagrams with a statically screened interaction gives the Bethe–Salpeter equation (Sect. 4.9). Near a band extremum it reduces to hydrogen with the reduced mass μ_x and the coupling $\alpha/\varepsilon_\infty$; in silicon the measured binding, 14.7 meV, fixes $\mu_x = 0.15\, m_e$ and with it a radius of 4.2 nm. In the model calculation for silicon the attraction restores the lower absorption peak, raises ε_∞ toward the measured value, and raises the loss function at 3 eV by a third; the spread across the completed hierarchy is the many-body uncertainty of a computed loss function.

Throughout, the nuclei were held fixed. They move, and their motion adds a second set of excitations, phonons, with energies of tens of meV, a second contribution to screening in polar crystals and a coupling between electrons and the lattice. Chap. 5 builds the lattice dynamics on which all three rest.

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Chapter 5

Lattice dynamics

Dark matter lighter than 0.33 MeV cannot carry an electron of silicon across the 1.17 eV gap (Sect. 3.3), yet the crystal still has excitations that this light dark matter can reach. The nuclei vibrate at energies up to 64 meV. By the bound $2\omega/v_{\mathrm{max}}^2$ of Sect. 1.3, a vibrational quantum of 64 meV is kinematically allowed for dark-matter masses down to about 18 keV, and the acoustic vibrations reach lower masses still. Chaps. 3 and 4 held the nuclei fixed and computed the response of the electrons alone, so the dynamic structure factor built there contains no vibrations of the nuclei. Letting the nuclei move raises questions that the static lattice could not ask. What potential do the nuclei move in, if the electrons that bind them also respond to their motion? How far does each nucleus stray from its site, even at a millikelvin, where zero-point motion is all there is? And what does a dark-matter particle that pushes on one nucleus set in motion in the whole crystal? The answers follow from the mass ratio $m_e/M \approx 2 \times 10^{-5}$, which separates the slow nuclei from the fast electrons and makes the vibrations of a crystal a nearly harmonic problem in a potential that density-functional theory computes. From the mass ratio we will build, in the steps of Fig. 5.1, the phonons of silicon and of polar crystals, with their quanta, correlations and lifetimes; every nuclear-recoil and phonon rate of the later chapters takes these quantities as input.

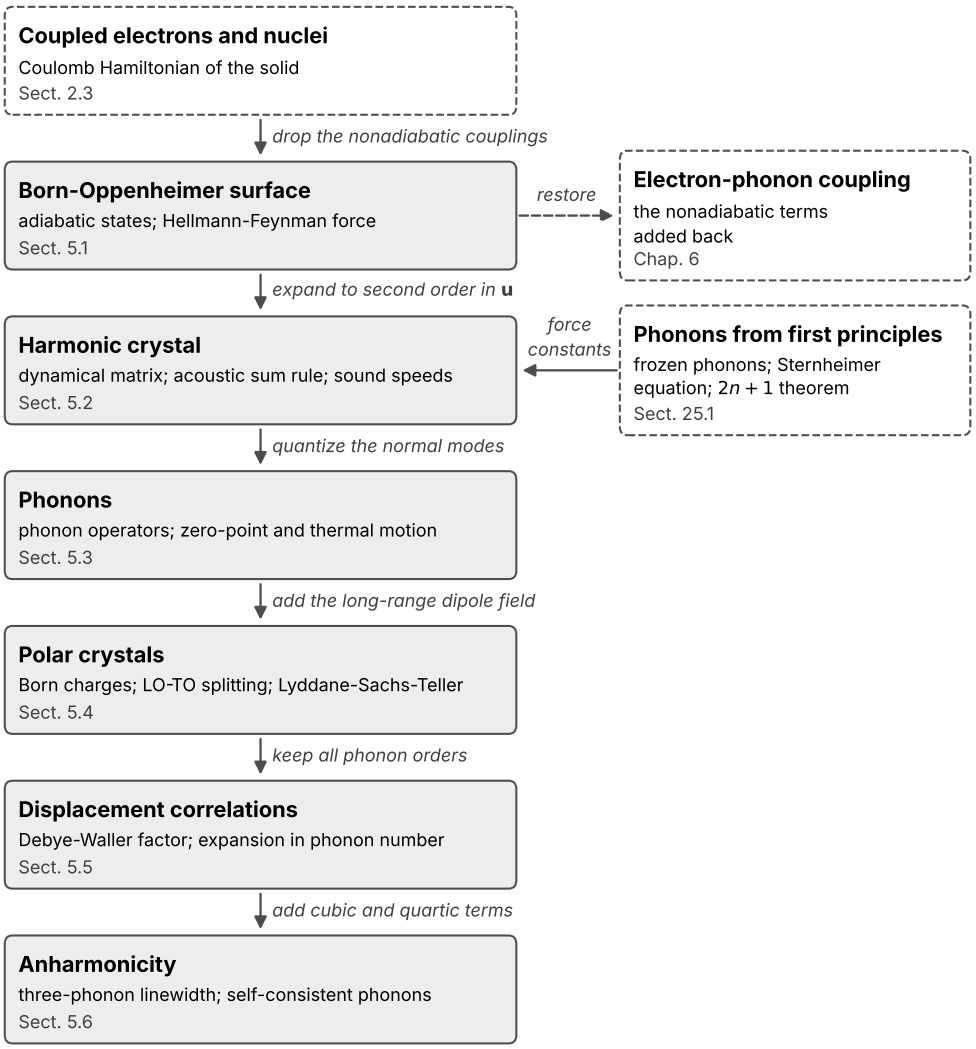


Figure 5.1: From coupled electrons and nuclei to the anharmonic crystal. Dropping the nonadiabatic couplings gives the Born–Oppenheimer surface (Sect. 5.1), and Chap. 6 restores the dropped couplings; expanding the surface to second order in the displacements gives the harmonic crystal (Sect. 5.2); quantizing its normal modes gives phonons (Sect. 5.3). The long-range dipole field adds the Born charges and LO–TO splitting of polar crystals (Sect. 5.4); keeping all phonon orders gives the displacement correlations, the Debye–Waller factor and the multiphonon expansion (Sect. 5.5); cubic and quartic terms give linewidths and self-consistent phonons (Sect. 5.6). Dashed box: the force constants computed as a density response (Chap. 25).

5.1 The Born–Oppenheimer approximation

The optical phonon of silicon at the zone center vibrates at 15.5 THz [1], an energy of 64.1 meV and a period of 64.5 fs. An electron excited across the gap of 1.17 eV evolves on the time $2\pi/E_{\rm g} = 3.5$ fs, eighteen times faster. During one swing of the nuclei the electrons have time to settle, many times over, into the lowest state compatible with where the nuclei happen to be. The crystal then behaves as the static lattice of Sect. 3.1 with the nuclei now allowed to move, slowly. This separation of the slow nuclei from the fast electrons is due to Born and Oppenheimer [2], in the form that Born and Huang gave it for crystals [3, App. VIII]; the derivations of this section are written out in Appendix D.11.

5.1.1 Adiabatic states

We start from the Coulomb Hamiltonian of electrons and point nuclei, Eq. (2.48). In a crystal it is natural to label the nucleus I of Sect. 2.3 by its cell and its place in the cell. In the cell at the Bravais vector $\boldsymbol{R}$, the nucleus of species κ sits at

$$\boldsymbol{X}_I = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}, \tag{5.1}$$

where $\boldsymbol{\tau}_\kappa$ is its basis position in the cell (Sect. 3.1) and $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ its displacement. We write u for the set of all displacements and $\nabla_{\boldsymbol{R}\kappa} = \partial/\partial \boldsymbol{u}_{\boldsymbol{R}\kappa}$. The equilibrium positions, about which u is measured, are fixed below. Of the terms of Eq. (2.48), only the kinetic energy of the nuclei contains $\nabla_{\boldsymbol{R}\kappa}$. We separate it from the rest,

$$H = -\sum_{\boldsymbol{R}\kappa} \frac{\nabla^2_{\boldsymbol{R}\kappa}}{2M_\kappa} + H_{\rm el}(u), \qquad H_{\rm el}(u) = T_e + V_{ee} + V_{eN}(u) + V_{NN}(u), \tag{5.2}$$

with T_e the kinetic energy of the electrons, the first term of Eq. (2.48), and V_{ee}, V_{eN} and V_{NN} the electron–electron, electron–nucleus and nucleus–nucleus terms of the bracket of Eq. (2.48). At fixed u, $H_{\rm el}(u)$ acts on the electron coordinates alone, which we collect in r, and the displacements u enter $H_{\rm el}(u)$ as a set of parameters. The eigenstates of $H_{\rm el}(u)$ are the *adiabatic states*,

$$H_{\rm el}(u)\, \Psi_i(r;u) = E_i(u)\, \Psi_i(r;u), \tag{5.3}$$

orthonormal at every u. Each is a state of all the electrons with the nuclei held at the positions u. In the absence of a magnetic field $H_{\rm el}$ is real and the Ψ_i can be chosen real, which we assume.

The adiabatic states are complete in r at each u, and an exact state of electrons and nuclei can be expanded in them with u-dependent coefficients,

$$\Psi(r,u) = \sum_i \Psi_i^{\rm nuc}(u)\, \Psi_i(r;u).$$

The coefficients $\Psi_i^{\rm nuc}$ are nuclear wavefunctions, one for each electronic state, and so far nothing has been approximated. Inserting the expansion into $H\Psi = E\Psi$ and projecting on Ψ_j at fixed u (Appendix D.11) leads to coupled equations for the nuclear wavefunctions,

$$\Big[-\sum_{\boldsymbol{R}\kappa} \frac{\nabla^2_{\boldsymbol{R}\kappa}}{2M_\kappa} + E_j(u)\Big]\Psi_j^{\rm nuc} + \sum_i \mathcal{N}_{ji}\, \Psi_i^{\rm nuc} = E\, \Psi_j^{\rm nuc}, \tag{5.4}$$

in which the electronic states couple to each other only through the operators

$$\mathcal{N}_{ji} = -\sum_{\boldsymbol{R}\kappa} \frac{1}{2M_\kappa}\Big[2\,\langle \Psi_j | \nabla_{\boldsymbol{R}\kappa} \Psi_i\rangle \cdot \nabla_{\boldsymbol{R}\kappa} + \langle \Psi_j | \nabla^2_{\boldsymbol{R}\kappa} \Psi_i\rangle\Big]. \tag{5.5}$$

These are the *nonadiabatic couplings*: the first-derivative (or derivative) coupling, which acts on the nuclear wavefunction through $\nabla_{\boldsymbol{R}\kappa}$, and the second-derivative (or scalar) coupling. The brackets are integrals over r and functions of u. Both couplings exist because the electronic state changes as the nuclei move, and both carry the factor $1/M_\kappa$.

The *Born–Oppenheimer approximation* drops the nonadiabatic couplings. The electrons then stay in one adiabatic state (for an insulator at low temperature, the ground state), and the nuclei move in its energy:

$$\Big[-\sum_{\boldsymbol{R}\kappa}\frac{\nabla^2_{\boldsymbol{R}\kappa}}{2M_\kappa}+E^{\mathrm{ad}}(u)\Big]\Psi_0^{\mathrm{nuc}}=E\,\Psi_0^{\mathrm{nuc}},\qquad E^{\mathrm{ad}}(u)\equiv E_0(u),\qquad \Psi\simeq\Psi_0^{\mathrm{nuc}}(u)\,\Psi_0(r;u). \quad (5.6)$$

The ground-state energy of the electrons at fixed nuclei, with the repulsion of the nuclei included, has become the potential energy of the nuclei. It is the *potential-energy surface*, or Born–Oppenheimer energy surface, $E^{\mathrm{ad}}(u)$, a function of the $3n_{\mathrm{at}}N$ nuclear coordinates for a crystal of N cells, each with n_{at} nuclei. The physical picture is a slow nucleus pulling a cloud of electrons that rearranges instantly: the electrons follow the nuclei adiabatically, never get excited, and pass to the nuclei only the energy of their ground state. The rest of this chapter treats phonons, from their frequencies to their anharmonic decay, and every result it obtains is a statement about the shape of E^{ad} and about quantum mechanics on that surface. Born and Huang keep the diagonal term $\mathcal{N}_{00}$ as a small correction to E^{ad} and call the result the adiabatic approximation [3, App. VIII]. The field uses "adiabatic" and "Born–Oppenheimer" interchangeably, and so do we, since $\mathcal{N}_{00}$ is negligible (Appendix D.11). The off-diagonal couplings $\mathcal{N}_{j\neq0,0}$ are what let a moving nucleus excite the electrons; they are the subject of Sect. 6.1.

In a density-functional code, E^{ad} at a given set of nuclear positions is the Kohn–Sham total energy plus the repulsion of the nuclei (evaluated as an Ewald sum), and every code prints it as its standard output. Evaluating E^{ad} at displaced positions is the frozen-phonon method, and density-functional perturbation theory (Sect. 25.1) computes the derivatives of E^{ad} at equilibrium directly. The Born–Oppenheimer approximation enters these calculations at one point only: the electrons are kept in their ground state at every configuration of the nuclei [4, Sect. III A].

5.1.2 Equilibrium positions and clamped nuclei

The nuclei of a crystal at rest sit at a minimum of the energy surface. We choose the equilibrium positions $\boldsymbol{R}+\boldsymbol{\tau}_\kappa$ so that $u=0$ is that minimum, $\nabla_{\boldsymbol{R}\kappa}E^{\mathrm{ad}}=0$ for every nucleus. In silicon, each atom of the diamond structure sits at a site of tetrahedral symmetry, and the only vector that every operation of the tetrahedral group leaves unchanged is zero, so no atom feels a force at any lattice constant. The only free parameter is the lattice constant itself. Minimizing E^{ad} per cell with respect to the lattice constant yields 5.401 Å in the local-density approximation [5, Sect. 10.2] and 5.477 Å in the PBE calculation of Sect. 3.4, against the measured 5.431 Å (Sect. 3.1). The lattice constant at the minimum of E^{ad} is not exactly the lattice constant that a diffraction experiment measures, since the zero-point spread of the nuclei on an anharmonic surface shifts the mean lattice constant away from the minimum (Sect. 5.6).

Chaps. 3 and 4 went one step further than Eq. (5.6), to the *clamped-nuclei approximation*, or static lattice. They kept the electronic problem at the single configuration $u=0$, where $H_{\mathrm{el}}(0)$ has the full periodicity of the lattice, and gave the nuclei no motion. The target is then $H_{\mathrm{el}}(0)$, its states are $\Psi_i(r;0)$, and the nuclei enter only as the fixed periodic potential of the band structure (Sect. 3.2). The Born–Oppenheimer approximation lets the electronic state follow the nuclei as they move; the clamped-nuclei approximation does not let them move.

The difference between the two approximations is the electron–phonon coupling. Expanding the electronic Hamiltonian about the clamped one gives

$$H_{\rm el}(u) = H_{\rm el}(0) + \sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \cdot \nabla_{\boldsymbol{R}\kappa} V_{eN}\Big|_{u=0} + \big[V_{NN}(u) - V_{NN}(0)\big] + O(u^2), \tag{5.7}$$

where the linear term in V_{eN} is a one-body operator on the electrons times a displacement, the bare form of the electron–phonon Hamiltonian of Sect. 6.2 (the second-quantized form is Eq. (6.4)). The adiabatic states are the clamped-nuclei states dressed by the electron–phonon coupling to all orders in u, with the nuclei held still (Sect. 6.1). The clamped-nuclei approximation therefore has no phonons, the only excitations below the gap. It has no phonon-assisted transitions, which let a slow dark-matter particle reach the indirect gap (Sect. 3.3), and no renormalization of the electronic energies by the zero-point motion of the nuclei, which in silicon closes the gap by about 60 meV (Sect. 6.3). The dynamic structure factor of Chap. 4, computed at clamped nuclei, contains no phonons, no phonon-assisted transitions and no zero-point renormalization.

5.1.3 The Born–Oppenheimer parameter

A nucleus and an electron in the same crystal feel forces of the same size, since the Coulomb interaction does not depend on mass. In the same potential, however, a heavy particle spreads much less than a light one, because confining a particle costs a kinetic energy that scales as $1/M$; the nuclei sit far more sharply at their sites than the electrons at theirs. The zero-point spread of the nuclei and the energy of their vibrations are then set by a balance that contains the mass ratio m_e/M, and the question is which power of the mass ratio.

Born and Oppenheimer measured lengths in units of the Bohr radius $a_0 = 1/(\alpha m_e) = 0.529\,\text{Å}$ and energies in units of $\alpha^2 m_e = 27.21\,\text{eV}$, the scales of the electronic problem. In these units $H_{\rm el}$ has no small parameter. Interatomic distances are a few a_0, electronic excitation energies are a fraction of $\alpha^2 m_e$, and $E^{\rm ad}$ changes by an amount of order $\alpha^2 m_e$ when a nucleus moves by a_0. Expanded about the minimum, the surface reads, schematically, $E^{\rm ad} = E_0^{\rm ad} + \frac{1}{2}\Phi^{(2)}u^2 + \frac{1}{6}\Phi^{(3)}u^3 + \cdots$, with force constants $\Phi^{(n)} \sim \alpha^2 m_e / a_0^n$ (Sect. 5.2). The nuclear kinetic energy is the only term that carries the mass ratio,

$$\frac{\nabla^2}{2M} = \alpha^2 m_e \, \frac{m_e}{M} \, \frac{a_0^2 \nabla^2}{2}.$$

Let a nuclear wavefunction be spread over a length $\ell_{\rm zp}$. Its kinetic energy is then $\alpha^2 m_e (m_e/M)(a_0/\ell_{\rm zp})^2$, and its potential energy is $\alpha^2 m_e (\ell_{\rm zp}/a_0)^2$. The two energies balance at

$$\frac{\ell_{\rm zp}}{a_0} \sim \Big(\frac{m_e}{M}\Big)^{1/4} = \kappa_{\rm BO}, \qquad \omega_{\rm ph} \sim \alpha^2 m_e \, \kappa_{\rm BO}^2, \tag{5.8}$$

where the vibrational energy $\omega_{\rm ph}$ is the common size of the two terms. The mass ratio of silicon is 2×10^{-5}, but because of the fourth root the zero-point amplitude is a fraction $\kappa_{\rm BO} = 0.067$ of the electronic length, and a phonon a fraction $\kappa_{\rm BO}^2 = 4.4 \times 10^{-3}$ of the electronic energy.

With the scale of the displacements fixed, every other term of Eq. (5.4) is counted by the same substitution, $u \sim \kappa_{\rm BO} a_0$ and $\nabla \sim 1/(\kappa_{\rm BO} a_0)$ on a nuclear wavefunction, and Appendix D.11 carries the counting through term by term. The cubic term of the surface is $\sim \alpha^2 m_e \, \kappa_{\rm BO}^3$, smaller than a phonon energy by one power of $\kappa_{\rm BO}$, which is the anharmonic small parameter of Sect. 5.6. Counted the same way, the derivative coupling is $\sim \alpha^2 m_e \, \kappa_{\rm BO}^3$ and the scalar one $\sim \alpha^2 m_e \, \kappa_{\rm BO}^4$, and since the derivative coupling connects the ground state to electronic states $\sim \alpha^2 m_e$ above it, the vibrational levels move first at $\alpha^2 m_e \, \kappa_{\rm BO}^6$.

Born–Oppenheimer hierarchy

With $\kappa_{\mathrm{BO}} = (m_e/M)^{1/4}$, the energies of electrons and nuclei are ordered as

$$\underbrace{\alpha^2 m_e}_{\text{electrons}} \;:\; \underbrace{\alpha^2 m_e\,\kappa_{\mathrm{BO}}^2}_{\text{phonons}} \;:\; \underbrace{\alpha^2 m_e\,\kappa_{\mathrm{BO}}^4}_{\text{anharmonic shifts}} \;:\; \underbrace{\alpha^2 m_e\,\kappa_{\mathrm{BO}}^6}_{\text{nonadiabatic shifts}} \;,$$

and the zero-point amplitude of the nuclei is $\ell_{\mathrm{zp}} \sim \kappa_{\mathrm{BO}} a_0$. The nonadiabatic correction to a vibrational energy is of relative order $\kappa_{\mathrm{BO}}^4 = m_e/M$.

This hierarchy is the result of Born and Oppenheimer [2]. Appendix D.11 checks it on an exactly solvable model, one electron bound by a spring to a nucleus that is itself bound to a fixed point. In this model the exact squared vibrational frequency differs from the Born–Oppenheimer one by the relative amount m_e/M, and the frequency itself by $m_e/2M$: the electron is dragged along and adds its mass to the nucleus, and nothing else happens at that order.

The counting assumed that every electronic excitation costs an energy $\Delta E \sim \alpha^2 m_e$. When some excitations cost less, the derivative coupling grows. Its exact form, derived below as Eq. (5.11), has ΔE in the denominator, and the amplitude with which a nuclear vibration mixes an excited electronic state j into the ground state is

$$\frac{\omega_{\mathrm{ph}}}{\Delta E}\,\frac{\ell_{\mathrm{zp}} F_{j0}}{\Delta E}, \tag{5.9}$$

with F_{j0} the matrix element of the force between the two states (Appendix D.11). The factor $\omega_{\mathrm{ph}}/\Delta E$ compares a phonon with an electronic excitation, and $\ell_{\mathrm{zp}} F_{j0}/\Delta E$ compares the work done by the force over a zero-point amplitude with the same excitation. At $\Delta E \sim \alpha^2 m_e$ they are κ_{BO}^2 and κ_{BO}, and we recover the κ_{BO}^3 of the derivative coupling. In an insulator the smallest ΔE is the gap, and the parameter that controls the adiabatic approximation is $\omega_{\mathrm{ph}}/E_{\mathrm{g}}$ rather than κ_{BO}^2. In a metal there are electronic excitations at every energy, the expansion in adiabatic states fails near the Fermi surface, and only a weaker statement holds, Migdal's adiabatic theorem, with the small parameter $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ (Sect. 6.3). Two adiabatic levels that cross, as they can in a defect or during a violent recoil, make ΔE vanish, and at the crossing the adiabatic approximation fails outright.

5.1.4 The Hellmann–Feynman force

The force on a nucleus is minus the gradient of the energy surface, and it takes a simple form. Write $E^{\mathrm{ad}} = \langle\Psi_0|H_{\mathrm{el}}|\Psi_0\rangle$ and differentiate with respect to the displacement of one nucleus. The derivative falls on the two states and on the Hamiltonian. Since $H_{\mathrm{el}}\Psi_0 = E^{\mathrm{ad}}\Psi_0$, the two terms in which the derivative falls on a state add up to $E^{\mathrm{ad}}\,\nabla_{\boldsymbol{R}\kappa}\langle\Psi_0|\Psi_0\rangle$, which is zero because the state stays normalized. The remaining term gives the *Hellmann–Feynman theorem* [6, 7],

$$\boldsymbol{F}_{\boldsymbol{R}\kappa} = -\nabla_{\boldsymbol{R}\kappa} E^{\mathrm{ad}} = -\big\langle\Psi_0\big|\nabla_{\boldsymbol{R}\kappa} H_{\mathrm{el}}\big|\Psi_0\big\rangle = -\big\langle\Psi_0\big|\nabla_{\boldsymbol{R}\kappa}(V_{eN} + V_{NN})\big|\Psi_0\big\rangle, \tag{5.10}$$

since only V_{eN} and V_{NN} depend on the positions of the nuclei. The derivative of the wavefunction does not enter. The operator V_{eN} is a sum of one-body terms, so its expectation value is fixed by the ground-state electron density $n(\boldsymbol{r})$, and the force follows from classical electrostatics. With Z_κ the charge number of the nucleus at $\boldsymbol{X}_I$,

$$\boldsymbol{F}_{\boldsymbol{R}\kappa} = Z_\kappa\, e^2\Big[\int \mathrm{d}^3 r\; n(\boldsymbol{r})\,\frac{\boldsymbol{r}-\boldsymbol{X}_I}{4\pi|\boldsymbol{r}-\boldsymbol{X}_I|^3} - \sum_{J\neq I} Z_{\kappa(J)}\,\frac{\boldsymbol{X}_J-\boldsymbol{X}_I}{4\pi|\boldsymbol{X}_J-\boldsymbol{X}_I|^3}\Big].$$

Each nucleus is pulled toward the electron cloud and pushed away from the other nuclei, and quantum mechanics enters only through the shape of the cloud. This form of the force is the electrostatic theorem, as Feynman stated it for molecules [7]. At equilibrium the attraction to the cloud and the repulsion from the other nuclei cancel. When a nucleus is displaced, the cloud is distorted, and the force that restores the nucleus comes from that distortion; computing the distortion to first order in the displacement is a linear-response problem.

The same differentiation, projected on an excited state Ψ_j instead of Ψ_0, yields the derivative coupling of Eq. (5.5) (Appendix D.11):

$$\langle \Psi_j | \nabla_{\boldsymbol{R}_\kappa} \Psi_i \rangle = \frac{\langle \Psi_j | \nabla_{\boldsymbol{R}_\kappa} V_{eN} | \Psi_i \rangle}{E_i(u) - E_j(u)}, \qquad j \neq i. \tag{5.11}$$

The numerator is the matrix element of the force that the nucleus exerts on the electrons, the bare electron–phonon vertex, and the denominator is an electronic excitation energy. Through this denominator the gap of the insulator enters the estimate Eq. (5.9).

In a periodic configuration the forces repeat from cell to cell. Displacing every nucleus of species κ by the same $\boldsymbol{u}_\kappa$ leaves the crystal periodic, and the force on each of those nuclei is the same in every cell. We write this force as

$$\boldsymbol{F}_\kappa = -\frac{\partial E^{\mathrm{ad}}_{\mathrm{cell}}}{\partial \boldsymbol{u}_\kappa}, \tag{5.12}$$

with $E^{\mathrm{ad}}_{\mathrm{cell}}$ the energy surface per cell for that uniform displacement of the sublattice. A calculation in the primitive cell returns this zone-center force, and the response of this force to a macroscopic electric field defines the Born effective charge of a polar crystal (Sect. 5.4).

The Hellmann–Feynman theorem also holds for the Kohn–Sham energy, so forces are cheap to compute. The Kohn–Sham energy is minimal with respect to the orbitals at self-consistency, so, as in the derivation above, the change of the orbitals drops out of its first derivative. The force is then the electrostatic expression above with the self-consistent Kohn–Sham density, or, with pseudopotentials, the derivative of the pseudopotential weighted by the density and the orbitals. With a plane-wave basis, which does not move with the nuclei, there is no other term. A basis of atom-centered functions moves with the nuclei and adds a term to the force, the Pulay force [8]. A code computes the energy and every force at the cost of one self-consistent calculation.

Fig. 5.2(c) tests Eq. (5.10) on silicon. A PBE calculation at the measured lattice constant moves the two atoms of the cell in opposite directions along [111], so that the bond pointing along [111] changes its length by u in every cell: the zone-center optical displacement. For $|u| \leq 0.2\,\text{Å}$, the Hellmann–Feynman force on the coordinate u agrees with $-\mathrm{d}E^{\mathrm{ad}}_{\mathrm{cell}}/\mathrm{d}u$, obtained by differentiating the computed energies, to better than 0.1 %. The curvature of $E^{\mathrm{ad}}_{\mathrm{cell}}$ at the minimum is $\Phi_{\mathrm{opt}} = 13.6\,\text{eV Å}^{-2}$ per cell. Since the two atoms move in opposite directions, the coordinate u carries the reduced mass $M_\kappa/2$, and the optical frequency is $\omega_{\mathrm{opt}} = (2\Phi_{\mathrm{opt}}/M_\kappa)^{1/2} = 63.7\,\text{meV}$, against the measured 64.1 meV. The surface is visibly asymmetric: compressing the bond costs more energy than stretching it by the same amount. The cubic term of the expansion produces this asymmetry. The cubic term is allowed for the optical displacement along [111] and vanishes by symmetry along [100], and Worked example 5.1 checks its size against the counting.

Worked example 5.1: The small parameter of silicon

For natural silicon, $M = 26.16\,\text{GeV}$ and $m_e/M = 1.953 \times 10^{-5}$, so that

$$\kappa_{\mathrm{BO}} = 0.0665, \qquad \kappa^2_{\mathrm{BO}} = 4.42 \times 10^{-3}, \qquad \kappa^3_{\mathrm{BO}} = 2.94 \times 10^{-4}.$$

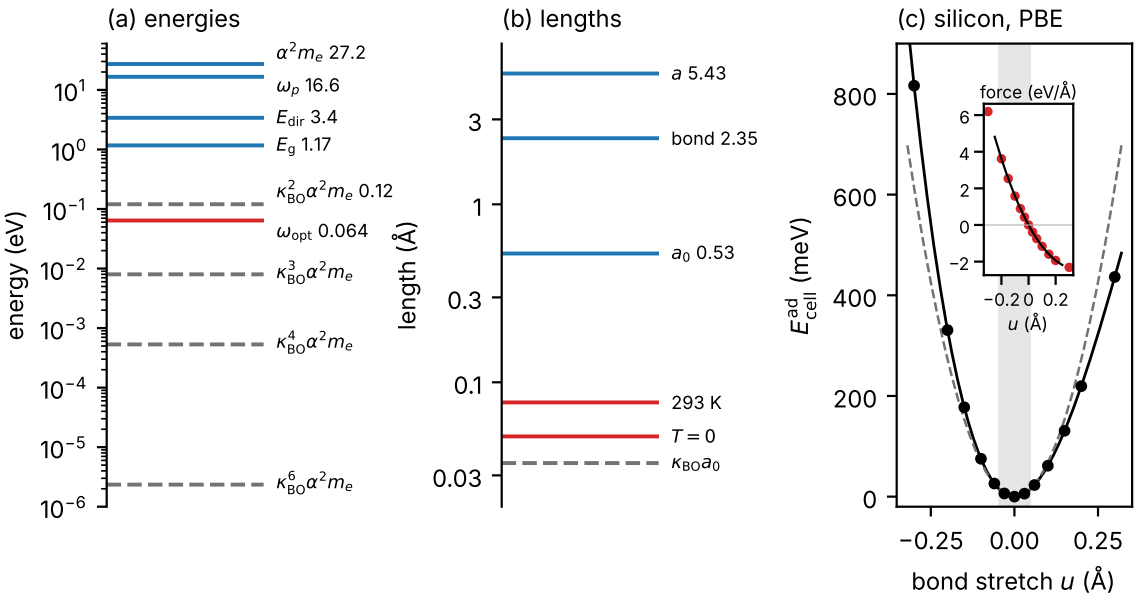


Figure 5.2: Scales of electrons and nuclei in silicon. (a) Energies on a logarithmic axis, in eV. Electronic (blue): the atomic unit $\alpha^2 m_e = 27.2\,\text{eV}$, the valence plasma energy $\omega_p = 16.6\,\text{eV}$, the direct gap $E_{\text{dir}} = 3.4\,\text{eV}$ and the indirect gap $E_{\text{g}} = 1.17\,\text{eV}$. Nuclear (red): the measured zone-center optical phonon, 64.1 meV (15.5 THz), the highest vibrational energy of the crystal. Gray dashed: the Born–Oppenheimer estimates $\kappa_{\text{BO}}^n \alpha^2 m_e$ for $n = 2$ (phonons), 3 (cubic anharmonicity), 4 (anharmonic level shifts) and 6 (nonadiabatic level shifts), with $\kappa_{\text{BO}} = (m_e/M)^{1/4} = 0.0665$ for ^{28}Si. (b) Lengths on a logarithmic axis, in Å: the lattice constant a, the bond length $\sqrt{3}a/4$ and the Bohr radius a_0 (blue). Red: the root-mean-square displacement of an atom along one Cartesian axis, $\langle u_x^2 \rangle^{1/2}$, at 293 K from a lattice-dynamical fit to neutron data and at $T = 0$ from the PBE phonon density of states. The PBE density of states uses the settings of panel (c) on a 4^3 grid of phonon wavevectors. Gray dashed: the estimate $\kappa_{\text{BO}} a_0$. (c) Energy per two-atom cell in a PBE calculation with an SG15 norm-conserving pseudopotential, a 50 Ry cutoff, an 8^3 grid of crystal momenta and $a = 5.431\,\text{Å}$. The circles are computed at displacements in which the two atoms move in opposite directions along [111] and the [111] bond changes its length by u. The curves are a sixth-degree polynomial fit (solid) and the harmonic term $\frac{1}{2}\Phi_{\text{opt}} u^2$, $\Phi_{\text{opt}} = 13.6\,\text{eV}\,\text{Å}^{-2}$ (dashed), from a quartic fit over $|u| \leq 0.1\,\text{Å}$. Shaded: $|u| < \ell_{\text{zp}} = (M_\kappa \omega_{\text{opt}})^{-1/2} = 0.048\,\text{Å}$, the zero-point length of the coordinate u. Inset: Hellmann–Feynman force on u (circles) and $-\text{d}E/\text{d}u$ from the sixth-degree fit (line).

The estimates Eq. (5.8) are $\kappa_{\mathrm{BO}} a_0 = 0.0352\,\text{Å}$ for the zero-point amplitude and $\kappa_{\mathrm{BO}}^2 \alpha^2 m_e = 0.120\,\mathrm{eV}$ for a phonon. We compare them with the crystal (Fig. 5.2).
Energies. The highest phonon of silicon, the optical mode at 64.1 meV, is $\omega_{\mathrm{opt}}/\alpha^2 m_e = 2.36 \times 10^{-3} = 0.53\,\kappa_{\mathrm{BO}}^2$: the counting, measured on the atomic scale for which it was built, is right to a factor of two. Against the gap the counting does much worse. The ratio is $\omega_{\mathrm{opt}}/E_{\mathrm{g}} = 0.0548 = 12.4\,\kappa_{\mathrm{BO}}^2$, because the gap of silicon is 23 times smaller than $\alpha^2 m_e$. By Eq. (5.9), $\omega_{\mathrm{ph}}/E_{\mathrm{g}}$ is the ratio that controls the mixing of excited electronic states into the ground state, so the adiabatic approximation in silicon is an order of magnitude less safe than the mass ratio alone suggests. It is still safe: the gap is eighteen phonon energies away. A semiconductor whose gap approaches the phonon energies, however, or a metal, is a different matter (Sect. 6.3).
Lengths. The zero-point amplitude of one Cartesian component is $\langle u_x^2 \rangle^{1/2} = 0.050\,\text{Å}$ in the PBE calculation of Fig. 5.2, 1.4 times $\kappa_{\mathrm{BO}} a_0$. At 293 K, where thermal phonons add to the zero-point motion, a lattice-dynamical fit to neutron data gives 0.0771 Å, from $\langle u_x^2 \rangle = 0.005\,941\,\text{Å}^2$ [9], and the PBE calculation yields 0.081 Å, 5 % more. This agreement suggests a similar accuracy for the zero-point value of the PBE calculation. Measured against the lattice constant, the zero-point amplitude is $\langle u_x^2 \rangle^{1/2}/a = 0.0091$, seven times smaller than κ_{BO} itself. The reason for the factor of seven is the unit of length: the counting compares the amplitude with a_0, and the lattice constant of silicon is $10.3\,a_0$. As a fraction of the bond length, $4.4\,a_0$, the amplitude is 0.021.
Anharmonicity. The fit of Fig. 5.2(c) has the cubic coefficient $\Phi_{\mathrm{opt}}^{(3)} = -41.5\,\mathrm{eV\,Å^{-3}}$. At $u = \ell_{\mathrm{zp}} = 0.048\,\text{Å}$ the ratio of the cubic to the harmonic term of the energy is $|\Phi_{\mathrm{opt}}^{(3)}| \ell_{\mathrm{zp}} / 3\Phi_{\mathrm{opt}} = 0.049$, against the estimate $\kappa_{\mathrm{BO}} = 0.067$.
Nonadiabatic shifts. Their relative size is $\kappa_{\mathrm{BO}}^4 = m_e/M = 2 \times 10^{-5}$, about 1 µeV on the optical phonon. A shift of 1 µeV is far below the anharmonic effects and below any resolution that matters for detection.
The powers of κ_{BO} order the effects correctly, with coefficients between one half and two when the comparison is made in atomic units. The coefficients become large when a ratio is taken to a scale of the crystal itself, such as the gap or the lattice constant, and a calculation should quote ratios to these scales of the crystal.

5.2 The harmonic crystal

Neutron scattering finds that the vibrations of silicon fill a band from zero up to 64.1 meV, organized in six branches over the Brillouin zone. Three of them start at zero energy at the zone center and rise linearly, with slopes between 4.7 and $9.4\,\mathrm{km\,s^{-1}}$: these are sound waves. Three start at the optical frequency and stay near it. The density of vibrational states piles up in sharp peaks, near 17 meV, where the sound waves stop rising, and near 60 meV. A dark-matter particle that deposits energy in single phonons sees these peaks directly in its spectrum. The branches, their slopes and the peaks of the density of states all follow from the shape of the energy surface $E^{\mathrm{ad}}(u)$ of Sect. 5.1 near its minimum.

5.2.1 Force constants and the dynamical matrix

The nucleus κ in the cell $\boldsymbol{R}$ is displaced from its equilibrium site by $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ (Eq. (5.1)), and the crystal has n_{at} nuclei per cell, two for silicon (Sect. 3.1). Expanded about its minimum, the

energy surface reads

$$E^{\mathrm{ad}}(u) = E^{\mathrm{ad}}(0) + \frac{1}{2}\sum_{\boldsymbol{R}\kappa\alpha}\sum_{\boldsymbol{R}'\kappa'\beta} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\, u^{\alpha}_{\boldsymbol{R}\kappa} u^{\beta}_{\boldsymbol{R}'\kappa'} + O(u^3),$$
$$\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}') = \left.\frac{\partial^2 E^{\mathrm{ad}}}{\partial u^{\alpha}_{\boldsymbol{R}\kappa}\,\partial u^{\beta}_{\boldsymbol{R}'\kappa'}}\right|_{u=0}, \tag{5.13}$$

with Cartesian indices α, β. The linear term is absent, since every force vanishes at equilibrium. The coefficients $\Phi^{(2)}$ are the *force constants*: $-\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\,u^{\beta}_{\boldsymbol{R}'\kappa'}$ is the force along α on the nucleus $(\boldsymbol{R},\kappa)$ when the nucleus $(\boldsymbol{R}',\kappa')$ alone is displaced. As second derivatives the force constants are symmetric under interchange of the two nuclei (the pair symmetry), $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}') = \Phi^{(2),\beta\alpha}_{\kappa'\kappa}(\boldsymbol{R}',\boldsymbol{R})$, and real. A lattice translation moves every nucleus to an equivalent site and leaves the surface unchanged. As a result the force constants depend on the two cells only through their separation, $\Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}') = \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R}'-\boldsymbol{R})$. Dropping the terms beyond second order is the *harmonic approximation*. By the counting of Sect. 5.1, the cubic term is smaller than the quadratic one by a factor $\sim \kappa_{\mathrm{BO}}$ at the zero-point amplitude, about 5 % in silicon (Worked example 5.1), and we return to it in Sect. 5.6.

The nuclei move in this quadratic potential. We treat their motion classically here. For a Hamiltonian quadratic in the coordinates and momenta, the Heisenberg equations of motion of the displacement operators are the same linear equations as the classical ones, so the normal modes we find are also the normal modes of the quantum problem; Sect. 5.3 quantizes them. Newton's law for each nucleus is

$$M_\kappa\, \ddot{u}^{\alpha}_{\boldsymbol{R}\kappa} = -\frac{\partial E^{\mathrm{ad}}}{\partial u^{\alpha}_{\boldsymbol{R}\kappa}} = -\sum_{\boldsymbol{R}'\kappa'\beta} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\, u^{\beta}_{\boldsymbol{R}'\kappa'}, \tag{5.14}$$

$3n_{\mathrm{at}}N$ coupled linear equations for a crystal of N cells. Two steps reduce them to a small matrix problem: the mass weighting that works for any set of coupled oscillators, and the translation symmetry of the crystal. For a set of coordinates x_i with masses m_i and potential $\frac{1}{2}\sum_{ij} K_{ij}x_i x_j$, the mass-weighted coordinates $y_i = \sqrt{m_i}\,x_i$ obey $\ddot{y}_i = -\sum_j (K_{ij}/\sqrt{m_i m_j})\,y_j$. The matrix on the right is real and symmetric, so it has real eigenvalues ω^2 and orthonormal eigenvectors, and each eigenvector oscillates on its own as $\mathrm{e}^{-\mathrm{i}\omega t}$: a normal mode. The crystal is a set of coupled oscillators with one more property: its matrix of force constants is invariant under lattice translations, and a translation-invariant matrix is diagonalized by plane waves in the cell index. We therefore try

$$u^{\alpha}_{\boldsymbol{R}\kappa}(t) = \frac{1}{\sqrt{M_\kappa}}\, e^{\alpha}_{\kappa}\, \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}\cdot\boldsymbol{R}-\omega t)}, \tag{5.15}$$

with a wavevector $\boldsymbol{Q}$ and $3n_{\mathrm{at}}$ amplitudes e^{α}_{κ} to be found. The physical displacement is the real part of Eq. (5.15), or the sum of the solutions at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$. Inserting Eq. (5.15) into Eq. (5.14), dividing by $\sqrt{M_\kappa}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{Q}\cdot\boldsymbol{R}-\omega t)}$ and writing $\boldsymbol{R}' = \boldsymbol{R} + \boldsymbol{R}''$ gives a sum over $\boldsymbol{R}''$ that no longer depends on $\boldsymbol{R}$. The $3n_{\mathrm{at}}$ amplitudes then obey an eigenvalue problem,

$$\sum_{\kappa'\beta} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q}) = \omega^2_{\boldsymbol{Q}\nu}\, e^{\alpha}_{\kappa\nu}(\boldsymbol{Q}), \qquad D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}}\sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{5.16}$$

the *dynamical matrix* $D^{\mathrm{dyn}}(\boldsymbol{Q})$, the Fourier transform of the mass-weighted force constants [3], [10, Chaps. 22 and 23], [4, Eq. (14)]; it is Eq. (A.22). Its eigenvalues are labeled by the *branch* index $\nu = 1, \ldots, 3n_{\mathrm{at}}$, and its eigenvectors $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$, with one three-vector for each nucleus of the cell, are the *polarization vectors*. The translation symmetry has turned $3n_{\mathrm{at}}N$ equations

into N independent problems of size $3n_{\mathrm{at}}$, one for each $\boldsymbol{Q}$: six by six for silicon. The phase in Eq. (5.16) contains the cell vector $\boldsymbol{R}$ and not the position $\boldsymbol{R}+\boldsymbol{\tau}_\kappa$ of the nucleus. Some codes and texts use the full position; the two choices differ by the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{\tau}_\kappa}$ on each component of $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ and give the same frequencies.

The properties of D^{dyn} follow from the properties of the force constants. By the pair symmetry of $\Phi^{(2)}$ and translation invariance, $\Phi^{(2),\beta\alpha}_{\kappa'\kappa}(\boldsymbol{0},\boldsymbol{R}) = \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},-\boldsymbol{R})$, and replacing $\boldsymbol{R}$ by $-\boldsymbol{R}$ in the sum shows that $D^{\mathrm{dyn}}(\boldsymbol{Q})$ is Hermitian. Its eigenvalues $\omega^2_{\boldsymbol{Q}\nu}$ are therefore real and its eigenvectors orthonormal and complete,

$$\sum_{\kappa\alpha} e^{\alpha}_{\kappa\nu}(\boldsymbol{Q})^*\, e^{\alpha}_{\kappa\nu'}(\boldsymbol{Q}) = \delta_{\nu\nu'}, \qquad \sum_{\nu} e^{\alpha}_{\kappa\nu}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q})^* = \delta_{\kappa\kappa'}\delta_{\alpha\beta}, \tag{5.17}$$

which contains the normalization $\sum_\kappa |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 = 1$ of Eq. (A.21). At a minimum of E^{ad} the quadratic form Eq. (5.13) is non-negative for every displacement, and so is $D^{\mathrm{dyn}}(\boldsymbol{Q})$ for every $\boldsymbol{Q}$; every $\omega^2_{\boldsymbol{Q}\nu}$ is then non-negative and $\omega_{\boldsymbol{Q}\nu}$ real. A computed $\omega^2_{\boldsymbol{Q}\nu} < 0$, usually printed as an imaginary frequency, signals that the structure is not at a minimum, or that the calculation is not converged. Since $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$, $D^{\mathrm{dyn}}(\boldsymbol{Q}+\boldsymbol{G}) = D^{\mathrm{dyn}}(\boldsymbol{Q})$, and $\boldsymbol{Q}$ can be taken in the first Brillouin zone: a displacement wave of wavevector $\boldsymbol{Q}+\boldsymbol{G}$ moves the nuclei exactly as a wave of wavevector $\boldsymbol{Q}$ does, because the nuclei sit only at the lattice sites. Under Born–von Karman conditions there are N allowed $\boldsymbol{Q}$ in the zone (Sect. 3.1), and the crystal has $3n_{\mathrm{at}}N$ modes, one for each degree of freedom. Finally, since the force constants are real, $D^{\mathrm{dyn}}(-\boldsymbol{Q}) = D^{\mathrm{dyn}}(\boldsymbol{Q})^*$. As a result $\omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}$, and the phases can be chosen so that $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$, the convention of Eq. (A.21).

A quick estimate fixes the scale of the phonon frequencies. The force constant between neighbors is $\Phi^{(2)} \sim \alpha^2 m_e/a_0^2 \sim 100\,\mathrm{eV\,\AA^{-2}}$ by the counting of Sect. 5.1, and the frequencies are $\omega \sim (\Phi^{(2)}/M_\kappa)^{1/2}$. For the zone-center optical mode of silicon, the PBE calculation of Fig. 5.2 gives $\Phi_{\mathrm{opt}} = 13.6\,\mathrm{eV\,\AA^{-2}}$ and $\omega = 63.7\,\mathrm{meV}$: the estimate is high by a factor of seven in the force constant, and the frequencies are tens of meV.

Each $\Phi^{(2)}$ is a second derivative of the Kohn–Sham total energy, and density-functional perturbation theory evaluates the whole matrix $D^{\mathrm{dyn}}(\boldsymbol{Q})$ at a given $\boldsymbol{Q}$ from the linear response of the electrons to a displacement wave. A calculation computes D^{dyn} on a coarse uniform grid of $\boldsymbol{Q}$, transforms back to real-space force constants, and evaluates Eq. (5.16) at any $\boldsymbol{Q}$ from them. The phonons of silicon shown below are computed in this way.

Worked example 5.2: The diatomic chain

The simplest crystal with both kinds of branch is a line of nuclei of masses M_1 and M_2, alternating, with period a_{c} and a spring of stiffness K between neighbors. The nucleus 1 of cell R is tied to the nucleus 2 of the same cell and to the nucleus 2 of the cell $R-a_{\mathrm{c}}$, and the energy is $\frac{1}{2}K\sum_R[(u_{R2}-u_{R1})^2 + (u_{R+a_{\mathrm{c}},1}-u_{R2})^2]$. Differentiating the energy twice gives the nonzero force constants from the cell at the origin, $\Phi^{(2)}_{11}(0,0) = \Phi^{(2)}_{22}(0,0) = 2K$, $\Phi^{(2)}_{12}(0,0) = \Phi^{(2)}_{12}(0,-a_{\mathrm{c}}) = -K$ and $\Phi^{(2)}_{21}(0,0) = \Phi^{(2)}_{21}(0,a_{\mathrm{c}}) = -K$. By Eq. (5.16),

$$D^{\mathrm{dyn}}(Q) = \begin{pmatrix} 2K/M_1 & -K(1+\mathrm{e}^{-\mathrm{i}Qa_{\mathrm{c}}})/\sqrt{M_1M_2} \\ -K(1+\mathrm{e}^{\mathrm{i}Qa_{\mathrm{c}}})/\sqrt{M_1M_2} & 2K/M_2 \end{pmatrix},$$

which is Hermitian, as it must be. With $|1+\mathrm{e}^{\mathrm{i}Qa_{\mathrm{c}}}|^2 = 4\cos^2(Qa_{\mathrm{c}}/2)$, the eigenvalues solve $\omega^4 - 2K(M_1^{-1}+M_2^{-1})\,\omega^2 + 4K^2\sin^2(Qa_{\mathrm{c}}/2)/(M_1M_2) = 0$:

$$\omega^2_{Q\pm} = K\Big(\frac{1}{M_1}+\frac{1}{M_2}\Big) \pm K\Big[\Big(\frac{1}{M_1}+\frac{1}{M_2}\Big)^2 - \frac{4\sin^2(Qa_{\mathrm{c}}/2)}{M_1M_2}\Big]^{1/2}.$$

The lower branch vanishes at $Q = 0$. For $Qa_c \ll 1$, expanding the square root to first order in its small term gives $\omega_{Q-}^2 = 2K \sin^2(Qa_c/2)/(M_1 + M_2) \simeq KQ^2a_c^2/[2(M_1 + M_2)]$. The frequency is linear in Q, and the lower branch is a sound wave with speed $c = a_c[K/2(M_1 + M_2)]^{1/2}$, in which the chain moves as a medium of mass $M_1 + M_2$ per length a_c. The eigenvector of the lower branch at $Q = 0$ is $\boldsymbol{e} \propto (\sqrt{M_1}, \sqrt{M_2})$, so that by Eq. (5.15) both nuclei move by the same amount: a rigid translation. The lower branch is the *acoustic* branch. The upper branch starts at $\omega_{0+}^2 = 2K/\mu_{\text{red}}$, with $\mu_{\text{red}} = M_1M_2/(M_1 + M_2)$, and its eigenvector at $Q = 0$ is $\boldsymbol{e} \propto (\sqrt{M_2}, -\sqrt{M_1})$: the two nuclei move against each other, $u_1/u_2 = -M_2/M_1$, and the center of mass of each cell stays at rest. The upper branch is the *optical* branch, so named because, when the two nuclei carry opposite charges, its oscillating dipole couples to light (Sect. 5.4). At the zone boundary, $Q = \pi/a_c$, the off-diagonal elements vanish. Each mode then moves one species only, with $\omega^2 = 2K/M_1$ and $2K/M_2$, and between these two frequencies lies a gap in which no wave propagates. For equal masses the gap closes. Then $\omega_{Q\pm} = (2K/M)^{1/2}[1 \pm \cos(Qa_c/2)]^{1/2}$, which is the single branch $\omega = 2(K/M)^{1/2}|\sin(Q'a_c/4)|$ of a chain of identical nuclei with period $a_c/2$, with $Q' = Q$ on the acoustic branch and $Q' = 2\pi/a_c - Q$ on the optical one. The doubled cell has folded the zone in half, and the optical branch is the far half of the acoustic one. Silicon contains exactly this chain. For a wave along [100] with longitudinal polarization, every atomic plane perpendicular to [100] moves rigidly, and the planes, 1.358 Å ($a/4$) apart, alternate between the two sublattices of the diamond structure. The two kinds of plane are related by a translation by $a/4$ combined with a rotation by 90° about [100], and the rotation does not affect a longitudinal motion. The planes therefore act as identical masses, and the longitudinal branches along [100] form a folded chain with $a_c = a/2$. The zone boundary of this folded chain is $Q = 2\pi/a$, the point X. In the folded chain the longitudinal acoustic and optical branches must meet at X, whatever the range of the interplanar forces, and in silicon they do, at 12.32 THz [1].

5.2.2 The acoustic sum rule

The acoustic branch of the chain vanished at $Q = 0$ because a rigid translation costs no energy. In every crystal, displacing all nuclei by the same vector $\boldsymbol{d}$ leaves the energy unchanged at any configuration, $E^{\text{ad}}(u+d) = E^{\text{ad}}(u)$, where $u+d$ means $\boldsymbol{u}_{\boldsymbol{R}\kappa}+\boldsymbol{d}$ for every nucleus. Differentiating with respect to $u^\alpha_{\boldsymbol{R}\kappa}$ and to d^β, and setting $u = d = 0$, yields $\sum_{\boldsymbol{R}'\kappa'} \partial^2 E^{\text{ad}}/\partial u^\alpha_{\boldsymbol{R}\kappa}\partial u^\beta_{\boldsymbol{R}'\kappa'} = 0$, that is

$$\sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R}) = 0 \qquad \text{for every } \kappa, \alpha, \beta. \tag{5.18}$$

This is the *acoustic sum rule*: a nucleus feels no force when all nuclei, itself included, are moved together. The consequence of the sum rule for the dynamical matrix follows at once. At $\boldsymbol{Q} = 0$, the vector $e^\beta_{\kappa'} = (M_{\kappa'}/M_{\text{cell}})^{1/2}\hat{e}^\beta$, with $M_{\text{cell}} = \sum_\kappa M_\kappa$ and any unit vector $\hat{\boldsymbol{e}}$, gives $\sum_{\kappa'\beta} D^{\text{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0})\, e^\beta_{\kappa'} = (M_\kappa M_{\text{cell}})^{-1/2} \sum_\beta [\sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R})]\hat{e}^\beta = 0$. Three independent choices of $\hat{\boldsymbol{e}}$ give three modes of zero frequency at the zone center, the *acoustic* branches, in which all nuclei move together, with the polarization vectors $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\text{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ of Appendix A.6. The other $3n_{\text{at}} - 3$ branches, three in silicon, are the *optical* branches, finite at $\boldsymbol{Q} = 0$. In the optical branches the nuclei of a cell move against each other with their center of mass at rest: orthogonality to the three acoustic vectors in Eq. (5.17) is $\sum_\kappa \sqrt{M_\kappa}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{0}) = 0$, and the displacement $\boldsymbol{e}_{\kappa\nu}/\sqrt{M_\kappa}$ of Eq. (5.15) then has $\sum_\kappa M_\kappa\, \boldsymbol{u}_\kappa = 0$. In field-theory language the acoustic phonons are the Goldstone modes of the translation symmetry that the crystal breaks [11, p. 69]. A uniform translation is a different ground state at the same energy, and a

translation that varies slowly in space costs an energy that vanishes as its wavelength grows.

A computed set of force constants violates Eq. (5.18) slightly, chiefly because the exchange–correlation potential is evaluated on a real-space grid that is not invariant under arbitrary translations. The acoustic frequencies at Γ then come out small but nonzero, 0.16 THz in the calculation of Fig. 5.3, and codes restore the sum rule by correcting the on-site force constants.

5.2.3 Sound speeds

Near $\boldsymbol{Q} = 0$ the acoustic frequencies rise from zero, and we want the rate at which they do. Expanding the exponential in Eq. (5.16) in powers of $\boldsymbol{Q}$,

$$\begin{aligned} D^{\mathrm{dyn}}(\boldsymbol{Q}) &= D_0 + D_1(\boldsymbol{Q}) + D_2(\boldsymbol{Q}) + O(Q^3), \\ D^{\alpha\beta}_{n,\kappa\kappa'}(\boldsymbol{Q}) &= \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, \frac{(\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R})^n}{n!}, \end{aligned} \tag{5.19}$$

and treating D_1 and D_2 as perturbations of $D_0 = D^{\mathrm{dyn}}(\boldsymbol{0})$, whose acoustic eigenvalue is triply degenerate at zero, leads to ω^2 as an expansion in Q. The acoustic frequencies are the eigenvalues of a 3×3 matrix in the space of the three acoustic vectors $\boldsymbol{a}_{\hat{\boldsymbol{e}}}$, with components $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$.

The first-order term would make ω^2 linear in Q, and $\omega \propto Q^{1/2}$, which no crystal shows. Rotational invariance makes the first-order term vanish. The matrix elements of D_1 are

$$\boldsymbol{a}^\dagger_{\hat{\boldsymbol{e}}} D_1\, \boldsymbol{a}_{\hat{\boldsymbol{e}}'} = \frac{\mathrm{i}Q_\gamma}{M_{\mathrm{cell}}}\, \hat{e}^\alpha \hat{e}'^\beta\, T^{\alpha\beta\gamma}, \qquad T^{\alpha\beta\gamma} = \sum_{\kappa\kappa'\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, R^\gamma,$$

with summation over repeated Cartesian indices. Two properties of T force it to zero. The first follows from the pair symmetry of $\Phi^{(2)}$. Writing $\Phi^{(2),\beta\alpha}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R}) = \Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{0},-\boldsymbol{R})$ and relabeling $\kappa \leftrightarrow \kappa'$ and $\boldsymbol{R} \to -\boldsymbol{R}$ in the sum shows $T^{\beta\alpha\gamma} = -T^{\alpha\beta\gamma}$. The second follows from rotations. A rigid infinitesimal rotation by the angle vector $\boldsymbol{\theta}$ moves every nucleus, at whatever position $\boldsymbol{X}_I$ it has, by $\boldsymbol{\theta} \times \boldsymbol{X}_I$ and leaves the energy unchanged. (Under Born–von Karman conditions a rotation is not a symmetry of the crystal, but the force constants are the same as in a large free crystal, for which a rotation is a symmetry.) Hence $\sum_I \nabla_I E^{\mathrm{ad}} \cdot (\boldsymbol{\theta} \times \boldsymbol{X}_I) = 0$ at every configuration: the total torque vanishes. Differentiating this identity with respect to the displacement of one nucleus and evaluating at equilibrium, where the forces vanish, leaves $\sum_I \Phi^{(2)}_{JI}\, (\boldsymbol{\theta} \times \boldsymbol{X}_I) = 0$ for every J and every $\boldsymbol{\theta}$. In components, $\sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, X^\gamma_{\boldsymbol{R}\kappa'}$ is symmetric in $\beta\gamma$, with $\boldsymbol{X}_{\boldsymbol{R}\kappa'} = \boldsymbol{R} + \boldsymbol{\tau}_{\kappa'}$. Summed over κ, this sum is $T^{\alpha\beta\gamma}$ plus a term $\sum_{\kappa\kappa'}[\sum_{\boldsymbol{R}} \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})]\tau^\gamma_{\kappa'}$, which vanishes because the sum rule Eq. (5.18) also holds with the roles of κ and κ' exchanged (by the same pair symmetry). So $T^{\alpha\beta\gamma}$ is symmetric in $\beta\gamma$ and antisymmetric in $\alpha\beta$, and a tensor with both properties is zero. Alternating the two properties six times returns to the start with a minus sign [3, Chap. V]:

$$T^{\alpha\beta\gamma} = -T^{\beta\alpha\gamma} = -T^{\beta\gamma\alpha} = T^{\gamma\beta\alpha} = T^{\gamma\alpha\beta} = -T^{\alpha\gamma\beta} = -T^{\alpha\beta\gamma} = 0.$$

At second order two terms contribute. One is the expectation value of D_2 in the acoustic space. The other is D_1 taken twice, through the optical modes: D_1 does connect an acoustic vector to an optical vector $\boldsymbol{o}$ at $\boldsymbol{Q} = 0$, of frequency ω_{o}, and second-order perturbation theory adds $-|\boldsymbol{o}^\dagger D_1 \boldsymbol{a}_{\hat{\boldsymbol{e}}}|^2/\omega_{\mathrm{o}}^2$. Both terms are quadratic in $\boldsymbol{Q}$, so

$$\omega^2_{\boldsymbol{Q}\nu} = \Lambda^{\mathrm{ac}}_\nu(\hat{\boldsymbol{Q}})\, Q^2 + O(Q^3), \qquad \omega_{\boldsymbol{Q}\nu} = c_\nu(\hat{\boldsymbol{Q}})\, Q, \qquad c_\nu(\hat{\boldsymbol{Q}}) = \Lambda^{\mathrm{ac}}_\nu(\hat{\boldsymbol{Q}})^{1/2}, \tag{5.20}$$

where $\Lambda^{\mathrm{ac}}_\nu(\hat{\boldsymbol{Q}})$ are the three eigenvalues of the 3×3 matrix

$$\Lambda^{\mathrm{ac},\alpha\beta}(\hat{\boldsymbol{Q}})\, Q^2 = -\frac{1}{2M_{\mathrm{cell}}} \sum_{\kappa\kappa'\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, (\boldsymbol{Q}\cdot\boldsymbol{R})^2 - \sum_{\mathrm{o}} \frac{(\boldsymbol{a}^\dagger_\alpha D_1 \boldsymbol{o})(\boldsymbol{o}^\dagger D_1 \boldsymbol{a}_\beta)}{\omega_{\mathrm{o}}^2},$$

with $\boldsymbol{a}_\alpha$ the acoustic vector polarized along the axis α and the sum over the optical modes at $\boldsymbol{Q} = 0$. The frequency of an acoustic mode is linear in Q, with a sound speed $c_\nu(\hat{\boldsymbol{Q}})$ that depends on the direction of propagation and on the branch, and the polarizations $\hat{\boldsymbol{e}}_\nu(\hat{\boldsymbol{Q}})$ are the eigenvectors of Λ^{ac}. The first term of Λ^{ac} is the stiffness of the crystal when every nucleus follows the long wave rigidly, the sum of the springs weighted by the squared lengths over which they are stretched. The second term lowers this stiffness. In a crystal with more than one nucleus per cell, a strain pushes the sublattices against each other, and the optical coordinate relaxes in the long wave; the second term describes this internal relaxation, which in the diamond structure reduces the shear stiffness [3, Chap. V]. In silicon the first-order coupling D_1 from acoustic to optical vectors vanishes for the longitudinal wave along [100], in which the planes of Worked example 5.2 move as identical masses and the two sublattices are not pushed against each other; for shear waves it is nonzero.

Since Λ^{ac} describes a long wave, it is a property of the crystal as an elastic continuum. Appendix C.18 derives the equation of motion of an elastic continuum, the Christoffel equation, with the elastic constants C_{ijkl} and the mass density ρ_T, and Eq. (5.20) is the same equation: $\rho_T \Lambda^{\mathrm{ac}}$, with $\rho_T = M_{\mathrm{cell}}/\Omega$, is the Christoffel matrix of the relaxed elastic constants. For a cubic crystal along [100] the Christoffel equation reads $\rho_T c_{\mathrm{LA}}^2 = C_{11}$ and $\rho_T c_{\mathrm{TA}}^2 = C_{44}$ for the two degenerate transverse branches. The sound speeds of the dark-matter kinematics of Sect. 1.3 are these slopes, and the acoustic edge $m_{\mathrm{min}}^{\mathrm{ac}}$ of Sect. 9.2 uses them. The longitudinal speed is the largest, since compressing a medium costs more than shearing it. The scale of the sound speeds is a vibrational frequency $(\Phi^{(2)}/M_\kappa)^{1/2}$ times a distance of the order of the spacing between atomic planes: for silicon, the optical frequency times $a/4$ is $13\,\mathrm{km\,s^{-1}}$, against measured speeds of 4.7 to $9.4\,\mathrm{km\,s^{-1}}$. Note that $c/v_{\mathrm{max}} \sim 10^{-2}$: sound is two orders of magnitude slower than the fastest dark matter, and for this reason an acoustic phonon takes only a small fraction of the kinetic energy of the dark-matter particle (Sect. 1.3).

5.2.4 The phonons of silicon

Fig. 5.3(a) shows the six branches of silicon along the three lines of the zone on which they were measured: from Γ to X along [100], from X to Γ along [110] through K, and from Γ to L along [111]. On these lines the symmetry of the crystal makes every polarization either parallel or perpendicular to $\boldsymbol{Q}$, and the branches are named accordingly: longitudinal and transverse acoustic (LA, TA), longitudinal and transverse optical (LO, TO). In a general direction the polarizations mix, and the names refer to the branch that is mostly longitudinal or mostly transverse. Along [100] and [111] the two transverse branches of each kind are degenerate by the symmetry of these axes; along [110] they split.

The measured frequencies at the high-symmetry points come from inelastic neutron scattering at room temperature [1, 12]. At Γ the three optical modes are degenerate, at 15.5 THz, or 64.1 meV. In a polar crystal the longitudinal optical mode would create an electric field, and that field would push the longitudinal frequency up by the LO–TO splitting of Sect. 5.4; silicon has no Born charges and no splitting. At X the longitudinal acoustic and optical branches meet, at 12.32 THz (50.95 meV), as the folded chain of Worked example 5.2 requires. Near the zone boundary the transverse acoustic branches bend over and become almost flat, at 4.49 THz (18.6 meV) at X and 3.43 THz (14.2 meV) at L. The reason for the flat transverse acoustic branches is the covalent bond. Stretching a bond is stiff, while changing the angle between two bonds is soft, and a short-wavelength transverse wave mostly bends bonds. Finally, the optical branches are narrow. From 15.5 THz at Γ they fall only to between 12.3 and 14.7 THz at the zone boundary. The reason is that the stiff bond between the two atoms of a cell controls how they move against each other, and the neighboring cells change that motion only weakly.

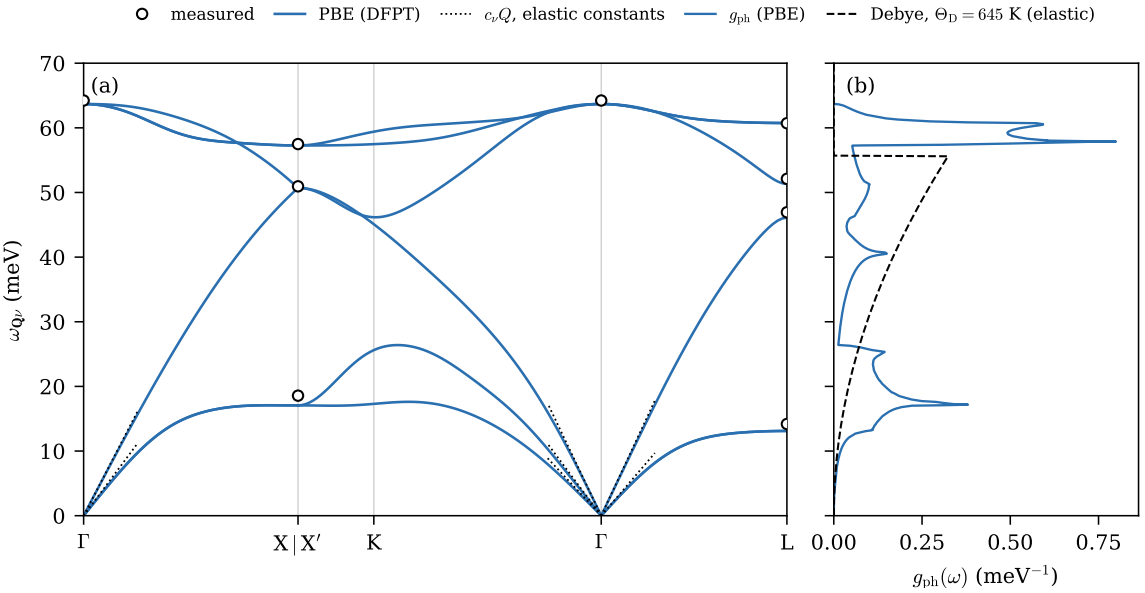


Figure 5.3: Phonons of silicon. (a) Dispersion along [100] from Γ to X, along [110] from X′ $= (1,1,0)2\pi/a$, which is equivalent to X, through K to Γ, and along [111] from Γ to L. Lines: density-functional perturbation theory with the PBE functional, an SG15 norm-conserving pseudopotential, a 60 Ry cutoff, a 12^3 grid of crystal momenta, the measured lattice constant $a = 5.431$ Å and the natural mass of silicon; force constants from a 6^3 grid of wavevectors, with the acoustic sum rule imposed, interpolated to the path. Circles: frequencies measured by inelastic neutron scattering at room temperature at Γ, X and L [1, 12], as tabulated in Ref. [14, Table I]. Dotted: $\omega = c_\nu Q$ with the sound speeds of the measured elastic constants (Worked example 5.3), drawn up to $Q = 0.25\,(2\pi/a)$. (b) Phonon density of states per two-atom cell, normalized to six modes, from the same force constants on a 48^3 grid (tetrahedron integration), and the Debye model Eq. (5.24) with $\bar{c}$ from the measured elastic constants, $\omega_{\mathrm{D}} = 55.6$ meV, in the convention of one wavevector per atom. One THz corresponds to 4.136 meV.

The lines in Fig. 5.3(a) are a density-functional calculation with the PBE functional at the measured lattice constant, from force constants computed by density-functional perturbation theory (DFPT) on a 6^3 grid of wavevectors. They follow the measurements closely. Of the eight measured frequencies at Γ, X and L, six agree to within 2 %, and the mean absolute difference over all eight is 2.6 %. The zone-center optical mode comes out at 15.40 THz (63.7 meV), 0.6 % below the measured 64.1 meV and the same as the frozen-phonon value of Fig. 5.2. The exceptions are the flat transverse acoustic modes, 8 % soft: 4.13 against 4.49 THz at X and 3.17 against 3.43 THz at L. PBE generally puts the phonons of silicon and germanium below the measured ones, a consequence of its underbinding [13, Sect. V]; in this calculation the deficit of the computed frequencies is largest on the branches set by bond bending, the weakest force constant of the crystal. The calculation describes a static lattice, while the measurement was made at room temperature, where the frequencies are lower by thermal expansion and by anharmonic shifts; these effects, and the widths $\gamma_{\boldsymbol{Q}\nu}$ that the harmonic crystal lacks entirely, are the subject of Sect. 5.6. A frequency also moves with the lattice constant at which it is computed, by an amount set by its Grüneisen parameter $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} = -\partial \ln \omega_{\boldsymbol{Q}\nu} / \partial \ln \Omega$. For most modes of silicon $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} > 0$, and the PBE lattice constant, 0.85 % too large, would soften them; the flat transverse acoustic modes have $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} < 0$ and would stiffen.

5.2.5 The density of states

For any quantity that sums over all phonons, such as a rate summed over final states or a thermal average, we need the number of modes at each frequency. The phonon density of states per cell is

$$g_{\mathrm{ph}}(\omega) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \delta(\omega - \omega_{\boldsymbol{Q}\nu}) = \frac{\Omega}{(2\pi)^3}\sum_{\nu}\int_{\mathrm{BZ}} \mathrm{d}^3Q\; \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \qquad \int_0^\infty \mathrm{d}\omega\, g_{\mathrm{ph}}(\omega) = 3n_{\mathrm{at}}, \tag{5.21}$$

where the sum over the N wavevectors of the zone became an integral with the density $\mathscr{V}/(2\pi)^3$ of Sect. 3.1, $\mathscr{V} = N\Omega$. The integral of g_{ph} counts the $3n_{\mathrm{at}}$ modes per cell, six for silicon. A density of states per atom, normalized to three, is $g_{\mathrm{ph}}/n_{\mathrm{at}}$. In silicon, Fig. 5.3(b), g_{ph} has a sharp peak at 17 meV from the flat transverse acoustic branches near the zone boundary. Above this peak lies a structured region between 25 and 52 meV from the longitudinal acoustic branch and the lower optical branches. The tallest peaks, at 58 and 61 meV, come from the flat transverse optical branches. The peaks are where the branches are flat. Near a frequency at which $\nabla_{\boldsymbol{Q}}\omega_{\boldsymbol{Q}\nu}$ vanishes, many wavevectors share nearly the same frequency, and in three dimensions the density of states has a kink with an infinite slope there, a Van Hove singularity [10, Chap. 23]. Because of the large peak of the transverse acoustic branches, silicon has many low-energy phonons: a dark-matter particle that can deposit 20 meV in single phonons finds a fifth of all modes of the crystal, 22 % in the calculation, below that energy.

At low frequency only the acoustic branches contribute, with $\omega_{\boldsymbol{Q}\nu} = c_\nu(\hat{\boldsymbol{Q}})Q$. Writing $\mathrm{d}^3Q = Q^2\mathrm{d}Q\,\mathrm{d}\hat{\boldsymbol{Q}}$ and integrating over Q with $\int Q^2\mathrm{d}Q\,\delta(\omega - c_\nu Q) = \omega^2/c_\nu^3$ gives

$$g_{\mathrm{ph}}(\omega) \simeq \frac{\Omega\,\omega^2}{2\pi^2}\sum_\nu \langle c_\nu^{-3}\rangle, \qquad \langle c_\nu^{-3}\rangle = \int \frac{\mathrm{d}\hat{\boldsymbol{Q}}}{4\pi}\, c_\nu(\hat{\boldsymbol{Q}})^{-3}, \tag{5.22}$$

an ω^2 law with a coefficient fixed by the sound speeds alone, averaged over directions. The ω^2 law holds as long as the dispersion is linear: in silicon, up to about 7 meV, where the computed g_{ph} exceeds the ω^2 law by 10 %.

5.2.6 The Debye model

Debye's model extends Eq. (5.22) to the whole spectrum. It replaces the $3n_{\mathrm{at}}$ branches by three, all with the linear dispersion $\omega = \bar{c}Q$ of a single averaged speed $\bar{c}$. It also replaces the zone by a sphere of radius Q_{D} that holds one wavevector per atom, $n_{\mathrm{at}}N$ in all, so that the three branches carry all $3n_{\mathrm{at}}N$ degrees of freedom [10, Chap. 23]. Counting with the density $\mathscr{V}/(2\pi)^3$, $(4\pi Q_{\mathrm{D}}^3/3)\,N\Omega/(2\pi)^3 = n_{\mathrm{at}}N$, so that

$$Q_{\mathrm{D}} = \Big(\frac{6\pi^2 n_{\mathrm{at}}}{\Omega}\Big)^{1/3}, \qquad \omega_{\mathrm{D}} = \bar{c}\,Q_{\mathrm{D}}, \qquad \Theta_{\mathrm{D}} = \omega_{\mathrm{D}}, \tag{5.23}$$

with the Debye frequency ω_{D} and the Debye temperature Θ_{D}, which in natural units are the same number. The density of states of the model follows from Eq. (5.22) with $c_\nu = \bar{c}$, cut off at ω_{D}:

$$g_{\mathrm{D}}(\omega) = \frac{3\Omega\,\omega^2}{2\pi^2\bar{c}^3}\,\Theta(\omega_{\mathrm{D}} - \omega) = \frac{9n_{\mathrm{at}}\,\omega^2}{\omega_{\mathrm{D}}^3}\,\Theta(\omega_{\mathrm{D}} - \omega), \tag{5.24}$$

where the second form uses Eq. (5.23); the integral of g_{D} is $3n_{\mathrm{at}}$, as it must be. Fixing $\bar{c}$ so that the model matches the low-frequency limit Eq. (5.22) exactly gives

$$\frac{3}{\bar{c}^3} = \sum_\nu \langle c_\nu^{-3}\rangle. \tag{5.25}$$

This is the definition of $\bar{c}$ that we use. It makes the model exact wherever only long acoustic waves matter, as in the heat capacity at low temperature, which then follows Debye's T^3 law with this Θ_D, and in a dark-matter rate dominated by the lowest phonon energies. Since c^{-3} is averaged, the slowest branches dominate $\bar{c}$. Note that the literature also uses a second convention. Some texts, particularly on phonon transport, put only the N acoustic wavevectors in the sphere, one per cell, and treat the optical branches separately; this convention lowers Q_D and Θ_D by the factor $n_\mathrm{at}^{1/3}$, $2^{1/3} = 1.26$ for silicon.

Above a few meV, however, the model is crude. It puts all modes below ω_D, and in silicon it misses both the peak of the flat transverse branches and the optical band at the top. The strength of the Debye model is that one number fixes the low-energy spectrum. Integrals over g_ph weighted toward low frequencies come out within tens of percent, for example the mean-square displacement at room temperature (Sect. 5.3) or the Debye–Waller exponent (Sect. 5.5). Worked example 5.3 carries the comparison through for silicon.

Worked example 5.3: Silicon sound speeds and Debye temperature

The elastic constants of silicon at room temperature are $C_{11} = 165.6\,\mathrm{GPa}$, $C_{12} = 63.9\,\mathrm{GPa}$ and $C_{44} = 79.5\,\mathrm{GPa}$, and its density is $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$ [15, Table I]. Along [100], $\rho_T c_\mathrm{LA}^2 = C_{11}$ and $\rho_T c_\mathrm{TA}^2 = C_{44}$, so

$$c_\mathrm{LA} = \Big(\frac{165.6\times10^9\,\mathrm{Pa}}{2330\,\mathrm{kg\,m^{-3}}}\Big)^{1/2} = 8.43\,\mathrm{km\,s^{-1}},$$

$$c_\mathrm{TA} = \Big(\frac{79.5\times10^9\,\mathrm{Pa}}{2330\,\mathrm{kg\,m^{-3}}}\Big)^{1/2} = 5.84\,\mathrm{km\,s^{-1}}\ \text{(twice)}.$$

In other directions the speeds are the eigenvalues of the Christoffel matrix (Appendix C.18): along [110], 9.13 for LA and 5.84 and $4.67\,\mathrm{km\,s^{-1}}$ for the two TA branches; along [111], 9.35 for LA and $5.09\,\mathrm{km\,s^{-1}}$ for both TA branches. The longitudinal speed varies by 11 % with direction and the transverse ones by 25 %. These speeds are the slopes of the acoustic branches at Γ in Fig. 5.3(a). The PBE calculation of that figure, interpolated from its 6^3 grid to small Q, has 8.34 and $5.44\,\mathrm{km\,s^{-1}}$ along [100], 1 % and 7 % below the speeds from the elastic constants. The transverse branches of the PBE calculation are soft at small Q as they are at X and L.

The Debye speed Eq. (5.25) needs the average of c_ν^{-3} over all directions, which we evaluate numerically by diagonalizing the Christoffel matrix on a fine grid of directions. With the three branches ordered by speed in each direction, the averages $\langle c_\nu^{-3}\rangle^{-1/3}$ are 9.00, 5.66 and $5.04\,\mathrm{km\,s^{-1}}$, and

$$\bar{c} = \Big[\frac{3}{\sum_\nu \langle c_\nu^{-3}\rangle}\Big]^{1/3} = 5.89\,\mathrm{km\,s^{-1}},$$

closer to the transverse speeds than to the longitudinal one: the longitudinal branch contributes only 9 % of $\sum_\nu\langle c_\nu^{-3}\rangle$. With $n_\mathrm{at}/\Omega = 8/a^3 = 4.994\times10^{22}\,\mathrm{cm^{-3}}$ (Sect. 3.1), Eq. (5.23) gives $Q_\mathrm{D} = 1.435\,\mathrm{Å}^{-1} = 1.24\,(2\pi/a)$, beyond the zone boundary since the sphere holds two wavevectors per cell, and

$$\omega_\mathrm{D} = \bar{c}Q_\mathrm{D} = 55.6\,\mathrm{meV}\ (13.4\,\mathrm{THz}), \qquad \Theta_\mathrm{D} = 645\,\mathrm{K}.$$

The Debye temperature measured from the heat capacity of silicon at low temperature is 645 K [16, Chap. 5, Table 1]. The agreement is expected: the heat capacity at low temperature measures the ω^2 coefficient of g_ph, and Eq. (5.25) takes this coefficient from the elastic constants. In the other convention, with one wavevector per cell in the sphere, the same speed gives $\Theta_\mathrm{D} = 512\,\mathrm{K}$.

Against the full spectrum, $\omega_\mathrm{D} = 55.6\,\mathrm{meV}$ sits 13 % below the optical phonon at 64.1 meV. Fig. 5.3(b) shows how the Debye density of states differs from the computed one. Below about 7 meV both follow ω^2, but with different coefficients: the computed g_ph is 1.3 to 1.4 times g_D between 5 and 7 meV, because the PBE sound speeds are softer than the measured ones. Above 7 meV the computed g_ph rises far above the Debye curve toward the transverse acoustic peak at 17 meV and falls below the Debye curve between 26 and 55 meV, and it leaves 38 % of all modes, most of the optical band, above ω_D, where the model has none. The low-frequency coefficient of the computed g_ph, fitted to ω^2 between 1 and 6 meV, implies $\bar{c} = 5.42\,\mathrm{km\,s^{-1}}$ and $\Theta_\mathrm{D} = 594\,\mathrm{K}$. The 8 % deficit of $\bar{c}$ comes from the transverse branches, which dominate $\bar{c}$ and are 7 % soft in the calculation. The deficit enters the ω^2 coefficient cubed, $(5.89/5.42)^3 = 1.28$; for a Debye temperature, the measured elastic constants are the better input.

5.3 Phonons

A silicon nucleus in a crystal cooled to a few millikelvin is not at rest. Its displacement along a cube axis has a root-mean-square value of 0.05 Å (Worked example 5.1), about a hundredth of the lattice constant, and at room temperature 0.077 Å. The classical normal modes of Sect. 5.2 cannot account for the zero-temperature value, since at zero temperature a classical crystal sits motionless at the bottom of its energy surface. They also miss that the energy of a mode comes in quanta. A dark-matter particle that scatters in a crystal at zero temperature cannot leave an arbitrary amount of energy in the lattice. It creates phonons, each with the energy $\omega_{\boldsymbol{Q}\nu}$ of one mode, out of a lattice in its ground state, and the amplitude for doing so is set by the zero-point motion of that mode. Quantizing the normal modes accounts for both the zero-point motion and the quanta of energy: it turns the crystal into a set of the harmonic oscillators of Sects. 2.3 and 2.5.

5.3.1 Normal coordinates

In the harmonic approximation the nuclei move on the quadratic energy surface Eq. (5.13), and the nuclear Hamiltonian of Eq. (5.6) becomes

$$H_\mathrm{N} = \sum_{\boldsymbol{R}\kappa} \frac{\boldsymbol{p}_{\boldsymbol{R}\kappa}^2}{2M_\kappa} + \frac{1}{2} \sum_{\boldsymbol{R}\kappa\alpha} \sum_{\boldsymbol{R}'\kappa'\beta} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R}, \boldsymbol{R}')\, u^\alpha_{\boldsymbol{R}\kappa} u^\beta_{\boldsymbol{R}'\kappa'}, \qquad [u^\alpha_{\boldsymbol{R}\kappa}, p^\beta_{\boldsymbol{R}'\kappa'}] = \mathrm{i}\,\delta_{\boldsymbol{R}\boldsymbol{R}'}\delta_{\kappa\kappa'}\delta_{\alpha\beta}, \tag{5.26}$$

where $\boldsymbol{p}_{\boldsymbol{R}\kappa} = -\mathrm{i}\nabla_{\boldsymbol{R}\kappa}$ is the momentum of the nucleus and the constant $E^\mathrm{ad}(0)$ has been dropped. The displacements and momenta are now operators. The classical solution of Sect. 5.2 already tells us which combinations of them decouple: mass-weighted plane waves in the cell index, projected on the polarization vectors. We define, for each $\boldsymbol{Q}$ in the first zone and each branch ν,

$$u_{\boldsymbol{Q}\nu} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}\kappa} \sqrt{M_\kappa}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^* \cdot \boldsymbol{u}_{\boldsymbol{R}\kappa}, \qquad p_{\boldsymbol{Q}\nu} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}\kappa} \frac{1}{\sqrt{M_\kappa}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot \boldsymbol{p}_{\boldsymbol{R}\kappa}, \tag{5.27}$$

the normal coordinate and the normal momentum of the mode $\boldsymbol{Q}\nu$. The transformation is inverted with the lattice sum Eq. (3.14), which gives $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}'-\boldsymbol{Q})\cdot\boldsymbol{R}} = N\delta_{\boldsymbol{Q}\boldsymbol{Q}'}$ for two wavevectors of the first zone, and with the completeness relation of Eq. (5.17):

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \frac{1}{\sqrt{NM_\kappa}} \sum_{\boldsymbol{Q}\nu} \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, u_{\boldsymbol{Q}\nu}, \qquad \boldsymbol{p}_{\boldsymbol{R}\kappa} = \sqrt{\frac{M_\kappa}{N}} \sum_{\boldsymbol{Q}\nu} \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\, \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, p_{\boldsymbol{Q}\nu}. \tag{5.28}$$

Inserting Eq. (5.28) into Eq. (5.27) returns $u_{\boldsymbol{Q}\nu}$, by the lattice sum and the orthonormality of the polarization vectors. The normal coordinates are not Hermitian. Since $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ is Hermitian, and $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ (Eq. (A.21)), the adjoint of Eq. (5.27) is the coordinate at $-\boldsymbol{Q}$, and the same holds for the momenta:

$$u^\dagger_{\boldsymbol{Q}\nu} = u_{-\boldsymbol{Q}\nu}, \qquad p^\dagger_{\boldsymbol{Q}\nu} = p_{-\boldsymbol{Q}\nu}.$$

The commutator of the normal coordinates and momenta follows from the commutator in Eq. (5.26). The masses cancel, the lattice sum sets $\boldsymbol{Q}' = \boldsymbol{Q}$, and orthonormality sets $\nu' = \nu$:

$$[u_{\boldsymbol{Q}\nu}, p_{\boldsymbol{Q}'\nu'}] = \frac{\mathrm{i}}{N}\sum_{\boldsymbol{R}\kappa} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}'-\boldsymbol{Q})\cdot\boldsymbol{R}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^* \cdot \boldsymbol{e}_{\kappa\nu'}(\boldsymbol{Q}') = \mathrm{i}\,\delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'},$$

while the coordinates commute among themselves, and so do the momenta. Each pair $(u_{\boldsymbol{Q}\nu}, p_{\boldsymbol{Q}\nu})$ is as canonical as the position and momentum of a single particle; the only novelty is that the pairs at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are adjoints of each other. In field theory the momentum conjugate to a mode of wavevector $\boldsymbol{k}$ carries $-\boldsymbol{k}$; here the phase $\mathrm{e}^{+\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ in the definition of $p_{\boldsymbol{Q}\nu}$ absorbs the sign, so that $p_{\boldsymbol{Q}\nu}$ is conjugate to $u_{\boldsymbol{Q}\nu}$ itself.

In these variables the Hamiltonian separates. In the kinetic term we write $\boldsymbol{p}^2_{\boldsymbol{R}\kappa} = \boldsymbol{p}^\dagger_{\boldsymbol{R}\kappa}\cdot\boldsymbol{p}_{\boldsymbol{R}\kappa}$ and insert Eq. (5.28); the sum over $\boldsymbol{R}$ sets the two wavevectors equal and the sum over κ the two branches. In the potential term we write the left displacement as $\boldsymbol{u}^\dagger_{\boldsymbol{R}\kappa}$ and expand both displacements. By translation invariance and Eq. (5.16), summing the force constants over $\boldsymbol{R}'$ with the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}'}$ gives $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}}\sqrt{M_\kappa M_{\kappa'}}\, D^{\mathrm{dyn}}_{\kappa\kappa'}(\boldsymbol{Q}')$. The dynamical matrix acting on its eigenvector returns $\omega^2_{\boldsymbol{Q}'\nu'}$, and the sums over $\boldsymbol{R}$ and κ again set $\boldsymbol{Q}' = \boldsymbol{Q}$ and $\nu' = \nu$, leaving

$$H_{\mathrm{N}} = \frac{1}{2}\sum_{\boldsymbol{Q}\nu}\Big(p^\dagger_{\boldsymbol{Q}\nu}p_{\boldsymbol{Q}\nu} + \omega^2_{\boldsymbol{Q}\nu}\, u^\dagger_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}\Big). \tag{5.29}$$

The $3n_{\mathrm{at}}N$ coupled nuclei have become $3n_{\mathrm{at}}N$ independent oscillators of unit mass, one for each mode. With Eq. (5.29) the Heisenberg equations are $\dot{u}_{\boldsymbol{Q}\nu} = p^\dagger_{\boldsymbol{Q}\nu}$ and $\dot{p}^\dagger_{\boldsymbol{Q}\nu} = -\omega^2_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}$, so that $\ddot{u}_{\boldsymbol{Q}\nu} = -\omega^2_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}$: the classical equation of each normal mode, as Sect. 5.2 anticipated. So far the calculation is classical; quantum mechanics enters through the commutator, which forbids $u_{\boldsymbol{Q}\nu}$ and $p_{\boldsymbol{Q}\nu}$ from vanishing together.

5.3.2 Phonon operators

For a single oscillator of unit mass the ladder operator is $(\omega/2)^{1/2}x + \mathrm{i}(2\omega)^{-1/2}p$. The normal modes need one change: since $u_{\boldsymbol{Q}\nu}$ and $p_{\boldsymbol{Q}\nu}$ are not Hermitian, the momentum enters through its adjoint,

$$b_{\boldsymbol{Q}\nu} = \Big(\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big)^{1/2}u_{\boldsymbol{Q}\nu} + \frac{\mathrm{i}}{(2\omega_{\boldsymbol{Q}\nu})^{1/2}}\,p^\dagger_{\boldsymbol{Q}\nu}, \qquad b^\dagger_{\boldsymbol{Q}\nu} = \Big(\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big)^{1/2}u_{-\boldsymbol{Q}\nu} - \frac{\mathrm{i}}{(2\omega_{\boldsymbol{Q}\nu})^{1/2}}\,p_{\boldsymbol{Q}\nu}. \tag{5.30}$$

With the commutator of the normal coordinates and its adjoint, $[u_{\boldsymbol{Q}\nu}, p^\dagger_{\boldsymbol{Q}'\nu'}] = \mathrm{i}\,\delta_{\boldsymbol{Q},-\boldsymbol{Q}'}\delta_{\nu\nu'}$, the two cross terms of $[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}]$ contribute $\frac{1}{2}\delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$ each. The two cross terms of $[b_{\boldsymbol{Q}\nu}, b_{\boldsymbol{Q}'\nu'}]$ are $\mp\frac{1}{2}\delta_{\boldsymbol{Q},-\boldsymbol{Q}'}\delta_{\nu\nu'}$ and cancel, because $\omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}$. Hence

$$[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = \delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}, \qquad [b_{\boldsymbol{Q}\nu}, b_{\boldsymbol{Q}'\nu'}] = [b^\dagger_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = 0,$$

the boson algebra Eq. (2.35) with $\gamma = (\boldsymbol{Q}, \nu)$. Adding Eq. (5.30) at $\boldsymbol{Q}$ to its adjoint at $-\boldsymbol{Q}$, and subtracting, inverts the definition:

$$u_{\boldsymbol{Q}\nu} = \frac{b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}}{(2\omega_{\boldsymbol{Q}\nu})^{1/2}}, \qquad p^\dagger_{\boldsymbol{Q}\nu} = -\mathrm{i}\Big(\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big)^{1/2}(b_{\boldsymbol{Q}\nu} - b^\dagger_{-\boldsymbol{Q}\nu}).$$

A normal coordinate contains the annihilator of the mode $\boldsymbol{Q}$ and the creator of the mode $-\boldsymbol{Q}$. Inserting these forms into Eq. (5.29) gives, for one bracket,

$$p^\dagger_{\boldsymbol{Q}\nu}p_{\boldsymbol{Q}\nu}+\omega^2_{\boldsymbol{Q}\nu}u^\dagger_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}=\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big[(b_{\boldsymbol{Q}\nu}-b^\dagger_{-\boldsymbol{Q}\nu})(b^\dagger_{\boldsymbol{Q}\nu}-b_{-\boldsymbol{Q}\nu})+(b^\dagger_{\boldsymbol{Q}\nu}+b_{-\boldsymbol{Q}\nu})(b_{\boldsymbol{Q}\nu}+b^\dagger_{-\boldsymbol{Q}\nu})\Big].$$

The products of two annihilators, $-b_{\boldsymbol{Q}\nu}b_{-\boldsymbol{Q}\nu}+b_{-\boldsymbol{Q}\nu}b_{\boldsymbol{Q}\nu}$, cancel since the two operators commute, and so do the products of two creators. The four remaining terms are $b_{\boldsymbol{Q}\nu}b^\dagger_{\boldsymbol{Q}\nu}+b^\dagger_{\boldsymbol{Q}\nu}b_{\boldsymbol{Q}\nu}$ and the same pair at $-\boldsymbol{Q}$, which in the sum over all $\boldsymbol{Q}$ repeats the pair at $\boldsymbol{Q}$. With $bb^\dagger=b^\dagger b+1$,

$$H_{\mathrm{N}}=\sum_{\boldsymbol{Q}\nu}\omega_{\boldsymbol{Q}\nu}\Big(b^\dagger_{\boldsymbol{Q}\nu}b_{\boldsymbol{Q}\nu}+\frac{1}{2}\Big). \tag{5.31}$$

H_{N} has the spectrum of independent oscillators. The eigenstates are labeled by one occupation number $n_{\boldsymbol{Q}\nu}=0,1,2,\dots$ per mode, $|\{n_{\boldsymbol{Q}\nu}\}\rangle=\prod_{\boldsymbol{Q}\nu}(n_{\boldsymbol{Q}\nu}!)^{-1/2}(b^\dagger_{\boldsymbol{Q}\nu})^{n_{\boldsymbol{Q}\nu}}|0\rangle$, with energy $\sum_{\boldsymbol{Q}\nu}\omega_{\boldsymbol{Q}\nu}(n_{\boldsymbol{Q}\nu}+\frac{1}{2})$. A *phonon* is one quantum of a normal mode: the state $b^\dagger_{\boldsymbol{Q}\nu}|0\rangle$ lies $\omega_{\boldsymbol{Q}\nu}$ above the ground state, and $\omega_{\boldsymbol{Q}\nu}$ of Sect. 5.2 is the energy of the phonon. Phonons are bosons, since any number of them can occupy a mode, and the number of phonons is not conserved, so their thermal occupation has no chemical potential (Sect. 2.5). Each phonon also carries a crystal momentum. A lattice translation $T_{\boldsymbol{R}}$, in the sense of Eq. (3.11), relabels the displacements, $\boldsymbol{u}_{\boldsymbol{R}'\kappa}\to\boldsymbol{u}_{\boldsymbol{R}'-\boldsymbol{R},\kappa}$. By Eqs. (5.27) and (5.30) the translation multiplies $b^\dagger_{\boldsymbol{Q}\nu}$ by $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, while the ground state is invariant. A state with one phonon in the mode $\boldsymbol{Q}\nu$ has crystal momentum $\boldsymbol{Q}$, defined up to a reciprocal-lattice vector, exactly as an electron in a Bloch state has $\boldsymbol{k}$ (Sect. 3.1).

The ground state $|0\rangle$ holds no phonons, but its energy is not zero. Every mode keeps $\frac{1}{2}\omega_{\boldsymbol{Q}\nu}$, and in silicon the zero-point energy $\frac{1}{2}\int\mathrm{d}\omega\,g_{\mathrm{ph}}(\omega)\,\omega$ is 123 meV per cell for the computed density of states of Fig. 5.3: $\frac{1}{2}\times 40.9\,\mathrm{meV}$, the mean phonon energy, times six modes. The zero-point energy depends on the cell volume through the frequencies, a dependence that Sect. 5.6 takes up. In the Heisenberg picture the phonon operators of Eq. (5.31) evolve as $b_{\boldsymbol{Q}\nu}(t)=b_{\boldsymbol{Q}\nu}\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$, and every displacement is a superposition of oscillations at the mode frequencies, as in the classical crystal.

The displacement of a nucleus follows from Eq. (5.28) once the normal coordinate is written with the phonon operators. For the momentum, the term $-\boldsymbol{Q}$ of the sum is relabeled with $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q})^*=\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$. The displacement and the momentum have the same structure:

Phonon expansion of the displacement

$$\boldsymbol{u}_{\boldsymbol{R}\kappa}=\sum_{\boldsymbol{Q}\nu}\frac{1}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\,\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,B_{\boldsymbol{Q}\nu},\qquad B_{\boldsymbol{Q}\nu}=b_{\boldsymbol{Q}\nu}+b^\dagger_{-\boldsymbol{Q}\nu}, \tag{5.32}$$

$$\boldsymbol{p}_{\boldsymbol{R}\kappa}=-\mathrm{i}\sum_{\boldsymbol{Q}\nu}\Big(\frac{M_\kappa\omega_{\boldsymbol{Q}\nu}}{2N}\Big)^{1/2}\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,(b_{\boldsymbol{Q}\nu}-b^\dagger_{-\boldsymbol{Q}\nu}),$$

with $[b_{\boldsymbol{Q}\nu},b^\dagger_{\boldsymbol{Q}'\nu'}]=\delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$, $\sum_\kappa|\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2=1$ and $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q})=\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$. The harmonic Hamiltonian is Eq. (5.31).

This expansion is Eq. (A.20), the convention of Ashcroft and Mermin [10, Chap. 23] and of Giustino [4, Eqs. (16)–(21)], with the phonon wavevector written $\boldsymbol{Q}$. The *displacement field* $B_{\boldsymbol{Q}\nu}$ obeys $B^\dagger_{\boldsymbol{Q}\nu}=B_{-\boldsymbol{Q}\nu}$, so that the terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ of Eq. (5.32) are adjoints of each other and the displacement is Hermitian. It is the operator that every coupling linear in the displacements contains: the electron–phonon Hamiltonian Eq. (A.24) and the single-phonon response of a nucleus to dark matter.

5.3.3 The zero-point length

Each term of Eq. (5.32) has three factors: the pattern $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ of Sect. 5.2, the operator $B_{\boldsymbol{Q}\nu}$, which changes the number of phonons by one, and a length that fixes how far one quantum moves the nuclei. We write that length as the *zero-point length* of the mode,

$$\ell_{\boldsymbol{Q}\nu} = \frac{1}{\sqrt{2M_{\mathrm{cell}}\,\omega_{\boldsymbol{Q}\nu}}}, \tag{5.33}$$

with the cell mass $M_{\mathrm{cell}} = \sum_\kappa M_\kappa$ and the factor $1/\sqrt{N}$ kept outside (Eq. (A.23)). For an acoustic mode at small $\boldsymbol{Q}$, where $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ (Sect. 5.2), the amplitude in Eq. (5.32) is $\ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ for every nucleus: the whole cell moves as one particle of mass M_{cell}. For other modes $\ell_{\boldsymbol{Q}\nu}$ sets the scale, and the polarization vector distributes the motion over the nuclei of the cell. The name refers to the ground state, in which $\langle 0|B_{\boldsymbol{Q}\nu}B^\dagger_{\boldsymbol{Q}\nu}|0\rangle = \langle 0|b_{\boldsymbol{Q}\nu}b^\dagger_{\boldsymbol{Q}\nu}|0\rangle = 1$: $\ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ is the zero-point amplitude that one mode contributes to each nucleus.

For silicon, with M_{cell} two silicon masses (52.33 GeV, Sect. 1.5), the zone-center optical phonon at 64.1 meV has $\ell_{\boldsymbol{Q}\nu} = 0.024\,\text{Å}$. This value is half the zero-point length $\ell_{\mathrm{zp}} = 0.048\,\text{Å}$ of the bond coordinate in Fig. 5.2(c), as it must be. The bond coordinate is the change of a bond length, and in the optical mode the two atoms of the cell move against each other with equal amplitude, $\boldsymbol{e}_{\kappa\nu} = \pm\hat{\boldsymbol{e}}/\sqrt{2}$, each by half of that change. The transverse acoustic phonon at X, 18.6 meV, has $\ell_{\boldsymbol{Q}\nu} = 0.045\,\text{Å}$, and an acoustic phonon of 5 meV has 0.086 Å: soft modes move the nuclei farther, as $\omega_{\boldsymbol{Q}\nu}^{-1/2}$. The factor $1/\sqrt{N}$ makes the motion of any single mode unobservably small in a macroscopic crystal. A kilogram of silicon has $N = 1.07\times10^{25}$ cells (Sect. 3.1), and one optical phonon moves each of its nuclei by $7\times10^{-15}\,\text{Å}$. The mean-square displacement sums shares of order $\ell^2_{\boldsymbol{Q}\nu}/N$ from all $3n_{\mathrm{at}}N$ modes; the sum is finite, and we compute it below.

5.3.4 Emission and absorption

The matrix elements of the displacement between phonon states follow from Eq. (5.32) and the boson relations Eq. (2.34). Only two terms of the sum connect a state with $n_{\boldsymbol{Q}\nu}$ phonons in the mode $\boldsymbol{Q}\nu$ to a state that differs from it in that mode alone. The term $\boldsymbol{Q}$, through $b_{\boldsymbol{Q}\nu}$, removes a phonon. The term $-\boldsymbol{Q}$, through $b^\dagger_{\boldsymbol{Q}\nu}$, adds one, with the pattern $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q})\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$. Hence

$$\begin{aligned}
\langle n_{\boldsymbol{Q}\nu}+1|\,\boldsymbol{u}_{\boldsymbol{R}\kappa}\,|n_{\boldsymbol{Q}\nu}\rangle &= \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\,\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\,\sqrt{n_{\boldsymbol{Q}\nu}+1},\\
\langle n_{\boldsymbol{Q}\nu}-1|\,\boldsymbol{u}_{\boldsymbol{R}\kappa}\,|n_{\boldsymbol{Q}\nu}\rangle &= \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\,\sqrt{n_{\boldsymbol{Q}\nu}},
\end{aligned} \tag{5.34}$$

with the occupations of all other modes unchanged. In thermal equilibrium the squared matrix elements are averaged over the Gibbs state of Sect. 2.5, and the occupation becomes the mean *phonon occupation*

$$n_{\boldsymbol{Q}\nu} = \langle b^\dagger_{\boldsymbol{Q}\nu}b_{\boldsymbol{Q}\nu}\rangle = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu}),$$

by Eq. (2.69). Emission then has the weight $n_{\boldsymbol{Q}\nu}+1$ and absorption the weight $n_{\boldsymbol{Q}\nu}$, the factors $n_{\boldsymbol{Q}\nu}+\frac{1}{2}\pm\frac{1}{2}$ of Eq. (A.27), and their ratio $\mathrm{e}^{\beta\omega_{\boldsymbol{Q}\nu}}$ is detailed balance for the mode. Each mode of the crystal is the oscillator of Worked example 2.7, with the coupling λ_{osc} replaced by the projection of the dark-matter vertex on $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})/\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}$. The numbers of that example carry over. At room temperature the optical phonon of silicon has $n_{\boldsymbol{Q}\nu} = 0.09$ and the transverse acoustic phonon at X has 0.95. In a cryogenic detector every mode above a few

tens of μeV is empty, absorption is absent, and emission into an empty mode, with weight one, is the whole signal.

Consider an interaction of dark matter with the nuclei of one species κ through the nuclear density $n_\kappa(\boldsymbol{q})$ of Sect. 2.3, which depends on the positions of the nuclei through $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, with $\boldsymbol{q}$ the momentum transfer. With $\boldsymbol{X} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$ (Eq. (5.1)),

$$n_\kappa(\boldsymbol{q}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}} \simeq \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}\big(1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}\big).$$

The expansion holds as long as $\langle(\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa})^2\rangle = q^2\langle u_x^2\rangle \ll 1$, that is, for q well below 26 keV in silicon at room temperature (with the value of $\langle u_x^2\rangle$ given below); the full exponential is the subject of Sect. 5.5. The term 1 leaves the lattice unchanged. The term linear in $\boldsymbol{u}$, with Eq. (5.34) and the lattice sum Eq. (3.14), emits one phonon with the amplitude

$$\langle n_{\boldsymbol{Q}\nu}+1|\, n_\kappa(\boldsymbol{q})\, |n_{\boldsymbol{Q}\nu}\rangle = \mathrm{i}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \frac{\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*}{\sqrt{2M_\kappa\omega_{\boldsymbol{Q}\nu}}}\, \sqrt{N(n_{\boldsymbol{Q}\nu}+1)}\, \sum_{\boldsymbol{G}} \delta_{\boldsymbol{q},\boldsymbol{Q}+\boldsymbol{G}}.$$

The lattice sum conserves crystal momentum, as every later phonon rate does: the emitted phonon carries the momentum transfer, up to a reciprocal-lattice vector, so that each $\boldsymbol{q}$ excites one wavevector $\boldsymbol{Q}$ in each branch. The squared amplitude is also proportional to N, as it must be for a target whose rate grows with its size. Per nucleus, the weight of the branch ν is $|\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2(n_{\boldsymbol{Q}\nu}+1)/(2M_\kappa\omega_{\boldsymbol{Q}\nu})$: the projection of the momentum transfer on the polarization, which favors longitudinal modes at small $\boldsymbol{Q}$, divided by the energy of the mode, which favors soft ones. Absorption of a phonon $-\boldsymbol{q}$ follows in the same way with $n_{\boldsymbol{Q}\nu}$, and with several species the amplitudes of the nuclei of a cell add with their phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$.

5.3.5 Zero-point and thermal motion

Each Cartesian component of a nuclear displacement behaves roughly like an oscillator of mass M_κ at a typical phonon frequency $\bar\omega$. In the ground state this oscillator has $\langle x^2\rangle = 1/(2M_\kappa\bar\omega)$, and at a temperature well above $\bar\omega$ equipartition, $M_\kappa\bar\omega^2\langle x^2\rangle = T$, leads to $T/(M_\kappa\bar\omega^2)$. For silicon with $\bar\omega \sim 30$ meV the zero-point estimate is $\langle u_x^2\rangle^{1/2} \sim 0.05$ Å, the number of Worked example 5.1, and room temperature, where $T = 25.9$ meV is close to $\bar\omega$, should roughly double $\langle u_x^2\rangle$. The question is what $\bar\omega$ means for a crystal with a whole spectrum of modes.

The exact statement is a sum over modes. In the Gibbs state the occupations are independent, and the only nonzero averages of two displacement fields pair a mode with its partner at $-\boldsymbol{Q}$:

$$\langle B_{\boldsymbol{Q}\nu}B_{\boldsymbol{Q}'\nu'}\rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\delta_{\nu\nu'}\big(\langle b_{\boldsymbol{Q}\nu}b^\dagger_{\boldsymbol{Q}\nu}\rangle + \langle b^\dagger_{-\boldsymbol{Q}\nu}b_{-\boldsymbol{Q}\nu}\rangle\big) = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\delta_{\nu\nu'}\,(2n_{\boldsymbol{Q}\nu}+1),$$

by Eq. (2.69) and $n_{-\boldsymbol{Q}\nu} = n_{\boldsymbol{Q}\nu}$. When Eq. (5.32) is inserted twice, the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ cancel, $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ supplies the conjugate, and the *displacement tensor* of the nucleus κ is

$$B_\kappa^{\alpha\beta} = \langle u^\alpha_{\boldsymbol{R}\kappa}u^\beta_{\boldsymbol{R}\kappa}\rangle = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{2n_{\boldsymbol{Q}\nu}+1}{2M_\kappa\,\omega_{\boldsymbol{Q}\nu}}\, e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa\nu}(\boldsymbol{Q})^*, \qquad 2n_{\boldsymbol{Q}\nu}+1 = \coth\frac{\omega_{\boldsymbol{Q}\nu}}{2T}, \tag{5.35}$$

the same in every cell. The terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are complex conjugates of each other, so B_κ is real and symmetric; it is also positive, since every term of its diagonal is positive. The average is over the thermal state, and at $T = 0$ over the ground state: B_κ then measures the *zero-point motion*, the spread of the nuclear wavefunction that no cooling removes. The two limits of B_κ are the two limits of the estimate. At $T = 0$, $2n+1 = 1$, and each mode contributes

$1/(2M_\kappa\omega_{\boldsymbol{Q}\nu})$ times the weight $|\boldsymbol{e}_{\kappa\nu}|^2/N$. For T above every phonon energy, $2n+1 \to 2T/\omega_{\boldsymbol{Q}\nu}$, and each mode contributes its equipartition value $T/(M_\kappa\omega_{\boldsymbol{Q}\nu}^2)$, growing linearly in T. The acoustic modes have $\omega_{\boldsymbol{Q}\nu} \to 0$, but they are few: in three dimensions their number per unit frequency falls as ω^2 (Eq. (5.22)), and the sum stays finite at every temperature.

At a site of cubic symmetry, such as the tetrahedral sites of the diamond and zinc-blende structures, a symmetric second-rank tensor that is invariant under the site group is a multiple of the identity. At these sites $B_\kappa = \langle u_x^2\rangle\,\mathbf{1}$, and the mean-square displacement $\langle u^2\rangle = \operatorname{tr} B_\kappa = 3\langle u_x^2\rangle$; crystallographers quote $\langle u_x^2\rangle$, or the isotropic displacement parameter $8\pi^2\langle u_x^2\rangle$. If, in addition, the n_{at} nuclei of the cell are of one species and related by symmetry, as in silicon and germanium, all n_{at} nuclei have the same trace of B_κ. We can then average the trace of Eq. (5.35) over κ, the normalization $\sum_\kappa |\boldsymbol{e}_{\kappa\nu}|^2 = 1$ removes the polarization vectors, and the sum over modes becomes an integral over the density of states Eq. (5.21):

$$\langle u^2\rangle = \int_0^\infty \mathrm{d}\omega\, \frac{g_{\mathrm{ph}}(\omega)}{n_{\mathrm{at}}}\, \frac{\coth(\omega/2T)}{2M_\kappa\,\omega}. \tag{5.36}$$

The density of states enters per atom, $g_{\mathrm{ph}}/n_{\mathrm{at}}$, normalized to three. For a crystal of several species, such as GaAs, the same steps keep the weight of each nucleus separately, and $g_{\mathrm{ph}}/n_{\mathrm{at}}$ is replaced by the atom-projected density of states $g_{\mathrm{ph},\kappa}(\omega) = N^{-1}\sum_{\boldsymbol{Q}\nu} |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2\delta(\omega-\omega_{\boldsymbol{Q}\nu})$, also normalized to three. Eq. (5.36) makes the estimate precise: the typical frequency at $T=0$ is the inverse of the mean of $1/\omega$ over the spectrum, $\langle u^2\rangle = (3/2M_\kappa)\langle\omega^{-1}\rangle$, and at high temperature the inverse root of the mean of $1/\omega^2$, $\langle u^2\rangle = (3T/M_\kappa)\langle\omega^{-2}\rangle$. Both means are weighted toward low frequencies, the second more strongly.

In the Debye model, $g_{\mathrm{ph}}/n_{\mathrm{at}} = 9\omega^2/\omega_{\mathrm{D}}^3$ below ω_{D} (Eq. (5.24)). Splitting $\coth = 1 + 2n_{\mathrm{B}}$ into its zero-point and thermal parts and writing $x = \omega/T$ gives

$$\langle u^2\rangle = \frac{9}{4M_\kappa\omega_{\mathrm{D}}}\Big[1 + 4\Big(\frac{T}{\omega_{\mathrm{D}}}\Big)^2 \int_0^{\omega_{\mathrm{D}}/T} \frac{x\,\mathrm{d}x}{\mathrm{e}^x - 1}\Big], \tag{5.37}$$

from which the Debye–Waller exponent Eq. (5.49) follows. The zero-point value $9/(4M_\kappa\omega_{\mathrm{D}})$ corresponds to $\bar\omega = 2\omega_{\mathrm{D}}/3$ in the estimate. At high temperature the integral is $y - y^2/4 + y^3/36 - \cdots$ with $y = \omega_{\mathrm{D}}/T$. The bracket then becomes $4T/\omega_{\mathrm{D}} + \omega_{\mathrm{D}}/(9T) + \cdots$, and $\langle u^2\rangle = (9T/M_\kappa\omega_{\mathrm{D}}^2)[1 + \omega_{\mathrm{D}}^2/36T^2 + \cdots]$, the equipartition value with the zero-point motion as its first correction. The thermal part equals the zero-point part when the bracket is two, at $T = 0.44\,\Theta_{\mathrm{D}}$. For silicon $0.44\,\Theta_{\mathrm{D}}$ is 284 K: at room temperature the nuclei of silicon move as much because of their zero-point motion as because of the heat.

For crystals of similar bonding, the zero-point value of Eq. (5.37) $\propto 1/(M_\kappa\omega_{\mathrm{D}})$ falls with the mass, since $\omega_{\mathrm{D}} \propto M_\kappa^{-1/2}$ at fixed force constants: heavier nuclei are better localized. The high-temperature value $\propto T/(M_\kappa\omega_{\mathrm{D}}^2)$ depends on the force constants alone: classically, the thermal stretch of a spring does not depend on the mass that hangs on it. A heavier and softer crystal has less zero-point motion and, at room temperature, more thermal motion, as germanium shows in Table 5.1.

In practice B_κ is computed from DFPT phonons. The sum Eq. (5.35) runs over a uniform grid of wavevectors, with the frequencies and polarization vectors interpolated from the force constants of a coarser grid, or Eq. (5.36) is integrated over a density of states computed on a fine grid. A finite grid misses the small acoustic frequencies near Γ, whose terms grow as $1/\omega$ at $T=0$ and $1/\omega^2$ at $T>0$, and Eq. (25.18) adds these terms back analytically. For silicon the 48^3 density of states of Fig. 5.3 leads to $\langle u_x^2\rangle^{1/2} = 0.0495$ Å at zero temperature, 0.0799 Å at 293 K and 0.0807 Å at 300 K. These values and the values from the 4^3 grid of Worked example 5.1 differ by 2 % at most, well inside the error that the functional itself makes (Worked example 5.4).

Worked example 5.4: Zero-point and room-temperature motion of silicon

We evaluate Eq. (5.37) for silicon with the natural isotope mixture, $M_\kappa = 28.0855\,\mathrm{u} = 26.16\,\mathrm{GeV}$. We take the Debye frequency from the measured elastic constants, $\omega_\mathrm{D} = 55.6\,\mathrm{meV}$ (Worked example 5.3). We convert with $\hbar c = 1973.27\,\mathrm{eV\,\AA}$.

Zero temperature. The product $M_\kappa\omega_\mathrm{D} = 1.455 \times 10^9\,\mathrm{eV}^2$, and

$$\langle u^2\rangle_{T=0} = \frac{9}{4M_\kappa\omega_\mathrm{D}} = 1.546 \times 10^{-9}\,\mathrm{eV}^{-2} = 6.02 \times 10^{-3}\,\mathrm{\AA}^2, \qquad \langle u_x^2\rangle_{T=0}^{1/2} = 0.0448\,\mathrm{\AA}.$$

Room temperature. At $T = 300\,\mathrm{K} = 25.85\,\mathrm{meV}$, $\omega_\mathrm{D}/T = 2.152$, and the integral of Eq. (5.37), evaluated numerically, is $\int_0^{2.152} x\,\mathrm{d}x/(\mathrm{e}^x - 1) = 1.259$. The bracket is $1 + 4 \times 0.4648^2 \times 1.259 = 2.088$, so that

$$\langle u^2\rangle_{300\,\mathrm{K}} = 1.257 \times 10^{-2}\,\mathrm{\AA}^2, \qquad \langle u_x^2\rangle_{300\,\mathrm{K}}^{1/2} = 0.0647\,\mathrm{\AA}.$$

Thermal phonons double the mean-square displacement, as the estimate suggested, and multiply the amplitude by $\sqrt{2.088} = 1.44$. Since T lies below $\omega_\mathrm{D}/2$, equipartition alone, $9T/(M_\kappa\omega_\mathrm{D}^2)$, falls 11 % short, at $\langle u_x^2\rangle = 3.73 \times 10^{-3}\,\mathrm{\AA}^2$; the first zero-point correction to equipartition, the factor $1 + \omega_\mathrm{D}^2/36T^2 = 1.129$, recovers the full value to 0.5 %.

Against the full spectrum and the measurement. A lattice-dynamical fit to the measured phonons, the reference of Worked example 5.1, gives $\langle u_x^2\rangle = 5.941 \times 10^{-3}\,\mathrm{\AA}^2$ at 293 K [9]. The Debye model predicts $4.11 \times 10^{-3}\,\mathrm{\AA}^2$ at that temperature. Eq. (5.36) with the computed density of states of Fig. 5.3 gives $\langle u_x^2\rangle = 2.45 \times 10^{-3}\,\mathrm{\AA}^2$ at zero temperature and $6.38 \times 10^{-3}\,\mathrm{\AA}^2$ at 293 K, 7 % above the measurement, as expected from the transverse acoustic branches, which are 8 % soft in this calculation (Sect. 5.2) and enter as $1/\omega^2$. Against this spectrum the Debye model is low by 18 % at zero temperature and by about a third at room temperature, and the reason is the flat transverse acoustic branches. In the computed spectrum the modes below 20 meV are 22 % of all modes but carry 45 % of the zero-point $\langle u^2\rangle$, since they enter with the weight $1/\omega$. In the Debye spectrum the modes below 20 meV are 5 % of all modes and carry 13 % of $\langle u^2\rangle$. The typical frequencies of the estimate make the same point. For the computed spectrum, $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$ and $\langle\omega^{-2}\rangle^{-1/2} = 25.2\,\mathrm{meV}$, against $2\omega_\mathrm{D}/3 = 37.1\,\mathrm{meV}$ and $\omega_\mathrm{D}/\sqrt{3} = 32.1\,\mathrm{meV}$ in the model. Elastic constants fix the low-frequency density of states exactly, but the displacement is weighted toward the zone-boundary modes of the transverse branches, which the model pushes up to ω_D.

The Debye model reproduces the zero-point and room-temperature motion of silicon to 20 % to 35 %, an accuracy that is enough for an estimate of where the Debye–Waller suppression sets in. A rate that depends on B_κ at the ten-percent level needs the full density of states, and at that level the softness of the computed transverse branches matters as well.

Table 5.1 puts silicon next to the two contrast materials of the book. Germanium has the same structure and bonding, weaker bonds and a nucleus 2.6 times heavier. The optical phonon of germanium, at 37.6 meV, lies below the 39.9 meV that the mass alone would give, 64.1 meV scaled by the square root of the mass ratio, and its elastic constants imply $\Theta_\mathrm{D} = 373\,\mathrm{K}$. Germanium has less zero-point motion than silicon and more room-temperature motion, as the trends of Eq. (5.37) require. At 300 K, $T = 0.80\,\Theta_\mathrm{D}$ for germanium, and the thermal part is 2.4 times the zero-point part. The transverse acoustic branches of germanium are flat near the zone boundary as well, at 9.9 meV at X, a third of ω_D as in silicon, and the Debye values in the table should be read as low by a similar fraction. GaAs, with nuclei of nearly

Table 5.1: Phonon energies and root-mean-square displacements $\langle u_x^2 \rangle^{1/2}$ along one Cartesian axis, in Å, of silicon, germanium and GaAs. Phonon energies: measured by inelastic neutron scattering, silicon at room temperature [1, 12], germanium at 80 K [17] as tabulated in Ref. [13, Table IV], and GaAs [18] as tabulated in Ref. [19, Table I]; for GaAs the transverse (TO) and longitudinal (LO) optical energies at Γ. Debye temperatures from the room-temperature elastic constants and lattice constants with the angular average of Eq. (5.25): silicon from Worked example 5.3, germanium from $C_{11} = 131$, $C_{12} = 49$, $C_{44} = 68$ GPa [20] (as tabulated in Ref. [13, Table III]) and the room-temperature $a = 5.658$ Å of Table 3.1. Debye columns: Eq. (5.37) with the natural isotope masses. DFPT columns: Eq. (5.36) at 0 and 300 K with the PBE density of states of Fig. 5.3 for silicon, on a 48^3 grid of wavevectors. Germanium has no DFPT entry because PBE leaves it nearly metallic at the measured lattice constant, with a direct gap of 0.05 eV and the wrong band order, and metallic at the PBE lattice constant. Entries marked † are computed for GaAs with the same method (PBE, $a = 5.638$ Å, a 4^3 grid of wavevectors and the dipole term of Sect. 5.4): the transverse acoustic energy at X, and the displacements of gallium (first) and arsenic (second) from the atom-projected densities of states $g_{\mathrm{ph},\kappa}$ on a 32^3 grid. The germanium energies are tabulated to 0.1 THz, 0.4 meV. A dash marks a value not given here.

	Phonon energy (meV)			Debye		DFPT	
	optical at Γ	TA at X	Θ_D (K)	0 K	300 K	0 K	300 K
Si	64.1	18.6	645	0.045	0.065	0.049	0.081
Ge	37.6	9.9	373	0.037	0.067	–	–
GaAs	33.6, 36.4	10.5†	–	–	–	0.040, 0.039†	0.080, 0.078†

equal mass, lies close to germanium, but it is polar: its longitudinal optical phonon at Γ lies 2.7 meV above the transverse one, the LO–TO splitting of Sect. 5.4.

5.4 Polar crystals

At the center of the zone the three optical phonons of silicon share one energy, 64.1 meV. In GaAs they do not. Two transverse optical (TO) modes lie at 33.6 meV and the longitudinal one (LO) at 36.4 meV [18], although the three move the nuclei in the same way, gallium against arsenic, and differ only in the direction of that motion relative to $\boldsymbol{Q}$. The three energies differ because gallium and arsenic are not equivalent. When the two sublattices move against each other, every cell acquires an electric dipole. In a longitudinal wave the dipoles add up to a macroscopic polarization, whose field pulls the ions back and stiffens the mode; in a transverse wave they produce no macroscopic field. For dark matter this dipole makes a polar crystal a useful target. A dark photon couples to the dipoles coherently, cell by cell, and so does a dark-matter particle that couples to charge through a light dark photon. The charge that sets the size of the dipole is the Born effective charge.

5.4.1 Born effective charges

The fields that matter here vary over many cells. The microscopic field $\boldsymbol{E}$ of Sect. 1.4 oscillates within each cell, and a long wave sees its average over a cell, the *macroscopic electric field* $\boldsymbol{\mathcal{E}}$, which is the $\boldsymbol{G} = 0$ component of the field in the language of Sect. 4.4. The density of dipole moment, averaged over a cell in the same way, is the *polarization* $\boldsymbol{P}$, and in the Heaviside–Lorentz units of the book the displacement field is $\boldsymbol{D} = \boldsymbol{\mathcal{E}} + \boldsymbol{P}$ (Appendix A.2), so that the electronic dielectric constant of Sect. 4.4 is $\varepsilon_\infty = 1 + \partial P / \partial \mathcal{E}$ at clamped nuclei.

Now let each sublattice κ of the crystal move rigidly by $\boldsymbol{u}_\kappa$, the zone-center displacement of Eq. (5.12), in a uniform field $\boldsymbol{\mathcal{E}}$. The electrons follow the displacements and the field adiabatically. The energy per cell at fixed field, $E^{\mathrm{ad}}_{\mathcal{E}}$, reduces to the energy surface $E^{\mathrm{ad}}_{\mathrm{cell}}$ of Sect. 5.1 at $\boldsymbol{\mathcal{E}} = 0$. To second order in the displacements and the field, $E^{\mathrm{ad}}_{\mathcal{E}}$ is

$$E^{\mathrm{ad}}_{\mathcal{E}}(u, \boldsymbol{\mathcal{E}}) = \frac{1}{2} \sum_{\kappa\kappa'} \boldsymbol{u}_\kappa \cdot C^{\mathrm{an}}_{\kappa\kappa'} \cdot \boldsymbol{u}_{\kappa'} - e \sum_{\kappa} \boldsymbol{u}_\kappa \cdot Z^*_\kappa \cdot \boldsymbol{\mathcal{E}} - \frac{\Omega}{2}\, \boldsymbol{\mathcal{E}} \cdot (\varepsilon_\infty - 1) \cdot \boldsymbol{\mathcal{E}}, \tag{5.38}$$

with Ω the cell volume. Each coefficient is defined as a second derivative of $E^{\mathrm{ad}}_{\mathcal{E}}$. $C^{\mathrm{an}}_{\kappa\kappa'} = \sum_{\boldsymbol{R}} \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R})$ is the zone-center block of the force constants of Sect. 5.2, taken at zero macroscopic field; ε_∞ is the electronic dielectric tensor; and the mixed derivative defines the *Born effective charge* tensor Z^*_κ of each sublattice. The first derivatives are the force on a nucleus and the polarization, $\boldsymbol{F}_\kappa = -\partial E^{\mathrm{ad}}_{\mathcal{E}}/\partial \boldsymbol{u}_\kappa$ and $\boldsymbol{P} = -\Omega^{-1} \partial E^{\mathrm{ad}}_{\mathcal{E}}/\partial \boldsymbol{\mathcal{E}}$:

$$\boldsymbol{F}_\kappa = -\sum_{\kappa'} C^{\mathrm{an}}_{\kappa\kappa'} \cdot \boldsymbol{u}_{\kappa'} + e\, Z^*_\kappa \cdot \boldsymbol{\mathcal{E}}, \qquad \boldsymbol{P} = \frac{e}{\Omega} \sum_{\kappa} \boldsymbol{u}_\kappa \cdot Z^*_\kappa + (\varepsilon_\infty - 1) \cdot \boldsymbol{\mathcal{E}}. \tag{5.39}$$

The force identifies the Born charge as the force on nucleus κ per unit uniform field. The polarization identifies the same tensor as the dipole per cell created by a unit displacement:

$$e\, Z^{*,\alpha\beta}_\kappa = \left. \frac{\partial F^\alpha_\kappa}{\partial \mathcal{E}^\beta} \right|_{\boldsymbol{u}=0} = \Omega \left. \frac{\partial P^\beta}{\partial u^\alpha_\kappa} \right|_{\boldsymbol{\mathcal{E}}=0}. \tag{5.40}$$

The two derivatives are equal because both are the same mixed second derivative of $E^{\mathrm{ad}}_{\mathcal{E}}$; the equality is a Maxwell relation. The first Cartesian index of Z^*_κ belongs to the force, or to the displacement, and the second to the field, or to the polarization. Some codes and papers store the transpose [21, Eqs. (89)–(91)]; the choice of convention matters only at sites of low symmetry, since at a cubic site Z^*_κ is a multiple of the identity.

The Born charge measures the charge that moves with a nucleus. A field acts on the bare nucleus with the force $Z_\kappa e \boldsymbol{\mathcal{E}}$, and on its electrons, which are pulled the other way and drag on the nucleus through the forces of Eq. (5.10). If the electrons of each atom followed their nucleus rigidly, Z^*_κ would be the net charge of the atom. They do not: as the sublattices move, the bonds change length and electronic charge flows along them, and this flow adds to the dipole. A Born charge is therefore a dynamical quantity. It need not be an integer, and it differs from any charge one might assign to an atom by dividing up the static density, a division that has no unique answer.

The Born charges of a neutral crystal add up to zero. Displacing all sublattices together by the same $\boldsymbol{u}$, with the electrons following, translates the charge distribution of every cell. The dipole moment of a neutral distribution does not depend on where it sits, and $\boldsymbol{P}$ does not change. By Eq. (5.39), an unchanged $\boldsymbol{P}$ requires $\boldsymbol{u} \cdot \sum_\kappa Z^*_\kappa = 0$ for every $\boldsymbol{u}$, that is

$$\sum_{\kappa} Z^*_\kappa = 0, \tag{5.41}$$

the charge neutrality of the Born charges, also called their acoustic sum rule. In GaAs each site has tetrahedral symmetry, the two Born-charge tensors are multiples of the identity, and neutrality leaves one number, $Z^*_{\mathrm{Ga}} = -Z^*_{\mathrm{As}} \equiv Z^*$. In silicon, inversion through the bond center exchanges the two atoms and forces $Z^*_1 = Z^*_2$; neutrality then forces both charges to vanish.

The polarization in Eq. (5.40) needs a definition. For a molecule the dipole moment is $\int \mathrm{d}^3 r\, \boldsymbol{r}\, \rho(\boldsymbol{r})$ with ρ the charge density. For a crystal the same integral over one cell depends on where the cell boundaries are drawn, and with the periodic boundary conditions of Sect. 3.1 the position operator is not even well defined. Only a change of polarization is measurable: the

charge per unit area that flows through the crystal while the nuclei are displaced slowly. King-Smith and Vanderbilt showed that this change is a Berry phase of the occupied Bloch states, accumulated as $\boldsymbol{k}$ crosses the zone [22]. Equivalently, the electronic polarization is $-eg_{\mathrm{deg}}/\Omega$ times the sum of the centers of the occupied Wannier functions of one cell (Sect. 27.1), with $g_{\mathrm{deg}} = 2$ for spin. This *Berry-phase theory* of polarization defines $\boldsymbol{P}$ only up to a quantum, a lattice vector times eg_{deg}/Ω, because a Wannier center can be assigned to any cell; the Born charge, a derivative, is free of that ambiguity.

The density-functional perturbation theory that yields the phonons also yields ε_∞ and Z^*_κ. A uniform field enters the Sternheimer equation as one more perturbation, through the matrix elements of $\boldsymbol{r}$ between occupied and empty states, and the self-consistent solution gives the induced polarization. Either side of Eq. (5.40) can then be evaluated, and Sect. 25.2 describes the practice. The sum rule Eq. (5.41) is a cheap convergence test. In the PBE calculation behind Fig. 5.4, the two charges of GaAs miss neutrality by 12 % on the 8^3 grid of crystal momenta used for the phonons. Only near a 28^3 grid does their sum vanish, with $Z^* = 2.094$, within 1 % of the value 2.11 that the measured LO–TO splitting implies (Worked example 5.5).

The dielectric constant converges more slowly, from 26.0 on the 8^3 grid to 13.7 on the 28^3 grid, where successive grids differ by less than one percent. In this calculation PBE puts the direct gap of GaAs at Γ at 0.57 eV, against the measured 1.52 eV (Table 23.1). The response to a field, built from the energy denominators of the interband transitions, is then dominated by the small region around Γ where the gap is smallest, and a coarse grid samples it poorly. The converged value is still 26 % above the measured $\varepsilon_\infty = 10.88$ [23]. That error belongs to the functional: a gap that is too small makes the electrons too polarizable, and no numerical parameter removes the error.

5.4.2 LO–TO splitting

In Eq. (5.39) the field is still an external parameter. In a lattice wave no one applies the field: the displaced ions create it, and it acts back on them. Let the displacements vary as $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$, with $|\boldsymbol{Q}|$ small compared with the reciprocal-lattice vectors, so that within many cells the displacement is the uniform $\boldsymbol{u}_\kappa$ of Eq. (5.38). We also take $|\boldsymbol{Q}|$ large compared with ω/c; this condition lets us neglect retardation and use electrostatics, and we will come back to it. Two equations then fix the field. A static field has no curl, so the field of a wave is longitudinal, $\boldsymbol{\mathcal{E}} = \hat{\boldsymbol{Q}}\,\mathcal{E}_\parallel$, and there is no free charge, so $\hat{\boldsymbol{Q}}\cdot\boldsymbol{D} = 0$. With Eq. (5.39) the second condition reads

$$\hat{\boldsymbol{Q}}\cdot\varepsilon_\infty\cdot\hat{\boldsymbol{Q}}\;\mathcal{E}_\parallel + \frac{e}{\Omega}\sum_{\kappa'}\boldsymbol{u}_{\kappa'}\cdot Z^*_{\kappa'}\cdot\hat{\boldsymbol{Q}} = 0.$$

Solving for $\mathcal{E}_\parallel$ and inserting it in the force adds a matrix $C^{\mathrm{na}}_{\kappa\kappa'}(\hat{\boldsymbol{Q}})$ to the force constants:

Nonanalytic term of the dynamical matrix

$$D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}\to 0) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}}\Big[C^{\mathrm{an},\alpha\beta}_{\kappa\kappa'} + \frac{e^2}{\Omega}\,\frac{(Z^*_\kappa\cdot\hat{\boldsymbol{Q}})^\alpha\,(Z^*_{\kappa'}\cdot\hat{\boldsymbol{Q}})^\beta}{\hat{\boldsymbol{Q}}\cdot\varepsilon_\infty\cdot\hat{\boldsymbol{Q}}}\Big], \tag{5.42}$$

with $(Z^*_\kappa\cdot\hat{\boldsymbol{Q}})^\alpha = \sum_\beta Z^{*,\alpha\beta}_\kappa\hat{Q}^\beta$, the analytic part C^{an} computed at zero macroscopic field, and the masses as in Eq. (5.16).

This is the Heaviside–Lorentz form of the result of Gonze and Lee [24] and of Ref. [21, Eq. (98)], after the substitution $4\pi e^2_{\mathrm{G}} \to e^2$ of Eq. (A.4). The second term, C^{na} with the masses divided out, is the *nonanalytic term*. The reason for the name is that the term depends on the direction $\hat{\boldsymbol{Q}}$ from which the zone center is approached but not on $|\boldsymbol{Q}|$. The macroscopic

field is the $\boldsymbol{G} = 0$ Coulomb interaction e^2/Q^2 acting between two dipole densities, each proportional to $\boldsymbol{Q}$, and the ratio of their product to Q^2 stays finite as $Q \to 0$ without having a limit. No Taylor series in $\boldsymbol{Q}$ can represent a function that depends only on the direction of $\boldsymbol{Q}$, and the dynamical matrix of a polar crystal at $\boldsymbol{Q} = 0$ is a family of matrices labeled by $\hat{\boldsymbol{Q}}$.

In a cubic crystal with two atoms per cell the consequences are explicit. There $Z_1^* = -Z_2^* = Z^*$ times the identity, ε_∞ is a scalar, and the acoustic sum rule of Sect. 5.2 leaves one zone-center constant, $C_{11}^{\text{an}} = C_{22}^{\text{an}} = -C_{12}^{\text{an}} = C_0$ times the identity. The equations of motion of the two sublattices, $M_1\ddot{\boldsymbol{u}}_1 = \boldsymbol{F}_1$ and $M_2\ddot{\boldsymbol{u}}_2 = \boldsymbol{F}_2$, then involve only the relative displacement $\boldsymbol{u}_{\text{rel}} = \boldsymbol{u}_1 - \boldsymbol{u}_2$. Dividing each equation by its mass and subtracting, with $\boldsymbol{F}_1 = -C_0\boldsymbol{u}_{\text{rel}} + eZ^*\boldsymbol{\mathcal{E}} = -\boldsymbol{F}_2$ from Eq. (5.39), gives

$$\mu_{\text{red}}\,\ddot{\boldsymbol{u}}_{\text{rel}} = -C_0\,\boldsymbol{u}_{\text{rel}} + eZ^*\boldsymbol{\mathcal{E}}, \qquad \boldsymbol{P} = \frac{eZ^*}{\Omega}\,\boldsymbol{u}_{\text{rel}} + (\varepsilon_\infty - 1)\boldsymbol{\mathcal{E}},$$

with $\mu_{\text{red}} = M_1M_2/(M_1 + M_2)$ the reduced mass of the pair, as in the diatomic chain of Worked example 5.2. In a transverse optical mode, $\boldsymbol{u}_{\text{rel}} \perp \hat{\boldsymbol{Q}}$, the dipoles create no longitudinal field, and the frequency comes from the springs alone, $\omega_{\text{T}}^2 = C_0/\mu_{\text{red}}$. In the longitudinal mode, $\boldsymbol{u}_{\text{rel}} \parallel \hat{\boldsymbol{Q}}$, the condition $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$ makes the field $\boldsymbol{\mathcal{E}} = -eZ^*\boldsymbol{u}_{\text{rel}}/(\Omega\varepsilon_\infty)$, which opposes the displacement. The longitudinal frequency ω_{L} is then higher,

$$\omega_{\text{L}}^2 = \omega_{\text{T}}^2 + \frac{e^2Z^{*2}}{\Omega\,\varepsilon_\infty\,\mu_{\text{red}}}, \tag{5.43}$$

and the difference $\omega_{\text{L}} - \omega_{\text{T}}$ is the *LO–TO splitting*. The longitudinal mode carries the electrostatic energy of its own polarization, screened by the electrons, and the Born charge enters squared, divided by ε_∞. A quick estimate shows the size of the splitting. With $Z^* = 2$, $\Omega = 45\,\text{Å}^3$, $\varepsilon_\infty = 11$ and $\mu_{\text{red}} = 36\,\text{u}$, the second term of Eq. (5.43) is $\sim 2 \times 10^{-4}\,\text{eV}^2$, small against $\omega_{\text{T}}^2 \approx 1.1 \times 10^{-3}\,\text{eV}^2$; the splitting is a few meV, less than a tenth of the optical energy.

Fig. 5.4 shows the computed phonons of GaAs. The transverse optical energy at Γ comes out at 34.22 meV, 1.8 % above the measured 33.62 meV. PBE usually makes phonons slightly soft, but here the measured lattice constant compresses the crystal relative to the PBE minimum at 5.746 Å (Table 23.1). The longitudinal branch meets the zone center at 36.34 meV from every direction. In a cubic crystal $Z_\kappa^* \cdot \hat{\boldsymbol{Q}} = Z_\kappa^*\hat{\boldsymbol{Q}}$ and $\hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}} = \varepsilon_\infty$ for every $\hat{\boldsymbol{Q}}$, and the nonanalytic term does not depend on the direction; in an anisotropic crystal such as sapphire it does, and so does the LO frequency at Γ [25]. The computed splitting is 2.12 meV against the measured 2.73 meV, 22 % short, and Eq. (5.43) shows why. The second term of Eq. (5.43) is $e^2Z^{*2}/(\Omega\varepsilon_\infty\mu_{\text{red}})$, the Born charge is right to a percent, and ε_∞ is 26 % too large. With the measured ε_∞ and the computed Z^* and ω_{T}, the same formula yields a splitting of 2.65 meV, within 3 % of the measurement (the diamond in Fig. 5.4(b)). The computed LO energy itself agrees with the measured one, but by accident: the transverse energy is high and the splitting short by nearly the same amount. The near-equal masses of gallium and arsenic leave almost no gap between the acoustic and optical bands: the longitudinal acoustic branch ends at 27.65 meV at X, and the optical band begins 0.2 meV higher.

Because of the nonanalytic term, Fourier interpolation of the force constants needs an extra step in a polar crystal. In real space the dipole–dipole interaction behind Eq. (5.42) falls off as the inverse cube of the distance, too slowly for any grid of wavevectors to capture, and the computed $D^{\text{dyn}}(\boldsymbol{Q})$ near Γ is not smooth. The analytic part C^{an} is the response with the $\boldsymbol{G} = 0$ term of the Coulomb interaction removed [26], and for this reason it can be computed at zero macroscopic field. Since the nonanalytic term is fixed by Z_κ^* and ε_∞ alone, it can be subtracted from the computed matrices in an Ewald-summed form that is periodic in $\boldsymbol{Q}$. The short-ranged difference between the computed matrices and this term is

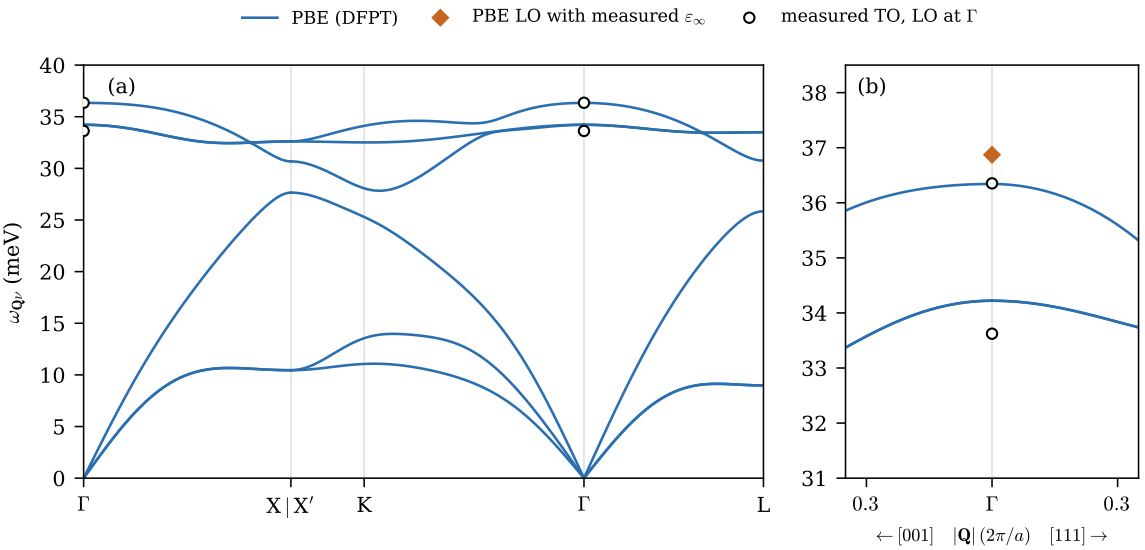


Figure 5.4: Phonon dispersion of GaAs. (a) Computed along Γ–X–K–Γ–L with density-functional perturbation theory: PBE functional, SG15 norm-conserving pseudopotentials with the gallium $3d$ shell in the valence, a 60 Ry cutoff, an 8^3 grid of crystal momenta, the static-lattice constant $a = 5.638$ Å of Table 23.1 (the measured value extrapolated to zero temperature with the zero-point expansion removed) and natural masses, on a 4^3 grid of wavevectors. The force constants are Fourier interpolated with the dipole term of Eq. (5.42) subtracted and added back, using $\varepsilon_\infty = 13.7$ and $Z^* = 2.094$ computed on a 28^3 grid of crystal momenta; acoustic sum rules imposed on the force constants and on the Born charges. Circles: measured TO and LO energies at Γ, 8.13 and 8.79 THz [18], as tabulated in Ref. [19, Table I]. (b) Optical branches near Γ along [001] (left) and [111] (right), with $|\boldsymbol{Q}|$ in units of $2\pi/a$. Diamond: LO energy at Γ from Eq. (5.43) with the computed TO energy and Born charge and the measured $\varepsilon_\infty = 10.88$ [23].

then interpolated, and the nonanalytic term is added back analytically at each target $\boldsymbol{Q}$ [24]. Fig. 5.4 was interpolated in this way from a 4^3 grid.

The electrostatic treatment assumed $|\boldsymbol{Q}| \gg \omega/c$. For wavevectors comparable with ω/c, the transverse optical phonon mixes with the photon of the same wavevector and frequency, and the two form the coupled modes called *phonon polaritons*. The crossover happens near $Q \sim \varepsilon_0^{1/2}\omega_\mathrm{T}$ in natural units, about 0.1 eV for $\omega_\mathrm{T} \sim 30$ meV and less than 10^{-4} of the size of the zone; at this wavevector the photon line $\omega = Q/\varepsilon_0^{1/2}$ meets the phonon. Dark-matter scattering transfers $q \sim m_\chi v \gtrsim 1$ eV for any mass above a keV, and for the phonons it excites the electrostatic result is exact. Absorption is different. A boson of mass m from the halo, moving at $v \sim 10^{-3}$, deposits $\omega \approx m$ with a momentum of only $mv \sim 10^{-3}\omega$, deep inside the polariton region. The crystal then sees a field that is uniform in space, and the coupling to a uniform field is derived below.

5.4.3 The Lyddane–Sachs–Teller relation

A uniform field oscillating at frequency ω, $\boldsymbol{\mathcal{E}}\,\mathrm{e}^{-\mathrm{i}\omega t}$, is the opposite limit, $Q \to 0$ at fixed ω, and it measures the dielectric function of the crystal with the lattice included. The field now drives the relative displacement without any longitudinal constraint, to the steady response $\boldsymbol{u}_\mathrm{rel} = eZ^*\boldsymbol{\mathcal{E}}/[\mu_\mathrm{red}(\omega_\mathrm{T}^2 - \omega^2)]$, and the electrons respond with ε_∞ as long as ω lies far below the gap, the adiabatic condition of Sect. 5.1. Inserting $\boldsymbol{u}_\mathrm{rel}$ into $\boldsymbol{P}$ gives the dielectric function $\varepsilon(\omega) = D/\mathcal{E} = 1 + P/\mathcal{E}$ as

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{e^2 Z^{*2}}{\Omega\,\mu_\mathrm{red}\,(\omega_\mathrm{T}^2 - \omega^2)} = \varepsilon_\infty\,\frac{\omega_\mathrm{L}^2 - \omega^2}{\omega_\mathrm{T}^2 - \omega^2}, \tag{5.44}$$

where the second form uses Eq. (5.43) to replace $e^2Z^{*2}/(\Omega\mu_\mathrm{red})$ by $\varepsilon_\infty(\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2)$. At frequencies well above ω_T the ions cannot follow, and $\varepsilon \to \varepsilon_\infty$; for this reason the electronic dielectric constant is also called the high-frequency dielectric constant (Sect. 4.4). At $\omega = 0$ the ions follow fully and add their polarization. The *static dielectric constant* $\varepsilon_0 = \varepsilon(0)$, lattice included, is $\varepsilon_\infty + e^2Z^{*2}/(\Omega\mu_\mathrm{red}\omega_\mathrm{T}^2)$, and the second form of Eq. (5.44) turns this expression into a relation between measurable quantities alone:

Lyddane–Sachs–Teller relation

$$\frac{\varepsilon_0}{\varepsilon_\infty} = \frac{\omega_\mathrm{L}^2}{\omega_\mathrm{T}^2}. \tag{5.45}$$

For several polar modes, $\varepsilon_0/\varepsilon_\infty = \prod_\nu \omega_{\mathrm{L}\nu}^2/\omega_{\mathrm{T}\nu}^2$ (Eq. (5.46)).

This is the relation of Lyddane, Sachs and Teller [27]. The Born charge, the masses and the cell volume have dropped out: the ratio of the two dielectric constants is fixed by the ratio of two frequencies. The general form, with one factor per polar mode, is due to Cochran and Cowley [28], and follows from the interlacing of the poles and zeros of $\varepsilon(\omega)$.

Those poles and zeros are the content of Eq. (5.44), and they hold beyond the model. The transverse frequency is a pole: a transverse mode creates no field, so it needs none, and it shows up where the response to a field diverges. The longitudinal frequency is a zero. A longitudinal mode has $\boldsymbol{\mathcal{E}} \parallel \hat{\boldsymbol{Q}}$ and needs $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = \varepsilon(\omega)\,\mathcal{E}_\parallel = 0$, which allows a nonzero field only where $\varepsilon(\omega)$ vanishes. A zero of $\varepsilon(\omega)$ also fixes the plasmon of Sect. 4.5, a self-sustained longitudinal oscillation of the electrons; the LO phonon is a self-sustained longitudinal oscillation of the ions. Between ω_T and ω_L, Eq. (5.44) is negative, no transverse wave propagates, and light in this band, the reststrahlen band, is reflected [10, Chap. 27]. The loss function $\mathrm{Im}[-1/\varepsilon(\omega)]$ of Sect. 4.4 peaks at ω_L, and the absorption ε_2 at ω_T, once the phonons are given a width.

With several polar modes the steps that led to Eq. (5.44) give the general form. Let $\omega_{\mathrm{T}\nu}$ and $\boldsymbol{e}_{\kappa\nu}$ be the frequencies and polarization vectors of the analytic problem at $\boldsymbol{Q}=0$, the eigenpairs of $C^{\mathrm{an}}_{\kappa\kappa'}/(M_\kappa M_{\kappa'})^{1/2}$, chosen real. With the expansion $\boldsymbol{u}_\kappa=\sum_\nu \boldsymbol{e}_{\kappa\nu}a_\nu/\sqrt{M_\kappa}$, each normal coordinate obeys $\ddot{a}_\nu=-\omega_{\mathrm{T}\nu}^2 a_\nu+e\,\boldsymbol{z}_\nu\cdot\boldsymbol{\mathcal{E}}$, with the mode charge vector $\boldsymbol{z}_\nu=\sum_\kappa \boldsymbol{e}_{\kappa\nu}\cdot Z^*_\kappa/\sqrt{M_\kappa}$, and it contributes $(e/\Omega)\,a_\nu\boldsymbol{z}_\nu$ to the polarization. In a cubic crystal, along any direction $\hat{\boldsymbol{e}}$,

$$\varepsilon(\omega)=\varepsilon_\infty+\frac{e^2}{\Omega}\sum_\nu\frac{(\boldsymbol{z}_\nu\cdot\hat{\boldsymbol{e}})^2}{\omega_{\mathrm{T}\nu}^2-\omega^2}.$$

This is a rational function of ω^2 that tends to ε_∞ at large ω, has simple poles at the polar transverse frequencies, and, by the argument above, vanishes at the longitudinal frequencies $\omega_{\mathrm{L}\nu}$. Between two consecutive poles it rises monotonically from $-\infty$ to $+\infty$, and above the highest pole it rises from $-\infty$ toward ε_∞, so there is exactly one zero above each pole and below the next. A rational function is fixed by its poles, its zeros and its limit at infinity, so $\varepsilon(\omega)=\varepsilon_\infty\prod_\nu(\omega_{\mathrm{L}\nu}^2-\omega^2)/(\omega_{\mathrm{T}\nu}^2-\omega^2)$, and at $\omega=0$

$$\frac{\varepsilon_0}{\varepsilon_\infty}=\prod_\nu\frac{\omega_{\mathrm{L}\nu}^2}{\omega_{\mathrm{T}\nu}^2},\tag{5.46}$$

which generalizes Eq. (5.45).

Worked example 5.5: The Lyddane–Sachs–Teller relation for GaAs

We test Eq. (5.45) on measured numbers. Neutron scattering puts the optical phonons of GaAs at $\omega_{\mathrm{T}}=8.13\,\mathrm{THz}=33.62\,\mathrm{meV}$ and $\omega_{\mathrm{L}}=8.79\,\mathrm{THz}=36.35\,\mathrm{meV}$ [18], as tabulated in Ref. [19, Table I], and infrared spectroscopy measured $\varepsilon_0=12.9$ and $\varepsilon_\infty=10.88$ [23], as quoted in Ref. [29].

The relation. The two sides are

$$\frac{\varepsilon_0}{\varepsilon_\infty}=\frac{12.9}{10.88}=1.186,\qquad\frac{\omega_{\mathrm{L}}^2}{\omega_{\mathrm{T}}^2}=\Big(\frac{8.79}{8.13}\Big)^2=1.169,$$

equal to 1.4 %. Read as a prediction, the relation puts the longitudinal phonon at $\omega_{\mathrm{L}}=\omega_{\mathrm{T}}(\varepsilon_0/\varepsilon_\infty)^{1/2}$, or 36.61 meV, 0.26 meV or 0.7 % above the measurement. The splitting itself comes out as 2.99 meV against 2.73 meV, 9 % high. The relative error grows because the ratio is close to one: $\omega_{\mathrm{L}}^2-\omega_{\mathrm{T}}^2$ is $0.17\,\omega_{\mathrm{T}}^2$, and a 1.4 % error in the ratio is a 10 % error in that difference. For a harmonic crystal with adiabatic electrons the relation is exact. The 1.4 % mismatch between the two sides therefore comes from the inputs. The four numbers come from different measurements on different samples, and the dielectric constants of GaAs change at the percent level between room temperature and liquid-helium temperature [25].

The Born charge. Solved for the Born charge, Eq. (5.43) reads $Z^{*2}=\Omega\varepsilon_\infty\mu_{\mathrm{red}}(\omega_{\mathrm{L}}^2-\omega_{\mathrm{T}}^2)/e^2$. With $a=5.638\,\text{Å}$, $\Omega=a^3/4=44.80\,\text{Å}^3=5.831\times10^{-9}\,\mathrm{eV}^{-3}$ in natural units, the masses 69.723 and 74.922 u, so that $\mu_{\mathrm{red}}=36.11\,\mathrm{u}=33.64\,\mathrm{GeV}$, $e^2=4\pi\alpha=0.0917$ and $\omega_{\mathrm{L}}^2-\omega_{\mathrm{T}}^2=1.910\times10^{-4}\,\mathrm{eV}^2$,

$$Z^*=\Big[\frac{(5.831\times10^{-9})\times10.88\times(3.364\times10^{10})\times(1.910\times10^{-4})}{0.0917}\Big]^{1/2}=2.11.$$

The static constant offers a second route, since Eq. (5.44) at $\omega=0$ gives $\varepsilon_0-\varepsilon_\infty=e^2Z^{*2}/(\Omega\mu_{\mathrm{red}}\omega_{\mathrm{T}}^2)$. The second route yields 2.21, 5 % more: the same 1.4 % inconsistency of the inputs magnified by the difference $\varepsilon_0-\varepsilon_\infty=2.02$. The splitting, measured directly by one experiment, is the better input.

Against the calculation. The converged PBE Born charge, 2.094, lies 0.7 % below the value just extracted, while the PBE ε_∞, 13.7, lies 26 % above the measurement; the error in ε_∞ is why the computed splitting of Fig. 5.4 falls short. Wherever a rate needs the combination $Z^{*2}/\varepsilon_\infty$, the measured splitting, or the computed Born charge with the measured ε_∞, is the input to use.

5.4.4 Coupling to a uniform field

The middle term of Eq. (5.38), $-e\sum_\kappa \boldsymbol{u}_\kappa \cdot Z^*_\kappa \cdot \boldsymbol{\mathcal{E}}$, is the energy of the dipoles of the displaced sublattices in the field. For a field uniform over the crystal and slow compared with the electrons, it adds a term to the nuclear Hamiltonian of Eq. (5.26):

$$H_{\mathcal{E}}(t) = -e\sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \cdot Z^*_\kappa \cdot \boldsymbol{\mathcal{E}}(t).$$

When the phonon expansion Eq. (5.32) is inserted, the lattice sum $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = N\delta_{\boldsymbol{Q},0}$ keeps only the modes at the zone center:

$$H_{\mathcal{E}}(t) = -e\sqrt{N}\sum_\nu \frac{\boldsymbol{z}_\nu \cdot \boldsymbol{\mathcal{E}}(t)}{\sqrt{2\omega_{0\nu}}}\, B_{0\nu}, \qquad \boldsymbol{z}_\nu = \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(0)\cdot Z^*_\kappa}{\sqrt{M_\kappa}}, \tag{5.47}$$

with $\boldsymbol{z}_\nu$ the mode charge vector and $\omega_{0\nu}$ the frequency of branch ν at Γ. The amplitude to create one phonon from the ground state, by Eq. (5.34), is $-e\sqrt{N}\,\boldsymbol{z}_\nu \cdot \boldsymbol{\mathcal{E}}/\sqrt{2\omega_{0\nu}}$.

In Eq. (5.47) the acoustic modes do not couple. At Γ their polarization vectors are $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$ (Sect. 5.2), so that $\boldsymbol{z}_\nu = \hat{\boldsymbol{e}}\cdot\sum_\kappa Z^*_\kappa/\sqrt{M_{\mathrm{cell}}}$, which vanishes by the charge neutrality Eq. (5.41): a uniform field cannot push a neutral crystal. In GaAs the optical modes at Γ have $\boldsymbol{e}_1 = (M_2/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$ and $\boldsymbol{e}_2 = -(M_1/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$, and the same algebra as in Appendix D.12 leads to $\boldsymbol{z}_\nu = Z^*\hat{\boldsymbol{e}}/\sqrt{\mu_{\mathrm{red}}}$: the field couples to the optical mode polarized along it, through the Born charge and the reduced mass. The squared amplitude is proportional to N. All cells are driven in phase, the whole crystal absorbs the quantum, and the rate grows with the target mass, as for the single-phonon amplitude of Sect. 5.3, without the momentum transfer that a scattering process has to supply.

Which resonance a probe excites depends on what it holds fixed. A field $\boldsymbol{\mathcal{E}}$ imposed inside the crystal drives the transverse resonance, the pole of ε at ω_{T}. A probe that fixes the source of $\boldsymbol{D}$, as a moving charge does, sees $1/\varepsilon$ and excites the longitudinal mode at the zero of ε, ω_{L}. A dark photon couples through its in-medium mixing with the photon. For a dark photon at rest, of mass $m = \omega$, the medium screens that mixing by $1/\varepsilon(\omega)$: the absorption rate is the photon absorption $\omega\,\mathrm{Im}\,\varepsilon$ times the screened coupling $1/|\varepsilon|^2$, and so proportional to the loss function $\mathrm{Im}[-1/\varepsilon(\omega)]$ [29, Eqs. (2) and (3)], which peaks at the zero of ε, ω_{L}. The absorption of dark-photon dark matter in GaAs therefore peaks at the LO energy, and Chap. 11 works out the rate. The longitudinal field also acts on the electrons. Its potential, which grows as $1/Q$ at small Q, is the Fröhlich coupling, with a strength set by ω_{L} and $1/\varepsilon_\infty - 1/\varepsilon_0$ (Eq. (6.15)).

The five numbers of this section, Z^*, ε_∞, ε_0, ω_{T} and ω_{L}, obey two relations, Eqs. (5.43) and (5.45), so that any three fix the other two.

5.5 Displacement correlations

A silicon nucleus in a cold crystal is spread over 0.05 Å by its zero-point motion. When a dark-matter particle transfers 50 keV of momentum to the nucleus, the phase $\boldsymbol{q}\cdot\boldsymbol{u}$ that the momentum transfer imprints varies by more than a radian across the nuclear wavefunction.

The expansion of $e^{i\boldsymbol{q}\cdot\boldsymbol{u}}$ to first order, which gave the single-phonon amplitude of Sect. 5.3, then no longer holds. In some events the nucleus takes up the momentum and the crystal is left in its ground state. In the other events the nucleus emits one phonon, or two, or more, and at large enough momentum transfer the sum over phonon numbers turns into the free recoil of a lone nucleus. To find the probability of each outcome, we need the thermal averages of exponentials of displacements, at one time and at two. For a harmonic crystal both averages are exponentials of quadratic forms in $\boldsymbol{q}$, the Debye–Waller exponent $2W_\kappa(\boldsymbol{q})$ and the displacement correlation $\mathcal{C}_{\kappa\kappa'}(t)$, and both follow from the tensor B_κ and the phonon modes of Sect. 5.3. The same algebra describes the optical spectrum of a localized electron whose transition moves the nuclei around it, and in that problem it gives the Franck–Condon factors and the Huang–Rhys factor.

5.5.1 The Bloch identity

A momentum transfer $\boldsymbol{q}$ acts on the nucleus κ in cell $\boldsymbol{R}$ through the operator $e^{i\boldsymbol{q}\cdot\boldsymbol{X}}$, with $\boldsymbol{X} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$ (Eq. (5.1)). The lattice position contributes a phase, and the displacement contributes the operator $e^{i\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}$. By Eq. (5.32), $\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}$ is a linear combination of the phonon operators $b_{\boldsymbol{Q}\nu}$ and $b^\dagger_{\boldsymbol{Q}\nu}$. The exponential $e^{i\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}$ contains every power of the phonon operators and connects the thermal state to states with any number of extra phonons. The first question is the thermal average of $e^{i\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}$.

For a classical Gaussian variable x with zero mean, the average of e^{iqx} is $e^{-q^2\langle x^2\rangle/2}$ (Eq. (C.43)). The same statement holds for operators in a harmonic crystal in thermal equilibrium. The density matrix of the crystal, $e^{-\beta H_\mathrm{N}}/\mathcal{Z}$, is Gaussian in the normal coordinates and momenta, because H_N of Eq. (5.31) is quadratic in them. For any operator $\hat{A}$ linear in the phonon operators,

$$\langle e^{\hat{A}}\rangle = e^{\langle\hat{A}^2\rangle/2},$$

which is Bloch's identity, Eq. (C.51). Expanded in powers of $\hat{A}$, it is Wick's theorem: the average of $\hat{A}^{2n}$ is the sum over the $(2n)!/(2^n n!)$ ways of pairing the $2n$ factors, each pair worth $\langle\hat{A}^2\rangle$.

To prove Bloch's identity, take one oscillator, $H = \omega_0(d^\dagger d + \frac{1}{2})$ as in Appendix C.6, in thermal equilibrium at temperature $T = 1/\beta$, with $n_\mathrm{B} = n_\mathrm{B}(\omega_0)$, and let $\hat{A} = \eta_1 d + \eta_2 d^\dagger$ with arbitrary complex η_1 and η_2, so that $\hat{A}$ need not be Hermitian. Consider $F(\theta) = \langle e^{\theta\hat{A}}\rangle$ as a function of a real parameter θ. The proof uses two facts. The commutators $[\hat{A}, d] = -\eta_2$ and $[\hat{A}, d^\dagger] = \eta_1$ are numbers. By Eq. (C.46), the transformation by $e^{\theta\hat{A}}$ therefore only shifts the ladder operators,

$$d\,e^{\theta\hat{A}} = e^{\theta\hat{A}}\,(d + \theta\eta_2), \qquad d^\dagger\,e^{\theta\hat{A}} = e^{\theta\hat{A}}\,(d^\dagger - \theta\eta_1).$$

The cyclic property of the trace, in turn, moves a ladder operator past the Boltzmann factor at a known cost, as in the proof of Wick's theorem in Appendix D.9: $d\,e^{-\beta H} = e^{-\beta\omega_0}e^{-\beta H}d$, since removing a quantum lowers H by ω_0, and $d^\dagger e^{-\beta H} = e^{+\beta\omega_0}e^{-\beta H}d^\dagger$. As a result, $\langle e^{\theta\hat{A}}d\rangle = \mathrm{Tr}[d\,e^{-\beta H}e^{\theta\hat{A}}]/\mathcal{Z} = e^{-\beta\omega_0}\langle d\,e^{\theta\hat{A}}\rangle$, and combining this relation with the shift gives

$$\langle d\,e^{\theta\hat{A}}\rangle = e^{-\beta\omega_0}\langle d\,e^{\theta\hat{A}}\rangle + \theta\eta_2 F(\theta), \qquad \langle d\,e^{\theta\hat{A}}\rangle = \frac{\theta\eta_2 F(\theta)}{1 - e^{-\beta\omega_0}} = \theta\eta_2\,(n_\mathrm{B} + 1)\,F(\theta).$$

The same steps for $d^\dagger$ give $\langle d^\dagger e^{\theta\hat{A}}\rangle = e^{\beta\omega_0}\langle d^\dagger e^{\theta\hat{A}}\rangle - \theta\eta_1 F(\theta)$, and $\langle d^\dagger e^{\theta\hat{A}}\rangle = \theta\eta_1\,n_\mathrm{B}\,F(\theta)$. The derivative of F is then

$$F'(\theta) = \eta_1\langle d\,e^{\theta\hat{A}}\rangle + \eta_2\langle d^\dagger e^{\theta\hat{A}}\rangle = \theta\,\eta_1\eta_2\,(2n_\mathrm{B} + 1)\,F(\theta) = \theta\,\langle\hat{A}^2\rangle\,F(\theta),$$

since $\langle \hat{A}^2 \rangle = \eta_1\eta_2\langle dd^\dagger + d^\dagger d\rangle = \eta_1\eta_2(2n_\mathrm{B}+1)$, the averages of d^2 and $d^{\dagger 2}$ being zero. With $F(0)=1$ the solution is $F(\theta) = \mathrm{e}^{\theta^2\langle\hat{A}^2\rangle/2}$, which at $\theta = 1$ is Bloch's identity. For several independent modes in thermal equilibrium, $\hat{A} = \sum_\gamma \hat{A}_\gamma$ is a sum of commuting pieces, one per mode, the exponential factorizes, and the average in the Gibbs state of Sect. 2.5 factorizes too. The exponents add, and $\sum_\gamma \langle \hat{A}_\gamma^2\rangle = \langle \hat{A}^2\rangle$, because the cross terms $\langle \hat{A}_\gamma \hat{A}_{\gamma'}\rangle = \langle \hat{A}_\gamma\rangle\langle \hat{A}_{\gamma'}\rangle$ vanish for $\gamma \neq \gamma'$. The identity therefore holds for any $\hat{A}$ linear in the operators of a set of harmonic modes.

The hypotheses are the ones the proof used. The operator must be linear in d and $d^\dagger$, so that its commutators are numbers, and the state must be the thermal state of a quadratic Hamiltonian, so that the trace moves each ladder operator at the cost of a number. In an anharmonic crystal neither hypothesis holds exactly. The logarithm of $\langle \mathrm{e}^{\hat{A}}\rangle$ is then a series in the cumulants of $\hat{A}$, of which the harmonic crystal has only the second. The anharmonic terms of Sect. 5.6 make the fourth cumulant nonzero, and the Debye–Waller exponent acquires a correction that grows with temperature faster than T.

With $\hat{A} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}$, $\langle \hat{A}^2\rangle = -\langle(\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa})^2\rangle = -\sum_{\alpha\beta} q^\alpha B_\kappa^{\alpha\beta} q^\beta$, with the displacement tensor B_κ of Eq. (5.35).

Debye–Waller factor

For a nucleus κ of a harmonic crystal in thermal equilibrium,

$$\langle \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}\rangle = \mathrm{e}^{-W_\kappa(\boldsymbol{q})}, \qquad 2W_\kappa(\boldsymbol{q}) = \boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{|\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2}{2M_\kappa\,\omega_{\boldsymbol{Q}\nu}} \coth\frac{\omega_{\boldsymbol{Q}\nu}}{2T}, \quad (5.48)$$

the same in every cell. At a site of cubic symmetry, $2W_\kappa(\boldsymbol{q}) = q^2\langle u_x^2\rangle$.

The factor e^{-W_κ} is an amplitude: the probability amplitude for the nucleus to absorb $\boldsymbol{q}$ and leave the lattice in the state it found it in, averaged over that state. The probability is its square, the *Debye–Waller factor* $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$, and $2W_\kappa(\boldsymbol{q})$ is the Debye–Waller exponent. In X-ray and neutron diffraction the Debye–Waller factor multiplies the intensity of every Bragg peak, and this reduction of the Bragg intensities is how B_κ is measured. In silicon at 293 K, with the reference value $\langle u_x^2\rangle = 5.94\times10^{-3}\,\text{Å}^2$ of Worked example 5.4, the Debye–Waller factor reduces the (111) reflection, at $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\mathrm{keV}$, by only 2.4 %. Note that the electron–phonon literature also uses the name "Debye–Waller" for the term of second order in the displacements of the electron–phonon interaction. That term shifts electronic energies and is a different object, the Allen–Heine term of Sect. 6.3 (Appendix A.6); in this book the name means Eq. (5.48) only.

Eq. (5.48) inherits the structure of B_κ. The exponent grows as q^2, and it is largest along directions in which the nucleus is soft. Each mode contributes in proportion to the projection of $\boldsymbol{q}$ on its polarization, divided by its energy, and weighted by $2n_{\boldsymbol{Q}\nu}+1$, which grows from one at $T=0$ to $2T/\omega_{\boldsymbol{Q}\nu}$ at high temperature. For one species at a cubic site, $2W_\kappa = q^2\langle u^2\rangle/3$, and in the Debye model Eq. (5.37) gives

$$2W_\kappa(\boldsymbol{q}) = \frac{3q^2}{4M_\kappa\omega_\mathrm{D}}\Big[1 + 4\Big(\frac{T}{\omega_\mathrm{D}}\Big)^2 \int_0^{\omega_\mathrm{D}/T} \frac{x\,\mathrm{d}x}{\mathrm{e}^x - 1}\Big]. \quad (5.49)$$

At $T = \omega_\mathrm{D}$ the bracket is 4.11: the exponent is four times its zero-point value. At $T=0$ the exponent is

$$2W_\kappa(\boldsymbol{q}) = \frac{3q^2}{4M_\kappa\omega_\mathrm{D}} = \frac{3}{2}\frac{E_R}{\omega_\mathrm{D}}, \qquad E_R = \frac{q^2}{2M_\kappa},$$

with E_R the recoil energy that a free nucleus would take (Sect. 1.3). The lattice stays in its ground state with high probability as long as the recoil energy is small compared with typical

phonon energies, and almost never once it is large. For numbers we use the computed spectrum, since the Debye model misses the flat transverse acoustic branches (Worked example 5.4).

5.5.2 The displacement correlation

A scattering rate needs more than an average. By Eq. (4.13), the dynamic structure factor of the nuclei is a Fourier transform in time: it transforms the correlation between the density operator at time t and its adjoint at time zero. For one nucleus the density is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$, and the static phases cancel between the two factors, leaving

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(0)} \big\rangle,$$

with the Heisenberg operators $b_{\boldsymbol{Q}\nu}(t) = b_{\boldsymbol{Q}\nu}\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$ of Sect. 5.3. The two exponents are still linear in the phonon operators, but they do not commute, and Bloch's identity has to be combined with the Baker–Campbell–Hausdorff formula. For two operators $\hat{A}$ and $\hat{B}$ linear in the phonon operators, the commutator is a number, $\mathrm{e}^{\hat{A}}\mathrm{e}^{\hat{B}} = \mathrm{e}^{\hat{A}+\hat{B}}\mathrm{e}^{[\hat{A},\hat{B}]/2}$ (Eq. (C.47)), and Bloch's identity applied to $\hat{A}+\hat{B}$ leads to

$$\langle \mathrm{e}^{\hat{A}}\mathrm{e}^{\hat{B}} \rangle = \exp\Big[\tfrac{1}{2}\langle \hat{A}^2 \rangle + \tfrac{1}{2}\langle \hat{B}^2 \rangle + \langle \hat{A}\hat{B} \rangle\Big],$$

Eq. (C.52). With $\hat{A} = -\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)$ and $\hat{B} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(0)$, the first two terms are $-W_\kappa(\boldsymbol{q})$ each, since the thermal state is stationary and the average of a square does not depend on time. The third term is a correlation of two displacements at different times. When Eq. (5.32) is inserted twice, the averages $\langle b_{\boldsymbol{Q}\nu} b^\dagger_{\boldsymbol{Q}\nu} \rangle = n_{\boldsymbol{Q}\nu} + 1$ and $\langle b^\dagger_{\boldsymbol{Q}\nu} b_{\boldsymbol{Q}\nu} \rangle = n_{\boldsymbol{Q}\nu}$ now come with the phases $\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$ and $\mathrm{e}^{+\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$, and the *displacement correlation* of the nuclei κ and κ' is

$$\mathcal{C}^{\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},t) = \big\langle u^\alpha_{\boldsymbol{R}\kappa}(t)\, u^\beta_{\boldsymbol{0}\kappa'}(0) \big\rangle = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q})^*\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{2\omega_{\boldsymbol{Q}\nu}\sqrt{M_\kappa M_{\kappa'}}} \Big[(n_{\boldsymbol{Q}\nu}+1)\,\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t} + n_{\boldsymbol{Q}\nu}\,\mathrm{e}^{\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}\Big], \tag{5.50}$$

For one nucleus, $\boldsymbol{R} = \boldsymbol{0}$ and $\kappa' = \kappa$, we write $\mathcal{C}_{\kappa\kappa}(t)$. Its value at $t = 0$ is the displacement tensor, $\mathcal{C}_{\kappa\kappa}(0) = B_\kappa$. The term with $n_{\boldsymbol{Q}\nu} + 1$ creates a phonon at time zero and removes it at time t, and oscillates at the positive frequency $\omega_{\boldsymbol{Q}\nu}$; the term with $n_{\boldsymbol{Q}\nu}$ absorbs a thermal phonon first, and it vanishes at $T = 0$. Unlike B_κ, the correlation is complex, because $\boldsymbol{u}(t)$ and $\boldsymbol{u}(0)$ do not commute. Its imaginary part, $\operatorname{Im}\mathcal{C}_{\kappa\kappa}(t) = -\sum_{\boldsymbol{Q}\nu} \boldsymbol{e}_{\kappa\nu}\boldsymbol{e}^*_{\kappa\nu} \sin(\omega_{\boldsymbol{Q}\nu}t)/(2NM_\kappa\omega_{\boldsymbol{Q}\nu})$, does not depend on temperature; we will see that it carries the recoil energy.

Generating formula

For a nucleus κ of a harmonic crystal in thermal equilibrium,

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(0)} \big\rangle = \mathrm{e}^{-2W_\kappa(\boldsymbol{q})}\, \mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}}, \tag{5.51}$$

with $2W_\kappa$ of Eq. (5.48) and $\mathcal{C}_{\kappa\kappa}(t)$ of Eq. (5.50). For two nuclei in different cells or sublattices, by the same steps,

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{0}\kappa'}(0)} \big\rangle = \mathrm{e}^{-W_\kappa(\boldsymbol{q})-W_{\kappa'}(\boldsymbol{q})}\, \mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}(\boldsymbol{R},t)\cdot\boldsymbol{q}}. \tag{5.52}$$

The generating formula is the starting point of the theory of neutron scattering from crystals [10, App. N], [30, Chaps. 3 and 4]. The left side equals one at $t = 0$, and so does the right side, since $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(0)\cdot\boldsymbol{q} = 2W_\kappa$. The single-nucleus form, Eq. (5.51), is the *self* part of the nuclear response, in which the same nucleus scatters at both times. The full structure factor of the crystal adds the pairs of different nuclei, whose phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa-\boldsymbol{\tau}_{\kappa'})}$ produce

the Bragg peaks and the crystal-momentum conservation of the single-phonon amplitude of Sect. 5.3; Sect. 11.3 builds the complete response from both parts.

Both W_κ and $\mathcal{C}_{\kappa\kappa'}(t)$ are sums over all phonons of the zone, computed after the phonon calculation from frequencies and polarization vectors that Fourier interpolation of the DFPT force constants provides on a fine grid of wavevectors. The cost of the sums is negligible next to the phonon calculation itself. The acoustic modes near Γ make the summand of $2W_\kappa$ diverge as $1/\omega$ or $1/\omega^2$, and a uniform grid needs the analytic correction of Sect. 5.3. The same small frequencies control $\mathcal{C}_{\kappa\kappa'}(t)$ at long times, and the long-time behavior sets the shape of the spectrum at small energies.

5.5.3 Expansion in phonon number

Expanding the second exponential of Eq. (5.51) in powers of $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}$ sorts the response by the number of phonons involved. The nth power is a sum of products of n factors $\mathrm{e}^{\mp\mathrm{i}\omega_{\boldsymbol{q}\nu}t}$, one per phonon created (upper sign) or absorbed (lower sign), and its Fourier transform puts weight at the sums of n phonon energies with these signs. The term $n=0$ is time independent and is the elastic weight e^{-2W_κ} at $\omega=0$. The term $n=1$ is the single-phonon response of Sect. 5.3, now multiplied by e^{-2W_κ}.

The bookkeeping is simplest at a site of cubic symmetry with one species of nucleus, as in silicon. In this case $\mathcal{C}_{\kappa\kappa}(t)$ is a multiple of the identity, like B_κ. Its trace reduces to an integral over the density of states as in Eq. (5.36):

$$\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}=\frac{q^2}{3}\int_0^\infty \mathrm{d}\omega\,\frac{g_{\mathrm{ph}}(\omega)}{n_{\mathrm{at}}}\,\frac{(n_{\mathrm{B}}(\omega)+1)\,\mathrm{e}^{-\mathrm{i}\omega t}+n_{\mathrm{B}}(\omega)\,\mathrm{e}^{\mathrm{i}\omega t}}{2M_\kappa\,\omega}=2W_\kappa(\boldsymbol{q})\int_{-\infty}^\infty \mathrm{d}\omega\;P_1(\omega)\,\mathrm{e}^{-\mathrm{i}\omega t}.$$

Here $P_1(\omega)$ is the single-phonon distribution, proportional to $g_{\mathrm{ph}}(|\omega|)[n_{\mathrm{B}}(\omega)+1]/\omega$ for both signs of ω and normalized to one; at negative ω, $n_{\mathrm{B}}(\omega)+1=-n_{\mathrm{B}}(|\omega|)$ gives the absorption weight. The prefactor is $2W_\kappa$, since at $t=0$ the left side is $\boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}$. Each power of the exponential is then a convolution in frequency, and the Fourier transform of Eq. (5.51) becomes

$$\int_{-\infty}^\infty \mathrm{d}t\;\mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{e}^{-2W_\kappa}\,\mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}}=2\pi\,\mathrm{e}^{-2W_\kappa}\sum_{n=0}^\infty\frac{(2W_\kappa)^n}{n!}\,P_n(\omega),\tag{5.53}$$

with $P_0(\omega)=\delta(\omega)$ and P_n the n-fold convolution of P_1 with itself, normalized to one. The shape of each term comes from the phonon spectrum; its weight does not. The probability that a momentum transfer $\boldsymbol{q}$ to one nucleus involves n phonons is the Poisson distribution

$$p_n^{\mathrm{ph}}=\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}\,\frac{[2W_\kappa(\boldsymbol{q})]^n}{n!},\tag{5.54}$$

with mean $2W_\kappa$. At $T=0$ every phonon is created, P_1 vanishes at negative ω, and n is the number of phonons emitted. At $T>0$, n counts creations and absorptions together, and a term $n=2$ that absorbs one phonon and emits another contributes near $\omega=0$. Eq. (5.54) is the oscillator result Eq. (C.49) for a momentum kick, distributed over all the modes of the crystal: a kick is a displacement in momentum, and the number of quanta it creates out of a Gaussian state is Poisson distributed.

The first-moment identity Eq. (4.28) expresses the mean energy deposited as a double commutator, and for $\hat{O}=\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$ only the kinetic energy $\boldsymbol{p}^2/2M_\kappa$ of the nucleus fails to commute with $\hat{O}$. With $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\,\boldsymbol{p}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}=\boldsymbol{p}-\boldsymbol{q}$, the commutator is $[\boldsymbol{p}^2/2M_\kappa,\hat{O}]=\hat{O}\,(\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa+q^2/2M_\kappa)$, and the double commutator is the number $[\hat{O}^\dagger,[H,\hat{O}]]=q^2/M_\kappa$. The self response is even in $\boldsymbol{q}$, since it depends on $\boldsymbol{q}$ only through quadratic forms, so the two terms on the right of

Eq. (4.28) are equal. Since the weights of Eq. (5.53) add up to one, the mean energy deposited is

$$\mathrm{e}^{-2W_\kappa} \sum_{n=0}^{\infty} \frac{(2W_\kappa)^n}{n!} \int_{-\infty}^{\infty} \mathrm{d}\omega\, \omega\, P_n(\omega) = \frac{q^2}{2M_\kappa} = E_R. \tag{5.55}$$

The average energy left in the lattice is the free recoil energy, at every q and every temperature, whatever the force constants. The derivation used only the kinetic energy of the nucleus, not the harmonic approximation or the phonon modes, so the identity holds on any potential-energy surface, anharmonic ones included. In Eq. (5.51) the sum rule appears as the slope at $t = 0$. The polarization vectors are complete at each $\boldsymbol{Q}$ (Eq. (5.17)), $\sum_\nu e^\alpha_{\kappa\nu}(\boldsymbol{Q}) e^\beta_{\kappa\nu}(\boldsymbol{Q})^* = \delta_{\alpha\beta}$, and the derivative of Eq. (5.50) at $t = 0$ is therefore $-\mathrm{i}\,\mathbf{1}/2M_\kappa$. The thermal factors cancel between creation and absorption, as for the single oscillator of Worked example 2.7, and $\mathrm{i}\,\mathrm{d}/\mathrm{d}t$ of Eq. (5.51) at $t = 0$ is $q^2/2M_\kappa$. The elastic weight e^{-2W_κ} and the first moment behave oppositely: heating the crystal shrinks the elastic weight, since $2W_\kappa$ grows with $\langle u_x^2 \rangle$, but leaves the first moment untouched, since the weight that leaves $\omega = 0$ goes to both signs of ω.

The Poisson weights and the sum rule together lay out the regimes of the nuclear response. While $2W_\kappa \ll 1$, the elastic term dominates the total weight, the single-phonon term carries nearly all the inelastic weight, and the first-order expansion of Sect. 5.3 holds. Once $2W_\kappa$ approaches one, two- and three-phonon terms carry a share comparable to the single-phonon term. For $2W_\kappa \gg 1$, the Poisson distribution peaks at $n \approx 2W_\kappa$ with relative width $(2W_\kappa)^{-1/2}$, the elastic weight is exponentially small, and the spectrum is a broad peak centered at E_R, the free recoil of the standard nuclear-recoil rate (Sect. 11.4). At $T = 0$ the mean energy per phonon, $E_R/2W_\kappa = 1/(2M_\kappa \langle u_x^2 \rangle)$, is the inverse of the mean of $1/\omega$ over the spectrum, the typical frequency $\langle \omega^{-1} \rangle^{-1}$ of Sect. 5.3. The ratio of the recoil energy to that typical phonon energy is the zero-temperature $2W_\kappa$ itself, and it decides which regime a given momentum transfer is in.

Worked example 5.6: The Debye–Waller exponent of silicon

We take the natural mass of silicon, $M_\kappa = 26.16\,\mathrm{GeV}$, and the values of $\langle u_x^2 \rangle$ from Worked example 5.4. The computed density of states gives $2.45 \times 10^{-3}\,\text{Å}^2$ at $T = 0$ and $6.38 \times 10^{-3}\,\text{Å}^2$ at 293 K; the lattice-dynamical fit to the measured phonons gives $5.94 \times 10^{-3}\,\text{Å}^2$ at 293 K [9]. With $\hbar c = 1973.27\,\mathrm{eV\,Å}$, $1\,\text{Å}^2$ is $2.568 \times 10^{-7}\,\mathrm{eV}^{-2}$.

$q = 10\,\mathrm{keV}$. The recoil energy is $E_R = q^2/2M_\kappa = 1.91\,\mathrm{meV}$, far below the phonon energies. At $T = 0$, $2W_\kappa = q^2 \langle u_x^2 \rangle = 1 \times 10^8 \times 2.45 \times 10^{-3} \times 2.568 \times 10^{-7} = 0.063$, the elastic weight is 0.939, and by Eq. (5.54) one phonon is created with probability 0.059 and two with probability 0.0019. At 293 K, $2W_\kappa = 0.164$ (0.153 with the reference value) and $\mathrm{e}^{-2W_\kappa} = 0.85$. A momentum transfer of 10 keV is in the single-phonon regime, and the expansion of Sect. 5.3 is accurate to the size of the two-phonon term, a few percent of the inelastic weight.

$q = 50\,\mathrm{keV}$. The recoil energy is 47.8 meV, comparable to the optical phonon. At $T = 0$, $2W_\kappa = 1.57$, and the Poisson weights for zero, one, two, three and four phonons are 0.21, 0.33, 0.26, 0.13 and 0.05: almost half of all scatterings, 0.47, create two or more phonons. The mean energy per phonon is $E_R/2W_\kappa = 30.4\,\mathrm{meV}$, the typical frequency $\langle \omega^{-1} \rangle^{-1}$ of the computed spectrum in Worked example 5.4. At 293 K, $2W_\kappa = 4.10$ (3.81 with the reference value), and the elastic weight has dropped to 0.017.

Where $2W_\kappa = 1$. The crossover from the single-phonon to the multiphonon regime lies at $q = \langle u_x^2 \rangle^{-1/2}$: 40 keV at $T = 0$ and 26 keV at 293 K with the reference value, the scale anticipated in Sect. 5.3. For a cryogenic detector the zero-temperature value applies. The crossover at 40 keV lies below the 56 keV of the single 60 meV oscillator of Appendix C.6

because the crystal's zero-point motion is set by $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, half the optical energy: the soft transverse acoustic modes move the nucleus most. A single oscillator of frequency 30.4 meV reproduces the zero-temperature $2W_\kappa$ exactly, since its $|\alpha_\mathrm{d}|^2 = E_R/\omega_0$ is then $1/(2M_\kappa\omega_0) \times q^2 = q^2\langle u_x^2\rangle$; it misses only the shape of the spectrum.
The Debye model. With $\omega_\mathrm{D} = 55.6\,\mathrm{meV}$, Eq. (5.49) gives $2W_\kappa = 1.29$ at 50 keV and $T = 0$, and 2.64 at 293 K, low by 18 % and 35 %. The elastic weight is correspondingly too large, 0.28 instead of 0.21 at $T = 0$, and at room temperature 0.071 instead of 0.017, a factor of four. A Debye estimate locates the crossover to within 10 % in q, but a rate that depends on the elastic or single-phonon weight at $2W_\kappa \gtrsim 1$ needs the full spectrum.

5.5.4 The displaced oscillator

An electron bound to a defect in silicon, or to the nitrogen–vacancy center of diamond, absorbs light at one energy and emits it at a lower one. The absorption and emission spectra both show a sharp line with a train of broader replicas spaced by a phonon energy. The replicas appear because the electronic transition changes the forces on the nearby nuclei. In the language of Sect. 5.1, each electronic state has its own potential-energy surface, and the minima of the two surfaces lie at different nuclear positions. The transition happens at fixed nuclei, and afterwards the nuclei relax to the new minimum and shed the relaxation energy as phonons. The momentum kick of Eq. (5.54) displaces the nuclear momentum; the electronic transition displaces the equilibrium position. In both cases the lattice ends up in a displaced Gaussian state, and the same Poisson statistics follow.

The simplest model keeps one vibrational mode, of mass M and frequency ω_0, with the same frequency in both electronic states. In the ground electronic state g and the excited state e the nuclear Hamiltonians are

$$H_\mathrm{g} = \frac{p^2}{2M} + \frac{1}{2}M\omega_0^2 x^2, \qquad H_\mathrm{e} = E_\mathrm{ZPL} + \frac{p^2}{2M} + \frac{1}{2}M\omega_0^2 (x - \Delta x)^2, \tag{5.56}$$

where Δx is the shift of the equilibrium position and E_ZPL the energy difference between the two minima, the zero-point energies being equal. The difference $H_\mathrm{e} - H_\mathrm{g}$ is linear in x; it is the linear coupling of the electron to the mode. In the Born–Oppenheimer approximation of Sect. 5.1, the vibronic states are products of an electronic state and an eigenstate of the oscillator that belongs to that electronic state. With $x = \ell_0(d + d^\dagger)$ and $\ell_0 = (2M\omega_0)^{-1/2}$ as in Appendix C.6, the displacement operator $\hat{D}(\alpha_\mathrm{d})$ of Eq. (C.48) with real $\alpha_\mathrm{d} = \Delta x/2\ell_0$ shifts the coordinate by Δx. Hence $H_\mathrm{e} = E_\mathrm{ZPL} + \hat{D}H_\mathrm{g}\hat{D}^\dagger$, and the oscillator states of the excited surface are $|m_\mathrm{e}\rangle = \hat{D}|m\rangle$, with energies $E_\mathrm{ZPL} + m\omega_0$ above the ground-state energy of the lower surface (we drop the common $\frac{1}{2}\omega_0$).

An optical transition is driven by the electronic dipole, whose matrix element between g and e depends weakly on the nuclear positions. The Condon approximation takes this matrix element as constant, and the transition from $|n_\mathrm{g}\rangle$ to $|m_\mathrm{e}\rangle$ then has a probability proportional to the *Franck–Condon factor* $|\langle m_\mathrm{e}|n_\mathrm{g}\rangle|^2$, the squared overlap of the two vibrational wavefunctions. From the vibrational ground state, $\langle m_\mathrm{e}|0\rangle = \langle m|\hat{D}^\dagger|0\rangle = \langle m|\hat{D}(-\alpha_\mathrm{d})|0\rangle$, and by Eq. (C.49)

$$|\langle m_\mathrm{e}|0_\mathrm{g}\rangle|^2 = \mathrm{e}^{-S_\mathrm{HR}}\,\frac{S_\mathrm{HR}^m}{m!}, \qquad S_\mathrm{HR} = \alpha_\mathrm{d}^2 = \frac{\Delta x^2}{4\ell_0^2} = \frac{M\omega_0\,\Delta x^2}{2} = \frac{\omega_0\,\Delta Q^2}{2}. \tag{5.57}$$

The *Huang–Rhys factor* S_HR [31] is written last with the mass-weighted displacement $\Delta Q = M^{1/2}\Delta x$, the form in which a first-principles calculation supplies it for each mode. It has a direct meaning. The relaxation energy that the lattice releases after a vertical transition,

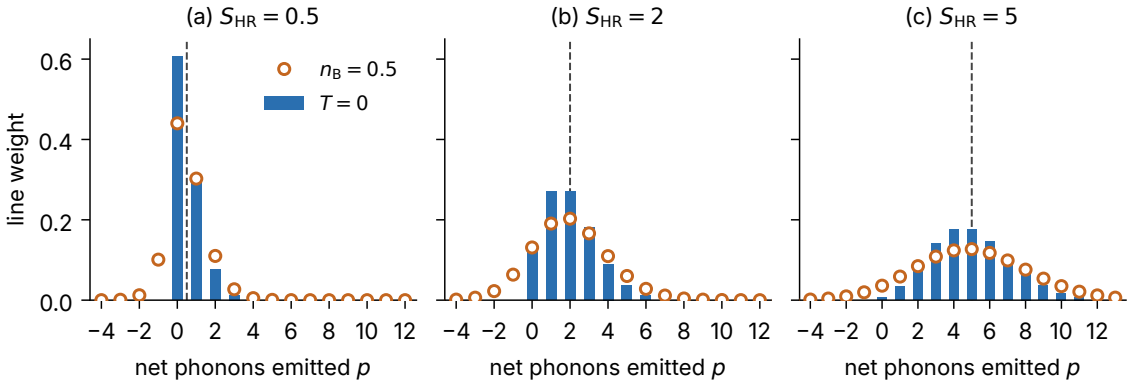


Figure 5.5: Franck–Condon progressions of a displaced oscillator, for Huang–Rhys factors $S_{\mathrm{HR}} = 0.5$, 2 and 5. Each panel shows the weights of the absorption lines at $E_{\mathrm{ZPL}} + p\,\omega_0$ against the net number p of phonons emitted, for one mode of frequency ω_0 that is the same in both electronic states, in the Condon approximation (dipole independent of the nuclear coordinate). Bars: zero temperature, Poisson weights $\mathrm{e}^{-S_{\mathrm{HR}}} S_{\mathrm{HR}}^{p}/p!$; the bar at $p = 0$ is the zero-phonon line. Open circles: mean occupation $n_{\mathrm{B}}(\omega_0) = 0.5$, that is $T = 0.91\,\omega_0$, from the Bessel-function weights of Eq. (5.59); the lines at $p < 0$ are anti-Stokes lines, which start from thermally excited vibrational states. Dashed line: the mean $p = S_{\mathrm{HR}}$, the same at both temperatures. The zero-phonon fractions at $T = 0$ are 0.61, 0.14 and 0.007.

$\frac{1}{2}M\omega_0^2\Delta x^2$, equals $S_{\mathrm{HR}}\,\omega_0$: S_{HR} is the mean number of phonons emitted in the transition. At $T = 0$ the absorption spectrum is a progression of lines at $E_{\mathrm{ZPL}} + m\omega_0$ with Poisson weights (Fig. 5.5). The line $m = 0$ is the *zero-phonon line*, with weight $\mathrm{e}^{-S_{\mathrm{HR}}}$; the lines $m \geq 1$ form the *phonon sideband*. Emission from the relaxed excited state is the mirror image, with lines at $E_{\mathrm{ZPL}} - m\omega_0$ and the same weights. The means of the two spectra lie at $E_{\mathrm{ZPL}} \pm S_{\mathrm{HR}}\omega_0$, and their separation $2S_{\mathrm{HR}}\omega_0$ is the Stokes shift. For $S_{\mathrm{HR}} \ll 1$ the zero-phonon line carries most of the weight; for $S_{\mathrm{HR}} \gg 1$ it is exponentially weak, and the band is a near-Gaussian of width $S_{\mathrm{HR}}^{1/2}\omega_0$ around its mean. Because the weight $\mathrm{e}^{-S_{\mathrm{HR}}}$ is the fraction of transitions that leave the lattice untouched, the color-center literature calls it the Debye–Waller factor of the line, by analogy with diffraction [32, Sect. 2.2].

At finite temperature the initial vibrational state is thermal, so the overlaps $|\langle m_{\mathrm{e}}|n_{\mathrm{g}}\rangle|^2$ for $n > 0$ are needed as well; the generating function sums them with Gibbs weights. The absorption line shape is

$$\begin{aligned} A(E) &= \sum_n p_n \sum_m |\langle m_{\mathrm{e}}|n_{\mathrm{g}}\rangle|^2\, \delta\big(E - E_{\mathrm{ZPL}} - (m-n)\,\omega_0\big) \\ &= \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{2\pi}\, \mathrm{e}^{\mathrm{i}(E-E_{\mathrm{ZPL}})t}\, \big\langle \mathrm{e}^{\mathrm{i}H_{\mathrm{g}}t}\hat{D}\,\mathrm{e}^{-\mathrm{i}H_{\mathrm{g}}t}\hat{D}^{\dagger}\big\rangle, \end{aligned}$$

where p_n are the Gibbs weights of the ground-state oscillator (Sect. 2.5). The second form writes each δ-function as the time integral of $\mathrm{e}^{\mathrm{i}(E-E_{\mathrm{ZPL}}-m\omega_0+n\omega_0)t}$, uses $\langle m_{\mathrm{e}}|n_{\mathrm{g}}\rangle = \langle m|\hat{D}^{\dagger}|n\rangle$, and lets $\mathrm{e}^{\pm\mathrm{i}H_{\mathrm{g}}t}$ supply the phases $\mathrm{e}^{\mathrm{i}n\omega_0 t}$ and $\mathrm{e}^{-\mathrm{i}m\omega_0 t}$; the sum over m is then a completeness sum. The operator $\mathrm{e}^{\mathrm{i}H_{\mathrm{g}}t}\hat{D}\mathrm{e}^{-\mathrm{i}H_{\mathrm{g}}t}$ is $\hat{D}$ with $d \to d\,\mathrm{e}^{-\mathrm{i}\omega_0 t}$, so both exponents are linear in d and $d^{\dagger}$, and Eq. (C.52) applies, with $\hat{A} = \alpha_{\mathrm{d}}(d^{\dagger}\mathrm{e}^{\mathrm{i}\omega_0 t} - d\,\mathrm{e}^{-\mathrm{i}\omega_0 t})$ and $\hat{B} = -\alpha_{\mathrm{d}}(d^{\dagger} - d)$. The averages are $\langle\hat{A}^2\rangle = \langle\hat{B}^2\rangle = -S_{\mathrm{HR}}(2n_{\mathrm{B}}+1)$ and $\langle\hat{A}\hat{B}\rangle = S_{\mathrm{HR}}[(n_{\mathrm{B}}+1)\mathrm{e}^{-\mathrm{i}\omega_0 t} + n_{\mathrm{B}}\mathrm{e}^{\mathrm{i}\omega_0 t}]$, with $n_{\mathrm{B}} = n_{\mathrm{B}}(\omega_0)$, so that

$$\big\langle \mathrm{e}^{\mathrm{i}H_{\mathrm{g}}t}\hat{D}\,\mathrm{e}^{-\mathrm{i}H_{\mathrm{g}}t}\hat{D}^{\dagger}\big\rangle = \mathrm{e}^{-S_{\mathrm{HR}}(2n_{\mathrm{B}}+1)}\,\exp\Big\{S_{\mathrm{HR}}\big[(n_{\mathrm{B}}+1)\,\mathrm{e}^{-\mathrm{i}\omega_0 t} + n_{\mathrm{B}}\,\mathrm{e}^{\mathrm{i}\omega_0 t}\big]\Big\}. \tag{5.58}$$

Eq. (5.58) has the form of the generating formula Eq. (5.51) for a crystal with a single mode: $2W_\kappa$ becomes $S_{\mathrm{HR}}(2n_{\mathrm{B}}+1)$, and $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}$ becomes the second exponent. A kick of momentum q to a single oscillator is the displacement $\hat{D}(\mathrm{i}q\ell_0)$ (Appendix C.6), and it leads to the same expression with $S_{\mathrm{HR}} \to E_R/\omega_0$. The second exponential expands with the generating function of the modified Bessel functions, $\mathrm{e}^{(z/2)(s+1/s)} = \sum_p s^p I_p(z)$, taken at $s = [(n_{\mathrm{B}}+1)/n_{\mathrm{B}}]^{1/2}\mathrm{e}^{-\mathrm{i}\omega_0 t}$. The coefficient of $\mathrm{e}^{-\mathrm{i}p\omega_0 t}$ is the weight of the line at $E_{\mathrm{ZPL}} + p\,\omega_0$, for every integer p:

$$w_p = \mathrm{e}^{-S_{\mathrm{HR}}(2n_{\mathrm{B}}+1)}\Big(\frac{n_{\mathrm{B}}+1}{n_{\mathrm{B}}}\Big)^{p/2} I_p\big(2S_{\mathrm{HR}}\sqrt{n_{\mathrm{B}}(n_{\mathrm{B}}+1)}\big). \tag{5.59}$$

At $T = 0$ only $p \geq 0$ survives, and w_p returns to the Poisson weights of Eq. (5.57). At $T > 0$ the lines with $p < 0$ appear, and since $I_{-p} = I_p$, $w_{-p}/w_p = \mathrm{e}^{-p\beta\omega_0}$, detailed balance between the Stokes and anti-Stokes sides. The mean of p is S_{HR} at every temperature; the slope of Eq. (5.58) at $t = 0$ fixes it, as the slope of Eq. (5.51) fixed the mean recoil energy in Eq. (5.55).

A real center couples to many modes k, each with its own factor $S_k = \omega_k \Delta Q_k^2/2$, and the generating function is the product of factors Eq. (5.58), one per mode. The total $S_{\mathrm{HR}} = \sum_k S_k$ is the mean number of phonons emitted, and at $T = 0$ the zero-phonon weight is $\mathrm{e}^{-S_{\mathrm{HR}}}$. With a continuum of modes, the replicas merge into a continuous sideband with the shape of the convolutions of Eq. (5.53), while the zero-phonon line stays sharp, since it involves no phonon. Huang and Rhys derived this structure for the F center of alkali halides, where S_{HR} is large and the band is broad [31]. Sect. 7.4 applies it to the color centers of silicon and diamond, with their measured Huang–Rhys factors.

Worked example 5.7: A displaced oscillator

As a toy model we take the bond coordinate of silicon from Sect. 5.1, an oscillator of frequency $\omega_0 = 64.1\,\mathrm{meV}$ whose zero-point length is $\ell_0 = \ell_{\mathrm{zp}} = 0.048\,\text{Å}$ (with the reduced mass $M/2$ of the pair of nuclei, $\ell_0 = (2\cdot\frac{M}{2}\cdot\omega_0)^{-1/2}$ is the ℓ_{zp} of Fig. 5.2). Suppose a localized electronic excitation lengthens the equilibrium bond by Δx.

The size of the shift. A Huang–Rhys factor of two needs $\Delta x = 2\ell_0\sqrt{S_{\mathrm{HR}}} = 0.136\,\text{Å}$, 5.8 % of the bond length $\sqrt{3}a/4 = 2.35\,\text{Å}$. A shift of a few times the zero-point length is enough to make the zero-phonon line a minority: the relaxation energy is $S_{\mathrm{HR}}\omega_0 = 128\,\mathrm{meV}$, and the Stokes shift between the means of absorption and emission is 256 meV.

The overlap directly. The ground-state wavefunction is $\psi_0(x) = (2\pi\ell_0^2)^{-1/4}\mathrm{e}^{-x^2/4\ell_0^2}$, and the overlap with the same Gaussian centered at Δx is

$$\langle 0_{\mathrm{e}}|0_{\mathrm{g}}\rangle = \frac{1}{\sqrt{2\pi}\,\ell_0}\int \mathrm{d}x\; \mathrm{e}^{-[x^2+(x-\Delta x)^2]/4\ell_0^2} = \frac{\mathrm{e}^{-\Delta x^2/4\ell_0^2}}{\sqrt{2\pi}\,\ell_0}\int \mathrm{d}x\; \mathrm{e}^{-x^2/2\ell_0^2 + x\Delta x/2\ell_0^2} = \mathrm{e}^{-\Delta x^2/8\ell_0^2},$$

by Eq. (C.41) with $\mu = 1/2\ell_0^2$ and $\eta = \Delta x/2\ell_0^2$, which evaluates the remaining integral to $\sqrt{2\pi}\,\ell_0\,\mathrm{e}^{\Delta x^2/8\ell_0^2}$. The squared overlap is $\mathrm{e}^{-\Delta x^2/4\ell_0^2} = \mathrm{e}^{-S_{\mathrm{HR}}}$, the zero-phonon weight of Eq. (5.57). For the first replica, $\psi_1(x) = (x/\ell_0)\psi_0(x)$, and the same integral with an extra factor $(x-\Delta x)/\ell_0$ gives $|\langle 1_{\mathrm{e}}|0_{\mathrm{g}}\rangle|^2 = S_{\mathrm{HR}}\mathrm{e}^{-S_{\mathrm{HR}}}$.

The progression. For $S_{\mathrm{HR}} = 2$ the weights of the lines with $p = 0$ to 5 phonons are 0.135, 0.271, 0.271, 0.180, 0.090 and 0.036, the middle panel of Fig. 5.5. The zero-phonon line carries 14 % of the absorption, and the strongest replicas are those with one and two phonons, which for an integer S_{HR} are equal.

Room temperature. At 300 K, $n_{\mathrm{B}}(\omega_0) = 0.091$, the value of Sect. 5.3 for the optical phonon. By Eq. (5.59), the first anti-Stokes line, at $E_{\mathrm{ZPL}} - \omega_0$, has weight 0.021, and the first Stokes replica 0.249; their ratio is $\mathrm{e}^{-\beta\omega_0} = 0.084$. The thermal population thus adds a weak line on the low-energy side of the zero-phonon line, while the mean of the spectrum

stays at $E_{\mathrm{ZPL}} + 2\omega_0$.

5.6 Anharmonicity

The Raman line of the zone-center optical phonon of silicon is not infinitely sharp. Near zero temperature its full width at half maximum is $1.45\,\mathrm{cm}^{-1}$, or $0.18\,\mathrm{meV}$ [33], so that a phonon created at 64.1 meV survives for about 3.7 ps, some sixty oscillations, before it disappears. The harmonic crystal of Sects. 5.2 and 5.3 cannot account for this width. Its phonons are exact eigenstates of Eq. (5.31) and have infinite lifetime. The lifetime is limited by the part of the energy surface that the harmonic approximation dropped, the cubic term already visible in Fig. 5.2(c), which lets one phonon turn into two. For detection the decay matters: when a dark-matter particle creates an optical phonon in a cryogenic crystal, the phonon does not stay where it was made, and within picoseconds it splits into acoustic phonons, which carry the energy to the sensors on the surface (Sect. 20.2). Every lattice response computed in this chapter is the leading term of an expansion in the anharmonic terms. Before trusting these responses, we need to know how large the anharmonic terms are.

5.6.1 Cubic and quartic force constants

The expansion Eq. (5.13) of the energy surface continues beyond second order. With the compact label $i = (\boldsymbol{R}_i \kappa_i \alpha_i)$ for a nucleus and a Cartesian direction, so that $u_i = u^{\alpha_i}_{\boldsymbol{R}_i\kappa_i}$,

$$\begin{aligned} E^{\mathrm{ad}}(u) = E^{\mathrm{ad}}(0) &+ \frac{1}{2!}\sum_{ij} \Phi^{(2)}_{ij}\, u_i u_j + \frac{1}{3!}\sum_{ijk} \Phi^{(3)}_{ijk}\, u_i u_j u_k \\ &+ \frac{1}{4!}\sum_{ijkl} \Phi^{(4)}_{ijkl}\, u_i u_j u_k u_l + \cdots , \\ \Phi^{(n)}_{i_1 \ldots i_n} &= \left.\frac{\partial^n E^{\mathrm{ad}}}{\partial u_{i_1} \cdots \partial u_{i_n}}\right|_{u=0} . \end{aligned} \tag{5.60}$$

The *third- and fourth-order force constants* $\Phi^{(3)}$ and $\Phi^{(4)}$ share the properties of $\Phi^{(2)}$ that Sect. 5.2 derived. As derivatives they are real and symmetric under any permutation of their labels. Since a lattice translation leaves the surface unchanged, they depend on the cells only through differences, $\Phi^{(3)}(\boldsymbol{R}, \boldsymbol{R}', \boldsymbol{R}'') = \Phi^{(3)}(\boldsymbol{0}, \boldsymbol{R}' - \boldsymbol{R}, \boldsymbol{R}'' - \boldsymbol{R})$. A rigid displacement of the whole crystal, $u_i \to u_i + d^{\alpha_i}$, also leaves the surface unchanged, so the total force $\sum_{\boldsymbol{R}\kappa} \partial E^{\mathrm{ad}}/\partial u^{\gamma}_{\boldsymbol{R}\kappa}$ vanishes at every configuration. Differentiating this identity $n-1$ times at $u = 0$ gives the sum rules

$$\sum_{\boldsymbol{R}''\kappa''} \Phi^{(n),\alpha\beta\cdots\gamma}_{\kappa\kappa'\cdots\kappa''}(\boldsymbol{R}, \boldsymbol{R}', \ldots, \boldsymbol{R}'') = 0, \tag{5.61}$$

of which Eq. (5.18) is the case $n = 2$. Physically, $\Phi^{(3)}$ says how the force constants change when the nuclei move, $\Phi^{(2)}_{ij}(u) = \Phi^{(2)}_{ij} + \sum_k \Phi^{(3)}_{ijk} u_k + \cdots$, and $\Phi^{(4)}$ how that change changes in turn. The asymmetric curve of Fig. 5.2(c) is a direct measurement of one combination of the cubic constants: along the [111] bond coordinate of silicon the cubic coefficient is $\Phi^{(3)}_{\mathrm{opt}} = -41.5\,\mathrm{eV\,\AA}^{-3}$ (Worked example 5.1), and the quartic one is about $40\,\mathrm{eV\,\AA}^{-4}$ (Worked example 25.1). A bond is stiffer under compression than under stretching: its force constant falls as it lengthens.

In practice the cubic constants come from the same calculations as the harmonic ones. By the $2n+1$ theorem of Sect. 25.1, the first-order density responses of DFPT fix the third derivatives of the energy, so that one set of linear-response calculations on a grid of wavevectors contains $\Phi^{(3)}$ as well [21, Eq. (123)]. The alternative is finite differences of forces in a supercell, with one calculation for each symmetry-distinct pair of displaced atoms and a cutoff on the range of the triplets, or a fit to forces sampled at temperature T. The fourth-order constants are rarely computed term by term; they enter through the effective harmonic theories at the end of this section.

To see what the cubic term does to phonons, we write it in the phonon operators of Sect. 5.3. When the phonon expansion Eq. (5.32) is inserted three times, each displacement brings a factor $(2NM_\kappa\omega)^{-1/2}$, a polarization vector, a phase and a displacement field. At fixed relative positions, summing the three phases over the first cell $\boldsymbol{R}$ gives $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}_1+\boldsymbol{Q}_2+\boldsymbol{Q}_3)\cdot\boldsymbol{R}} = N\sum_{\boldsymbol{G}} \delta_{\boldsymbol{Q}_1+\boldsymbol{Q}_2+\boldsymbol{Q}_3,\boldsymbol{G}}$, by the lattice sum Eq. (3.14). With the phonon labels written $1 = (\boldsymbol{Q}_1\nu_1)$ and so on,

$$H_3 = \frac{1}{3!}\frac{1}{\sqrt{N}} \sum_{123} \sum_{\boldsymbol{G}} \delta_{\boldsymbol{Q}_1+\boldsymbol{Q}_2+\boldsymbol{Q}_3,\boldsymbol{G}}\, V^{(3)}(1,2,3)\, B_1 B_2 B_3, \tag{5.62}$$

with the *three-phonon vertex*

$$\begin{aligned} V^{(3)}(1,2,3) = {} & \frac{1}{\sqrt{8\,\omega_1\omega_2\omega_3}} \sum_{\kappa} \sum_{\boldsymbol{R}'\kappa'} \sum_{\boldsymbol{R}''\kappa''} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}_2\cdot\boldsymbol{R}'+\boldsymbol{Q}_3\cdot\boldsymbol{R}'')} \\ & \times \frac{\Phi^{(3),\alpha\beta\gamma}_{\kappa\kappa'\kappa''}(\boldsymbol{0},\boldsymbol{R}',\boldsymbol{R}'')\, e^{\alpha}_{\kappa\nu_1}(\boldsymbol{Q}_1)\, e^{\beta}_{\kappa'\nu_2}(\boldsymbol{Q}_2)\, e^{\gamma}_{\kappa''\nu_3}(\boldsymbol{Q}_3)}{\sqrt{M_\kappa M_{\kappa'} M_{\kappa''}}}, \end{aligned} \tag{5.63}$$

summed over Cartesian indices, and $\omega_i = \omega_{\boldsymbol{Q}_i\nu_i}$. The vertex has the dimension of an energy and is symmetric in its three labels, and $V^{(3)}(-1,-2,-3) = V^{(3)}(1,2,3)^*$ by Eq. (A.21). The factor $1/\sqrt{N}$ comes from spreading three phonons over the whole crystal, and the Kronecker delta is the conservation of crystal momentum: three phonons can meet only if their wavevectors add up to a reciprocal-lattice vector. Since each B_i creates or destroys one phonon, H_3 turns one phonon into two, two into one, or creates or destroys three at once. The quartic term, written in the same way, has four fields and the factor $1/N$.

5.6.2 Three-phonon decay and the linewidth

Consider a crystal in thermal equilibrium, except for a small excess population in one mode $\boldsymbol{Q}\nu$. Since $\omega_{\boldsymbol{Q}\nu} > 0$, the processes of H_3 that conserve energy are of two kinds. The phonon can decay into two others, $\boldsymbol{Q}\nu \to 1+2$, or merge with a thermal phonon into a third, $\boldsymbol{Q}\nu + 1 \to 2$. For the decay, the matrix element of Eq. (5.62) between $|n_{\boldsymbol{Q}\nu}, n_1, n_2\rangle$ and $|n_{\boldsymbol{Q}\nu}-1, n_1+1, n_2+1\rangle$ comes from the terms that contain $b_{\boldsymbol{Q}\nu}$, $b_1^\dagger$ and $b_2^\dagger$. These three operators are in $B_{\boldsymbol{Q}\nu}$, B_{-1} and B_{-2}, with $-1 = (-\boldsymbol{Q}_1\nu_1)$, and there are 3! ways of assigning the three fields to the three slots of H_3, all equal by the symmetry of the vertex. The 3! cancels, and the boson factors of Eq. (2.34) give

$$\langle n_{\boldsymbol{Q}\nu}-1, n_1+1, n_2+1 | H_3 | n_{\boldsymbol{Q}\nu}, n_1, n_2\rangle = \frac{V^{(3)}(\boldsymbol{Q}\nu,-1,-2)}{\sqrt{N}} \sqrt{n_{\boldsymbol{Q}\nu}(n_1+1)(n_2+1)},$$

with $\boldsymbol{Q}_1 + \boldsymbol{Q}_2 = \boldsymbol{Q}$ up to a reciprocal-lattice vector. By the golden rule Eq. (2.5), the rate at which the mode loses phonons this way is $(2\pi/N)\sum |V^{(3)}|^2 n_{\boldsymbol{Q}\nu}(n_1+1)(n_2+1)\,\delta(\omega_{\boldsymbol{Q}\nu} - \omega_1 - \omega_2)$, summed over distinct final pairs, a sum that is half the sum over $\boldsymbol{Q}_1$, ν_1 and ν_2 with $\boldsymbol{Q}_2$ fixed. The inverse process, in which the pair merges back, feeds the mode at the rate with $(n_{\boldsymbol{Q}\nu}+1)n_1 n_2$ in place of $n_{\boldsymbol{Q}\nu}(n_1+1)(n_2+1)$. The difference of the two

occupation factors is $n_{\boldsymbol{Q}\nu}(1+n_1+n_2)-n_1n_2$. At equilibrium this difference vanishes, since $n_{\mathrm{B}}(\omega_1+\omega_2)(1+n_1+n_2)=n_1n_2$ for Bose functions, the detailed-balance condition for the process (Sect. 2.5). For an excess δn in the mode the difference is $(1+n_1+n_2)\,\delta n$, so the excess decays exponentially. The merging processes contribute in the same way, with the occupation difference $n_{\boldsymbol{Q}\nu}n_1(1+n_2)-(n_{\boldsymbol{Q}\nu}+1)(n_1+1)n_2=(n_1-n_2)\,\delta n$, and with no factor one half, since the two partners now play different roles. After the relabeling $\boldsymbol{Q}_i\to-\boldsymbol{Q}_i$, which leaves ω_i and n_i unchanged, the decay and merging terms add, and the excess decays as $\mathrm{e}^{-\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}t}$ with

Three-phonon linewidth

$$\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}=\frac{\pi}{N}\sum_{\boldsymbol{Q}_1\nu_1\nu_2}\left|V^{(3)}(\boldsymbol{Q}\nu,\boldsymbol{Q}_1\nu_1,\boldsymbol{Q}_2\nu_2)\right|^2 \times\Big[(1+n_1+n_2)\,\delta(\omega_{\boldsymbol{Q}\nu}-\omega_1-\omega_2)+2(n_1-n_2)\,\delta(\omega_{\boldsymbol{Q}\nu}+\omega_1-\omega_2)\Big], \tag{5.64}$$

with $\boldsymbol{Q}_2=-\boldsymbol{Q}-\boldsymbol{Q}_1$ up to a reciprocal-lattice vector, $\omega_i=\omega_{\boldsymbol{Q}_i\nu_i}$ and $n_i=n_{\mathrm{B}}(\omega_i)$. The first term is the decay of the phonon into the pair $(-\boldsymbol{Q}_1\nu_1,-\boldsymbol{Q}_2\nu_2)$, the second its merging with $\boldsymbol{Q}_1\nu_1$ into $-\boldsymbol{Q}_2\nu_2$.

This is the result of Maradudin and Fein [34, 35, Eq. (6)]. The rate at which an excess population relaxes is also the width of the line. A state that decays with probability $\mathrm{e}^{-\gamma t}$ has an amplitude that decays as $\mathrm{e}^{-\gamma t/2}$, and its spectrum is a Lorentzian with full width γ at half maximum. The *phonon linewidth* $\gamma_{\boldsymbol{Q}\nu}$ is this full width, the inverse lifetime of the phonon, and $\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}$ is its anharmonic part; in the language of the phonon self-energy of Sect. 6.4, defined there in units of a squared frequency, $\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}$ is $-\operatorname{Im}\Pi_\nu/\omega_{\boldsymbol{Q}\nu}$ for the bubble with two cubic vertices. Other mechanisms add to the linewidth: the creation of electron–hole pairs, $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$, which vanishes in an insulator at zero temperature (Sect. 6.4), and the scattering by isotopes and defects (Sect. 11.6).

Eq. (5.64) has two ingredients, the strength of the vertex and the number of pairs that conserve energy and crystal momentum, and the second is often the more restrictive. Temperature enters only through the occupations. At $T=0$ only spontaneous decay remains, with weight one: a phonon of a cryogenic crystal can split, but it finds no thermal phonon to merge with. At high temperature $n_i\approx T/\omega_i$, and the width grows linearly with T.

The zone-center optical phonon of silicon is a concrete case. Since its wavevector is zero, it decays into pairs with opposite wavevectors, $(\boldsymbol{Q}_1,-\boldsymbol{Q}_1)$, and their energies must add to $\omega_{\mathrm{opt}}=\omega_{\Gamma,\mathrm{opt}}$. It is also the highest phonon of the crystal and has no partner to merge with, so only the decay term of Eq. (5.64) contributes, at any temperature. Fig. 5.6(a) shows the allowed pairs on the computed dispersion as crossings between a branch and the mirror image $\omega_{\mathrm{opt}}-\omega_{\boldsymbol{Q}\nu}$ of another. Klemens proposed the simplest channel: two acoustic phonons of the same branch, each with energy $\omega_{\mathrm{opt}}/2$ [36]. With $\omega_1=\omega_2=\omega_{\mathrm{opt}}/2$ in Eq. (5.64) the occupation factor is the same for every pair, and the width follows the *Klemens law*

$$\gamma^{\mathrm{anh}}_{\Gamma,\mathrm{opt}}(T)=\gamma^{\mathrm{anh}}_{\Gamma,\mathrm{opt}}(0)\left[1+2n_{\mathrm{B}}\Big(\frac{\omega_{\mathrm{opt}}}{2}\Big)\right]. \tag{5.65}$$

In silicon the Klemens channel is only partly open. The transverse acoustic branches never reach $\omega_{\mathrm{opt}}/2$: in the PBE calculation of Fig. 5.6 they top out at 26.4 meV, below $\omega_{\mathrm{opt}}/2=31.8$ meV, and two transverse phonons cannot share the energy. Two longitudinal acoustic phonons can. However, few modes lie near $\omega_{\mathrm{opt}}/2$, where the phonon density of states is small. Most of the phase space is in asymmetric pairs, a transverse acoustic phonon near 22 meV with a

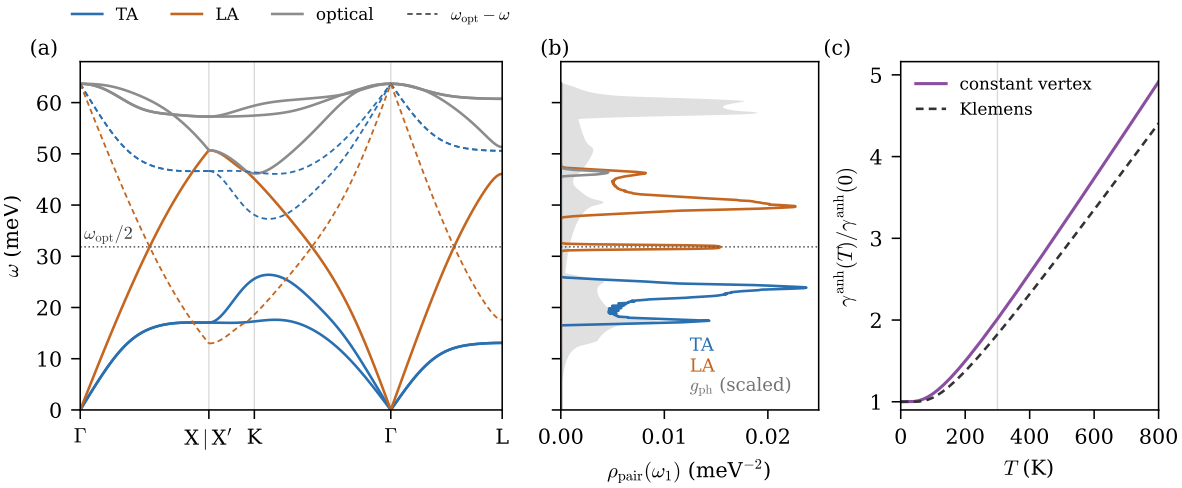


Figure 5.6: Three-phonon decay of the zone-center optical phonon of silicon, $\omega_{\mathrm{opt}} = 63.7\,\mathrm{meV}$ in this calculation. Phonons from the PBE force constants of Fig. 5.3 (DFPT on a 6^3 grid of wavevectors, $a = 5.431\,\text{Å}$), Fourier-interpolated; branches are labeled by energy order, the lowest two transverse acoustic (TA, blue), the third longitudinal acoustic (LA, orange), the upper three optical (gray). (a) Dispersion (solid) and the mirror curves $\omega_{\mathrm{opt}} - \omega_{\boldsymbol{Q}\nu}$ of the acoustic branches (dashed). A solid branch crossing a dashed one marks a pair $(\boldsymbol{Q}, -\boldsymbol{Q})$ whose energies add to ω_{opt}; dotted: $\omega_{\mathrm{opt}}/2$. (b) Two-phonon density of final states resolved by the energy ω_1 of one member of the pair and by its branch, $\rho_{\mathrm{pair}}(\omega_1) = N^{-1}\sum_{\boldsymbol{Q}\nu_1\nu_2}\delta(\omega_1 - \omega_{\boldsymbol{Q}\nu_1})\,\delta(\omega_{\mathrm{opt}} - \omega_{\boldsymbol{Q}\nu_1} - \omega_{\boldsymbol{Q}\nu_2})$, on an 80^3 grid with Gaussians of width 0.25 meV and a 0.9 meV running mean; it is symmetric about $\omega_{\mathrm{opt}}/2$, each TA phonon (blue) pairing with an LA phonon (orange) at the mirror energy. Shaded: the one-phonon density of states g_{ph}, scaled. LA+TA pairs make up 89 % of the final states, LA+LA 7 %, and TA+TA none; the remaining 4 % are optical+TA pairs from a small region near K. (c) Temperature dependence of the width relative to $T = 0$: Klemens's law Eq. (5.65) (dashed) and the average of $1 + n_1 + n_2$ over the final states of panel (b) (solid), which assumes a constant vertex; the vertical line marks 300 K.

longitudinal one near 42 meV, which make up 89 % of the density of final states in Fig. 5.6(b), against 7 % for the pairs of longitudinal phonons. A full DFPT calculation of the vertex finds the same ordering, 94.3 % and 5.7 % [37, Table 1]. For asymmetric pairs the occupation factor $1 + n_1 + n_2$ is larger than Klemens's at the same temperature, because the softer partner is populated first (Fig. 5.6(c)).

Worked example 5.8: Klemens decay of the optical phonon of silicon

We compare the width of the zone-center optical phonon with the Klemens law Eq. (5.65), using the measured $\omega_{\mathrm{opt}} = 64.1\,\mathrm{meV}$.

The measured width. At low temperature the Raman line of the mode has a full width of $1.45\,\mathrm{cm}^{-1}$ [33], as tabulated in Ref. [37, Table 1]. With $1\,\mathrm{cm}^{-1} = 0.123\,98\,\mathrm{meV}$, this is $\gamma_{\Gamma,\mathrm{opt}} = 0.180\,\mathrm{meV}$, and

$$\frac{\gamma_{\Gamma,\mathrm{opt}}}{\omega_{\mathrm{opt}}} = 2.80 \times 10^{-3}, \qquad \frac{1}{\gamma_{\Gamma,\mathrm{opt}}} = 3.66\,\mathrm{ps},$$

57 periods of 64.5 fs. Other measurements range from 1.24 to $2.1\,\mathrm{cm}^{-1}$ [37], a spread that reflects sample quality and spectral resolution. Silicon is an insulator, with $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu} = 0$ at

low temperature, and isotope scattering is weak for a mode at the top of the spectrum. The width is anharmonic. A DFPT calculation of the full vertex Eq. (5.63), inserted in Eq. (5.64), yields $1.43\,\mathrm{cm}^{-1}$ [37, Table 1], 99 % of the measured value, so three-phonon decay accounts for the whole width.

The temperature dependence. At the 10 mK of a detector, $\omega_{\mathrm{opt}}/2T = 3.7 \times 10^4$, the occupation $n_{\mathrm{B}}(\omega_{\mathrm{opt}}/2)$ is e^{-37000}, and the width is its zero-temperature value: the optical phonon decays spontaneously, with no help from thermal phonons. At 77 K, $n_{\mathrm{B}}(\omega_{\mathrm{opt}}/2) = 0.008$ and the Klemens factor is 1.016. At 300 K, $n_{\mathrm{B}}(\omega_{\mathrm{opt}}/2) = 0.407$, and Eq. (5.65) predicts

$$\gamma_{\Gamma,\mathrm{opt}}(300\,\mathrm{K}) = 1.45\,\mathrm{cm}^{-1} \times (1 + 2 \times 0.407) = 2.63\,\mathrm{cm}^{-1}.$$

With a constant vertex, averaging $1 + n_1 + n_2$ over the dominant asymmetric pairs gives a factor 2.01 instead of 1.82 and a width of $2.92\,\mathrm{cm}^{-1}$ (Fig. 5.6(c)). The Klemens law thus gets the form of the temperature dependence right: a constant plus a term that grows as the Bose function of the daughters. Its single daughter energy, $\omega_{\mathrm{opt}}/2$, misses the true pairs of silicon, and at room temperature the Klemens law gives a width about 10 % below the estimate from those pairs. With the full vertex and the true daughters, Eq. (5.64) follows the measured widths up to temperatures of the order of $\omega_{\mathrm{opt}} \approx 740\,\mathrm{K}$ [37].

The Klemens law is a good parametrization of the temperature dependence once its zero-temperature width is taken from experiment or from a calculation. It does not predict that width, and its picture of the channel is wrong in silicon.

5.6.3 The anharmonic small parameter

How large are the anharmonic terms compared with the harmonic ones? The natural measure is the ratio of the cubic to the harmonic energy at the zero-point amplitude of a mode. Take a bond-like coordinate with curvature $\Phi^{(2)}$, cubic coefficient $\Phi^{(3)}$ and zero-point length ℓ_{zp}. The ratio of $\frac{1}{6}\Phi^{(3)}\ell_{\mathrm{zp}}^3$ to $\frac{1}{2}\Phi^{(2)}\ell_{\mathrm{zp}}^2$ defines the *anharmonic small parameter*

$$\eta_{\mathrm{anh}} = \frac{|\Phi^{(3)}|\,\ell_{\mathrm{zp}}}{3\,\Phi^{(2)}}. \tag{5.66}$$

Equivalently, η_{anh} is the zero-point amplitude measured in units of the length $3\Phi^{(2)}/|\Phi^{(3)}|$ over which the force constants change by their own size. The counting of Sect. 5.1 estimates it at once. Force constants of every order scale as $\Phi^{(n)} \sim \alpha^2 m_e/a_0^n$, and the zero-point amplitude as $\ell_{\mathrm{zp}} \sim \kappa_{\mathrm{BO}} a_0$, so

$$\eta_{\mathrm{anh}} \sim \frac{\alpha^2 m_e/a_0^3}{\alpha^2 m_e/a_0^2}\,\kappa_{\mathrm{BO}} a_0 = \kappa_{\mathrm{BO}}.$$

The anharmonic parameter is the Born–Oppenheimer parameter itself: the cubic term is smaller than a phonon energy by one power of κ_{BO}, as Sect. 5.1 found. The quartic term, $\frac{1}{24}\Phi^{(4)}\ell_{\mathrm{zp}}^4$, is smaller than the harmonic one by η_{anh}^2, since $\Phi^{(4)}\ell_{\mathrm{zp}}^2/\Phi^{(2)} \sim (\ell_{\mathrm{zp}}/a_0)^2$, and each further order brings one more power.

The physical effects follow by counting powers of η_{anh}. The cubic term changes no energy at first order, since it is a product of an odd number of displacements and its average over any harmonic state vanishes. At second order it shifts the levels by $(\eta_{\mathrm{anh}}\omega_{\mathrm{ph}})^2/\omega_{\mathrm{ph}} = \eta_{\mathrm{anh}}^2\omega_{\mathrm{ph}}$, since the energy denominators of second-order perturbation theory are themselves phonon energies. The quartic term shifts them at first order by the same amount, $\eta_{\mathrm{anh}}^2\omega_{\mathrm{ph}}$. The decay rate Eq. (5.64) is the square of a cubic matrix element of size $\eta_{\mathrm{anh}}\omega_{\mathrm{ph}}$ times a density of final

states of order $1/\omega_{\mathrm{ph}}$, again $\eta_{\mathrm{anh}}^2\omega_{\mathrm{ph}}$. The three effects are of one order:

$$\frac{\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}}{\omega_{\boldsymbol{Q}\nu}} \sim \frac{\Delta\omega_{\boldsymbol{Q}\nu}^{\mathrm{anh}}}{\omega_{\boldsymbol{Q}\nu}} \sim \eta_{\mathrm{anh}}^2 \sim \kappa_{\mathrm{BO}}^2, \tag{5.67}$$

where $\Delta\omega_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$ is the anharmonic shift of the frequency. In absolute terms the shift and the width are $\alpha^2 m_e\,\kappa_{\mathrm{BO}}^4$, the anharmonic level of the Born–Oppenheimer hierarchy of Sect. 5.1. In the labeling of Sect. 8.5 the order η_{anh}^2 is anharmonic order $n_A = 2$, where n_A counts powers of the anharmonic expansion parameter in the response rather than vertices, and the leading anharmonic phonon self-energy is of that order. In thermal equilibrium the zero-point length is replaced by the thermal amplitude, $\ell_{\mathrm{zp}}^2 \to \ell_{\mathrm{zp}}^2(2n_{\mathrm{B}}+1)$, as in Eq. (5.35), and at high temperature η_{anh}^2 grows in proportion to T, as the widths do.

Silicon bears out this counting. Along the bond coordinate of Fig. 5.2(c), $\Phi^{(2)} = 13.6\,\mathrm{eV\,\AA^{-2}}$, $\Phi^{(3)} = -41.5\,\mathrm{eV\,\AA^{-3}}$ and $\ell_{\mathrm{zp}} = 0.048\,\mathrm{\AA}$ give $\eta_{\mathrm{anh}} = 0.049$, against $\kappa_{\mathrm{BO}} = 0.067$; the length over which the force constants change is $3\Phi^{(2)}/|\Phi^{(3)}| = 0.98\,\mathrm{\AA}$, about 40 % of the bond. Then $\eta_{\mathrm{anh}}^2 = 2.4\times10^{-3}$. The quartic-to-harmonic ratio at the same amplitude, $\Phi^{(4)}\ell_{\mathrm{zp}}^2/12\Phi^{(2)}$ with $\Phi^{(4)} \approx 40\,\mathrm{eV\,\AA^{-4}}$, is 0.6×10^{-3}, a quarter of η_{anh}^2, and the measured width of Worked example 5.8 is $\gamma/\omega_{\mathrm{opt}} = 2.8\times10^{-3}$, 1.2 times η_{anh}^2. The coefficients of the counting are close to one, between a quarter and two here and in the Born–Oppenheimer hierarchy of Worked example 5.1. Silicon at detector temperatures is a weakly anharmonic crystal, and its harmonic phonons are correct to a few parts in a thousand, well below the few-percent error of the functional (Sect. 5.2).

The anharmonic parameter is not always small, and it grows with ℓ_{zp}. Light atoms have larger amplitudes, since $\ell_{\mathrm{zp}} \propto M^{-1/4}$ at fixed force constants, and hydrogen has $\kappa_{\mathrm{BO}} = 0.15$. The parameter also grows for soft modes near a structural instability, where $\Phi^{(2)} \to 0$ and the harmonic expansion about the minimum loses its meaning. The parameter is also large in quantum crystals such as solid helium, where the zero-point motion is a sizable fraction of the spacing, and in any crystal near melting. At low energy phonon decay is controlled by phase space rather than by η_{anh}. The width of a long-wavelength acoustic phonon at $T = 0$ scales as the fifth power of its energy, and this width sets how fast the phonons that a dark-matter event leaves behind down-convert. For the nuclear response to dark matter, η_{anh} measures how far the harmonic results of this chapter can be trusted, and Sect. 11.5 turns it into a correction.

5.6.4 Effective harmonic theories

When temperature, light atoms or soft modes make the expansion Eq. (5.60) converge slowly, the usual remedy is not to add orders but to find a better harmonic problem. Three effective harmonic theories are in common use, and they differ in what the harmonic force constants are fitted to.

The *quasi-harmonic approximation* fits the force constants to the volume. It keeps every mode harmonic but computes the frequencies at the cell volume Ω that minimizes the free energy at temperature T, Eq. (25.24). The volume dependence of the frequencies is itself a cubic effect. A homogeneous strain ϵ displaces every nucleus by $u_i = \sum_\beta \epsilon^{\alpha_i\beta} X_i^\beta$, with $\boldsymbol{X}_i$ its equilibrium position. The force constants about the strained configuration then change, to first order, by

$$\Phi_{ij}^{(2)}(\epsilon) - \Phi_{ij}^{(2)} = \sum_k \Phi_{ijk}^{(3)} \sum_\beta \epsilon^{\alpha_k\beta} X_k^\beta. \tag{5.68}$$

The Grüneisen parameter $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{G}} = -\partial \ln\omega_{\boldsymbol{Q}\nu}/\partial\ln\Omega$ of Sect. 5.2 is therefore a projection of $\Phi^{(3)}$, and a harmonic crystal, with $\Phi^{(3)} = 0$, neither expands when heated nor shifts its phonons

under pressure. The zero-point energy of Sect. 5.3 also depends on the volume, and the resulting zero-point pressure expands the lattice at $T = 0$ beyond the static minimum of Sect. 5.1. The quasi-harmonic approximation keeps the cubic term at the level of the volume. It drops the shifts at fixed volume, which Eq. (5.67) puts at the same order, and it has no widths.

The *self-consistent harmonic approximation* (SCHA) fits the force constants to the distribution of the nuclei, by a variational principle [38]. For any trial density matrix $\tilde{\varrho}$, the functional $\mathcal{F}[\tilde{\varrho}] = \langle H \rangle_{\tilde{\varrho}} - TS_{\text{ent}}[\tilde{\varrho}]$ is an upper bound on the free energy (the Gibbs–Bogoliubov inequality). The SCHA takes for $\tilde{\varrho}$ the thermal state of a harmonic Hamiltonian H_{trial} with centroids $\bar{X}_a$ and auxiliary force constants $\Phi^{(2)}_{\text{aux}}$, Eq. (25.26), with a running over the Cartesian coordinates X_a of all nuclei and $\delta X_a = X_a - \bar{X}_a$. The distribution of positions in this state is a Gaussian centered at $\bar{X}$. With $\mathcal{F}_{\text{trial}}$ the free energy of H_{trial} and $V_{\text{trial}} = \frac{1}{2}\sum_{ab} \delta X_a \Phi^{(2)}_{\text{aux},ab} \delta X_b$ its potential, the bound reads

$$\mathcal{F}[\tilde{\varrho}] = \mathcal{F}_{\text{trial}} + \big\langle E^{\text{ad}} - V_{\text{trial}} \big\rangle_{\tilde{\varrho}}.$$

We minimize the bound. A change of $\Phi^{(2)}_{\text{aux},ab}$ changes $\mathcal{F}_{\text{trial}}$ by $\frac{1}{2}\langle \delta X_a \delta X_b \rangle\, \delta\Phi^{(2)}_{\text{aux},ab}$, the thermal form of the Hellmann–Feynman theorem of Sect. 5.1, and the average of V_{trial} by the same amount plus $\frac{1}{2}\sum_{cd} \Phi^{(2)}_{\text{aux},cd}\, \delta\langle \delta X_c \delta X_d \rangle$. The first contributions cancel. The average of E^{ad} over a Gaussian depends on the force constants only through the covariance $\langle \delta X_c \delta X_d \rangle$, and a Gaussian obeys the diffusion equation in its covariance, so that $\delta\langle E^{\text{ad}} \rangle = \frac{1}{2}\sum_{cd} \langle \partial_c \partial_d E^{\text{ad}} \rangle\, \delta\langle \delta X_c \delta X_d \rangle$. Altogether

$$\delta\mathcal{F} = \frac{1}{2}\sum_{cd} \Big[\big\langle \partial_c \partial_d E^{\text{ad}} \big\rangle_{\tilde{\varrho}} - \Phi^{(2)}_{\text{aux},cd} \Big]\, \delta\langle \delta X_c \delta X_d \rangle,$$

and the centroids enter only through $\langle E^{\text{ad}} \rangle$, whose derivative is the average force. Both variations vanish at the minimum:

Self-consistent harmonic approximation

$$\Phi^{(2)}_{\text{aux},ab} = \Big\langle \frac{\partial^2 E^{\text{ad}}}{\partial X_a\, \partial X_b} \Big\rangle_{\tilde{\varrho}}, \qquad \Big\langle \frac{\partial E^{\text{ad}}}{\partial X_a} \Big\rangle_{\tilde{\varrho}} = 0. \tag{5.69}$$

The auxiliary force constants are the curvature of the true energy surface averaged over the Gaussian distribution that they themselves generate, and the centroids lie where the average force vanishes.

These are the stationarity conditions of Ref. [38, Eqs. (21) and (22)], which writes them as averages of forces. The first condition is self-consistent because the width of the distribution, $\langle \delta X_a \delta X_b \rangle$, is computed from the auxiliary frequencies with the zero-point and thermal factors of Eq. (5.35). Expanded in the force constants, the average curvature is $\Phi^{(2)} + \frac{1}{2}\sum_{cd} \Phi^{(4)}_{abcd} \langle \delta X_c \delta X_d \rangle + \cdots$: its first correction is the first-order quartic shift of Eq. (5.67), and the SCHA sums this shift to all orders, together with the corrections from the higher even force constants. The cubic term enters only through the shift of the centroids and, in the physical phonon frequencies, through an added bubble term. For soft modes the SCHA is the right starting point even when the harmonic frequencies are imaginary, since the average curvature can be positive where the curvature at the minimum is not.

The *temperature-dependent effective potential* fits the force constants to forces. It samples nuclear configurations at temperature T, by molecular dynamics, computes the forces in each, and finds the effective constants $\tilde{\Phi}^{(2)}$ that reproduce them best in the least-squares sense,

Eq. (25.27) [39]. For a Gaussian sample the least-squares fit returns exactly the average curvature of Eq. (5.69). The reason is that a Gaussian average of the force times a displacement is the average of the force's derivative times the covariance. The temperature-dependent effective potential and the SCHA differ in the distribution they sample, a classical one in molecular dynamics and a quantum Gaussian in the SCHA. That difference matters at detector temperatures, where a classical sample sees an almost static lattice and misses the zero-point motion. The same fit extends to effective third-order constants, from which Eq. (5.64) gives temperature-dependent widths [40].

For silicon at 10 mK, all three effective harmonic theories reduce to the harmonic phonons of Sect. 5.2 with corrections of a few parts in a thousand, so the rest of the book can work with the harmonic phonons.

Summary

Electrons are lighter than nuclei by a factor of about 2×10^{-5} in silicon, and they settle into their ground state many times during one vibration of the lattice (Sect. 5.1). The Born–Oppenheimer approximation turns their ground-state energy at fixed nuclei, with the repulsion of the nuclei added, into the potential-energy surface $E^{\mathrm{ad}}(u)$ on which the nuclei move. The small parameter is the fourth root of the mass ratio, $\kappa_{\mathrm{BO}} = (m_e/M)^{1/4} = 0.067$; in an insulator the ratio that controls the approximation is the phonon energy over the gap, 0.0548 in silicon.

Expanded to second order about its minimum, the energy surface defines the force constants $\Phi^{(2)}$ and the dynamical matrix $D^{\mathrm{dyn}}(\boldsymbol{Q})$, whose eigenvalues are the squared phonon frequencies $\omega^2_{\boldsymbol{Q}\nu}$ and whose eigenvectors are the polarizations $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ (Sect. 5.2). Silicon has six branches up to 64.1 meV, flat transverse acoustic branches that pile up states near 17 meV, and no LO–TO splitting.

Quantized, each normal mode is an oscillator and a phonon is one of its quanta (Sect. 5.3). Emitting a phonon has the weight $n_{\boldsymbol{Q}\nu} + 1$ and absorbing one the weight $n_{\boldsymbol{Q}\nu}$, and in a cryogenic detector every mode is empty, so emission is the whole signal. Zero-point motion alone spreads a silicon nucleus over 0.05 Å per Cartesian component.

In a polar crystal the sublattices carry Born effective charges Z^*_κ, which add up to zero; Z^*_κ is the force on a nucleus per unit macroscopic field (Sect. 5.4). The long-range field of a longitudinal optical wave splits the longitudinal from the transverse modes, in GaAs 36.4 meV against 33.6 meV, and the splitting gives $Z^* = 2.11$. A uniform field couples only to the polar optical modes at the zone center, through Z^*_κ, coherently in every cell.

At finite momentum transfer the response of a nucleus is controlled by thermal averages of exponentials of displacements (Sect. 5.5). For a harmonic crystal the probability that the lattice absorbs $\boldsymbol{q}$ and stays in its state is the Debye–Waller factor e^{-2W_κ}, with $2W_\kappa = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}$, and the number of phonons created is a Poisson distribution with mean $2W_\kappa$. In silicon at zero temperature $2W_\kappa = 1$ at $q = 40$ keV, the crossover from single phonons to multiphonon emission.

The cubic term of the energy surface lets a phonon decay into two (Sect. 5.6); the zone-center optical phonon of silicon decays mainly into a transverse and a longitudinal acoustic phonon of unequal energies. The anharmonic small parameter η_{anh}, the ratio of the cubic to the harmonic energy at the zero-point amplitude, is of order κ_{BO}, 0.049 in silicon, and widths and anharmonic shifts are both of relative order η^2_{anh}.

The phonons, their correlations and their lifetimes are all properties of the energy surface of electrons that never leave their ground state. The couplings that the Born–Oppenheimer approximation dropped let a moving nucleus excite the electrons, and a passing electron excite

the lattice; Chap. 6 derives them.

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Chapter 6

Electron–phonon coupling

The band gap of silicon is 1.17 eV at low temperature and 1.12 eV at room temperature (Sect. 3.3), and the zero-point motion of the nuclei already lowers it (Sect. 6.3). The static lattice of Chaps. 3 and 4 has a gap that changes with neither the temperature nor the nuclear mass, and the harmonic crystal of Chap. 5 treats the phonons as a system of their own. The zero-point shift of the gap and the change of the gap with temperature come from the coupling between electrons and nuclei: a displaced nucleus changes the potential the electrons feel, and an electron in turn changes the forces between nuclei. For dark-matter detection this coupling matters well beyond the gap. A phonon can supply the crystal momentum that an electron needs to cross the indirect gap and that a slow incoming particle lacks. The same coupling limits how long an excited carrier keeps its energy before the energy ends up as heat in the lattice. And when a dark-matter particle strikes a nucleus instead of an electron, the coupling decides whether the electrons follow the nucleus or stay behind, excited or ionized; the second outcome is the Migdal effect. All of these effects follow from adding back the couplings that the Born–Oppenheimer approximation of Chap. 5 dropped. We will start from those couplings and end, along Fig. 6.1, with the absorption of light, and of dark matter, at the indirect gap of silicon.

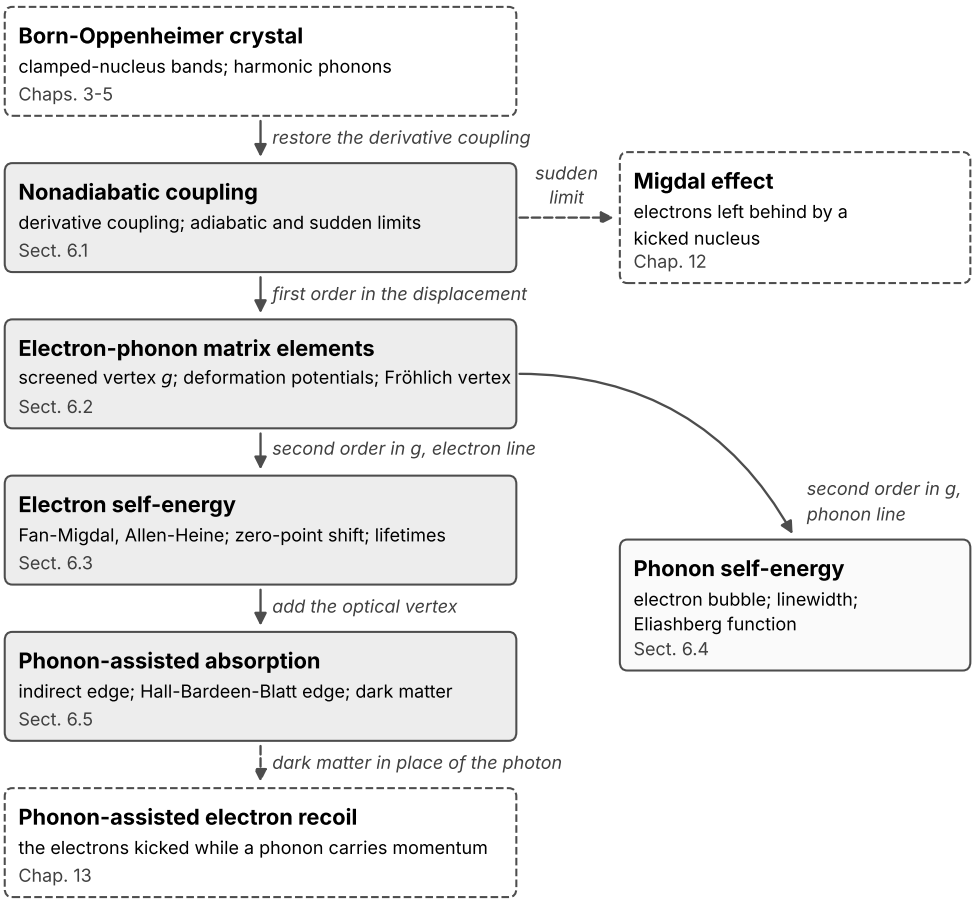


Figure 6.1: Adding back to the Born–Oppenheimer crystal (Chaps. 3 and 5) what it dropped. The derivative coupling, with its adiabatic and sudden limits (Sect. 6.1), gives at first order in the displacement the electron–phonon vertex g (Sect. 6.2). At second order in g it gives the electron self-energy, with the zero-point shift of the gap and carrier lifetimes (Sect. 6.3), and the phonon self-energy and Eliashberg function (Sect. 6.4). With an optical vertex the derivative coupling gives phonon-assisted absorption at the indirect gap (Sect. 6.5). Dashed boxes: the sudden limit of a kicked nucleus becomes the Migdal effect, and dark matter in place of the photon becomes phonon-assisted electron recoil.

6.1 Nonadiabatic coupling

When a dark-matter particle of 1 GeV scatters off a silicon nucleus at the maximum momentum transfer, $q = 2\mu_{\chi N} v_{\max} = 5.10\,\mathrm{MeV}$ (Sect. 1.3), the nucleus recoils at $v_N = q/M = 1.95\times10^{-4}$, or $58\,\mathrm{km\,s^{-1}}$. The outer electrons of the atom move at typical speeds $\sim \alpha = 7.3\times10^{-3}$, 37 times faster, and the Born–Oppenheimer picture of Sect. 5.1 says that they follow the nucleus. The electrons follow the nucleus closely but not exactly. Seen from the recoiling nucleus, every electron has suddenly acquired the velocity $-v_N$ and the momentum $m_e v_N = 100\,\mathrm{eV}$, and an electron cloud bound with momenta $\sim \alpha m_e = 3.7\,\mathrm{keV}$ absorbs this small kick elastically most of the time, but not always. For an electron bound as in hydrogen, the probability that it is left excited or ionized is $(v_N/\alpha)^2 = 7\times10^{-4}$. This small probability is the origin of the Migdal effect, which turns a nuclear recoil too small to detect into an electronic signal (Chap. 12). At the slow speeds of thermal and zero-point motion, the Born–Oppenheimer approximation drops the couplings that excite the electrons of a recoiling nucleus.

6.1.1 Derivative coupling

The two brackets of Eq. (5.5) are not independent, since both come from the change of the adiabatic states $\Psi_i(r;u)$ with the displacements. Both brackets can be written through the *derivative-coupling vector*

$$\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}(u) = \langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle, \qquad \boldsymbol{d}_{ij}^{\boldsymbol{R}\kappa} = -\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}. \tag{6.1}$$

The antisymmetry follows from differentiating the orthonormality $\langle\Psi_j|\Psi_i\rangle = \delta_{ji}$ with respect to a displacement, $\langle\nabla\Psi_j|\Psi_i\rangle + \langle\Psi_j|\nabla\Psi_i\rangle = 0$, together with the reality of the states; in particular $\boldsymbol{d}_{ii} = 0$. For the matrix element of the second derivative, we move one gradient to the left. Then we insert the completeness of the adiabatic states at fixed u, $\sum_k |\Psi_k\rangle\langle\Psi_k| = 1$:

$$\langle\Psi_j|\nabla^2\Psi_i\rangle = \nabla\cdot\langle\Psi_j|\nabla\Psi_i\rangle - \langle\nabla\Psi_j|\nabla\Psi_i\rangle = \nabla\cdot\boldsymbol{d}_{ji} - \sum_k \boldsymbol{d}_{kj}\cdot\boldsymbol{d}_{ki} = \nabla\cdot\boldsymbol{d}_{ji} + \sum_k \boldsymbol{d}_{jk}\cdot\boldsymbol{d}_{ki},$$

where we dropped the label $\boldsymbol{R}\kappa$, which every gradient and every $\boldsymbol{d}$ carries, and used the antisymmetry in the last step. Inserted in Eq. (5.5), the scalar coupling contributes $\nabla\cdot\boldsymbol{d}_{ji} + (\boldsymbol{d}\cdot\boldsymbol{d})_{ji}$, a product of matrices in the state labels, and the derivative coupling $2\boldsymbol{d}_{ji}\cdot\nabla$. These two contributions are exactly the cross terms of a square. With the nuclear wavefunctions Ψ_i^{nuc} written as a column, the nuclear kinetic energy and the couplings of Eq. (5.4) combine into

$$-\sum_{\boldsymbol{R}\kappa}\frac{\nabla_{\boldsymbol{R}\kappa}^2}{2M_\kappa}\,\delta_{ji} + \mathcal{N}_{ji} = \sum_{\boldsymbol{R}\kappa}\frac{1}{2M_\kappa}\Big[\big(-\mathrm{i}\nabla_{\boldsymbol{R}\kappa} - \mathrm{i}\boldsymbol{d}^{\boldsymbol{R}\kappa}\big)^2\Big]_{ji}, \tag{6.2}$$

since $(\nabla+\boldsymbol{d})^2 = \nabla^2 + 2\boldsymbol{d}\cdot\nabla + (\nabla\cdot\boldsymbol{d}) + \boldsymbol{d}\cdot\boldsymbol{d}$, with ∇ acting on everything to its right. In the adiabatic basis the momentum of a nucleus, $\boldsymbol{p}_{\boldsymbol{R}\kappa} = -\mathrm{i}\nabla_{\boldsymbol{R}\kappa}$ of Sect. 5.3, becomes the matrix $-\mathrm{i}(\nabla_{\boldsymbol{R}\kappa} + \boldsymbol{d}^{\boldsymbol{R}\kappa})$. The nucleus carries its electronic state along, and the derivative-coupling vector enters the momentum of the nucleus much as a vector potential enters the momentum of a charge; it measures how much of the electronic state changes per unit of displacement. It is a Berry connection, in the space of nuclear displacements instead of crystal momenta. The Born–Oppenheimer approximation keeps only the diagonal part of the matrix in Eq. (6.2), in which $\boldsymbol{d}_{00} = 0$ and only the small potential $\mathcal{N}_{00}$ of Appendix D.11 survives.

The derivative coupling has a simple reading if we take $\boldsymbol{p}/M_\kappa$ to be a velocity, heuristically at first. The off-diagonal, first-order part of Eq. (6.2) is $-(\mathrm{i}/M_\kappa)\,\boldsymbol{d}_{ji}\cdot\boldsymbol{p}_{\boldsymbol{R}\kappa}$, and on a nuclear wavefunction moving with velocity $\dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}$ it acts as $-\mathrm{i}\,\dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}\cdot\boldsymbol{d}_{ji}$. The same coupling appears when the nuclei are treated as classical particles that follow a trajectory $u(t)$. The electrons

then obey $\mathrm{i}\partial_t\Psi = H_{\rm el}(u(t))\Psi$, and we expand $\Psi(t) = \sum_i c_i(t)\,\Psi_i(r;u(t))$ in the adiabatic states at the instantaneous positions. The time derivative falls on the coefficients and, through $u(t)$, on the states, $\partial_t\Psi_i = \sum_{\boldsymbol{R}\kappa} \dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}\cdot\nabla_{\boldsymbol{R}\kappa}\Psi_i$. Projecting on Ψ_j gives

$$\mathrm{i}\dot{c}_j = E_j(u(t))\,c_j - \mathrm{i}\sum_i \sum_{\boldsymbol{R}\kappa} \dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}\cdot\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}\,c_i. \tag{6.3}$$

The electrons are driven out of their adiabatic state by the velocity of the nuclei times the derivative coupling, and by nothing else. By Eq. (5.11) the coupling is $\dot{\boldsymbol{u}}\cdot\boldsymbol{F}_{ji}/(E_i-E_j)$, with $\boldsymbol{F}_{ji} = \langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}V_{eN}|\Psi_i\rangle$ the matrix element of the force. The numerator is the power delivered by the nucleus to the transition, and the denominator is the energy the transition costs.

The scalar coupling has no counterpart in Eq. (6.3). Roughly speaking, it is the quantum correction to a classical trajectory, and compared with the derivative coupling it is suppressed by $|\boldsymbol{d}|/(Mv)$, with $|\boldsymbol{d}| \sim 1/a_0$ for an electronic state of size a_0. A nucleus in its zero-point motion has a momentum $Mv \sim 1/\ell_{\rm zp}$, and the suppression factor $|\boldsymbol{d}|/(Mv)$ is $\ell_{\rm zp}/a_0 \sim \kappa_{\rm BO}$, which is the extra power of $\kappa_{\rm BO}$ that separates the scalar from the derivative coupling in Sect. 5.1. A nucleus kicked by dark matter has $Mv_N = q$, and the suppression factor is $1/(qa_0)$, small for every momentum transfer well above $1/a_0 = 3.7\,\mathrm{keV}$: the kicked nucleus is a classical projectile as far as its electrons are concerned.

The velocity form of the derivative coupling also makes the counting of Sect. 5.1 transparent. In zero-point motion a nucleus moves at $v \sim \omega_{\rm ph}\ell_{\rm zp} \sim (\alpha^2 m_e\kappa_{\rm BO}^2)(\kappa_{\rm BO}a_0) = \alpha\,\kappa_{\rm BO}^3$, while the electrons move at $\sim\alpha$. The nuclei are slower than the electrons by $\kappa_{\rm BO}^3 = (m_e/M)^{3/4}$, which is 2.9×10^{-4} in silicon, and the derivative coupling $v\,|\boldsymbol{d}| \sim \alpha^2 m_e\,\kappa_{\rm BO}^3$ is the $\kappa_{\rm BO}^3$ of the Born–Oppenheimer hierarchy. The kicked nucleus of the opening is 91 times faster than this zero-point speed, and correspondingly more disruptive.

In a metal the expansion in adiabatic states breaks down at the Fermi surface (Sect. 5.1). The counting then becomes a statement about the electrons that cannot follow the nuclei: those within $\omega_{\rm ph}$ of the Fermi energy, which make up a fraction $\omega_{\rm ph}/E_{\rm F}$ of the conduction electrons. Since $E_{\rm F} \sim \alpha^2 m_e$ in a typical metal, this fraction is $\sim \kappa_{\rm BO}^2 = (m_e/M)^{1/2}$, between 10^{-3} and 10^{-2}. The vertex corrections to the electron–phonon interaction come from these electrons and are suppressed by the same fraction. This suppression of the vertex corrections is Migdal's adiabatic theorem [1]; the precise statement of the theorem with diagrams, and its form for insulators, are given in Sect. 6.3.

6.1.2 The electron–phonon Hamiltonian

The adiabatic basis follows the nuclei and acquires the derivative couplings. The basis of clamped-nuclei states $\Psi_i(r;0)$ does not move, and the coupling appears in the Hamiltonian instead, as the change $V_{eN}(u) - V_{eN}(0)$ of the electron–nucleus potential (Eq. (5.7)). This change of potential is the form in which the coupling of electrons to phonons is computed and used. Let $v_\kappa(\boldsymbol{r})$ be the potential energy of an electron at $\boldsymbol{r}$ due to a nucleus of species κ at the origin: $-Z_\kappa e^2/(4\pi r)$ for a bare nucleus, or a pseudopotential (Sect. 24.1). With the electron density operator $\hat{n}_e(\boldsymbol{r}) = \sum_e \delta(\boldsymbol{r}-\boldsymbol{r}_e)$, whose Fourier transform is Eq. (2.38),

$$V_{eN}(u) = \int \mathrm{d}^3r\;\hat{n}_e(\boldsymbol{r}) \sum_{\boldsymbol{R}\kappa} v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa-\boldsymbol{u}_{\boldsymbol{R}\kappa}).$$

To first order in the displacements, $V_{eN}(u) - V_{eN}(0) = \sum_{\boldsymbol{R}\kappa}\boldsymbol{u}_{\boldsymbol{R}\kappa}\cdot\int\mathrm{d}^3r\,\hat{n}_e(\boldsymbol{r})\,\nabla_{\boldsymbol{R}\kappa}V$, with $\nabla_{\boldsymbol{R}\kappa}V = -\nabla v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa)$, the bare force field of one nucleus. We now insert the phonon expansion of the displacement, Eq. (5.32),

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\,B_{\boldsymbol{Q}\nu}.$$

The sum over cells collects the phase factors into a displacement wave of one sublattice,

$$\partial_{\kappa,\boldsymbol{Q}} V(\boldsymbol{r}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} \nabla_{\boldsymbol{R}\kappa} V(\boldsymbol{r}), \qquad \partial_{\kappa,\boldsymbol{Q}} V(\boldsymbol{r}+\boldsymbol{R}') = \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}'}\, \partial_{\kappa,\boldsymbol{Q}} V(\boldsymbol{r}),$$

where the second relation follows from relabeling $\boldsymbol{R} - \boldsymbol{R}' \to \boldsymbol{R}$ in the sum. The potential of a displacement wave with wavevector $\boldsymbol{Q}$ is a Bloch function with crystal momentum $\boldsymbol{Q}$. Its matrix element between Bloch states $\psi_{m\boldsymbol{k}'}$ and $\psi_{n\boldsymbol{k}}$ (normalized as in Eq. (A.14)) is therefore zero unless $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{Q}$ up to a reciprocal-lattice vector: a lattice translation by $\boldsymbol{R}'$ multiplies the integrand by $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{Q}-\boldsymbol{k}')\cdot\boldsymbol{R}'}$, and the integral cannot change. Writing the one-body operator in second quantization, Eq. (2.36), we arrive at the *electron–phonon Hamiltonian*,

$$\begin{aligned} H_{\mathrm{ep}} &= \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu}, \\ g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) &= \frac{1}{\sqrt{2\omega_{\boldsymbol{Q}\nu}}} \sum_{\kappa} \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{M_\kappa}} \cdot \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa,\boldsymbol{Q}} V | \psi_{n\boldsymbol{k}} \rangle, \end{aligned} \tag{6.4}$$

with the spin carried in the band labels. This Hamiltonian is Eqs. (A.24) and (A.25), with the bare potential in place of the Kohn–Sham one [2, Eqs. (31)–(38)], and it has a simple reading. An electron in state $n\boldsymbol{k}$ is scattered to $m, \boldsymbol{k}+\boldsymbol{Q}$ while a phonon of crystal momentum $\boldsymbol{Q}$ is absorbed, or one of $-\boldsymbol{Q}$ is emitted, through $B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}$. The vertex g is the force of the displacement wave on the electron, weighted by the zero-point amplitude $(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}$ of each nucleus in the mode.

The term of second order in u, left out of Eq. (6.4), gives a vertex quadratic in B, the Allen–Heine vertex g^{AH}, which matters as much as the first order squared for the shifts of electronic levels (Sect. 6.3). The change of V_{NN} acts on the nuclei only and belongs to the energy surface E^{ad} that already fixes the phonons. Most important, the bare vertex ignores the response of the other electrons. When a nucleus moves, the electron cloud around it moves too and screens most of the force field of the nucleus, and the potential an electron actually feels is the change of the self-consistent potential. In density-functional theory, $\partial_{\kappa,\boldsymbol{Q}} V$ is replaced by $\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}}$, computed by density-functional perturbation theory at the same time as the phonons themselves (Sect. 25.1).

The electron–phonon Hamiltonian and the nonadiabatic couplings describe one interaction in two bases. To first order, Eq. (6.4) mixes the clamped states with an amplitude $\boldsymbol{u}\cdot\boldsymbol{F}_{ji}/(E_i - E_j)$, proportional to the displacement. By Eq. (5.11) this amplitude is just $\boldsymbol{u}\cdot\boldsymbol{d}_{ji}$, the first term in the Taylor series of the adiabatic state $\Psi_i(r;u)$. This first-order mixing is the electron cloud deforming with the lattice; it is already in the adiabatic states and involves no real transition. Real transitions need more than a displacement: Eq. (6.3) shows that they are driven by the velocity, and their amplitude carries one more factor $\omega_{\mathrm{ph}}/\Delta E$. This extra factor is why a phonon alone, far below every electronic excitation in energy, cannot excite the electrons of an insulator. The vertex g readily scatters electrons that are already excited, or takes part in a second-order process that supplies the excitation energy (Sect. 6.5).

6.1.3 Adiabatic and sudden limits

Whether a moving nucleus leaves its electrons excited depends on how its velocity changes in time. Take one nucleus, or a whole atom, that is set in motion along a prescribed path $\boldsymbol{X}(t)$, from rest to the final velocity $\boldsymbol{v}_N$, and let its electrons start in the ground state. Translation invariance makes the electronic energies and the derivative couplings independent of the position of the nucleus. Let j be an excited state at $\Delta E_j = E_j - E_0$ above the ground state.

In the interaction picture of Eq. (2.49), Eq. (6.3) gives its amplitude to first order,

$$c_j(t) = -\int_{-\infty}^{t} \mathrm{d}t'\; \dot{\boldsymbol{X}}(t') \cdot \boldsymbol{d}_{j0}\, \mathrm{e}^{\mathrm{i}\Delta E_j t'}.$$

An integration by parts separates two kinds of response,

$$c_j(t) = -\frac{\dot{\boldsymbol{X}}(t)\cdot\boldsymbol{d}_{j0}}{\mathrm{i}\Delta E_j}\,\mathrm{e}^{\mathrm{i}\Delta E_j t} + \frac{\boldsymbol{d}_{j0}}{\mathrm{i}\Delta E_j}\cdot\int_{-\infty}^{t}\mathrm{d}t'\;\ddot{\boldsymbol{X}}(t')\,\mathrm{e}^{\mathrm{i}\Delta E_j t'}. \tag{6.5}$$

The first term follows the instantaneous velocity and disappears when the nucleus stops. It describes the electron cloud of a uniformly moving atom: the cloud carries the momentum of the electrons along and is, for that reason, not the ground state of the atom at rest. The second term remains after the motion: a real excitation, driven by the acceleration and weighted by the Fourier component of the acceleration at the excitation energy. If the nucleus is accelerated over a time τ_{acc} long compared with $1/\Delta E_j$, the integrand oscillates many times during the acceleration and the Fourier component is small; for a smooth velocity profile it falls faster than any power of $\Delta E_j \tau_{\mathrm{acc}}$. Slow acceleration of this kind is the *adiabatic limit*, in which the electrons follow the nuclei and end in the ground state. If instead $\Delta E_j \tau_{\mathrm{acc}} \ll 1$, the phase stays fixed while the nucleus accelerates. The integral is then just $\boldsymbol{v}_N$, and

$$c_j = \frac{\boldsymbol{v}_N \cdot \boldsymbol{d}_{j0}}{\mathrm{i}\Delta E_j}. \tag{6.6}$$

This case is the *sudden limit*: the electrons have no time to react while the nucleus is accelerated, and they find themselves in a state that is no longer an eigenstate around the moving nucleus.

A model that can be solved exactly shows the crossover between the sudden and the adiabatic limit (Fig. 6.2). One electron of mass m_e is bound to a nucleus by a harmonic force of frequency ω_{e}, the electronic oscillator of the solvable model of Appendix D.11, and the nucleus follows the velocity profile $\dot{X}(t) = \frac{1}{2}v_N[1+\tanh(t/\tau_{\mathrm{acc}})]$. Let $y = x - X(t)$ be the position of the electron relative to the nucleus. In the frame of the nucleus the electron is an oscillator pushed by the inertial force $-m_e\ddot{X}(t)$,

$$H = \frac{p_y^2}{2m_e} + \frac{1}{2}m_e\omega_{\mathrm{e}}^2 y^2 + m_e\ddot{X}(t)\, y.$$

A linear force acting on an oscillator in its ground state leaves it in a coherent state, the displaced ground state $\hat{D}(\alpha_{\mathrm{d}})|0\rangle$ of Eq. (C.48). The coherent-state amplitude follows from the classical equation of motion, or from the exact form of Eq. (6.5) for this model, as $\alpha_{\mathrm{d}} = -\mathrm{i}(m_e/2\omega_{\mathrm{e}})^{1/2}\int \mathrm{d}t\,\ddot{X}\,\mathrm{e}^{\mathrm{i}\omega_{\mathrm{e}}t}$, and the number of quanta is Poisson distributed (Eq. (C.49)). The electron stays in the ground state with probability $P_0 = \mathrm{e}^{-|\alpha_{\mathrm{d}}|^2}$. With $\int \mathrm{d}s\,\mathrm{sech}^2 s\;\mathrm{e}^{\mathrm{i}ks} = \pi k/\sinh(\pi k/2)$ the Fourier integral is elementary, and

$$|\alpha_{\mathrm{d}}|^2 = \frac{m_e v_N^2}{2\omega_{\mathrm{e}}}\left[\frac{\pi\omega_{\mathrm{e}}\tau_{\mathrm{acc}}/2}{\sinh(\pi\omega_{\mathrm{e}}\tau_{\mathrm{acc}}/2)}\right]^2. \tag{6.7}$$

The prefactor is the sudden value: the kinetic energy $\frac{1}{2}m_e v_N^2$ of the electron relative to the nucleus after the kick, measured in quanta of the oscillator. The bracket is the adiabatic suppression: it is 0.47 at $\omega_{\mathrm{e}}\tau_{\mathrm{acc}} = 1$ and 0.074 at $\omega_{\mathrm{e}}\tau_{\mathrm{acc}} = 2$, and at larger $\omega_{\mathrm{e}}\tau_{\mathrm{acc}}$ it is close to $\pi^2(\omega_{\mathrm{e}}\tau_{\mathrm{acc}})^2\,\mathrm{e}^{-\pi\omega_{\mathrm{e}}\tau_{\mathrm{acc}}}$. The dots in Fig. 6.2(c) come from a direct solution of the time-dependent Schrödinger equation. Panel (b) shows the sudden and the adiabatic behavior in time. A sudden kick leaves the electron oscillating about the nucleus with the amplitude v_N/ω_{e}; a

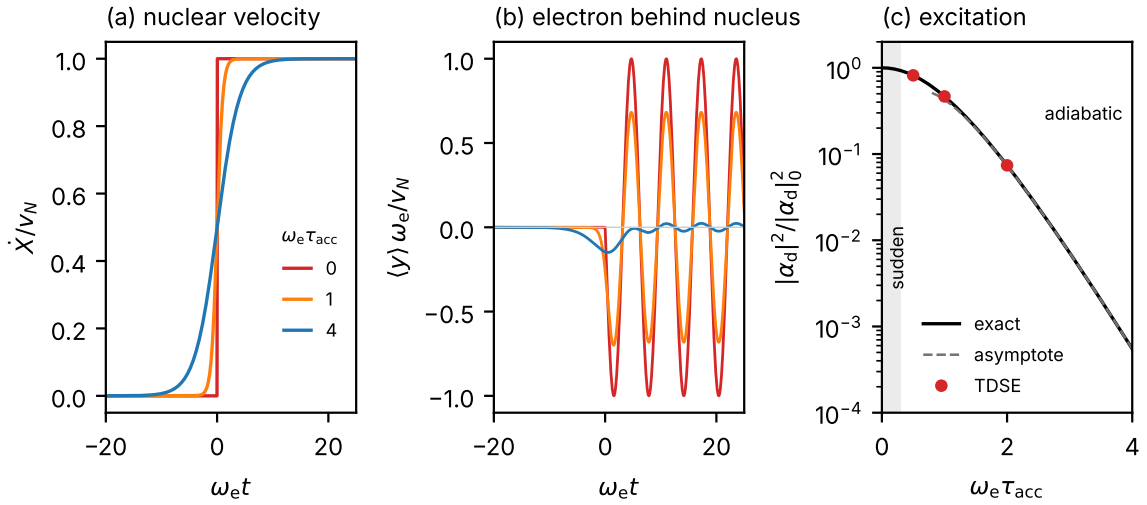


Figure 6.2: Adiabatic and sudden responses of an electron to a moving nucleus, in a model in which one electron is bound to the nucleus by a harmonic force of frequency ω_e and the nucleus follows a prescribed classical path. (a) The velocity of the nucleus, $\dot{X} = \frac{1}{2}v_N[1+\tanh(t/\tau_{acc})]$, for acceleration times $\omega_e\tau_{acc} = 0$ (an instantaneous kick), 1 and 4. (b) The mean position $\langle y\rangle$ of the electron relative to the nucleus, in units of v_N/ω_e, for the same three cases. After a kick the electron oscillates about the nucleus with the full amplitude; after a slow acceleration it lags behind while the force acts and returns to rest. (c) The excitation $|\alpha_d|^2 = -\ln P_0$, with P_0 the probability that the electron ends in the ground state, divided by its value $|\alpha_d|_0^2 = m_e v_N^2/2\omega_e$ for an instantaneous kick, against $\omega_e\tau_{acc}$. Solid: the exact result Eq. (6.7). Dashed: its adiabatic asymptote $\pi^2(\omega_e\tau_{acc})^2 e^{-\pi\omega_e\tau_{acc}}$. Dots: $-\ln P_0$ from a split-operator solution of the time-dependent Schrödinger equation for $m_e v_N^2/2\omega_e = 1/2$. Shaded: the sudden region $\omega_e\tau_{acc} < 0.3$, where the excitation is within 10 % of its sudden value.

slow acceleration displaces it behind the nucleus by $\ddot{X}/\omega_e^2$ while the force acts and then lets it return to rest at the center. The displacement $\ddot{X}/\omega_e^2$ is the first term of Eq. (6.5).

Two times matter for a nucleus in a crystal: how long the kick lasts, and how soon the neighbors start pulling the nucleus back. The kick itself is fast. A classical estimate of its duration, a heuristic only, is the time the dark matter takes to cross the region where it interacts with the nucleus. For a contact interaction and a silicon nucleus of radius 3.5 fm this crossing time is 4×10^{-21} s at $v_{\max}$, while $1/E_g = 0.56$ fs. The deceleration is slow. After the kick the nucleus is pulled back by the lattice over a time $\sim 1/\omega_{\rm ph}$, long enough to put every excitation across the gap in the adiabatic regime. In the oscillator model, take a nucleus that is kicked at $t = 0$ and then oscillates at $\omega_{\rm ph}$ in a harmonic well, $X(t) = (v_N/\omega_{\rm ph}) \sin \omega_{\rm ph} t$ for $t > 0$. Its acceleration is $\ddot{X} = v_N \delta(t) - v_N \omega_{\rm ph} \sin(\omega_{\rm ph} t)\, \Theta(t)$, and $\int \mathrm{d}t\, \ddot{X}\, \mathrm{e}^{\mathrm{i}\omega_e t} = v_N/[1 - (\omega_{\rm ph}/\omega_e)^2]$, so $|\alpha_{\rm d}|^2$ exceeds the free-recoil value by the factor $[1 - (\omega_{\rm ph}/\omega_e)^2]^{-2}$. The zone-center optical phonon of silicon has $\omega_{\rm opt} = 64.1$ meV, and the ratio of this energy to the gap is $\omega_{\rm opt}/E_g = 0.055$. With $\omega_{\rm ph}/\omega_e$ set to this ratio, the factor is 1.006. A kicked nucleus in an insulator therefore excites its electrons as if it were free, with corrections of relative order $(\omega_{\rm ph}/\Delta E)^2$, the square of the small ratio that controls the adiabatic approximation. The sudden picture fails where that ratio is not small, that is, for excitation energies comparable to phonon energies, as at the Fermi surface of a metal. The sudden picture also fails where the classical path itself fails, for recoil energies $q^2/2M$ that do not exceed a phonon energy (Sect. 5.5). At such low recoil energies the nucleus does not recoil, the kick creates a few phonons, and the electron–lattice response of Sect. 12.1 replaces the trajectory.

6.1.4 The sudden kick

In the sudden limit the excitation probability does not need perturbation theory. Immediately after the kick the electronic wavefunction is what it was before, $\Psi_0(r; u)$, but the nucleus moves with velocity $\boldsymbol{v}_N$. In the frame that moves with the nucleus, the electrons have all acquired the velocity $-\boldsymbol{v}_N$. A Galilean boost of every electron by $-\boldsymbol{v}_N$ multiplies the wavefunction by a plane wave:

$$\Psi_0 \;\to\; \exp\Big(-\mathrm{i} m_e \boldsymbol{v}_N \cdot \sum_e (\boldsymbol{r}_e - \boldsymbol{X})\Big)\Psi_0, \qquad P_0 = \Big|\Big\langle \Psi_0 \Big| \exp\Big(-\mathrm{i} m_e \boldsymbol{v}_N \cdot \sum_e \boldsymbol{r}_e\Big)\Big|\Psi_0\Big\rangle\Big|^2, \quad (6.8)$$

where P_0 is the probability that the electrons are found in the ground state around the moving nucleus, and the phase $\mathrm{e}^{\mathrm{i} m_e \boldsymbol{v}_N \cdot \boldsymbol{X}}$ drops out of P_0. This boost is Migdal's formulation [3, 4, Eqs. (1) and (2)]. To first order in v_N the amplitude of an excited state is $-\mathrm{i} m_e \boldsymbol{v}_N \cdot \langle \Psi_j | \sum_e \boldsymbol{r}_e | \Psi_0 \rangle$, a dipole transition driven by the kick, in agreement with Eq. (6.6). For an atom the adiabatic states depend on the nuclear position only through $\boldsymbol{r}_e - \boldsymbol{X}$, so that $\nabla_{\boldsymbol{X}} \Psi_0 = -\sum_e \nabla_e \Psi_0$, and the total electron momentum is $-\mathrm{i} \sum_e \nabla_e = \mathrm{i} m_e [H_{\rm el}, \sum_e \boldsymbol{r}_e]$. The matrix element between Ψ_j and Ψ_0 gives $\boldsymbol{d}_{j0} = m_e \Delta E_j \langle \Psi_j | \sum_e \boldsymbol{r}_e | \Psi_0 \rangle$, and Eq. (6.6) becomes the dipole amplitude. The derivative coupling of an atom is its dipole matrix element times the excitation energy and the electron mass.

For one electron in hydrogen the sudden result is exact. The Hamiltonian $p_e^2/2m_e + p_p^2/2m_p - \alpha/|\boldsymbol{r}_e - \boldsymbol{X}|$ separates into the motion of the center of mass $\boldsymbol{R}_{\rm cm} = (m_p \boldsymbol{X} + m_e \boldsymbol{r}_e)/M_{\rm H}$, with $M_{\rm H} = m_p + m_e$, and the relative motion in $\boldsymbol{\rho} = \boldsymbol{r}_e - \boldsymbol{X}$, whose ground state is the 1s state with the reduced mass $\mu_{ep} = m_e m_p / M_{\rm H}$ and the Bohr radius $a_{ep} = 1/(\alpha \mu_{ep})$. Dark matter that couples to the proton acts on the atom, to first order in its coupling, through $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$ (Sect. 2.2); a coupling of the dark matter to the electron itself is a separate process, which we leave aside. Since $\boldsymbol{X} = \boldsymbol{R}_{\rm cm} - (m_e/M_{\rm H})\boldsymbol{\rho}$,

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\rm cm}}\; \mathrm{e}^{-\mathrm{i}(m_e/M_{\rm H})\boldsymbol{q}\cdot\boldsymbol{\rho}}.$$

The first factor gives the whole atom the momentum $\boldsymbol{q}$. The second acts on the internal state like a momentum transfer $-(m_e/M_{\rm H})\boldsymbol{q}$ to the electron, and its ground-state expectation value is the elastic form factor Eq. (2.21) with a_{ep} in place of a_0. The argument of the form factor is $(m_e q/M_{\rm H})\, a_{ep} = q/(\alpha m_p) = v_N/\alpha$, where $v_N = q/m_p$ is the velocity given to the proton, and the reduced-mass corrections cancel. With $a_0 = 1/(\alpha m_e)$ we may write the argument as $m_e v_N a_0$:

Sudden kick of a hydrogen atom

A hydrogen atom in its ground state whose nucleus is given the velocity $v_N = q/m_p$ remains in its ground state with probability

$$P_0 = \left|F_{1s}(m_e v_N)\right|^2 = \left[\frac{16}{(4 + v_N^2/\alpha^2)^2}\right]^2 \simeq 1 - \frac{v_N^2}{\alpha^2} \quad (v_N \ll \alpha), \tag{6.9}$$

and is excited or ionized with probability $1 - P_0$. The controlling ratio is that of the nuclear velocity to the orbital velocity α of the electron.

The derivation used no assumption about time. In a first-order amplitude the momentum transfer is a single act, and the dark matter couples to the nuclear coordinate only. Nothing in a free atom decelerates the nucleus afterward, and the separation of the center of mass replaces the suddenness assumed in Eq. (6.8). Ibe et al. treat the general atom in this way, with the amplitude of every electronic transition given by Eq. (6.8) and the energy of the atom conserved overall [4, Eqs. (55)–(58)]. For more than one electron the separation is not exact, and the argument $m_e v_N$ is corrected at relative order m_e/M; Sect. 12.2 makes the correction precise.

The *Migdal effect* is the fraction $1 - P_0$ of nuclear recoils that leave the atom excited or ionized. For dark-matter searches its interest is the electronic energy it adds to the recoil. A recoil of a few hundred eV in silicon may be below the threshold of a phonon or charge readout. An electron–hole pair from the same event, carrying at least the gap, is easy to count. In a crystal the electrons are shared between atoms, and the excitation is an electron–hole pair rather than an atomic transition.

Worked example 6.1: Ground-state survival of a kicked hydrogen atom

Dark matter of $m_\chi = 1\,\text{GeV}$ scatters elastically off a hydrogen atom at rest, at the largest laboratory speed $v_{\rm max} = 2.650 \times 10^{-3}$ (Eq. (1.9)). The atom has the mass $M_{\rm H} = 938.783\,\text{MeV}$, the reduced mass is $\mu_{\chi\rm H} = (1000 \times 938.783/1938.783)\,\text{MeV} = 484.2\,\text{MeV}$, and the largest momentum transfer is $q = 2\mu_{\chi\rm H} v_{\rm max} = 2.566\,\text{MeV}$, at which the atom recoils with $q^2/2M_{\rm H} = 3.51\,\text{keV}$.

The proton, of mass $m_p = 938.272\,\text{MeV}$, is kicked to $v_N = q/m_p = 2.735 \times 10^{-3}$, or $820\,\text{km}\,\text{s}^{-1}$; dark matter as heavy as the target gives the proton about the speed of the dark matter. In units of the orbital velocity, $v_N/\alpha = 0.375$. Then $F_{1s}(m_e v_N) = 16/(4 + 0.1405)^2 = 0.933$, and Eq. (6.9) gives

$$P_0 = 0.871, \qquad 1 - P_0 = 0.129.$$

One recoil in eight at this momentum transfer leaves the atom excited or ionized, with at least the $10.2\,\text{eV}$ of the $1s \to 2p$ transition in its electrons. The small-v_N form, $v_N^2/\alpha^2 = 0.140$, is 9 % high at this velocity. The atom survives with probability one half at $v_N/\alpha = 0.87$, where $F_{1s}^2 = \frac{1}{2}$ (Worked example 2.2): $v_N = 6.35 \times 10^{-3}$, $q = 5.96\,\text{MeV}$ and a recoil of $18.9\,\text{keV}$, beyond the reach of dark matter at this mass.

Is the kick sudden? The excitation energies start at $\Delta E = \frac{3}{8}\alpha^2 m_e = 10.2\,\mathrm{eV}$, so the electron responds on the time $1/\Delta E = 0.065\,\mathrm{fs}$. Dark matter with a contact interaction crosses the proton, of charge radius 0.84 fm [5], in $1.1 \times 10^{-21}\,\mathrm{s}$, and $\Delta E\,\tau_{\mathrm{acc}} = 1.6 \times 10^{-5}$: on the scale of Fig. 6.2(c) the kick is instantaneous.
For the kick of the opening, $v_N = 1.95 \times 10^{-4}$ for a silicon nucleus, the same formula gives $1 - P_0 = 7.1 \times 10^{-4}$ for an electron bound as in hydrogen. The outer electrons of silicon are bound more weakly and the inner ones far more strongly, so this value is an order-of-magnitude estimate.

6.2 Electron–phonon matrix elements

An electron at the bottom of the conduction band of GaAs absorbs a longitudinal optical phonon of wavevector $0.01\,\text{Å}^{-1}$ with a vertex $|g| = 3.3\,\mathrm{eV}$. An electron in a conduction valley of silicon absorbs a longitudinal sound wave of the same wavevector, running along the valley axis, with $|g| = 26\,\mathrm{meV}$, more than a hundred times less. Ten times longer wavelengths make the GaAs vertex ten times larger and the silicon one $\sqrt{10}$ times smaller. The two vertices of 3.3 eV and 26 meV follow from the definition Eq. (6.4) once the bare potential of a displaced nucleus is replaced by the potential that the electrons actually feel, screened by all the other electrons. The coupling to long-wavelength sound waves controls how an excited carrier in silicon sheds its last few meV. The Fröhlich coupling of GaAs comes from a macroscopic field, and a dark photon drives the optical phonons of a polar crystal through the same field (Sect. 5.4). The phonon that lets a slow dark-matter particle cross the indirect gap of silicon couples through the screened vertex as well, at large wavevector (Sect. 6.5).

6.2.1 The screened vertex

The bare potential of a displaced nucleus is long-ranged. A pseudo-ion of valence charge Z_κ has the potential energy $v_\kappa(k) \to -Z_\kappa e^2/k^2$ for an electron at small wavevector k, the Coulomb tail of Eq. (A.3). The gradient of v_κ, the force field of Sect. 6.1, has the Fourier components $-\mathrm{i}\boldsymbol{k}\, v_\kappa(k)$. A displacement wave of wavevector $\boldsymbol{Q}$ therefore has a bare potential whose component at $\boldsymbol{Q}$ grows as $Z_\kappa e^2/Q$, for every crystal. The electrons screen out most of this bare potential. They flow toward a displaced positive ion and away from the region it left, and an electron near the ion feels the change of the self-consistent potential, $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}}$ of Eq. (25.5), not the change of the bare one.

The screening can be written in one line. The first-order density of Eq. (25.8) is the static Kohn–Sham response to the screened potential, $\partial n = R^0\, \partial V_{\mathrm{KS}}$, with R^0 the independent-particle response of Sect. 4.6 at $\omega = 0$, and Eq. (25.5) adds back the Hartree and exchange–correlation potentials of that density, $\partial V_{\mathrm{KS}} = \partial V_{\mathrm{ext}} + f_{\mathrm{Hxc}}\, \partial n$ with $f_{\mathrm{Hxc}} = v_{\mathrm{C}} + f_{\mathrm{xc}}$. Eliminating ∂n,

$$\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}} = \left(1 - f_{\mathrm{Hxc}} R^0\right)^{-1} \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{ext}} = \left(1 + f_{\mathrm{Hxc}} R\right) \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{ext}}, \tag{6.10}$$

where products are integral operators in space, or matrices in the reciprocal-lattice vectors $\boldsymbol{Q} + \boldsymbol{G}$ at fixed $\boldsymbol{Q}$, and $R = R^0(1 - f_{\mathrm{Hxc}} R^0)^{-1}$ is the full static response of Eq. (4.70); the form with $1 + f_{\mathrm{Hxc}} R$ follows from expanding the inverse in powers of $f_{\mathrm{Hxc}} R^0$. With f_{xc} dropped, $1 + v_{\mathrm{C}} R$ is the inverse dielectric matrix $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q}, 0)$ of Eq. (4.43). With f_{xc} kept, $1 + f_{\mathrm{Hxc}} R$ is the inverse dielectric function of a *test electron*: the potential that a second electron feels includes the exchange–correlation hole that the response builds around that electron, while the test charge of Sect. 4.4 feels only the Hartree part. The screening is static. A phonon of tens of meV is slow compared with the electronic response of an insulator, whose scale is

the gap, and the static limit is the adiabatic approximation of Sect. 6.1. Below the gap a response function is even in the frequency, so the static screening is accurate to relative order $(\omega_{\rm ph}/E_{\rm g})^2$, as are the phonon frequencies (Sect. 6.4). The first power of $\omega_{\rm ph}/E_{\rm g}$ in Sect. 6.1 measures the amplitude for exciting the electrons, not the change of their response. In a metal, whose response has no gap, Eq. (6.10) holds with the response evaluated at the phonon frequency; the result is the dynamically screened vertex of Bardeen and Pines [6].

How much of the bare $Z_\kappa e^2/Q$ survives at long wavelength depends on the material. In a metal, the Thomas–Fermi limit of Eq. (4.53) divides the bare $Z_\kappa e^2/Q$ by $\varepsilon_{\rm TF}(Q) = 1 + k_{\rm TF}^2/Q^2$, and the screened vertex becomes $Z_\kappa e^2 Q/(Q^2 + k_{\rm TF}^2)$: the divergence has turned into a coupling that vanishes linearly in Q. The electrons have neutralized the displaced ion completely. In an insulator the head of the dielectric matrix tends to ε_∞, which reduces the bare $1/Q$ tail by a factor of 11.7 in silicon but does not remove it. The tail is removed by the local fields of Sect. 4.4, the off-diagonal elements $\varepsilon^{-1}_{\boldsymbol{0G}}$ that describe electrons moving with the ion on the scale of a bond. Their effect is to replace the ionic charge by the Born effective charge of Sect. 5.4. A displaced ion drags part of its bond charge along, and the induced density at the wavevectors $\boldsymbol{Q} + \boldsymbol{G}$ with $\boldsymbol{G} \neq 0$ is the comoving cloud of bond charge. Its dipole per unit displacement is $-e(Z_\kappa - Z^*_\kappa)$, so that ion and cloud together carry eZ^*_κ. At long wavelength a displaced, screened ion is a dipole $eZ^*_\kappa \cdot \boldsymbol{u}$ in a medium of dielectric constant ε_∞, as Pick, Cohen and Martin showed from the structure of the dielectric matrix [7]. In silicon $Z^* = 0$. The electron cloud follows each nucleus so well that no dipole is left, and the long-range part of the screened potential starts at the quadrupole. In GaAs $Z^* = 2.1$, the dipole survives, and its field gives the $1/Q$ vertex of the opening, the Fröhlich coupling.

Eq. (6.4) with the screened potential is the vertex that density-functional perturbation theory computes. Since $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\rm KS}$ is a Bloch function of crystal momentum $\boldsymbol{Q}$, we write it as $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}\, \partial_{\kappa\alpha,\boldsymbol{Q}} v_{\rm KS}(\boldsymbol{r})$, with $\partial_{\kappa\alpha,\boldsymbol{Q}} v_{\rm KS}$ periodic in the lattice. With the Bloch states of Eq. (A.14), the phases $\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{Q})\cdot\boldsymbol{r}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ cancel, as they did for the density vertex Eq. (A.16), the factor $1/N$ cancels the N cells, and

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = \ell_{\boldsymbol{Q}\nu} \left\langle u_{m,\boldsymbol{k}+\boldsymbol{Q}} \right| \Delta_{\boldsymbol{Q}\nu} v_{\rm KS} \left| u_{n\boldsymbol{k}} \right\rangle_\Omega, \qquad \Delta_{\boldsymbol{Q}\nu} v_{\rm KS} = \sum_{\kappa\alpha} \Big(\frac{M_{\rm cell}}{M_\kappa} \Big)^{1/2} e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, \partial_{\kappa\alpha,\boldsymbol{Q}} v_{\rm KS}, \tag{6.11}$$

where the bracket is an integral over one cell. We have written the prefactor $(2M_\kappa \omega_{\boldsymbol{Q}\nu})^{-1/2}$ of Eq. (A.25) as the zero-point length $\ell_{\boldsymbol{Q}\nu}$ of Eq. (A.23) times $(M_{\rm cell}/M_\kappa)^{1/2}$. The same normalization is used in Refs. [2, Eqs. (33)–(38)] and [8, Eq. (2)]. The zero-point length, typically a few hundredths of an Å for an optical phonon, carries all the mass and frequency dependence. The other factor, the matrix element of $\Delta_{\boldsymbol{Q}\nu} v_{\rm KS}$, is the change of the potential per unit mass-weighted displacement of the atoms, in eV per Å. For an acoustic mode at small Q, where $(M_{\rm cell}/M_\kappa)^{1/2} \boldsymbol{e}_{\kappa\nu} = \hat{\boldsymbol{e}}_\nu$ for every κ (Sect. 5.3), it is the potential change per unit displacement of the whole crystal. The vertex itself has the dimension of energy, and $|g|^2/N$ is the squared amplitude per mode. The vertex depends on the phases of the two Bloch states, as ρ_{mn} does in Eq. (A.18); only sums of $|g|^2$ over degenerate partners can be compared between calculations.

In practice one run of density-functional perturbation theory per phonon wavevector solves the Sternheimer equation Eq. (25.7) for the three displacement directions of every atom and returns $\partial_{\kappa\alpha,\boldsymbol{Q}} v_{\rm KS}$ together with the phonon frequencies and polarization vectors. The vertex for any pair of bands and any $\boldsymbol{k}$ then costs just one matrix element, Eq. (6.11), one application of a known potential to a known state. The vertex needs no change to the run: the screened potential is the quantity whose self-consistency already produced the phonon.

6.2.2 Acoustic phonons and deformation potentials

Moving the whole crystal changes no band energy. If every nucleus is displaced by the same vector $\boldsymbol{t}$ and the electrons follow, the self-consistent potential is translated with them, $V_{\mathrm{KS}}(\boldsymbol{r}; u+t) = V_{\mathrm{KS}}(\boldsymbol{r}-\boldsymbol{t}; u)$, or $\hat{T}_{\boldsymbol{t}} V_{\mathrm{KS}} \hat{T}_{\boldsymbol{t}}^{\dagger}$ with the translation operator $\hat{T}_{\boldsymbol{t}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{t}\cdot\hat{\boldsymbol{p}}}$ when the potential is a nonlocal pseudopotential. Differentiating at $\boldsymbol{t}=0$ collects the derivatives with respect to all nuclei into the derivative of the potential with respect to the electron coordinate. The sum over all cells and all species is the displacement wave at $\boldsymbol{Q}=0$ summed over κ, so $\sum_\kappa \partial_{\kappa\alpha,\boldsymbol{0}} V_{\mathrm{KS}} = -\mathrm{i}[\hat{p}_\alpha, V_{\mathrm{KS}}] = -\mathrm{i}[\hat{p}_\alpha, H_{\mathrm{KS}}]$, where the last step uses that the kinetic energy commutes with the momentum. Between two Bloch states at the same $\boldsymbol{k}$, eigenstates of H_{KS},

$$\sum_\kappa \langle \psi_{m\boldsymbol{k}} |\, \partial_{\kappa\alpha,\boldsymbol{0}} V_{\mathrm{KS}} \,| \psi_{n\boldsymbol{k}} \rangle = -\mathrm{i}\,(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\, p^\alpha_{mn}(\boldsymbol{k}), \tag{6.12}$$

with p^α_{mn} the momentum matrix element of Sect. 3.2. The diagonal element vanishes: a rigid translation does not shift band n at first order. Eq. (6.12) is the acoustic sum rule of the vertex, the counterpart of Eq. (5.18) for the force constants, and like that sum rule it follows from translation invariance alone.

An acoustic phonon at small Q approaches this rigid translation. Its polarization vectors tend to $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ (Sect. 5.3), so that $\Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}}$ in Eq. (6.11) becomes $\hat{\boldsymbol{e}}_\nu \cdot \sum_\kappa \partial_{\kappa,\boldsymbol{Q}} v_{\mathrm{KS}}$, and Eq. (6.12) makes the diagonal matrix element of $\Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}}$ vanish as $Q \to 0$. The term of first order in Q remains, multiplied by a zero-point length that grows as $\ell_{\boldsymbol{Q}\nu} = (2M_{\mathrm{cell}} c_\nu Q)^{-1/2}$. The intraband vertex therefore vanishes as

$$g_{nn\nu}(\boldsymbol{k}, \boldsymbol{Q}) \propto Q\, \ell_{\boldsymbol{Q}\nu} \propto Q^{1/2} \qquad (\nu \text{ acoustic},\ Q \to 0),$$

the statement that $g_{nn\nu}(\boldsymbol{k}, \boldsymbol{0}) = 0$ for acoustic modes. The interband elements of Eq. (6.12) do not vanish. They describe the rigid translation of the electronic state itself: shifted by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu$, a state changes by $-\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu \cdot \nabla\psi_{n\boldsymbol{k}}$. Expanded in the other bands at the same $\boldsymbol{k}$, this change has as coefficients exactly the interband elements divided by $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}$. No energy is exchanged in the rigid translation, and in the energy shifts of Sect. 6.3 its contribution cancels against the second-order Allen–Heine term, as translation invariance demands. Codes therefore set the vertices of the three acoustic modes at $\boldsymbol{Q}=0$ to zero [2, below Eq. (38)].

The term of first order in Q is the deformation potential. Consider a region short compared with the wavelength $2\pi/Q$ but containing many cells. In such a region one long acoustic mode displaces every nucleus by the same amount, $\boldsymbol{u}(\boldsymbol{r}) = (\ell_{\boldsymbol{Q}\nu}/\sqrt{N})\,\hat{\boldsymbol{e}}_\nu\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}$, and the crystal is locally translated and uniformly strained, with strain tensor

$$S_{\alpha\beta}(\boldsymbol{r}) = \tfrac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha) = \frac{\mathrm{i}}{2}(Q_\alpha \hat{e}_{\nu\beta} + Q_\beta \hat{e}_{\nu\alpha})\, \frac{\ell_{\boldsymbol{Q}\nu}}{\sqrt{N}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}.$$

The local translation does nothing, by Eq. (6.12). A uniform strain shifts the band energies at first order, $\delta\varepsilon_{n\boldsymbol{k}} = \sum_{\alpha\beta} D^n_{\alpha\beta} S_{\alpha\beta}$, and this defines the *deformation-potential tensor* $D^n_{\alpha\beta}$ of band n near the extremum of interest. The tensor is symmetric, and it is taken with the nuclei inside each cell relaxed, since a long acoustic wave carries the internal relaxation of Sect. 5.2 with it. An electron near the band edge, in a wavepacket much longer than the cell but short compared with $2\pi/Q$, feels the slowly varying shift $\delta\varepsilon_n(\boldsymbol{r})$ as a potential energy. This argument is due to Bardeen and Shockley [9]. The matrix element of that potential between $n\boldsymbol{k}$ and $n, \boldsymbol{k}+\boldsymbol{Q}$ is the amplitude of the shift times $\rho_{nn}(\boldsymbol{k}, \boldsymbol{Q}) \simeq 1$. Comparison with Eq. (6.4) gives

$$g_{nn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu} \sum_{\alpha\beta} Q_\alpha D^n_{\alpha\beta}\, \hat{e}_{\nu\beta} \qquad (\nu \text{ acoustic},\ Q \to 0). \tag{6.13}$$

For a band extremum at Γ in a cubic crystal the tensor is a multiple of the identity, $D^n_{\alpha\beta} = E_1\delta_{\alpha\beta}$, and $E_1 = \Omega\,\partial\varepsilon_n/\partial\Omega$, since the trace of the strain is the relative change of volume [2, Eq. (6)]. Only the longitudinal mode couples, through $\boldsymbol{Q}\cdot\hat{\boldsymbol{e}}_\nu$: a transverse wave shears the crystal without changing its volume. With $\omega_{\boldsymbol{Q}\nu} = c_{\mathrm{LA}}Q$, $|g|^2 = E_1^2 Q/(2M_{\mathrm{cell}}c_{\mathrm{LA}})$.

The conduction valleys of silicon are not at Γ, and each valley is symmetric only about its own axis $\hat{\boldsymbol{k}}_{\mathrm{c}}$ (Sect. 3.3). The most general symmetric tensor with this axial symmetry has two constants, $D_{\alpha\beta} = \Xi_d\,\delta_{\alpha\beta} + \Xi_u\,\hat{k}_{\mathrm{c}\alpha}\hat{k}_{\mathrm{c}\beta}$, the dilatation and uniaxial deformation potentials of Herring and Vogt [10]. With θ the angle between $\boldsymbol{Q}$ and the valley axis, Eq. (6.13) gives the effective constants $\Xi_d + \Xi_u\cos^2\theta$ for the longitudinal mode and $\Xi_u\sin\theta\cos\theta$ for the transverse mode polarized in the plane of $\boldsymbol{Q}$ and $\hat{\boldsymbol{k}}_{\mathrm{c}}$; the other transverse mode does not couple [11, Eqs. (13) and (14)]. Li, Graziosi and Neophytou extracted the two constants from the slopes of DFPT vertices of silicon at small Q, and found $\Xi_d = 1.01\,\mathrm{eV}$ and $\Xi_u = 8.84\,\mathrm{eV}$ [11]. A longitudinal phonon of $Q = 0.01\,\mathrm{Å}^{-1}$ along the valley axis has an energy of 0.55 meV and a zero-point length $\ell_{\boldsymbol{Q}\nu} = 0.26\,\mathrm{Å}$, and Eq. (6.13) yields $|g| = 9.85\times 0.01\times 0.26\,\mathrm{eV} = 26\,\mathrm{meV}$, the number of the opening. Across the valley axis the coupling is almost ten times weaker, 2.6 meV. Fig. 6.3 shows the coupling along and across the valley axis against Q. The form of Eq. (6.13) holds for intraband scattering near a band edge, for Q small compared with the zone; the computed matrix elements are linear in Q over the whole computed range, up to $0.1\,\mathrm{Å}^{-1}$ [11, Fig. 5].

In a crystal without inversion symmetry, such as GaAs, a strain also polarizes the cell, $P_\alpha = \sum_{\beta\gamma} e^{\mathrm{pz}}_{\alpha\beta\gamma} S_{\beta\gamma}$ with the piezoelectric tensor e^{pz}, and an acoustic wave then carries a macroscopic field. Appendix D.12 turns the polarization of a longitudinal optical wave into the Fröhlich vertex, and its steps apply unchanged. The field obeys $\hat{\boldsymbol{Q}}\cdot(\varepsilon_\infty\boldsymbol{\mathcal{E}} + \boldsymbol{P}) = 0$, an electron feels the potential energy $V = e\hat{\boldsymbol{Q}}\cdot\boldsymbol{\mathcal{E}}/(\mathrm{i}Q)$, and with the strain above

$$g^{\mathrm{pz}}_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = -\frac{e}{\varepsilon_\infty}\,\ell_{\boldsymbol{Q}\nu}\sum_{\alpha\beta\gamma}\hat{Q}_\alpha\,e^{\mathrm{pz}}_{\alpha\beta\gamma}\,\hat{Q}_\beta\,\hat{e}_{\nu\gamma}\,\rho_{mn}(\boldsymbol{k},\boldsymbol{Q}),$$

for a cubic crystal. This vertex is the *piezoelectric coupling*. The factor Q of the strain cancels the $1/Q$ of the field, and the vertex depends on the magnitude of $\boldsymbol{Q}$ only through $\ell_{\boldsymbol{Q}\nu}\propto Q^{-1/2}$, and otherwise on the direction of $\boldsymbol{Q}$. It grows at long wavelength, more gently than the Fröhlich vertex. In a first-principles calculation it has two contributions. One comes from the dynamical quadrupoles of the displaced atoms, the next term after the Born charge in the long-wavelength expansion of the charge that a displaced atom induces. The other comes from the Born charges, through the relative sublattice displacement that the strain induces. In GaAs the piezoelectric coupling dominates the scattering of electrons by acoustic phonons [12].

6.2.3 Optical deformation potential

An optical phonon at $\boldsymbol{Q} = 0$ moves the sublattices against each other. Nothing in the translation argument forces its vertex to vanish, and in general the vertex does not vanish: bond lengths and angles inside the cell change at first order in the displacement, and so do the band energies. The vertex is finite at $\boldsymbol{Q} = 0$, and it is conventionally written as

$$g_{nn\nu}(\boldsymbol{k},\boldsymbol{0}) = \ell_{\boldsymbol{0}\nu}\,d^n_\nu, \qquad d^n_\nu = \langle\psi_{n\boldsymbol{k}}|\sum_\kappa\Big(\frac{M_{\mathrm{cell}}}{M_\kappa}\Big)^{1/2}\boldsymbol{e}_{\kappa\nu}(\boldsymbol{0})\cdot\partial_{\kappa,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{n\boldsymbol{k}}\rangle, \tag{6.14}$$

with d^n_ν the *optical deformation potential* of band n, in eV per Å. Since $\partial_{\kappa,\boldsymbol{0}}V_{\mathrm{KS}}$ is the potential change when a whole sublattice is displaced, d^n_ν is, by the Hellmann–Feynman theorem of Sect. 5.1, the shift of $\varepsilon_{n\boldsymbol{k}}$ per unit mass-weighted displacement of the sublattices in the pattern

of mode ν. In silicon the two atoms have equal masses, $\boldsymbol{e}_{1\nu} = -\boldsymbol{e}_{2\nu} = \hat{\boldsymbol{e}}/\sqrt{2}$, and the pattern moves the two atoms of every cell by $+u\hat{\boldsymbol{e}}$ and $-u\hat{\boldsymbol{e}}$; d^n_ν is then $\mathrm{d}\varepsilon_{n\boldsymbol{k}}/\mathrm{d}u$. Compared with the acoustic constant D^n, which enters with an extra factor Q, d^n_ν is of zeroth order in Q, and D^n and d^n_ν are often called the first-order and zero-order deformation potentials.

Symmetry can still make d^n_ν vanish, and for the conduction valleys of silicon it does: within a valley the electrons scatter off acoustic phonons only, and DFPT finds the intravalley optical vertices small near $\boldsymbol{Q} = 0$ [11]. The valence maximum is threefold degenerate at Γ (Sect. 3.3), and the optical phonon splits it. For a degenerate level the coupling is a matrix in the partners; Li et al. combine the elements of this matrix into a single value, $10.45\,\mathrm{eV\,\AA^{-1}}$, the mean of two combinations of the elements [11, Eq. (12) and the text below it]. With $\omega_{\boldsymbol{0}\nu} = 64.1\,\mathrm{meV}$ the zero-point length of the optical mode is $\ell_{\boldsymbol{0}\nu} = 0.024\,\text{Å}$. This zero-point length is half the $0.048\,\text{Å}$ of the bond coordinate in Sect. 5.1, because ℓ refers the motion to the mass of the whole cell. The vertex of the holes is then $|g| \simeq 0.25\,\mathrm{eV}$, independent of Q near the zone center (Fig. 6.3). The hole vertex is ten times the acoustic vertex of the electrons at $Q = 0.01\,\text{Å}^{-1}$, but emitting the optical phonon takes 64 meV, an energy that a hole near the band edge does not have. Electrons in silicon meet the same zero-order coupling when they scatter between valleys. The crystal momentum that separates two valleys is a sizable fraction of the zone, the phonons that supply it are near the zone boundary, and the vertex is finite there; these intervalley processes cool hot carriers in a detector. In a polar crystal the optical deformation potential is present as well, but the longitudinal optical mode carries in addition the macroscopic field.

6.2.4 The Fröhlich vertex

In a longitudinal optical wave of GaAs the gallium and arsenic sublattices move against each other along $\boldsymbol{Q}$, every cell acquires a dipole $eZ^*\boldsymbol{u}_{\mathrm{rel}}$, with $\boldsymbol{u}_{\mathrm{rel}} = \boldsymbol{u}_1 - \boldsymbol{u}_2$ the relative displacement of the two sublattices (Sect. 5.4), and the dipoles of a long wave add up to a macroscopic polarization. Its field reaches every electron in the crystal, however far from the displaced atoms. Appendix D.12 computes that field, and the potential energy of an electron in it, from the condition $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$ of electrostatics, in the same way as the LO–TO splitting of Sect. 5.4. Written as a vertex in the normalization of Eq. (6.4), this potential energy is the *long-range vertex*

$$g^{\mathrm{L}}_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = \mathrm{i}\,\frac{e^2}{\Omega}\sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\cdot Z^*_\kappa\cdot\boldsymbol{Q}}{(2M_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\,\boldsymbol{Q}\cdot\varepsilon_\infty\cdot\boldsymbol{Q}}\,\rho_{mn}(\boldsymbol{k},\boldsymbol{Q}),$$

Eq. (D.51), the generalization, due to Verdi and Giustino [13, Eq. (4)], of Fröhlich's coupling to any crystal and any number of polar modes. Only longitudinal modes couple, since only they have $Z^*_\kappa\cdot\boldsymbol{Q}$ along their displacement, and the vertex grows as $1/Q$, the Fourier transform of a field that falls off as the inverse square of the distance from a dipole. The density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{Q})$ tends to δ_{mn} as $Q \to 0$, so the divergence is intraband. Between different bands ρ_{mn} vanishes linearly in Q, by Eq. (A.17), and the interband long-range vertex tends to a finite limit that depends on the direction of $\boldsymbol{Q}$.

For a cubic crystal with two atoms per cell the Born charges and the masses drop out. The longitudinal polarization vectors, the reduced mass and the LO–TO relation Eq. (5.43) combine with the Lyddane–Sachs–Teller relation Eq. (5.45) into a vertex that contains only macroscopic, measurable constants (Appendix D.12).

Fröhlich vertex

In a cubic polar crystal with two atoms per cell, the longitudinal optical phonon of small wavevector $\boldsymbol{Q}$ couples to the electrons with

$$g^{\mathrm{L}}_{mn}(\boldsymbol{k},\boldsymbol{Q}) = \frac{\mathrm{i}\,C_{\mathrm{F}}}{Q}\,\rho_{mn}(\boldsymbol{k},\boldsymbol{Q}), \qquad C_{\mathrm{F}}^2 = \frac{e^2\omega_{\mathrm{L}}}{2\Omega}\Big(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\Big), \tag{6.15}$$

with ω_{L} the longitudinal optical frequency at Γ and ε_∞ and ε_0 the electronic and static dielectric constants. The sign is that of the Born charge of the first atom; the coupling constant C_{F} has the dimension of energy squared.

Eq. (6.15) is Fröhlich's vertex [14], generalized to the density vertex ρ_{mn} in place of the single-band overlap [13, Eq. (1)]. The combination $1/\varepsilon_\infty - 1/\varepsilon_0$ measures the part of the screening that comes from the ions. A static charge in the crystal is screened by electrons and ions together, through ε_0. The field of the displaced ions, as seen by the electrons, is screened by the electrons alone, through ε_∞. The difference of the two inverse constants is the field that the ions contribute. In silicon $\varepsilon_0 = \varepsilon_\infty$ and $C_{\mathrm{F}} = 0$. The vertex is the Coulomb propagator $1/Q^2$ acting on a dipole source proportional to Q, the bound charges of the polarization wave, and the square of the vertex falls as $1/Q^2$. A light mediator weights a rate toward small momentum transfers more steeply, as $1/q^4$, the squared propagator acting on a point charge (Sect. 1.4).

The strength of the Fröhlich coupling is measured by a dimensionless number. Take an electron at rest at the bottom of an isotropic parabolic band of mass m^*, at $T = 0$. At second order in g^{L} it emits a phonon $-\boldsymbol{Q}$ and moves to the state $\boldsymbol{Q}$, at an energy $\omega_{\mathrm{L}} + Q^2/2m^*$ above its own, and comes back. The energy of the electron is lowered by

$$\Delta E = -\frac{1}{N}\sum_{\boldsymbol{Q}}\frac{|g^{\mathrm{L}}|^2}{\omega_{\mathrm{L}}+Q^2/2m^*} = -\Omega\int\frac{\mathrm{d}^3Q}{(2\pi)^3}\,\frac{C_{\mathrm{F}}^2/Q^2}{\omega_{\mathrm{L}}+Q^2/2m^*} = -\frac{\Omega C_{\mathrm{F}}^2}{2\pi^2}\,\frac{\pi}{2}\Big(\frac{2m^*}{\omega_{\mathrm{L}}}\Big)^{1/2},$$

where we used $\rho_{nn} = 1$, replaced the sum over the zone by an integral (Appendix A.3) and extended it to infinity, and integrated $\int_0^\infty \mathrm{d}Q\,(\omega_{\mathrm{L}} + Q^2/2m^*)^{-1} = (\pi/2)(2m^*/\omega_{\mathrm{L}})^{1/2}$. The integral is dominated by $Q \sim (2m^*\omega_{\mathrm{L}})^{1/2}$, the wavevector at which the kinetic energy of the electron equals the phonon energy. Writing $\Delta E = -\alpha_{\mathrm{F}}\,\omega_{\mathrm{L}}$ and inserting C_{F}^2 from Eq. (6.15), with $e^2 = 4\pi\alpha$,

$$\alpha_{\mathrm{F}} = \frac{\Omega\,C_{\mathrm{F}}^2}{4\pi\,\omega_{\mathrm{L}}^2}\,(2m^*\omega_{\mathrm{L}})^{1/2} = \alpha\,\Big(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\Big)\Big(\frac{m^*}{2\omega_{\mathrm{L}}}\Big)^{1/2}, \tag{6.16}$$

the *Fröhlich coupling constant* [14]. It is the fine-structure constant, reduced by the ionic part of the screening and enhanced by the square root of the ratio of the band mass to twice the phonon energy. For $\alpha_{\mathrm{F}} \ll 1$ the electron carries a thin cloud of virtual phonons and the second order is accurate; the electron dressed by this cloud is the Fröhlich polaron, and Sect. 6.3 treats the energy shift as part of the electron self-energy. The cloud extends over the length $(2m^*\omega_{\mathrm{L}})^{-1/2}$, many cells for a light electron, and this extent is why the continuum description of the crystal, ε_∞ and ε_0 in place of the atoms, is adequate.

Worked example 6.2: The Fröhlich constant of GaAs

We take the measured constants of Worked example 5.5: $\omega_{\mathrm{L}} = 8.79\,\mathrm{THz} = 36.35\,\mathrm{meV}$ [15], $\varepsilon_\infty = 10.88$ and $\varepsilon_0 = 12.9$ [16], the cell volume $\Omega = a^3/4 = 44.80\,\text{Å}^3 = 5.831\times10^{-9}\,\mathrm{eV}^{-3}$ with the static-lattice $a = 5.638\,\text{Å}$ of Table 23.1, and the conduction mass $m^* = 0.067\,m_e =$

34.24 keV (Sect. 3.3) [17]. The ionic part of the screening is

$$\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0} = 0.09191 - 0.07752 = 0.01439,$$

a small difference of two larger numbers. With $e^2 = 0.0917$, Eq. (6.15) gives $C_\mathrm{F}^2 = 0.0917 \times 0.036\,35\,\mathrm{eV} \times 0.01439/(2 \times 5.831 \times 10^{-9}\,\mathrm{eV}^{-3}) = 4.11 \times 10^3\,\mathrm{eV}^4$, or $C_\mathrm{F} = 64.1\,\mathrm{eV}^2$. A phonon of $Q = 0.01\,\text{Å}^{-1} = 19.7\,\mathrm{eV}$ couples to an electron at the band bottom with $|g^\mathrm{L}| = C_\mathrm{F}/Q = 3.25\,\mathrm{eV}$, the number of the opening; at $0.1\,\text{Å}^{-1}$ the vertex has fallen to 0.33 eV.

The Fröhlich constant follows from Eq. (6.16). The square root is $(34{,}240/(2 \times 0.03635))^{1/2} = 686$, and

$$\alpha_\mathrm{F} = \frac{0.01439 \times 686}{137.036} = 0.072.$$

The coupling is weak. An electron at the bottom of the band is lowered by $\alpha_\mathrm{F}\omega_\mathrm{L} = 2.6\,\mathrm{meV}$, and its phonon cloud extends over $(2m^*\omega_\mathrm{L})^{-1/2} = 39.6\,\text{Å}$, seven lattice constants. The wavevectors that matter are near $(2m^*\omega_\mathrm{L})^{1/2} = 0.025\,\text{Å}^{-1}$, about a fortieth of the distance to the zone boundary; at this wavevector $|g^\mathrm{L}| = 1.29\,\mathrm{eV}$.

How well is α_F known? The measured constants obey the Lyddane–Sachs–Teller relation only to 1.4 % (Worked example 5.5). If we eliminate ε_0 with that relation, $1/\varepsilon_\infty - 1/\varepsilon_0 = (1 - \omega_\mathrm{T}^2/\omega_\mathrm{L}^2)/\varepsilon_\infty$, the measured $\omega_\mathrm{T} = 33.62\,\mathrm{meV}$ yields 0.01328, 8 % less, and $\alpha_\mathrm{F} = 0.067$. The small difference amplifies the residual of the inputs fivefold. For GaAs the Fröhlich constant is therefore 0.07 to within several percent. The same amplification is why Eq. (6.15) needs dielectric constants from one consistent source: the PBE value $\varepsilon_\infty = 13.7$ of Sect. 5.4, combined with the measured ε_0, would even reverse the sign of the difference.

6.2.5 Wannier interpolation of the vertex

A phonon-assisted rate in silicon with an energy resolution of 14 meV needs the vertex at about 4×10^9 pairs $(\boldsymbol{k}, \boldsymbol{Q})$, for every pair of bands and every branch. Density-functional perturbation theory costs about one ground-state calculation per phonon wavevector, and a calculation can afford a few hundred wavevectors, a 6^3 or 8^3 grid. The step from such a *coarse grid* to the *fine grids* of a rate is an interpolation, and it works for the same reason as the interpolation of bands in Sect. 27.1: in a suitable real-space representation the vertex is short-ranged.

The representation uses Wannier functions for the electrons and single displaced atoms for the phonons. We insert the Bloch sums of Wannier functions of one gauge, $\psi_{n\boldsymbol{k}} = N^{-1/2} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \sum_{n'} w_{n'}(\boldsymbol{r} - \boldsymbol{R})\, U^\dagger_{n'n}(\boldsymbol{k})$ (Eq. (A.19) inverted), into the matrix element of $\partial_{\kappa\alpha,\boldsymbol{Q}} V_\mathrm{KS} = \sum_{\boldsymbol{R}_p} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}_p}\, \partial V_\mathrm{KS}/\partial u^\alpha_{\boldsymbol{R}_p\kappa}$. We then translate every term by the cell of the bra and sum over that cell, and the sum cancels the factor $1/N$. The matrix element becomes

$$\langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa\alpha,\boldsymbol{Q}} V_\mathrm{KS} | \psi_{n\boldsymbol{k}} \rangle = \sum_{\boldsymbol{R}_e \boldsymbol{R}_p} \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{R}_e + \boldsymbol{Q}\cdot\boldsymbol{R}_p)} \left[U(\boldsymbol{k}+\boldsymbol{Q})\, g^W_{\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p)\, U^\dagger(\boldsymbol{k}) \right]_{mn},$$

$$g^W_{m'n'\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p) = \Big\langle w_{m'}(\boldsymbol{r}) \Big| \frac{\partial V_\mathrm{KS}}{\partial u^\alpha_{\boldsymbol{R}_p\kappa}} \Big| w_{n'}(\boldsymbol{r} - \boldsymbol{R}_e) \Big\rangle. \tag{6.17}$$

The real-space vertex g^W is the matrix element of the screened potential change caused by moving a single atom in cell $\boldsymbol{R}_p$, taken between a Wannier function at the origin and one in cell $\boldsymbol{R}_e$. Multiplying by the polarization factors of Eq. (A.25) gives Eq. (27.15), and on

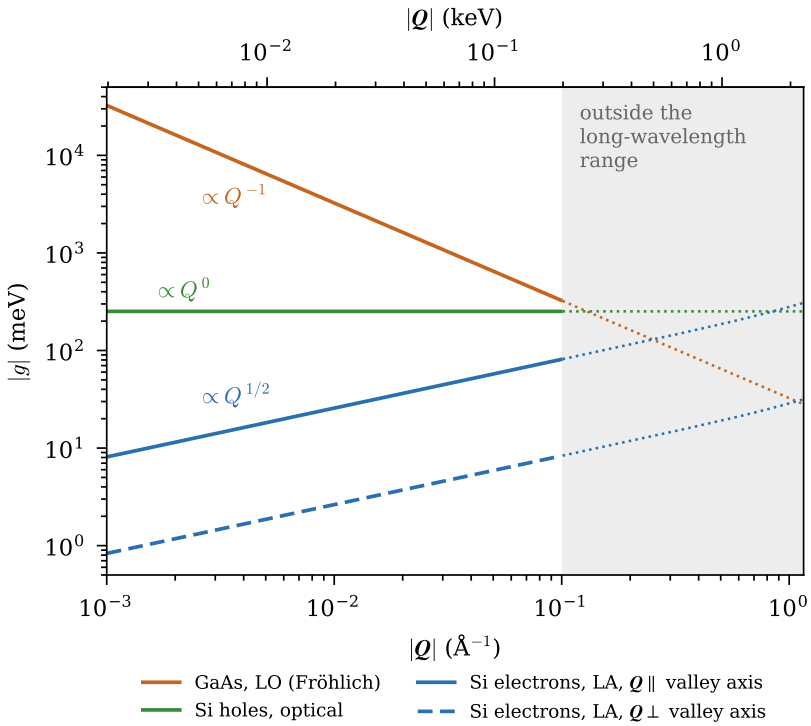


Figure 6.3: Magnitude of the electron–phonon vertex $|g|$, in the normalization in which H_{ep} carries $N^{-1/2}$, against the phonon wavevector $|\boldsymbol{Q}|$ (bottom axis in Å^{-1}, top axis in keV), for the three long-wavelength forms of this section. Orange: Fröhlich vertex Eq. (6.15) of GaAs for an electron at the conduction-band minimum, with $\rho_{nn} = 1$ and the measured $\omega_{\text{L}} = 36.35\,\text{meV}$, $\varepsilon_\infty = 10.88$, $\varepsilon_0 = 12.9$. Green: optical deformation potential Eq. (6.14) for the holes at the valence maximum of silicon, with the DFPT value $10.45\,\text{eV}\,\text{Å}^{-1}$ combined over the degenerate bands [11] and $\omega = 64.1\,\text{meV}$. Blue: acoustic deformation potential Eq. (6.13) for an electron in a conduction valley of silicon and the longitudinal acoustic phonon, with $\boldsymbol{Q}$ along the valley axis (solid, $\Xi_d + \Xi_u = 9.85\,\text{eV}$) and perpendicular to it (dashed, $\Xi_d = 1.01\,\text{eV}$). The blue curves use the DFPT constants of Ref. [11] and the DFPT longitudinal acoustic branch along Γ–X of Fig. 5.3, whose small-Q slope is $8.34\,\text{km}\,\text{s}^{-1}$ (measured: $8.43\,\text{km}\,\text{s}^{-1}$). The curves are these limiting forms, not vertices computed point by point. They are drawn solid up to $0.1\,\text{Å}^{-1}$, the end of the range over which the acoustic vertices of silicon were computed and found linear in Q [11, Fig. 5]. Beyond $0.1\,\text{Å}^{-1}$ they are dotted, in the shaded region up to the zone boundary of silicon at $2\pi/a = 1.157\,\text{Å}^{-1}$. On logarithmic axes the slopes -1, 0 and $\frac{1}{2}$ distinguish the polar, optical and acoustic couplings.

the coarse grids the relation is inverted by a discrete Fourier transform, as for $H^W(\boldsymbol{R})$ in Eq. (27.11). This interpolation is the scheme of Giustino, Cohen and Louie [18], reviewed in Ref. [2, Eqs. (177)–(179)] and implemented in the Electron–Phonon Wannier (EPW) code [8].

The real-space vertex decays in both of its lattice vectors. In $\boldsymbol{R}_e$ it decays as the overlap of two Wannier functions does, exponentially for smooth gauges, because both must reach the region where the potential changes. In $\boldsymbol{R}_p$ it decays as the screened potential of one displaced atom, seen from the Wannier function at the origin. In a nonpolar insulator this potential has neither a monopole, since the displaced atom is screened to neutrality, nor a dipole, since $Z^* = 0$. It still has a quadrupole, and the vertex falls off as R_p^{-3}, a tail of no finite range whose Fourier transform carries a term of order Q^0 that depends on the direction of $\boldsymbol{Q}$. In silicon this quadrupolar part is small, and it can be subtracted like the dipole of a polar crystal below. By the argument that led to Eq. (27.14), a coarse grid whose supercell contains the short-ranged remainder of g^W reproduces the vertex everywhere, and the tails beyond the supercell set the error. For diamond and silicon a 6^3 or 8^3 coarse grid of phonon wavevectors is enough.

A polar crystal breaks the argument. The displaced atom carries the dipole $eZ^*_\kappa \cdot \boldsymbol{u}$ screened by ε_∞, its potential falls off as R_p^{-2}, and the Fourier transform of that tail is the $1/Q$ singularity of the Fröhlich vertex. No finite sum of Fourier components reproduces a singularity. The remedy is the one that Sect. 5.4 used for the nonanalytic part of the dynamical matrix. The long-range vertex is known in closed form from Z^*_κ and ε_∞ alone, Eq. (D.51) or its version with the sum over reciprocal-lattice vectors, Eq. (27.16). One subtracts it from the coarse-grid vertices, interpolates the short-range remainder $g^{\mathrm{S}} = g - g^{\mathrm{L}}$, which is as smooth as the vertex of a nonpolar crystal, and adds g^{L} back analytically at every fine-grid point [13]. The split $g = g^{\mathrm{S}} + g^{\mathrm{L}}$ is not unique, since any smooth function can be moved from one part to the other, but every choice puts the whole singularity into g^{L}, and the interpolation needs nothing more. The density vertex in g^{L} is needed at the fine-grid points too, and at small $\boldsymbol{Q}$ it is $[U(\boldsymbol{k}+\boldsymbol{Q})U^\dagger(\boldsymbol{k})]_{mn}$ in the Wannier gauge, so that g^{S} and g^{L} carry the same phases (Appendix A.5). The quadrupolar terms are present in silicon as in GaAs, and together with the Born charges they give the piezoelectric vertex of GaAs. The quadrupolar terms are long-ranged in the same way as the dipole and are subtracted with it where the code allows.

Fig. 6.4 summarizes the chain. Its pitfalls are practical: a Wannier manifold that misses the states a rate needs, a coarse grid that misplaces the conduction minimum of silicon, a gauge that differs between g and ρ. Sect. 27.2 discusses each, with the grid sizes, the counts of matrix elements and the checks against direct DFPT at points off the coarse grid. The result is a vertex at any $(\boldsymbol{k}, \boldsymbol{Q})$ for about the cost of a few hundred phonon calculations.

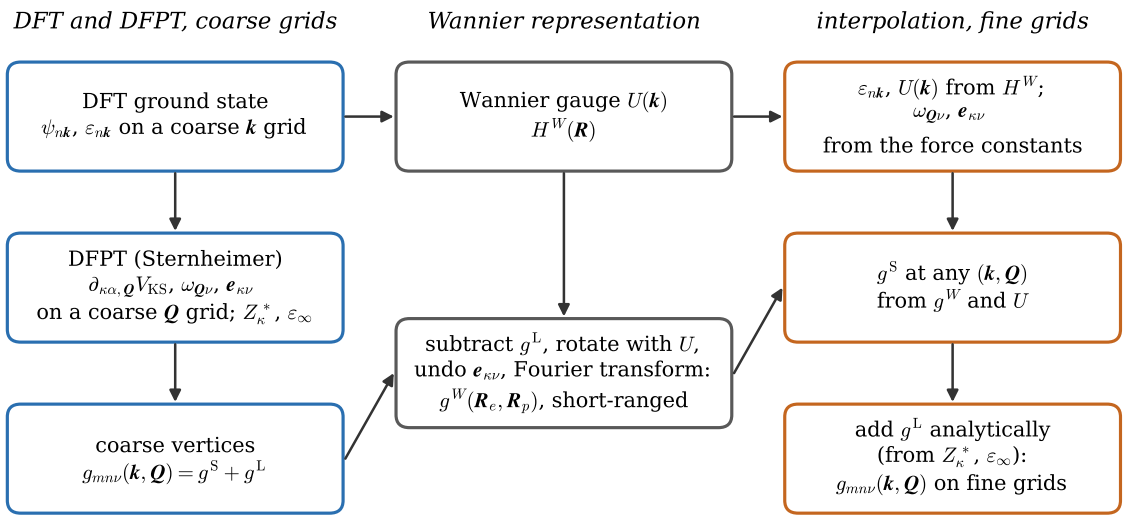


Figure 6.4: From first principles to interpolated electron–phonon vertices. Left: a density-functional calculation on a coarse grid of crystal momenta gives the Bloch states and energies. Density-functional perturbation theory on a coarse grid of phonon wavevectors gives the screened potential changes $\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\mathrm{KS}}$, the phonons, and in a polar crystal Z^*_κ and ε_∞. The matrix elements of the potential changes are the coarse vertices, Eq. (6.11). Middle: the Bloch states define the Wannier gauge $U(\boldsymbol{k})$; the long-range part g^{L} is subtracted, and the rotated remainder is transformed to the real-space vertex $g^W(\boldsymbol{R}_e,\boldsymbol{R}_p)$ of Eq. (6.17), short-ranged in both lattice vectors. Right: the real-space Hamiltonian and force constants give bands, gauge matrices and phonons at any point; g^W is transformed back, Eq. (27.15), and g^{L}, Eq. (27.16), is added analytically. Every quantity in one amplitude uses the same $U(\boldsymbol{k})$.

6.3 Electron self-energy from phonons

At 10 mK a silicon crystal holds no thermal phonons, yet its band gap is about 60 meV smaller than the gap of the same crystal with its nuclei frozen at their equilibrium positions. The zero-point motion of Sect. 5.3, with a root-mean-square amplitude of 0.05 Å along each axis, moves the bands by an amount that matters near threshold: the lightest dark-matter mass a detector can see scales with the gap (Eq. (1.24)). The electron–phonon coupling moves the bands, and it also decides how long a conduction electron with a few meV of excess energy keeps that energy before losing it to the lattice. The shift of the bands and the lifetime of the excited electron both come from the electron self-energy of Sect. 4.7, with phonons as the boson.

6.3.1 Fan–Migdal and Allen–Heine terms

A displaced nucleus changes the potential that the electrons feel, at first and at second order in the displacement. We expand the self-consistent potential V_{KS} about equilibrium:

$$V_{\mathrm{KS}} = V_{\mathrm{KS}}^{(0)} + \sum_a \frac{\partial V_{\mathrm{KS}}}{\partial u_a}\, u_a + \frac{1}{2}\sum_{ab} \frac{\partial^2 V_{\mathrm{KS}}}{\partial u_a\, \partial u_b}\, u_a u_b,$$

with $a = (\boldsymbol{R}\kappa\alpha)$ running over cells, species and Cartesian directions. The linear term is the electron–phonon Hamiltonian H_{ep} of Eq. (6.4), with the screened vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ of Sect. 6.2. The quadratic term, with each displacement expanded in normal modes by Eq. (A.20), is a vertex with two phonon operators,

$$\begin{aligned} H_{\mathrm{ep}}^{(2)} &= \frac{1}{N}\sum_{\boldsymbol{k}\boldsymbol{Q}\boldsymbol{Q}'}\sum_{mn\nu\nu'} g^{\mathrm{AH}}_{mn\nu\nu'}(\boldsymbol{k},\boldsymbol{Q},\boldsymbol{Q}')\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}+\boldsymbol{Q}'} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu} B_{\boldsymbol{Q}'\nu'}, \\ g^{\mathrm{AH}}_{mn\nu\nu'}(\boldsymbol{k},\boldsymbol{Q},\boldsymbol{Q}') &= \frac{1}{2}\sum_{\kappa\alpha,\kappa'\beta} \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa'\nu'}(\boldsymbol{Q}')}{\sqrt{2M_\kappa\omega_{\boldsymbol{Q}\nu}\, 2M_{\kappa'}\omega_{\boldsymbol{Q}'\nu'}}} \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}+\boldsymbol{Q}'} | \partial^2_{\kappa\alpha\boldsymbol{Q},\kappa'\beta\boldsymbol{Q}'} V_{\mathrm{KS}} | \psi_{n\boldsymbol{k}} \rangle, \end{aligned} \tag{6.18}$$

where $\partial^2_{\kappa\alpha\boldsymbol{Q},\kappa'\beta\boldsymbol{Q}'} V_{\mathrm{KS}}$ is the second derivative with respect to two displacement waves, the sum over $\boldsymbol{R}$ and $\boldsymbol{R}'$ of $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}+\mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}'}$ times the second derivative with respect to $u^\alpha_{\boldsymbol{R}\kappa}$ and $u^\beta_{\boldsymbol{R}'\kappa'}$ (Appendix D.13). The coefficient g^{AH} is the Allen–Heine vertex (Appendix A.6). In the energy of an electron, H_{ep} enters at second order and $H_{\mathrm{ep}}^{(2)}$ at first order, and both carry two factors of the zero-point length $\ell_{\boldsymbol{Q}\nu}$: they are of the same order, $\langle u^2\rangle$, and a consistent theory keeps both. We will see below that dropping either term violates translation invariance.

The term of second order in H_{ep} is the boson-exchange diagram of Sect. 4.8 (Fig. 4.14c): the electron emits or absorbs a phonon, propagates in another band and wavevector, and reabsorbs or re-emits it. Compared with Eq. (4.98), the vertex carries band indices and the factor $1/\sqrt{N}$ instead of $1/\sqrt{\mathcal{V}}$, and the intermediate electron can be in any band m. The term of first order in $H_{\mathrm{ep}}^{(2)}$ needs no propagator. Its two phonon operators are contracted with each other into a phonon line that leaves and returns to the same point of the electron line, and the pair is replaced by its thermal average $\langle B_{\boldsymbol{Q}\nu} B_{-\boldsymbol{Q}\nu}\rangle = 2n_{\boldsymbol{Q}\nu} + 1$. Appendix D.13 carries out the Matsubara sum and the continuation for both diagrams.

Fan–Migdal and Allen–Heine self-energies

For a nondegenerate band n at wavevector $\boldsymbol{k}$, with energies ω and $\xi_{m\boldsymbol{p}} = \varepsilon_{m\boldsymbol{p}} - \mu_{\mathrm{ch}}$

measured from the chemical potential, $f_{m\boldsymbol{p}} = n_{\mathrm{F}}(\xi_{m\boldsymbol{p}})$ and $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$,

$$\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \Big[\frac{n_{\boldsymbol{Q}\nu} + 1 - f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} - \omega_{\boldsymbol{Q}\nu} + \mathrm{i}0^+} + \frac{n_{\boldsymbol{Q}\nu} + f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} + \omega_{\boldsymbol{Q}\nu} + \mathrm{i}0^+} \Big], \tag{6.19}$$

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q})\,(2n_{\boldsymbol{Q}\nu}+1). \tag{6.20}$$

The Fan–Migdal term depends on frequency and has an imaginary part; the Allen–Heine term is static and real.

Eq. (6.19) is the second-order self-energy of Fan [19], called the Migdal self-energy in the theory of metals [1], and Eq. (6.20) is the term added by Allen and Heine [20]; together they are Ref. [2, Eqs. (157), (158), (160) and (161)]. The thermal factors read as in Sect. 4.8. In the emission term, $n+1$ is stimulated plus spontaneous emission and $-f$ blocks an occupied intermediate state. In the absorption term, n counts the phonons present, and f adds the process in which an electron already in $(m, \boldsymbol{k}+\boldsymbol{Q})$ emits a phonon and drops into the state $n\boldsymbol{k}$. The sum over m now runs over occupied bands as well. For a degenerate level, such as the threefold valence maximum of silicon (Sect. 3.3), the self-energy is a matrix in the partners, and the terms between partners, whose denominators vanish, are left out, as is the term $m = n$ at $\boldsymbol{Q} = 0$ [21, Sect. III A]. Summed over the whole zone, the matrix commutes with the symmetry operations that leave $\boldsymbol{k}$ unchanged. On a level whose partners form one irreducible representation it is therefore a multiple of the identity, and every partner shifts by the average of the diagonal elements over the level. The shifts of degenerate band edges quoted below are this average. At $T = 0$ the zero-point motion enters Σ^{FM} through the 1 of $n+1$ and through f, and Σ^{AH} through the 1 of $2n+1$. One more diagram is of the same order, a phonon line at zero wavevector ending on a closed electron loop; it vanishes because the expansion is made about the minimum of the Born–Oppenheimer surface (Appendix D.13).

The pole of the Dyson equation Eq. (4.81) then lies at the renormalized band energy

$$\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} + \mathrm{Re}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}) + \Sigma^{\mathrm{AH}}_{n\boldsymbol{k}}, \tag{6.21}$$

with ω now measured from the same origin as ε, the quasiparticle condition of Eq. (4.83). The value of $\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}$ at $T = 0$, less $\varepsilon_{n\boldsymbol{k}}$, is the *zero-point renormalization* (ZPR) of the state, and the ZPR of a gap is the ZPR of the conduction state minus the ZPR of the valence state. Allen–Heine–Cardona (AHC) theory is the theory built on the Fan–Migdal and Allen–Heine terms, usually in the adiabatic form below and with the rigid-ion approximation of Allen and Cardona [22]. In practice Σ^{FM} is often evaluated at $\omega = \varepsilon_{n\boldsymbol{k}}$ rather than at the solution of Eq. (6.21); for the Einstein boson of Worked example 4.13 the two choices yield -9.14 meV and -8.73 meV, a difference set by $1 - z_{\mathrm{qp}}$.

6.3.2 The adiabatic limit

Where the phonon energies are small next to the electronic energy differences that dominate the sum over m, they can be dropped from the denominators of Eq. (6.19). If we also set $\omega = \varepsilon_{n\boldsymbol{k}}$, the two terms acquire the same denominator, and their numerators add to $(n+1-f)+(n+f) = 2n+1$: the occupations of the intermediate states drop out. The

result is the formula of Allen and Heine [20],

$$\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}(2n_{\boldsymbol{Q}\nu}+1)\left[\sum_m \frac{|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}} + g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q})\right], \tag{6.22}$$

Ref. [2, Eq. (167)]. Every mode now enters with the weight $2n_{\boldsymbol{Q}\nu}+1$ of its mean-square displacement (Eq. (5.35)). Allen and Heine derived Eq. (6.22) without diagrams. With the nuclei frozen at displacements u_a, a band energy is a function $\varepsilon_{n\boldsymbol{k}}(\{u\})$, and second-order perturbation theory gives its curvature: the Fan sum is the second-order term of H_{ep}, the Allen–Heine vertex the first-order term of the quadratic potential (Eq. (D.53)). The nuclei are not frozen, however, but spread over a Gaussian distribution of width $\langle u_a u_b\rangle$. Averaged over this distribution, the linear term vanishes and the energy shifts by half the curvature times the width, $\frac{1}{2}\sum_{ab}\langle u_a u_b\rangle\, \partial^2\varepsilon_{n\boldsymbol{k}}/\partial u_a \partial u_b$. In the adiabatic limit the AHC shift is the average of the band energy over the zero-point and thermal smearing of the lattice.

The adiabatic form fails where the dropped phonon energies are not small next to the electronic energy differences. The electronic energy differences become small for $m = n$ at small Q, where $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{n,\boldsymbol{k}+\boldsymbol{Q}}$ vanishes with Q, and wherever another band is degenerate with the state $n\boldsymbol{k}$. Both cases occur at band edges, the states that set a detector's threshold. The Fröhlich vertex of a polar crystal shows how badly the adiabatic form can fail. Take an electron at the bottom of a parabolic band of mass m^* at $T = 0$, coupled to the longitudinal optical phonon by $|g^{\mathrm{L}}|^2 = C_{\mathrm{F}}^2/Q^2$ (Eq. (6.15)). In the adiabatic form the intraband Fan term of this electron is

$$\frac{1}{N}\sum_{\boldsymbol{Q}} \frac{|g^{\mathrm{L}}|^2}{-Q^2/2m^*} = -\Omega\int \frac{\mathrm{d}^3Q}{(2\pi)^3}\,\frac{2m^*C_{\mathrm{F}}^2}{Q^4} = -\frac{\Omega\, m^* C_{\mathrm{F}}^2}{\pi^2}\int_0 \frac{\mathrm{d}Q}{Q^2},$$

which diverges at small Q. The divergence is an artifact. With the phonon energy kept, only the emission term of Eq. (6.19) survives for an empty band at $T = 0$, its denominator at $\omega = \varepsilon_{n\boldsymbol{k}}$ is $-Q^2/2m^* - \omega_{\mathrm{L}}$, and $\mathrm{Re}\,\Sigma^{\mathrm{FM}}$ is the finite second-order shift of Sect. 6.2, $-\alpha_{\mathrm{F}}\omega_{\mathrm{L}} = -2.6\,\mathrm{meV}$ for the conduction band of GaAs (Worked example 6.2). The Fröhlich polaron shift is the Fan–Migdal self-energy at the band bottom. Every infrared-active crystal has this $1/Q^2$ coupling and therefore an infinite adiabatic ZPR, and the nonadiabatic form, with the phonon energies in the denominators, is needed to get a number [21]. In silicon the vertex has no $1/Q$ term (Sect. 6.2), both forms converge, and they differ by 8 meV: -64.3 meV adiabatic and -56.2 meV nonadiabatic for the indirect gap in the local-density approximation [21, Table VI].

6.3.3 Translation invariance and the rigid-ion approximation

Moving the whole crystal by a constant vector shifts no band energy. For long-wavelength acoustic phonons this translation invariance ties the Fan and Allen–Heine terms together. As $Q \to 0$ an acoustic mode becomes a rigid translation, with $\boldsymbol{e}_{\kappa\nu} \to (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ and every nucleus displaced by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu/\sqrt{N}$ per unit of $B_{\boldsymbol{Q}\nu}$ (Appendix A.6). Its vertex then follows from Eq. (6.11) and the acoustic sum rule Eq. (6.12),

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) \to \ell_{\boldsymbol{Q}\nu}\sum_\kappa \hat{\boldsymbol{e}}_\nu \cdot \langle\psi_{m\boldsymbol{k}}|\partial_{\kappa,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{n\boldsymbol{k}}\rangle = -\mathrm{i}\,\ell_{\boldsymbol{Q}\nu}\,(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}(\boldsymbol{k}).$$

The intraband element vanishes, as Sect. 6.2 showed; the interband elements do not. In the Fan sum of Eq. (6.22) the energy difference is squared in the numerator and appears once in the denominator, and the Fan sum becomes

$$\sum_{m\neq n}\frac{|g_{mn\nu}|^2}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}} \to \ell^2_{\boldsymbol{Q}\nu}\sum_m(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\,|\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}|^2.$$

For the Allen–Heine vertex of the same mode, the same polarization vectors turn the prefactor of Eq. (6.18) into $\ell^2_{\boldsymbol{Q}\nu}\hat{e}_{\nu\alpha}\hat{e}_{\nu\beta}$, and, summed over κ and κ', the second derivative with respect to two displacement waves of zero wavevector becomes the second derivative with respect to a uniform translation $\boldsymbol{t}$ of the whole crystal. The self-consistent potential moves with the crystal, $V_{\mathrm{KS}} \to \hat{T}_{\boldsymbol{t}} V_{\mathrm{KS}} \hat{T}_{\boldsymbol{t}}^\dagger$ with $\hat{T}_{\boldsymbol{t}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{t}\cdot\hat{\boldsymbol{p}}}$ (Sect. 6.2). Differentiating twice along $\hat{\boldsymbol{e}}_\nu$ gives $-[\hat{p}_\nu,[\hat{p}_\nu,V_{\mathrm{KS}}]]$ with $\hat{p}_\nu = \hat{\boldsymbol{e}}_\nu\cdot\hat{\boldsymbol{p}}$, and since the kinetic energy commutes with $\hat{p}_\nu$, V_{KS} may be replaced by H_{KS}. Expanding the double commutator in the eigenstates of H_{KS} at $\boldsymbol{k}$, the states that $\hat{p}_\nu$ connects to $\psi_{n\boldsymbol{k}}$, gives

$$g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q}) \to -\frac{\ell^2_{\boldsymbol{Q}\nu}}{2}\langle\psi_{n\boldsymbol{k}}|[\hat{p}_\nu,[\hat{p}_\nu,H_{\mathrm{KS}}]]|\psi_{n\boldsymbol{k}}\rangle = -\ell^2_{\boldsymbol{Q}\nu}\sum_m(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}})\,|\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{mn}|^2.$$

The two contributions cancel term by term:

$$\sum_{m\neq n}\frac{|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}} + g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q}) \to 0 \qquad (\nu \text{ acoustic},\ Q\to 0). \tag{6.23}$$

The physical picture is the rigid translation of Sect. 6.2. The interband vertex translates the electronic state along with the crystal, and at second order it shifts the energy by $\ell^2_{\boldsymbol{Q}\nu}\sum_m(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}})|\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{mn}|^2$, a sum of either sign: bands above n push the energy down, bands below push it up. The Allen–Heine term measures the curvature of the potential under the same translation, and it shifts the energy back by exactly as much. Neither term means anything alone, and a calculation must treat both on the same grid of wavevectors. In the language of scalar quantum electrodynamics (QED) the Fan and Allen–Heine terms are the rainbow and the seagull diagrams, and their cancellation for a rigid translation is the analogue of the cancellation that makes the scattering of low-frequency light by a bound electron vanish. The same double commutator underlies the f-sum rule (Eq. (4.28)). The three translational modes at exactly $\boldsymbol{Q}=0$, with zero frequency, are left out of both sums; Eq. (6.23) states that the integrand is continuous there. For small Q the first correction to the cancellation is linear in $\boldsymbol{Q}$ and drops out of the sum over directions. The interband remainder is then smaller than either term by a factor $\sim Q^2$, and long-wavelength acoustic phonons, which carry a large share of the thermal $\langle u^2\rangle$ of silicon (Worked example 5.4), barely shift the bands. At small Q the intraband term $m=n$ remains; its denominator vanishes with Q, and this term needs the phonon energy kept.

The Allen–Heine vertex needs second derivatives of the self-consistent potential, a second-order perturbation calculation that few codes perform. Translation invariance offers a way around this calculation. A rigid translation leaves every band energy unchanged at every configuration of the nuclei, so the sum over all nuclei of $\partial\varepsilon_{n\boldsymbol{k}}/\partial u_{\boldsymbol{R}\kappa\alpha}$ vanishes everywhere, and so does its derivative. At equilibrium the vanishing derivative gives an exact sum rule, Eq. (D.55): the diagonal element of the second derivative, summed over the sites of the whole crystal, equals a product of first derivatives at $\boldsymbol{Q}=0$. The sum rule fixes a sum over sites, while g^{AH} needs the on-site element. The *rigid-ion approximation* supplies the step from the sum over sites to the on-site element. It writes the potential as a sum of atom-centered terms that move rigidly with their nuclei, $V_{\mathrm{KS}} = \sum_{\boldsymbol{R}\kappa} v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa-\boldsymbol{u}_{\boldsymbol{R}\kappa})$. Second derivatives with respect to two different atoms then vanish, the sum rule determines the on-site element (Eq. (D.56)), and the Allen–Heine term becomes

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = -\frac{1}{2}\sum_{\kappa\kappa'}\sum_{\alpha\beta}(B^{\alpha\beta}_\kappa + B^{\alpha\beta}_{\kappa'})\ \mathrm{Re}\sum_m{}' \frac{\langle\psi_{n\boldsymbol{k}}|\partial_{\kappa\alpha,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{m\boldsymbol{k}}\rangle\,\langle\psi_{m\boldsymbol{k}}|\partial_{\kappa'\beta,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{n\boldsymbol{k}}\rangle}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}}}, \tag{6.24}$$

where $B_\kappa = \langle\boldsymbol{u}_\kappa\boldsymbol{u}_\kappa\rangle$ is the displacement tensor of Eq. (5.35) at temperature T and the prime excludes the states degenerate with $(n,\boldsymbol{k})$ (Appendix D.13). Eq. (6.24) is the form of Allen

and Cardona [22], Ref. [2, Eqs. (195)–(198)]. The rigid-ion form needs just the first-order potentials at $\boldsymbol{Q} = 0$, which every phonon calculation already supplies. It has the structure of a Fan term at zero wavevector with the opposite sign, weighted by the mean-square displacement of each atom; the cancellation of Eq. (6.23) is built in. The rigid-ion approximation neglects the change of screening when two atoms move together rather than one at a time. Poncé et al. find the effect of this change large in molecules [21]; for crystals the rigid-ion form is the standard choice.

In practice the AHC shift is computed from DFPT vertices on a grid of phonon wavevectors. The sum over bands is split into an active window, summed explicitly, and the remaining bands, whose contribution comes from a Sternheimer equation. The sum over $\boldsymbol{Q}$ converges slowly near the vanishing intraband denominators and needs grids of 10^5 to 10^6 points, or Wannier interpolation of the vertex (Sect. 6.2), with a small imaginary part in the denominators extrapolated to zero. Sect. 25.3 describes the calculation, its convergence parameters and its pitfalls, among them the small residual Born charges of a nonpolar DFPT calculation, which must be removed, since otherwise the sum over $\boldsymbol{Q}$, or the integral that replaces it, diverges at a band extremum.

6.3.4 Smeared ionic potentials

The literature calls Σ^{AH} the Debye–Waller self-energy, and in the rigid-ion approximation the name has a reason. Average the rigid-ion potential over the Gaussian distribution of the displacements of each nucleus. A Fourier component $v_\kappa(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa)}$ acquires the factor $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{u}_\kappa}\rangle = \mathrm{e}^{-\boldsymbol{G}\cdot B_\kappa\cdot\boldsymbol{G}/2}$, and to first order in B_κ

$$\big\langle v_\kappa(\boldsymbol{r}-\boldsymbol{u}_\kappa)\big\rangle - v_\kappa(\boldsymbol{r}) = \frac{1}{2}\sum_{\alpha\beta} B_\kappa^{\alpha\beta}\,\partial_\alpha\partial_\beta v_\kappa(\boldsymbol{r}) = -\sum_{\boldsymbol{G}} W_\kappa(\boldsymbol{G})\,v_\kappa(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \tag{6.25}$$

for the atom at the origin, with $2W_\kappa(\boldsymbol{G}) = \boldsymbol{G}\cdot B_\kappa\cdot\boldsymbol{G}$, the exponent of Eq. (5.48) evaluated at a reciprocal-lattice vector. Summed over all the sites of every species, the diagonal matrix element of the middle expression is the rigid-ion Allen–Heine term (Appendix D.13). So Σ^{AH} is the first-order change of a band energy when each ionic potential is smeared out by the zero-point and thermal motion of its nucleus, the "fuzziness" of the nuclear charge described in Ref. [2, below Eq. (154)].

It is nonetheless a different object from the Debye–Waller factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$ of Sect. 5.5, a probability that multiplies a rate at the momentum transfer of the scattering. The smearing of Eq. (6.25) involves no momentum transfer and no rate. It is evaluated at the reciprocal-lattice vectors of the static potential, enters an energy at first order in B_κ, and is one of the two terms of the second-order shift. The Fan term is of the same order and partly cancels it, exactly so for acoustic phonons at small Q (Eq. (6.23)). Allen and Heine showed that a theory of the band shifts with the smeared potential alone is incomplete [20]. The smearing term and the Debye–Waller factor share the displacement tensor B_κ, and nothing else.

6.3.5 Zero-point renormalization

For the indirect gap of silicon, AHC theory in the local-density approximation gives a zero-point renormalization of $-56.2\,\mathrm{meV}$ in the nonadiabatic form [21, Table VI]. The valence maximum rises and the conduction minimum drops, by 35 meV and 22 meV in a calculation by a different method [23, Sect. VII B], and the gap closes from both sides. That calculation used *special displacements*, a non-perturbative route that avoids both the vertices and the sums over wavevectors. Since the adiabatic shift is the average of the band energy over the Gaussian distribution of the nuclei, it can be computed directly: displace the nuclei of a large

supercell and compute the bands. Zacharias and Giustino showed that a single configuration suffices, one in which every normal coordinate has its root-mean-square amplitude and the signs alternate between modes so that the cross terms cancel [24]. In an $8 \times 8 \times 8$ supercell of 1024 atoms the special-displacement method yields $-57\,\mathrm{meV}$ [23, Sect. VII B]. The method includes all orders in the displacements but is adiabatic by construction.

The renormalization extracted from experiment is $-62\,\mathrm{meV}$ or $-64\,\mathrm{meV}$, depending on the route of extraction [25], as tabulated in Ref. [21, Table VI]. A ZPR cannot be measured directly, since no experiment freezes the nuclei. Both routes rest on the dependence of the gap on temperature and on the nuclear mass, and the simplest model makes this dependence explicit.

Suppose every phonon has the same energy ω_{Ein}, an Einstein crystal. The weight $2n_{\boldsymbol{Q}\nu}+1 = \coth(\omega_{\mathrm{Ein}}/2T)$ is then the same for every mode and comes out of the sum in Eq. (6.22). The shift of any band at temperature T is its zero-point value times one function of temperature,

$$\Delta\varepsilon_{n\boldsymbol{k}}(T) = \Delta\varepsilon_{n\boldsymbol{k}}(0)\,\coth\frac{\omega_{\mathrm{Ein}}}{2T}. \tag{6.26}$$

At $T \gg \omega_{\mathrm{Ein}}$, $\coth x \simeq 1/x + x/3$, and the shift becomes linear in T with the slope $2\Delta\varepsilon(0)/\omega_{\mathrm{Ein}}$. Since this line extrapolates to zero shift at $T = 0$, extrapolating a measured gap back from high temperature gives the gap of the clamped lattice, and the low-temperature gap lies below that intercept by the ZPR. The high-temperature extrapolation is one of the two extractions; the other uses the nuclear mass. At fixed force constants $\omega_{\mathrm{Ein}} \propto M^{-1/2}$, the squared vertex and the Allen–Heine term are both proportional to $\ell^2 \propto 1/(M\omega_{\mathrm{Ein}}) \propto M^{-1/2}$, and so is the ZPR. Then $\mathrm{ZPR} = -2M\,\partial E_{\mathrm{g}}/\partial M$, and the gaps of isotopically enriched crystals measure it.

Worked example 6.3: The Einstein model of the silicon gap

The Einstein boson of Worked example 4.13 shifted the band bottom by $-9.14\,\mathrm{meV}$ at $T = 0$ and by $-11.09\,\mathrm{meV}$ at 300 K, a ratio of 1.213 against $2n_{\mathrm{B}} + 1 = 1.218$. The small difference came from the boson energy in the denominators. In the adiabatic limit Eq. (6.26) makes the ratio exact, and the model reduces to two numbers: the ZPR and ω_{Ein}.

For silicon we choose ω_{Ein} by the zero-point motion. At $T = 0$ every term of the ZPR carries a factor $1/\omega_{\boldsymbol{Q}\nu}$ from $\ell^2_{\boldsymbol{Q}\nu}$, as the mean-square displacement does, and the Einstein energy that reproduces the zero-point $\langle u^2\rangle$ of the computed phonon spectrum is $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, or 353 K (Worked example 5.4). For the ZPR we take the nonadiabatic AHC value, $-56.2\,\mathrm{meV}$. Applying the adiabatic weight of Eq. (6.26) to a nonadiabatic value is an approximation; the two forms differ by 8 meV at $T = 0$.

At 300 K, $T = 25.85\,\mathrm{meV}$, $\omega_{\mathrm{Ein}}/2T = 0.588$ and $\coth(0.588) = 1.891$. The model predicts that the gap closes between zero and room temperature by

$$E_{\mathrm{g}}(300\,\mathrm{K}) - E_{\mathrm{g}}(0) = -56.2\,\mathrm{meV} \times (1.891 - 1) = -50.1\,\mathrm{meV}.$$

The measured gap falls by 46.2 meV, from 1.1692 eV to 1.1230 eV, according to the fit $E_{\mathrm{g}}(T) = E_{\mathrm{g}}(0) - aT^2/(T+b)$ with $E_{\mathrm{g}}(0) = 1.1692\,\mathrm{eV}$, $a = 4.9\times10^{-4}\,\mathrm{eV\,K^{-1}}$ and $b = 655\,\mathrm{K}$ to photoluminescence up to 1000 K [26, 27, Eq. (11)]. The model is 8 % high; with the adiabatic ZPR, $-64.3\,\mathrm{meV}$, it would predict $-57.3\,\mathrm{meV}$. Its high-temperature slope is $2 \times 56.2\,\mathrm{meV}/353\,\mathrm{K} = 0.318\,\mathrm{meV\,K^{-1}}$, between the slopes of the measured curve at 300 K and 500 K, 0.26 and $0.33\,\mathrm{meV\,K^{-1}}$.

Had we assumed that the gap follows the full mean-square displacement of the computed spectrum, the shift would have scaled with $\langle u^2\rangle(T)/\langle u^2\rangle(0) = 2.661$, and the gap would have closed by $56.2\,\mathrm{meV} \times 1.661 = 93.3\,\mathrm{meV}$ at 300 K, twice the measured value. That

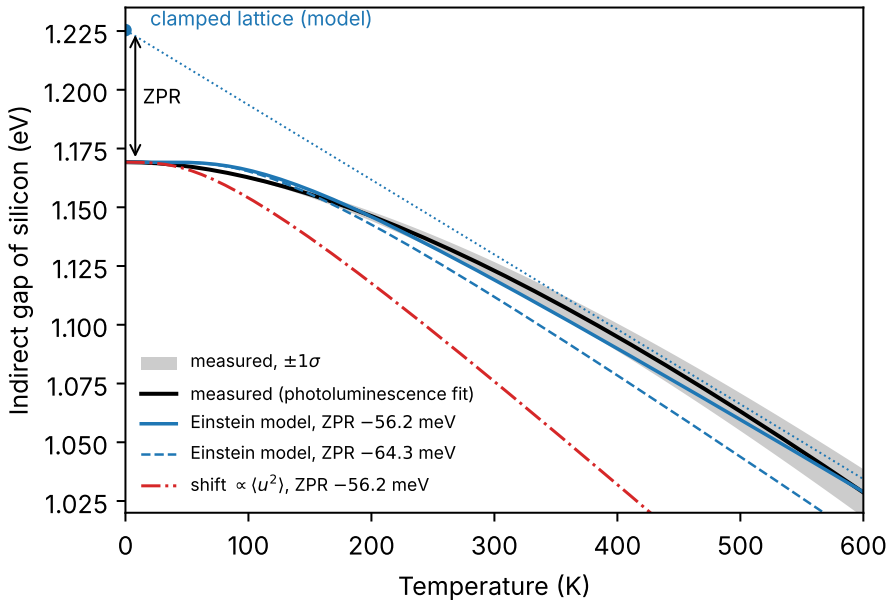


Figure 6.5: The indirect gap of silicon against temperature. Black: measured, the fit $E_g(T) = 1.1692\,\text{eV} - aT^2/(T+b)$ with $a = 4.9 \times 10^{-4}\,\text{eV K}^{-1}$ and $b = 655\,\text{K}$ to photoluminescence data [26, 27]; the gray band is the envelope of the fits with a and b varied independently by their quoted errors of $0.2 \times 10^{-4}\,\text{eV K}^{-1}$ and 40 K. Blue: Einstein model $E_g(0) + \text{ZPR}\,[\coth(\omega_{\text{Ein}}/2T) - 1]$ with $\omega_{\text{Ein}} = 30.4\,\text{meV}$, the Einstein energy that reproduces the zero-point mean-square displacement of the computed phonon spectrum, and the computed zero-point renormalization, −56.2 meV (solid, nonadiabatic) or −64.3 meV (dashed, adiabatic) [21]. Red, dash-dotted: a shift proportional to the full mean-square displacement $\langle u^2\rangle(T)$ of the computed spectrum, with the same zero-point value. Blue dotted line and dot: the high-temperature asymptote of the solid model, which extrapolates to the model's clamped-lattice gap, 1.2254 eV, at $T = 0$; the arrow marks the ZPR. All model curves are anchored at the measured gap at $T = 0$ and leave out thermal expansion.

assumption gives every mode a shift in proportion to its mean-square displacement, and the thermal $\langle u^2\rangle$ is weighted toward the softest modes, which enter with $1/\omega^2$: the long-wavelength acoustic phonons and the flat transverse acoustic branches near the zone boundary (Worked example 5.4). The measured curve says that these modes move the gap less than their share of $\langle u^2\rangle$, and for the acoustic modes at small Q Eq. (6.23) explains why: they are nearly rigid translations. The gap responds to the displacements of atoms relative to their neighbors, not to their displacement as such.

The model's clamped-lattice gap, the intercept of its high-temperature line, is $1.1692\,\text{eV} + 0.0562\,\text{eV} = 1.2254\,\text{eV}$; the model is anchored at the measured $E_g(0)$, which contains the zero-point expansion of the lattice, and it leaves out thermal expansion. Replacing natural silicon, $M = 28.09\,\text{u}$, by ^{30}Si scales the ZPR by $(28.09/29.97)^{1/2}$ and raises the gap by 1.8 meV. At detector temperatures the thermal part of the shift vanishes. At 1 K, $\omega_{\text{Ein}}/T = 353$ and $2n_B(\omega_{\text{Ein}}) \sim 10^{-153}$; the only phonons still occupied are acoustic modes at small Q, and their contribution vanishes by Eq. (6.23) and with their phase space. A detector at 10 mK therefore sees $E_g(0)$.

In Fig. 6.5 the Einstein model follows the measured gap to within about 5 meV from zero

to 600 K, with a computed ZPR and an Einstein energy fixed by the zero-point motion. Below 150 K it falls slightly too slowly, since a single energy of 30 meV cannot represent the softer modes that the real spectrum starts to populate first. The curve that follows $\langle u^2 \rangle$ fails from the start.

The values -62 meV and -64 meV extracted from experiment come from the isotope dependence of the gap and from the linear extrapolation of its high-temperature part [25], and they need no Einstein energy. They contain the whole zero-point shift of the gap, including a part that is not electron–phonon coupling: the zero-point motion also expands the lattice, and this expansion by itself opens the gap by 9 meV [28, Table S2]. The electron–phonon part implied by experiment is therefore -71 to -73 meV. That is the number to compare with the computed -56.2 meV, a fixed-volume electron–phonon value, and the calculation falls about 20 % short of it, as does the special-displacement value of -57 meV. The shortfall is consistent with the overscreening of the vertex by the small Kohn–Sham gap: a calculation of the same kind on *GW* bands gives 66 meV [23, Sect. VII B].

6.3.6 Phonon-limited lifetimes

The imaginary part of the self-energy is a decay rate. The Allen–Heine term is real and contributes nothing to it. For the Fan–Migdal term, the Sokhotski–Plemelj formula Eq. (2.58) turns each denominator of Eq. (6.19) into a delta function,

$$\begin{aligned} \operatorname{Im}\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) = -\frac{\pi}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \Big[&(n_{\boldsymbol{Q}\nu} + 1 - f_{m,\boldsymbol{k}+\boldsymbol{Q}})\, \delta(\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} - \omega_{\boldsymbol{Q}\nu}) \\ &+ (n_{\boldsymbol{Q}\nu} + f_{m,\boldsymbol{k}+\boldsymbol{Q}})\, \delta(\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} + \omega_{\boldsymbol{Q}\nu}) \Big], \end{aligned} \tag{6.27}$$

and by Eq. (4.83) the state decays at the rate $\Gamma_{n\boldsymbol{k}} = -2 z_{\mathrm{qp}} \operatorname{Im}\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\xi_{n\boldsymbol{k}})$; its *phonon-limited lifetime* is $1/\Gamma_{n\boldsymbol{k}}$. The cutting rule of Sect. 4.8 reads Eq. (6.27) as golden-rule rates: the cut runs through the intermediate electron and the phonon, and puts both on shell. For an electron in an empty band the out-scattering rates are all that remain, emission of a phonon with the weight $n + 1$ and absorption with the weight n of Eq. (A.27).

At $T = 0$ only emission is left, and an electron can emit only if a state of lower energy is available. An electron at the bottom of the conduction band of an insulator has none, and its width is zero: band edges are sharp at zero temperature, as Sect. 4.7 anticipated. A width found at a band extremum at $T = 0$ in a calculation is the imaginary part added to the denominators for numerical reasons, not a lifetime. At 10 mK the absorption term is absent for every phonon above about 10 µeV, and a hot carrier in a detector loses its energy by spontaneous emission only. How fast, and down to what energy?

For a conduction electron near a band minimum, acoustic phonons set both the rate of the emission and the energy at which it stops. Take a parabolic band of mass m^*, an electron of wavevector $\boldsymbol{k}$ measured from the minimum, and the deformation-potential vertex $|g|^2 = \Xi^2 Q/(2M_{\mathrm{cell}}c)$ of Eq. (6.13), for a branch of sound speed c and an effective constant Ξ. At $T = 0$, with $(1/N)\sum_{\boldsymbol{Q}} \to \Omega \int \mathrm{d}^3Q/(2\pi)^3$,

$$\Gamma(\boldsymbol{k}) = 2\pi\,\Omega \int \frac{\mathrm{d}^3Q}{(2\pi)^3}\, \frac{\Xi^2 Q}{2M_{\mathrm{cell}}c}\, \delta\Big(\frac{k^2 - |\boldsymbol{k}+\boldsymbol{Q}|^2}{2m^*} - cQ\Big).$$

The delta function fixes the angle θ between $\boldsymbol{Q}$ and $\boldsymbol{k}$ at $\cos\theta = -(Q/2 + m^*c)/k$, and the integral over $\cos\theta$ gives $m^*/(kQ)$. A solution exists only for $Q \le Q_{\max} = 2(k - m^*c)$, which requires $k > m^*c$: the electron must move faster than sound. The emission is Cherenkov radiation of sound, and the phonons go out on a cone about the direction of motion. The

remaining integral over Q gives

$$\Gamma(\boldsymbol{k}) = \frac{\Omega\,\Xi^2\,m^*\,Q_{\max}^3}{12\pi M_{\text{cell}}\,c\,k}, \qquad Q_{\max} = 2(k - m^*c), \tag{6.28}$$

which grows linearly with the kinetic energy $k^2/2m^*$ far above the threshold.

For silicon we estimate the rate with three simplifications. The valley is taken spherical, with the density-of-states mass $m^* = m_{\text{d}}^* = (m_\parallel^* m_\perp^{*2})^{1/3} = 0.322\,m_e$ from the valley masses of Sect. 3.3. The valley constants of Sect. 6.2, $\Xi_d = 1.01\,\text{eV}$ and $\Xi_u = 8.84\,\text{eV}$, enter with their angular factors averaged over the angle between $\boldsymbol{Q}$ and the valley axis, as for valleys of random orientation: $\Xi^2 = \Xi_d^2 + \frac{2}{3}\Xi_d\Xi_u + \frac{1}{5}\Xi_u^2 = (4.75\,\text{eV})^2$ for the longitudinal branch and $\frac{2}{15}\Xi_u^2 = (3.23\,\text{eV})^2$ for the one transverse branch that couples. And the sound speeds are the measured ones along [100], 8.43 and $5.84\,\text{km}\,\text{s}^{-1}$ (Sect. 5.2). The thresholds $m^*c^2/2$ are 0.065 meV for longitudinal and 0.031 meV for transverse emission. At 10 meV of kinetic energy the two branches give $\Gamma = 7.2 + 5.2\,\mu\text{eV} = 12.3\,\mu\text{eV}$, a lifetime of 53 ps; at 1 meV the lifetime is 0.9 ns, and at 30 meV 16 ps. Above the optical phonon energy, 64 meV, and through intervalley scattering (Sect. 6.2) the rates are far higher. The estimate fixes the scale. An electron created with a few meV of kinetic energy in a silicon detector sheds this energy as acoustic phonons over a fraction of a nanosecond, and then stops emitting once its kinetic energy falls below a few tens of μeV, under 1 K. In a biased detector the field keeps the carriers above that threshold, and the phonons emitted along the drift are the Luke phonons of Sect. 20.1. A quantitative rate uses Eq. (6.27) with DFPT vertices on fine grids and the real anisotropic valleys.

6.3.7 Migdal's adiabatic theorem

The Fan–Migdal self-energy keeps the vertex g bare at both ends and the propagators free. The exact self-energy has the same structure with every piece dressed: the full electron propagator, the full phonon propagator, and a vertex function in place of one of the two g's [2, Eq. (156)]. At fourth order in g two new diagrams appear. In one, a second phonon line is emitted and reabsorbed inside the first, on the intermediate electron line; it is a self-energy insertion, and replacing the free propagator in Eq. (6.19) by the full one sums all such insertions. In the other, the two phonon lines cross; this crossed diagram is a vertex correction, and no self-consistency of the propagators includes it. Migdal's question was how large the crossed diagram is [1].

In a metal, the counting of Sect. 6.1 answers Migdal's question. Consider an electron at the Fermi surface. In the self-energy insertion the inner phonon carries a frequency of at most ω_{D}. The intermediate electron can be anywhere on the Fermi surface, within ω_{D} of it in energy. The insertion is then as large as the self-energy itself, a fraction λ_{ep} of ω_{D}, with λ_{ep} the dimensionless electron–phonon coupling, defined so that the mass of an electron at the Fermi surface is enhanced by $1 + \lambda_{\text{ep}}$ (Sect. 6.4). In the crossed diagram, two intermediate electron lines carry momenta that differ by the momentum of the outer phonon. Both lines lie within ω_{D} of the Fermi surface only where two thin shells intersect. The shells have the thickness $\omega_{\text{D}}/v_{\text{F}}$, with v_{F} the Fermi velocity, and are shifted against each other by the momentum of the outer phonon; their intersection is a fraction $\sim \omega_{\text{D}}/E_{\text{F}}$ of the momenta summed over. Elsewhere one of the two lines is off shell by an energy of order E_{F}.

Migdal's adiabatic theorem

In a metal, the vertex corrections to the electron–phonon self-energy are of relative order $\lambda_{\text{ep}}\,\omega_{\text{D}}/E_{\text{F}}$. To that accuracy, the self-energy is the Fan–Migdal diagram with the

screened vertex g, and the electron propagator inside it may be made self-consistent when λ_{ep} is not small.

The small ratio is the $\omega_{\mathrm{D}}/E_{\mathrm{F}} \sim \kappa_{\mathrm{BO}}^2$ of Sect. 6.1. The theorem does not require weak coupling. Because the theorem allows λ_{ep} of order one, the theory of the electron–phonon coupling in metals and superconductors is built on the single diagram Eq. (6.19) and its phonon counterpart (Sect. 6.4). The theorem fails where its counting fails. Small Fermi surfaces, as in a doped semiconductor, bring E_{F} down toward the phonon energy. In nearly forward scattering the two shells are parallel, and the counting breaks down for $v_{\mathrm{F}}Q \lesssim \omega_{\boldsymbol{Q}\nu}$. The counting fails for the same reason in flat bands, where the Fermi velocity is small, and at strong coupling once the lattice distorts around each electron.

An insulator has no Fermi surface, and the role of E_{F} passes to the gap. Every virtual excitation of the filled bands costs at least E_{g}, and the electrons of the filled bands follow the nuclei. Their response to a phonon is the static screening of the vertex (Sect. 6.2) together with the Allen–Heine term, both computed with the nuclei at rest. Vertex corrections through interband excitations are part of this static response. The theorem for insulators states that the electron–phonon vertex may be computed statically, by DFPT with frozen nuclei. Since the screening is even in the frequency (Sect. 6.2), the corrections to the static vertex are of relative order $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2$, 0.003 for the optical phonon of silicon. The theorem says nothing about a carrier moving within its own band near an edge. There the electron has kinetic energies comparable to ω_{ph}, the ratio $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ is not small, and the only expansion parameter is the coupling itself. For the Fröhlich polaron we use the same counting, with the coupling as the only parameter, and estimate the crossed diagram to be smaller than the Fan–Migdal diagram by a relative order $\sim \alpha_{\mathrm{F}}$, a few percent for GaAs with $\alpha_{\mathrm{F}} = 0.072$ (Worked example 6.2). In strongly polar crystals with α_{F} of order one the carrier becomes a polaron that no second-order self-energy describes.

The gap at 10 mK is the measured low-temperature gap, which already contains the full zero-point renormalization, so a calculation that starts from it must not add a computed renormalization again. The two vertices g of the Fan–Migdal diagram, joined the other way into an electron loop, shift and broaden the phonons, as Sect. 6.4 shows.

6.4 Phonon self-energy and the Eliashberg function

A longitudinal phonon of 35 meV in aluminum lasts a few tens of periods before it disappears into an electron–hole pair at the Fermi surface. The zone-center optical phonon of silicon, at 64.1 meV, cannot decay that way at low temperature: the cheapest pair costs the 1.17 eV gap, and the phonon lives until anharmonicity breaks it into two phonons, after about 57 periods (Sect. 5.6). The vertex g of Sect. 6.2 acts in both crystals; the difference between the two crystals is whether real pairs exist at the phonon's energy. Electron–hole pairs created virtually also shift the phonon frequency, and in a metal they leave a kink in the dispersion wherever the Fermi surface can just span the phonon wavevector. For dark-matter detection, the superconducting sensors of Chap. 7 owe their gap to a phonon-mediated attraction, whose strength is a single number λ_{ep} built from the electron–phonon widths of the phonons. Every phonon energy used in a scattering rate is a frequency from density-functional perturbation theory, and adding a self-energy correction to it counts the static response twice unless the static part of the correction is removed (Eq. (6.31)).

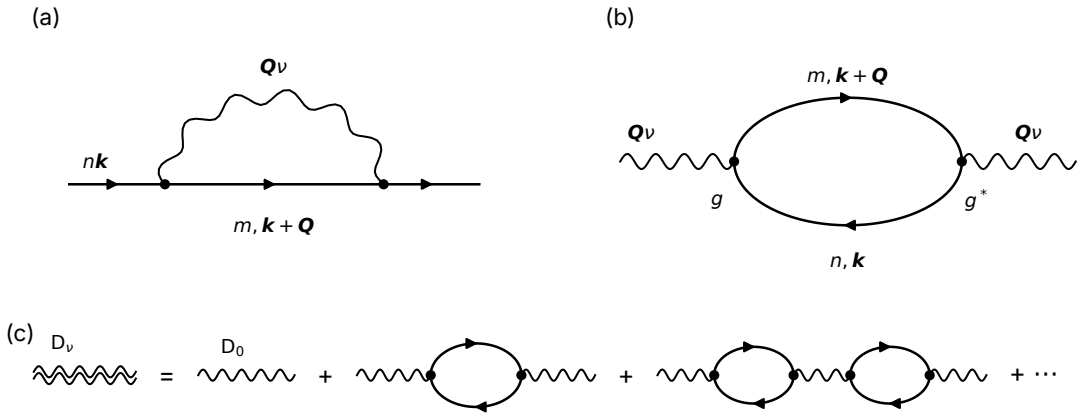


Figure 6.6: The two second-order diagrams of the electron–phonon interaction. Solid lines with arrows are electrons, wavy lines phonons, dots the vertex g. (a) The electron self-energy: an electron in $(n,\boldsymbol{k})$ exchanges a phonon $(\boldsymbol{Q},\nu)$ with the intermediate state $(m,\boldsymbol{k}+\boldsymbol{Q})$. (b) The phonon self-energy: a phonon creates an electron in $(m,\boldsymbol{k}+\boldsymbol{Q})$ and a hole in $(n,\boldsymbol{k})$ through g, and the pair recombines through g^*. (c) The Dyson series for the dressed phonon propagator D_ν (double wavy line): the free propagator D_0 with any number of bubbles inserted.

6.4.1 The electron bubble

A phonon $(\boldsymbol{Q},\nu)$ can be absorbed by an electron in $(n,\boldsymbol{k})$, which jumps to $(m,\boldsymbol{k}+\boldsymbol{Q})$ and leaves a hole behind; the pair can later recombine and emit the phonon again. At second order in $H_{\rm ep}$ of Eq. (A.24), this absorption and re-emission is the only correction to the propagation of a phonon; it is the bubble of Fig. 6.6(b), with the vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ where the pair is created and the complex conjugate of the vertex where the pair is destroyed. The electron self-energy of Sect. 6.3, Fig. 6.6(a), is the same pair of vertices joined the other way: there the electron line runs through and the phonon line closes on itself. The bubble is the density bubble of Eq. (4.93) with the density vertex replaced by g, and for this reason a phonon is screened by the same transitions that screen a Coulomb potential.

The phonon Green's function in imaginary time is $\mathsf{D}_\nu(\boldsymbol{Q},\tau) = -\langle \mathrm{T}_\tau\, B_{\boldsymbol{Q}\nu}(\tau)\, B_{-\boldsymbol{Q}\nu}(0)\rangle$, with the free form $\mathsf{D}_0(\boldsymbol{Q}\nu,\mathrm{i}\omega_j^{\rm B}) = 2\omega_{\boldsymbol{Q}\nu}/[(\mathrm{i}\omega_j^{\rm B})^2-\omega_{\boldsymbol{Q}\nu}^2]$ of Eq. (D.37). Its expansion in $H_{\rm ep}$ is Eq. (D.32) with $B_{\boldsymbol{Q}\nu}$ and $B_{-\boldsymbol{Q}\nu}$ as the external operators. The first nonzero correction is of second order. The expansion supplies $(-1)^2/2!$. The external $B_{\boldsymbol{Q}\nu}(\tau)$ is contracted with the boson of one of the two factors $H_{\rm ep}$, and $B_{-\boldsymbol{Q}\nu}(0)$ with the other; the two choices are equal after relabeling and cancel the $1/2!$. Each boson contraction is $-\mathsf{D}_0$. The vertex at τ_1 must carry $B_{-\boldsymbol{Q}\nu}$, and the one at τ_2 carries $B_{\boldsymbol{Q}\nu}$, so the vertices are $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^*$ and $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ by Eq. (A.26), with the factor $1/N$ from the two $1/\sqrt{N}$. The two electron bilinears close into one loop, which gives -1 (rule 3 of Appendix D.9) and a sum over two spin orientations. With the -1 of the definition, the total sign is $(-1)(-1)^2(-1) = +1$. In frequency space the correction is $\mathsf{D}_0\,\Pi^{\rm b}\,\mathsf{D}_0$ with

$$\Pi^{\rm b}_\nu(\boldsymbol{Q},\mathrm{i}\omega_j^{\rm B}) = \frac{2}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, T\sum_{j'} \mathsf{G}_0(n,\boldsymbol{k},\mathrm{i}\omega_{j'}^{\rm F})\,\mathsf{G}_0(m,\boldsymbol{k}+\boldsymbol{Q},\mathrm{i}\omega_{j'}^{\rm F}+\mathrm{i}\omega_j^{\rm B}).$$

The frequency sum is Eq. (D.31) with band labels, and

$$\Pi^{\mathrm{b}}_{\nu}(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}_{j}) = \frac{2}{N} \sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})|^2 \, \frac{f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\mathrm{i}\omega^{\mathrm{B}}_{j} + \varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}.$$

Repeating the bubble along the phonon line gives the geometric series of Fig. 6.6(c), $\mathsf{D} = \mathsf{D}_0 + \mathsf{D}_0\Pi^{\mathrm{b}}\mathsf{D}_0 + \cdots = \mathsf{D}_0 + \mathsf{D}_0\Pi^{\mathrm{b}}\mathsf{D}$, so that $\mathsf{D}^{-1} = \mathsf{D}_0^{-1} - \Pi^{\mathrm{b}}$, a Dyson equation like Eq. (4.81) for electrons. With $\mathsf{D}_0^{-1} = [(\mathrm{i}\omega^{\mathrm{B}}_j)^2 - \omega^2_{\boldsymbol{Q}\nu}]/2\omega_{\boldsymbol{Q}\nu}$ it is natural to define the self-energy in units of a squared frequency, $\Pi_\nu = 2\omega_{\boldsymbol{Q}\nu}\Pi^{\mathrm{b}}_\nu$. The continuation to $\omega + \mathrm{i}0^+$ gives

Phonon self-energy and Dyson equation

$$\mathsf{D}_{\nu}(\boldsymbol{Q}, \omega) = \frac{2\omega_{\boldsymbol{Q}\nu}}{\omega^2 - \omega^2_{\boldsymbol{Q}\nu} - \Pi_{\nu}(\boldsymbol{Q}, \omega)},$$
$$\Pi_{\nu}(\boldsymbol{Q}, \omega) = \frac{4\omega_{\boldsymbol{Q}\nu}}{N} \sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})|^2 \, \frac{f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega + \varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} + \mathrm{i}0^+}, \tag{6.29}$$

with $f_{n\boldsymbol{k}} = n_{\mathrm{F}}(\xi_{n\boldsymbol{k}})$ the occupations and Π_ν in eV^2. The factor 4 is the 2 of D_0^{-1} times the two spin orientations of the loop. Note that we have put the screened vertex g at both ends, with noninteracting electrons between them. Strictly, one end should carry the bare vertex [2, Eqs. (145) and (146)]; the choice of vertex at that end matters for the static part only. We have dropped the bubbles that connect two branches $\nu \neq \nu'$ at the same $\boldsymbol{Q}$. They mix the modes, and they matter where two branches are nearly degenerate.

The poles of D_ν are the dressed phonons, $\omega^2 = \omega^2_{\boldsymbol{Q}\nu} + \Pi_\nu(\boldsymbol{Q}, \omega)$. When $|\Pi_\nu| \ll \omega^2_{\boldsymbol{Q}\nu}$ the root lies close to $\omega_{\boldsymbol{Q}\nu}$. Expanding the denominator there, $\omega^2 - \omega^2_{\boldsymbol{Q}\nu} \simeq 2\omega_{\boldsymbol{Q}\nu}(\omega - \omega_{\boldsymbol{Q}\nu})$, turns D_ν into a Lorentzian, $1/(\omega - \tilde{\omega}_{\boldsymbol{Q}\nu} + \mathrm{i}\gamma_{\boldsymbol{Q}\nu}/2)$, with

$$\tilde{\omega}_{\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu} + \frac{\mathrm{Re}\,\Pi_\nu(\boldsymbol{Q}, \omega_{\boldsymbol{Q}\nu})}{2\omega_{\boldsymbol{Q}\nu}}, \qquad \gamma_{\boldsymbol{Q}\nu} = -\frac{\mathrm{Im}\,\Pi_\nu(\boldsymbol{Q}, \omega_{\boldsymbol{Q}\nu})}{\omega_{\boldsymbol{Q}\nu}}. \tag{6.30}$$

The real part shifts the phonon, and the imaginary part gives it a full width at half maximum $\gamma_{\boldsymbol{Q}\nu}$, the quantity of Sect. 5.6. The electron–phonon part of this width is $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$, and the anharmonic bubble of Sect. 5.6 adds $\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}$ to it.

The sign of the static part $\Pi_\nu(\boldsymbol{Q}, 0)$ is fixed. In an insulator at $T = 0$, $f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}} = +1$ for n occupied and m empty, and then $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} < 0$; the term with the roles exchanged has both signs reversed. Every term of $\Pi_\nu(\boldsymbol{Q}, 0)$ is therefore negative. The virtual pairs soften the lattice, as second-order perturbation theory always lowers a ground-state energy. In a metal the static bubble inherits the structure of the Lindhard function (Sect. 4.5). For a spherical Fermi surface the density of pairs with $\varepsilon_{n\boldsymbol{k}} = \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} = E_{\mathrm{F}}$ changes abruptly when Q passes $2k_{\mathrm{F}}$, the largest wavevector that joins two points of the surface, and the derivative of the Lindhard function diverges there logarithmically (Appendix C.8). The phonon frequencies carry the singularity as a kink at $Q = 2k_{\mathrm{F}}$, or at whatever wavevectors span flat, parallel pieces of a real Fermi surface. This kink is the Kohn anomaly [29], and its position in a measured dispersion is an image of the Fermi surface.

6.4.2 Adiabatic and nonadiabatic phonons

The static part of Π_ν is already in the phonons we start from. Density-functional perturbation theory computes second derivatives of the Born–Oppenheimer energy with the electrons relaxed to their ground state at every nuclear configuration. The relaxation is the electronic

response at zero frequency, which at this order is $\Pi_\nu(\boldsymbol{Q},0)$. The first-order density of Eq. (25.8) is precisely that response, and the Hartree and exchange–correlation potential it generates turns the bare vertex at one end of the bubble into the screened one. Starting from the frequencies of such a calculation and adding the full $\Pi_\nu(\boldsymbol{Q},\omega)$ would count the static response twice. Only the difference between the dynamic and the static response may be added [30, Eqs. (22) and (23)],

$$\left(\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}\right)^2 = \omega^2_{\boldsymbol{Q}\nu} + \mathrm{Re}\left[\Pi_\nu(\boldsymbol{Q},\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}) - \Pi_\nu(\boldsymbol{Q},0)\right], \tag{6.31}$$

where $\omega_{\boldsymbol{Q}\nu}$ is now the adiabatic frequency of density-functional perturbation theory and $\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}$ the nonadiabatic one; Eq. (6.31) is the subtraction $W_e(\omega) - W_e(0)$ of Ref. [2, Eq. (140)]. The imaginary part needs no subtraction, since $\Pi_\nu(\boldsymbol{Q},0)$ is real.

How large is the difference between the dynamic and the static response? In an insulator every denominator of Eq. (6.29) is at least E_g in magnitude, and the pair of terms with roles exchanged makes Π_ν even in ω. The difference $\Pi_\nu(\omega) - \Pi_\nu(0)$ is then of relative order $(\omega_{\boldsymbol{Q}\nu}/E_\mathrm{g})^2$, which is 0.003 for the optical phonon of silicon. The accuracy of the adiabatic phonons of an insulator is the phonon-side statement of the counting of Sect. 6.1, where $\omega_\mathrm{ph}/E_\mathrm{g}$ was the small parameter. A metal has no gap. Its intraband pairs with $|\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{k}}| \lesssim \omega_{\boldsymbol{Q}\nu}$ respond differently at $\omega = 0$ and at $\omega = \omega_{\boldsymbol{Q}\nu}$, and for them the difference is of order one. Such pairs dominate the bubble at small wavevector. For a band with Fermi velocity v_F, the energy difference of a pair is $v_\mathrm{F} Q\cos\theta$, and the adiabatic and nonadiabatic frequencies differ appreciably only for $Q \lesssim \omega_{\boldsymbol{Q}\nu}/v_\mathrm{F}$ [30]. In aluminum, with $v_\mathrm{F} = 2.0\times10^6\,\mathrm{m\,s^{-1}}$, the condition is $Q \lesssim 2\times10^{-3}\,\text{Å}^{-1}$ for a 30 meV phonon, a thousandth of the zone. Near the zone center of metals with few carriers and stiff optical modes the nonadiabatic correction can, however, be large. Calandra, Profeta and Mauri note corrections of up to 30 % of the phonon frequency there [30]; graphite intercalation compounds and doped graphene are the standard examples. Everywhere else the adiabatic frequencies are the right starting point.

The bubble with undressed vertices is itself an approximation. The counting of Migdal's theorem (Sect. 6.3) makes the vertex corrections to it of relative order $\lambda_\mathrm{ep}\omega_\mathrm{D}/E_\mathrm{F}$, with the coupling constant λ_ep defined below; for aluminum, with phonon energies up to $\omega_\mathrm{max} = 40\,\mathrm{meV}$ and $E_\mathrm{F} = 11.7\,\mathrm{eV}$, this relative order is at most 0.0015.

6.4.3 The electron–phonon linewidth

The width follows from the imaginary part of Eq. (6.29). With $1/(x+\mathrm{i}0^+) = \mathrm{PV}(1/x) - \mathrm{i}\pi\delta(x)$ from Eq. (2.58) in Eq. (6.30),

Electron–phonon linewidth

$$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu} = -\frac{\mathrm{Im}\,\Pi_\nu(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu})}{\omega_{\boldsymbol{Q}\nu}} = \frac{4\pi}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \left(f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}\right)\delta\left(\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{k}} - \omega_{\boldsymbol{Q}\nu}\right), \tag{6.32}$$

a full width, with the spin sum in the factor $4\pi = 2\times2\pi$. Ref. [2, Eq. (146)] writes the same rate as an inverse lifetime, with the spin factor 2 written explicitly, and its $1/\tau$ equals $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$. The cutting rule of Sect. 4.8 tells us how to read Eq. (6.32). The delta function puts the pair on shell with the energy of the phonon, and $2\pi|g|^2/N$ times the delta function is the golden-rule rate Eq. (2.67) for absorbing the phonon into the pair $(n\boldsymbol{k} \to m, \boldsymbol{k}+\boldsymbol{Q})$. The occupation factor is the rate of that process, $f_n(1-f_m)$, minus the rate of the inverse one, $f_m(1-f_n)$, in which a pair recombines and emits the phonon; the difference is $f_n - f_m$.

In an undoped insulator at $T = 0$ the width vanishes below the gap. The delta function needs an occupied state n and an empty state m with $\varepsilon_m - \varepsilon_n = \omega_{\boldsymbol{Q}\nu}$, and every such energy difference is at least E_g, while the largest phonon energy of silicon is 64 meV. No term survives. The phonons of an insulator are, however, still broadened by anharmonicity (Sect. 5.6) and by isotopes and defects (Sect. 11.6). Since the electron–phonon width vanishes, the measured 0.18 meV width of the silicon Raman line could be compared with the three-phonon formula alone. At $T > 0$ thermally excited electrons and holes absorb phonons by intraband transitions, with a width proportional to their density, which in an intrinsic crystal is suppressed by $\mathrm{e}^{-E_\mathrm{g}/2T}$: 4×10^{-10} in silicon at room temperature. Doping supplies free carriers without the exponential suppression, and a doped semiconductor behaves for this purpose like a dilute metal.

In a metal the electronic states near the Fermi level change on scales of eV, far larger than $\omega_{\boldsymbol{Q}\nu}$. For $T \ll E_\mathrm{F}$ the occupation difference at $\varepsilon_m = \varepsilon_n + \omega_{\boldsymbol{Q}\nu}$ is $f(\varepsilon_n) - f(\varepsilon_n + \omega_{\boldsymbol{Q}\nu}) \simeq -\omega_{\boldsymbol{Q}\nu}\, \partial f/\partial\varepsilon$, which tends to $\omega_{\boldsymbol{Q}\nu}\delta(\xi_{n\boldsymbol{k}})$ for $T \to 0$. Dropping $\omega_{\boldsymbol{Q}\nu}$ also in the remaining delta function, we arrive at Allen's formula [31],

$$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu} \simeq \frac{4\pi\omega_{\boldsymbol{Q}\nu}}{N} \sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \delta(\xi_{n\boldsymbol{k}})\, \delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}), \tag{6.33}$$

which is positive term by term. Both electrons now lie on the Fermi surface, and the width is proportional to the phonon frequency. It no longer depends on temperature: thermal smearing removes states from below the Fermi level and adds them above, and the net number of pairs that can absorb $\omega_{\boldsymbol{Q}\nu}$ stays the same. The approximation requires bands that are smooth on the scale of $\omega_{\boldsymbol{Q}\nu}$ and a temperature small on the electronic scale. It fails for a band edge within $\omega_{\boldsymbol{Q}\nu}$ of the Fermi level, and for $Q \lesssim \omega_{\boldsymbol{Q}\nu}/v_\mathrm{F}$, the same small-$Q$ region where the nonadiabatic correction is appreciable.

The electron gas makes the passage from Eq. (6.32) to Eq. (6.33) explicit. For one parabolic band and a constant vertex, the angular integral in $(1/\mathcal{V})\sum_{\boldsymbol{k}}(f_{\boldsymbol{k}} - f_{\boldsymbol{k}+\boldsymbol{Q}})\,\delta(\varepsilon_{\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{\boldsymbol{k}} - \omega)$ gives $m_e/(kQ)$ wherever the delta function can be satisfied, and with $k\,\mathrm{d}k = m_e\,\mathrm{d}\varepsilon$ the radial integral becomes $\int \mathrm{d}\varepsilon\,[f(\varepsilon) - f(\varepsilon+\omega)] = \omega$. The result, $m_e^2\omega/(4\pi^2 Q)$ per spin and volume, holds at any $T \ll E_\mathrm{F}$ for $2k_\mathrm{F} > Q \gg \omega/v_\mathrm{F}$, and it is exactly what the double delta function of Eq. (6.33) gives, $\omega \int \mathrm{d}^3k\,(2\pi)^{-3}\delta(\xi_{\boldsymbol{k}})\,\delta(\xi_{\boldsymbol{k}+\boldsymbol{Q}}) = m_e^2\omega/(4\pi^2 Q)$.

6.4.4 The Eliashberg function

The double sum in Eq. (6.33) is a Fermi-surface average of $|g|^2$ for one phonon. Superconductivity, the electron mass enhancement and the resistivity of a metal all involve the same average, weighted over all phonons at a given energy. This weighted average is the Eliashberg function [32],

$$\alpha^2F(\omega) = \frac{1}{\mathcal{N}_\mathrm{F}\Omega\, N^2} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \delta(\xi_{n\boldsymbol{k}})\, \delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})\, \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \tag{6.34}$$

with $\mathcal{N}_\mathrm{F}$ the density of states per spin and volume at the Fermi level (Sect. 2.5), so that $\mathcal{N}_\mathrm{F}\Omega$, per spin and per cell, satisfies $N\mathcal{N}_\mathrm{F}\Omega = \sum_{n\boldsymbol{k}} \delta(\xi_{n\boldsymbol{k}})$. The prefactor makes α^2F an average over initial states on the Fermi surface and a sum over final states, and α^2F is dimensionless [2, Eq. (205)]. The symbol α^2F is read as one. If $|g|^2$ were the same for every pair on the Fermi surface, α^2F would be that constant times $\mathcal{N}_\mathrm{F}\Omega$ times the phonon density of states $g_\mathrm{ph}(\omega)$ of Eq. (5.21). The historical notation splits α^2F in this way into a coupling $\alpha^2(\omega)$ and a density $F(\omega)$. Comparing Eqs. (6.33) and (6.34) term by term turns the Eliashberg function into a sum of linewidths. The coupling constant is defined as the inverse moment of the Eliashberg function [2, Eq. (206)]:

Eliashberg function, coupling constant and linewidths

$$\alpha^2F(\omega) = \frac{1}{4\pi\mathcal{N}_\mathrm{F}\Omega\,N}\sum_{\boldsymbol{Q}\nu}\frac{\gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}}{\omega_{\boldsymbol{Q}\nu}}\,\delta(\omega-\omega_{\boldsymbol{Q}\nu}),$$
$$\lambda_\mathrm{ep} = 2\int_0^\infty \mathrm{d}\omega\,\frac{\alpha^2F(\omega)}{\omega} = \frac{1}{2\pi\mathcal{N}_\mathrm{F}\Omega\,N}\sum_{\boldsymbol{Q}\nu}\frac{\gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}}{\omega^2_{\boldsymbol{Q}\nu}}. \tag{6.35}$$

With a full width and a density of states per spin the factors are as written; a half width [2, Eq. (185)] or a two-spin $\mathcal{N}_\mathrm{F}$ changes them by factors of two. The coupling constant is a zone average of mode contributions $\lambda_{\boldsymbol{Q}\nu} = \gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}/(2\pi\mathcal{N}_\mathrm{F}\Omega\,\omega^2_{\boldsymbol{Q}\nu})$. The coupling of a mode can therefore be read off its measured linewidth, which was the purpose of Allen's relation [31], and the factor $1/\omega^2$ makes soft modes with large widths dominate the coupling constant.

The name "coupling constant" comes from the electrons. Average the Fan–Migdal self-energy Eq. (6.19) of a state $(n,\boldsymbol{k})$ over the Fermi surface, and insert $1 = \int \mathrm{d}\omega'\,\delta(\omega' - \omega_{\boldsymbol{Q}\nu})\int \mathrm{d}\varepsilon\,\delta(\varepsilon-\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})$. Suppose that the density of final states and the vertices change little over the energy range ω_D of the phonons. Then $\delta(\varepsilon-\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})$ may be replaced by $\delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})$ under the average, and the weight of each (ω',ε) is $\alpha^2F(\omega')$ by Eq. (6.34):

$$\langle\Sigma^\mathrm{FM}\rangle_\mathrm{FS}(\omega) = \int_0^\infty \mathrm{d}\omega'\,\alpha^2F(\omega')\int_{-E_\mathrm{band}}^{E_\mathrm{band}}\mathrm{d}\varepsilon\left[\frac{n_\mathrm{B}(\omega')+1-n_\mathrm{F}(\varepsilon)}{\omega-\varepsilon-\omega'+\mathrm{i}0^+}+\frac{n_\mathrm{B}(\omega')+n_\mathrm{F}(\varepsilon)}{\omega-\varepsilon+\omega'+\mathrm{i}0^+}\right],$$

where the cutoff $E_\mathrm{band} \gg \omega_\mathrm{D}$ is of the order of the band width. At $T=0$ only $1-\Theta(-\varepsilon) = \Theta(\varepsilon)$ survives in the first numerator and $\Theta(-\varepsilon)$ in the second. The ε integrals are logarithms, $-\ln[(E_\mathrm{band}+\omega'-\omega)/(\omega'-\omega)]$ and $\ln[(E_\mathrm{band}+\omega'+\omega)/(\omega'+\omega)]$, whose derivatives at $\omega=0$ are $-1/\omega'+1/(E_\mathrm{band}+\omega')$ each and tend to $-1/\omega'$ for $E_\mathrm{band}\to\infty$; a finite band reduces each by the fraction $\omega'/(E_\mathrm{band}+\omega')$, 1 % for $E_\mathrm{band} = 100\,\omega'$. The Allen–Heine term of Eq. (6.20) does not depend on ω and does not contribute. The slope of the averaged self-energy at the Fermi level is therefore $-2\int_0^\infty \mathrm{d}\omega'\,\alpha^2F(\omega')/\omega'$, and by Eq. (6.35)

$$-\frac{\partial\,\mathrm{Re}\langle\Sigma^\mathrm{FM}\rangle_\mathrm{FS}}{\partial\omega}\bigg|_{\omega=0} = \lambda_\mathrm{ep}. \tag{6.36}$$

By Eq. (4.83) the quasiparticles within about ω_D of the Fermi level have the residue $z_\mathrm{qp} = 1/(1+\lambda_\mathrm{ep})$, their velocity is reduced by the same factor, and their density of states, and with it the electronic specific heat, is enhanced by $1+\lambda_\mathrm{ep}$. The free electron gas has the specific heat $(\pi^2/3)\,2\mathcal{N}_\mathrm{F}\,T$ per volume [33, Chap. 2]. For aluminum, for example, first-principles calculations give the density of states $2\mathcal{N}_\mathrm{F}\Omega = 5.49$ states per Ry and cell for both spins [34, Table II], which corresponds to $0.951\,\mathrm{mJ\,K^{-2}\,mol^{-1}}$, while the measured coefficient is $1.36\,\mathrm{mJ\,K^{-2}\,mol^{-1}}$ [34, Table II]. The ratio of the measured to the free-electron coefficient is $1+\lambda_\mathrm{ep} = 1.43$. The specific heat alone measures $\lambda_\mathrm{ep} = 0.43$, without any phonon calculation.

The Eliashberg function has a different shape from the phonon density of states, and Fig. 6.7 shows by how much for aluminum. The model phonons come from force constants between first and second neighbors of the fcc lattice, with $a = 4.05\,\text{Å}$ and $M = 26.98\,\mathrm{u}$. A neighbor at the lattice vector $\boldsymbol{R}$ contributes its spring tensor times $1-\cos(\boldsymbol{Q}\cdot\boldsymbol{R})$ to the dynamical matrix, divided by M. The tensor of the first neighbor at $(a/2)(1,1,0)$ has the elements 0.648 ($xx=yy$), −0.187 (zz) and 0.574 (xy), and that of the second neighbor at $a(1,0,0)$ has 0.038 (xx) and −0.052 ($yy=zz$), all in $\mathrm{eV\,Å^{-2}}$. These five numbers are a least-squares fit to the measured zone-boundary frequencies, 9.69 and 5.78 THz (longitudinal and transverse) at X and 9.69 and 4.19 THz at L [34, Table I], and to the elastic constants

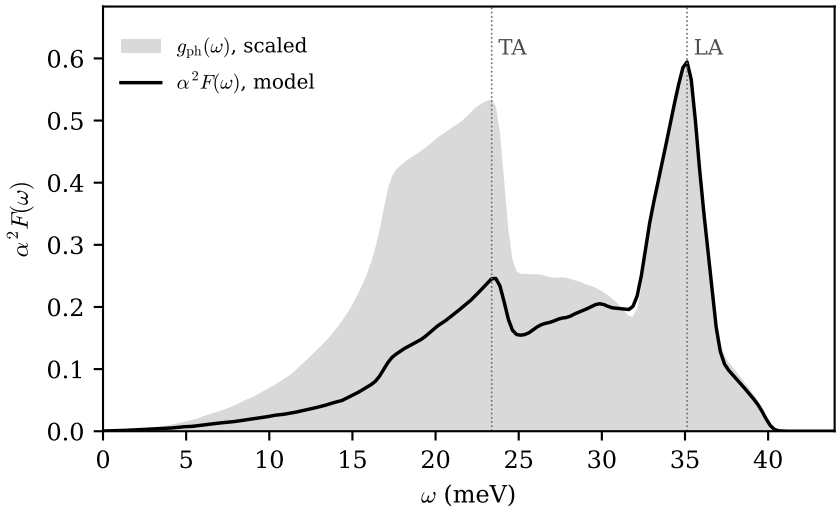


Figure 6.7: Eliashberg function of aluminum (black line) in a model of free electrons scattered by screened empty-core ions, Eq. (6.37). The core radius, 0.56 Å, is taken where $\lambda_{\rm ep}$ is stationary with respect to it, and the phonons come from force constants between first and second neighbors fitted to measured zone-boundary frequencies and elastic constants. The gray area is the phonon density of states $g_{\rm ph}(\omega)$ of the same force constants, scaled to the height of the α^2F peak. Dotted lines mark the transverse (TA) and longitudinal (LA) maxima of $g_{\rm ph}$. In α^2F the transverse peak is suppressed, although three quarters of the modes lie below 32 meV, because a transverse displacement couples to an electron only when the momentum transfer leaves the first zone. Both curves are histograms in bins of 0.25 meV smoothed by a Gaussian of 0.3 meV.

$C_{11} = 106.8$, $C_{12} = 60.7$ and $C_{44} = 28.2$ GPa at a density of $2.70\,\mathrm{g\,cm^{-3}}$ [35], the room-temperature values of that source. They reproduce the transverse frequencies at X and L to 2 % and the longitudinal one at L to 8 %, and the largest phonon energy is 40 meV.

The electrons are a free gas of three electrons per atom, with $k_{\rm F} = 1.75\,\text{Å}^{-1}$ and $E_{\rm F} = 11.7$ eV. The potential of each ion is a Coulomb potential $-3e^2/(4\pi r)$ outside a core radius $r_{\rm c}$ and zero inside, screened by the Lindhard function of Sect. 4.5. Its Fourier transform per cell is $w(q) = -3e^2\cos(qr_{\rm c})/[\Omega q^2\varepsilon_{\rm Lin}(q,0)]$. For plane waves and one atom per cell, Eq. (A.25) becomes $g_\nu = (2M\omega_{\boldsymbol{Q}\nu})^{-1/2}(\boldsymbol{q}\cdot\boldsymbol{e}_\nu)\,w(q)$, where $\boldsymbol{q} = \boldsymbol{Q} + \boldsymbol{G}$ is the full momentum transfer, umklapp included. The sums over $\boldsymbol{k}$ and over the reciprocal-lattice vectors become integrals, $N^{-1}\sum_{\boldsymbol{k}} \to \Omega\int \mathrm{d}^3k/(2\pi)^3$ and $N^{-1}\sum_{\boldsymbol{Q}}\sum_{\boldsymbol{G}} \to \Omega\int \mathrm{d}^3q/(2\pi)^3$. The double delta function integrates to $\int \mathrm{d}^3k\,(2\pi)^{-3}\delta(\xi_{\boldsymbol{k}})\delta(\xi_{\boldsymbol{k}+\boldsymbol{q}}) = m_e^2/(4\pi^2 q)$ for $q < 2k_{\rm F}$, as for the linewidth of the electron gas, and with $\mathcal{N}_{\rm F} = m_e k_{\rm F}/2\pi^2$, Eq. (6.34) becomes

$$\alpha^2F(\omega) = \frac{m_e\Omega}{2k_{\rm F}}\int_{q<2k_{\rm F}}\frac{\mathrm{d}^3q}{(2\pi)^3}\sum_\nu \frac{(\hat{\boldsymbol{q}}\cdot\boldsymbol{e}_\nu)^2\,q\,|w(q)|^2}{2M\omega_{\boldsymbol{q}\nu}}\,\delta(\omega-\omega_{\boldsymbol{q}\nu}). \tag{6.37}$$

The core radius is the only free parameter, a choice of the model. As a function of $r_{\rm c}$, $\lambda_{\rm ep}$ has a flat minimum at $r_{\rm c} = 0.56$ Å, where the zero of $\cos(qr_{\rm c})$ lies among the umklapp momentum transfers between $|\boldsymbol{G}_{111}| = 2.69\,\text{Å}^{-1}$ and $2k_{\rm F} = 3.50\,\text{Å}^{-1}$. We take $r_{\rm c}$ at this minimum, where $\lambda_{\rm ep}$ depends least on the core radius; between 0.55 and 0.59 Å the coupling stays between 0.44 and 0.46. The model uses no measured coupling.

The model gives $\lambda_{\rm ep} = 0.436$, against 0.44 from a linear-response calculation and 0.42 from tunneling spectra [34, Table II], and 0.43 from the specific heat above. Savrasov and Savrasov

found their first-principles α^2F practically identical to an empirical rigid-ion pseudopotential result of this kind [34], and aluminum is the metal where the free-electron picture is expected to work. Compared with the density of states, α^2F has a strongly suppressed transverse peak near 23 meV. The reason is the factor $(\hat{\boldsymbol{q}}\cdot\boldsymbol{e}_\nu)^2$: within the first zone a transverse wave moves the ions perpendicular to $\boldsymbol{q}$, and an electron feels it only through an umklapp process, where $\boldsymbol{q}=\boldsymbol{Q}+\boldsymbol{G}$ has a component along the displacement. The model gives the logarithmic average frequency $\omega_{\log}=\exp[(2/\lambda_{\mathrm{ep}})\int \mathrm{d}\omega\,\alpha^2F(\omega)\ln\omega/\omega]$ as 250 K, against 270 K from the linear-response calculation [34, Table III]; $\omega_{\log}$ sets the prefactor of the critical temperature. A constant α^2 with the same total weight would give a coupling constant of 0.50 and a logarithmic average frequency of 228 K. Summed over the zone with Eq. (6.35), the model widths average $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}/\omega_{\boldsymbol{Q}\nu}=4.4\times10^{-3}$ over all modes, or 0.12 meV: a phonon in aluminum decays into electron–hole pairs after about 36 periods, much as the optical phonon of silicon decays anharmonically.

Worked example 6.4: Coupling constant of aluminum from two Einstein peaks

Replace the model α^2F of Fig. 6.7 by two delta functions at the maxima of the phonon density of states, $\omega_{\mathrm{TA}}=23.4$ meV and $\omega_{\mathrm{LA}}=35.1$ meV, each carrying the weight of α^2F on its side of the minimum of g_{ph} at 31.9 meV. The weights are $I_{\mathrm{TA}}=\int_0^{31.9}\mathrm{d}\omega\,\alpha^2F=3.13$ meV and $I_{\mathrm{LA}}=2.26$ meV. Below the minimum lie 2.24 of the three modes per atom and above it 0.76, a ratio of 2.9; the weights of α^2F stand in the ratio 1.4, so each transverse mode couples half as strongly as a longitudinal one.
With $\alpha^2F=I_{\mathrm{TA}}\delta(\omega-\omega_{\mathrm{TA}})+I_{\mathrm{LA}}\delta(\omega-\omega_{\mathrm{LA}})$, Eq. (6.35) gives

$$\lambda_{\mathrm{ep}}=2\Big(\frac{I_{\mathrm{TA}}}{\omega_{\mathrm{TA}}}+\frac{I_{\mathrm{LA}}}{\omega_{\mathrm{LA}}}\Big)=2\,(0.134+0.064)=0.396,$$

9 % below the full model's 0.436 and 6 to 8 % below the measured 0.42–0.43. The shortfall of the two-peak model is all in the transverse peak. The transverse weight, spread over the low-frequency tail of Fig. 6.7, counts for more, since $1/\omega$ is convex: the transverse part of λ_{ep} in the full model is 0.306 instead of $2\times0.134=0.268$, and the longitudinal part is 0.130 in both models. Although the longitudinal peak is the taller, the transverse branches carry 70 % of the coupling. For the two-peak model $\omega_{\log}=\exp[(2/\lambda_{\mathrm{ep}})(I_{\mathrm{TA}}\ln\omega_{\mathrm{TA}}/\omega_{\mathrm{TA}}+I_{\mathrm{LA}}\ln\omega_{\mathrm{LA}}/\omega_{\mathrm{LA}})]=26.7$ meV $=310$ K, too high for the same reason.

In practice $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$ and α^2F are computed from Eqs. (6.32) and (6.34) with the vertices of density-functional perturbation theory, Wannier-interpolated to fine grids of $\boldsymbol{k}$ and $\boldsymbol{Q}$ as described in Sect. 6.2. The two delta functions that pin both electrons to the Fermi surface make these the slowest-converging integrals of electron–phonon theory. Sect. 28.1 takes the resulting α^2F into the Eliashberg equations for the gap of aluminum; the gap comes out at the measured 0.18 meV once the Coulomb repulsion between paired electrons is fixed by the measured critical temperature.

The phonon-mediated attraction between the paired electrons is the exchange of the phonons whose widths we have just summed. Sect. 7.2 reduces it to the model interaction of Bardeen, Cooper and Schrieffer, with λ_{ep} as its strength and ω_{D} as its range in energy, and finds the gap that sets the threshold of a superconducting detector.

6.5 Phonon-assisted optical absorption

Silicon solar cells absorb the whole visible spectrum, from about 1.6 to 3.3 eV, and silicon photodiodes respond down to its gap of 1.17 eV. Yet the smallest vertical gap of silicon, E_{dir}, is 3.4 eV (Sect. 3.3). A photon carries almost no crystal momentum, and on its own it can lift an electron only vertically, from $(v, \boldsymbol{k})$ to $(c, \boldsymbol{k})$. Every photon that silicon absorbs between 1.17 and 3.4 eV is therefore absorbed together with a phonon, which carries the crystal momentum $\boldsymbol{k}_{\mathrm{c}}$ from the valence maximum at Γ to a conduction valley. The phonon can be created or destroyed, and it takes a few tens of meV from the absorbed energy or adds as much. Phonon-assisted absorption is the simplest composite process: two vertices, the photon's and the electron–phonon vertex g of Sect. 6.2, joined through virtual intermediate states with the energy denominators of Sect. 2.4. An absorbed dark-matter boson has even less momentum for its energy than a photon, and between the indirect and the direct gap it is absorbed only with a phonon (Sect. 13.8). A scattering dark-matter particle just above its threshold mass delivers its energy at momentum transfers too small to reach $\boldsymbol{k}_{\mathrm{c}}$, and a phonon again supplies the crystal momentum that the momentum transfer lacks (Sect. 13.1).

6.5.1 The optical vertex

A uniform optical field $\boldsymbol{\mathcal{E}}(t) = -\mathcal{E}\,\hat{\boldsymbol{e}}_\gamma \cos\omega t$, with unit polarization vector $\hat{\boldsymbol{e}}_\gamma$, is described by the vector potential $\boldsymbol{A}(t) = (\mathcal{E}/\omega)\,\hat{\boldsymbol{e}}_\gamma \sin\omega t$. To first order in $\boldsymbol{A}$ the field couples to the electrons through

$$H_A = e\,\boldsymbol{A}\cdot\hat{\boldsymbol{V}}, \qquad \hat{\boldsymbol{V}} = \sum_{\boldsymbol{k}}\sum_{mn} \boldsymbol{v}_{mn}(\boldsymbol{k})\, c^\dagger_{m\boldsymbol{k}} c_{n\boldsymbol{k}}, \tag{6.38}$$

where $\hat{\boldsymbol{V}}$ is the total velocity of the electrons, written over Bloch states with the velocity matrix elements $\boldsymbol{v}_{mn}(\boldsymbol{k})$ of Eq. (3.38), and the spin is summed implicitly. For a local potential $\hat{\boldsymbol{V}} = \sum_j \hat{\boldsymbol{p}}_j/m_e$, and the sign of the charge is irrelevant for rates. The velocity operator conserves $\boldsymbol{k}$, because the velocity commutes with lattice translations. The part of H_A that oscillates as $\mathrm{e}^{-\mathrm{i}\omega t}$ absorbs the energy ω, with the amplitude $(e\mathcal{E}/2\omega)\,\hat{\boldsymbol{e}}_\gamma\cdot\hat{\boldsymbol{V}}$ up to a phase.

Eq. (6.38) is the *velocity form* of the optical vertex. Its alternative, the length form $e\boldsymbol{\mathcal{E}}\cdot\sum_j \boldsymbol{r}_j$, is awkward in a crystal, since the position has no matrix elements between extended Bloch states except through the commutator of Eq. (3.39). The velocity form needs just $\boldsymbol{v} = \mathrm{i}[H_{\mathrm{e}}, \boldsymbol{r}]$, which stays well defined when the potential is nonlocal (Eq. (3.40)). It is also the vertex that dark matter shares with light. By Eq. (3.43), the density vertex at small momentum transfer is $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) \to \boldsymbol{q}\cdot\boldsymbol{v}_{mn}/(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})$. On the energy shell the denominator is ω, and a longitudinal field coupled through the density and a transverse field coupled through the velocity carry the same matrix element, $\hat{\boldsymbol{e}}_\gamma\cdot\boldsymbol{v}_{mn}/\omega$. Replacing the photon by dark matter replaces this optical limit by the full density vertex, and nothing else in the structure changes.

At first order the golden rule Eq. (2.67) with H_A gives the absorption rate between many-body states i and f, and the absorbed power is ω times the net rate, absorption minus stimulated emission. Equating the time-averaged power per volume with $\sigma_1\mathcal{E}^2/2$ and using $\sigma_1 = \omega\varepsilon_2$ (Eq. (4.50)) gives

$$\varepsilon_2(\omega) = \frac{\pi e^2}{\omega^2 \mathcal{V}} \sum_{i,f} (p_i - p_f)\, \left|\hat{\boldsymbol{e}}_\gamma\cdot\langle f|\hat{\boldsymbol{V}}|i\rangle\right|^2 \delta(E_f - E_i - \omega). \tag{6.39}$$

For independent electrons at $T \ll E_{\mathrm{g}}$ the final states are one electron in $(c, \boldsymbol{k})$ and one hole in $(v, \boldsymbol{k})$, and Eq. (6.39) is the direct absorption of Sect. 4.4, the familiar sum over vertical

transitions,

$$\varepsilon_2(\omega) = \frac{g_{\mathrm{deg}}\,\pi e^2}{\omega^2 \mathscr{V}} \sum_{\boldsymbol{k}} \sum_{cv} \left|\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cv}(\boldsymbol{k})\right|^2 \delta(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega), \tag{6.40}$$

with $g_{\mathrm{deg}} = 2$ for the spin [36, Chap. 6]. In a frozen lattice this direct ε_2 vanishes below E_{dir}, and below E_{dir} light needs something else to supply the crystal momentum. A phonon can, and so can an impurity or a free carrier; in a pure, cold crystal only the phonon remains. The absorption coefficient, the inverse length over which the intensity falls by e, follows from ε_2 and the refractive index $n_r(\omega)$ as

$$\alpha_{\mathrm{abs}}(\omega) = \frac{\omega\,\varepsilon_2(\omega)}{n_r(\omega)}, \tag{6.41}$$

since, with $\varepsilon = (n_r + \mathrm{i}\kappa_{\mathrm{ext}})^2$ and κ_{ext} the extinction coefficient, the intensity decays as $\mathrm{e}^{-2\omega\kappa_{\mathrm{ext}}x}$ and $\varepsilon_2 = 2n_r\kappa_{\mathrm{ext}}$.

6.5.2 The phonon-assisted amplitude

Add the electron–phonon Hamiltonian H_{ep} of Eq. (A.24) and count the phonons in the states. The initial state $|i\rangle$ has filled valence bands, empty conduction bands and $n_{\boldsymbol{Q}\nu}$ phonons in each mode. The final state has an electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$, a hole in $(v, \boldsymbol{k})$, and one phonon fewer in the mode $(\boldsymbol{Q}, \nu)$ or one more in $(-\boldsymbol{Q}, \nu)$. No single vertex connects the initial and the final state. H_A conserves $\boldsymbol{k}$ and does not touch the phonons, and H_{ep} changes $\boldsymbol{k}$ but cannot create a pair across a gap of an electronvolt with tens of meV. The second-order amplitude of Eq. (2.51), linear in H_A and linear in H_{ep}, both changes $\boldsymbol{k}$ and creates the pair. Its long-time limit is the rate $2\pi|T_{fi}|^2\delta(E_f - E_i - \omega)$ of Eq. (2.52), with the two time orderings of Eq. (2.54):

$$T_{fi} = \frac{e\mathcal{E}}{2\omega} \sum_m \left[\frac{\langle f|H_{\mathrm{ep}}|m\rangle\langle m|\hat{\boldsymbol{e}}_\gamma \cdot \hat{\boldsymbol{V}}|i\rangle}{E_i + \omega - E_m} + \frac{\langle f|\hat{\boldsymbol{e}}_\gamma \cdot \hat{\boldsymbol{V}}|m\rangle\langle m|H_{\mathrm{ep}}|i\rangle}{E_i - E_m} \right].$$

The photon energy appears in the denominator of the photon-first ordering, where it has been absorbed in the intermediate state, and not in the phonon-first one.

Four kinds of intermediate state contribute; we drop the common factors $e\mathcal{E}/2\omega$, $N^{-1/2}$ from Eq. (A.24) and the phonon matrix element $(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of Eq. (A.27), with the upper sign for emission throughout.

1. The photon lifts the electron from $(v, \boldsymbol{k})$ to an empty state $(p, \boldsymbol{k})$, and the phonon scatters it to $(c, \boldsymbol{k}+\boldsymbol{Q})$. The intermediate energy exceeds $E_i + \omega$ by $\varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega$. The operator string is $c_c^\dagger c_p\, c_p^\dagger c_v$, and since p is empty it equals $c_c^\dagger c_v$. The term is $g_{cp\nu}(\boldsymbol{k}, \boldsymbol{Q})\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})/(\omega - \varepsilon_{p\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}})$.

2. The phonon scatters the electron from $(v, \boldsymbol{k})$ to an empty $(p, \boldsymbol{k}+\boldsymbol{Q})$, and the photon lifts it to $(c, \boldsymbol{k}+\boldsymbol{Q})$. The intermediate state holds the pair and the changed phonon, $E_m - E_i = \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, and the term is $\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\, g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q})/(\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu})$.

3. The photon creates the electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$ from an occupied $(p, \boldsymbol{k}+\boldsymbol{Q})$, and the phonon then moves an electron from $(v, \boldsymbol{k})$ into the hole. The string $c_p^\dagger c_v\, c_c^\dagger c_p$ reorders to $-c_c^\dagger\, c_p^\dagger c_p\, c_v$, which is $-c_c^\dagger c_v$ on a state with p occupied. The term is $-\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\, g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q})/(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} + \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}})$.

4. The phonon creates the electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$ from an occupied $(p, \boldsymbol{k})$, and the photon moves an electron from $(v, \boldsymbol{k})$ into the hole. The same reordering gives -1, and the term is $-g_{cp\nu}(\boldsymbol{k}, \boldsymbol{Q})\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})/(\varepsilon_{p\boldsymbol{k}} - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu})$.

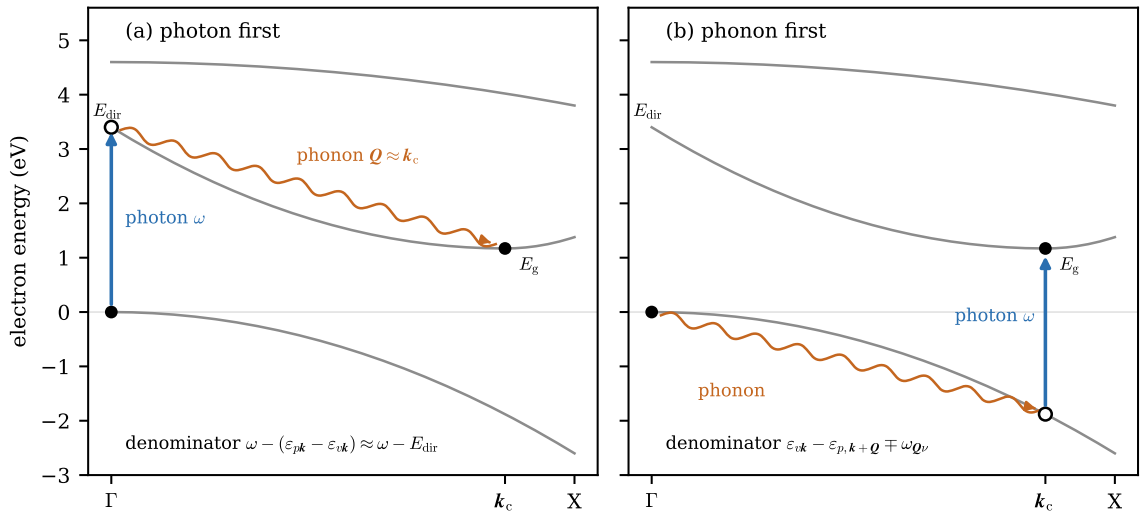


Figure 6.8: The two time orderings of phonon-assisted absorption across the indirect gap of silicon, on schematic bands along Γ–X. Filled dots: the initial electron at the valence maximum at Γ and the final electron at the conduction minimum near $\boldsymbol{k}_{\mathrm{c}} = 0.85\,(2\pi/a)$, 1.17 eV higher. Open dots: virtual intermediate states. Straight arrows: the vertical optical vertex $\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}$; wavy arrows: the electron–phonon vertex g, which changes the crystal momentum by $\boldsymbol{Q} \approx \boldsymbol{k}_{\mathrm{c}}$ and the energy by just $\pm\omega_{\boldsymbol{Q}\nu}$, tens of meV. Energy is not conserved at the intermediate state; each panel gives the energy denominator with which its amplitude enters Eq. (6.42). (a) Photon first: the intermediate electron is at Γ, and the denominator vanishes when ω reaches the vertical gap $E_{\mathrm{dir}} = 3.4$ eV. (b) Phonon first: the intermediate state is a valence state at $\boldsymbol{k}_{\mathrm{c}}$, which is occupied. The process then runs through hole orderings, in which an electron from an occupied band is promoted first and the valence electron fills the hole it leaves (sign -1), and the denominator of these orderings never vanishes. Band curves drawn by hand; only E_{g}, E_{dir} and $\boldsymbol{k}_{\mathrm{c}}$ are measured values.

In the first two paths the electron propagates in the intermediate state, in the last two the hole; the hole paths are the Z graphs of old-fashioned perturbation theory. Fig. 6.8 draws the two electron paths for silicon. Near the absorption edge of silicon, with $\omega \approx E_{\mathrm{g}}$ and the lowest vertical transition at 3.4 eV, the intermediate electron of path 1 is off the energy shell by about 2 eV. In silicon the natural intermediate state of path 2 is a valence state at $\boldsymbol{k}_{\mathrm{c}}$, which is occupied. The electron cannot enter a band p that is already full; Sect. 2.4 anticipated this Pauli blocking, and paths 3 and 4 take the place of path 2.

On the energy shell, $\omega = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, the denominator of path 3 becomes $\varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, and its term takes the form of path 2 with an occupied p. The denominator of path 4 becomes $\varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega$, and its term takes the form of path 1 with an occupied p. The fermion sign of the hole paths is what makes their terms match the electron paths. The four paths therefore combine into a single one-electron amplitude, summed over every band p, occupied or empty:

$$M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q}) = \sum_p \left[\frac{g_{cp\nu}(\boldsymbol{k},\boldsymbol{Q})\; \hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})}{\omega - \varepsilon_{p\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}}} + \frac{\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\; g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}} \right]. \tag{6.42}$$

The sum includes $p = v$ and $p = c$, in which the photon acts as an intraband velocity and the phonon does all the interband work. Refs. [2, Eq. (200)] and [37, Eqs. (2) and (3)] write the same sum with both denominators of opposite sign, a change that leaves $|M^{\pm}|^2$ unchanged.

Note that the phonon energy enters twice: in the denominator of the phonon-first term, where against electronic energies of electronvolts it is usually negligible, and in the energy balance $\omega = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, where it shifts the threshold.

A rough size of the amplitude follows from Eq. (6.42) before any calculation. Compared to a direct transition with a velocity matrix element of similar size, the phonon-assisted amplitude carries an extra factor of the vertex over a denominator. Typical optical-phonon vertices in silicon are a few tenths of an eV (Sect. 6.2) and the denominators near the silicon edge are 2 to 3 eV; together they put $|M^{\pm}|^2$ at $\sim 10^{-2}$ of the squared optical vertex for each phonon mode that couples. The phonon-assisted ε_2 near the indirect edge is then $\sim 10^{-2}$ of the direct one above E_{dir}, before phase space and occupations are counted. This factor of $\sim 10^{-2}$ is why silicon absorbs weakly between its two gaps, and why, for dark matter, the phonon-assisted rate dominates only where the zero-phonon rate is kinematically forbidden.

The photon-first denominator vanishes when $\omega = \varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}$, that is, when the photon alone could create a real pair at $\boldsymbol{k}$. In silicon the denominator first vanishes at E_{dir}. There the intermediate electron becomes real, and Sect. 2.4 gives the treatment. A real electron decays, by phonon emission among other channels, with a width Γ_{w} (Sect. 6.3), and the denominator acquires $+\mathrm{i}\Gamma_{\mathrm{w}}/2$ (Eq. (2.57)). By Eq. (2.60) the squared resonant term then splits into an on-shell part and a finite part. The on-shell part is a cascade: direct absorption at $\boldsymbol{k}$, followed by the scattering of the real electron by a phonon, weighted by the branching ratio of the phonon scattering. Direct absorption already counts the cascade once, in Eq. (6.40) with a broadened delta function. Adding the full phonon-assisted ε_2 to the direct one above E_{dir} counts the cascade twice. Below E_{dir} the double counting does not arise: every intermediate state is off shell, by electronvolts at the indirect edge. First-principles calculations in silicon give the intermediate states a constant width of 100 meV and find the spectrum between the gaps insensitive to this width [37]; the insensitivity is the numerical form of the statement that every intermediate state is off shell.

6.5.3 The phonon-assisted dielectric function

The rest of the derivation is bookkeeping, done as for Eq. (6.40): T_{fi} goes into the rate, the rate into the absorbed power, and the power into ε_2. Summing over the final states, labeled by $\boldsymbol{k}$, $\boldsymbol{Q}$, ν, c, v, the spin and the sign, we obtain

Phonon-assisted dielectric function

$$\varepsilon_2^{\mathrm{PA}}(\omega) = \frac{g_{\mathrm{deg}}\,\pi e^2}{\omega^2\,\Omega} \left\langle\!\!\left\langle \sum_{\nu} \sum_{cv} \sum_{\pm} \left(n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2}\right) \left|M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q})\right|^2 \delta\left(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega\right) \right\rangle\!\!\right\rangle, \tag{6.43}$$

where $\langle\!\langle \cdots \rangle\!\rangle$ is the average over the electron wavevector $\boldsymbol{k}$ and the phonon wavevector $\boldsymbol{Q}$ in the zone of Eq. (A.10), and $g_{\mathrm{deg}} = 2$ counts the spin. The formula holds for $T \ll E_{\mathrm{g}}$, where thermally excited pairs, which would run the process backwards, are absent. Together with Eq. (6.41), Eq. (6.43) gives the absorption coefficient used in first-principles work. Ref. [37, Eq. (1)] writes it in Gaussian units, $e_{\mathrm{G}}^2 = e^2/4\pi$ (Appendix A.2), with the spin factor explicit, and Ref. [2, Eq. (200)] writes it with the spin sum implicit. Eq. (6.43) adds to the direct ε_2 of Eq. (6.40), and below E_{dir} it is the whole absorption.

The formula contains three sums, and each sets a cost. The sum over intermediate bands p, inside $M^{\pm}$, runs over occupied and empty bands alike, and it converges no faster than its denominators grow; a calculation must include bands several electronvolts above the conduction minimum. The sum over phonon branches is short, six in silicon. The average

over $\boldsymbol{k}$ and $\boldsymbol{Q}$ is the expensive one. The two wavevectors range independently over the whole zone, linked by the energy delta function alone, and the onset region needs both on fine grids. An energy resolution of 14 meV in silicon took 40^3 points for each, 4×10^9 pairs, each with its own vertices for every pair of bands [37].

In practice no calculation computes so many vertices directly. The energies, velocities and vertices are computed by density-functional perturbation theory on coarse grids, rotated to Wannier functions, and interpolated to the fine grids, with every quantity in one gauge (Sect. 6.2). Two corrections enter the optical vertex. With a nonlocal pseudopotential the velocity includes the commutator of the nonlocal part (Eq. (3.40)), and when the gap is corrected the velocities are rescaled by the ratio of corrected to Kohn–Sham transition energies (Eq. (23.19)). A different route avoids the double sum altogether. Freezing the nuclei of a supercell in a special displaced configuration breaks translation symmetry; the photon then connects states of different crystal momentum, and one static calculation of ε_2 in that supercell contains the phonon-assisted absorption at a given temperature (Sect. 6.3). Being adiabatic, the special-displacement calculation does not distinguish the emission of a phonon from its absorption, and it cannot resolve the two thresholds $E_\mathrm{g} \pm \omega_{\boldsymbol{Q}\nu}$ that the next subsection derives.

6.5.4 The Hall–Bardeen–Blatt edge

Just above the threshold, Eq. (6.43) simplifies enough to be done by hand. The final hole lies near the valence maximum at $\boldsymbol{k} = 0$, and the final electron near one of the conduction minima $\boldsymbol{k}_\mathrm{c}$. We measure their crystal momenta from the extrema, $\boldsymbol{k} = \boldsymbol{\kappa}_\mathrm{h}$ and $\boldsymbol{k} + \boldsymbol{Q} = \boldsymbol{k}_\mathrm{c} + \boldsymbol{\kappa}_\mathrm{e}$, and take the bands as parabolic, with kinetic energies $E_\mathrm{h} = \kappa_\mathrm{h}^2/2m_\mathrm{h}$ and $E_\mathrm{e} = \kappa_\mathrm{e}^2/2m_\mathrm{d}^*$. Here m_h is the mass of one valence band and $m_\mathrm{d}^* = (m_\parallel^* m_\perp^{*2})^{1/3}$ the density-of-states mass of one valley, $0.322\, m_e$ in silicon with the valley masses of Table 3.1. Two approximations make the sums elementary. First, the phonon wavevector $\boldsymbol{Q} = \boldsymbol{k}_\mathrm{c} + \boldsymbol{\kappa}_\mathrm{e} - \boldsymbol{\kappa}_\mathrm{h}$ stays within the small pockets around $\boldsymbol{k}_\mathrm{c}$, and the phonon energies and vertices may be taken at $\boldsymbol{Q} = \boldsymbol{k}_\mathrm{c}$; we write $\omega_\nu \equiv \omega_{\boldsymbol{k}_\mathrm{c}\nu}$ and $n_\nu = n_\mathrm{B}(\omega_\nu)$. Second, the denominators of Eq. (6.42) are electronic energy differences of an electronvolt or more, far from any resonance, and they barely change across the pockets. The squared amplitude $|M^\pm|^2$ is then a constant, its value $|M_\nu^\pm|^2$ at the extrema.

Then only phase space remains. The averages over $\boldsymbol{k}$ and $\boldsymbol{Q}$ become independent integrals over $\boldsymbol{\kappa}_\mathrm{h}$ and $\boldsymbol{\kappa}_\mathrm{e}$, each of which turns into an energy integral with the density of states of a parabolic band, $\int \mathrm{d}^3\kappa\, (2\pi)^{-3} = \int \mathrm{d}E\, (2m)^{3/2} E^{1/2}/4\pi^2$. The delta function fixes $E_\mathrm{e} + E_\mathrm{h} = x$, the energy above the threshold, and

$$\begin{aligned} \Omega^2 \int \frac{\mathrm{d}^3\kappa_\mathrm{h}}{(2\pi)^3} \frac{\mathrm{d}^3\kappa_\mathrm{e}}{(2\pi)^3}\, \delta(x - E_\mathrm{e} - E_\mathrm{h}) &= \frac{\Omega^2\, (2m_\mathrm{d}^*)^{3/2} (2m_\mathrm{h})^{3/2}}{16\pi^4} \int_0^x \mathrm{d}E_\mathrm{e}\, E_\mathrm{e}^{1/2} (x - E_\mathrm{e})^{1/2} \\ &= \frac{\Omega^2\, (m_\mathrm{d}^* m_\mathrm{h})^{3/2}}{16\pi^3}\, x^2, \end{aligned} \tag{6.44}$$

for $x \geq 0$, since the last integral is $\pi x^2/8$. Two densities of states, each growing as $\varepsilon^{1/2}$, have a convolution that grows as x^2. A direct edge has one free momentum, because the photon ties the electron to the hole, and its joint density of states grows as $x^{1/2}$; the phonon frees the second momentum and adds a power 3/2. With all factors collected, each valley, valence band and branch adds a term

Hall–Bardeen–Blatt edge

$$\alpha_{\mathrm{abs}}(\omega) \simeq \sum_{\nu} A_\nu \Big[n_\nu \, (\omega - E_{\mathrm{g}} + \omega_\nu)^2 \, \Theta(\omega - E_{\mathrm{g}} + \omega_\nu) + (n_\nu + 1) \, (\omega - E_{\mathrm{g}} - \omega_\nu)^2 \, \Theta(\omega - E_{\mathrm{g}} - \omega_\nu) \Big], \tag{6.45}$$

with

$$A_\nu = \frac{g_{\mathrm{deg}} \, e^2 \, \Omega \, (m_{\mathrm{d}}^* m_{\mathrm{h}})^{3/2}}{16\pi^2 \, \omega \, n_r} \sum_{\text{valleys}} |M_\nu|^2,$$

constant across the edge. In A_ν we have set $|M_\nu^+| = |M_\nu^-| \equiv |M_\nu|$, which holds when ω_ν is negligible in the denominators, and treated ω and n_r as constant over the width of the edge, a few times ω_ν or about 0.1 eV. Heavy and light holes enter as separate valence bands, each with its own m_{h}. Eq. (6.45) is the form of Hall, Bardeen and Blatt [38].

Eq. (6.45) turns the edge into a spectroscopy. Each branch that couples contributes two components with the same coefficient A_ν: one starting at $E_{\mathrm{g}} - \omega_\nu$, where the photon is helped by a phonon absorbed from the lattice, and one at $E_{\mathrm{g}} + \omega_\nu$, where a phonon is emitted. The midpoint of each pair of thresholds is E_{g}, and their separation is $2\omega_\nu$. Each component is a straight line in a plot of $\sqrt{\alpha_{\mathrm{abs}}}$ against ω, with slope $\sqrt{A_\nu n_\nu}$ or $\sqrt{A_\nu (n_\nu + 1)}$. The sum of several components is not a straight line, and the measured curve is decomposed by subtracting, at each threshold, the extrapolation of the components that started below it.

The quadratic law assumes that M_ν is finite at the extrema, that is, that the transition is *allowed* there. If symmetry forces M_ν to vanish at the extrema, it grows linearly with κ_{e} or κ_{h}, an extra power of energy enters the integrand of Eq. (6.44), and the forbidden edge grows as x^3. The quadratic law also neglects the attraction between the electron and the hole. In silicon the bound pair, the exciton, lies $E_x^b = 14.7\,\mathrm{meV}$ below the gap at total crystal momentum $\boldsymbol{k}_{\mathrm{c}}$ (Sect. 4.9), and a photon reaches the exciton with a phonon just as it reaches a free pair. Each threshold of Eq. (6.45) therefore has an excitonic companion E_x^b lower, with its own onset shape, and the kinks of a measured edge are exciton thresholds at $E_{\mathrm{g}} - E_x^b \pm \omega_\nu$.

6.5.5 Temperature dependence

All of the temperature dependence of Eq. (6.45) at a fixed gap is in the occupations. The component that absorbs a phonon is proportional to n_ν, the number of phonons available; the one that emits is proportional to $n_\nu + 1$, spontaneous plus stimulated emission. Their ratio,

$$\frac{n_\nu}{n_\nu + 1} = \mathrm{e}^{-\omega_\nu / T}, \tag{6.46}$$

is detailed balance (Eq. (2.68)) for the phonon system alone: the two components have the same $|M_\nu|^2$, and the lattice gives up an energy ω_ν with a Boltzmann penalty relative to taking it. The ratio of the slopes of the two components in $\sqrt{\alpha_{\mathrm{abs}}}$ is $\mathrm{e}^{-\omega_\nu / 2T}$, and the edge is a thermometer of the lattice that needs no knowledge of A_ν. At high temperature both components grow as T/ω_ν, and the total weight of a branch, proportional to $2n_\nu + 1$, rises with T.

As $T \to 0$ the absorption components vanish. A cold crystal has no phonons to give, and every photon absorbed at the indirect edge must create one. The edge then lies at $E_{\mathrm{g}} + \omega_\nu$ for the softest branch that couples, and the energy ω_ν of the emitted phonon goes into the lattice, not into the pair. At the 10 mK of a dark-matter detector the emission-only edge is exact for every practical purpose: the transverse-acoustic occupation in silicon is vanishingly small (Worked example 6.5).

The gap itself also moves with temperature. The phonon-assisted formula contains the occupations explicitly; the bands in Eq. (6.43) are the bands the electrons actually occupy, renormalized by the same electron–phonon coupling (Sect. 6.3). In silicon the gap shrinks from 1.17 eV at low temperature to 1.12 eV at 300 K (Sect. 3.3), and all four thresholds of each branch move down with it. First-principles spectra that include the occupations alone must be shifted rigidly to match the measured onsets, by 0.15 to 0.23 eV in silicon, a shift that absorbs both the remaining error of the computed gap and the electron–phonon renormalization of the gap [37].

6.5.6 The silicon edge

Macfarlane, McLean, Quarrington and Roberts measured the absorption edge of silicon between 4.2 and 415 K, with enough resolution to see the thresholds of Eq. (6.45) [39]. They found four series of thresholds and quoted the phonon energies as temperatures θ, which in natural units are the energies themselves. Two of these phonon energies, $\theta = 212$ K and 670 K, belong to the transverse-acoustic and transverse-optical phonons at $\boldsymbol{k}_c$ [36, Chap. 6]. With 1 K = 86.17 µeV (Sect. 2.5) the two energies are

$$\omega_{TA} = 18.27\,\text{meV}, \qquad \omega_{TO} = 57.73\,\text{meV}.$$

Neutron scattering measures the same branches at the zone boundary X, at 4.49 THz and 13.90 THz, or 18.57 meV and 57.49 meV (Sect. 5.2). Both branches are flat between $\boldsymbol{k}_c = 0.85\,\text{X}$ and X: in the computed dispersion of Sect. 5.2 they change by 0.1 % (TA) and 0.3 % (TO) over that interval. The optical and the neutron values measure the same phonons, and they agree to 2 % for the transverse-acoustic branch and to 0.4 % for the transverse-optical one. The measured thresholds were exciton thresholds, a binding energy below the free-pair ones, as Sect. 4.9 anticipated.

Fig. 6.9 shows Eq. (6.45) with these energies. Panel (b) separates the four components at 300 K and shows that each is a straight line in $\sqrt{\alpha_{abs}}$ with its own threshold, while their sum bends at every threshold. Panel (a) follows the sum down in temperature. The components that absorb a phonon fade first, the transverse-optical one long before the transverse-acoustic one, and at 77 K the edge is already dominated by emission.

Worked example 6.5: The phonon thresholds of silicon

With $E_g = 1.17$ eV and the phonon energies above, the four free-pair thresholds of Eq. (6.45) are

$$E_g - \omega_{TO} = 1.1123\,\text{eV}, \qquad E_g - \omega_{TA} = 1.1517\,\text{eV},$$
$$E_g + \omega_{TA} = 1.1883\,\text{eV}, \qquad E_g + \omega_{TO} = 1.2277\,\text{eV}.$$

The exciton binding $E_x^b = 14.7$ meV moves each of them down by that amount. The pairs of thresholds are symmetric about E_g, separated by $2\omega_{TA} = 36.5$ meV and $2\omega_{TO} = 115.5$ meV. The slopes depend on the occupations alone, not on the gap. At 300 K, $T = 25.85$ meV, and

$$n_{TA} = \frac{1}{e^{18.27/25.85} - 1} = 0.973, \qquad n_{TO} = \frac{1}{e^{57.73/25.85} - 1} = 0.120.$$

In units of $\sqrt{A_\nu}$, the transverse-acoustic components of $\sqrt{\alpha_{abs}}$ rise with slopes $\sqrt{n_{TA}} = 0.987$ (absorption) and $\sqrt{n_{TA}+1} = 1.405$ (emission), the transverse-optical ones with 0.346 and 1.058. The ratios of the slopes in each pair, 0.702 and 0.327, are $e^{-\omega_\nu/2T}$ for the two branches, independent of A_ν. Read backwards, the transverse-acoustic ratio gives $-2T \ln 0.702 = 18.27$ meV: the slope ratio of one branch returns its phonon energy if the temperature is known, or the temperature if the phonon energy is. At 77 K, $T = 6.635$ meV,

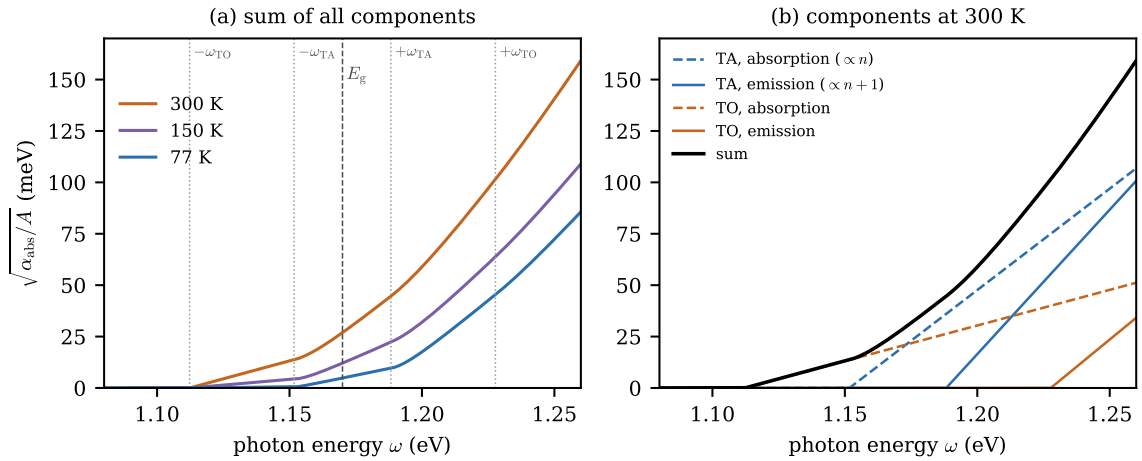


Figure 6.9: The Hall–Bardeen–Blatt edge of silicon, Eq. (6.45), plotted as $\sqrt{\alpha_{\mathrm{abs}}/A}$ against the photon energy, with $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$ and the phonon energies $\omega_{\mathrm{TA}} = 18.27\,\mathrm{meV}$ and $\omega_{\mathrm{TO}} = 57.73\,\mathrm{meV}$ measured from the optical edge [39]. Dotted lines mark the free-pair thresholds $E_{\mathrm{g}} \mp \omega_{\mathrm{TO}}$ and $E_{\mathrm{g}} \mp \omega_{\mathrm{TA}}$; the dashed line marks E_{g}, their common midpoint. (a) Sum of the four components at 300 K, 150 K and 77 K. (b) The components at 300 K: dashed, phonon absorption, proportional to n_ν; solid, phonon emission, proportional to $n_\nu + 1$. Each component is a straight line; the sum (black) is not. The two branches are given equal coefficients $A_{\mathrm{TA}} = A_{\mathrm{TO}} = A$, so their relative heights are illustrative. The gap is held at its low-temperature value at every temperature, to isolate the phonon occupations (Sect. 6.3 treats its shift). The excitonic thresholds, 14.7 meV lower, are omitted.

the occupations fall to $n_{\mathrm{TA}} = 0.068$ and $n_{\mathrm{TO}} = 1.7 \times 10^{-4}$, and the slope ratios to 0.252 and 0.013. At 20 K the transverse-acoustic absorption component has $n_{\mathrm{TA}} = 2.5 \times 10^{-5}$, and at the 10 mK of a cryogenic detector $\omega_{\mathrm{TA}}/T = 2.1 \times 10^{4}$: the occupation is e^{-21200}, and only the emission thresholds exist. The lowest of them is the true optical edge of cold silicon, 18 meV above the gap. With the measured low-temperature gap $E_{\mathrm{g}}(0) = 1.1692\,\mathrm{eV}$ of Sect. 6.3 in place of the rounded value, this edge lies at $E_{\mathrm{g}}(0) + \omega_{\mathrm{TA}} = 1.1875\,\mathrm{eV}$ for a free pair, and the excitonic edge at 1.1728 eV, since ω_{TA} and E_x^b nearly cancel.

6.5.7 Phonon-assisted dark-matter scattering

The derivation of Eq. (6.42) used only two properties of the photon: it couples to the electrons through a one-body operator, and it carries negligible momentum. A dark-matter particle that interacts with the electron density keeps the first property and loses the second. Its vertex is the dark-matter potential $\tilde{V}(\boldsymbol{q})$ times the matrix element of $n_e(\boldsymbol{q}) = \int \mathrm{d}^3 r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, \hat{n}_e(\boldsymbol{r})$, the Fourier component of the density operator. Between Bloch states this matrix element is the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ of Eq. (3.34). The optical vertex $\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})$ is replaced by $\rho_{pv}(\boldsymbol{k}, \boldsymbol{q})$, which is no longer vertical: it takes the electron from $\boldsymbol{k}$ to $\boldsymbol{k} + \boldsymbol{q}$. The phonon then supplies the rest of the crystal momentum, $\boldsymbol{Q} = \boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q}$ up to a reciprocal-lattice vector for a transition from the valence maximum to a valley. The amplitude has the same orderings and the same hole paths, but its vertices and energies are evaluated at three crystal momenta, $\boldsymbol{k}$, $\boldsymbol{k} + \boldsymbol{q}$ and $\boldsymbol{k} + \boldsymbol{q} + \boldsymbol{Q}$, instead of two.

The energy denominators take the energy deposited ω in place of the photon energy, and

they resonate wherever ω matches a real pair at momentum transfer $\boldsymbol{q}$, with the same risk of double counting as at E_{dir}. Since ρ_{mn} reduces to the optical vertex as $q \to 0$ (Eq. (3.43)), the measured silicon edge is the zero-momentum limit of the dark-matter amplitude and a test of any calculation of it.

Absorbed bosons and scattering particles meet this amplitude in different limits. An absorbed boson, a kinetically mixed dark photon for instance, deposits its whole mass with a momentum $m_\phi v \approx 10^{-3} m_\phi$, a smaller momentum for its energy than a photon carries. Its absorption between E_{g} and E_{dir} is Eq. (6.43) with the photon's coupling replaced by its own and screened by the medium (Sect. 8.4.5). A scattering particle transfers $\boldsymbol{q}$ and ω together, with $\omega \leq q v_{\max} - q^2/2m_\chi$ (Eq. (1.25)). Far above its threshold mass it reaches $\boldsymbol{k}_{\mathrm{c}}$ at first order, but just above the threshold mass the energy needed to cross the gap is available at momentum transfers well below k_{c}, where the zero-phonon transition is forbidden (Sect. 3.3). At the threshold mass, $\omega = E_{\mathrm{g}}$ is reached just at $q = 2E_{\mathrm{g}}/v_{\max} = 0.88\,\mathrm{keV}$, and for heavier dark matter the smallest such q falls toward $E_{\mathrm{g}}/v_{\max} = 0.44\,\mathrm{keV}$; both lie well below $k_{\mathrm{c}} = 0.85\,(2\pi/a) = 1.9\,\mathrm{keV}$. At these momentum transfers, as for light below E_{dir}, the phonon-assisted amplitude is the leading process, and it opens a window of masses in which silicon sees dark matter only through the phonon-assisted process. In a cold detector, however, only the emission branch contributes: the phonon takes part of the energy deposited, and the pair receives $\omega - \omega_{\boldsymbol{Q}\nu}$.

Summary

The Born–Oppenheimer approximation keeps only the diagonal of the coupling between electrons and nuclei, and the off-diagonal coupling follows from the derivative-coupling vector $\boldsymbol{d}_{ji} = \langle \Psi_j | \nabla \Psi_i \rangle$ (Sect. 6.1). To first order in the displacements the same physics is the electron–phonon Hamiltonian H_{ep}, in which a phonon of wavevector $\boldsymbol{Q}$ moves an electron from $\boldsymbol{k}$ to $\boldsymbol{k} + \boldsymbol{Q}$ with the vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$. Whether the electrons follow a moving nucleus depends on how fast the nucleus is accelerated. If the acceleration is slow compared with $1/\Delta E$, the electrons stay in their ground state. If it is sudden, they keep their state while the nucleus acquires a velocity v_N, and the complement of the overlap of the boosted ground state with the original one is the Migdal effect.

The vertex that density-functional perturbation theory computes is screened (Sect. 6.2). Acoustic phonons couple through deformation potentials, with $g \propto Q^{1/2}$, optical phonons in nonpolar crystals through optical deformation potentials, and longitudinal optical phonons in polar crystals through the Fröhlich vertex $g^{\mathrm{L}} = \mathrm{i} C_{\mathrm{F}} \rho_{mn}/Q$, which in GaAs gives a polaron constant $\alpha_{\mathrm{F}} = 0.072$.

Second order in the displacements yields the electron self-energy, the Fan–Migdal and Allen–Heine terms, which in the adiabatic limit combine into the average of the band energy over the zero-point and thermal smearing of the lattice (Sect. 6.3). For the indirect gap of silicon the computed zero-point renormalization is $-56.2\,\mathrm{meV}$; the measured electron–phonon part, -71 to $-73\,\mathrm{meV}$, is about 20 % larger. A detector at 10 mK sees the low-temperature gap, which already contains the full renormalization.

Photons between the indirect gap of silicon and its direct gap of 3.4 eV are absorbed only with a phonon (Sect. 6.5). Summed over electron and phonon wavevectors in the whole zone, the squared second-order amplitude $|M^{\pm}|^2$ gives the phonon-assisted dielectric function $\varepsilon_2^{\mathrm{PA}}$. Near the edge it reduces to the Hall–Bardeen–Blatt law, each phonon branch adding components growing as $(\omega - E_{\mathrm{g}} \pm \omega_\nu)^2$, and in a cold detector only emission survives. Replacing the optical vertex by the density vertex turns phonon-assisted absorption of light into the phonon-assisted response to dark matter, which governs the absorption of light bosons

(Sect. 13.8) and the threshold window of electron recoils in silicon.

A phonon decays into electron–hole pairs only where real pairs exist at its energy (Sect. 6.4). Its self-energy is the electron bubble, whose static part is already contained in the frequencies of density-functional perturbation theory. In a metal the imaginary part leads to Allen's formula, the Eliashberg function $\alpha^2F(\omega)$ and the coupling constant λ_{ep}. For aluminum a free-electron model with one core radius yields $\lambda_{\mathrm{ep}} = 0.436$, against 0.42 to 0.43 from tunneling and the specific heat, and λ_{ep} sets the strength of the pairing in the superconducting sensors of Chap. 7.

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Chapter 7

Magnets, superconductors, Dirac materials and defects

Silicon offers dark matter two kinds of excitation: an electron lifted across the 1.17 eV gap, and phonons up to 64 meV. A particle that couples to the spin of the electron finds no excitation in silicon below the gap, since every spin is paired in a bond and flipping one costs the gap. Other solids have excitations below these energies or of other kinds. In NiO the spins of the nickel ions order, and a particle that couples to spin can create a quantum of the spin precession, a magnon. In a superconductor such as aluminum the electrons at the Fermi surface bind into pairs, and breaking a pair costs only twice the superconducting gap. In Dirac materials the electrons obey a Dirac equation with a velocity far below that of light, and the pair excitations lie across a gap much smaller than that of silicon. Donors and color centers put bound levels inside the gap of silicon itself, shallow for a donor and deep for a color center, and a struck atom can leave a defect that stores energy. The theory of each of these targets is built largely from pieces already in hand: the exchange term of the Coulomb interaction (Sect. 2.3), the electron–phonon coupling (Chap. 6), the $\boldsymbol{k} \cdot \boldsymbol{p}$ expansion (Sect. 3.3) and the hydrogenic envelope (Sect. 4.9). Beyond these pieces, the targets need physics that silicon does not have, above all spontaneously broken symmetry. A broken symmetry brings with it collective modes and quasiparticles whose energies are set by an interaction rather than by a band gap. We will derive, for each target, the excitation spectrum and the response that the rate formulas of the later chapters take as input.

7.1 Magnetic order and spin waves

Each nickel ion in NiO carries a spin, and below 523 K, the measured Néel temperature [1, Table IV], the spins order: those in one (111) plane of the crystal point one way, those in the next plane the opposite way. A disturbance of that order travels through the crystal as a spin wave. The quanta of spin waves, the *magnons*, have energies up to ∼117 meV. Inelastic neutron scattering at 78 K measured their dispersion, with an initial slope of ∼250 meV Å [2]. In an ordered magnet a particle that couples to the spin of the electron can create a magnon [3]. The energy deposits then run from a few meV to about a tenth of an eV. To compute the rate of magnon creation we need the interaction that orders the spins, the spectrum of the spin waves, and the operator through which a probe creates magnons.

7.1.1 Exchange and the Heisenberg model

The Coulomb interaction does not act on spin, yet it makes the energy of two electrons depend on how their spins are aligned. Put one electron in each of two orthogonal orbitals φ_a and φ_b. In the spin triplet the spatial wavefunction is antisymmetric, $[\varphi_a(\boldsymbol{r})\varphi_b(\boldsymbol{r}') - \varphi_b(\boldsymbol{r})\varphi_a(\boldsymbol{r}')]/\sqrt{2}$, and in the singlet it is symmetric. The expectation value of the Coulomb repulsion $v_{\rm C}(\boldsymbol{r}-\boldsymbol{r}')$ is then $C \mp K_{\rm ex}$, upper sign for the triplet, with the direct integral C and the exchange integral

$$K_{\rm ex} = \int \mathrm{d}^3r\, \mathrm{d}^3r'\, \varphi_a^*(\boldsymbol{r})\varphi_b^*(\boldsymbol{r}')\, v_{\rm C}(\boldsymbol{r}-\boldsymbol{r}')\, \varphi_b(\boldsymbol{r})\varphi_a(\boldsymbol{r}'). \tag{7.1}$$

The exchange integral $K_{\rm ex}$ is the exchange term of Eq. (2.46), written for localized orbitals instead of plane waves. It is the Coulomb self-energy of the charge distribution $\varphi_a^*\varphi_b$, and since the Fourier transform $v_{\rm C}(k) = e^2/k^2$ is positive, $K_{\rm ex} > 0$. The triplet therefore lies $2K_{\rm ex}$ below the singlet: parallel spins keep the electrons apart and save Coulomb energy. The resulting preference for parallel spins is Hund's rule, and within one Ni^{2+} ion it aligns the two holes of the $3d$ shell into a spin $|\boldsymbol{S}| = 1$.

The four spin states can be described without any reference to orbitals. With $\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 = \frac{1}{2}[(\boldsymbol{S}_1+\boldsymbol{S}_2)^2 - \boldsymbol{S}_1^2 - \boldsymbol{S}_2^2]$, which is $-\frac{3}{4}$ in the singlet and $\frac{1}{4}$ in the triplet, the operator

$$H_{12} = E_{\rm orb} + C - \tfrac{1}{2}K_{\rm ex} - 2K_{\rm ex}\, \boldsymbol{S}_1 \cdot \boldsymbol{S}_2 \tag{7.2}$$

has the singlet energy $E_{\rm orb} + C + K_{\rm ex}$ and the triplet energy $E_{\rm orb} + C - K_{\rm ex}$, with $E_{\rm orb}$ the sum of the two one-electron orbital energies. The Coulomb interaction depends only on the positions of the electrons, and through exchange it has become a coupling between their spins, of strength $-2K_{\rm ex}$ and ferromagnetic sign.

Kinetic exchange. Between electrons on neighboring ions of an insulator, however, a second mechanism usually dominates, and it has the opposite sign. Take two sites with one orbital each and the hopping $t_{\rm hop} > 0$ of the tight-binding model of Sect. 3.2. Putting two electrons on the same site costs the on-site or Hubbard repulsion $U_{\rm H}$:

$$H = -t_{\rm hop} \sum_s (c_{1s}^\dagger c_{2s} + c_{2s}^\dagger c_{1s}) + U_{\rm H} \sum_{i=1,2} c_{i\uparrow}^\dagger c_{i\uparrow}\, c_{i\downarrow}^\dagger c_{i\downarrow}. \tag{7.3}$$

With two electrons and $t_{\rm hop} = 0$, the four states with one electron on each site are degenerate at zero energy, and the two doubly occupied states lie at $U_{\rm H}$. The hopping connects the singly occupied states with the doubly occupied ones. A triplet cannot hop, since the Pauli principle forbids two parallel spins in one orbital: for the state $c_{1\uparrow}^\dagger c_{2\uparrow}^\dagger|0\rangle$ the hop is blocked directly, and the other two members of the triplet follow by rotational symmetry. The triplet stays at zero energy to all orders. The singlet couples to the symmetric combination of the doubly occupied states, through two hops (from site 1 or from site 2) that add, with a matrix element of magnitude $2t_{\rm hop}$. To second order in $t_{\rm hop}/U_{\rm H}$ its energy is $-(2t_{\rm hop})^2/U_{\rm H}$. As Eq. (7.2) shows, the coefficient of $\boldsymbol{S}_1 \cdot \boldsymbol{S}_2$ is the triplet energy minus the singlet energy. With both mechanisms present the coefficient is

$$J^{\rm ex} = \frac{4t_{\rm hop}^2}{U_{\rm H}} - 2K_{\rm ex}. \tag{7.4}$$

The picture behind the first term is that an electron lowers its kinetic energy by a virtual hop onto its neighbor and back, and it can do so only if the neighbor's spin is antiparallel. The virtual hop costs $U_{\rm H}$, so the kinetic-energy gain is second order in $t_{\rm hop}$. The second-order expansion holds as long as $t_{\rm hop} \ll U_{\rm H}$. The same condition keeps the charge frozen at one

electron per site, and in this two-site picture an insulator with frozen charge of this kind is a *Mott insulator*. The corrections to Eq. (7.4) are of relative order $(t_{\mathrm{hop}}/U_{\mathrm{H}})^2$.

In NiO the nickel ions do not touch; between two of them sits an oxygen ion. Anderson's mechanism of *superexchange* [4] applies Eq. (7.4) with an effective hopping that runs through the oxygen p orbital, so that t_{hop} is itself second order in the nickel–oxygen hopping. The geometry of the nickel–oxygen–nickel bond then decides the sign and the size of the coupling. Next-nearest nickel neighbors sit on a straight line through one oxygen ion. The oxygen p orbital along that line overlaps strongly with a $3d$ orbital of each nickel ion, and the coupling of next-nearest neighbors is large and antiferromagnetic. Nearest neighbors see each other at 90° through two different p orbitals, and their coupling is weak and ferromagnetic. The measured values are [2]

$$J_2 = 19.01\,\mathrm{meV}, \qquad J_1 = -1.37\,\mathrm{meV}, \tag{7.5}$$

in the convention of Eq. (7.6) below. With several $3d$ orbitals and $|\boldsymbol{S}| = 1$, Eq. (7.4) acquires factors of order one, and we use it as an estimate, $J^{\mathrm{ex}} \sim t_{\mathrm{hop}}^2/U_{\mathrm{H}}$. The Hubbard repulsion of the nickel $3d$ shell is several eV; the linear-response values of Sect. 28.1 lie between 4.6 and 8.2 eV. Inverting the estimate for J_2 gives an effective hopping $(J_2 U_{\mathrm{H}})^{1/2} \approx 0.3$ to 0.4 eV, so that $t_{\mathrm{hop}}/U_{\mathrm{H}} \approx 0.05$ to 0.06. Since the electrons of NiO hop little compared with the cost of a double occupancy, only their spins remain free at low energy. The estimate is rough in one more respect. NiO is a *charge-transfer insulator*: the cheapest charge excitation moves an electron from oxygen to nickel, at the charge-transfer energy Δ_{CT}, which is smaller than U_{H}. Superexchange through the oxygen depends on both Δ_{CT} and U_{H}, and the effective hopping above absorbs the two energies into one number.

The Heisenberg model. In a crystal of ions with frozen charge, each carrying a spin of length $|\boldsymbol{S}|$ fixed by Hund's rule, the low-energy states are the spin configurations. To leading order in $t_{\mathrm{hop}}/U_{\mathrm{H}}$ the coupling of the spins is a sum over pairs, and the only coupling of two spins that is invariant under a common rotation and bilinear in each is their scalar product. The result is the *Heisenberg model*,

$$H_{\mathrm{H}} = \sum_{\langle ij \rangle} J_{ij}^{\mathrm{ex}}\, \boldsymbol{S}_i \cdot \boldsymbol{S}_j, \tag{7.6}$$

with each pair of ions counted once and the *exchange constant* $J_{ij}^{\mathrm{ex}} > 0$ antiferromagnetic. The ionic spins obey $[S_i^a, S_j^b] = \mathrm{i}\delta_{ij}\epsilon_{abc}S_i^c$ and $\boldsymbol{S}_i^2 = |\boldsymbol{S}|(|\boldsymbol{S}|+1)$. The literature uses every combination of sign, of single or double counting and of spin length, and a quoted J^{ex} is meaningful only together with its Hamiltonian.

If every $J_{ij}^{\mathrm{ex}} < 0$, the ground state has all spins parallel, and the magnet is a *ferromagnet*. If the antiferromagnetic bonds connect two sublattices, the classical ground state has the spins of one sublattice up and those of the other down, and the magnet is an *antiferromagnet*. In NiO, J_2 connects the ions of adjacent (111) planes, and the order is the alternation of ferromagnetic (111) planes described above, called type II (Fig. 7.1a). Eq. (7.6) leaves out the coupling of the spins to the lattice through spin–orbit coupling. The spin–orbit coupling fixes the direction of the moments in the crystal and opens small gaps in the magnon spectrum, below 5 meV in NiO [5, Sect. III], and we neglect it.

The exchange constants of a real magnet are computed from the electronic structure. The standard method is the density-functional theory (DFT) of Sect. 3.4 with a Hubbard correction on the $3d$ shell, called DFT $+U$. Three routes are common: total energies of several collinear orders mapped onto Eq. (7.6), the energies of spin spirals, and a Green's-function formula for small rotations of two moments. All three inherit the dependence of superexchange on U_{H}.

For NiO the density-functional values of J_2 in one comparison range from 11 to 27 meV [1, Table III], and linear spin-wave theory with the best of them overestimates the measured maximum magnon energy by about 14 % [5, Sect. III]. For a target without neutron data, this spread is the theory uncertainty of its magnon spectrum.

7.1.2 Spin waves of a ferromagnet

The ferromagnetic ground state is known exactly, and its excitations are the simplest place to see what a magnon is. Order the spins along $+\hat{\boldsymbol{z}}$. A state with a spin deviation at site j, one unit of spin removed there, is not an eigenstate, because the transverse part of $\boldsymbol{S}_i \cdot \boldsymbol{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)$ moves the deviation to the neighbors. The eigenstates, the spin waves, are spin deviations spread over all sites with a definite crystal momentum.

Holstein and Primakoff turned the problem of spin deviations into a boson problem [6]. On each site write

$$S_j^z = |\boldsymbol{S}| - d_j^\dagger d_j, \qquad S_j^+ = (2|\boldsymbol{S}| - d_j^\dagger d_j)^{1/2} d_j, \qquad S_j^- = d_j^\dagger (2|\boldsymbol{S}| - d_j^\dagger d_j)^{1/2}, \tag{7.7}$$

with boson operators $[d_i, d_j^\dagger] = \delta_{ij}$ of the kind introduced in Sect. 2.3. The number $n_j = d_j^\dagger d_j$ counts the deviation of the site from full polarization. The spin algebra holds on the physical states $n_j \le 2|\boldsymbol{S}|$. To check $[S^+, S^-] = 2S^z$ on one site, use $f(n)d = d\,f(n-1)$ for any function f. Then $S^+S^- = (2|\boldsymbol{S}| - n)(n+1)$ and $S^-S^+ = d^\dagger(2|\boldsymbol{S}| - n)d = n(2|\boldsymbol{S}| - n + 1)$, and the difference of the two products is $2|\boldsymbol{S}| - 2n = 2S^z$; the relations $[S^z, S^\pm] = \pm S^\pm$ follow since S^+ lowers n by one. The square roots make Eq. (7.7) exact and nonlinear. *Linear spin-wave theory* is an expansion in $1/|\boldsymbol{S}|$, valid when the number of deviations per site is small compared with $2|\boldsymbol{S}|$; it replaces the square roots by $(2|\boldsymbol{S}|)^{1/2}$ and keeps the Hamiltonian to second order in the bosons. For one bond,

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j \simeq |\boldsymbol{S}|^2 - |\boldsymbol{S}|(n_i + n_j) + |\boldsymbol{S}|(d_i^\dagger d_j + d_j^\dagger d_i),$$

where the dropped quartic term $n_i n_j$ comes from $S_i^z S_j^z$. The deviations hop from site to site with amplitude $J_{ij}^{\mathrm{ex}}|\boldsymbol{S}|$.

On a Bravais lattice of N magnetic ions at the sites $\boldsymbol{R}_j$, the hopping is diagonalized by plane waves,

$$d_j = N^{-1/2} \sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_j}\, a_{\boldsymbol{k}},$$

with $\boldsymbol{k}$ in the Brillouin zone. The operators $a_{\boldsymbol{k}}$ are bosons, the *magnon operators*; with several magnetic ions per cell they carry a branch index, $a_{\boldsymbol{k}\nu}$, which we drop for a single branch. In the sum over pairs, each site has the bonds $\sum_{\boldsymbol{R}\neq 0} J_{0\boldsymbol{R}}^{\mathrm{ex}}$, which multiply $-|\boldsymbol{S}|n_j$, and the hopping terms collect into $|\boldsymbol{S}| \sum_{\boldsymbol{k}} J^{\mathrm{ex}}(\boldsymbol{k})\, a_{\boldsymbol{k}}^\dagger a_{\boldsymbol{k}}$ with the lattice Fourier transform of the exchange constants. The result is a set of independent oscillators:

Linear spin waves of a ferromagnet

$$H_{\mathrm{H}} \simeq E_{\mathrm{cl}} + \sum_{\boldsymbol{k}} \omega_{\boldsymbol{k}}^{\mathrm{M}}\, a_{\boldsymbol{k}}^\dagger a_{\boldsymbol{k}}, \qquad \omega_{\boldsymbol{k}}^{\mathrm{M}} = |\boldsymbol{S}|\big[J^{\mathrm{ex}}(\boldsymbol{k}) - J^{\mathrm{ex}}(0)\big], \qquad J^{\mathrm{ex}}(\boldsymbol{k}) = \sum_{\boldsymbol{R}\neq 0} J_{0\boldsymbol{R}}^{\mathrm{ex}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}, \tag{7.8}$$

with the classical energy $E_{\mathrm{cl}} = \frac{1}{2}N|\boldsymbol{S}|^2 J^{\mathrm{ex}}(0)$, each pair counted once.

For inversion-symmetric lattices $J^{\mathrm{ex}}(\boldsymbol{k})$ is real and even. With all $J_{0\boldsymbol{R}}^{\mathrm{ex}} < 0$, $J^{\mathrm{ex}}(0)$ is the minimum of $J^{\mathrm{ex}}(\boldsymbol{k})$ and every $\omega_{\boldsymbol{k}}^{\mathrm{M}} \ge 0$. The mode $\omega_0^{\mathrm{M}} = 0$ tilts all spins together. A common

tilt costs nothing because Eq. (7.6) is invariant under a common rotation, and the mode $\omega_0^{\mathrm{M}} = 0$ is the Goldstone mode of the broken rotational symmetry, the analog of the acoustic phonon of a crystal. A particle physicist would count two Goldstone modes for the two broken rotations about $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$. The ferromagnet has one, with a quadratic dispersion. Its two broken generators do not commute in the ground state, $\langle [S^x_{\mathrm{tot}}, S^y_{\mathrm{tot}}] \rangle = \mathrm{i}\langle S^z_{\mathrm{tot}} \rangle \neq 0$, so they act as a coordinate and its conjugate momentum and together describe a single mode. In the antiferromagnet $\langle S^z_{\mathrm{tot}} \rangle = 0$, and the two Goldstone modes of the particle physicist's count appear, both with a linear dispersion. For the ferromagnet at small k, expanding $J^{\mathrm{ex}}(\boldsymbol{k})$ to second order gives $\omega^{\mathrm{M}}_{\boldsymbol{k}} \simeq \frac{1}{2}|\boldsymbol{S}| \sum_{\boldsymbol{R}} (-J^{\mathrm{ex}}_{0\boldsymbol{R}})(\boldsymbol{k}\cdot\boldsymbol{R})^2$, quadratic in k like the energy of a free particle. Physically, the magnon $a^\dagger_{\boldsymbol{k}}|0\rangle$ is one unit of spin deviation shared by all ions: each spin precesses on a narrow cone about $\hat{\boldsymbol{z}}$, and the phase of the precession advances by $\boldsymbol{k}\cdot\boldsymbol{R}$ from ion to ion.

Worked example 7.1: The ferromagnetic chain

Take a ring of N spins at spacing a with ferromagnetic nearest-neighbor exchange $J^{\mathrm{ex}} < 0$. Then $J^{\mathrm{ex}}(k) = 2J^{\mathrm{ex}} \cos ka$, and Eq. (7.8) gives

$$\omega^{\mathrm{M}}_k = 2|J^{\mathrm{ex}}||\boldsymbol{S}|\,(1 - \cos ka),$$

which rises as $|J^{\mathrm{ex}}||\boldsymbol{S}|a^2k^2$ from $k = 0$ and reaches $4|J^{\mathrm{ex}}||\boldsymbol{S}|$ at the zone boundary $k = \pi/a$. For one magnon this linear result is exact. Let $|F\rangle$ be the fully polarized state, of energy $E_F = NJ^{\mathrm{ex}}|\boldsymbol{S}|^2$, and $|j\rangle = (2|\boldsymbol{S}|)^{-1/2} S^-_j |F\rangle$ the normalized state with one unit of deviation on site j. The bonds that do not touch j keep their energy $J^{\mathrm{ex}}|\boldsymbol{S}|^2$. The two bonds that do touch site j have $S^z S^z = |\boldsymbol{S}|(|\boldsymbol{S}| - 1)$, and together they raise $\langle j|H_{\mathrm{H}}|j\rangle$ to $E_F - 2J^{\mathrm{ex}}|\boldsymbol{S}|$, above E_F since $J^{\mathrm{ex}} < 0$. The transverse term $\frac{1}{2}J^{\mathrm{ex}} S^+_j S^-_{j\pm1}$ moves the deviation to a neighbor with amplitude $\frac{1}{2}J^{\mathrm{ex}} \cdot 2|\boldsymbol{S}| = J^{\mathrm{ex}}|\boldsymbol{S}|$, since each of the two operators contributes $(2|\boldsymbol{S}|)^{1/2}$: $S^-_{j\pm1}$ acting on a full site and S^+_j acting on a site with one deviation. Hence

$$H_{\mathrm{H}}|j\rangle = (E_F - 2J^{\mathrm{ex}}|\boldsymbol{S}|)\,|j\rangle + J^{\mathrm{ex}}|\boldsymbol{S}|\,(|j+1\rangle + |j-1\rangle),$$

an exact statement for any $|\boldsymbol{S}|$, because the states with one deviation are closed under H_{H}. The Bloch combination $N^{-1/2}\sum_j \mathrm{e}^{\mathrm{i}kja}|j\rangle$ is an eigenstate with energy $E_F - 2J^{\mathrm{ex}}|\boldsymbol{S}| + 2J^{\mathrm{ex}}|\boldsymbol{S}|\cos ka = E_F + \omega^{\mathrm{M}}_k$. The quartic terms dropped in linear spin-wave theory act only on states with two or more deviations, where they produce an attraction between magnons of relative order $1/|\boldsymbol{S}|$.

For dark matter the quadratic dispersion matters. A magnon of energy ω has $k = [\omega/(|J^{\mathrm{ex}}||\boldsymbol{S}|a^2)]^{1/2}$, and its group velocity $2(\omega|J^{\mathrm{ex}}||\boldsymbol{S}|)^{1/2}a$ vanishes at small ω. At low energy a ferromagnet offers excitations of small momentum and small velocity, and their kinematic match differs from that of the linear branches of phonons and of the antiferromagnet below.

7.1.3 Spin waves of an antiferromagnet

In an antiferromagnet, however, the classical state with sublattice A up and sublattice B down is not an eigenstate of Eq. (7.6). For an antiparallel pair the transverse term $S^+_i S^-_j$, with i on A and j on B, lowers a spin that already points down and raises one that points up: it creates two deviations at once. Because the transverse term creates deviations in pairs, the

antiferromagnetic spin waves are mixtures of deviations and their absence, and the Bogoliubov transformation of Appendix C.7 diagonalizes the spin-wave Hamiltonian.

The magnetic ions form two interpenetrating sublattices A and B with N_s sites each, related by a translation or an inversion that exchanges them. On B the reference spin points along $-\hat{\boldsymbol{z}}$, so the roles of raising and lowering are exchanged: $S^z_j = -|\boldsymbol{S}| + d^\dagger_{\mathrm{B}j} d_{\mathrm{B}j}$ and $S^-_j \simeq (2|\boldsymbol{S}|)^{1/2} d_{\mathrm{B}j}$, while A uses Eq. (7.7) with bosons $d_{\mathrm{A}i}$. At linear order an antiparallel bond becomes

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j \simeq -|\boldsymbol{S}|^2 + |\boldsymbol{S}|(n_{\mathrm{A}i} + n_{\mathrm{B}j}) + |\boldsymbol{S}|(d_{\mathrm{A}i} d_{\mathrm{B}j} + d^\dagger_{\mathrm{A}i} d^\dagger_{\mathrm{B}j}),$$

with $n_{\mathrm{A}i} = d^\dagger_{\mathrm{A}i} d_{\mathrm{A}i}$ and $n_{\mathrm{B}j}$ likewise. Each deviation raises the energy of the bond, and the pair term $d_{\mathrm{A}i} d_{\mathrm{B}j} + \mathrm{h.c.}$ creates or destroys deviations in pairs. Bonds within a sublattice keep the ferromagnetic form. Exchange of any range enters through two lattice sums, over the neighbors of a site of A on its own sublattice and on the other,

$$J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}\in\mathrm{A},\,\boldsymbol{\delta}\neq 0} J^{\mathrm{ex}}_{0\boldsymbol{\delta}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}, \qquad J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}\in\mathrm{B}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}, \tag{7.9}$$

where $\boldsymbol{\delta}$ runs over the positions of the neighbors relative to the site. The symmetry that exchanges A and B gives the same sums for a site of B. When the set of neighbors is symmetric under $\boldsymbol{\delta} \to -\boldsymbol{\delta}$, as it is for every lattice of this section, both $J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k})$ and $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k})$ are real and even in $\boldsymbol{k}$.

Every site of A has the antiparallel bonds of total strength $J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ and the parallel bonds $J^{\mathrm{ex}}_{\mathrm{AA}}(0)$, and likewise every site of B. Introduce

$$d_{\mathrm{A}i} = N_\mathrm{s}^{-1/2} \sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_i}\, d_{\mathrm{A}\boldsymbol{k}}, \qquad d_{\mathrm{B}j} = N_\mathrm{s}^{-1/2} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_j}\, d_{\mathrm{B}\boldsymbol{k}},$$

with $\boldsymbol{k}$ in the Brillouin zone of one sublattice. The opposite sign on B is a convenience: $d_{\mathrm{B}\boldsymbol{k}}$ destroys a deviation of crystal momentum $-\boldsymbol{k}$, so that the pair terms couple $d_{\mathrm{A}\boldsymbol{k}}$ with $d_{\mathrm{B}\boldsymbol{k}}$. With $\sum_i \mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{R}_i} = N_\mathrm{s}\delta_{\boldsymbol{k}\boldsymbol{k}'}$, the hopping terms on A give $|\boldsymbol{S}| \sum_{\boldsymbol{k}} J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k})\, d^\dagger_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{A}\boldsymbol{k}}$, where each pair counted once contributes $J^{\mathrm{ex}}_{0\boldsymbol{\delta}}$ to both $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}$ and $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}$. The hopping terms on B give the same with $\boldsymbol{k} \to -\boldsymbol{k}$, a substitution that changes nothing. The pair terms give $\sum_i \sum_{\boldsymbol{\delta}\in\mathrm{B}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}}\, d_{\mathrm{A}i} d_{\mathrm{B},i+\boldsymbol{\delta}} = \sum_{\boldsymbol{k}} J^{\mathrm{ex}}_{\mathrm{AB}}(-\boldsymbol{k})\, d_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}$. Collecting all terms turns the Hamiltonian into a set of two-mode problems,

$$\begin{gathered} H_\mathrm{H} \simeq E_\mathrm{cl} + |\boldsymbol{S}| \sum_{\boldsymbol{k}} \Big[A_{\boldsymbol{k}} (d^\dagger_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{A}\boldsymbol{k}} + d^\dagger_{\mathrm{B}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}) + B_{\boldsymbol{k}} (d_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}} + d^\dagger_{\mathrm{B}\boldsymbol{k}} d^\dagger_{\mathrm{A}\boldsymbol{k}}) \Big], \\ A_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(0) - J^{\mathrm{ex}}_{\mathrm{AA}}(0) + J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k}), \qquad B_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}). \end{gathered} \tag{7.10}$$

The coefficient $A_{\boldsymbol{k}}$ is the cost of one deviation: the exchange field of the antiparallel neighbors minus that of the parallel ones, plus the hopping on its own sublattice. Without $B_{\boldsymbol{k}}$, Eq. (7.10) would describe a ferromagnet on each sublattice. At each $\boldsymbol{k}$, Eq. (7.10) is the two-mode Hamiltonian Eq. (C.59), with the diagonal energy $|\boldsymbol{S}|A_{\boldsymbol{k}}$ and the pairing amplitude $g_0 = |\boldsymbol{S}|B_{\boldsymbol{k}}$. The new operators

$$a_{\boldsymbol{k}1} = u^{\mathrm{M}}_{\boldsymbol{k}}\, d_{\mathrm{A}\boldsymbol{k}} + v^{\mathrm{M}}_{\boldsymbol{k}}\, d^\dagger_{\mathrm{B}\boldsymbol{k}}, \qquad a_{\boldsymbol{k}2} = u^{\mathrm{M}}_{\boldsymbol{k}}\, d_{\mathrm{B}\boldsymbol{k}} + v^{\mathrm{M}}_{\boldsymbol{k}}\, d^\dagger_{\mathrm{A}\boldsymbol{k}}, \tag{7.11}$$

with $u^{\mathrm{M}}_{\boldsymbol{k}} = \cosh\theta_{\boldsymbol{k}}$ and $v^{\mathrm{M}}_{\boldsymbol{k}} = \sinh\theta_{\boldsymbol{k}}$, are bosons because $(u^{\mathrm{M}}_{\boldsymbol{k}})^2 - (v^{\mathrm{M}}_{\boldsymbol{k}})^2 = 1$, and the pair terms vanish for $\tanh 2\theta_{\boldsymbol{k}} = B_{\boldsymbol{k}}/A_{\boldsymbol{k}}$. The functions $u^{\mathrm{M}}_{\boldsymbol{k}}$ and $v^{\mathrm{M}}_{\boldsymbol{k}}$ are the *magnon Bogoliubov coefficients*. Equivalently,

$$(u^{\mathrm{M}}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(\frac{A_{\boldsymbol{k}}}{\sqrt{A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}}} + 1\Big), \qquad (v^{\mathrm{M}}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(\frac{A_{\boldsymbol{k}}}{\sqrt{A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}}} - 1\Big), \qquad u^{\mathrm{M}}_{\boldsymbol{k}} v^{\mathrm{M}}_{\boldsymbol{k}} = \frac{B_{\boldsymbol{k}}}{2\sqrt{A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}}},$$

from $\cosh 2\theta = A/(A^2 - B^2)^{1/2}$ and $\sinh 2\theta = B/(A^2 - B^2)^{1/2}$. Eq. (C.60) then yields:

Linear spin waves of a two-sublattice antiferromagnet

$$H_{\mathrm{H}} \simeq E_{\mathrm{gs}} + \sum_{\boldsymbol{k}} \sum_{\nu=1,2} \omega^{\mathrm{M}}_{\boldsymbol{k}}\, a^{\dagger}_{\boldsymbol{k}\nu} a_{\boldsymbol{k}\nu}, \qquad \omega^{\mathrm{M}}_{\boldsymbol{k}} = |\boldsymbol{S}|\big(A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}\big)^{1/2}, \tag{7.12}$$

with the ground-state energy $E_{\mathrm{gs}} = E_{\mathrm{cl}} + \sum_{\boldsymbol{k}} (\omega^{\mathrm{M}}_{\boldsymbol{k}} - |\boldsymbol{S}| A_{\boldsymbol{k}})$ and $E_{\mathrm{cl}} = N_{\mathrm{s}} |\boldsymbol{S}|^2 [J^{\mathrm{ex}}_{\mathrm{AA}}(0) - J^{\mathrm{ex}}_{\mathrm{AB}}(0)]$.

The two branches are degenerate without anisotropy, and they differ in the spin they carry: $a^{\dagger}_{\boldsymbol{k}1} = u^{\mathrm{M}}_{\boldsymbol{k}} d^{\dagger}_{\mathrm{A}\boldsymbol{k}} + v^{\mathrm{M}}_{\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}$ adds a deviation on A or removes one from B, and each of the two lowers the total S^z by one, while $a^{\dagger}_{\boldsymbol{k}2}$ raises it by one. Since $\omega^{\mathrm{M}}_{\boldsymbol{k}} \le |\boldsymbol{S}| A_{\boldsymbol{k}}$, the ground state lies below the classical Néel energy: in the true ground state each site of A carries the zero-point deviation $(v^{\mathrm{M}}_{\boldsymbol{k}})^2$ per mode, which reduces the ordered moment. Linear spin-wave theory is consistent only if $A_{\boldsymbol{k}} \ge |B_{\boldsymbol{k}}|$ at every $\boldsymbol{k}$; otherwise $\omega^{\mathrm{M}}_{\boldsymbol{k}}$ is imaginary and the assumed order is unstable. The quartic terms dropped in the expansion in $1/|\boldsymbol{S}|$ change the magnon energies by a relative amount of order $1/(2|\boldsymbol{S}|)$ times a lattice sum of $(v^{\mathrm{M}}_{\boldsymbol{k}})^2$, an amount that is small for three-dimensional lattices and for $|\boldsymbol{S}| \ge 1$. The classical precession equations $\dot{\boldsymbol{S}}_i = -\boldsymbol{S}_i \times \sum_j J^{\mathrm{ex}}_{ij} \boldsymbol{S}_j$, linearized about the Néel state, give the same frequencies, Eq. (7.12).

The antiferromagnet differs from the ferromagnet at long wavelengths. At $\boldsymbol{k} = 0$, $A_0 = B_0 = J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ and $\omega^{\mathrm{M}}_0 = 0$, the Goldstone mode again. For small k, write $A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}} = (A_{\boldsymbol{k}} - B_{\boldsymbol{k}})(A_{\boldsymbol{k}} + B_{\boldsymbol{k}})$. The sum $A_{\boldsymbol{k}} + B_{\boldsymbol{k}}$ tends to $2J^{\mathrm{ex}}_{\mathrm{AB}}(0)$, and the difference is quadratic, $A_{\boldsymbol{k}} - B_{\boldsymbol{k}} = [J^{\mathrm{ex}}_{\mathrm{AB}}(0) - J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k})] - [J^{\mathrm{ex}}_{\mathrm{AA}}(0) - J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k})] \simeq \frac{1}{2}\sum_{\boldsymbol{\delta}\in\mathrm{B}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}} (\boldsymbol{k}\cdot\boldsymbol{\delta})^2 - \frac{1}{2}\sum_{\boldsymbol{\delta}\in\mathrm{A}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}} (\boldsymbol{k}\cdot\boldsymbol{\delta})^2$. Hence

$$\omega^{\mathrm{M}}_{\boldsymbol{k}} \simeq |\boldsymbol{S}| \Big\{ J^{\mathrm{ex}}_{\mathrm{AB}}(0) \Big[\sum_{\boldsymbol{\delta}\in\mathrm{B}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}} (\boldsymbol{k}\cdot\boldsymbol{\delta})^2 - \sum_{\boldsymbol{\delta}\in\mathrm{A}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}} (\boldsymbol{k}\cdot\boldsymbol{\delta})^2 \Big] \Big\}^{1/2}, \tag{7.13}$$

linear in k. The magnon velocity is the geometric mean of the large exchange field $|\boldsymbol{S}| J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ and the small stiffness in $A_{\boldsymbol{k}} - B_{\boldsymbol{k}}$. Physically, a long-wavelength deviation on one sublattice drags the other along through the pair term, and every cycle of the precession works against the full exchange field. For type-II order on the fcc lattice with J_2 alone, the 6 next-nearest neighbors at $\pm a\hat{\boldsymbol{x}}$, $\pm a\hat{\boldsymbol{y}}$, $\pm a\hat{\boldsymbol{z}}$, with a the cubic lattice constant, are all antiparallel, $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}) = 2J_2(\cos k_x a + \cos k_y a + \cos k_z a)$, and $J^{\mathrm{ex}}_{\mathrm{AA}} = 0$. Then

$$\omega^{\mathrm{M}}_{\boldsymbol{k}} = 6J_2|\boldsymbol{S}|\big(1 - \zeta^2_{\boldsymbol{k}}\big)^{1/2}, \qquad \zeta_{\boldsymbol{k}} = \tfrac{1}{3}(\cos k_x a + \cos k_y a + \cos k_z a), \tag{7.14}$$

with the maximum $6J_2|\boldsymbol{S}|$ where $\zeta_{\boldsymbol{k}} = 0$. With $\sum_{\boldsymbol{\delta}} (\boldsymbol{k}\cdot\boldsymbol{\delta})^2 = 2a^2k^2$, the small-$k$ law Eq. (7.13) becomes $\omega^{\mathrm{M}}_{\boldsymbol{k}} \simeq 2\sqrt{3}\, J_2|\boldsymbol{S}|\, ak$, the same in every direction. The nearest-neighbor J_1 adds 6 parallel bonds, in the same (111) plane, and 6 antiparallel ones, in the adjacent planes, and makes the magnon velocity depend on direction. Fig. 7.1 shows the magnon spectrum of NiO.

Worked example 7.2: The magnon velocity of NiO

With the exchange constants Eq. (7.5), $|\boldsymbol{S}| = 1$ for Ni^{2+}, and the measured lattice constant $a = 4.17$ Å as tabulated by Fischer et al. [1, Table I], Eq. (7.14) gives the magnon velocity for J_2 alone

$$2\sqrt{3}\, J_2|\boldsymbol{S}|\, a = 2\sqrt{3} \times 19.01\,\mathrm{meV} \times 4.17\,\text{Å} = 274.6\,\mathrm{meV\,Å}.$$

Dividing by $\hbar c = 1973.27\,\mathrm{eV\,Å}$ turns this velocity into 1.39×10^{-4} in natural units, or 41.7 km/s. The nearest-neighbor exchange modifies the velocity. Along [111], the direction normal to the ferromagnetic planes, the 6 parallel neighbors lie in a plane perpendicular to $\boldsymbol{k}$ and drop out of Eq. (7.13). Each of the 6 antiparallel ones has $(\boldsymbol{k}\cdot\boldsymbol{\delta})^2 = a^2k^2/3$, the same as each next-nearest neighbor contributes on average, and $J^{\mathrm{ex}}_{\mathrm{AB}}(0) = 6(J_1 + J_2)$.

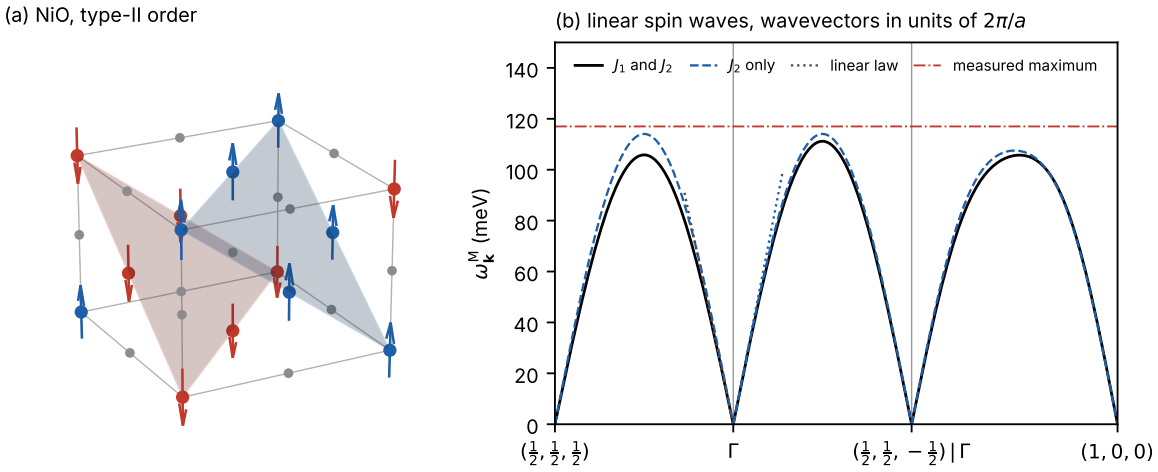


Figure 7.1: Magnetic order and spin waves of NiO. (a) Conventional cubic cell of the rock-salt structure: nickel ions (colored) on the fcc sites, oxygen ions (gray) between them. In the type-II order the spins of each (111) plane are parallel and alternate in sign from plane to plane; two of these planes are shaded. The arrows show only the sign of each moment; in NiO the moments lie within the planes, along a direction set by the weak anisotropy. (b) Magnon energies $\omega^{\mathrm{M}}_{\boldsymbol{k}}$ (meV) from linear spin-wave theory, Eq. (7.12), along three lines from the zone center Γ, with wavevectors in units of $2\pi/a$ and $a = 4.17\,\text{Å}$. Model: undistorted fcc lattice, Heisenberg exchange with $J_1 = -1.37\,\text{meV}$ and $J_2 = 19.01\,\text{meV}$ (each pair counted once, positive antiferromagnetic) measured by neutron scattering, $|\boldsymbol{S}| = 1$, no anisotropy. Solid: J_1 and J_2; dashed: J_2 alone, Eq. (7.14). Dotted: the linear laws $2\sqrt{3}(J_1 + J_2)|\boldsymbol{S}|ak$ along [111], and $2\sqrt{3}J_2|\boldsymbol{S}|ak$ for J_2 alone. Dash-dotted: the measured maximum, $\sim$117 meV. The point $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, along the stacking direction of the planes, is equivalent to Γ in the magnetic structure, and the energy returns to zero there. The zeros at $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ and $(1, 0, 0)$ are soft points of the undistorted model; in real NiO the small rhombohedral distortion lifts them to a few meV. The largest energy in the zone, $|\boldsymbol{S}|(6J_2 - 3J_1) = 118.2\,\text{meV}$, lies off these lines.

The bracket of Eq. (7.13) is $2a^2k^2(J_1 + J_2)$, and

$$\omega_{\boldsymbol{k}}^{\mathrm{M}} \simeq 2\sqrt{3}\,(J_1 + J_2)|\boldsymbol{S}|\,ak = 254.8\,\mathrm{meV\,\AA} \times k \qquad (\boldsymbol{k} \parallel [111]).$$

In other directions the parallel bonds no longer drop out, and evaluating Eq. (7.12) numerically at small k gives slopes between 255 and 269 meV Å. The measured initial slope is ∼250 meV Å.

The largest magnon energy follows from a bound. With $J_1 < 0$, the in-plane sum $J_{\mathrm{AA}}^{\mathrm{ex}}(\boldsymbol{k}) = J_1 \sum_{\boldsymbol{\delta}} \cos \boldsymbol{k} \cdot \boldsymbol{\delta}$ over the 6 parallel neighbors is largest when the cosines sum to their minimum, -3, the value of a triangular lattice at the corner of its zone. Then $A_{\boldsymbol{k}} \le J_{\mathrm{AB}}^{\mathrm{ex}}(0) - J_{\mathrm{AA}}^{\mathrm{ex}}(0) - 3J_1 = 6J_2 - 3J_1$, and $\omega_{\boldsymbol{k}}^{\mathrm{M}} \le |\boldsymbol{S}|A_{\boldsymbol{k}}$ with equality where $B_{\boldsymbol{k}} = 0$. The cosines reach their minimum and $B_{\boldsymbol{k}}$ vanishes at the same $\boldsymbol{k}$, where $\omega_{\mathrm{max}}^{\mathrm{M}} = |\boldsymbol{S}|(6J_2 - 3J_1) = 118.2$ meV, against $6J_2|\boldsymbol{S}| = 114.1$ meV for J_2 alone and a measured maximum of ∼117 meV. Linear spin-wave theory with two exchange constants thus reproduces both ends of the measured spectrum to a few percent. The agreement is partly by construction, since the exchange constants were fitted to the same data, and it does not test the $1/|\boldsymbol{S}|$ corrections, which a fit absorbs into J_1 and J_2.

The magnon velocity, 1.39×10^{-4}, is 19 times smaller than the fastest dark-matter speed in the laboratory, $v_{\mathrm{max}} = 2.650 \times 10^{-3}$ (Sect. 1.3), and five times larger than the longitudinal sound speed of silicon, 8.43 km/s along [100] (Appendix C.18). A dark-matter particle moves faster than the magnons it can emit. Emitting a magnon near the zone center is kinematically allowed in the same way as emitting an acoustic phonon, and at the same momentum transfer magnon emission deposits about five times more energy than phonon emission.

7.1.4 Spin-density response

A probe that couples to the spin of the electron, such as a slow neutron or a dark-matter particle with a spin-dependent interaction, couples to the *electron spin density*

$$\boldsymbol{s}_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\,\frac{\boldsymbol{\sigma}_i}{2}, \tag{7.15}$$

the spin analog of the density $n_e(\boldsymbol{q})$ of Sect. 2.3, with the Pauli matrices $\boldsymbol{\sigma}_i$ of electron i. Like $n_e(\boldsymbol{q})$, the spin density $\boldsymbol{s}_e(\boldsymbol{q})$ adds the momentum $\boldsymbol{q}$ to the target. For dark matter of spin $\frac{1}{2}$ with a spin-dependent coupling, the interaction takes the form $-\sum_a \hat{O}_{\chi a}(\boldsymbol{q})\, s_{e,a}(\boldsymbol{q})$ [3, Eq. (4)], with an operator $\hat{O}_{\chi a}$ that acts on the spin of the dark-matter particle; Sect. 8.1 lists and derives these operators. The three components $s_{e,a}(\boldsymbol{q})$ are then target operators O_a of the response matrix S_{ab} of Eq. (4.13), as the density was for a charge coupling.

In a magnetic insulator the spin density is localized on the ions. The filled shells carry no spin, and the unpaired electrons of ion j occupy an open shell whose spin, at low energy, is $\boldsymbol{S}_j$. Within the ground multiplet of the ion every vector operator is proportional to $\boldsymbol{S}_j$, by the Wigner–Eckart theorem: the matrix elements of any vector operator within a multiplet of fixed total spin are those of the spin itself times one constant. The spin density near ion j is therefore $\boldsymbol{S}_j\, \varrho_{\mathrm{s}}(\boldsymbol{r} - \boldsymbol{R}_j)$, with ϱ_{s} the normalized density of the unpaired electrons. The Fourier transform of the spin density is

$$\boldsymbol{s}_e(\boldsymbol{q}) = F^{\mathrm{mag}}(q) \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{S}_j, \qquad F^{\mathrm{mag}}(q) = \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\varrho_{\mathrm{s}}(\boldsymbol{r}), \tag{7.16}$$

where $F^{\mathrm{mag}}(q)$, the *magnetic form factor* of the ion, is spherically averaged and normalized to $F^{\mathrm{mag}}(0)=1$. It is the spin analog of the atomic form factor of Sect. 2.2. For a $3d$ shell it stays close to one up to the inverse shell radius, a few Å^{-1} or several keV (Appendix C.19), above most momentum transfers of sub-MeV dark matter. Appendix C.19 adds the orbital moment of shells that have one and derives the neutron cross-section, which measures the same S_{ab}. Eq. (7.16) reduces the response of the electrons to the response of the ionic spins, a quantity that linear spin-wave theory supplies.

For the ferromagnet, at zero temperature, the transverse spin operators create single magnons. Define $S^{\pm}(\boldsymbol{q})=\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}S_j^{\pm}$. At linear order $S_j^- \simeq (2|\boldsymbol{S}|)^{1/2}d_j^\dagger$, and with $d_j^\dagger = N^{-1/2}\sum_{\boldsymbol{k}}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_j}a_{\boldsymbol{k}}^\dagger$ and the lattice sum Eq. (3.14),

$$S^-(\boldsymbol{q}) \simeq (2|\boldsymbol{S}|N)^{1/2}\, a_{\boldsymbol{q}_r}^\dagger,$$

where $\boldsymbol{q}=\boldsymbol{q}_r+\boldsymbol{G}_{\boldsymbol{q}}$ is the split of Eq. (3.16) into a wavevector in the zone and a reciprocal-lattice vector. Crystal momentum is conserved: the magnon carries $\boldsymbol{q}_r$, and the lattice takes $\boldsymbol{G}_{\boldsymbol{q}}$. The operator $S^+(\boldsymbol{q})$ annihilates the ground state, which is fully polarized, and S^z is quadratic in the bosons and creates no single magnon. With $s_{e,x}=F^{\mathrm{mag}}(S^+ + S^-)/2$ and $s_{e,y}=F^{\mathrm{mag}}(S^+ - S^-)/2\mathrm{i}$, the one-magnon matrix elements are $\langle \boldsymbol{q}_r|s_{e,x}|0\rangle = F^{\mathrm{mag}}(2|\boldsymbol{S}|N)^{1/2}/2$ and $\langle \boldsymbol{q}_r|s_{e,y}|0\rangle = \mathrm{i}F^{\mathrm{mag}}(2|\boldsymbol{S}|N)^{1/2}/2$. Inserting them into Eq. (4.13) with $\mathscr{V}=N\Omega$, where Ω is the volume per magnetic ion, gives:

One-magnon spin response of a ferromagnet

$$S_{ab}(\boldsymbol{q},\omega) = \frac{\pi|\boldsymbol{S}|}{\Omega}\,|F^{\mathrm{mag}}(q)|^2\,(\delta_{ab}-\hat{z}_a\hat{z}_b+\mathrm{i}\epsilon_{abc}\hat{z}_c)\,\delta(\omega-\omega^{\mathrm{M}}_{\boldsymbol{q}_r}), \tag{7.17}$$

for $T=0$, $\omega>0$, and spins ordered along $\hat{\boldsymbol{z}}$, with a,b the Cartesian components of $\boldsymbol{s}_e$.

The matrix in brackets has the eigenvalues 2, for the circular combination $s_{e,x}-\mathrm{i}s_{e,y}$ that lowers the spin, and 0 for the other two: Eq. (7.17) is Hermitian and positive semidefinite, as Eq. (4.20) requires. A ferromagnet absorbs only the sense of rotation in which its spins precess, and this sense of rotation is the polarization vector $(|\boldsymbol{S}|/2)^{1/2}(1,\mathrm{i},0)$ of Ref. [3, Eq. (6)]. Apart from $|F^{\mathrm{mag}}|^2$ the one-magnon weight does not depend on $\boldsymbol{q}$. Through the fluctuation–dissipation theorem Eq. (4.18), the same delta function is a pole of the transverse spin response R_{ab}, the susceptibility that magnetic resonance and neutron scattering measure.

In the antiferromagnet a transverse spin operator again creates one magnon, now with the amplitudes of Eq. (7.11). The part of $S^+(\boldsymbol{k})$ on A is to linear order $S^+_{\mathrm{A}}(\boldsymbol{k})\equiv\sum_{i\in\mathrm{A}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_i}S_i^+ = (2|\boldsymbol{S}|N_{\mathrm{s}})^{1/2}d_{\mathrm{A},-\boldsymbol{k}}$, by the lattice sum $\sum_i \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{k}')\cdot\boldsymbol{R}_i}=N_{\mathrm{s}}\delta_{\boldsymbol{k},-\boldsymbol{k}'}$. Inverting Eq. (7.11) as in Appendix C.7 gives

$$d_{\mathrm{A}\boldsymbol{k}} = u^{\mathrm{M}}_{\boldsymbol{k}}\,a_{\boldsymbol{k}1} - v^{\mathrm{M}}_{\boldsymbol{k}}\,a^\dagger_{\boldsymbol{k}2}, \qquad d_{\mathrm{B}\boldsymbol{k}} = u^{\mathrm{M}}_{\boldsymbol{k}}\,a_{\boldsymbol{k}2} - v^{\mathrm{M}}_{\boldsymbol{k}}\,a^\dagger_{\boldsymbol{k}1}. \tag{7.18}$$

On B the raising operator is $S_j^+\simeq(2|\boldsymbol{S}|)^{1/2}d^\dagger_{\mathrm{B}j}$, and the same lattice sum turns $\sum_{j\in\mathrm{B}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_j}d^\dagger_{\mathrm{B}j}$ into $N_{\mathrm{s}}^{1/2}d^\dagger_{\mathrm{B},-\boldsymbol{k}}$. Since $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$, and with them $u^{\mathrm{M}}_{\boldsymbol{k}}$ and $v^{\mathrm{M}}_{\boldsymbol{k}}$, are even in $\boldsymbol{k}$, the sum of the contributions of both sublattices at small $\boldsymbol{k}$ is $S^+(\boldsymbol{k})\simeq(2|\boldsymbol{S}|N_{\mathrm{s}})^{1/2}(d_{\mathrm{A},-\boldsymbol{k}}+d^\dagger_{\mathrm{B},-\boldsymbol{k}}) = (2|\boldsymbol{S}|N_{\mathrm{s}})^{1/2}(u^{\mathrm{M}}_{\boldsymbol{k}}-v^{\mathrm{M}}_{\boldsymbol{k}})(a_{-\boldsymbol{k}1}+a^\dagger_{-\boldsymbol{k}2})$. Near the ordering wavevector, where the two sublattices enter with opposite phases, the one-magnon amplitude is $(2|\boldsymbol{S}|N_{\mathrm{s}})^{1/2}(u^{\mathrm{M}}_{\boldsymbol{k}}+v^{\mathrm{M}}_{\boldsymbol{k}})$. From the expressions of $\cosh 2\theta_{\boldsymbol{k}}$ and $\sinh 2\theta_{\boldsymbol{k}}$ in Appendix C.7,

$$(u^{\mathrm{M}}_{\boldsymbol{k}}\mp v^{\mathrm{M}}_{\boldsymbol{k}})^2 = \frac{A_{\boldsymbol{k}}\mp B_{\boldsymbol{k}}}{(A^2_{\boldsymbol{k}}-B^2_{\boldsymbol{k}})^{1/2}} = \frac{|\boldsymbol{S}|(A_{\boldsymbol{k}}\mp B_{\boldsymbol{k}})}{\omega^{\mathrm{M}}_{\boldsymbol{k}}}. \tag{7.19}$$

Near Γ, $A_{\boldsymbol{k}} - B_{\boldsymbol{k}} \propto k^2$ while $\omega^{\mathrm{M}}_{\boldsymbol{k}} \propto k$, so the weight near Γ vanishes linearly in k: a uniform field cannot tilt two opposite sublattices without working against the exchange field. Near the ordering wavevector the weight grows as the inverse distance from that wavevector. Unlike the ferromagnet, the antiferromagnet thus distributes its one-magnon weight unevenly: at the same $\boldsymbol{k}$ the weights near Γ and near the ordering wavevector differ by the factor $(A_{\boldsymbol{k}}+B_{\boldsymbol{k}})/(A_{\boldsymbol{k}}-B_{\boldsymbol{k}})$. Dark matter samples the suppressed weight near Γ unless its momentum transfer reaches the ordering wavevector, $(\pi/a)(1,1,1)$ in NiO, of length $\sqrt{3}\pi/a = 1.30\,\text{Å}^{-1} = 2.6\,\text{keV}$. Since $q \leq 2m_\chi v_{\max}$, reaching the ordering wavevector takes $m_\chi \gtrsim 0.5\,\text{MeV}$. The longitudinal component S^z creates magnons in pairs.

7.2 Superconductivity

Aluminum loses its electrical resistance below 1.18 K, and at the lowest temperatures tunneling measures an energy gap of 0.18 meV at its Fermi surface [7, Table III]. Unlike a band gap, the superconducting gap opens because the electrons within a few tens of meV of the Fermi surface bind into pairs, and an excitation that conserves the number of electrons must break a pair, at a cost of $2\Delta_{\mathrm{BCS}} \approx 0.36\,\text{meV}$. The pair-breaking energy $2\Delta_{\mathrm{BCS}}$ is the smallest quantum of Table 1.4, and by the bound $m_{\min} = 2E_{\mathrm{th}}/v^2_{\max}$ of Sect. 1.3 it would let dark matter far below a keV deposit detectable energy (Worked example 7.3) [8]. Superconducting films are also the sensors that read out most phonon-based detectors (Sect. 1.5). Where does the attraction that binds the pairs come from, and how strong is it? What are the excitations of the paired state, and how do the density and the spin density of the electrons create them? And does the answer respect charge conservation, which the simplest treatment of the paired state does not? We start from the electron–phonon vertex of Sect. 6.2.

7.2.1 The phonon-mediated attraction

Two electrons repel each other through the Coulomb interaction, yet in a metal they can attract each other by exchanging a phonon. Appendix D.9 showed how. In the diagram rules, a phonon line replaces the Coulomb line, and with the normalization of Eq. (A.24), in which the vertex g comes with $1/\sqrt{N}$, the rule Eq. (D.38) reads $-v_{\mathrm{C}}/\mathscr{V} \to -|g|^2 \mathsf{D}_0(\boldsymbol{Q},\omega)/N$. Two electrons that exchange energy ω through a phonon $(\boldsymbol{Q},\nu)$ therefore feel the interaction

$$\frac{1}{N}\,|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2\,\mathsf{D}_0(\boldsymbol{Q}\nu,\omega) = \frac{1}{N}\,\frac{2\omega_{\boldsymbol{Q}\nu}\,|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2}{\omega^2-\omega^2_{\boldsymbol{Q}\nu}}, \tag{7.20}$$

with g_ν the vertex of a single band at the Fermi level, whose band indices we drop. For $|\omega| < \omega_{\boldsymbol{Q}\nu}$ the interaction Eq. (7.20) is negative, an attraction. The attraction is retarded. An electron moving through the crystal pulls the ions toward its path, and the ions, which respond on the scale $1/\omega_{\boldsymbol{Q}\nu}$, are still displaced after the electron has left. A second electron that arrives within that time is attracted by the positive charge the first one left behind. The screened Coulomb repulsion between the two electrons acts only when they are close at the same moment, and when the second electron arrives the first one is already far away. Bardeen and Pines combined the two interactions in the electron gas and found the net interaction attractive for energy transfers below the phonon energy [9]. In a scattering of two electrons near the Fermi surface the energy transfer is $\omega = \xi_{\boldsymbol{k}'} - \xi_{\boldsymbol{k}}$, with $\xi_{\boldsymbol{k}}$ the band energy measured from the Fermi level. The attraction acts between electrons whose energies lie within about the Debye frequency ω_{D} of the Fermi surface, the scale of the largest phonon energies (Sect. 5.2).

The pairing strength. Take two electrons of opposite momentum and spin, $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$, and let them scatter into $(\boldsymbol{k}'\uparrow, -\boldsymbol{k}'\downarrow)$ with $\boldsymbol{Q} = \boldsymbol{k}' - \boldsymbol{k}$. For energy transfers small compared

with the phonon energy, Eq. (7.20) takes its static value $-2|g_\nu|^2/(N\omega_{\boldsymbol{Q}\nu})$. In the interaction Hamiltonian $(1/2N)\sum_{\boldsymbol{Q}} v(\boldsymbol{Q})\, n_e(\boldsymbol{Q}) n_e(-\boldsymbol{Q})$ the pair process appears twice, once with the spin-up electron in $n_e(\boldsymbol{Q})$ and once with the spin-down electron there, and the two appearances cancel the $\frac{1}{2}$. Time reversal makes the vertex of the spin-down electron, $g_\nu(-\boldsymbol{k}, -\boldsymbol{Q})$, the complex conjugate of $g_\nu(\boldsymbol{k}, \boldsymbol{Q})$, so the product of the two vertices is $|g_\nu(\boldsymbol{k}, \boldsymbol{Q})|^2$. The pair term is therefore

$$-\frac{2}{N}\sum_{\nu} \frac{|g_\nu(\boldsymbol{k}, \boldsymbol{Q})|^2}{\omega_{\boldsymbol{Q}\nu}}\; c^\dagger_{\boldsymbol{k}'\uparrow} c^\dagger_{-\boldsymbol{k}'\downarrow} c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}.$$

The model of Bardeen, Cooper and Schrieffer [10] replaces the coefficient by a constant $-V_{\rm BCS}/\mathcal{V}$ for all $\boldsymbol{k}$ and $\boldsymbol{k}'$ within $\omega_{\rm D}$ of the Fermi surface, and by zero outside that shell; here $\mathcal{V} = N\Omega$ is the volume of the crystal and Ω the unit-cell volume. The natural constant is the Fermi-surface average, $V_{\rm BCS} = 2\Omega\langle \sum_\nu |g_\nu|^2/\omega_{\boldsymbol{Q}\nu}\rangle_{\rm FS}$, where $\langle X\rangle_{\rm FS} = \sum_{\boldsymbol{k}\boldsymbol{k}'} X\, \delta(\xi_{\boldsymbol{k}})\delta(\xi_{\boldsymbol{k}'})/[\sum_{\boldsymbol{k}} \delta(\xi_{\boldsymbol{k}})]^2$ averages over both electrons on the Fermi surface. With $\sum_{\boldsymbol{k}} \delta(\xi_{\boldsymbol{k}}) = N\Omega\mathcal{N}_{\rm F}$ for one spin orientation and the Eliashberg function Eq. (6.34),

$$\mathcal{N}_{\rm F} V_{\rm BCS} = \frac{2}{\mathcal{N}_{\rm F}\Omega\, N^2} \sum_{\boldsymbol{k}\boldsymbol{Q}\nu} \frac{|g_\nu(\boldsymbol{k}, \boldsymbol{Q})|^2}{\omega_{\boldsymbol{Q}\nu}}\, \delta(\xi_{\boldsymbol{k}})\, \delta(\xi_{\boldsymbol{k}+\boldsymbol{Q}}) = 2\int_0^\infty \mathrm{d}\omega\, \frac{\alpha^2 F(\omega)}{\omega}.$$

The right side is the coupling constant of Eq. (6.35):

Phonon-mediated pairing strength

$$\mathcal{N}_{\rm F} V_{\rm BCS} = \lambda_{\rm ep}. \tag{7.21}$$

The dimensionless strength of the attraction is the same number that enhances the electron mass and that the phonon linewidths of Sect. 6.4 measure.

Eq. (7.21) leaves out the Coulomb repulsion and the quasiparticle renormalization, both of the same order as $\lambda_{\rm ep}$ itself. The screened Coulomb repulsion acts at all energy transfers, while the attraction acts only within $\omega_{\rm D}$ of the Fermi surface. The Fermi-surface average of the Coulomb repulsion is reduced by this mismatch of time scales to the Coulomb pseudopotential $\mu^*_{\rm C}$ of Sect. 28.1, of order 0.1, which subtracts from $\lambda_{\rm ep}$. The pairing electrons are, in addition, the quasiparticles of Sect. 6.4, with the residue $1/(1+\lambda_{\rm ep})$. The Eliashberg equations contain both the Coulomb pseudopotential and the quasiparticle renormalization. At low frequencies the renormalization function of Eq. (28.8) is $1+\lambda_{\rm ep}$, the mass enhancement. Suppose we go further and cut the frequency-dependent coupling Eq. (28.10) off at about $\omega_{\rm D}$, replacing it by $\lambda_{\rm ep}$ below and by zero above. The Eliashberg gap equation Eq. (28.9) then reduces to the BCS gap equation derived below, Eq. (7.29), written in Matsubara form, with the effective coupling

$$\mathcal{N}_{\rm F} V_{\rm BCS} \simeq \frac{\lambda_{\rm ep} - \mu^*_{\rm C}}{1+\lambda_{\rm ep}}. \tag{7.22}$$

For aluminum, with $\lambda_{\rm ep} = 0.44$ and $\mu^*_{\rm C} = 0.12$, the effective coupling is 0.22, half of the value in Eq. (7.21). In practice the gap of a real superconductor is computed by solving the Eliashberg equations with the α^2F of Sect. 6.4, which comes from density-functional perturbation theory and Wannier interpolation.

The reduced Hamiltonian. The attraction scatters pairs of any total momentum, but pairs of zero total momentum have the most final states. For every $\boldsymbol{k}$ in the shell, every $\boldsymbol{k}'$ in the shell is a possible final state of the pair $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$. A pair with total momentum $\boldsymbol{P} \neq 0$, by contrast, needs both $\boldsymbol{k}$ and $\boldsymbol{P}-\boldsymbol{k}$ in the shell, a condition that removes most of the phase

space once $|\boldsymbol{P}|$ exceeds $\omega_{\rm D}/v_{\rm F}$. The *reduced* BCS Hamiltonian keeps only the zero-momentum pairs:

$$H_{\rm BCS} = \sum_{\boldsymbol{k}\sigma} \xi_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma} - \frac{V_{\rm BCS}}{\mathscr{V}} {\sum_{\boldsymbol{k}\boldsymbol{k}'}}' c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}'\downarrow} c_{\boldsymbol{k}'\uparrow}, \tag{7.23}$$

with the prime restricting $|\xi_{\boldsymbol{k}}|$ and $|\xi_{\boldsymbol{k}'}|$ to less than $\omega_{\rm D}$. The pairs of finite momentum return when we ask how the paired state responds to a probe at wavevector $\boldsymbol{q}$.

7.2.2 Cooper pairs and the BCS state

Mean field. The paired state follows from the mean-field theory of Appendix D.14. Define the pair amplitude $b_{\boldsymbol{k}} = \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}\rangle$, which is zero in any state with a definite number of electrons, and the gap

$$\Delta_{\rm BCS} = \frac{V_{\rm BCS}}{\mathscr{V}} {\sum_{\boldsymbol{k}}}' b_{\boldsymbol{k}}, \tag{7.24}$$

real by the choice of an overall phase. In Eq. (7.23) each pair operator interacts with all $\sim N$ others with a strength of order $1/\mathscr{V}$, and replacing the sum of the other pair operators by its average, Eq. (7.24), makes an error that vanishes as $\mathscr{V}\to\infty$. The interaction then becomes the pair field $-\Delta_{\rm BCS}(c^\dagger_{\boldsymbol{k}\uparrow}c^\dagger_{-\boldsymbol{k}\downarrow} + c_{-\boldsymbol{k}\downarrow}c_{\boldsymbol{k}\uparrow})$ acting on each pair of modes, and for each $\boldsymbol{k}$ the mean-field Hamiltonian $H_{\rm MF}$ is the two-mode fermion problem Eq. (C.61) with $\xi_0 = \xi_{\boldsymbol{k}}$ and $g_0 = \Delta_{\rm BCS}$. The mean-field Hamiltonian is diagonalized by the fermionic Bogoliubov transformation of Appendix C.7, which uses sines and cosines where the bosonic transformation of the antiferromagnet in Sect. 7.1 used hyperbolic functions. The quasiparticle operators are

$$\gamma_{\boldsymbol{k}1} = u^{\rm B}_{\boldsymbol{k}}\, c_{\boldsymbol{k}\uparrow} - v^{\rm B}_{\boldsymbol{k}}\, c^\dagger_{-\boldsymbol{k}\downarrow}, \qquad \gamma_{\boldsymbol{k}2} = u^{\rm B}_{\boldsymbol{k}}\, c_{-\boldsymbol{k}\downarrow} + v^{\rm B}_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\uparrow}, \tag{7.25}$$

and by Eq. (C.62),

Bogoliubov quasiparticles

$$\begin{aligned} &H_{\rm MF} = {\sum_{\boldsymbol{k}}}' E_{\boldsymbol{k}}(\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1} + \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2}) + \text{const}, \qquad E_{\boldsymbol{k}} = \left(\xi^2_{\boldsymbol{k}} + \Delta^2_{\rm BCS}\right)^{1/2}, \\ &(u^{\rm B}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(1 + \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\Big), \qquad (v^{\rm B}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(1 - \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\Big), \qquad 2u^{\rm B}_{\boldsymbol{k}}v^{\rm B}_{\boldsymbol{k}} = \frac{\Delta_{\rm BCS}}{E_{\boldsymbol{k}}}. \end{aligned} \tag{7.26}$$

The coefficients $u^{\rm B}_{\boldsymbol{k}}$ and $v^{\rm B}_{\boldsymbol{k}}$ are the *coherence factors* of the BCS state. The quasiparticle $\gamma^\dagger_{\boldsymbol{k}1}$ carries momentum $\boldsymbol{k}$ and spin up. Far above the Fermi surface $u^{\rm B}_{\boldsymbol{k}}\to 1$ and it is an added electron; far below, $v^{\rm B}_{\boldsymbol{k}}\to 1$ and it is a removed electron of momentum $-\boldsymbol{k}$ and spin down, a hole; at the Fermi surface it is half of each, with energy $\Delta_{\rm BCS}$. The ground state, annihilated by every γ, is the BCS state

$$|{\rm BCS}\rangle = \prod_{\boldsymbol{k}} (u^{\rm B}_{\boldsymbol{k}} + v^{\rm B}_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow})|0\rangle, \tag{7.27}$$

in which each pair state $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$ is occupied with probability $(v^{\rm B}_{\boldsymbol{k}})^2$. The step of the Fermi sea is smeared over an energy $\Delta_{\rm BCS}$, and the coherent superposition of empty and full pair states is what makes $b_{\boldsymbol{k}} = u^{\rm B}_{\boldsymbol{k}}v^{\rm B}_{\boldsymbol{k}}$ nonzero. The state $|{\rm BCS}\rangle$ has no definite electron number, and its relative number fluctuations, of order $N^{-1/2}$, are negligible for energies but spoil the density response, which the last subsection of this section repairs.

Quasiparticle density of states. Since $\mathrm{d}E_{\boldsymbol{k}}/\mathrm{d}\xi_{\boldsymbol{k}} = \xi_{\boldsymbol{k}}/E_{\boldsymbol{k}}$, each energy $E > \Delta_{\mathrm{BCS}}$ is reached from two values of ξ, an electronlike state above the Fermi level and a holelike one below. With $\mathcal{N}_{\mathrm{F}}$ constant over the shell, the quasiparticles of one spin orientation have the density of states per volume

$$\mathcal{N}_{\mathrm{qp}}(E) = 2\mathcal{N}_{\mathrm{F}} \frac{E}{\sqrt{E^2 - \Delta_{\mathrm{BCS}}^2}} \Theta(E - \Delta_{\mathrm{BCS}}). \tag{7.28}$$

The factor 2 counts the two branches, not the spin; both spin orientations together have $g_{\mathrm{deg}}\mathcal{N}_{\mathrm{qp}}$, with $g_{\mathrm{deg}} = 2$. In the normal metal, $\mathcal{N}_{\mathrm{qp}} \to 2\mathcal{N}_{\mathrm{F}}$, the density of electron and hole excitations at energy E above the ground state. Pairing removes the states from $|E| < \Delta_{\mathrm{BCS}}$ and piles them up at the gap edge, where the density of states diverges as $(E - \Delta_{\mathrm{BCS}})^{-1/2}$ (Fig. 7.2a). The number of states is conserved: $\int_{\Delta_{\mathrm{BCS}}}^{\omega} \mathrm{d}E\, \mathcal{N}_{\mathrm{qp}} = 2\mathcal{N}_{\mathrm{F}}(\omega^2 - \Delta_{\mathrm{BCS}}^2)^{1/2}$, which approaches the normal-state count $2\mathcal{N}_{\mathrm{F}}\omega$ as ω grows.

The gap equation. The gap must reproduce itself through Eq. (7.24). In thermal equilibrium the quasiparticles are independent fermions, occupied with $n_{\mathrm{F}}(E_{\boldsymbol{k}})$, and inverting Eq. (7.25) yields $b_{\boldsymbol{k}} = u_{\boldsymbol{k}}^{\mathrm{B}} v_{\boldsymbol{k}}^{\mathrm{B}} [1 - 2n_{\mathrm{F}}(E_{\boldsymbol{k}})] = (\Delta_{\mathrm{BCS}}/2E_{\boldsymbol{k}}) \tanh(E_{\boldsymbol{k}}/2T)$. Inserting this pair amplitude into Eq. (7.24), dividing by Δ_{BCS} and replacing $\mathcal{V}^{-1}\sum_{\boldsymbol{k}}$ by $\mathcal{N}_{\mathrm{F}}\int \mathrm{d}\xi$ gives

BCS gap equation and its weak-coupling solutions

$$\begin{gathered} \frac{1}{\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}} = \int_0^{\omega_{\mathrm{D}}} \mathrm{d}\xi\, \frac{\tanh(\sqrt{\xi^2 + \Delta_{\mathrm{BCS}}^2}/2T)}{\sqrt{\xi^2 + \Delta_{\mathrm{BCS}}^2}}, \\ \Delta_{\mathrm{BCS}}(0) \simeq 2\omega_{\mathrm{D}}\, \mathrm{e}^{-1/\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}}, \qquad T_c = \frac{2\mathrm{e}^{\gamma_{\mathrm{E}}}}{\pi}\, \omega_{\mathrm{D}}\, \mathrm{e}^{-1/\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}}, \\ \frac{\Delta_{\mathrm{BCS}}(0)}{T_c} = \pi \mathrm{e}^{-\gamma_{\mathrm{E}}} = 1.764, \end{gathered} \tag{7.29}$$

with $\gamma_{\mathrm{E}} = 0.5772$ Euler's constant. The right side of the gap equation diverges logarithmically as Δ_{BCS} and T go to zero, and because of this logarithmic divergence a solution exists for any attraction. A weak attraction does not bind two particles in three-dimensional vacuum, where the density of pair states vanishes as $\xi^{1/2}$ and the integral converges; above a Fermi surface that density stays finite down to zero energy. At $T = 0$ the integral is $\operatorname{arsinh}(\omega_{\mathrm{D}}/\Delta_{\mathrm{BCS}})$, and the gap follows. At T_c the gap closes, and the integral fixes the critical temperature; Appendix D.14 evaluates this integral by expanding the hyperbolic tangent over Matsubara frequencies. The gap is nonanalytic in the coupling and invisible at any finite order of perturbation theory. Both $\Delta_{\mathrm{BCS}}(0)$ and T_c depend exponentially on $\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}$ and linearly on the cutoff, but their ratio depends on neither, and it is the same for every material within the model. Deviations from 1.764 measure the strength and the retardation of the real interaction. Aluminum, with the measured gap of 0.18 meV and critical temperature of 1.18 K, has $2\Delta_{\mathrm{BCS}}/T_c = 3.54$, against $2\pi\mathrm{e}^{-\gamma_{\mathrm{E}}} = 3.53$.

The pair-breaking threshold. A dark-matter particle or a phonon couples to the electrons through operators that conserve their number, such as the density $n_e(\boldsymbol{q})$ and the spin density. Expressed through Eq. (7.25), a number-conserving operator either scatters a quasiparticle that is already present or creates two at once, one from each branch; it never creates a single one, which would change the electron number by one. At $T = 0$ no quasiparticles are present, and the lowest excitation that a probe can make is a pair of quasiparticles, at

$$\omega \geq E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} \geq 2\Delta_{\mathrm{BCS}}, \tag{7.30}$$

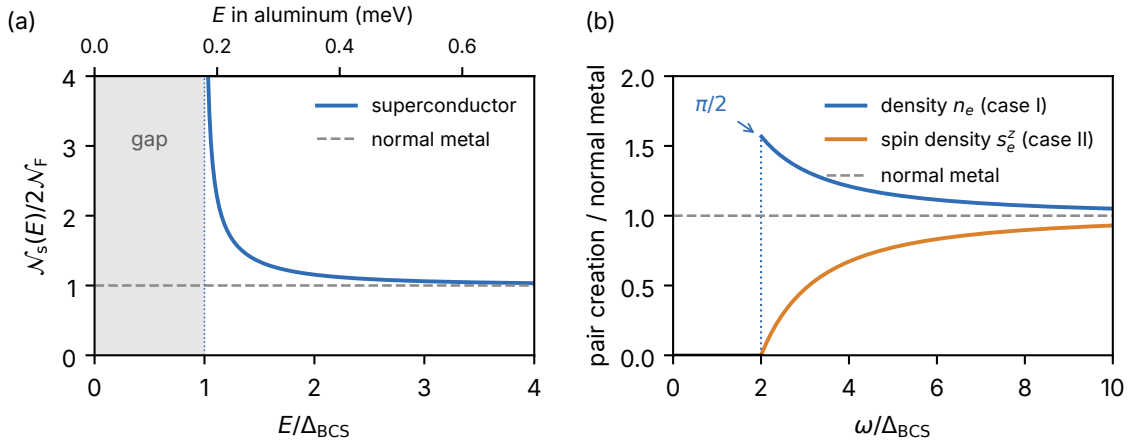


Figure 7.2: Quasiparticles of a BCS superconductor at $T = 0$. (a) Density of states per spin orientation, Eq. (7.28), in units of its normal-state value $2\mathcal{N}_F$: no states below Δ_{BCS} (gray), an integrable divergence at the edge. The top axis shows the energy for aluminum, $\Delta_{BCS} = 1.764\,T_c = 0.179\,\text{meV}$ with the measured $T_c = 1.18\,\text{K}$. (b) Rate of pair creation by a probe of energy ω coupled to the electron density (case I) or to the z component of the spin density (case II), divided by its normal-state value, Eq. (7.33). Both curves are bare quasiparticle bubbles for a linearized band with $q \ll k_F$ and $qv_F \geq \omega$, the regime of dark-matter scattering in aluminum, where the ratio does not depend on q. The density curve jumps to $\pi/2$ at the threshold $2\Delta_{BCS}$; the spin curve starts from zero and is the zero-temperature conductivity ratio of Mattis and Bardeen. Both curves tend to one for $\omega \gg \Delta_{BCS}$.

the *pair-breaking threshold*. It is reached when both quasiparticles lie at the Fermi surface, and for $q < 2k_F$ this configuration is always possible. In the language of Sect. 1.3, $E_{th} = 2\Delta_{BCS}$. At $T > 0$ thermally excited quasiparticles can absorb energy in smaller amounts, but their number is suppressed by $e^{-\Delta_{BCS}/T}$, and in aluminum at 10 mK the exponent is -208.

7.2.3 Coherence factors

The amplitude to break a pair depends on what the probe couples to, because each quasiparticle is part electron and part hole and the two parts can interfere. Take the density, $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}\sigma} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma} c_{\boldsymbol{k}\sigma}$, and the z component of the spin density of Eq. (7.15), $s^z_e(\boldsymbol{q}) = \frac{1}{2}\sum_{\boldsymbol{k}}(c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow} c_{\boldsymbol{k}\uparrow} - c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow} c_{\boldsymbol{k}\downarrow})$. Write $u = u^B_{\boldsymbol{k}}$, $v = v^B_{\boldsymbol{k}}$, and primes for $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q}$. Substituting the inverse of Eq. (7.25) into the spin-up term gives $c^\dagger_{\boldsymbol{k}'\uparrow} c_{\boldsymbol{k}\uparrow} = u'u\,\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1} + u'v\,\gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2} + \ldots$, and the spin-down term, after relabeling $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$, contributes $-vv'$ to quasiparticle scattering and uv' to pair creation (Appendix D.14). The density adds the spin-up and spin-down terms and the spin density subtracts them, so that the amplitudes are

$$\begin{aligned} n_e(\boldsymbol{q}):&\quad \text{pair creation } u'v + v'u, &&\quad \text{quasiparticle scattering } uu' - vv', \\ s^z_e(\boldsymbol{q}):&\quad \text{pair creation } \tfrac{1}{2}(u'v - v'u), &&\quad \text{quasiparticle scattering } \tfrac{1}{2}(uu' + vv'). \end{aligned} \tag{7.31}$$

Squaring the amplitudes and expressing them through the energies with Eq. (7.26) gives

Coherence factors of pair creation and quasiparticle scattering

$$(uv' \pm vu')^2 = \frac{1}{2}\Big(1 - \frac{\xi\xi' \mp \Delta_{\mathrm{BCS}}^2}{EE'}\Big), \qquad (uu' \mp vv')^2 = \frac{1}{2}\Big(1 + \frac{\xi\xi' \mp \Delta_{\mathrm{BCS}}^2}{EE'}\Big), \quad (7.32)$$

with the upper signs for the density and the lower signs for the spin density. The two operators differ in their behavior under time reversal: the density is even and the spin density odd. The even and the odd operators form the two classes that Tinkham calls case I and case II [11]. Every one-body probe is a sum of a part even and a part odd under time reversal, one from each class; the current, which drives the absorption of light, is odd and belongs to case II with the spin.

At the threshold both quasiparticles lie at the Fermi surface, $\xi = \xi' = 0$, and the creation factor is $\frac{1}{2}(1 \pm 1)$: one for the density, zero for the spin density. In the density the electron and hole parts of the pair add; in the spin density they cancel. Away from the Fermi surface the factors depend on the signs of ξ and ξ', and the dark-matter kinematics decide how the values of ξ and ξ' are sampled. For a linearized band, $\xi' = \xi + v_{\mathrm{F}} q \cos\theta$ with θ the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. Two quasiparticles with $E + E' = \omega$ have $|\xi' - \xi| \le |\xi| + |\xi'| < \omega$, so as long as $v_{\mathrm{F}} q \ge \omega$ every pair of values (ξ, ξ') in that range is reached at some angle. Since $\cos\theta$ is uniformly distributed over the Fermi sphere, each pair of values is reached with the same weight.

The regime $v_{\mathrm{F}} q \ge \omega$ is the regime of dark-matter scattering in a metal. A dark-matter particle that deposits ω transfers at least the momentum $q = \omega / v_{\max}$ (Sect. 1.3), and in aluminum $v_{\mathrm{F}} = 6.75 \times 10^{-3}$, the free-electron value of Sect. 6.4, is 2.5 times $v_{\max}$. Then ξ and ξ' are independent and uniformly distributed, and the terms odd in either one average to zero. The mean creation factor of the density becomes $\frac{1}{2}(1 + \Delta_{\mathrm{BCS}}^2 / EE')$, and that of the spin density $\frac{1}{2}(1 - \Delta_{\mathrm{BCS}}^2 / EE')$.

The pair-creation spectrum. With these factors the absorption spectrum of the paired state follows in closed form. At $T = 0$ the dynamic structure factor of the density is the golden-rule sum over pairs, $S(\boldsymbol{q}, \omega) = (2\pi/\mathcal{V}) \sum_{\boldsymbol{k}} (uv' + vu')^2 \, \delta(\omega - E_{\boldsymbol{k}} - E_{\boldsymbol{k}+\boldsymbol{q}})$; the sum over $\boldsymbol{k}$ with both branches covers both spin orientations, as the normal limit below confirms. Replace $\mathcal{V}^{-1} \sum_{\boldsymbol{k}}$ by $\mathcal{N}_{\mathrm{F}} \int \mathrm{d}\xi \int_{-1}^{1} \mathrm{d}\cos\theta/2$, trade the angle for ξ' at $\mathrm{d}\xi'/2v_{\mathrm{F}}q$, and change variables from ξ and ξ' to E and E' with the density of states Eq. (7.28) for each; this density of states absorbs the sum over the four sign combinations. In the normal state, $\Delta_{\mathrm{BCS}} = 0$, the result is $S_{\mathrm{n}}(\boldsymbol{q}, \omega) = 2\pi \mathcal{N}_{\mathrm{F}} \, \omega / v_{\mathrm{F}} q = m_e^2 \omega / \pi q$, which is the electron-gas response of Sect. 4.5 below the curve $\omega = q v_{\mathrm{F}} - q^2 / 2m_e$. In the superconductor,

Bare pair-creation spectra for $qv_{\mathrm{F}} \ge \omega$

$$\frac{S_{\mathrm{I,II}}(\boldsymbol{q}, \omega)}{S_{\mathrm{n}}(\boldsymbol{q}, \omega)} = \frac{1}{\omega} \int_{\Delta_{\mathrm{BCS}}}^{\omega - \Delta_{\mathrm{BCS}}} \mathrm{d}E \, \frac{E(\omega - E) \pm \Delta_{\mathrm{BCS}}^2}{\sqrt{E^2 - \Delta_{\mathrm{BCS}}^2} \, \sqrt{(\omega - E)^2 - \Delta_{\mathrm{BCS}}^2}} \, \Theta(\omega - 2\Delta_{\mathrm{BCS}}), \quad (7.33)$$

with the upper sign for the density (case I) and the lower sign for the spin density (case II), each divided by its own normal-state value. Fig. 7.2(b) shows both spectra. Near the threshold $E(\omega - E) \to \Delta_{\mathrm{BCS}}^2$, so the numerator of the integrand tends to $\Delta_{\mathrm{BCS}}^2 \pm \Delta_{\mathrm{BCS}}^2$, that is $2\Delta_{\mathrm{BCS}}^2$ for the density and zero for the spin density, while the denominator tends to $2\Delta_{\mathrm{BCS}}$ times the product of the two edge divergences of Eq. (7.28). The integral of that product over the shrinking interval stays finite, $\int \mathrm{d}E \, [(E - \Delta_{\mathrm{BCS}})(\omega - \Delta_{\mathrm{BCS}} - E)]^{-1/2} = \pi$. For the density the

integral is therefore $\Delta_{\mathrm{BCS}}\pi$, and with the prefactor $1/\omega = 1/2\Delta_{\mathrm{BCS}}$ the spectrum jumps at $\omega = 2\Delta_{\mathrm{BCS}}$ to $\pi/2$ times the normal value. Since $E(\omega - E) - \Delta_{\mathrm{BCS}}^2$ vanishes at the threshold, the spin spectrum instead rises continuously from zero. Far above the gap both spectra approach the normal metal: at $\omega = 4\Delta_{\mathrm{BCS}}$ the ratios are 1.21 for the density and 0.67 for the spin density. The case-II integral is the one that Mattis and Bardeen found for the absorption of light by a superconductor at zero temperature [12], in the limit of short mean free path, where ξ and ξ' are again independent. The case-II integral reproduces their closed form in elliptic integrals. For a dark-matter particle that couples to the electron density, the gap therefore enhances the bare rate just above threshold instead of suppressing it. The pile-up of states at the gap edge outweighs the narrow range of quasiparticle energies, from Δ_{BCS} to $\omega - \Delta_{\mathrm{BCS}}$, that is available just above threshold.

Quasiparticle scattering, the second process in Eq. (7.31), needs quasiparticles in the initial state. It is weighted by $n_{\mathrm{F}}(E)[1 - n_{\mathrm{F}}(E')]$ and absorbs energies below $2\Delta_{\mathrm{BCS}}$. Its coherence factors are the mirror images of the creation factors: the density scattering factor averages to $\frac{1}{2}(1 - \Delta_{\mathrm{BCS}}^2/EE')$ and vanishes at the gap edge, so the attenuation of sound in a superconductor falls with temperature as $2n_{\mathrm{F}}(\Delta_{\mathrm{BCS}})$ instead of showing the pile-up [11]. In a detector at millikelvin temperatures the quasiparticle-scattering channel is closed by the factor $\mathrm{e}^{-\Delta_{\mathrm{BCS}}/T}$.

Eq. (7.33) for the density is the bare quasiparticle bubble, the response of independent Bogoliubov quasiparticles, and not yet the density response of the superconductor.

7.2.4 Charge conservation and the Nambu formalism

At $\boldsymbol{q} = 0$ the density is the total electron number $\hat{N}_e$, which commutes with Eq. (7.23) and cannot excite anything. Yet by Eq. (7.31) its pair-creation amplitude at $\boldsymbol{q} = 0$ is $2uv = \Delta_{\mathrm{BCS}}/E_{\boldsymbol{k}}$, not zero. The error comes from the mean field. H_{MF} creates and destroys pairs, so it does not commute with $\hat{N}_e$, and its eigenstates have no definite electron number. A response computed from independent quasiparticles then puts spectral weight $\Delta_{\mathrm{BCS}}^2/E_{\boldsymbol{k}}^2$ at $\omega = 2E_{\boldsymbol{k}}$ into the uniform density response. With the gap equation at $T = 0$, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}}' 1/E_{\boldsymbol{k}} = 2/V_{\mathrm{BCS}}$, the first moment of this bare density response is

$$\int_0^\infty \mathrm{d}\omega\, \omega\, S^0(0,\omega) = \frac{4\pi}{\mathscr{V}}\sum_{\boldsymbol{k}}{}' \frac{\Delta_{\mathrm{BCS}}^2}{E_{\boldsymbol{k}}} = \frac{8\pi\Delta_{\mathrm{BCS}}^2}{V_{\mathrm{BCS}}}. \tag{7.34}$$

All bubbles here carry the restriction of the pair states to the shell, as in Eq. (7.23). The f-sum rule Eq. (4.30) requires the first moment to equal $\pi n_e q^2/m_e$, which vanishes at $q = 0$. The bare bubble violates it, and not only at $q = 0$: as long as $v_{\mathrm{F}}q \ll \omega_{\mathrm{D}}$ the spurious first moment keeps the value Eq. (7.34), since it comes from pairs spread over the whole shell, at frequencies up to $2\omega_{\mathrm{D}}$. For a linearized band the moment stays within 0.2 % of this value up to $v_{\mathrm{F}}q = 20\Delta_{\mathrm{BCS}}$. With $n_e = \frac{4}{3}\mathcal{N}_{\mathrm{F}}E_{\mathrm{F}}$ for free electrons, the spurious moment relative to the f-sum value is $12(\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q)^2/\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}}$. At the aluminum coupling $\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}} = 0.164$ of Worked example 7.3, obtained with a 40 meV cutoff, this ratio exceeds one for $v_{\mathrm{F}}q < 8.6\,\Delta_{\mathrm{BCS}}$. The bare bubble is a poor guide to the total spectral weight even at the momenta of dark-matter scattering.

Nambu form. The cure for this violation is easiest to see in the notation of Nambu [13], which treats the two operators that pairing mixes as one spinor, $\Psi_{\boldsymbol{k}} = (c_{\boldsymbol{k}\uparrow}, c_{-\boldsymbol{k}\downarrow}^\dagger)^{\mathrm{T}}$. The physical operators are the bilinears $O_A(\boldsymbol{q}) = \sum_{\boldsymbol{k}} \Psi_{\boldsymbol{k}+\boldsymbol{q}}^\dagger A\, \Psi_{\boldsymbol{k}}$ with the 2×2 matrices

$$\boldsymbol{\eta} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathsf{s}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{s}_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} \tag{7.35}$$

and for $\boldsymbol{q} \neq 0$, $O_{\boldsymbol{\eta}}(\boldsymbol{q}) = n_e(\boldsymbol{q})$ after anticommuting the spin-down term, while O_{s_1} and O_{s_2} create and destroy pairs of total momentum $\boldsymbol{q}$ (Appendix D.15). For a real gap, s_1 changes the magnitude of the pair field and s_2 its phase. Appendix C.7 uses the same matrix $\boldsymbol{\eta}$. The three matrices obey the algebra of the Pauli matrices, but we do not call them σ_i, because the σ_i act on the electron spin. In this notation $H_{\mathrm{MF}} = \sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}} \Psi_{\boldsymbol{k}} + \text{const}$ with $h_{\boldsymbol{k}} = \xi_{\boldsymbol{k}} \boldsymbol{\eta} - \Delta_{\mathrm{BCS}} \mathsf{s}_1$, whose eigenvalues are $\pm E_{\boldsymbol{k}}$, and the Matsubara Green's function of Sect. 4.7 becomes the 2×2 matrix $\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}) = (\mathrm{i}\omega^{\mathrm{F}} - h_{\boldsymbol{k}})^{-1}$, whose off-diagonal elements are the pair amplitudes. The bare bubble with vertices A and B is the loop of Sect. 4.8 with a trace over the Nambu index, and the trace for $A = B = \boldsymbol{\eta}$ reproduces the case-I factor of Eq. (7.32).

Since $[\mathsf{s}_1, \boldsymbol{\eta}] = -2\mathrm{i}\mathsf{s}_2$,

$$[H_{\mathrm{MF}}, n_e(\boldsymbol{q})] = \boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q}) + 2\mathrm{i}\Delta_{\mathrm{BCS}}\, O_{\mathsf{s}_2}(\boldsymbol{q}). \tag{7.36}$$

The kinetic term gives the current of the continuity equation Eq. (4.29), and the pair field adds a term that turns density into fluctuations of the pair phase. Under H_{MF}, charge leaks into the condensate and does not come back. The commutator of the full Hamiltonian with $n_e(\boldsymbol{q})$ has no pair-phase term, because the attraction $-V_{\mathrm{BCS}} \psi^\dagger_\uparrow \psi^\dagger_\downarrow \psi_\downarrow \psi_\uparrow$ is a function of the local densities and commutes with $n_e(\boldsymbol{q})$, up to corrections of relative order $v_{\mathrm{F}} q / \omega_{\mathrm{D}}$ from the shell (Appendix D.15).

The conserving response. The bare bubble misses that the pair field is itself made of electrons. A density perturbation at wavevector $\boldsymbol{q}$ changes the pair amplitudes, and the changed amplitudes act back on the quasiparticles as a pair field oscillating at $\boldsymbol{q}$ and ω. Anderson [14] included this back-action as the linear response of the time-dependent mean field. To let the pair field respond at $\boldsymbol{q}$, the attraction must scatter pairs of every total momentum, not only the zero-momentum pairs of Eq. (7.23); written in the pair channels it is $-(V_{\mathrm{BCS}}/4\mathcal{V}) \sum_{\boldsymbol{P}} \sum_{a=1,2} O_{\mathsf{s}_a}(\boldsymbol{P}) O_{\mathsf{s}_a}(-\boldsymbol{P})$ plus one-body terms. The mean field of this attraction at wavevector $\boldsymbol{q}$ couples the induced $\delta\langle O_{\mathsf{s}_a}\rangle$ back to O_{s_a}. The response is then the random-phase approximation of Sect. 4.5, with the pair channels $(\boldsymbol{\eta}, \mathsf{s}_1, \mathsf{s}_2)$ in place of the Coulomb channel and the kernel $K = \mathrm{diag}(0, -V_{\mathrm{BCS}}/2, -V_{\mathrm{BCS}}/2)$:

Conserving density response

$$R = R^0 + R^0 K R, \qquad R_{\boldsymbol{\eta}\boldsymbol{\eta}} = R^0_{\boldsymbol{\eta}\boldsymbol{\eta}} + \sum_{a,b=1,2} R^0_{\boldsymbol{\eta}\mathsf{s}_a} \Big[(1 - K R^0)^{-1} K \Big]_{ab} R^0_{\mathsf{s}_b \boldsymbol{\eta}}, \tag{7.37}$$

with the bare Nambu bubbles R^0_{AB} and the inverse taken in the block of the two pair channels. In diagrams the second term of Eq. (7.37) is the bare bubble with a ladder of pair propagators between its two vertices. That the response Eq. (7.37) conserves charge follows from a general property of mean-field dynamics (Appendix D.15). In a state for which Wick's theorem holds, the time derivative of any one-body average is the same whether it is computed with the self-consistent mean field or with the full Hamiltonian, and the full Hamiltonian obeys the continuity equation. The term $2\mathrm{i}\Delta_{\mathrm{BCS}}\, \delta\langle O_{\mathsf{s}_2}\rangle$ that Eq. (7.36) produces is cancelled by the commutator of n_e with the induced pair field. A response that obeys the continuity equation satisfies the f-sum rule by the double-commutator argument of Sect. 4.3.

At $\boldsymbol{q} = 0$ the cancellation can be seen in closed form. If the density of states is symmetric about the Fermi level over the shell, the bubbles that connect s_1 to $\boldsymbol{\eta}$ or to s_2 are odd in ξ and vanish, and the magnitude of the pair field decouples. We assume this particle–hole symmetry here; real metals break it weakly, and the general argument above does not need it.

The remaining bubbles are, with $F(\omega) = \mathscr{V}^{-1}\sum_{\boldsymbol{k}}'[E_{\boldsymbol{k}}(\omega^2 - 4E_{\boldsymbol{k}}^2)]^{-1}$,

$$R^0_{\boldsymbol{\eta\eta}} = 4\Delta^2_{\mathrm{BCS}}F, \qquad R^0_{\mathsf{s}_2\mathsf{s}_2} = \omega^2 F - \frac{2}{V_{\mathrm{BCS}}}, \qquad R^0_{\mathsf{s}_2\boldsymbol{\eta}} = -R^0_{\boldsymbol{\eta}\mathsf{s}_2} = 2\mathrm{i}\Delta_{\mathrm{BCS}}\,\omega F, \tag{7.38}$$

where the gap equation was used in $R^0_{\mathsf{s}_2\mathsf{s}_2}$ (Appendix D.15). The phase channel has the denominator $1 + \frac{1}{2}V_{\mathrm{BCS}}R^0_{\mathsf{s}_2\mathsf{s}_2} = \frac{1}{2}V_{\mathrm{BCS}}\omega^2 F$, and Eq. (7.37) gives

$$R_{\boldsymbol{\eta\eta}}(0,\omega) = 4\Delta^2_{\mathrm{BCS}}F + \frac{(-2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)(-\frac{1}{2}V_{\mathrm{BCS}})(2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)}{\frac{1}{2}V_{\mathrm{BCS}}\,\omega^2 F} = 4\Delta^2_{\mathrm{BCS}}F - 4\Delta^2_{\mathrm{BCS}}F = 0 \tag{7.39}$$

at every frequency. The phase fluctuations of the condensate absorb exactly the spurious pair-breaking weight. The cancellation used the gap equation twice, once in $R^0_{\mathsf{s}_2\mathsf{s}_2}$ and once through the coefficient V_{BCS} of the kernel: a response built on a gap that does not solve the gap equation does not conserve charge.

The Goldstone mode. The phase-channel denominator vanishes at $\omega = 0$ for $\boldsymbol{q} = 0$, because rotating the phase of Δ_{BCS} uniformly costs no energy, just as rotating all spins of the ferromagnet of Sect. 7.1 cost none. At small $\boldsymbol{q}$ the zero of the denominator moves to $\omega = c_{\mathrm{AB}}q$, and the response has a pole there: a sound wave of the condensate phase, the Anderson–Bogoliubov mode, with $c_{\mathrm{AB}} = v_{\mathrm{F}}/\sqrt{3}$ [14]. In a neutral superconductor this mode carries the density response below $2\Delta_{\mathrm{BCS}}$ and exhausts the f-sum rule as $q \to 0$. Electrons are charged, however. When the Coulomb interaction is added to Eq. (7.37) as the random-phase kernel $v_{\mathrm{C}}(q)$ in the $\boldsymbol{\eta}$ channel, it pushes the Anderson–Bogoliubov mode up to the plasma frequency of Sect. 4.5. The density response of a superconductor at low frequency is then screened as in the normal metal [14]. In the language of field theory the shift of the Goldstone mode to the plasma frequency is the Anderson–Higgs mechanism: a Goldstone mode coupled to a long-range gauge field becomes massive. Since a density–density interaction commutes with the density, the response stays conserving. The magnitude channel s_1, which decoupled above, carries a second collective mode at the pair-breaking edge $2\Delta_{\mathrm{BCS}}$, the amplitude or Higgs mode of the condensate. In a particle–hole-symmetric metal the density does not excite the amplitude mode.

The Ward identity. Charge conservation also constrains the building blocks of diagrams, the *vertices*: the amplitudes with which the density or the current scatters an electron from $(\boldsymbol{k},\nu)$ to $(\boldsymbol{k}+\boldsymbol{q},\nu+\omega)$, with ν the frequency of the incoming electron. The bare density vertex is the matrix $\boldsymbol{\eta}$ of the operator $O_{\boldsymbol{\eta}}$; the interactions dress it, as the ladder of Eq. (7.37) dresses the bubble. To find the constraint, start from the correlation function of the density with the two Nambu fields of the incoming and outgoing electron. The time derivative of this correlation function turns the density into the divergence of the current, by the continuity equation, and leaves two extra terms from the equal-time commutators of the density with the two fields, each with a single Green's function. Multiplying by G^{-1} on the left and on the right then removes the propagation of the incoming and the outgoing electron and leaves the dressed vertices alone, the dressed density vertex $\mathsf{V}_{\boldsymbol{\eta}}$ and current vertex $\mathbf{V}_{\boldsymbol{j}}$. They obey

Density Ward identity in Nambu form

$$\boldsymbol{q}\cdot\mathbf{V}_{\boldsymbol{j}}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k}) - \omega\,\mathsf{V}_{\boldsymbol{\eta}}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k}) = \boldsymbol{\eta}\,\mathsf{G}^{-1}(\boldsymbol{k},\nu) - \mathsf{G}^{-1}(\boldsymbol{k}+\boldsymbol{q},\nu+\omega)\,\boldsymbol{\eta}. \tag{7.40}$$

In words, the continuity equation tests the combination of vertices on the left, and the Green's functions of the two electrons fix that combination. In the normal state, with

$\mathsf{G}^{-1} = \nu - \xi_{\boldsymbol{k}}\boldsymbol{\eta}$, the right side is $-\omega\boldsymbol{\eta} + (\xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}})$, and the bare vertices $\mathsf{V}_{\boldsymbol{\eta}} = \boldsymbol{\eta}$ and $\mathbf{V}_{\boldsymbol{j}} = (\boldsymbol{k} + \boldsymbol{q}/2)/m_e$ satisfy the identity for a parabolic band. In the superconductor, $\mathsf{G}^{-1} = \nu - h_{\boldsymbol{k}}$ adds $\Delta_{\mathrm{BCS}}(\boldsymbol{\eta}\mathsf{s}_1 - \mathsf{s}_1\boldsymbol{\eta}) = 2\mathrm{i}\Delta_{\mathrm{BCS}}\mathsf{s}_2$ to the right side, the counterpart in the Ward identity of the pair-field term of Eq. (7.36). No bare vertex supplies the s_2 term, but the ladder of Eq. (7.37) does. The dressed density vertex of the ladder is $\mathsf{V}_{\boldsymbol{\eta}} = \boldsymbol{\eta} + \mathsf{c}_2\mathsf{s}_2$, and at $\boldsymbol{q} = 0$ the phase-channel coefficient is $\mathsf{c}_2 = -2\mathrm{i}\Delta_{\mathrm{BCS}}/\omega$, so that $-\omega\mathsf{V}_{\boldsymbol{\eta}} = -\omega\boldsymbol{\eta} + 2\mathrm{i}\Delta_{\mathrm{BCS}}\mathsf{s}_2$ is exactly the right side. The pole of c_2 at $\omega = 0$ is the Goldstone mode; at finite $\boldsymbol{q}$ it moves to $\omega = c_{\mathrm{AB}}q$, and the vertex keeps satisfying the identity.

The spin density needs no ladder correction. At $\boldsymbol{q} = 0$ it has the creation factor $\frac{1}{2}(uv - vu) = 0$, and its Nambu vertex is the unit matrix, which commutes with $h_{\boldsymbol{k}}$. Total S^z commutes with H_{MF}, and the bare quasiparticle response to a spin probe already conserves spin. The case-II curve of Fig. 7.2(b) is therefore final at this level. Eq. (7.40) is the identity for the density alone, local in the electron's momentum and frequency. Sect. 8.3 derives the identities that translation invariance imposes on every dark-matter operator.

For dark matter the size of the conserving correction is set by $\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q$, which is $1/\pi$ times the ratio of the probed wavelength $1/q$ to the *coherence length* $v_{\mathrm{F}}/\pi\Delta_{\mathrm{BCS}}$, the size of a pair. Since a deposit $\omega \geq 2\Delta_{\mathrm{BCS}}$ needs $q \geq \omega/v_{\mathrm{max}}$, the ratio $\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q$ is at most $v_{\mathrm{max}}/2v_{\mathrm{F}} = 0.20$ in aluminum. That bound is approached only for heavy particles, which can deposit $2\Delta_{\mathrm{BCS}}$ at the smallest momentum; the typical momentum transfer, $q \sim m_\chi v$, grows with the mass, and the typical ratio falls. The rate of pair breaking uses the conserving response of Eq. (7.37), screened by the Coulomb interaction, and Sects. 15.1 and 15.2 evaluate how the conserving response modifies Eq. (7.33) near the edge.

Worked example 7.3: The aluminum gap and the threshold mass

The measured critical temperature of aluminum, $T_c = 1.18\,\mathrm{K}$ [7, Table III], is $0.1017\,\mathrm{meV}$ in energy units. The weak-coupling ratio of Eq. (7.29) yields

$$\Delta_{\mathrm{BCS}} = 1.764 \times 0.1017\,\mathrm{meV} = 0.179\,\mathrm{meV}, \qquad 2\Delta_{\mathrm{BCS}} = 0.36\,\mathrm{meV},$$

in agreement with the measured gap of $0.18\,\mathrm{meV}$ [7, Table III]. Aluminum is a weak-coupling superconductor, and the ratio needs neither the cutoff nor the coupling.
The threshold mass follows from Eq. (1.24) with $E_{\mathrm{th}} = 2\Delta_{\mathrm{BCS}}$ and $v_{\mathrm{max}} = 2.650 \times 10^{-3}$:

$$m_{\mathrm{min}} = \frac{2 \times 0.36\,\mathrm{meV}}{(2.650 \times 10^{-3})^2} = 0.10\,\mathrm{keV}.$$

The threshold mass above holds for bulk aluminum. Thin aluminum films have higher critical temperatures, and dark-matter projections adopt the film convention $2\Delta_{\mathrm{BCS}} = 0.6\,\mathrm{meV}$ (Sect. 1.5), which moves the threshold mass to $0.17\,\mathrm{keV}$. At the threshold mass the particle moves at v_{max} and gives up its whole kinetic energy, with the momentum transfer $q = m_{\mathrm{min}}v_{\mathrm{max}} = 2 \times (2\Delta_{\mathrm{BCS}})/v_{\mathrm{max}} = 0.27\,\mathrm{eV}$, so $1/q = 0.73\,\mu\mathrm{m}$. With the free-electron $v_{\mathrm{F}} = 6.75 \times 10^{-3}$, the coherence length, the size of a pair, is $v_{\mathrm{F}}/\pi\Delta_{\mathrm{BCS}} = 2.4\,\mu\mathrm{m}$, three times larger than $1/q$, and $v_{\mathrm{F}}q = 10\,\Delta_{\mathrm{BCS}}$. The condition $v_{\mathrm{F}}q \geq \omega$ of Eq. (7.33) holds with room to spare, and the ratio $\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q = 0.10$ that controls the conserving correction is small but not negligible.
The coupling of Sect. 6.4 alone does much worse. The gap formula of Eq. (7.29) needs a cutoff, and with ω_{D} replaced by the largest phonon energy of aluminum, $40\,\mathrm{meV}$, the measured gap requires $\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}} = 1/\operatorname{arsinh}(\omega_{\mathrm{D}}/\Delta_{\mathrm{BCS}}) = 0.164$; with the logarithmic average phonon energy of Sect. 6.4, $\omega_{\log} = 270\,\mathrm{K} = 23.3\,\mathrm{meV}$ [7, Table III], as the cutoff the measured gap requires 0.18. The bare identification Eq. (7.21), $\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}} =$

$\lambda_{\rm ep} = 0.44$, would predict $T_c = 54\,\mathrm{K}$. The quasiparticle residue reduces the coupling to $\lambda_{\rm ep}/(1+\lambda_{\rm ep}) = 0.31$ and T_c to 20 K; the Coulomb pseudopotential, Eq. (7.22) with $\mu^*_{\rm C} = 0.12$, brings them to 0.22 and 5.8 K, still a factor 4.9 above the measured value. Since $\mathrm{d}\ln T_c/\mathrm{d}(\mathcal{N}_{\rm F}V_{\rm BCS}) = (\mathcal{N}_{\rm F}V_{\rm BCS})^{-2}$, a coupling a third too large produces the factor $\exp(1/0.164 - 1/0.222) = 4.9$, and the choice of cutoff between 40 meV and $\omega_{\log}$ changes the prefactor $1.134\,\omega_{\rm D}$ by another factor of 1.7. The Eliashberg equations with the full α^2F reproduce the measured T_c only after $\mu^*_{\rm C}$ has been fixed to it (Worked example 28.1). For a dark-matter target the gap is taken from measurement, and theory supplies its relation to T_c.

7.3 Dirac and narrow-gap materials

The pentatelluride $ZrTe_5$ has a band gap of about 24 meV, and near the gap its electrons move at about 10^{-3} of the speed of light whatever their energy. Dark matter as light as about 8 keV can carry an electron across this gap; silicon requires 0.333 MeV (Sect. 1.3), about forty times more. Materials like $ZrTe_5$ are called Dirac materials, because near the gap the electrons obey the Dirac equation of a relativistic particle, with a velocity $v_{\rm D}$ in place of the speed of light and half the gap in place of the mass. As in the superconductor of Sect. 7.2, the cheapest excitation is a pair whose energy starts at twice a half-gap, here an electron and a hole on either side of a band gap. Pairs are cheap in momentum too: the bands are linear, and slow enough, $v_{\rm D} < v_{\max}$, for a dark-matter particle to outrun them. And the same electrons screen the interaction in a way that is fixed by two numbers, $v_{\rm D}$ and the dielectric constant of everything else in the crystal.

7.3.1 The Dirac Hamiltonian from $\boldsymbol{k}\cdot\boldsymbol{p}$ theory

The $\boldsymbol{k}\cdot\boldsymbol{p}$ expansion of Sect. 3.3 assumed a band well separated from all others. Near a point where two bands nearly touch, the second-order formula Eq. (3.50) breaks down, because its energy denominator becomes small, and the two bands have to be treated together. Take a time-reversal-invariant momentum $\boldsymbol{k}_0$, such as Γ, in a crystal with inversion symmetry. Time reversal combined with inversion maps a Bloch state at $\boldsymbol{k}$ onto a different state at the same $\boldsymbol{k}$, and because time reversal squares to -1 for electrons, every band is then doubly degenerate at every $\boldsymbol{k}$: the Kramers degeneracy. Suppose that at $\boldsymbol{k}_0$ two of these Kramers doublets lie close to each other and far from all other bands: a conduction doublet at energy $+\Delta_{\rm D}$ and a valence doublet at $-\Delta_{\rm D}$, measured from the middle of the gap, with opposite parity under inversion. We write $\Delta_{\rm D}$ for this *half-gap*. Keep these four states exactly and drop the remote bands. At first order in $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_0$, the Bloch Hamiltonian of Sect. 3.3 acts in the four-state space through the matrix elements of $\delta\boldsymbol{k}\cdot\boldsymbol{v}$, with $\boldsymbol{v}$ the velocity operator of Eq. (3.40). The velocity operator is odd under inversion, and its matrix elements between two states of the same parity vanish. Only the block between the doublets survives:

$$H_{\boldsymbol{k}_0+\delta\boldsymbol{k}} = \begin{pmatrix} \Delta_{\rm D}\,\mathbb{1} & M(\delta\boldsymbol{k}) \\ M(\delta\boldsymbol{k})^\dagger & -\Delta_{\rm D}\,\mathbb{1} \end{pmatrix}, \qquad M(\delta\boldsymbol{k}) = \sum_{i=1}^{3} \delta k_i M_i, \tag{7.41}$$

where $\mathbb{1}$ is the 2×2 unit matrix and M_i is the 2×2 matrix of the velocity component v^i between the conduction and the valence doublet. The free-electron term $\delta k^2/2m_e$ and the second-order couplings to the remote bands are of order δk^2; they bend the bands at large δk, and we return to them when we fix the cutoff.

The Kramers degeneracy pins down the block $M(\delta\boldsymbol{k})$. Square Eq. (7.41). The off-diagonal blocks of H^2 are $\Delta_{\rm D}M - M\Delta_{\rm D}$, which cancel because $\Delta_{\rm D}$ multiplies the unit matrix, and H^2 is block diagonal, with the blocks $\Delta_{\rm D}^2 + MM^\dagger$ and $\Delta_{\rm D}^2 + M^\dagger M$. The eigenvalues of H come in pairs $\pm E$. If each is doubly degenerate at every $\delta\boldsymbol{k}$, as the Kramers degeneracy requires, then E^2 is a fourfold eigenvalue of H^2, and the 2×2 matrix $MM^\dagger$ must be a multiple of the unit matrix for every direction of $\delta\boldsymbol{k}$. The requirement on $MM^\dagger$ is a strong condition on the three matrices M_i. A general complex 2×2 matrix is a combination of the unit matrix and the three Pauli matrices, with eight real coefficients. The Pauli matrices here act within a doublet, not on the electron spin of Eq. (7.15), and we write them $\sigma_i^{\rm d}$, with d for doublet. Appendix D.16 shows step by step what the condition leaves. Up to a common unitary factor, which a choice of basis in the valence doublet absorbs, each M_i is a real combination of the four matrices $\not{1}$, $\mathrm{i}\sigma_1^{\rm d}$, $\mathrm{i}\sigma_2^{\rm d}$ and $\mathrm{i}\sigma_3^{\rm d}$: four real numbers per direction instead of eight. A unitary change of basis within each doublet turns the three four-vectors of real coefficients, one for each M_i, rigidly, by any rotation of four-dimensional space. A suitable rotation brings M to the form $M = \sum_i v_i\,\delta k_i\,\sigma_i^{\rm d}$ with three real velocities v_i along three principal axes. In that basis we define the 4×4 Dirac matrices

$$\alpha_{\rm D}^i = \begin{pmatrix} 0 & \sigma_i^{\rm d} \\ \sigma_i^{\rm d} & 0 \end{pmatrix}, \qquad \beta_{\rm D} = \begin{pmatrix} \not{1} & 0 \\ 0 & -\not{1} \end{pmatrix},$$

the matrices of Appendix C.15 with the subscript D that keeps them apart from the fine-structure constant and the inverse temperature. They anticommute pairwise and square to one. The four-state Hamiltonian then takes the Dirac form.

Dirac Hamiltonian

Near a time-reversal-invariant momentum $\boldsymbol{k}_0$ of a crystal with inversion symmetry, two Kramers doublets of opposite parity at $\pm\Delta_{\rm D}$ are described, to first order in $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_0$, by

$$H_{\boldsymbol{k}_0+\delta\boldsymbol{k}} = \sum_{i=1}^{3} v_i\,\delta k_i\,\alpha_{\rm D}^i + \Delta_{\rm D}\,\beta_{\rm D}, \qquad \varepsilon_{\pm,\delta\boldsymbol{k}} = \pm\Big(\sum_i v_i^2\delta k_i^2 + \Delta_{\rm D}^2\Big)^{1/2}, \tag{7.42}$$

with each band doubly degenerate. The velocities v_i are the singular values of the interband velocity matrix elements, and $\Delta_{\rm D}$ is half the energy separation of the two doublets at $\boldsymbol{k}_0$.

Eq. (7.42) is the Dirac Hamiltonian of a particle of mass $\Delta_{\rm D}$, with the speed of light replaced by v_i along the principal axis i; it is the form used for $ZrTe_5$ in Refs. [15, Eq. (1.1)] and [16]. When the crystal symmetry makes the three velocities equal to a common *Dirac velocity*, $v_i = v_{\rm D}$, the Hamiltonian becomes $H = v_{\rm D}\boldsymbol{\alpha}_{\rm D}\cdot\delta\boldsymbol{k} + \Delta_{\rm D}\beta_{\rm D}$ with $E_{\boldsymbol{k}} = (v_{\rm D}^2\delta k^2 + \Delta_{\rm D}^2)^{1/2}$, in the notation of Appendix D.16. The literature usually writes $v_{\rm F}$ for the Dirac velocity; in this book $v_{\rm F}$ is the Fermi velocity of the electron gas. One of the v_i may come out negative in the last rotation (Appendix D.16). The sign of that velocity does not enter the spectrum or any other result of this section, since each of them depends on v_i^2, and we take all three velocities positive.

The two-level Kane model of Sect. 3.3 has the same form. There the conduction state of GaAs couples to the valence state $|X\rangle$ through $\delta k_x v_{cX}^x$ across the gap $E_{\rm g}$, and the two levels repel as $\pm[(E_{\rm g}/2)^2 + |v_{cX}^x|^2\delta k_x^2]^{1/2}$, which is Eq. (7.42) with $\Delta_{\rm D} = E_{\rm g}/2$ and $v_{\rm D} = |v_{cX}^x|$. Near its minimum the conduction band is parabolic, with mass $\Delta_{\rm D}/v_{\rm D}^2 = m_e E_{\rm g}/E_P$: the Kane mass Eq. (3.52) without the free-electron term. With $E_P = 22.9\,\mathrm{eV}$ ($|v_{cX}^x| = 4.7\times 10^{-3}$) and the GaAs gap of $1.52\,\mathrm{eV}$, this mass is $0.066\,m_e$, within 2 % of the measured $0.067\,m_e$.

A narrow-gap semiconductor is thus a massive Dirac material, and since the Kane energy changes little from one III–V compound to the next, its Dirac velocity stays near 4.7×10^{-3} as the gap closes. A Dirac velocity of 4.7×10^{-3} is $1.8\, v_{\max}$, and we will see below that it is too fast for dark-matter scattering.

At $\Delta_{\mathrm{D}} = 0$ the conduction and valence doublets are degenerate at $\boldsymbol{k}_0$, and Eq. (7.42) separates. In the basis of the sums and differences of the conduction and valence states, it becomes two 2×2 blocks, $\pm \sum_i v_i\, \delta k_i\, \sigma_i^{\mathrm{d}}$. Each block is a *Weyl node*, a crossing of two nondegenerate bands that uses all three Pauli matrices, so that no small perturbation of the block alone can open a gap. The chirality of a node is the sign in front of its block, and the two nodes have opposite chirality; only a perturbation that couples them, such as the mass term $\Delta_{\mathrm{D}}\beta_{\mathrm{D}}$, gaps the spectrum. A crystal is a Dirac semimetal when its two Weyl nodes lie at the same momentum and a symmetry keeps $\Delta_{\mathrm{D}} = 0$. Na_3Bi and Cd_3As_2 are examples, although their nodes lie away from the time-reversal-invariant momenta, on a rotation axis where a crystal rotation protects them. If inversion or time reversal is broken, the Kramers degeneracy is lifted, the nodes can move apart in momentum, and the material is a Weyl semimetal [17]. The sign of Δ_{D} distinguishes an ordinary from an inverted band order, and pressure, strain or composition can tune Δ_{D} through zero. Because $ZrTe_5$ lies close to this sign change, its gap is small and first-principles values of the gap disagree: they range from 24 to 35 meV [15, Table 1] and [16, Table 1], and the gap is sensitive to the lattice constant used.

The linear form Eq. (7.42) holds while the neglected terms of order δk^2 are small compared with $v_i \delta k_i$. Beyond some momentum the bands bend away, and the states there behave like those of an ordinary insulator. We model the bending by a hard *momentum cutoff* Λ_{D} on $|\delta\boldsymbol{k}|$, inside which the Dirac form is used and outside which the states are counted with the remote bands. The value of Λ_{D} is read off a computed band structure, as the momentum at which the bands deviate from linear; for $ZrTe_5$ it is 0.1 to $0.23\,\text{Å}^{-1}$ depending on the direction, or 200 to 450 eV. Finally, let g_{deg} count the degenerate states per band and wavevector, summed over all Dirac points in the zone: $g_{\mathrm{deg}} = 2$ for one four-component cone, and twice the number of cones in general.

7.3.2 Density of states

How many states does a Dirac cone offer at energy E? For the isotropic cone the number of states per volume in one band with energy below E is g_{deg} times the volume of a sphere of radius $\delta k = (E^2 - \Delta_{\mathrm{D}}^2)^{1/2}/v_{\mathrm{D}}$, divided by $(2\pi)^3$. Differentiating with $\delta k\, \mathrm{d}(\delta k) = E\, \mathrm{d}E/v_{\mathrm{D}}^2$ gives the density of states per volume of the conduction band,

$$\mathcal{N}_{\mathrm{D}}(E) = \frac{g_{\mathrm{deg}} \delta k^2}{2\pi^2} \frac{\mathrm{d}(\delta k)}{\mathrm{d}E} = \frac{g_{\mathrm{deg}}}{2\pi^2 v_{\mathrm{D}}^3}\, E\, (E^2 - \Delta_{\mathrm{D}}^2)^{1/2}\, \Theta(E - \Delta_{\mathrm{D}}), \tag{7.43}$$

and the valence band mirrors it at $-E$. For an anisotropic cone, the substitution $\delta k_i \to \delta k_i v_{\mathrm{D}}/v_i$ shows that v_{D}^3 is replaced by $v_1 v_2 v_3$. Just above the edge, $E - \Delta_{\mathrm{D}} \ll \Delta_{\mathrm{D}}$, the band is parabolic with mass $\Delta_{\mathrm{D}}/v_{\mathrm{D}}^2$ and $\mathcal{N}_{\mathrm{D}}$ grows as $(E - \Delta_{\mathrm{D}})^{1/2}$, as at any band edge. Far above it, or at any energy in a gapless cone, $\mathcal{N}_{\mathrm{D}} = g_{\mathrm{deg}} E^2/(2\pi^2 v_{\mathrm{D}}^3)$, the density of states of massless particles, which vanishes at the node.

The vanishing density of states at $E = 0$ means that an undoped gapless cone has no Fermi surface. Thomas–Fermi screening (Eq. (4.53)), which needs a finite density of states at the Fermi level, is then absent, and whatever screening the Dirac electrons provide comes from interband pairs, which we compute below. The factor $1/v_{\mathrm{D}}^3$ means that slow cones have many states: at fixed E, halving the velocity multiplies the number of states by eight. The density of thermal carriers follows from the parabolic band edge. The symmetry of the two bands puts the chemical potential in the middle of the gap. The conduction-electron density

is then the intrinsic density Eq. (3.48) with both density-of-states masses equal to $\Delta_{\mathrm{D}}/v_{\mathrm{D}}^2$, the spin factor 2 replaced by g_{deg} and $E_{\mathrm{g}} = 2\Delta_{\mathrm{D}}$:

$$n_{\mathrm{i}} = g_{\mathrm{deg}} \Big(\frac{\Delta_{\mathrm{D}} T}{2\pi v_{\mathrm{D}}^2} \Big)^{3/2} \mathrm{e}^{-\Delta_{\mathrm{D}}/T}, \qquad T \ll \Delta_{\mathrm{D}}. \tag{7.44}$$

For the $ZrTe_5$-like cone of Worked example 7.4, with $\Delta_{\mathrm{D}} = 11.8\,\mathrm{meV}$, $v_{\mathrm{D}} = 7.9 \times 10^{-4}$ and $g_{\mathrm{deg}} = 2$, the band-edge mass $\Delta_{\mathrm{D}}/v_{\mathrm{D}}^2$ is $0.037\,m_e$. At room temperature $\Delta_{\mathrm{D}}/T = 0.46$ and the material is full of carriers. At 4 K, $\Delta_{\mathrm{D}}/T = 34$ and Eq. (7.44) gives $n_{\mathrm{i}} \approx 0.4\,\mathrm{cm}^{-3}$, within 5 % of the exact integral over Eq. (7.43); at the millikelvin temperatures of a phonon or charge readout the thermal population is zero for all purposes. A gap of tens of meV is small enough for light dark matter and large enough to freeze out the carriers.

7.3.3 Interband pair threshold

At $T = 0$, with the chemical potential in the gap, the lower band is full and the upper band empty. The excitations are pairs: an electron at $\boldsymbol{k} + \boldsymbol{q}$ in the upper band and a hole at $\boldsymbol{k}$ in the lower one, with energy $\omega = E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}}$. For given $\boldsymbol{q}$, the cheapest pair has $\boldsymbol{k} = -\boldsymbol{q}/2$. The reason is that $E_{\boldsymbol{k}}$ is even and convex in $\boldsymbol{k}$, so that $E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} = E_{-\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} \geq 2E_{\boldsymbol{q}/2}$, with equality at $\boldsymbol{k} = -\boldsymbol{q}/2$. The pair continuum is therefore

$$\omega \geq (v_{\mathrm{D}}^2 q^2 + 4\Delta_{\mathrm{D}}^2)^{1/2}, \tag{7.45}$$

the shaded region of Fig. 7.3b. The edge is the dispersion of a relativistic particle of mass $2\Delta_{\mathrm{D}}$ and momentum q, with v_{D} in place of the speed of light. In a gapless cone the edge is the line $\omega = v_{\mathrm{D}} q$. Compare the electron gas of Sect. 4.5, whose continuum reaches down to $\omega = 0$ at every $q < 2k_{\mathrm{F}}$: pairs in a Fermi sea can be made at no energy cost, while in a Dirac cone every pair costs at least $v_{\mathrm{D}} q$.

Dark matter of mass m_χ can deposit at most $\omega = q v_{\max} - q^2/2m_\chi$ at momentum q (Eq. (1.25)), and a pair is kinematically allowed where this curve enters the continuum. Fig. 7.3b shows that the Dirac velocity decides whether a cone can be reached at all, and the gap decides at which mass. The dark-matter curve lies below the line $\omega = v_{\max} q$, and the continuum edge lies above the line $\omega = v_{\mathrm{D}} q$, so that a cone with $v_{\mathrm{D}} > v_{\max}$ is out of reach of scattering at any mass, with or without a gap. The requirement $v_{\mathrm{D}} < v_{\max}$ rules out the III–V narrow-gap semiconductors, with $v_{\mathrm{D}} \approx 1.8\, v_{\max}$, as scattering targets (absorption at $q \approx 0$ is another matter), and it is the reason the slow cones of $ZrTe_5$, at $v_{\mathrm{D}} = 0.30\, v_{\max}$, are of interest. In the limit $v_{\mathrm{D}} \ll v_{\max}$ the edge is flat at $2\Delta_{\mathrm{D}}$, and the threshold mass is that of Eq. (1.24) with $E_{\mathrm{th}} = 2\Delta_{\mathrm{D}}$, $m_{\min} = 4\Delta_{\mathrm{D}}/v_{\max}^2 = 6.7\,\mathrm{keV}$ for $\Delta_{\mathrm{D}} = 11.8\,\mathrm{meV}$. The finite Dirac velocity raises the threshold mass: the smallest m_χ for which $q v_{\max} - q^2/2m_\chi \geq (v_{\mathrm{D}}^2 q^2 + 4\Delta_{\mathrm{D}}^2)^{1/2}$ at some q, found numerically, is 8.0 keV at $v_{\mathrm{D}} = 7.9 \times 10^{-4}$. In an anisotropic cone, v_{D} in Eq. (7.45) is replaced by $(\sum_i v_i^2 \hat{q}_i^2)^{1/2}$ for $\boldsymbol{q}$ along the unit vector $\hat{\boldsymbol{q}}$; for $ZrTe_5$ the threshold mass is 7.1 keV along its slowest axis and 9.6 keV along its fastest. The dark-matter wind arrives from a fixed direction in the Galaxy. In the laboratory this direction turns once a day with the rotation of the Earth, so the direction dependence of the threshold mass is a source of daily modulation [15].

7.3.4 Screening by Dirac electrons

The remote bands that we dropped from Eq. (7.41) still contribute to the response. They are the core and the bonding states of the crystal, separated from the Dirac bands by an eV or more, and at the energies $\omega \lesssim 0.1\,\mathrm{eV}$ and momenta $q \lesssim \Lambda_{\mathrm{D}}$ that matter here they respond instantly and uniformly, like a dielectric medium. We describe them by a constant,

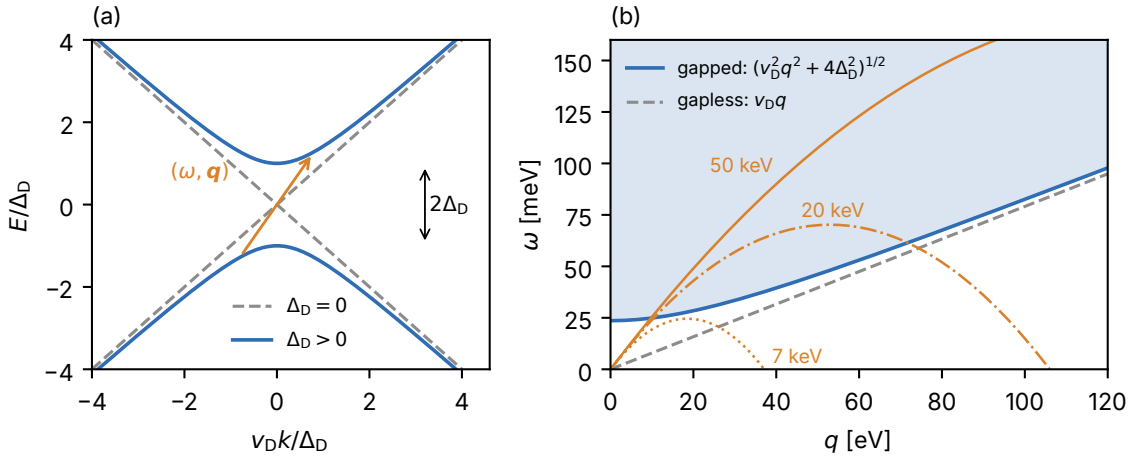


Figure 7.3: Dirac cones and their pairs. (a) Bands $\pm(v_D^2 k^2 + \Delta_D^2)^{1/2}$ of a gapped cone (solid) and of a gapless one (dashed), in units of the half-gap Δ_D; the arrow is an interband pair with energy ω and momentum $\boldsymbol{q}$. (b) Energy ω against momentum q of the pairs of an isotropic cone with the $ZrTe_5$-like parameters $v_D = 7.9 \times 10^{-4}$ (geometric mean of the three principal velocities) and $\Delta_D = 11.8\,\text{meV}$ of Ref. [15, Table 1 and text below it]. Shaded: the pair continuum $\omega \geq (v_D^2 q^2 + 4\Delta_D^2)^{1/2}$; dashed: its edge $\omega = v_D q$ for $\Delta_D = 0$. Orange: the largest energy that dark matter of mass 7, 20 and 50 keV can deposit at momentum q, $\omega = q v_{\max} - q^2/2m_\chi$ with $v_{\max} = 2.650 \times 10^{-3}$. The 7 keV curve stays just below the continuum; the threshold mass is 8.0 keV.

the *background dielectric constant* ε_{bg}: the electronic dielectric constant ε_∞ of Sect. 4.4 that the crystal would have without its Dirac electrons. In $ZrTe_5$ the remote interband transitions and the polar lattice make ε_{bg} large, between about 20 and 300 depending on the direction. Two charges among the Dirac electrons then interact through $e^2/(\varepsilon_{bg} q^2)$, not e^2/q^2, and the Dirac electrons screen on top of the background. In the random-phase approximation of Sect. 4.5, with R_D^0 the independent-pair response of the Dirac electrons,

$$\varepsilon(\boldsymbol{q}, \omega) = \varepsilon_{bg} - v_C(q)\, R_D^0(\boldsymbol{q}, \omega) = \varepsilon_{bg}\Big[1 - \frac{e^2}{\varepsilon_{bg} q^2} R_D^0(\boldsymbol{q}, \omega)\Big]. \tag{7.46}$$

The bare Coulomb interaction $v_C = e^2/q^2$ multiplies the Dirac response, and the background enters as an additive constant. Inside the bracket the Dirac electrons see the coupling e^2/ε_{bg}. This placement of ε_{bg} fixes the overall size of every rate in a Dirac target.

How strong is that coupling? A pair of Dirac electrons at distance r has Coulomb energy $\alpha/(\varepsilon_{bg} r)$ and kinetic energy of order v_D/r, since a particle localized within r has momentum $\sim 1/r$ and energy v_D/r. The ratio of the Coulomb energy to the kinetic energy,

$$\tilde{\alpha} = \frac{\alpha}{\varepsilon_{bg} v_D}, \tag{7.47}$$

is the same at every distance. The *effective coupling* $\tilde{\alpha}$ is the analog of the fine-structure constant of quantum electrodynamics, where $v_D = 1$ and $\varepsilon_{bg} = 1$. A small velocity makes it large, and a large background constant makes it small again. For $ZrTe_5$ the Dirac velocity and the background constant compete: $1/v_D \approx 1300$ and $\varepsilon_{bg} \approx 50$ on average, and $\tilde{\alpha} = 0.18$ to 0.20 with $g_{deg} = 2$ (Worked example 7.4). Part of the literature folds the degeneracy into the coupling and writes $g_{deg}\alpha/(\varepsilon_{bg} v_D)$ for what we write $g_{deg}\tilde{\alpha}$ [16, Eq. (2.5)].

Appendix D.16 evaluates R_{D}^{0} for one isotropic cone at $T=0$ with the chemical potential in the gap. Only interband pairs contribute, and each enters with the overlap of the band spinors, Eq. (D.59), which vanishes at $q=0$, because the two bands are orthogonal, and grows as q^2. That factor q^2 cancels the $1/q^2$ of the Coulomb interaction. In a metal the Thomas–Fermi term k_{TF}^2/q^2 diverges as $q\to 0$; in a Dirac material the Dirac contribution to ε stays finite, as it does in an insulator. The sum over $\boldsymbol{k}$ diverges logarithmically at large $\boldsymbol{k}$, for the same reason as the vacuum polarization of quantum electrodynamics, and the cutoff $|\boldsymbol{k}|<\Lambda_{\mathrm{D}}$ makes it finite. The result is

$$\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_{\mathrm{bg}}} = 1+\frac{g_{\mathrm{deg}}\tilde{\alpha}}{6\pi}\Big[\ln\frac{4v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2}{\Delta_{\mathrm{D}}^2}-\frac{5}{3} - 6\int_0^1 \mathrm{d}x\, x(1-x)\ln\Big(1-x(1-x)\frac{\omega^2-v_{\mathrm{D}}^2q^2}{\Delta_{\mathrm{D}}^2}-\mathrm{i}0^+\,\mathrm{sgn}\,\omega\Big)\Big], \tag{7.48}$$

valid for q, ω/v_{D} and $\Delta_{\mathrm{D}}/v_{\mathrm{D}}$ small compared with Λ_{D}. The response depends on $\boldsymbol{q}$ and ω only through $\omega^2-v_{\mathrm{D}}^2q^2$, the invariant of a Lorentz transformation with v_{D} in place of the speed of light. The Dirac response is the one-loop vacuum polarization of quantum electrodynamics with an electron of mass Δ_{D} [16, Eq. (2.6)]. That reference fixes the constant inside the bracket by a subtraction at the cutoff instead of a hard cutoff, and the cutoff of that reference corresponds to $\mathrm{e}^{-5/6}\Lambda_{\mathrm{D}}=0.43\,\Lambda_{\mathrm{D}}$ in our convention.

The imaginary part, which counts the pairs, needs no cutoff. For $\omega>0$ the logarithm in Eq. (7.48) has a negative argument inside the continuum Eq. (7.45), and with $s=\omega^2-v_{\mathrm{D}}^2q^2$,

$$\frac{\varepsilon_2(\boldsymbol{q},\omega)}{\varepsilon_{\mathrm{bg}}} = \frac{g_{\mathrm{deg}}\tilde{\alpha}}{6}\Big(1-\frac{4\Delta_{\mathrm{D}}^2}{s}\Big)^{1/2}\Big(1+\frac{2\Delta_{\mathrm{D}}^2}{s}\Big)\Theta(s-4\Delta_{\mathrm{D}}^2). \tag{7.49}$$

The imaginary part ε_2 rises from zero at the continuum edge as the square root of the distance to the edge, like the density of states of a band edge, and for $s\gg\Delta_{\mathrm{D}}^2$ it tends to the constant $\varepsilon_{\mathrm{bg}}g_{\mathrm{deg}}\tilde{\alpha}/6$, the same at every q and ω. The flat ε_2 of a gapless cone, $\varepsilon_{\mathrm{bg}}g_{\mathrm{deg}}\tilde{\alpha}/6$ above $\omega=v_{\mathrm{D}}q$, is characteristic of linear bands; optical measurements on the Dirac semimetal $\mathrm{Na_3Bi}$ agree with it to about 10 % [16].

The real part gives the screening. At $q\to 0$ and $\omega=0$ the integral in Eq. (7.48) vanishes. The static dielectric constant of a gapped Dirac material is then

$$\varepsilon(0,0)=\varepsilon_{\mathrm{bg}}\Big[1+\frac{g_{\mathrm{deg}}\tilde{\alpha}}{3\pi}\Big(\ln\frac{2v_{\mathrm{D}}\Lambda_{\mathrm{D}}}{\Delta_{\mathrm{D}}}-\frac{5}{6}\Big)\Big]. \tag{7.50}$$

The Dirac electrons add to the background a term proportional to $g_{\mathrm{deg}}\tilde{\alpha}$ times a logarithm of the ratio of the band width $v_{\mathrm{D}}\Lambda_{\mathrm{D}}$ to the gap. Pairs of every energy between $2\Delta_{\mathrm{D}}$ and $2v_{\mathrm{D}}\Lambda_{\mathrm{D}}$ contribute equally per logarithmic interval of energy, and this equal weighting produces the logarithm. The dependence on the cutoff is mild: doubling Λ_{D} changes $\varepsilon(0,0)/\varepsilon_{\mathrm{bg}}$ by $g_{\mathrm{deg}}\tilde{\alpha}\ln 2/3\pi$, under 3 % for $\mathrm{ZrTe_5}$. The split between $\varepsilon_{\mathrm{bg}}$ and the Dirac term depends on the cutoff: states beyond Λ_{D} belong to the background, so a larger cutoff moves weight from $\varepsilon_{\mathrm{bg}}$ into the logarithm, and only the total ε is measurable.

As q grows at $\omega=0$, the argument of the logarithm in Eq. (7.48) grows and the screening weakens. Once $v_{\mathrm{D}}q\gg 2\Delta_{\mathrm{D}}$ the gap no longer matters, and the static dielectric function approaches the dielectric function of a gapless cone. The gapless formula [16, Eq. (2.7)] follows from Eq. (7.48) by expanding the logarithm for $|s|\gg\Delta_{\mathrm{D}}^2$: $\ln[1-x(1-x)s/\Delta_{\mathrm{D}}^2]\to\ln[-x(1-x)s/\Delta_{\mathrm{D}}^2]+\Delta_{\mathrm{D}}^2/[-x(1-x)s]$, and with $\int_0^1 x(1-x)\ln[x(1-x)]\,\mathrm{d}x=-5/18$ the constants $-\frac{5}{3}$ cancel and Δ_{D} drops out of the logarithm. The dielectric function becomes

$$\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_{\mathrm{bg}}} = 1+\frac{g_{\mathrm{deg}}\tilde{\alpha}}{6\pi}\ln\frac{4v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2}{|v_{\mathrm{D}}^2q^2-\omega^2|}+\mathrm{i}\,\frac{g_{\mathrm{deg}}\tilde{\alpha}}{6}\,\Theta(\omega-v_{\mathrm{D}}q). \tag{7.51}$$

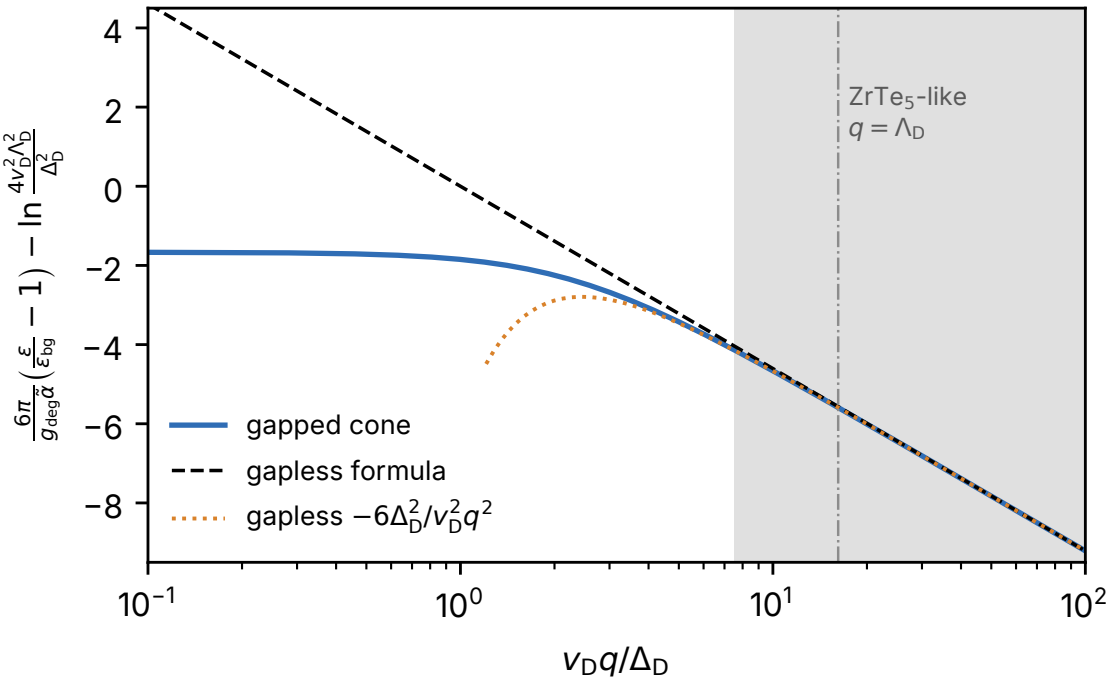


Figure 7.4: Static dielectric function of a gapped Dirac cone against the gapless formula. Plotted is the induced part $\varepsilon(q,0)/\varepsilon_\mathrm{bg}-1$ in units of $g_\mathrm{deg}\tilde{\alpha}/6\pi$, less the cutoff term $\ln(4v_\mathrm{D}^2\Lambda_\mathrm{D}^2/\Delta_\mathrm{D}^2)$; the subtraction makes the curves independent of the cutoff (hard cutoff on $|\boldsymbol{k}|$). Solid: Eq. (7.48) at $\omega=0$; dashed: the gapless formula, $-2\ln(v_\mathrm{D}q/\Delta_\mathrm{D})$ on this scale; dotted: the gapless formula with its leading correction $-6\Delta_\mathrm{D}^2/v_\mathrm{D}^2q^2$. Gray: $v_\mathrm{D}q>7.5\,\Delta_\mathrm{D}$, where the gapless formula is within 0.1 of the full result; with the correction the gapless formula is within 0.1 from $3.7\,\Delta_\mathrm{D}$ on, and at $v_\mathrm{D}q=10\,\Delta_\mathrm{D}$ its residual is 3×10^{-3}. Dash-dotted: $q=\Lambda_\mathrm{D}$ for the ZrTe$_5$-like cone of Worked example 7.4, where $v_\mathrm{D}\Lambda_\mathrm{D}=16.2\,\Delta_\mathrm{D}$. The Dirac region ends there.

The next term of the expansion contributes $-6\Delta_\mathrm{D}^2/(v_\mathrm{D}^2q^2-\omega^2)$ to the bracket. Fig. 7.4 compares the gapless formula with Eq. (7.48) at $\omega=0$. Below $v_\mathrm{D}q\approx2\Delta_\mathrm{D}$ the gapless formula diverges logarithmically, while the gapped cone saturates at Eq. (7.50): the gap cuts off the pairs that would screen at long wavelength. Static screening in a gapless cone, $\varepsilon(q,0)=\varepsilon_\mathrm{bg}[1+(g_\mathrm{deg}\tilde{\alpha}/3\pi)\ln(2\Lambda_\mathrm{D}/q)]$, grows without bound as $q\to0$, but only logarithmically. There is no screening length, since the density of states vanishes at the node; the electron gas screens a charge completely beyond $1/k_\mathrm{TF}$, whereas a gapless Dirac material only reduces it by a slowly growing factor.

The logarithms also set the validity of the calculation. The random-phase approximation keeps the first term in $\tilde{\alpha}$ of the polarization, and the induced part of Eq. (7.48) is of order $g_\mathrm{deg}\tilde{\alpha}\ln(v_\mathrm{D}\Lambda_\mathrm{D}/\max(\Delta_\mathrm{D},v_\mathrm{D}q))$, up to the numerical factor $1/3\pi$. We take this product as the expansion parameter: perturbation theory is reliable when the product is small compared with one. The screened interaction $e^2/\varepsilon q^2$, read as a coupling that depends on q, grows weaker at smaller q because of the screening, as the fine-structure constant does in quantum electrodynamics, and the theory becomes weakly coupled at long wavelength [16]. For $\tilde{\alpha}\gtrsim1$ the one-loop result is an estimate; Sect. 16.1 takes up the open question of what replaces it.

Anisotropy enters through a rescaling. With $\delta k_i\to\delta k_i/v_i$, the Hamiltonian Eq. (7.42) becomes isotropic with unit velocity, the sum over $\boldsymbol{k}$ acquires the Jacobian $1/(v_1v_2v_3)$, and q^2 in the overlap becomes $\sum_i v_i^2q_i^2$ [16, Eq. (B.1)]. The cutoff becomes an ellipsoid in $\boldsymbol{k}$, a sphere

of radius $\Lambda_{\mathrm{D}} v_{\mathrm{D}}$ in energy units for the rescaled momenta. For $\boldsymbol{q}$ along the principal axis i, the Coulomb factor e^2/q^2 then multiplies $v_i^2/(v_1 v_2 v_3)$ in place of $1/v_{\mathrm{D}}$, and the background constant becomes its component $\varepsilon_{\mathrm{bg},ii}$ along that axis, provided the background and the cone share principal axes. The static constant along i is

$$\varepsilon_{ii}(0,0) = \varepsilon_{\mathrm{bg},ii} + \frac{g_{\mathrm{deg}}\alpha}{3\pi}\,\frac{v_i^2}{v_1 v_2 v_3}\Big(\ln\frac{2\tilde{\Lambda}}{\Delta_{\mathrm{D}}} - \frac{5}{6}\Big), \tag{7.52}$$

with $\tilde{\Lambda}$ the cutoff on the energy $(\sum_i v_i^2 \delta k_i^2)^{1/2}$. Along each axis the effective coupling is $\alpha v_i^2/(\varepsilon_{\mathrm{bg},ii} v_1 v_2 v_3)$, which reduces to Eq. (7.47) for an isotropic cone.

The four inputs of Eq. (7.48), v_i, Δ_{D}, Λ_{D} and $\varepsilon_{\mathrm{bg}}$, come from first-principles calculations. The velocities and the cutoff are fitted to a computed band structure near the node. The half-gap is the hardest to compute: semilocal functionals misplace small gaps and can invert the band order, and quasiparticle corrections (Sect. 23.4) or the measured gap are needed. The background constant comes from density-functional perturbation theory (Sect. 25.2) or from the random-phase sum over bands. Both methods compute the dielectric constant of the whole crystal, which includes the Dirac pairs, and they do so at the calculated gap, to which the Dirac term is sensitive through $\ln(1/\Delta_{\mathrm{D}})$. A consistent $\varepsilon_{\mathrm{bg}}$ leaves the Dirac bands out of the sum over transitions, or subtracts Eq. (7.50) from the computed total.

Worked example 7.4: Effective coupling and static screening of $ZrTe_5$

$ZrTe_5$ has one gapped cone at Γ, $g_{\mathrm{deg}} = 2$. We take the first-principles values for the measured crystal structure [15, Table 1 and text below it]. They are the principal velocities $v_x, v_y, v_z = 1.1 \times 10^{-3}$, 4.5×10^{-4}, 1.0×10^{-3}, a gap of 23.6 meV ($\Delta_{\mathrm{D}} = 11.8$ meV), and cutoffs $\Lambda_x, \Lambda_y, \Lambda_z = 0.23$, 0.215, 0.1 Å^{-1}. A relaxed cell volume gives the same v_x, slower $v_y = 4.4 \times 10^{-4}$ and $v_z = 9.1 \times 10^{-4}$, and a gap of 31.2 meV ($\Delta_{\mathrm{D}} = 15.6$ meV). The dielectric tensor from density-functional perturbation theory has the diagonal components 308.4, 20.75 and 126.1 [15, Table 2].

Cutoff. With $1\ \text{Å}^{-1} = 1.973$ keV, the band energies at the three cutoffs are $v_i\Lambda_i = 0.499$, 0.191 and 0.197 eV. The cone is linear to a much higher energy along x than along y. We keep the states with $(\sum_i v_i^2 \delta k_i^2)^{1/2} < \tilde{\Lambda} = 0.191$ eV, the smallest of the three band energies [15, Eq. (2.21)]. Taking the smallest band energy is conservative, and a better fit of the bands would replace the cutoff by a larger ellipsoid. The dielectric function of Ref. [15, Eq. (3.10)] carries the same constant $-\frac{5}{3}$ as Eq. (7.48), so that $\tilde{\Lambda}$ is a hard cutoff in our sense, $\tilde{\Lambda} = v_{\mathrm{D}}\Lambda_{\mathrm{D}}$.

Isotropic estimate. To get a single number, replace the cone by an isotropic one with the same density of states, $v_{\mathrm{D}} = (v_x v_y v_z)^{1/3} = 7.9 \times 10^{-4}$, and the dielectric tensor by the harmonic mean of its components, $\varepsilon_{\mathrm{bg}} = 3/(\sum_i 1/\varepsilon_{\mathrm{bg},ii}) = 50.5$, the average used in Ref. [16]. Then Eq. (7.47) gives $\tilde{\alpha} = 0.18$ and $g_{\mathrm{deg}}\tilde{\alpha} = 0.37$. The cutoff in momentum is $\Lambda_{\mathrm{D}} = \tilde{\Lambda}/v_{\mathrm{D}} = 241$ eV $= 0.12\ \text{Å}^{-1}$. An earlier calculation [16, Table 1], with faster velocities (2.9×10^{-3}, 5.0×10^{-4}, 2.1×10^{-3}) and smaller dielectric constants (187.5, 9.8 and 90.9, harmonic mean 25.3), yields a similar $\tilde{\alpha} = 0.20$, quoted there as $g_{\mathrm{deg}}\tilde{\alpha} = 0.80$ with $g_{\mathrm{deg}} = 4$ states per band.

Static screening. With $\ln(2\tilde{\Lambda}/\Delta_{\mathrm{D}}) = \ln 32.4 = 3.48$, Eq. (7.50) yields

$$\frac{\varepsilon(0,0)}{\varepsilon_{\mathrm{bg}}} - 1 = \frac{0.37}{3\pi}\,(3.48 - 0.83) = 0.102,$$

so that $\varepsilon(0,0) = 55.7$. The Dirac electrons add 10 % to the screening of the background. At finite q, the integral in Eq. (7.48) reduces the induced term to 0.091 at $q = 2\Delta_{\mathrm{D}}/v_{\mathrm{D}} =$

30 eV and to 0.052 at $q = \Lambda_D/2 = 121$ eV, momenta at which sub-MeV dark matter scatters (Fig. 7.3b). The expansion parameter at long wavelength, $g_{\mathrm{deg}}\tilde{\alpha}\ln(v_D\Lambda_D/\Delta_D) = 0.37\ln 16.2 = 1.0$, is about one. The induced term itself, 0.102, is small mainly through the factor $1/3\pi$, and the one-loop result is a fair estimate of it. Since $v_D\Lambda_D = 16.2\,\Delta_D$, $ZrTe_5$ reaches the gapless regime of Fig. 7.4 only above $v_Dq = 7.5\,\Delta_D$, or $q = 112$ eV: below that momentum the gap matters.
Anisotropy. Along the principal axes, Eq. (7.52) with $\ln(2\tilde{\Lambda}/\Delta_D) - \frac{5}{6} = 2.64$ adds 10.0, 1.67 and 8.3 to the background components 308.4, 20.75 and 126.1, that is 3.2, 8.1 and 6.6 %. The effective couplings along the three axes, $\alpha v_i^2/(\varepsilon_{\mathrm{bg},ii}v_xv_yv_z)$, are 0.058, 0.144 and 0.117: the slow direction y has the smallest background constant as well, and it is the most strongly coupled. The averaged $\tilde{\alpha} = 0.18$ is a fair summary, though it overstates the coupling along x threefold.
Size of the Dirac screening. The Dirac contribution is 3 to 8 % of the dielectric constant, smaller than the spread between the two sets of computed parameters. Both sets of dielectric constants are totals for the whole crystal, computed at a gap of 24 to 35 meV, and already contain a Dirac term of this size; at the few-percent level the included Dirac term is harmless. The coupling that matters for a rate is the total ε, dominated here by the background: with $\varepsilon \approx 56$, every Coulomb-like interaction inside $ZrTe_5$ is reduced about fifty-fold.

7.4 Defects and dopants

A phosphorus atom in silicon binds its fifth electron by 45.6 meV, a twenty-fifth of the gap. Dark matter of 13 keV carries enough kinetic energy to set that electron free, whereas the gap of the perfect crystal needs dark matter of 0.333 MeV (Sect. 1.3). A carbon–hydrogen complex in the same crystal, the T center, emits light at 935 meV, and a quarter of its photons leave without exciting a single phonon. And a silicon atom struck with a few tens of eV leaves its site for good, storing several eV in the vacancy and the interstitial it leaves behind. Defects therefore enter a dark-matter experiment as targets with their own low thresholds, and as final states or readouts of an energy deposit. The theory of each defect builds on formalism already in hand. The shallow donor is the hydrogenic envelope of Sect. 4.9 with the valley masses of Sect. 3.3, the color center is the displaced oscillator of Sect. 5.5, and the displacement threshold is a property of the energy surface of Sect. 5.1.

7.4.1 Shallow donors

An atom from group V on a silicon site, such as phosphorus, arsenic or antimony, uses four of its five valence electrons in the bonds. The fifth valence electron is bound to the ion core and its one extra positive charge, in a hydrogenic donor level just below the conduction band, built from conduction-band states. An impurity of this kind is a *donor*; a group-III atom such as boron, one electron short, binds a hole instead and is an *acceptor*. Donors and acceptors both bind their carrier weakly, because the host screens the Coulomb attraction by the static dielectric constant ε_0 and the carrier moves with the band mass m^* rather than m_e.

The hydrogen atom with these two substitutions already sets the scale. Appendix D.17 derives the envelope equation of Kohn and Luttinger [18] and [19, Chap. 4]. The donor potential varies slowly on the scale of the lattice and mixes only conduction states within $\sim 1/a^*$ of a band minimum $\boldsymbol{k}_j$. The donor state is then the Bloch function at the minimum

modulated by an envelope,

$$\psi(\boldsymbol{r}) \simeq F_j(\boldsymbol{r})\, \psi_{c\boldsymbol{k}_j}(\boldsymbol{r}), \qquad \Big[-\frac{\partial_x^2+\partial_y^2}{2m_\perp^*} - \frac{\partial_z^2}{2m_\parallel^*} - \frac{\alpha}{\varepsilon_0 r}\Big] F_j(\boldsymbol{r}) = -E_\mathrm{b}^\mathrm{D}\, F_j(\boldsymbol{r}), \tag{7.53}$$

with z along the axis of the valley and E_b^D the *donor binding energy*, measured down from the conduction-band minimum. Eq. (7.53) is the envelope equation of *effective-mass theory*, and it is the Wannier–Mott equation Eq. (4.106) with two changes. The electron is bound to a fixed ion rather than to a hole, so the mass is the band mass and not a reduced mass. And the mass is a tensor, because the valleys of silicon are ellipsoids (Sect. 3.3).

Effective-mass donor levels

For an isotropic mass m^* the solution of Eq. (7.53) is hydrogenic, with the *effective Bohr radius* a^*,

$$E_\mathrm{b}^\mathrm{D} = \frac{m^*\alpha^2}{2\varepsilon_0^2} = \frac{m^*/m_e}{\varepsilon_0^2}\,\mathrm{Ry}, \qquad a^* = \frac{\varepsilon_0}{m^*\alpha} = \varepsilon_0\,\frac{m_e}{m^*}\,a_0, \tag{7.54}$$

and excited levels at $E_\mathrm{b}^\mathrm{D}/n^2$ below the band edge.

With $\varepsilon_0 = 11.7$ and masses of a few tenths of m_e, the binding energy is tens of meV and the orbit tens of Å, several lattice constants. The assumption behind Eq. (7.53), an envelope spread over many cells, is then satisfied, and the estimate is self-consistent. For silicon the estimate brackets the answer without giving it: the transverse mass $m_\perp^* = 0.191\, m_e$ puts the binding energy at 19.0 meV and $a^* = 32.4$ Å, the longitudinal mass $m_\parallel^* = 0.916\, m_e$ at 91.0 meV and 6.8 Å.

The anisotropic problem has one parameter, the mass ratio $\gamma_m = m_\perp^*/m_\parallel^* = 0.209$. We use the transverse Rydberg $\mathrm{Ry}_\perp^* = m_\perp^*\alpha^2/2\varepsilon_0^2 = 18.98$ meV and the radius $a_\perp^* = 32.4$ Å as units. The variational envelope of Kohn and Luttinger is a hydrogen $1s$ function squeezed along the heavy axis, and it binds the electron by $E_\mathrm{b}^\mathrm{D} = 1.564\,\mathrm{Ry}_\perp^* = 29.7$ meV, with the radii $r_\perp = 24.3$ Å across the valley axis and $r_\parallel = 14.0$ Å along it (Eq. (D.62)). The orbit is an oblate spheroid, flattened along the heavy direction, where the electron is less mobile, and the binding energy lies between the two isotropic values, closer to the transverse-mass value, because two of the three directions are light. The full spectrum of Eq. (7.53) needs a numerical solution. Scaling z by $\gamma_m^{1/2}$ makes the kinetic energy isotropic and turns the Coulomb potential into $-2/[r\,(1-(1-\gamma_m)\cos^2\theta)^{1/2}]$ in these units. We expand that angular factor in Legendre polynomials and the envelope in spherical harmonics of fixed m and parity, and solve the resulting coupled radial equations on a grid. The ground state comes out at $1.566\,\mathrm{Ry}_\perp^* = 29.7$ meV, so the variational envelope is within 0.002 $\mathrm{Ry}_\perp^*$ of the exact effective-mass ground state. The excited states are labeled by their hydrogenic parents; the anisotropy splits each np level into np_0, with the node across the valley axis, and $np_\pm$. Fig. 7.5 shows these levels.

The measured spectra test effective-mass theory state by state [20, Fig. 1]. The p levels of phosphorus and arsenic agree with each other to 0.02 meV: $2p_0$ at 11.48 and 11.50 meV, $2p_\pm$ at 6.40 meV for both. They are independent of the chemistry of the donor, as effective-mass theory predicts. With $\varepsilon_0 = 11.7$, however, effective-mass theory puts these two levels 5 % too shallow, at 10.92 and 6.09 meV. Since every level of Eq. (7.53) scales as $1/\varepsilon_0^2$, the measured $2p_\pm$ fixes $\varepsilon_0 = 11.7 \times (6.09/6.40)^{1/2} = 11.4$. The value 11.4 is the static dielectric constant at liquid-helium temperature, where donor spectra are taken, 2.6 % below the room-temperature value 11.7, and calculations of donor levels use it [21, 22]. With 11.4, effective-mass theory

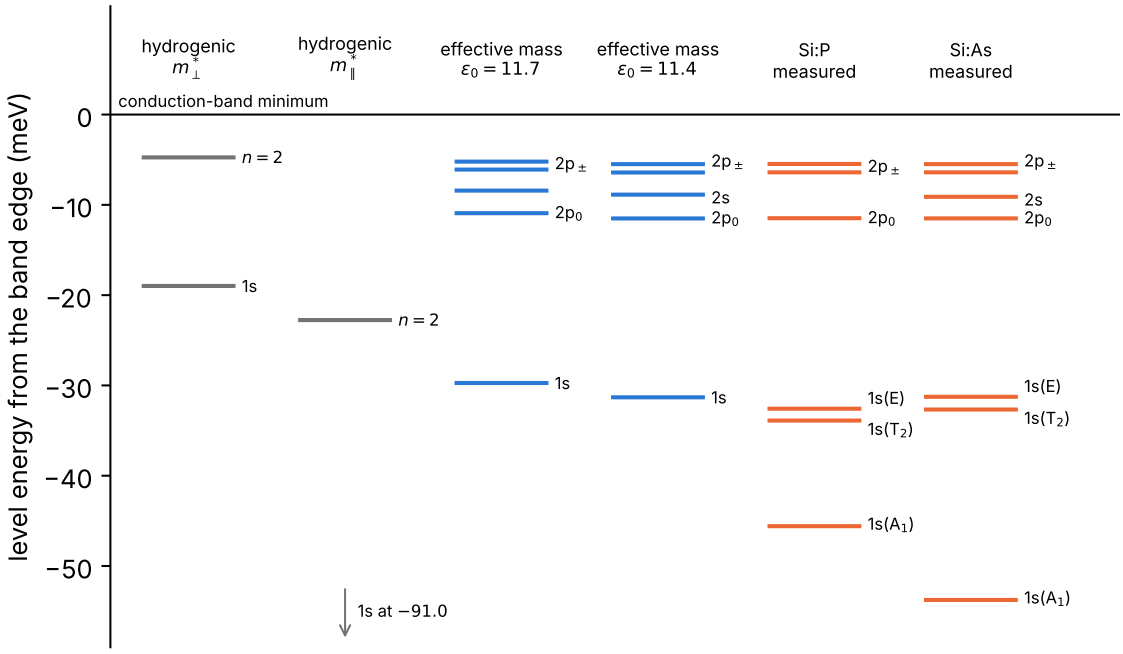


Figure 7.5: Bound levels of a shallow donor in silicon, measured from the conduction-band minimum (meV). Gray: the hydrogenic model, Eq. (7.54), with $\varepsilon_0 = 11.7$ and the transverse mass $0.191\,m_e$ ($1s$ at 19.0 meV) or the longitudinal mass $0.916\,m_e$ ($1s$ at 91.0 meV, off the scale), each with its $n = 2$ level. Blue: the effective-mass equation Eq. (7.53) with both masses, solved numerically (coupled radial equations in spherical harmonics after scaling z by $\gamma_m^{1/2}$, $l \leq 14$), for $\varepsilon_0 = 11.7$ (room temperature) and 11.4 (liquid-helium temperature); from the top, $3p_0$, $2p_\pm$, $2s$, $2p_0$ and $1s$. Orange: measured levels of phosphorus and arsenic donors [20, Fig. 1]. The p levels agree with effective-mass theory at $\varepsilon_0 = 11.4$ to 0.4 %; the $1s$ level splits into A_1, T_2 and E states, and the A_1 ground state lies below the effective-mass value by 14 meV for phosphorus and 22 meV for arsenic.

puts $2p_0$, $2p_\pm$ and $3p_0$ at 11.51, 6.42 and 5.49 meV, within 0.4 % of the measured 11.48, 6.40 and 5.47 meV, and the $1s$ state at 31.3 meV, the value Faulkner obtained [21, 22]. The p envelopes vanish at the donor site and are insensitive to the ion core.

The ground states depend on the donor. Phosphorus binds its electron by 45.59 meV [23], arsenic by 53.76 meV [20, Fig. 1], and antimony by 42.7 meV [24, Table 3], all deeper than 31.3 meV and different from one another. The differences between the donors arise near the ion core. Within a lattice constant of the donor the screening by ε_0 is incomplete, and the potential carries the chemistry of the ion; the envelope equation replaces the true potential near the core by $-\alpha/\varepsilon_0 r$. The $1s$ envelope is largest near the core, and the part of the potential that Eq. (7.53) leaves out shifts the $1s$ level. This shift is the *central-cell correction*.

The central cell also lifts a degeneracy that the effective-mass equation leaves intact. Silicon has six valleys, on the ⟨100⟩ axes, and each carries its own envelope, rotated with its axis; at the order of Eq. (7.53), any combination $\sum_j c_j F_j \psi_{c\boldsymbol{k}_j}$ is a ground state, sixfold degenerate without spin. The short-range part of the donor potential couples the valleys, because a potential concentrated in one cell has Fourier components at the large momenta $\boldsymbol{k}_j - \boldsymbol{k}_{j'}$ between the valleys. The donor site has the symmetry of a regular tetrahedron, the point group T_d, which leaves three matrix elements of the short-range potential, Δ_0 within a valley, Δ_g between opposite valleys and Δ_f between perpendicular ones. The six valley states

split into a singlet A_1, a doublet E and a triplet T_2. The labels A_1, E and T_2 are the standard names of the irreducible representations of T_d: each label says how a set of states transforms under the rotations and reflections of the tetrahedron, and here it fixes the degeneracy, one, two or three. The levels lie at

$$E_{A_1} = \Delta_0 + \Delta_g + 4\Delta_f, \qquad E_E = \Delta_0 + \Delta_g - 2\Delta_f, \qquad E_{T_2} = \Delta_0 - \Delta_g, \tag{7.55}$$

relative to the effective-mass level (Eq. (D.63)). In the A_1 state all six valleys add in phase at the donor site; it has the largest density on the core and an attractive central cell lowers it most. The measured spectra show this *valley-orbit splitting*: the phosphorus $1s$ level is a singlet at 45.59 meV, with the triplet T_2 at 33.89 meV and the doublet E at 32.58 meV, close to the effective-mass value [20, Fig. 1]. The central-cell correction is a property of the s-like ground state, and it is large, a third of the binding energy of phosphorus.

The size of the donor orbit limits the density at which donors stay isolated. The *defect density* n_D is the number of donors (or of any defect) per unit volume, and it has to be small enough that orbits do not overlap, $n_D a^{*3} \ll 1$. The spheroidal orbit of radii $r_\perp$, $r_\perp$ and $r_\parallel$ has a volume of $3.5 \times 10^{-20}\,\mathrm{cm}^3$, and one donor per orbit means $n_D \approx 3 \times 10^{19}\,\mathrm{cm}^{-3}$. In a lightly doped crystal, with $n_D = 10^{15}\,\mathrm{cm}^{-3}$, donors lie 100 nm apart and each is an isolated hydrogen atom of the crystal. At higher densities the electrons hop between donors and form an impurity band, and the isolated-donor picture fails before the orbits touch, at the Mott metal–insulator transition.

Acceptors are bound by similar amounts, tens of meV in silicon, but their effective-mass problem is harder. The valence maximum is degenerate, with heavy and light holes and a split-off band (Sect. 3.3), and Eq. (7.53) becomes a set of coupled equations for several envelopes, the same complication that the heavy–light mixing brings to excitons (Sect. 4.9). Donors are the cleaner case, and we keep to them.

Effective-mass theory is the computational method of choice for shallow donors, because no supercell is large enough to hold the envelope. An orbit of radius 2.4 nm spans thousands of atoms, and a supercell calculation of the donor (Sect. 28.2) has to extrapolate in the supercell size. The supercell calculation also inherits the error of its functional, magnified: a semilocal functional spreads the donor electron too far and binds phosphorus by only 19.2 meV after extrapolation, against the measured 45.6 meV [24, Table 3]. A supercell describes the potential within a cell or two of the core accurately. Combined with the envelope equation, in multivalley effective-mass theory, that potential gives the central-cell correction and the valley-orbit splitting; Appendix D.17 sets up this scheme.

Worked example 7.5: The binding of phosphorus in silicon

We compare the effective-mass binding of a phosphorus donor with the measured 45.59 meV and extract the valley-orbit matrix elements, with the masses $m_\perp^* = 0.191\,m_e$ and $m_\parallel^* = 0.916\,m_e$ of Table 3.1.

Units. With $\varepsilon_0 = 11.7$, $\mathrm{Ry}_\perp^* = 0.191 \times 13\,606\,\mathrm{meV}/11.7^2 = 18.98\,\mathrm{meV}$, $a_\perp^* = 11.7 \times 0.529\,\text{Å}/0.191 = 32.4\,\text{Å}$ and $\gamma_m = 0.209$.

Effective-mass ground state. The variational envelope of Appendix D.17 binds the electron by 29.7 meV, and the numerical solution of Eq. (7.53) gives $1.566\,\mathrm{Ry}_\perp^*$, also 29.7 meV. At the liquid-helium value $\varepsilon_0 = 11.4$, which the p levels require, $\mathrm{Ry}_\perp^* = 20.00\,\mathrm{meV}$ and the effective-mass $1s$ level lies at 31.3 meV.

Central-cell correction. The measured ground state is $45.59\,\mathrm{meV} - 31.3\,\mathrm{meV} = 14.3\,\mathrm{meV}$ deeper than the effective-mass level, 31 % of the binding. The measured binding itself depends on the theory: the spectroscopy measures transitions, and the absolute ground-state energy follows by adding the effective-mass binding of a high p state, 3.12 meV for

$3p_\pm$, to the measured $1s(A_1) \to 3p_\pm$ line [22]. The p levels are accurate enough for this step, as Fig. 7.5 shows.
Valley-orbit matrix elements. Relative to the effective-mass level at $-31.3\,\mathrm{meV}$, the three measured $1s$ levels of phosphorus, A_1 at -45.59, T_2 at -33.89 and E at $-32.58\,\mathrm{meV}$, lie at -14.29, -2.59 and $-1.28\,\mathrm{meV}$. Eq. (7.55) gives $E_{A_1} - E_E = 6\Delta_\mathrm{f}$, so $\Delta_\mathrm{f} = -13.01/6 = -2.17\,\mathrm{meV}$; then $\Delta_0 + \Delta_\mathrm{g} = E_E + 2\Delta_\mathrm{f} = -5.62\,\mathrm{meV}$ and $\Delta_0 - \Delta_\mathrm{g} = E_{T_2} = -2.59\,\mathrm{meV}$, so that $\Delta_0 = -4.10\,\mathrm{meV}$ and $\Delta_\mathrm{g} = -1.51\,\mathrm{meV}$. All three matrix elements are negative, as an attractive core requires, and the coupling between perpendicular valleys is the larger of the two intervalley terms. The E and T_2 states lie only 1.3 and 2.6 meV below the effective-mass value, and the A_1 state is lowered by 14.3 meV, because only A_1 collects all six valleys in phase at the core.
Size. The variational envelope of Appendix D.17 has $|F(0)|^2 = 1/(\pi r_\perp^2 r_\parallel)$, with the radii $r_\perp = 24.3\,\text{Å}$ and $r_\parallel = 14.0\,\text{Å}$ found above. The volume of one silicon atom is $a^3/8$, with the lattice constant $a = 5.43\,\text{Å}$, and the probability of finding the electron within one such atomic volume around the donor is $|F(0)|^2 a^3/8 = (5.43\,\text{Å})^3/(8\pi \times 24.3^2 \times 14.0\,\text{Å}^3) = 7.7 \times 10^{-4}$ per valley envelope. The central cell is a small region of the orbit, and still it moves the ground state by a third: the potential there is that of an unscreened ion, many eV deep.
Kinematics. Dark matter can ionize the donor if $\frac{1}{2} m_\chi v_\mathrm{max}^2 \geq E_\mathrm{b}^\mathrm{D}$, that is for $m_\chi \geq 2 \times 45.59\,\mathrm{meV}/7.023 \times 10^{-6} = 13.0\,\mathrm{keV}$, a mass twenty-six times smaller than the $0.333\,\mathrm{MeV}$ that the silicon gap requires. At threshold the momentum transfer is $m_\chi v_\mathrm{max} = 34.4\,\mathrm{eV}$. The envelope spreads over momenta up to $1/r_\perp = 81\,\mathrm{eV}$ across the valley axis and $1/r_\parallel = 141\,\mathrm{eV}$ along it, so $q r_\perp \approx 0.42$: the dark matter that can just ionize the donor transfers momenta that the bound electron has. In the language of Sect. 1.3, the donor is a good kinematic match to dark matter of a few tens of keV.

7.4.2 Color centers

The T center of silicon emits at 935 meV, 1326 nm in the telecommunications O band, and the nitrogen–vacancy center of diamond at 1.945 eV, 637 nm. Both lines are sharp, and both lie on a broad band of lower-energy light. The T center and the nitrogen–vacancy center are *color centers*, named after the F centers that color the alkali halides. Their levels lie deep in the gap, and their electrons are confined within a cell or two of the defect, unlike the donor electron of the previous subsection. A localized electron that changes its state exerts large forces on the few atoms around it. The nuclei settle at different positions in the ground and the excited state, and an optical transition, which is fast compared with the nuclear motion, leaves them displaced from their new equilibrium. The displaced oscillator of Sect. 5.5 describes this displacement, and its parameters are measured on color centers.

The *configuration-coordinate diagram* draws the two Born–Oppenheimer surfaces of Sect. 5.1, one for each electronic state, along the single coordinate Q that connects their minima. With Q mass-weighted, as $\Delta Q = M^{1/2}\Delta x$ in Sect. 5.5, and one effective mode of energy ω_0 that is the same in both states,

$$E_\mathrm{g}^\mathrm{ad}(Q) = \tfrac{1}{2}\omega_0^2 Q^2, \qquad E_\mathrm{e}^\mathrm{ad}(Q) = E_\mathrm{ZPL} + \tfrac{1}{2}\omega_0^2 (Q - \Delta Q)^2, \qquad S_\mathrm{HR} = \tfrac{1}{2}\omega_0 \Delta Q^2. \tag{7.56}$$

The *zero-phonon-line energy* E_ZPL is the energy difference between the two minima; it is the energy of the transition that leaves the vibrational state unchanged. The Huang–Rhys factor S_HR measures the displacement ΔQ in units of the zero-point spread of the mode. A vertical transition from the ground minimum lands $\frac{1}{2}\omega_0^2 \Delta Q^2 = S_\mathrm{HR}\omega_0$ above the excited minimum,

so absorption peaks near $E_{\mathrm{ZPL}} + S_{\mathrm{HR}}\omega_0$; emission from the relaxed excited state peaks near $E_{\mathrm{ZPL}} - S_{\mathrm{HR}}\omega_0$, and the lattice takes up the relaxation energy $S_{\mathrm{HR}}\omega_0$ each way. At $T = 0$ the emission lines at $E_{\mathrm{ZPL}} - p\,\omega_0$ carry the Poisson weights $\mathrm{e}^{-S_{\mathrm{HR}}} S_{\mathrm{HR}}^p/p!$ of Eq. (5.57), and the zero-phonon line carries

$$w_{\mathrm{ZPL}} = \mathrm{e}^{-S_{\mathrm{HR}}}. \tag{7.57}$$

A real center couples to a continuum of crystal modes and to a few local modes of its own atoms; S_{HR} is then the sum of the Huang–Rhys factors of all the modes, the phonon replicas of the line merge into a sideband, and the zero-phonon weight is still $\mathrm{e}^{-S_{\mathrm{HR}}}$ (Sect. 5.5). The color-center literature calls this weight the Debye–Waller factor of the line; in this book that name belongs to e^{-2W} of Eq. (5.48), and we write *zero-phonon fraction*.

The *T center* is a pair of carbon atoms sharing one silicon site, with a hydrogen atom on one of them [25]. Its ground state has one unpaired electron. Its luminescent state is a bound exciton: a second electron pairs with the first into a singlet, and a hole is bound to the negatively charged complex by only $(34.78 \pm 0.15)\,\mathrm{meV}$, out of a total pair binding of about 235 meV. The hole binding comes from the series of excited hole levels seen in photoluminescence excitation, extrapolated to the ionization limit of the hole [25, Sect. IV]. The loosely bound hole is a shallow, effective-mass-like particle, while the electrons are deep and localized. In measurements on an ensemble in isotopically pure ^{28}Si between 1.4 and 4.2 K, the exciton lives 0.94 µs, and 23 % of the emitted photons are in the zero-phonon line [26].

The *nitrogen–vacancy center* of diamond is a substitutional nitrogen next to a vacant site along [111]. In its negative charge state six electrons occupy orbitals built from the dangling bonds around the vacancy, and its spin-triplet ground state is optically read out at room temperature [27, Sect. 1.1]. A fit of its measured emission band at 8 K returns $S_{\mathrm{HR}} = 3.85 \pm 0.05$ with an effective phonon energy of 64 meV [28, Sects. 3 and 4.3].

Fig. 7.6 draws both centers in units of their own phonon energy; in these units each diagram depends on S_{HR} alone. The T center's minima are displaced by $(2S_{\mathrm{HR}})^{1/2} = 1.7$ zero-point units, the nitrogen–vacancy center's by 2.8. The difference between the two displacements becomes a factor of ten in the zero-phonon weight. The measurements are taken at temperatures where the coupled modes are empty: at 4.2 K, a phonon of 10 meV has a thermal occupation of $\mathrm{e}^{-\omega/T} = 1 \times 10^{-12}$, and the zero-temperature weights of Eq. (7.57) apply.

Comparing Eq. (7.57) with a luminescence spectrum needs one correction. The rate of spontaneous emission into a line of energy E grows as $E^3|\boldsymbol{d}|^2$, with $\boldsymbol{d}$ the transition dipole, so a luminescence spectrum counts photons with the Franck–Condon weight times E^3 [28, Eq. (2)]. The sideband lies at lower energy and is suppressed by this factor, and the zero-phonon line takes a larger share of the photons than of the Franck–Condon weight:

$$\frac{\text{photons in the zero-phonon line}}{\text{all photons}} = \frac{\mathrm{e}^{-S_{\mathrm{HR}}} E_{\mathrm{ZPL}}^3}{\sum_p \mathrm{e}^{-S_{\mathrm{HR}}} S_{\mathrm{HR}}^p (E_{\mathrm{ZPL}} - p\,\omega_0)^3/p!}, \tag{7.58}$$

in the single-mode model. The E^3 correction is of order $3S_{\mathrm{HR}}\omega_0/E_{\mathrm{ZPL}}$, large when the relaxation energy is a sizable fraction of the line energy.

Computing E_{ZPL} and S_{HR} is a supercell problem. E_{ZPL} is the difference of two total energies, each at its own relaxed geometry, with the excited state obtained by constraining the occupations of the defect orbitals; S_{HR} follows from the displacement between the two geometries projected on the vibrational modes, $S_{\mathrm{HR}} = \sum_k \omega_k \Delta Q_k^2/2$. A hybrid functional in a 512-atom cell puts the T-center line at 985 meV, 50 meV above the measurement [29], and the converged Huang–Rhys factor of the nitrogen–vacancy center is 3.67 against the measured 3.85 [28, Sects. 4.2 and 4.3].

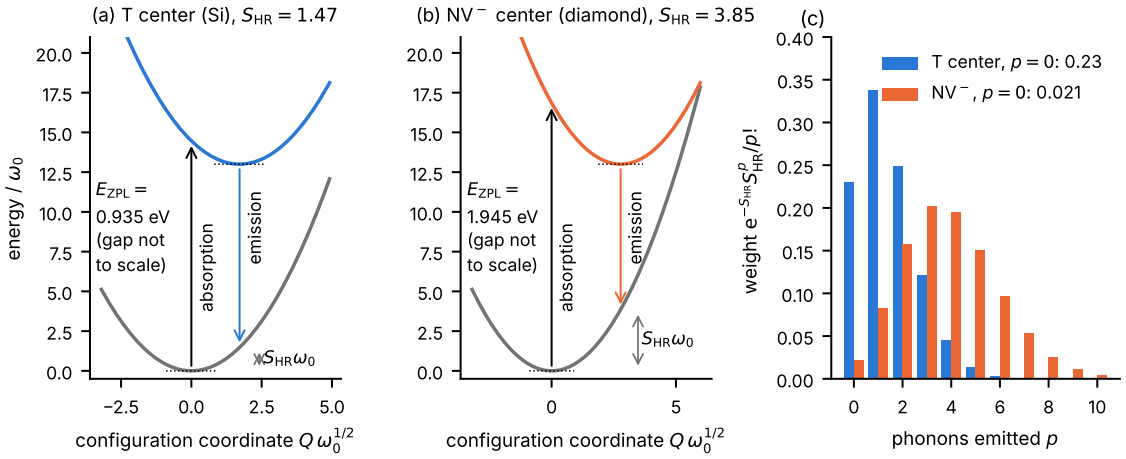


Figure 7.6: Configuration-coordinate diagrams of (a) the T center in silicon, $E_{ZPL} = 0.935\,\text{eV}$, $S_{HR} = 1.47$ (from the measured zero-phonon fraction 0.23 [26]), and (b) the negatively charged nitrogen–vacancy center in diamond, $E_{ZPL} = 1.945\,\text{eV}$, $S_{HR} = 3.85$ [28]. Each center is modeled by one effective mode of energy ω_0, the same in both electronic states, in the Condon approximation; energies are in units of ω_0 (64 meV for the nitrogen–vacancy center) and the coordinate in units of $\omega_0^{-1/2}$, Eq. (7.56). Gray: ground-state surface; colored: excited state, drawn with a reduced gap E_{ZPL} (not to scale). Arrows: vertical absorption from the ground minimum and emission from the excited minimum; the double arrow is the relaxation energy $S_{HR}\omega_0$. (c) Zero-temperature weights of the emission lines with p phonons emitted, $e^{-S_{HR}} S_{HR}^p/p!$; the zero-phonon line is $p = 0$.

Worked example 7.6: Zero-phonon fractions

We compute the zero-phonon fractions of the T center and the nitrogen–vacancy center and check the nitrogen–vacancy value against its measured photon count.

T center. Through Eq. (7.57), the measured photon fraction 0.23 means $S_{HR} = -\ln 0.23 = 1.47$. The weights of the lines with 0, 1, 2 and 3 phonons are 0.23, 0.34, 0.25 and 0.12 (Fig. 7.6c): one photon in four is in the zero-phonon line, and one in three has emitted exactly one phonon. Read through Eq. (7.58), the photon fraction slightly overstates the Franck–Condon weight. Since ω_0 is at most the 64 meV of the silicon optical phonon (Sect. 5.3), the correction $3S_{HR}\omega_0/E_{ZPL}$ is bounded by $3 \times 1.47 \times 0.064/0.935 = 0.30$, so the photon count shifts S_{HR} by at most 0.30. The true Huang–Rhys factor lies between 1.47 and about 1.8, and the zero-phonon weight between 0.23 and 0.17.

Nitrogen–vacancy center. With $S_{HR} = 3.85$, Eq. (7.57) gives $w_{ZPL} = e^{-3.85} = 0.021$, ten times less than for the T center. The line with three phonons is the strongest, at 0.20. The relaxation energy is $S_{HR}\omega_0 = 3.85 \times 64\,\text{meV} = 0.246\,\text{eV}$, so absorption peaks near 2.19 eV and emission near 1.70 eV, a Stokes shift of 0.49 eV.

The photon check. In Eq. (7.58), with $\omega_0/E_{ZPL} = 0.064/1.945 = 0.0329$, the Poisson average of $(1 - 0.0329\,p)^3$ is

$$1 - 3(0.0329)S_{HR} + 3(0.0329)^2(S_{HR} + S_{HR}^2) - (0.0329)^3(S_{HR} + 3S_{HR}^2 + S_{HR}^3) = 0.677,$$

from the first three moments of the Poisson distribution. The photon fraction is then $0.0213/0.677 = 0.031$. The measured share of the zero-phonon line in the emission is 0.032 [28, Sect. 3]. If the measured share is taken at face value, $-\ln 0.032$ gives

$S_{\mathrm{HR}} = 3.44$, and the single-mode model attributes this underestimate by 0.4 entirely to the E^3 weighting.
Detection fraction. A detector that reads a color center through its zero-phonon line collects a quarter of the T center's photons and three percent of the nitrogen–vacancy center's. The other photons come out in the sideband, spread over several phonon energies below the line, where a narrow filter rejects them together with the background.

7.4.3 Displacement thresholds and Frenkel pairs

A silicon atom struck along $\langle 111\rangle$ with 12.5 eV can leave its site for good; struck along $\langle 100\rangle$, it needs 20 eV [30, Table IV]. Below these energies the atom swings out, hands its energy to its neighbors and returns. The difference between the two outcomes is a statement about the energy surface $E^{\mathrm{ad}}(u)$ of Sect. 5.1. The perfect crystal is its lowest minimum, but the surface has many other local minima, separated from the perfect-crystal minimum by barriers of several eV. A *vacancy* is an empty lattice site. An *interstitial* is an extra atom between the sites; in silicon it can occupy the tetrahedral or the hexagonal interstitial site or share a site with a lattice atom as a split $\langle 110\rangle$ dumbbell. An atom moved from its site into an interstitial position leaves a vacancy behind, and the interstitial and the vacancy together form a *Frenkel pair*. The formation energy of each configuration is the height of its minimum above the minimum of the perfect crystal. Density-functional calculations put the isolated vacancy at 3.4 to 3.9 eV, the interstitials at 4.7 to 5.8 eV, and the most stable close Frenkel pair, a tetrahedral interstitial 4.7 Å from its vacancy, at 6.8 to 7.5 eV [30, Table III]. A cheaper minimum exists without any vacancy: the *bond defect*, a local rearrangement of bonds, at 2.6 to 3.0 eV.

The *displacement threshold energy* $E_{\mathrm{d}}(\hat{\boldsymbol{r}})$ is the smallest kinetic energy that, given to one atom in the direction $\hat{\boldsymbol{r}}$, leaves the crystal in a minimum other than the perfect one once the motion has died out,

$$E_{\mathrm{d}}(\hat{\boldsymbol{r}}) = \min\{E_R : \text{ the trajectory on } E^{\mathrm{ad}} \text{ from } (E_R, \hat{\boldsymbol{r}}) \text{ ends in a defect}\}. \tag{7.59}$$

It is always larger than the formation energy of the pair it creates. The struck atom must squeeze between its neighbors, over a barrier that lies on its path and not at the end points. It loses energy to every atom it pushes. And if it stops too close to its own vacancy, the pair recombines and the crystal returns to the perfect-crystal minimum. E_{d} therefore depends on the direction, through the open and closed channels of the lattice, and on the path of the whole collision. The threshold E_{d} is computed by molecular dynamics: a single trajectory on E^{ad} for each direction and energy, repeated over many directions (Sect. 29.1).

For silicon, density-functional molecular dynamics finds the lowest threshold, 12.5 eV, along $\langle 111\rangle$, where the struck atom pushes its neighbor into the tetrahedral site in a short replacement sequence; 20 eV along $\langle 100\rangle$; and 36 eV for a Frenkel pair averaged over all directions [30, Table IV]. Electron-irradiation experiments see damage set in at 12.9 ± 0.6 eV [31], in agreement with the lowest computed threshold. In most directions a bond defect forms at a lower energy than a Frenkel pair, and the average threshold for a defect of either kind is 24 eV [30, Table IV]. Which number is quoted depends on what is counted: a count of vacant and doubly occupied lattice sites sees the Frenkel pair but not the bond defect. The classical Stillinger–Weber potential, used for the large statistical scans, overshoots the density-functional thresholds along $\langle 111\rangle$ by 4 to 6 eV [30, Sect. IV], and its thresholds range from 17.5 to 63.5 eV over all directions [32].

Only part of the threshold energy stays in the crystal. A close Frenkel pair stores 6.8 eV to 7.5 eV, about half of the minimum threshold and a fifth of the average one; the rest of the

recoil energy becomes heat within picoseconds. The stored energy is invisible to a calorimeter that measures the heat, and it is missing from the measured deposit; Sect. 19.4.4 turns this missing energy into the defect fraction of a recoil and the bounds that calibration data put on it. Only heavy enough dark-matter particles reach these thresholds. By Eq. (1.20), with the silicon nuclear mass $M = 26.16\,\mathrm{GeV}$ and $v_{\max} = 2.650 \times 10^{-3}$, a recoil of 12.5 eV needs $m_\chi \geq 0.153\,\mathrm{GeV}$, and one of 36 eV needs 0.26 GeV. Since E_d depends on the direction of the recoil relative to the crystal axes, the rate of defect creation near threshold carries the direction of the dark-matter wind, and directional detection can exploit this dependence [32].

7.4.4 Defects as targets and readouts

A shallow donor is itself a target, with the threshold and the kinematic match of Worked example 7.5. Its weakness is the small number of donor electrons. A donor density of $10^{15}\,\mathrm{cm}^{-3}$ is five parts in 10^9 of the valence electrons of silicon, $2.0 \times 10^{23}\,\mathrm{cm}^{-3}$, and the interaction with the bound electron is screened by the same ε_0 that makes the donor shallow. The donor rate requires the response of the donor to the dark-matter operators, with its valleys and central cell, which Sect. 16.5 derives.

A color center is a target with a level inside the gap and a readout with an optical line. As a target, the T center needs $m_\chi \geq 2 \times 0.935\,\mathrm{eV}/v_{\max}^2 = 0.266\,\mathrm{MeV}$, below the 0.333 MeV of the silicon gap, because its excitation lies 0.24 eV below the band edge. The momentum at threshold is $2E_{\mathrm{ZPL}}/v_{\max} = 0.71\,\mathrm{keV}$, and a transition at that momentum needs states localized to about $1/0.71\,\mathrm{keV} = 2.8\,\text{Å}$, and the deep levels of the center are localized on that scale. The nitrogen–vacancy center needs a mass of 0.554 MeV and a momentum of 1.47 keV. For a center used as a readout, the zero-phonon fraction sets how many of a center's photons survive a narrow filter (Worked example 7.6), and the spins of both centers can be polarized and read optically. As a final state, a color center can be created by a nuclear recoil, by a displacement or by the capture of a vacancy, and the threshold E_d sets the recoil energy at which this creation begins (Sect. 16.6).

A displaced atom, finally, is a long-lived final state. The Frenkel pair or the bond defect left by a recoil persists as long as its barriers allow. At 10 mK the Boltzmann factor for crossing a barrier of 0.1 eV is $\mathrm{e}^{-1.2\times10^5}$, and only tunneling or the phonons of a later event move the defect.

Summary

The Coulomb interaction does not act on spin, yet it couples spins, and in NiO it orders them below 523 K (Sect. 7.1). At low energy only the spins remain, coupled by the Heisenberg model, and in NiO one exchange constant, the measured $J_2 = 19.01\,\mathrm{meV}$ between next-nearest nickel neighbors, sets the scale. In an antiferromagnet the exchange creates spin deviations in pairs, a bosonic Bogoliubov transformation removes the pair terms, and the magnon dispersion becomes linear, with a measured maximum of ∼117 meV in NiO. Density-functional values of J_2 spread from 11 to 27 meV, and that spread is the theory uncertainty for a magnet without neutron data. A dark-matter particle that couples to the electron spin acts through the spin density $\boldsymbol{s}_e(\boldsymbol{q})$, and the one-magnon response of the antiferromagnet follows from the Bogoliubov coefficients.

Phonon exchange makes electrons near the Fermi surface attract each other, with the dimensionless strength $\mathcal{N}_\mathrm{F} V_\mathrm{BCS} = \lambda_\mathrm{ep}$, reduced by the quasiparticle residue and the Coulomb pseudopotential (Sect. 7.2). The excitations of the BCS state are Bogoliubov quasiparticles of energy $(\xi_{\boldsymbol{k}}^2 + \Delta_\mathrm{BCS}^2)^{1/2}$, and a probe that conserves the electron number creates them in pairs, so the threshold is $2\Delta_\mathrm{BCS}$. The response of independent quasiparticles, however, does

not conserve charge; the conserving response of the Nambu formalism removes the violation through the phase mode of the condensate and satisfies the density Ward identity. For aluminum the measured critical temperature fixes $\Delta_{\mathrm{BCS}} = 0.179\,\mathrm{meV}$, a threshold of 0.36 meV and a threshold mass of 0.10 keV.

Near a point where two bands nearly touch, the $\boldsymbol{k}\cdot\boldsymbol{p}$ expansion becomes a Dirac Hamiltonian, with the velocity v_{D} in place of the speed of light and the half-gap Δ_{D} in place of the mass (Sect. 7.3). A cone with $v_{\mathrm{D}} > v_{\max}$ cannot be excited by scattering at any mass, while a slow cone is reached by particles as light as 8.0 keV for $ZrTe_5$-like parameters. The Dirac electrons screen on top of a background dielectric constant $\varepsilon_{\mathrm{bg}}$, with the effective coupling $\tilde{\alpha} = \alpha/(\varepsilon_{\mathrm{bg}} v_{\mathrm{D}})$, and a gapless cone has no screening length.

Defects put levels inside the gap of silicon itself (Sect. 7.4). Effective-mass theory binds every shallow donor by 31.3 meV at the liquid-helium dielectric constant 11.4, and the central-cell correction of the s-like ground state, 14.3 meV, brings phosphorus to its measured 45.6 meV, so that dark matter of 13 keV can ionize the donor. A color center couples its electronic transition to local vibrations, and the measured zero-phonon fraction $\mathrm{e}^{-S_{\mathrm{HR}}} = 0.23$ of the T center of silicon means $S_{\mathrm{HR}} = 1.47$. A recoiling atom that receives more than the displacement threshold, 12.5 eV along $\langle 111\rangle$ in silicon, leaves a Frenkel pair or a bond defect that stores energy no calorimeter sees. Chap. 8 gathers the responses of all these targets, and those of the earlier chapters, into one framework for the rate.

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Chapter 8

The response framework

A particle with a magnetic moment couples to the charge of a solid and also to the spins and currents of the solid, and a fermion that is absorbed leaves a light partner behind. Neither the magnetic coupling nor the fermion absorption fits the single product of Eq. (1.1), a halo factor times a coupling $|\tilde{V}|^2$ times one dynamic structure factor. So far Chap. 4 took the electron density of silicon, Chap. 5 the nuclei of a harmonic crystal, Chap. 6 electrons and lattice together, and Chap. 7 the spins of NiO and the paired electrons of aluminum. A new model or a new target should not mean starting again. What do these calculations have in common, and what does each of them leave out?

Every model reduces to a tensor that weights the response matrix of a few target operators, and every target to that one matrix. A new model then needs a new tensor but not a new matrix, and a new target needs a new matrix but not a new tensor. Because the matrix belongs to the target alone, three questions about it can be settled once, for every coupling at the same time: what is exact about the matrix, how the medium screens it, and where a calculation truncates it. We will build this framework (Fig. 8.1) from the separate results of Chaps. 1 to 7, up to the final state that the dark matter leaves in the target. Each truncation on the way leaves in the rate a theory uncertainty of known size and shape.

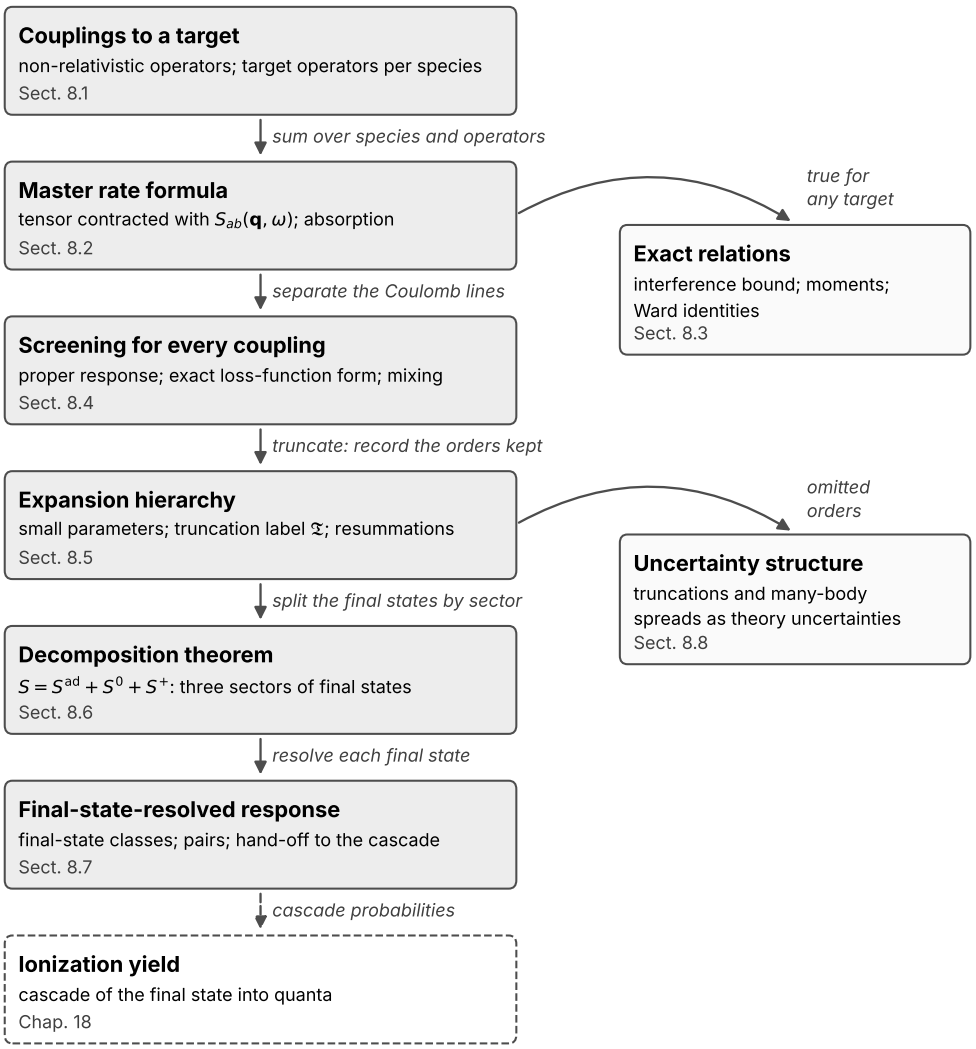


Figure 8.1: The response framework from couplings to final states. The non-relativistic operators of every model reduce to target operators per species (Sect. 8.1), and the master rate formula contracts a dark-matter tensor with the response matrix $S_{ab}(\boldsymbol{q}, \omega)$ (Sect. 8.2), which obeys exact relations for any target (Sect. 8.3). Separating the Coulomb lines screens every coupling (Sect. 8.4); the truncation label records the orders a calculation keeps (Sect. 8.5). The decomposition theorem splits the final states into three sectors (Sect. 8.6), and the final-state-resolved response divides each sector by the class of its final state (Sect. 8.7). Theory uncertainties come from the omitted orders and many-body spreads of the truncation label (Sect. 8.8).

8.1 Dark-matter couplings to a target

A dark photon couples only to the charge of a solid. In the language of Sect. 4.2 it needs one target operator per species, the electron density $n_e(\boldsymbol{q})$ and the density $n_\kappa(\boldsymbol{q})$ of each kind of nucleus, with potentials of opposite sign. A dark-matter particle with a magnetic dipole moment does more: it feels the spins of the electrons, and, since a charge moving past a magnetic moment feels a force, the velocities of the electrons as well [1, Table II]. Particle physics offers many models of dark-matter couplings, and at first sight each would need its own calculation of the target. A slow dark-matter particle, however, sees an electron or a nucleus through three things only: the momentum it hands over, the velocity of the two particles relative to each other, and their spins. As a result every model reduces to the same short list of target operators per species: the density, the spin density, the current, the spin current and a kinetic-energy density. The model enters only through a dark-matter tensor that weights the response matrix of these operators.

8.1.1 The non-relativistic operators

Take a dark-matter particle of mass m_χ and spin operator $\boldsymbol{S}_\chi$ that scatters on a free electron, with spin $\boldsymbol{S}_e = \boldsymbol{\sigma}/2$. The dark matter goes from $\boldsymbol{p}$ to $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$ and the electron from $\boldsymbol{k}$ to $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q}$, with $\boldsymbol{q}$ the momentum transfer of Sect. 1.3. Momentum conservation leaves three of the four momenta independent. Galilean invariance is the freedom to add a common velocity to both particles, and it allows an interaction to depend on these three momenta only through $\boldsymbol{q}$ and one velocity difference [2, Sect. 2.2]. The Hermitian velocity difference is symmetric between the incoming and outgoing particles and reads

$$\boldsymbol{v}^\perp = \bar{\boldsymbol{v}}_\chi - \frac{\boldsymbol{k}+\boldsymbol{k}'}{2m_e}, \qquad \bar{\boldsymbol{v}}_\chi \equiv \frac{\boldsymbol{p}+\boldsymbol{p}'}{2m_\chi} = \boldsymbol{v} - \frac{\boldsymbol{q}}{2m_\chi}, \tag{8.1}$$

the mean velocity of the dark matter minus the mean velocity of the electron, with $\boldsymbol{v} = \boldsymbol{p}/m_\chi$ the incoming velocity. For an electron initially at rest, $\boldsymbol{v}^\perp = \boldsymbol{v} - \boldsymbol{q}/2\mu_{\chi e}$, and energy conservation in the form $\boldsymbol{q}\cdot\boldsymbol{v} = q^2/2\mu_{\chi e}$ (Sect. 1.3) makes $\boldsymbol{q}\cdot\boldsymbol{v}^\perp = 0$, so that $\boldsymbol{v}^\perp$ is transverse to $\boldsymbol{q}$. The transversality is the origin of the name [2, Eq. (14)], and Appendix D.18 shows that it holds for any free target. The momentum enters as $\mathrm{i}\boldsymbol{q}$. Hermitian conjugation exchanges the incoming and outgoing particles; the exchange reverses $\boldsymbol{q}$ and leaves $\boldsymbol{v}^\perp$ unchanged, so that $\mathrm{i}\boldsymbol{q}$, $\boldsymbol{v}^\perp$ and the spins are all Hermitian.

An interaction is then a scalar built from $\mathrm{i}\boldsymbol{q}$, $\boldsymbol{v}^\perp$, $\boldsymbol{S}_\chi$ and $\boldsymbol{S}_e$. Powers of q^2 can be absorbed into a momentum-dependent coefficient, as the mediator propagator of Eq. (1.31) already is, and products of more than one spin of the same particle reduce to at most one by the algebra of the Pauli matrices [2, Sect. 2.2]. The result is a finite list of operators, which Fitzpatrick and collaborators wrote down and Anand, Fitzpatrick and Haxton completed [2, Eqs. (19)–(24) and (45), 3, Eqs. (12) and (13)]. They are listed in Table 8.1, which writes $\boldsymbol{q}/m_e$ in place of $\boldsymbol{q}$, so that each operator is dimensionless. The list is complete for dark matter of spin $\frac{1}{2}$, since the Pauli algebra reduces its spin products. For spin 1 a product of two dark-matter spins leaves a symmetric tensor built from the polarization vectors, and two more operators appear, $\boldsymbol{q}$ contracted through that tensor with $\boldsymbol{v}^\perp$ or with $\boldsymbol{S}_e$ [4, Eq. (18)]; for spin 0 only the operators free of $\boldsymbol{S}_\chi$ remain. The target side does not change: the new operators are linear in the electron's velocity or spin, and reduce to the same target operators (Eq. (8.3)) as the operators for spin $\frac{1}{2}$. The potential between the dark matter and the electron is a sum

$$\tilde{V}(\boldsymbol{q}) = \sum_i c_i^{(e)}(q)\, \mathcal{O}_i^{(e)}, \tag{8.2}$$

an operator in the spin spaces of both particles, whose matrix element between plane waves is the transition amplitude in the first Born approximation (Sect. 2.2). The coefficients $c_i^{(e)}(q)$ carry the dimension eV^{-2} of $\tilde{V}$ and the momentum dependence of the mediator. For the Yukawa potential of Sect. 1.4, $c_1^{(e)}(q) = \tilde{V}(q) = y_\chi y_e/(q^2 + m_V^2)$, and every other coefficient vanishes. Papers that normalize their states relativistically quote $4m_\chi m_e\, c_i^{(e)}$ instead, the factor of Eq. (A.45). Refs. [2, 3] define $\boldsymbol{q}$ as the momentum the dark matter gains, the opposite of the $\boldsymbol{q}$ of this book, so a coefficient quoted there changes sign for the operators odd in $\boldsymbol{q}$, $i = 3, 5, 9, 10, 11, 13, 14$; $\boldsymbol{v}^\perp$ is the same in both conventions.

The non-relativistic limit of the couplings of a model fixes which operators the model produces. The limits of the four Dirac bilinears are derived in Appendix C.15 and, between plane waves, in Appendix D.18. The scalar $\bar{\psi}\psi$ and the time component of the vector $\bar{\psi}\gamma^\mu\psi$ both reduce to the density, 1. The spatial vector reduces to the velocity plus a spin term, $(\boldsymbol{k} + \boldsymbol{k}')/2m_e - \mathrm{i}\boldsymbol{q} \times \boldsymbol{S}_e/m_e$, the spatial axial vector $\bar{\psi}\gamma^\mu\gamma^5\psi$ to twice the spin, $2\boldsymbol{S}_e$, and the pseudoscalar $\bar{\psi}\mathrm{i}\gamma^5\psi$ to $-\mathrm{i}\boldsymbol{q} \cdot \boldsymbol{S}_e/m_e$. The coefficients are the product of a dark-matter bilinear, an electron bilinear and the mediator propagator. A vector mediator coupled to both vector currents leads to $\mathcal{O}_1$, and an axial-vector mediator coupled to both axial currents to $\mathcal{O}_4$, the standard spin-independent and spin-dependent interactions. A scalar mediator with a pseudoscalar coupling to the electron produces $\mathcal{O}_{10}$, and with pseudoscalar couplings on both sides $\mathcal{O}_6$. Dark matter with an electric dipole moment produces $\mathcal{O}_{11}$ when it couples through a photon-like mediator; a magnetic dipole moment, $\mathcal{O}_1$, $\mathcal{O}_4$, $\mathcal{O}_5$ and $\mathcal{O}_6$; and an anapole moment, $\mathcal{O}_8$ and $\mathcal{O}_9$ [1, Table II]. The dipole and anapole couplings bring extra powers of q, in the operator or in the coefficient, and with a light mediator these couplings are strongly weighted toward small q. Operator $\mathcal{O}_2 = (\boldsymbol{v}^\perp)^2$ arises in none of these models at leading order [3, Sect. 2.1]. It appears as a correction, of relative size v^2, to $\mathcal{O}_1$, and many lists omit it [5, Table I]; we keep it, because its reduction needs the kinetic-energy density, a target operator that the scalar absorption of Sect. 8.1.5 needs anyway.

8.1.2 Target operators per species

Inside a crystal the electron that the dark matter strikes is one of 10^{23}, bound by the nuclei and by the other electrons, and its bare momentum $\boldsymbol{k}$ is not conserved; only its crystal momentum is (Chap. 3). The structure of Eq. (8.2) survives. Each operator acts on the target particle through three ingredients: the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ that adds the momentum $\boldsymbol{q}$, the velocity $(\boldsymbol{k} + \boldsymbol{k}')/2m_e$ inside $\boldsymbol{v}^\perp$, and the spin $\boldsymbol{S}_e$. Each ingredient has a one-body operator whose matrix elements between plane waves reproduce it. The phase is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ itself. The velocity is the symmetrized product $\frac{1}{2}\{\boldsymbol{p}/m_e, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}$, whose element from $\boldsymbol{k}$ to $\boldsymbol{k}'$ is $(\boldsymbol{k} + \boldsymbol{k}')/2m_e$, since $\boldsymbol{p}$ acting to the left gives $\boldsymbol{k}'$ and to the right gives $\boldsymbol{k}$. The square of the velocity is the doubly symmetrized $\{\boldsymbol{p}, \{\boldsymbol{p}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}\}/4m_e^2$. Summing over the particles, as in Eq. (2.36), leads to five target operators, the rows of the response matrix. Let s be a species, either the electrons ($s = e$) or the nuclei of species κ ($s = \kappa$), with mass m_s. Its particles j have positions $\boldsymbol{r}_j$, momenta $\boldsymbol{p}_j$ and spins $\boldsymbol{S}_j$, and

Target operators per species

$$n_s(\boldsymbol{q}) = \sum_{j\in s} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}, \qquad \boldsymbol{s}_s(\boldsymbol{q}) = \sum_{j\in s} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\,\boldsymbol{S}_j,$$
$$\boldsymbol{j}_s(\boldsymbol{q}) = \sum_{j\in s} \frac{1}{2}\Big\{\frac{\boldsymbol{p}_j}{m_s}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\Big\}, \quad \mathcal{J}_s^{\alpha\beta}(\boldsymbol{q}) = \sum_{j\in s} \frac{1}{2}\Big\{\frac{p_j^\alpha}{m_s}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\Big\}\, S_j^\beta, \tag{8.3}$$
$$\mathcal{T}_s(\boldsymbol{q}) = \sum_{j\in s} \frac{1}{8m_s}\{\boldsymbol{p}_j, \{\boldsymbol{p}_j, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\}\} = \sum_{j\in s} \frac{\boldsymbol{p}_j\cdot \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\,\boldsymbol{p}_j}{2m_s} + \frac{q^2}{8m_s}\, n_s(\boldsymbol{q}).$$

These are the density, the spin density, the paramagnetic current, the spin current (velocity index α, spin index β) and the kinetic-energy density. Their matrix elements from a plane wave $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$ are 1, $\boldsymbol{S}$, $(\boldsymbol{k}+\boldsymbol{k}')/2m_s$, $(k^\alpha+k'^\alpha)S^\beta/2m_s$ and $(\boldsymbol{k}+\boldsymbol{k}')^2/8m_s$.

For the electrons, the density $n_e(\boldsymbol{q})$ is Eq. (2.38), the spin density $\boldsymbol{s}_e(\boldsymbol{q})$ is Eq. (7.15), and the current $\boldsymbol{j}_e(\boldsymbol{q})$ is the current of the continuity equation Eq. (4.29). The spin current and the kinetic-energy density are new. At $\boldsymbol{q}=0$ the kinetic-energy density is the kinetic energy $\sum_j p_j^2/2m_s$ of the species, and at finite $\boldsymbol{q}$ it is the kinetic energy of the mean of the incoming and outgoing momenta. This mean produces the term $q^2/8m_s$ beside the familiar $\boldsymbol{p}\cdot\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{p}/2m_s$. For nuclei the spin $\boldsymbol{S}_j$ is the nuclear spin, and Sect. 8.1.3 turns couplings to nucleons into couplings to these rows.

With the rows in hand the reduction is mechanical. Substitute $\boldsymbol{v}^\perp = \bar{\boldsymbol{v}}_\chi - (\boldsymbol{k}+\boldsymbol{k}')/2m_e$ in each operator, expand, and replace every monomial in the electron's velocity and spin by its row: $1 \to n_e$, $\boldsymbol{S}_e \to \boldsymbol{s}_e$, the velocity $\to \boldsymbol{j}_e$, the velocity times the spin $\to \mathcal{J}_e$, and the square of the velocity $\to (2/m_e)\mathcal{T}_e$. The dark-matter velocity $\bar{\boldsymbol{v}}_\chi$ and spin $\boldsymbol{S}_\chi$ are not target operators; they stay in the coefficients. For instance,

$$\mathcal{O}_8 = \boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp \;\to\; (\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}}_\chi)\, n_e(\boldsymbol{q}) - \boldsymbol{S}_\chi\cdot\boldsymbol{j}_e(\boldsymbol{q}), \qquad \mathcal{O}_7 = \boldsymbol{S}_e\cdot\boldsymbol{v}^\perp \;\to\; \bar{\boldsymbol{v}}_\chi\cdot\boldsymbol{s}_e(\boldsymbol{q}) - \mathcal{J}_e^{\alpha\alpha}(\boldsymbol{q}), \tag{8.4}$$

with a sum over the repeated Cartesian index. Table 8.1 lists all fifteen reductions, and Appendix D.18 derives them. Every operator except $\mathcal{O}_2$ is at most linear in the electron's velocity and in its spin, and needs at most the four rows n_e, $\boldsymbol{s}_e$, $\boldsymbol{j}_e$ and $\mathcal{J}_e$ [1, Table III]; $\mathcal{O}_2$ adds $\mathcal{T}_e$.

In a solid the electron moves too. The velocity-dependent operators split into a piece in which the dark-matter velocity multiplies a velocity-independent row and a piece that needs the target's own velocity. For a free electron at rest the second piece vanishes. In a solid it does not vanish, and it is often the larger piece: in a free-gas model at the valence density, the valence electrons of silicon move at $v_\mathrm{F} = k_\mathrm{F}/m_e = 7.0\times10^{-3}$ (Sect. 4.5), faster than any dark-matter particle in the laboratory. In the language of Ref. [1, Table III], the target velocity turns a coupling to the charge or spin of the electron into a coupling to its orbital motion. The rows also overlap, in the longitudinal direction. The longitudinal current is not independent of the density: by the continuity equation Eq. (4.29), $\langle f|\boldsymbol{q}\cdot\boldsymbol{j}_e|i\rangle = (E_f-E_i)\langle f|n_e|i\rangle$ between eigenstates of any Hamiltonian whose potentials are local. On the dark-matter side, $\boldsymbol{q}\cdot\bar{\boldsymbol{v}}_\chi = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi = \omega + \Delta_\chi$ by Eq. (1.21). In every matrix element that enters a rate, therefore,

$$\boldsymbol{q}\cdot\boldsymbol{v}^\perp \;\to\; \Delta_\chi\, n_e(\boldsymbol{q}), \tag{8.5}$$

which vanishes for elastic dark matter (Appendix D.18). The transversality of $\boldsymbol{v}^\perp$ is thus not a property of free particles alone: as long as the target's potentials are local, only the components of $\bar{\boldsymbol{v}}_\chi$ and $\boldsymbol{j}_e$ transverse to $\boldsymbol{q}$ enter the rate of an elastic process. The same argument applies to the spin current, $q^\alpha\mathcal{J}_e^{\alpha\beta} \to \omega\, s_e^\beta$, provided the Hamiltonian conserves each electron's spin; spin–orbit coupling and the hyperfine interaction violate this conservation.

Table 8.1: Non-relativistic operators of dark-matter scattering and absorption, and the target operators each needs. Upper block: the operators $\mathcal{O}_i$ of Refs. [2, Eqs. (19)–(24) and (45)] and [3, Eqs. (12) and (13)], written for an electron with $\boldsymbol{q}$ the momentum the dark matter loses; $\boldsymbol{v}^\perp$ is Eq. (8.1), $\bar{\boldsymbol{v}} \equiv \bar{\boldsymbol{v}}_\chi = \boldsymbol{v} - \boldsymbol{q}/2m_\chi$, and $m = m_e$. The rows are those of Eq. (8.3) for $s = e$, all at the same $\boldsymbol{q}$; repeated Cartesian indices are summed, and $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbol. For nucleons, replace e by N and m_e by the nucleon mass; the nuclear rows follow from Eq. (8.8). Lower block: absorption of a boson ϕ of mass m_ϕ at $\boldsymbol{q} = \boldsymbol{p}_\phi$, and fermionic absorption with an outgoing fermion $\tilde{\nu}$ (Sect. 8.1.5). Last column: what each row excites in an insulator. Through the nuclei the density row also excites phonons, and in an ordered magnet the spin rows excite magnons (Sect. 7.1).

	Operator	Target operators	Excites
$\mathcal{O}_1$	1	n	pairs, plasmons
$\mathcal{O}_2$	$(\boldsymbol{v}^\perp)^2$	$\bar{v}^2 n - 2\bar{\boldsymbol{v}}\cdot\boldsymbol{j} + (2/m)\mathcal{T}$	pairs
$\mathcal{O}_3$	$\mathrm{i}\boldsymbol{S}_e\cdot(\boldsymbol{q}/m\times\boldsymbol{v}^\perp)$	$(\mathrm{i}/m)\big[(\boldsymbol{q}\times\bar{\boldsymbol{v}})\cdot\boldsymbol{s} - \epsilon_{\alpha\beta\gamma}q^\beta\mathcal{J}^{\gamma\alpha}\big]$	spin flips
$\mathcal{O}_4$	$\boldsymbol{S}_\chi\cdot\boldsymbol{S}_e$	$\boldsymbol{S}_\chi\cdot\boldsymbol{s}$	spin flips
$\mathcal{O}_5$	$\mathrm{i}\boldsymbol{S}_\chi\cdot(\boldsymbol{q}/m\times\boldsymbol{v}^\perp)$	$(\mathrm{i}/m)\,\boldsymbol{S}_\chi\cdot\big[(\boldsymbol{q}\times\bar{\boldsymbol{v}})\,n - \boldsymbol{q}\times\boldsymbol{j}\big]$	pairs
$\mathcal{O}_6$	$(\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m)(\boldsymbol{S}_e\cdot\boldsymbol{q}/m)$	$(\boldsymbol{S}_\chi\cdot\boldsymbol{q})(\boldsymbol{q}\cdot\boldsymbol{s})/m^2$	spin flips
$\mathcal{O}_7$	$\boldsymbol{S}_e\cdot\boldsymbol{v}^\perp$	$\bar{\boldsymbol{v}}\cdot\boldsymbol{s} - \mathcal{J}^{\alpha\alpha}$	spin flips
$\mathcal{O}_8$	$\boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp$	$(\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}})\,n - \boldsymbol{S}_\chi\cdot\boldsymbol{j}$	pairs
$\mathcal{O}_9$	$\mathrm{i}\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{q}/m)$	$(\mathrm{i}/m)\,\boldsymbol{S}_\chi\cdot(\boldsymbol{s}\times\boldsymbol{q})$	spin flips
$\mathcal{O}_{10}$	$\mathrm{i}\boldsymbol{S}_e\cdot\boldsymbol{q}/m$	$(\mathrm{i}/m)\,\boldsymbol{q}\cdot\boldsymbol{s}$	spin flips
$\mathcal{O}_{11}$	$\mathrm{i}\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m$	$(\mathrm{i}/m)(\boldsymbol{S}_\chi\cdot\boldsymbol{q})\,n$	pairs
$\mathcal{O}_{12}$	$\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)$	$\boldsymbol{S}_\chi\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}) - \epsilon_{\alpha\beta\gamma}S_\chi^\alpha\mathcal{J}^{\gamma\beta}$	spin flips
$\mathcal{O}_{13}$	$\mathrm{i}(\boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp)(\boldsymbol{S}_e\cdot\boldsymbol{q}/m)$	$(\mathrm{i}/m)\big[(\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}})(\boldsymbol{q}\cdot\boldsymbol{s}) - S_\chi^\alpha q^\beta\mathcal{J}^{\alpha\beta}\big]$	spin flips
$\mathcal{O}_{14}$	$\mathrm{i}(\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m)(\boldsymbol{S}_e\cdot\boldsymbol{v}^\perp)$	$(\mathrm{i}/m)(\boldsymbol{S}_\chi\cdot\boldsymbol{q})\big(\bar{\boldsymbol{v}}\cdot\boldsymbol{s} - \mathcal{J}^{\alpha\alpha}\big)$	spin flips
$\mathcal{O}_{15}$	$-(\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m)\big[(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)\cdot\boldsymbol{q}/m\big]$	$-(\boldsymbol{S}_\chi\cdot\boldsymbol{q})\big[\boldsymbol{q}\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}) - \epsilon_{\alpha\beta\gamma}q^\alpha\mathcal{J}^{\gamma\beta}\big]/m^2$	spin flips
Dark photon	$-\zeta e\,A'_\mu J^\mu_{\mathrm{em}}$	$-\boldsymbol{j}_e + \sum_\kappa Z_\kappa\boldsymbol{j}_\kappa$	pairs; polar phonons
ALP	$g_{aee}\,\phi\,\bar{\psi}\mathrm{i}\gamma^5\psi$	$\boldsymbol{s}_e$ ($\propto|\boldsymbol{p}_\phi|$); $\mathcal{J}_e^{\alpha\alpha}$ ($\propto m_\phi$)	pairs
Scalar	$g_{\phi ee}\,\phi\,\bar{\psi}\psi$	$n_e - \mathcal{T}_e/m_e$	pairs
Fermionic	$\Lambda_F^{-2}(\bar{\tilde{\nu}}\Gamma\chi)(\bar{\psi}\Gamma'\psi)$	n, $\boldsymbol{j}$, $\boldsymbol{s}$, $\mathcal{J}^{\alpha\alpha}$ or $n - \mathcal{T}/m$, by Γ'	pairs; nuclear recoils

When we collect the coefficients, the interaction between a dark-matter particle and the target takes the form assumed in Sect. 4.2. From dark-matter momentum $\boldsymbol{p}$ to $\boldsymbol{p}-\boldsymbol{q}$ the interaction is

$$\langle \boldsymbol{p}-\boldsymbol{q}|H_{\text{int}}|\boldsymbol{p}\rangle = \frac{1}{\mathcal{V}}\sum_a \tilde{V}_a(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)\,O_a(\boldsymbol{q}), \tag{8.6}$$

with the index a running over the rows of every species. The only change from Sect. 4.2 is that the potentials $\tilde{V}_a$ now depend on the dark-matter velocity and are matrices in the spin space of the dark matter. For $\mathcal{O}_8$, for instance, $\tilde{V}_{n_e} = c_8^{(e)}\,\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}}_\chi$ and $\tilde{V}_{j_e^\alpha} = -c_8^{(e)}S_\chi^\alpha$. An unpolarized beam averages over the initial spin of the dark matter and sums over its final spin, and the square of the amplitude $\mathcal{V}^{-1}\sum_b\langle f|O_b|i\rangle\tilde{V}_b$ becomes a trace over that spin space:

Dark-matter-side tensor

$$\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v}) = \frac{1}{2|\boldsymbol{S}_\chi|+1}\,\text{Tr}_\chi[\tilde{V}_a(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)^\dagger\,\tilde{V}_b(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)], \tag{8.7}$$

with $|\boldsymbol{S}_\chi|$ the spin quantum number of the dark matter. The rate of Eq. (4.14) holds with $\tilde{V}_a^*\tilde{V}_b \to \mathfrak{D}_{ab}$. The tensor is Hermitian and positive semidefinite, and for a coupling that does not involve $\boldsymbol{S}_\chi$ it is $\tilde{V}_a^*\tilde{V}_b$.

Since $\text{Tr}_\chi\,\boldsymbol{S}_\chi = 0$, a coefficient linear in $\boldsymbol{S}_\chi$ never interferes with one that is independent of it, and the operators split into two classes that add in the rate. For spin $\frac{1}{2}$, $\text{Tr}_\chi(S_\chi^\alpha S_\chi^\beta)/2 = \delta_{\alpha\beta}/4$.

Responses of the density are what Chap. 4 computed. The other rows need the same calculation with other vertices. For the spin density and the spin current, the vertex is the density vertex or the current vertex multiplied by a Pauli matrix. For the current and the kinetic-energy density the vertex needs the momentum operator. When the pseudopotential is nonlocal, the matrix elements of the momentum operator change and must be corrected for the nonlocality. In practice a calculation computes the independent-particle response matrix of all the electron rows at once and screens it with one Dyson equation (Sect. 8.4).

8.1.3 Nuclei

A model specifies its couplings to protons and neutrons, $c_i^{(p)}$ and $c_i^{(n)}$, with operators $\mathcal{O}_i^{(N)}$ built from the nucleon spin $\boldsymbol{S}_N$, the nucleon mass m_N and the nucleon velocity. The solid, however, contains nuclei. Nucleus I of species κ is at $\boldsymbol{X}_I$, and its A_κ nucleons are at $\boldsymbol{X}_I+\boldsymbol{y}_j$, within a radius of a few fm. The nucleon operators reduce to nuclear rows because the nucleus is small and the energy deposited is small. Light dark matter transfers a momentum of at most $2m_\chi v_{\max}\approx 5\,\text{MeV}$, against $1/R\approx 57\,\text{MeV}$ for silicon (Sect. 1.4), so $\boldsymbol{q}\cdot\boldsymbol{y}_j$ is small. The energy deposited on a nucleus is small as well: at most 0.50 keV on silicon at $m_\chi = 1\,\text{GeV}$ (Sect. 1.3), far below the excitation energies of nuclei [2, Sect. 2.2], so the nucleus stays in its ground multiplet. The small value of $\boldsymbol{q}\cdot\boldsymbol{y}_j$ turns the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{X}_I+\boldsymbol{y}_j)}$ into $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ times the nuclear form factor $F_N(q)$ of Eq. (2.26), and the nucleon couplings add coherently, as in Sect. 1.4. The nucleon velocity is the velocity of the nucleus, $\boldsymbol{P}_I/M_\kappa$, plus an internal velocity; the internal velocity averages to zero in the ground state and contributes at higher order in $\boldsymbol{q}\cdot\boldsymbol{y}_j$ [2, Eqs. (26) and (27)], and we drop it. Each nucleon spin carries its coupling, and the weighted sum of the nucleon spins is a vector operator. Since the nucleus stays in its ground multiplet, the Wigner–Eckart theorem makes this weighted sum proportional to the nuclear spin $\boldsymbol{J}_I^{\text{nuc}}$ within the ground multiplet of spin J_{nuc}. The proportionality constant follows from the state with $M = J_{\text{nuc}}$, in which the proton and neutron spins have the expectation values $\langle S_p\rangle$ and $\langle S_n\rangle$ along the nuclear spin. Together the coherent sum and the Wigner–Eckart

theorem give, for nucleus species κ (and, for spin couplings, a single isotope),

$$c_i^{(\kappa)}(q) = [Z_\kappa c_i^{(p)} + (A_\kappa - Z_\kappa)\, c_i^{(n)}] F_N(q) \quad \text{or} \quad c_i^{(\kappa)}(q) = \frac{c_i^{(p)}\langle S_p\rangle + c_i^{(n)}\langle S_n\rangle}{J_\text{nuc}}, \tag{8.8}$$

the first for operators without $\boldsymbol{S}_N$ and the second, with $\boldsymbol{S}_N \to \boldsymbol{J}_I^\text{nuc}$, for those linear in it. The nuclear operators then reduce as in Table 8.1, to the rows of Eq. (8.3) with $s = \kappa$, $m_s = M_\kappa$ and $\boldsymbol{S}_j = \boldsymbol{J}_I^\text{nuc}$; in particular the nuclear spin density is $\boldsymbol{s}_\kappa(\boldsymbol{q}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \boldsymbol{J}_I^\text{nuc}$. The operators keep $\boldsymbol{q}/m_N$ with the nucleon mass, since the nucleon coefficients are defined with $\boldsymbol{q}/m_N$.

The spin coupling is far weaker than the coherent one. Silicon has one isotope with a nuclear spin, ^{29}Si, with $J_\text{nuc} = \frac{1}{2}$ [6, Fig. 5] and an abundance of 4.685 % in natural silicon [7]. Shell-model calculations give $\langle S_n\rangle = 0.156$ and $\langle S_p\rangle = 0.016$ [6, Table III]. For a coupling to neutron spins alone, Eq. (8.8) gives $c_4^{(\kappa)} = 0.312\, c_4^{(n)}$. Averaged over the initial orientations of the nucleus and summed over the final ones, the squared spin operator is $J_\text{nuc}(J_\text{nuc}+1) = \frac{3}{4}$. The strength of the spin coupling is then $(0.312)^2 \times \frac{3}{4} = 0.073$ in units of $|c_4^{(n)}|^2$, against $\frac{3}{4}$ for a free neutron. A ^{29}Si nucleus thus responds with 10 % of the strength of one neutron, whereas the spin-independent coupling of a ^{29}Si nucleus is $A^2 = 841$ times the coupling of one nucleon. In natural silicon, moreover, one nucleus in 21 carries a nuclear spin. A sample of unpolarized nuclear spins has no coherent spin response, because the spins of two different nuclei are uncorrelated; only the incoherent part of $S_{\boldsymbol{s}_\kappa \boldsymbol{s}_\kappa}$ survives (Sect. 11.6).

Worked example 8.1: A heavy dark photon reduces to the density rows

Take the thermal benchmark of Worked example 1.5: $m_\chi = 10\,\text{MeV}$, $m_V = 30\,\text{MeV}$, $y_\chi = 0.3$ and $\zeta = 10^{-4}$. The dark photon couples to the dark-matter current with strength y_χ and to J^μ_em with strength ζe (Sect. 1.4). Exchanged between a dark-matter particle and a target fermion with the currents J^μ_χ and J^μ, the dark photon gives, for $\omega \ll q$, the potential $y_\chi y\,(J^0_\chi J^0 - \boldsymbol{J}_\chi\cdot\boldsymbol{J})/(q^2 + m_V^2)$, with y the coupling of the target fermion (Appendix D.18). The time components reduce to 1 on both sides, and the leading term is $\mathcal{O}_1$ with

$$c_1^{(e)} = \frac{y_\chi y_e}{q^2 + m_V^2} = \tilde{V}(q), \qquad c_1^{(p)} = -c_1^{(e)}, \qquad c_1^{(n)} = 0,$$

since $y_e = -\zeta e$ and a proton has $+\zeta e$. By Eq. (8.8), $c_1^{(\kappa)} = -Z_\kappa \tilde{V} F_N(q)$, and with $F_N = 1$ to better than 0.1 % at these momenta the interaction is $\mathscr{V}^{-1}\tilde{V}(q)[n_e(\boldsymbol{q}) - \sum_\kappa Z_\kappa n_\kappa(\boldsymbol{q})] = -\mathscr{V}^{-1}\tilde{V}(q)\, n_Z(\boldsymbol{q})$, with n_Z the charge-number density of Sect. 2.3. The tensor is $\mathfrak{D}_{ab} = \tilde{V}_a^* \tilde{V}_b$ over the rows n_e and n_κ, and the rate is $|\tilde{V}|^2$ times the structure factor of the total charge. The electron block of this structure factor is the S of Eq. (1.1); the nuclear block and the cross terms, of opposite sign, matter where the nuclei move, at phonon energies (Sect. 8.4).

What is dropped? At $m_V = 30\,\text{MeV}$ the mediator is heavy at every momentum transfer, $q \le 2m_\chi v_\text{max} = 53\,\text{keV}$, and the propagator is constant to $(q/m_V)^2 \le 3\times10^{-6}$. The largest correction to the density vertex comes from the factor $1 - q^2/8m_e^2$ of Eq. (C.132); the correction is 7×10^{-6} at q_0 and 1.3×10^{-3} at 53 keV. The spatial currents, $\boldsymbol{J}_\chi\cdot\boldsymbol{J}$, are products of velocities and spin terms. Their spin-independent piece, $\bar{\boldsymbol{v}}_\chi\cdot(\boldsymbol{k}+\boldsymbol{k}')/2m_e$, is at most $v_\text{max} v_\text{F} = 1.9\times10^{-5}$ for valence electrons. The pieces linear in one spin do not interfere with the density coupling in a nonmagnetic target and enter the rate squared, and the product of the two spin terms is at most $q^2/(4m_\chi m_e) = 1.4\times10^{-4}$. To a part in 10^3, therefore, a heavy dark photon couples only to the density rows.

8.1.4 The velocity-squared row

The operator $\mathcal{O}_2 = (\boldsymbol{v}^\perp)^2$ is the only one that is quadratic in a velocity. Expanding the square,

$$(\boldsymbol{v}^\perp)^2 = \bar{v}_\chi^2 - 2\bar{\boldsymbol{v}}_\chi \cdot \frac{\boldsymbol{k}+\boldsymbol{k}'}{2m_e} + \frac{(\boldsymbol{k}+\boldsymbol{k}')^2}{4m_e^2} \;\to\; \bar{v}_\chi^2\, n_e(\boldsymbol{q}) - 2\bar{\boldsymbol{v}}_\chi \cdot \boldsymbol{j}_e(\boldsymbol{q}) + \frac{2}{m_e}\,\mathcal{T}_e(\boldsymbol{q}). \tag{8.9}$$

The first two terms use the density and current rows. The third needs the square of the electron's velocity, and since $\mathcal{O}_2$ is a scalar, only the trace of the velocity tensor $v^\alpha v^\beta$ appears. The trace is $2/m_e$ times the kinetic-energy density of Eq. (8.3). One row per species therefore suffices.

How large is each term? A naive estimate sets $\boldsymbol{v}^\perp \sim \boldsymbol{v} \sim 10^{-3}$ and makes $\mathcal{O}_2$ a correction of about 10^{-6} to $\mathcal{O}_1$. That estimate, however, treats the electron as being at rest, and Eq. (8.9) shows what it misses. By Eq. (8.5), the components along $\boldsymbol{q}$ of the dark-matter and electron velocities cancel for elastic scattering, and $\boldsymbol{v}^\perp$ reduces to the difference between the transverse velocities of the dark matter and the electron. The dark matter's transverse velocity obeys $v_\mathrm{T}^2 = v^2 - v_\mathrm{min}^2 \le v_\mathrm{max}^2$, since its longitudinal part is v_min at $\Delta_\chi = 0$. The transverse velocity of the electron is set by the motion of the electrons in the solid, about v_F for valence states and larger for the core.

Worked example 8.2: The velocity-squared operator in silicon

We model the valence electrons of silicon as the free electron gas of Sect. 4.5, at $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, with $k_\mathrm{F} = 3.57\,\mathrm{keV}$ and $v_\mathrm{F} = 6.99 \times 10^{-3}$, at $T = 0$. Take dark matter of mass 100 MeV moving at $v = 2.0 \times 10^{-3}$ that transfers $q = q_0 = 3.73\,\mathrm{keV}$ and deposits $\omega = 5\,\mathrm{eV}$. A pair takes an electron from $\boldsymbol{k}$ inside the Fermi sphere to $\boldsymbol{k}+\boldsymbol{q}$ outside it, with $\omega = \boldsymbol{q}\cdot(2\boldsymbol{k}+\boldsymbol{q})/2m_e$, and this energy condition fixes the component of $\boldsymbol{k}$ along $\boldsymbol{q}$ at $m_e\omega/q - q/2 = -1.18\,\mathrm{keV}$. The two Pauli conditions then confine the transverse momentum k_T to an annulus, $k_\mathrm{F}^2 - (2.55\,\mathrm{keV})^2 < k_\mathrm{T}^2 < k_\mathrm{F}^2 - (1.18\,\mathrm{keV})^2$, with 2.55 keV the longitudinal momentum after the momentum transfer; that is, $2.50\,\mathrm{keV} < k_\mathrm{T} < 3.37\,\mathrm{keV}$. Every pair of this $(\boldsymbol{q},\omega)$ fills the annulus uniformly in area. The electron's transverse velocity, $(\boldsymbol{k}+\boldsymbol{k}')_\mathrm{T}/2m_e = \boldsymbol{k}_\mathrm{T}/m_e$, has $\langle v_{e\mathrm{T}}^2\rangle = 3.37 \times 10^{-5}$ and $\langle v_{e\mathrm{T}}^4\rangle = 1.17 \times 10^{-9}$ over the annulus.

On the dark-matter side, $v_\mathrm{min} = \omega/q + q/2m_\chi = 1.36 \times 10^{-3}$ and $v_\mathrm{T}^2 = v^2 - v_\mathrm{min}^2 = 2.15 \times 10^{-6}$, smaller than the electron's by a factor of 16; at $v = v_\mathrm{max} = 2.650 \times 10^{-3}$ the factor is still 6.5. Per pair, $(\boldsymbol{v}^\perp)^2 = |\boldsymbol{v}_\mathrm{T} - \boldsymbol{v}_{e\mathrm{T}}|^2$, and the azimuth of $\boldsymbol{v}_{e\mathrm{T}}$ about $\boldsymbol{q}$ is uniform. With $A = v_\mathrm{T}^2$ and $B = v_{e\mathrm{T}}^2$, the square of $(\boldsymbol{v}^\perp)^2$ averages over the azimuth to $(A+B)^2 + 2AB$, and over the annulus

$$\big\langle (\boldsymbol{v}^\perp)^4 \big\rangle = A^2 + 4A\langle B\rangle + \langle B^2\rangle = (0.0046 + 0.290 + 1.167) \times 10^{-9} = 1.46 \times 10^{-9}.$$

The naive estimate keeps the term A^2, which is 0.3 % of the total. The fourth power of the electron's transverse velocity gives the term $\langle B^2\rangle$, which carries 80 %. The cross term $4A\langle B\rangle$ carries the remaining 20 %; in the cross term the two transverse velocities meet. The three shares belong to velocities, not to rows: every row of Eq. (8.9) also carries longitudinal velocity, which cancels in the sum of the three rows. At equal coefficients the rate of $\mathcal{O}_2$ is 1.5×10^{-9} times the rate of $\mathcal{O}_1$ at this $(\boldsymbol{q},\omega)$, about 90 times the naive $v^4 = 1.6 \times 10^{-11}$.

The electron gas understates the electron velocities of a real crystal. The valence wavefunctions of silicon have Fourier components at the reciprocal-lattice vectors (Sect. 3.2), so the electron velocities spread beyond v_F. The $1s$ electrons move at $Z\alpha \approx 0.10$, and for

them $(\boldsymbol{v}^\perp)^2 \approx 10^{-2}$ (Appendix C.15); their absorption edge lies at 1.84 keV (Sect. 4.3). A velocity-dependent coupling therefore weights the core and the high-momentum tails of the valence states far more than the density coupling does. In practice, a calculation must get the target term $\mathcal{T}_e$ right.

8.1.5 Absorption operators

A bosonic dark-matter field ϕ of mass m_ϕ oscillates as $\mathrm{e}^{-\mathrm{i}m_\phi t + \mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$, and absorbing one quantum deposits $\omega = m_\phi$ and $\boldsymbol{q} = \boldsymbol{p}_\phi$ (Eq. (1.37)). Its coupling to the target is a bilinear of the target fields multiplied by ϕ or a derivative of ϕ, and the spatial integral over $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$ turns each bilinear into a row of Eq. (8.3) at $\boldsymbol{q} = \boldsymbol{p}_\phi$. The rows are the same as for scattering; what differs is the point $(\boldsymbol{q}, \omega)$ at which their response is needed, $|\boldsymbol{p}_\phi| \approx 10^{-3}\omega$ instead of $\omega \leq qv$. The three bosons of Sect. 1.4 need different rows (Appendix D.18).

The dark photon couples to J^μ_{em} like a photon, with $e \to \zeta e$. For a dark photon A'_μ of mass $m_V = m_\phi$ at rest, the time component is smaller than the spatial ones by $|\boldsymbol{p}_\phi|/m_\phi$, by the Lorenz condition, and the coupling is to the spatial current of all charges, $-\boldsymbol{j}_e(\boldsymbol{p}_\phi) + \sum_\kappa Z_\kappa \boldsymbol{j}_\kappa(\boldsymbol{p}_\phi)$. At $\boldsymbol{p}_\phi \to 0$ the response of this current is the optical conductivity (Sect. 8.2.3). The nuclear currents carry the absorption into polar optical phonons below the electronic gap.

The axion-like particle couples through Eq. (1.39), in which we write ϕ, $\nabla\phi$ and $\partial_t\phi$ for the amplitudes of the plane wave. The leading term of the coupling is $(g_{aee}/m_e)(\nabla\phi)\cdot\boldsymbol{s}_e$, the spin-density row with a coefficient proportional to $|\boldsymbol{p}_\phi|$. The second term contains $\hat\psi^\dagger\boldsymbol{\sigma}\cdot\overleftrightarrow{\nabla}\hat\psi$, whose matrix element between plane waves is $\mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{k}+\boldsymbol{k}')$; this operator is the trace of the spin current, and the interaction becomes

$$H_{aee} = \frac{g_{aee}}{m_e}\Big[(\nabla\phi)\cdot\boldsymbol{s}_e(\boldsymbol{p}_\phi) + (\partial_t\phi)\,\mathcal{J}_e^{\alpha\alpha}(\boldsymbol{p}_\phi)\Big]. \tag{8.10}$$

The trace couples $\boldsymbol{S}_j$ to the velocity of the same electron. With $|\partial_t\phi| = m_\phi|\phi|$ and $|\nabla\phi| = |\boldsymbol{p}_\phi||\phi|$, the second term exceeds the first by the ratio of the electron velocity to the dark-matter velocity, as Sect. 1.4 found. When spin–orbit coupling is negligible the spin and the orbit of each electron separate, and the response of $\mathcal{J}_e^{\alpha\alpha}$ reduces to the response of the current. The rate of absorption of the axion-like particle therefore again follows from the conductivity [8, Sect. III B].

The scalar couples through Eq. (1.40). The operator $\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi$ has the plane-wave matrix element $-(\boldsymbol{k}+\boldsymbol{k}')^2$, so $(1/8m_e^2)\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi$ is $-\mathcal{T}_e/m_e$, and

Scalar absorption through the kinetic-energy density

$$H_{\phi ee} = -g_{\phi ee}\,\phi\Big[n_e(\boldsymbol{p}_\phi) - \frac{\mathcal{T}_e(\boldsymbol{p}_\phi)}{m_e}\Big]. \tag{8.11}$$

The bracket is the non-relativistic limit of $\bar\psi\psi$: the number density times $1 - v^2/2$, the factor $(1-v^2)^{1/2}$ by which a moving particle's scalar density falls below its number density. Matrix elements of $n_e(\boldsymbol{p}_\phi)$ between different states vanish as $\boldsymbol{p}_\phi \to 0$, and the matrix elements of $\mathcal{T}_e$ do not, so the kinetic-energy term dominates absorption [8, Eq. (18) and Sect. III C].

Scalar absorption needs the same kinetic-energy row as $\mathcal{O}_2$, so a single new row per species covers both $\mathcal{O}_2$ and scalar absorption. The density term is suppressed twice: by orthogonality, since $\langle f|n_e(\boldsymbol{p}_\phi)|i\rangle \simeq \mathrm{i}\boldsymbol{p}_\phi\cdot\langle f|\sum_j \boldsymbol{r}_j|i\rangle$, and by the screening of a density coupling in the medium [8, Sect. III C]. The kinetic energy does not commute with the crystal Hamiltonian and has matrix elements between different states already at $\boldsymbol{p}_\phi = 0$. Unlike the absorption

rates of the dark photon and the axion-like particle, the rate of scalar absorption is not fixed by the optical conductivity, and Sect. 10.7 computes it.

Fermionic dark matter can be absorbed too. The operators of Refs. [9, Eq. (1), 10, Eqs. (1) and (2)] are $\Lambda_F^{-2}(\bar{\tilde{\nu}}\Gamma\chi)(\bar{\psi}\Gamma'\psi)$, with a heavy scale Λ_F and Dirac matrices Γ, Γ'. In these operators the dark-matter fermion χ is destroyed and a light fermion $\tilde{\nu}$ of mass $m_{\tilde{\nu}} \ll m_\chi$ is emitted. The target bilinear $\bar{\psi}\Gamma'\psi$ reduces as for scattering: the vector to the density and current rows, the axial vector to the spin density and the trace of the spin current, the scalar to the bracket of Eq. (8.11). The dark-matter side is new; Sect. 8.2.3 builds it into the rate, and Sect. 8.1.6 fixes the kinematics.

8.1.6 Energy deposited

In every process of this section the target receives the energy that the dark-matter side gives up. A particle of mass m_χ and momentum $\boldsymbol{p}$ enters, and either nothing or a particle of momentum $\boldsymbol{p} - \boldsymbol{q}$ leaves, so that

$$\omega = E_{\text{in}}(\boldsymbol{p}) - E_{\text{out}}(\boldsymbol{p} - \boldsymbol{q}). \tag{8.12}$$

For elastic scattering, E_{out} is the kinetic energy of the same particle and Eq. (8.12) is $\omega_{\boldsymbol{q}}$ of Eq. (1.21) at $\Delta_\chi = 0$. For inelastic dark matter the outgoing state χ_2 has mass $m_\chi + \Delta_\chi$. With the energy of χ_2 expanded to first order in $(\boldsymbol{p} - \boldsymbol{q})^2$, Eq. (8.12) becomes

$$\omega = \omega_{\boldsymbol{q}} + \frac{\Delta_\chi}{m_\chi + \Delta_\chi} \frac{(\boldsymbol{p} - \boldsymbol{q})^2}{2m_\chi}, \tag{8.13}$$

with $\omega_{\boldsymbol{q}}$ including the term $-\Delta_\chi$ (Appendix D.18). Eq. (1.21) drops the second term, which corrects the recoil energy by the fraction Δ_χ/m_χ; endothermic and exothermic scattering are then exactly the cases of Sect. 1.3. The outgoing particle can also be a light fermion. For $m_{\tilde{\nu}} \to 0$ and $v \to 0$, $E_{\text{in}} = m_\chi$ and $E_{\text{out}} = |\boldsymbol{q}|$, so

$$\omega = m_\chi - q, \tag{8.14}$$

and nearly all of the rest energy goes to $\tilde{\nu}$: the target is probed along a line in the (q, ω) plane at $q \approx m_\chi$, far above the $q \leq 2m_\chi v$ of scattering. On a free nucleus of mass M the recoil is then $\omega \approx m_\chi^2/2M$ [9, Eq. (2)], larger than an elastic recoil by $1/v^2$. Bosonic absorption is the limit in which nothing leaves, with $\omega = m_\phi$ at $\boldsymbol{q} = \boldsymbol{p}_\phi$. Scattering fills a region of the plane, fermionic absorption a line and bosonic absorption a point, and Sect. 8.2 shows that one rate formula, with the rows of Eq. (8.3) and the tensor of Eq. (8.7), covers all three.

8.2 The master rate formula

Consider a dark-matter particle that scatters in silicon, a dark photon of a few eV that is absorbed in it, and an excited dark-matter state that drops to its ground state on the way through. All three probe one property of the crystal: how readily the crystal takes up a momentum $\boldsymbol{q}$ together with an energy ω. For scattering through several target operators at once, Eq. (4.14) already gives the rate, for one dark-matter particle of fixed velocity and for potentials that are plain numbers. Three things are missing before Eq. (4.14) can serve every model of Sect. 8.1. The potentials are matrices in the spin space of the dark matter, and the rate needs the trace $\mathfrak{D}_{ab}$ of Eq. (8.7). The halo supplies a distribution of velocities, and experiments quote rates per unit target mass. And two kinds of event look at first sight like different processes: absorption, in which the dark matter disappears, and transitions in which the target gives energy away. All these processes follow from one golden-rule calculation and land in different regions of the $(\boldsymbol{q}, \omega)$ plane (Fig. 8.3). The master rate formula is exact in the target and first order in the coupling to the dark matter.

8.2.1 Scattering

Take a dark-matter particle in the plane wave $\boldsymbol{p} = m_\chi \boldsymbol{v}$, in spin state s, and a target in the eigenstate $|i\rangle$. By Eq. (8.6) the amplitude to reach $\boldsymbol{p} - \boldsymbol{q}$, spin s' and $|f\rangle$ is $\mathscr{V}^{-1} \sum_b \langle s'|\tilde{V}_b|s\rangle \langle f|O_b(\boldsymbol{q})|i\rangle$. The golden rule squares it, the spin sums turn the product of two potentials into the trace of Eq. (8.7), the thermal average weights $|i\rangle$ with p_i, and the sum over final dark-matter momenta becomes $\mathscr{V} \int \mathrm{d}^3q/(2\pi)^3$, as in Sect. 2.5. Each step is written out in Appendix D.19. The target then enters through the response matrix Eq. (4.13), and the dark matter through $\mathfrak{D}_{ab}$:

Master rate formula: scattering

$$\Gamma(\boldsymbol{v}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{a,b} \underbrace{\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})}_{\text{coupling}} \underbrace{S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}})}_{\text{target}}, \tag{8.15}$$

the rate of one dark-matter particle of velocity $\boldsymbol{v}$, with $\omega_{\boldsymbol{q}}$ of Eq. (1.21), and per unit target mass

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{1}{\rho_T} \underbrace{\frac{\rho_\chi}{m_\chi} \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})}_{\text{halo}} \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{a,b} \underbrace{\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})}_{\text{coupling}} \times \underbrace{S_{ab}(\boldsymbol{q},\omega)}_{\text{target}} \underbrace{\delta(\omega - \omega_{\boldsymbol{q}})}_{\text{kinematics}}. \tag{8.16}$$

The rows a, b run over the target operators of every species (Eq. (8.3)), the target is in thermal equilibrium at temperature T, and ω takes either sign.

The integrand is a contraction of two Hermitian positive semidefinite matrices, $\mathfrak{D}$ from the dark-matter side and S from the target, and is therefore never negative. For potentials without dark-matter spin, Eq. (8.15) is the rate Eq. (4.14), and for the density row alone it is Eq. (2.67). In the same form, Trickle, Zhang and Zurek write the scattering rate with an operator-valued potential [1, Eq. (1)], and Trickle and collaborators write the total rate per unit target mass [11, Eqs. (14)–(16)]; both papers take a target that starts in one eigenstate.

The order of the indices matters once both matrices are complex. Since the golden rule pairs the conjugated potential $\tilde{V}_a^\dagger$ with the conjugated operator $O_a^\dagger$, Eq. (8.15) contracts $\mathfrak{D}_{ab}$ with S_{ab}, element by element; this element-wise contraction is the trace $\mathrm{Tr}[\mathfrak{D} S^{\mathrm{T}}]$. If we split each Hermitian matrix into a real symmetric and a real antisymmetric part, $\mathfrak{D} = \mathfrak{D}_{\mathrm{R}} + \mathrm{i}\mathfrak{D}_{\mathrm{I}}$ and $S = S_{\mathrm{R}} + \mathrm{i}S_{\mathrm{I}}$, the other order, $\mathrm{Tr}[\mathfrak{D} S]$, differs from $\mathrm{Tr}[\mathfrak{D} S^{\mathrm{T}}]$ by $2\,\mathrm{Tr}[\mathfrak{D}_{\mathrm{I}} S_{\mathrm{I}}]$ (Appendix D.19). The two orders agree whenever one of the matrices is real. They differ when both are complex. An example is an operator with an explicit i, such as $\mathcal{O}_3$ of Table 8.1, acting on a target whose current–spin response has an imaginary antisymmetric part, as spin–orbit coupling produces.

Eq. (8.16) reduces exactly to the rates of Chap. 1. For an interaction through the electron density alone, $\mathfrak{D}_{ab} = |\tilde{V}|^2$ for $a = b = n_e$ and zero otherwise, and the contraction is $|\tilde{V}(\boldsymbol{q})|^2 S(\boldsymbol{q},\omega)$. With $\Delta_\chi = 0$, Eq. (8.16) and Eq. (1.1) have the same delta function, and the two formulas coincide term by term, including the normalization $\rho_\chi/m_\chi\rho_T$; the common formula is Eq. (A.42). For a structure factor that depends only on q, or for one averaged over the directions of $\boldsymbol{q}$, the velocity integral and the angular integral act on the delta function alone. They give $\eta(v_{\min})/2q$ per unit $\mathrm{d}^3q/(2\pi)^3$ (Eq. (A.41)), and with $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ the result is Eq. (1.46). For a crystal S_{ab} depends on the direction of $\boldsymbol{q}$, and the full velocity kernel is kept, as Appendix A.8 describes for the daily modulation.

8.2.2 Assumptions of the formula

The derivation uses four assumptions, and each can fail in a known place.

First Born approximation. The interaction with the dark matter acts once. The next term in the amplitude passes through intermediate dark-matter states and is smaller by roughly the ratio of the scattering amplitude $|f| = \mu_{\chi T}|\tilde{V}|/2\pi$ (Sect. 2.2) to the range of the potential [12, Chap. 6]. For the heavy dark photon of Worked example 8.1, at $m_\chi = 10\,\mathrm{MeV}$ and $m_V = 30\,\mathrm{MeV}$, $|f|\,m_V = 2 \times 10^{-8}$. At the cross-sections used in this book the correction from this next term is below 10^{-4} for m_χ from 1 keV to 1 GeV and m_V up to 10 GeV (Limit Check 8.1). The interactions inside the target are not expanded: S_{ab} is exact, and any approximation in it is an approximation to the target. Sect. 8.5 records the approximations to the target in a truncation label $\mathfrak{T}$, the list of orders that a calculation keeps in each small parameter of the target. A truncation of the target changes only S_{ab}, never the master formula.

Plane waves. The halo is an incoherent mixture of momenta, distributed as f_χ, and the final dark-matter state is not observed. Averaging plane-wave rates over f_χ is then exact. For bosons light enough that the halo is a classical wave, the ensemble average still holds for a long enough observation, and Sect. 8.2.3 gives the condition on the observation time.

Gibbs target. The target is in thermal equilibrium before each event, with the weights $p_i = \mathrm{e}^{-\beta E_i}/\mathcal{Z}$. A detector qualifies when events are rare compared with its relaxation. Eq. (8.15) still holds for a target prepared out of equilibrium, such as an ensemble of polarized nuclear spins, with the target's own populations in S_{ab}, provided the density operator of the target is diagonal in energy. Detailed balance and the fluctuation–dissipation theorem then fail. Two extensions are open: to a density operator with coherences between energy eigenstates, and to a target driven during the exposure.

Unpolarized dark matter. The factor $1/(2|\boldsymbol{S}_\chi| + 1)$ in $\mathfrak{D}_{ab}$ assumes that the halo has no preferred spin direction. For a polarized flux, with spin density matrix $\hat{\varrho}_\chi$ of unit trace, the dark-matter tensor is

$$\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v}) = \mathrm{Tr}_\chi\big[\hat{\varrho}_\chi\, \tilde{V}_a(\boldsymbol{q}, \bar{\boldsymbol{v}}_\chi)^\dagger\, \tilde{V}_b(\boldsymbol{q}, \bar{\boldsymbol{v}}_\chi)\big], \tag{8.17}$$

which is Eq. (8.7) for $\hat{\varrho}_\chi = 1/(2|\boldsymbol{S}_\chi| + 1)$.

Limit Check 8.1. First Born approximation for a Yukawa potential

For a Yukawa potential of range $1/m_V$ and strength $y_\chi y_T$, with $y_T = y_e$ for an electron and $A y_N$ for a nucleus, the second Born term changes the cross-section by the fraction

$$\delta_{\mathrm{Born}} \simeq \pm\frac{\mu_{\chi T}\, y_\chi y_T}{2\pi(m_V + p_{\mathrm{rel}})} = \pm|f(0)|\, m_V\, \frac{m_V}{m_V + p_{\mathrm{rel}}}, \tag{8.18}$$

upper sign for attraction [12, Chap. 6]. At $p_{\mathrm{rel}} \ll m_V$ Eq. (8.18) is exact: the zero-energy radial equation gives the scattering length $a = a_{\mathrm{Born}}(1 \pm |f(0)|m_V/2)$. At $p_{\mathrm{rel}} \gg m_V$, partial waves give a correction below $y_\chi y_T/4\pi v$ at every $q \le 2p_{\mathrm{rel}}$ (0.28 of this bound at $q = 2p_{\mathrm{rel}} = 20\,m_V$). The correction falls as $m_V/p_{\mathrm{rel}} \to 0$, the limit in which the Born cross-section of the Coulomb potential is exact; Eq. (8.18) then overestimates the correction.

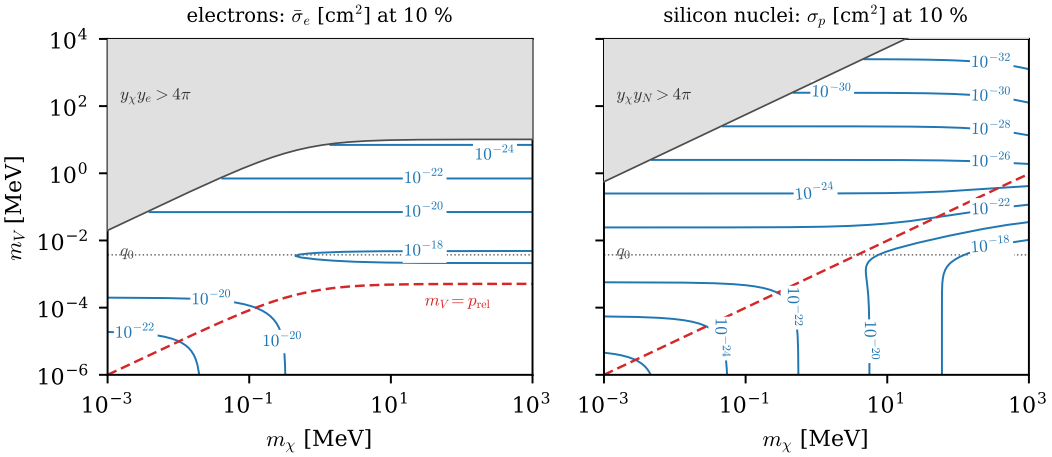


Figure 8.2: Cross-section at which $|\delta_{\rm Born}|$ of Eq. (8.18) reaches 0.1, against m_χ and m_V, at $v = 10^{-3}$. Left: electrons, $\bar{\sigma}_e$ with the form factor normalized at $q_0 = \alpha m_e$ (Eq. (1.34)). Right: silicon nuclei, coherent ($A = 28$), σ_p normalized in the same way. With the coupling fixed by the cross-section, m_χ enters only through $p_{\rm rel} = \mu_{\chi T} v$ (dashed, $m_V = p_{\rm rel}$; dotted, $m_V = q_0$). Gray: for $m_V + p_{\rm rel} > 20 A \mu_{\chi T}$ ($A = 1$ for electrons), the 10 % point needs $y_\chi y_e$ or $y_\chi y_N$ above 4π. At $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ the largest correction in the left panel is 4.5×10^{-5}, and at $\sigma_p = 10^{-40}\,\mathrm{cm}^2$ the largest correction in the right panel is 8.0×10^{-5}, both at $m_V = 10\,\mathrm{GeV}$. With perturbative couplings, 10 % needs $\bar{\sigma}_e \geq 4.9 \times 10^{-25}\,\mathrm{cm}^2$ (at $m_V \approx 10\,\mathrm{MeV}$) and $\sigma_p \geq 1.6 \times 10^{-34}\,\mathrm{cm}^2$ (at $m_\chi = 1\,\mathrm{GeV}$, $m_V = 10\,\mathrm{GeV}$).

A correction of order one needs couplings near 4π with $m_V \lesssim 20 A \mu_{\chi T}$, where the attractive potential binds and the cross-section has resonances that only the full partial-wave solution describes.

8.2.3 Absorption of a bosonic field

An absorbed boson ϕ of mass m_ϕ leaves nothing behind, and its whole energy and momentum go into the target. By Sect. 8.1.5 its coupling is linear in the field and turns each target bilinear into a row at $\boldsymbol{q} = \boldsymbol{p}_\phi$. We write the coefficient of row a per unit field amplitude as g_a^ϕ, so that for the scalar of Eq. (8.11), $g_{n_e}^\phi = -g_{\phi ee}$ and $g_{\mathcal{T}_e}^\phi = g_{\phi ee}/m_e$. For a vector field, g_a^ϕ contains the polarization vector of the absorbed quantum. Quantize the field in the box, with quanta of energy $\omega_\phi = m_\phi(1 + v^2/2)$. Removing one quantum from a mode with N_ϕ quanta has the matrix element $\sqrt{N_\phi/2\omega_\phi \mathcal{V}}\ \sum_b g_b^\phi \langle f|O_b(\boldsymbol{p}_\phi)|i\rangle$, and the golden rule, summed over f and averaged over i, gives the response matrix again (Appendix D.19). Per absorbed particle,

Master rate formula: absorption

$$\Gamma_{\rm abs} = \frac{1}{2\omega_\phi} \sum_{a,b} \underbrace{g_a^{\phi *}\, g_b^\phi}_{\text{coupling}}\ \underbrace{S_{ab}(\boldsymbol{p}_\phi, \omega_\phi)}_{\text{target}},$$
$$\Gamma_{\rm net} = (1 - \mathrm{e}^{-\beta\omega_\phi})\,\Gamma_{\rm abs} = -\frac{1}{\omega_\phi}\ \mathrm{Im} \sum_{a,b} g_a^{\phi *}\, g_b^\phi\, R_{ab}(\boldsymbol{p}_\phi, \omega_\phi), \tag{8.19}$$

with $\omega_\phi \simeq m_\phi$. The event rate per unit target mass is $(\rho_\chi/m_\phi\rho_T)\langle\Gamma_{\rm abs}\rangle$, averaged over the directions and magnitudes of $\boldsymbol{p}_\phi = m_\phi\boldsymbol{v}$ with f_χ.

The absorption rate is the scattering formula evaluated at a single point, with $\mathfrak{D}_{ab}$ replaced by $g_a^{\phi\,*}g_b^\phi/2\omega_\phi$. The factor $1/2\omega_\phi$ is the relativistic normalization of one quantum. The event rate is set by the number of quanta per unit volume, ρ_χ/m_ϕ, and not by their flux: the speed of the dark matter enters only through the small momentum $\boldsymbol{p}_\phi$, as Sect. 2.1 found for a classical field.

The two rates in Eq. (8.19) answer different questions. $\Gamma_{\rm abs}$ counts the quanta the target absorbs, each depositing ω_ϕ, and it is what a detector records. The field also stimulates the target to emit quanta into the occupied modes, at the rate $\mathrm{e}^{-\beta\omega_\phi}\Gamma_{\rm abs}$ by detailed balance, and each stimulated emission takes energy out of the target. $\Gamma_{\rm net}$ is the difference between the absorption rate and the stimulated-emission rate, that is, the rate at which the field loses quanta, and the fluctuation–dissipation theorem Eq. (4.18) writes it through the imaginary part of the response. Mitridate and collaborators write absorption in the same way, as the imaginary part of the self-energy of ϕ, $\Gamma = -\operatorname{Im}\Pi_{\phi\phi}/\omega$ [8, Eqs. (20)–(22)], with $\Pi_{\phi\phi} = \sum_{ab} g_a^{\phi\,*}g_b^\phi R_{ab}$ in our normalization. For ω_ϕ of an eV in a detector at 10 mK, however, $\mathrm{e}^{-\beta\omega_\phi}$ is zero for every purpose, and the two rates coincide.

The number of quanta in a volume $(1/m_\phi v)^3$ is $\rho_\chi/(m_\phi^4 v^3)$, which exceeds one for m_ϕ below about 7 eV at $v = 10^{-3}$, and grows as m_ϕ^{-4} below. Lighter bosons form a classical field rather than a gas of particles. The quantized calculation still applies, because the rate is linear in N_ϕ, and in the emission matrix element the difference between N_ϕ and $N_\phi + 1$ is negligible; this difference is the spontaneous emission that a classical field lacks (Sect. 2.1). The same rate also follows from the classical field directly. The amplitude of the field is a sum of many modes with random phases, and by the central limit theorem the field is a Gaussian random field. We average the transition probability over this Gaussian ensemble; this *Gaussian field average* removes the cross terms between different modes and leaves $\langle|\phi_{\boldsymbol{k}}|^2\rangle$ for each mode, a quantity that the energy density fixes. Appendix D.19 shows that the result is Eq. (8.19) term by term.

The ensemble average needs an observation time long compared with the shorter of two times: the coherence time $t_{\rm coh}$ of the field, of order $1/m_\phi v_0^2$ and about 1 ns at $m_\phi = 1$ eV, and the inverse width of the target's response. A single record reproduces the ensemble average only when the record is also long compared with $t_{\rm coh}$. When the coherence time and the inverse width both exceed the length of the record, as for a field of neV mass or below acting on a sharp resonance such as a nuclear spin, the field drives the target coherently. The transition amplitude rather than the probability then grows linearly in time, and the golden rule is replaced by the coherent classical-field treatment of Sect. 14.5.

Eq. (8.19) reduces exactly to the dark-photon rate Eq. (1.38). A dark photon at rest couples to the current of the electrons with strength ζe times its polarization $\hat{\boldsymbol{e}}$ (Sect. 8.1.5), and the only row is $\hat{\boldsymbol{e}}\cdot\boldsymbol{j}_e$, with $g^\phi = \zeta e$. With the proper current response $\tilde{R}^{\rm T}_{jj}$ of Sect. 4.4, Eq. (8.19) gives $\Gamma_{\rm net} = -(\zeta^2 e^2/m_\phi)\operatorname{Im}\tilde{R}^{\rm T}_{jj}(m_\phi)$. The diamagnetic term of the current is real and does not contribute. By the transverse dielectric function of Sect. 4.4, $\varepsilon_2 = -(e^2/\omega^2)\operatorname{Im}\tilde{R}^{\rm T}_{jj}$, and with $\sigma_1 = \omega\varepsilon_2$ (Eq. (4.50)) the rate is $\zeta^2\sigma_1(m_\phi)$, the same for every polarization in a cubic crystal (Eq. (4.52)). The full response also contains the chain in which the induced current radiates an ordinary photon that the target absorbs again. Appendix D.19 sums this chain and shows that the summed chain turns ζ^2 into $\zeta^2_{\rm eff}$ of Sect. 1.4. Multiplying by $\rho_\chi/m_\phi\rho_T$ gives Eq. (1.38).

Fermionic absorption (Sect. 8.1.5) fits the same formula. The dark matter disappears and a light fermion $\tilde{\nu}$ leaves with momentum $\boldsymbol{p} - \boldsymbol{q}$. The sum over the momenta of $\tilde{\nu}$ replaces

the sum over final dark-matter momenta that produced Eq. (8.15), and energy conservation sets ω to Eq. (8.12), which is $m_\chi - q$ for a massless $\tilde{\nu}$. The rows and S_{ab} are the same as for scattering, now sampled along a line at $q \approx m_\chi$. Only $\mathfrak{D}_{ab}$ changes: the spinors of the relativistic $\tilde{\nu}$ enter it with relativistic normalization, and Sect. 10.9 constructs the resulting $\mathfrak{D}_{ab}$.

8.2.4 Negative energies and exothermic dark matter

The master formula holds for either sign of ω. The kinematics of scattering allows ω to be negative: a dark-matter particle gains energy, $\omega_{\boldsymbol{q}} < 0$, when it moves against $\boldsymbol{q}$ or recoils with more kinetic energy than it brings. The target alone decides whether a transfer with $\omega < 0$ happens. At $T = 0$ the target starts in its ground state, $S_{ab}(\boldsymbol{q}, \omega)$ vanishes for $\omega < 0$, and the half-plane below the axis in Fig. 8.3 is closed. A warm target opens it, with the weight fixed by detailed balance. In matrix form, $S_{ab}(-\boldsymbol{q}, -\omega) = \mathrm{e}^{-\beta\omega} S_{ba}(\boldsymbol{q}, \omega)$ (Sect. 4.2), so a transfer of $\omega < 0$ is suppressed relative to the reverse transfer by $\mathrm{e}^{-\beta|\omega|}$, the Boltzmann factor of the excitation the target gives up. For an electronic excitation across the gap of silicon, this factor is smaller than any power of the temperature, and an expansion in the phonon occupations (the thermal order n_T of Sect. 8.5) drops it at every order. For a phonon it is the occupation $n_{\boldsymbol{Q}\nu}$ itself: the negative-ω part of a one-phonon rate is proportional to $n_{\boldsymbol{Q}\nu}$, of first order in the occupations. Detailed balance relates absorption in the same way to its emission partner, at the point $(-\boldsymbol{p}_\phi, -\omega_\phi)$.

Exothermic dark matter moves the band of allowed $(\boldsymbol{q}, \omega)$ instead of reweighting it. A particle with $\Delta_\chi < 0$ arrives in an excited state, leaves in the lower one and adds $|\Delta_\chi|$ to every $\omega_{\boldsymbol{q}}$ of Eq. (1.21). The band shifts up by $|\Delta_\chi|$, reaches $q \to 0$ at $\omega = |\Delta_\chi|$, and there looks like the absorption of a boson of mass $|\Delta_\chi|$, with $\boldsymbol{q}$ set by the velocity rather than fixed at $m_\phi \boldsymbol{v}$ (Fig. 8.3). Since the energy now comes from the dark matter, the target needs no temperature, and the response is sampled at positive ω even for the slowest particles. An endothermic splitting, $\Delta_\chi > 0$, shifts the band down, and part of it can then be reached only with energy taken from a warm target.

Detailed balance ties the two directions of each transition, for the dark matter as well as for the target. Let $\boldsymbol{p}$ and $\boldsymbol{p}'$ be two dark-matter states, of any masses, and $\omega = E(\boldsymbol{p}) - E(\boldsymbol{p}')$ the energy that the transition $\boldsymbol{p} \to \boldsymbol{p}'$ deposits. Then, for unpolarized dark-matter states of equal spin,

$$\Gamma(\boldsymbol{p}' \to \boldsymbol{p}) = \mathrm{e}^{-\beta\omega}\, \Gamma(\boldsymbol{p} \to \boldsymbol{p}'), \tag{8.20}$$

for the rate into one final momentum (Appendix D.19). The reverse transition is driven by the conjugated interaction, which exchanges $\mathfrak{D}_{ab}$ for $\mathfrak{D}_{ba}$ and O_b for $O_b^\dagger$, and the two exchanges cancel in the contraction, leaving the ratio of the Gibbs weights of the two target states. Two transitions are therefore one process read in two directions: the down-scattering of an exothermic particle, $\chi_2 \to \chi_1$, and the up-scattering $\chi_1 \to \chi_2$ that a warm target can drive. Given long enough contact with the target, a population of dark matter would relax to the temperature of the target.

8.2.5 Conditions for off-diagonal terms

The off-diagonal terms of the contraction are the part of a rate that no single-operator calculation contains. They are nonzero only when the dark-matter side couples two rows with a nonzero product $\mathfrak{D}_{ab}$ and the target correlates the same two rows, $S_{ab} \neq 0$. Each of the two factors, $\mathfrak{D}_{ab}$ and S_{ab}, vanishes for a reason of symmetry in the commonest cases. On the dark-matter side, a coefficient linear in $\boldsymbol{S}_\chi$ never interferes with one that is independent of it (Sect. 8.1). On the target side, without magnetic order, the density and the spin density are

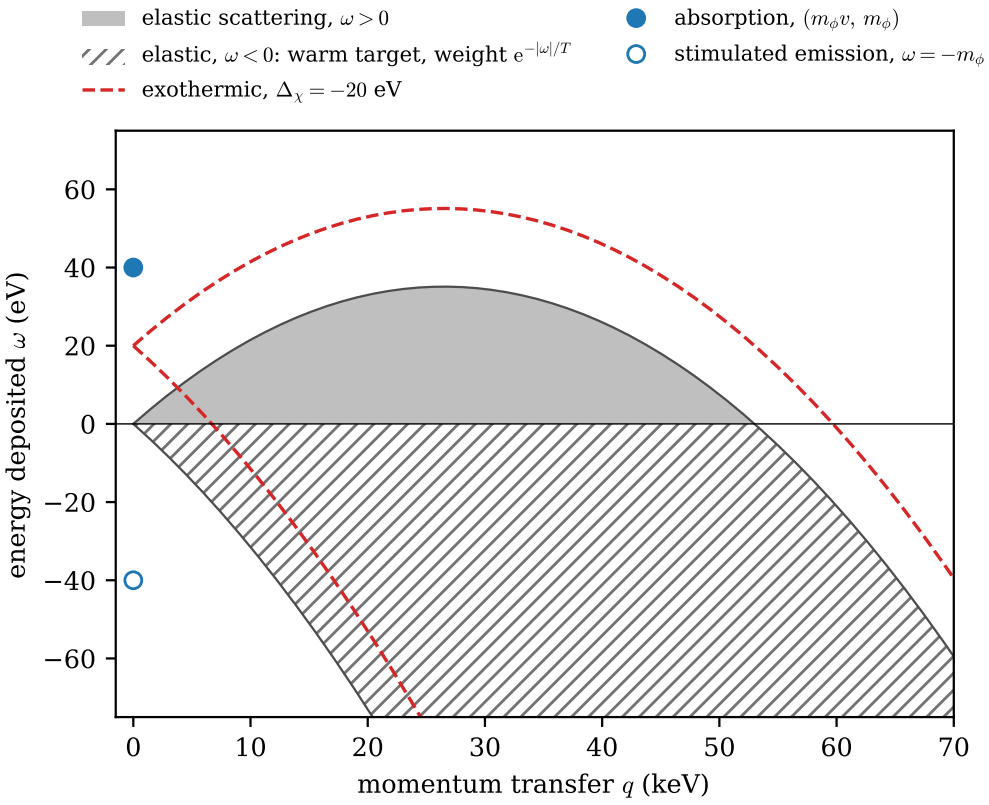


Figure 8.3: Where the master rate formula samples the response, in the plane of momentum transfer q and energy deposited ω, for $m_\chi = 10\,\mathrm{MeV}$ and speeds up to $v_{\max} = 2.650 \times 10^{-3}$. Elastic scattering reaches the band $-qv_{\max} - q^2/2m_\chi \le \omega \le qv_{\max} - q^2/2m_\chi$, the extreme values of $\omega_{\boldsymbol{q}}$ over all directions of $\boldsymbol{v}$. The upper part of the band (gray, $\omega > 0$, maximum $35\,\mathrm{eV}$ at $q = m_\chi v_{\max} = 26.5\,\mathrm{keV}$) is open at any temperature. The lower part (hatched, $\omega < 0$) is open only for a target at $T > 0$, with weight $e^{-|\omega|/T}$ relative to the reverse transfer. Exothermic dark matter with $\Delta_\chi = -20\,\mathrm{eV}$ shifts the band up by $|\Delta_\chi|$ (dashed). Absorption of a boson of mass $m_\phi = 40\,\mathrm{eV}$ is a point at $q = m_\phi v \approx 40\,\mathrm{meV}$, $\omega = m_\phi$ (filled circle); stimulated emission into the occupied field is the point $\omega = -m_\phi$ (open circle). Fermionic absorption with a massless outgoing fermion, $\omega = m_\chi - q$, is a line at $q \approx m_\chi$, far to the right.

uncorrelated, since $\mathrm{Tr}\,\boldsymbol{\sigma} = 0$. The worked example takes a target with magnetic order and a coupling that survives the trace over the dark-matter spin.

Worked example 8.3: Density and spin couplings in a polarized electron gas

Take the free electron gas at the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, fully polarized along $\hat{\boldsymbol{z}}$ by a uniform exchange field strong enough to lift the minority band above the Fermi level, a toy half-metal. All electrons have $S_z = +\frac{1}{2}$ and fill one Fermi sphere of radius $k_{\mathrm{F}\uparrow} = 2^{1/3} k_{\mathrm{F}} = 4.50\,\mathrm{keV}$, with $v_{\mathrm{F}\uparrow} = 8.80 \times 10^{-3}$ and $E_{\mathrm{F}\uparrow} = 19.8\,\mathrm{eV}$; the exchange splitting must exceed $E_{\mathrm{F}\uparrow}$. We work at $T = 0$ with two rows, the density n_e and the spin density s_e^z of Eq. (8.3).

The Hamiltonian conserves the spin of each electron, so every state that n_e or s_e^z reaches from the ground state has all spins up, and on those states $s_e^z(\boldsymbol{q}) = \frac{1}{2} n_e(\boldsymbol{q})$ exactly. The response matrix is then

$$\begin{pmatrix} S_{nn} & S_{ns} \\ S_{sn} & S_{ss} \end{pmatrix} = S_\uparrow(\boldsymbol{q}, \omega) \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{pmatrix},$$

with $S_\uparrow$ the structure factor of one spin, half of Eq. (4.22) with $k_{\mathrm{F}} \to k_{\mathrm{F}\uparrow}$. The determinant of the response matrix vanishes: the two rows are one operator, and the matrix saturates the bound Eq. (4.20). At $q = q_0 = 3.73\,\mathrm{keV}$ and $\omega = 5\,\mathrm{eV}$ we are in the linear region of the sphere, below $q v_{\mathrm{F}\uparrow} - q^2/2m_e = 19.2\,\mathrm{eV}$, and $S_\uparrow = m_e^2 \omega / 2\pi q = 5.57 \times 10^7\,\mathrm{eV}^2$, half the structure factor of the unpolarized gas at the same density.

For potentials $\tilde{V}_n$ and $\tilde{V}_s$ on the two rows, real and without dark-matter spin, the contraction is

$$\tilde{V}_n^2 S_{nn} + 2\tilde{V}_n \tilde{V}_s S_{ns} + \tilde{V}_s^2 S_{ss} = S_\uparrow (\tilde{V}_n + \tfrac{1}{2}\tilde{V}_s)^2 .$$

Every electron carries $S_z = \frac{1}{2}$, so a coupling to the spin acts as an extra density potential and adds to $\tilde{V}_n$ in the amplitude. The sum of the two single-operator rates is $S_\uparrow(\tilde{V}_n^2 + \tilde{V}_s^2/4)$. It misses the term $\tilde{V}_n \tilde{V}_s S_\uparrow$, which is first order in the weaker coupling, and at $\tilde{V}_s = -2\tilde{V}_n$, where the true rate vanishes, it is wrong by the whole of $2\tilde{V}_n^2 S_\uparrow$.

Which models produce a $\tilde{V}_s$ without dark-matter spin? The standard pair $\mathcal{O}_1$ and $\mathcal{O}_4 = \boldsymbol{S}_\chi \cdot \boldsymbol{S}_e$ does not: $\tilde{V}_s = c_4^{(e)} \boldsymbol{S}_\chi$, and the trace over the unpolarized dark-matter spin removes the cross term. A spin coupling free of $\boldsymbol{S}_\chi$ is $\mathcal{O}_7 = \boldsymbol{S}_e \cdot \boldsymbol{v}^\perp$, which reduces to $\bar{\boldsymbol{v}}_\chi \cdot \boldsymbol{s}_e - \mathcal{J}_e^{\alpha\alpha}$ (Table 8.1). No standard model produces $\mathcal{O}_7$ at leading order (Sect. 8.1); we take it as the simplest spin coupling that survives the trace. Take it together with $\mathcal{O}_1$, with coefficients c_1 and c_7. The rows s_e^x, s_e^y, $\mathcal{J}_e^{xx}$ and $\mathcal{J}_e^{yy}$ flip the spin and reach states orthogonal to the states that n_e reaches. The terms of first order in c_7 are therefore

$$2 c_1 c_7 \big[\bar{v}_{\chi,z}\, S_{ns} - S_{n\mathcal{J}}\big], \qquad S_{n\mathcal{J}} = \tfrac{1}{2} \Big\langle \frac{(2\boldsymbol{k} + \boldsymbol{q})_z}{2m_e} \Big\rangle S_\uparrow ,$$

since the element of $\mathcal{J}_e^{zz}$ between $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{q}$ is $\frac{1}{2}(2\boldsymbol{k} + \boldsymbol{q})_z / 2m_e$, and the brackets average over the pairs at this $(\boldsymbol{q}, \omega)$. Since the pairs fill an annulus symmetric about $\boldsymbol{q}$ (Worked example 8.2), the transverse part of $\boldsymbol{k}$ averages away, and $(2\boldsymbol{k} + \boldsymbol{q}) \cdot \hat{\boldsymbol{q}} / 2m_e = \omega/q$ for every pair. As a result, $S_{n\mathcal{J}} = \frac{1}{2}(\omega/q) \hat{q}_z S_\uparrow$. On the dark-matter side, $\hat{\boldsymbol{q}} \cdot \bar{\boldsymbol{v}}_\chi = \omega/q$ for elastic scattering, so $\bar{v}_{\chi,z} = (\omega/q)\hat{q}_z + \boldsymbol{v}_{\mathrm{T}} \cdot \hat{\boldsymbol{z}}$, with $\boldsymbol{v}_{\mathrm{T}}$ the part of $\boldsymbol{v}$ transverse to $\boldsymbol{q}$. The longitudinal parts cancel, as Eq. (8.5) requires, and the integrand of Eq. (8.15) becomes

$$c_1^2\, S_\uparrow + c_1 c_7\, (\boldsymbol{v}_{\mathrm{T}} \cdot \hat{\boldsymbol{z}})\, S_\uparrow + O(c_7^2).$$

For the dark matter of Worked example 8.2, $m_\chi = 100\,\mathrm{MeV}$ and $v = 2.0 \times 10^{-3}$, $v_{\min} = 1.36 \times 10^{-3}$ and $v_{\mathrm{T}} = 1.47 \times 10^{-3}$. With $\boldsymbol{v}_{\mathrm{T}}$ along the magnetization, the cross term is

$1.5 \times 10^{-3}\, c_7/c_1$ of the $\mathcal{O}_1$ rate. The rate of $\mathcal{O}_7$ alone is second order in c_7 and quadratic in the velocities. The squared matrix elements of $\mathcal{O}_7$, summed over the final spin, are $|\boldsymbol{v}^\perp|^2/4 \le (v_{\max} + v_{\mathrm{F}\uparrow} + q/2m_e)^2/4 = 5.7 \times 10^{-5}$. The structure factors of $\mathcal{O}_7$ are at most $S_\uparrow$ and $m_e k_{\mathrm{F}\uparrow}^2/4\pi q = 4.0\, S_\uparrow$, the second for spin flips into the empty minority band, which an exchange splitting above $E_{\mathrm{F}\uparrow} + \omega$ closes; the bound on the rate of $\mathcal{O}_7$ is generous. At $c_7 = c_1$ the cross term is therefore at least five times the rate of $\mathcal{O}_7$ alone.
The cross term is odd in $\boldsymbol{v}_\mathrm{T}$. At fixed $\boldsymbol{q}$ the halo integral runs over the velocities with $\hat{\boldsymbol{q}} \cdot \boldsymbol{v} = v_{\min}$, and for the laboratory Maxwellian $f_\chi \propto \mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_\mathrm{E}|^2/v_0^2}$ of Appendix A.8 the exponent splits into a longitudinal and a transverse Gaussian. Apart from the escape cut, the transverse Gaussian is centered on $-\boldsymbol{v}_{\mathrm{E,T}}$, the part of $-\boldsymbol{v}_\mathrm{E}$ transverse to $\boldsymbol{q}$, and the averaged cross term is $-c_1 c_7\, (\boldsymbol{v}_{\mathrm{E,T}} \cdot \hat{\boldsymbol{z}})\, S_\uparrow$, at most $8.4 \times 10^{-4}\, c_7/c_1$ of the $\mathcal{O}_1$ rate for $|\boldsymbol{v}_\mathrm{E}| = 250.5\,\mathrm{km\,s^{-1}}$. It vanishes for an isotropic halo, follows the angle between the magnetization and the dark-matter wind, and changes sign when the magnetization is reversed.
A partially polarized gas has two spheres and $S_{ns} = \frac{1}{2}(S_\uparrow - S_\downarrow)$. In the linear region, however, the structure factor of one spin, $m_e^2\omega/2\pi q$, does not depend on the radius of its sphere, and S_{ns} vanishes there even though the gas is magnetized. At a polarization of 0.5 and $q = q_0$, $S_{ns} = 0$ at 5 eV; at 10 eV, where the minority sphere has left its linear region (the edge of the linear region is at 7.07 eV), $S_{ns}/S_{nn} = 0.068$.

A cross term needs a coupling and a target that break the same symmetry, here time reversal, and when both do, it enters the rate at first order in the weaker coupling, whereas each single-operator rate enters at second order. Positivity bounds the size of a cross term, and the polarized gas reaches the bound; Sect. 8.3 derives the bound together with the other relations that hold for every target.

8.3 Exact relations

A computed response of silicon can be wrong in ways that no convergence test reveals. Take an electron–nucleus interference spectrum that keeps one sign, an energy-weighted spectrum that counts thirteen electrons per atom, or a coupling proportional to mass that excites optical phonons at vanishing momentum transfer. Each of these three results is an error, whatever the method and however well converged, because each violates a relation that holds for every target. Sects. 4.2 and 4.3 derived the first of these relations: detailed balance, the positivity of the response matrix, the fluctuation–dissipation theorem, the Kramers–Kronig relations and the f-sum rule of every species. The multi-operator rate of Sect. 8.2 needs three more: an interference bound, higher moments and a Ward identity. Positivity limits how strongly two couplings can interfere. The higher frequency moments of the response matrix are fixed by commutators, and the third moment of the electron–nucleus interference is set by the electron density at the nuclei. Translation invariance links the densities of all species to the momentum density, and as a result a coupling proportional to mass can only make sound at long wavelengths. None of these relations depends on an approximation to the target, so a response that violates one is wrong, not merely truncated.

8.3.1 Interference of two couplings

In the polarized electron gas of Worked example 8.3, a density coupling and a spin coupling together gave the rate $S_\uparrow(\tilde{V}_n + \frac{1}{2}\tilde{V}_s)^2$, which can be anything from zero to twice the sum of the two single-operator rates. How far can a cross term go in general? Split the rows of

Eq. (8.15) into two groups, A and B, for instance the electron rows and the nuclear rows of a dark photon, or the density and the current rows of Sect. 8.1. The rate is then

$$\Gamma = \Gamma_A + \Gamma_{AB} + \Gamma_B,$$

where the rates Γ_A and Γ_B of each group alone keep only the terms with both indices in one group. The cross term Γ_{AB} collects the terms with one index in each group, $2\,\mathrm{Re}\sum_{a\in A,b\in B}\mathfrak{D}_{ab}S_{ab}$ under the integral.

A bound on Γ_{AB} follows from a family of models. Multiply every coupling of group B by a real number λ. Each element $\mathfrak{D}_{ab}$ is multiplied by λ once for every index in B. The new tensor is $\mathfrak{D}$ multiplied on both sides by a real diagonal matrix, so it is still Hermitian and positive semidefinite. Its contraction with the positive semidefinite S_{ab} of Eq. (4.20) is therefore never negative, at every $(\boldsymbol{q},\omega)$ and for every velocity. After any integration over $\boldsymbol{q}$, ω and the halo, the rate is then never negative either:

$$\Gamma(\lambda) = \Gamma_A + \lambda\,\Gamma_{AB} + \lambda^2\,\Gamma_B \geq 0 \qquad \text{for every real } \lambda.$$

A quadratic that never goes negative has a discriminant that is not positive, $\Gamma_{AB}^2 \leq 4\Gamma_A\Gamma_B$.

Interference bound

For any two groups of rows, any target and any dark-matter model,

$$|\Gamma_{AB}| \leq 2\,(\Gamma_A\Gamma_B)^{1/2}, \qquad \text{equivalently} \qquad \left(\Gamma_A^{1/2}-\Gamma_B^{1/2}\right)^2 \leq \Gamma \leq \left(\Gamma_A^{1/2}+\Gamma_B^{1/2}\right)^2. \tag{8.21}$$

The cross term is at most twice the geometric mean of the separate rates. The bound holds pointwise in $(\boldsymbol{q},\omega)$, for the spectrum $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ in any bin, and for the total rate.

For one row in each group the pointwise statement is the Cauchy–Schwarz inequality $|S_{ab}|^2 \leq S_{aa}S_{bb}$ of Eq. (4.20) combined with $|\mathfrak{D}_{ab}|^2 \leq \mathfrak{D}_{aa}\mathfrak{D}_{bb}$ from Eq. (8.7), and the integrated statement then follows from the Cauchy–Schwarz inequality for integrals, $\int(xy)^{1/2} \leq (\int x \int y)^{1/2}$. The argument through λ gives the pointwise and the integrated statements at once, and for blocks of any size.

The bound is reached when $\mathfrak{D}$ and S both have rank one on the two groups and their phases line up, as in the polarized gas: there $S_{ns}^2 = S_{nn}S_{ss}$, and at $\tilde{V}_s = -2\tilde{V}_n$ the rate is on the lower edge of Eq. (8.21), zero. The bound is not an estimate of the cross term: for an unpolarized target and couplings that are blind to the dark-matter spin, the cross terms between density and spin vanish by symmetry (Sect. 8.2.5), far inside the band of Eq. (8.21). The bound puts a rigorous band on a rate whose interference term has not been computed: with $r = \Gamma_B/\Gamma_A$, the total rate lies between $(\sqrt{r}-1)^2$ and $(\sqrt{r}+1)^2$ times Γ_A. The same inequality bounds the Migdal interference, $|S_{e\kappa}|^2 \leq S_{ee}S_{\kappa\kappa}$, and constrains the theory uncertainties of Sect. 8.8.

8.3.2 Moment matrices

The f-sum rule fixed the first frequency moment of every density by a double commutator (Eqs. (4.28) and (4.34)). Higher moments weight the spectrum toward high energies, so they are sensitive to an approximation to the core electrons or to the electron–nucleus interaction. We collect them into matrices,

$$\mathscr{M}^k_{ab}(\boldsymbol{q}) \equiv \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}\,\omega^k\,[S_{ab}(\boldsymbol{q},\omega) + S_{ba}(-\boldsymbol{q},\omega)] = \frac{1}{\mathscr{V}}\Big\langle [O_a(\boldsymbol{q})^\dagger,\, [H, O_b(\boldsymbol{q})]_k]_{\mp}\Big\rangle, \tag{8.22}$$

for operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$ and any stationary state, where $[H,O]_k$ is O commuted k times with H, as in Sect. 4.3, and $[X,Y]_\mp$ is the commutator $XY - YX$ for odd k and the anticommutator $XY + YX$ for even k. The derivation of Eq. (4.27) goes through unchanged. The term $O_a^\dagger [H,O_b]_k$ gives the k-th moment of $\mathcal{V} S_{ab}(\boldsymbol{q},\omega)/2\pi$, the term $[H,O_b]_k O_a^\dagger$ gives $(-1)^k$ times the moment of $\mathcal{V} S_{ba}(-\boldsymbol{q},\omega)/2\pi$, and the sign of the bracket makes the two terms add for every k. For $k = 1$, Eq. (8.22) is Eq. (4.28), so $\mathcal{M}^1$ is the first-moment matrix of Sect. 4.3. Since S_{ab} has the units of eV^2 for densities, $\mathcal{M}^k$ has units of eV^{k+3}.

The odd and the even moments carry different information. An odd moment is a commutator. It needs only the algebra of H and equal-time averages of simpler operators, and for the densities of electrons and nuclei the first one is a number, Eq. (4.34). An even moment is an equal-time correlation function. On the diagonal, with $X = [H,O_a]_{k/2}$, it reads $\mathcal{V}^{-1}\langle X^\dagger X + X X^\dagger\rangle$; for instance $\mathcal{M}^0_{ee} = 2\boldsymbol{n}_e S^{\mathrm{st}}$ for a target with inversion symmetry, by Eq. (4.17). An even moment depends on the state, its temperature and its correlations, but no dynamics is needed to compute it. The moments of the nuclear self response can therefore be taken from a static path-integral simulation (Sect. 30.2). In the same way as for $k = 1$ in Sect. 4.3, an odd $\mathcal{M}^k$ is the coefficient of $z^{-(k+1)}$ in the expansion of $R_{ab}(\boldsymbol{q},z)$ at large z (Appendix D.20).

Positivity constrains the moments among themselves. The spectral weight of one row, summed over $\pm\boldsymbol{q}$, is never negative. It defines a positive measure on the real line, and the moments $\mathcal{M}^k_{aa}$ of this measure form a positive semidefinite Hankel matrix, with elements $\mathcal{M}^{i+j}_{aa}$:

$$\sum_{i,j=0}^{n} x_i\, x_j\, \mathcal{M}^{i+j}_{aa} = \int \frac{\mathrm{d}\omega}{2\pi} \Big(\sum_i x_i \omega^i\Big)^2 \left[S_{aa}(\boldsymbol{q},\omega) + S_{aa}(-\boldsymbol{q},\omega)\right] \geq 0. \tag{8.23}$$

The smallest case is $(\mathcal{M}^1_{aa})^2 \leq \mathcal{M}^0_{aa}\mathcal{M}^2_{aa}$. At $T = 0$ the measure is confined to $\omega \geq 0$, and the shifted matrix $\mathcal{M}^{i+j+1}_{aa}$ is positive semidefinite too; this second condition gives for instance $(\mathcal{M}^2_{aa})^2 \leq \mathcal{M}^1_{aa}\mathcal{M}^3_{aa}$. The first few moments of a spectrum determine the spectrum within a band, the Markov–Krein band of Sect. 11.5, which these inequalities make precise.

How many moments exist? For electrons and point nuclei with Coulomb interactions, $\mathcal{M}^3$ is finite and $\mathcal{M}^4$ is not. The fourth moment of a density is the equal-time average of $|[H,[H,n_s]]|^2$, and $[H,[H,n_s]]$ contains the force on each particle (Appendix D.20). The Coulomb force between an electron and a nucleus has the mean square $\int \mathrm{d}^3y\, \boldsymbol{n}(\boldsymbol{y})\,(Z_\kappa\alpha/y^2)^2$, with $\boldsymbol{n}$ finite at the nucleus, and this integral diverges as $\int \mathrm{d}y/y^2$. The spectrum shows the same divergence at large ω. The oscillator strength of the hydrogen continuum falls as $\omega^{-7/2}$ (Worked example 4.5), and at small q, where $S \propto q^2\,\omega^{-1}\mathrm{d}f^{\mathrm{osc}}/\mathrm{d}\omega$, the structure factor falls as $\omega^{-9/2}$: $\omega^3 S$ is integrable and $\omega^4 S$ is not. In the Born–Oppenheimer approximation the nuclei move on a smooth energy surface, and every moment of their self response is finite; this approximation is the order $n_B = 0$ of the expansion in the derivative coupling (Sect. 8.5).

8.3.3 The mixed third moment

The first moment of the electron–nucleus interference vanishes, Eq. (4.34), and at $T = 0$ the real part of $S_{e\kappa}$ must change sign along the energy axis. Which sign wins at high energy, in the range of the Migdal effect and of core excitations? The third moment weights the spectrum by ω^3 and answers the question.

The calculation takes three steps, written out in Appendix D.20. First, in a stationary state the average of $[X,[H,Y]]$ equals the average of $[[X,H],Y]$, since the average of a commutator with H vanishes; this identity moves one commutator onto the electron density, $\mathcal{M}^3_{e\kappa} = \mathcal{V}^{-1}\langle[[n_e^\dagger,H],[H,[H,n_\kappa]]]\rangle$. Second, the continuity equations of the two species turn the two inner commutators into currents: Eq. (4.29) and its nuclear analogue give

$[n_e(\boldsymbol{q})^\dagger, H] = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})^\dagger$ and $[H, n_\kappa(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_\kappa(\boldsymbol{q})$. Third, the commutator of H with the nuclear current contains the kinetic energy of the nuclei and the gradient of the potential energy at each nucleus, $\mathrm{i}M_\kappa^{-1}\sum_{I\in\kappa}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\,\boldsymbol{q}\cdot\nabla_I U$, with U the potential energy. The electron current commutes with every nuclear operator. It fails to commute only with the electron positions inside the force. Of the terms of U, only the electron–nucleus Coulomb energy depends on both the electron and the nuclear positions; this energy is $\sum_{j,I} w_\kappa(\boldsymbol{r}_j - \boldsymbol{X}_I)$, with $w_\kappa(\boldsymbol{y}) = -Z_\kappa\alpha/y$. The electron–electron and nucleus–nucleus interactions, and every kinetic term, drop out. The result involves one function, the mean electron density $\boldsymbol{n}_e^{(\kappa)}(\boldsymbol{y})$ at the displacement $\boldsymbol{y}$ from a nucleus of species κ, averaged over the nuclei of that species and over their motion. Its value at $\boldsymbol{y}=0$ is the contact density $\boldsymbol{n}_e^{(\kappa)}$, the electron density at the nucleus, and far from the nucleus it averages to the mean density $\boldsymbol{n}_e$.

Mixed electron–nucleus third moment

For point nuclei, nonrelativistic electrons and local potentials, at any $\boldsymbol{q}$, temperature and coupling strength,

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q}) = -\frac{\boldsymbol{n}_\kappa}{m_e M_\kappa}\int \mathrm{d}^3y\; \boldsymbol{n}_e^{(\kappa)}(\boldsymbol{y})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}\,(\boldsymbol{q}\cdot\nabla)^2 w_\kappa(\boldsymbol{y}). \tag{8.24}$$

At sites of cubic symmetry, for q small compared with the shortest reciprocal-lattice vector $G_{\min}$ but large compared with the inverse size of the sample,

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q}) = -\frac{Z_\kappa e^2 q^2\,\boldsymbol{n}_\kappa}{3\,m_e M_\kappa}\big[\boldsymbol{n}_e^{(\kappa)} + 2\boldsymbol{n}_e\big]. \tag{8.25}$$

The high-frequency interference of electrons and nuclei is negative, and its weight is fixed by the electron density at the nuclei.

The third moment of a single species is classical, and Puff wrote it for an interacting Bose fluid [13]. The two terms of the bracket have different origins, as the identity

$$(\boldsymbol{q}\cdot\nabla)^2\frac{1}{y} = -\frac{4\pi}{3}\,q^2\,\delta^3(\boldsymbol{y}) + q^2\,\frac{3(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{y}})^2 - 1}{y^3}$$

shows. The contact part picks out the density at the nucleus and gives $\boldsymbol{n}_e^{(\kappa)}/3$ at every q, with the phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}$ equal to one on its support. The quadrupolar part averages to zero against any density that has cubic symmetry about the nucleus and varies on scales short compared with $1/q$, the core and the valence bonds alike. Against the uniform part of the density, however, the quadrupolar part does not vanish. Its integral against the uniform density is dominated by distances $y \sim 1/q$, where the phase matters, and since the kernel scales as $1/y^3$ the integral is independent of q; it contributes $2\boldsymbol{n}_e/3$. This uniform-density term is the Coulomb field of the bare nucleus acting on the electrons of the whole crystal at distances $\sim 1/q$. The order of limits is the physical one: the sample is infinite at fixed q, and only then is q small.

The negative sign of the mixed third moment has a simple reading. For an electron bound to a nucleus by a spring of constant K, Eq. (8.24) gives $\mathscr{M}^3_{e\kappa} = -Kq^2/(m_e M_\kappa \mathscr{V})$ at small q. The relative coordinate makes the sign obvious: in the mode in which the two particles oscillate against each other, at $\omega^2 = K(m_e^{-1} + M_\kappa^{-1})$, a displacement of one is accompanied by an opposite displacement of the other (Appendix D.20). The Coulomb attraction is a spring whose stiffness is the curvature $\nabla^2 w_\kappa = Z_\kappa e^2\delta^3(\boldsymbol{y})$, concentrated at the nucleus. The contact density measures how much of the electron cloud feels this stiffness. Together with the vanishing first moment, the negative third moment says that the interference is positive at

low energies, where the electrons follow the nucleus (Sect. 8.6), and negative at high energies, where they are left behind. The positive part of the interference is in the phonon final states, on the acoustic branch, at energies that vanish with q, and on the optical phonons, wherever the electrons move with the nuclei (Sect. 8.6). Every electronic final state carries the negative interference $-(m_e/M_\kappa)|\langle f|n_e|0\rangle|^2$ at small q, as Eq. (8.32) below shows.

Eq. (8.25) has a limited range in q, and three regimes follow from the exact form Eq. (8.24), which Appendix D.20 writes as a sum over reciprocal-lattice vectors. For $q \ll G_{\min}$, Eq. (8.25) holds. Its real part is corrected at relative order $(q/G_{\min})^2$ at every cubic site. At sites without inversion symmetry, such as the tetrahedral sites of silicon, there is also a correction of first order in $q/G_{\min}$. This correction is purely imaginary: it comes from the part of the density odd about the nucleus, whose Fourier components are imaginary, and it enters $\operatorname{Im} \mathscr{M}^3_{e\kappa}$ alone. In silicon $G_{\min} = |\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$ (Sect. 3.1). Once q passes $G_{\min}$ the valence structure starts to matter, and the bracket depends on q, although the contact term still dominates. Once q reaches the inverse size $1/a_{\mathrm{core}}$ of the innermost shell, the core itself is resolved: the quadrupolar kernel then acts on the density near the nucleus as on a uniform one, and the bracket climbs from $n_e^{(\kappa)}/3$ toward $n_e^{(\kappa)}$. For a density with spherical symmetry about the nucleus, which is exact for a free atom and a good approximation for the core in a cubic crystal, the angular integral of Eq. (8.24) can be done once and for all (Appendix D.20):

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q}) = -\frac{Z_\kappa e^2 q^2\, n_\kappa}{m_e M_\kappa}\Big[\frac{n_e^{(\kappa)}}{3} + 2\int_0^\infty \mathrm{d}y\, \frac{\bar{n}_e^{(\kappa)}(y)\, j_2(qy)}{y}\Big], \tag{8.26}$$

with $\bar{n}_e^{(\kappa)}(y)$ the spherically averaged density at distance y from the nucleus and j_2 the spherical Bessel function of order two. Since $\int_0^\infty \mathrm{d}x\, j_2(x)/x = 1/3$, a density that is uniform over the range $y \sim 1/q$ contributes $2/3$ of its value through the integral, and in this way the mean density n_e gives the $2n_e/3$ of the small-q limit.

The contact density is a material quantity, computed from all-electron densities. Pseudopotential calculations replace the cores and cannot supply it; the projector-augmented-wave reconstruction or a full-potential code restores the density near the nucleus. The spherical atomic part comes from a calculation of the free atom, and the crystal adds a small anisotropy of the valence (Sect. 30.1). Relativity changes the density at the nucleus more than anything else. For a finite nucleus it raises the contact density by about 8 % in silicon. The relativistic increase grows roughly as $(Z\alpha)^2$ in heavier atoms, and Sect. 12.5 adds it to Eq. (8.25).

Worked example 8.4: The mixed third moment of silicon

We evaluate Eq. (8.26) for silicon at $q = 10\,\mathrm{keV}$, a typical momentum transfer for the Migdal effect. For the density we take the free atom in its ground state from an all-electron Hartree–Fock calculation, nonrelativistic and with a point nucleus. The total energy of the atom, −288.854 hartree, is the Roothaan–Hartree–Fock value quoted in Sect. 4.3 [14]. The contact density is $n_e^{(\kappa)} = 1765\, a_0^{-3} = 1.19 \times 10^{28}\,\mathrm{cm}^{-3}$, converged to 0.1 % in the Gaussian basis. The ten core electrons contribute $1757.7\, a_0^{-3}$ of it, and the four valence electrons, through their 3s orbital, $7.6\, a_0^{-3}$, or 0.43 %. Against the contact density, the mean density of all fourteen electrons, $7.00 \times 10^{23}\,\mathrm{cm}^{-3}$, is tiny: $2n_e/n_e^{(\kappa)} = 1.2 \times 10^{-4}$. In silicon the uniform-density term is a correction at the level of 10^{-4}.
At $q = 10\,\mathrm{keV} = 2.68\, a_0^{-1}$ the kernel $j_2(qy)/y$ samples distances $y \sim 1/q = 0.20\,\text{Å}$. This momentum transfer is above $G_{\min} = 3.95\,\mathrm{keV}$, in the middle regime, and below the inverse size $\sim 14/a_0 = 52\,\mathrm{keV}$ of the 1s shell. The bracket of Eq. (8.26) comes out at $0.3353\, n_e^{(\kappa)}$, against 1/3 in the limit $q \to 0$: Eq. (8.25) is 0.6 % low at 10 keV. The core supplies 99.56 % of the bracket and the valence 0.44 %, almost all of the valence part from the 3s density at

the nucleus. The free atom misses the crystal valence, which enters at most through the mean density, 1.8×10^{-4} of the bracket. As q grows the 1s shell is resolved; the bracket is $0.347\,\boldsymbol{n}_e^{(\kappa)}$ at 30 keV and $0.426\,\boldsymbol{n}_e^{(\kappa)}$ at 100 keV.
The mixed moment is best read against the nuclear f-sum $\mathscr{M}^1_{\kappa\kappa}=q^2\boldsymbol{n}_\kappa/M_\kappa$. The ratio of the mixed moment to the nuclear f-sum is a squared energy,

$$-\frac{\mathscr{M}^3_{e\kappa}}{\mathscr{M}^1_{\kappa\kappa}}=\frac{Z_\kappa e^2}{m_e}\times(\text{bracket})=(8.78\,\text{keV})^2,$$

with $(8.76\,\text{keV})^2$ in the limit $q\to0$. This energy is on the scale of the K shell: the 1s orbital energy of the same calculation is 1.87 keV, and the 1s electrons move with kinetic energies near 2.7 keV (Sect. 4.3). The third moment of the electron–nucleus interference is a property of the core electrons of silicon, to better than a percent, and a calculation gets the third moment wrong by construction if it freezes the core or replaces the core by a pseudopotential.

The diagonal third moments follow from the same algebra. For species s of mass m_s,

$$\mathscr{M}^3_{ss}(\boldsymbol{q})=\frac{\boldsymbol{n}_s q^2}{m_s}\Big[\Big(\frac{q^2}{2m_s}\Big)^2+\frac{3q^2\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2\rangle_s}{m_s^2}\Big]+\frac{1}{m_s^2\mathscr{V}}\Big\langle\sum_{i,j\in s}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)}(\boldsymbol{q}\cdot\nabla_i)(\boldsymbol{q}\cdot\nabla_j)\,U\Big\rangle, \tag{8.27}$$

with $\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2\rangle_s$ the mean square of one particle's momentum along $\hat{\boldsymbol{q}}$, $2m_s\bar{K}/3$ for electrons in a cubic crystal, with the $\bar{K}$ of Eq. (4.33) (Appendix D.20). The kinetic bracket is the free recoil, broadened by the motion of the particles. The potential term measures the curvature of the energy surface on which the particles move, and for electrons in jellium it reduces to $\boldsymbol{n}_e q^2\omega_p^2/m_e$: the third moment of the electron gas is the plasma frequency squared times the f-sum. In a crystal with mobile nuclei, however, the electron–nucleus curvature at the nuclei dominates the potential term, and the next subsection shows that this term is tied to Eq. (8.25) exactly.

8.3.4 Translation Ward identities

Consider a dark-matter coupling that is the same per unit mass for every particle, for instance a heavy scalar coupled to nucleons in a crystal of one element. This coupling pushes every part of the crystal in the same proportion. At long wavelength the push is uniform, like a gravitational field, and by the equivalence principle it accelerates every particle equally: it can move the crystal, or at finite q launch a sound wave, but it cannot excite the crystal's internal motion. Translation invariance gives the identities that make the equivalence-principle argument precise; in quantum mechanics it is the conservation of total momentum.

Add the continuity equations of all species, each weighted by its mass. The mass-density operator is $n_{\mathrm{M}}(\boldsymbol{q})=\sum_s m_s n_s(\boldsymbol{q})$, with s running over the electrons and every nuclear species, and it then obeys

$$[H,n_{\mathrm{M}}(\boldsymbol{q})]=\boldsymbol{q}\cdot\boldsymbol{g}(\boldsymbol{q}),\qquad \boldsymbol{g}(\boldsymbol{q})=\sum_s m_s\boldsymbol{j}_s(\boldsymbol{q})=\sum_i\frac{1}{2}\{\boldsymbol{p}_i,\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\}, \tag{8.28}$$

where the last sum runs over all particles, electrons and nuclei, with their positions and momenta. The operator $\boldsymbol{g}(\boldsymbol{q})$ is the momentum density, and at $\boldsymbol{q}=0$ it is the total momentum. Between eigenstates, $[H,X]$ becomes $\omega_{fi}X$, so $\omega_{fi}\langle f|n_{\mathrm{M}}(\boldsymbol{q})|i\rangle=\boldsymbol{q}\cdot\langle f|\boldsymbol{g}(\boldsymbol{q})|i\rangle$. Inserted into Eq. (4.13), this amplitude relation constrains every response that involves the mass density.

Translation Ward identity

For any operator O_B and $\omega \neq 0$, with S_{Bs} the response matrix of O_B and n_s,

$$\sum_s m_s\, S_{Bs}(\boldsymbol{q},\omega) = \frac{\boldsymbol{q}}{\omega} \cdot S_{B\boldsymbol{g}}(\boldsymbol{q},\omega). \tag{8.29}$$

In amplitudes, $\omega_{fi} \sum_s m_s \langle f|n_s(\boldsymbol{q})|i\rangle = \boldsymbol{q} \cdot \langle f|\boldsymbol{g}(\boldsymbol{q})|i\rangle$ for every pair of states.

The name follows field theory, where a Ward identity is a conservation law written for matrix elements. In quantum electrodynamics the Ward identities express the conservation of charge, and so does the density identity Eq. (7.40) of the superconductor; the identities here express the conservation of momentum. Eq. (8.29) holds under the hypothesis of the f-sum rule, that every interaction depends on positions and spins only.

So far the identity only adds up continuity equations. Its content comes from the next commutator. The total momentum $\boldsymbol{g}(\boldsymbol{0})$ commutes with H, since the crystal holds itself together and no external potential acts. For interactions between pairs of particles, the Coulomb interactions among them, $[H, \boldsymbol{g}(\boldsymbol{q})]$ is then proportional to $\boldsymbol{q}$ (Appendix D.20):

$$[H, g^\alpha(\boldsymbol{q})] = q_\beta\, \mathsf{T}^{\alpha\beta}(\boldsymbol{q}), \tag{8.30}$$

with $\mathsf{T}^{\alpha\beta}$ the momentum-flux tensor, the kinetic flux of momentum plus the stress carried by the forces between pairs. Eq. (8.30) is Newton's third law in operator form: the forces inside the crystal add up to zero, and momentum can only flow from one region to the next. Combined with Eq. (8.29), it gives each factor of $\boldsymbol{q}/\omega$ a partner.

Soft acoustic limit

For pairwise, translation-invariant interactions and $\omega \neq 0$,

$$\sum_s m_s\, S_{Bs}(\boldsymbol{q},\omega) = \frac{q_\alpha q_\beta}{\omega^2}\, S_{B\mathsf{T}^{\alpha\beta}}(\boldsymbol{q},\omega), \qquad S_{n_\mathrm{M} n_\mathrm{M}}(\boldsymbol{q},\omega) = \frac{q_\alpha q_\beta q_\gamma q_\delta}{\omega^4}\, S_{\mathsf{T}^{\alpha\beta}\mathsf{T}^{\gamma\delta}}(\boldsymbol{q},\omega). \tag{8.31}$$

At fixed $\omega \neq 0$ the response of the mass density vanishes as q^4, two powers faster than the response of any single species. As $q \to 0$ its weight collects on the acoustic branches, where $\omega = c_\nu q$ vanishes with q.

On the acoustic branch $q^4/\omega^4 = c_\nu^{-4}$ stays finite, and the mass density makes sound. Everywhere else, at an optical phonon, a pair of phonons with a finite total energy, or an electronic excitation, its amplitude is $O(q^2)$, against $O(q)$ for the density of one species. For a coupling proportional to mass, the structure factors are therefore $\propto q^4$ for the single optical phonon, the two-phonon final states and the anharmonic channel, and $\propto q$ for the single acoustic phonon [15, Table I and Sect. V B], anharmonic terms included. In the harmonic lattice the soft acoustic limit is the acoustic sum rule Eq. (5.18): a rigid displacement exerts no force; equivalently, the potential part of $[H, \boldsymbol{g}(\boldsymbol{q})]$ vanishes at $\boldsymbol{q} = 0$.

The Ward identity sorts any coupling to the densities. Write $\sum_s c_s n_s = \lambda n_\mathrm{M} + \sum_s (c_s - \lambda m_s) n_s$ for any λ. The first term reaches internal excitations at order q^2 alone. At long wavelengths, as a result, the internal motion of the crystal is excited only by the part of a coupling that is not proportional to mass. A coupling to nucleon number differs from a coupling to mass only by the electron masses and the nuclear binding energies. For a coupling to nucleon number, the optical phonons of a polar crystal are suppressed accordingly at small q.

The electron masses give the most instructive case. In a crystal of one element, the density of all nuclei is $\sum_\kappa n_\kappa = (n_\mathrm{M} - m_e n_e)/M_\kappa$. For every final state f at a finite energy ω_{f0} above the ground state, Eq. (8.31) then makes the mass density drop out to the order that matters:

$$\Big\langle f \Big| \sum_\kappa n_\kappa(\boldsymbol{q}) \Big| 0 \Big\rangle = -\frac{m_e}{M_\kappa} \langle f | n_e(\boldsymbol{q}) | 0 \rangle + O(q^2). \tag{8.32}$$

A kicked nucleus excites the electrons as if they had been kicked with the momentum $m_e\boldsymbol{q}/M_\kappa$, in the opposite direction. Eq. (8.32) is the boost of the Migdal effect, Eq. (8.53) summed over all nuclei, now as an exact consequence of momentum conservation (Appendix D.20). At small q the electronic part of the nuclear response is $(m_e/M_\kappa)^2 S_{ee}$, 3.8×10^{-10} times the electron structure factor in silicon, and the interference $S_{e\kappa}$ summed over the nuclei is $-(m_e/M_\kappa)S_{ee}$ there: negative, as the third moment requires.

The third moments obey the same identity. Multiplying Eq. (8.29) by ω^3 and integrating leaves one commutator with $n_e^\dagger$ acting on $q_\alpha q_\beta \mathsf{T}^{\alpha\beta}$, so $m_e \mathscr{M}^3_{ee} + \sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa}$ is $O(q^3)$ at least. The kinetic part of Eq. (8.27) is of order q^4, and the terms of order q^2 must cancel between the electron self moment and the mixed moments. They do (Appendix D.20). The potential term of Eq. (8.27) for electrons is, at small q and cubic sites,

$$\mathscr{M}^3_{ee}(\boldsymbol{q}) \to \frac{e^2 q^2}{3 m_e^2} \Big[\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa \boldsymbol{n}_e^{(\kappa)} + 2\boldsymbol{n}_e^2 \Big], \tag{8.33}$$

which by charge neutrality, $\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa = \boldsymbol{n}_e$, is exactly $-\sum_\kappa (M_\kappa/m_e)$ times Eq. (8.25). The electron self moment carries the same contact densities. In jellium, where the nuclei are replaced by a fixed uniform background and translation invariance is lost, the contact density is replaced by $\boldsymbol{n}_e$, and the bracket $3\boldsymbol{n}_e^2$ returns Puff's $\boldsymbol{n}_e q^2 \omega_p^2/m_e$. For silicon at small q the potential term divided by the f-sum is $e^2[\boldsymbol{n}_e^{(\kappa)} + 2\boldsymbol{n}_e]/3m_e = (2.34\,\mathrm{keV})^2$, far above the kinetic terms $(q^2/2m_e)^2 = (98\,\mathrm{eV})^2$ and $2q^2\bar{K}/m_e = (469\,\mathrm{eV})^2$ at $q = 10\,\mathrm{keV}$.

Worked example 8.5: The Ward identity of a harmonic chain

Take a chain of identical nuclei of mass M, joined by springs of constant K at spacing d, with positions $x_l = ld + u_l$, momenta P_l and periodic boundary conditions over N sites. Its single branch is $\omega_k = 2(K/M)^{1/2} |\sin(kd/2)|$. For numbers we use the longitudinal [100] chain of silicon planes of Sect. 5.2: $M = 26.16\,\mathrm{GeV}$, $d = a/4 = 1.358\,\text{Å}$ and a maximum frequency $2(K/M)^{1/2} = 64.1\,\mathrm{meV}$, which give the sound speed $c = d(K/M)^{1/2} = 6.61\,\mathrm{km\,s^{-1}}$. The two operators of Eq. (8.29) are $n(q) = \sum_l \mathrm{e}^{\mathrm{i}qx_l}$ and $g(q) = \sum_l \frac{1}{2}\{P_l, \mathrm{e}^{\mathrm{i}qx_l}\}$, and the identity reads $M\omega_{fi}\langle f|n|i\rangle = q\langle f|g|i\rangle$.

One phonon. With $u_l = \sum_k (2NM\omega_k)^{-1/2}(a_k \mathrm{e}^{\mathrm{i}kld} + a_k^\dagger \mathrm{e}^{-\mathrm{i}kld})$, the term of first order in u_l creates the phonon $k = q$ from the ground state with $\langle 1_q|n|0\rangle = \mathrm{i}q\,(N/2M\omega_q)^{1/2}$. The momentum density to the same order is $\sum_l \mathrm{e}^{\mathrm{i}qld} P_l$, and $\langle 1_q|P_l|0\rangle = \mathrm{i}M\omega_q \langle 1_q|u_l|0\rangle$ gives $\langle 1_q|g|0\rangle = \mathrm{i}(NM\omega_q/2)^{1/2}$. The ratio of the density element to the momentum-density element is $q/M\omega_q$, as the identity demands. At $q = 1\,\mathrm{keV} = 0.507\,\text{Å}^{-1}$, $qd = 0.688$ and $\omega_q = 21.6\,\mathrm{meV}$.

Two phonons. The second-order terms are $-\frac{1}{2}q^2 \sum_l \mathrm{e}^{\mathrm{i}qld} u_l^2$ in n and $\frac{1}{2}\mathrm{i}q \sum_l \mathrm{e}^{\mathrm{i}qld}\{P_l, u_l\}$ in g. For the pair $|k, q-k\rangle$, with $k \neq q-k$, they give

$$\langle k, q-k|n|0\rangle = -\frac{q^2}{2M(\omega_k\omega_{q-k})^{1/2}}, \qquad \langle k, q-k|g|0\rangle = -\frac{q\,(\omega_k + \omega_{q-k})}{2(\omega_k\omega_{q-k})^{1/2}},$$

and $M(\omega_k + \omega_{q-k})$ times the first is q times the second. The pair has a finite energy as $q \to 0$, near $2\omega_k$, and its amplitude is of order q^2: the soft acoustic limit, here visible term

Table 8.2: Relations that hold for every target, with the hypotheses they need. "Stationary" means any state diagonal in the energy basis, the Gibbs state at any temperature included; "local" means that every interaction depends on positions and spins only; "translation invariant" means pairwise interactions, no external potential.

Relation	Statement	Hypotheses
Detailed balance	$S_{ab}(-\boldsymbol{q},-\omega)=\mathrm{e}^{-\beta\omega}S_{ba}(\boldsymbol{q},\omega)$, Eq. (A.33)	Gibbs state
Fluctuation–dissipation	S from $\operatorname{Im}R$, Eq. (4.18)	Gibbs state
Kramers–Kronig	$\operatorname{Re}R$ from $\operatorname{Im}R$, Eq. (4.24)	causality
Positivity	S_{ab} positive semidefinite, Eq. (4.20)	stationary
Interference bound	$\lvert\Gamma_{AB}\rvert\le 2(\Gamma_A\Gamma_B)^{1/2}$, Eq. (8.21)	stationary
Species f-sum	$\mathscr{M}^1_{ss'}=\delta_{ss'}q^2\boldsymbol{n}_s/m_s$, Eq. (4.34)	stationary, local
Moment matrices	Eq. (8.22), Hankel positivity Eq. (8.23)	stationary
Mixed third moment	$\mathscr{M}^3_{e\kappa}$, Eqs. (8.24) and (8.25)	stationary, local, point nuclei
Ward identity	$\sum_s m_s S_{Bs}=(\boldsymbol{q}/\omega)\cdot S_{B\boldsymbol{g}}$, Eq. (8.29)	stationary, local
Soft acoustic limit	$S_{n_\mathrm{M}n_\mathrm{M}}=O(q^4)$ at $\omega\neq 0$, Eq. (8.31)	translation invariant

by term. Measured against the one-phonon weight, the two-phonon weight carries a factor of order $q^2/(2M\omega)$, the recoil energy over a phonon energy, 8.8×10^{-4} at 1 keV.
The flux. The forces enter through $[H,g(q)]$. Each spring pulls its two ends with opposite forces, $\pm K\Delta_l$ with $\Delta_l=u_{l+1}-u_l$, and the potential part of the commutator is $\mathrm{i}K\sum_l\Delta_l(\mathrm{e}^{\mathrm{i}qx_{l+1}}-\mathrm{e}^{\mathrm{i}qx_l})$. At lowest order the phase difference between the two ends of a spring is $\mathrm{e}^{\mathrm{i}qld}(\mathrm{e}^{\mathrm{i}qd}-1)$, and it carries the factor q of Eq. (8.30): Newton's third law makes the force on the chain as a whole vanish, the one-dimensional form of Eq. (5.18). The one-phonon element of the potential part is $4\mathrm{i}K\sin^2(qd/2)\,(N/2M\omega_q)^{1/2}$, and with $\omega_q\langle 1_q|g|0\rangle=\langle 1_q|[H,g]|0\rangle$ we recover $M\omega_q^2=4K\sin^2(qd/2)$, the dispersion of the chain, up to terms of relative order $q^2/(M\omega_q)$ from the kinetic part of the flux. The Ward identity, read at one phonon, is the equation of motion.
A cubic term in the springs does not change the order q^2 of the two-phonon amplitude. It adds to the two-phonon amplitude a path through the one-phonon state $|1_q\rangle$, whose amplitude from n is of order q. The anharmonic path alone is of order q times the cubic coupling. The flux Eq. (8.30) still carries its factor q, because the cubic springs are still forces between pairs, and the total amplitude to the pair therefore remains of order q^2: the terms of order q cancel between the paths.

8.3.5 Table of exact relations

Table 8.2 collects the exact relations of Sects. 4.2 and 4.3 and of this section with their hypotheses. Each follows from stationarity, the positivity of probabilities, causality, or the conservation of particle number or momentum, and together they constrain the uncertainties that Sect. 8.8 assigns to truncations.

None of these relations involves the coupling to the dark matter beyond the positivity of $\mathfrak{D}_{ab}$, and none involves screening.

8.4 Screening for every coupling

In GaAs the dielectric function vanishes at 36.35 meV, the energy of the longitudinal optical phonon (Sect. 5.4). Sect. 4.4 wrote the rate for a probe of the electron density as the unscreened rate divided by $|\varepsilon|^2$, and a rate of this form would be infinite at 36.35 meV. For a scalar that couples to nucleons the rate is finite, and the phonon it excites is at 36.35 meV rather than at the 33.62 meV of the transverse mode, because the Born charges shift the pole and leave the coupling unchanged (Worked example 8.6). The rule $1/|\varepsilon|^2$ belongs to couplings that see the charge, and so does the loss function behind it. A dark photon sees the total charge of electrons and nuclei, an electrophilic mediator the electrons alone, a hadrophilic scalar the nucleons, and a spin coupling no charge at all, and each is screened differently.

8.4.1 The proper response of a channel

Electrons and nuclei interact through one thing, the Coulomb energy $H_{\rm C}$ of Eq. (2.48), which couples them only through the charge-number density $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ of Sect. 2.3. Phonons, electron–phonon vertices and screening are consequences of the Coulomb energy. Through a momentum transfer $\boldsymbol{q}$, a dark-matter particle couples to the rows of Eq. (8.3), and in a crystal the rows it excites induce charge at $\boldsymbol{q}+\boldsymbol{G}$ for every reciprocal-lattice vector (Sect. 4.4). Here $\boldsymbol{q}$ is in the first zone, and the dark-matter vertex is at the reciprocal-lattice vector $\boldsymbol{G}_q$ of Eq. (A.12). The set of momenta $\{\boldsymbol{q}+\boldsymbol{G}\}$ is the *channel* of the external field. Only the Coulomb lines with a momentum of the channel screen the dark-matter vertex. All others, at $\boldsymbol{q}' \neq \boldsymbol{q}+\boldsymbol{G}$, belong to the material: they bind the electrons, make the phonons and dress the electron–phonon vertex.

Follow the argument of Sect. 4.4 with every species at once. Let the external field couple to row b with amplitude ϕ_b, and write $\delta O_a = \delta\langle O_a(\boldsymbol{q})^\dagger\rangle/\mathscr{V}$ for the induced change of row a, so that the Kubo formula Eq. (4.3) reads $\delta O_a = R_{ab}\phi_b$. The induced charge-number density $\delta n_{Z,\boldsymbol{G}}$ at $\boldsymbol{q}+\boldsymbol{G}$ produces the Hartree potential energy $v_{\rm C}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta n_{Z,\boldsymbol{G}}$ on a unit charge, and this potential acts on the target as an extra field on the row $n_Z(\boldsymbol{q}+\boldsymbol{G})$. With electrons alone, $n_Z = -n_e$ and the two signs cancel; the result is Eq. (4.38). The *proper response of the channel*, $\tilde{R}$, is the response to the external and Hartree fields together,

$$\delta O_a = \sum_b \tilde{R}_{ab}\,\phi_b + \sum_{\boldsymbol{G}} \tilde{R}_{aZ_{\boldsymbol{G}}}\, v_{\rm C}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta n_{Z,\boldsymbol{G}}, \tag{8.34}$$

where the subscript $Z_{\boldsymbol{G}}$ stands for the row $n_Z(\boldsymbol{q}+\boldsymbol{G})$, so that $\tilde{R}_{aZ_{\boldsymbol{G}}} = -\tilde{R}_{ae_{\boldsymbol{G}}} + \sum_\kappa Z_\kappa \tilde{R}_{a\kappa_{\boldsymbol{G}}}$. Electrons and nuclei enter on the same footing, as two kinds of charge. In the language of Sect. 4.8, $\tilde{R}$ is the sum of the diagrams that cannot be cut by removing a single Coulomb line of the channel. By the argument of Eq. (4.97) it is also the response of the Hamiltonian with the Coulomb terms of the channel deleted; this reduced Hamiltonian has the same equilibrium state in the thermodynamic limit. Every exact property of Chap. 4 therefore holds for $\tilde{R}$, the Lehmann form and positivity included.

Taking $a = Z_{\boldsymbol{G}}$ in Eq. (8.34) closes the equations for the induced charge, and the result for any pair of rows is (Appendix D.21)

Screened response for any coupling

$$\begin{aligned} R_{ab}(\boldsymbol{q},\omega) &= \tilde{R}_{ab} + \sum_{\boldsymbol{G},\boldsymbol{G}'} \tilde{R}_{aZ_{\boldsymbol{G}}}\,(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}\, v_{\rm C}(|\boldsymbol{q}+\boldsymbol{G}'|)\,\tilde{R}_{Z_{\boldsymbol{G}'}b}, \\ \varepsilon_{\boldsymbol{G}\boldsymbol{G}'} &= \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\rm C}(|\boldsymbol{q}+\boldsymbol{G}|)\,\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}, \end{aligned} \tag{8.35}$$

with every function at $(\boldsymbol{q},\omega)$. The matrix ε is the total dielectric matrix of electrons

and nuclei. A row is screened only through its proper correlation with the charge, $\tilde{R}_{aZ}$.

Without local fields the sums reduce to $\boldsymbol{G} = 0$, and

$$R_{ab} = \tilde{R}_{ab} + v_{\mathrm{C}}(q)\,\frac{\tilde{R}_{aZ}\,\tilde{R}_{Zb}}{\varepsilon}, \qquad \varepsilon = 1 - v_{\mathrm{C}}(q)\,\tilde{R}_{ZZ}. \tag{8.36}$$

Eq. (8.36) reads as in Sect. 4.4, with the vertex made explicit. The dark matter acts on row b; the target answers with a proper charge $\tilde{R}_{Zb}$; the whole medium screens the field of this charge to $\varepsilon^{-1}v_{\mathrm{C}}$, and the screened field acts back on row a through $\tilde{R}_{aZ}$. The dielectric function contains the polarization of both the electrons and the nuclei; it is the static ε_0 of Sect. 5.4 at low frequency and ε_∞ between the phonons and the gap. With the nuclei clamped it reduces to the electronic ε of Sect. 4.4. For time-dependent density-functional theory the same algebra holds with f_{xc} inside the proper part (Sect. 4.6).

The proper cross responses are material quantities. The charge–charge block is the dielectric matrix that codes compute in the random-phase approximation with local fields or with an exchange–correlation kernel (Sect. 26.1), and its ionic part at long wavelength is fixed by the Born charges and phonons of density-functional perturbation theory (Sect. 25.2). The blocks $\tilde{R}_{aZ}$ for a current, spin or kinetic-energy row are computed from the same Kohn–Sham states as the response of row a to a scalar potential.

8.4.2 Four couplings

The rows of a density coupling are the densities of the species, weighted by the coefficients $c_1^{(s)}$ of Sect. 8.1: the dark matter acts through $\sum_s c_1^{(s)} n_s(\boldsymbol{q})$, with $c_1^{(\kappa)} = [Z_\kappa c_1^{(p)} + (A_\kappa - Z_\kappa)c_1^{(n)}]F_N$ from Eq. (8.8). Three choices of the coefficients cover the standard density couplings. A fourth coupling, to the spin, carries no charge, and Eq. (8.35) treats each of the four in one line.

Total charge. A dark photon, heavy or light, couples to $-\tilde{V}n_Z$ (Worked example 8.1), and so does any millicharged particle. With $a = b = Z$, Eq. (8.35) reduces to $v_{\mathrm{C}}R_{ZZ} = \varepsilon^{-1} - 1$ (Appendix D.21), the relation Sect. 4.4 stated, now with the nuclei moving. The fluctuation–dissipation theorem then gives $S_{ZZ} = (2q^2/e^2)\,L/(1 - \mathrm{e}^{-\beta\omega})$, with $L = \mathrm{Im}[-(\varepsilon^{-1})_{00}]$ the loss function of the total dielectric matrix, exactly and at every energy. Below the gap L contains the longitudinal optical phonons of a polar crystal, which Sect. 11.2 turns into a rate.

Electron density. A scalar or vector mediator acts through $c_1^{(e)} n_e$ when it couples to electrons and not to nucleons, and Eq. (8.35) gives

$$R_{ee} = \tilde{R}_{ee} + \tilde{R}_{eZ}\,\varepsilon^{-1}v_{\mathrm{C}}\,\tilde{R}_{Ze}, \qquad \tilde{R}_{eZ} = -\tilde{R}_{ee} + \sum_\kappa Z_\kappa \tilde{R}_{e\kappa}. \tag{8.37}$$

With the nuclei clamped, n_κ is a static Bragg sum, $\tilde{R}_{e\kappa} = 0$, and Eq. (8.37) is Eq. (4.41) again. With moving nuclei the second term of $\tilde{R}_{eZ}$ is the electron density that follows a nucleus, and it is large at phonon energies, where the electrons of an atom move with their nucleus (Sect. 8.6). The exact statement for this coupling is the electron-channel form of Appendix A.7, $S_{ee} = (2q^2/e^2)\,L_e/(1 - \mathrm{e}^{-\beta\omega})$ with $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}}R_{ee}$, but ε_e is not the dielectric function of any experiment. Its inverse has a pole at every acoustic phonon, weighted by the electrons that follow the nuclei, while the total ε^{-1} at small q has none, since a moving neutral cell carries no charge.

Nucleon number. A scalar coupled equally to protons and neutrons acts through $\sum_\kappa A_\kappa n_\kappa$, times its coupling per nucleon. Its vertex has no electronic part, and its screening runs entirely through $\tilde{R}_{NZ}$, the proper charge that moves when the nucleon density does. At phonon

energies this proper charge is the Born charge of each sublattice; at electronic energies the nuclei do not move and the coupling is unscreened. The Born charge never enters the vertex itself. It enters the pole: the proper response has its optical pole at the transverse frequency and the screened response at the longitudinal frequency, with the same f-sum weight (Worked example 8.6).

Spin. Without spin–orbit coupling and without magnetic order, the spin density $\boldsymbol{s}_e$ of a target is not screened. The Hamiltonian of this target commutes with the total spin, the charge density is invariant under spin rotations and $\boldsymbol{s}_e$ is a vector under them, and the equilibrium state is invariant too. The cross response $\tilde{R}_{sZ}$ then vanishes, and $R_{ss} = \tilde{R}_{ss}$. Spin–orbit coupling or magnetic order breaks the argument; the polarized gas of Worked example 8.3 has $S_{ns} \neq 0$ and screens its spin rows with its density. The nuclear spin density $\boldsymbol{s}_\kappa$ is unscreened for the same reason. Beyond independent particles a spin-flip pair still feels the attraction between its electron and hole, the direct kernel of Sect. 4.9, but not the exchange kernel that carries the local fields: a spin-flip pair has no charge to exchange.

The rows that are not densities follow the same rule. The longitudinal current is screened as the density. The continuity equation in the form of Eq. (8.5) makes $\hat{\boldsymbol{q}} \cdot \boldsymbol{j}_s$ act as $(\omega/q)\, n_s$ on the energy shell, and the Ward identity Eq. (4.51) holds for proper and full responses alike. The transverse current and the kinetic-energy density $\mathcal{T}_e$ are screened where their proper correlation with the charge is nonzero. The spin current is a vector under spin rotations like $\boldsymbol{s}_e$, so it has no such correlation without spin–orbit coupling or magnetic order and is unscreened. The kinetic-energy density is a scalar under spin rotations and the point group and correlates with the charge already at $\boldsymbol{G} = 0$, so it is screened at the head, by a term of order q^2 at small q in a crystal without a polar axis. In a cubic crystal, with $\boldsymbol{q}$ along a symmetry axis, $\tilde{R}_{\boldsymbol{j}_\mathrm{T} Z}$ vanishes at $\boldsymbol{G} = 0$ by symmetry, and the transverse current is screened through the local fields at $\boldsymbol{G} \neq 0$. In a crystal of lower symmetry, or off the symmetry axes, longitudinal and transverse parts mix already at $\boldsymbol{G} = 0$. Exchange of a transverse photon between currents is a separate and smaller effect at the momentum transfers of scattering. The propagator of the transverse photon is $1/(\omega^2 - q^2 - \Pi_\mathrm{em})$, with Π_em the photon self-energy of the medium (Sect. 8.4.5); it differs from $-1/q^2$ by $\Pi_\mathrm{em}/q^2 \sim \omega_p^2/q^2$, which is 2×10^{-5} in silicon at q_0. Every row with $\tilde{R}_{aZ} \neq 0$ is screened through Eq. (8.35). For every coupling except the charge, a rate is wrong if it contracts Eq. (8.15) with the proper response $\tilde{R}$ alone, or with a factor $1/|\varepsilon|^2$ in place of the cross terms.

8.4.3 Exactness of the loss function

The loss function is measured for many materials, optically at $q \to 0$ and by electron energy loss at finite q, and it is the quantity that a computed response is checked against. The screened response fixes which couplings the loss function describes exactly.

Exactness of the loss-function form

Let a dark-matter particle couple to $\sum_s c_1^{(s)} n_s(\boldsymbol{q})$ in a crystal, a target whose long-wavelength longitudinal acoustic phonon moves it locally as a whole. Let nuclei of different atomic number have different masses, and set the nuclear form factors to one. Then

$$S_{cc}(\boldsymbol{q}, \omega) = \lambda_Z^2 \, \frac{2q^2}{e^2} \, \frac{L(\boldsymbol{q}, \omega)}{1 - \mathrm{e}^{-\beta\omega}} \quad \text{at every } \boldsymbol{q} \text{ and every } \omega \neq 0, \tag{8.38}$$

with S_{cc} the structure factor of the coupling and L the loss function of the total dielectric matrix. Eq. (8.38) holds for some constant λ_Z if and only if the coupling is proportional to the charge, $c_1^{(e)} = -\lambda_Z$ and $c_1^{(\kappa)} = \lambda_Z Z_\kappa$, or the nuclei are clamped and $c_1^{(e)} = \pm\lambda_Z$.

A coupling to the electron density is exact only at a static lattice.

Sufficiency is immediate. If the coupling differs from $\lambda_Z n_Z$ by an operator with no dynamics, a sum of static Bragg terms when the nuclei are clamped, the coupling and $\lambda_Z n_Z$ have the same transitions and $S_{cc} = \lambda_Z^2 S_{ZZ}$ at every $\omega \neq 0$. By the total-charge relation above, this equality is Eq. (8.38).

The converse takes two limits in which electrons and nuclei come apart, and Appendix D.21 gives each step. At large momentum transfer, above the inverse size of every electron orbital and in the regime where a nucleus recoils freely from its site (the impulse regime of Sect. 8.5), each particle recoils independently. The structure factor of the coupling is then a sum of recoil peaks, at $q^2/2m_e$ for the electrons and $q^2/2M_\kappa$ for each nuclear species, with weights $(c_1^{(s)})^2 \boldsymbol{n}_s$, and the structure factor of the charge has the weights $\boldsymbol{n}_e$ and $Z_\kappa^2 \boldsymbol{n}_\kappa$. Equality peak by peak requires $|c_1^{(e)}| = |\lambda_Z|$ and $|c_1^{(\kappa)}| = |\lambda_Z| Z_\kappa$ for every species: the coupling has the magnitudes of the charge coefficients, but the signs are still free. The signs are fixed at the opposite end, $\boldsymbol{q} \to 0$ at the frequency of a longitudinal acoustic phonon. There the crystal moves as a whole, electrons included, and the amplitude to excite the phonon is proportional to the coupling carried by a unit volume, $\sum_s c_1^{(s)} \boldsymbol{n}_s$. For the charge this sum is the net charge density, zero. Equality at small q needs $c_1^{(e)} \boldsymbol{n}_e + \sum_\kappa c_1^{(\kappa)} \boldsymbol{n}_\kappa = 0$, and with the magnitudes already fixed this condition can hold only if every nuclear coefficient has the sign opposite to the electron coefficient. The coupling is then $\pm\lambda_Z n_Z$. With the nuclei clamped, the nuclear peaks and the acoustic phonon are absent, and only $|c_1^{(e)}| = |\lambda_Z|$ survives.

The theorem leaves no exact use of the loss function for an electrophilic mediator once the lattice moves, and none for a nucleon or isospin-violating coupling. Its practical content is a bound on where the error lies. For the electron density, the difference between S_{ee} and S_{ZZ} comes from the nuclear rows of Eq. (8.37). Above the phonon band, however, the nuclei reach the electronic final states only through the Migdal amplitudes, by which a moving nucleus excites the electrons (Sect. 8.6). These amplitudes are suppressed by powers of κ_{BO}, and to that accuracy the loss function gives the electron-only rate, as Sect. 4.4 stated. Inside the phonon band S_{ee} and S_{ZZ} differ at leading order: S_{ee} has an acoustic peak carrying the electrons that follow the nuclei, a fraction $Z_\kappa m_e/M_\kappa$ of its f-sum (Eq. (8.54)), while S_{ZZ} has none at small q. Sect. 10.2 uses the theorem for the electronic rate, and this acoustic peak gives the relaxed-lattice correction to the sum-rule bound on the electronic rate.

Table 8.3 collects the couplings with the screening each takes.

Worked example 8.6: The nucleon-number coupling of a heavy scalar in GaAs

Take a heavy scalar that couples to protons and neutrons with the same coefficient, $c_1^{(p)} = c_1^{(n)}$, and to no electron. By Eq. (8.8) its row is $c_1^{(p)} \sum_\kappa A_\kappa n_\kappa(\boldsymbol{q})$, with $F_N = 1$ at the momenta of phonon scattering. We use the measured GaAs numbers of Worked example 5.5: $\omega_{\mathrm{T}} = 33.62\,\mathrm{meV}$, $\omega_{\mathrm{L}} = 36.35\,\mathrm{meV}$, $\varepsilon_\infty = 10.88$, $\Omega = 44.80\,\text{Å}^3$, $\mu_{\mathrm{red}} = 36.11\,\mathrm{u}$ and $Z^* = 2.11$. Take $\boldsymbol{q}$ small against the zone and ω near the optical phonons, where the macroscopic field is the whole of the long-range Coulomb line and Eq. (8.36) applies.

Vertices. In an optical mode at small $\boldsymbol{q}$ the two sublattices move against each other with the center of mass at rest, $\boldsymbol{u}_{\mathrm{Ga}} = (\mu_{\mathrm{red}}/M_{\mathrm{Ga}})\boldsymbol{u}_{\mathrm{rel}}$ and $\boldsymbol{u}_{\mathrm{As}} = -(\mu_{\mathrm{red}}/M_{\mathrm{As}})\boldsymbol{u}_{\mathrm{rel}}$, with $\boldsymbol{u}_{\mathrm{rel}} = \boldsymbol{u}_{\mathrm{Ga}} - \boldsymbol{u}_{\mathrm{As}}$ the relative coordinate of Sect. 5.4. When $n_\kappa(\boldsymbol{q}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa+\boldsymbol{u}_{\boldsymbol{R}\kappa})}$ is expanded to first order in the displacements, the nucleon row creates the mode through

Table 8.3: External couplings of dark matter and the screening each takes, from the screened response Eq. (8.35). "Exact L" states whether the structure factor of the coupling equals $(2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$, with L the loss function of the total dielectric matrix of electrons and nuclei, at every momentum transfer and energy (Eq. (8.38)). SOC: spin–orbit coupling.

Coupling	Example	Screened by	Exact L
Total charge n_Z	dark photon, heavy or light; millicharge	ε^{-1} of electrons and nuclei; $v_\mathrm{C}R_{ZZ} = \varepsilon^{-1} - 1$	yes, any lattice
Electron density n_e	electrophilic scalar or vector mediator	$\tilde{R}_{eZ}\varepsilon^{-1}v_\mathrm{C}\tilde{R}_{Ze}$: electrons, and the nuclei they follow	clamped nuclei, $\omega \neq 0$; above the phonon band to Migdal order
Nucleon number $\sum_\kappa A_\kappa n_\kappa$	hadrophilic scalar	ionic polarization through the Born charges; it moves the poles (TO to LO), not the vertex	no
Other density mixtures	isospin-violating couplings	$\tilde{R}_{cZ}\varepsilon^{-1}v_\mathrm{C}\tilde{R}_{Zc}$	only if proportional to n_Z
Spin density $\boldsymbol{s}_e$, $\boldsymbol{s}_\kappa$	$\mathcal{O}_4$, $\mathcal{O}_6$, $\mathcal{O}_{10}$	none without SOC or magnetic order; the triplet exchange–correlation kernel remains	no
Longitudinal current $\hat{\boldsymbol{q}}\cdot\boldsymbol{j}$	$\mathcal{O}_8$, $\mathcal{O}_2$ (parts along $\boldsymbol{q}$)	as the density, (ω/q) times its row	as the density
Transverse current $\boldsymbol{j}_\mathrm{T}$	$\mathcal{O}_5$, $\mathcal{O}_8$ (part across $\boldsymbol{q}$)	only through local fields ($\boldsymbol{G} \neq 0$) on symmetry axes; at $\boldsymbol{G} = 0$ only through longitudinal–transverse mixing	no
Spin current $\mathcal{J}_e^{\alpha\beta}$	$\mathcal{O}_3$, $\mathcal{O}_7$, $\mathcal{O}_{12}$–$\mathcal{O}_{15}$; ALP absorption	none without SOC or magnetic order, as the spin density	no
Kinetic-energy density $\mathcal{T}_e$	$\mathcal{O}_2$ (part in $(\boldsymbol{v}^\perp)^2$); scalar absorption	head and local fields; head term $O(q^2)$ at small q in crystals without a polar axis	no

$\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\mathrm{rel}}$ times an amplitude per cell. At leading order in $\boldsymbol{q}$ this amplitude is

$$a_N = \mu_{\mathrm{red}}\Big(\frac{A_{\mathrm{Ga}}}{M_{\mathrm{Ga}}} - \frac{A_{\mathrm{As}}}{M_{\mathrm{As}}}\Big).$$

At finite $\boldsymbol{q}$ the vertex combines the site phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ of n_κ with the zone-center displacements, and it then changes when another arsenic atom is assigned to the cell. Only with the eigenvector of the mode at $\boldsymbol{q}$ is the vertex independent of which arsenic atom is assigned to the cell, $a_N(\boldsymbol{q}) = \mu_{\mathrm{red}}^{1/2}\sum_\kappa A_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\,\hat{\boldsymbol{q}}\cdot\boldsymbol{e}^*_{\kappa\mathrm{L}}(\boldsymbol{q})/M_\kappa^{1/2}$, with the conventions of Eq. (11.1). The charge row creates the mode through the Born charges, $\sum_\kappa Z^*_\kappa\,\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_\kappa = Z^*\,\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\mathrm{rel}}$ at leading order in $\boldsymbol{q}$, since $\mu_{\mathrm{red}}(1/M_{\mathrm{Ga}} + 1/M_{\mathrm{As}}) = 1$. The Born charge is the right vertex for the proper response: Sect. 5.4 defines it at zero macroscopic field, and removing the Coulomb line at $\boldsymbol{q}$ imposes this condition. For the same reason the relative coordinate of the proper response oscillates at ω_{T}, for longitudinal as well as transverse motion, with the response $1/[\mu_{\mathrm{red}}(\omega^2 - \omega_{\mathrm{T}}^2)]$ per cell. With $N/\mathcal{V} = 1/\Omega$,

$$\tilde{R}_{NN} = \frac{q^2|a_N|^2}{\mu_{\mathrm{red}}\Omega\,(\omega^2-\omega_{\mathrm{T}}^2)}, \qquad \tilde{R}_{NZ} = \frac{q^2 a_N Z^*}{\mu_{\mathrm{red}}\Omega\,(\omega^2-\omega_{\mathrm{T}}^2)}, \qquad \varepsilon = \varepsilon_\infty + \frac{e^2 Z^{*2}}{\Omega\,\mu_{\mathrm{red}}(\omega_{\mathrm{T}}^2-\omega^2)},$$

where ε is Eq. (5.44): the electrons give ε_∞, and the ionic part of $\tilde{R}_{ZZ}$ the rest.
Screening. Insert $\tilde{R}_{NN}$, $\tilde{R}_{NZ}$ and ε into Eq. (8.36). By Eqs. (5.43) and (5.44), $\varepsilon = \varepsilon_\infty(\omega_{\mathrm{L}}^2-\omega^2)/(\omega_{\mathrm{T}}^2-\omega^2)$ and $e^2Z^{*2}/(\Omega\mu_{\mathrm{red}}\varepsilon_\infty) = \omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2$, and the screening term is

$$v_{\mathrm{C}}\frac{\tilde{R}_{NZ}\tilde{R}_{ZN}}{\varepsilon} = \frac{q^2|a_N|^2}{\mu_{\mathrm{red}}\Omega}\,\frac{\omega_{\mathrm{L}}^2-\omega_{\mathrm{T}}^2}{(\omega^2-\omega_{\mathrm{T}}^2)(\omega^2-\omega_{\mathrm{L}}^2)},$$

and added to $\tilde{R}_{NN}$ it cancels the transverse pole:

$$R_{NN}(\boldsymbol{q},\omega) = \frac{q^2|a_N|^2}{\mu_{\mathrm{red}}\Omega\,(\omega^2-\omega_{\mathrm{L}}^2)}, \qquad S_{NN}(\boldsymbol{q},\omega) = \frac{\pi q^2|a_N|^2}{\mu_{\mathrm{red}}\Omega\,\omega_{\mathrm{L}}}\,\delta(\omega-\omega_{\mathrm{L}}),$$

at $T = 0$ and $\omega > 0$. The screened response is the proper one with its pole moved from ω_{T} to ω_{L}, and its f-sum weight $\int \mathrm{d}\omega\,\omega S_{NN}/2\pi$ is unchanged; the area of the peak, $\int \mathrm{d}\omega\, S_{NN}/2\pi$, falls by $\omega_{\mathrm{T}}/\omega_{\mathrm{L}} = 0.925$. The Born charges and ε_∞ enter through ω_{L} and nowhere else; the vertex is a_N, a property of the nuclei and their masses. The rule $1/|\varepsilon|^2$ makes two errors here. The proper structure factor has its weight at ω_{T}, where ε has a pole and $1/|\varepsilon|^2 = 0$, so the naive rate would vanish; the true rate lies at ω_{L}, where $1/|\varepsilon|^2$ is infinite. The naive rule drops the cross term of Eq. (8.36), and this cross term resolves both errors.
The size of the vertex. At $\boldsymbol{q}\to 0$ the phases drop out and $a_N \to \mu_{\mathrm{red}}(A_{\mathrm{Ga}}/M_{\mathrm{Ga}} - A_{\mathrm{As}}/M_{\mathrm{As}})$, which would vanish if atomic masses were proportional to nucleon numbers. The atomic masses are nearly proportional to the nucleon numbers. With the atomic masses of the isotopes [16] and the natural abundances of gallium, 60.108 % ^{69}Ga and 39.892 % ^{71}Ga [7], the mean nucleon number per mass is 1.001072 per u for gallium and 1.001046 per u for arsenic, and $a_N = 36.11 \times 2.59 \times 10^{-5} = 9.4 \times 10^{-4}$. A nucleon-number coupling barely moves the center of nucleon number when the center of mass stays put. At finite $\boldsymbol{q}$ the center of mass no longer stays put. In zincblende a shear strain displaces the two sublattices against each other; this internal strain is behind the piezoelectric charge of Sect. 6.2. By the same internal-strain coupling, an optical wave drags the center of mass along wherever a longitudinal wave contains shear. The nucleon row then sees the mass density of the cell, $A_\kappa/M_\kappa \approx 1$ per u, and a coupling to the mass density alone gives

nearly the same vertex. Along [111], the direction of the Ga–As bond, a longitudinal wave contains shear, and the drag grows linearly in q and overtakes the mass term near $q \approx 0.05\,\mathrm{eV}$. Scattering reaches ω_L only for $q \geq \omega_\mathrm{L}/v_\mathrm{max} = 13.7\,\mathrm{eV}$, where, with the phonons of Fig. 5.4 at that $\boldsymbol{q}$, $|a_N| = 0.24$, 250 times the long-wavelength value; at 100 eV, $|a_N| = 1.74$. Along [100] a longitudinal wave contains no shear, the drag is absent at first order in q, and $|a_N| = 2.1 \times 10^{-3}$ at 13.7 eV, close to its value at $q \to 0$. The longitudinal acoustic phonon has the vertex $\sum_\kappa A_\kappa = 144.8$ per cell, which is not screened at leading order because $\sum_\kappa Z^*_\kappa = 0$. The energy-weighted optical share is $|a_N|^2 M_\mathrm{cell}/\mu_\mathrm{red}$ divided by the squared acoustic vertex; it grows as q^2, and along [111] it is 1.1×10^{-5} at 13.7 eV and 5.8×10^{-4} at 100 eV. This small optical share is the suppression of optical phonons by destructive interference within the cell that Griffin and collaborators found for couplings proportional to the mass number [17, Eqs. (30) and (33)]. For a dark photon the same mode couples through $Z^*/\varepsilon_\infty = 0.19$ at any small q, and it is the polar phonon of Sect. 11.2.

8.4.4 Coulomb-line classes

Below the direct gap of silicon, dark matter can excite an electron–hole pair only together with a phonon (Sect. 6.5). The amplitude contains three objects that the medium screens. The dark-matter vertex is at $(\boldsymbol{q}, \omega)$. The electron–phonon vertex is at the phonon wavevector $\boldsymbol{Q}$, and density-functional perturbation theory returns it already screened (Sect. 6.2). The outgoing pair has momentum $\boldsymbol{q} + \boldsymbol{Q}$ and energy $\omega \mp \omega_{\boldsymbol{Q}\nu}$, upper sign for emission. If a calculation divides the dark-matter vertex by $\varepsilon(\boldsymbol{q}, \omega)$ and uses the computed vertex g, it might count one Coulomb line twice, or miss one. The question is which Coulomb lines each factor contains, and the diagrams of Sect. 4.8 answer it.

Work at the random-phase level, where every Coulomb line joins two different fermion loops and nothing runs inside a loop, at second order in the electron–phonon vertex g in the rate, with one phonon in the final state. Draw a diagram of the response as loops joined by Coulomb lines, with one phonon line between two ion vertices and the two dark-matter vertices on loops.

Coulomb-line classes

In such a diagram every Coulomb line belongs to exactly one of three classes.

- *External*, at $(\boldsymbol{q} + \boldsymbol{G}, \omega)$: a line whose removal cuts the diagram in two. These lines form the chains that Eq. (8.35) resums into $(\varepsilon^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ on the dark-matter vertex, local fields included.
- *Vertex*, at $\boldsymbol{Q} + \boldsymbol{G}$ and frequencies of order $\omega_{\boldsymbol{Q}\nu}$: lines between an ion vertex and the first loop that carries a dark-matter vertex. They make the screened electron–phonon vertex g.
- *Pair*, at $(\boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}, \omega \mp \omega_{\boldsymbol{Q}\nu})$: lines between the two loops that carry the dark-matter vertices. They resum into the screened interaction $W^\mathrm{scr}(\boldsymbol{q} + \boldsymbol{Q}, \omega \mp \omega_{\boldsymbol{Q}\nu})$ of Sect. 4.8, acting on the final pair.

A phonon line that itself cuts the diagram in two is not a Coulomb line. It is the ionic part of the total dielectric matrix, included by using the ε of Eq. (8.35) in the external and pair chains.

The proof is by the topology of the diagram (Appendix D.21). With the phonon line removed, the loops and Coulomb lines form a tree, since at the random-phase level every Coulomb line is a bridge; a closed cycle of Coulomb lines would need a line inside a loop. Putting the phonon line back closes exactly one cycle. Lines off the cycle remain bridges, and momentum conservation sets them to $\boldsymbol{q}+\boldsymbol{G}$; these are the external lines. On the cycle, the segments next to the ion vertices carry $\boldsymbol{Q}+\boldsymbol{G}$, and the segment between the two loops with dark-matter vertices carries $\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}$. When both dark-matter vertices lie on one loop, the pair segment is absent. The class of a line follows from its position, not from its momentum label, and the classes stay distinct where the labels coincide. At $\boldsymbol{q}\to 0$ the pair segment carries the momentum $\boldsymbol{Q}$ of the vertex segments but an energy of order eV, against energies of order $\omega_{\boldsymbol{Q}\nu}$ on the vertex segments, and at $\boldsymbol{Q}\to 0$ the pair segment lies on the cycle while the external lines do not.

First, three factors together count every random-phase diagram once: the screened vertex of density-functional perturbation theory, an external $\varepsilon^{-1}(\boldsymbol{q},\omega)$ and the pair interaction W^{scr}. The computed g already contains the vertex lines, and an external $1/\varepsilon(\boldsymbol{Q})$ on top of it would count them twice; leaving out W^{scr} is an omission, not a cure for a double count. Second, the vertex lines are evaluated by density-functional perturbation theory at zero frequency, while they carry frequencies of order $\omega_{\boldsymbol{Q}\nu}$. Below the gap, time reversal makes $\varepsilon(\boldsymbol{Q},\omega')$ an even and analytic function of the frequency ω' for $|\omega'|<E_{\mathrm{g}}$ (Appendix D.21). The error of the static vertex is then of order $(\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}})^2$, about 3×10^{-3} in silicon, the number of Worked example 8.7; in a magnetic insulator, without time reversal, it can be of order $\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}}$. Third, the ionic insertion is small at electronic energies. Far above the optical phonons, Eq. (5.44) gives an ionic part $\varepsilon-\varepsilon_\infty\approx-(\varepsilon_0-\varepsilon_\infty)\,\omega_{\mathrm{T}}^2/\omega^2$, which is -0.9×10^{-3} in GaAs at 1.5 eV, 10^{-4} of ε_∞; in silicon, whose Born charges vanish, it is smaller still.

The pair lines are the new element. They screen the final pair at its own momentum and energy: once the phonon has taken $\boldsymbol{Q}$, the charge induced by the dark-matter vertex oscillates at $\boldsymbol{q}+\boldsymbol{Q}$, and the medium screens it there, not at $\boldsymbol{q}$. The final pair is on shell, the screened interaction at its momentum and energy is complex, and it acts between the two halves of the amplitude. Sect. 13.4 writes out this screening of the final pair, with its size in silicon. Lines inside a loop, the electron–hole ladder of Sect. 4.9, lie outside the classification; an adiabatic exchange–correlation kernel adds a source term of its own. With local fields in the supercell basis, a frozen-phonon supercell calculation contains all three classes at once, since every $\boldsymbol{Q}+\boldsymbol{G}$ is then a reciprocal vector of the supercell.

8.4.5 In-medium mixing

At 3 eV silicon has $\varepsilon_2=2.380$ and a loss function $L=3.127\times10^{-3}$ (Sect. 4.4), so $|\varepsilon|^2=\varepsilon_2/L=761$. When a dark photon is absorbed at 3 eV, it couples as if its squared mixing were $\zeta^2/761$ rather than ζ^2. This reduction of the squared mixing is the in-medium mixing ζ_{eff} of Sect. 1.4, and it comes from the photon-reducible part of the response. A dark photon couples to the electromagnetic current with strength ζe (Sect. 1.4). The current it induces radiates an ordinary photon, which the medium absorbs again, and the chain of emission and reabsorption repeats. Split the current response as the density response was split: a part $\tilde{R}$ that is proper with respect to one photon line, longitudinal or transverse, and the photon-reducible chains. For absorption at rest, Appendix D.19 summed the chain and found that the summed chain turns ζ^2 into

$$\zeta_{\mathrm{eff}}^2=\frac{\zeta^2 m_\phi^4}{|m_\phi^2-\Pi_{\mathrm{em}}|^2},\tag{8.39}$$

with Π_{em} the photon self-energy, the proper current response times e^2, diamagnetic term included. In the language of Mitridate and collaborators, the dark-matter self-energy acquires a mixing term $\Pi_{\phi A}\Pi_{A\phi}/(m_\phi^2 - \Pi_{AA})$ beside its proper part [8, Eqs. (20)–(22)]; the mixing term is the photon-reducible response.

Both polarizations follow from one relation. The transverse self-energy is $\Pi_{\text{T}} = \omega^2(1-\varepsilon^{\text{T}})$ (Sect. 4.4), and the longitudinal one is $\Pi_{\text{L}} = (\omega^2-k^2)(1-\varepsilon)$ [8, Eq. (34)], with k the momentum of the dark photon and ε the longitudinal function of Eq. (4.41). On shell, $\omega^2 - k^2 = m_\phi^2$, and $m_\phi^2 - \Pi_{\text{L}} = m_\phi^2\varepsilon$: a longitudinal dark photon mixes with $\zeta_{\text{eff}} = \zeta/\varepsilon$ exactly. A transverse dark photon has the same mixing at $k \to 0$, where $\varepsilon^{\text{T}} = \varepsilon$ by Eq. (4.52) and $\omega = m_\phi$. The net rate per quantum, Eq. (8.19) with the full response, is then

$$\Gamma_{\text{net}} = -\frac{\zeta^2 m_\phi^2}{\omega}\,\text{Im}\,\frac{\Pi_{\text{em}}}{m_\phi^2 - \Pi_{\text{em}}} = -\zeta^2 m_\phi\,\text{Im}\Big(\frac{1}{\varepsilon} - 1\Big) = \zeta^2 m_\phi\, L(m_\phi),$$

which is $\zeta_{\text{eff}}^2\sigma_1$ with $\sigma_1 = \omega\varepsilon_2$ [8, Eq. (40)]. A dark photon at rest measures the loss function of the target, not its absorption, and as a result the absorption of a dark photon in GaAs peaks at the longitudinal phonon (Sect. 5.4).

Scattering through a dark-photon mediator is the same construction at $\omega \ll q$. The interaction of Worked example 8.1 uses the potential of the dark matter in vacuum, $\tilde{V}$, and the full response R_{ZZ}, which carries the chain through ε^{-1}. We can instead put the chain into the mediator: the dark photon then mixes with the longitudinal photon of the medium, its propagator acquires $1/\varepsilon(\boldsymbol{q},\omega)$, and it acts on the proper response. The two bases agree [18, Eq. (8) and the text below it]: with $\varepsilon = 1 - v_{\text{C}}\tilde{R}_{ZZ}$, $\tilde{R}_{ZZ}/\varepsilon = \tilde{R}_{ZZ}\varepsilon^*/|\varepsilon|^2$, and $\text{Im}(\tilde{R}_{ZZ}\varepsilon^*) = \text{Im}\,\tilde{R}_{ZZ}$ because $v_{\text{C}}|\tilde{R}_{ZZ}|^2$ is real, so that $\text{Im}\,R_{ZZ} = \text{Im}\,\tilde{R}_{ZZ}/|\varepsilon|^2$. In scattering, the in-medium mixing is the factor $1/\varepsilon(\boldsymbol{q},\omega)$, and it is the screening of the total-charge row of Table 8.3.

The bookkeeping follows the Coulomb lines. Either the vacuum mixing ζ multiplies the full response, or ζ_{eff} multiplies the proper one. The loss function already contains the screening, and multiplying a loss-function rate by $\zeta_{\text{eff}}^2/\zeta^2$ counts the screening twice. The optical conductivity σ_1 is a proper response, and Eq. (1.38) multiplies it by ζ_{eff}^2 once. Other bosons mix in the same way through their own rows. The scalar of Eq. (8.11) couples to n_e and $\mathcal{T}_e$, both of which correlate with the charge, and its absorption carries a photon-reducible part through $\tilde{R}_{n_eZ}$ and $\tilde{R}_{\mathcal{T}_eZ}$. An axion-like particle in a static field $\boldsymbol{B}_0$ acts as a source current along $\boldsymbol{B}_0$ and mixes as a dark photon does; the spin rows of the axioelectric coupling do not mix without spin–orbit coupling. Sect. 10.7 works out these rates.

8.5 The expansion hierarchy

The phonon-assisted absorption of silicon, Eq. (6.43), accounts for the measured indirect edge so well that the phonon energies that are read off its thresholds agree with the neutron values to 2 % (Sect. 6.5), and yet it is a truncated formula. It keeps the electron–phonon vertex to second order and drops the fourth. It takes the phonons as harmonic, the electrons as independent particles, the thermal occupations to first order, and the nuclei as slow. Each of these choices drops the next term of an expansion in some small number. In a silicon detector at 10 mK each small number has a value: the Born–Oppenheimer parameter is 0.067, the anharmonic parameter 0.049, the electron–phonon parameter at the edge $\sim 10^{-2}$, the ratio of temperature to phonon energy $\sim 10^{-5}$. Every rate in this book is built in the same way, and so every rate can be labeled by the orders it keeps. The label says where the next correction comes from and, with the small parameter, roughly how large it is.

8.5.1 Small parameters

Nuclear mass. The Born–Oppenheimer parameter $\kappa_{\mathrm{BO}} = (m_e/M)^{1/4}$ of Sect. 5.1 is 0.0665 for natural silicon (Worked example 5.1). Through the hierarchy Eq. (5.8) it sets the zero-point amplitude in units of the Bohr radius, the phonon energy in units of $\alpha^2 m_e$ (as κ_{BO}^2), the anharmonic level shifts (κ_{BO}^4) and the nonadiabatic ones (κ_{BO}^6). The Born–Oppenheimer approximation drops the non-adiabatic couplings $\mathcal{N}_{ji}$ of Eq. (5.5), whose first-derivative and second-derivative terms are both built from the derivative coupling $\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}$ of Eq. (6.1) (Eq. (6.2)). The effect of the non-adiabatic couplings on a given process depends on how fast the nuclei move. In zero-point motion a nucleus mixes excited electronic states into the ground state with an amplitude $\kappa_{\mathrm{BO}}^3 = 2.9 \times 10^{-4}$, and in an insulator the ratio that controls the mixing is $\omega_{\mathrm{ph}}/\Delta E$, which is $\omega_{\mathrm{opt}}/E_{\mathrm{g}} = 0.055 \approx 12\,\kappa_{\mathrm{BO}}^2$ in silicon (Sects. 5.1 and 6.1). A nucleus kicked by dark matter moves far faster, and the same coupling then produces the sudden amplitude Eq. (6.6), which is the Migdal effect.

Electron–phonon coupling. A second-order amplitude is smaller than a first-order one by the coupling at the added vertex, divided by the off-shell distance of the intermediate state (Sect. 2.4). For the electron–phonon coupling the square of that ratio is the *off-shell parameter*

$$\theta_{\mathrm{ep}} = \frac{\langle |g|^2 \rangle}{\delta_{\mathrm{off}}^2}, \tag{8.40}$$

where δ_{off} is the distance of the intermediate electronic state from the energy shell and $\langle |g|^2 \rangle$ is the mean squared vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$ of Eq. (6.11) over the phonon modes and intermediate states that the amplitude visits. Each pair of vertices added to a rate costs one power of θ_{ep}. In silicon the optical vertices of the holes are about 0.25 eV (Sect. 6.2), and at the absorption edge the intermediate states lie 2 to 2.5 eV off shell (Sect. 6.5). These values give $\theta_{\mathrm{ep}} = 0.010$ to 0.016, the "$\sim 10^{-2}$ per mode" that Sect. 6.5 estimated for the phonon-assisted amplitude. Acoustic vertices near the zone center are ten times smaller (26 meV at $Q = 0.01\,\text{Å}^{-1}$) and their θ_{ep} a hundred times smaller still. The parameter is not small everywhere. When an intermediate state approaches the shell, at the direct gap for light or near the edge $E_{\mathrm{min}}(q)$ for dark-matter scattering, δ_{off} falls toward $|g|$ and θ_{ep} reaches one at $\delta_{\mathrm{off}} \approx 0.25$ eV. Closer still the expansion fails, the widths of Eq. (2.57) enter the energy denominators, and the resonant part becomes a cascade of real transitions (Eq. (2.60)). Away from the shell, where $\theta_{\mathrm{ep}} \ll 1$, lies the off-shell window in which Sect. 8.6 separates the zero-phonon from the phonon-dressed final states. Its edge is not where θ_{ep} reaches one: at the margin of 0.3 eV that Sect. 13.6 fixes for silicon, $\theta_{\mathrm{ep}} \approx 0.7$ bounds the correction from the next pair of vertices, whose size there is about 0.2.

Anharmonicity. The anharmonic parameter $\eta_{\mathrm{anh}} = |\Phi^{(3)}|\ell_{\mathrm{zp}}/3\Phi^{(2)}$ of Eq. (5.66) measures the zero-point amplitude against the length over which the force constants change, $\eta_{\mathrm{anh}} \sim \langle u \rangle / a$. By the counting of Sect. 5.6 it is of the size of κ_{BO}, and in silicon it is 0.049. The cubic force constants are of first order in it and the quartic ones of second, so the three-phonon width and the first-order quartic shift are both of relative size $\eta_{\mathrm{anh}}^2 = 2.4 \times 10^{-3}$ (Eq. (5.67)). The silicon Raman line confirms this size: its measured width is $\gamma/\omega_{\mathrm{opt}} = 2.8 \times 10^{-3}$.

Temperature. A detector at 10 mK has $T = 0.862\,\mu\text{eV}$. The softest phonon that assists the silicon edge is the transverse acoustic phonon at X, $\omega_{\mathrm{TA}} = 18.57$ meV. Against this phonon, $T/\omega_{\boldsymbol{Q}\nu} = 4.6 \times 10^{-5}$ and $n_{\boldsymbol{Q}\nu} = \mathrm{e}^{-21550}$; against the gap, $T/E_{\mathrm{g}} = 7.4 \times 10^{-7}$. The ratio $T/\omega_{\boldsymbol{Q}\nu}$ is not small for every mode, since acoustic phonons near the zone center have energies below T. These phonons are, however, a fraction $(T/\omega_{\mathrm{D}})^3 = 4 \times 10^{-15}$ of the modes, with $\omega_{\mathrm{D}} = 55.6$ meV (Worked example 5.3), and the expansion is in the occupations $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$ that weight each mode, not in T itself. The occupations enter every thermal formula of Sect. 2.5, and they matter wherever a measured input was taken warm: the absorption edge

of Worked example 6.5 at 300 K, the gap at room temperature, the phonon widths at 77 K.

Recoil. A nucleus given the momentum $\boldsymbol{q}$ would recoil freely with the energy $E_R = q^2/2M_\kappa$, and the *recoil parameter*

$$x_R = \frac{q^2}{2M_\kappa \langle\omega\rangle} = \frac{E_R}{\langle\omega\rangle}, \tag{8.41}$$

compares that energy with the mean phonon energy $\langle\omega\rangle$ of the atom. The mean is taken over the phonon density of states per atom, normalized to three,

$$\langle\omega^k\rangle = \frac{1}{3}\int_0^\infty \mathrm{d}\omega\, \omega^k\, \frac{g_{\mathrm{ph}}(\omega)}{n_{\mathrm{at}}},$$

and for the density of states of Sect. 5.2 it is $\langle\omega\rangle = 40.9\,\mathrm{meV}$. At $q = 10\,\mathrm{keV}$, $E_R = 1.91\,\mathrm{meV}$ and $x_R = 0.047$; x_R reaches one at 46 keV. Together with the Debye–Waller exponent $2W_\kappa$ of Eq. (5.48), x_R decides whether a momentum transfer creates one phonon, several, or a free recoil.

Suddenness. A nucleus accelerated over a time τ_{acc} excites an electronic state of energy ΔE with the sudden amplitude when $\Delta E\,\tau_{\mathrm{acc}} \ll 1$, and with an amplitude that falls faster than any power of $\Delta E\,\tau_{\mathrm{acc}}$ when that product is large (Sect. 6.1). The *suddenness ratio* $\Delta E\,\tau_{\mathrm{acc}}$ is the small parameter of the sudden limit. A silicon nucleus is kicked in about $4\times10^{-21}\,\mathrm{s}$, while $1/E_{\mathrm{g}} = 0.56\,\mathrm{fs}$, so that $E_{\mathrm{g}}\tau_{\mathrm{acc}} = 7\times10^{-6}$ across the gap. The lattice pulls the nucleus back on the slower time $1/\omega_{\mathrm{ph}}$, and Sect. 6.1 found that the pull-back changes the excitation probability by the factor $[1-(\omega_{\mathrm{ph}}/\Delta E)^2]^{-2}$, 1.006 for silicon: the deceleration enters through the square of the adiabatic ratio $\omega_{\mathrm{ph}}/\Delta E$ of the first paragraph.

Multipoles. The density vertex $\mathrm{e}^{\mathrm{i}\boldsymbol{q}_e\cdot\boldsymbol{r}}$ that excites an electron can be expanded in powers of the *multipole parameter* $\xi_D = q_e r_{\mathrm{orb}}$, the momentum given to the electron times the size r_{orb} of the orbital it leaves or enters. The first nonvanishing term between orthogonal states is the dipole term of Sect. 2.2. Three dark-matter processes give the electrons momenta that span six orders of magnitude. In electron scattering $q_e = q$, and at the typical $q_0 = \alpha m_e = 1.89\,\text{Å}^{-1}$ an orbital of 1 Å gives $\xi_D \approx 1.9$: there is no multipole expansion, and the crystal form factors of Chap. 10 keep the plane wave whole. In the Migdal effect the electrons receive only the boost $m_e v_N$, 100 eV for a silicon nucleus kicked by 1 GeV dark matter at the largest momentum transfer (Sect. 6.1), and $\xi_D \approx 0.05$. An absorbed boson of mass m_ϕ brings $q_e = m_\phi v \approx 10^{-3} m_\phi$, and $\xi_D = 5\times10^{-6}$ at $m_\phi = 10\,\mathrm{eV}$.

Table 8.4 collects the seven parameters. Each is an estimate with a coefficient of order one in front: the counting of Sect. 5.6 found coefficients between a quarter and two in silicon. Each is also small for silicon at the conditions of a dark-matter search, with three exceptions: θ_{ep} near resonances, x_R at large momentum transfer, and ξ_D in electron scattering. Where a parameter is of order one the corresponding expansion is not used, and the label below records that the labeled quantity is computed to all orders in that parameter.

8.5.2 Lattice regimes

The multiphonon expansion Eq. (5.53) has a simple statistical reading. The number of phonons is Poisson distributed with mean $2W_\kappa$, and each phonon carries an energy drawn from the single-phonon distribution $P_1(\omega)$. For this compound Poisson distribution the cumulants of the energy deposited are $2W_\kappa$ times the moments of P_1, $c_k = 2W_\kappa \int \mathrm{d}\omega\, \omega^k P_1(\omega)$. At $T = 0$ and on a cubic site, $P_1(\omega) \propto g_{\mathrm{ph}}(\omega)/\omega$ for $\omega > 0$, and $2W_\kappa = E_R\langle\omega^{-1}\rangle$ from Eq. (5.48). The factors of $\langle\omega^{-1}\rangle$ cancel, and

$$c_k = E_R\,\langle\omega^{k-1}\rangle, \qquad c_1 = E_R, \qquad c_2 = E_R\langle\omega\rangle. \tag{8.42}$$

Table 8.4: Small parameters of the expansion hierarchy, with their values for silicon in a detector at 10 mK. Each controls one component of the truncation label $\mathfrak{T}$ (Eq. (8.44)). "Introduced" is the section where the physics first appears; "Number" is where it becomes a correction to a dark-matter rate. The values are estimates, with coefficients of order one. The inputs are $M = 26.16\,\mathrm{GeV}$ (natural silicon) for κ_{BO}, the optical hole vertex of 0.25 eV and off-shell distances of 2 to 2.5 eV at the absorption edge for θ_{ep}, the bond coordinate of Sect. 5.1 for η_{anh}, the transverse acoustic phonon at X (18.57 meV) for $T/\omega_{\boldsymbol{Q}\nu}$, the density of states of Sect. 5.2 ($\langle\omega\rangle = 40.9\,\mathrm{meV}$) and the natural isotope mixture for x_R, a kick of $4 \times 10^{-21}\,\mathrm{s}$ across the 1.17 eV gap for $\Delta E\,\tau_{\mathrm{acc}}$, and an orbital of 1 Å for ξ_D.

Parameter	Definition	Silicon	Component	Introduced	Number
κ_{BO}	$(m_e/M)^{1/4}$	0.0665; $\omega_{\mathrm{opt}}/E_{\mathrm{g}} = 0.055$, so $(\omega_{\mathrm{opt}}/E_{\mathrm{g}})^2 = 3{\times}10^{-3}$ per step of n_B in a lattice at rest	n_B	Sect. 5.1	Chaps. 12 and 13
θ_{ep}	$\langle\lvert g\rvert^2\rangle/\delta_{\mathrm{off}}^2$, Eq. (8.40)	0.010–0.016 at the edge; 1 at $\delta_{\mathrm{off}} \approx 0.25\,\mathrm{eV}$	n_g	Sect. 6.2	Chap. 13
η_{anh}	$\lvert\Phi^{(3)}\rvert\ell_{\mathrm{zp}}/3\Phi^{(2)}$, Eq. (5.66)	0.049 ($\eta_{\mathrm{anh}}^2 = 2.4 \times 10^{-3}$)	n_A	Sect. 5.6	Sect. 11.5
$T/\omega_{\boldsymbol{Q}\nu}$	temperature over phonon energy	4.6×10^{-5}; $n_{\boldsymbol{Q}\nu} = \mathrm{e}^{-21550}$	n_T	Sect. 2.5	Sect. 13.7
x_R	$q^2/2M_\kappa\langle\omega\rangle$, Eq. (8.41)	0.047 at 10 keV; 1 at 46 keV	$\mathcal{K}$	Sect. 5.5	Sects. 11.3 and 11.4
$\Delta E\,\tau_{\mathrm{acc}}$	excitation energy times kick duration	7×10^{-6}	$\mathfrak{M}$	Sect. 6.1	Chap. 12
ξ_D	$q_e r_{\mathrm{orb}}$	1.9 (electron scattering at q_0); 0.05 (Migdal); 5×10^{-6} (absorption at 10 eV)	n_D	Sect. 2.2	Sects. 10.6 and 12.2

The mean deposit is the free recoil energy, the sum rule Eq. (5.55). The variance is $E_R\langle\omega\rangle$. The width of the spectrum relative to its mean is

$$\frac{c_2^{1/2}}{c_1} = \Big(\frac{\langle\omega\rangle}{E_R}\Big)^{1/2} = x_R^{-1/2}. \tag{8.43}$$

Eq. (8.43) is the reason for defining x_R with the arithmetic mean $\langle\omega\rangle$, while the Debye–Waller exponent carries the inverse mean: $2W_\kappa = x_R\,\langle\omega\rangle\langle\omega^{-1}\rangle \geq x_R$, with $\langle\omega\rangle\langle\omega^{-1}\rangle = 1.35$ in silicon ($\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, Worked example 5.6). The exponent $2W_\kappa$ counts phonons; the recoil parameter says whether the deposit is concentrated near E_R.

Together, the exponent and the recoil parameter fix the lattice regime. For $2W_\kappa \ll 1$, and therefore $x_R \ll 1$, most scatterings leave the lattice unchanged and a fraction $2W_\kappa$ creates one phonon, which carries an energy of the order of $\langle\omega\rangle$, far above E_R. In this *one-phonon* regime the coherent response of the crystal (Sect. 11.3) decides the rate. In silicon the one-phonon regime holds below about 10 keV, where $2W_\kappa \leq 0.063$ and $x_R \leq 0.047$. For $2W_\kappa$ of order one, several terms of Eq. (5.53) compete and the deposit comes in a few whole phonons: the *multiphonon* regime, with $2W_\kappa = 1$ at 40 keV and $x_R = 1$ at 46 keV. For $x_R \gg 1$ the spectrum narrows around E_R, and by the central limit theorem it approaches a Gaussian of width $(E_R\langle\omega\rangle)^{1/2}$: the free recoil, broadened by the motion of the nucleus in its well. In this *impulse* regime the spectrum is the elastic recoil of Sect. 1.3, with the width as its leading correction [19, Sect. IV A, Eqs. (90)–(92)]. The width falls below $E_R/3$ for $x_R \geq 9$, at $q \geq 139\,\mathrm{keV}$ or $E_R \geq 0.37\,\mathrm{eV}$ in silicon. The Einstein oscillator of Sect. 1.3, at 60 meV, puts the same boundary at 0.54 eV; the computed density of states has a lower mean phonon energy and moves the boundary down by a third.

Temperature acts differently on the exponent $2W_\kappa$ and on the recoil parameter. Thermal phonons add the factor $\coth(\omega/2T)$ to $2W_\kappa$ (Eq. (5.48)), raising the number of phonons absorbed and emitted, while the mean deposit stays at E_R at any temperature. At 10 mK the factor is one for every mode that matters, and the regimes are the same as at $T = 0$. The boundaries between the regimes are gradual, and a formula is placed in a regime by what it assumes, not by the momentum at which it is evaluated.

8.5.3 The truncation label

Each small parameter has an order to which a formula keeps it. Together with three labels that are not orders, the orders form the *truncation label*

$$\mathfrak{T} = (n_B,\, n_g,\, n_A,\, \mathcal{X},\, n_T,\, \mathcal{K},\, \mathfrak{M},\, n_D). \tag{8.44}$$

The five integers are the Born–Oppenheimer, electron–phonon, anharmonic, thermal and multipole orders. One rule fixes them all: *an order is the power of its expansion parameter carried by the quantity being labeled*, counted in the response or the rate, not in the number of vertices. The rule generalizes the counting of Sect. 5.6, where a width with two cubic vertices and a shift with one quartic vertex are of one order. The three remaining components are labels: the many-body approximation $\mathcal{X}$, the lattice regime $\mathcal{K}$ and the Migdal regime $\mathfrak{M}$. The Born–Oppenheimer order n_B always appears with its value, as in $n_B = 0$; the value distinguishes it from the Bose function $n_\mathrm{B}(\omega)$.

Born–Oppenheimer order n_B, the power of the derivative coupling $\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}$ in the quantity. At $n_B = 0$ the electrons are in the adiabatic state of the instantaneous nuclear configuration. The order $n_B = 0$ covers the clamped-nucleus band structures of Chap. 3, the response functions of Chap. 4, and the phonons on the Born–Oppenheimer surface of Chap. 5, all of which are evaluated without any derivative coupling. The sudden amplitude Eq. (6.6) is linear in $\boldsymbol{d}$, so

the probability $1 - P_0$ that a kick excites the electrons is at $n_B = 2$, and so is the Migdal rate. The odd order $n_B = 1$ appears only in interference, between an amplitude that contains the coupling and one that does not; Sect. 8.6 meets such a term. The cost of a step depends on how fast the nuclei move, because each power of $\boldsymbol{d}$ enters with the nuclear velocity, as $\boldsymbol{v}_N \cdot \boldsymbol{d}/\Delta E$ (Eq. (6.6)). For a lattice in its vibrational ground state the nuclear velocity is the velocity of zero-point motion. The step $n_B \to n_B + 2$ then multiplies an adiabatic quantity by $1 + O\big((\omega_{\mathrm{ph}}/\Delta E)^2\big)$, and the correction is 3×10^{-3} for the optical phonon across the gap of silicon. The amplitude κ_{BO}^3 above is the same ratio in the atomic counting, with $\Delta E \sim \alpha^2 m_e$. For a nucleus kicked by dark matter the nuclear velocity is $\boldsymbol{q}/M_\kappa$, and the step from $n_B = 0$ to $n_B = 2$ is the sudden probability itself. The electrons that follow a nucleus carry a share $\kappa_{\mathrm{BO}}^4 = m_e/M$ of an f-sum (Sect. 8.6); this share is fixed by the masses alone and is not a step of this kind.

Electron–phonon order n_g, the power of the vertex g. Direct absorption, Eq. (6.40), is at $n_g = 0$; phonon-assisted absorption, Eq. (6.43), has one vertex in the amplitude and is at $n_g = 2$, as are the one-phonon emission rates and the Fan–Migdal widths of Sect. 6.3. The vertex itself is of first order in the displacement and is at $n_g = 1$. Raising n_g by two costs one power of θ_{ep}.

Anharmonic order n_A, the power of η_{anh}. The harmonic crystal of Sects. 5.2, 5.3 and 5.5 is at $n_A = 0$, and so is the multiphonon formula. The three-phonon width Eq. (5.64) and the first-order quartic shift are both at $n_A = 2$, and so is the leading anharmonic phonon self-energy; at $n_A = 1$ the energies do not change, since the average of the cubic term over a harmonic state vanishes. First-principles electron–phonon theory starts from the point $n_B = 0$, $n_A = 0$: its lattice Hamiltonian rests only on the harmonic and the Born–Oppenheimer approximations [20, Sect. III A, text after Eq. (19)].

Many-body approximation $\mathcal{X}$, one of five: the independent-particle bubble Eq. (4.63), IP (the Lindhard function of Sect. 4.5 in the electron gas); the random-phase approximation (RPA) Eq. (4.67) with the head alone, or with local fields (LF) Eq. (4.68), written RPA + LF; an exchange–correlation kernel of time-dependent density-functional theory (TDDFT) Eq. (4.70); and the Bethe–Salpeter equation (BSE) Eq. (4.104). $\mathcal{X}$ is not an order. Density-functional perturbation theory supplies the phonons, Born charges and ε_∞ as static self-consistent responses with the Hartree and exchange–correlation kernel of the functional, and these inputs carry $\mathcal{X}$ = TDDFT at $\omega = 0$, whatever the approximation of the electronic response they are combined with.

Thermal order n_T, the power of the phonon occupations $n_{\boldsymbol{Q}\nu}$. At $n_T = 0$ there is no thermal occupation, though zero-point motion is kept wherever phonons enter; at $n_T = 1$ a rate is linear in the occupations, as Eq. (6.43) is through $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$. The thermal occupation of electrons across a gap, $\mathrm{e}^{-E_{\mathrm{g}}/T}$, is smaller than any power and is dropped at every order.

Lattice regime $\mathcal{K}$, one of one-phonon, multiphonon and impulse: the phonon content of the final state. For a coupling to the nuclei it is read from x_R and $2W_\kappa$ as above. The single term $n = 1$ of Eq. (5.53) is one-phonon; the full sum is multiphonon; its Gaussian limit around E_R is impulse. A process is one-phonon as well when it creates its phonon through one electron–phonon vertex in the amplitude, as phonon-assisted absorption does. Without phonons in the final state the label does not apply.

Migdal regime $\mathfrak{M}$, sudden or adiabatic. The boost Eq. (6.8) and the hydrogen survival probability Eq. (6.9) are sudden; the other value is the adiabatic limit of Sect. 6.1, in which the electrons follow the nucleus and end in their ground state. The label applies to a formula that treats the struck nucleus as a classical trajectory. Where the phonon final states are kept, at small x_R, the same physics is carried by n_B and $\mathcal{K}$ instead (Sect. 12.1).

Multipole order n_D, the power of ξ_D. At small momentum transfer the dipole transition $1s \to 2p$ of hydrogen has $|f|^2 \simeq 0.555\, q^2 a_0^2$ (Sect. 2.2) and is at $n_D = 2$, the lowest order at

which a transition between orthogonal states occurs. The full plane wave, kept whole, has no multipole order.

The *base point* of the label is

$$\mathfrak{T}_0 = (n_B = 0,\, n_g = 0,\, n_A = 0,\, \mathcal{X} = \mathrm{IP},\, n_T = 0;\ \mathcal{K},\, \mathfrak{M},\, n_D \text{ not applicable}) : \tag{8.45}$$

clamped nuclei, no electron–phonon vertex, independent particles, zero temperature, and no expansion of the plane wave, with no phonon in the final state and no nuclear recoil to classify. The unscreened crystal form factor of Chap. 10 is at $\mathfrak{T}_0$. When a formula takes its phonons or Born charges from density-functional perturbation theory, it keeps $\mathcal{X}$ for its electronic response and names those inputs as static TDDFT. A formula elsewhere on the label is written by listing only the components that differ from $\mathfrak{T}_0$. The direct dielectric function Eq. (6.40) is $\mathfrak{T}_0$ in this notation, a three-phonon linewidth is "$n_A = 2$", and the screened loss function with local fields is "$\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$". Qualifiers that are not orders, such as "unscreened" or "with spin–orbit coupling", are added in words. Sect. 9.4 places the literature formulas on the label.

Several approximations are deliberately left off the label, because none of them is an expansion in a small number with a next order that could be computed. Of this kind are the exchange–correlation functional, the frozen core of a pseudopotential, the *GW* quasiparticle energies, and the numerical parameters of a calculation (grids, cutoffs, Wannier interpolation); their errors are measured by tests and convergence. Methods that sum an expansion to all orders lie off the finite values of an axis and are described in words. The self-consistent harmonic approximation of Sect. 5.6 resums the anharmonic expansion, molecular dynamics on the full Born–Oppenheimer surface is exact in n_A at $n_B = 0$, and a classical cascade calculation is off the perturbative axes altogether.

8.5.4 Many-body approximations as resummations

The approximations of Sects. 4.6 and 4.9 are ordered by what they include, and each is a limit of the next, but they are not successive orders in a small number. Each is a resummation. The random-phase approximation already sums the Coulomb interaction to all orders through the Dyson equation, the kernel of time-dependent density-functional theory adds a correction inside that sum, and the Bethe–Salpeter equation adds a ladder of electron–hole attractions. An expansion would need the kernel to be small against the Coulomb interaction it corrects, and the kernel is not small. In silicon the local fields lower ε_∞ by 11 %, from 11.45 to 10.22; the adiabatic kernel restores about half of this reduction, to 10.81; and adding the attraction to the local fields raises the local-field value by 17 %, to ≈ 11.9 (Worked example 4.11 and Sect. 4.9). Each correction moves ε_∞ by about as much as the last, and not always in the same direction, so there is nothing to expand in. At the lower absorption peak the approximations disagree by a factor of more than two, while above 4 eV they agree within the accuracy of the bands.

The error of a many-body approximation is therefore estimated by computing the next one, not by a power. The difference between two computed approximations is the many-body uncertainty of a rate: for the screening factor $1/\varepsilon_\infty^2$ of silicon it is up to a third across the hierarchy, and for light dark matter near the gap it is the leading uncertainty. Sect. 8.8 turns these computed differences, and the next-order terms of the other components, into the theory uncertainty of a rate.

Worked example 8.7: The label of phonon-assisted absorption

We read the label of the phonon-assisted dielectric function Eq. (6.43) off its assumptions, component by component, and estimate the size of each first omitted term in silicon at the indirect edge.

The phonon frequencies and vertices come from density-functional perturbation theory on the Born–Oppenheimer surface, with the electrons responding statically, so the formula is at $n_B = 0$, and these inputs are at $\mathcal{X} =$ TDDFT in its static limit. The static screening of the vertex misses corrections of relative order $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2$ (Sect. 6.2), 3×10^{-3} for the optical phonon. The phonons are harmonic, $n_A = 0$; their anharmonic widths, $\gamma/\omega = 2.8 \times 10^{-3}$ for the optical phonon, would broaden each threshold by a fraction of a meV.
The amplitude $M^{\pm}_{cv\nu}$ contains one optical vertex and one electron–phonon vertex, so the rate is quadratic in g: $n_g = 2$. The first omitted order, $n_g = 4$, adds processes with two phonons and the dressing of the one-phonon process by a second. It is smaller by $\theta_{\mathrm{ep}} \approx 10^{-2}$ below E_{dir}, where every intermediate state lies electronvolts off shell, and it is not small near E_{dir}, where the intermediate electron becomes real and Sect. 6.5 replaced the expansion by the width Γ_{w}.
The occupation factor $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$ is linear in $n_{\boldsymbol{Q}\nu}$, so the formula is at $n_T = 1$. Linearity is exact at $n_g = 2$ for harmonic phonons: a term quadratic in the occupations needs two phonons and therefore $n_g = 4$. At 10 mK the occupations are e^{-21550} and smaller, and only the emission branch survives; the label still records $n_T = 1$, because the formula holds at the 300 K and 77 K of the measured edge in Worked example 6.5. The electrons are thermally unoccupied across the gap, $T \ll E_{\mathrm{g}}$, as at every order.
The bands are independent-particle bands with a rigid shift, and the electron and hole do not attract each other: $\mathcal{X} =$ IP. Each event creates or absorbs exactly one phonon, through the vertex g, so $\mathcal{K} =$ one-phonon; the photon vertex acts on the electrons and no nucleus recoils, so $\mathfrak{M}$ does not apply. The dielectric function ε_2 is defined in the limit $\boldsymbol{q} \to 0$, where no multipole expansion is made, so n_D does not apply either. In full,

$$\mathfrak{T}[\varepsilon_2^{\mathrm{PA}}] = (n_B = 0,\, n_g = 2,\, n_A = 0,\, \mathcal{X} = \mathrm{IP},\, n_T = 1,\, \mathcal{K} = \text{one-phonon};\ \mathfrak{M},\, n_D \text{ not applicable}),$$

written as "$n_g = 2$, $n_T = 1$, $\mathcal{K} =$ one-phonon" against the base point, with $\mathcal{X}$ at its base value IP. A calculation for a detector at millikelvin temperatures drops the absorption branch and so puts the same formula at $n_T = 0$; the phonon-assisted loss function of Sect. 31.1 is labeled that way.
The label also ranks the corrections. The next electron–phonon order costs 10^{-2}, the static screening of the vertex a few 10^{-3}, and the anharmonic widths less. The excitonic correction, however, is not a percent effect at the edge. The electron–hole attraction moves every threshold down by the exciton binding energy, 14.7 meV (Sect. 4.9), comparable to the energy of the transverse acoustic phonon that assists the transition, and the measured kinks of Worked example 6.5 are exciton thresholds. Near the edge, then, the largest error of Eq. (6.43) lies on the axis that has no small parameter, $\mathcal{X} = \mathrm{IP} \to$ BSE, and it is estimated by solving the Bethe–Salpeter equation.

8.6 The decomposition theorem

When dark matter kicks a silicon nucleus, the electrons follow most of the time, and the recoil energy ends up in the lattice as phonons. In a small fraction of events (Sect. 6.1) the electrons are left behind, and the same kick leaves an electron–hole pair. At 50 keV the nucleus does not recoil freely: it leaves the lattice untouched in 21 % of the events and creates two or more phonons in 47 % (Worked example 5.6). The electron density offers the same variety. Through a coupling to the electrons, a slow particle excites a pair with no phonon (Chap. 4), a pair

with a phonon below the direct gap (Sect. 6.5), or no pair, since the electrons that follow a vibrating nucleus scatter as well (Sect. 5.4). The literature computes each of these final states with its own formula and its own code. The sum of these separate rates is the total rate only if no final state is counted twice and none is left out. Worked example 2.6 and Sect. 6.5 have already met a double count: above the direct gap, the phonon-assisted absorption of light contains a cascade that direct absorption counts once more.

To make the sum count each final state once we split the final states of every response into three sectors. The derivations are in Appendix D.22.

8.6.1 Three sectors of final states

The Born–Oppenheimer approximation of Sect. 5.1 already suggests the first cut. Every state of the form $\Psi^{\mathrm{nuc}}(u)\,\Psi_0(r;u)$ lies in the range of the projector

$$\mathcal{P}_{\mathrm{BO}} = \int \mathrm{d}u\; |\Psi_0(u)\rangle\langle\Psi_0(u)| \otimes |u\rangle\langle u|, \tag{8.46}$$

where $|\Psi_0(u)\rangle$ acts on the electrons and $|u\rangle$ on the nuclei. In a state of this form the electrons are in the adiabatic ground state at each configuration u of the nuclei, and $\Psi^{\mathrm{nuc}}(u)$ is any nuclear wavefunction. Phonons, multiphonon states and a freely recoiling nucleus are all in this subspace; electron–hole pairs are not. The subspace is not exactly invariant under the Hamiltonian, however. The nuclear kinetic energy differentiates $\Psi_0(r;u)$ with respect to u, and the nonadiabatic couplings Eq. (5.5) carry part of the state into the excited adiabatic states. For zero-point motion the leak out of the subspace is the derivative coupling $\alpha^2 m_e\,\kappa_{\mathrm{BO}}^3$ of the Born–Oppenheimer hierarchy (Sect. 6.1).

A final state must have a definite sector, and a projector that leaks does not give it one. Panati, Spohn and Teufel showed how to repair the leak order by order [21, Thm. 2.1]. Their small parameter is $(m_e/M)^{1/2} = \kappa_{\mathrm{BO}}^2$, the ratio of a nuclear to an electronic velocity at equal kinetic energy. Built as a power series in κ_{BO}^2, a unitary U_n rotates $\mathcal{P}_{\mathrm{BO}}$ into a *superadiabatic* projector

$$\mathcal{P}_{\mathrm{ad}} = U_n\,\mathcal{P}_{\mathrm{BO}}\,U_n^{\dagger}, \qquad \big\|(1-\mathcal{P}_{\mathrm{ad}})\,H\,\mathcal{P}_{\mathrm{ad}}\big\| = O\big(\kappa_{\mathrm{BO}}^{2(n+1)}\big)\,\alpha^2 m_e, \tag{8.47}$$

on states below any fixed energy [21, Sect. 4.5]. The order $n = 0$ is Eq. (8.46) itself. Each further order removes one more power of κ_{BO}^2. The first correction tilts the electronic state in the direction of the nuclear velocity, $\Psi_0 \to \Psi_0 + \sum_{j\neq 0} \Psi_j\, \mathrm{i}\,\boldsymbol{d}_{j0}\cdot\boldsymbol{p}/[M_\kappa(E_j - E_0)]$, with $\boldsymbol{d}_{j0}$ the derivative-coupling vector Eq. (6.1) and $\boldsymbol{p}$ the momentum of the nucleus: the electron cloud of a moving nucleus lags behind it and is slightly polarized. Appendix D.22 gives the construction and its hypotheses.

The range of $\mathcal{P}_{\mathrm{ad}}$ is the *electron-following sector*. Its complement $1-\mathcal{P}_{\mathrm{ad}}$ contains every state with an electronic excitation, and it splits once more by what happens to the lattice. Measured from the initial state, the number of phonons can stay the same or change. We write

$$1 = \mathcal{P}_{\mathrm{ad}} + \mathcal{P}_0 + \mathcal{P}_+, \tag{8.48}$$

with $\mathcal{P}_0$ the *zero-phonon electronic sector*, an electronic excitation with no net change of the phonon number, and $\mathcal{P}_+$ the *phonon-dressed electronic sector*, an electronic excitation with a nonzero net change of the phonon number. At $T = 0$ phonons can only be emitted, and the change is at least one phonon created; a warm target can also give up phonons, as at the absorption thresholds of Sect. 6.5. The phonon number here is the phonon number of the harmonic reference lattice of the excited electronic state. The excited surfaces are displaced, and a zero-phonon state of an excited surface is a coherent superposition of phonons of the ground surface, as the displaced oscillator of Sect. 5.5 showed. At $T = 0$, $\mathcal{P}_0$ holds

the zero-phonon line of every electronic transition and $\mathcal{P}_+$ its sideband. The three projectors are orthogonal and sum to one, so every final state has one label. The lattice regime $\mathcal{K}$ of Sect. 8.5 labels the phonon content within a sector: one phonon, several or a free recoil inside $\mathcal{P}_{\mathrm{ad}}$, and the phonons that dress the excitation inside $\mathcal{P}_+$. The sector and the lattice regime are nested labels.

The response of each sector follows from Eq. (4.13) by inserting the projector on both sides of the energy delta function,

$$S^A_{ab}(\boldsymbol{q},\omega) = \frac{2\pi}{\mathcal{V}} \sum_i p_i \, \langle i|O_a(\boldsymbol{q})^\dagger \, \mathcal{P}_A \, \delta(\omega + E_i - H) \, \mathcal{P}_A \, O_b(\boldsymbol{q})|i\rangle, \qquad A \in \{\mathrm{ad}, 0, +\}, \quad (8.49)$$

with the initial states in the electron-following sector, as they are at $T \ll E_{\mathrm{g}}$. If $\mathcal{P}_A$ commutes with H, the delta function can be written in eigenstates that each lie in one sector, S^A_{ab} is the sum over the final states of sector A, and $\sum_A S^A_{ab} = S_{ab}$ exactly. Each S^A_{ab} is then a positive semidefinite matrix by the argument of Sect. 4.2, because it is again a sum of outer products with positive weights. If $\mathcal{P}_A$ commutes with H approximately, the cross terms $\mathcal{P}_A \, \delta(\omega + E_i - H) \, \mathcal{P}_B$ with $A \neq B$ do not vanish. They are small on energy bins wider than the leakage norm, and not on narrower ones. For a bin of width $\Delta\omega_{\mathrm{bin}}$ they are bounded by the leakage norm of Eq. (8.47) divided by $\Delta\omega^2_{\mathrm{bin}}$, times the norms of $O_a|i\rangle$ and $O_b|i\rangle$ (Eq. (D.84)).

Decomposition theorem

For a target with a gap, with the nuclear charges smeared over the finite size of the nuclei, at temperatures $T \ll E_{\mathrm{g}}$, and with the electronic ground state nondegenerate at every nuclear configuration that the vibrational state visits,

$$S_{ab}(\boldsymbol{q},\omega) = S^{\mathrm{ad}}_{ab}(\boldsymbol{q},\omega) + S^0_{ab}(\boldsymbol{q},\omega) + S^+_{ab}(\boldsymbol{q},\omega) + \delta S_{ab}, \qquad \frac{|\delta S_{ab}|}{S^{\mathrm{bin}}} = O\Big(\kappa_{\mathrm{BO}}^{2(n+1)} \, \frac{\alpha^2 m_e}{\Delta\omega_{\mathrm{bin}}}\Big), \quad (8.50)$$

for every pair of target operators, with S^{bin} the response averaged over an energy bin of width $\Delta\omega_{\mathrm{bin}}$ and n the order of the superadiabatic projector. The electron-following sector and the sum of the two electronic sectors are positive semidefinite matrices. The split between S^0 and S^+ is unique to relative order θ_{ep} inside the off-shell window, where each is positive; outside it the split is fixed by the Dyson convention of Eq. (8.59), and near a resonance only the sum of the two is positive.

For bins of 1 meV, the scale of phonon spectra, the factor $\alpha^2 m_e/\Delta\omega_{\mathrm{bin}}$ is 2.7×10^4, and the bound becomes useful from $n = 2$ on, where it is 2×10^{-3}. The finite size of the nuclei keeps the electronic Hamiltonian a smooth function of the nuclear positions. The adiabatic construction requires this smoothness, and point charges violate it at coincidence [22, Sect. 2]; the nuclear charge radius changes nothing else in this book. The gap excludes metals and level crossings, and the temperature condition keeps the initial state in the electron-following sector. The theorem is proved for a finite crystal. It is open whether the theorem extends to the infinite crystal with constants that do not grow with the number of nuclei; this problem is stated with the other open problems at the end of this section. The theorem does not depend on which operators O_a appear: the final state alone decides the sector.

8.6.2 Densities in each sector

Inside the electron-following sector an electronic operator acts on the nuclei alone. The electrons are in $\Psi_0(r;u)$, and the electron density reaches the sector only through its expectation value

at each configuration,

$$\mathcal{P}_{\rm ad}\, n_e(\boldsymbol{q})\, \mathcal{P}_{\rm ad} \simeq \bar{\rho}_e(\boldsymbol{q};u) \equiv \langle \Psi_0(u)|n_e(\boldsymbol{q})|\Psi_0(u)\rangle, \tag{8.51}$$

up to corrections of relative order $\kappa_{\rm BO}^2$ from the tilt of the state. The function $\bar{\rho}_e(\boldsymbol{q};u)$ is the Fourier component of the ground-state electron density of the crystal frozen in configuration u, an operator on nuclear wavefunctions like the nuclear density $n_\kappa(\boldsymbol{q})$ itself. At $u=0$ it is the static density, nonzero only at reciprocal-lattice vectors: Bragg scattering. Its term linear in u creates one phonon, and its higher terms several. Through the electrons that the vibrating nuclei carry along, a coupling to the electrons alone therefore excites phonons. If each nucleus dragged a rigid cloud with form factor $f_\kappa(\boldsymbol{q})$, we would have $\bar{\rho}_e = \sum_I f_{\kappa(I)}(\boldsymbol{q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, and the electron channel would see the lattice with the weight $|f_\kappa|^2$ of the nuclear channel. In a real crystal the cloud deforms. At long wavelength and zero macroscopic field, the electrons that follow nucleus κ carry the charge $Z_\kappa \mathbf{1} - Z_\kappa^*$, a tensor, so that nucleus and electrons together carry the Born effective charge of Eq. (5.40). In silicon $Z_\kappa^* = 0$ and all fourteen electrons follow; in GaAs a fraction is left behind, and the lattice is seen by a dark photon through Z_κ^*, screened by ε_∞ (Sect. 8.4).

The electronic sectors are reached through the transition densities between adiabatic states. Take a final state $\Psi^{\rm nuc\prime}\Psi_j$ on the excited surface j, with $\Psi^{\rm nuc\prime}$ one of its vibrational states. The amplitude to reach this final state from $\Psi^{\rm nuc}\Psi_0$ is

$$\begin{aligned} \langle \Psi^{\rm nuc\prime}\Psi_j|n_e(\boldsymbol{q})|\Psi^{\rm nuc}\Psi_0\rangle &= \langle \Psi^{\rm nuc\prime}|\rho_{j0}(\boldsymbol{q};u)|\Psi^{\rm nuc}\rangle, \\ \rho_{j0}(\boldsymbol{q};u) &= \rho_{j0}(\boldsymbol{q};0) + \sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa}\cdot\nabla_{\boldsymbol{R}\kappa}\rho_{j0} + \cdots , \end{aligned} \tag{8.52}$$

where $\rho_{j0}(\boldsymbol{q};u) = \langle \Psi_j(u)|n_e(\boldsymbol{q})|\Psi_0(u)\rangle$ is the transition density of the frozen crystal. The constant term is the Condon approximation of Sect. 5.5. It weights the vibrational states by the Franck–Condon overlaps $|\langle \Psi^{\rm nuc\prime}|\Psi^{\rm nuc}\rangle|^2$: the zero-phonon line, in $\mathcal{P}_0$, with the weight $\mathrm{e}^{-S_{\rm HR}}$ of Eq. (5.57), and the replicas, in $\mathcal{P}_+$. For an electron–hole pair in a crystal the surfaces of the ground and excited states are displaced by almost nothing, since one pair spread over many cells exerts a negligible force on each nucleus, and the zero-phonon line carries nearly all of the Condon term. The linear term creates a phonon even without a displacement; in molecular spectroscopy it is the Herzberg–Teller term. In a crystal it does more: at $u \neq 0$ the frozen lattice is not periodic, and ρ_{j0} connects pairs of different crystal momentum that the static lattice keeps apart. The linear term is phonon-assisted electron excitation in adiabatic form, the term that the special-displacement supercell of Sect. 6.5 computes. Being adiabatic, the linear term misses the phonon energy in the final state, a correction of relative order $\omega_{\boldsymbol{Q}\nu}/\delta_{\rm off}$; the electron–phonon form Eq. (6.43) restores this energy.

The nuclear density cannot reach the electronic sectors in the Born–Oppenheimer approximation. Its amplitude between the same two states is $\int \mathrm{d}u\, \Psi^{\rm nuc\prime *}(u)\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\, \Psi^{\rm nuc}(u)\, \langle \Psi_j(u)|\Psi_0(u)\rangle$, and the electronic overlap vanishes at every u because the adiabatic states are orthogonal. The amplitude comes from the tilt of the initial and final states by the superadiabatic correction. At its first order (Appendix D.22),

$$\langle \widetilde{\Psi^{\rm nuc\prime}\Psi_j}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}|\widetilde{\Psi^{\rm nuc}\Psi_0}\rangle = -\mathrm{i}\,\frac{\boldsymbol{q}\cdot\boldsymbol{d}_{j0}^I}{M_\kappa(E_j-E_0)}\,\langle \Psi^{\rm nuc\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}|\Psi^{\rm nuc}\rangle, \tag{8.53}$$

with $\boldsymbol{d}_{j0}^I$ the derivative coupling with respect to the position of nucleus I of species κ, and the tildes marking the tilted states. The initial and the final tilt each give a term proportional to the gradient of the nuclear wavefunctions; the two terms cancel, and only the momentum $\boldsymbol{q}$ of the kick survives. The electronic factor is the sudden amplitude $\boldsymbol{v}_N\cdot\boldsymbol{d}_{j0}/(\mathrm{i}\Delta E_j)$ of Eq. (6.6),

with $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$: the electrons see a nucleus that suddenly moves. The nuclear factor is the kick of Sect. 5.5, with its Debye–Waller and multiphonon structure. The Migdal effect is purely nonadiabatic. Its zero-phonon term lies in $\mathcal{P}_0$, and $\mathcal{P}_+$ holds the terms in which the kick also creates phonons, or a free recoil. In the truncation label of Sect. 8.5, which counts powers in the rate, a Migdal rate carries the derivative coupling squared and is at $n_B = 2$. Where a Migdal amplitude and an electron recoil reach the same final state, their interference is linear in the derivative coupling, at $n_B = 1$.

The cross response $S_{e\kappa}$ is nonzero in every sector whenever the dark matter couples to electrons and nuclei at once, as a dark photon does, and the next subsection shows that it cannot be dropped as small without violating a sum rule.

In practice each sector comes from a different chain of codes, and the three calculations must use one set of phonon energies, vertices and band energies, or the sum of the three sector responses is the response of no single Hamiltonian.

8.6.3 Partial sum rules

The first-moment identity Eq. (4.28) holds sector by sector, because a projector that commutes with H can be carried through the double commutator. Restricted to the electron-following sector, both the electron and the nuclear density are functions of the nuclear positions, and of the terms of the Hamiltonian only the kinetic energy of the nuclei fails to commute with them, exactly as for n_κ in Sect. 4.3. For the electron density, Eq. (8.51) turns the double commutator into a sum over nuclei of squared gradients (Appendix D.22),

$$\mathscr{M}^1_{ee}\big|_{\mathrm{ad}}(\boldsymbol{q}) = \frac{1}{\mathscr{V}}\Big\langle \sum_{\boldsymbol{R}\kappa} \frac{|\nabla_{\boldsymbol{R}\kappa}\bar{\rho}_e(\boldsymbol{q};u)|^2}{M_\kappa}\Big\rangle [1 + O(\kappa_{\mathrm{BO}}^2)] \;\simeq\; q^2 \sum_\kappa \frac{n_\kappa\, |f_\kappa(\boldsymbol{q})|^2}{M_\kappa}, \tag{8.54}$$

where the second form holds for rigid clouds. Measured against the f-sum $n_e q^2/m_e$, the electron-following first moment is of relative size $Z_\kappa m_e/M_\kappa$ at small q, where the whole cloud follows. The relative size is 2.73×10^{-4} for silicon with all fourteen electrons, the number that Sect. 4.3 found for the nuclear share of the f-sum of the total charge, and for the same reason: either way, Z_κ units of charge are attached to a mass M_κ. In powers of the small parameter it is $\kappa_{\mathrm{BO}}^4 = m_e/M$. The two electronic sectors together therefore carry the f-sum up to a part in 10^3, and the f-sum test of a clamped-nuclei calculation (Sect. 4.3) applies to them unchanged.

For the nuclear density the electronic sectors carry the Migdal events, and their share follows from Eq. (8.53). Summed over the vibrational states $\Psi^{\mathrm{nuc}\prime}$, which form a complete set, the probability that a kick of nucleus I leaves the electrons excited is

$$P_{\mathrm{Mig}}(\boldsymbol{q}) = \sum_{j\neq 0} \frac{|\boldsymbol{q}\cdot\boldsymbol{d}^I_{j0}|^2}{M_\kappa^2 (E_j - E_0)^2}, \tag{8.55}$$

at leading order, the sudden probability of Sect. 6.1 with $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$. Each Migdal event deposits the electronic energy $E_j - E_0$ plus, on average, the recoil energy $E_R = q^2/2M_\kappa$ of Eq. (5.55). With $\boldsymbol{d}_{j0} = \boldsymbol{F}_{j0}/(E_0 - E_j)$ from Eq. (5.11), the electronic part of the deposit adds up to $\boldsymbol{q}\cdot\delta M_\kappa\cdot\boldsymbol{q}/2M_\kappa^2$, and the partial sum rules per nucleus read (Appendix D.22)

$$\begin{aligned} \mathscr{M}^1_{\kappa\kappa}\big|_{0+} &= n_\kappa\Big[P_{\mathrm{Mig}}\,\frac{q^2}{M_\kappa} + \frac{\boldsymbol{q}\cdot\delta M_\kappa\cdot\boldsymbol{q}}{M_\kappa^2}\Big], \qquad \mathscr{M}^1_{\kappa\kappa}\big|_{\mathrm{ad}} = n_\kappa\,(1-P_{\mathrm{Mig}})\;\boldsymbol{q}\cdot(M_\kappa+\delta M_\kappa)^{-1}\cdot\boldsymbol{q}, \\ \delta M_\kappa &= 2\sum_{j\neq 0}\frac{\mathrm{Re}[\boldsymbol{F}_{0j}\,\boldsymbol{F}_{j0}]}{(E_j-E_0)^3}, \qquad \boldsymbol{F}_{j0} = \langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}V_{eN}|\Psi_0\rangle, \end{aligned} \tag{8.56}$$

to first order in P_{Mig} and $\delta M_\kappa/M_\kappa$, with 0+ marking the two electronic sectors together. The tensor δM_κ is the *electron drag mass*: a slowly moving nucleus drags part of its electron cloud along and vibrates as if it were heavier. The solvable model of Appendix D.11 found this effect at order m_e/M. The same term appears in the second-order Born–Oppenheimer Hamiltonian of Panati, Spohn and Teufel [21, Eq. (40), last term]. For an isolated atom the sum is fixed by the Thomas–Reiche–Kuhn rule Eq. (4.37), and $\delta M = N_e m_e$: the atom recoils with its electrons, as it must (Appendix D.22). In a crystal the f-sum rule fixes the sum of the drag masses over all nuclei, exactly $14\,m_e$ per silicon atom (Eq. (12.39)), and this sum gives $\delta M_\kappa/M_\kappa \approx 2.7\times10^{-4}$. A kick on one nucleus probes instead the on-site drag mass, which no sum rule fixes: Sect. 12.5.2 evaluates its valence part as $18.6\,m_e$, between 11.8 and $24.0\,m_e$ in the approximations behind that evaluation.

Eq. (8.56) refines Eq. (5.55). That rule says that the mean energy given to the lattice is E_R on any potential-energy surface, and it is a statement within the Born–Oppenheimer approximation. Beyond the Born–Oppenheimer approximation the total first moment is still E_R per nucleus, by Eq. (4.34), but the lattice receives less. The electrons take $P_{\mathrm{Mig}}E_R + \boldsymbol{q}\cdot\delta M_\kappa\cdot\boldsymbol{q}/2M_\kappa^2$, and the lattice the rest of E_R, which is the recoil of a nucleus of mass $M_\kappa+\delta M_\kappa$ in the events that leave the electrons alone. In silicon the two electronic terms stay below 10^{-3} of E_R for recoil energies up to about 16 eV, a range that covers the phonon and multiphonon regime. Corrections below 10^{-3} are far below the accuracy of any multiphonon calculation; at 500 eV the weight $1-P_{\mathrm{Mig}}$ is a correction of 3 %. The bookkeeping, however, is exact, and Sect. 12.5 carries it through.

The total first moment of the electron–nucleus interference vanishes, Eq. (4.34), but the first moments of the individual sectors do not. In the electron-following sector the same steps that led to Eq. (8.54) give

$$\mathscr{M}^1_{e\kappa}\big|_{\mathrm{ad}}(\boldsymbol{q}) = \frac{1}{\mathscr{V}}\Big\langle \sum_{I\in\kappa} \frac{\nabla_I \bar\rho_e(\boldsymbol{q};u)^\dagger\cdot \mathrm{i}\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}}{M_\kappa}\Big\rangle \simeq \frac{q^2\,\boldsymbol{n}_\kappa\,f_\kappa(\boldsymbol{q})^*}{M_\kappa}, \qquad \mathscr{M}^1_{e\kappa}\big|_{0+} = -\,\mathscr{M}^1_{e\kappa}\big|_{\mathrm{ad}}, \tag{8.57}$$

to leading order in κ_{BO}. Electrons that follow the nucleus move in phase with it and carry a positive energy-weighted interference. The electronic sectors carry the same weight with the opposite sign. The opposite signs of the two parts are the sign change that Sect. 4.3 required of the interference spectrum, now located: the electron-following and the electronic parts of $S_{e\kappa}$ are mirror images in their first moment. If a model keeps the Born-charge interference and drops the electronic parts of $S_{e\kappa}$, it violates the sum rule by the full Born-charge term.

These rules give prescriptions for adding rates. If a Migdal rate is added to a multiphonon $S_{\kappa\kappa}$ normalized to the full nuclear f-sum, the electronic share is counted twice; the electron-following part must be depleted by P_{Mig} and computed with the mass $M_\kappa+\delta M_\kappa$. A resonant sideband is never added to a zero-phonon line that already contains its cascade (Sects. 2.4 and 6.5); the next subsection makes this rule precise. A measured loss function or optical conductivity contains both electronic sectors, and a computed sideband is never added to it. And when the dark matter couples to both species, the electron and nuclear amplitudes to the same final state interfere in every sector, with the interference bounded by the positivity of Eq. (4.20).

8.6.4 Zero-phonon line and phonon sideband

The superadiabatic projector commutes with H to the order of Eq. (8.47) because a gap separates the electronic states. Nothing comparable, however, separates $\mathcal{P}_0$ from $\mathcal{P}_+$. A zero-phonon pair at energy E and a one-phonon state at the same energy are both electronic excitations, and the electron–phonon coupling H_{ep} of Sect. 6.2 turns one into the other. The

split between $\mathcal{P}_0$ and $\mathcal{P}_+$ may or may not mean anything, depending on where the energy deposited ω lies relative to the zero-phonon states at the same momentum transfer.

Below the zero-phonon states the answer is simple. Take ω inside the off-shell window, below the bottom $E_{\min}(q)$ of the zero-phonon pair continuum by a margin δ_{w} large compared with the phonon energies, the widths and $|g|$ (the window is made precise in Sect. 13.6). No zero-phonon state exists at that energy, and every final state contains a phonon. An exact final state $|\tilde{f}\rangle$ is a one-phonon state $|f\rangle$ dressed by a virtual admixture of zero-phonon pairs m, each off shell by at least δ_{w},

$$|\tilde{f}\rangle = |f\rangle + \sum_m |m\rangle \frac{\langle m|H_{\mathrm{ep}}|f\rangle}{E_f - E_m} + \cdots, \qquad \|\mathcal{P}_0^{(0)}|\tilde{f}\rangle\|^2 = O(\theta_{\mathrm{ep}}),$$

with $\mathcal{P}_0^{(0)}$ the zero-phonon projector of the uncoupled reference and θ_{ep} the off-shell parameter Eq. (8.40). The perturbation series converges, since no denominator comes close to zero, and the state can be followed continuously from $g = 0$. Every reasonable definition of its label agrees: the phonon number of its leading component, the number obtained after a Schrieffer–Wolff transformation that removes H_{ep} to a given order, and the number of phonons that eventually leave. The three definitions differ by the weight $O(\theta_{\mathrm{ep}})$ of the virtual admixture, and the whole response at ω belongs to $\mathcal{P}_+$ to that accuracy (Appendix D.22). In silicon, at the absorption edge, $\theta_{\mathrm{ep}} = 0.010$ to 0.016 (Sect. 8.5). The bare projector would not do. Inserted in Eq. (8.49), $\mathcal{P}_0^{(0)}$ would pick out the virtual admixture through which the whole phonon-assisted amplitude passes, and assign to the zero-phonon sector a weight as large as the rate itself. Sector projectors must commute with H, and here commuting with H requires dressed projectors.

Above $E_{\min}(q)$ the zero-phonon pair is real, and the label becomes a matter of time. A pair m at energy ω lies inside the continuum of one-phonon states f and decays into it by emitting a phonon. Its amplitude follows from the level-and-band model of Worked example 4.12, with the zero-phonon pair as the level and the one-phonon states as the band. Let $\rho_m = \langle m|O_b|i\rangle$ be the zero-phonon vertex, $g_{fm} = \langle f|H_{\mathrm{ep}}|m\rangle$ the electron–phonon matrix element built from the $g_{mn\nu}$ of Eq. (6.11), and $g_{fm}\mathcal{A}_f^{\mathrm{off}}$ the amplitude with which the probe reaches f by every path that does not pass through m on shell: the phonon emitted first, hole orderings, and nonresonant pairs. The amplitude into f at $\omega = E_f$ is then exactly (Appendix D.22)

$$A_f(\omega) = g_{fm}\Big[\mathcal{A}_f^{\mathrm{off}} + \mathsf{G}_m(\omega)\big(\rho_m + \Upsilon_m(\omega)\big)\Big], \qquad \mathsf{G}_m(\omega) = \frac{1}{\omega - E_m - \Sigma_m(\omega)}, \tag{8.58}$$

with the self-energy of the pair $\Sigma_m(\omega) = \sum_f |g_{fm}|^2/(\omega - E_f + \mathrm{i}0^+)$ and the vertex correction $\Upsilon_m(\omega) = \sum_f |g_{fm}|^2\mathcal{A}_f^{\mathrm{off}}/(\omega - E_f + \mathrm{i}0^+)$. The width of the pair is $\Gamma_m(\omega) = -2\,\mathrm{Im}\,\Sigma_m(\omega)$, its part due to phonon branch ν is $\Gamma_m^{(\nu)}$, and the quasiparticle energy $\tilde{E}_m$ and residue $z_{\mathrm{qp}}(m)$ follow from Σ_m as in Eq. (4.83). The response is $(2\pi/\mathcal{V})\sum_f |A_f|^2\delta(\omega - E_f)$. Near $\tilde{E}_m$ the squared propagator $|\mathsf{G}_m|^2$ is a Lorentzian of width Γ_m, and Eq. (2.60) splits it into a delta function and a finite part. We adopt the convention that assigns the delta function to the zero-phonon sector and everything else to the phonon-dressed sector.

Dyson convention

For Γ_m small compared with the detuning δ_{F}, the distance from $\tilde{E}_m$ to the nearest

structure of the one-phonon continuum,

$$S^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathcal{V}} \sum_m z_{\mathrm{qp}}(m)\, |\tilde{\rho}_m|^2\, \delta(\omega - \tilde{E}_m),$$
$$S^+(\boldsymbol{q},\omega) = \frac{2\pi}{\mathcal{V}} \sum_m \sum_f |g_{fm}|^2\, \delta(\omega - E_f) \Big[|\rho_m|^2\, \mathrm{fp}\, \frac{1}{(\omega - \tilde{E}_m)^2} + 2\,\mathrm{Re}(\mathcal{A}_f^{\mathrm{off}*} \rho_m)\, \mathrm{PV}\, \frac{1}{\omega - \tilde{E}_m} + |\mathcal{A}_f^{\mathrm{off}}|^2 \Big], \tag{8.59}$$

where $|\tilde{\rho}_m|^2 = |\rho_m|^2 + 2\,\mathrm{Re}[\rho_m^*\, \mathrm{PV}\, \Upsilon_m(\tilde{E}_m)]$ contains the virtual-phonon correction of the vertex. Inside the off-shell window no $\tilde{E}_m$ lies at ω, the delta term is absent, and S^+ is the ordinary second-order rate.

The delta part is a real zero-phonon pair, which is handed to the cascade of Chap. 18 as a real pair; the cascade lets the pair emit its first phonon (Sect. 8.7). The finite-part and principal-value terms are the sideband, a pair and a phonon created together. The finite part is negative against a narrow positive test function, as Sect. 2.4 found, so S^+ alone is not a positive spectrum near $\tilde{E}_m$.

The convention preserves the sum of the sectors. Against any test function smooth on the scale δ_{F}, such as a detector resolution or the integral over $\boldsymbol{q}$ and the velocity distribution, $S^0 + S^+$ reproduces the exact response up to terms of order g^4 and $\Gamma_m^2/\delta_{\mathrm{F}}^2$ (Appendix D.22). A per-pair zeroth-moment identity makes the weights add up. Differentiating the Kramers–Kronig form of $\mathrm{Re}\,\Sigma_m$ with Eq. (C.30) gives $\partial_\omega\, \mathrm{Re}\,\Sigma_m = -(2\pi)^{-1}\, \mathrm{fp}\!\int \mathrm{d}\omega'\, \Gamma_m(\omega')/(\omega' - \omega)^2$, and since $z_{\mathrm{qp}} = 1 + \partial_\omega\, \mathrm{Re}\,\Sigma_m + O(g^4)$,

$$z_{\mathrm{qp}}(m) + \frac{1}{2\pi}\, \mathrm{fp} \int \mathrm{d}\omega'\, \frac{\Gamma_m(\omega')}{(\omega' - \tilde{E}_m)^2} = 1 + O(g^4). \tag{8.60}$$

The zero-phonon line loses weight to its residue, and the lost weight is exactly the finite-part weight of the sideband, the first term of S^+ integrated over ω. The level of Worked example 4.12 shows that the residue is not a probability. There $z_{\mathrm{qp}} = 1.0064$, and the finite part of the flat band is $-2\Gamma/(\pi E_{\mathrm{w}}) = -0.0064$: the line carries more than the full weight and the off-shell part returns the excess. Inside a continuum whose width grows with energy, the residue also exceeds one, and a negative finite-part weight in a computed sideband is no error. The identity concerns the self-energy only; the vertex correction in $|\tilde{\rho}_m|^2$ redistributes weight that no zeroth-moment rule fixes, and there is no global zeroth-moment rule for n_e.

The split itself is a convention, and its ambiguity has a size. Three choices of convention are natural. The first is the power of the residue, which a time-domain definition by the surviving amplitude raises from z_{qp} to z_{qp}^2. The second is the shape of the line, a Lorentzian or any normalized kernel no wider than Γ_m in place of the delta function. The third is a window $|\omega - \tilde{E}_m| < x_{\mathrm{w}}$, with $\Gamma_m \ll x_{\mathrm{w}} \le \delta_{\mathrm{F}}$, all of whose weight is assigned to the line. If two conventions differ from Eq. (8.59) only in these ways, they assign to a bin around $\tilde{E}_m$ zero-phonon weights w_0 that differ by (Appendix D.22)

$$\frac{|\Delta w_0|}{w_0} \le \frac{5}{2\pi}\, \frac{\hat{\Gamma}_m}{\delta_{\mathrm{F}}} + \frac{\hat{\Gamma}_m}{\pi x_{\mathrm{w}}} + \Big(1 + \frac{|\mathcal{A}^{\mathrm{off}}|\, \delta_{\mathrm{F}}}{|\rho_m|}\Big)^2 \frac{\hat{\Gamma}_m x_{\mathrm{w}}}{\delta_{\mathrm{F}}^2}, \tag{8.61}$$

with $\hat{\Gamma}_m$ the largest width within δ_{F} of the line. At $x_{\mathrm{w}} \approx \delta_{\mathrm{F}}$ the first two terms add up to $1.1\, \hat{\Gamma}_m/\delta_{\mathrm{F}}$, and the bound is of order $\hat{\Gamma}_m/\delta_{\mathrm{F}}$ times a number of order one when the off-shell amplitude is no larger than $|\rho_m|/\delta_{\mathrm{F}}$. The split is well defined where the pair lives long

compared with $1/\delta_{\mathrm{F}}$ and becomes meaningless as Γ_m approaches δ_{F}. The bound does not extend to every convention. If the split is defined by the number of phonons present after an infinite time, every pair inside the continuum eventually decays, the zero-phonon sector is empty, and the split moves by an amount of order one. The sector label belongs to a time window between $1/\delta_{\mathrm{F}}$ and $1/\Gamma_m$, and the Dyson convention is the choice that makes the totals and the cascade consistent.

Two mixed conventions are common and wrong. If a Lorentzian zero-phonon line is added to the finite-part sideband, the off-shell weight $(\Gamma_m/2\pi)|\rho_m|^2\,\mathrm{fp}(1/x^2)$ is counted twice, an error of order g^2. If a Lorentzian line is added to the resummed sideband $|x+\mathrm{i}\Gamma_m/2|^{-2}$, the delta function is counted twice, an error of order one; this second error is the double counting of the cascade in Sect. 2.4. Within a few widths of $E_{\min}(q)$, or of a singularity of the joint density of states, no test function is smooth on the scale δ_{F}, and only totals computed from the unexpanded amplitude Eq. (8.58) are quoted.

There is one more term, and it changes no total. When the pair decays, its first decay product can interfere with the same product made directly, and this interference shifts the branching of the first decay among the channels. The delta function receives, besides $z_{\mathrm{qp}}|\tilde{\rho}_m|^2$, a contribution from the interference between the off-shell path $\mathcal{A}_f^{\mathrm{off}}$ and the decay of the real pair. Summed over the final states of one decay channel ν, the probability that a zero-phonon event decays first into channel ν is (Appendix D.22)

$$\frac{\Gamma_m^{(\nu)}}{\Gamma_m^{\mathrm{tot}}}\bigl(1+\bar{c}^{\mathrm{F}}-c_\nu^{\mathrm{F}}\bigr), \qquad c_\nu^{\mathrm{F}}=\frac{\Gamma_m^{\mathrm{tot}}\,\mathrm{Im}(\rho_m^{*}\mathcal{A}_\nu^{\mathrm{off}})}{|\rho_m|^2}, \qquad \bar{c}^{\mathrm{F}}=\sum_\nu \frac{\Gamma_m^{(\nu)}}{\Gamma_m^{\mathrm{tot}}}\,c_\nu^{\mathrm{F}}, \tag{8.62}$$

with ν running over every decay channel of the pair, each phonon branch and the others, such as impact ionization, $\mathcal{A}_\nu^{\mathrm{off}}$ the off-shell amplitude on the shell of channel ν, and $\Gamma_m^{\mathrm{tot}}=\sum_\nu \Gamma_m^{(\nu)}$ the total width. The vertex correction Υ_m then sums over all channels as well. The deviations $\bar{c}^{\mathrm{F}}-c_\nu^{\mathrm{F}}$ of the *Fano coefficients* from their mean have a weighted average of zero, with the weights $\Gamma_m^{(\nu)}$, because ν runs over all channels. The deviations therefore change no total and no number of zero-phonon events; the coefficients themselves need not average to zero. The deviations change the branching of the first decay, at relative order $\Gamma_m|\mathcal{A}^{\mathrm{off}}|/|\rho_m| \sim \Gamma_m/\delta_{\mathrm{F}}$, which is the order of the cascade itself. The Fano term is the interference of a discrete state with a continuum that Fano analyzed for atomic autoionization [23]. It is unrelated to the Fano factor of the number of electron–hole pairs (Sect. 18.2), which measures the variance of a count. An incoherent cascade cannot generate the term, since the term is a coherence between production and decay, and the response calculation must supply it.

In many crystals the Fano term vanishes. Let the target be invariant under inversion combined with time reversal, and let its number of electrons be even. The antiunitary operator PT then squares to one, and every state of the electrons and phonons can be chosen invariant under it, with the phases of the phonon polarization vectors fixed to match. The density $n_e(\boldsymbol{q})$ and H_{ep} are invariant too, so every matrix element and every off-shell amplitude is real, and $\mathrm{Im}(\rho_m^{*}\mathcal{A}_\nu^{\mathrm{off}})=0$ (Appendix D.22). Silicon and germanium satisfy the hypothesis, with or without spin–orbit coupling, at the level of one-body self-energies. In silicon and germanium the level-shift matrix of a one-body self-energy is a multiple of the identity on each degenerate zero-phonon manifold, a Kramers pair or an irreducible representation, so the invariant basis also diagonalizes the level-shift matrix. Level shifts from the electron–hole vertex can split such a manifold in a basis that is not invariant; then only the sum of the Fano terms over the branches vanishes, not each one. GaAs has no center of inversion, and its Fano coefficients are generically nonzero.

8.6.5 Targets without a gap

Aluminum has electron–hole pairs at every energy down to zero, and its phonons, up to 40 meV (Sect. 6.4), lie inside that continuum. The adiabatic construction of Eq. (8.47) needs a gap between the electronic ground state and the excited states at every nuclear configuration, and a metal has none: as Sect. 5.1 found, the expansion in adiabatic states fails at the Fermi surface. A superconductor has a gap, but $2\Delta_{\mathrm{BCS}} = 0.36\,\mathrm{meV}$ in aluminum lies far below the phonon energies, and the adiabatic hierarchy is reversed: the electrons are the slow degrees of freedom near the Fermi surface. A Dirac semimetal is gapless at its nodes. Magnetic insulators are a fourth case. Their charge excitations are gapped, but at clamped nuclei the magnons are electronic excitations whose gaps, set by the magnetic anisotropy, lie below most phonon energies, so that no gap separates them from the lattice. The construction then goes through with the whole low-energy spin manifold in place of the electronic ground state, so that magnons and phonons share the electron-following sector (the magnetic targets of Chap. 14).

For the gapless targets the sectors must count quasiparticles instead. The low-energy excitations of a metal are the quasiparticles of Sect. 4.7 and the phonons, and Migdal's theorem (Sect. 6.3) separates them with the small parameter $\omega_{\mathrm{D}}/E_{\mathrm{F}} \sim \kappa_{\mathrm{BO}}^2$. Suppose that a dressing unitary U_{qp} maps bare electrons and phonons onto quasiparticles and dressed phonons with number operators $\hat{N}_{\mathrm{qp}}$ and $\hat{N}_{\mathrm{ph}}$; Landau's theory and the Migdal–Eliashberg equations construct a unitary of this kind order by order. Let $\mathcal{P}^{\mathrm{qp}}_{n,m}$ be the joint spectral projectors of $U_{\mathrm{qp}}\hat{N}_{\mathrm{qp}}U_{\mathrm{qp}}^{\dagger}$ and $U_{\mathrm{qp}}\hat{N}_{\mathrm{ph}}U_{\mathrm{qp}}^{\dagger}$, with n and m the net numbers of quasiparticles and phonons created. These projectors are orthogonal and sum to one. They do not commute with H, since quasiparticles have a width Γ_{qp} and phonons decay into pairs. Split $H = H_{\mathrm{d}} + H_{\mathrm{off}}$, with $H_{\mathrm{d}} = \sum_{n,m} \mathcal{P}^{\mathrm{qp}}_{n,m} H \mathcal{P}^{\mathrm{qp}}_{n,m}$, and define the sector responses with H_{d}, which commutes with every $\mathcal{P}^{\mathrm{qp}}_{n,m}$. Each is then positive, and they sum exactly to the response of H_{d}. Take the response of H from one initial state $|i\rangle$, without the factor $2\pi/\mathcal{V}$. On a Gaussian bin of standard deviation $\Delta\omega_{\mathrm{bin}}$, this response differs from the response of H_{d} by at most (Appendix D.22)

$$|C_{ab}^{\mathrm{cross}}| \le \frac{\mathfrak{h}_{\mathrm{off}}}{\pi\,\Delta\omega_{\mathrm{bin}}^2}\,\big\|O_a|i\rangle\big\|\,\big\|O_b|i\rangle\big\|, \qquad \mathfrak{h}_{\mathrm{off}} = \sup_{t'} \frac{\big\|H_{\mathrm{off}}\,\mathrm{e}^{-\mathrm{i}H_{\mathrm{d}}t'}\,O_b|i\rangle\big\|}{\big\|O_b|i\rangle\big\|}. \tag{8.63}$$

With the dressing in place, the bound needs only the few-quasiparticle state that the probe creates, not the whole crystal, and it holds for any orthogonal sectors; with the leakage of Eq. (8.47) in place of $\mathfrak{h}_{\mathrm{off}}$ it is the remainder of the decomposition theorem. Relative to a typical binned value, $\|O|i\rangle\|^2/\Delta\omega_{\mathrm{bin}}$, the cross terms are of order $\mathfrak{h}_{\mathrm{off}}/\Delta\omega_{\mathrm{bin}}$. For a state that decays by the golden rule into a continuum of width ΔE_{c}, $\mathfrak{h}_{\mathrm{off}}^2 \approx \Gamma\,\Delta E_{\mathrm{c}}/2\pi$, which is much larger than Γ^2. The bound therefore guarantees small cross terms only on bins wide compared with $(\Gamma\,\Delta E_{\mathrm{c}})^{1/2}$. Without any dressing, $U_{\mathrm{qp}} = 1$, the bound fails: H_{off} acts on the filled Fermi sea as well as on the quasiparticles the probe creates, and this action on the Fermi sea makes $\mathfrak{h}_{\mathrm{off}}$ grow with the size of the crystal (Appendix D.22). The action on the Fermi sea is the disconnected part. A bound has to control the connected part, in which H_{off} acts on the probe's excitations. Acting on a quasiparticle, H_{off} has two parts: the electron–phonon coupling, which adds or removes a phonon, and the part of the Coulomb interaction that turns the quasiparticle into two particles and a hole at fixed phonon number. The final states of the two parts differ in phonon number, so their squared norms add. The square of the connected part is then at most the largest eigenvalue of a band matrix, formed by the zeroth moments of the Fan–Migdal spectral function and by the analogous electron–electron sums over final states with two particles and a hole (Appendix D.22). The electron–phonon entries are the sums $N^{-1}\sum |g_{mn\nu}|^2(2n_{\boldsymbol{Q}\nu}+1)$ of Sect. 6.3 without their energy delta function. In aluminum, with $|g_{mn\nu}|^2$ taken constant over the band, the electron–phonon matrix alone has

the eigenvalue $\int \mathrm{d}\omega\, \alpha^2 F/\mathcal{N}_\mathrm{F}\Omega = 5.39\,\mathrm{meV}/0.193\,\mathrm{eV}^{-1} \approx 0.03\,\mathrm{eV}^2$ (Sect. 6.4), so the electron–phonon contribution to the connected part is at most about 0.17 eV. The electron–electron entries cannot lower this eigenvalue, and they are not estimated.

The sectors of a superconductor follow the counting of Sect. 7.2. Acting on the BCS state, a density or spin operator creates Bogoliubov quasiparticles in pairs, so $n \in \{0, 2\}$ at leading order. Phonons with no broken pair, $(0, m)$, are the analogue of the electron-following sector, and they are stable only below $2\Delta_\mathrm{BCS}$, since a phonon above it breaks a pair. Pair breaking, $(2, 0)$, is the analogue of the zero-phonon sector and is computed with the conserving response of Sect. 7.2; phonon-assisted pair breaking, $(2, m \geq 1)$, is the analogue of the phonon-dressed sector. The collective modes leave no sub-gap line in the density response: the phase mode is pushed to the plasma frequency, and the amplitude mode is not excited by the density in a particle–hole-symmetric metal (Sect. 7.2). These collective modes are composites of pairs, and assigning them to a sector is a convention. In a normal metal n counts electron–hole pairs, and a phonon decays into pairs at the rate $\gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}$ of Eq. (6.32); the sectors $(0, m)$ and $(2, m-1)$ mix within this decay width. In a Dirac semimetal the spectrum is linear and the effective fine-structure constant $\tilde{\alpha}$ of Sect. 7.3 is dimensionless, so the only energy scale near the node is ω itself. By dimensional analysis the quasiparticle width is then ω times a function of $\tilde{\alpha}$, which starts at $\tilde{\alpha}^2$ because a decay needs two Coulomb vertices. The split is therefore meaningful only for $\tilde{\alpha}^2 \ll 1$, and near $\tilde{\alpha} \sim 1$ only totals are quoted. The totals are safe in every case if the response is conserving. The f-sum then holds for the sum of the sectors, while the share of each sector is not fixed. No phonon sector can be paired with a nonconserving pair-breaking response, such as the bare bubble that violates the f-sum in Sect. 7.2.

Worked example 8.8: A two-level electron on a vibrating bond

We take the bond coordinate x of silicon from Worked example 5.7, an oscillator of frequency $\omega_0 = 64.1\,\mathrm{meV}$, zero-point length $\ell_0 = 0.048\,\text{Å}$ and mass $M/2$, and attach to it an electron with two levels, a lower one $|\mathrm{lo}\rangle$ and an upper one $|\mathrm{up}\rangle$, split by $\Delta_\mathrm{tl} = 1.17\,\mathrm{eV}$, the gap of silicon. With $x = \ell_0(d + d^\dagger)$,

$$H = \omega_0\, d^\dagger d + [\Delta_\mathrm{tl} + g_\mathrm{d}(d + d^\dagger)]\, |\mathrm{up}\rangle\langle\mathrm{up}| + g_\mathrm{n}(d + d^\dagger)(|\mathrm{up}\rangle\langle\mathrm{lo}| + |\mathrm{lo}\rangle\langle\mathrm{up}|).$$

The diagonal coupling $g_\mathrm{d} = -\sqrt{2}\,\omega_0$ displaces the excited surface as in Worked example 5.7, with $S_\mathrm{HR} = 2$. The off-diagonal coupling $g_\mathrm{n} = 20\,\mathrm{meV}$ is a model choice; it mixes the two levels as the bond moves and is the only source of nonadiabatic coupling, with the mixing $g_\mathrm{n}/\Delta_\mathrm{tl} = 0.017$ at the zero-point amplitude. The probes are the electronic transition operator $\sigma_x = |\mathrm{up}\rangle\langle\mathrm{lo}| + |\mathrm{lo}\rangle\langle\mathrm{up}|$ and a kick $\mathrm{e}^{\mathrm{i}qx}$ with $q\ell_0 = 1$, that is $q = 41.1\,\mathrm{keV}$, whose recoil energy on the coordinate, q^2/M, equals ω_0. We diagonalize H exactly in 160 oscillator states; 90 give the same numbers to the digits quoted.

Labels. The two probes reach forty eigenstates with a weight above 10^{-10}. Each has at least 0.996 of its norm on a single Born–Oppenheimer product state, $\Psi^\mathrm{nuc}_{0,m}(x)\Psi_0(x)$ on the ground surface or $\Psi^\mathrm{nuc}_{1,m}(x)\Psi_1(x)$ on the excited one, with m the vibrational level. We assign the ground surface to the electron-following sector, $m = 0$ on the excited surface to the zero-phonon sector and $m \geq 1$ there to the phonon-dressed one. The sector projectors are sums of exact eigenprojectors, they commute with H, and the split is exact.

The electron probe. The weight of σ_x, one in total, splits as 1.25×10^{-3}, 0.152 and 0.847 among the three sectors. The electron-following part is a line at 63.4 meV: the bond's ground-state surface is softened by the mixing and carries an electronic dipole $\langle\sigma_x\rangle_0 \simeq -2g_\mathrm{n}x/(\ell_0\Delta_\mathrm{tl})$ that follows the bond, with the one-phonon weight $(2g_\mathrm{n}/\Delta_\mathrm{tl})^2 = 1.17\times10^{-3}$ at leading order. The zero-phonon line, at 1.0463 eV, has the weight 0.152, the Franck–Condon factor of the two computed surfaces (0.1519 with the dependence of the transition

dipole on x). It exceeds the $\mathrm{e}^{-2} = 0.135$ of Worked example 5.7 because the mixing softens the ground surface to 63.4 meV and stiffens the excited one to 65.5 meV. The first moment of σ_x is fixed by its double commutator at $-\langle(\Delta_{\mathrm{tl}} + g_{\mathrm{d}} x/\ell_0)\sigma_z\rangle = 1.169\,15\,\mathrm{eV}$, with $\sigma_z = |\mathrm{up}\rangle\langle\mathrm{up}| - |\mathrm{lo}\rangle\langle\mathrm{lo}|$, and the electron-following sector takes 8.32×10^{-5} eV of it, a fraction 7.1×10^{-5}. At leading order the fraction is $4g_{\mathrm{n}}^2\omega_0/\Delta_{\mathrm{tl}}^3 = 6.4 \times 10^{-5}$, which is also the drag ratio $\delta M/(M/2)$ of Eq. (8.56) for this model, for the reason that made $Z_\kappa m_e/M_\kappa$ appear twice in silicon.

The kick. The electrons are left excited with probability $P_{\mathrm{Mig}} = 3.94 \times 10^{-6}$, of which 0.38×10^{-6} is zero-phonon and 3.56×10^{-6} phonon-dressed. Evaluated at $x = 0$ with the vertical gap, the boost formula Eq. (8.55) gives $(2\omega_0 q\ell_0 g_{\mathrm{n}}/\Delta_{\mathrm{tl}}^2)^2 = 3.51 \times 10^{-6}$; the exact probability exceeds the boost-formula probability by 12 %, a difference of the order $S_{\mathrm{HR}}\omega_0/\Delta_{\mathrm{tl}} = 0.11$ of the displacement of the excited surface, which the leading formula ignores. As $g_{\mathrm{n}} \to 0$ the probability falls as g_{n}^2: the kick reaches the electronic sectors through the nonadiabatic coupling alone. The electronic sectors take 4.72 μeV of the mean recoil energy of 64.1 meV, against $P_{\mathrm{Mig}}\, q^2/M + (q^2/M)\,\delta M/(M/2) = 0.25 + 4.10\,\mu\mathrm{eV}$ from Eq. (8.56), and the lattice keeps the rest. The exact and the predicted energies differ by 8 %, for the same reason as the 12 % in P_{Mig}: with the displacement removed, $g_{\mathrm{d}} = 0$, the exact values are 3.52×10^{-6} and 4.34 μeV, within 0.3 % of the leading formulas. The energy taken by the electronic sectors is the correction to Eq. (5.55) in its smallest form.

The interference. The energy-weighted interference of the two probes is $(2.17 - 13.22\,\mathrm{i}) \times 10^{-4}$ eV in the electron-following sector and exactly the opposite in the two electronic sectors together. Its total vanishes because σ_x commutes with $[H, \mathrm{e}^{\mathrm{i}qx}]$, the toy form of Eq. (8.57).

In this model every zero-phonon level is discrete and nothing decays, so the labels are exact and no convention is needed. Couple the excited level to a continuum of phonons, as the band of Worked example 4.12 couples to its level, and the zero-phonon line acquires a width; the split then follows Eq. (8.59), with the residue 1.0064 and the finite-part weight −0.0064 of that example.

The decomposition rests on three statements that are not yet theorems. First, two results are derived at the level of perturbation theory in g with distributional limits: the split between the zero-phonon and the phonon-dressed sectors outside the off-shell window, and the convention independence of the totals. A proof needs a limiting-absorption principle for the coupled electron–phonon Hamiltonian at fixed total crystal momentum, on the energies outside the window, uniform in the coupling and in the number of cells. The same estimate would sharpen the gapless bound Eq. (8.63) from $(\Gamma\,\Delta E_{\mathrm{c}})^{1/2}$ to the width Γ. Second, the constants of the superadiabatic construction grow with the number of nuclei, so Eq. (8.47) is proved for a finite crystal. The missing step is a superadiabatic generator built from exponentially localized terms, so that the leak per unit volume stays finite in the infinite crystal. Third, for gapless targets it is open whether the dressing U_{qp} exists beyond perturbation theory, with bounds, since without a gap there is no adiabatic theorem to provide it. The bin bound Eq. (8.63) needs this dressing and, with it, a bound on the connected part of $\mathfrak{h}_{\mathrm{off}}$ through a commutator estimate uniform in the number of cells. Until these three statements are proved, the decomposition is exact for a finite insulator and controlled, with the stated error terms, for everything else.

8.7 Final-state-resolved response

When a dark-matter particle deposits 10 eV in silicon, it leaves an electron and a hole with about 9 eV of kinetic energy between them. The split of the 9 eV between the two carriers decides how many further pairs they make on their way down. The rate of impact ionization rises steeply with the energy of a carrier, so a lopsided split makes more pairs than an even one, and an even split is the one that minimizes impact ionization [24, Sect. II A]. The step model of Eq. (1.42) assigns three pairs to every 10 eV deposit, and the rate Eq. (8.16) cannot do better, because the structure factor sums over all final states with the same ω and keeps no record of which one was made. The detector, however, counts the products of a particular final state. What must the response keep, so that the cascade of Chaps. 18 to 20 receives the state the dark matter actually produced, and nothing is counted twice or lost on the way?

The answer is a response matrix resolved by the class of the final state, $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$, together with a rule for handing it over.

8.7.1 Final-state classes

The cascade needs more than the sector of Sect. 8.6 in which a final state lies. It needs the state itself, at the moment the dark matter has finished acting and before the target starts to relax. Two time scales bracket that moment. Within a time of order $1/\delta_\mathrm{F}$, about a femtosecond for a pair in silicon, the electrons and phonons around the final state dress it; here δ_F is the distance from the state to the nearest structure of the continuum it can decay into. The final state decays, by emitting its first phonon or by impact ionization, within $1/\Gamma_m$, with Γ_m the width of Sect. 8.6. Between the two times, the state is a well-defined *prompt* final state. The interval between $1/\delta_\mathrm{F}$ and $1/\Gamma_m$ is the time window in which Sect. 8.6 found the split between $\mathcal{P}_0$ and $\mathcal{P}_+$ meaningful, and the Dyson convention Eq. (8.59) is the rule that labels a state inside it.

We group the prompt final states into *final-state classes* Λ, each described by the quantum numbers the cascade needs. The commonest are these. An electron–hole pair is labeled by the conduction band c and crystal momentum $\boldsymbol{k} + \boldsymbol{q}$ of the electron and the valence band v and crystal momentum $\boldsymbol{k}$ of the hole. These four labels fix the valleys the carriers occupy and their kinetic energies E_e and E_h, measured from the band edges as in Sect. 6.5. A spin label is added when the coupling acts on the spin. A pair with phonons carries in addition the branch, momentum and energy of each phonon. A core hole is labeled by its shell, since it relaxes by emitting an Auger electron (Sect. 10.6). In the electron-following sector the class is the phonon content, or, for a freely recoiling nucleus, its energy E_R and direction. Each class lies in one sector, so the classes refine the partition Eq. (8.48) rather than cut across it.

The plasmon of silicon is not a class of its own. At the zero of ε_1 the measured ε_2 is 0.224 (Worked example 4.10), interband pairs exist at the plasmon energy, and the collective oscillation decays into them. In the random-phase approximation, the screened pair weights already contain the plasmon peak of the loss function, distributed over the pairs into which the plasmon decays (Sect. 8.7.2). A plasmon needs a label only where it cannot decay into pairs, below the cutoff q_c of the electron gas of Sect. 4.5, or where it can leave the target or convert into another excitation, as at the surfaces of films (Sect. 16.2). It then has its own decay, which the cascade must treat.

For each class we write P_Λ for the projector on its states. The classes are orthogonal and exhaust the final states, $P_\Lambda P_{\Lambda'} = \delta_{\Lambda\Lambda'} P_\Lambda$ and $\sum_\Lambda P_\Lambda = 1$. The response into one class follows from inserting P_Λ around the energy delta function of Eq. (4.13), as Eq. (8.49) did for the sectors.

Final-state-resolved response

$$S_{ab}(\boldsymbol{q},\omega;\Lambda) = \frac{2\pi}{\mathscr{V}} \sum_i p_i \,\langle i|O_a(\boldsymbol{q})^\dagger \, P_\Lambda \, \delta(\omega + E_i - H) \, P_\Lambda \, O_b(\boldsymbol{q})|i\rangle,$$
$$\sum_\Lambda S_{ab}(\boldsymbol{q},\omega;\Lambda) = S_{ab}(\boldsymbol{q},\omega). \tag{8.64}$$

The sum rule holds exactly when each P_Λ commutes with H, and each $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ is then a positive semidefinite matrix.

Both statements follow as for the sectors. When P_Λ commutes with H, the delta function can be written in eigenstates of H that each lie in one class, the cross terms between classes vanish, and each $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ is a sum of outer products with positive weights. Since the prompt classes are defined on the window between $1/\delta_\mathrm{F}$ and $1/\Gamma_m$, they commute with H only to the accuracy of that window. Inside the off-shell window of Sect. 8.6, where the labels are unique to relative order θ_ep, each resolved response is positive. Near a zero-phonon resonance, the classes of $\mathcal{P}_+$ are not separately positive, and only their sum with the zero-phonon pair is, as Eq. (8.59) showed for the sectors. The delta part of the Dyson convention is a zero-phonon pair, weighted by its residue $z_\mathrm{qp}(m)$, and it enters $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ as such. When this pair later emits a phonon, the emission is the first step of the cascade, not a second final state.

The rate into one class follows from the steps that gave Eq. (8.15), with P_Λ inserted before and after the delta function. For one dark-matter particle of velocity $\boldsymbol{v}$,

$$\Gamma(\boldsymbol{v};\Lambda) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v}) \, S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}};\Lambda), \tag{8.65}$$

and $\sum_\Lambda \Gamma(\boldsymbol{v};\Lambda) = \Gamma(\boldsymbol{v})$. For absorption, $\mathfrak{D}_{ab}$ is replaced by $g_a^{\phi\,*} g_b^\phi / 2\omega_\phi$ at the single point $(\boldsymbol{p}_\phi, \omega_\phi)$, as in Eq. (8.19). In a crystal the resolved response depends on the direction of $\boldsymbol{q}$, as the total does, and so does the mix of classes a given particle produces.

8.7.2 Pairs in an insulator

For independent electrons the classes of the zero-phonon sector are single pairs. The ground state of a clamped insulator at $T = 0$ is the Slater determinant with every valence state filled. The density $n_e(\boldsymbol{q})$ of Eq. (3.37) moves one electron from $v\boldsymbol{k}$ to $c, \boldsymbol{k}+\boldsymbol{q}$, with the amplitude $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ of Eq. (3.34) and the energy $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$. Each final state of Eq. (4.13) is therefore one pair. Summed over the two spins of the pair,

$$S^0(\boldsymbol{q},\omega;cv\boldsymbol{k}) = \frac{4\pi}{\mathscr{V}} \left|\rho_{cv}(\boldsymbol{k},\boldsymbol{q})\right|^2 \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}). \tag{8.66}$$

Eq. (8.66) is the term of the independent-particle structure factor that belongs to one pair; summed over c, v and $\boldsymbol{k}$ it is the crystal form of Eq. (2.67). Here $\boldsymbol{q}$ is not reduced to the first zone, and the folding vector $\boldsymbol{G}_\mathrm{U}$ of the final state stays in the vertex.

In the random-phase approximation, or in time-dependent density-functional theory, the external potential at $\boldsymbol{q}$ induces a density, and the density adds its own Hartree and exchange–correlation potentials at every $\boldsymbol{q}+\boldsymbol{G}$. The electrons, independent in the Kohn–Sham sense, respond to the total potential. The total potential is the external potential times the column

$$(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega) = \delta_{\boldsymbol{G}0} + \sum_{\boldsymbol{G}'} (f_\mathrm{Hxc})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) \, R_{\boldsymbol{G}'0}(\boldsymbol{q},\omega),$$

the inverse dielectric matrix of a test electron, which Eq. (6.10) met for the static potential of a phonon. Without the exchange–correlation kernel it is the inverse dielectric matrix Eq. (4.43).

The index 0 is the row $n_e(\boldsymbol{q})$ itself, and R is the retarded response. It is natural to guess that the screened pair weight replaces $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ in Eq. (8.66) by the vertex of the total potential, $\sum_{\boldsymbol{G}}(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})$. The guess is right, and the algebra that confirms it admits a second answer.

Let $f = f_{\mathrm{Hxc}}$ be Hermitian as a matrix in the reciprocal-lattice vectors, as it is in the random-phase approximation and for any real kernel without frequency dependence, such as the adiabatic local-density kernel of Sect. 4.6. Write $R^\dagger$ for the Hermitian conjugate matrix, the advanced response. The Dyson equation Eq. (4.70) reads $R = R^0 + R^0 f R$, and its conjugate $R^\dagger = R^{0\dagger} + R^\dagger f R^{0\dagger}$. The Dyson equation gives $R^0(1+fR) = R$, its conjugate gives $(1+R^\dagger f)R^{0\dagger} = R^\dagger$, and together

$$(1+R^\dagger f)\,(R^0 - R^{0\dagger})\,(1+fR) = (R + R^\dagger f R) - (R^\dagger + R^\dagger f R) = R - R^\dagger, \tag{8.67}$$

for every target. The fluctuation–dissipation theorem Eq. (4.18) turns both differences into structure factors: at $T = 0$ and $\omega > 0$, $S_{\boldsymbol{G}\boldsymbol{G}'} = \mathrm{i}(R - R^\dagger)_{\boldsymbol{G}\boldsymbol{G}'}$, and the same holds for R^0. In the bubble Eq. (4.63) only the resonant term has an imaginary part at $\omega > 0$, and $\mathrm{i}[1/(x+\mathrm{i}0^+) - 1/(x-\mathrm{i}0^+)] = 2\pi\delta(x)$, so

$$\mathrm{i}(R^0 - R^{0\dagger})_{\boldsymbol{G}\boldsymbol{G}'} = \frac{4\pi}{\mathscr{V}} \sum_{cv\boldsymbol{k}} \rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')\,\delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}).$$

Take the element 00 of Eq. (8.67). Since f is Hermitian, $(1+R^\dagger f)_{0\boldsymbol{G}} = [(1+fR)_{\boldsymbol{G}0}]^* = [(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}]^*$, and the double sum over $\boldsymbol{G}$ and $\boldsymbol{G}'$ closes into a squared modulus for each pair.

Screened pair-resolved response

$$S(\boldsymbol{q},\omega;cv\boldsymbol{k}) = \frac{4\pi}{\mathscr{V}} \Big|\sum_{\boldsymbol{G}} (\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})\Big|^2 \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}), \tag{8.68}$$

with the retarded $\tilde{\varepsilon}^{-1}$, in the random-phase approximation and in time-dependent density-functional theory with a Hermitian kernel. The sum over c, v and $\boldsymbol{k}$ is the screened structure factor $S(\boldsymbol{q},\omega) = (q^2/2\pi\alpha)L(\boldsymbol{q},\omega)$ of Eq. (4.45), local fields included.

Each term is positive, the terms add up to the full screened response, and the plasmon peak of L is in the sum, spread over the pairs into which the plasmon decays. The pair weight is the bare vertex of the total potential, evaluated at the energy deposited. In the deviation notation of Sect. 8.5, Eq. (8.66) is at $\mathcal{X} = \mathrm{IP}$ and Eq. (8.68) at $\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$ or TDDFT; with the BSE the final states are eigenstates of the pair Hamiltonian, excitons among them, and these eigenstates become the classes (Sect. 10.5).

The second answer comes from the same algebra in the other order. The Dyson equation can equally be written $R = R^0 + RfR^0$, with conjugate $R^\dagger = R^{0\dagger} + R^{0\dagger} f R^\dagger$, and then the identity $(1+Rf)(R^0 - R^{0\dagger})(1+fR^\dagger) = R - R^\dagger$ holds as well, and it distributes the same total over the pairs with the advanced weight $\delta_{\boldsymbol{G}0} + \sum_{\boldsymbol{G}'} f_{\boldsymbol{G}\boldsymbol{G}'} R^*_{0\boldsymbol{G}'}$ in place of $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}$. The two distributions have the same sum for every target, and pair by pair they need not agree. Time decides which of the two distributions labels the final state. The pair that the cascade receives is the state found at late times, an *out-state* of scattering theory, and the retarded resolvent enters through the overlap of a free final state with the evolving one: in Lippmann–Schwinger form, $\langle\psi_f^{(-)}| = \langle\phi_f|(1 + VG^{(+)})$ [12, Chap. 6]. In the time-dependent mean-field picture the choice of the retarded weight is plain causality. The external potential acts first and induces the total potential $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}\phi_{\mathrm{ext}}$; the total potential creates bare pairs, and the bare pairs are the asymptotic states of the mean-field evolution. The advanced form labels each pair by the state it is evolved from, an in-state, and the in-state is not the label the cascade needs.

The two labelings agree pair by pair whenever $R(\boldsymbol{q},\omega)$ is Hermitian, since then the two weights coincide. Inside the off-shell window R is Hermitian up to terms of order g^2: the zero-phonon pairs are closed, and the anti-Hermitian part of the external response comes from the phonon-assisted absorption itself; the difference in a pair-resolved rate is then of order g^4. The two labelings also agree in every target that has an even number of electrons and is invariant under the product PT of inversion and time reversal, for the reason that made the Fano coefficients of Eq. (8.62) vanish. The PT argument needs a kernel that is real symmetric, $f_{\boldsymbol{G}\boldsymbol{G}'} = f_{\boldsymbol{G}'\boldsymbol{G}} = f^*_{\boldsymbol{G}\boldsymbol{G}'}$, as the random-phase and adiabatic local-density kernels are; a kernel that is merely Hermitian does not suffice. Every state can then be chosen invariant under PT, and with the origin at an inversion center $n_e(\boldsymbol{q}+\boldsymbol{G})$ is invariant too, since inversion reverses $\boldsymbol{q}+\boldsymbol{G}$ and complex conjugation restores it. Every amplitude $\langle f|n_e(\boldsymbol{q}+\boldsymbol{G})|i\rangle$ is then real, $R_{\boldsymbol{G}\boldsymbol{G}'} = R_{\boldsymbol{G}'\boldsymbol{G}}$, and the advanced weight is the complex conjugate of the retarded one. The symmetry of R depends on that choice of origin, since a shift multiplies $R_{\boldsymbol{G}\boldsymbol{G}'}$ by phases; the pair weights do not. A real vertex sum has the same modulus with the retarded and with the advanced weight, so each pair, or each set of degenerate pairs, gets the same weight. Silicon and germanium satisfy the hypothesis. GaAs has no center of inversion, and there the choice of the out-state matters.

The pair weights need nothing beyond the ingredients of the structure factor itself. The vertices $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})$ come from the plane-wave coefficients of the Bloch states (Sect. 26.2), and the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}$ from a dielectric-matrix calculation at the same $(\boldsymbol{q},\omega)$. A structure-factor calculation and a resolved calculation differ in what they keep. A code that returns $S(\boldsymbol{q},\omega)$ sums the pairs and discards them. The resolved response stores their weights, binned in the carrier energies and valleys, for each $\boldsymbol{q}$, ω and coupling row, and a response library must say which binning it keeps.

Worked example 8.9: Pairs made at one momentum transfer in silicon

Which pairs does a dark-matter kick make, and how do they differ from the pairs that a photon of the same energy makes? We take the empirical pseudopotential of Cohen and Bergstresser for silicon [25], the model of Sects. 3.3 and 4.6: three form factors fitted to measured optical transitions, $a = 5.43\,\text{Å}$, 113 plane waves and no spin–orbit coupling. Its indirect gap is 0.82 eV, below the measured 1.17 eV, with the conduction minimum at 0.853 of the way to X. The momentum transfer is $q_0 = \alpha m_e = 3.73\,\text{keV}$ along [100], a typical value for electron scattering, which is $1.633\,(2\pi/a)$ and lies beyond the first zone; its Umklapp partner $\boldsymbol{q}+\boldsymbol{G}$ with $\boldsymbol{G} = (-2,0,0)\,2\pi/a$ has $|\boldsymbol{q}+\boldsymbol{G}| = 0.84\,\text{keV}$. The energy deposited is $\omega = 10\,\text{eV}$. The sums run over a shifted 24^3 grid of $\boldsymbol{k}$ over the full zone, with 4 valence and 26 conduction bands. The local-field matrix has the 51 vectors with $|\boldsymbol{q}+\boldsymbol{G}|^2 \le 12\,(2\pi/a)^2$, and the $\mathrm{i}0^+$ of the bubble is a Lorentzian of half-width 0.10 eV. The energy delta function of each pair is a Gaussian of standard deviation 0.10 eV instead. A Lorentzian has long tails, and in the optical limit below, whose weights carry $1/(\varepsilon_c-\varepsilon_v)^2$, the tails would admit strong transitions several eV below ω and break the energy balance of the pairs. The numbers quoted change by at most 3 % between 20^3 and 24^3 points and between Gaussians of 0.10 and 0.05 eV.

Screening. The independent-particle structure factor is $S^{\text{IP}}(\boldsymbol{q}_0,\omega) = 1.7\times10^8\,\text{eV}^2$. The random-phase approximation with local fields gives $(\varepsilon^{-1})_{00} = 0.55 - 0.16\,\mathrm{i}$, and summing Eq. (8.68) over pairs gives the screened structure factor $4.9\times10^7\,\text{eV}^2$, a factor of 0.29 of the unscreened one. The head of the column does most of the screening, since $|(\varepsilon^{-1})_{00}|^2 = 0.33$; the other 50 elements are at most 0.05 each, the Umklapp partner among the small ones, and together they lower the rate by a further 12 %. For comparison, the free gas at the valence density gives $m_e^2\omega/\pi q_0 = 2.2\times10^8\,\text{eV}^2$ at the same point, the unpolarized form of

the one-spin structure factor of Worked example 8.3.
Which pairs. Every pair at this ω shares $\omega - E_\mathrm{g} = 9.2\,\mathrm{eV}$ between the kinetic energies E_e and E_h of its carriers. With the screened weights, the mean electron energy is 4.8 eV and the mean hole energy 4.4 eV; the electron takes more than half in 64 % of the weight, and one carrier takes more than three quarters in 45 %. The lowest valence band supplies 4 % of the holes and the other three about a third each. The independent-particle weights give the same picture, each of these fractions within three percentage points. Screening lowers the number of pairs made by a factor of three and leaves their distribution nearly unchanged.
Other seeds at the same energy. A photon of 10 eV, or a dark photon, makes vertical pairs with the weights $|\hat{\boldsymbol{e}} \cdot \boldsymbol{v}_{cv}|^2/(\varepsilon_c - \varepsilon_v)^2$ of the optical limit Eq. (3.43). Its electrons are more energetic, 5.7 eV on average, and its holes less, 3.5 eV, about 0.9 eV below the holes of the dark-matter kick; the electron takes more than half in 85 % of the weight. The transverse current $\hat{\boldsymbol{y}} \cdot \boldsymbol{j}_e(\boldsymbol{q}_0)$ at the same momentum transfer lies in between, with 5.3 eV and 3.8 eV, and puts half its weight on lopsided pairs with one carrier above three quarters. Every seed shares the same 9.2 eV, and yet the three seeds hand the cascade three different starting distributions.
In-states and out-states. In this model of silicon, with the origin at the bond center, an inversion center, the two labelings give every pair the same weight, and $R_{\boldsymbol{G}\boldsymbol{G}'} = R_{\boldsymbol{G}'\boldsymbol{G}}$, as the PT argument requires for the real symmetric kernel of the random-phase approximation. The same model of GaAs has no center of inversion. At the same q_0 and 10 eV, the pair-by-pair differences between the two labelings add up, in absolute value, to 15 % of the screened total, although the mean carrier energies of the two labelings differ by 0.02 eV at most. The choice of the out-state matters for momentum- and valley-resolved seeds, and much less for the energies alone.

8.7.3 Cascade probabilities of a final state

Once the dark matter has acted, the target evolves under its own Hamiltonian. A pair of class Λ emits phonons, ionizes further pairs and slows down, and the event ends with some number N_eh of pairs, some energy in phonons and some energy stored in defects. We write $P_\mathrm{casc}(N_\mathrm{eh}|\Lambda)$ for the probability that the cascade started by a final state of class Λ ends with N_eh pairs; Sect. 18.3 computes it from a branching process. We call *seeding* the step that turns the prompt state into the initial condition of the cascade. The one assumption here is that the class carries everything the later evolution depends on. The prompt state is reached before the first decay, so the history of how it was made can enter only through coherences between production and decay. These are the Fano coefficients of Eq. (8.62), and they vanish in silicon and germanium.

Under that assumption, the probability of an event with N_eh pairs is the probability of each class times the probability that its cascade ends with N_eh pairs, summed over the classes. With the rate into a class, Eq. (8.65), and the halo average of Eq. (8.16):

Hand-off to the cascade

$$\mathcal{R}_{N_\mathrm{eh}} = \frac{\rho_\chi}{m_\chi\,\rho_T} \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v}) \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_\Lambda \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\, S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}};\Lambda)\, P_\mathrm{casc}(N_\mathrm{eh}|\Lambda), \quad (8.69)$$

the rate per unit target mass of events with N_eh pairs. The coupling tensor is contracted inside the sum over classes, and the sum over classes is inside the integrals over $\boldsymbol{q}$ and

$\boldsymbol{v}$. Equivalently, at fixed $(\boldsymbol{q}, \omega, \boldsymbol{v})$ the pair number is distributed as

$$P(N_{\mathrm{eh}}|\boldsymbol{q}, \omega, \boldsymbol{v}) = \frac{\sum_{\Lambda} \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})\, S_{ab}(\boldsymbol{q}, \omega; \Lambda)\, P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)}{\sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})\, S_{ab}(\boldsymbol{q}, \omega)}. \tag{8.70}$$

The same form holds for any other count of the cascade, the phonon energy or the energy stored in defects, with P_{casc} replaced by the distribution of that quantity, and for absorption, with $\mathfrak{D}_{ab}$ replaced as in Eq. (8.65). Eq. (8.69) contains the pair bins of Eq. (1.45) as a special case. If $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ depended on the class only through its energy ω, Eq. (8.70) would reduce to $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\omega)$, the sum over Λ would close into $S_{ab}(\boldsymbol{q}, \omega)$ by Eq. (8.64), and the integrals would collapse to $\int \mathrm{d}\omega\, (\mathrm{d}\mathcal{R}/\mathrm{d}\omega)\, P_{\mathrm{casc}}(N_{\mathrm{eh}}|\omega)$. The step model is the further choice of a distribution that puts all its weight on $1 + \lfloor(\omega - E_{\mathrm{g}})/\varepsilon_{\mathrm{eh}}\rfloor$.

That reduction is the common shortcut, and Eq. (8.69) says when it fails.

The momentum transfer stays inside. At one ω, different $\boldsymbol{q}$ make different pairs. Worked example 8.9 shows that at 10 eV in silicon the hole carries about 0.9 eV more, on average, for a dark-matter kick at q_0 than for an absorbed photon. This difference of 0.9 eV is close to the gap, the energy scale on which impact ionization switches on. If the spectrum is integrated over $\boldsymbol{q}$ first and the cascade is seeded afterwards with a distribution of pairs chosen at fixed ω, the pairs of one momentum are attached to the rate of another. The resolved response depends on the direction of $\boldsymbol{q}$ as well, so in a crystal the mix of seeds follows the dark-matter wind through the day, and a directional analysis needs the seeds at every direction it uses.

The velocity and the model stay inside. With one coupling row a, the weights of the classes in Eq. (8.70) are $S_{aa}(\boldsymbol{q}, \omega; \Lambda)/S_{aa}(\boldsymbol{q}, \omega)$, independent of $\boldsymbol{v}$ and of the model, and a table of $P(N_{\mathrm{eh}}|\boldsymbol{q}, \omega)$ per row serves every model that couples through that row. With several rows a table per row does not serve every model. The weights then depend on $\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})$, so on the coefficients of the model and on the velocity, and the off-diagonal elements $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$ populate the classes too. If single-row tables are mixed in proportion to the single-row rates, the mixture misses the interference of Sect. 8.2.5 class by class, even where the interference cancels in the total.

The operator structure stays inside. Different rows probe different parts of the electron's state and make different pairs at the same $(\boldsymbol{q}, \omega)$. A spin row reaches spin-flipped pairs that the density cannot. A current row weights each pair by the velocity of its carriers, not by the overlap of their periodic parts. The longitudinal current is the exception: the continuity equation Eq. (4.29) gives $\langle f|\hat{\boldsymbol{q}} \cdot \boldsymbol{j}_e(\boldsymbol{q})|i\rangle = (\omega/q)\langle f|n_e(\boldsymbol{q})|i\rangle$ for every pair of eigenstates, so the longitudinal current makes exactly the pairs the density makes. The transverse current, the spin density and the spin current make their own pairs, and the velocity-dependent operators of Table 8.1 reach the target through them.

The cascade depends on $\boldsymbol{q}$ mainly through its first generation. The electron and hole of the prompt pair carry the crystal momenta $\boldsymbol{k} + \boldsymbol{q}$ and $\boldsymbol{k}$. Where collisions randomize the momentum of a carrier faster than they change its energy, its later history depends on its energy alone, and the secondary carriers inherit nothing of $\boldsymbol{q}$; Sect. 18.1 states this reduction and the energies at which it fails. The difference between dark-matter and photon seeds then enters once, through the distribution of $(E_{\mathrm{e}}, E_{\mathrm{h}})$ and of the valleys at the start. This one difference is enough to change the pair count at the few-pair level, where a single impact ionization more or less is a whole bin. The ionization model of Ramanathan and Kurinsky describes the split of the energy between the carriers by one distribution of the electron's energy at each total, fitted to the ionization yields of photon absorption [24, Eqs. (5)–(7)]. This distribution is the $\boldsymbol{q} \to 0$ marginal of the seeds of Eq. (8.70), and using it for dark-matter scattering replaces the seeds of one momentum by the seeds of another; Sect. 18.3 sizes the

error. For absorption the momentum is $p_\phi = m_\phi v \approx 10^{-3} m_\phi$, the transitions are vertical, and a dark photon is absorbed through the same current as a photon. Its seeds are the photon's, and a photon calibration applies to it exactly.

We have derived the identification of the prompt pair with the out-state of Eq. (8.68) only within time-dependent mean-field linear response, where the bare pair is the asymptotic state of the evolution. The exact final states of an interacting insulator are not labeled by pairs, and it is not established whether the retarded weights give the distribution of the carriers found at late times beyond mean field. The assumption behind Eq. (8.69) is that the class fixes the statistics of the cascade. This assumption holds when the only coherence between production and decay is the one the Fano coefficients carry; the widths of the prompt states would set a bound on what else the history of an event can contribute, but this bound is not known.

8.8 Uncertainty structure

The local fields of silicon raise its screening factor at small momentum transfer by a quarter (Sect. 4.6), and at a momentum transfer of 7.5 keV they lower the screened structure factor by 22 % to 44 % between 2 and 10 eV (Worked example 8.10). If a limit is computed with the random-phase approximation alone, it carries the local-field difference whether or not anyone states the difference. Sect. 1.5 called such errors the theory uncertainty of a limit: they come from approximations, and a longer exposure does not reduce them. Sect. 8.5 gave every approximation of the response a place on the truncation label $\mathfrak{T}$, and with it a next term and, where the axis has a small parameter, a size. Each truncation still has to become a number that a likelihood can use. A theory uncertainty built this way is computed from the physics, and two analyses that use the same truncation share it.

8.8.1 The uncertainty of a truncation

A formula lies at a point $\mathfrak{T}$ of Eq. (8.44). Let $\mathfrak{T}_k^+$ be the label one step further along the axis k and nowhere else. On the integer axes the step goes to the next order at which the labeled quantity changes, by the counting rule of Sect. 8.5: $n_g = 0 \to 2$, $n_B = 0 \to 2$, $n_A = 0 \to 2$ and $n_T = 0 \to 1$. On the many-body axis it is the next approximation, from IP to RPA, to RPA + LF, to TDDFT or BSE. The lattice regime is not an order, and its step is the next term of the multiphonon series Eq. (5.53), which is known exactly. The difference between the two responses and its size relative to $S^{\mathfrak{T}}$ are

$$\Delta S_k(\boldsymbol{q},\omega) = S^{\mathfrak{T}_k^+}(\boldsymbol{q},\omega) - S^{\mathfrak{T}}(\boldsymbol{q},\omega), \qquad \delta_k(\boldsymbol{q},\omega) = \frac{\Delta S_k(\boldsymbol{q},\omega)}{S^{\mathfrak{T}}(\boldsymbol{q},\omega)}. \tag{8.71}$$

In the likelihood of a search (Sect. 21.4) the truncation is represented by a parameter ϑ_k that interpolates between the two responses. For a single response, such as S_{ee} or the contraction $\sum_{ab} \mathfrak{D}_{ab} S_{ab}$ of one model,

$$S(\boldsymbol{q},\omega;\vartheta) = S^{\mathfrak{T}}(\boldsymbol{q},\omega) \prod_k \left[1 + \delta_k(\boldsymbol{q},\omega)\right]^{\vartheta_k}, \tag{8.72}$$

so that $\vartheta_k = 0$ is the formula, $\vartheta_k = 1$ is its next order, and ϑ_k is given a standard normal constraint in the likelihood. To first order, $S(\vartheta) = S^{\mathfrak{T}} + \sum_k \vartheta_k \ln(1+\delta_k)\, S^{\mathfrak{T}}$, the familiar additive form. The power form differs from it at second order, and the difference matters: $S^{\mathfrak{T}}(1 + \vartheta_k \delta_k)$ turns negative at $\vartheta_k = -1/\delta_k$, two standard deviations out for $\delta_k = 0.5$, while Eq. (8.72) is positive for every ϑ.

The parameter ϑ_k does not vary from bin to bin: a truncation is a property of the calculation, and the same ϑ_k acts at every $\boldsymbol{q}$ and ω, in every bin and on every final-state class Λ of Sect. 8.7. It is also shared by every target computed with the same approximation and by every experiment that uses the result. Only the template $\delta_k(\boldsymbol{q},\omega)$ varies. The single shared value distinguishes ϑ_k from an error bar attached to each bin. If a truncation lowers the spectrum near the gap and raises it at higher energy, it is one parameter; treating the two regions as independent errors would let the data fit a shape that no calculation can produce.

When is ΔS_k computable? The next order may have been computed, it may be estimated from a small parameter, or neither may be possible. In the first case the next order is known, as for the next many-body approximation of Sect. 4.6, the zero-point renormalization of a gap at $n_g = 2$ (Sect. 6.3) or the next term of Eq. (5.53). The template is then a number at every point, and the better result usually becomes the central value, with the uncertainty carried by the step after it. For the many-body approximations that step is again a computed difference, and like every many-body spread it is an estimate, not a bound (Sect. 4.9). In the second case the next order has not been computed, but the axis has a small parameter whose size Table 8.4 gives. Then δ_k is the parameter raised to the power of the step, times a coefficient of order one, between a quarter and two in the silicon counting of Sect. 5.6. The uncertainty is then an estimate and is labeled as such. In the third case there is neither: no computed next order and no small parameter. The size of the truncation is then not known, and no width is assigned to it. The pair-channel screening of the phonon-assisted sideband is the example of this chapter. Sect. 8.4 showed that dropping it is an omission, not a choice of convention, and Sect. 13.4 gives only its two solvable limits, which bracket it loosely and do not fix its sign. No calculation has yet sized the pair-channel screening. Until one does, a rate that omits the pair-channel screening is reported at its nominal value with the omission named, because a width invented for the omission would hide that its size is unknown.

Approximations off the label carry theory uncertainties too. The exchange–correlation functional, the frozen core and the numerical parameters of a calculation have no next order, and their sizes come from the spreads and convergence tests of Chaps. 23 to 27. These approximations enter the likelihood in the same way, with the spread in place of ΔS_k.

For a response matrix S_{ab} the power form needs care. Raising each element to its own power leaves the cone of positive semidefinite matrices, and $\delta_{k,ab}$ is undefined wherever $S^{\mathfrak{T}}_{ab} = 0$. The matrix analog of Eq. (8.72) is

$$S(\vartheta_k) = S^{1/2}\big(S^{-1/2}S^{+}S^{-1/2}\big)^{\vartheta_k}S^{1/2}, \qquad S \equiv S^{\mathfrak{T}}(\boldsymbol{q},\omega),\ S^{+} \equiv S^{\mathfrak{T}^{+}_{k}}(\boldsymbol{q},\omega), \tag{8.73}$$

which reduces to Eq. (8.72) for one row. It equals S at $\vartheta_k = 0$ and S^{+} at $\vartheta_k = 1$. It is positive semidefinite for every $\vartheta_k \geq 0$: $S^{-1/2}S^{+}S^{-1/2}$ is positive semidefinite, so is any nonnegative power of it, and the congruence $X \to S^{1/2}XS^{1/2}$ preserves the property. The inverse square root is taken on the subspace where S is nonzero, and this restriction requires S^{+} to have no weight outside that subspace. A profile explores negative ϑ_k as well, and a negative power exists only if S^{+} is positive definite on that subspace: the two truncations must have the same support, $\operatorname{supp} S^{+} = \operatorname{supp} S$, and then Eq. (8.73) is positive semidefinite for every real ϑ_k. Where the next order opens new support, Eq. (8.75) below applies instead. Several truncations are applied one after another; their order matters only at second order in the δ_k.

8.8.2 Scale, tilt and edge shifts

A truncation uncertainty scales a spectrum, tilts it or moves its edge, and each of the three effects moves a limit differently.

A *normalization* uncertainty has a template δ_k that is constant over the energies a rate samples at fixed $\boldsymbol{q}$. It scales the spectrum and leaves its shape unchanged, and by Eq. (1.53)

it moves the limit by the same fraction. It also leaves the weights of the final-state classes unchanged, so it passes to the pair-number spectrum of Eq. (8.70) as an overall factor (Sect. 8.7).

A *tilt* has a template that varies across the energies a rate samples. It moves weight between bins, and its effect on a limit depends on which bins the dark-matter spectrum fills, so on the mass and the mediator. A tilt changes the mix of final-state classes as well, and with it the pair-number spectrum.

A *threshold* uncertainty moves an edge,

$$S(\boldsymbol{q},\omega;\vartheta_k) = S^{\mathfrak{T}}(\boldsymbol{q},\omega - \vartheta_k\,\delta E_k), \tag{8.74}$$

with δE_k the shift that the next order produces. A shift acts on the limit through the edge of Sect. 1.3.5, $\delta \ln m_{\min} = \delta \ln E_{\text{th}}$, and near the edge it moves the limit by a far larger factor than its fractional size, because $\eta(v_{\min})$ falls steeply there (Sect. 1.5.5).

Some truncations do more than deform the spectrum: their next order puts weight where the truncated formula has none. Examples are the exciton lines of silicon below the pair continuum, which dark matter with $\boldsymbol{q} \approx \boldsymbol{k}_{\text{c}}$ can create directly (Sect. 4.9), and the two-phonon response above the top of the one-phonon band. No power of $1 + \delta_k$ can create weight from zero. When the formula drops a positive piece, the omission is an error of known sign, so once computed the piece belongs in the central value. The parameter ϑ_k of the new piece then acts on the size of that piece,

$$S(\boldsymbol{q},\omega;\vartheta_k) = S^{\mathfrak{T}}(\boldsymbol{q},\omega) + \Delta S_k(\boldsymbol{q},\omega)\,(1+\delta'_k)^{\vartheta_k}, \tag{8.75}$$

where δ'_k is the fractional size of the next correction to the new piece. The rate stays positive for every ϑ_k as long as ΔS_k is itself a positive spectrum. ΔS_k is positive for new lines and for a sideband inside the off-shell window of Sect. 8.6. Outside the window positivity fails, since the finite-part terms of Eq. (8.59) are not separately positive, and there ϑ_k acts on the zero-phonon and phonon-dressed sectors together.

Table 8.5 gives an example of each type in each sector of Eq. (8.48), for silicon wherever the size is known. Two entries share one parameter. The quasiparticle residue z_{qp} lowers the zero-phonon pair weight, and Eq. (8.60) returns exactly that weight to the finite part of the sideband. One parameter acts on both sectors with opposite signs, and the sum of the two sector weights does not move. Treated as two independent parameters, the residue and the sideband would double the uncertainty of a quantity that the identity holds fixed.

The Debye–Waller entry uses $2W_\kappa = 0.063$ at 10 keV (Sect. 8.5): the correct one-phonon rate is lower by $1 - \mathrm{e}^{-0.063} = 6.1\,\%$ than a rate that sets $\mathrm{e}^{-2W_\kappa} = 1$, at that momentum and uniformly in ω. The template is exact, and its value grows with q. The two-phonon term of the new-support row is the other half of the same step of Eq. (5.53): e^{-2W_κ} removes weight from the one-phonon term and passes it to the two-phonon term and to the elastic line, and the two entries carry one ϑ. In the gap entry the central value moves. A clamped-nucleus gap is at $n_g = 0$, and its computed next order is the zero-point renormalization, -56.2 meV in the nonadiabatic form (Sect. 6.3). Once the renormalization is included, the remaining uncertainty is the step after it; two numbers give the size of this step, the 8 meV between the adiabatic and nonadiabatic forms and the difference to the measured -62 meV to -64 meV. The anharmonic shift uses the counting $\eta^2_{\text{anh}}\omega_{\text{ph}}$ of Sect. 5.6, with $\eta^2_{\text{anh}} = 2.4 \times 10^{-3}$ and $\omega_{\text{opt}} = 64.1$ meV. The thermal entry is an uncertainty whose template has been computed and found to vanish: at 10 mK the occupations of Sect. 8.5 make it smaller than any number a likelihood can resolve.

Table 8.5: Types of truncation uncertainty, with one truncation of each type in each sector of the decomposition Eq. (8.48): $\mathcal{P}_{\mathrm{ad}}$ (electrons following the nuclei; phonons and recoils), $\mathcal{P}_0$ (zero-phonon electronic excitations) and $\mathcal{P}_+$ (electronic excitations with phonons). The axis of the truncation label is given in parentheses. Sizes are for silicon, and the mark "est." denotes an estimate from a small parameter of Table 8.4. The residue entry in $\mathcal{P}_0$ and the finite-part entry in $\mathcal{P}_+$ carry one parameter, tied by Eq. (8.60); the two $\mathcal{K}$ entries of $\mathcal{P}_{\mathrm{ad}}$, the two parts of one step of Eq. (5.53), also carry one parameter.

Type	Form	$\mathcal{P}_{\mathrm{ad}}$	$\mathcal{P}_0$	$\mathcal{P}_+$
Normalization	Eq. (8.72), δ_k constant in ω	Debye–Waller factor e^{-2W_κ} dropped from a one-phonon rate ($\mathcal{K}$): $\delta = -0.061$ at 10 keV	residue z_{qp} of the pair ($n_g = 2$); est. of order $\theta_{\mathrm{ep}} \sim 10^{-2}$	static screening of the vertex (n_B): est. $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2 = 3 \times 10^{-3}$; finite part tied to the residue
Tilt	Eq. (8.72), $\delta_k(\omega)$	anharmonic reshaping of the recoil spectrum at fixed area and mean ($n_A = 2$): Eq. (8.77)	local-field correction, RPA $\to$ RPA + LF ($\mathcal{X}$): -0.44 to $+0.25$ across q and ω, Worked example 8.10	two more vertices ($n_g = 4$): est. $\theta_{\mathrm{ep}} \approx 10^{-2}$ below E_{dir}, rising to 1 near it
Threshold	Eq. (8.74)	anharmonic shift of the optical phonon ($n_A = 2$): est. $\eta^2_{\mathrm{anh}}\omega_{\mathrm{opt}} = 0.15$ meV	zero-point renormalization of the gap ($n_g = 2$): $\delta E = -56.2$ meV; adiabatic against nonadiabatic 8 meV	exciton binding ($\mathcal{X} = \mathrm{IP} \to \mathrm{BSE}$): every phonon-assisted threshold 14.7 meV lower
New support	Eq. (8.75)	two-phonon response above the one-phonon band ($\mathcal{K}$)	exciton lines below the pair continuum at $\boldsymbol{q} \approx \boldsymbol{k}_{\mathrm{c}}$ ($\mathcal{X} = \mathrm{BSE}$)	phonon-absorption branch ($n_T = 1$): weight e^{-21550} at 10 mK

8.8.3 Exact relations as constraints

The relations of Sect. 8.3 hold for the exact response and for every truncation that respects them. If a value of ϑ_k breaks one of them, the value describes no possible calculation, and the relations are therefore constraints on the parameters ϑ_k, not further sources of uncertainty.

Positivity. The response matrix is positive semidefinite at every $(\boldsymbol{q}, \omega)$ (Eq. (4.20)), and $S(\vartheta)$ must be positive semidefinite for every ϑ that the likelihood explores. Three forms satisfy this requirement by construction: the power form Eq. (8.72), the matrix form Eq. (8.73) and the new-support form Eq. (8.75) with a positive ΔS_k. The additive form $S^{\mathfrak{T}} + \vartheta_k \Delta S_k$ does not, wherever $|\Delta S_k|$ is comparable to $S^{\mathfrak{T}}$, which is common near edges.

The interference bound. The cross term of two groups of rows can never exceed twice the geometric mean of their separate rates, Eq. (8.21). A parameter on the cross term, however, is not free. If a truncation scales the separate rates by $(1+\delta_A)^{\vartheta}$ and $(1+\delta_B)^{\vartheta}$, the largest cross term allowed scales by $[(1+\delta_A)(1+\delta_B)]^{\vartheta/2}$, and a truncation describes no target if it grows the cross term faster than the separate rates. In practice the parameters are assigned to the matrix, through Eq. (8.73), and never to an interference term alone; the bound is then kept automatically. The same holds for the mixed electron–nucleus response of the Migdal effect, where $|S_{e\kappa}|^2 \leq S_{ee} S_{\kappa\kappa}$.

Sum rules. The first moment of the density response is fixed by the f-sum rule (Eq. (4.28)) for any approximation that conserves particle number with local potentials. Every approximation of Sect. 4.6 is of that kind, from the bubble through the random-phase approximation with or without local fields to the adiabatic kernel: each adds static, local terms that leave the tail coefficient of the response unchanged. For these truncations

$$\int_{-\infty}^{\infty} \mathrm{d}\omega \; \omega \, \Delta S_k(\boldsymbol{q}, \omega) = 0. \tag{8.76}$$

A many-body uncertainty moves weight in energy but cannot add or remove first moment, and its template must change sign: what it lowers in one range it raises in another. A normalization uncertainty on the whole spectrum would violate Eq. (8.76), so the many-body uncertainties are tilts. The relation is not vacuous for a rate, which samples only energies below $\frac{1}{2} m_\chi v_{\max}^2$. Within that window a many-body uncertainty may look like a normalization, and Eq. (8.76) says the compensating weight lies above the window. Sect. 4.6 saw the local fields push weight up in the optical spectrum. Some truncations do not conserve the f-sum: a cut in the band sum or in q, a nonlocal pseudopotential, a scissor shift of the conduction bands. For them the violation of the sum rule measures the error, and the template can be calibrated against it. The partial sum rules of Sect. 8.6 act in the same way between sectors, and the residue identity Eq. (8.60) is one of them.

Detailed balance. In a warm target the response at negative energies is the Boltzmann-weighted image of the response at positive energies (Eq. (2.68)), and the dark-matter rates in the two directions are tied by Eq. (8.20). A parameter applied to $\omega > 0$ alone would break the tie. The same ϑ_k must act on both signs of ω, with the template at $(-\boldsymbol{q}, -\omega)$ fixed by the one at $(\boldsymbol{q}, \omega)$.

Moments of the recoil spectrum. The self response of a nucleus has a fixed zeroth moment, the norm $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} \rangle = 1$, which no Hamiltonian changes, and a fixed first moment, the recoil energy E_R of Eq. (5.55). The derivation of Eq. (5.55) used only the kinetic energy of the nucleus, so both moments hold on any energy surface, anharmonic ones included. The anharmonic truncation $n_A = 0 \to 2$ can only reshape the recoil spectrum at fixed area and fixed mean. The power form keeps both moments only at $\vartheta = 0$ and 1. A *constrained parameter* keeps them at every ϑ. With $y = \omega - E_R$ and $\psi = \ln(1+\delta_A)$, the template of the

anharmonic step, the constrained form reads

$$S(\boldsymbol{q},\omega;\vartheta) = S^{\mathfrak{T}}(\boldsymbol{q},\omega)\,\exp\bigl[\vartheta\,\psi(\omega) + \lambda_0(\vartheta) + \lambda_1(\vartheta)\,y\bigr], \tag{8.77}$$

where λ_0 and λ_1 are fixed at each ϑ by requiring the zeroth and first moments to equal the moments of $S^{\mathfrak{T}}$. Positivity is manifest. One convex function shows that λ_0 and λ_1 exist and are unique. Write $m_n(\vartheta) = \int \mathrm{d}\omega\, y^n S(\boldsymbol{q},\omega;\vartheta)$ for the moments in the variable y, and let $F(\lambda_0,\lambda_1) = \int \mathrm{d}\omega\, S^{\mathfrak{T}} \exp(\vartheta\psi + \lambda_0 + \lambda_1 y) - \lambda_0 m_0(0) - \lambda_1 m_1(0)$. Its gradient vanishes exactly where $m_0(\vartheta) = m_0(0)$ and $m_1(\vartheta) = m_1(0)$, and its matrix of second derivatives is the moment matrix $\left(\begin{smallmatrix} m_0 & m_1 \\ m_1 & m_2 \end{smallmatrix}\right)$ of the deformed spectrum. This moment matrix is the Hankel matrix of Eq. (8.23), positive definite for any spectrum with more than one point of support. As a result F is strictly convex and has at most one stationary point, which exists when the prescribed mean lies inside the support, as the free-recoil energy does. Since $S^{\mathfrak{T}_A^+}$ obeys the same two moment conditions, the solution at $\vartheta = 1$ is $\lambda = 0$, and Eq. (8.77) passes through the next order exactly. The second moment is not protected. It contains the mean kinetic energy of the nucleus along $\boldsymbol{q}$, which anharmonicity changes, and the change of the second moment is a separate parameter with its own computed size. The tilt of Eq. (8.77) is not limited to the nuclear self response: any truncation that must preserve known linear moments uses it, with one λ per moment.

Worked example 8.10: The local-field uncertainty of the random-phase rate in silicon

If a rate is computed with the random-phase approximation of the head element alone, Eq. (4.67), it is at $\mathcal{X} = \mathrm{RPA}$. The next approximation adds the local fields, $\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$ of Eq. (4.68), and the template of the local-field uncertainty is $\delta_{\mathcal{X}} = S^{\mathrm{RPA+LF}}/S^{\mathrm{RPA}} - 1$ for the density structure factor. We compute both approximations in the empirical pseudopotential model of silicon used in Worked example 8.9, with the same bands, the same 51 local-field vectors and the same Lorentzian half-width of 0.10 eV, at $q_0 = 3.73\,\mathrm{keV}$ and $2q_0 = 7.46\,\mathrm{keV}$ along [100] and at energies from 0.5 to 100 eV. The model gap is 0.82 eV, and we quote energies from 2 eV up. The quoted values change by 0.01 or less between 16^3 and 20^3 points of $\boldsymbol{k}$.

At q_0 the template is small and has structure. It is -0.06 at 2 and 3 eV, passes through zero at 4 eV, is -0.03 at 5 to 7 eV and -0.06 at 10 eV, and reaches its minimum, -0.15, near 17.5 eV. It crosses zero at 25.6 eV and is $+0.02$ at 30 eV. At $2q_0$ the local fields matter more: $\delta_{\mathcal{X}} = -0.44$ at 2 eV, -0.30 at 4 eV, -0.22 at 10 eV and -0.06 at 20 eV, with the zero near 29 eV. At small q, finally, the local-field reduction of ε_∞ alone raises the screening factor $1/\varepsilon_\infty^2$ by $(11.45/10.22)^2 \simeq 1.25$ (Sect. 4.6), so that $\delta_{\mathcal{X}} \approx +0.25$ at the bottom of the spectrum.

The local-field uncertainty is a tilt, in ω and in q. Weighted with the random-phase spectrum between 1.5 and 30 eV, the mean of the template is -0.06 at q_0 and -0.09 at $2q_0$. A normalization uncertainty of that size would understate the template near the gap at $2q_0$ by a factor of about five and would have the wrong sign at small q. The f-sum rule explains the change of sign. Over 0.5 to 100 eV the first moments of the two approximations agree to 1×10^{-4} at q_0 and 2×10^{-4} at $2q_0$, a residual difference that the Lorentzian tails and the finite grid leave, while the zeroth moments fall by 2.1 % and 3.0 %. The local fields move weight upward, as Eq. (8.76) demands and as Sect. 4.6 found for the optical spectrum, and the weight they remove below 25 eV reappears above it. A particle of mass m_χ deposits at most $\frac{1}{2} m_\chi v_{\max}^2 = 3.5\,\mathrm{eV}$ per MeV, so dark matter lighter than about 7 MeV samples only energies where $\delta_{\mathcal{X}} \le 0$ at these momenta and never meets the compensating weight.

In the likelihood a single $\vartheta_{\mathcal{X}}$ multiplies $\ln(1+\delta_{\mathcal{X}})$ at every point. At $\vartheta_{\mathcal{X}}=1$ the rate is that of the random-phase approximation with local fields. Once that approximation is computed it becomes the central value, and the uncertainty moves to the step to the adiabatic kernel or to the Bethe–Salpeter equation. The adiabatic kernel restores about half of the local-field reduction of ε_∞ (Sect. 4.6), so the template of the kernel step has the opposite sign at small q, and near the gap the Bethe–Salpeter step is the larger one (Worked example 8.7).

Summary

A slow dark-matter particle sees an electron or a nucleus only through the momentum it hands over, their relative velocity and their spins (Sect. 8.1). Every nonrelativistic interaction therefore reduces to five target operators per species: the density, the spin density, the current, the spin current and a kinetic-energy density. The model survives only in the dark-matter-side tensor $\mathfrak{D}_{ab}$, a Hermitian positive semidefinite matrix that weights the response matrix of these operators.

Integrated over the momentum transfer and the halo, the contraction of $\mathfrak{D}_{ab}$ with S_{ab} is the master rate formula (Sect. 8.2). It holds at any temperature and for energies of either sign, and its integrand is never negative. Scattering fills a region of the (q,ω) plane, absorption a point, and the absorption of a fermion a line.

Exact relations constrain every target (Sect. 8.3). Positivity of the response bounds any interference term by twice the geometric mean of the separate rates. The frequency moments of the response form matrices, the odd ones fixed by commutators, the even ones by equal-time correlations, and all of them obey Hankel positivity.

Screening is a property of the coupling, not of the target alone (Sect. 8.4). Only a coupling to the total charge reproduces the loss function at every $(\boldsymbol{q},\omega)$, and an absorbed dark photon in silicon at 3 eV is suppressed by the factor $|\varepsilon|^2=761$ of in-medium mixing.

Every rate is a truncation of one expansion in seven small parameters (Sect. 8.5), among them, in silicon at 10 mK, the Born–Oppenheimer parameter 0.0665, the off-shell parameter 0.010 to 0.016 at the edge and the anharmonic parameter 0.049. The truncation label $\mathfrak{T}$ records the order that a formula keeps in each small parameter, together with the many-body approximation, the lattice regime and the Migdal regime that the formula uses.

In a target with a gap the final states split into three sectors (Sect. 8.6): electrons following the nuclei, electronic excitations without phonons and electronic excitations with phonons. The split is unique to the order of the off-shell parameter inside the off-shell window.

The final-state-resolved response sums to the full response (Sect. 8.7). At q_0 and 10 eV in silicon, dark matter leaves the electron with 4.8 eV and the hole with 4.4 eV on average, while a photon of the same energy gives 5.7 eV and 3.5 eV. The cascade must therefore start from the pair distribution at the momentum transfer that made the pairs.

Each truncation becomes a parameter ϑ_k whose size is the computed step to its next order (Sect. 8.8). These parameters act as normalizations, tilts or threshold shifts, or open new support, and the exact relations constrain them. In silicon the step from the random-phase approximation to local fields is such a tilt, from -0.44 to $+0.25$ across momentum and energy.

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Appendix A

Conventions and units

A.1 Natural units

We set $\hbar = c = k_\mathrm{B} = 1$. Energy, mass, momentum, frequency and temperature are then all measured in electronvolts, length and time in eV^{-1}, and speeds are pure numbers, fractions of the speed of light. A dark-matter speed of $250.5\,\mathrm{km\,s^{-1}}$ is 8.356×10^{-4}, and a particle of mass m_χ at this speed carries the energy $\frac{1}{2} m_\chi v^2$ and the momentum $m_\chi v$, both in eV. Natural units suit a subject in which a particle-physics mass, a crystal momentum, a phonon frequency and a temperature enter one energy-conservation condition. Particle physics uses natural units routinely, and Ref. [1, Appendix A, Table III] tabulates the astrophysical units they are mixed with. Condensed-matter texts often keep $\hbar$ and k_B, and their formulas are converted by restoring the constants as described below.

A.1.1 Restoring the constants

A formula in natural units is converted to conventional units by inserting the powers of $\hbar$, c and k_B that make every term dimensionally consistent. Four constants suffice; their values follow from the 2022 adjustment of CODATA (the Committee on Data of the International Science Council) [2], in which c, h, e and k_B are exact:

$$\begin{aligned} \hbar c &= 1.973\,270 \times 10^{-5}\,\mathrm{eV\,cm} = 197.3270\,\mathrm{MeV\,fm}, & \hbar &= 6.582\,120 \times 10^{-16}\,\mathrm{eV\,s}, \\ k_\mathrm{B} &= 8.617\,333 \times 10^{-5}\,\mathrm{eV\,K^{-1}}, & c &= 2.997\,925 \times 10^{5}\,\mathrm{km\,s^{-1}}. \end{aligned} \tag{A.1}$$

A length L becomes $L/(\hbar c)$ in eV^{-1}, a time t becomes $t/\hbar$, a temperature T becomes $k_\mathrm{B}T$ and a speed u becomes u/c. A density per unit volume picks up $(\hbar c)^3$, a cross-section $(\hbar c)^{-2}$, and a rate per unit time $\hbar$.

Table A.1 lists the factors. Each entry is a power of one of the products in Eq. (A.1). For example, $1\,\mathrm{cm^{-3}} = (\hbar c)^3/\mathrm{cm^3} = (1.973\,270 \times 10^{-5}\,\mathrm{eV})^3 = 7.6835 \times 10^{-15}\,\mathrm{eV^3}$, and $1\,\mathrm{b} = 10^{-24}\,\mathrm{cm^2}$ divided by $(\hbar c)^2 = 3.8938 \times 10^{-10}\,\mathrm{eV^2\,cm^2}$ gives $2.5682 \times 10^{-15}\,\mathrm{eV^{-2}}$. A mass density in $\mathrm{g\,cm^{-3}}$ first becomes an energy density through $1\,\mathrm{g} = 5.610 \times 10^{23}\,\mathrm{GeV}$ and then picks up $(\hbar c)^3$. Spectroscopic frequencies need care: a frequency ν quoted in THz corresponds to the energy $h\nu = 2\pi\hbar\nu$, while an angular frequency ω in $\mathrm{rad\,s^{-1}}$ corresponds to $\hbar\omega$. Phonon tables usually quote ν, so the optical phonon of silicon at the zone center, at 15.5 THz [3], has $\omega_{Q\nu} = 64.1\,\mathrm{meV}$.

A.1.2 Checking dimensions

In natural units every quantity has a single dimension, a power of energy, and a formula is checked by counting powers of eV. The rate per unit target mass of Eq. (1.46) is the

Table A.1: Conversion factors between conventional units and natural units ($\hbar = c = k_B = 1$), computed from CODATA 2022 values [2]. Examples in the last column are values used in the book.

Quantity	Conversion	Example
length	$1\,\mathrm{cm} = 5.0677 \times 10^4\,\mathrm{eV}^{-1}$, $1\,\text{Å} = 5.0677 \times 10^{-4}\,\mathrm{eV}^{-1}$	Bohr radius $1/(\alpha m_e) = 0.5292\,\text{Å}$
time	$1\,\mathrm{s} = 1.5193 \times 10^{15}\,\mathrm{eV}^{-1}$	$1\,\mathrm{yr} = 3.156 \times 10^7\,\mathrm{s} = 4.795 \times 10^{22}\,\mathrm{eV}^{-1}$
temperature	$1\,\mathrm{K} = 8.6173 \times 10^{-5}\,\mathrm{eV}$, $1\,\mathrm{eV} = 11\,605\,\mathrm{K}$	$300\,\mathrm{K} = 25.85\,\mathrm{meV}$
speed	$1\,\mathrm{km\,s^{-1}} = 3.3356 \times 10^{-6}$	$v_{\max} = 794.5\,\mathrm{km\,s^{-1}} = 2.650 \times 10^{-3}$
number density	$1\,\mathrm{cm}^{-3} = 7.6835 \times 10^{-15}\,\mathrm{eV}^3$	$\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}} = 2.31 \times 10^{-6}\,\mathrm{eV}^4$
mass density	$1\,\mathrm{g\,cm^{-3}} = 4.310 \times 10^{18}\,\mathrm{eV}^4$	silicon, $2.33\,\mathrm{g\,cm^{-3}} = 1.004 \times 10^{19}\,\mathrm{eV}^4$
mass	$1\,\mathrm{kg} = 5.610 \times 10^{26}\,\mathrm{GeV}$	–
cross-section	$1\,\mathrm{cm}^2 = 2.5682 \times 10^9\,\mathrm{eV}^{-2}$, $1\,\mathrm{b} = 2.5682 \times 10^{-15}\,\mathrm{eV}^{-2}$	$1\,\mathrm{GeV}^{-2} = 0.3894\,\mathrm{mb} = 3.894 \times 10^{-28}\,\mathrm{cm}^2$
spectroscopy	$1\,\mathrm{THz} = 4.1357\,\mathrm{meV}$ $(h\nu)$, $1\,\mathrm{cm}^{-1} = 0.123\,98\,\mathrm{meV}$	–

standard example: it is dimensionless, as a rate (eV) per unit mass (eV) must be (Sect. 1.5). The dimensions of the main objects are fixed by their definitions: the dynamic structure factor $S(q,\omega)$ has dimension eV^2 (Sect. 4.2), the transform $\tilde{V}(\boldsymbol{q})$ of a potential has eV^{-2}, the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ and the dielectric function are dimensionless, and $\bar{\sigma}_e$ has eV^{-2}. Final numbers are always converted back: rates to $\mathrm{kg^{-1}\,yr^{-1}}$, cross-sections to cm^2, momenta to keV, and energies to eV or meV. A rate per unit mass in natural units is converted with $1\,\mathrm{kg^{-1}\,yr^{-1}} = 3.718 \times 10^{-59}$, the inverse of the product of 1 kg in eV and 1 yr in eV^{-1}.

The metric signature is $(+,-,-,-)$, with $x^\mu = (t,\boldsymbol{x})$ and $\partial_\mu\partial^\mu = \partial_t^2 - \nabla^2$, so that a free particle obeys $p^\mu p_\mu = E^2 - |\boldsymbol{p}|^2 = m^2$. Most particle-physics texts use the same signature, and so does Appendix C.9. Physical values in the text are quoted in conventional units with a thin space, 1.17 eV or $0.3\,\mathrm{GeV\,cm^{-3}}$, and natural-unit values are written without units only for speeds.

A.2 Charge conventions

The elementary charge is $e > 0$ with

$$e^2 = 4\pi\alpha, \qquad \alpha = 1/137.035999, \tag{A.2}$$

the rationalized Heaviside–Lorentz convention of particle physics [2]. The electron carries charge $-e$ and a nucleus of species κ carries $+Z_\kappa e$. The charge density operator is therefore e times the difference between the Z_κ-weighted sum of the nuclear densities and the electron density. Two electrons at distance r have the interaction energy $e^2/(4\pi r) = \alpha/r$, and Gauss's law states that $\nabla\cdot\boldsymbol{E}$ equals the charge density, without a factor 4π (Appendix C.9). The Coulomb interaction in momentum space, with the Fourier convention of Appendix A.3, is the kernel

$$v_C(k) \equiv \frac{e^2}{k^2} = \frac{4\pi\alpha}{k^2}, \tag{A.3}$$

the transform of $e^2/(4\pi r)$ by Eq. (C.79) at $m_V = 0$. In a crystal the Coulomb kernel is a diagonal matrix in reciprocal-lattice vectors, $(v_C)_{\boldsymbol{G}\boldsymbol{G}'} = v_C(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta_{\boldsymbol{G}\boldsymbol{G}'}$ (Sect. 4.4).

A.2.1 Heaviside–Lorentz units and the loss function

Dark-matter couplings are written in field theory, where Heaviside–Lorentz units are standard: the dark-photon coupling to the electron, $y_e = -\zeta e$, is the electromagnetic one scaled by ζ (Sect. 1.4), and the Yukawa potential of Eq. (1.30) has the same $1/(4\pi r)$ as Coulomb's law. The dielectric response of a solid is expressed in the same units, so the relation Eq. (A.36) between the structure factor and the loss function carries no factors of 4π [4, Eq. (6), 5, Eq. (S.7)]. In a dark-photon model the factor $1/e^2$ of Eq. (A.36) cancels against $y_e^2 = \zeta^2 e^2$, and the rate depends only on ζ^2 and the loss function [5, below Eq. (S.8)].

A.2.2 Converting from Gaussian and SI units

Condensed-matter sources usually use Gaussian units, in which the electron charge e_G obeys $e_\mathrm{G}^2 = \alpha$ (with $\hbar = c = 1$), the Coulomb energy is e_G^2/r and the Coulomb kernel is $4\pi e_\mathrm{G}^2/k^2$. SI sources write $e^2/(4\pi\varepsilon_0^\mathrm{SI} r)$, with $\varepsilon_0^\mathrm{SI}$ the vacuum permittivity; with $\hbar = c = 1$, $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI} = 4\pi\alpha$. The three charges are therefore related by

$$e^2 = 4\pi e_\mathrm{G}^2 = \frac{e_\mathrm{SI}^2}{\varepsilon_0^\mathrm{SI}} = 4\pi\alpha. \tag{A.4}$$

A formula is converted by replacing every e^2 of this book with $4\pi e_\mathrm{G}^2$ (to Gaussian) or $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI}$ (to SI), and the reverse for the other direction [6, Appendix]. Dimensionless response functions, ε, ε^{-1} and L, have the same value in every system; only the formulas that express them through charges change.

The plasma frequency is

$$\omega_p^2 = \frac{e^2 n_e}{m_e} = \frac{4\pi e_\mathrm{G}^2 n_e}{m_e} = \frac{e_\mathrm{SI}^2 n_e}{\varepsilon_0^\mathrm{SI} m_e},$$

in the three systems; all three give 16.6 eV for the four valence electrons per atom of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ at $a = 5.431\,\text{Å}$ (Sect. 4.3). The dielectric matrix is $\varepsilon_{\boldsymbol{GG'}} = \delta_{\boldsymbol{GG'}} - v_\mathrm{C}(|\boldsymbol{q}+\boldsymbol{G}|)\tilde{R}_{\boldsymbol{GG'}}$ (Appendix A.7). In the Gaussian sources of the local-field literature [7, 8] the same expression appears with $4\pi e_\mathrm{G}^2/|\boldsymbol{q}+\boldsymbol{G}|^2$, and the polarizability of those sources is our $\tilde{R}$ up to a sign and a factor $1/\mathcal{V}$, both fixed by each source in its own definition. For the optical conductivity of Appendix A.7, $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}/\omega$, the Gaussian form is $\varepsilon = 1 + 4\pi\mathrm{i}\sigma_\mathrm{cond}/\omega$ and the SI form $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}/(\varepsilon_0^\mathrm{SI}\omega)$, so a measured conductivity σ_1^SI in $\mathrm{S\,m^{-1}}$ gives $\varepsilon_2 = \sigma_1^\mathrm{SI}/(\varepsilon_0^\mathrm{SI}\omega)$. The prefactor $8\pi^2\alpha/(\Omega\omega^2)$ of the independent-particle ε_2 in the optical limit (Sect. 4.4) reads $8\pi^2 e_\mathrm{G}^2/(\Omega\omega^2)$ in the Gaussian sources, and the two prefactors agree under Eq. (A.4).

Fields convert as $\boldsymbol{E} = \boldsymbol{E}_\mathrm{G}/\sqrt{4\pi} = (\varepsilon_0^\mathrm{SI})^{1/2}\boldsymbol{E}_\mathrm{SI}$, so that $e\boldsymbol{E} = e_\mathrm{G}\boldsymbol{E}_\mathrm{G} = e_\mathrm{SI}\boldsymbol{E}_\mathrm{SI}$ and the force on a charge is the same in every system. Magnetic fields convert in the same way, $\boldsymbol{B} = \boldsymbol{B}_\mathrm{G}/\sqrt{4\pi} = (\varepsilon_0^\mathrm{SI})^{1/2}\boldsymbol{B}_\mathrm{SI}$ with $c = 1$. The Bohr magneton is then $\mu_\mathrm{B} = e/(2m_e)$, and the spin coupling of an electron is $g_e\mu_\mathrm{B}\,\boldsymbol{s}\cdot\boldsymbol{B}$, with $\boldsymbol{s}$ its spin and $g_e \simeq 2$; the product of μ_B and the field strength is an energy with the same value in every system, 5.788×10^{-5} eV per tesla.

The symbol ε_0 alone is the static dielectric constant of a crystal, lattice included (Sect. 5.4); the vacuum permittivity appears only as $\varepsilon_0^\mathrm{SI}$, inside quoted SI formulas.

A.3 Fourier transforms and normalization

A function of position is transformed with a plus sign in the exponent,

$$O(\boldsymbol{q}) = \int \mathrm{d}^3 r\; \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, O(\boldsymbol{r}), \qquad O(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\; \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, O(\boldsymbol{q}), \tag{A.5}$$

and a function of time with the same sign,

$$f(\omega) = \int \mathrm{d}t\, \mathrm{e}^{+\mathrm{i}\omega t} f(t), \qquad f(t) = \int \frac{\mathrm{d}\omega}{2\pi}\, \mathrm{e}^{-\mathrm{i}\omega t} f(\omega). \tag{A.6}$$

The factor $(2\pi)^{-1}$ per dimension is in the inverse transform. Applied to the density of point electrons, $n_e(\boldsymbol{r}) = \sum_i \delta^3(\boldsymbol{r} - \boldsymbol{r}_i)$, Eq. (A.5) gives the density operator

$$n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}, \tag{A.7}$$

and applied to the dark-matter–electron potential it gives $\tilde{V}(\boldsymbol{q}) = \int \mathrm{d}^3x\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} V(\boldsymbol{x})$, the form used in Eq. (1.31).

A.3.1 Sign of the spatial transform

The momentum transfer $\boldsymbol{q}$ is the momentum the dark matter gives to the target (Appendix A.4), and the direction of $\boldsymbol{q}$ fixes the sign of the spatial transform. Take dark-matter plane waves $\langle \boldsymbol{r}_\chi | \boldsymbol{p} \rangle = \mathscr{V}^{-1/2} \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}_\chi}$ in a box of volume $\mathscr{V}$, and a potential $V(\boldsymbol{r}_\chi - \boldsymbol{r}_e)$ between the dark matter and one electron. The matrix element of the potential follows by shifting the integration variable to $\boldsymbol{x} = \boldsymbol{r}_\chi - \boldsymbol{r}_e$:

$$\langle \boldsymbol{p} - \boldsymbol{q} | V(\boldsymbol{r}_\chi - \boldsymbol{r}_e) | \boldsymbol{p} \rangle = \frac{1}{\mathscr{V}} \int \mathrm{d}^3 r_\chi\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_\chi} V(\boldsymbol{r}_\chi - \boldsymbol{r}_e) = \frac{1}{\mathscr{V}}\, \tilde{V}(\boldsymbol{q})\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_e}. \tag{A.8}$$

With Eq. (A.5), the transform of the potential and the target operator $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_e}$ both carry $+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$, and summed over electrons, the target operator is $n_e(\boldsymbol{q})$ of Eq. (A.7). With the opposite spatial convention, $\tilde{V}$ would be evaluated at $-\boldsymbol{q}$ and the target operator would be $n_e(-\boldsymbol{q})$. Single-particle form factors follow the same rule: $f_{i\to f}(\boldsymbol{q}) = \langle f | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | i \rangle$ [9, Eq. (3.10)]. Ref. [5, Eq. (S.4)] uses the other spatial convention with the same direction of $\boldsymbol{q}$, and writes the density operator as $\sum_i \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$; Ref. [10, Eq. (4)] keeps $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ on the target by writing the potential's transform at $-\boldsymbol{q}$. Appendix A.4 gives the condition under which the choice of convention matters.

The time convention fixes where retarded functions are analytic. A response that starts at $t = 0$ and oscillates at a transition energy Ω transforms as $-\mathrm{i}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega-\Omega)t - 0^+ t} = 1/(\omega - \Omega + \mathrm{i}0)$. The pole of $1/(\omega - \Omega + \mathrm{i}0)$ lies in the lower half of the complex ω plane, so retarded functions are analytic for $\mathrm{Im}\,\omega > 0$, and their denominators carry $\omega - (E_f - E_i) + \mathrm{i}0$ (Sect. 4.1). A time-dependent perturbation of frequency $\omega > 0$ is $\propto \mathrm{e}^{-\mathrm{i}\omega t}$, the convention of quantum mechanics; engineering texts use $\mathrm{e}^{+\mathrm{i}\omega t}$ (written $\mathrm{e}^{j\omega t}$), and their dielectric functions are the complex conjugates of ours, with $\mathrm{Im}\,\varepsilon < 0$ for absorption. Here $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$ with $\varepsilon_2 > 0$ for an absorbing medium at $\omega > 0$.

A.3.2 Box normalization

States are normalized in a box of volume $\mathscr{V}$ with periodic boundary conditions, and the limit $\mathscr{V} \to \infty$ is taken at the end. Allowed wavevectors form a grid of density $\mathscr{V}/(2\pi)^3$, so sums and Kronecker deltas become integrals and delta functions by

$$\sum_{\boldsymbol{q}} \longleftrightarrow \mathscr{V} \int \frac{\mathrm{d}^3 q}{(2\pi)^3}, \qquad \delta_{\boldsymbol{q}\boldsymbol{q}'} \longleftrightarrow \frac{(2\pi)^3}{\mathscr{V}}\, \delta^3(\boldsymbol{q} - \boldsymbol{q}'). \tag{A.9}$$

The second correspondence follows from the first, since $\sum_{\boldsymbol{q}} \delta_{\boldsymbol{q}\boldsymbol{q}'} = 1$. The box contains N unit cells of volume Ω, so $\mathscr{V} = N\Omega$, and a uniform grid of crystal momenta in the first Brillouin

zone has N points (Sect. 3.1). Sums over the zone become

$$\frac{1}{N}\sum_{\boldsymbol{k}\in\mathrm{BZ}} \longleftrightarrow\ \Omega\int_{\mathrm{BZ}}\frac{\mathrm{d}^3k}{(2\pi)^3}, \qquad \langle\!\langle\cdots\rangle\!\rangle \equiv \frac{1}{N^2}\sum_{\boldsymbol{k},\boldsymbol{Q}}(\cdots) = \Omega^2\int_{\mathrm{BZ}}\frac{\mathrm{d}^3k\,\mathrm{d}^3Q}{(2\pi)^6}\,(\cdots), \tag{A.10}$$

where the double bracket is the zone average over an electron momentum $\boldsymbol{k}$ and a phonon momentum $\boldsymbol{Q}$, used for phonon-assisted processes (Sect. 6.5).

Every intensive quantity must come out independent of N, and counting the powers of N and $\mathscr{V}$ checks this independence mechanically. The structure factor of Sect. 4.2 has a prefactor $2\pi/\mathscr{V}$, an independent-particle sum $\sum_{\boldsymbol{k}}$ contributes a factor N, and the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is of order one (Appendix A.5). The product of the three factors is $2\pi N/\mathscr{V} = 2\pi/\Omega$, independent of the size of the crystal. A factor of N or $\sqrt{N}$ that survives this count signals an error in a normalization. In the same way the rate per unit target mass contains $1/\rho_T$ and no N (Appendix A.8).

A.3.3 Lattice-periodic functions

A function with the periodicity of the lattice, such as the crystal potential or the periodic part of a Bloch state, has components only at reciprocal-lattice vectors. With Eq. (A.5) restricted to one cell,

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\int_{\Omega}\mathrm{d}^3r\ \mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}f(\boldsymbol{r}), \qquad f(\boldsymbol{r}) = \sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \tag{A.11}$$

and the transform over the whole crystal is

$$f(\boldsymbol{q}) = \mathscr{V}\sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\delta_{\boldsymbol{q},\boldsymbol{G}}\ \longrightarrow\ (2\pi)^3\sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\delta^3(\boldsymbol{q}-\boldsymbol{G}),$$

by Eq. (A.9). Plane-wave codes use the opposite sign, with coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ in $u_{n\boldsymbol{k}}(\boldsymbol{r})$ [9, Eq. (3.14)]. A code coefficient $c(\boldsymbol{G})$ is therefore our component at $-\boldsymbol{G}$, and a formula that combines code coefficients with the components of Eq. (A.11) needs $\boldsymbol{G}\to-\boldsymbol{G}$ in one of the two sets. For a real function, $f_{-\boldsymbol{G}} = f^*_{\boldsymbol{G}}$, and the sign flip only conjugates the component; for a Bloch function this relation does not hold. Ref. [11, Appendix A] states the transforms between the direct and the reciprocal lattice that the electron–phonon literature uses.

A.3.4 Crystal momenta and reciprocal-lattice matrices

In a crystal, a momentum transfer $\boldsymbol{q}$ is split into a first-zone part and a reciprocal-lattice vector, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ with $\boldsymbol{q}_r$ in the first Brillouin zone (Sect. 3.1). Response functions are matrices in reciprocal-lattice vectors, such as $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}_r,\omega)$: the transform of a function of two positions $\boldsymbol{r}$ and $\boldsymbol{r}'$ with $\mathrm{e}^{+\mathrm{i}(\boldsymbol{q}_r+\boldsymbol{G})\cdot\boldsymbol{r}}$ and $\mathrm{e}^{-\mathrm{i}(\boldsymbol{q}_r+\boldsymbol{G}')\cdot\boldsymbol{r}'}$. The book indexes such a matrix by the unreduced $\boldsymbol{q}$ at $\boldsymbol{G}=0$:

$$(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega) \equiv (\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega), \tag{A.12}$$

and likewise for every reciprocal-lattice matrix. The unreduced and the reduced form both appear in the literature, often for the same object; Ref. [4, Eq. (24)] writes the reduced form with an explicit selection $\delta_{\boldsymbol{k},|\boldsymbol{q}+\boldsymbol{G}|}$. The momentum transfer $\boldsymbol{q}$ itself is never folded: it is the momentum the dark matter loses, and the dark-matter kinematics of Sect. 1.3 depend on $|\boldsymbol{q}|$, not on $|\boldsymbol{q}_r|$.

A.4 Kinematic signs

The momentum $\boldsymbol{q}$ is transferred from the dark matter to the target: a dark-matter particle of momentum $\boldsymbol{p}$ leaves with $\boldsymbol{p}-\boldsymbol{q}$. The energy ω is the energy given to the target, so a final target state has $E_f = E_i + \omega$. For a dark-matter particle of laboratory velocity $\boldsymbol{v}$ that changes its mass by $\Delta_\chi = m_{\chi 2} - m_{\chi 1}$ in the collision, energy conservation gives

$$\omega = \omega_{\boldsymbol{q}} \equiv \boldsymbol{q}\cdot\boldsymbol{v} - \frac{q^2}{2m_\chi} - \Delta_\chi, \tag{A.13}$$

as in Eq. (1.21). The minimum velocity needed for a momentum transfer $\boldsymbol{q}$ and an energy deposit ω is $v_{\min}(q,\omega) = |(\omega+\Delta_\chi)/q + q/(2m_\chi)|$ (Eq. (1.22)). The scattering is endothermic for $\Delta_\chi > 0$ and exothermic for $\Delta_\chi < 0$; elastic scattering has $\Delta_\chi = 0$. The energy-conserving delta function of a rate selects the component $(\omega+\Delta_\chi)/q + q/(2m_\chi)$ of the dark-matter velocity along $\hat{\boldsymbol{q}}$, and the speed integral of every rate depends on $(\boldsymbol{q},\omega)$ only through this component and through $\hat{\boldsymbol{q}}$ (Sect. 1.3).

A.4.1 Negative energy transfer

At zero temperature the target starts in its ground state and $\omega \geq 0$. At $T > 0$ it starts in a thermal mixture and can give energy to the dark matter, so $\omega < 0$ is allowed. Energy transfers with $\omega < 0$ are part of the structure factor, which at negative ω is fixed by its value at positive ω through detailed balance, $S(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q},\omega)$ (Appendix A.7). A negative ω is therefore suppressed by the Boltzmann factor of the target excitation that is de-excited. For electronic excitations across a gap of 1 eV at the 0.1 K or less of a cryogenic detector, $\beta\omega$ exceeds 10^5, and $\omega < 0$ is dropped. For phonons, magnons and other excitations of a few meV in a target at a few kelvin, the $\omega < 0$ part is kept. The book does not restrict ω to positive values; an expression that holds only for $\omega > 0$ says so.

A.4.2 Direction of the momentum transfer

The direction convention for $\boldsymbol{q}$ differs between papers. Some take $\boldsymbol{q}$ as the momentum gained by the dark matter; Ref. [10, below Eq. (12)] lists papers on each side. Others keep our direction but transform with the opposite sign and write the target vertex as $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ [5, Eq. (S.4)]. Each choice replaces $S(\boldsymbol{q},\omega)$ by $S(-\boldsymbol{q},\omega)$.

The conventions agree whenever the target responds in the same way at $\boldsymbol{q}$ and $-\boldsymbol{q}$. Inversion symmetry gives the same response at $\boldsymbol{q}$ and $-\boldsymbol{q}$ directly, since inversion maps $n_e(\boldsymbol{q})$ to $n_e(-\boldsymbol{q})$ and leaves the Hamiltonian unchanged. Time-reversal symmetry gives the same result: the antiunitary time-reversal operator maps $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ to $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and the thermal ensemble of a time-reversal-invariant target is mapped to itself. Each term $|\langle f|n_e(\boldsymbol{q})|i\rangle|^2$ of the structure factor then equals $|\langle\tilde{f}|n_e(-\boldsymbol{q})|\tilde{i}\rangle|^2$, where $|\tilde{i}\rangle$ and $|\tilde{f}\rangle$ are the time-reversed states, which have the same energies as the original states. The sum over the time-reversed states runs over the same set of states as the original sum, so $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$. Silicon, germanium and GaAs are time-reversal invariant, and silicon and germanium also have inversion symmetry. A crystal with neither symmetry, for example a magnetically ordered crystal without an inversion center, can have $S(\boldsymbol{q},\omega) \neq S(-\boldsymbol{q},\omega)$, and a daily modulation computed with the wrong sign of $\boldsymbol{q}$ is then wrong. A formula written in the other convention becomes ours under $\boldsymbol{q} \to -\boldsymbol{q}$ in every target quantity.

The first-zone part $\boldsymbol{q}_r$ of the momentum transfer labels the crystal-momentum change of the target, and its reciprocal-lattice part $\boldsymbol{G}_q$ labels an Umklapp component (Appendix A.3 and Sect. 3.1); both parts are needed, because $\omega_{\boldsymbol{q}}$ and $v_{\min}$ depend on the full $\boldsymbol{q}$. The phonon wavevector is a different object, written $\boldsymbol{Q}$ and always in the first zone (Appendix A.6).

A.5 Bloch and Wannier conventions

A Bloch state is normalized to one over the whole crystal, and its periodic part to one over a unit cell:

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \int_{\mathscr{V}} \mathrm{d}^3r\,|\psi_{n\boldsymbol{k}}|^2 = 1, \qquad \int_{\Omega} \mathrm{d}^3r\,|u_{n\boldsymbol{k}}|^2 = 1. \tag{A.14}$$

The two normalizations are consistent: $|\psi_{n\boldsymbol{k}}|^2 = |u_{n\boldsymbol{k}}|^2/N$ is periodic, and $\mathscr{V}$ holds N cells. Ref. [11, Eq. (29)] uses the normalizations of Eq. (A.14). Ref. [9, Eqs. (3.14) and (3.15)] writes the same states as $\mathscr{V}^{-1/2}\sum_{\boldsymbol{G}} u_n(\boldsymbol{k}+\boldsymbol{G})\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}$ with $\sum_{\boldsymbol{G}}|u_n(\boldsymbol{k}+\boldsymbol{G})|^2 = 1$; the plane-wave coefficients $u_n(\boldsymbol{k}+\boldsymbol{G})$ are the Fourier components of our $u_{n\boldsymbol{k}}$ times $\Omega^{1/2}$.

A.5.1 Periodic gauge

A Bloch state is fixed only up to a phase, $\psi_{n\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$, and for degenerate or entangled bands up to a unitary mixing [12, Eqs. (6), (7) and (10)]. We fix one part of this freedom everywhere: states at $\boldsymbol{k}$ and at $\boldsymbol{k}+\boldsymbol{G}$ are the same state,

$$\psi_{n,\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r}) = \psi_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \text{equivalently} \qquad u_{n,\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}\,u_{n\boldsymbol{k}}(\boldsymbol{r}). \tag{A.15}$$

This *periodic gauge* lets a crystal momentum outside the first zone be used directly, without bookkeeping of the reciprocal-lattice vector it differs by. It is the gauge in which Wannier functions are built [12, Sect. II A, footnote 1, and Eq. (10)].

A.5.2 The density vertex

The matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between Bloch states, the density vertex, is

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \equiv \langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle = \int_{\Omega} \mathrm{d}^3r\; u^*_{m,\boldsymbol{k}+\boldsymbol{q}}(\boldsymbol{r})\,u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \rho_{nn}(\boldsymbol{k},0) = 1. \tag{A.16}$$

The second form follows from Eq. (A.14): the phases $\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ cancel, the integrand is periodic, and the factor $1/N$ cancels the N cells. When $\boldsymbol{k}+\boldsymbol{q}$ lies outside the first zone it is folded back, $\boldsymbol{k}+\boldsymbol{q} = \boldsymbol{k}'+\boldsymbol{G}$ with $\boldsymbol{k}'$ in the zone, and by Eq. (A.15) the vertex becomes $\int_\Omega u^*_{m\boldsymbol{k}'}\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$: the Umklapp vector stays in the plane wave. The vertex is of order one and carries no factor of N (Appendix A.3). In the notation of Ref. [9, Eq. (3.16)], $\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = f_{[n\boldsymbol{k},m\boldsymbol{k}',\boldsymbol{G}]}$ with $\boldsymbol{q} = \boldsymbol{k}'-\boldsymbol{k}+\boldsymbol{G}$; Ref. [4, Eq. (B2)] defines the same integral over the unit cell, divided by Ω, for u normalized to Ω rather than to one.

At small q and $m \neq n$, the vertex reduces to a velocity matrix element. The position matrix element $\langle m|\boldsymbol{r}|n\rangle = -\mathrm{i}\,\boldsymbol{v}_{mn}/(\varepsilon_m-\varepsilon_n)$ follows from $\mathrm{i}[H,\boldsymbol{r}] = \boldsymbol{v}$ for $m \neq n$. Expanding $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \simeq 1+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$ and inserting the position matrix element gives

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \xrightarrow{q\to 0} \frac{\boldsymbol{q}\cdot\boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}}}, \qquad m \neq n, \tag{A.17}$$

with $\boldsymbol{v}_{mn}(\boldsymbol{k}) = \langle\psi_{m\boldsymbol{k}}|\boldsymbol{v}|\psi_{n\boldsymbol{k}}\rangle$ (Sect. 3.2). The velocity is $\boldsymbol{p}/m_e$ for a local potential; a nonlocal pseudopotential adds $\mathrm{i}[V_{\mathrm{NL}},\boldsymbol{r}]$, where V_{NL} is the nonlocal part of the pseudopotential, and the momentum matrix elements of a code must then be corrected before they are used as velocities (Sect. 26.2).

A.5.3 Gauge consistency

Under a change of gauge $\psi_{n\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$, the vertex and the velocity transform as

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \to \mathrm{e}^{\mathrm{i}[\theta_n(\boldsymbol{k})-\theta_m(\boldsymbol{k}+\boldsymbol{q})]}\rho_{mn}(\boldsymbol{k},\boldsymbol{q}), \qquad \boldsymbol{v}_{mn}(\boldsymbol{k}) \to \mathrm{e}^{\mathrm{i}[\theta_n(\boldsymbol{k})-\theta_m(\boldsymbol{k})]}\boldsymbol{v}_{mn}(\boldsymbol{k}), \tag{A.18}$$

consistently with Eq. (A.17). A first-order rate contains $|\rho_{mn}|^2$ and does not depend on the gauge. The squared modulus of a second-order amplitude is gauge independent only under a condition: the amplitude sums products such as $g_{cm\nu}\rho_{mv}$ over intermediate bands m (Sect. 6.5), and each product is gauge-covariant only if both factors were computed from the same set of states. Every quantity that enters one amplitude is therefore computed in one gauge, with one set of gauge matrices $U(\boldsymbol{k})$: the vertex ρ, the electron–phonon vertex g of Appendix A.6 and any tensor interpolated with Wannier functions. Codes that diagonalize at each $\boldsymbol{k}$ independently assign random phases, and independent runs for ρ and g do not share these phases. Projector-augmented-wave and ultrasoft inputs need the augmentation term in ρ at every $\boldsymbol{q}$, or $\rho_{nn}(\boldsymbol{k},0) = 1$ fails (Sect. 24.1).

A.5.4 Wannier functions and band labels

Wannier functions are built from the Bloch states of one gauge [12, Eqs. (10) and (14)],

$$w_n(\boldsymbol{r}-\boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\in\mathrm{BZ}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \sum_m U_{mn}(\boldsymbol{k})\, \psi_{m\boldsymbol{k}}(\boldsymbol{r}). \tag{A.19}$$

They are orthonormal over the crystal, and w_n has dimension $\Omega^{-1/2}$ when it is localized within a few cells. The real-space Hamiltonian $H^W(\boldsymbol{R})$ and the interpolated vertices are the matrix elements between Wannier functions (Sect. 27.1).

With spin–orbit coupling, Bloch states are two-component spinors, $u_{n\boldsymbol{k}}$ carries a spin index, and Eq. (A.16) includes the sum over it; each band then holds one electron per $\boldsymbol{k}$, and the spin factor 2 of the independent-particle formulas is dropped (Appendix A.7). Without spin–orbit coupling the states are spin-degenerate, the band index runs over spatial bands only, and the factor 2 is written explicitly. A degenerate subspace has no preferred basis, so only quantities summed over it, such as $\sum_m |\rho_{mn}|^2$ over the degenerate partners m, are compared between codes.

Band indices are n for a generic band and m for an intermediate one. The letters v and c label occupied (valence) and empty (conduction) bands and appear only as subscripts, $\varepsilon_{v\boldsymbol{k}}$ or ρ_{cv}; bare v is the dark-matter velocity, and c is the speed of light, set to one.

A.6 Phonon conventions

The displacement of nucleus κ in the cell at $\boldsymbol{R}$ is expanded in normal modes as

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} \frac{1}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, B_{\boldsymbol{Q}\nu}, \qquad B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}, \tag{A.20}$$

with $\boldsymbol{Q}$ in the first Brillouin zone, ν the branch, M_κ the nuclear mass and $[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = \delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$. The polarization vectors are normalized over the basis and obey

$$\sum_\kappa |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 = 1, \qquad \boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*, \qquad \omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}. \tag{A.21}$$

Refs. [11, Eqs. (16)–(18)] and [10, Eq. (29)] use the same conventions. The relation $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$, a choice of phase, makes $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ Hermitian: the terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are conjugates of

each other. Ref. [11] and most of the phonon literature write the wavevector as $\boldsymbol{q}$ and the operators as $a_{\boldsymbol{q}\nu}$. In this book $\boldsymbol{q}$ is the dark-matter momentum transfer and $a_{\boldsymbol{k}\nu}$ is reserved for magnons (Sect. 7.1), so sources are converted by $\boldsymbol{q} \to \boldsymbol{Q}$ and $a \to b$.

The frequencies and polarization vectors are the eigenvalues and eigenvectors of the dynamical matrix,

$$\sum_{\kappa'\beta} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q}) = \omega^2_{\boldsymbol{Q}\nu}\, e^{\alpha}_{\kappa\nu}(\boldsymbol{Q}), \qquad D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{A.22}$$

where $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})$ is the second derivative of the Born–Oppenheimer energy with respect to the displacements $u^{\alpha}_{\boldsymbol{0}\kappa}$ and $u^{\beta}_{\boldsymbol{R}\kappa'}$, and α, β are Cartesian indices (Sect. 5.2) [11, Eq. (14)]. The sign of the exponent matches Eq. (A.20): a displacement pattern $\propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ is an eigenvector of $D^{\mathrm{dyn}}(\boldsymbol{Q})$, not of $D^{\mathrm{dyn}}(-\boldsymbol{Q})$. A code that uses the opposite sign exchanges $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$; by Eq. (A.21) the exchange amounts to mapping $\boldsymbol{Q}$ to $-\boldsymbol{Q}$, and it must be undone before polarization vectors from such a code are combined with vertices from another code.

A.6.1 Zero-point length

The scale of the displacement is the zero-point length of a mode,

$$\ell_{\boldsymbol{Q}\nu} \equiv \frac{1}{\sqrt{2M_{\mathrm{cell}}\,\omega_{\boldsymbol{Q}\nu}}}, \tag{A.23}$$

with $M_{\mathrm{cell}} = \sum_\kappa M_\kappa$ and with the factor $1/\sqrt{N}$ of Eq. (A.20) kept outside $\ell_{\boldsymbol{Q}\nu}$. For an acoustic mode at small Q all nuclei move together, and the polarization vector is $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ with a unit vector $\hat{\boldsymbol{e}}_\nu$; this polarization vector satisfies Eq. (A.21). The amplitude in Eq. (A.20) is then $(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}(M_\kappa/M_{\mathrm{cell}})^{1/2} = \ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ for every κ, so one mode displaces each nucleus by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu/\sqrt{N}$ per unit of $B_{\boldsymbol{Q}\nu}$. Keeping $1/\sqrt{N}$ outside $\ell_{\boldsymbol{Q}\nu}$ matches the vertex normalization below, in which all factors of $1/\sqrt{N}$ are explicit. Ref. [11, Eq. (34)] writes the same scale with an arbitrary reference mass M_0 and the ratio $(M_0/M_\kappa)^{1/2}$.

A.6.2 The electron–phonon vertex

The first-order coupling of electrons to phonons has the form

$$H_{\mathrm{ep}} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\, c^{\dagger}_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu}. \tag{A.24}$$

The vertex is

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = \frac{1}{\sqrt{2\omega_{\boldsymbol{Q}\nu}}} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{M_\kappa}} \cdot \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} | \psi_{n\boldsymbol{k}} \rangle. \tag{A.25}$$

Here $\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \partial V_{\mathrm{KS}}/\partial \boldsymbol{u}_{\boldsymbol{R}\kappa}$ is the change of the self-consistent Kohn–Sham potential for a displacement wave of nucleus κ. It is computed with density-functional perturbation theory and therefore screened by the electrons (Sects. 6.2 and 25.2). The matrix element, normalized as in Eq. (A.14), is of order one in N, and so is g; the single factor $1/\sqrt{N}$ of Eq. (A.24) carries the size dependence. Refs. [11, Eqs. (37) and (38)] and [13, Eq. (2)] normalize the vertex in the same way, and so does the EPW code in its output; computed vertices therefore enter the formulas of this book without rescaling. The vertex has dimension of energy.

Hermiticity of H_{ep} fixes the conjugate vertex. The Hermitian conjugate of the term $(\boldsymbol{k},\boldsymbol{Q},m,n)$ is $g^{*}_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\,c^{\dagger}_{n\boldsymbol{k}}c_{m,\boldsymbol{k}+\boldsymbol{Q}}B^{\dagger}_{\boldsymbol{Q}\nu}$, and $B^{\dagger}_{\boldsymbol{Q}\nu}=B_{-\boldsymbol{Q}\nu}$. The Hermitian conjugate must equal the term $(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q},n,m)$ of the sum, so

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^{*}=g_{nm\nu}(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q}), \tag{A.26}$$

which also follows from Eq. (A.25) with $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q})=\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^{*}$ and $(\partial_{\kappa,\boldsymbol{Q}}V_{\mathrm{KS}})^{\dagger}=\partial_{\kappa,-\boldsymbol{Q}}V_{\mathrm{KS}}$. The vertex is split into a short-range and a long-range part, $g=g^{\mathrm{S}}+g^{\mathrm{L}}$, in polar crystals, where g^{L} is the Fröhlich term (Sect. 6.2). Both parts carry the gauge of the electronic states, and the rule of Appendix A.5 applies: g and ρ in one amplitude come from one set of states.

A.6.3 Emission and absorption

In Eq. (A.24) the electron gains crystal momentum $\boldsymbol{Q}$, either by absorbing a phonon $(\boldsymbol{Q},\nu)$, through $b_{\boldsymbol{Q}\nu}$, or by emitting a phonon $(-\boldsymbol{Q},\nu)$, through $b^{\dagger}_{-\boldsymbol{Q}\nu}$; both processes use the same vertex $g(\cdot,\boldsymbol{Q})$. With $n_{\boldsymbol{Q}\nu}=n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$ phonons in the mode, $n_{\mathrm{B}}(x)=(\mathrm{e}^{\beta x}-1)^{-1}$, the squared matrix elements $|\langle n+1|b^{\dagger}|n\rangle|^{2}=n+1$ and $|\langle n-1|b|n\rangle|^{2}=n$ give the thermal weights

$$n_{\boldsymbol{Q}\nu}+\tfrac{1}{2}\pm\tfrac{1}{2}, \tag{A.27}$$

with the upper sign for emission of $-\boldsymbol{Q}$ and the lower for absorption of $+\boldsymbol{Q}$. The energy given to the lattice is $\pm\omega_{\boldsymbol{Q}\nu}$ with the same sign convention. At $T=0$ only emission survives.

A.6.4 Second order and the Debye–Waller factor

The coupling at second order in the displacements has a vertex quadratic in B, which we call the Allen–Heine vertex g^{AH}; its self-energy is Σ^{AH} (Sect. 6.3). Ref. [11, Eqs. (39) and (40)] calls the same vertex g^{DW}, a "Debye–Waller" vertex. This book reserves "Debye–Waller" for one object only: the factor $\mathrm{e}^{-2W_{\kappa}(\boldsymbol{q})}$ that reduces the zero-phonon weight when nucleus κ takes up a momentum transfer $\boldsymbol{q}$, with

$$2W_{\kappa}(\boldsymbol{q})=\boldsymbol{q}\cdot B_{\kappa}\cdot\boldsymbol{q},\qquad B_{\kappa}=\langle\boldsymbol{u}_{\kappa}\boldsymbol{u}_{\kappa}\rangle, \tag{A.28}$$

where B_{κ} is the thermal displacement tensor, isotropic only at a site of cubic symmetry (Sect. 5.5). The Allen–Heine vertex and the Debye–Waller factor share B_{κ}, and both appear, in different roles, next to the double commutator of the first-moment sum rule Eq. (4.28). In Sect. 5.5 the factor $\mathrm{e}^{-2W_{\kappa}(\boldsymbol{q})}$ is the elastic share of the scattering weight, and the sum rule fixes the mean energy of the scattering weight at the free recoil energy. In Sect. 6.3 the commutator, taken for a rigid translation, makes Σ^{AH} cancel the Fan–Migdal term for long-wavelength acoustic phonons.

A.7 Response conventions

Response functions are written R, never χ, because χ is the dark-matter particle. For two target operators $O_a(\boldsymbol{q})$ and $O_b(\boldsymbol{q})$, the retarded response and the response matrix are

$$R_{ab}(\boldsymbol{q},\omega)=-\frac{\mathrm{i}}{\mathscr{V}}\int_{0}^{\infty}\mathrm{d}t\;\mathrm{e}^{\mathrm{i}\omega t}\,\langle[O_a(\boldsymbol{q},t)^{\dagger},O_b(\boldsymbol{q},0)]\rangle, \tag{A.29}$$

$$S_{ab}(\boldsymbol{q},\omega)=\frac{2\pi}{\mathscr{V}}\sum_{i,f}p_i\,\langle i|O_a(\boldsymbol{q})^{\dagger}|f\rangle\langle f|O_b(\boldsymbol{q})|i\rangle\,\delta(\omega-E_f+E_i), \tag{A.30}$$

with Heisenberg operators $O(t) = \mathrm{e}^{\mathrm{i}Ht} O \mathrm{e}^{-\mathrm{i}Ht}$, thermal averages with Gibbs weights $p_i = \mathrm{e}^{-\beta E_i}/\mathcal{Z}$, and the time convention of Appendix A.3. The dynamic structure factor of the electron density is the diagonal element $S \equiv S_{ee}$ with $O_e = n_e(\boldsymbol{q})$ (Sects. 4.1 and 4.2). R_{ab} and S_{ab} both carry the factor $1/\mathcal{V}$, so both are intensive, and for density operators both have dimension eV^2. Ref. [4, Eqs. (2) and (4)] defines the same objects for the density, with χ for R, $n_{-\boldsymbol{k}}$ for the operator inside S and $\boldsymbol{k}$ for the momentum transfer. The literature's χ, Π and W for response, polarization and loss are listed in Appendix B.2.

Inserting eigenstates of the target into Eq. (A.29) gives the spectral (Lehmann) form of R_{ab},

$$R_{ab}(\boldsymbol{q},\omega) = \frac{1}{\mathcal{V}} \sum_{i,f} p_i \left[\frac{\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\omega - (E_f - E_i) + \mathrm{i}0} - \frac{\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle}{\omega + (E_f - E_i) + \mathrm{i}0} \right], \tag{A.31}$$

with every pole below the real axis, as Appendix A.3 requires. For a diagonal element at $T = 0$ and $\omega > 0$ only the resonant term has an imaginary part, $-\pi\delta(\omega - E_f + E_i)$ times a positive weight, so $\operatorname{Im} R_{aa} < 0$ for $\omega > 0$. The sign of $\operatorname{Im} R_{aa}$ is a consequence of the time convention: with $\mathrm{e}^{+\mathrm{i}\omega t}$ in the inverse transform it would be reversed. Analyticity in the upper half plane gives the Kramers–Kronig relations, and they hold for R, for $\varepsilon^{-1} - 1$ and, in a crystal with a gap, for $\varepsilon - 1$ (Sects. 4.1 and 4.4).

A.7.1 Fluctuation–dissipation theorem and detailed balance

Inserting energy eigenstates into Eqs. (A.29) and (A.30) and using $\int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \langle O_a^\dagger(t) O_b\rangle = \mathcal{V} S_{ab}(\boldsymbol{q},\omega)$ and $\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \langle O_b O_a^\dagger(t)\rangle = \mathrm{e}^{-\beta\omega} \mathcal{V} S_{ab}(\boldsymbol{q},\omega)$ gives

$$(1 - \mathrm{e}^{-\beta\omega})\, S_{ab}(\boldsymbol{q},\omega) = \mathrm{i}\big[R_{ab}(\boldsymbol{q},\omega) - R_{ba}(\boldsymbol{q},\omega)^*\big], \qquad S_{aa} = -\frac{2\,\operatorname{Im} R_{aa}}{1 - \mathrm{e}^{-\beta\omega}}, \tag{A.32}$$

the fluctuation–dissipation theorem (Sect. 4.2). The diagonal form is Ref. [4, Eq. (3)]. For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, such as densities, exchanging i and f in Eq. (A.30) gives detailed balance,

$$S_{ab}(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega}\, S_{ba}(\boldsymbol{q},\omega). \tag{A.33}$$

At $T \ll \omega$ the factor $1 - \mathrm{e}^{-\beta\omega}$ is one.

A.7.2 Dielectric function and loss

The Coulomb-proper response $\tilde{R}$ is the part of the response that is irreducible with respect to one Coulomb line (Sect. 4.4). The longitudinal dielectric matrix of a crystal, with local fields, is built from $\tilde{R}$:

$$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, \tilde{R}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega), \tag{A.34}$$

with $\tilde{R}$ the response of the total charge in units of e and v_{C} from Eq. (A.3). Eq. (A.34) is the unsymmetrized form; the symmetrized form uses $v_{\mathrm{C}}^{1/2} \tilde{R}\, v_{\mathrm{C}}^{1/2}$, has the same diagonal and differs off the diagonal [4, footnote 3]. The macroscopic dielectric function and the loss function follow from the inverse matrix:

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega) = \frac{1}{(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)}, \qquad L(\boldsymbol{q},\omega) = \operatorname{Im}\big[-(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)\big], \tag{A.35}$$

with the unreduced-$\boldsymbol{q}$ indexing of Eq. (A.12), $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$, and $L \geq 0$ for $\omega > 0$. For the electron channel alone, $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$, and ε_e^{-1} equals ε^{-1} when the nuclei are clamped. The Coulomb-proper response gives the absorptive part of the dielectric function directly, $\varepsilon_2 = -v_{\mathrm{C}} \operatorname{Im} \tilde{R}$.

Combining $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$ with Eq. (A.32) gives the relation between the structure factor and the loss function,

$$S(\boldsymbol{q},\omega) = \frac{2q^2}{e^2}\,\frac{L_e(\boldsymbol{q},\omega)}{1-\mathrm{e}^{-\beta\omega}}, \qquad \varepsilon_2(\boldsymbol{q},\omega) = \frac{e^2}{2q^2}\,(1-\mathrm{e}^{-\beta\omega})\,S^{\mathrm{prop}}(\boldsymbol{q},\omega), \tag{A.36}$$

where L_e is the loss of the electron channel and S^{prop} the structure factor built from $\tilde{R}_{ee}$. The first relation is Ref. [4, Eq. (6)], with $e^2 = 4\pi\alpha$. It holds with the total loss L at clamped nuclei, and for a dark photon, which couples to the total charge, it holds exactly with the total ε (Sect. 4.4). The optical conductivity is defined by $\varepsilon = 1 + \mathrm{i}\sigma_{\mathrm{cond}}/\omega$ in the transverse, long-wavelength limit, so that $\sigma_1 = \omega\varepsilon_2$ (Appendix A.2 gives the Gaussian and SI forms).

A.7.3 Spin factor

The structure factor and ε_2 are summed over spin. For a spin-independent vertex without spin–orbit coupling, each band is doubly occupied, and the independent-particle form carries an explicit factor 2:

$$S(\boldsymbol{q},\omega) = \frac{4\pi}{\mathscr{V}} \sum_{v,c,\boldsymbol{k}} |\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2\,\delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}) \qquad (T=0), \tag{A.37}$$

where $4\pi/\mathscr{V}$ is $2\times 2\pi/\mathscr{V}$ and the sums run over spatial bands. With spin–orbit coupling the factor 2 is dropped and the sums run over spinor bands. Refs. [9, Eq. (A.29)] and [14, Eq. (6)] write the same factor outside their band sums, and Ref. [4, Appendix B, above Eq. (B7)] states that the factor is implicit in its main-text formulas. A factor of 2 between two published rates for the same material is often the spin factor.

A.7.4 The first frequency moment

The first frequency moment of the density structure factor is fixed by the f-sum rule. For a target with $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega\;\omega\,S(\boldsymbol{q},\omega) = \frac{\pi\,\boldsymbol{n}_e\,q^2}{m_e}, \tag{A.38}$$

at any temperature, with $\boldsymbol{n}_e$ the mean electron density (Sect. 4.3 and Appendix D.4). Splitting the integral at $\omega = 0$ and using Eq. (A.33) turns the left side into $\int_0^\infty \mathrm{d}\omega\,\omega(1-\mathrm{e}^{-\beta\omega})S$, and Eq. (A.36) then gives [5, Eq. (S.10)]

$$\int_0^{\infty} \mathrm{d}\omega\;\omega\,L_e(\boldsymbol{q},\omega) = \frac{\pi}{2}\,\omega_p^2, \qquad \omega_p^2 = \frac{e^2\boldsymbol{n}_e}{m_e}. \tag{A.39}$$

At $T = 0$, where S vanishes for $\omega < 0$, dividing Eq. (A.38) by $2\pi\boldsymbol{n}_e$ gives the per-particle form of many-body texts, $\int_0^\infty \mathrm{d}\omega\,\omega\,S(\boldsymbol{q},\omega)/(2\pi\boldsymbol{n}_e) = q^2/(2m_e)$.

A.8 Halo conventions and rate normalization

Every rate in this book is a rate per unit target mass, $\mathcal{R}$ in $\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$. Every halo average uses the standard halo model with the parameters of Ref. [15, Table 1]: $\rho_\chi = 0.3\,\mathrm{GeV\,cm}^{-3}$, $v_0 = 238\,\mathrm{km\,s}^{-1}$, $v_{\mathrm{esc}} = 544\,\mathrm{km\,s}^{-1}$, and the solar velocity $\boldsymbol{v}_\odot = (11.1, 250.2, 7.3)\,\mathrm{km\,s}^{-1}$ in galactic coordinates (Eq. (1.5), Table 1.1). The laboratory distribution is $f_\chi(\boldsymbol{v},t) = N_{\mathrm{esc}}^{-1}\mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}}(t)|^2/v_0^2}\,\Theta(v_{\mathrm{esc}} - |\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}}(t)|)$ with $\boldsymbol{v}_{\mathrm{E}}(t) = \boldsymbol{v}_\odot + \boldsymbol{v}_\oplus(t)$ (Eq. (1.8)). These halo parameters are the standard values of the field; the measured ranges are given in Sect. 1.2.

A.8.1 Two Earth speeds

The Earth's speed enters in two roles. Halo integrals, such as $\eta(v_{\rm min})$ and the daily and annual modulation, use the full time-dependent $\boldsymbol{v}_{\rm E}(t)$, averaged over the exposure when a time-averaged rate is quoted. Reference scalars, such as kinematic edges, threshold masses and the windows in (q,ω) that a mass can reach, use one fixed speed,

$$v_{\rm E}^{\rm ref} \equiv |\boldsymbol{v}_\odot| = 250.5\,{\rm km\,s^{-1}}, \qquad v_{\rm max} \equiv v_{\rm esc} + v_{\rm E}^{\rm ref} = 794.5\,{\rm km\,s^{-1}} = 2.650\times10^{-3}, \quad \text{(A.40)}$$

as in Eq. (1.9). The fixed value keeps every threshold time-independent. Over a year, $v_{\rm esc} + |\boldsymbol{v}_{\rm E}(t)|$ ranges from 781 to 810 km s^{-1} (Sect. 1.2); this spread is part of the uncertainty of the halo. The annual mean of $|\boldsymbol{v}_{\rm E}(t)|$ differs from $|\boldsymbol{v}_\odot|$ at second order in $v_\oplus/v_\odot$, and this difference is also counted in the uncertainty of the halo. Papers that quote "$v_{\rm E}$" mean either the annual mean of $|\boldsymbol{v}_{\rm E}(t)|$ or $|\boldsymbol{v}_\odot|$.

A.8.2 Speed integrals

The halo enters through the velocity kernel $\int {\rm d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}})$, which depends on $\hat{\boldsymbol{q}}$ and on $v_{\rm min}(q,\omega)$. For a structure factor that depends only on $|\boldsymbol{q}|$, or after averaging S over the directions of $\boldsymbol{q}$, the angular integral over $\hat{\boldsymbol{q}}$ can be done first, and

$$\int \frac{{\rm d}^3q}{(2\pi)^3}\,S(q,\omega)\int {\rm d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}}) = \int \frac{q\,{\rm d}q}{4\pi^2}\,\eta(v_{\rm min}(q,\omega))\,S(q,\omega), \quad \text{(A.41)}$$

with $\eta(v_{\rm min}) = \langle\Theta(v-v_{\rm min})/v\rangle$ (Eq. (1.27)). Per unit ${\rm d}^3q/(2\pi)^3$ the kernel is $\eta(v_{\rm min})/(2q)$; the factor 1/2 comes from the angular integral $\frac{1}{2}\int_{-1}^{1}{\rm d}\cos\theta\,\delta(\cdots - qv\cos\theta) = \Theta(v-v_{\rm min})/(2qv)$ (Sect. 1.3). The reduction to Eq. (A.41) requires only that the target response be independent of the direction of $\boldsymbol{q}$; the halo itself need not be isotropic, since the integral over $\hat{\boldsymbol{q}}$ at fixed $\boldsymbol{v}$ is the same for every direction of $\boldsymbol{v}$. A second speed moment, $\eta_2(v_{\rm min}) = 2\int_{v_{\rm min}}^{\infty}{\rm d}v'\,v'\eta(v')$, appears for interactions whose strength grows with the dark-matter velocity. Halo-independent analyses absorb the particle physics into $\tilde\eta(v_{\rm min}) = \rho_\chi\bar\sigma_e\eta(v_{\rm min})/m_\chi$ [16, Eq. (2.8)].

For a crystal the structure factor depends on the direction of $\boldsymbol{q}$, and Eq. (A.41) does not apply, so the full kernel is kept. The halo distribution is written in the laboratory frame at the time of the event, the crystal axes are rotated into that frame with the Earth's rotation (Eq. (1.14)), and the integral over $\hat{\boldsymbol{q}}$ is done with $S(\boldsymbol{q},\omega)$ evaluated in crystal axes. Keeping the full kernel gives the daily modulation of Sect. 1.2. A time-averaged rate averages the full kernel over the exposure; it is not the rate computed with the angle-averaged $\bar S$ unless the exposure samples all orientations of $\boldsymbol{v}_{\rm E}$ relative to the crystal equally, and a fixed detector does not sample them equally.

A.8.3 Rate per unit target mass

For an interaction with potential transform $\tilde V(\boldsymbol{q})$ that couples to the electron density, a dark-matter particle in the box of Appendix A.3 scatters at the rate $\int {\rm d}^3q\,(2\pi)^{-3}|\tilde V(\boldsymbol{q})|^2 S(\boldsymbol{q},\omega_{\boldsymbol{q}})$. By Eq. (A.8) the matrix element is $\mathcal{V}^{-1}\tilde V\langle f|n_e(\boldsymbol{q})|i\rangle$, the golden rule squares it, the sum over final dark-matter momenta gives $\mathcal{V}\int {\rm d}^3q/(2\pi)^3$, and Eq. (A.30) absorbs the remaining $2\pi/\mathcal{V}$ (Eq. (2.67)). The box holds $\rho_\chi\mathcal{V}/m_\chi$ dark-matter particles and a target mass $\rho_T\mathcal{V}$, so the spectrum per unit target mass is

$$\frac{{\rm d}\mathcal{R}}{{\rm d}\omega} = \frac{\rho_\chi}{m_\chi\,\rho_T}\int {\rm d}^3v\, f_\chi(\boldsymbol{v})\int \frac{{\rm d}^3q}{(2\pi)^3}\,|\tilde V(\boldsymbol{q})|^2\,S(\boldsymbol{q},\omega)\,\delta(\omega-\omega_{\boldsymbol{q}}), \quad \text{(A.42)}$$

in which the volume has cancelled. With $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_{\rm DM}^2/\mu_{\chi e}^2$ (Eq. (1.33)) and Eq. (A.41),

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e}{4\pi\, m_\chi\, \rho_T\, \mu_{\chi e}^2} \int q\,\mathrm{d}q\; F_{\rm DM}(q)^2\, \eta(v_{\rm min}(q,\omega))\, \bar{S}(q,\omega), \tag{A.43}$$

with $\bar{S}$ the average of S over the directions of $\boldsymbol{q}$. This spectrum is Eq. (1.46), and it agrees with Refs. [4, Eq. (10)] and [5, Eq. (S.9)], the latter after Eq. (A.36). For operators other than the density, $|\tilde{V}|^2 S$ is replaced by a contraction of a dark-matter-side tensor with the response matrix S_{ab} (Sect. 4.2).

Other papers normalize per target or per unit cell. Ref. [9, Eq. (3.13)] gives the rate of a whole target of $N_{\rm cell}$ cells, and Ref. [14, Eq. (36)] the same with $n_\chi = \rho_\chi/m_\chi$. Dividing by the target mass $N_{\rm cell}M_{\rm cell}$, with $\rho_T = M_{\rm cell}/\Omega$, and comparing with Eq. (A.43) identifies the crystal form factor of Ref. [9] with the angle-averaged structure factor,

$$|f_{\rm crys}(q,\omega)|^2 = \frac{q^3\,\Omega}{4\pi\alpha\, m_e^2}\,\bar{S}(q,\omega) = \frac{q^5\,\Omega}{8\pi^2\alpha^2 m_e^2}\,\varepsilon_2(q,\omega), \tag{A.44}$$

where the second form uses Eq. (A.36) at $T = 0$ without screening and agrees with Ref. [4, Eq. (16)]. The product $\Omega\bar{S}$, of dimension eV^{-1}, is the natural per-cell quantity to tabulate.

A.9 Cross-section conventions

The strength of a dark-matter–electron interaction is quoted as the reference cross-section $\bar{\sigma}_e$ and its momentum dependence as the form factor $F_{\rm DM}(q)$ of Eq. (1.32), normalized at $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$. For a mediator of mass m_V, $F_{\rm DM}(q) = (m_V^2 + \alpha^2 m_e^2)/(m_V^2 + q^2)$, which is 1 for a heavy and $(q_0/q)^2$ for a light mediator [4, Eqs. (11) and (12)]. The form factor multiplies the amplitude, so it enters every rate as $F_{\rm DM}^2$. Papers define $\bar{\sigma}_e$ and $F_{\rm DM}(q)$ with different reference points, media and amplitude normalizations.

The momentum $q_0 = \alpha m_e$ is the inverse Bohr radius, typical of bound electrons. A paper that normalizes at another momentum $q_{\rm ref}$ quotes $\sigma_{\rm ref} = \bar{\sigma}_e\, F_{\rm DM}(q_{\rm ref})^2$; for a light mediator this is $\bar{\sigma}_e\,(q_0/q_{\rm ref})^4$, so a limit quoted at $q_{\rm ref} = 2q_0$ is smaller by a factor 16 for the same physics.

Here $\bar{\sigma}_e$ is defined in vacuum: $\tilde{V}$ is the bare potential between a dark-matter particle and a free electron, and screening, local fields and every other effect of the target are in the response S (Sect. 4.4). The same convention is used in Refs. [4, Eq. (8) and the paragraph after it] and [5, below Eq. (S.8)]. Some papers put the in-medium dielectric factor into the coupling, as is common for dark photons and for Dirac materials with a background dielectric constant, and quote a cross-section that depends on the target. A target-dependent cross-section of this kind is converted by removing the dielectric factor from the coupling and putting the corresponding screening into the response.

Ref. [9, Eq. (3.8)] defines

$$\bar{\sigma}_e = \frac{\mu_{\chi e}^2\,|\mathcal{M}_{\rm free}(q_0)|^2}{16\pi m_\chi^2 m_e^2},$$

with the relativistically normalized free amplitude $\mathcal{M}_{\rm free}$, averaged over initial and summed over final spins. Comparison with Eq. (1.32) gives the dictionary

$$|\mathcal{M}_{\rm free}(q)|^2 = 16\, m_\chi^2 m_e^2\, |\tilde{V}(q)|^2, \tag{A.45}$$

the factor $4m_\chi m_e$ per amplitude being the ratio of relativistic to nonrelativistic state normalizations. A nonrelativistic effective operator with coefficient c_1 in the relativistic normalization therefore has $\tilde{V} = c_1/(4m_\chi m_e)$ and $\bar{\sigma}_e = \mu_{\chi e}^2 |c_1|^2/(16\pi m_\chi^2 m_e^2)$.

The spin average is part of the definition of $\bar{\sigma}_e$ in Ref. [9, Eq. (3.8)]. The amplitude squared in Eq. (A.45) is averaged over the initial spins of the dark matter and the electron and summed over the final spins [9, below Eq. (3.8)]. For a spin-independent interaction this average equals the squared amplitude of a single spin state, so $\bar{\sigma}_e$ is the same for scalar and fermionic dark matter with the same $\tilde{V}$ [9, below Eq. (2.2)]. The electron spin sum is then not repeated in the target response: there the factor 2 of Appendix A.7 counts occupied spin states, not final spins of a free electron [9, below Eqs. (A.28) and (A.29)].

A.9.1 Differential cross-section

The differential cross-section of free elastic scattering in the first Born approximation is Eq. (2.15),

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}q} = \frac{q\,\bar{\sigma}_e}{2\mu_{\chi e}^2 v^2}\, F_{\mathrm{DM}}(q)^2, \qquad 0 \le q \le 2\mu_{\chi e} v.$$

For $F_{\mathrm{DM}} = 1$ the differential cross-section integrates to $\bar{\sigma}_e$, so for a heavy mediator $\bar{\sigma}_e$ is the total cross-section of free elastic scattering. The same normalization holds for nuclei, $\mathrm{d}\sigma_N/\mathrm{d}q = q\sigma_N F_N(q)^2/(2\mu_{\chi N}^2 v^2)$ (Eq. (2.27)). A differential cross-section defined per unit q^2 is $\mathrm{d}\sigma_T/\mathrm{d}q^2 = (2q)^{-1}\mathrm{d}\sigma_T/\mathrm{d}q$; a factor of 2 between two papers often traces to the choice between q and q^2.

A.9.2 Nucleons and nuclei

For a mediator coupled to nucleons, the dark-matter–proton cross-section σ_p is the analogue of $\bar{\sigma}_e$ with $\mu_{\chi p}$, and the spin-independent nuclear cross-section at small q follows from σ_p as $\sigma_N = \sigma_p(\mu_{\chi N}/\mu_{\chi p})^2 A^2$ (Eq. (1.36)). For unequal proton and neutron couplings, $A^2 \to [Z + (A-Z)y_n/y_p]^2$; Ref. [1, Eq. (104)] writes the equivalent rescaling relative to the neutron coupling. For a dark photon, which couples to protons only, $A^2 \to Z^2$. At larger q the nuclear form factor $F_{\mathrm{Helm}}(q)$ multiplies the amplitude (Sect. 2.2). For a dark photon the electron and proton couplings have equal magnitude ζe, so for a heavy dark photon, $m_V \gg q_0$, $\sigma_p/\bar{\sigma}_e = (\mu_{\chi p}/\mu_{\chi e})^2$; this ratio is how electron- and nuclear-scattering limits on the same model are compared.

A.9.3 Kinetic mixing

The dark photon mixes with the photon through the term $+(\zeta/2)F_{\mu\nu}F'^{\mu\nu}$ in the Lagrangian density, with $\mathcal{L} \supset -eA_\mu J^\mu_{\mathrm{em}}$ for the photon (Eq. (C.83)). The sign of the mixing term follows Ref. [1, Eq. (70)], which writes κ for ζ and $+eA_\mu J^\mu_{\mathrm{em}}$ for the photon coupling; Ref. [9, Eq. (2.1)] writes ε and mixes with hypercharge, $(\varepsilon/2\cos\theta_W)F_{Y\mu\nu}F'^{\mu\nu}$, where θ_W is the weak mixing angle. With our signs, the dark photon couples to electrons with $y_e = -\zeta e$ and to protons with $+\zeta e$ (Sect. 1.4). Reversing either the sign of the mixing term or the sign of the photon coupling flips the sign of y_e and of $\tilde{V}$, and no rate changes, because every rate contains ζ^2. The symbols used for ζ in the literature are listed in Appendix B.2.

A.9.4 Relativistic dark matter

The form factor of Eq. (1.32) is the nonrelativistic limit, $\omega \ll q$, of the mediator propagator $1/(q^2 - \omega^2 + m_V^2)$ (Appendix C.9). For a light mediator in scattering, $\omega \le q v_{\max}$ and the correction from ω^2 in the propagator is below 10^{-5}. Dark matter boosted to relativistic speeds is outside this limit: its rates need the full propagator and the coupling to the transverse current, and $F_{\mathrm{DM}}(q)$ alone does not describe them.

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Appendix B

Notation

Every symbol first used in this volume is listed in Appendix B.1 with its meaning, its units and the section of its first use, and every acronym in Appendix B.3 with its expansion. Appendix B.2 gives the symbols that common papers and textbooks use for the quantities of this book; Appendix A gives the sign and normalization conventions in which these papers and textbooks differ from this book.

B.1 List of symbols

The list holds the symbols first used in Volume I; a symbol first used in another volume is listed in the notation appendix of that volume. Each symbol is listed once, under the chapter or appendix of its first use; the last column gives the section of its first use. The Units column gives the dimension in natural units ($\hbar = c = k_\mathrm{B} = 1$, Appendix A.1) and, after a semicolon, the conventional unit in which values are quoted. The entry 1 marks a dimensionless quantity, and a dash an object without a dimension of its own, such as an operator, a state or a label.

Symbol	Meaning	Units	Sect.
Chapter 1: Light dark matter			
$\hbar, c, k_\mathrm{B}$	Set to 1	—	1.1
$v_\mathrm{c}(r)$	Circular velocity of a galactic orbit	$\mathrm{km\,s^{-1}}$	1.1
h	Dimensionless Hubble parameter, the Hubble rate today in units of 100 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$	1	1.1
$\Omega_\mathrm{c} h^2$	Cosmological dark-matter density parameter	1	1.1
$\langle \sigma v \rangle$	Thermally averaged annihilation cross-section times relative speed	$\mathrm{cm^3\,s^{-1}}$	1.1
χ	The dark-matter particle	—	1.1
m_χ	Dark-matter mass	eV	1.1
n_χ	Dark-matter number density ρ_χ / m_χ	$\mathrm{eV^3}$; $\mathrm{cm^{-3}}$	1.1
ρ_χ	Local dark-matter mass density, 0.3 $\mathrm{GeV\,cm^{-3}}$	$\mathrm{eV^4}$; $\mathrm{GeV\,cm^{-3}}$	1.1
$f_\chi(v)$	Dark-matter velocity distribution in the laboratory frame	$(\mathrm{velocity})^{-3}$	1.1
v	Dark-matter velocity in the laboratory frame	1; $\mathrm{km\,s^{-1}}$	1.1
q	Momentum transferred from the dark matter to the target	eV	1.1
ω	Energy given to the target (may be negative at $T > 0$)	eV	1.1
v_min	Minimum dark-matter speed $v_\mathrm{min} = \lvert(\omega + \Delta_\chi)/q + q/2m_\chi\rvert$ for a transfer (q, ω)	1; $\mathrm{km\,s^{-1}}$	1.1
$\eta(v_\mathrm{min})$	Mean inverse speed above v_min, $\int \mathrm{d}^3 v f_\chi \Theta(v - v_\mathrm{min})/v$	$(\mathrm{velocity})^{-1}$	1.1

Symbol	Meaning	Units	Sect.
$\mu_{\chi e}$, $\mu_{\chi N}$, $\mu_{\chi p}$	Dark-matter–electron, –nucleus, –proton reduced masses	eV	1.1
$\tilde{V}(q)$	Fourier transform of the dark-matter–electron potential, $\int \mathrm{d}^3x \mathrm{e}^{+\mathrm{i}q\cdot x}V(x)$	eV^{-2}	1.1
$\bar{\sigma}_e$	Reference dark-matter–electron cross-section, $\mu_{\chi e}^2\lvert\tilde{V}(q_0)\rvert^2/\pi$, in vacuum; for spin couplings $\mu_{\chi e}^2\langle\lvert\tilde{V}\rvert^2\rangle/\pi$ with the average over both spins	eV^{-2}; cm^2	1.1
$F_{\mathrm{DM}}(q)$	Dark-matter form factor $(m_V^2+\alpha^2m_e^2)/(m_V^2+q^2)$	1	1.1
ϕ, m_ϕ	Absorbed bosonic dark-matter field and its mass	eV	1.1
$\mathcal{R}$, $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$	Event rate per unit target mass, and its spectrum	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.1
ρ_T	Target mass density	$\mathrm{g\,cm}^{-3}$	1.1
$S(q,\omega)$	Dynamic structure factor of the electron density	eV^2	1.1
$\rho_{\mathrm{halo}}(r)$	Dark-matter halo mass density at galactocentric radius r (singular isothermal sphere, $\rho_{\mathrm{halo}}\propto r^{-2}$)	eV^4; $\mathrm{GeV\,cm}^{-3}$	1.2
v_0	Circular speed parameter of the standard halo model, 238 km/s	$\mathrm{km\,s}^{-1}$	1.2
v_{esc}	Galactic escape speed, 544 km/s	$\mathrm{km\,s}^{-1}$	1.2
$v_\odot$	Solar velocity in the galactic frame, $\lvert v_\odot\rvert = 250.55$ km/s	$\mathrm{km\,s}^{-1}$	1.2
$v_\oplus(t)$	Earth's orbital velocity relative to the Sun	$\mathrm{km\,s}^{-1}$	1.2
$v_{\mathrm{E}}(t)$	Earth velocity in the galactic frame, $v_\odot + v_\oplus(t)$	$\mathrm{km\,s}^{-1}$	1.2
$v_{\mathrm{E}}^{\mathrm{ref}}$	Reference Earth speed for thresholds and edges: the adopted value of $\lvert v_\odot\rvert$, 250.5 km/s	$\mathrm{km\,s}^{-1}$	1.2
v_{max}	Maximum laboratory speed $v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}} = 794.5$ km/s	$\mathrm{km\,s}^{-1}$	1.2
N_{esc}	Normalization of the truncated halo distribution	$(\mathrm{velocity})^3$	1.2
Φ_χ	Dark-matter number flux	$\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$	1.2
v_{gal}, $f_{\mathrm{gal}}(v_{\mathrm{gal}})$	Dark-matter velocity in the galactic rest frame; standard-halo distribution in that frame	$\mathrm{km\,s}^{-1}$; $(\mathrm{velocity})^{-3}$	1.2
$F_\chi(v)$	Laboratory speed distribution, $\int \mathrm{d}vF_\chi = 1$	$(\mathrm{velocity})^{-1}$	1.2
ω_{yr}, ω_{sid}	Orbital angular frequency of the Earth, 0.0172 d^{-1}; sidereal angular frequency $2\pi/(23\ \mathrm{h}\ 56\ \mathrm{min}\ 4\ \mathrm{s})$	d^{-1}; s^{-1}	1.2
t_{peak}, t_{E}	Date of largest Earth speed; time at which a laboratory axis and v_{E} share a meridian	d; h	1.2
θ_{E}	Angle between $v_\odot$ and the Earth's rotation axis, 42.5°	rad	1.2
$\hat{n}$	Unit vector along a laboratory-fixed axis	1	1.2
$\theta_{\mathrm{lab}}(t)$, $\theta_{\hat{n}}$	Angle between v_{E} and a laboratory axis $\hat{n}$; polar angle of $\hat{n}$ from the celestial pole	rad	1.2
α_{E}, α_{z}, $\alpha_{\hat{n}}$; δ_{E}; h_{E}	Right ascensions of $v_\odot$, of the zenith and of $\hat{n}$; declination of $v_\odot$; its hour angle	rad	1.2
ℓ_{lat}	Geographic latitude	rad	1.2
u_{esc}	Local escape speed of an attracting body (Sun, Earth), focusing estimate only	$\mathrm{km\,s}^{-1}$	1.2
ϑ, ϑ_{cut}	Angle between v and v_{E} in the derivation of F_χ; its cut-off value	rad	1.2
x_{eq}, y_{eq}, z_{eq}; x_{gal}, y_{gal}, z_{gal}	Equatorial and galactic Cartesian components of a unit vector	1	1.2
$\hat{v}_{\mathrm{east}}$, $\hat{v}_{\mathrm{north}}$, $\hat{v}_{\mathrm{up}}$	Components of the unit vector $\hat{v}_\odot$ along the local east, north and vertical axes of a laboratory	1	1.2
$\Theta(x)$	Heaviside step function	1	1.2
α	Fine-structure constant	1	1.3
m_e	Electron mass	eV	1.3
$\omega_{\boldsymbol{q}}$	Kinematic energy transfer $q\cdot v - q^2/2m_\chi - \Delta_\chi$	eV	1.3
Δ_χ	Dark-matter mass splitting (exothermic for $\Delta_\chi < 0$)	eV	1.3
E_{th}	Threshold energy	eV	1.3
m_{min}	Threshold mass $2E_{\mathrm{th}}/v_{\mathrm{max}}^2$	eV	1.3
q_0	Reference momentum αm_e	eV	1.3

Symbol	Meaning	Units	Sect.
a	Cubic lattice constant; silicon 5.431 Å	Å	1.3
$\boldsymbol{G}_{111}$	Shortest reciprocal-lattice vector of fcc silicon, $\lvert\boldsymbol{G}_{111}\rvert = \sqrt{3}\cdot 2\pi/a = 3.95\ \mathrm{keV} \approx 1.06 q_0$	eV	1.3
θ	Angle between q and v; an integration angle elsewhere, defined where it is used (the scattering angle is θ_s)	rad	1.3
χ_1, χ_2; $m_{\chi 1}$, $m_{\chi 2}$	Lighter and heavier states of inelastic dark matter and their masses, $\Delta_\chi = m_{\chi 2} - m_{\chi 1}$	—; eV	1.3
E_R	Nuclear recoil energy	eV	1.3
E_g	Indirect (fundamental) band gap	eV	1.3
M	Generic nuclear mass in κ_BO	eV	1.3
$\mu_{\chi T}$	Reduced mass of dark matter and a free target of mass M	eV	1.3
G	Reciprocal-lattice vector	eV	1.3
c_ν	Sound speed of branch ν	$\mathrm{km\,s^{-1}}$	1.3
$\mu_{\chi n}$	Dark-matter–nucleon reduced mass (as $\mu_{\chi N}$ with the nucleon)	eV	1.3
e	Elementary charge, $e > 0, e^2 = 4\pi\alpha$; the electron charge is $-e$	1	1.4
m_V	Mediator mass	eV	1.4
y_χ, y_e, y_N	Mediator couplings to dark matter, electrons, nucleons	1	1.4
$V(x)$	Dark-matter–electron potential in position space	eV	1.4
ζ	Dark-photon kinetic mixing	1	1.4
g_{aee}, $g_{a\gamma\gamma}$	Axion-like-particle couplings to electrons and photons	1; $\mathrm{eV^{-1}}$	1.4
$\boldsymbol{p}_\phi$	Momentum of an absorbed bosonic dark-matter particle, $\lvert\boldsymbol{p}_\phi\rvert \approx m_\phi v$	eV	1.4
σ_p, σ_N	Dark-matter–proton and dark-matter–nucleus cross-sections	$\mathrm{cm^2}$	1.4
A, Z	Mass number, atomic number	1	1.4
A'_μ, $F'_{\mu\nu}$	Vector-mediator (dark-photon) field and its field strength	eV; $\mathrm{eV^2}$	1.4
A_μ, $F_{\mu\nu}$, $\tilde{A}_\mu$; $\tilde{F}^{\mu\nu}$	Photon field and field strength; photon after the mixing redefinition; dual field strength $\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$	eV; $\mathrm{eV^2}$	1.4
ϕ_V	Scalar-mediator field	eV	1.4
J^μ_χ, J^μ_e, J^μ_N, $\boldsymbol{J}_\chi$	Number currents of dark matter, electrons and nucleons; $J^0 = \boldsymbol{n}$	$\mathrm{eV^3}$	1.4
J^μ_em	Electromagnetic current in units of e (an electron counts -1)	$\mathrm{eV^3}$	1.4
$\boldsymbol{n}_N$, $\boldsymbol{n}_N(r)$	Nucleon number density; of a nucleus, $\int \mathrm{d}^3 r \boldsymbol{n}_N(r) = A$	$\mathrm{eV^3}$	1.4
y_p, y_n	Mediator couplings to protons and neutrons when they differ from y_N	1	1.4
$g_{\phi ee}$	Scalar-dark-matter–electron Yukawa coupling, $\mathcal{L} \supset g_{\phi ee}\phi\bar{\psi}\psi$	1	1.4
ζ_eff	In-medium effective kinetic mixing for absorption	1	1.4
Π_em	In-medium photon self-energy	$\mathrm{eV^2}$	1.4
$\boldsymbol{\sigma}$	Pauli-matrix vector of an electron	1	1.4
$\mathcal{L}$, $\mathcal{L}_\mathrm{int}$, $\mathcal{L}_{aee}$, $\mathcal{L}_{\phi ee}$	Lagrangian density	$\mathrm{eV^4}$	1.4
$\boldsymbol{E}$, $\boldsymbol{B}$, $\boldsymbol{A}$	Microscopic electric and magnetic fields and vector potential	$\mathrm{eV^2}$	1.4
ψ, $\bar{\psi}$, γ^μ, γ^5	Relativistic (Dirac) electron field and Dirac matrices	$\mathrm{eV^{3/2}}$; 1	1.4
$\boldsymbol{B}_0$	Static laboratory magnetic field	T	1.4
N_eh	Number of electron–hole pairs (ionization yield)	1	1.5
ε_eh	Mean energy per electron–hole pair	eV	1.5
M_det, t_exp	Detector (target) mass and live time; exposure $M_\mathrm{det}t_\mathrm{exp}$	kg; yr	1.5

Symbol	Meaning	Units	Sect.
N_{sig}, N_{obs}, N_{bkg}	Expected signal count, observed count, expected background count	1	1.5
N_{90}	90% confidence upper limit on N_{sig} (2.303 for $N_{\mathrm{obs}} = 0$)	1	1.5
$\mathcal{R}_{\mathrm{bkg}}$	Background rate per unit exposure	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.5
$r(m_\chi)$	Rate per unit cross-section, $\mathcal{R} = \bar{\sigma}_e r$	$\mathrm{cm}^{-2}\,\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.5
$\bar{\sigma}_e^{\mathrm{lim}}(m_\chi)$	Cross-section limit	cm^2	1.5
$\langle F_{\mathrm{DM}}^2 \rangle$	Average of F_{DM}^2 over (q, ω) weighted by η and S	1	1.5
$\mathcal{R}_n$	Rate of events with n electron–hole pairs	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.5
$\mathcal{R}_{\mathrm{toy}}$	Toy rate of the exclusion-limit worked example	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.5
V_{bias}	Detector bias voltage	V	1.5
E_{ph}, E_{prim}	Phonon energy measured after Luke amplification, $E_{\mathrm{prim}} + N_{\mathrm{eh}} e V_{\mathrm{bias}}$; energy deposited directly into primary phonons	eV	1.5
A_{mod}	Annual-modulation amplitude	1	1.5
$\lfloor x \rfloor$	Floor function	—	1.5
M_κ, M_{cell}, Z_κ	Nuclear mass, cell mass, nuclear charge number	eV; 1	1.5
Δ_{BCS}	Superconducting gap	meV	1.5
T_{c}	Critical temperature	K	1.5
Chapter 2: Transition rates			
$\Gamma_{i\to f}$, Γ	Transition rate; rate per dark-matter particle	eV	2.1
H', H_{int}	Perturbation; dark-matter–target interaction	eV	2.1
$\mathscr{V}$	Normalization (crystal) volume	eV^{-3}	2.1
$\mathcal{D}_f(E)$	Density of final states	eV^{-1}	2.1
E_i, E_f	Energies of initial and final states	eV	2.1
H_0, H	Unperturbed Hamiltonian; full Hamiltonian $H_0 + H'$	eV	2.1
$c_n(t)$, $c_f(t)$, $c_i(t)$	Expansion amplitudes in the interaction picture	1	2.1
$P_{i\to f}(t)$	Transition probability	1	2.1
ξ	Energy mismatch $E_f - E_i$	eV	2.1
Γ_i	Total rate out of state i, $2\pi\langle\lvert H'\rvert^2\rangle\mathcal{D}_f(E_i)$	eV	2.1
E_{w}	Energy width of a continuum	eV	2.1
$\hat{O}$, O_{fi}	Operator of a harmonic perturbation $H' = \hat{O}\mathrm{e}^{-\mathrm{i}\omega t} + \hat{O}^\dagger \mathrm{e}^{\mathrm{i}\omega t}$; its matrix element $\langle f\lvert\hat{O}\rvert i\rangle$	eV	2.1
O, $O(t)$	Generic operator on the target, unhatted, in formulas valid for any operator; $O(t)$ its Heisenberg form	—	2.1
N_f	Number of final states	1	2.1
τ_{s}	Switching time	eV^{-1}	2.1
ϕ_0	Amplitude of an oscillating bosonic dark-matter field, $\rho_\chi = m_\phi^2 \phi_0^2/2$	eV	2.1
σ_T, m_T	Cross-section on, and mass of, one target particle	cm^2; eV	2.1
$\mathcal{M}$	Invariant (relativistic) amplitude; $\lvert\mathcal{M}\rvert^2 = 16 m_\chi^2 m_e^2 \lvert\tilde{V}\rvert^2$ non-relativistically	1	2.1
$\langle p\lvert p'\rangle_{\mathrm{rel}}$, $E_{\boldsymbol{p}}$	Relativistic normalization $2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p} - \boldsymbol{p}')$; relativistic energy of momentum $\boldsymbol{p}$	—; eV	2.1
$\mathrm{Si}(z)$	Sine integral	1	2.1
$\hat{T}$	T-matrix operator in $\langle f\lvert\mathrm{i}\hat{T}\rvert i\rangle = \mathrm{i}(2\pi)^4\delta^4\mathcal{M}$	—	2.1
$\boldsymbol{P}_i$, $\boldsymbol{P}_f$	Total initial and final momenta	eV	2.1
$\boldsymbol{p}_\chi$, $\boldsymbol{p}_e$; $\boldsymbol{x}_\chi$, $\boldsymbol{x}_e$	Momenta and positions of the dark matter and the electron in a two-particle state	eV; eV^{-1}	2.1
$\lvert n\rangle$	Generic unperturbed eigenstate	—	2.1
$w(\xi)$, Δ_{w}	Test function in the δ-limit and its energy scale	—; eV	2.1
$\mathrm{d}\Omega_{\boldsymbol{q}}$	Solid-angle element of the scattered momentum	sr	2.2
$f_{i\to f}(q)$	Single-particle transition form factor $\langle f\lvert\mathrm{e}^{\mathrm{i}q\cdot r}\rvert i\rangle$	1	2.2
$F_{1s}(q)$	Hydrogen 1s elastic form factor $16/(4 + q^2 a_0^2)^2$	1	2.2

Symbol	Meaning	Units	Sect.
a_0	Bohr radius	Å	2.2
$F_{\mathrm{Helm}}(q)$	Helm nuclear form factor	1	2.2
θ_{s}	Scattering angle between initial and final relative momenta	rad	2.2
p, p'; $\boldsymbol{P}_{\mathrm{tot}}$, $\boldsymbol{p}_{\mathrm{rel}}$	Relative momenta before and after a two-body collision; total and relative momenta of a two-body system	eV	2.2
n_T	Number density of target atoms or nuclei	eV^3	2.2
ψ_i, ψ_f, ψ_{1s}, ψ_{2p}	Single-particle wavefunctions	$\mathrm{eV}^{3/2}$	2.2
$\tilde{\psi}_i(k)$, $\tilde{\psi}_{nl}(k)$	Momentum-space wavefunction $\int \mathrm{d}^3 r \mathrm{e}^{-\mathrm{i}k\cdot r}\psi_i$; radial momentum wavefunction of shell nl, $\int k^2 dk \lvert\tilde{\psi}_{nl}\rvert^2 = (2\pi)^3$	$\mathrm{eV}^{-3/2}$	2.2
$\tilde{\psi}_{plm}(r)$, $\tilde{\psi}_{p_{\mathrm{ej}}l'm'}(r)$	Outgoing continuum partial wave of an ejected electron of momentum $p(p_{\mathrm{ej}})$, normalized to $(2\pi)^3\delta_{l'l}\delta_{m'm}\delta(p'-p)/p^2$	$\mathrm{eV}^{-3/2}$	2.2
n, l, m	Principal, orbital and magnetic quantum numbers, as subscripts of atomic orbitals only	1	2.2
Y_{lm}	Spherical harmonics	1	2.2
p_{ej}, E_{ej}	Momentum and kinetic energy of an ionized electron	eV	2.2
E_B	Binding energy of an atomic shell, > 0	eV	2.2
$\lvert f_{\mathrm{ion}}(p_{\mathrm{ej}}, q)\rvert^2$	Atomic ionization form factor of a shell	1	2.2
Z_{eff}, $F_{\mathrm{Fermi}}(p)$	Effective charge in the Fermi factor; the Fermi factor	1	2.2
ξ_{C}	Coulomb (Sommerfeld) parameter $Z_{\mathrm{eff}}\alpha m_e/p$	1	2.2
$j_1(x)$	Spherical Bessel function of order one, $(\sin x - x\cos x)/x^2$	1	2.2
λ_{1s}, λ_{2p}	Radial decay constants $2/a_0$ and $3/(2a_0)$ of the hydrogen 1s density and of the $1s \to 2p$ radial integrand	eV	2.2
$V_N(x)$, $\tilde{V}_N(q)$	Dark-matter–nucleon potential and its transform	eV; eV^{-2}	2.2
$F_N(q)$	Nuclear form factor $(1/A)\int \mathrm{d}^3 r \mathrm{e}^{\mathrm{i}q\cdot r} n_N(r)$, $F_N(0) = 1$; F_{Helm} is its model	1	2.2
$\boldsymbol{R}_{\mathrm{cm}}$, $\boldsymbol{r}'_j$	Nuclear center of mass and internal nucleon coordinates	eV^{-1}	2.2
R_{nuc}, s_{skin}	Helm radius and skin thickness	fm	2.2
c_{nuc}, a_{nuc}	Woods–Saxon half-density radius and diffuseness in the Lewin–Smith R_{nuc}	fm	2.2
q_{max}	Largest elastic transfer $2\mu v$	eV	2.2
$c^\dagger$, c	Fermion creation and annihilation operators ($c_{n\boldsymbol{k}}$ for Bloch states)	—	2.3
$\hat{\psi}(r)$	Fermion field operator	$\mathrm{eV}^{3/2}$	2.3
$n_e(q)$	Electron density operator $\sum_i \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$	1	2.3
$v_{\mathrm{C}}(k)$	Coulomb kernel e^2/k^2	eV^{-2}	2.3
γ, γ', $\gamma_1 \ldots \gamma_4$, $\tilde{\gamma}$	Single-particle state labels (spin included; $\tilde{\gamma}$ labels a second basis); orbitals $\psi_\gamma(\boldsymbol{r}, s) = \langle \boldsymbol{r}s\lvert\gamma\rangle$	—	2.3
s, s'	Spin projection $\uparrow$, $\downarrow$	—	2.3
$x = (r, s)$	Position and spin of one particle	—	2.3
n_γ	Occupation number of state γ	1	2.3
$\hat{N}_e$, N_e	Electron number operator $\sum c^\dagger c$ and its value	1	2.3
d_γ, $d^\dagger_\gamma$	Generic boson operators (phonons are $b_{\boldsymbol{Q}\nu}$, magnons $a_{\boldsymbol{k}\nu}$)	—	2.3
E_γ	Single-particle (orbital) energy of state γ; the Hartree–Fock or Kohn–Sham eigenvalue ($\varepsilon_{n\boldsymbol{k}}$ in a crystal)	eV	2.3
S_γ	Number of occupied states before γ in the fixed order	1	2.3
$\hat{O}_1$, $\hat{O}_2$, $\hat{o}$	One- and two-body operators; one-particle operator	—	2.3
$\Psi(x_1, \ldots, x_N)$, $\Psi_{\gamma_1 \ldots \gamma_N}$	N-particle wavefunction of identical particles; the Slater determinant of the orbitals γ_1, …, γ_N	$\mathrm{eV}^{3N/2}$	2.3

Symbol	Meaning	Units	Sect.
o_γ	Eigenvalue of the one-particle operator $\hat{o}$ on the orbital γ (in the basis that diagonalizes $\hat{o}$)	as $\hat{o}$	2.3
$U_1(r)$, $U_2(r)$	One-body potential; pair potential	eV	2.3
$:\cdots:$	Normal ordering	—	2.3
k_{reg}	Infrared regulator of the Coulomb kernel, $v_{\mathrm{C}} \to e^2/(k^2 + k_{\mathrm{reg}}^2)$	eV	2.3
X_I, I	Position of nucleus I; nucleus index ($\kappa(I)$ its species)	Å	2.3
$n_Z(k)$	Charge-number density $-n_e(\boldsymbol{k}) + \sum_\kappa Z_\kappa n_\kappa(\boldsymbol{k})$	1	2.3
H_{ee}, H_{jel}, H_C, H_{Coul}	Electron–electron Coulomb term; jellium Hamiltonian; Coulomb Hamiltonian of electrons and nuclei	eV	2.3
d	Mean electron spacing $\boldsymbol{n}_e^{-1/3}$	Å	2.3
$\lvert\Phi_0\rangle$	A reference Slater determinant	—	2.3
e_{G}, e_{SI}, $\varepsilon_0^{\mathrm{SI}}$	Gaussian elementary charge, $e_{\mathrm{G}}^2 = \alpha$; SI charge and vacuum permittivity	1; —	2.3
r_s	Electron-gas density parameter: Wigner–Seitz radius per electron in units of a_0, $(3/4\pi\boldsymbol{n}_e)^{1/3}/a_0$; 2.0 for silicon's valence density	1	2.3
κ	Nuclear species (basis-site) index	—	2.3
$n_\kappa(q)$	Nuclear density operator of species κ, $\sum_{I\in\kappa} \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_I}$	1	2.3
T_{fi}	Transition (T-matrix) element	eV	2.4
Γ_{w}	Decay width of an intermediate state	eV	2.4
PV, fp	Principal value; Hadamard finite part	—	2.4
$\lvert\psi_{\mathrm{I}}(t)\rangle, H'_{\mathrm{I}}(t), U_{\mathrm{I}}(t,\, t_0)$	Interaction-picture state, perturbation and evolution operator	—; eV; 1	2.4
H'_{mn}	Matrix element $\langle m\lvert H'\rvert n\rangle$	eV	2.4
$\lvert m\rangle, E_m$	Intermediate state and its energy	—; eV	2.4
$c_f^{(2)}(t)$	Second-order amplitude	1	2.4
$\mathrm{i}0^+$	Positive infinitesimal in an energy denominator	eV	2.4
ΔE_i, ΔE_m	Level shift of a state coupled to a continuum, PV $\sum \lvert H'\rvert^2/(E_i - E_f)$ (decorated Δ)	eV	2.4
$\tilde{E}_m$	Shifted energy $E_m + \Delta E_m$	eV	2.4
i_{T}, f_{T}; $E_{i_{\mathrm{T}}}$, $E_{f_{\mathrm{T}}}$	Initial and final target states of a second-order process, in kets and subscripts only; their energies	—; eV	2.4
ω_1, ω_2	Energies of the absorbed and emitted (or two absorbed) quanta	eV	2.4
H_1, H_2; $\hat{O}_a$, $\hat{O}_b$	The two couplings of a second-order process; their target operators	eV	2.4
$w(x)$	Test function of the real denominator variable x	—	2.4
x_{c}	Symmetric cut-off half-width in PV and fp	eV	2.4
δ_1, δ_2, $\mathcal{A}(\omega_1)$	Detunings $\omega_j - (E_m - E_{i_{\mathrm{T}}})$ and two-ordering amplitude	eV; eV^{-1}	2.4
$\Gamma_{m\to f_{\mathrm{T}}}$, $\mathcal{D}_{f_{\mathrm{T}}}$	Rate at which a beam empties m into the band; density of final target states	eV; eV^{-1}	2.4
$\hat{\varrho}$	Density matrix	1	2.5
T, β	Temperature, $\beta = 1/\mathrm{T}$	eV	2.5
p_i	Gibbs weight $\mathrm{e}^{-\beta E_i}/\mathcal{Z}$	1	2.5
$\mathcal{Z}$	Partition function	1	2.5
μ_{ch}	Chemical potential	eV	2.5
$\mathcal{S}_O(\omega)$	Transition sum $\sum_{if} p_i \lvert\langle f\lvert O\rvert i\rangle\rvert^2 \delta(\omega - E_f + E_i)$ of an operator O; identified with S_{ab}	eV^{-1}	2.5
$n_{\mathrm{B}}(x)$, $n_{\mathrm{F}}(x)$	Bose and Fermi functions	1	2.5
$\boldsymbol{n}_e$	Mean electron number density $N_e/\mathcal{V}$	$\mathrm{Å}^{-3}$	2.5
k_{F}, E_{F}	Fermi wavevector and Fermi energy of a free electron gas	eV	2.5

Symbol	Meaning	Units	Sect.
v_{F}	Fermi velocity of a metal, k_{F}/m_e for the free electron gas	1	2.5
$\mathcal{N}_{\mathrm{F}}$	Density of states per spin and volume at the Fermi level	eV^2	2.5
Δ_{tl}	Splitting of a two-level system	eV	2.5
p_0, p_1, O_{10}	Two-level populations and matrix element $\langle 1\|\hat{O}\|0\rangle$	1; eV	2.5
ω_0, λ_{osc}	Oscillator frequency and coupling of the thermal-oscillator worked example	eV	2.5
$U(t)$	Evolution operator	—	2.5
N_e	Number of electrons in the volume $\mathcal{V}$, $n_e = N_e/\mathcal{V}$	1	2.5
Chapter 3: Electrons in crystals			
$\boldsymbol{a}_i$, $\boldsymbol{b}_i$	Primitive vectors of the direct and reciprocal lattices	Å; $\mathrm{Å}^{-1}$	3.1
R	Bravais lattice vector	Å	3.1
Ω	Unit-cell volume	$\mathrm{Å}^3$	3.1
N	Number of unit cells (= number of k points on a uniform grid)	1	3.1
k, k'	Electron crystal momenta in the first zone	eV	3.1
$\boldsymbol{q}_r$, $\boldsymbol{G}_q$	First-zone part and reciprocal vector of $\boldsymbol{q}$, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$	eV	3.1
Γ, X, L, K	High-symmetry points of the fcc zone; also the point W	—	3.1
$X_{\boldsymbol{R}\kappa}$	Equilibrium nuclear site $\boldsymbol{R} + \tau_\kappa$ of the static lattice	Å	3.1
n_{at}	Number of atoms in the primitive cell (N_{at} counts the atoms of a target or a calculation)	1	3.1
$S^{\mathrm{bas}}(G)$	Basis structure factor $\sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\tau_\kappa}$	1	3.1
$T_{\boldsymbol{R}}$	Lattice translation operator acting on all electrons	—	3.1
$\boldsymbol{G}_{hkl}$	Reciprocal-lattice vector $(2\pi/a)(h,k,l)$ of a cubic lattice	eV	3.1
$\tilde{v}_\kappa(G)$, $\tilde{v}(G)$	Transform of the atomic term of species κ in a lattice-periodic sum, $f_{\boldsymbol{G}} = \Omega^{-1}\sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\tau_\kappa}\tilde{v}_\kappa(\boldsymbol{G})$ ($\tilde{v}_\kappa = -Z_\kappa v_{\mathrm{C}}$ for a clamped nucleus); its one-species form	as $f_{\boldsymbol{G}}\ \Omega$	3.1
τ_κ	Position of basis site (atom) κ in the unit cell, bold $\boldsymbol{\tau}_\kappa$	Å	3.1
$\psi_{n\boldsymbol{k}}(r)$, $u_{n\boldsymbol{k}}(r)$	Bloch state normalized over $\mathcal{V}$; periodic part normalized over Ω	$\mathcal{V}^{-1/2}$; $\Omega^{-1/2}$	3.2
n, m	Band indices: generic, intermediate	—	3.2
$\varepsilon_{n\boldsymbol{k}}$	Band energy	eV	3.2
$U_{\boldsymbol{G}}$	Fourier component of the periodic potential	eV	3.2
t_{hop}	Tight-binding hopping amplitude	eV	3.2
$v_n^{\mathrm{g}}(k)$	Group velocity $\nabla_k\varepsilon_{n\boldsymbol{k}}(= v_{nn}(\boldsymbol{k}))$	1	3.2
$\boldsymbol{p}_{mn}(k)$, $v_{mn}(k)$	Momentum and velocity matrix elements between Bloch states; $v_{mn} = \boldsymbol{p}_{mn}/m_e$ for a local potential	eV; 1	3.2
$\rho_{mn}(k,q)$	Density vertex $\langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}\|\mathrm{e}^{\mathrm{i}q\cdot\boldsymbol{r}}\|\psi_{n\boldsymbol{k}}\rangle$, $\rho_{nn}(\boldsymbol{k},0) = 1$; optical limit $q\cdot v_{mn}/(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})$	1	3.2
$f_{n\boldsymbol{k}}$	Occupation of state nk	1	3.2
H_{e}	One-electron Hamiltonian $\hat{p}^2/2m_e + U(\boldsymbol{r})$	eV	3.2
$H_{\boldsymbol{k}}$	Bloch Hamiltonian $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}H_{\mathrm{e}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ acting on $u_{n\boldsymbol{k}}$ (also at shifted arguments, $H_{\boldsymbol{k}+\boldsymbol{q}}$, $H_{\boldsymbol{k}_0+\delta\boldsymbol{k}}$)	eV	3.2
$U(r)$	Effective lattice-periodic one-electron potential	eV	3.2
$\hat{p}$, $\boldsymbol{v}$	Momentum operator $-\mathrm{i}\nabla$; velocity operator $i[H_{\mathrm{e}}, \boldsymbol{r}]$	eV; 1	3.2
$\varepsilon_{\boldsymbol{k}}^0$	Free-electron energy $k^2/2m_e$	eV	3.2
ϕ_j^{at}, $\phi_{j\boldsymbol{R}}^{\mathrm{at}}$	Atomic orbital j of tight binding, and its copy in cell $\boldsymbol{R}$	$\mathrm{Å}^{-3/2}$	3.2
$C_{nj}(k)$	LCAO coefficients of band n on the Bloch sum of orbital j	1	3.2
$H^{\mathrm{TB}}(k)$, $S^{\mathrm{ov}}(k)$	Tight-binding Hamiltonian and overlap matrices between Bloch sums	eV; 1	3.2

Symbol	Meaning	Units	Sect.
E_s, E_p	On-site s and p energies of an atom in tight binding (E_p in Sect. 3.2)	eV	3.2
V_{cov}	Covalent energy between two bond hybrids	eV	3.2
$c_{n\boldsymbol{k}}(G)$	Plane-wave coefficients of a Bloch state, of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ in $\Omega^{1/2}u_{n\boldsymbol{k}}$	1	3.2
$\boldsymbol{G}_{\mathrm{U}}$, N_{FFT}	Umklapp vector of the fold $\boldsymbol{k}+q=\boldsymbol{k}'+\boldsymbol{G}_{\mathrm{U}}$ with $\boldsymbol{k}'$ in the first zone; number of FFT grid points	eV (quoted in $\mathrm{Å}^{-1}$); 1	3.2
v, c	Occupied (valence) and empty (conduction) band labels, only as subscripts	—	3.3
E_{dir}	Minimum direct gap	eV	3.3
$\boldsymbol{k}_{\mathrm{c}}$	Crystal momentum of the conduction-band minimum	eV	3.3
m^*, $m^*_{\parallel}$, $m^*_{\perp}$	Effective-mass tensor and its valley components	m_e	3.3
Δ_{SO}	Spin–orbit splitting of the valence-band maximum	eV	3.3
$E_{\mathrm{min}}(q)$	Minimum electron–hole pair energy at fixed q	eV	3.3
δk	Crystal momentum measured from a band extremum k_0, $\delta k=\boldsymbol{k}-k_0$	eV	3.3
E_P, v^x_{cX}	Kane energy $2m_e\lvert v^x_{cX}\rvert^2$; velocity matrix element between the conduction state and the orbital state $\lvert X\rangle$	eV; 1	3.3
ξ_{SO}, $\xi_{\mathrm{SO},\kappa}$, ξ_{eff}	Radial spin–orbit constant of an atomic level, of atom κ; its p-weighted crystal average	eV	3.3
$U_{\mathrm{at}}(r)$	Central potential energy of an electron in an atom, in the spin–orbit term H_{so}	eV	3.3
m_{hh}, m_{lh}, m_{h}	Heavy- and light-hole masses; a generic hole mass	eV; m_e	3.3
m_{dc}, m_{dv}	Density-of-states masses of the conduction and valence bands	m_e	3.3
n_{i}	Intrinsic carrier density	cm^{-3}	3.3
γ_1^{Lu}, γ_2^{Lu}, γ_3^{Lu}, $\bar{\gamma}^{\mathrm{Lu}}$	Luttinger parameters; their spherical average $(2\gamma_2^{\mathrm{Lu}}+3\gamma_3^{\mathrm{Lu}})/5$	1	3.3
$\boldsymbol{L}$, $\boldsymbol{S}$; j, l	Orbital and spin angular momentum of an electron; total and orbital angular-momentum quantum numbers	1	3.3
$n(r)$	Electron number density (Kohn–Sham density)	$\mathrm{Å}^{-3}$	3.4
V_{KS}, v_{xc}, E_{xc}	Kohn–Sham potential, exchange–correlation potential and energy	eV	3.4
Δ_{scis}	Scissor shift	eV	3.4
E_{N_e}, E_{N_e+1}, E_{N_e-1}	Ground-state energy of $N_e(N_e\pm1)$ electrons; $E^r_{N_e+1}, E^a_{N_e-1}$ are the frozen-orbital removal and addition energies	eV	3.4
$P_{\mathrm{c}}(k)$, $H^{\mathrm{s}}_{\boldsymbol{k}}$, v^{s}_{cv}	Projector $\sum_c \lvert u_{c\boldsymbol{k}}\rangle\langle u_{c\boldsymbol{k}}\rvert$ on the conduction bands; scissor Bloch Hamiltonian $H_{\boldsymbol{k}}+\Delta_{\mathrm{scis}}P_{\mathrm{c}}$; scissor-rescaled velocity matrix element	1; eV; 1	3.4
$v_{\mathrm{H}}(r)$	Hartree potential $\int e^2 n(\boldsymbol{r}')/4\pi\lvert\boldsymbol{r}-\boldsymbol{r}'\rvert$	eV	3.4
$\mathcal{F}[n]$	Universal (Levy–Lieb) functional $\min_{\Psi\to n}\langle H_{\mathrm{kin}}+H_{ee}\rangle$	eV	3.4
H_{kin}	Kinetic-energy operator $\sum_i p_i^2/2m_e$ (decorated H)	eV	3.4
$K_{\mathrm{s}}[n]$	Independent-electron (Kohn–Sham) kinetic energy	eV	3.4
$E_{\mathrm{H}}[n]$	Hartree energy	eV	3.4
I, A	Ionization energy, electron affinity	eV	3.4
$E_{\mathrm{g}}^{\mathrm{KS}}$, Δ_{xc}	Kohn–Sham gap; derivative discontinuity, $E_{\mathrm{g}}=E_{\mathrm{g}}^{\mathrm{KS}}+\Delta_{\mathrm{xc}}$	eV	3.4
$\varepsilon_{\mathrm{xc}}(n)$, $\varepsilon_{\mathrm{x}}(n)$	Exchange–correlation and exchange energy per electron of the uniform gas	eV	3.4
f_{x}, v_{x}	Exchange-only kernel $-e^2/4k_{\mathrm{F}}^2$ and exchange potential $(4/3)\varepsilon_{\mathrm{x}}$ of the gas	eV^{-2}; eV	3.4
$E_{\mathrm{qp}}(k)$, $\Gamma_{\mathrm{qp}}(k)$	Quasiparticle energy and full width	eV	3.4

Symbol	Meaning	Units	Sect.
$\Sigma_{\rm x}$, $\Sigma_{\rm c}$	Exchange and correlation parts of the self-energy Σ; $\Sigma_{\rm x}(x,x')$ is the nonlocal Fock term, $\Sigma_{\rm x}(\boldsymbol{k}) = -(e^2 k_{\rm F}/2\pi^2) F_{\rm L}(\boldsymbol{k}/k_{\rm F})$ its electron-gas value	eV	3.4
Chapter 4: Linear response			
$\phi_{\rm ext}(t)$	External potential coupled to O_b in the Kubo formula	eV	4.1
$R_{ab}(q,\omega)$	Retarded response function of operators O_a, O_b	eV^2 for density operators ($1/\mathscr{V}$ included)	4.1
$O_a(q)$	Target operator coupled to an external field	—	4.1
$H_{\rm tot}(t)$, $H_{\rm ext}$	Hamiltonian of the target in an external field, $H + O_b\phi_{\rm ext}(t)$; the coupling term alone	eV	4.1
$\hat{\varrho}_{\rm I}(t)$	Interaction-picture density matrix with respect to the full target H	1	4.1
z, $z_{\rm p}$	Complex frequency ($\mathrm{Re}\, z, \mathrm{Im}\, z$) of a response or Green's function continued off the real axis; a pole of the continuation in the lower half plane	eV	4.1
A	Generic responding operator of the Kubo derivation	—	4.1
P	Mean power absorbed from a harmonic field, $-(\omega\phi_0^2/2)\mathscr{V}\,\mathrm{Im}\, R_{bb}$	eV^2	4.1
$M_{\rm osc}$	Mass of the oscillator in the classical check	eV	4.1
$\eta_{\rm b}$	Finite broadening of an energy denominator, $\eta_{\rm b} \to 0^+$ in exact forms; also the adiabatic switching rate	eV	4.1
$S_{ab}(q,\omega)$	Response matrix $(2\pi/\mathscr{V})\sum p_i \langle i\vert O_a^\dagger\vert f\rangle\langle f\vert O_b\vert i\rangle \delta(\omega - E_f + E_i)$	eV^2 for density operators	4.2
$\mathcal{G}_{\rm VH}(r,t)$	Van Hove correlation function	$\mathrm{Å}^{-3}$	4.2
$S^{\rm st}(q)$	Static structure factor	1	4.2
$\tilde{V}_b(q)$, $\tilde{V}_e$, $\tilde{V}_{n_e}$	Potential with which the dark matter couples to the target operator $O_b(q)$; $\tilde{V}_e = \tilde{V}_{n_e} = \tilde{V}$ for the density	eV^{-2}	4.2
O_x	Combined operator $\sum_b x_b O_b$ with coefficients x_b in the positivity proof	—	4.2
$S^0(q,\omega)$	Independent-particle structure factor	eV^2	4.2
$\hat{\boldsymbol{n}}_e(r)$	Position-space electron density operator $\sum_j \delta^3(\boldsymbol{r} - \boldsymbol{r}_j)$	$\mathrm{Å}^{-3}$	4.2
$g_{\rm pair}(r)$	Pair distribution function	1	4.2
$\boldsymbol{R}_\ell$, $\boldsymbol{r}_\ell$	Nuclear and electron positions of atom ℓ in the hydrogen-gas example	Å	4.2
$\omega_\pm(q)$	Upper and lower edges of the particle–hole continuum of the free gas, $qv_{\rm F} + q^2/2m_e$ and $\max(0, q^2/2m_e - qv_{\rm F})$	eV	4.2
u	Cosine of the angle between k (or v) and q in angular integrals	1	4.2
$N_{\rm at}$	Number of atoms of a sample or of a (super)cell	1	4.2
$\mathscr{M}^1$	First-moment (double-commutator) matrix	eV^4	4.3
$j_e(q)$	Paramagnetic electron current $\sum \frac{1}{2}\{p/m_e, \mathrm{e}^{\mathrm{i}q\cdot\boldsymbol{r}}\}$	1	4.3
ω_p	Electron plasma frequency, $\omega_p^2 = e^2 n_e/m_e$	eV	4.3
$f_{n0}^{\rm osc}$	Oscillator strength of the transition $0 \to n$	1	4.3
$\bar{K}$	Mean kinetic energy per electron (the relativistic f-sum estimate)	eV	4.3
n_κ, $j_\kappa(q)$	Number density of the nuclei of species κ; their paramagnetic current $\sum_{I\in\kappa} \frac{1}{2}\{P_I/M_\kappa, \mathrm{e}^{\mathrm{i}q\cdot X_I}\}$	$\mathrm{Å}^{-3}$; 1	4.3
$\mathscr{M}^1_{ss'}$, $\mathscr{M}^1_{ee}$, $\mathscr{M}^1_{ZZ}$	Species elements of the first-moment matrix $\mathscr{M}^1$ (electrons e, nuclei κ); its charge-number element	eV^4	4.3
$S_{e\kappa}$, $S_{\kappa e}$	Electron–nucleus elements of the response matrix S_{ab}	eV^2	4.3
$[H,O]_k$	O commuted k times with H	—	4.3
$f_2(\omega)$	Imaginary part of the atomic scattering factor, $(\pi/2)\omega \mathrm{d}f^{\rm osc}/\mathrm{d}\omega$	1	4.3

Symbol	Meaning	Units	Sect.
$\Omega_{\rm arc}$	Radius of the large semicircle of the Kramers–Kronig contour	eV	4.3
$\tilde{R}$	Coulomb-proper (irreducible) response	as R	4.4
$\varepsilon(q,\omega)$, ε_1, ε_2	Longitudinal dielectric function, its real and imaginary parts	1	4.4
$\varepsilon^{\rm T}(q,\omega)$	Transverse dielectric function; $\varepsilon^{\rm T}\to\varepsilon$ as $q\to 0$	1	4.4
$\varepsilon_{\boldsymbol{GG'}}(q,\omega)$	Dielectric matrix with local fields	1	4.4
$\varepsilon_{\rm mac}$	Macroscopic dielectric function $1/(\varepsilon^{-1})_{\mathbf{00}}$	1	4.4
ε_∞	Electronic (static-lattice) dielectric constant	1	4.4
$L(q,\omega)$	Loss function $\mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$	1	4.4
$k_{\rm TF}$	Thomas–Fermi wavevector	eV	4.4
$\sigma_{\rm cond}(\omega)$, σ_1, σ_2	Optical conductivity and its real and imaginary parts; $\sigma_1=\omega\varepsilon_2$	eV	4.4
$-dE/\mathrm{d}x$	Stopping power (energy loss per unit path)	eV Å^{-1}	4.4
$\phi_{\rm tot}(q,\omega)$	Total (external + Hartree) potential energy acting on an electron	eV	4.4
$\delta n_e(q,\omega)$, $\delta n_{e,\boldsymbol{G}}$	Amplitude of the induced electron density $\delta\langle n_e(q)^\dagger\rangle/\mathscr{V}$; its component at $q+\boldsymbol{G}$	eV^3	4.4
$R_{\boldsymbol{GG'}}(q,\omega)$, $\tilde{R}_{\boldsymbol{GG'}}$	Density-response matrix in reciprocal-lattice vectors $(O_a=n_e(q+\boldsymbol{G}), O_b=n_e(q+\boldsymbol{G'}))$; its proper part	eV^2	4.4
$\bar{R}$, $\bar{R}_{\mathbf{00}}$	Response with the $\boldsymbol{G}=0$ Coulomb term removed; $\varepsilon_{\rm mac}=1-v_{\rm C}(q)\bar{R}_{\mathbf{00}}$	eV^2	4.4
R_{ZZ}, S_{ZZ}	Response and structure factor of the charge-number density n_Z	eV^2	4.4
$R^{\alpha\beta}_{jj}$, $\tilde{R}^{\alpha\beta}_{jj}$	Current–current response and its proper part	eV^2	4.4
$\varepsilon^{\alpha\beta}(q,\omega)$, $\varepsilon^{\rm s}$	Dielectric tensor ($\varepsilon^{\rm T}$ and ε are its projections); symmetrized dielectric matrix	1	4.4
$\boldsymbol{J}$	Induced current density, $\boldsymbol{J}=\sigma_{\rm cond}\boldsymbol{E}$	eV^3	4.4
$\varepsilon_{\rm TF}(q)$	Thomas–Fermi dielectric function $1+k_{\rm TF}^2/q^2$	1	4.4
Z_1, M_1	Charge number and mass of a heavy charged projectile in dielectric stopping, $M_1\to\infty$	1; eV	4.4
$\omega_{\rm res}(q)$	Resonance frequency of the single-oscillator loss function	eV	4.4
$\bar{E}$	Mean excitation energy of the loss function, $\bar{E}^2=\int\omega L/\int(L/\omega)$ at $q\to 0$	eV	4.4
R^0	Independent-particle bubble: of the gas at any T, of a crystal, $R^0_{\boldsymbol{GG'}}$, in imaginary frequency $R^0(q,\mathrm{i}\omega_j^{\rm B})$	as R	4.5
$\varepsilon_{\rm Lin}$, $\varepsilon_{\rm Mer}$	Lindhard and Mermin dielectric functions	1	4.5
$\omega_{\rm pl}(q)$, $q_{\rm c}$	Plasmon dispersion; Landau-damping onset	eV	4.5
$\gamma_{\rm rel}$	Relaxation rate	eV	4.5
$\alpha_{\rm pl}$	Plasmon dispersion coefficient, $\omega_{\rm pl}\simeq\omega_p+\alpha_{\rm pl}q^2/m_e, \alpha_{\rm pl}=(3/5)E_{\rm F}/\omega_p$	1	4.5
$\omega_{p,i}$, γ_i, $\omega_{{\rm edge},i}$, $A_i(q)$	Parameters of the i-th term of the Mermin-oscillator fit	eV; eV; eV; 1	4.5
ν, $\nu_\pm$	Reduced frequency $z/qv_{\rm F}$ and $\nu\pm q/2k_{\rm F}$ of the Lindhard function (ν in Sect. 4.5 and Appendix D.6)	1	4.5
y	$q/2k_{\rm F}$ in the Lindhard function	1	4.5
$F_{\rm L}(y)$	Static Lindhard function $\frac{1}{2}+[(1-y^2)/4y]\ln\lvert(1+y)/(1-y)\rvert$	1	4.5
$\ell(w)$	Lindhard logarithm $\ln[(w+1)/(w-1)]$ on the principal branch	1	4.5
$R^{\rm Mer}$	Mermin (relaxation-time, number-conserving) density response; $\varepsilon_{\rm Mer}=1-v_{\rm C}R^{\rm Mer}$	eV^2	4.5
$g_{\rm deg}$	Degeneracy factor (number of internal states): spin degeneracy 2 of the electron–hole bubble and of the singlet exciton	1	4.6

Symbol	Meaning	Units	Sect.
f_{xc}, f_{Hxc}	Exchange–correlation kernel; Hartree-plus-xc kernel $v_{\mathrm{C}} + f_{\mathrm{xc}}$	eV^{-2}	4.6
$G_{\mathrm{LF}}(q)$	Static local-field factor	1	4.6
$\Delta_{cv\boldsymbol{k}}$	Vertical transition energy $\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}$	eV	4.6
$\varepsilon_2^{\mathrm{IP}}$, $\varepsilon_1^{\mathrm{IP}}$	Independent-particle dielectric function at $q \to 0$	1	4.6
$n_{\mathrm{gs}}(r)$	Ground-state density at which f_{xc} is evaluated	$\mathrm{Å}^{-3}$	4.6
G	Single-particle Green's function	eV^{-1}	4.7
$\mathsf{A}(k,\omega)$	Spectral function	eV^{-1}	4.7
Σ	Electron self-energy	eV	4.7
τ	Imaginary time, $0 \le \tau \le \beta$	eV^{-1}	4.7
ω_j^{F}, ω_j^{B}	Fermionic $(2j+1)\pi T$ and bosonic $2j\pi T$ Matsubara frequencies	eV	4.7
z_{qp}	Quasiparticle residue; $z_{\mathrm{qp}}(n,\boldsymbol{k})$ the residue of the one-electron state $n\boldsymbol{k}$	1	4.7
H_μ, H_{int}	Grand-canonical Hamiltonian $H - \mu_{\mathrm{ch}}\hat{N}_e$; its interaction part	eV	4.7
G^{R}, G^{A}, G, G_0	Retarded, advanced and time-ordered Green's functions; subscript 0 without interactions; $\mathsf{G}(\boldsymbol{k},\mathrm{i}\omega_j^{\mathrm{F}})$, $\mathsf{G}(\boldsymbol{k},\tau)$ Matsubara forms	eV^{-1}	4.7
$\mathsf{G}^{\mathrm{R}}_{dd}$, $\mathsf{G}^{\mathrm{R}}_{\boldsymbol{k}d}$, A_{dd}, Σ^{R}_d	Green's functions of the level and the mixed level–band function, the spectral function and the self-energy of the level	eV^{-1}; eV^{-1}; eV^{-1}; eV	4.7
$\mathsf{F}^{\mathrm{R}}(k,\omega)$	Retarded correlation of $[c_{\boldsymbol{k}}, H_{\mathrm{int}}]$ with $c^\dagger_{\boldsymbol{k}}$ in the equation of motion	1	4.7
$\Sigma^{\mathrm{R}}(k,\omega)$, $\Sigma(k,\mathrm{i}\omega_j^{\mathrm{F}})$	Retarded and Matsubara self-energies	eV	4.7
T_τ, T	Imaginary-time and real-time ordering symbols	—	4.7
d, $d^\dagger$, E_{lev}, E_{w}, $V_{\boldsymbol{k}}$	Level operators (a fermion), level energy, band width and hybridization	—; eV; eV; eV	4.7
$\xi_{\boldsymbol{k}}$, $E_{\boldsymbol{k}}$	Energy of orbital $\boldsymbol{k}$ from the chemical potential, $E - \mu_{\mathrm{ch}}$; Bogoliubov energy $(\xi_{\boldsymbol{k}}^2 + \Delta_{\mathrm{BCS}}^2)^{1/2}$	eV	4.7
W^{scr}, $W^{\mathrm{scr,R}}$	Screened Coulomb interaction $v_{\mathrm{C}} + v_{\mathrm{C}}Rv_{\mathrm{C}} = \varepsilon^{-1}v_{\mathrm{C}}$, in a crystal the matrix $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}v_{\mathrm{C}}(\lvert\boldsymbol{q}+G'\rvert)$; $W^{\mathrm{scr,R}}$ its retarded form; time-ordered when written with a real argument and no further superscript; evaluated at any $(\boldsymbol{q},\omega)$, complex where pairs are open, e.g	eV^{-2}	4.8
ω_{Ein}	Energy of a dispersionless (Einstein) boson; the Einstein energy $\langle\omega^{-1}\rangle^{-1}$ of a phonon spectrum, 30.4 meV for silicon	eV	4.8
$d_{\boldsymbol{Q}}$, $\omega_{\boldsymbol{Q}}$, $\phi_{\boldsymbol{Q}}$, $g(Q)$	Generic boson mode $\boldsymbol{Q}$, its energy, its field $d_{\boldsymbol{Q}} + d^\dagger_{-Q}$ and its coupling to electrons, $g(-\boldsymbol{Q}) = g(\boldsymbol{Q})^*$	—; eV; —; eV	4.8
$\mathsf{D}_0(Q,\mathrm{i}\omega_j^{\mathrm{B}})$, $\mathsf{D}_0(Q,\omega)$	Free boson propagator $2\omega_{\boldsymbol{Q}}/((\mathrm{i}\omega_j^{\mathrm{B}})^2 - \omega_{\boldsymbol{Q}}^2)$; time-ordered $2\omega_{\boldsymbol{Q}}/(\omega^2 - \omega_{\boldsymbol{Q}}^2 + \mathrm{i}0^+)$	eV^{-1}	4.8
Γ_{em}, Γ_{out}, Γ_{in}	Golden-rule boson-emission rate; out- and in-scattering rates of a fermion state, $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}} = \Gamma_{\mathrm{out}} + \Gamma_{\mathrm{in}}$	eV	4.8
E_{c}, $\ell_{\mathrm{b}}(x)$	Conduction-band bottom of the model band and band-integral logarithm $\ln[(x - E_{\mathrm{c}} + \mathrm{i}0^+)/(x - E_{\mathrm{c}} - E_{\mathrm{w}} + \mathrm{i}0^+)]$	eV; 1	4.8
ξ'	ξ_{k-Q}, energy of the intermediate electron in the boson self-energy	eV	4.8
$B(q,\omega)$	Spectral weight of $W^{\mathrm{scr}} - v_{\mathrm{C}}$, $-\,\mathrm{Im}\,W^{\mathrm{scr,R}}/\pi = v_{\mathrm{C}}L/\pi$	eV^{-2}	4.8
E_{cs}	Total energy of the lines crossed by a cut	eV	4.8
$\bar{H}$	Hamiltonian without the Coulomb terms at $\pm\boldsymbol{q}$	eV	4.8
$A^\lambda_{cv\boldsymbol{k}}$, E_λ, λ	Exciton envelope, exciton energy, exciton label: eigenpairs of the transition-basis BSE with K^{x} (no $\boldsymbol{G} = 0$ term), which depend on $\boldsymbol{q}$	1; eV	4.9

Symbol	Meaning	Units	Sect.
E_x^b, a_x, μ_x	Exciton binding energy, radius, reduced mass of the Wannier–Mott (hydrogenic) exciton	eV; Å; m_e	4.9
$F_\lambda(r)$, F_{ns}	Exciton envelope function in the relative coordinate, $\mathscr{V}^{-1/2}\sum_k A_{\boldsymbol{k}}^\lambda \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot r}$, $\int \lvert F\rvert^2 = 1$; hydrogenic envelope	$\text{Å}^{-3/2}$	4.9
K^{x}, K^{d}, H^{exc}	Bethe–Salpeter exchange kernel without its $\boldsymbol{G}=0$ term and direct kernel; resonant (Tamm–Dancoff) BSE Hamiltonian $\Delta + g_{\mathrm{deg}}K^{\mathrm{x}} - K^{\mathrm{d}}$ in the transition basis	eV	4.9
Chapter 5: Lattice dynamics			
$E^{\mathrm{ad}}(u)$	Born–Oppenheimer energy surface $E^{\mathrm{ad}}(u) \equiv E_0(u)$, the ground-state eigenvalue of $H_{\mathrm{el}}(u)$ at fixed nuclei, nuclear repulsion V_{NN} included	eV	5.1
κ_{BO}	Born–Oppenheimer small parameter $(m_e/M)^{1/4}$	1	5.1
F_κ, $F_{\boldsymbol{R}\kappa}$	Zone-center Hellmann–Feynman force on sublattice κ, $-\partial E_{\mathrm{cell}}^{\mathrm{ad}}/\partial u_\kappa$; force on the nucleus κ of cell $\boldsymbol{R}$	eV Å^{-1}	5.1
$u_{\boldsymbol{R}\kappa}$, u	Nuclear displacement from equilibrium, $X_I = \boldsymbol{R} + \tau_\kappa + u_{\boldsymbol{R}\kappa}$; the set of all displacements	Å	5.1
$\Phi^{(n)}$	n-th order force constants	eV Å^{-n}	5.1
$E_{\mathrm{cell}}^{\mathrm{ad}}$	Energy surface per cell for a uniform displacement of each sublattice	eV	5.1
$H_{\mathrm{el}}(u)$	Electronic Hamiltonian at fixed nuclei, $T_e + V_{ee} + V_{eN} + V_{NN}$	eV	5.1
T_e, V_{ee}, V_{eN}, V_{NN}	Electron kinetic energy; electron–electron, electron–nucleus and nucleus–nucleus interaction energies	eV	5.1
$\Psi_i(r;u)$, $E_i(u)$	Adiabatic electronic states (real) and energies at fixed nuclei, eigenstates of $H_{\mathrm{el}}(u)$; $E_0(u) = E^{\mathrm{ad}}(u)$	—; eV	5.1
$\Psi_i^{\mathrm{nuc}}(u)$	Nuclear wavefunctions of the adiabatic expansion	—	5.1
$\mathcal{N}_{ji}$	Nonadiabatic coupling operator between adiabatic states j and i, first- and second-derivative (derivative and scalar) couplings, $\propto 1/M_\kappa$	eV	5.1
ℓ_{zp}	Zero-point length $(M_\kappa\omega)^{-1/2}$ of the bond coordinate (reduced mass $M_\kappa/2$), and the estimate $\kappa_{\mathrm{BO}}a_0$ in the power counting	Å	5.1
ω_{ph}	Generic phonon energy in estimates	meV	5.1
ΔE, F_{j0}	Excitation energy $E_j - E_0$ of the adiabatic states in estimates; matrix element of the force between Ψ_j and Ψ_0 (two state labels)	eV; eV Å^{-1}	5.1
ω_{opt}, $\omega_{\Gamma,\mathrm{opt}}$	Zone-center optical frequency of silicon	meV	5.1
Φ_{opt}, $\Phi_{\mathrm{opt}}^{(3)}$, $\Phi_{\mathrm{opt}}^{(4)}$	Curvature, cubic and quartic coefficients of $E_{\mathrm{cell}}^{\mathrm{ad}}$ along the [111] bond coordinate of silicon	eV Å^{-n}	5.1
$\langle u_x^2\rangle$, $\langle u^2\rangle$	Mean-square displacement of an atom along one Cartesian axis ($= B_\kappa^{xx}$); its trace tr B_κ, $\langle u^2\rangle = 3\langle u_x^2\rangle$ at a cubic site	Å^2	5.1
$D^{\mathrm{dyn}}(Q)$	Dynamical matrix	eV^2	5.2
Q	Phonon wavevector (first zone)	eV	5.2
ν	Phonon branch	—	5.2
$\omega_{\boldsymbol{Q}\nu}$	Phonon frequency	meV	5.2
$e_{\kappa\nu}(Q)$	Phonon polarization vector, $\sum_\kappa \lvert e_{\kappa\nu}\rvert^2 = 1$	1	5.2
$g_{\mathrm{ph}}(\omega)$	Phonon density of states $N^{-1}\sum_{\boldsymbol{Q}\nu}\delta(\omega - \omega_{\boldsymbol{Q}\nu})$, normalized to $3n_{\mathrm{at}}$ modes per cell	eV^{-1}	5.2
Θ_{D}, ω_{D}	Debye temperature and Debye frequency ($\omega_{\mathrm{D}} = \Theta_{\mathrm{D}}$ in natural units)	K; meV	5.2
$\Lambda^{\mathrm{ac},\alpha\beta}(\hat{Q})$, $\Lambda_\nu^{\mathrm{ac}}(\hat{Q})$	Acoustic sound-speed matrix, $\omega^2 = \Lambda^{\mathrm{ac}}Q^2$ at long wavelength, and its eigenvalues c_ν^2	$\text{km}^2\,\text{s}^{-2}$	5.2

Symbol	Meaning	Units	Sect.
$D_n(Q)$, $T^{\alpha\beta\gamma}$, $\boldsymbol{a}_{\hat{e}}$, $\boldsymbol{a}'_{\hat{e}}$, $\boldsymbol{o}$, ω_{o}	Order-n terms ($n = 0, 1, 2$) of D^{dyn} in powers of Q; first moment $\sum_R \Phi^{(2)} R^\gamma$ of the force constants; acoustic and optical vectors at Γ and an optical frequency at Γ	eV^2; $\mathrm{eV\,\AA}^{-1}$; 1; meV	5.2
a_{c}, K, M_1, M_2, $\omega_{Q\pm}$, ω_{0+}	Period, spring constant and masses of the diatomic chain; its two branches and the upper branch at $Q = 0$	Å; $\mathrm{eV\,\AA}^{-2}$; eV; meV	5.2
Q_{D}	Debye wavevector $(6\pi^2 n_{\mathrm{at}}/\Omega)^{1/3}$, one wavevector per atom	$\mathrm{\AA}^{-1}$	5.2
$\bar{c}$	Debye-averaged sound speed, $3/\bar{c}^3 = \sum_\nu \langle c_\nu^{-3} \rangle$	$\mathrm{km\,s}^{-1}$	5.2
$g_{\mathrm{D}}(\omega)$	Debye density of states $9 n_{\mathrm{at}} \omega^2/\omega_{\mathrm{D}}^3$ per cell below ω_{D}	eV^{-1}	5.2
μ_{red}	Reduced mass: of the diatomic chain, of a polar mode	eV	5.2
$\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}$	Grüneisen parameter $-\partial \ln \omega_{\boldsymbol{Q}\nu}/\partial \ln \Omega$ of mode $Q\nu$	1	5.2
$b_{\boldsymbol{Q}\nu}$, $b^\dagger_{\boldsymbol{Q}\nu}$	Phonon operators	—	5.3
$B_{\boldsymbol{Q}\nu}$	Displacement field $b_{\boldsymbol{Q}\nu} + b^\dagger_{-Q\nu}$	—	5.3
$n_{\boldsymbol{Q}\nu}$	Phonon occupation $n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$	1	5.3
$\ell_{\boldsymbol{Q}\nu}$	Zero-point length $(2M_{\mathrm{cell}}\omega_{\boldsymbol{Q}\nu})^{-1/2}$, with the $1/\sqrt{N}$ kept outside	Å	5.3
B_κ	Displacement tensor $\langle u_\kappa u_\kappa \rangle$	$\mathrm{\AA}^2$	5.3
H_{N}	Harmonic nuclear Hamiltonian	eV	5.3
$\boldsymbol{p}_{\boldsymbol{R}\kappa}$	Momentum $-\mathrm{i}\nabla_{\boldsymbol{R}\kappa}$ of the nucleus κ in cell $\boldsymbol{R}$	eV	5.3
$u_{\boldsymbol{Q}\nu}$, $p_{\boldsymbol{Q}\nu}$	Normal coordinate and normal momentum of mode $Q\nu$	$\mathrm{eV}^{-1/2}$; $\mathrm{eV}^{1/2}$	5.3
$g_{\mathrm{ph},\kappa}(\omega)$	Atom-projected phonon density of states $N^{-1} \sum_{\boldsymbol{Q}\nu} \lvert e_{\kappa\nu}(\boldsymbol{Q}) \rvert^2 \delta(\omega - \omega_{\boldsymbol{Q}\nu})$, normalized to three	eV^{-1}	5.3
$\bar{\omega}$	Typical phonon frequency of the estimate	meV	5.3
$\mathcal{E}$, $\boldsymbol{P}$	Macroscopic electric field; macroscopic polarization	eV^2; eV^2	5.4
Z^*_κ	Born effective-charge tensor $\partial F_\kappa/\partial \mathcal{E}$ (in units of e)	1	5.4
ε_0	Static dielectric constant $\varepsilon(0, \omega \to 0)$, lattice included: a pure number, never the vacuum permittivity	1	5.4
ω_{L}, ω_{T}	Longitudinal and transverse optical frequencies at $\boldsymbol{Q} \to 0$	meV	5.4
$\boldsymbol{D}$	Electric displacement field $\mathcal{E} + \boldsymbol{P}$	eV^2	5.4
$H_{\mathcal{E}}(t)$	Coupling $-e \sum u \cdot Z^* \cdot \mathcal{E}$ of the nuclei to a uniform macroscopic field	eV	5.4
Z^*_{Ga}, Z^*_{As}, Z^*	Born charges of GaAs by species; their common magnitude in a two-atom cubic crystal, $Z^*_{\mathrm{Ga}} = -Z^*_{\mathrm{As}} = Z^*$	1	5.4
$\boldsymbol{u}_{\mathrm{rel}}$	Relative displacement $\boldsymbol{u}_1 - \boldsymbol{u}_2$ of the two sublattices of a diatomic polar crystal	Å	5.4
$C^{\mathrm{an}}_{\kappa\kappa'}$, $C^{\mathrm{na}}_{\kappa\kappa'}(\hat{Q})$, C_0	Analytic (zero-field) and nonanalytic (dipole) zone-center force constants; the single constant of the cubic diatomic crystal	$\mathrm{eV\,\AA}^{-2}$	5.4
$\boldsymbol{z}_\nu$	Mode charge vector $\sum_\kappa e_{\kappa\nu} \cdot Z^*_\kappa/\sqrt{M}_\kappa$	$\mathrm{eV}^{-1/2}$	5.4
$E^{\mathrm{ad}}_{\mathcal{E}}(u, \mathcal{E})$	Energy per cell at fixed macroscopic field; E^{ad} at $\mathcal{E} = 0$	eV	5.4
$\omega_{\mathrm{T}\nu}$, $\omega_{\mathrm{L}\nu}$, $\omega_{0\nu}$	Transverse and longitudinal zone-center frequencies of polar mode ν (generalized Lyddane–Sachs–Teller relation); frequency of branch ν at Γ without the field ($\omega_{\mathrm{T}}, \omega_{\mathrm{L}}$ for a single polar mode)	meV	5.4
$W_\kappa(q)$	Half the Debye–Waller exponent: the exponent is $2W_\kappa(\boldsymbol{q}) = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}$	1	5.5
$\mathcal{C}_{\kappa\kappa'}(R, t)$, $\mathcal{C}_{\kappa\kappa'}(t)$	Displacement correlation function $\langle u_{\boldsymbol{R}\kappa}(t) u_{\boldsymbol{0}\kappa'}(0) \rangle$; $\mathcal{C}_{\kappa\kappa'}(t)$ where the cell is clear	$\mathrm{\AA}^2$	5.5
S_{HR}	Huang–Rhys factor	1	5.5
$P_1(\omega)$, $P_n(\omega)$	Single-phonon distribution of the self response ($\propto g_{\mathrm{ph}}(\lvert\omega\rvert)[n_{\mathrm{B}}(\omega) + 1]/\omega$, unit norm); its n-fold convolution, the n-phonon distribution	eV^{-1}	5.5

Symbol	Meaning	Units	Sect.
p_n^{ph}	Probability that a momentum transfer involves n phonons	1	5.5
H_{g}, H_{e}	Nuclear Hamiltonians on the ground and excited electronic surfaces of a displaced oscillator	eV	5.5
Δx, $A(E)$, w_p	Shift of the equilibrium coordinate between the two surfaces; absorption line shape of the displaced oscillator; weight of the line $E_{\mathrm{ZPL}} + p\omega_0$	Å; eV^{-1}; 1	5.5
$I_p(z)$	Modified Bessel function of the first kind	1	5.5
E_{ZPL}	Zero-phonon-line energy of a color center	eV	5.5
ℓ_0	Zero-point length $(2M\omega_0)^{-1/2}$ of a single oscillator of mass M and frequency ω_0	Å	5.5
$\gamma_{\boldsymbol{Q}\nu}$, $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$	Phonon linewidth (full width); its anharmonic part	meV	5.6
η_{anh}	Anharmonic small parameter	1	5.6
$V^{(3)}(1,2,3)$	Three-phonon vertex, an energy; the labels $i = (Q_i\nu_i)$	meV	5.6
H_3	Cubic part of the nuclear Hamiltonian in phonon fields (H_N is the harmonic part)	eV	5.6
$\Delta\omega_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$	Anharmonic frequency shift	meV	5.6
$\gamma_{\Gamma,\mathrm{opt}}$, $\gamma_{\Gamma,\mathrm{opt}}^{\mathrm{anh}}$	Width (FWHM) and anharmonic width of the zone-center optical phonon of silicon ($\gamma_{\boldsymbol{Q}\nu}$ at $\boldsymbol{Q} = \Gamma, \nu = \mathrm{opt}$)	meV	5.6
$\mathcal{F}_{\mathrm{trial}}$, V_{trial}, δX_a	Free energy and potential of the SCHA trial Hamiltonian H_{trial}; displacement $X_a - \bar{X}_a$ from the centroid	eV; eV; Å	5.6
$\varepsilon^{\alpha\beta}$	Homogeneous strain tensor of the quasi-harmonic argument	1	5.6
$\rho_{\mathrm{pair}}(\omega_1)$	Two-phonon density of final states resolved by the energy of one daughter, $N^{-1}\sum\delta(\omega_1 - \omega_{\boldsymbol{Q}\nu_1})\delta(\omega_{\mathrm{opt}} - \omega_{\boldsymbol{Q}\nu_1} - \omega_{\boldsymbol{Q}\nu_2})$	eV^{-2}	5.6
S_{ent}	Entropy	1	5.6
$\tilde{\varrho}$, H_{trial}	Trial density matrix of the SCHA; trial harmonic Hamiltonian	1; eV	5.6
X_a, $\bar{X}_a$, $\bar{X}$	Cartesian nuclear coordinates of a supercell; their centroids	Å	5.6
$\Phi_{\mathrm{aux}}^{(2)}$, $\Phi_{\mathrm{aux},ab}^{(2)}$, $\tilde{\Phi}^{(2)}$, $\tilde{\Phi}_{ij}^{(2)}$	SCHA auxiliary force constants; TDEP effective second-order force constants	$\mathrm{eV\,Å}^{-2}$	5.6
Chapter 6: Electron–phonon coupling			
H_{ep}	Electron–phonon Hamiltonian, general bare form	eV	6.1
v_N	Nuclear velocity after a kick, q/M	1	6.1
$\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}$	Derivative-coupling vector $\langle\Psi_j\vert\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle = \boldsymbol{F}_{ji}/(E_i - E_j)$, antisymmetric, $\boldsymbol{d}_{ii} = 0$, a Berry connection	eV (quoted in $\mathrm{Å}^{-1}$)	6.1
ΔE_j, $\boldsymbol{F}_{ji}$	Excitation energy $E_j - E_0$ of the adiabatic state j; matrix element $\langle\Psi_j\vert\nabla_{\boldsymbol{R}\kappa}V_{eN}\vert\Psi_i\rangle$ of the force	eV; $\mathrm{eV\,Å}^{-1}$	6.1
τ_{acc}	Acceleration time of a kicked nucleus	eV^{-1} (quoted in s)	6.1
ω_{e}	Frequency of the bound-electron oscillator of the solvable model	eV	6.1
P_0	Probability that the electrons stay in the ground state after a kick, $\mathrm{e}^{-\lvert\alpha_{\mathrm{d}}\rvert^2}$ in the model, $\lvert F_{1s}\rvert^2$ for hydrogen	1	6.1
M_{H}, μ_{ep}, a_{ep}	Hydrogen-atom mass $m_p + m_e$; electron–proton reduced mass $m_e m_p/M_{\mathrm{H}}$; its Bohr radius $1/(\alpha\mu_{ep})$	eV; eV; Å	6.1
$\mu_{\chi\mathrm{H}}$	Dark-matter–hydrogen-atom reduced mass	eV	6.1
$\boldsymbol{\rho}$, $\boldsymbol{R}_{\mathrm{cm}}$	Relative coordinate $\boldsymbol{r}_e - \boldsymbol{X}$ and center of mass $(m_p\boldsymbol{X} + m_e\boldsymbol{r}_e)/M_{\mathrm{H}}$ of the hydrogen atom; $\boldsymbol{\rho}_e$ and $\boldsymbol{R}_{\mathrm{cm}}$ of a many-electron atom with the mass m_A	Å	6.1

Symbol	Meaning	Units	Sect.
$g_{mn\nu}(k,Q)$	Electron–phonon vertex	eV	6.2
g^{S}, g^{L}	Short- and long-range parts of g	eV	6.2
D^n, $D^n_{\alpha\beta}$, d^n_ν	Acoustic deformation-potential tensor and its Cartesian components; optical deformation potential of band n and zone-center mode ν	eV; eV Å^{-1}	6.2
C_{F}, α_{F}	Fröhlich coupling strength ($g^{\mathrm{L}} = \mathrm{i}C_{\mathrm{F}}\rho_{mn}/Q, C_{\mathrm{F}}^2 = (e^2\omega_{\mathrm{L}}/2\Omega)(1/\varepsilon\infty - 1/\varepsilon_0)$); dimensionless Fröhlich constant $\alpha(1/\varepsilon\infty - 1/\varepsilon_0)(m^*/2\omega_{\mathrm{L}})^{1/2}$	eV^2 (natural units); 1	6.2
$\partial_{\kappa\alpha,Q}v_{\mathrm{KS}}$, v_{KS}	Lattice-periodic part of the potential change, $\partial_{\kappa\alpha,Q}V_{\mathrm{KS}} = \mathrm{e}^{\mathrm{i}Q\cdot r}\partial_{\kappa\alpha,Q}v_{\mathrm{KS}}$ (lower-case v marks the periodic part)	eV Å^{-1}	6.2
$\Delta_{\boldsymbol{Q}\nu}v_{\mathrm{KS}}$	Mass-weighted mode potential $\sum_{\kappa\alpha}(M_{\mathrm{cell}}/M_\kappa)^{1/2}\mathrm{e}^\alpha_{\kappa\nu}\partial_{\kappa\alpha,Q}v_{\mathrm{KS}}$; $g_{mn\nu} = \ell_{\boldsymbol{Q}\nu}\langle u_{m,\boldsymbol{k}+\boldsymbol{Q}}\vert\Delta_{\boldsymbol{Q}\nu}v_{\mathrm{KS}}\vert u_{n\boldsymbol{k}}\rangle_\Omega$	eV Å^{-1}	6.2
$S_{\alpha\beta}$, E_1	Strain tensor $\frac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha)$ of a long acoustic wave; isotropic deformation potential $\Omega\partial\varepsilon_n/\partial\Omega$ (Bardeen–Shockley)	1; eV	6.2
Ξ_d, Ξ_u, $\hat{k}_{\mathrm{c}}$	Herring–Vogt dilatation and uniaxial deformation potentials of a conduction valley; unit vector along the valley axis	eV; 1	6.2
$e^{\mathrm{pz}}_{\alpha\beta\gamma}$, g^{pz}	Piezoelectric tensor; piezoelectric vertex, $\propto Q^{-1/2}$	eV^2 (quoted in C m^{-2}); eV	6.2
$\boldsymbol{t}$, $\hat{T}_{\boldsymbol{t}}$	Rigid translation of the whole crystal; its translation operator $\mathrm{e}^{-\mathrm{i}\boldsymbol{t}\cdot\hat{p}}$	Å; —	6.2
g^W, $\boldsymbol{R}_e$, $\boldsymbol{R}_p$	Real-space (Wannier-gauge) electron–phonon matrix $g^W_{m'n'\kappa\alpha}(\boldsymbol{R}_e,\boldsymbol{R}_p)$; lattice vectors of the electron hop and of the displaced atom	eV Å^{-1}; Å	6.2
Σ^{FM}, Σ^{AH}, g^{AH}	Fan–Migdal and Allen–Heine self-energies; second-order (Allen–Heine) vertex	eV	6.3
$\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}$	Phonon-renormalized band energy	eV	6.3
$H^{(2)}_{\mathrm{ep}}$	Second-order electron–phonon Hamiltonian $(1/N)\sum g^{\mathrm{AH}}c^\dagger cBB$	eV	6.3
ZPR	Zero-point renormalization $\varepsilon^{\mathrm{ren}}(T=0) - \varepsilon$ of a state; of a gap, conduction minus valence	meV	6.3
$\Delta\varepsilon_{n\boldsymbol{k}}(T)$	Phonon-induced shift $\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}(T) - \varepsilon_{n\boldsymbol{k}}$	eV	6.3
$\hat{p}_\nu$	$\hat{e}_\nu\cdot\hat{p}$, momentum along a polarization	eV	6.3
Ξ, $Q_{\max}$	Effective deformation constant of one branch, Ξ^2 averaged over directions; largest emitted wavevector $2(\boldsymbol{k} - m^*c)$	eV; eV	6.3
m^*_{d}	Density-of-states mass $(m^*_\parallel m^{*2}_\perp)^{1/3}$ of one valley, 0.322 m_e in Si	eV (quoted in m_e)	6.3
$\Pi_\nu(Q,\omega)$	Phonon self-energy of branch ν	eV^2	6.4
$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$	Electron–phonon part of the phonon linewidth, $-\operatorname{Im}\Pi_\nu(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu})/\omega_{\boldsymbol{Q}\nu}$, a full width	meV	6.4
$\alpha^2F(\omega)$	Eliashberg spectral function	1	6.4
λ_{ep}	Electron–phonon coupling constant $2\int \mathrm{d}\omega\alpha^2F(\omega)/\omega$	1	6.4
$\mathsf{D}_\nu(Q,\omega)$, D	Dressed phonon propagator $2\omega_{\boldsymbol{Q}\nu}/(\omega^2 - \omega^2_{\boldsymbol{Q}\nu} - \Pi_\nu)$	eV^{-1}	6.4
$\Pi^{\mathrm{b}}_\nu(Q,\omega)$, Π^{b}	Phonon bubble in energy units, $\Pi_\nu = 2\omega_{\boldsymbol{Q}\nu}\Pi^{\mathrm{b}}_\nu$	eV	6.4
$\tilde{\omega}_{\boldsymbol{Q}\nu}$, $\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}$	Pole $\omega_{\boldsymbol{Q}\nu} + \operatorname{Re}\Pi_\nu/2\omega_{\boldsymbol{Q}\nu}$ of the dressed phonon; nonadiabatic (renormalized) frequency, $\omega_{\boldsymbol{Q}\nu}$ being the adiabatic DFPT one	meV	6.4
$\lambda_{\boldsymbol{Q}\nu}$	Mode coupling $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}/(2\pi\mathcal{N}_{\mathrm{F}}\Omega\omega^2_{\boldsymbol{Q}\nu})$, λ_{ep} its zone average	1	6.4
$\langle\ldots\rangle_{\mathrm{FS}}$	Fermi-surface average	—	6.4
$w(q)$	Screened empty-core ionic form factor per cell $-3e^2\cos(qr_{\mathrm{c}})/[\Omega q^2\varepsilon_{\mathrm{Lin}}(q,0)]$ of the Al model	eV	6.4

Symbol	Meaning	Units	Sect.
ω_{TA}, ω_{LA}, ω_{TO}, I_{TA}, I_{LA}	Peak energies and weights of the two-peak α^2F model of Al	meV	6.4
E_{band}	Band-width cutoff of the Fermi-surface average	eV	6.4
$\varepsilon_2^{\mathrm{PA}}$	Phonon-assisted imaginary dielectric function; at finite $\boldsymbol{q}$, $\varepsilon_2^{\mathrm{PA}}(\boldsymbol{q},\omega)$ is the phonon-assisted ε_2 whose $\boldsymbol{q}\to 0$ limit is the optical one	1	6.5
$\alpha_{\mathrm{abs}}(\omega)$	Optical absorption coefficient; $\alpha_{\mathrm{abs},\nu}(\omega;T)$ its branch-ν component at temperature T	cm^{-1}	6.5
$\hat{e}_\gamma$	Unit polarization vector of a photon	1	6.5
$\hat{\boldsymbol{V}}$, H_A	Total electron velocity operator $\sum_j \boldsymbol{v}_j = \sum \boldsymbol{v}_{mn} c^\dagger c$; coupling $e\boldsymbol{A}\cdot\hat{\boldsymbol{V}}$ of a uniform vector potential to the electrons	1; eV	6.5
$M^{\pm}_{cv\nu}(k,Q)$, $M^{\pm}_{cv\nu}(k,q,Q)$	One-electron phonon-assisted amplitude summed over all intermediate bands p, upper sign emission: with two arguments the optical amplitude of the velocity vertex, with three the dimensionless density amplitude at momentum transfer $\boldsymbol{q}$; $M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q};\hat{e})$ the optical amplitude along the polarization $\hat{e}$, the $\boldsymbol{q}\to 0$ limit of $(\omega/q)M^{\pm}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})$ along $\hat{e}=\hat{q}$; the argument count is part of the name	1 (a velocity); 1	6.5
κ_{ext}	Extinction coefficient $\varepsilon_2/2n_r, \varepsilon = (n_r + \mathrm{i}\kappa_{\mathrm{ext}})^2$	1	6.5
ω_ν, n_ν, A_ν, A_{TA}, A_{TO}, n_{TA}, n_{TO}	Energy and occupation of the phonon of branch ν at $\boldsymbol{Q}=\boldsymbol{k}_{\mathrm{c}}$; Hall–Bardeen–Blatt coefficient of branch ν	meV; 1; $\mathrm{cm}^{-1}\,\mathrm{eV}^{-2}$	6.5
$\boldsymbol{\kappa}_{\mathrm{e}}$, $\boldsymbol{\kappa}_{\mathrm{h}}$	Electron and hole crystal momenta measured from the band extrema; their kinetic energies are $E_{\mathrm{e}}, E_{\mathrm{h}}$	eV	6.5
$n_r(\omega)$	Refractive index of the medium	1	6.5
M_ν	Value of $M^{\pm}_{cv\nu}$ at the band extrema	1	6.5
Chapter 7: Magnets, superconductors, Dirac materials and defects			
J_{ij}^{ex}	Exchange constant	meV	7.1
U_{H}	On-site (Hubbard) repulsion	eV	7.1
$\boldsymbol{S}_j$	Ionic spin operator of site j	1	7.1
$a_{\boldsymbol{k}\nu}$, $a^\dagger_{\boldsymbol{k}\nu}$	Magnon operators	—	7.1
$u^{\mathrm{M}}_{\boldsymbol{k}}$, $v^{\mathrm{M}}_{\boldsymbol{k}}$	Magnon Bogoliubov coefficients	1	7.1
$F^{\mathrm{mag}}(q)$	Magnetic form factor	1	7.1
$s_e(q)$	Electron spin-density operator $\sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\sigma_i/2$	1	7.1
K_{ex}	Exchange integral of two orthogonal orbitals, the Coulomb self-energy of $\phi_a^*\phi_b$, positive	eV	7.1
E_{orb}	Sum of the two one-electron orbital energies in the Dirac identity of the two-electron problem	eV	7.1
Δ_{CT}	Charge-transfer energy of a charge-transfer insulator, $\Delta_{\mathrm{CT}} < U_{\mathrm{H}}$ in NiO	eV	7.1
d_j, $d_{\mathrm{A}i}$, $d_{\mathrm{B}j}$, $d_{\mathrm{A}\boldsymbol{k}}$, $d_{\mathrm{B}\boldsymbol{k}}$	Holstein–Primakoff bosons of the ferromagnet; of sublattices A and B, and their transforms at wavevector $\boldsymbol{k}$; special cases of the generic d_γ	—	7.1
$J^{\mathrm{ex}}_{\mathrm{AA}}(k)$, $J^{\mathrm{ex}}_{\mathrm{AB}}(k)$	Intra- and inter-sublattice exchange sums of a two-sublattice magnet	meV	7.1
$A_{\boldsymbol{k}}$, $B_{\boldsymbol{k}}$	Diagonal and pair coefficients of the two-sublattice spin-wave Hamiltonian; $\lvert\boldsymbol{S}\rvert A_{\boldsymbol{k}}$ is the diagonal energy of the boson problem	meV	7.1
$\omega^{\mathrm{M}}_{\boldsymbol{k}}$	Magnon energy	meV	7.1
N_{s}	Number of sites per sublattice	1	7.1
E_{cl}, E_{gs}	Classical (fully polarized or Néel) energy of the Heisenberg model; ground-state energy of the linear spin-wave Hamiltonian, $E_{\mathrm{cl}} + \sum_k(\omega^{\mathrm{M}}_{\boldsymbol{k}} - \lvert\boldsymbol{S}\rvert A_{\boldsymbol{k}})$	eV	7.1

Symbol	Meaning	Units	Sect.
$\theta_{\boldsymbol{k}}$	Bogoliubov angle: of the magnon problem, $\tanh 2\theta_{\boldsymbol{k}} = B_{\boldsymbol{k}}/A_{\boldsymbol{k}}$; of the BCS problem	1	7.1
$\zeta_{\boldsymbol{k}}$	Type-II J_2-only structure factor $(\cos k_x a + \cos k_y a + \cos k_z a)/3$	1	7.1
$S^{\pm}(q)$, $S^{\pm}_j$	Transverse ionic spin at wavevector $\boldsymbol{q}$, $\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j} S^{\pm}_j$; raising and lowering components of $\boldsymbol{S}_j$	1	7.1
S^a_{tot}, S^-_{tot}	Total spin components $\sum_j S^a_j (a = x, y, z)$ and total lowering operator	1	7.1
$\hat{O}_{\chi a}(q)$	Dark-matter-side operator of a spin-dependent coupling $-\sum_a \hat{O}_{\chi a}(\boldsymbol{q}) s_{e,a}(\boldsymbol{q})$, as Trickle, Zhang and Zurek write it	—	7.1
$\lvert\boldsymbol{S}\rvert$	Spin length of a magnetic ion, $\boldsymbol{S}^2 = \lvert\boldsymbol{S}\rvert(\lvert\boldsymbol{S}\rvert + 1)$	1	7.1
$\boldsymbol{S}_{\mathrm{tot}}$, $S_{\mathrm{tot},x}$, $S_{\mathrm{tot},y}$, S^z_{tot}, $S^{\pm}_{\mathrm{tot}}$	Total spin $\sum_j \boldsymbol{S}_j$ of a magnet and its components	1	7.1
H_{H}, J_1, J_2, $J^{\mathrm{ex}}(Q)$	Heisenberg Hamiltonian $\sum_{\langle ij\rangle} J^{\mathrm{ex}}_{ij} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$; NiO nearest- and next-nearest-shell constants in that convention; lattice Fourier transform $\sum_R J^{\mathrm{ex}}_{0\boldsymbol{R}} \mathrm{e}^{\mathrm{i}Q\cdot\boldsymbol{R}}$	eV; meV; meV	7.1
$u^{\mathrm{B}}_{\boldsymbol{k}}$, $v^{\mathrm{B}}_{\boldsymbol{k}}$	BCS coherence factors	1	7.2
V_{BCS}	BCS pairing constant, a volume times an energy, acting within the shell $\lvert\xi_{\boldsymbol{k}}\rvert < \omega_{\mathrm{D}}$; $\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}} = \lambda_{\mathrm{ep}}$	eV^{-2} ($\mathrm{eV\,\AA}^3$)	7.2
H_{BCS}, H_{MF}	Reduced BCS Hamiltonian; its mean-field form	eV	7.2
$\gamma_{\boldsymbol{k}1}$, $\gamma_{\boldsymbol{k}2}$	Bogoliubov quasiparticle operators	—	7.2
$b_{\boldsymbol{k}}$	Pair amplitude $\langle c_{-k\downarrow} c_{\boldsymbol{k}\uparrow}\rangle$, $(\Delta_{\mathrm{BCS}}/2E_{\boldsymbol{k}})\tanh(E_{\boldsymbol{k}}/2T)$	1	7.2
$\lvert BCS\rangle$	BCS ground state $\Pi_k(u^{\mathrm{B}}_{\boldsymbol{k}} + v^{\mathrm{B}}_{\boldsymbol{k}} c^{\dagger}_{\boldsymbol{k}\uparrow} c^{\dagger}_{-k\downarrow})\lvert 0\rangle$	—	7.2
γ_{E}	Euler's constant 0.5772	1	7.2
$\mathcal{N}_{\mathrm{qp}}(E)$	BCS quasiparticle density of states per volume and spin orientation	eV^2	7.2
S_{I}, S_{II}, S_{n}	Bare pair-creation spectra of the density (case I) and of s^z_e (case II) and their normal-state value	as S(q, ω)	7.2
$\Psi_{\boldsymbol{k}}$	Nambu spinor $(c_{\boldsymbol{k}\uparrow}, c^{\dagger}_{-k\downarrow})^{\mathsf{T}}$	—	7.2
η, s_1, s_2, s_a, s_3	Nambu matrices: $\eta = \mathrm{diag}(1, -1)$ and the off-diagonal $\mathsf{s}_1, \mathsf{s}_2$; s_a either of them $(a = 1, 2)$; $\mathsf{s}_3 \equiv \eta$ in the trace algebra	1	7.2
$h_{\boldsymbol{k}}$	Nambu Bloch matrix $\xi_{\boldsymbol{k}}\eta - \Delta_{\mathrm{BCS}}\mathsf{s}_1$	eV	7.2
$O_A(q)$	Nambu bilinear $\sum_k \Psi^{\dagger}_{\boldsymbol{k}+\boldsymbol{q}} A \Psi_{\boldsymbol{k}}$ of a 2×2 matrix A; $O_{\boldsymbol{\eta}}(q) = n_e(q)$	1	7.2
K	Channel kernel $\mathrm{diag}(0, -V_{\mathrm{BCS}}/2, -V_{\mathrm{BCS}}/2)$ of the conserving response in the channels $\eta, \mathsf{s}_1, \mathsf{s}_2$; its Coulomb entry $K_{\boldsymbol{\eta}\boldsymbol{\eta}} = v_{\mathrm{C}}(q)$	eV^{-2}	7.2
$F(\omega)$	$q = 0$ pair sum $\mathscr{V}^{-1} \sum'_k [E_{\boldsymbol{k}}(\omega^2 - 4E^2_{\boldsymbol{k}})]^{-1}$	1	7.2
$\mathsf{V}_{\boldsymbol{\eta}}$, $\mathsf{V}_{\boldsymbol{j}}$	Dressed density and current vertices in Nambu form	1; 1	7.2
c_2	Phase-channel coefficient of the dressed density vertex, $\mathsf{V}_{\boldsymbol{\eta}} = \eta + \mathsf{c}_2\mathsf{s}_2$, equal to $-2\mathrm{i}\Delta_{\mathrm{BCS}}/\omega$ at $q = 0$	1	7.2
c_{AB}	Speed of the Anderson–Bogoliubov mode, $v_{\mathrm{F}}/\sqrt{3}$ in a neutral superconductor	1	7.2
$c_{\boldsymbol{k}\sigma}$, $c^{\dagger}_{\boldsymbol{k}\sigma}$, $\psi_\sigma(r)$	Electron operators and field operator with the spin label $\sigma = \uparrow, \downarrow$	—; $\Omega^{-1/2}$	7.2
v_{D}	Dirac velocity	1	7.3
Δ_{D}	Dirac half-gap	meV	7.3
$\varepsilon_{\mathrm{bg}}$, $\varepsilon_{\mathrm{bg},ii}$	Background (remote-band) dielectric constant; principal components of the background dielectric tensor of an anisotropic material	1	7.3
$\tilde{\alpha}$	Effective Dirac coupling $\alpha/(\varepsilon_{\mathrm{bg}} v_{\mathrm{D}})$	1	7.3
Λ_{D}	Dirac momentum cutoff	eV	7.3
σ^{d}_i, $\boldsymbol{\sigma}^{\mathrm{d}}$	Pauli matrices acting within a Kramers doublet	1	7.3
$\mathbb{1}$	2×2 unit matrix	1	7.3

Symbol	Meaning	Units	Sect.
$M(\delta k)$, M_i	Interband velocity block of the four-state $\boldsymbol{k}\cdot p$ Hamiltonian, $M=\sum_i \delta k_i M_i$	1	7.3
v_i	Principal Dirac velocities of an anisotropic cone, the singular values of the interband velocity block (v_{D} when equal)	1	7.3
$\mathcal{N}_{\mathrm{D}}(E)$	Density of states per volume of one Dirac conduction band	eV^2	7.3
$\tilde{\Lambda}$, Λ_x, Λ_y, Λ_z	Energy cutoff on $(\sum_i v_i^2\delta k_i^2)^{1/2}$ of an anisotropic cone; per-axis momentum cutoffs	eV	7.3
R^0_{D}	Independent-pair (interband) response of the Dirac electrons ($\varepsilon=\varepsilon_{\mathrm{bg}}-v_{\mathrm{C}}R^0_{\mathrm{D}}$)	eV^2	7.3
a^*, $E_{\mathrm{b}}^{\mathrm{D}}$	Effective Bohr radius $\varepsilon_0 m_e a_0/m^*$; donor binding energy	Å; meV	7.4
$\boldsymbol{n}_{\mathrm{D}}$	Defect or dopant number density	cm^{-3}	7.4
$E_{\mathrm{d}}(\hat{r})$	Displacement threshold energy of a recoil in the direction $\hat{r}$	eV	7.4
$r_\perp$, $r_\parallel$, e_{ecc}	Radii of the Kohn–Luttinger variational envelope across and along the valley axis; eccentricity $(1-r_\parallel^2/r_\perp^2)^{1/2}$ of the envelope spheroid	Å; 1	7.4
$F_j(r)$, c_j	Donor envelope of valley j; valley coefficients of a multivalley donor state	$\mathrm{Å}^{-3/2}$; 1	7.4
γ_m	Valley mass ratio $m^*_\perp/m^*_\parallel$	1	7.4
Δ_0, Δ_{g}, Δ_{f}	Valley-orbit matrix elements of the donor 1s manifold: intravalley, opposite valleys, perpendicular valleys	meV	7.4
E_{A_1}, E_E, E_{T_2}	Valley-orbit levels of the donor 1s manifold, labelled by the irreducible representations of T_d	meV	7.4
$E^{\mathrm{ad}}_{\mathrm{g}}(Q)$, $E^{\mathrm{ad}}_{\mathrm{e}}(Q)$	Ground- and excited-state Born–Oppenheimer surfaces of a color center along the configuration coordinate	eV	7.4
Q, ΔQ	Mass-weighted configuration coordinate of a color center; separation of the minima of the two surfaces	$\mathrm{eV}^{-1/2}$	7.4
w_{ZPL}	Zero-phonon fraction $\mathrm{e}^{-S_{\mathrm{HR}}}$ of a color-center line at $T=0$	1	7.4
$\boldsymbol{d}$	Transition dipole of a color center	eV^{-1} (e Å)	7.4
Chapter 8: The response framework			
$\mathcal{O}_i$	Non-relativistic effective operators ($\mathcal{O}_2=(v^\perp)^2$)	—	8.1
$v^\perp$	Transverse velocity of the effective theory	1	8.1
$s_\kappa(q)$, $\boldsymbol{s}_s(\boldsymbol{q})$, $\boldsymbol{j}_s(\boldsymbol{q})$	Nuclear spin density of species κ, $\sum_{I\in\kappa}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\boldsymbol{J}^{\mathrm{nuc}}_I$; spin density and paramagnetic current of species s	1	8.1
$\mathcal{T}_e(q)$, $\mathcal{T}_s(q)$, $\mathcal{T}_\kappa$, $\mathcal{T}$	Kinetic-energy density operator of the electrons, of species s, of nuclei κ (mass M_κ)	eV	8.1
$\mathfrak{D}_{ab}(q,v)$, $\mathfrak{D}$, $\mathfrak{D}_{\mathrm{R}}$, $\mathfrak{D}_{\mathrm{I}}$	Dark-matter-side tensor $\mathrm{Tr}_\chi[\tilde{V}_a^\dagger\tilde{V}_b]/(2\lvert\boldsymbol{S}_\chi\rvert+1)$, Hermitian and positive semidefinite, contracting S_{ab}; the matrix; its real symmetric and imaginary antisymmetric parts	model-dependent	8.1
$\bar{v}_\chi$, $\bar{v}_{\chi,z}$	Mean dark-matter velocity $(\boldsymbol{p}+\boldsymbol{p}')/2m_\chi=\boldsymbol{v}-\boldsymbol{q}/2m_\chi$; its component along the magnetization	1	8.1
$c_i^{(s)}(q)$	Coefficient of $\mathcal{O}_i$ for target particle $s\in\{e,p,n\}$ or nucleus κ, non-relativistic normalization, $c_1^{(e)}=\tilde{V}$	eV^{-2}	8.1
$\boldsymbol{S}_\chi$, $\lvert\boldsymbol{S}_\chi\rvert$	Dark-matter spin operator; its spin quantum number	1	8.1
$\boldsymbol{S}_e$, $\boldsymbol{S}_N$	Spin of a target electron ($\boldsymbol{\sigma}/2$) and of a nucleon in the operator basis	1	8.1
$\mathcal{J}_s^{\alpha\beta}(\boldsymbol{q})$, $\mathcal{J}_e$, $\mathcal{J}$	Spin-current operator of species s, $\sum_j \frac{1}{2}\{\boldsymbol{p}_j^\alpha/m_s, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\}S_j^\beta$	1	8.1
$\boldsymbol{J}^{\mathrm{nuc}}_I$, J_{nuc}	Spin operator of nucleus I; its quantum number	1	8.1

Symbol	Meaning	Units	Sect.
$\langle S_p\rangle$, $\langle S_n\rangle$	Proton and neutron spin expectation values in the nuclear state $M = J_{\rm nuc}$	1	8.1
H_{aee}, $H_{\phi ee}$	Interaction Hamiltonians of axion-like and scalar absorption on electrons	eV	8.1
Λ_F	Scale of a fermionic-absorption operator $\Lambda_F^{-2}(\bar\tilde\nu\Gamma\chi)(\bar\psi\Gamma'\psi)$, neutral or charged-current	eV	8.1
$m_{\tilde\nu}$	Mass of the outgoing fermion $\tilde\nu$	eV	8.1
$E_{\rm out}$, $m_{\rm out}$	Energy and mass of the outgoing dark-sector particle ($m_{\rm out} = m_\chi + \Delta_\chi$ for inelastic dark matter; $E_{\rm out} = \lvert\boldsymbol{p}-\boldsymbol{q}\rvert$ for a massless $\tilde\nu$)	eV	8.1
Tr_χ	Trace over the dark-matter spin space	—	8.1
$\boldsymbol{y}_j$	Position of nucleon j relative to the center of its nucleus	fm	8.1
m_N	Nucleon mass (normalizes $\boldsymbol{q}/m_N$ in the nucleon operators)	eV	8.1
$v_{\rm T}$, $v_{e\rm T}$	Dark-matter velocity transverse to $\boldsymbol{q}$ and the electron velocity transverse to $\boldsymbol{q}$	1	8.1
$\tilde\nu$, $\bar\tilde\nu$	Final-state fermion of fermionic absorption; its Dirac conjugate	—	8.1
g_a^ϕ, $g_{n_e}^\phi$, $g_{\mathcal{T}_e}^\phi$, $g_{\mathcal{J}}^\phi$	Coefficient of target row $O_a(\boldsymbol{p}_\phi)$ per unit amplitude of an absorbed bosonic field, derivative and polarization factors included; for a scalar $g_{n_e}^\phi = -g_{\phi ee}, g_{\mathcal{T}_e}^\phi = g_{\phi ee}/m_e$; for an axion-like particle $g_{\mathcal{J}}^\phi = g_{aee}m_\phi/m_e$	model-dependent	8.2
$\Gamma_{\rm abs}$, $\Gamma_{\rm net}$	Absorption rate per quantum of a bosonic field (what a detector counts); the net rate $(1-\mathrm{e}^{-\beta\omega_\phi})\Gamma_{\rm abs}$ at which the field loses quanta	eV	8.2
ω_ϕ	Energy of one quantum of the absorbed field, $m_\phi(1+v^2/2) \simeq m_\phi$	eV	8.2
N_ϕ	Number of quanta in one mode of the dark-matter field	1	8.2
$\delta_{\rm Born}$	Relative second-Born correction to a Yukawa cross-section, $\pm\mu_{\chi T}y_\chi y_T/[2\pi(m_V+p_{\rm rel})] = \pm\lvert f(0)\rvert m_V\cdot m_V/(m_V+p_{\rm rel}), +$ for attraction	1	8.2
$a_{\rm Born}$	First-Born value of the zero-energy scattering length of the Yukawa potential, of magnitude $\lvert f(0)\rvert = \mu_{\chi T}y_\chi y_T/(2\pi m_V^2)$	eV^{-1}	8.2
y_T	Mediator coupling to the target particle, y_e for an electron, Ay_N for a coherent nucleus	1	8.2
$\hat\varrho_\chi$	Spin density matrix of the dark-matter flux, unit trace; $1/(2\lvert\boldsymbol{S}_\chi\rvert+1)$ when unpolarized	1	8.2
$t_{\rm coh}$	Coherence time of an ultralight dark-matter field	s	8.2
$\mathscr{M}^k$, $\mathscr{M}^k_{ab}$	k-th moment matrix $\int(\mathrm{d}\omega/2\pi)\omega^k[S_{ab}(q,\omega)+S_{ba}(-q,\omega)]$, a commutator for odd k and an anticommutator for even k	eV^{k+3} (density rows)	8.3
$\boldsymbol{g}(q)$	Momentum-density operator	eV	8.3
Γ_A, Γ_B, Γ_{AB}	Rates of two groups A, B of rows alone and their cross term	eV	8.3
$n_{\rm M}(\boldsymbol{q})$	Mass-density operator $\sum_s m_s n_s(\boldsymbol{q})$	eV	8.3
$\mathsf{T}^{\alpha\beta}(\boldsymbol{q})$	Momentum-flux (stress) operator, $[\mathrm{H}, g^\alpha(\boldsymbol{q})] = q_\beta\mathsf{T}^{\alpha\beta}$	eV^2	8.3
$\bar{\boldsymbol{n}}_e^{(\kappa)}(y)$	Spherical average of the conditional electron density $\boldsymbol{n}_e^{(\kappa)}(\boldsymbol{y})$	cm^{-3}	8.3
$G_{\rm min}$	Length of the shortest nonzero reciprocal-lattice vector	keV	8.3
$a_{\rm core}$	Size of the innermost electron shell	Å	8.3
$w_\kappa(\boldsymbol{y})$, $\tilde w_\kappa(\boldsymbol{k})$, $w_{ee}(\boldsymbol{r})$, $\tilde w_{ee}(q)$	Electron–nucleus pair potential $-Z_{\kappa\alpha}/y$ and electron–electron pair potential; their Fourier transforms	eV; eV^{-2}	8.3

Symbol	Meaning	Units	Sect.
$\boldsymbol{n}_e^{(\kappa)}$, $\boldsymbol{n}_e^{(\kappa)}(\boldsymbol{y})$	Electron number density at the site of nucleus κ (contact density, mixed third moment); with an argument, the conditional density at displacement $\boldsymbol{y}$ from a nucleus of species κ, averaged over those nuclei and their motion	cm^{-3}	8.3
$\tilde{R}_{aZ}$, $\tilde{R}_{Zb}$, $\tilde{R}_{ZZ}$, $\tilde{R}_{aZ_{\boldsymbol{G}}}$, $\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}$, $\tilde{R}_{eZ}$, $\tilde{R}_{Ze}$, $\tilde{R}_{e\kappa}$, $\tilde{R}_{NZ}$	Proper (channel-irreducible) responses of rows a, b to and from the charge row $n_Z(\boldsymbol{q}+\boldsymbol{G})$; the proper charge–charge block, $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}$ with nuclei	eV^2	8.4
$\delta\boldsymbol{n}_{Z,\boldsymbol{G}}$, $\boldsymbol{n}_Z$	Induced charge-number density at $\boldsymbol{q}+\boldsymbol{G}$, $\delta\langle n_Z(\boldsymbol{q}+\boldsymbol{G})^\dagger\rangle/\mathcal{V}$; mean charge-number density	eV^3	8.4
ϕ_b	Amplitude of an external field on target row b	eV	8.4
λ_Z	Proportionality constant of the loss-function theorem, $S_{cc} = \lambda_Z^2(2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$	1	8.4
S_{cc}, O_c	Structure factor and operator of a density coupling $O_c = \sum_s c_1^{(s)} n_s$	eV^2 (per eV^{-4} of coupling); —	8.4
$a_N(\boldsymbol{q})$	Nucleon-number amplitude per cell of the optical mode	1	8.4
Π_{T}, Π_{L}, $\Pi_{\mathrm{T,L}}$	Transverse and longitudinal photon self-energies, $\omega^2(1-\varepsilon^{\mathrm{T}})$ and $(\omega^2-k^2)(1-\varepsilon)$ (the two polarizations of Π_{em})	eV^2	8.4
ω'	Frequency on a vertex segment of a diagram	eV	8.4
$\mathfrak{T}$	Truncation label $(n_B, n_g, n_A, \mathcal{X}, n_T, \mathcal{K}, \mathfrak{M}, n_D)$ of the expansion hierarchy	—	8.5
n_B, n_g, n_A, n_T, n_D	Born–Oppenheimer, electron–phonon, anharmonic, thermal and multipole orders	1	8.5
$\mathcal{X}$, $\mathcal{K}$, $\mathfrak{M}$	Many-body approximation; lattice regime (one-phonon, multiphonon, impulse); Migdal regime (sudden, adiabatic)	—	8.5
θ_{ep}, δ_{off}	Off-shell parameter $\langle\lvert g\rvert^2\rangle/\delta_{\mathrm{off}}^2$; off-shell distance of an intermediate state	1; eV	8.5
x_R	Recoil parameter $q^2/(2M_\kappa\langle\omega\rangle) = E_R/\langle\omega\rangle$, $\langle\omega\rangle$ the arithmetic mean of the per-atom phonon density of states	1	8.5
ξ_D	Multipole parameter (momentum times the size of the excited orbital)	1	8.5
$\langle\omega^k\rangle$	Moment $(1/3)\int \mathrm{d}\omega\omega^k g_{\mathrm{ph}}(\omega)/n_{\mathrm{at}}$ of the per-atom phonon density of states; $\langle\omega\rangle = 40.9$ meV, $\langle\omega^{-1}\rangle^{-1} = 30.4$ meV in Si	meV^k	8.5
c_k	k-th cumulant of the energy deposited in the self response, $c_k = E_R\langle\omega^{k-1}\rangle$ at $T=0$	eV^k	8.5
$\mathfrak{T}_0$	Base point of $\mathfrak{T}: n_B=0, n_g=0, n_A=0, \mathcal{X}=\mathrm{IP}$, $n_T=0$; $\mathcal{K}, \mathfrak{M}, n_D$ not applicable	—	8.5
q_e, r_{orb}	Momentum given to the electrons; size of the excited orbital	keV; Å	8.5
$\mathcal{P}_{\mathrm{ad}}$, $\mathcal{P}_0$, $\mathcal{P}_+$	Projectors on the electron-following, zero-phonon electronic and phonon-dressed electronic sectors	1	8.6
$\mathcal{P}_{\mathrm{BO}}$	Born–Oppenheimer projector $\int \mathrm{d}u\lvert\Psi_0(u)\rangle\langle\Psi_0(u)\rvert \otimes \lvert u\rangle\langle u\rvert$	1	8.6
$\mathcal{P}_A$, $S_{ab}^A(\boldsymbol{q},\omega)$	Generic sector projector, $A \in \{\mathrm{ad}, 0, +\}$; the sector response	1; eV^2	8.6
δS_{ab}, S^{bin}	Remainder of the decomposition; a response averaged over a bin of width $\Delta\omega_{\mathrm{bin}}$	eV^2	8.6
$\bar{\rho}_e(\boldsymbol{q};u)$, $\bar{n}_\kappa$	Electron density following the nuclei, $\langle\Psi_0(u)\rvert n_e(\boldsymbol{q})\lvert\Psi_0(u)\rangle$ (an operator on nuclear wavefunctions); the nuclear density in the electron-following sector	1	8.6

Symbol	Meaning	Units	Sect.
$\rho_{j0}(\boldsymbol{q};u)$	Transition density of the frozen crystal between adiabatic states j and 0	1	8.6
$f_\kappa(q)$	Form factor of a rigid electron cloud attached to nucleus $\kappa(Z_\kappa$ at $\boldsymbol{q}\to 0$ for a neutral atom)	1	8.6
ρ_m, $\tilde{\rho}_m$	Zero-phonon vertex $\langle m\|O_b\|i\rangle$ of pair state m; with its virtual-phonon correction	as O	8.6
g_{fm}	Electron–phonon matrix element $\langle f\|H_{\mathrm{ep}}\|m\rangle$ between a pair and a one-phonon state	eV	8.6
$\mathcal{A}_f^{\mathrm{off}}$, $\mathcal{A}_\nu^{\mathrm{off}}$	Off-shell amplitude per unit coupling into one-phonon state f; on the shell of decay channel ν	$eV^{-1}\cdot[O]$	8.6
$\mathsf{G}_m(\omega)$, $\Sigma_m(\omega)$, $\Upsilon_m(\omega)$	Green's function, self-energy and vertex correction of a zero-phonon pair m(Υ_m over all decay channels)	eV^{-1}; eV; [O]	8.6
Γ_m, $\Gamma_m^{(\nu)}$, Γ_m^{tot}, $\hat{\Gamma}_m$	Width of pair m, its part from decay channel ν, total width $\sum_\nu \Gamma_m^{(\nu)}$, largest width within δ_{F}; $\hat{\Gamma}$ without index: the largest width of the zero-phonon pairs between the edge and δ_{w} above it	eV	8.6
$\tilde{E}_m$, $z_{\mathrm{qp}}(m)$	Quasiparticle energy and residue of a pair state (z_{qp} widened to pairs)	eV; 1	8.6
δ_{F}	Detuning: distance from $\tilde{E}_m$ to the nearest structure of the one-phonon continuum	eV	8.6
c_ν^{F}, $\bar{c}^{\mathrm{F}}$	Fano coefficients of the first emission, $\Gamma_m^{\mathrm{tot}}\,\mathrm{Im}(\rho_m^*\mathcal{A}_\nu^{\mathrm{off}})/\lvert\rho_m\rvert^2, \nu$ over all decay channels; their width-weighted mean	1	8.6
w_0, x_{w}	Zero-phonon weight in a bin; half-width of a line window	as S; eV	8.6
$\mathcal{P}_{n,m}^{\mathrm{qp}}$, U_{qp}, $\hat{N}_{\mathrm{qp}}$, $\hat{N}_{\mathrm{ph}}$	Quasiparticle- and phonon-number sector projectors of gapless targets; dressing unitary; quasiparticle and phonon number operators	—	8.6
H_{d}, H_{off}, $\mathfrak{h}_{\mathrm{off}}$	Sector-diagonal and off-diagonal parts of H; norm of the off-diagonal leak out of a state	eV	8.6
C_{ab}^{cross}, $\Delta\omega_{\mathrm{bin}}$, w_{b}, f_Δ, $\mathfrak{c}_{\mathrm{b}}$	Cross term between sectors; bin width; bin shape, its kernel and the bin constant ($1/\pi$ for a Gaussian bin)	eV^2; eV; 1; 1; 1	8.6
ΔE_{c}	Width of the decay continuum in the $\mathfrak{h}_{\mathrm{off}}$ estimate	eV	8.6
g_{d}, g_{n}, $\lvert lo\rangle, \lvert up\rangle$	Diagonal and off-diagonal couplings of the two-level electron to the bond coordinate; its lower and upper level	eV; eV; —	8.6
δM_κ	Electron drag mass of species κ	eV	8.6
P_{Mig}	Migdal probability	1	8.6
Λ, P_Λ, $P_{\Lambda'}$	Final-state class and its projector (classes refine the sectors $\mathcal{P}_{\mathrm{ad}}, \mathcal{P}_0, \mathcal{P}_+$; Λ' a second class)	—	8.7
$S(q,\omega;\Lambda)$	Final-state-resolved response	eV^2	8.7
$(\tilde{\varepsilon}^{-1})_{G0}(\boldsymbol{q},\omega)$, $\tilde{\varepsilon}^{-1}$	Inverse dielectric matrix of a test electron, $\delta_{\boldsymbol{G}0}+\sum_{\boldsymbol{G}'}(f_{\mathrm{Hxc}})_{\boldsymbol{G}\boldsymbol{G}'}R_{\boldsymbol{G}'0}$	1	8.7
$\Gamma(v;\Lambda)$	Rate of one dark-matter particle into final-state class Λ	eV	8.7
$S^{\mathrm{IP}}(\boldsymbol{q},\omega)$, $S(\boldsymbol{q},\omega;cv\boldsymbol{k})$	Independent-particle structure factor; screened pair-resolved response (class $\Lambda=cv\boldsymbol{k}$, spin summed)	eV^2	8.7
$\mathcal{R}_{N_{\mathrm{eh}}}$	Rate per unit target mass of events with N_{eh} pairs	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	8.7
$P(N_{\mathrm{eh}}\|q,\omega,v)$	Pair-number distribution at fixed momentum transfer, energy and velocity	1	8.7
E_{e}, E_{h}	Kinetic energies of the electron and the hole of a prompt pair, measured from the band edges	eV	8.7
$P_{\mathrm{casc}}(N_{\mathrm{eh}}\|\Lambda)$	Cascade distribution of the pair number for a seed class Λ	1	8.7

Symbol	Meaning	Units	Sect.
ϑ_k, $\vartheta_{\mathcal{X}}$	Nuisance parameter of truncation k : $\vartheta_k = 0$ the formula, $\vartheta_k = 1$ its next order, standard-normal constraint $\mathrm{e}^{-\vartheta_k^2/2}$; $\vartheta_{\mathcal{X}}$ the nuisance parameter of the many-body approximation (θ is an angle, Θ the Heaviside function)	1	8.8
$\mathfrak{T}_k^+$	Label one step further along axis k of $\mathfrak{T}$ (next nonvanishing order, next many-body approximation, next multiphonon term)	—	8.8
ΔS_k, δ_k, $\delta_{\mathcal{X}}$	Next-order difference $S^{\mathfrak{T}_k^+} - S^{\mathfrak{T}}$ and its fractional size, the template of nuisance parameter k; $\delta_{\mathcal{X}}$ the fractional size for the many-body approximation, the template of $\vartheta_{\mathcal{X}}$	eV^2; 1	8.8
δE_k	Edge shift produced by the next order of truncation k (threshold nuisance parameter)	meV	8.8
δ'_k	Fractional size of the next correction to a new-support piece	1	8.8
$\boldsymbol{r}_j(m_\chi;\vartheta)$	Rate per unit cross-section into detector bin j at nuisance parameters ϑ	$\mathrm{cm}^{-2}\,\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	8.8
$N_{\mathrm{obs},j}$, $N_{\mathrm{sig},j}$, $N_{\mathrm{bkg},j}$	Observed, signal and background counts in bin j	1	8.8
ψ, λ_0, λ_1, $m_n(\vartheta)$	Log-template $\ln(1+\delta_A)$, Lagrange multipliers and moments (in $y = \omega - E_R$) of the constrained nuisance parameter (ψ in Sect. 8.8)	1; 1, eV^{-1}; eV^{n+2}	8.8
t^*, N_{nom}, $N_{\mathrm{nom}}^{\mathrm{lim}}$	Threshold of the zero-count cost, nominal signal and its limit value	1	8.8
L, t, t_{p}	Likelihood; $-2\ln$ of the likelihood ratio, the test statistic; its profiled form	1	8.8
Appendix A: Conventions and units			
μ_{B}, g_e	Bohr magneton $e/2m_e$; electron g-factor	eV^{-1}; 1	A.2
$\boldsymbol{E}_{\mathrm{G}}$, $\boldsymbol{B}_{\mathrm{G}}$; $\boldsymbol{E}_{\mathrm{SI}}$, $\boldsymbol{B}_{\mathrm{SI}}$	Electric and magnetic fields in Gaussian and in SI units	as quoted	A.2
$f_{\boldsymbol{G}}$	Fourier component of a lattice-periodic function, $\Omega^{-1}\int_\Omega \mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f$	as f	A.3
$\langle\langle\cdot\rangle\rangle$	Zone average $N^{-2}\sum_{kQ}$	—	A.3
$\boldsymbol{r}_\chi$, $\boldsymbol{r}_e$	Positions of the dark matter and of one electron	eV^{-1}	A.3
$\tilde{i}$, $\tilde{f}$	Time-reversed initial and final states, as kets $\lvert\tilde{i}\rangle, \lvert\tilde{f}\rangle$	—	A.4
V_{NL}	Nonlocal part of a pseudopotential	eV	A.5
$D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}$, $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}$	Cartesian components of the dynamical matrix and of the force constants, written as superscripts	as $D^{\mathrm{dyn}}, \Phi^{(2)}$	A.6
$\hat{e}_\nu$	Unit polarization vector of an acoustic branch ν at small $\boldsymbol{Q}$	1	A.6
b, $b^\dagger$	Annihilation and creation operators of a single phonon mode, written without labels	—	A.6
L_e	Electron-channel loss function $\mathrm{Im}[-(\varepsilon_e^{-1})_{\mathbf{oo}}]$	1	A.7
S^{prop}	Structure factor of $\tilde{R}_{ee}$	eV^2	A.7
O_e, $\varepsilon_e(q,\omega)$	Electron density operator $n_e(\boldsymbol{q})$ as an operator O_a; electron-channel dielectric function, $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$ (equals ε at clamped nuclei)	—; 1	A.7
$R_{ee}(q,\omega)$, $\tilde{R}_{ee}(q,\omega)$, $S_{ee}(q,\omega)$	Density–density response function, its proper part, and the structure factor of the electron channel ($S \equiv S_{ee}$)	eV^2; eV^2; eV^2	A.7
$\bar{S}(q,\omega)$	S averaged over directions of q	eV^2	A.8
$\eta_2(v_{\mathrm{min}})$	Second speed moment $2\int_{v_{\mathrm{min}}}^\infty \mathrm{d}v' v' \eta(v')$	1	A.8
$\tilde{\eta}(v_{\mathrm{min}})$	$\rho_\chi \bar{\sigma}_e \eta / m_\chi$	—	A.8
N_{cell}	Number of unit cells of a target of mass $M_{\mathrm{det}}, M_{\mathrm{det}}/M_{\mathrm{cell}}$	1	A.8
$\mathcal{M}_{\mathrm{free}}$	Relativistic free dark-matter–electron amplitude	1	A.9

Symbol	Meaning	Units	Sect.
q_{ref}, σ_{ref}	Reference momentum transfer of a cross-section conversion and the cross-section $\bar{\sigma}_e F_{\mathrm{DM}}(q_{\mathrm{ref}})^2$ quoted there	eV; cm^2	A.9
Appendix C: Mathematical tools			
$\mathcal{F}_{\boldsymbol{R}}$, $\mathcal{F}(z)$	Generic function in a Poisson sum and its lattice components; Cauchy integral	—	C.2
σ_{br}	Broadening width of a delta function on a $\boldsymbol{k}$ grid	eV	C.2
$\mathrm{d}\Omega_{\hat{n}}$, $\mathrm{d}\Omega_{\hat{\boldsymbol{Q}}}$, n^α	Solid-angle element of the direction $\hat{n}$; Cartesian components of a unit vector	sr; 1	C.3
$P_l(u)$	Legendre polynomial	1	C.3
$Y_l^{\mathrm{c}}(\hat{n})$	Cubic harmonic of order l	1	C.3
$\delta_\Gamma(x)$	Normalized Lorentzian $(\Gamma_{\mathrm{w}}/2\pi)/(x^2+\Gamma_{\mathrm{w}}^2/4)$	eV^{-1}	C.4
$\varrho_R(\omega)$	Spectral density $-\operatorname{Im} R/\pi$ of a response	as R eV^{-1}	C.5
$\hat{D}(\alpha_{\mathrm{d}})$, α_{d}	Displacement operator $\exp(\alpha_{\mathrm{d}} d^\dagger - \alpha_{\mathrm{d}}^* d)$ and its complex amplitude	1	C.6
$\hat{A}$, $\hat{B}$, $\hat{X}$, $\hat{Y}$	Generic operators in the exponential identities	—	C.6
$\boldsymbol{M}$, $\boldsymbol{A}$, …	Generic matrices, bold upright: blocks $\boldsymbol{A},\boldsymbol{B},\boldsymbol{C},\boldsymbol{D}$ of $\boldsymbol{M}$, the Schur complement $\boldsymbol{M}/\boldsymbol{D}$ and $\boldsymbol{N}=(\boldsymbol{M}/\boldsymbol{D})^{-1}$; also $\boldsymbol{U},\boldsymbol{W},\boldsymbol{K},\boldsymbol{X},\boldsymbol{T},\boldsymbol{h},\boldsymbol{\eta}$	—	C.7
g_0, ξ_0	Pairing amplitude and diagonal energy of a two-mode quadratic Hamiltonian	eV	C.7
E_{b}, E_{f}, θ_{b}, θ_{f}, u^{b}, v^{b}, u^{f}, v^{f}	Boson and fermion Bogoliubov energies, angles and coefficients	eV; 1	C.7
$\tilde{d}_j$, $\tilde{c}_j$	Bogoliubov-transformed boson and fermion operators	—	C.7
$K(s)$, $E(s)$, s, s'	Complete elliptic integrals; modulus and complementary modulus (s in Appendix C.8)	1	C.8
μ_0, μ_1	Coefficients of $\mu_0+\mu_1\cos\phi$ in the elliptic reductions	—	C.8
$\rho_{\mathrm{s}}(x)$	Scalar source density	eV^3	C.9
$G_{\mathrm{Y}}(x)$, $\tilde{G}_{\mathrm{Y}}(q)$	Static Green's function $\mathrm{e}^{-mr}/4\pi r$ of the Klein–Gordon operator and its transform $1/(q^2+m^2)$	eV	C.9
$U(x)$	Interaction energy of two static sources	eV	C.9
$\bar{F}_{\mu\nu}$, $\bar{F}^2$	Field strength $\partial_\mu\tilde{A}_\nu-\partial_\nu\tilde{A}_\mu$ of the photon after the mixing redefinition; its square $\bar{F}_{\mu\nu}\bar{F}^{\mu\nu}$	eV^2; eV^4	C.9
H_{s}, H_{v}	Static field energy of the scalar and vector configurations	eV	C.9
ϱ_{MK}	Markov–Krein band width	1	C.10
μ_k, $\bar{\mu}_k$	k-th moment and central moment of a normalized distribution	as x^k	C.10
$F_P(\xi)$	Cumulative distribution of P (decorated F)	1	C.10
$\boldsymbol{H}$, $\boldsymbol{H}_1$	Hankel matrix μ_{i+j} and shifted Hankel matrix μ_{i+j+1}	—	C.10
$\mathsf{K}(x,\xi)$	Reproducing kernel $\boldsymbol{v}(x)^{\mathsf{T}}\boldsymbol{H}^{-1}\boldsymbol{v}(\xi)$	—	C.10
w_ξ, F_{model}	Weight of the canonical three-point representation at its node ξ, equal to $\varrho_{\mathrm{MK}}(\xi)$; cumulative distribution of a model with the same moments as P	1; 1	C.10
$\mathfrak{G}_N(s)$, $\mathfrak{g}(s)$	Probability generating function of N; offspring generating function	1	C.11
$\bar{m}$	Mean offspring number $\mathfrak{g}'(1)$ of a Galton–Watson process, $\langle N_n\rangle=\bar{m}^n$	1	C.11
P_{ext}	Extinction probability	1	C.11
F_{loss}, $\langle X\rangle$, σ_X	Loss distribution of a renewal chain; its mean and width	1; eV; eV	C.11
E_{res}	Undershoot (residual energy) of a renewal chain	eV	C.11
$\bar{Y}(x)$	Mean $\langle Y\vert x\rangle$ of an additive cascade observable $Y=d_c+Y_1+\ldots+Y_k$ started in state x	as Y	C.11

Symbol	Meaning	Units	Sect.
S_n, $S_{N(E)}$	Partial sum $X_1 + \ldots + X_n$ of the losses of a renewal chain, $S_0 = 0$; sum of the $N(E)$ losses that fit into E	eV	C.11
Ξ, $\mathcal{B}$	Nonnegative velocity measure (in Appendices C.12 and E.16 and Sect. 21.2); its compact domain	—	C.12
λ_i, N_i^{up}, κ_{v}	Dual variables; count bounds; visibility constant	1	C.12
$\mathcal{B}_h$, Φ_h, $\boldsymbol{\lambda}_h$, g_{λ_h}, $\boldsymbol{w}_h$	Grid of directions and speeds with covering radius h; grid primal optimum, a lower bound on the optimum of the full program; grid dual solution; its dual function, nonnegative on the grid only; nearest grid point of $\boldsymbol{w}$	—; as the objective; as λ_i; as g_λ; as $\boldsymbol{w}$	C.12
$\succ$, $h(a)$	Resolved-above relation of an interval order; height	—	C.13
I_a, ℓ_a, u_a	Interval $[\ell_a, u_a]$ of an element of an interval order, oriented so that larger values are better; its lower and upper ends	as the ranked quantity	C.13
$\hat{\vartheta}$, $\hat{\hat{\vartheta}}$	Best-fit nuisance parameter; its value maximizing L at fixed signal	1	C.14
$\hat{N}$, $\sigma_{\hat{N}}$, σ_ϑ	Gaussian estimate of N_{sig} and its width; constraint width of a nuisance parameter	1	C.14
$\Phi(z)$	Standard normal distribution function	1	C.14
z_{CL}	Standard-normal quantile at the confidence level of a one-sided upper limit	1	C.14
E, O, H_{D}, H_{FW}	Even and odd parts of H_{D}; Dirac Hamiltonian in a potential; Foldy–Wouthuysen Hamiltonian to $O(m_e v^4)$(H_{Dar} is the Darwin term)	eV	C.15
$d_{\boldsymbol{k}j}$, $\boldsymbol{\epsilon}_{\boldsymbol{k}j}$	Photon annihilation operator and polarization vector of mode $(\boldsymbol{k}, j), j = 1, 2$	1	C.16
$\mathfrak{a}_{\boldsymbol{k}j}$	Classical amplitude of radiation mode $(\boldsymbol{k}, j)$, normalized so that the mode energy is $\omega_k \lvert\mathfrak{a}_{\boldsymbol{k}j}\rvert^2$; quantization replaces it by $d_{\boldsymbol{k}j}$	1	C.16
e_{c}	Charge of a radiating particle	1	C.16
n_{grp}	Group index $dk/\mathrm{d}\omega$ of a medium	1	C.16
H_{rad}, $\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}$, $\tilde{\boldsymbol{J}}(\boldsymbol{k}, \omega)$	Hamiltonian of the free radiation field, with the photon number states as eigenstates; solid-angle element of the photon direction; space-time Fourier transform of a prescribed classical current $\boldsymbol{J}(\boldsymbol{r}, t)$	eV; 1; eV^{-1}	C.16
$\{\mathsf{R}\lvert\boldsymbol{t}\}$, $\mathcal{P}$	Space-group operation; point group	—	C.17
$c^{(a)}(\mathsf{R})$, c_{V}, c_{A}	Irreducible, polar-vector and axial-vector characters	1	C.17
γ_{ijl}, g_{ml}, N_{gyr}	Linear-$\boldsymbol{q}$ coefficient of the current response; gyration tensor; number of gyration invariants	—	C.17
η	Eigenvalue sign of a glide on Bloch states	1	C.17
s_Q	Direction invariant $\sum \hat{q}_i^2 \hat{q}_j^2 (i < j)$	1	C.18
$\boldsymbol{u}(\boldsymbol{r}, t)$	Continuum displacement field, long-wavelength limit of $u_{\boldsymbol{R}\kappa}$	Å	C.18
u_{ij}, w_{el}	Strain tensor $\frac{1}{2}(\partial_i u_j + \partial_j u_i)$; elastic energy per unit volume $\frac{1}{2} C_{ijkl} u_{ij} u_{kl}$	1; GPa	C.18
σ_{ij}	Stress tensor $C_{ijkl} u_{kl}$	GPa	C.18
ε_I	Voigt strain vector $(u_{11}, u_{22}, u_{33}, 2u_{23}, 2u_{13}, 2u_{12})$	1	C.18
$\mathsf{M}_{ik}(\hat{Q})$	Christoffel matrix $C_{ijkl} \hat{Q}_j \hat{Q}_l$	GPa	C.18
Z_{ac}	Acoustic impedance ρc of a medium	$\mathrm{kg\,m^{-2}\,s^{-1}}$	C.18
ρ_1, ρ_2	Mass densities of the two media of an interface	$\mathrm{g\,cm^{-3}}$	C.18
r_{12}, t_{12}^{amp}, $t_{12}(\theta_1)$	Displacement reflection and transmission amplitudes at normal incidence; energy transmission from medium 1 to 2 at incidence angle θ_1	1	C.18
m_n, γ_n, μ_N	Neutron mass; neutron magnetic moment in nuclear magnetons, -1.913 (in Appendix C.19); nuclear magneton $e/2m_p$	eV; 1; eV^{-1}	C.19

Symbol	Meaning	Units	Sect.
$\boldsymbol{\sigma}_n$, $\boldsymbol{p}_n$, $\boldsymbol{p}'_n$	Neutron Pauli matrices; incident and outgoing neutron momenta	1; eV	C.19
$\boldsymbol{B}_e$, $\boldsymbol{M}_e$	Magnetic field and spin magnetization of the electrons	eV^2	C.19
$\hat{f}_{\mathrm{mag}}$, $\hat{f}_{\mathrm{nuc}}$, $\boldsymbol{s}_\perp$	Magnetic and nuclear neutron amplitude operators; transverse spin density $P_T s_e(\boldsymbol{q})$	fm; 1	C.19
$\varrho_{\mathrm{s}}(r)$, $R_{\mathrm{sh}}(r)$, $\langle j_l(q)\rangle$	Normalized unpaired-spin density of an ion; radial function of its shell; radial integrals of j_l	$\mathrm{Å}^{-3}$; $\mathrm{Å}^{-3/2}$; 1	C.19
$\boldsymbol{J}_j$	Total angular momentum of magnetic ion j	1	C.19
σ_x, σ_z	Pseudo-spin Pauli matrices of a two-level system in the localized basis $\lvert L\rangle, \lvert R\rangle$	1	C.20
Δ_0, Δ_{as}, Δ_{r}	Tunneling amplitude; asymmetry; renormalized tunneling amplitude	eV	C.20
d_k, ω_k, λ_k	Bath oscillators, their energies and couplings in the spin–boson model (ω_k in Appendix C.20)	—; eV; eV	C.20
H_{L}, H_{R}	Spin–boson Hamiltonians in the configurations L and R, each with the oscillator equilibria shifted by $\pm\lambda_k/2\omega_k$	eV	C.20
$J_{\mathrm{s}}(\omega)$, ω_{cut}	Bath spectral density $\pi\sum_k \lambda_k^2\delta(\omega-\omega_k)$; its high-energy cutoff	eV	C.20
$\mathcal{Q}(t)$, $C(t)$	Bath correlation exponent and overlap $\mathrm{e}^{-\mathcal{Q}}$	1	C.20
Γ_{LR}, Γ_{RL}	Incoherent hopping rates between the configurations	eV	C.20
$\Gamma(z)$	Euler's gamma function	1	C.20
a_{ch}, K_{ch}, V_{s}	Spacing, spring constant and substrate-potential amplitude of a Frenkel–Kontorova chain	Å; $\mathrm{eV\,Å}^{-2}$; eV	C.20
H_{FK}	Frenkel–Kontorova Hamiltonian of a row of atoms in a periodic substrate potential	eV	C.20
c_{ch}, ω_{s}, d_{kink}	Sound speed of the free chain; single-atom frequency in the substrate; kink width $c_{\mathrm{ch}}/\omega_{\mathrm{s}}$	$\mathrm{km\,s}^{-1}$; meV; Å	C.20
E_{kink}, M_{kink}, E_{PN}	Kink rest energy; kink mass $E_{\mathrm{kink}}/c_{\mathrm{ch}}^2$; Peierls–Nabarro barrier	eV	C.20
Appendix D: Long derivations			
$\mathcal{U}_{\gamma\tilde{\gamma}}$, $\mathcal{U}$	Basis-change matrix $\langle\gamma\lvert\tilde{\gamma}\rangle$; the matrix itself, unitary, with complex conjugate $\mathcal{U}^*$ and adjoint $\mathcal{U}^\dagger$	1	D.2
$K(t-t')$, $\tilde{K}(s)$, $R(z)$	Memory kernel; its Laplace transform, $\sum_f \lvert H'_{fi}\rvert^2/(s+\mathrm{i}\xi_f)$ for $\mathrm{Re}\,s>0$; level-shift function	eV^2; eV; eV	D.3
r_{p}	Residue of the decaying pole, $1/(1-R'(z_{\mathrm{p}}))$	1	D.3
v_0, $\bar{v}$	Head and remainder ($\boldsymbol{G}\neq 0$) of the Coulomb matrix (v_0 in Appendices D.5 and E.7)	eV^{-2}	D.5
$\tilde{k}$	k/k_{F}, radial integration variable	1	D.6
$\mathcal{I}(w)$	Radial Lindhard integral $w+\frac{1}{2}(1-w^2)\ell(w)$	1	D.6
$\delta\varrho_{\boldsymbol{k}}(t)$, $\delta\varrho_{\boldsymbol{k}}^{\mathrm{loc}}$	Coherence $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}\rangle$ and its local-equilibrium value	1	D.7
$D_{\boldsymbol{k}}$, $w_{\boldsymbol{k}}$, w_{tot}, ς, Φ_ϱ	$\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}}$; weight $(n_{\boldsymbol{k}}-n_{\boldsymbol{k}+\boldsymbol{q}})/(\xi_{\boldsymbol{k}+\boldsymbol{q}}-\xi_{\boldsymbol{k}})$ and its sum; $\sum_k \delta\varrho_{\boldsymbol{k}}$; the functional $\sum\lvert\delta\varrho_{\boldsymbol{k}}\rvert^2/w_{\boldsymbol{k}}\geq 0$	eV; eV^{-1}; eV^{-1}; 1; eV	D.7
$\phi_{\mathrm{loc}}(t)$, $\delta\mu$	Local-equilibrium potential and local chemical-potential shift in Mermin's construction	eV	D.7
R^0_γ, R^0_{st}	Shorthands $R^0(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\mathrm{rel}})$ and $R^0(\boldsymbol{q},0)$	eV^2	D.7
F	Number of closed fermion loops of a diagram	1	D.9
$U_{\mathrm{I}}(\tau,\tau')$, $O_{\mathrm{I}}(\tau)$, $H_{\mathrm{int,I}}$	Imaginary-time interaction-picture evolution $\mathrm{e}^{\tau H_0}\mathrm{e}^{-(\tau-\tau')H_\mu}\mathrm{e}^{-\tau' H_0}$, operator and interaction	1; —; eV	D.9
$\langle\cdot\rangle_0$, $\mathcal{Z}_0$, $\lvert\Phi_0\rangle$	Free grand-canonical average and partition function; Fermi sea	—; 1; —	D.9
A_j, $\hat{A}_j$	Operators of Wick's theorem; the hat marks an operator removed from the string	—	D.9
$K^{\mathrm{x}}_{\mathrm{full}}$, K^{x}_0	Exchange kernel with all $\boldsymbol{G}$; its $\boldsymbol{G}=0$ term, $K^{\mathrm{x}}_{\mathrm{full}}=K^{\mathrm{x}}+K^{\mathrm{x}}_0$	eV	D.10

Symbol	Meaning	Units	Sect.
$H^{\mathrm{exc}}_{\mathrm{full}}$; $E^{\mathrm{full}}_{\lambda}$, $A^{\lambda}_{\mathrm{full}}$	BSE Hamiltonian with $K^{\mathrm{x}}_{\mathrm{full}}$ and its eigenpairs	eV; eV; 1	D.10
$\mathfrak{L}$, $\mathfrak{L}^0$, $\bar{\mathfrak{L}}$	Pair propagator $-\langle T c^{\dagger}_n c_m(\tau) c^{\dagger}_{m'} c_{n'}(0)\rangle$; its independent-quasiparticle value; its solution with K^{x} in place of $K^{\mathrm{x}}_{\mathrm{full}}$	eV^{-1}	D.10
$K_{\mathrm{r}\bar{\mathrm{r}}}$	Coupling (resonant–antiresonant) block of the BSE kernel	eV	D.10
$v_{ab,cd}$	Coulomb matrix element of four orbitals $\psi_a, \psi_b, \psi_c, \psi_d$	eV	D.10
$V^{\mathrm{L}}(r)$	Potential energy of an electron in the macroscopic field of a longitudinal phonon, $\nabla V^{\mathrm{L}} = e\mathcal{E}$	eV	D.12
$\mathcal{F}(k,k')$, $\rho(u)$, b_U	Interband overlap of the Dirac cone per state; pair spectral function $(1-4\Delta_{\mathrm{D}}^2/u)^{1/2}(1+2\Delta_{\mathrm{D}}^2/u)$ in the invariant u; its square-root variable at the cutoff	1	D.16
s	Lorentz invariant $\omega^2 - v_{\mathrm{D}}^2 q^2$ of the Dirac polarization	eV^2	D.16
N_i, z_i, $\boldsymbol{c}_i$, U_{c}, U_{v}	Reduced velocity blocks $M_i M_1^{-1}$, their scalar and vector parts, and the basis changes in the conduction and valence doublets of the uniqueness argument	1	D.16
$Ry^*_{\perp}$, $a^*_{\perp}$	Transverse effective Rydberg and Bohr radius of a donor, with the mass $m^*_{\perp}$	meV; Å	D.17
$n_{\boldsymbol{k}}$, $\phi_{\boldsymbol{k}}$	Number density of quanta in mode $\boldsymbol{k}$; complex amplitude of mode $\boldsymbol{k}$ of a classical field	eV^3; eV	D.19
$\boldsymbol{P}_e$	Total momentum of the electrons, $im_e[H_e, \sum_i \boldsymbol{r}_i]$	eV	D.20
H_{ad}, h_{BO}, $\bar{O}_a$, $\bar{O}$	$\mathcal{P}_{\mathrm{ad}} H \mathcal{P}_{\mathrm{ad}}$; the Born–Oppenheimer Hamiltonian of the nuclei it becomes after the rotation U_n; restricted operator $\mathcal{P}_{\mathrm{ad}} O_a \mathcal{P}_{\mathrm{ad}}$	eV; eV; as O	D.22

B.2 Literature notation

The dark-matter literature and the condensed-matter literature use the same letters for different objects. In dark-matter papers the letter χ is the dark-matter particle and q the momentum transfer; in condensed-matter papers χ is a susceptibility and q the phonon wavevector. The letter W is the loss function, the screened interaction or the Debye–Waller exponent, depending on the paper. This book gives each letter one meaning, so several common symbols are renamed. Table B.2 lists the renamed symbols.

Table B.2: Symbols of the literature and of this book. Where a paper's symbol differs from the book's only by a factor or a sign, the factor is given in the Meaning column.

Literature	Here	Meaning	Section
Response and screening			
χ_{AB}, χ, C^R_{AB}	R_{ab}	retarded response function; the literature usually omits the factor $1/\mathcal{V}$	4.1
Π	$\tilde{R}$	polarization, the Coulomb-proper (irreducible) response	4.4
W, $\mathrm{Im}(-1/\varepsilon)$	L	loss function $\mathrm{Im}[-(\varepsilon^{-1})_{00}]$	4.4
W (as in GW)	W^{scr}	screened Coulomb interaction	4.8
$-2\,\mathrm{Im}\,G^{\mathrm{R}}$	A	spectral function; the book's A integrates to one, the literature's to 2π	4.7
$G(\boldsymbol{r},t)$	$\mathcal{G}_{\mathrm{VH}}(\boldsymbol{r},t)$	Van Hove correlation function	4.2
λ_{TF}	$1/k_{\mathrm{TF}}$	Thomas–Fermi screening length	4.4
κ	$\varepsilon_{\mathrm{bg}}$	background (remote-band) dielectric constant	7.3

Literature	Here	Meaning	Section
$\alpha(\omega)$	$\alpha_{\mathrm{abs}}(\omega)$	optical absorption coefficient	6.5
Dark matter and kinematics			
ε, κ, χ	ζ	kinetic mixing of the dark photon	1.4
g_χ, g_e	y_χ, y_e	mediator couplings to dark matter and electrons	1.4
n_χ	n_χ	dark-matter number density ρ_χ/m_χ	1.1
$\boldsymbol{k}$	$\boldsymbol{q}$	momentum transferred from the dark matter to the target	1.3
$\mathrm{d}\Omega$	$\mathrm{d}\Omega_{\boldsymbol{q}}$	solid-angle element of the scattered momentum	2.2
σ_e (fermionic absorption)	σ_F	absorption cross-section of fermionic dark matter	10.9
Q	N_{eh}	number of electron–hole pairs (ionization yield)	1.5
Nuclei and phonons			
$\boldsymbol{q}$, $\boldsymbol{k}$	$\boldsymbol{Q}$	phonon wavevector in the first zone	5.2
r_n, R_1; s	R_{nuc}; s_{skin}	Helm radius; skin thickness	2.2
c, a	c_{nuc}, a_{nuc}	Woods–Saxon half-density radius and diffuseness	2.2
λ, μ (Lamé)	C_{12}, C_{44}	elastic constants of an isotropic medium	C.18
λ	λ_{ep}	electron–phonon coupling constant	6.4
μ^*	μ_{C}^*	Coulomb pseudopotential	28.1
Z, Δ	Z^{El}, Δ^{El}	Eliashberg renormalization function and gap	28.1
Electrons in crystals			
μ	μ_{ch}	chemical potential	2.5
$N(0)$, $N(E_{\mathrm{F}})$	$\mathcal{N}_{\mathrm{F}}$	density of states per spin and volume at the Fermi level	2.5
v_{F} (Dirac material)	v_{D}	Dirac velocity; v_{F} is the Fermi velocity of the electron gas	7.3
χ_{nl}	$\tilde{\psi}_{nl}$	radial momentum-space wavefunction of an atomic shell	2.2
T_{s}	K_{s}	Kohn–Sham (independent-electron) kinetic energy	3.4
V_2	V_{cov}	covalent energy between two bond hybrids	3.2
Ω; Ω_{I}; $\Omega_{\mathrm{D}}+\Omega_{\mathrm{OD}}$	Ω_W; Ω_W^{I}; $\tilde{\Omega}_W$	Wannier spread and its gauge-invariant and gauge-dependent parts	27.1
V_{cell}	Ω	unit-cell volume	3.1
χ, χ^0, α	R^U, $R^{U,0}$, δV^U	interacting and bare occupation responses, and the potential shift, of linear-response Hubbard U	28.1
α, K	α_{ohm}	ohmic coupling of a tunneling defect to conduction electrons	16.8
e (Gaussian)	e_{G}	Gaussian elementary charge, $e_{\mathrm{G}}^2=\alpha$; the book's e has $e^2=4\pi\alpha$	A.2
k (elliptic modulus)	s	modulus of the complete elliptic integrals	C.8

Four conventions differ from parts of the literature by a sign or a factor rather than by a letter: the Fourier sign (Appendix A.3), the sign of the mixing term (Appendix A.9), the units of charge (Appendix A.2) and the normalization of the structure factor (Appendix A.7). Before a formula taken from a paper is combined with the formulas of this book, it must be converted to the book's choice of these four conventions.

B.3 Acronyms

The last column gives the section or appendix in which the acronym first appears.

Acronym	Expansion	First use
ACE	atomic cluster expansion	32.1
AHC	Allen–Heine–Cardona	6.3
ALDA	adiabatic local-density approximation	4.6
ALP	axion-like particle	1.4
ANO	atomic natural orbital	30.2
BCS	Bardeen–Cooper–Schrieffer	1.5
BSE	Bethe–Salpeter equation	8.5
CASSCF	complete-active-space self-consistent field	30.2
CBM	conduction-band minimum	3.3
CCD	charge-coupled device	1.5
CDMS	Cryogenic Dark Matter Search	18.5
CMB	cosmic microwave background	1.1
CNT	carbon nanotube	30.4
CODATA	Committee on Data of the International Science Council	A.1
COHSEX	Coulomb-hole plus screened-exchange	23.4
CPU	central processing unit	29.2
DAMIC	Dark Matter in CCDs	10.9
DFPT	density-functional perturbation theory	5.2
DFT	density-functional theory	7.1
DLMF	Digital Library of Mathematical Functions	C.8
EPW	Electron–Phonon Wannier	6.2
FFT	fast Fourier transform	23.3
FIRE	fast inertial relaxation engine	32.1
FP	Frenkel pair	19.1
FT	fast transverse	29.2
G4CMP	Geant4 Condensed Matter Physics	20.1
GGA	generalized-gradient approximation	3.4
GIPAW	gauge-including projector-augmented-wave	31.5
GPAW	grid-based projector-augmented wave	12.5
GPU	graphics processing unit	30.5
GRASP	General-purpose Relativistic Atomic Structure Package	30.3
HOMO	highest occupied molecular orbital	23.2
HSE	hybrid functional of Heyd, Scuseria and Ernzerhof	3.4
HSE06	hybrid functional of Heyd, Scuseria and Ernzerhof, 2006 parametrization	10.6
IP	independent particles	4.6
JARVIS	Joint Automated Repository for Various Integrated Simulations	17.6
KK	Kramers–Kronig	30.3
KKR	Korringa–Kohn–Rostoker	30.5
LA	longitudinal acoustic	5.2
LCAO	linear combination of atomic orbitals	3.2
LDA	local-density approximation	3.4
LF	local fields	8.5
LO	longitudinal optical	4.9
MC	Monte Carlo	32.1
MD	molecular dynamics	30.5
ML	machine-learned	30.4
MLIP	machine-learned interatomic potential	32.1
NEB	nudged elastic band	30.5
OPW	orthogonalized plane wave	16.3
PAW	projector-augmented-wave	10.6

Acronym	Expansion	First use
PBE	generalized-gradient functional of Perdew, Burke and Ernzerhof	3.3
PBE0	hybrid functional built on the functional of Perdew, Burke and Ernzerhof	30.4
QED	quantum electrodynamics	6.3
RAVE	Radial Velocity Experiment	1.2
RCC	relativistic core correlation	31.3
RPA	random-phase approximation	8.5
SALMON	Scalable Ab initio Light–Matter simulator for Optics and Nanoscience (code)	32.1
SCAN	strongly constrained and appropriately normed	30.1
SCF	self-consistent-field	16.6
SCHA	self-consistent harmonic approximation	5.6
SH	transverse polarization perpendicular to the plane of incidence (shear horizontal)	20.3
SHM	standard halo model	1.2
SOC	spin–orbit coupling	8.4
ST	slow transverse	29.2
SV	transverse polarization in the plane of incidence (shear vertical)	20.3
TA	transverse acoustic	5.2
TB09	meta-generalized-gradient functional of Tran and Blaha (2009)	33.1
TDDFT	time-dependent density-functional theory	8.5
TDEP	temperature-dependent effective potential	25
TO	transverse optical	4.9
TRIM	Transport of Ions in Matter	19.4.4
VBM	valence-band maximum	3.3
X2C	exact two-component (scalar-relativistic Hamiltonian)	31.3
XCOM	photon cross-section database of the National Institute of Standards and Technology	10.8
YIG	yttrium iron garnet	14.1
ZBL	Ziegler, Biersack and Littmark	29.1
ZG	Zacharias–Giustino (special displacement)	25.3
ZPR	zero-point renormalization	6.3

Appendix C

Mathematical tools

C.1 Fourier transforms and delta functions

Fermi's golden rule rests on one limit: the function $\sin^2(\xi t/2)/\xi^2$ of Eq. (2.3) grows in height as t^2 and shrinks in width as $1/t$, and at long times it acts on smooth functions as $(\pi t/2)\,\delta(\xi)$ (Eq. (2.4)). Sect. 2.1 used the limit and the integral $\int \mathrm{d}u\, \sin^2 u/u^2 = \pi$ behind it; Sect. 2.2 used the rule for a delta function of a function. This section proves the limit, the integral and the rule for a delta function of a function, together with the Fourier identities that the rest of the book uses, in the conventions of Appendix A.3.

C.1.1 Delta functions as limits

A delta function is defined by what it does under an integral. For every test function $w(x)$, a smooth function that falls off fast enough for the integrals below to converge,

$$\int_{-\infty}^{\infty} \mathrm{d}x\; \delta(x)\, w(x) = w(0). \tag{C.1}$$

No ordinary function does this; δ is a distribution, a linear rule that assigns a number to each test function [1, Sect. 1.16]. In physics it always appears as the limit of ordinary functions that become tall and narrow, and the useful question is when and how fast the limit is reached.

The basic construction takes any integrable $f(u)$ with $\int \mathrm{d}u\, f(u) = 1$ and squeezes it, $f_t(x) = t\, f(tx)$. The area of f_t stays one while its width shrinks as $1/t$. With $u = tx$,

$$\int \mathrm{d}x\, f_t(x)\, w(x) = \int \mathrm{d}u\, f(u)\, w(u/t) \;\longrightarrow\; w(0) \int \mathrm{d}u\, f(u) = w(0) \qquad (t \to \infty),$$

because $w(u/t) \to w(0)$ at every fixed u, and $|f(u)\, w(u/t)| \le |f(u)| \max|w|$ lets the limit pass under the integral (dominated convergence). The family f_t therefore tends to δ. Three members recur in this book: the Gaussian $f(u) = \mathrm{e}^{-u^2/2}/\sqrt{2\pi}$, which describes a broadened line on a computational grid (Appendix C.2); the Lorentzian $f(u) = \pi^{-1}/(1+u^2)$, which describes a decaying state (Sect. 2.4 and Appendix C.4); and the golden-rule function, $f(u) = (2/\pi)\sin^2(u/2)/u^2$.

How fast the limit is reached depends on the tails of f. For an even f with a finite second moment $\langle u^2\rangle_f$, expanding $w(u/t)$ in powers of u/t gives $w(0) + \frac{1}{2}w''(0)\langle u^2\rangle_f/t^2$, and the correction to $w(0)$ is of relative order $1/t^2$. The Gaussian is of this kind. The Lorentzian and the golden-rule function fall off only as $1/u^2$, and their second moments diverge. The expansion then fails. For the Lorentzian and the golden-rule function the correction is of order $1/t$, and it is set by the values of w far from the origin, not by the curvature of w at the origin. Eq. (C.3) below gives the correction for the golden-rule function and Appendix C.4 the one for the Lorentzian.

C.1.2 The golden-rule limit

The golden-rule limit needs two integrals, the Dirichlet integral $\int_0^\infty \mathrm{d}u\, \sin u/u = \pi/2$ and $\int \mathrm{d}u\, \sin^2 u/u^2 = \pi$. The Dirichlet integral follows from a convergence factor. Since $\int_0^\infty \mathrm{d}u\, \mathrm{e}^{-su} \sin u = 1/(1+s^2)$ for $s > 0$, and $1/u = \int_0^\infty \mathrm{d}s\, \mathrm{e}^{-su}$ for $u > 0$,

$$\int_0^\infty \mathrm{d}u\, \frac{\sin u}{u} = \int_0^\infty \mathrm{d}s \int_0^\infty \mathrm{d}u\, \mathrm{e}^{-su} \sin u = \int_0^\infty \frac{\mathrm{d}s}{1+s^2} = \frac{\pi}{2}.$$

The exchange of the two integrals is legitimate because the double integral converges absolutely once a factor $\mathrm{e}^{-s_0 u}$ with $s_0 > 0$ is included, and the result is continuous as $s_0 \to 0$. The integrand is even, so $\int_{-\infty}^\infty \mathrm{d}u\, \sin u/u = \pi$. Integrating by parts, with $\sin^2 u/u \to 0$ at both ends and $\mathrm{d}(\sin^2 u)/\mathrm{d}u = \sin 2u$,

$$\int_{-\infty}^\infty \mathrm{d}u\, \frac{\sin^2 u}{u^2} = \Big[-\frac{\sin^2 u}{u}\Big]_{-\infty}^\infty + \int_{-\infty}^\infty \mathrm{d}u\, \frac{\sin 2u}{u} = \pi, \tag{C.2}$$

where the last integral is the Dirichlet integral with $u \to 2u$. Eq. (C.2) is the integral quoted in Sect. 2.1. The central lobe of the integrand, $|u| < \pi$, holds 90.3 % of the integral.

Write $\sin^2(\xi t/2) = (1 - \cos \xi t)/2$ and split off $w(0)$:

$$\int \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2}\, w(\xi) = w(0)\, \frac{\pi t}{2} + \frac{1}{2} \int \mathrm{d}\xi\, \frac{w(\xi) - w(0)}{\xi^2} - \frac{1}{2} \int \mathrm{d}\xi\, \cos(\xi t)\, \frac{w(\xi) - w(0)}{\xi^2}.$$

The first term uses Eq. (C.2) with $u = \xi t/2$. In the second and third terms, $w(\xi) - w(0) = w'(0)\xi + O(\xi^2)$ near the origin. The odd part $w'(0)\xi$ makes the integrand odd and singular as $1/\xi$, and the contributions from the two sides of $\xi = 0$ cancel if the integral is taken symmetrically about $\xi = 0$; the $O(\xi^2)$ term leaves an integrable integrand. The second term is then a constant, the Hadamard finite part of $\int \mathrm{d}\xi\, w(\xi)/\xi^2$ defined in Eq. (2.59). The third term tends to zero as $t \to \infty$: the integral of a rapidly oscillating cosine against an integrable function vanishes in the limit (the Riemann–Lebesgue lemma [1, Eq. (1.8.10)]), and the singular odd part contributes nothing by symmetry. Hence

$$\int \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2}\, w(\xi) = \frac{\pi t}{2}\, w(0) + \frac{1}{2}\, \mathrm{fp} \int \mathrm{d}\xi\, \frac{w(\xi)}{\xi^2} + o(1) \qquad (t \to \infty), \tag{C.3}$$

where $o(1)$ stands for terms that vanish as $t \to \infty$. Dividing by $\pi t/2$ proves Eq. (2.4). For a w that varies on an energy scale δ_w, the finite part is of size $w(0)/\delta_w$, so the correction relative to the leading term is of order $1/(t\,\delta_w)$, the estimate quoted in Sect. 2.1. A constant w has finite part zero, and then the area $\pi t/2$ is exact at every t.

The time derivative of Eq. (2.3) brings in $\sin(\xi t)/\xi$, which has area π at every $t > 0$ by the Dirichlet integral. The function $\sin(\xi t)/\xi$ is not absolutely integrable, so the dominated-convergence argument does not apply, but splitting off $w(0)$ as above still works. With $w(\xi) = w(0) + [w(\xi) - w(0)]$, the first part gives $\pi w(0)$. The second part gives $\int \mathrm{d}\xi\, \sin(\xi t)\, [w(\xi) - w(0)]/\xi$, whose integrand is a sine times a function that is finite at the origin. This integral vanishes as $t \to \infty$ by the Riemann–Lebesgue lemma (at large $|\xi|$, where $w(0)/\xi$ is not integrable, the oscillation of the sine still makes the contribution vanish). Hence

$$\frac{\sin \xi t}{\xi} \;\longrightarrow\; \pi\, \delta(\xi) \qquad (t \to \infty). \tag{C.4}$$

The convergence is weaker than for $\sin^2(\xi t/2)/\xi^2$. The function $\sin(\xi t)/\xi$ changes sign, its absolute area grows as $\ln t$, and the limit holds only under an integral against a smooth function. Because of this weaker convergence, Sect. 2.1 reads the rate off the probability, whose kernel is positive, as well as off the time derivative of the probability.

C.1.3 Delta functions of functions

Let $g(x)$ be smooth with simple roots x_i, that is $g(x_i) = 0$ and $g'(x_i) \neq 0$. Near each root $g(x) \approx g'(x_i)(x - x_i)$, and a delta function of g is supported only at the roots. Changing the variable to $y = g(x)$ in a small interval around x_i, with $\mathrm{d}x = \mathrm{d}y/|g'(x)|$ (the absolute value appears because the orientation of the interval is restored after the substitution),

$$\int_{x_i-\epsilon}^{x_i+\epsilon} \mathrm{d}x\, \delta(g(x))\, w(x) = \int \mathrm{d}y\, \delta(y)\, \frac{w(x(y))}{|g'(x(y))|} = \frac{w(x_i)}{|g'(x_i)|}.$$

Summing over the roots,

$$\delta(g(x)) = \sum_i \frac{\delta(x - x_i)}{|g'(x_i)|}. \tag{C.5}$$

Two cases appear in Chap. 2. The scaling rule $\delta(cx) = \delta(x)/|c|$ is the case of one root. Energy conservation in elastic scattering, $\delta(p'^2/2\mu_{\chi e} - p^2/2\mu_{\chi e})$ with $p, p' \geq 0$, has the root $p' = p$ with $g'(p) = p/\mu_{\chi e}$ and gives $(\mu_{\chi e}/p)\, \delta(p' - p)$, as used in Sect. 2.2. A double root, where $g'(x_i) = 0$, makes the right side of Eq. (C.5) infinite; it signals that the delta function must be kept inside a larger integral, as happens at the van Hove singularities of a density of states (Appendix C.2).

In three dimensions $\delta^3(\boldsymbol{x}) = \delta(x_1)\delta(x_2)\delta(x_3)$. The analogue of Eq. (C.5) for a scalar function of a vector is an integral over the surface on which the function vanishes,

$$\int \mathrm{d}^3k\, \delta(g(\boldsymbol{k}))\, w(\boldsymbol{k}) = \int_{g(\boldsymbol{k})=0} \frac{\mathrm{d}S}{|\nabla g(\boldsymbol{k})|}\, w(\boldsymbol{k}), \tag{C.6}$$

which follows from Eq. (C.5) applied along the normal to the surface at each point, with $\mathrm{d}S$ the element of area. Appendix C.2 uses Eq. (C.6) for densities of states.

C.1.4 Fourier identities

The transforms of Eqs. (A.5) and (A.6) are inverse to each other because the plane waves are complete,

$$\int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} = 2\pi\, \delta(\omega), \qquad \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = (2\pi)^3\, \delta^3(\boldsymbol{q}). \tag{C.7}$$

The first identity is the golden-rule limit once more. The integral over a finite window, $\int_{-T/2}^{T/2} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} = 2\sin(\omega T/2)/\omega$, tends to $2\pi\delta(\omega)$ as $T \to \infty$ by Eq. (C.4) with $\xi = \omega$ and $t = T/2$. The second identity is the product of three copies of the first. Inserting the inverse transform of $O(\boldsymbol{q})$ into the forward transform of Eq. (A.5) and using Eq. (C.7) returns $O(\boldsymbol{q})$, so the two transforms are indeed inverse to each other. The transform of $\delta^3(\boldsymbol{r} - \boldsymbol{r}_0)$ is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_0}$, and the density operator Eq. (A.7) of point particles is a sum of these phases. The transform of a derivative is $\int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \nabla O = -\mathrm{i}\boldsymbol{q}\, O(\boldsymbol{q})$ after an integration by parts. Hence $-\nabla^2 \to q^2$, and this replacement solves the static field equations of Appendix C.9.

Parseval's theorem and the convolution theorem follow from Eq. (C.7). Inserting the inverse transforms of f and g into $\int \mathrm{d}^3r\, f^*(\boldsymbol{r})\, g(\boldsymbol{r})$ and doing the $\boldsymbol{r}$ integral first gives Parseval's theorem,

$$\int \mathrm{d}^3r\, f^*(\boldsymbol{r})\, g(\boldsymbol{r}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\, f^*(\boldsymbol{q})\, g(\boldsymbol{q}), \tag{C.8}$$

and in one dimension of time $\int \mathrm{d}t\, f^*(t)g(t) = \int \mathrm{d}\omega\, f^*(\omega)g(\omega)/2\pi$. Let $(f * g)(\boldsymbol{r}) = \int \mathrm{d}^3r'\, f(\boldsymbol{r} - \boldsymbol{r}')\, g(\boldsymbol{r}')$ be the convolution. Writing $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'}$ and shifting the variable of the outer integral gives the convolution theorem,

$$(f * g)(\boldsymbol{q}) = f(\boldsymbol{q})\, g(\boldsymbol{q}). \tag{C.9}$$

The Helm form factor of Sect. 2.2 is an example: a uniform sphere smeared by a Gaussian is the convolution of the density of the uniform sphere with the Gaussian, and the transform of the smeared sphere is the product of $3j_1(qR_{\rm nuc})/(qR_{\rm nuc})$ (Appendix C.3) and the Gaussian factor $\mathrm{e}^{-q^2 s_{\rm skin}^2/2}$ (Appendix C.6).

The time form of Eq. (C.9) explains the structure of response functions. A causal response is a convolution in time of a response kernel with the applied field, and in frequency it becomes a product. The kernel vanishes for negative times, and its transform $\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}(\cdots)$ converges only with a factor $\mathrm{e}^{-0^+ t}$, a decay slower than any physical time scale. For a kernel that oscillates at a single transition energy $\Delta E = E_f - E_i$, as in Appendix A.3,

$$-\mathrm{i}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega-\Delta E)t - 0^+ t} = \frac{1}{\omega - \Delta E + \mathrm{i}0^+}, \tag{C.10}$$

which is the origin of the infinitesimal $\mathrm{i}0^+$ in every energy denominator of this book (Appendix A.3 and Sect. 2.4). Appendix C.4 gives the meaning of the right side of Eq. (C.10) as a distribution, and Appendix C.5 the analyticity that Eq. (C.10) implies.

C.2 Box normalization and Brillouin-zone sums

A crystal of 10^{23} cells has 10^{23} allowed crystal momenta per band, and no calculation sums over them one by one. Every rate in this book replaces the sum over crystal momenta by an integral, Eqs. (2.10), (A.9) and (A.10), and every first-principles calculation then replaces the integral by a sum again, over a grid of perhaps 10^3 to 10^6 points. The Poisson summation formula gives the error of both replacements; it also gives the infinite-crystal form of the lattice sum of Sect. 3.1.

C.2.1 From sums to integrals

In a periodic box of side L the allowed momenta in one dimension are $p = 2\pi n/L$ with integer n. The replacement of Eq. (2.10) is $\sum_n g(2\pi n/L) \to (L/2\pi)\int \mathrm{d}p\, g(p)$. Its exact form follows from the Dirac comb, the function $\sum_n \delta(y-n)$. The comb is periodic in y with period one, and its Fourier coefficients over one period are $\int_{-1/2}^{1/2} \mathrm{d}y\, \delta(y)\, \mathrm{e}^{-2\pi \mathrm{i} m y} = 1$ for every integer m, so

$$\sum_{n=-\infty}^{\infty} \delta(y-n) = \sum_{m=-\infty}^{\infty} \mathrm{e}^{2\pi \mathrm{i} m y}. \tag{C.11}$$

Multiplying by $\mathcal{F}(y)$ and integrating gives the Poisson summation formula, $\sum_n \mathcal{F}(n) = \sum_m \int \mathrm{d}y\, \mathcal{F}(y)\, \mathrm{e}^{2\pi \mathrm{i} m y}$ [1, Eq. (1.8.14)]. With $\mathcal{F}(y) = g(2\pi y/L)$ and $p = 2\pi y/L$,

$$\sum_p g(p) = \frac{L}{2\pi} \sum_{m=-\infty}^{\infty} \tilde{g}(mL), \qquad \tilde{g}(x) \equiv \int \mathrm{d}p\, \mathrm{e}^{\mathrm{i}px}\, g(p). \tag{C.12}$$

The term $m = 0$ is the integral. The terms with nonzero m are the Fourier transform of g evaluated at multiples of the box size: in position space they are the contributions of the periodic images of the system, which the boundary conditions put at distances mL. If g varies on a momentum scale Δp, its transform is confined to $|x| \lesssim 1/\Delta p$, and the images do not matter once $L\,\Delta p \gg 1$, the condition stated in Sect. 2.1.

How small the image terms are depends on the smoothness of g. A Gaussian of width Δp has $\tilde{g}(L) \propto \mathrm{e}^{-L^2 \Delta p^2/2}$, and the error is exponentially small. A function with a jump, such as the occupation of a Fermi sea with its step at $k_{\rm F}$, has a transform that falls off only as $1/x$, and the error then decreases only as a power of $1/(L\,\Delta p)$. In three dimensions the comb is

a product of three combs, the images lie on the lattice of vectors $L\boldsymbol{n}$, and Eq. (C.12) holds with $L/2\pi \to \mathscr{V}/(2\pi)^3$ and a sum over all image vectors. A delta function in g, such as the energy delta of the golden rule, has no meaning on a discrete set of momenta. It must first be broadened by more than the level spacing, as Sect. 2.1 required of the time window, and the limit $\mathscr{V} \to \infty$ is taken before the broadening is removed.

The density of final states is the main application. With Eq. (C.6), the number of states per unit energy of a band $\varepsilon(\boldsymbol{p})$ is

$$\mathcal{D}_f(E) = \sum_{\boldsymbol{p}} \delta(E - \varepsilon(\boldsymbol{p})) = \mathscr{V} \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, \delta(E - \varepsilon(\boldsymbol{p})) = \frac{\mathscr{V}}{(2\pi)^3} \int_{\varepsilon(\boldsymbol{p})=E} \frac{\mathrm{d}S}{|\nabla_{\boldsymbol{p}}\varepsilon|}, \tag{C.13}$$

per spin state. For a free particle the surface $\varepsilon(\boldsymbol{p}) = E$ is a sphere of area $4\pi p^2$ with $|\nabla_{\boldsymbol{p}}\varepsilon| = p/m$, which gives Eq. (2.11). In a crystal $\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}}$ vanishes at points of every band, by periodicity; there the integrand of Eq. (C.13) diverges. In three dimensions the divergence is integrable and produces square-root kinks in the density of states, the van Hove singularities; in two dimensions it produces a logarithm (Appendix C.8).

C.2.2 The lattice sum

On the Born–von Karman grid of a block of N cells, Eq. (3.13), the lattice sum is $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N \sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}}$ (Eq. (3.14)). In an infinite crystal, with $\delta_{\boldsymbol{kG}} \to [(2\pi)^3/\mathscr{V}]\, \delta^3(\boldsymbol{k} - \boldsymbol{G})$ by Eq. (A.9) and $\mathscr{V} = N\Omega$, it becomes $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = [(2\pi)^3/\Omega] \sum_{\boldsymbol{G}} \delta^3(\boldsymbol{k} - \boldsymbol{G})$, the three-dimensional form of Eq. (C.11).

A sum over all momenta in the box decomposes as $\boldsymbol{q} = \boldsymbol{k} + \boldsymbol{G}$ with $\boldsymbol{k}$ in the zone,

$$\sum_{\boldsymbol{q}} \mathcal{F}(\boldsymbol{q}) = \sum_{\boldsymbol{k}\in\mathrm{BZ}} \sum_{\boldsymbol{G}} \mathcal{F}(\boldsymbol{k} + \boldsymbol{G}) \;\longrightarrow\; \mathscr{V} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3 k}{(2\pi)^3} \sum_{\boldsymbol{G}} \mathcal{F}(\boldsymbol{k} + \boldsymbol{G}), \tag{C.14}$$

which is how an integral over the unreduced momentum transfer $\boldsymbol{q}$ of the dark matter is matched to quantities indexed by $\boldsymbol{q}_r$ and $\boldsymbol{G}_q$ (Appendix A.3).

C.2.3 Zone averages and grids

The zone average of a function of crystal momentum is $\langle\mathcal{F}\rangle_{\mathrm{BZ}} = N^{-1} \sum_{\boldsymbol{k}} \mathcal{F}(\boldsymbol{k})$, which by Eq. (A.10) equals $\Omega \int_{\mathrm{BZ}} \mathrm{d}^3 k\, \mathcal{F}(\boldsymbol{k})/(2\pi)^3$. A band energy, a squared matrix element or an occupation is periodic in $\boldsymbol{k}$, $\mathcal{F}(\boldsymbol{k} + \boldsymbol{G}) = \mathcal{F}(\boldsymbol{k})$. It therefore has a Fourier series over the direct lattice,

$$\mathcal{F}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} \mathcal{F}_{\boldsymbol{R}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}, \qquad \langle\mathcal{F}\rangle_{\mathrm{BZ}} = \mathcal{F}_{\boldsymbol{R}=0},$$

where the zone average picks out $\boldsymbol{R} = 0$ by Eq. (3.14) read with the roles of $\boldsymbol{k}$ and $\boldsymbol{R}$ exchanged. A calculation evaluates $\mathcal{F}$ on a uniform grid of $n_k = n_1 n_2 n_3$ points, the allowed momenta of a fictitious block of n_k cells. By the same identity, the grid average of $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$ is one when $\boldsymbol{R}$ is a lattice vector of that block, $\boldsymbol{R} = \sum_i l_i n_i \boldsymbol{a}_i$ with integer l_i, and zero otherwise. The grid average therefore equals

$$\frac{1}{n_k} \sum_{\boldsymbol{k}\in\mathrm{grid}} \mathcal{F}(\boldsymbol{k}) = \langle\mathcal{F}\rangle_{\mathrm{BZ}} + \sum_{(l_1,l_2,l_3)\neq 0} \mathcal{F}_{\boldsymbol{R}=\sum_i l_i n_i \boldsymbol{a}_i}. \tag{C.15}$$

The error of the grid average consists of the Fourier components $\mathcal{F}_{\boldsymbol{R}}$ at the nonzero lattice vectors of the fictitious block. This error is the crystal version of the image terms of Eq. (C.12), and it is small when $\mathcal{F}_{\boldsymbol{R}}$ has decayed over n_i cells.

In an insulator, the quantities summed over occupied bands are analytic functions of $\boldsymbol{k}$, their components $\mathcal{F}_{\boldsymbol{R}}$ decay exponentially with $|\boldsymbol{R}|$, on the scale of the localization length of the Wannier functions (Sect. 27.1), and the grid average converges exponentially in n_i. In a metal, the occupation jumps at the Fermi surface, $\mathcal{F}_{\boldsymbol{R}}$ decays only as a power of $|\boldsymbol{R}|$, and convergence is slow; a smearing of the step, or an interpolation, is needed. Grids displaced from the origin, introduced by Monkhorst and Pack [2], have the same error structure, Eq. (C.15) with a phase on each term (a sign alternating with l_i for a shift of half a grid step), and need fewer symmetry-distinct points.

Energy delta functions on a grid need one more step. The transition sum $N^{-1}\sum_{\boldsymbol{k}}\delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}})$ on n_k points is a set of spikes. The spikes merge into a curve only after each is broadened by a width σ_{br} larger than the change of the transition energy between neighboring grid points, about $|\nabla_{\boldsymbol{k}}(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})|\,|\boldsymbol{b}_i|/n_i$. A Gaussian broadening is preferred to a Lorentzian of the same width. A Gaussian gives an error of order σ_{br}^2 times the curvature of the spectrum, while the slow tails of a Lorentzian give an error of order σ_{br} that depends on the whole spectrum (Appendices C.1 and C.4). The width σ_{br} must then be reduced together with the grid spacing. The alternative is the tetrahedron method, which interpolates the band energies linearly inside small tetrahedra filling the zone and integrates the delta function exactly within each [3]. With broadening or with tetrahedra, a computed spectrum has a resolution set by the grid, and features narrower than the resolution, such as the onset of absorption at a gap, need a denser grid or an analytic treatment (Sect. 3.3).

C.2.4 Normalization checks

Counting powers of N and $\mathcal{V}$ catches most normalization errors. Each sum over an independent crystal momentum brings a factor N, or $\mathcal{V}/(2\pi)^3$ times an integral, and each plane wave or Bloch state normalized over the crystal brings $\mathcal{V}^{-1/2}$; an intensive quantity must have the powers of N cancel. Each Kronecker delta in crystal momentum that a matrix element produces removes one sum. In the Bloch basis, the product of two lattice sums, Eq. (3.14) for $\boldsymbol{k}$ and again for $\boldsymbol{k}'$, gives $N^2\delta_{\boldsymbol{k}\boldsymbol{G}}\delta_{\boldsymbol{k}'\boldsymbol{G}'}$, and the square of one Kronecker delta is the same Kronecker delta. In the continuum, the square of $\delta^3(\boldsymbol{k})$ is $\delta^3(0)\,\delta^3(\boldsymbol{k})$ with $\delta^3(0) = \mathcal{V}/(2\pi)^3$. The continuum rule is the Kronecker rule in another form, and it is the reason why rates are worked out in the box (Sect. 2.1).

C.3 Angular averages and spherical functions

A dark-matter rate integrates over the direction of the momentum transfer, and an atomic form factor integrates over the direction of the electron. Sect. 2.2 needed three such integrals: the average of a plane wave over directions, $4\pi\sin(qr)/(qr)$; its first moment, $4\pi\mathrm{i}\,j_1(qr)$; and the sum rule $\sum_m |Y_{lm}|^2 = (2l+1)/4\pi$ that makes a closed shell spherical. The three integrals are special cases of the expansion of a plane wave in spherical harmonics, and this section derives the expansion in full.

C.3.1 Plane waves averaged over directions

We write $\hat{\boldsymbol{n}}$ for a unit vector, $\mathrm{d}\Omega_{\hat{n}}$ for the element of solid angle of its direction, and $\int \mathrm{d}\Omega_{\hat{n}} = 4\pi$. With the polar axis along $\boldsymbol{q}$ and $u = \cos\theta$ the cosine of the angle between $\boldsymbol{q}$ and $\hat{\boldsymbol{n}}$, $\mathrm{d}\Omega_{\hat{n}} = \mathrm{d}\phi\,\mathrm{d}u$ and the azimuth gives 2π. For $\boldsymbol{r} = r\hat{\boldsymbol{n}}$,

$$\int \mathrm{d}\Omega_{\hat{n}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = 2\pi\int_{-1}^{1}\mathrm{d}u\,\mathrm{e}^{\mathrm{i}qru} = 4\pi\,\frac{\sin qr}{qr} = 4\pi\,j_0(qr). \tag{C.16}$$

The first moment follows by differentiating with respect to $a = qr$ under the integral, $\int_{-1}^{1} \mathrm{d}u\, u\, \mathrm{e}^{\mathrm{i}au} = -\mathrm{i}\,\mathrm{d}(2\sin a/a)/\mathrm{d}a = 2\mathrm{i}(\sin a - a\cos a)/a^2$, so

$$2\pi \int_{-1}^{1} \mathrm{d}u\, u\, \mathrm{e}^{\mathrm{i}qru} = 4\pi\mathrm{i}\, j_1(qr), \qquad j_1(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z}. \tag{C.17}$$

The functions j_0 and j_1 are the spherical Bessel functions of orders zero and one [1, Eq. (10.49.3)]. A spherical density $\boldsymbol{n}(r)$ therefore has the transform $4\pi \int_0^\infty \mathrm{d}r\, r^2 j_0(qr)\, \boldsymbol{n}(r)$, a one-dimensional integral; for a density proportional to $\mathrm{e}^{-\beta r}$ the remaining integrals are Laplace transforms of powers times sin and cos. For instance, $\int_0^\infty \mathrm{d}r\, r\, \mathrm{e}^{-\beta r} \sin qr = 2\beta q/(\beta^2+q^2)^2$ and $\int_0^\infty \mathrm{d}r\, r^2 \mathrm{e}^{-\beta r} \cos qr = 2\beta(\beta^2 - 3q^2)/(\beta^2+q^2)^3$. With Eq. (C.17) the two integrals combine into

$$\int_0^\infty \mathrm{d}r\, r^3 \mathrm{e}^{-\beta r} j_1(qr) = \frac{1}{q^2}\,\frac{2\beta q}{(\beta^2+q^2)^2} - \frac{1}{q}\,\frac{2\beta(\beta^2-3q^2)}{(\beta^2+q^2)^3} = \frac{8\beta q}{(\beta^2+q^2)^3},$$

the radial integral of the hydrogen $1s \to 2p$ form factor in Sect. 2.2.

Averages of products of components of $\hat{\boldsymbol{n}}$ follow from symmetry. We write n^α for the Cartesian components, with $\alpha, \beta, \gamma, \delta$ as superscripts. The average $\langle n^\alpha n^\beta \rangle = (4\pi)^{-1} \int \mathrm{d}\Omega_{\hat{n}}\, n^\alpha n^\beta$ is a tensor that does not change under rotations, so it is proportional to $\delta^{\alpha\beta}$; contracting with $\delta^{\alpha\beta}$ and using $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}} = 1$ fixes the constant. In the same way the fourth moment is a combination of the three products of Kronecker deltas, with equal weights by the symmetry in its indices. Contracting two indices reduces it to the second moment:

$$\langle n^\alpha n^\beta \rangle = \tfrac{1}{3}\delta^{\alpha\beta}, \qquad \langle n^\alpha n^\beta n^\gamma n^\delta \rangle = \tfrac{1}{15}(\delta^{\alpha\beta}\delta^{\gamma\delta} + \delta^{\alpha\gamma}\delta^{\beta\delta} + \delta^{\alpha\delta}\delta^{\beta\gamma}). \tag{C.18}$$

Odd moments vanish, since $\hat{\boldsymbol{n}} \to -\hat{\boldsymbol{n}}$ leaves the measure unchanged. As a result, the direction average of $(\boldsymbol{q}\cdot\boldsymbol{v})^2$ is $q^2v^2/3$, and the direction average of a quadratic form $\boldsymbol{q}\cdot B\cdot\boldsymbol{q}$ is $q^2\,\mathrm{Tr}\,B/3$. In general, $\langle (n^x)^{2a}(n^y)^{2b}(n^z)^{2c}\rangle = (2a-1)!!\,(2b-1)!!\,(2c-1)!!/(2a+2b+2c+1)!!$, with $(-1)!! = 1$; the formula gives $\langle (n^x)^4 \rangle = 1/5$, in agreement with Eq. (C.18).

C.3.2 Spherical harmonics

The spherical harmonics $Y_{lm}(\hat{\boldsymbol{n}})$, with $l = 0, 1, 2, \ldots$ and $m = -l, \ldots, l$, are the eigenfunctions of the squared angular momentum with eigenvalue $l(l+1)$ and of its z component with eigenvalue m. We use the convention, with the Condon–Shortley phase, of most quantum-mechanics texts [1, Eq. (14.30.1)]: $Y_{00} = (4\pi)^{-1/2}$, $Y_{10} = (3/4\pi)^{1/2}\cos\theta$, $Y_{1,\pm1} = \mp(3/8\pi)^{1/2}\sin\theta\,\mathrm{e}^{\pm\mathrm{i}\phi}$, and $Y_{l,-m} = (-1)^m Y_{lm}^*$. The spherical harmonics are orthonormal on the sphere and complete,

$$\int \mathrm{d}\Omega_{\hat{n}}\, Y_{lm}^*(\hat{\boldsymbol{n}})\, Y_{l'm'}(\hat{\boldsymbol{n}}) = \delta_{ll'}\delta_{mm'}, \tag{C.19}$$

so any function of direction expands as $\sum_{lm} c_{lm} Y_{lm}$ [1, Eq. (14.30.8)]. The harmonics of order l are the restrictions to the sphere of the homogeneous polynomials of degree l in n^x, n^y, n^z that solve Laplace's equation, and under inversion $\hat{\boldsymbol{n}} \to -\hat{\boldsymbol{n}}$ they change by $(-1)^l$.

The $m = 0$ harmonics are Legendre polynomials, $Y_{l0} = [(2l+1)/4\pi]^{1/2} P_l(\cos\theta)$, with Rodrigues' formula $P_l(u) = (2^l l!)^{-1}\,\mathrm{d}^l(u^2-1)^l/\mathrm{d}u^l$ and $\int_{-1}^{1} \mathrm{d}u\, P_l P_{l'} = 2\delta_{ll'}/(2l+1)$. The addition theorem connects the Legendre polynomials to the full set of harmonics [1, Eq. (14.30.9)],

$$P_l(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{n}}') = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}^*(\hat{\boldsymbol{n}})\, Y_{lm}(\hat{\boldsymbol{n}}'). \tag{C.20}$$

Setting $\hat{\boldsymbol{n}}' = \hat{\boldsymbol{n}}$ and using $P_l(1) = 1$ gives Unsöld's theorem,

$$\sum_{m=-l}^{l} |Y_{lm}(\hat{\boldsymbol{n}})|^2 = \frac{2l+1}{4\pi}, \tag{C.21}$$

independent of direction. A filled shell, with every m occupied equally, therefore has a spherical density whatever the shape of its individual orbitals; Sect. 2.2 uses this result.

C.3.3 The plane-wave expansion

The function $\mathrm{e}^{\mathrm{i}zu}$ of $u \in [-1,1]$ expands in Legendre polynomials as $\sum_l c_l(z) P_l(u)$, with $c_l(z) = \frac{1}{2}(2l+1)\int_{-1}^{1} \mathrm{d}u\, P_l(u)\,\mathrm{e}^{\mathrm{i}zu}$ by their orthogonality. We evaluate the integral with Rodrigues' formula. Integrating by parts l times moves the derivatives onto the exponential, and the boundary terms vanish because $(u^2-1)^l$ has zeros of order l at $u = \pm 1$:

$$\int_{-1}^{1} \mathrm{d}u\, P_l(u)\,\mathrm{e}^{\mathrm{i}zu} = \frac{(-\mathrm{i}z)^l}{2^l l!}\int_{-1}^{1} \mathrm{d}u\,(u^2-1)^l \mathrm{e}^{\mathrm{i}zu} = \frac{(\mathrm{i}z)^l}{2^l l!}\, I_l(z), \qquad I_l(z) \equiv \int_{-1}^{1} \mathrm{d}u\,(1-u^2)^l \mathrm{e}^{\mathrm{i}zu}.$$

The integrals I_l obey a recursion. Differentiating under the integral and integrating by parts once, with $u(1-u^2)^l = -\mathrm{d}(1-u^2)^{l+1}/\mathrm{d}u\,/[2(l+1)]$,

$$\frac{\mathrm{d}I_l}{\mathrm{d}z} = \mathrm{i}\int_{-1}^{1} \mathrm{d}u\, u(1-u^2)^l \mathrm{e}^{\mathrm{i}zu} = \frac{\mathrm{i}\cdot\mathrm{i}z}{2(l+1)}\, I_{l+1}(z) = -\frac{z}{2(l+1)}\, I_{l+1}(z).$$

Starting from $I_0 = 2\sin z/z$ and iterating $I_{l+1} = -2(l+1)\, z^{-1}\,\mathrm{d}I_l/\mathrm{d}z$ gives

$$I_l(z) = 2^{l+1} l!\left(-\frac{1}{z}\frac{\mathrm{d}}{\mathrm{d}z}\right)^l \frac{\sin z}{z} = 2^{l+1} l!\, z^{-l} j_l(z),$$

where the second step uses $j_l(z) = z^l(-z^{-1}\mathrm{d}/\mathrm{d}z)^l(\sin z/z)$ [1, Eq. (10.49.14)]. Therefore $\int_{-1}^{1} \mathrm{d}u\, P_l\,\mathrm{e}^{\mathrm{i}zu} = 2\mathrm{i}^l j_l(z)$. Hence $c_l = (2l+1)\,\mathrm{i}^l j_l(z)$, and with $z = qr$, $u = \hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{r}}$ and Eq. (C.20),

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \sum_{l=0}^{\infty}(2l+1)\,\mathrm{i}^l j_l(qr)\, P_l(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{r}}) = 4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{l} \mathrm{i}^l j_l(qr)\, Y_{lm}^*(\hat{\boldsymbol{q}})\, Y_{lm}(\hat{\boldsymbol{r}}), \tag{C.22}$$

the Rayleigh expansion [1, Eq. (10.60.7)]. Integrating over the directions of $\hat{\boldsymbol{r}}$ keeps only $l = 0$ and returns Eq. (C.16); multiplying by Y_{10} before integrating keeps only $l = 1$ and returns Eq. (C.17).

The expansion is the multipole expansion of a form factor. A transition between orbitals of angular momenta l_i and l_f, through $\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$, picks out the terms with $|l_i - l_f| \le l \le l_i + l_f$, and $l_i + l + l_f$ even by parity. The spherical Bessel functions behave as

$$j_l(z) \simeq \frac{z^l}{(2l+1)!!} \quad (z \ll 1), \qquad j_l(z) \simeq \frac{\sin(z - l\pi/2)}{z} \quad (z \gg l), \tag{C.23}$$

[1, Eqs. (10.52.1) and (10.52.3)]. For an orbital of size r_{orb}, the multipole l of the form factor is therefore suppressed by $(qr_{\mathrm{orb}})^l$ when $qr_{\mathrm{orb}} \ll 1$, and the lowest allowed multipole dominates: the monopole for elastic scattering, the dipole for $s \to p$ transitions. When $qr_{\mathrm{orb}} \gtrsim 1$ many multipoles contribute with comparable weight, and the transition sees the electron as a particle rather than as a charge distribution. For an atomic valence orbital, r_{orb} is about an Ångström, and $qr_{\mathrm{orb}} = 1$ at $q \simeq 2\,\mathrm{keV}$, which is $\sim q_0/2$ (Sect. 1.3); the momentum transfers of light dark matter lie in the transition region.

C.3.4 Cubic harmonics

A cubic crystal comes close to spherical symmetry. The point group of silicon and germanium, O_h, contains all the rotations and reflections that map a cube onto itself, and at fixed $|\boldsymbol{q}|$ the response of a crystal with this point group is a function of $\hat{\boldsymbol{q}}$ invariant under these

operations. The point group of gallium arsenide, T_d, has only half of these operations and lacks inversion. A response that is even in $\boldsymbol{q}$, such as $S(\boldsymbol{q},\omega)$ in a target with time-reversal symmetry (Appendix A.4), is invariant under $\hat{\boldsymbol{q}} \to -\hat{\boldsymbol{q}}$ even though T_d lacks inversion, and T_d together with this inversion is O_h. The functions of direction we need are then those invariant under all permutations of (n^x, n^y, n^z) and all changes of their signs.

A homogeneous polynomial invariant under the sign changes contains only even powers of each component, so odd l is absent. At degree two, invariance under permutations leaves $(n^x)^2 + (n^y)^2 + (n^z)^2 = 1$, which is the $l = 0$ harmonic, so $l = 2$ is absent too. At degree four the invariants are $\sum_\alpha (n^\alpha)^4$ and 1, and the combination orthogonal to the constant is new. At degree six there is one more invariant, built from $(n^x n^y n^z)^2$. The first cubic harmonics are therefore

$$Y_0^{\mathrm{c}} = \frac{1}{\sqrt{4\pi}}, \qquad Y_4^{\mathrm{c}}(\hat{\boldsymbol{n}}) = \frac{5}{8}\sqrt{\frac{21}{\pi}}\Big[(n^x)^4 + (n^y)^4 + (n^z)^4 - \frac{3}{5}\Big], \qquad \ldots, \tag{C.24}$$

normalized as in Eq. (C.19). The constant $-3/5$ makes Y_4^{c} orthogonal to Y_0^{c}, since $\langle \sum_\alpha (n^\alpha)^4 \rangle = 3/5$. The prefactor follows from the moment formula above: $\langle (\sum_\alpha (n^\alpha)^4)^2 \rangle = 3\langle (n^x)^8 \rangle + 6\langle (n^x)^4 (n^y)^4 \rangle = 3/9 + 6/105 = 41/105$, the variance of $\sum_\alpha (n^\alpha)^4$ is $41/105 - 9/25 = 16/525$, and $4\pi \times (5/8)^2 (21/\pi) \times 16/525 = 1$. In spherical harmonics with the polar axis along a cube edge, $Y_4^{\mathrm{c}} = (7/12)^{1/2} Y_{40} + (5/24)^{1/2}(Y_{44} + Y_{4,-4})$, the combination given by Von der Lage and Bethe [4], who introduced these functions for electronic states in cubic crystals. The next invariant orders are $l = 6, 8, 10$ with one function each, and $l = 12$ with two.

Consider a property of a cubic crystal that is a second-rank tensor in $\hat{\boldsymbol{q}}$. Examples are the dielectric tensor at $q \to 0$, the displacement tensor B_κ at a site of cubic symmetry (Appendix A.6) or the effective-mass tensor of a nondegenerate band extremum at Γ. Each is a multiple of the unit tensor, since an anisotropy would need an $l = 2$ component, and none exists. The direction dependence of a cubic response at fixed $|\boldsymbol{q}|$ starts at $l = 4$. In silicon the anisotropy of $S(\boldsymbol{q},\omega)$ is a combination of $Y_4^{\mathrm{c}}(\hat{\boldsymbol{q}})$, $Y_6^{\mathrm{c}}(\hat{\boldsymbol{q}})$ and higher terms, with coefficients that depend on $|\boldsymbol{q}|$ and ω, and it is small where the momentum transfer is small compared with the reciprocal-lattice vectors. Sect. 11.7 expands rates in these functions to compute the daily modulation of Sect. 1.2 for cubic targets.

C.4 Singular denominators

An energy denominator $1/(E_i - E_m + \mathrm{i}0^+)$ is the transform of a causal time evolution, Eq. (C.10), and it is not an ordinary function at $E_m = E_i$. Sect. 2.4 split it into a principal value and a delta function, Eq. (2.58), by taking the limit of a finite width. Sect. 2.4 also gave the squared denominator a meaning through the Hadamard finite part, Eq. (2.59), with the expansion Eq. (2.60) of a Lorentzian in powers of its width. Here we prove the splitting for any smooth test function, collect the rules for manipulating these distributions, and give the correction terms of the Lorentzian limits, which control how a broadened spectrum approaches its zero-width limit.

C.4.1 The Sokhotski–Plemelj formula

Let $w(x)$ be a test function, bounded, integrable and differentiable at $x = 0$; a Lipschitz condition $|w(x) - w(0)| \le c|x|$ near the origin is enough. The principal value is the symmetric limit

$$\mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x} = \lim_{x_{\mathrm{c}} \to 0} \int_{|x| > x_{\mathrm{c}}} \mathrm{d}x\, \frac{w(x)}{x} = \int \mathrm{d}x\, \frac{w(x) - w(0)\,\Theta(X_{\mathrm{c}} - |x|)}{x}, \tag{C.25}$$

where the second form holds for any fixed $X_\mathrm{c} > 0$. On the symmetric interval $|x| < X_\mathrm{c}$ the subtracted term integrates to zero because $1/x$ is odd, and the remaining integrand is bounded near the origin by the Lipschitz condition. The second form shows that the principal value is finite. Folding the negative half line onto the positive one gives $\mathrm{PV}\int \mathrm{d}x\, w(x)/x = \int_0^\infty \mathrm{d}x\, [w(x) - w(-x)]/x$, whose integrand is finite at the origin: a principal-value integral over a tabulated spectrum is computed by folding, not by cutting out an interval around the pole.

For a finite width $\Gamma_\mathrm{w} > 0$, the real and imaginary parts of $1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)$ act on w separately. The imaginary part is $-\pi$ times the normalized Lorentzian $(\Gamma_\mathrm{w}/2\pi)/(x^2 + \Gamma_\mathrm{w}^2/4)$, which is the squeezed family $t\, f(tx)$ of Appendix C.1 with $f(u) = \pi^{-1}/(1 + u^2)$ and $t = 2/\Gamma_\mathrm{w}$. The integral of the normalized Lorentzian against w tends to $w(0)$ by the dominated-convergence argument of Appendix C.1. The real part is $x/(x^2 + \Gamma_\mathrm{w}^2/4)$. Subtracting $w(0)\,\Theta(X_\mathrm{c} - |x|)$, which integrates to zero against this odd kernel,

$$\int \mathrm{d}x\, \frac{x\, w(x)}{x^2 + \Gamma_\mathrm{w}^2/4} = \int \mathrm{d}x\, \frac{x^2}{x^2 + \Gamma_\mathrm{w}^2/4}\, \frac{w(x) - w(0)\,\Theta(X_\mathrm{c} - |x|)}{x}.$$

The first factor lies between 0 and 1 and tends to 1 at every $x \neq 0$; the second factor is integrable and independent of Γ_w. By dominated convergence the integral tends to the second form of Eq. (C.25). Together,

$$\lim_{\Gamma_\mathrm{w} \to 0^+} \int \mathrm{d}x\, \frac{w(x)}{x - x_0 \pm \mathrm{i}\Gamma_\mathrm{w}/2} = \mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x - x_0} \mp \mathrm{i}\pi\, w(x_0), \tag{C.26}$$

after a shift of the variable by x_0. With the lower sign, Eq. (C.26) is the complex conjugate of the statement with the upper sign. Written as distributions, $1/(x \pm \mathrm{i}0^+) = \mathrm{PV}(1/x) \mp \mathrm{i}\pi\delta(x)$, the Sokhotski–Plemelj formula, Eq. (2.58). The proof used only that w is integrable and Lipschitz at x_0; a sum over a dense set of final states, as in the level shift of Eq. (2.55), satisfies these conditions once the levels are replaced by a smooth density of states (Appendix C.2).

The formula has an equivalent form for functions of a complex variable. Let

$$\mathcal{F}(z) = \frac{1}{2\pi\mathrm{i}} \int \mathrm{d}x\, \frac{w(x)}{x - z}, \tag{C.27}$$

a Cauchy integral, which is analytic in z above and below the real axis. For z approaching a point x_0 of the axis from above, $z = x_0 + \mathrm{i}0^+$, and from below, Eq. (C.26) gives

$$\mathcal{F}(x_0 + \mathrm{i}0^+) - \mathcal{F}(x_0 - \mathrm{i}0^+) = w(x_0), \qquad \mathcal{F}(x_0 + \mathrm{i}0^+) + \mathcal{F}(x_0 - \mathrm{i}0^+) = \frac{1}{\pi\mathrm{i}}\, \mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x - x_0}. \tag{C.28}$$

The jump across the axis is the density w, and the average of the two sides is the principal-value transform of w. Every response function of the book has this structure: its imaginary part on the real axis is the density of transitions, and its real part is a principal-value integral over that density. Appendix C.5 derives the Kramers–Kronig relations from this structure.

C.4.2 Rules for the distributions

Since a distribution is defined by its action on test functions, identities among distributions are proved by integrating against w.

The product of $\delta(x)$ with a smooth function $h(x)$ is $h(0)\delta(x)$; in particular $x\,\delta(x) = 0$ and $x\,\mathrm{PV}(1/x) = 1$, since $\mathrm{PV}\int \mathrm{d}x\, x\, w(x)/x = \int \mathrm{d}x\, w(x)$. The two boundary values $1/(x \pm \mathrm{i}0^+)$ differ by a delta function,

$$\frac{1}{x - \mathrm{i}0^+} - \frac{1}{x + \mathrm{i}0^+} = 2\pi\mathrm{i}\, \delta(x), \tag{C.29}$$

which is Eq. (C.26) with both signs. A derivative acts on a distribution T by moving onto the test function, $\int \mathrm{d}x\, T'(x)\, w(x) = -\int \mathrm{d}x\, T(x)\, w'(x)$, the definition that makes integration by parts valid for distributions [1, Eq. (1.16.8)]. Hence $\mathrm{d}\ln|x|/\mathrm{d}x = \mathrm{PV}(1/x)$, as follows from $\int \mathrm{d}x\, \ln|x|\, w'(x)$ split at $\pm x_\mathrm{c}$, where the boundary terms $\ln x_\mathrm{c}\,[w(x_\mathrm{c}) - w(-x_\mathrm{c})]$ vanish as $x_\mathrm{c} \to 0$.

The derivative of the principal value is minus the finite part. Splitting at $\pm x_\mathrm{c}$ and integrating by parts,

$$-\int_{|x|>x_\mathrm{c}} \mathrm{d}x\, \frac{w'(x)}{x} = \frac{w(x_\mathrm{c}) + w(-x_\mathrm{c})}{x_\mathrm{c}} - \int_{|x|>x_\mathrm{c}} \mathrm{d}x\, \frac{w(x)}{x^2},$$

and as $x_\mathrm{c} \to 0$ the right side becomes minus the bracket of Eq. (2.59), because $w(x_\mathrm{c}) + w(-x_\mathrm{c}) = 2w(0) + O(x_\mathrm{c}^2)$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\,\mathrm{PV}\,\frac{1}{x} = -\,\mathrm{fp}\,\frac{1}{x^2}. \tag{C.30}$$

The finite part is thus the derivative of a distribution we already have, and it obeys the usual rules: $x^2\,\mathrm{fp}(1/x^2) = 1$, $x\,\mathrm{fp}(1/x^2) = \mathrm{PV}(1/x)$, and $\mathrm{fp}\int \mathrm{d}x\, w(x)/x^2 = 0$ for constant w, the last one by the explicit form of Eq. (2.59).

The square of a denominator follows from the derivative. Since $1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)^2 = -\mathrm{d}[1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)]/\mathrm{d}x$ at every finite width, and derivatives commute with the limit when both act on a test function, Eqs. (2.58) and (C.30) give

$$\frac{1}{(x + \mathrm{i}0^+)^2} = \mathrm{fp}\,\frac{1}{x^2} + \mathrm{i}\pi\,\delta'(x). \tag{C.31}$$

A double pole on one side of the axis is a well-defined distribution. The modulus squared $|x + \mathrm{i}\Gamma_\mathrm{w}/2|^{-2}$ is a different object, because it is the product of a denominator and its complex conjugate. The two factors have poles on opposite sides of the axis, and the poles pinch the axis as $\Gamma_\mathrm{w} \to 0$. The limit of the modulus squared does not exist; its expansion in Γ_w starts with $2\pi\delta(x)/\Gamma_\mathrm{w}$, Eq. (2.60). Products of distributions are in general undefined, and the squared delta function of a rate always comes from two poles that pinch the real axis.

C.4.3 Corrections to the Lorentzian limit

The normalized Lorentzian of full width Γ_w at half maximum is

$$\delta_\Gamma(x) = \frac{1}{\pi}\,\frac{\Gamma_\mathrm{w}/2}{x^2 + \Gamma_\mathrm{w}^2/4} = -\frac{1}{\pi}\,\mathrm{Im}\,\frac{1}{x + \mathrm{i}\Gamma_\mathrm{w}/2}, \tag{C.32}$$

and Eq. (2.60), multiplied by $\Gamma_\mathrm{w}/2\pi$, states how δ_Γ approaches the delta function. The first correction is linear in the width and involves the whole of w, through the finite part. It comes from the slow $1/x^2$ tails, which give weight $\Gamma_\mathrm{w}/(\pi|x|)$ to the region beyond any $|x|$. A Gaussian of the same width has no linear term: its first correction is proportional to $\Gamma_\mathrm{w}^2\, w''(0)$, and it depends on w only near the origin (Appendix C.1). For $w(x) = 1/(1 + x^2)$, the Lorentzian integral is exactly $1/(1 + \Gamma_\mathrm{w}/2)$, as closing the contour in either half plane shows. The finite part of $\int \mathrm{d}x\, w(x)/x^2$ is $-\pi$, since $1/[x^2(1 + x^2)] = 1/x^2 - 1/(1 + x^2)$ and the finite part of $\int \mathrm{d}x/x^2$ vanishes. The second derivative is $w''(0) = -2$. The expansion of Eq. (2.60) then reads $1 - \Gamma_\mathrm{w}/2 + \Gamma_\mathrm{w}^2/4 + \ldots$, the expansion of the exact result. At $\Gamma_\mathrm{w} = 0.2$ the linear term is a 10 % correction, and a Gaussian of the same full width at half maximum, whose correction is $\frac{1}{2}w''(0)$ times its variance $(\Gamma_\mathrm{w}/2.355)^2$, would make an error of 0.7 %.

The Lorentzian is the line shape of exponential decay. A state that decays as $\mathrm{e}^{-\Gamma_\mathrm{w}|t|/2}$ in amplitude has, by Eq. (A.6), the spectrum $\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}xt}\mathrm{e}^{-\Gamma_\mathrm{w}|t|/2} = 2\pi\delta_\Gamma(x)$. The convolution

theorem, Eq. (C.9) in its time form, then fixes the shape of combined lines. Two independent decay processes multiply in time and convolve in energy, $\mathrm{e}^{-\Gamma_1|t|/2}\mathrm{e}^{-\Gamma_2|t|/2} = \mathrm{e}^{-(\Gamma_1+\Gamma_2)|t|/2}$. The convolution of Lorentzians is a Lorentzian whose width is the sum of the two widths, $\delta_{\Gamma_1} * \delta_{\Gamma_2} = \delta_{\Gamma_1+\Gamma_2}$; widths of independent channels add, as Sect. 2.4 assumed for the total width of an intermediate state. A Lorentzian convolved with a Gaussian, the Voigt profile, keeps the $1/x^2$ tails of the Lorentzian, so the linear correction, the finite-part term of Eq. (2.60), survives any additional Gaussian broadening. A width that depends on energy, $\Gamma_\mathrm{w}(x)$, keeps the delta limit of Eq. (C.26) at its value at x_0, provided $\Gamma_\mathrm{w}(x)$ varies little over a range Γ_w around the pole; the slope of $\Gamma_\mathrm{w}(x)$ enters the finite-width corrections.

C.5 Complex analysis for response functions

A response cannot precede its cause. Causality alone fixes the real part of every response function once the imaginary part is known at all frequencies. The dielectric function of silicon at zero frequency, for example, follows from its measured absorption spectrum. Contour integration and the residue theorem are assumed; Sects. 4.1 and 4.7 use the results.

C.5.1 Causality and analyticity

Let $R(t)$ be a retarded response: the change of an observable at time t caused by a unit impulse at $t = 0$, so that $R(t) = 0$ for $t < 0$. The response function of Eq. (A.29) is of this kind. With the time convention of Appendix A.3, the transform of $R(t)$ is

$$R(z) = \int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}zt} R(t), \qquad z = \omega + \mathrm{i}y. \tag{C.33}$$

For $y > 0$ the factor $\mathrm{e}^{\mathrm{i}zt} = \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{-yt}$ decays, and the integral converges for any $R(t)$ that grows no faster than a polynomial. Differentiating under the integral with respect to z is then allowed, so $R(z)$ is analytic in the upper half plane. The physical response at real ω is the boundary value $R(\omega + \mathrm{i}0^+)$, which is the 0^+ prescription of Eq. (C.10). For $y < 0$ the integral diverges, and the function on the lower half plane is defined, if at all, by analytic continuation. In the spectral (Lehmann) form Eq. (A.31), each term is $1/(z - \Delta E)$ with real ΔE, analytic everywhere except on the real axis. With $z = \omega + \mathrm{i}0^+$, every pole of the Lehmann form lies below the axis of real ω, as Appendix A.3 requires.

C.5.2 Spectral representation

The Lehmann form makes the structure of $R(z)$ explicit. By Eq. (C.26), the imaginary part of each term on the real axis is a delta function, and R is recovered from $\operatorname{Im} R$ by

$$R(z) = \int \mathrm{d}\omega' \, \frac{\varrho_R(\omega')}{z - \omega'}, \qquad \varrho_R(\omega) = -\frac{1}{\pi} \operatorname{Im} R(\omega + \mathrm{i}0^+), \tag{C.34}$$

for a diagonal response, $O_a = O_b$, where R is real on the real axis apart from its delta functions. The spectral density ϱ_R is the density of transitions weighted by their squared matrix elements, positive for $\omega > 0$ and negative for $\omega < 0$ (Appendix A.7). At large $|z|$ the denominator can be expanded, $1/(z - \omega') = z^{-1} + \omega' z^{-2} + \ldots$, and

$$R(z) = \frac{1}{z} \int \mathrm{d}\omega' \, \varrho_R(\omega') + \frac{1}{z^2} \int \mathrm{d}\omega' \, \omega' \varrho_R(\omega') + O(z^{-3}). \tag{C.35}$$

For the density response the zeroth moment vanishes, since ϱ_R is odd in ω for $S(\boldsymbol{q}, \omega) = S(-\boldsymbol{q}, \omega)$ (Appendix A.4), and the first moment is fixed by the f-sum rule Eq. (A.38). With

$-2\,\mathrm{Im}\,R = (1 - \mathrm{e}^{-\beta\omega})S$ (Eq. (A.32)) and detailed balance, $\int \mathrm{d}\omega'\,\omega' \varrho_R = (2\pi)^{-1} \cdot 2 \int \mathrm{d}\omega\,\omega\,S = \boldsymbol{n}_e q^2/m_e$, so that $R(z) \simeq \boldsymbol{n}_e q^2/(m_e z^2)$: at high frequency every electron responds as a free particle. The derivations below assume only that the response function falls off at least as $1/|z|$ in the upper half plane.

C.5.3 Kramers–Kronig relations

Take z in the upper half plane and integrate $R(z')/(z' - z)$ over a closed contour made of the real axis and a large semicircle above it. The only singularity inside is the pole at $z' = z$, and the semicircle contributes nothing when R falls off as $1/|z'|$, since its length grows as $|z'|$ and the integrand falls off as $|z'|^{-2}$. Cauchy's formula gives $R(z) = (2\pi\mathrm{i})^{-1} \int \mathrm{d}\omega'\,R(\omega')/(\omega' - z)$, the Cauchy integral Eq. (C.27) of the boundary values. Let $z \to \omega + \mathrm{i}0^+$. By Eq. (C.26), $1/(\omega' - \omega - \mathrm{i}0^+) = \mathrm{PV}[1/(\omega' - \omega)] + \mathrm{i}\pi\delta(\omega' - \omega)$, so

$$R(\omega) = \frac{1}{2\pi\mathrm{i}}\,\mathrm{PV}\int \mathrm{d}\omega'\,\frac{R(\omega')}{\omega' - \omega} + \frac{1}{2}R(\omega),$$

where $R(\omega)$ means the boundary value from above. Moving the last term to the left and separating real and imaginary parts,

$$\mathrm{Re}\,R(\omega) = \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Im}\,R(\omega')}{\omega' - \omega}, \qquad \mathrm{Im}\,R(\omega) = -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Re}\,R(\omega')}{\omega' - \omega}. \tag{C.36}$$

These are the Kramers–Kronig relations [5, 6]. Toll showed that they are equivalent to causality, under mild conditions on the response [7].

Most responses have a definite symmetry under $\omega \to -\omega$. If the response in time is real, as for the response of a Hermitian observable to a real field, then $R(-\omega) = R(\omega)^*$ for real ω: the real part is even and the imaginary part odd. The diagonal density response of a target with $S(\boldsymbol{q}, \omega) = S(-\boldsymbol{q}, \omega)$ and the dielectric function are of this kind. Folding the negative frequencies onto the positive ones, as in Appendix C.4, gives

$$\mathrm{Re}\,R(\omega) = \frac{2}{\pi}\,\mathrm{PV}\int_0^{\infty} \mathrm{d}\omega'\,\frac{\omega'\,\mathrm{Im}\,R(\omega')}{\omega'^2 - \omega^2}, \qquad \mathrm{Im}\,R(\omega) = -\frac{2\omega}{\pi}\,\mathrm{PV}\int_0^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Re}\,R(\omega')}{\omega'^2 - \omega^2}. \tag{C.37}$$

For the first relation, the negative half axis contributes

$$\int_0^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Im}\,R(-\omega')}{-\omega' - \omega} = \int_0^{\infty} \mathrm{d}\omega'\,\frac{\mathrm{Im}\,R(\omega')}{\omega' + \omega},$$

and $1/(\omega' - \omega) + 1/(\omega' + \omega) = 2\omega'/(\omega'^2 - \omega^2)$; the second relation follows in the same way with the even real part.

C.5.4 Direct and inverse dielectric functions

The relations hold for any function that is analytic in the upper half plane and falls off at large $|z|$. Three functions of Sect. 4.4 are candidates, and the conditions differ.

The inverse dielectric function obeys the Kramers–Kronig relations in the form $\varepsilon^{-1} - 1$, because $\varepsilon^{-1} - 1 = v_C R_{ee}$ for the electron channel (Appendix A.7) is itself a retarded response: the induced potential caused by an external one. With $L = -\,\mathrm{Im}\,\varepsilon^{-1}$ and Eq. (C.37) at $\omega = 0$,

$$1 - \mathrm{Re}\,\varepsilon^{-1}(\boldsymbol{q}, 0) = \frac{2}{\pi}\int_0^{\infty} \mathrm{d}\omega'\,\frac{L(\boldsymbol{q}, \omega')}{\omega'}. \tag{C.38}$$

In a metal, static screening is perfect at long wavelength, $\varepsilon^{-1}(\boldsymbol{q} \to 0, 0) = 0$, and the loss function then obeys $\int_0^{\infty} \mathrm{d}\omega\,L/\omega = \pi/2$, the perfect-screening sum rule checked on the electron gas in Sect. 4.5.

The dielectric function itself obeys the relations in the form $\varepsilon - 1$ only if ε has no zeros in the upper half plane, since $1/\varepsilon$ must be analytic there for $\varepsilon - 1 = 1/\varepsilon^{-1} - 1$ to be analytic. Zeros of ε are self-sustained oscillations, and in the upper half plane they would grow in time; a stable medium has none. In a crystal with a gap ε is also finite at $\omega = 0$, and

$$\varepsilon_1(\omega) - 1 = \frac{2}{\pi}\,\mathrm{PV}\int_0^\infty \mathrm{d}\omega'\,\frac{\omega'\varepsilon_2(\omega')}{\omega'^2 - \omega^2}, \tag{C.39}$$

which at $\omega = 0$ gives the static electronic dielectric constant from the absorption spectrum, $\varepsilon_\infty - 1 = (2/\pi)\int_0^\infty \mathrm{d}\omega'\,\varepsilon_2(\omega')/\omega'$ at $q \to 0$ with the lattice clamped. In a metal ε has a pole at $\omega = 0$, from the conductivity, $\varepsilon \simeq \mathrm{i}\sigma_{\mathrm{cond}}(0)/\omega$ by Appendix A.7, and the pole must be subtracted before the relations are applied.

At large frequency every electron responds as if free, $\varepsilon(\omega) \to 1 - \omega_p^2/\omega^2$ by Eq. (C.35) and the f-sum rule. Comparing with the large-ω limit of Eq. (C.39), $\varepsilon_1 - 1 \to -(2/\pi\omega^2)\int_0^\infty \mathrm{d}\omega'\,\omega'\varepsilon_2(\omega')$, gives $\int_0^\infty \mathrm{d}\omega\,\omega\,\varepsilon_2 = (\pi/2)\,\omega_p^2$, the sum rule for ε_2 of Sect. 4.4.

C.5.5 Matsubara frequencies and analytic continuation

At finite temperature, many-body perturbation theory computes response and Green's functions on the imaginary axis, at the bosonic Matsubara frequencies $\mathrm{i}\omega_j^{\mathrm{B}} = 2\pi\mathrm{i}jT$ or the fermionic ones $\mathrm{i}\omega_j^{\mathrm{F}} = (2j+1)\pi\mathrm{i}T$, with integer j (Sect. 4.7). Frequency sums and analytic continuation connect the values at these frequencies with the real axis.

The Bose function $n_{\mathrm{B}}(z) = 1/(\mathrm{e}^{\beta z} - 1)$, continued to complex z, has poles at $z = \mathrm{i}\omega_j^{\mathrm{B}}$; near each, $\mathrm{e}^{\beta z} \simeq 1 + \beta(z - \mathrm{i}\omega_j^{\mathrm{B}})$, so the residue is $1/\beta = T$. For a function $g(z)$ that is analytic near the imaginary axis and falls off faster than $1/|z|$, the sum $T\sum_j g(\mathrm{i}\omega_j^{\mathrm{B}})$ is therefore the integral of $n_{\mathrm{B}}(z)g(z)/2\pi\mathrm{i}$ around small circles enclosing the poles of n_{B}. Merging the circles and opening the contour to infinity, where the integrand vanishes, leaves the poles z_{p} of g, traversed in the opposite sense:

$$T\sum_j g(\mathrm{i}\omega_j^{\mathrm{B}}) = -\sum_{z_{\mathrm{p}}} \mathrm{Res}_{z=z_{\mathrm{p}}}\big[n_{\mathrm{B}}(z)\,g(z)\big], \qquad T\sum_j g(\mathrm{i}\omega_j^{\mathrm{F}}) = \sum_{z_{\mathrm{p}}} \mathrm{Res}_{z=z_{\mathrm{p}}}\big[n_{\mathrm{F}}(z)\,g(z)\big]. \tag{C.40}$$

The fermionic form uses $n_{\mathrm{F}}(z) = 1/(\mathrm{e}^{\beta z} + 1)$, whose residues at $\mathrm{i}\omega_j^{\mathrm{F}}$ are $-T$. For an oscillator of energy ω_0, take $g(z) = 1/(\omega_0^2 - z^2)$, so that $g(\mathrm{i}\omega_j^{\mathrm{B}}) = 1/(\omega_0^2 + (\omega_j^{\mathrm{B}})^2) > 0$. The poles of g at $z = \pm\omega_0$ have residues $\mp 1/(2\omega_0)$, and with $n_{\mathrm{B}}(-\omega_0) = -1 - n_{\mathrm{B}}(\omega_0)$,

$$T\sum_j \frac{1}{\omega_0^2 + (\omega_j^{\mathrm{B}})^2} = \frac{n_{\mathrm{B}}(\omega_0)}{2\omega_0} + \frac{1 + n_{\mathrm{B}}(\omega_0)}{2\omega_0} = \frac{2n_{\mathrm{B}}(\omega_0) + 1}{2\omega_0}.$$

Up to the factor $1/M$, this sum is the thermal mean-square displacement of an oscillator of mass M. The growth of the displacement with temperature, by the factor $2n_{\mathrm{B}} + 1$, was found directly in Worked example 2.7. Appendix D.8 uses Eq. (C.40) for the sums that diagrams produce.

Suppose a function is computed at the Matsubara frequencies, $R(\mathrm{i}\omega_j^{\mathrm{B}})$. It is turned into the retarded function by finding the function $R(z)$ that is analytic off the real axis, agrees with the computed values at every $\mathrm{i}\omega_j^{\mathrm{B}}$ with $j > 0$, and falls off at infinity, and then setting $z = \omega + \mathrm{i}0^+$. For an expression in closed form the continuation is a substitution, $\mathrm{i}\omega_j^{\mathrm{B}} \to \omega + \mathrm{i}0^+$, made after every occupation factor has been simplified with $\mathrm{e}^{\mathrm{i}\beta\omega_j^{\mathrm{B}}} = 1$. The order of these steps matters. The function $(\mathrm{e}^{\beta z} - 1)\,h(z)$, for any analytic h, vanishes at every bosonic Matsubara frequency, so infinitely many analytic functions take the same values on the Matsubara points. The condition that $R(z)$ falls off at large $|z|$ removes all the functions $(\mathrm{e}^{\beta z} - 1)\,h(z)$, since

$e^{\beta z}$ grows along the real axis, and the continuation is unique [8]. The spectral representation Eq. (C.34) is the form in which uniqueness is manifest: it defines one function on the whole plane off the real axis, equal to the Matsubara function on the imaginary axis and to the retarded function just above the real axis. Numerical data on the imaginary axis are a harder case. Continuation then amplifies small errors exponentially, and a rational (Padé) fit to the Matsubara values, the usual tool, must be checked against sum rules such as Eq. (A.38).

C.6 Gaussian integrals and oscillator identities

When dark matter scatters off a nucleus in a crystal, the target operator is $e^{i\boldsymbol{q}\cdot\boldsymbol{u}}$, the exponential of a displacement that is a sum of harmonic-oscillator coordinates. Its thermal average gives the Debye–Waller factor, and its matrix elements give the probabilities of creating zero, one or many phonons; the average and the matrix elements reduce to Gaussian integrals or to their operator analogues. This section collects the identities that Sects. 5.3, 5.5 and 6.1 use: Gaussian integrals in one and many variables, the Baker–Campbell–Hausdorff formula in the form needed for linear operators, the displacement operator and coherent states, and the statement $\langle e^{\hat{A}}\rangle = e^{\langle\hat{A}^2\rangle/2}$ for operators $\hat{A}$ linear in the oscillator variables.

C.6.1 Gaussian integrals

For $\mathrm{Re}\,\mu > 0$ and any complex η,

$$\int_{-\infty}^{\infty} \mathrm{d}x\; e^{-\mu x^2+\eta x} = \sqrt{\frac{\pi}{\mu}}\, e^{\eta^2/4\mu}. \tag{C.41}$$

For $\eta = 0$ and real μ, squaring the integral and changing to polar coordinates in the plane gives $\int \mathrm{d}x\,\mathrm{d}y\, e^{-\mu(x^2+y^2)} = 2\pi\int_0^\infty \mathrm{d}r\, r\, e^{-\mu r^2} = \pi/\mu$. For $\eta \neq 0$, completing the square, $-\mu x^2 + \eta x = -\mu(x-\eta/2\mu)^2 + \eta^2/4\mu$, and shifting the variable by $\eta/2\mu$ gives the result. For complex η the shift moves the path of integration off the real axis; the integrand is analytic and decays at the ends of the strip between the real axis and the shifted path, so by Cauchy's theorem the path can be moved back. The same argument extends the result to complex μ with $\mathrm{Re}\,\mu > 0$, with the branch of $\sqrt{\mu}$ that is positive for real μ. The limit $\mathrm{Re}\,\mu \to 0^+$ gives the Fresnel integral, $\int \mathrm{d}x\, e^{i\mu x^2} = \sqrt{\pi/\mu}\, e^{i\pi/4}$ for real $\mu > 0$, which appears in the time integrals of a freely recoiling nucleus.

With $\eta = iq$ and $\mu = 1/(2s^2)$, the Fourier transform of a normalized Gaussian of width s is $e^{-q^2s^2/2}$, in any number of dimensions, as for the skin factor of the Helm form factor (Sect. 2.2). Differentiating Eq. (C.41) at $\eta = 0$ with respect to μ gives the moments of the normalized distribution $\sqrt{\mu/\pi}\, e^{-\mu x^2}$,

$$\langle x^{2n}\rangle = (2n-1)!!\,\langle x^2\rangle^n, \qquad \langle x^2\rangle = \frac{1}{2\mu}, \tag{C.42}$$

with odd moments zero. Reading Eq. (C.41) divided by $\sqrt{\pi/\mu}$ as an average gives the generating function

$$\langle e^{\eta x}\rangle = e^{\eta^2\langle x^2\rangle/2}, \tag{C.43}$$

which is the classical form of the operator identity of Eq. (C.51) below. Expanding both sides in powers of η reproduces Eq. (C.42).

Let $\mathbf{M}$ be a real symmetric positive-definite $n \times n$ matrix and $\boldsymbol{s}$ a vector. In n variables,

$$\int \mathrm{d}^n x\; e^{-\frac{1}{2}\boldsymbol{x}\cdot\mathbf{M}\cdot\boldsymbol{x}+\boldsymbol{s}\cdot\boldsymbol{x}} = \frac{(2\pi)^{n/2}}{(\det\mathbf{M})^{1/2}}\, e^{\frac{1}{2}\boldsymbol{s}\cdot\mathbf{M}^{-1}\cdot\boldsymbol{s}}. \tag{C.44}$$

The proof rotates to the eigenvectors of $\mathbf{M}$; the rotation leaves $\mathrm{d}^n x$ unchanged and turns the integral into a product of n copies of Eq. (C.41) with μ equal to half an eigenvalue. The product of the eigenvalues is $\det \mathbf{M}$, and the exponents combine into $\frac{1}{2}\boldsymbol{s}\cdot\mathbf{M}^{-1}\cdot\boldsymbol{s}$. For the normalized distribution, differentiating twice with respect to $\boldsymbol{s}$ at $\boldsymbol{s}=0$ gives the covariance $\langle x_i x_j\rangle = (\mathbf{M}^{-1})_{ij}$, and differentiating four times gives

$$\langle x_i x_j x_k x_l\rangle = \langle x_i x_j\rangle\langle x_k x_l\rangle + \langle x_i x_k\rangle\langle x_j x_l\rangle + \langle x_i x_l\rangle\langle x_j x_k\rangle, \tag{C.45}$$

the sum over the three ways of pairing the four variables [9]. Every higher even moment is the sum over all pairings in the same way, because the generating function is the exponential of a quadratic form in $\boldsymbol{s}$. In quantum many-body theory this sum over pairings is Wick's theorem (Sect. 4.8).

C.6.2 Operator identities

Products of exponentials of operators need care, because $\mathrm{e}^{\hat X}\mathrm{e}^{\hat Y} \neq \mathrm{e}^{\hat X+\hat Y}$ when $\hat X$ and $\hat Y$ do not commute. We first need the transformation of an operator $\hat Y$ by $\mathrm{e}^{\hat X}$. The function $f(\lambda) = \mathrm{e}^{\lambda\hat X}\hat Y\mathrm{e}^{-\lambda\hat X}$ obeys $f'(\lambda) = \mathrm{e}^{\lambda\hat X}[\hat X,\hat Y]\mathrm{e}^{-\lambda\hat X}$, and repeating the step gives the Taylor series

$$\mathrm{e}^{\hat X}\hat Y\mathrm{e}^{-\hat X} = \hat Y + [\hat X,\hat Y] + \frac{1}{2!}[\hat X,[\hat X,\hat Y]] + \frac{1}{3!}[\hat X,[\hat X,[\hat X,\hat Y]]] + \cdots . \tag{C.46}$$

We also need the Baker–Campbell–Hausdorff formula for the case in which the commutator $[\hat X,\hat Y]$ commutes with both $\hat X$ and $\hat Y$, as it does when $\hat X$ and $\hat Y$ are linear in boson operators and $[\hat X,\hat Y]$ is a number. Let $g(\lambda) = \mathrm{e}^{\lambda\hat X}\mathrm{e}^{\lambda\hat Y}$. Then $g'(\lambda) = (\hat X + \mathrm{e}^{\lambda\hat X}\hat Y\mathrm{e}^{-\lambda\hat X})\,g(\lambda) = (\hat X+\hat Y+\lambda[\hat X,\hat Y])\,g(\lambda)$, where the series Eq. (C.46) stops after the commutator. The operators $\hat X+\hat Y$ and $[\hat X,\hat Y]$ commute, so the solution with $g(0)=1$ is $g(\lambda)=\exp[\lambda(\hat X+\hat Y)+\frac{1}{2}\lambda^2[\hat X,\hat Y]]$, and at $\lambda=1$

$$\mathrm{e}^{\hat X}\mathrm{e}^{\hat Y} = \mathrm{e}^{\hat X+\hat Y}\,\mathrm{e}^{[\hat X,\hat Y]/2}, \qquad \mathrm{e}^{\hat X}\mathrm{e}^{\hat Y} = \mathrm{e}^{\hat Y}\mathrm{e}^{\hat X}\,\mathrm{e}^{[\hat X,\hat Y]}. \tag{C.47}$$

The second form follows from the first by exchanging $\hat X$ and $\hat Y$ and dividing. For general $\hat X$ and $\hat Y$ the exponent contains nested commutators to all orders [10]; they are not needed here.

C.6.3 Displacement operators and coherent states

We write the oscillator as in Worked example 2.7, $H=\omega_0(d^\dagger d+\frac{1}{2})$ with $[d,d^\dagger]=1$, and its coordinate as $x=\ell_0(d+d^\dagger)$ with the zero-point length $\ell_0=(2M\omega_0)^{-1/2}$ for a mass M; for a phonon mode the zero-point length is $\ell_{\boldsymbol{Q}\nu}$ of Eq. (A.23). The displacement operator with a complex amplitude α_d is

$$\hat D(\alpha_\mathrm{d}) = \exp(\alpha_\mathrm{d} d^\dagger - \alpha_\mathrm{d}^* d) = \mathrm{e}^{-|\alpha_\mathrm{d}|^2/2}\,\mathrm{e}^{\alpha_\mathrm{d} d^\dagger}\mathrm{e}^{-\alpha_\mathrm{d}^* d}, \tag{C.48}$$

where the second form is Eq. (C.47) with $\hat X=\alpha_\mathrm{d}d^\dagger$, $\hat Y=-\alpha_\mathrm{d}^* d$ and $[\hat X,\hat Y]=|\alpha_\mathrm{d}|^2$. The operator $\hat D$ is unitary, and by Eq. (C.46) it shifts the ladder operators, $\hat D^\dagger d\,\hat D = d+\alpha_\mathrm{d}$, so it shifts the coordinate by $2\ell_0\,\mathrm{Re}\,\alpha_\mathrm{d}$ and the momentum in proportion to $\mathrm{Im}\,\alpha_\mathrm{d}$. A momentum kick is the special case of a real coordinate: $\mathrm{e}^{\mathrm{i}qx} = \mathrm{e}^{\mathrm{i}q\ell_0(d+d^\dagger)} = \hat D(\mathrm{i}q\ell_0)$, which displaces the momentum by q. Dark matter acts on a bound nucleus through $\mathrm{e}^{\mathrm{i}qx}$.

On the ground state, $\mathrm{e}^{-\alpha_\mathrm{d}^* d}|0\rangle = |0\rangle$, and $\mathrm{e}^{\alpha_\mathrm{d} d^\dagger}|0\rangle = \sum_n \alpha_\mathrm{d}^n (d^\dagger)^n|0\rangle/n!$, which is $\sum_n \alpha_\mathrm{d}^n|n\rangle/\sqrt{n!}$ by $(d^\dagger)^n|0\rangle=\sqrt{n!}\,|n\rangle$ (Sect. 2.3). The result is a coherent state,

$$\hat D(\alpha_\mathrm{d})|0\rangle = \mathrm{e}^{-|\alpha_\mathrm{d}|^2/2}\sum_{n=0}^{\infty}\frac{\alpha_\mathrm{d}^n}{\sqrt{n!}}\,|n\rangle, \qquad |\langle n|\hat D(\alpha_\mathrm{d})|0\rangle|^2 = \mathrm{e}^{-|\alpha_\mathrm{d}|^2}\frac{|\alpha_\mathrm{d}|^{2n}}{n!}. \tag{C.49}$$

The number of quanta in a displaced ground state is Poisson distributed with mean $|\alpha_\mathrm{d}|^2$. For a kick, $|\alpha_\mathrm{d}|^2 = q^2\ell_0^2 = q^2/(2M\omega_0)$, the recoil energy $q^2/2M$ in units of the quantum ω_0. The probability of leaving the oscillator in its ground state is $\mathrm{e}^{-q^2\ell_0^2}$, and the mean energy deposited is $\omega_0|\alpha_\mathrm{d}|^2 = q^2/2M$, the free recoil energy, at every q. For a 60 meV mode of a silicon nucleus, $M = 26.16\,\mathrm{GeV}$ (the mean atomic mass of natural silicon, as in Sect. 5.3), $q^2\ell_0^2 = 1$ at $q = (2M\omega_0)^{1/2} = 56\,\mathrm{keV}$. Below 56 keV a kick mostly creates no quantum or one, and far above it the Poisson distribution is broad and the nucleus recoils almost as if free. The same distribution, with $|\alpha_\mathrm{d}|^2$ the Huang–Rhys factor S_HR, describes an electronic transition that displaces the equilibrium position of an oscillator (Sect. 5.5).

Ordering an exponential so that all annihilation operators stand to the right makes thermal and ground-state averages easy. For a real or complex η, Eq. (C.47) with $\hat{X} = \eta d^\dagger$, $\hat{Y} = \eta d$ and $[\hat{X}, \hat{Y}] = -\eta^2$ gives

$$\mathrm{e}^{\eta(d+d^\dagger)} = \mathrm{e}^{\eta d^\dagger}\mathrm{e}^{\eta d}\,\mathrm{e}^{\eta^2/2}. \tag{C.50}$$

In the ground state, $\langle 0|\mathrm{e}^{\eta d^\dagger}\mathrm{e}^{\eta d}|0\rangle = 1$, so $\langle 0|\mathrm{e}^{\eta(d+d^\dagger)}|0\rangle = \mathrm{e}^{\eta^2/2}$. For a kick, $\eta = \mathrm{i}q\ell_0$ and the ground-state average is $\mathrm{e}^{-q^2\ell_0^2/2}$, the square root of the ground-state survival probability. Since $\langle 0|(d+d^\dagger)^2|0\rangle = 1$, the exponent is $\frac{1}{2}\langle 0|(\eta(d+d^\dagger))^2|0\rangle$: the ground-state average of the exponential is the exponential of half the average of the square, as for the classical Gaussian of Eq. (C.43).

C.6.4 Thermal averages of exponentials

The ground state of an oscillator is a Gaussian in x, and so is its thermal state: the density matrix $\mathrm{e}^{-\beta H}/\mathcal{Z}$ is Gaussian in the coordinate and the momentum. The identity of the last paragraph therefore extends to any temperature and to any number of independent oscillators. For an operator $\hat{A}$ that is a linear combination of the d_γ and $d_\gamma^\dagger$ of a set of harmonic modes in thermal equilibrium,

$$\langle \mathrm{e}^{\hat{A}}\rangle = \mathrm{e}^{\langle \hat{A}^2\rangle/2}, \tag{C.51}$$

which is Bloch's identity, proved in Sect. 5.5; Mermin gave a short alternative derivation [11]. In the classical limit $T \gg \omega_0$ it is Eq. (C.43), since the coordinates are then classical Gaussian variables. For one mode and $\hat{A} = \mathrm{i}qx$, it gives $\langle \mathrm{e}^{\mathrm{i}qx}\rangle = \mathrm{e}^{-q^2\langle x^2\rangle/2}$ with $\langle x^2\rangle = \ell_0^2(2n_\mathrm{B}(\omega_0)+1)$, the zero-point value multiplied by the factor found in Worked example 2.7 and in the Matsubara sum of Appendix C.5. For a crystal, $\hat{A} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_\kappa$ and the result is $\mathrm{e}^{-W_\kappa(\boldsymbol{q})}$ with $2W_\kappa(\boldsymbol{q}) = \boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}$, the Debye–Waller factor of Eq. (A.28) at the level of the amplitude.

Combined with Eq. (C.47), Bloch's identity gives the average of a product. For $\hat{A}$ and $\hat{B}$ both linear, $[\hat{A},\hat{B}]$ is a number, $\mathrm{e}^{\hat{A}}\mathrm{e}^{\hat{B}} = \mathrm{e}^{\hat{A}+\hat{B}}\mathrm{e}^{[\hat{A},\hat{B}]/2}$, and $\langle \mathrm{e}^{\hat{A}+\hat{B}}\rangle = \mathrm{e}^{\langle(\hat{A}+\hat{B})^2\rangle/2}$. Expanding $(\hat{A}+\hat{B})^2 = \hat{A}^2+\hat{B}^2+\hat{A}\hat{B}+\hat{B}\hat{A}$ and adding $\frac{1}{2}[\hat{A},\hat{B}] = \frac{1}{2}\langle\hat{A}\hat{B}-\hat{B}\hat{A}\rangle$ to the exponent,

$$\langle \mathrm{e}^{\hat{A}}\mathrm{e}^{\hat{B}}\rangle = \exp\Big[\tfrac{1}{2}\langle\hat{A}^2\rangle + \tfrac{1}{2}\langle\hat{B}^2\rangle + \langle\hat{A}\hat{B}\rangle\Big]. \tag{C.52}$$

Sect. 5.5 builds the multiphonon response from Eq. (C.52).

C.7 Matrix tools

The response of a crystal is a matrix: in reciprocal-lattice vectors for the dielectric function, in operators for the response matrix S_{ab}, in bands and crystal momenta for excitons. In Chaps. 4 and 7, positive semidefiniteness bounds the interference between two couplings, block inversion turns the inverse dielectric matrix into a macroscopic dielectric function, Dyson-type equations sum geometric series of interactions, and Bogoliubov transformations diagonalize Hamiltonians that create excitations in pairs. We use bold upright capitals, **M**, **A**, **B**, for

generic matrices, $\mathbf{1}$ for the unit matrix and $\boldsymbol{x}$ for a column vector, and we write $\boldsymbol{x}^\dagger$ for its conjugate transpose.

C.7.1 Positive semidefinite matrices

A Hermitian matrix $\mathbf{M} = \mathbf{M}^\dagger$ is positive semidefinite if $\boldsymbol{x}^\dagger \mathbf{M} \boldsymbol{x} \geq 0$ for every vector $\boldsymbol{x}$. Positive semidefiniteness is equivalent to each of the following conditions. All eigenvalues are nonnegative: take $\boldsymbol{x}$ to be an eigenvector, and conversely expand any $\boldsymbol{x}$ in the orthonormal eigenvectors. $\mathbf{M}$ is a sum of outer products with nonnegative weights, $\mathbf{M} = \sum_f w_f \, \boldsymbol{c}_f \boldsymbol{c}_f^\dagger$ with $w_f \geq 0$: the eigen-decomposition is such a sum, and conversely $\boldsymbol{x}^\dagger \mathbf{M} \boldsymbol{x} = \sum_f w_f |\boldsymbol{c}_f^\dagger \boldsymbol{x}|^2 \geq 0$. Every principal submatrix has nonnegative determinant, where a principal submatrix keeps a subset of the rows and the same subset of the columns. The last condition must include all principal submatrices, not only the leading ones: $\mathrm{diag}(0, -1)$ has leading minors 0 and 0 and is not positive semidefinite.

The diagonal elements are then nonnegative, and the 2×2 principal minors give

$$|M_{ab}|^2 \leq M_{aa} M_{bb}, \tag{C.53}$$

the Cauchy–Schwarz inequality for the matrix. The response matrix of Eq. (A.30) has the outer-product form, with one term for each pair of states (i, f), weight $2\pi p_i \delta(\omega - E_f + E_i)/\mathcal{V} \geq 0$ and vector components $c_b = \langle f | O_b | i \rangle$, since $\langle i | O_a^\dagger | f \rangle = c_a^*$. Hence $S_{ab}(\boldsymbol{q}, \omega)$ is positive semidefinite at every $\boldsymbol{q}$ and ω (Sect. 4.2). For a probe that couples to the operators O_a with amplitudes x_a, the rate is proportional to $\sum_{ab} x_a^* S_{ab} x_b \geq 0$, and Eq. (C.53) bounds the interference term between two channels by the geometric mean of their separate rates. A computed S_{ab} with a negative eigenvalue signals an error.

C.7.2 Block inversion

Partition a square matrix into blocks, with $\mathbf{A}$ and $\mathbf{D}$ square and $\mathbf{D}$ invertible:

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{1} & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{0} & \mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{1} \end{pmatrix}.$$

The middle factor contains the Schur complement of $\mathbf{D}$, $\mathbf{M}/\mathbf{D} \equiv \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$. Multiplying the first two factors gives $\left(\begin{smallmatrix} \mathbf{M}/\mathbf{D} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{smallmatrix}\right)$ with $\mathbf{M}/\mathbf{D}$ the middle upper block, and multiplying by the third adds $\mathbf{B}\mathbf{D}^{-1}\mathbf{C}$ to the upper-left block and puts $\mathbf{C}$ in the lower-left one, so that the product is $\mathbf{M}$ again. The outer factors have unit determinant and inverses obtained by changing the sign of their off-diagonal block. Inverting the three factors in reverse order, and writing $\mathbf{N} \equiv (\mathbf{M}/\mathbf{D})^{-1}$,

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{N} & -\mathbf{N}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{N} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{N}\mathbf{B}\mathbf{D}^{-1} \end{pmatrix}, \tag{C.54}$$

and the determinants of the three factors give $\det \mathbf{M} = \det \mathbf{D} \, \det(\mathbf{M}/\mathbf{D})$. The upper-left block of the inverse is the inverse of the Schur complement. When $\mathbf{A}$ is a single element, the head of a matrix in reciprocal-lattice vectors with $\boldsymbol{G} = \boldsymbol{G}' = 0$, the relation between the inverse and the Schur complement reads

$$\frac{1}{(\mathbf{M}^{-1})_{00}} = M_{00} - \sum_{\boldsymbol{G}, \boldsymbol{G}' \neq 0} M_{0\boldsymbol{G}} \left[(\mathbf{M}_{\neq 0})^{-1} \right]_{\boldsymbol{G}\boldsymbol{G}'} M_{\boldsymbol{G}'0}, \tag{C.55}$$

where $\mathbf{M}_{\neq 0}$ is the block with $\boldsymbol{G}, \boldsymbol{G}' \neq 0$. Applied to the dielectric matrix, Eq. (C.55) gives the macroscopic dielectric function $\varepsilon_{\mathrm{mac}} = 1/(\varepsilon^{-1})_{00}$ as the head ε_{00} corrected by the local fields that the off-diagonal elements carry (Sect. 4.4 and Appendix D.5).

The same algebra solves eigenvalue problems in a subspace. Split a Hilbert space into a part P of interest and the rest Q, and write the equation $(E-H)\psi = 0$ in blocks. Eliminating $\psi_Q = (E - H_{QQ})^{-1} H_{QP} \psi_P$ leaves

$$[H_{PP} + H_{PQ}(E - H_{QQ})^{-1} H_{QP}]\psi_P = E\,\psi_P, \tag{C.56}$$

an effective Hamiltonian in P alone, whose energy dependence carries the coupling to the eliminated states. The bracket is E minus the Schur complement of $E - H_{QQ}$ in $E - H$, and the equation is exact. To second order in the coupling, with H_{QQ} replaced by its diagonal, the term $H_{PQ}(E - H_{QQ})^{-1} H_{QP}$ is the level shift $\sum_f |H'_{fi}|^2/(E_i - E_f)$ of Sect. 2.4; with $E \to E + \mathrm{i}0^+$ and a continuum in Q it acquires the width. Sect. 8.6 uses Eq. (C.56) to separate the electronic and the phonon-dressed sectors of a target.

C.7.3 Dyson-type equations

Many response objects obey an equation of the form

$$\mathbf{X} = \mathbf{X}_0 + \mathbf{X}_0 \mathbf{K} \mathbf{X}, \tag{C.57}$$

with a known bare object $\mathbf{X}_0$ and a kernel $\mathbf{K}$. Its solution is $\mathbf{X} = (\mathbf{1} - \mathbf{X}_0\mathbf{K})^{-1}\mathbf{X}_0$, as follows by moving $\mathbf{X}_0\mathbf{K}\mathbf{X}$ to the left and inverting. When $\mathbf{X}_0$ and $\mathbf{K}$ are invertible this is $\mathbf{X}^{-1} = \mathbf{X}_0^{-1} - \mathbf{K}$, and multiplying out shows that $\mathbf{X} = \mathbf{X}_0 + \mathbf{X}\mathbf{K}\mathbf{X}_0$ holds too. Iterating Eq. (C.57) gives the geometric series $\mathbf{X} = \mathbf{X}_0 + \mathbf{X}_0\mathbf{K}\mathbf{X}_0 + \mathbf{X}_0\mathbf{K}\mathbf{X}_0\mathbf{K}\mathbf{X}_0 + \cdots$, which converges when every eigenvalue of $\mathbf{X}_0\mathbf{K}$ has modulus below one. The closed form remains valid where the series diverges, as the analytic continuation of the series, and the poles of the closed form, where $\det(\mathbf{1} - \mathbf{X}_0\mathbf{K}) = 0$, are the collective modes that no finite order of the series contains.

In Chap. 4 this form appears for the density response, for the time-dependent density-functional response and for the single-particle Green's function. The density response, written through its Coulomb-proper part, $R = \tilde{R} + \tilde{R} v_{\mathrm{C}} R$, has $\mathbf{X}_0 = \tilde{R}$ and $\mathbf{K} = v_{\mathrm{C}}$; for a scalar it gives $R = \tilde{R}/(1 - v_{\mathrm{C}}\tilde{R}) = \tilde{R}/\varepsilon$, whose poles at $\varepsilon = 0$ are the plasmons (Sects. 4.4 and 4.5). The time-dependent density-functional response $R = R^0 + R^0 f_{\mathrm{Hxc}} R$ has the kernel f_{Hxc} (Sect. 4.6). The single-particle Green's function obeys $\mathsf{G} = \mathsf{G}_0 + \mathsf{G}_0 \Sigma \mathsf{G}$, so that $\mathsf{G}^{-1} = \mathsf{G}_0^{-1} - \Sigma$ and the self-energy shifts and broadens the poles (Sect. 4.7). In a crystal each is a matrix equation in reciprocal-lattice vectors or bands, and the order of the factors matters.

When the kernel has low rank, a smaller inversion suffices. For $\mathbf{K} = \mathbf{U}\mathbf{W}\mathbf{U}^\dagger$ with $\mathbf{U}$ of size $n \times r$ and $r \ll n$, the Woodbury identity

$$(\mathbf{A} + \mathbf{U}\mathbf{W}\mathbf{U}^\dagger)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\big(\mathbf{W}^{-1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U}\big)^{-1}\mathbf{U}^\dagger\mathbf{A}^{-1}, \tag{C.58}$$

reduces the inversion to an inversion of size r. When the right side of Eq. (C.58) is multiplied by $\mathbf{A} + \mathbf{U}\mathbf{W}\mathbf{U}^\dagger$, the terms combine into $\mathbf{1} + \mathbf{A}^{-1}\mathbf{U}[\mathbf{W} - (\mathbf{W}^{-1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U})^{-1}(\mathbf{1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U}\mathbf{W})]\mathbf{U}^\dagger$, and the bracket vanishes because $\mathbf{1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U}\mathbf{W} = (\mathbf{W}^{-1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U})\mathbf{W}$. The Woodbury identity is Eq. (C.54) in another guise, and it is the efficient route when one element, such as the long-range Coulomb head, is treated apart from the other elements.

C.7.4 Bogoliubov transformations

A quadratic Hamiltonian that creates excitations in pairs is diagonalized by mixing creation and annihilation operators. Take two boson modes, d_1 and d_2, as for a magnon at $\boldsymbol{k}$ and at

$-\boldsymbol{k}$ (Sect. 7.1), with

$$H = \omega_0(d_1^\dagger d_1 + d_2^\dagger d_2) + g_0(d_1 d_2 + d_2^\dagger d_1^\dagger), \qquad \omega_0 > |g_0|. \tag{C.59}$$

New operators $\tilde{d}_1 = u^{\rm b} d_1 + v^{\rm b} d_2^\dagger$ and $\tilde{d}_2 = u^{\rm b} d_2 + v^{\rm b} d_1^\dagger$, with real coefficients, are bosons if $[\tilde{d}_1, \tilde{d}_1^\dagger] = (u^{\rm b})^2 - (v^{\rm b})^2 = 1$; the other commutators vanish identically. We set $u^{\rm b} = \cosh\theta_{\rm b}$ and $v^{\rm b} = \sinh\theta_{\rm b}$. The inverse is $d_1 = u^{\rm b}\tilde{d}_1 - v^{\rm b}\tilde{d}_2^\dagger$ and the same with $1 \leftrightarrow 2$. Substituting and normal ordering with $[\tilde{d}_j, \tilde{d}_j^\dagger] = 1$, with $\tilde{N} = \tilde{d}_1^\dagger\tilde{d}_1 + \tilde{d}_2^\dagger\tilde{d}_2$ and $\tilde{P} = \tilde{d}_1\tilde{d}_2 + \tilde{d}_2^\dagger\tilde{d}_1^\dagger$,

$$\begin{aligned} d_1^\dagger d_1 + d_2^\dagger d_2 &= \cosh 2\theta_{\rm b}\,\tilde{N} + 2\sinh^2\theta_{\rm b} - \sinh 2\theta_{\rm b}\,\tilde{P}, \\ d_1 d_2 + d_2^\dagger d_1^\dagger &= \cosh 2\theta_{\rm b}\,\tilde{P} - \sinh 2\theta_{\rm b}\,(\tilde{N}+1). \end{aligned}$$

The pair term $\tilde{P}$ drops out of H if $\omega_0 \sinh 2\theta_{\rm b} = g_0 \cosh 2\theta_{\rm b}$, that is $\tanh 2\theta_{\rm b} = g_0/\omega_0$. Then $\cosh 2\theta_{\rm b} = \omega_0/E_{\rm b}$ and $\sinh 2\theta_{\rm b} = g_0/E_{\rm b}$ with

$$H = E_{\rm b}(\tilde{d}_1^\dagger\tilde{d}_1 + \tilde{d}_2^\dagger\tilde{d}_2 + 1) - \omega_0, \qquad E_{\rm b} = (\omega_0^2 - g_0^2)^{1/2}. \tag{C.60}$$

The pairing lowers the mode energy from ω_0 to $E_{\rm b}$ and the ground-state energy by $\omega_0 - E_{\rm b}$. The new ground state, annihilated by $\tilde{d}_1$ and $\tilde{d}_2$, is not empty of the old quanta: $\langle d_1^\dagger d_1\rangle = (v^{\rm b})^2 = \sinh^2\theta_{\rm b}$. In an antiferromagnet this occupation is the zero-point reduction of the ordered moment. For $|g_0| \geq \omega_0$ no real $\theta_{\rm b}$ exists, $E_{\rm b}$ is imaginary and the system is unstable. For $\omega_0 = 1$ and $g_0 = 0.6$, Eq. (C.60) gives $E_{\rm b} = 0.8$, a ground state at -0.2 and $\sinh^2\theta_{\rm b} = 0.125$.

The matrix form generalizes to many modes. With $\Psi = (d_1, d_2^\dagger)^{\rm T}$, Eq. (C.59) is $\Psi^\dagger \mathbf{h} \Psi$ minus a constant, with $\mathbf{h}$ the 2×2 matrix with ω_0 on the diagonal and g_0 off it. A linear transformation $\Psi = \mathbf{T}\tilde{\Psi}$ keeps the commutation relations only if $\mathbf{T}\boldsymbol{\eta}\mathbf{T}^\dagger = \boldsymbol{\eta}$ with $\boldsymbol{\eta} = {\rm diag}(1, -1)$, because the second component of Ψ is a creation operator; such a $\mathbf{T}$ is called para-unitary. The excitation energies are the eigenvalues of $\boldsymbol{\eta}\mathbf{h}$, here $\pm(\omega_0^2 - g_0^2)^{1/2}$, which come in pairs of opposite sign. For any number of modes, a para-unitary diagonalization with positive excitation energies exists if $\mathbf{h}$ is positive definite [12], the matrix condition for stability.

Fermions differ in one sign. For $c_1 = c_{\boldsymbol{k}\uparrow}$ and $c_2 = c_{-\boldsymbol{k}\downarrow}$, as in a superconductor (Sect. 7.2), take

$$H = \xi_0(c_1^\dagger c_1 + c_2^\dagger c_2) - g_0(c_1^\dagger c_2^\dagger + c_2 c_1). \tag{C.61}$$

The operators $\tilde{c}_1 = u^{\rm f} c_1 - v^{\rm f} c_2^\dagger$ and $\tilde{c}_2 = u^{\rm f} c_2 + v^{\rm f} c_1^\dagger$ are fermions if $\{\tilde{c}_1, \tilde{c}_1^\dagger\} = (u^{\rm f})^2 + (v^{\rm f})^2 = 1$, and we set $u^{\rm f} = \cos\theta_{\rm f}$, $v^{\rm f} = \sin\theta_{\rm f}$. The anticommutator adds the two squares where the boson commutator subtracted them, the transformation is unitary instead of para-unitary, and the hyperbolic functions become trigonometric. The inverse is $c_1 = u^{\rm f}\tilde{c}_1 + v^{\rm f}\tilde{c}_2^\dagger$, $c_2 = u^{\rm f}\tilde{c}_2 - v^{\rm f}\tilde{c}_1^\dagger$. Substituting and ordering with $\tilde{c}_j\tilde{c}_j^\dagger = 1 - \tilde{c}_j^\dagger\tilde{c}_j$ and $\tilde{c}_2\tilde{c}_1 = -\tilde{c}_1\tilde{c}_2$, with $\tilde{N} = \tilde{c}_1^\dagger\tilde{c}_1 + \tilde{c}_2^\dagger\tilde{c}_2$ and $\tilde{P} = \tilde{c}_1^\dagger\tilde{c}_2^\dagger + \tilde{c}_2\tilde{c}_1$,

$$\begin{aligned} c_1^\dagger c_1 + c_2^\dagger c_2 &= \cos 2\theta_{\rm f}\,\tilde{N} + 2\sin^2\theta_{\rm f} + \sin 2\theta_{\rm f}\,\tilde{P}, \\ c_1^\dagger c_2^\dagger + c_2 c_1 &= \cos 2\theta_{\rm f}\,\tilde{P} - \sin 2\theta_{\rm f}\,(\tilde{N} - 1). \end{aligned}$$

For the first line, $c_1^\dagger c_1 = (u^{\rm f})^2\tilde{c}_1^\dagger\tilde{c}_1 + u^{\rm f}v^{\rm f}\tilde{P} + (v^{\rm f})^2(1 - \tilde{c}_2^\dagger\tilde{c}_2)$, and $c_2^\dagger c_2 = (u^{\rm f})^2\tilde{c}_2^\dagger\tilde{c}_2 + u^{\rm f}v^{\rm f}\tilde{P} + (v^{\rm f})^2(1 - \tilde{c}_1^\dagger\tilde{c}_1)$, where the minus sign in the transformation of c_2 cancels against the sign of an anticommuting reordering. For the second line, $c_1^\dagger c_2^\dagger = (u^{\rm f})^2\tilde{c}_1^\dagger\tilde{c}_2^\dagger - (v^{\rm f})^2\tilde{c}_2\tilde{c}_1 + u^{\rm f}v^{\rm f}(1 - \tilde{N})$ and $c_2 c_1 = (u^{\rm f})^2\tilde{c}_2\tilde{c}_1 - (v^{\rm f})^2\tilde{c}_1^\dagger\tilde{c}_2^\dagger + u^{\rm f}v^{\rm f}(1 - \tilde{N})$. The pair term drops out of Eq. (C.61) if $\xi_0 \sin 2\theta_{\rm f} = g_0 \cos 2\theta_{\rm f}$, that is $\tan 2\theta_{\rm f} = g_0/\xi_0$. Then $\cos 2\theta_{\rm f} = \xi_0/E_{\rm f}$, $\sin 2\theta_{\rm f} = g_0/E_{\rm f}$, the

coefficient of $\tilde{N}$ is $\xi_0 \cos 2\theta_\mathrm{f} + g_0 \sin 2\theta_\mathrm{f} = E_\mathrm{f}$, the constant is $\xi_0(1-\cos 2\theta_\mathrm{f}) - g_0 \sin 2\theta_\mathrm{f} = \xi_0 - E_\mathrm{f}$, and

$$H = E_\mathrm{f}(\tilde{c}_1^\dagger \tilde{c}_1 + \tilde{c}_2^\dagger \tilde{c}_2) + \xi_0 - E_\mathrm{f}, \qquad E_\mathrm{f} = \left(\xi_0^2 + g_0^2\right)^{1/2}. \tag{C.62}$$

The pairing now raises the excitation energy, from $|\xi_0|$ to $E_\mathrm{f} \geq |g_0|$: a gap opens even where $\xi_0 = 0$. With $\xi_0 \to \xi_{\boldsymbol{k}}$ and $g_0 \to \Delta_\mathrm{BCS}$, E_f is the Bogoliubov energy of Sect. 7.2, and $u^\mathrm{f}, v^\mathrm{f}$ become the coherence factors $u^\mathrm{B}_{\boldsymbol{k}}, v^\mathrm{B}_{\boldsymbol{k}}$; in Sect. 7.1 the boson coefficients become $u^\mathrm{M}_{\boldsymbol{k}}, v^\mathrm{M}_{\boldsymbol{k}}$.

C.8 Special functions

The standard halo model cuts a Gaussian off at the escape speed, and every one of its speed integrals is a Gaussian over a finite interval: an error function. The normalization N_esc of Sect. 1.2 and the three branches of $\eta(v_\mathrm{min})$ in Sect. 1.3 and Appendix D.1 are differences of error functions.

C.8.1 The error function

The error function and its complement are [1, Eqs. (7.2.1) and (7.2.2)]

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \mathrm{d}y\, \mathrm{e}^{-y^2}, \qquad \mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty \mathrm{d}y\, \mathrm{e}^{-y^2}. \tag{C.63}$$

The error function is odd, $\mathrm{erf}(-x) = -\mathrm{erf}(x)$, it tends to one as $x \to \infty$ by the Gaussian integral Eq. (C.41), and its derivative is $(2/\sqrt{\pi})\mathrm{e}^{-x^2}$. Rescaling the variable gives the Gaussian over a finite interval,

$$\int_{y_1}^{y_2} \mathrm{d}y\, \mathrm{e}^{-y^2/v_0^2} = \frac{\sqrt{\pi}\, v_0}{2} \Big[\mathrm{erf}\Big(\frac{y_2}{v_0}\Big) - \mathrm{erf}\Big(\frac{y_1}{v_0}\Big)\Big], \tag{C.64}$$

which is Eq. (D.1). Integrals with higher even powers of y reduce to Eq. (C.64) by parts. Since $y\,\mathrm{e}^{-y^2} = -\frac{1}{2}\mathrm{d}\mathrm{e}^{-y^2}/\mathrm{d}y$,

$$\int_0^x \mathrm{d}y\, y^2 \mathrm{e}^{-y^2} = -\frac{x}{2}\,\mathrm{e}^{-x^2} + \frac{1}{2}\int_0^x \mathrm{d}y\, \mathrm{e}^{-y^2} = \frac{\sqrt{\pi}}{4}\,\mathrm{erf}(x) - \frac{x}{2}\,\mathrm{e}^{-x^2}, \tag{C.65}$$

which, with $x = v_\mathrm{esc}/v_0$ and a factor $4\pi v_0^3$, is the normalization Eq. (1.4). Odd powers need no error function: $\int \mathrm{d}y\, y\, \mathrm{e}^{-y^2} = -\frac{1}{2}\mathrm{e}^{-y^2}$.

A Taylor series and an asymptotic series give numbers without tables. For small and moderate x the Taylor series [1, Eq. (7.6.1)]

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!\,(2n+1)} = \frac{2}{\sqrt{\pi}}\Big(x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots\Big), \tag{C.66}$$

converges for every x, after the terms have grown for $n \lesssim x^2$. At $x = v_\mathrm{E}^\mathrm{ref}/v_0 = 1.0525$, the value of Sect. 1.2, the partial sums oscillate about the limit and settle to five digits after nine terms, $\mathrm{erf}(1.0525) = 0.86338$. For large x the complement has the asymptotic expansion [1, Eq. (7.12.1)]

$$\mathrm{erfc}(x) \sim \frac{\mathrm{e}^{-x^2}}{x\sqrt{\pi}}\Big(1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \frac{15}{8x^6} + \cdots\Big), \tag{C.67}$$

obtained by integrating $\mathrm{e}^{-y^2} = -(2y)^{-1}\mathrm{d}\mathrm{e}^{-y^2}/\mathrm{d}y$ by parts repeatedly. The series diverges for every x, since the nth coefficient grows as $(2n-1)!!/2^n$, but its terms first decrease, and truncated at the smallest term it is accurate to about the size of that term. At $x = v_\mathrm{esc}/v_0 =$

$544/238 = 2.2857$ the leading term is $\mathrm{e}^{-5.224}/(2.286\sqrt{\pi}) = 1.329 \times 10^{-3}$, the next two terms bring the sum to 1.238×10^{-3}, and the exact value is 1.227×10^{-3}, so $\mathrm{erf}(2.286) = 0.99877$. Since $\mathrm{erfc}(x) \simeq \mathrm{e}^{-x^2}/(x\sqrt{\pi})$ up to a relative correction $1/(2x^2)$, the fraction of a Gaussian halo beyond the escape speed is small for $v_{\mathrm{esc}} \gg v_0$. The second term of Eq. (C.65), the Gaussian evaluated at the cut, sets the size of the correction that the cut makes to the normalization. The second term is 1.4 % of the normalization, and together with the erfc it gives the 1.5 % of Sect. 1.2.

C.8.2 Complete elliptic integrals

Averaged over an angle ϕ, the square root of a linear function of $\cos\phi$ is not elementary. Let s be the modulus, $0 \le s < 1$, and $s' = (1 - s^2)^{1/2}$. The complete elliptic integrals of the first and second kind are [1, Eq. (19.2.8)]

$$\mathrm{K}(s) = \int_0^{\pi/2} \frac{\mathrm{d}\theta}{(1 - s^2 \sin^2\theta)^{1/2}}, \qquad \mathrm{E}(s) = \int_0^{\pi/2} \mathrm{d}\theta\, (1 - s^2 \sin^2\theta)^{1/2}. \tag{C.68}$$

The Digital Library of Mathematical Functions (DLMF) writes k for the modulus, which in this book is a crystal momentum; numerical libraries often take the parameter $m = s^2$ as the argument, so that $\mathrm{K}(s)$ is computed as a function of s^2. $\mathrm{K}(s)$ and $\mathrm{E}(s)$ both equal $\pi/2$ at $s = 0$. We expand the integrand in powers of $s^2 \sin^2\theta$ and use $\int_0^{\pi/2} \mathrm{d}\theta\, \sin^{2n}\theta = (\pi/2)(2n-1)!!/(2n)!!$. The expansion gives [1, Eqs. (19.5.1) and (19.5.2)]

$$\mathrm{K}(s) = \frac{\pi}{2}\Big(1 + \frac{s^2}{4} + \frac{9s^4}{64} + \cdots\Big), \qquad \mathrm{E}(s) = \frac{\pi}{2}\Big(1 - \frac{s^2}{4} - \frac{3s^4}{64} - \cdots\Big). \tag{C.69}$$

As $s \to 1$, $\mathrm{E}(s) \to 1$, while $\mathrm{K}(s)$ diverges logarithmically [1, Eq. (19.12.1)],

$$\mathrm{K}(s) = \ln\frac{4}{s'} + O\Big(s'^2 \ln\frac{1}{s'}\Big). \tag{C.70}$$

The divergence comes from the end of the interval, $\theta \to \pi/2$, where the integrand behaves as $1/[s'^2 + (\pi/2 - \theta)^2]^{1/2}$ and its integral over a range of order one is $\ln(1/s')$ plus a constant. At $s' = 0.1$, Eq. (C.70) gives 3.689 against the exact 3.696; at $s' = 0.01$ the approximation and the exact value differ by 1×10^{-4}.

The integrals of the form we need reduce to K and E. For $\mu_0 > |\mu_1|$, the substitution $\phi = 2\theta$ turns $\cos\phi$ into $1 - 2\sin^2\theta$ and $\mu_0 + \mu_1\cos\phi$ into $(\mu_0 + \mu_1)(1 - s^2\sin^2\theta)$ with $s^2 = 2\mu_1/(\mu_0 + \mu_1)$; for $\mu_1 < 0$ the substitution $\phi = \pi - 2\theta$ does the same with $\mu_1 \to -\mu_1$. Hence, with $s = [2|\mu_1|/(\mu_0 + |\mu_1|)]^{1/2}$,

$$\int_0^{\pi} \frac{\mathrm{d}\phi}{(\mu_0 + \mu_1\cos\phi)^{1/2}} = \frac{2\,\mathrm{K}(s)}{(\mu_0 + |\mu_1|)^{1/2}}, \qquad \int_0^{\pi} \mathrm{d}\phi\, (\mu_0 + \mu_1\cos\phi)^{1/2} = 2(\mu_0 + |\mu_1|)^{1/2}\,\mathrm{E}(s). \tag{C.71}$$

Integrals of this kind arise when a quantity depends on the distance between a fixed point and a point moving on a circle. The angle between the Earth's galactic velocity and a laboratory axis, Eq. (1.16), has exactly the form $\cos\theta_{\mathrm{lab}} = \mu_0 + \mu_1\cos\phi$ with ϕ the sidereal phase. For a rate that varies as $|\boldsymbol{v}_{\mathrm{E}} - v\hat{\boldsymbol{n}}|^{\pm 1}$ at a fixed speed v along the laboratory axis, the daily average is an elliptic integral. Logarithmic singularities of the same origin appear in the density of states of a two-dimensional band at a saddle point (Appendix C.2), where K diverges as in Eq. (C.70).

C.8.3 The Lindhard logarithms

The independent-particle response of the electron gas (Sect. 4.5) contains logarithms of ratios such as $(u+1)/(u-1)$, where u is a combination of ω and q in units set by the Fermi momentum k_F and the Fermi velocity v_F (Sect. 2.5). We need the behavior of these logarithms on the real axis, where they are real, and their imaginary part when the frequency acquires the $+\mathrm{i}0^+$ of a retarded function.

For real u the function $\ln|(u+1)/(u-1)|$ is finite except at $u=\pm1$, where it diverges logarithmically but integrably. The static response contains it in the combination

$$F_\mathrm{L}(y) = \frac{1}{2} + \frac{1-y^2}{4y}\ln\Big|\frac{1+y}{1-y}\Big|, \qquad y = \frac{q}{2k_\mathrm{F}}, \tag{C.72}$$

whose limits are elementary. For $y\to0$, $\ln[(1+y)/(1-y)] = 2(y+y^3/3+y^5/5+\dots)$ gives $F_\mathrm{L} = 1-y^2/3-y^4/15+O(y^6)$, the long-wavelength limit in which the response tends to the density of states at the Fermi level and Thomas–Fermi screening follows (Sect. 4.4). At $y=1$, the momentum $2k_\mathrm{F}$ that spans the Fermi sphere, $F_\mathrm{L} = \frac{1}{2}$, but its derivative diverges as $\frac{1}{2}\ln|1-y|$: the logarithm has a kink there, the origin of the Kohn anomaly and of Friedel oscillations. For $y\to\infty$, expanding in $1/y$ gives $F_\mathrm{L} = 1/(3y^2)+O(y^{-4})$.

At finite frequency the argument becomes complex, and the logarithm needs a branch. Take the principal branch, $\ln z = \ln|z| + \mathrm{i}\arg z$ with $-\pi < \arg z \le \pi$, which is analytic off the negative real axis. Let u acquire a small positive imaginary part, $u \to u+\mathrm{i}0^+$, as the retarded prescription $\omega\to\omega+\mathrm{i}0^+$ requires when u increases with ω. The ratio $(u+1+\mathrm{i}0^+)/(u-1+\mathrm{i}0^+)$ is real and positive for $|u|>1$. For $|u|<1$ it is negative: the numerator is positive and the denominator negative, with a small positive imaginary part, so the inverse of the denominator has a small negative imaginary part and the ratio lies just below the negative real axis. Hence

$$\ln\frac{u+1+\mathrm{i}0^+}{u-1+\mathrm{i}0^+} = \ln\Big|\frac{u+1}{u-1}\Big| - \mathrm{i}\pi\,\Theta(1-|u|). \tag{C.73}$$

The imaginary part is a step. It is nonzero exactly where the frequency lies inside the band of particle–hole energies that a momentum transfer q can create, and its value $-\pi$ produces the negative imaginary part that a retarded response has for $\omega>0$ (Appendix A.7). The same function follows from Eq. (C.26): $\ln[(u+1)/(u-1)] = \int_{-1}^{1}\mathrm{d}u'/(u-u')$, and the delta-function term of $1/(u-u'+\mathrm{i}0^+)$ contributes $-\mathrm{i}\pi$ when u lies inside the interval of integration. Equivalently, Im of the logarithm counts the on-shell transitions, and the real part of the logarithm is the principal-value transform of the density of these transitions, as Eq. (C.28) states for any Cauchy integral.

C.9 Classical fields and static propagators

Sect. 1.4 needs two results from classical field theory: the potential that a massive mediator produces around a static source, and the field redefinition that removes kinetic mixing between two vector fields. Both results follow from Lagrangian densities and their Euler–Lagrange equations, without quantization. We use natural units, Heaviside–Lorentz charge units (Appendix A.2) and the metric signature $(+,-,-,-)$, with $x^\mu = (t,\boldsymbol{x})$, $\partial_\mu = (\partial_t, \nabla)$ and $\partial^\mu\partial_\mu = \partial_t^2 - \nabla^2$.

C.9.1 Scalar and vector fields with sources

We start with a real scalar field $\phi_V(x)$ of mass m_V. Coupled to a source density $\rho_\mathrm{s}(x)$, it has the Lagrangian density

$$\mathcal{L}_\mathrm{s} = \tfrac{1}{2}\,\partial_\mu\phi_V\,\partial^\mu\phi_V - \tfrac{1}{2}\,m_V^2\phi_V^2 - \phi_V\,\rho_\mathrm{s}. \tag{C.74}$$

Its Euler–Lagrange equation is $\partial_\mu[\partial\mathcal{L}/\partial(\partial_\mu\phi_V)] = \partial\mathcal{L}/\partial\phi_V$. Here it reads $\partial_\mu\partial^\mu\phi_V = -m_V^2\phi_V - \rho_\mathrm{s}$, the Klein–Gordon equation with a source,

$$(\partial_\mu\partial^\mu + m_V^2)\phi_V = -\rho_\mathrm{s}. \tag{C.75}$$

The vector mediator is a massive (Proca) field $A'_\mu(x)$ with field strength $F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu$. Coupled to a current $J^\mu(x)$, it has

$$\mathcal{L}_\mathrm{v} = -\tfrac{1}{4}\,F'_{\mu\nu}F'^{\mu\nu} + \tfrac{1}{2}\,m_V^2 A'_\mu A'^\mu - A'_\mu J^\mu. \tag{C.76}$$

Here $\partial\mathcal{L}_\mathrm{v}/\partial(\partial_\mu A'_\nu) = -F'^{\mu\nu}$, since $\partial(F'_{\alpha\beta}F'^{\alpha\beta})/\partial(\partial_\mu A'_\nu) = 4F'^{\mu\nu}$ (each of $F'_{\mu\nu}$ and $F'_{\nu\mu} = -F'_{\mu\nu}$ contributes $2F'^{\mu\nu}$), and $\partial\mathcal{L}_\mathrm{v}/\partial A'_\nu = m_V^2 A'^\nu - J^\nu$. The field equation is then

$$\partial_\mu F'^{\mu\nu} + m_V^2 A'^\nu = J^\nu. \tag{C.77}$$

Since $F'^{\mu\nu}$ is antisymmetric, $\partial_\nu\partial_\mu F'^{\mu\nu} = 0$, and the divergence of Eq. (C.77) leaves $m_V^2\,\partial_\nu A'^\nu = \partial_\nu J^\nu$. For a conserved current and $m_V \neq 0$ the field obeys $\partial_\nu A'^\nu = 0$; then $\partial_\mu F'^{\mu\nu} = \partial_\mu\partial^\mu A'^\nu$, and every component satisfies $(\partial_\mu\partial^\mu + m_V^2)A'^\nu = J^\nu$. At $m_V = 0$, Eq. (C.77) is Maxwell's equation in Heaviside–Lorentz units: its $\nu = 0$ component is $\nabla\cdot\boldsymbol{E} = J^0$, with $\boldsymbol{E} = -\nabla A^0 - \partial_t\boldsymbol{A}$.

For non-relativistic point particles the sources are number densities and number currents. A particle at $\boldsymbol{x}_i(t)$ contributes $\delta^3(\boldsymbol{x} - \boldsymbol{x}_i)$ to J^0 and $\dot{\boldsymbol{x}}_i\,\delta^3(\boldsymbol{x} - \boldsymbol{x}_i)$ to $\boldsymbol{J}$; a particle that couples with strength y contributes these terms multiplied by y. For the scalar, ρ_s is the coupling-weighted number density, which is the non-relativistic limit of the scalar density $\bar\psi\psi$ [13, Eq. (7)].

C.9.2 The static Green's function

For time-independent sources ∂_t drops out and $\partial_\mu\partial^\mu \to -\nabla^2$. A static source has $\boldsymbol{J} = 0$, so only A'^0 is sourced. Eqs. (C.75) and (C.77) become

$$(-\nabla^2 + m_V^2)\phi_V = -\rho_\mathrm{s}, \qquad (-\nabla^2 + m_V^2)A'^0 = J^0. \tag{C.78}$$

Both equations are solved by the Green's function $G_\mathrm{Y}(\boldsymbol{x})$, defined by $(-\nabla^2 + m_V^2)\,G_\mathrm{Y}(\boldsymbol{x}) = \delta^3(\boldsymbol{x})$. With the Fourier convention of Appendix A.3, $\tilde G_\mathrm{Y}(\boldsymbol{q}) = \int \mathrm{d}^3x\,\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}G_\mathrm{Y}(\boldsymbol{x})$, the operator $-\nabla^2$ becomes q^2 and the delta function becomes 1:

$$\tilde G_\mathrm{Y}(\boldsymbol{q}) = \frac{1}{q^2 + m_V^2}. \tag{C.79}$$

The inverse transform is $G_\mathrm{Y}(\boldsymbol{x}) = \int \mathrm{d}^3q\,(2\pi)^{-3}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}/(q^2{+}m_V^2)$. We take the polar axis along $\boldsymbol{x}$ and write $r = |\boldsymbol{x}|$. The angular integral is $\int \mathrm{d}\Omega\,\mathrm{e}^{-\mathrm{i}qr\cos\theta} = 2\pi\int_{-1}^{1}\mathrm{d}u\,\mathrm{e}^{-\mathrm{i}qru} = 4\pi\sin(qr)/(qr)$, which leaves

$$G_\mathrm{Y}(\boldsymbol{x}) = \frac{1}{2\pi^2 r}\int_0^\infty \mathrm{d}q\,\frac{q\sin(qr)}{q^2 + m_V^2} = \frac{1}{4\pi^2 r}\,\mathrm{Im}\int_{-\infty}^{\infty}\mathrm{d}q\,\frac{q\,\mathrm{e}^{\mathrm{i}qr}}{(q - \mathrm{i}m_V)(q + \mathrm{i}m_V)}.$$

The second form holds because the integrand of the first form is even in q, so the half-line integral is half the full-line one, and because $\sin(qr) = \mathrm{Im}\,\mathrm{e}^{\mathrm{i}qr}$ for real q. Since for $r > 0$ the factor $\mathrm{e}^{\mathrm{i}qr}$ decays in the upper half plane, by Jordan's lemma the integral equals the integral over a contour closed there. The contour encloses the pole at $q = \mathrm{i}m_V$, with residue $\mathrm{i}m_V\,\mathrm{e}^{-m_V r}/(2\mathrm{i}m_V) = \mathrm{e}^{-m_V r}/2$. The full-line integral is $2\pi\mathrm{i}\times\mathrm{e}^{-m_V r}/2 = \mathrm{i}\pi\,\mathrm{e}^{-m_V r}$, its imaginary part is $\pi\,\mathrm{e}^{-m_V r}$, and

$$G_\mathrm{Y}(\boldsymbol{x}) = \frac{\mathrm{e}^{-m_V r}}{4\pi r}. \tag{C.80}$$

A direct check confirms Eq. (C.80): for $r > 0$, $\nabla^2 G_{\rm Y} = r^{-1}\,\mathrm{d}^2(rG_{\rm Y})/\mathrm{d}r^2 = m_V^2 G_{\rm Y}$, and $(-\nabla^2 + m_V^2)G_{\rm Y}$ vanishes away from the origin. Near the origin $G_{\rm Y} \to 1/(4\pi r)$, and $-\nabla^2(1/4\pi r) = \delta^3(\boldsymbol{x})$ supplies the source. Eq. (C.80) is the Yukawa form: the field falls off over the range $1/m_V$ and becomes the Coulomb potential $1/(4\pi r)$ as $m_V \to 0$.

By superposition the static fields are $A'^0(\boldsymbol{x}) = \int \mathrm{d}^3x'\, G_{\rm Y}(\boldsymbol{x}-\boldsymbol{x}')J^0(\boldsymbol{x}')$ and $\phi_V(\boldsymbol{x}) = -\int \mathrm{d}^3x'\, G_{\rm Y}(\boldsymbol{x}-\boldsymbol{x}')\rho_{\rm s}(\boldsymbol{x}')$. A point particle with coupling y_χ at the origin produces

$$A'^0(\boldsymbol{x}) = y_\chi \frac{\mathrm{e}^{-m_V r}}{4\pi r}, \qquad \phi_V(\boldsymbol{x}) = -y_\chi \frac{\mathrm{e}^{-m_V r}}{4\pi r}. \tag{C.81}$$

C.9.3 Validity of the static limit

Dark matter that scatters moves, and so does the electron it scatters on; the sources are not strictly static. With the time convention $f(\omega) = \int \mathrm{d}t\, \mathrm{e}^{+\mathrm{i}\omega t} f(t)$ of Appendix A.3, ∂_t^2 becomes $-\omega^2$, and the operator $\partial_\mu\partial^\mu + m_V^2$ of Eq. (C.75) becomes $q^2 - \omega^2 + m_V^2$ for a field component of momentum $\boldsymbol{q}$ and energy ω. The full Green's function in momentum space is then $1/(q^2 - \omega^2 + m_V^2)$, which reduces to Eq. (C.79) when $\omega^2 \ll q^2 + m_V^2$. In a scattering event the mediator carries the energy ω and momentum $\boldsymbol{q}$ given to the target, and kinematics bounds them by $\omega \le q\, v_{\rm max}$ (Sect. 1.3), with $v_{\rm max} = 2.65 \times 10^{-3}$. The neglected term ω^2 is then smaller than $v_{\rm max}^2 = 7 \times 10^{-6}$ relative to q^2, and the static potential is accurate to better than a part in 10^5 for every purpose of this book.

Absorption, however, has no static limit. There the boson is the dark matter itself, and its field obeys the source-free equation $(\partial_\mu\partial^\mu + m_\phi^2)\phi = 0$ with $\omega^2 = |\boldsymbol{p}_\phi|^2 + m_\phi^2$: an oscillating, on-shell field. Sect. 1.4 therefore treats scattering and absorption separately.

C.9.4 Interaction energy of two sources

Since a static configuration has no time derivatives, its energy is $-\int \mathrm{d}^3x\, \mathcal{L}$, written $H_{\rm s}$ for the scalar and $H_{\rm v}$ for the vector. For the scalar, Eq. (C.74) with $\partial_t\phi_V = 0$ reads $\mathcal{L}_{\rm s} = -\frac{1}{2}(\nabla\phi_V)^2 - \frac{1}{2}m_V^2\phi_V^2 - \phi_V\rho_{\rm s}$. Integrating $(\nabla\phi_V)^2$ by parts and using Eq. (C.78),

$$\int \mathrm{d}^3x\, \left[\tfrac{1}{2}(\nabla\phi_V)^2 + \tfrac{1}{2}m_V^2\phi_V^2\right] = \tfrac{1}{2}\int \mathrm{d}^3x\, \phi_V(-\nabla^2 + m_V^2)\phi_V = -\tfrac{1}{2}\int \mathrm{d}^3x\, \phi_V\, \rho_{\rm s},$$

so $H_{\rm s} = \frac{1}{2}\int \mathrm{d}^3x\, \phi_V\rho_{\rm s} = -\frac{1}{2}\int \mathrm{d}^3x\, \mathrm{d}^3x'\, \rho_{\rm s}(\boldsymbol{x})\, G_{\rm Y}(\boldsymbol{x}-\boldsymbol{x}')\, \rho_{\rm s}(\boldsymbol{x}')$. For the vector with only A'^0 nonzero, $F'_{0i} = -\partial_i A'^0$ and $-\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} = -\frac{1}{2}F'_{0i}F'^{0i} = +\frac{1}{2}(\nabla A'^0)^2$, and $\mathcal{L}_{\rm v} = \frac{1}{2}(\nabla A'^0)^2 + \frac{1}{2}m_V^2(A'^0)^2 - A'^0J^0$. The same steps lead to $H_{\rm v} = +\frac{1}{2}\int \mathrm{d}^3x\, \mathrm{d}^3x'\, J^0(\boldsymbol{x})\, G_{\rm Y}(\boldsymbol{x}-\boldsymbol{x}')\, J^0(\boldsymbol{x}')$. At $m_V = 0$, $H_{\rm v}$ is $\frac{1}{2}\int \mathrm{d}^3x\, \boldsymbol{E}^2$, the positive electrostatic energy; this limit fixes the overall sign.

Now take two point particles, one with coupling y_χ at $\boldsymbol{x}_\chi$ and one with y_e at $\boldsymbol{x}_e$. Each double integral contains two self-energy terms, independent of the separation, and two equal cross terms. The cross terms give the interaction energy

$$U(\boldsymbol{x}_e - \boldsymbol{x}_\chi) = \pm\, y_\chi y_e\, G_{\rm Y}(\boldsymbol{x}_e - \boldsymbol{x}_\chi) = \pm\, y_\chi y_e \frac{\mathrm{e}^{-m_V|\boldsymbol{x}_e - \boldsymbol{x}_\chi|}}{4\pi\,|\boldsymbol{x}_e - \boldsymbol{x}_\chi|}, \tag{C.82}$$

with the upper sign for the vector and the lower for the scalar. Vector exchange repels sources whose couplings have the same sign, as like electric charges repel; scalar exchange attracts them. The same result follows from treating the second particle as a test particle: its term $-y_eA'^0(\boldsymbol{x}_e)$ or $-y_e\phi_V(\boldsymbol{x}_e)$ in the Lagrangian is a potential energy $+y_eA'^0$ or $+y_e\phi_V$, evaluated with the fields of Eq. (C.81). The Fourier transform of Eq. (C.82) is $\pm\, y_\chi y_e/(q^2 + m_V^2)$ by Eq. (C.79). The sign drops out of every rate, since a rate depends on the square of the interaction energy.

C.9.5 Kinetic mixing

Let the photon A_μ, with field strength $F_{\mu\nu}$, couple to the electromagnetic current $eJ^\mu_{\rm em}$, where $J^\mu_{\rm em}$ counts each particle with its charge in units of e (an electron counts -1). Let a second vector A'_μ of mass m_V couple to the dark-matter number current J^μ_χ with strength y_χ. Gauge invariance allows a term that mixes the two field strengths, with a dimensionless coefficient ζ:

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tfrac{1}{4}F'_{\mu\nu}F'^{\mu\nu} + \tfrac{\zeta}{2}F_{\mu\nu}F'^{\mu\nu} + \tfrac{1}{2}m_V^2 A'_\mu A'^\mu - eA_\mu J^\mu_{\rm em} - y_\chi A'_\mu J^\mu_\chi. \qquad \text{(C.83)}$$

The sign of the mixing term follows Ref. [14, Eq. (70)]; other papers use the opposite sign or write ϵ, κ or χ for ζ (Appendix B.2). Only ζ^2 enters a rate.

The kinetic terms are not diagonal, so A_μ and A'_μ are not independent propagating fields. We shift the photon, $A_\mu = \tilde{A}_\mu + \zeta A'_\mu$, so that $F_{\mu\nu} = \bar{F}_{\mu\nu} + \zeta F'_{\mu\nu}$, with $\bar{F}_{\mu\nu} = \partial_\mu\tilde{A}_\nu - \partial_\nu\tilde{A}_\mu$. The photon kinetic term becomes $-\frac{1}{4}\bar{F}^2 - \frac{\zeta}{2}\bar{F}F' - \frac{\zeta^2}{4}F'^2$, with $\bar{F}F' \equiv \bar{F}_{\mu\nu}F'^{\mu\nu}$. The mixing term becomes $\frac{\zeta}{2}\bar{F}F' + \frac{\zeta^2}{2}F'^2$. The cross terms cancel, and

$$\mathcal{L} = -\tfrac{1}{4}\bar{F}_{\mu\nu}\bar{F}^{\mu\nu} - \tfrac{1}{4}(1-\zeta^2)F'_{\mu\nu}F'^{\mu\nu} + \tfrac{1}{2}m_V^2 A'_\mu A'^\mu - e\tilde{A}_\mu J^\mu_{\rm em} - \zeta e\, A'_\mu J^\mu_{\rm em} - y_\chi A'_\mu J^\mu_\chi, \quad \text{(C.84)}$$

which differs from Ref. [14, Eq. (72)] only in the sign convention for the current. Shifting the photon, and not A'_μ, leaves the mass term untouched: $\tilde{A}_\mu$ stays massless and the mass matrix stays diagonal. Rescaling A'_μ by $(1-\zeta^2)^{-1/2}$ restores its canonical normalization and changes m_V and the couplings at order ζ^2; we drop these changes.

For $m_V = 0$ the alternative shift $A'_\mu \to A'_\mu + \zeta A_\mu$ also removes the mixing; the photon then couples to dark matter with strength ζy_χ, and dark matter appears millicharged [14, Eq. (71)]. For $m_V \neq 0$, however, that shift would mix the mass term, and the photon shift is the one to use. In the new basis the photon couples only to charged particles, and the massive vector couples to dark matter with strength y_χ and to the electromagnetic current with strength ζe. For electrons, $J^\mu_{\rm em}$ contains $-J^\mu_e$, with J^μ_e the electron number current, so the electron coupling in the form $-y_e A'_\mu J^\mu_e$ is $y_e = -\zeta e$; protons have $+\zeta e$.

The redefinition is exact in vacuum. In a medium the photon acquires a self-energy from the charges it polarizes, the two kinetic terms are no longer the whole quadratic Lagrangian, and mixing reappears at the level of the in-medium propagators (Sects. 1.4 and 4.4).

C.10 Moment problems

The first five energy moments of a nuclear response at fixed $\boldsymbol{q}$ are equal-time averages, which path-integral simulations give for any anharmonicity, while the response itself needs real-time dynamics. Five moments fix the distribution of energy deposits only up to a band around its cumulative distribution (Sect. 11.5). The width of the band is fixed by the moments alone; the band is wide at the peak and narrow in the tails.

C.10.1 The Hamburger and Stieltjes problems

Let P be a probability distribution on the real line, with moments

$$\mu_k = \int x^k \,\mathrm{d}P(x), \qquad k = 0, 1, \ldots, 4, \qquad \mu_0 = 1. \qquad \text{(C.85)}$$

For the nuclear response, x is the energy transfer ω and $\mathrm{d}P = S(\boldsymbol{q},\omega)\,\mathrm{d}\omega / \int S\,\mathrm{d}\omega$. The *Hamburger moment problem* asks which numbers μ_k are the moments of some distribution on the real line, and how many distributions share them. The *Stieltjes problem* asks the same for distributions on the half line $x \geq 0$, the case of a response at zero temperature, which has no

weight at $\omega < 0$ (Appendix A.4). We treat the truncated problems, in which only $\mu_0, \dots, \mu_4$ are given.

A necessary condition on the moments comes from positivity. For any real polynomial $p(x) = c_0 + c_1 x + c_2 x^2$, written $p(x) = \mathbf{c}^{\mathsf{T}}\mathbf{v}(x)$ with $\mathbf{v}(x) = (1, x, x^2)^{\mathsf{T}}$,

$$0 \le \int p(x)^2 \, \mathrm{d}P(x) = \mathbf{c}^{\mathsf{T}}\mathbf{H}\,\mathbf{c}, \qquad \mathbf{H} = \int \mathbf{v}(x)\mathbf{v}(x)^{\mathsf{T}} \mathrm{d}P(x) = \begin{pmatrix} \mu_0 & \mu_1 & \mu_2 \\ \mu_1 & \mu_2 & \mu_3 \\ \mu_2 & \mu_3 & \mu_4 \end{pmatrix}. \tag{C.86}$$

The *Hankel matrix* $\mathbf{H}$, whose entries depend only on the sum of their indices, $H_{ij} = \mu_{i+j}$, is therefore positive semidefinite (Appendix C.7). It is positive definite unless P is concentrated on the two or fewer roots of some quadratic p, and we assume from now on that P has at least three points of support, so that $\mathbf{H}$ is positive definite and invertible. On the half line, $\int x\, p_1(x)^2 \, \mathrm{d}P \ge 0$ for every linear p_1 adds a second condition: the shifted matrix $\mathbf{H}_1$, with entries μ_{i+j+1} for $i, j = 0, 1$, is positive semidefinite as well. For the truncated problems these conditions are also sufficient, up to boundary cases in which a matrix is singular [15]. If all moments of P are given, they fix P uniquely when they grow slowly enough (Carleman's condition, which a Gaussian satisfies); five moments never do, and infinitely many distributions share them. The question is how far apart these distributions can be.

C.10.2 Christoffel function and canonical representation

One function of a point ξ bounds how far apart two distributions with the same five moments can be. Define the kernel

$$\mathsf{K}(x, \xi) = \mathbf{v}(x)^{\mathsf{T}}\mathbf{H}^{-1}\mathbf{v}(\xi), \tag{C.87}$$

a quadratic polynomial in x for fixed ξ. It reproduces every quadratic $q(x) = \mathbf{a}^{\mathsf{T}}\mathbf{v}(x)$ under the integral: by Eq. (C.86),

$$\int q(x)\, \mathsf{K}(x, \xi) \, \mathrm{d}P(x) = \mathbf{a}^{\mathsf{T}}\mathbf{H}\mathbf{H}^{-1}\mathbf{v}(\xi) = q(\xi). \tag{C.88}$$

The *Christoffel function* is

$$\varrho_{\mathrm{MK}}(\xi) = \frac{1}{\mathsf{K}(\xi, \xi)} = \frac{1}{\mathbf{v}(\xi)^{\mathsf{T}}\mathbf{H}^{-1}\mathbf{v}(\xi)}. \tag{C.89}$$

It is the smallest value of $\int p^2 \, \mathrm{d}P$ over quadratics with $p(\xi) = 1$: minimizing $\mathbf{c}^{\mathsf{T}}\mathbf{H}\mathbf{c}$ subject to $\mathbf{c}^{\mathsf{T}}\mathbf{v}(\xi) = 1$ with a Lagrange multiplier gives $\mathbf{c} \propto \mathbf{H}^{-1}\mathbf{v}(\xi)$, and the minimum is Eq. (C.89). Hence no distribution with these moments can put more than $\varrho_{\mathrm{MK}}(\xi)$ at the single point ξ: if it puts w there, then $w = w\, p(\xi)^2 \le \int p^2 \, \mathrm{d}P$ for the minimizing p, so $w \le \varrho_{\mathrm{MK}}(\xi)$.

The bound is attained by a three-point distribution. Let $\mathsf{K}(\cdot, \xi)$ have nonzero leading coefficient, a condition that fails only at the at most two roots of a quadratic in ξ. Then the cubic $r(x) = (x - \xi)\, \mathsf{K}(x, \xi)$ has three distinct real roots. For every linear s, $\int r(x) s(x) \, \mathrm{d}P = 0$, by Eq. (C.88) applied to the quadratic $(x - \xi)s(x)$, which vanishes at ξ. A polynomial that is orthogonal to all linear polynomials changes sign at least twice on the support of P: if it changed sign at most once, at x_*, the product with $s(x) = \pm(x - x_*)$ would have one sign and a nonzero integral. A real cubic with two sign changes has three real roots. One root is ξ, and it is simple, since $\mathsf{K}(\xi, \xi) > 0$. The other two roots are the roots of $\mathsf{K}(\cdot, \xi)$, and they are distinct, because a double root would not change sign. Call the three roots $y_1 < y_2 < y_3$; one of them is ξ.

The three points carry a quadrature that is exact for every polynomial f of degree up to four. Divide f by r: $f = r\, s + t$, with s linear and t the quadratic that interpolates f

at y_1, y_2, y_3. The term $\int r\, s\, \mathrm{d}P$ vanishes, as shown above. The quadratic t is a combination of the three Lagrange polynomials ℓ_i, with $\ell_i(y_j) = \delta_{ij}$, so that $\int t\, \mathrm{d}P = \sum_i w_i f(y_i)$ with $w_i = \int \ell_i\, \mathrm{d}P$. Therefore

$$\int f(x)\, \mathrm{d}P(x) = \sum_{i=1}^{3} w_i\, f(y_i) \qquad \text{for every } f \text{ of degree} \le 4. \tag{C.90}$$

The weights are positive: ℓ_i^2 has degree four, and Eq. (C.90) gives $\int \ell_i^2\, \mathrm{d}P = w_i$. The weight at ξ follows from $f = \mathsf{K}(\cdot, \xi)^2$, which vanishes at the two other nodes. By Eq. (C.88) with $q = \mathsf{K}(\cdot, \xi)$ the left side of Eq. (C.90) is $\mathsf{K}(\xi, \xi)$, and the right side is $w_\xi\, \mathsf{K}(\xi, \xi)^2$. So $w_\xi = \varrho_{\mathrm{MK}}(\xi)$. The three-point distribution with nodes y_i and weights w_i has the same five moments as P (take $f = x^k$) and puts the largest admissible mass at ξ. It is the *canonical representation* through ξ.

C.10.3 The Markov–Krein band

Write $F_P(\xi) = P((-\infty, \xi])$ for the cumulative distribution and $F_P(\xi - 0)$ for its limit from below. The band is

Markov–Krein band

$$\sum_{y_i < \xi} w_i \;\le\; F_P(\xi - 0) \;\le\; F_P(\xi) \;\le\; \sum_{y_i \le \xi} w_i, \tag{C.91}$$

with y_i, w_i the canonical representation through ξ. Every distribution with the moments $\mu_0, \ldots, \mu_4$ satisfies Eq. (C.91), and the two bounds differ by $\varrho_{\mathrm{MK}}(\xi)$ of Eq. (C.89).

The result goes back to Chebyshev, Markov and Stieltjes, and the general form is in Refs. [15] and [16].

The proof compares the indicator function of $(-\infty, \xi]$ with a polynomial U of degree four that lies above it and touches it at the nodes. If $U(x) \ge \Theta(\xi - x)$ for all x, with $\Theta(0) = 1$, and $U(y_i) = \Theta(\xi - y_i)$ at the three nodes, then by Eq. (C.90)

$$F_P(\xi) = \int \Theta(\xi - x)\, \mathrm{d}P(x) \le \int U\, \mathrm{d}P = \sum_i w_i\, U(y_i) = \sum_{y_i \le \xi} w_i.$$

We construct U for the three positions of ξ among the nodes. If $\xi = y_3$, the largest node, $U = 1$ works and the bound is the trivial $F_P \le 1$. If $\xi = y_1$, the smallest, take

$$U(x) = \frac{(x - y_2)^2 (x - y_3)^2}{(y_1 - y_2)^2 (y_1 - y_3)^2}.$$

It is nonnegative everywhere, equals 1 at y_1 and 0 at y_2, y_3, and for $x < y_1$ both factors $|x - y_2|$ and $|x - y_3|$ exceed their values at y_1, so $U \ge 1$ there. If $\xi = y_2$, let U be the quartic with $U(y_1) = U(y_2) = 1$, $U(y_3) = 0$ and $U'(y_1) = U'(y_3) = 0$: five conditions, which fix a polynomial of degree four. Between y_1 and y_2, where U takes the value 1 twice, Rolle's theorem places a root c of U'. The cubic U' then has the three roots $y_1 < c < y_3$ and no other, $U' = A(x - y_1)(x - c)(x - y_3)$. On (y_2, y_3), U falls from 1 to 0 with no critical point, so $U' < 0$ there; since $(x - y_1)(x - c)(x - y_3) < 0$ on that interval, $A > 0$. The signs of U' now fix the shape: U decreases towards $U(y_1) = 1$ for $x < y_1$, rises on (y_1, c), falls on (c, y_3) through $U(y_2) = 1$ to $U(y_3) = 0$, and rises for $x > y_3$. So $U \ge 1$ for $x \le y_2$ and $U \ge 0$ beyond. The lower bound in Eq. (C.91) is the upper bound applied to the mirrored distribution of $-x$, whose canonical representation through $-\xi$ is the mirror image of the canonical representation

through ξ: the mass of $[\xi,\infty)$ is at most $\sum_{y_i\ge\xi} w_i$, so $F_P(\xi-0)\ge\sum_{y_i<\xi} w_i$. If the leading coefficient of $\mathsf{K}(\cdot,\xi)$ vanishes, one node runs off to infinity with vanishing weight; since F_P is monotone and the bounds are continuous in ξ, Eq. (C.91) holds there by continuity.

For a standard Gaussian, $\mu_0,\dots,\mu_4 = 1,0,1,0,3$. The inverse of $\mathbf{H}$ is

$$\mathbf{H}^{-1} = \begin{pmatrix} 3/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1/2 \end{pmatrix},$$

as multiplication by $\mathbf{H}$ confirms. With $\mathbf{v}(\xi) = (1,\xi,\xi^2)^{\mathsf{T}}$, the quadratic form is $\frac{3}{2} + 2(-\frac{1}{2})\xi^2 + \xi^2 + \frac{1}{2}\xi^4$. The two terms in ξ^2 cancel, $\mathsf{K}(\xi,\xi) = \frac{3}{2} + \frac{1}{2}\xi^4$, and

$$\varrho_{\mathrm{MK}}(\xi) = \frac{2}{3+\xi^4} \qquad \text{(standard Gaussian)}. \tag{C.92}$$

At the mean, $\varrho_{\mathrm{MK}}(0) = 2/3$: the canonical representation through 0 has nodes 0 and $\pm\sqrt{3}$ with weights $\frac{2}{3}$ and $\frac{1}{6}$, and Eq. (C.91) reads $\frac{1}{6} \le F_P(0) \le \frac{5}{6}$, so five moments leave the median almost undetermined.

C.10.4 Tails of the band

Far from the mean, $\mathbf{v}(\xi)$ is dominated by its last entry, ξ^2, and $\mathsf{K}(\xi,\xi) \simeq (\mathbf{H}^{-1})_{22}\,\xi^4$, with indices running over 0, 1, 2. For a distribution shifted to zero mean, with central moments $\bar\mu_2, \bar\mu_3, \bar\mu_4$, the corner element of the inverse is the 2×2 minor over the determinant, $(\mathbf{H}^{-1})_{22} = \bar\mu_2/\det\mathbf{H}$, and $\det\mathbf{H} = \bar\mu_2\bar\mu_4 - \bar\mu_3^2 - \bar\mu_2^3$ by expansion along the first row. Hence

$$\varrho_{\mathrm{MK}}(\xi) \simeq \frac{\bar\mu_4 - \bar\mu_2^2 - \bar\mu_3^2/\bar\mu_2}{(\xi-\mu_1)^4}, \qquad |\xi-\mu_1| \to \infty. \tag{C.93}$$

The numerator is positive because $\mathbf{H}$ is positive definite. For the Gaussian it is $3-1=2$, and Eq. (C.92) behaves as $2/\xi^4$. For a skewed example, the gamma distribution of shape 3, the exact ϱ_{MK} exceeds Eq. (C.93) by 4.1 % at 58 standard deviations above the mean and by 1.3 % at 173; at 58 standard deviations below the mean the exact ϱ_{MK} falls 3.9 % short of Eq. (C.93). The next term of the expansion is smaller than Eq. (C.93) by one power of $\xi-\mu_1$.

The band is narrow in the tails in absolute terms, but it narrows only as a power law. A true threshold tail falls much faster: like a Gaussian for a harmonic multiphonon response, or exponentially. At three standard deviations a Gaussian has $1-F_P = 1.35\times10^{-3}$, while the canonical representation through $\xi=3$ leaves $1-F_P$ anywhere in $[0, 2.38\times10^{-2}]$, eighteen times the true value; at four standard deviations the ratio is 240. The band therefore bounds $|F_P - F_{\mathrm{model}}|$ for any model distribution with the right moments, everywhere, by a number that is small in the tail. It does not bound the ratio $(1-F_P)/(1-F_{\mathrm{model}})$, which is what a rate near threshold needs: when the threshold lies far in the tail, the whole predicted rate is smaller than the band width, and five moments leave it undetermined.

C.10.5 Distributions on the half line

At zero temperature the response has no weight below $\omega=0$, and P is a Stieltjes distribution. Every such distribution is also a Hamburger distribution, so Eq. (C.91) remains valid. It is no longer sharp, because a canonical representation with a node below zero is not admissible, and the positivity of $\mathbf{H}_1$ restricts the distributions further. The bound on the mass at a single point $\xi>0$ shows how. Since $x\ge0$ on the support, a mass w at ξ obeys $w\,\xi\,p_1(\xi)^2 \le \int x\,p_1(x)^2\,\mathrm{d}P$ for every linear p_1. Minimizing over p_1 as before gives

$$w \le \min\Big\{\varrho_{\mathrm{MK}}(\xi),\ \big[\xi\,\mathbf{u}(\xi)^{\mathsf{T}}\mathbf{H}_1^{-1}\mathbf{u}(\xi)\big]^{-1}\Big\}, \qquad \mathbf{u}(\xi) = (1,\xi)^{\mathsf{T}}. \tag{C.94}$$

The second entry diverges as $\xi \to 0$ and falls only as ξ^{-3} in the tail, so it is weaker than ϱ_{MK} near the origin and far out. It is tighter only at intermediate ξ, where a node of the canonical representation through ξ lies below zero: for silicon at $q = 100\,\mathrm{keV}$ the second entry lies below ϱ_{MK} by at most 6.8 %, near $\xi = 0.74\,E_R$ (Sect. 11.5). The sharp Stieltjes band is built from canonical representations whose nodes all lie in $x \geq 0$ [15]. For $\xi < 0$ the band collapses to $F_P = 0$.

C.11 Generating functions, branching and renewal

A 100 eV electron in silicon ends as about 27 electron–hole pairs, each made by a carrier that was itself made by an earlier one. Sects. 18.1 and 18.2 treat this cascade as a branching process and read the mean energy per pair $\varepsilon_{\mathrm{eh}}$ and the Fano factor F_{eh} off its first two moments. Probability generating functions turn sums of independent counts into products. The branching recursion builds the statistics of a whole tree from the statistics of one step. Renewal theory says why the mean pair number becomes linear in the deposited energy, and how much of the energy is left over at the end.

C.11.1 Probability generating functions

For a random count $N = 0, 1, 2, \ldots$ with probabilities p_n, the *probability generating function* is

$$\mathfrak{G}_N(s) = \langle s^N \rangle = \sum_{n \geq 0} p_n\, s^n, \qquad 0 \leq s \leq 1. \tag{C.95}$$

It is a power series with nonnegative coefficients, so it is increasing and convex on $[0, 1]$, with $\mathfrak{G}_N(0) = p_0$ and $\mathfrak{G}_N(1) = 1$. Derivatives at $s = 1$ give factorial moments: $\mathfrak{G}'_N(1) = \langle N \rangle$ and $\mathfrak{G}''_N(1) = \langle N(N-1) \rangle$. Hence

$$\operatorname{Var} N = \mathfrak{G}''_N(1) + \mathfrak{G}'_N(1) - \mathfrak{G}'_N(1)^2. \tag{C.96}$$

The Poisson distribution of Eq. (1.50) with mean $\bar{N}$ has $\mathfrak{G}(s) = \sum_n \mathrm{e}^{-\bar{N}} (\bar{N}s)^n / n! = \mathrm{e}^{\bar{N}(s-1)}$. Then $\mathfrak{G}'(1) = \bar{N}$, $\mathfrak{G}''(1) = \bar{N}^2$, and $\operatorname{Var} N = \bar{N}$: the Fano factor $\operatorname{Var} N / \langle N \rangle$ of a Poisson count is 1, the reference against which the value 0.1–0.15 of silicon (Sect. 1.5) is sub-Poissonian.

If N_1 and N_2 are independent, $\langle s^{N_1 + N_2} \rangle = \langle s^{N_1} \rangle \langle s^{N_2} \rangle$: the generating function of a sum is the product. If N items are each kept independently with probability f, the kept number N_{k} is a sum of N independent Bernoulli counts, each with generating function $1 - f + fs$. At fixed N, N_{k} has the generating function $(1 - f + fs)^N$, and averaging over N,

$$\mathfrak{G}_{N_{\mathrm{k}}}(s) = \mathfrak{G}_N(1 - f + fs). \tag{C.97}$$

By the chain rule $\langle N_{\mathrm{k}} \rangle = f \langle N \rangle$ and $\mathfrak{G}''_{N_{\mathrm{k}}}(1) = f^2 \mathfrak{G}''_N(1) = f^2 (\operatorname{Var} N - \langle N \rangle + \langle N \rangle^2)$. Inserting $\langle N_{\mathrm{k}} \rangle$ and $\mathfrak{G}''_{N_{\mathrm{k}}}(1)$ into Eq. (C.96) and dividing by $f \langle N \rangle$ gives the Fano factor of the kept items,

$$F_{\mathrm{k}} = f\, F_N + 1 - f, \qquad F_N = \operatorname{Var} N / \langle N \rangle. \tag{C.98}$$

Random loss moves any Fano factor towards the Poisson value 1. By Eq. (C.98), a detector that collects a fraction f_{det} of the pairs measures $f_{\mathrm{det}} F_{\mathrm{eh}} + 1 - f_{\mathrm{det}}$.

C.11.2 The branching recursion and extinction

In the simplest branching process, the Galton–Watson process, each individual of generation n has a random number of offspring, independently of all others and with one distribution, of

generating function $\mathfrak{g}(s)$ and mean $\bar{m} = \mathfrak{g}'(1)$. Let N_n be the size of generation n, with $N_0 = 1$, and $\mathfrak{G}_n$ its generating function. Given N_n, the next generation is a sum of N_n independent offspring counts, so $\langle s^{N_{n+1}} \mid N_n \rangle = \mathfrak{g}(s)^{N_n}$, and averaging over N_n,

$$\mathfrak{G}_{n+1}(s) = \mathfrak{G}_n(\mathfrak{g}(s)) = \mathfrak{g}(\mathfrak{G}_n(s)), \qquad \mathfrak{G}_0(s) = s. \tag{C.99}$$

The first form is what we derived; it makes $\mathfrak{G}_n$ the n-fold composition of $\mathfrak{g}$ with itself, so the second follows. The second form is a *first-step* statement: condition on the offspring of the ancestor, and each child starts an independent copy of the process with n generations to go. Differentiating at $s = 1$ gives $\langle N_n \rangle = \bar{m}^{\,n}$.

The probability that the line has died out by generation n is $\mathfrak{G}_n(0)$. By Eq. (C.99), $\mathfrak{G}_{n+1}(0) = \mathfrak{g}(\mathfrak{G}_n(0))$. Since $\mathfrak{g}$ is increasing and $\mathfrak{G}_1(0) = \mathfrak{g}(0) \geq 0 = \mathfrak{G}_0(0)$, the sequence increases, and being bounded by 1 it converges to a limit P_{ext} with $\mathfrak{g}(P_{\text{ext}}) = P_{\text{ext}}$, by continuity of $\mathfrak{g}$. The equation $\mathfrak{g}(s) = s$ can have two roots in $[0, 1]$, and the iteration selects the smaller: for any root s_*, $0 \leq s_*$ implies $\mathfrak{G}_n(0) \leq \mathfrak{G}_n(s_*) = s_*$ for all n, so $P_{\text{ext}} \leq s_*$. Hence

Extinction probability

The extinction probability P_{ext} is the smallest root in $[0, 1]$ of $\mathfrak{g}(s) = s$. It is 1 for $\bar{m} \leq 1$ (excluding the trivial $\mathfrak{g}(s) = s$) and smaller than 1 for $\bar{m} > 1$ [17, Chap. I].

The dependence on $\bar{m}$ follows from convexity. The function $\mathfrak{g}(s) - s$ is convex, vanishes at $s = 1$ and has the slope $\bar{m} - 1$ there. For $\bar{m} \leq 1$ a convex function with nonpositive slope at its zero $s = 1$ lies above its tangent there, $(\bar{m} - 1)(s - 1) \geq 0$ on $[0, 1)$. A root $s_0 < 1$ would therefore need $\bar{m} = 1$. Convexity would then hold $\mathfrak{g}(s) - s$ between the zero tangent and the zero chord from s_0 to 1, so it would vanish on $[s_0, 1]$. A power series that vanishes on an interval vanishes identically, and $\mathfrak{g}(s) = s$ is the trivial case excluded above. For $\bar{m} > 1$ the function is negative just below $s = 1$ and equals $p_0 \geq 0$ at $s = 0$, so a root lies in $[0, 1)$. For a concrete case let each individual have no offspring with probability $1 - p$ and two with probability p, $\mathfrak{g}(s) = 1 - p + ps^2$. The roots of $ps^2 - s + 1 - p = 0$ are $s = 1$ and $s = (1 - p)/p$. For $p \leq \frac{1}{2}$ the smaller root is 1 and the line dies out with certainty; for $p = 0.6$ it survives with probability $1 - \frac{2}{3} = \frac{1}{3}$, and iterating Eq. (C.99) from $s = 0$ converges to $\frac{2}{3}$, the smaller root.

In a cascade, the fixed-point equation of a branching process has more solutions than the physical one, and the physical solution is the *minimal* one, obtained by iterating from the terminal data. In a carrier cascade every collision lowers the energy of the particles it produces, every branch stops below the impact-ionization threshold, and termination has probability 1; the recursion is then solved from low energy upward and the solutions other than the physical one never arise.

C.11.3 Branching with types and deposits

A cascade particle carries more than a generation index: a type (electron or hole) and an energy x. Let a particle in state x undergo its next collision in channel c with probability $p_c(x)$. The collision produces particles in states $x'_1, \ldots, x'_k$, drawn from a distribution that depends on x and c, and leaves an additive deposit d_c: a pair, an emitted phonon energy, or a residual kinetic energy at the end. Let Y be the total deposit of the tree started by the particle, and $\mathfrak{G}_x(s) = \langle s^Y \mid x \rangle$ for an integer deposit. If the subtrees of the k products evolve independently, the first-step argument that gave Eq. (C.99) gives

$$\mathfrak{G}_x(s) = \sum_c p_c(x) \left\langle s^{d_c} \prod_{j=1}^{k} \mathfrak{G}_{x'_j}(s) \right\rangle_c, \tag{C.100}$$

with $\langle\cdots\rangle_c$ the average over the products of channel c. Moments follow directly from $Y = d_c + Y_1 + \cdots + Y_k$, without differentiating. Write $\bar{Y}(x) = \langle Y \,|\, x\rangle$ and $\overline{Y^2}(x) = \langle Y^2 \,|\, x\rangle$. Averaging Y and Y^2 over the first collision, and using $\langle Y_i Y_j\rangle = \bar{Y}(x_i')\bar{Y}(x_j')$ for $i \neq j$ (independent subtrees),

$$\bar{Y}(x) = \sum_c p_c(x)\Big\langle d_c + \sum_j \bar{Y}(x_j')\Big\rangle_c, \tag{C.101}$$

$$\overline{Y^2}(x) = \sum_c p_c(x)\Big\langle d_c^2 + 2d_c\sum_j \bar{Y}(x_j') + \sum_j \overline{Y^2}(x_j') + 2\sum_{i<j}\bar{Y}(x_i')\bar{Y}(x_j')\Big\rangle_c. \tag{C.102}$$

Eqs. (C.101) and (C.102) hold for any additive deposit, integer or not. Every term on the right refers to products of lower energy than x, so a single sweep upward in x solves both equations, starting from the terminal states, where $\bar{Y}$ and $\overline{Y^2}$ are known. The variance is $\overline{Y^2} - \bar{Y}^2$, and covariances of two deposits follow from the second moment of their sum.

C.11.4 Renewal theory

For deposits large compared with the gap, the mean pair number is linear in the deposited energy. Strip the cascade to a single chain: a quantity E, the energy, is reduced by independent losses $X_1, X_2, \ldots$ with one distribution F_{loss} on $x > 0$, mean $\langle X\rangle$ and second moment $\langle X^2\rangle$. Let $N(E)$ be the number of losses that fit, the largest n with $S_n = X_1 + \cdots + X_n \le E$, and $\bar{N}(E) = \langle N(E)\rangle$. Condition on the first loss $X_1 = x$: either $x > E$ and $N = 0$, or the chain restarts with $E - x$ and one more count. The two cases give the *renewal equation*

$$\bar{N}(E) = F_{\text{loss}}(E) + \int_0^E \bar{N}(E - x)\,\mathrm{d}F_{\text{loss}}(x). \tag{C.103}$$

Let $Z(E) = z(E) + \int_0^E Z(E - x)\,\mathrm{d}F_{\text{loss}}(x)$, with z nonnegative, nonincreasing and integrable (more generally, directly Riemann integrable).

Key renewal theorem

If F_{loss} is not concentrated on a lattice $\{0, d, 2d, \ldots\}$ and has finite mean, then

$$\lim_{E\to\infty} Z(E) = \frac{1}{\langle X\rangle}\int_0^\infty z(y)\,\mathrm{d}y. \tag{C.104}$$

The proof is in Ref. [18, Chap. XI.1].

We first apply the theorem to the energy left over at the end, the *undershoot* $E_{\text{res}}(E) = E - S_{N(E)}$, the part of E that no further loss fits into. Fix $y \ge 0$ and let $Z(E) = P(E_{\text{res}}(E) > y)$. If the first loss already exceeds E, the residual is E itself, which exceeds y when $E > y$; otherwise the chain restarts with $E - x$. So $Z = z + Z * F_{\text{loss}}$ with $z(E) = \Theta(E - y)\,[1 - F_{\text{loss}}(E)]$, nonincreasing beyond y. By Eq. (C.104),

$$\lim_{E\to\infty} P(E_{\text{res}} > y) = \frac{1}{\langle X\rangle}\int_y^\infty [1 - F_{\text{loss}}(x)]\,\mathrm{d}x, \qquad \langle E_{\text{res}}\rangle \to \frac{\langle X^2\rangle}{2\langle X\rangle}. \tag{C.105}$$

The limiting density of the residual is $[1 - F_{\text{loss}}(y)]/\langle X\rangle$, and the mean residual follows by integrating y against this density by parts: $\int_0^\infty y[1 - F_{\text{loss}}(y)]\,\mathrm{d}y = \frac{1}{2}\int_0^\infty y^2\,\mathrm{d}F_{\text{loss}}(y) = \frac{1}{2}\langle X^2\rangle$. A residual is not a typical loss: long losses cover more of the energy axis, so the gap in which the chain ends is biased towards them.

The mean count follows from energy bookkeeping. With the convention $S_0 = 0$, the loss that first overshoots E is X_{N+1}, and $N + 1$ is a stopping time: whether it has occurred is decided by $X_1, \ldots, X_{N+1}$. Wald's identity then gives $\langle S_{N+1} \rangle = \langle X \rangle (\bar{N} + 1)$ [18]. The overshoot $S_{N+1} - E$ obeys a renewal equation of the same kind: if the first loss exceeds E, the overshoot is $X_1 - E$, which exceeds y with probability $1 - F_{\mathrm{loss}}(E + y)$; otherwise the chain restarts. With $z(E) = 1 - F_{\mathrm{loss}}(E + y)$, Eq. (C.104) gives the limit $\langle X \rangle^{-1} \int_y^\infty [1 - F_{\mathrm{loss}}(x)]\, \mathrm{d}x$, the law of Eq. (C.105), with mean $\langle X^2 \rangle / (2\langle X \rangle)$. Hence

$$\bar{N}(E) = \frac{E}{\langle X \rangle} + \frac{\langle X^2 \rangle}{2\langle X \rangle^2} - 1 + o(1). \tag{C.106}$$

The leading term is the linear yield, $\bar{N}(E)/E \to 1/\langle X \rangle$; in the cascade the mean loss per pair is $\varepsilon_{\mathrm{eh}}$, and Sect. 18.2 derives $\bar{n}(\omega) = \omega/\varepsilon_{\mathrm{eh}} + o(\omega)$ in this way. The fluctuations follow from the central limit theorem for S_n. $N(E) < n$ means $S_n > E$, and S_n is approximately Gaussian with mean $n\langle X \rangle$ and variance $n\sigma_X^2$, $\sigma_X^2 = \langle X^2 \rangle - \langle X \rangle^2$. Setting $n = E/\langle X \rangle + u\,(E\sigma_X^2/\langle X \rangle^3)^{1/2}$ and keeping the leading order in E gives $P(N < n) = P(S_n > E) \to \Phi(u)$, the standard normal distribution function. So $N(E)$ is Gaussian at large E with

$$\mathrm{Var}\, N(E) \simeq \frac{\sigma_X^2}{\langle X \rangle^3}\, E, \qquad \frac{\mathrm{Var}\, N}{\bar{N}} \to \frac{\sigma_X^2}{\langle X \rangle^2}. \tag{C.107}$$

The Fano factor of a single renewal chain is the squared relative width of one loss. A deterministic loss gives zero, and an exponential loss, for which $\sigma_X = \langle X \rangle$, gives the Poisson value 1. For losses uniform on $[0, 1]$, $\langle X \rangle = \frac{1}{2}$ and $\langle X^2 \rangle = \frac{1}{3}$. Eqs. (C.105) to (C.107) then give $\langle E_{\mathrm{res}} \rangle = \frac{1}{3}$, $\bar{N} - 2E = -\frac{1}{3}$ and a Fano factor $\frac{1}{3}$. A numerical solution of Eq. (C.103) at $E = 20$ reproduces the offset to 10^{-6}, and a simulation of 2×10^5 chains at $E = 50$ reproduces $\langle E_{\mathrm{res}} \rangle$ and the Fano factor to 1 %.

Each hypothesis of the key renewal theorem fails somewhere in a cascade. A stationary loss law, independent of E, fails when the losses depend on the carrier energy; the theorem then applies only to the high-energy part of the chain. A non-lattice law fails when only a few loss values occur. For a deterministic loss d, $N(E) = \lfloor E/d \rfloor$ and $\bar{N} - E/d = -\{E/d\}$, the negative fractional part, which oscillates forever between -1 and 0. Its average over a period, $-\frac{1}{2}$, is what Eq. (C.106) predicts with $\langle X^2 \rangle = d^2$, but the limit does not exist. A model whose losses are integer combinations of a gap and one phonon energy has a lattice loss law, and its yield carries a periodic term. A finite second moment fails for a fast primary that loses energy in knock-on collisions with a tail $\mathrm{d}F_{\mathrm{loss}}/\mathrm{d}x \propto x^{-2}$ up to $E/2$. Then $\langle X^2 \rangle$ grows in proportion to E, the residual of Eq. (C.105) grows without bound, and the corrections to linearity can be logarithmic in E rather than constant.

C.12 Linear programming and duality

Every rate of the book is linear in the dark-matter velocity distribution: double the number of particles moving at one velocity, and their contribution to every bin doubles. Sect. 21.2 needs a limit that should hold for every velocity distribution. Because every rate is linear, this limit is the largest value of a linear quantity over all nonnegative distributions that keep the predicted counts below the observed bounds. The maximization is a linear program with infinitely many variables. Any feasible dual solution certifies an upper bound, the optimum is attained by a halo made of a few streams, and a grid solution errs by an amount that a Lipschitz constant bounds.

C.12.1 Finite linear programs

A linear program in standard form maximizes $\mathbf{c}^\mathsf{T}\mathbf{x}$ over vectors $\mathbf{x} \in \mathbb{R}^M$ with $\mathbf{x} \geq 0$ and $\mathbf{A}\mathbf{x} \leq \mathbf{b}$, where $\mathbf{A}$ is an $n \times M$ matrix and inequalities between vectors hold componentwise. Its *dual* minimizes $\mathbf{b}^\mathsf{T}\boldsymbol{\lambda}$ over $\boldsymbol{\lambda} \in \mathbb{R}^n$ with $\boldsymbol{\lambda} \geq 0$ and $\mathbf{A}^\mathsf{T}\boldsymbol{\lambda} \geq \mathbf{c}$. For any feasible pair,

$$\mathbf{c}^\mathsf{T}\mathbf{x} \leq (\mathbf{A}^\mathsf{T}\boldsymbol{\lambda})^\mathsf{T}\mathbf{x} = \boldsymbol{\lambda}^\mathsf{T}(\mathbf{A}\mathbf{x}) \leq \boldsymbol{\lambda}^\mathsf{T}\mathbf{b}. \tag{C.108}$$

The first inequality uses $\mathbf{x} \geq 0$ and $\mathbf{A}^\mathsf{T}\boldsymbol{\lambda} \geq \mathbf{c}$, the second $\boldsymbol{\lambda} \geq 0$ and $\mathbf{A}\mathbf{x} \leq \mathbf{b}$. Eq. (C.108) is *weak duality*: every dual-feasible $\boldsymbol{\lambda}$ bounds the primal maximum from above, whether or not anyone has found the maximum. *Strong duality*, the equality of the two optima, holds for finite programs whenever the primal and the dual are both feasible.

The optimum of a finite program is attained at a point with at most n nonzero components, one per constraint. Suppose an optimal $\mathbf{x}$ has more than n positive components. The corresponding columns of $\mathbf{A}$, more than n vectors in $\mathbb{R}^n$, are linearly dependent, so there is a nonzero $\mathbf{d}$, supported on those components, with $\mathbf{A}\mathbf{d} = 0$. Then $\mathbf{x} \pm t\mathbf{d}$ satisfies every constraint for small $t > 0$, and since $\mathbf{x}$ is optimal, $\mathbf{c}^\mathsf{T}\mathbf{d} = 0$. Increasing t until the first component of $\mathbf{x} + t\mathbf{d}$ or $\mathbf{x} - t\mathbf{d}$ reaches zero yields another optimal point with one positive component fewer. Repeating the step leaves at most n positive components.

C.12.2 Semi-infinite programs

Now let the variable be a nonnegative measure Ξ on a compact set $\mathcal{B}$, which for a halo-independent limit is the set of velocities $\boldsymbol{w}$ with $|\boldsymbol{w}| \leq v_\mathrm{esc}$ in the galactic frame. The number of constraints stays finite. Bin i of an experiment predicts $\int \mathsf{k}_i \, \mathrm{d}\Xi$ events, with a continuous kernel $\mathsf{k}_i(\boldsymbol{w}) \geq 0$, the count from a single stream of unit density at $\boldsymbol{w}$; the observations allow at most $N_i^\mathrm{up} > 0$. The quantity to bound is $\int \mathsf{f} \, \mathrm{d}\Xi$ with a continuous $\mathsf{f} \geq 0$: a halo integral above some speed, or the rate in a second experiment. The program and its dual are

$$\begin{aligned} \Phi_\mathrm{max} &= \sup_{\Xi \geq 0} \Big\{ \int_\mathcal{B} \mathsf{f} \, \mathrm{d}\Xi \;:\; \int_\mathcal{B} \mathsf{k}_i \, \mathrm{d}\Xi \leq N_i^\mathrm{up},\ i = 1, \ldots, n \Big\}, \\ &\min_{\boldsymbol{\lambda} \geq 0} \Big\{ \sum_i \lambda_i N_i^\mathrm{up} \;:\; \mathsf{g}_\lambda(\boldsymbol{w}) \geq 0 \quad \forall \boldsymbol{w} \in \mathcal{B} \Big\}, \end{aligned} \tag{C.109}$$

with $\mathsf{g}_\lambda(\boldsymbol{w}) = \sum_i \lambda_i \mathsf{k}_i(\boldsymbol{w}) - \mathsf{f}(\boldsymbol{w})$. The primal has infinitely many variables, the values of Ξ everywhere on $\mathcal{B}$; the dual has infinitely many constraints, one per point. The primal and the dual are therefore called *semi-infinite* programs.

Weak duality carries over in one line. For feasible Ξ and $\boldsymbol{\lambda}$,

$$\int \mathsf{f} \, \mathrm{d}\Xi \leq \int \sum_i \lambda_i \mathsf{k}_i \, \mathrm{d}\Xi = \sum_i \lambda_i \int \mathsf{k}_i \, \mathrm{d}\Xi \leq \sum_i \lambda_i N_i^\mathrm{up}, \tag{C.110}$$

by $\mathsf{g}_\lambda \geq 0$ and $\Xi \geq 0$, then by $\lambda_i \geq 0$ and the primal constraints. A vector $\boldsymbol{\lambda}$ with $\mathsf{g}_\lambda \geq 0$ on all of $\mathcal{B}$ is a *dual certificate*: it proves $\Phi_\mathrm{max} \leq \sum_i \lambda_i N_i^\mathrm{up}$ for every halo at once. Checking a certificate requires only the evaluation of one function on $\mathcal{B}$, so a halo-independent bound can be verified independently of the optimizer that produced it.

Strong duality needs hypotheses. Let $\mathcal{B}$ be compact, k_i and f continuous, and let every point where $\mathsf{f} > 0$ be seen by some bin, $\kappa_\mathrm{v} \equiv \min_{\boldsymbol{w} \in \operatorname{supp} \mathsf{f}} \max_i \mathsf{k}_i(\boldsymbol{w}) > 0$. Mass outside $\operatorname{supp} \mathsf{f}$ does not raise the objective and, since $\mathsf{k}_i \geq 0$, only uses up the bounds, so it can be removed. On $\operatorname{supp} \mathsf{f}$, $\sum_i \mathsf{k}_i \geq \kappa_\mathrm{v}$, and the constraints give

$$\kappa_\mathrm{v} \, \Xi(\operatorname{supp} \mathsf{f}) \leq \int \sum_i \mathsf{k}_i \, \mathrm{d}\Xi \leq \sum_i N_i^\mathrm{up} \equiv N_\mathrm{tot}, \tag{C.111}$$

so the relevant halos have bounded total mass. The feasible set is then compact in the weak sense in which $\int \mathsf{f}\,\mathrm{d}\Xi$ is continuous, and $\Xi = 0$ satisfies every constraint strictly (the Slater condition). Under these conditions the two optima of Eq. (C.109) are equal and both are attained [19].

C.12.3 Extreme points and streams

The optimal halo is a sum of at most n streams, $\Xi^* = \sum_{a=1}^{n'} \xi_a\, \delta_{\boldsymbol{w}_a}$ with $n' \le n$ and $\xi_a > 0$. On a finite grid of velocities the bound on the number of streams is the statement proved above for finite programs, with one variable per grid point. On the continuum, Rogosinski's theorem applied to the $n+1$ functionals $\mathsf{k}_1, \dots, \mathsf{k}_n, \mathsf{f}$ replaces any optimal Ξ by a measure with the same counts and objective on at most $n+1$ points [20], and the finite argument above, applied to those points, removes one more. The positions of the streams follow from equality in Eq. (C.110), which strong duality enforces at the optimum. The first inequality is an equality only if Ξ^* puts no mass where $\mathsf{g}_\lambda > 0$: the streams lie at zeros of g_λ, which, since $\mathsf{g}_\lambda \ge 0$, are its minima. The second inequality is an equality only if $\lambda_i = 0$ for every bin whose bound is not saturated. The two equality conditions are the *complementary-slackness* conditions, and they make the optimum checkable: the streams touch the dual function from below, and only saturated bins carry weight.

With one bin the problem is solved in closed form. Then $\mathsf{g}_\lambda \ge 0$ means $\lambda_1 \ge \mathsf{f}(\boldsymbol{w})/\mathsf{k}_1(\boldsymbol{w})$ wherever $\mathsf{f} > 0$, the smallest certificate is $\lambda_1 = \sup_{\boldsymbol{w}} \mathsf{f}/\mathsf{k}_1$, and the dual value is $N_1^{\mathrm{up}} \sup_{\boldsymbol{w}} \mathsf{f}/\mathsf{k}_1$. A single stream at the maximizer $\boldsymbol{w}_*$, with $\xi_1 = N_1^{\mathrm{up}}/\mathsf{k}_1(\boldsymbol{w}_*)$, saturates the bin and achieves the same value. Hence

$$\Phi_{\max} = N_1^{\mathrm{up}} \sup_{\boldsymbol{w} \in \mathcal{B}} \frac{\mathsf{f}(\boldsymbol{w})}{\mathsf{k}_1(\boldsymbol{w})} \qquad (n = 1). \tag{C.112}$$

The bound is set by the velocity at which the experiment is least sensitive relative to the quantity being bounded; every other halo does better. For a target whose response depends only on speed, k_1 depends only on $|\boldsymbol{w}|$, and Eq. (C.112) becomes the familiar statement that a halo-independent bound is set by the single speed shell that the bin sees worst.

C.12.4 Discretization error

In practice the program is solved on a grid $\mathcal{B}_h \subset \mathcal{B}$ of directions and speeds, with covering radius h: every $\boldsymbol{w} \in \mathcal{B}$ lies within h of a grid point. The grid gives a lower and an upper bound on $\Phi_{\max}$. Restricting Ξ to the grid shrinks the primal feasible set, so the grid primal optimum Φ_h obeys $\Phi_h \le \Phi_{\max}$. The grid dual solution $\boldsymbol{\lambda}_h$ satisfies $\mathsf{g}_{\lambda_h} \ge 0$ only on $\mathcal{B}_h$, and is no certificate on $\mathcal{B}$. The violation of $\mathsf{g}_{\lambda_h} \ge 0$ between grid points is bounded by the Lipschitz constant L_h of g_{λ_h}: for any $\boldsymbol{w}$ and its nearest grid point $\boldsymbol{w}_h$, $\mathsf{g}_{\lambda_h}(\boldsymbol{w}) \ge \mathsf{g}_{\lambda_h}(\boldsymbol{w}_h) - L_h|\boldsymbol{w} - \boldsymbol{w}_h| \ge -L_h h$. Repeating Eq. (C.110) with $\mathsf{f} \le \sum_i \lambda_i \mathsf{k}_i + L_h h$ on $\operatorname{supp}\mathsf{f}$, and bounding the mass there with Eq. (C.111),

$$\Phi_h \;\le\; \Phi_{\max} \;\le\; \sum_i \lambda_{h,i} N_i^{\mathrm{up}} + \frac{L_h\, h\, N_{\mathrm{tot}}}{\kappa_{\mathrm{v}}}. \tag{C.113}$$

The gap between the lower and the upper bound is the discretization error, and it is computable: L_h follows from the derivatives of the kernels and κ_{v} from their values. It falls linearly with h, and faster in practice, because the optimal streams lie at minima of g_λ, where the function is flat. Refining the grid only near the few streams of the grid optimum, and checking $\mathsf{g}_\lambda \ge 0$ on the refined grid, is the standard way to close the gap. A bound reported from a grid computation is conservative only if it uses the right side of Eq. (C.113).

The positivity of the kernels was used twice, to discard mass outside $\operatorname{supp}\mathsf{f}$ and in Eq. (C.111). If a truncated response makes some k_i negative at some velocities, Eq. (C.110)

still holds for every feasible pair, but the mass bound fails, and with it the control of Eq. (C.113); Sect. 21.2 bounds the resulting leakage separately.

C.13 Interval orders

Two materials whose figures of merit have overlapping 68% intervals cannot be put in order, and a third may be resolved from one of them but not from the other. Sect. 17.6 groups materials into tiers that respect exactly what the intervals resolve. The construction uses an interval order, a partial order built from intervals, and the heights of its elements.

C.13.1 Partial orders from intervals

A *strict partial order* $\succ$ on a finite set is a relation that is irreflexive (never $a \succ a$) and transitive ($a \succ b$ and $b \succ c$ imply $a \succ c$). Together the two properties imply asymmetry: $a \succ b$ and $b \succ a$ would give $a \succ a$. Two elements with neither $a \succ b$ nor $b \succ a$ are *incomparable*. A partial order differs from a ranking in that incomparability is allowed and is not transitive.

Let each element a carry an interval $I_a = [\ell_a, u_a]$, oriented so that larger values are better; for a figure of merit where smaller is better, use its negative logarithm. Define

$$a \succ b \quad \Longleftrightarrow \quad \ell_a > u_b, \tag{C.114}$$

that is, a is resolved above b when the whole interval of a lies above the whole interval of b. The relation is irreflexive, since $\ell_a \le u_a$, and transitive: $\ell_a > u_b \ge \ell_b > u_c$ gives $\ell_a > u_c$. It is therefore a strict partial order, called an *interval order*. Two elements are incomparable exactly when their intervals overlap.

Incomparability is not transitive. With $I_a = [3, 5]$, $I_b = [2, 4]$ and $I_c = [1, 2.5]$, a overlaps b and b overlaps c, but $\ell_a = 3 > 2.5 = u_c$, so $a \succ c$. A ranking that declares a tied with b and b tied with c must then either tie a with c, which discards a resolved difference, or break one of the ties, which asserts a difference the intervals do not show.

Interval orders have one property that general partial orders lack: if $a \succ b$ and $c \succ d$, then $a \succ d$ or $c \succ b$. To prove the property, suppose $a \not\succ d$, so $\ell_a \le u_d$. Then $\ell_c > u_d \ge \ell_a > u_b$, and $c \succ b$. Conversely, every finite partial order with this property can be represented by intervals as in Eq. (C.114) [21]. When all intervals have the same length the order is called a semiorder; intervals of different widths, which arise when inputs of different provenance carry different uncertainties, give the general case.

C.13.2 Heights as tiers

A *chain* above a is a sequence $b_1 \succ b_2 \succ \cdots \succ b_k \succ a$. The *height* of a is

$$h(a) = 1 + (\text{length } k \text{ of the longest chain above } a), \tag{C.115}$$

so that $h(a) = 1$ when nothing is resolved above a. The *tier* k is the set of elements of height k.

Each tier is an *antichain*: no two of its members are comparable. If $a \succ b$, any chain above a extended by a is a chain above b, one element longer, so $h(b) \ge h(a) + 1$ and a, b lie in different tiers. Members of one tier are therefore unresolved from each other, and calling them tied within the tier is exact. Every member a of tier $k > 1$ is resolved below at least one member of tier $k - 1$. A longest chain above a has $k - 1$ elements, and its last element b has height exactly $k - 1$. The chain above b formed by the first $k - 2$ elements shows $h(b) \ge k - 1$, while a longer chain above b would extend through b to a chain above a of more than $k - 1$ elements.

For interval orders the heights are computed by sorting. Process the elements in decreasing order of u_a. When a is reached, every b with $b \succ a$ has $\ell_b > u_a$, so $u_b \geq \ell_b > u_a$ and b has already been processed; then $h(a) = 1 + \max\{h(b) : \ell_b > u_a\}$, with the maximum over an empty set equal to zero. The cost is at most quadratic in the number of elements.

C.13.3 Interpretation of tiers

Take four materials, with intervals $I_A = [8, 10]$, $I_B = [5, 9]$, $I_C = [6, 7]$ and $I_D = [1, 9.5]$. Nothing lies above A, B or D, since no lower end exceeds their upper ends ($\ell_A = 8 < 9 \leq u_B, u_D$, and $\ell_B, \ell_C, \ell_D < 9$), so all three have height 1. Only A is resolved above C ($8 > 7$), so C has height 2. The tiers are $\{A, B, D\}$ and $\{C\}$. Material D, whose interval is the widest, is in the first tier although its midpoint, 5.25, is below the midpoint of C, 6.5: nothing is resolved above D. C is in the second tier although B, which is in the first tier, is not resolved above C.

A tier does not assert that every member of a higher tier beats every member of a lower one: B does not beat C. Nor does it assert that a member of a higher tier is likely to be better than one of a lower tier, since a wide interval rises to the first tier because it is wide. Tiers also depend on the confidence level of the intervals and on the widths assigned to the inputs; at a higher level, the intervals widen and the tiers merge.

C.14 Likelihoods, intervals and profiling

A search that sees no events with no expected background excludes, at 90 % confidence, every signal expectation above $N_{90} = 2.303$ (Eq. (1.51)). Real searches add backgrounds, several bins and uncertain inputs, and Sects. 21.4 and 21.5 handle them with likelihoods: the Poisson and Gaussian likelihoods, the likelihood ratio and the intervals it defines, nuisance parameters and their profiling, and, in Sect. 17.6, Monte Carlo propagation of input widths. A confidence level keeps the meaning given in Sect. 1.5: a statement about the procedure in repeated experiments.

C.14.1 Poisson and Gaussian likelihoods

The *likelihood* L is the probability of the observed data, read as a function of the parameters. A counting experiment that observes N_{obs} events, with expected signal N_{sig} and background N_{bkg}, has by Eq. (1.50)

$$\mathsf{L}(N_{\text{sig}}) = \frac{(N_{\text{sig}} + N_{\text{bkg}})^{N_{\text{obs}}}}{N_{\text{obs}}!}\, \mathrm{e}^{-(N_{\text{sig}}+N_{\text{bkg}})}. \tag{C.116}$$

Independent bins multiply, so a spectrum in bins j has the product of factors of this form, one per bin, with expectations $N_{\text{sig},j} + N_{\text{bkg},j}$ [22, Eq. (6)]. Only ratios of likelihoods enter what follows, and factors independent of the parameters, such as $1/N_{\text{obs}}!$, drop out.

When a parameter is measured with a Gaussian error, for example an estimate $\hat{N}$ of N_{sig} with standard deviation $\sigma_{\hat{N}}$ from many events, the likelihood is Gaussian,

$$\mathsf{L}(N_{\text{sig}}) \propto \exp\Big[-\frac{(\hat{N} - N_{\text{sig}})^2}{2\sigma_{\hat{N}}^2}\Big]. \tag{C.117}$$

The Poisson likelihood takes this form at large counts. Expanding $\ln \mathsf{L}$ of Eq. (C.116) to second order about its maximum, at $N_{\text{sig}} + N_{\text{bkg}} = N_{\text{obs}}$, gives $\ln \mathsf{L} = \text{const} - (N_{\text{sig}} + N_{\text{bkg}} - N_{\text{obs}})^2/(2N_{\text{obs}})$, a Gaussian of width $N_{\text{obs}}^{1/2}$. The cubic term of the expansion is smaller than

the quadratic term by a factor of order $N_{\text{obs}}^{-1/2}$ at one standard deviation from the maximum, so the Gaussian form needs tens of events.

C.14.2 The likelihood ratio and intervals

The *likelihood ratio* compares a hypothesized value with the best one,

$$\mathsf{t}(N_{\text{sig}}) = -2 \ln \frac{\mathsf{L}(N_{\text{sig}})}{\mathsf{L}(\hat{N}_{\text{sig}})} \geq 0, \tag{C.118}$$

where $\hat{N}_{\text{sig}}$ maximizes L [22, Eq. (8)]. Large t means poor agreement. For the Gaussian likelihood Eq. (C.117), $\hat{N}_{\text{sig}} = \hat{N}$ and $\mathsf{t} = (\hat{N} - N_{\text{sig}})^2/\sigma^2_{\hat{N}}$ exactly. If the data are repeated, $\hat{N}$ scatters as a Gaussian about the true value, and t at the true value is the square of a standard normal variable, a χ^2 variable with one degree of freedom. The set of N_{sig} with $\mathsf{t}(N_{\text{sig}}) \leq z^2$, the interval $\hat{N} \pm z\sigma_{\hat{N}}$, then covers the true value with probability $2\Phi(z) - 1$, with Φ the standard normal distribution function. For a likelihood that is not Gaussian, Wilks' theorem states that t at the true value approaches the same χ^2 distribution as the sample grows [23], [22, Sect. 3.1], and the same rule applies. The thresholds are

$$\mathsf{t} \leq 1.00,\ 2.71,\ 3.84 \qquad \text{for } 68.27\,\%,\ 90\,\%,\ 95\,\% \text{ (two-sided)}, \tag{C.119}$$

the squares of $z = 1$, 1.645 and 1.960. A 1σ interval is the 68.27 % interval, written "68 %" throughout the book.

An upper limit is one-sided. Data with $\hat{N}_{\text{sig}}$ above the tested value are not counted as evidence against it, so the statistic is set to zero there [22, Eq. (14)], and all of the rejected probability lies on one side. For a Gaussian likelihood the upper limit at confidence level CL is $\hat{N} + z_{\text{CL}}\sigma_{\hat{N}}$ with $\Phi(z_{\text{CL}}) = \text{CL}$: $z = 1.282$ at 90 % and $z = 1.645$ at 95 %, which correspond to $\mathsf{t} = 1.64$ and 2.71. The same number, 2.71, thus marks a two-sided 90 % interval and a one-sided 95 % limit.

The asymptotic rule fails at small counts, the regime of dark-matter limits. With $N_{\text{obs}} = 0$ and $N_{\text{bkg}} = 0$, Eq. (C.116) gives $\mathsf{L} = \mathrm{e}^{-N_{\text{sig}}}$, maximal at $\hat{N}_{\text{sig}} = 0$, and $\mathsf{t} = 2N_{\text{sig}}$. The one-sided 90 % threshold 1.64 would give $N_{\text{sig}} < 0.82$, while the exact construction of Sect. 1.5 gives 2.303. Limits from a few events must therefore be built from the exact Poisson probabilities, by the classical construction of Sect. 1.5 or by its generalization that orders outcomes by the likelihood ratio and yields one- or two-sided intervals as the data demand [24].

C.14.3 Nuisance parameters and the profile likelihood

The predicted counts depend on quantities of no direct interest that are known only approximately: a background normalization, an efficiency, or the size of a truncation in the theory (Sect. 8.8). Each becomes a *nuisance parameter* ϑ in the likelihood, usually with a constraint term that encodes what is known about it from a calibration or from a computed next order. The *profile likelihood* replaces each nuisance parameter by its best value at fixed N_{sig},

$$\mathsf{t}_{\text{p}}(N_{\text{sig}}) = -2 \ln \frac{\mathsf{L}\big(N_{\text{sig}}, \hat{\hat{\vartheta}}(N_{\text{sig}})\big)}{\mathsf{L}(\hat{N}_{\text{sig}}, \hat{\vartheta})}, \tag{C.120}$$

where $\hat{\hat{\vartheta}}(N_{\text{sig}})$ maximizes L at fixed N_{sig} and the denominator is the overall maximum [22, Eq. (7)]. Wilks' theorem holds for t_{p} as well, with one degree of freedom per parameter of interest, so Eq. (C.119) still applies.

A Gaussian example shows what profiling does. Let the estimate be shifted by an unknown offset, so that $\hat{N}$ is Gaussian with mean $N_{\mathrm{sig}}+\vartheta$ and width $\sigma_{\hat{N}}$, and let a calibration constrain the offset by a Gaussian of mean 0 and width σ_ϑ. Then

$$-2\ln \mathsf{L}(N_{\mathrm{sig}},\vartheta) = \frac{(\hat{N}-N_{\mathrm{sig}}-\vartheta)^2}{\sigma_{\hat{N}}^2} + \frac{\vartheta^2}{\sigma_\vartheta^2} + \text{const.}$$

Write $r=\hat{N}-N_{\mathrm{sig}}$. Setting the derivative with respect to ϑ to zero, $-(r-\vartheta)/\sigma_{\hat{N}}^2+\vartheta/\sigma_\vartheta^2=0$, gives $\hat{\hat{\vartheta}} = r\sigma_\vartheta^2/(\sigma_{\hat{N}}^2+\sigma_\vartheta^2)$. With $\hat{\hat{\vartheta}}$ inserted, the two terms become $r^2\sigma_{\hat{N}}^2/(\sigma_{\hat{N}}^2+\sigma_\vartheta^2)^2$ and $r^2\sigma_\vartheta^2/(\sigma_{\hat{N}}^2+\sigma_\vartheta^2)^2$, and their sum is $r^2/(\sigma_{\hat{N}}^2+\sigma_\vartheta^2)$. The overall minimum is zero, at $N_{\mathrm{sig}}=\hat{N}$ and $\vartheta=0$. Hence

$$\mathsf{t}_{\mathrm{p}}(N_{\mathrm{sig}}) = \frac{(\hat{N}-N_{\mathrm{sig}})^2}{\sigma_{\hat{N}}^2+\sigma_\vartheta^2}. \tag{C.121}$$

The profiled interval is the interval of a Gaussian of width $(\sigma_{\hat{N}}^2+\sigma_\vartheta^2)^{1/2}$: statistical and systematic widths add in quadrature. A nuisance parameter that scales the normalization, $N_{\mathrm{sig}}\to N_{\mathrm{sig}}(1+\vartheta)$, gives the same result for the relative widths to first order in ϑ. Profiling broadens the likelihood, and the broadening is the information about N_{sig} lost to the nuisance parameter. When the data constrain a nuisance parameter better than the calibration does, profiling pulls the parameter towards the value the data prefer; a fixed-width envelope would miss this pull.

C.14.4 Monte Carlo propagation

When a derived quantity depends on many inputs through a formula that is not linear, its distribution is obtained by sampling. Each input x_j is given a distribution, for example lognormal, with $\ln x_j$ Gaussian of mean $\ln x_j^{(0)}$ and width σ_j, a width set by how the input was obtained. We draw N_{MC} sets of inputs, evaluate the output y on each, and read off percentiles. The 16th and 84th percentiles bound the central 68 % interval, because $\Phi(-1)=0.159$; for a Gaussian output they are the mean minus and plus one standard deviation, and for any output they define an interval of the same coverage that need not be symmetric.

Comparisons follow from the same draws. The probability that material A outperforms B is the fraction of draws in which $y_A>y_B$. For lognormal outputs with log-widths σ_A, σ_B and correlation r between their errors, $\ln y_A-\ln y_B$ is Gaussian with mean $\Delta=\ln y_A^{(0)}-\ln y_B^{(0)}$ and variance $\sigma_A^2+\sigma_B^2-2r\sigma_A\sigma_B$, and

$$P(A\succ B) = \Phi\Big(\frac{\Delta}{(\sigma_A^2+\sigma_B^2-2r\sigma_A\sigma_B)^{1/2}}\Big). \tag{C.122}$$

With $\Delta=0.3$, $\sigma_A=0.2$, $\sigma_B=0.25$ and $r=0$, $P(A\succ B)=\Phi(0.94)=0.83$. A positive correlation shrinks the variance of the difference and sharpens the comparison; it arises when both materials are computed with the same method and share its error. A ratio of two positively correlated outputs is more certain than either. The sampling error of an estimated probability p is $[p(1-p)/N_{\mathrm{MC}}]^{1/2}$, at most 0.0035 for $N_{\mathrm{MC}}=2\times10^4$. All such probabilities are conditional on the input distributions: they propagate the assigned widths, but they cannot be more reliable than the widths themselves.

C.15 Dirac equation and Foldy–Wouthuysen reduction

A 1s electron of germanium moves at $Z\alpha=0.23$ of the speed of light, and its relativistic corrections are a few percent of its binding energy. Sect. 10.6 bounds what relativistic

corrections do to the response at large q, Sect. 12.5 needs the electron density at a nucleus with its relativistic correction, and Sect. 10.9 writes the spinor algebra of a fermion absorbed on an electron. All three start from the Dirac equation and its expansion in $1/m_e$. We use natural units, Heaviside–Lorentz charges with the electron charge $-e$ (Appendix A.2) and the metric $(+,-,-,-)$ of Appendix C.9.

C.15.1 The Dirac equation in a potential

The free Dirac field ψ, a four-component spinor, obeys $(\mathrm{i}\gamma^\mu\partial_\mu - m_e)\psi = 0$, with the Dirac matrices γ^μ satisfying $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$. We work in the Dirac representation,

$$\gamma^0 = \beta_\mathrm{D} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \gamma^i = \beta_\mathrm{D}\alpha^i_\mathrm{D}, \qquad \alpha^i_\mathrm{D} = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}, \qquad \gamma^5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{C.123}$$

in 2×2 blocks, with σ^i the Pauli matrices. The subscript D keeps $\boldsymbol{\alpha}_\mathrm{D}$ and β_D apart from the fine-structure constant and the inverse temperature. The matrices obey $(\alpha^i_\mathrm{D})^2 = \beta^2_\mathrm{D} = 1$, and any two of $\alpha^1_\mathrm{D}, \alpha^2_\mathrm{D}, \alpha^3_\mathrm{D}, \beta_\mathrm{D}$ anticommute.

The electron couples to the electromagnetic potential through $\mathcal{L} \supset -eA_\mu J^\mu_\mathrm{em}$ with $J^\mu_\mathrm{em} = -\bar\psi\gamma^\mu\psi$ for the electron (Eq. (C.83)), $\bar\psi = \psi^\dagger\gamma^0$. The Lagrangian density $\bar\psi(\mathrm{i}\gamma^\mu\partial_\mu - m_e)\psi + eA_\mu\bar\psi\gamma^\mu\psi$ gives the field equation $[\mathrm{i}\gamma^\mu(\partial_\mu - \mathrm{i}eA_\mu) - m_e]\psi = 0$. Multiplying by γ^0 and solving for the time derivative, with $\boldsymbol{p} = -\mathrm{i}\nabla$, $A_\mu = (A^0, -\boldsymbol{A})$ and $\gamma^0\gamma^i = \alpha^i_\mathrm{D}$,

$$\mathrm{i}\partial_t\psi = H_\mathrm{D}\psi, \qquad H_\mathrm{D} = \boldsymbol{\alpha}_\mathrm{D}\cdot\boldsymbol{\pi} + \beta_\mathrm{D}m_e + V, \qquad \boldsymbol{\pi} = \boldsymbol{p} + e\boldsymbol{A}, \quad V = -eA^0. \tag{C.124}$$

For an electron in the field of a nucleus of charge Ze, $A^0 = Ze/(4\pi r)$ and $V = -Z\alpha/r$, with $e^2 = 4\pi\alpha$. A particle of charge $+e$ has $\boldsymbol{\pi} = \boldsymbol{p} - e\boldsymbol{A}$ and $V = +eA^0$.

The free equation has plane-wave solutions $\psi = u\,\mathrm{e}^{-\mathrm{i}Et + \mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}$. With $u = (u_\mathrm{A}, u_\mathrm{B})$ written in upper and lower two-component blocks, Eq. (C.124) with $\boldsymbol{A} = 0$, $V = 0$ reads $(E - m_e)u_\mathrm{A} = \boldsymbol{\sigma}\cdot\boldsymbol{p}\,u_\mathrm{B}$ and $(E + m_e)u_\mathrm{B} = \boldsymbol{\sigma}\cdot\boldsymbol{p}\,u_\mathrm{A}$. With $(\boldsymbol{\sigma}\cdot\boldsymbol{p})^2 = p^2$, nonzero solutions need $E^2 = p^2 + m_e^2$. For $E = +(p^2 + m_e^2)^{1/2}$ and a two-component spinor ξ_s of spin s,

$$u(\boldsymbol{p}, s) = \sqrt{E + m_e}\begin{pmatrix} \xi_s \\ \dfrac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{E + m_e}\,\xi_s \end{pmatrix}, \qquad u^\dagger u = 2E, \qquad \bar u u = 2m_e. \tag{C.125}$$

The lower, or small, component is smaller than the upper by $p/(E + m_e) \simeq v/2$. The solutions with $E = -(p^2 + m_e^2)^{1/2}$ describe positrons in field theory; in a bound-state calculation they are the states that a no-pair projection removes. The relativistic 1s state of a point nucleus behaves as $r^{\gamma_\mathrm{D}-1}$ near the origin, with $\gamma_\mathrm{D} = [1 - (Z\alpha)^2]^{1/2}$ [25], so its density diverges at $r = 0$, weakly: $\gamma_\mathrm{D} - 1 = -0.005$ for silicon and -0.028 for germanium. A nucleus of finite size removes the divergence.

C.15.2 Bilinears and their non-relativistic limits

Couplings of dark matter to electrons are written with four bilinears, as in Sect. 1.4: the scalar $\bar\psi\psi$, the vector $\bar\psi\gamma^\mu\psi$, the axial vector $\bar\psi\gamma^\mu\gamma^5\psi$ and the pseudoscalar $\bar\psi\mathrm{i}\gamma^5\psi$. Their non-relativistic forms follow from $\psi = (\psi_\mathrm{A}, \psi_\mathrm{B})$ with the small component $\psi_\mathrm{B} \simeq (\boldsymbol{\sigma}\cdot\boldsymbol{p}/2m_e)\psi_\mathrm{A}$, the low-energy limit of Eq. (C.125). With the blocks of Eq. (C.123), $\gamma^0\gamma^0 = 1$, $\gamma^0\gamma^i = \alpha^i_\mathrm{D}$, $\gamma^0\gamma^i\gamma^5 = \mathrm{diag}(\sigma^i, \sigma^i)$ and $\gamma^0\mathrm{i}\gamma^5 = \mathrm{i}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so

$$\begin{aligned} \bar\psi\psi &= \psi^\dagger_\mathrm{A}\psi_\mathrm{A} - \psi^\dagger_\mathrm{B}\psi_\mathrm{B} \simeq \psi^\dagger_\mathrm{A}\psi_\mathrm{A}, & \bar\psi\gamma^0\psi &= \psi^\dagger_\mathrm{A}\psi_\mathrm{A} + \psi^\dagger_\mathrm{B}\psi_\mathrm{B} \simeq \psi^\dagger_\mathrm{A}\psi_\mathrm{A}, \\ \bar\psi\gamma^i\gamma^5\psi &= \psi^\dagger_\mathrm{A}\sigma^i\psi_\mathrm{A} + \psi^\dagger_\mathrm{B}\sigma^i\psi_\mathrm{B} \simeq \psi^\dagger_\mathrm{A}\sigma^i\psi_\mathrm{A}, & \bar\psi\mathrm{i}\gamma^5\psi &= \mathrm{i}(\psi^\dagger_\mathrm{A}\psi_\mathrm{B} - \psi^\dagger_\mathrm{B}\psi_\mathrm{A}) \simeq \frac{1}{2m_e}\nabla\cdot(\psi^\dagger_\mathrm{A}\boldsymbol{\sigma}\psi_\mathrm{A}). \end{aligned} \tag{C.126}$$

The last form uses $\boldsymbol{\sigma}\cdot\boldsymbol{p} = -\mathrm{i}\boldsymbol{\sigma}\cdot\nabla$: $\mathrm{i}[\psi_\mathrm{A}^\dagger(-\mathrm{i}\boldsymbol{\sigma}\cdot\nabla\psi_\mathrm{A}) - (-\mathrm{i}\boldsymbol{\sigma}\cdot\nabla\psi_\mathrm{A})^\dagger\psi_\mathrm{A}]/2m_e = [\psi_\mathrm{A}^\dagger\boldsymbol{\sigma}\cdot\nabla\psi_\mathrm{A} + (\nabla\psi_\mathrm{A}^\dagger)\cdot\boldsymbol{\sigma}\psi_\mathrm{A}]/2m_e$. The scalar and the time component of the vector both reduce to the number density, the limit used in Sect. 1.4 and Appendix C.9; they differ at order v^2, where the scalar acquires the kinetic-energy term of Sect. 1.4. The axial vector reduces to the spin density. The pseudoscalar is suppressed by $1/m_e$ and becomes the divergence of the spin density, so a coupling $g\phi\bar\psi\mathrm{i}\gamma^5\psi$ becomes, after integration by parts, $-(g/2m_e)(\nabla\phi)\cdot\psi_\mathrm{A}^\dagger\boldsymbol{\sigma}\psi_\mathrm{A}$, the first term of the axion-like coupling in Sect. 1.4. The spatial vector $\bar\psi\gamma^i\psi = \psi_\mathrm{A}^\dagger\sigma^i\psi_\mathrm{B} + \psi_\mathrm{B}^\dagger\sigma^i\psi_\mathrm{A}$ reduces, by $\sigma^i\sigma^j = \delta^{ij} + \mathrm{i}\epsilon^{ijk}\sigma^k$, to the convection current $(2\mathrm{i}m_e)^{-1}(\psi_\mathrm{A}^\dagger\nabla\psi_\mathrm{A} - \nabla\psi_\mathrm{A}^\dagger\,\psi_\mathrm{A})$ plus the spin current $(2m_e)^{-1}\nabla\times(\psi_\mathrm{A}^\dagger\boldsymbol{\sigma}\psi_\mathrm{A})$, the Gordon decomposition.

C.15.3 The Foldy–Wouthuysen transformation

The large and small components are coupled by $\boldsymbol{\alpha}_\mathrm{D}\cdot\boldsymbol{\pi}$, the only part of H_D that mixes them. Foldy and Wouthuysen removed $\boldsymbol{\alpha}_\mathrm{D}\cdot\boldsymbol{\pi}$ order by order in $1/m_e$ by unitary transformations [26]. Call an operator *even* if it commutes with β_D (block diagonal) and *odd* if it anticommutes (block off-diagonal). We take static fields and $\boldsymbol{A} = 0$, so that

$$H_\mathrm{D} = \beta_\mathrm{D}m_e + \mathsf{E} + \mathsf{O}, \qquad \mathsf{E} = V, \qquad \mathsf{O} = \boldsymbol{\alpha}_\mathrm{D}\cdot\boldsymbol{p}. \tag{C.127}$$

The rules are $\beta_\mathrm{D}\mathsf{O} = -\mathsf{O}\beta_\mathrm{D}$, $\beta_\mathrm{D}\mathsf{E} = \mathsf{E}\beta_\mathrm{D}$, and O^2, a product of two odd operators, is even. Let $X = \beta_\mathrm{D}\mathsf{O}/(2m_e)$, which is anti-Hermitian, so e^X is unitary, and transform $H' = \mathrm{e}^X H_\mathrm{D}\mathrm{e}^{-X} = H_\mathrm{D} + [X, H_\mathrm{D}] + \frac{1}{2!}[X,[X,H_\mathrm{D}]] + \frac{1}{3!}[X,[X,[X,H_\mathrm{D}]]] + \ldots$ (Appendix C.6). The commutators of $\beta_\mathrm{D}\mathsf{O}$ with the parts of H_D follow from these rules:

$$[\beta_\mathrm{D}\mathsf{O},\beta_\mathrm{D}] = \beta_\mathrm{D}\mathsf{O}\beta_\mathrm{D} - \mathsf{O} = -2\mathsf{O}, \qquad [\beta_\mathrm{D}\mathsf{O},\mathsf{O}] = \beta_\mathrm{D}\mathsf{O}^2 - \mathsf{O}\beta_\mathrm{D}\mathsf{O} = 2\beta_\mathrm{D}\mathsf{O}^2,$$
$$[\beta_\mathrm{D}\mathsf{O},\mathsf{E}] = \beta_\mathrm{D}[\mathsf{O},\mathsf{E}].$$

In each, β_D was moved to the left with its anticommutation or commutation rule. The first-order term is then

$$[X, H_\mathrm{D}] = -\mathsf{O} + \frac{\beta_\mathrm{D}\mathsf{O}^2}{m_e} + \frac{\beta_\mathrm{D}[\mathsf{O},\mathsf{E}]}{2m_e}.$$

Its first term cancels the odd operator of H_D; the cancellation fixes the choice of X. The commutator of X with each of the three new terms follows from the same rules, using that O^2 is even and $[\mathsf{O},\mathsf{E}]$ is odd: $[X,-\mathsf{O}] = -\beta_\mathrm{D}\mathsf{O}^2/m_e$, $[X,\beta_\mathrm{D}\mathsf{O}^2/m_e] = (\beta_\mathrm{D}\mathsf{O}\beta_\mathrm{D}\mathsf{O}^2 - \beta_\mathrm{D}\mathsf{O}^2\beta_\mathrm{D}\mathsf{O})/(2m_e^2) = -\mathsf{O}^3/m_e^2$, and $[X,\beta_\mathrm{D}[\mathsf{O},\mathsf{E}]/2m_e] = -[\mathsf{O},[\mathsf{O},\mathsf{E}]]/(4m_e^2)$. The same rules give the next orders, $[X,-\beta_\mathrm{D}\mathsf{O}^2/m_e] = \mathsf{O}^3/m_e^2$, $[X,-\mathsf{O}^3/m_e^2] = -(\beta_\mathrm{D}\mathsf{O}^4 - \mathsf{O}^3\beta_\mathrm{D}\mathsf{O})/(2m_e^3) = -\beta_\mathrm{D}\mathsf{O}^4/m_e^3$ and $[X,\mathsf{O}^3/m_e^2] = \beta_\mathrm{D}\mathsf{O}^4/m_e^3$, so that, up to terms of higher order in $1/m_e$, $\frac{1}{3!}[X,[X,[X,H_\mathrm{D}]]] = \mathsf{O}^3/(6m_e^2) - \beta_\mathrm{D}\mathsf{O}^4/(6m_e^3)$ and $\frac{1}{4!}[X,[X,[X,[X,H_\mathrm{D}]]]] = \beta_\mathrm{D}\mathsf{O}^4/(24m_e^3)$. Collecting,

$$H' = \beta_\mathrm{D}\Big(m_e + \frac{\mathsf{O}^2}{2m_e} - \frac{\mathsf{O}^4}{8m_e^3}\Big) + \mathsf{E} - \frac{[\mathsf{O},[\mathsf{O},\mathsf{E}]]}{8m_e^2} + \mathsf{O}', \qquad \mathsf{O}' = \frac{\beta_\mathrm{D}[\mathsf{O},\mathsf{E}]}{2m_e} - \frac{\mathsf{O}^3}{3m_e^2}. \tag{C.128}$$

The coefficients are $1 - \frac{1}{2} = \frac{1}{2}$ for $\beta_\mathrm{D}\mathsf{O}^2/m_e$, $-\frac{1}{6} + \frac{1}{24} = -\frac{1}{8}$ for $\beta_\mathrm{D}\mathsf{O}^4/m_e^3$, and $-\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$ for O^3/m_e^2.

The power counting is in the electron velocity v: $\mathsf{O}\sim p\sim m_e v$ and $\mathsf{E}\sim m_e v^2$. The even terms of Eq. (C.128) are of order m_e, $m_e v^2$ and $m_e v^4$, while the remaining odd operator O' is of order $m_e v^3$. A second transformation with $X' = \beta_\mathrm{D}\mathsf{O}'/(2m_e)$ removes O' in the same way. By the first-order formula it adds $\beta_\mathrm{D}\mathsf{O}'^2/(2m_e)$, of order $m_e v^6$, to the even part, and leaves an odd remainder $\beta_\mathrm{D}[\mathsf{O}',\mathsf{E}]/(2m_e)$ of order $m_e v^5$, which a third transformation removes with even changes of still higher order. To order $m_e v^4$, the even part of Eq. (C.128) is the final Hamiltonian, and it acts on the upper components alone.

C.15.4 Mass–velocity, Darwin and spin–orbit terms

The three operators of Eq. (C.128) are evaluated with the identity $\alpha_{\rm D}^i\alpha_{\rm D}^j = \delta^{ij} + \mathrm{i}\epsilon^{ijk}\Sigma^k$, where $\boldsymbol{\Sigma} = \mathrm{diag}(\boldsymbol{\sigma},\boldsymbol{\sigma})$ is twice the spin. The identity gives $\mathsf{O}^2 = p^2$ and $\mathsf{O}^4 = p^4$. For the double commutator, let $\boldsymbol{W} = \nabla V$. Then $[\mathsf{O}, V] = \alpha_{\rm D}^j[p^j, V] = -\mathrm{i}\alpha_{\rm D}^j W^j$, and

$$[\mathsf{O},[\mathsf{O},V]] = -\mathrm{i}\sum_{ij}(\alpha_{\rm D}^i\alpha_{\rm D}^j\, p^i W^j - \alpha_{\rm D}^j\alpha_{\rm D}^i\, W^j p^i).$$

With p^i moved to the right, $p^iW^j = W^jp^i - \mathrm{i}\,\partial_i W^j$, the bracket becomes $(\alpha_{\rm D}^i\alpha_{\rm D}^j - \alpha_{\rm D}^j\alpha_{\rm D}^i)W^jp^i - \mathrm{i}\alpha_{\rm D}^i\alpha_{\rm D}^j\partial_i W^j$. The antisymmetric combination is $2\mathrm{i}\epsilon^{ijk}\Sigma^k$, and the sum $\sum_{ij}\alpha_{\rm D}^i\alpha_{\rm D}^j\partial_iW^j = \nabla^2V + \mathrm{i}\boldsymbol{\Sigma}\cdot(\nabla\times\nabla V) = \nabla^2V$. With $\epsilon^{ijk}W^jp^i = -(\boldsymbol{W}\times\boldsymbol{p})^k$,

$$[\mathsf{O},[\mathsf{O},V]] = -\nabla^2V - 2\boldsymbol{\Sigma}\cdot(\nabla V\times\boldsymbol{p}). \tag{C.129}$$

On the upper components, with $\boldsymbol{\Sigma}\to\boldsymbol{\sigma} = 2\boldsymbol{S}$, the Hamiltonian to order m_ev^4 is therefore

$$H_{\rm FW} = m_e + \frac{p^2}{2m_e} + V - \frac{p^4}{8m_e^3} + \frac{\nabla^2V}{8m_e^2} + \frac{1}{4m_e^2}\,\boldsymbol{\sigma}\cdot(\nabla V\times\boldsymbol{p}). \tag{C.130}$$

The fourth term is the *mass–velocity* correction, the next term of $(p^2+m_e^2)^{1/2}$. The fifth is the *Darwin* term, a smearing of the potential over the Compton wavelength $1/m_e$. The last is the *spin–orbit* coupling; for a central potential, $\nabla V = (\boldsymbol{r}/r)\,\mathrm{d}V/\mathrm{d}r$ and $\boldsymbol{r}\times\boldsymbol{p} = \boldsymbol{L}$, so it equals $(2m_e^2)^{-1}(r^{-1}\mathrm{d}V/\mathrm{d}r)\,\boldsymbol{S}\cdot\boldsymbol{L}$. Eq. (C.130) is the result of Refs. [25, 26] at $\boldsymbol{A} = 0$.

C.15.5 The contact density at a nucleus

For the Coulomb potential $V = -Z\alpha/r$, the identity $\nabla^2(1/r) = -4\pi\delta^3(\boldsymbol{r})$ gives $\nabla^2V = 4\pi Z\alpha\,\delta^3(\boldsymbol{r})$. The Darwin term becomes a contact interaction,

$$H_{\rm Dar} = \frac{\pi Z\alpha}{2m_e^2}\,\delta^3(\boldsymbol{r}), \qquad \langle H_{\rm Dar}\rangle = \frac{\pi Z\alpha}{2m_e^2}\,|\psi(0)|^2. \tag{C.131}$$

Its expectation value is fixed by the *contact density* $|\psi(0)|^2$, the electron density at the nucleus, which only s states have. For a hydrogenic ns state of nuclear charge Z, $|\psi_{ns}(0)|^2 = (Z\alpha m_e)^3/(\pi n^3)$, the $n=1$ value being $1/(\pi a_0^3)$ at $Z=1$ (Sect. 2.2). The 1s Darwin shift is then $m_e(Z\alpha)^4/2$, a fraction $(Z\alpha)^2$ of the hydrogenic binding energy $m_e(Z\alpha)^2/2$: 0.72 meV, or 5.3×10^{-5} of the binding, for hydrogen; 28 eV, or 0.010, for $Z=14$; and 0.76 keV, or 0.055, for $Z=32$. Each relativistic term has the relative size $(Z\alpha)^2$. The mass–velocity term, $-\frac{5}{8}m_e(Z\alpha)^4$ for 1s, nearly cancels the Darwin term, and the net 1s shift is about a quarter of $(Z\alpha)^2$ times the binding energy, 0.26 % for silicon and 1.46 % for germanium in all-electron calculations of the atoms (Worked example 24.2). Corrections of this relative size, $(Z\alpha)^2$, from the core shells enter every operator of the response that probes the electron density near the nucleus, such as the mixed third moment of Sect. 8.3.3.

The density vertex itself changes under the transformation. Since $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ is even, its transform is $\mathrm{e}^X\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{-X} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \beta_{\rm D}[\mathsf{O},\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]/(2m_e) - [\mathsf{O},[\mathsf{O},\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]]/(8m_e^2) + \ldots$, by the same algebra with V replaced by $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$. The odd term connects positive- and negative-energy states and contributes to positive-energy matrix elements only at higher order. Eq. (C.129) with $\nabla\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \mathrm{i}\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ and $\nabla^2\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = -q^2\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ gives, on the upper components,

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \;\to\; \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\Big[1 - \frac{q^2}{8m_e^2} + \frac{\mathrm{i}}{4m_e^2}\,\boldsymbol{\sigma}\cdot(\boldsymbol{q}\times\boldsymbol{p})\Big]. \tag{C.132}$$

The scalar correction is $q^2/(8m_e^2) = 4.8\times10^{-5}$ at $q = 10\,\mathrm{keV}$. The spin-dependent term is of order qp/m_e^2, larger for core electrons, whose momenta reach $Z\alpha m_e$; Sect. 10.6 bounds the scalar correction and the spin-dependent term for silicon and germanium.

C.16 The quantized radiation field

The 2p state of hydrogen decays to 1s in 1.60 ns by emitting one photon, and a silicon nucleus kicked by dark matter radiates in the same way, more weakly. Sect. 12.6 computes such emission rates in a crystal. The calculation needs the electromagnetic field as a set of quantized oscillators, the coupling of a charge to it, the one-photon emission rate with its dipole limit, and the change of all three in a transparent medium of refractive index $n_r(\omega)$. We use Heaviside–Lorentz units with $e^2 = 4\pi\alpha$ (Appendix A.2) and the box normalization of Appendix C.2, with volume $\mathcal{V}$.

C.16.1 The field as oscillators

In the Coulomb gauge, $\nabla \cdot \boldsymbol{A} = 0$ and A^0 is the instantaneous Coulomb potential of the charges, already contained in the Coulomb interaction of Sect. 2.3. The radiation field is the transverse vector potential. In the periodic box it is a sum over wavevectors $\boldsymbol{k}$ and two polarization vectors $\boldsymbol{\epsilon}_{\boldsymbol{k}j}$, $j = 1, 2$, orthonormal and perpendicular to $\boldsymbol{k}$; the perpendicularity makes $\nabla \cdot \boldsymbol{A} = 0$ term by term. Without sources each term obeys the wave equation with $\omega_k = k$:

$$\boldsymbol{A}(\boldsymbol{r}, t) = \sum_{\boldsymbol{k}j} \big[c_{\boldsymbol{k}j}\, \boldsymbol{\epsilon}_{\boldsymbol{k}j}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r} - \mathrm{i}\omega_k t} + \text{c.c.} \big].$$

The fields are $\boldsymbol{E} = -\partial_t \boldsymbol{A}$ and $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, so each mode contributes $\mathrm{i}\omega_k c_{\boldsymbol{k}j} \boldsymbol{\epsilon}_{\boldsymbol{k}j}$ to $\boldsymbol{E}$ and $\mathrm{i}\boldsymbol{k} \times \boldsymbol{\epsilon}_{\boldsymbol{k}j} c_{\boldsymbol{k}j}$ to $\boldsymbol{B}$, times the plane wave. The energy $\frac{1}{2}\int \mathrm{d}^3 r\, (\boldsymbol{E}^2 + \boldsymbol{B}^2)$ is evaluated with $\int \mathrm{d}^3 r\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}} = \mathcal{V}\delta_{\boldsymbol{k}\boldsymbol{k}'}$. Terms that pair $\boldsymbol{k}$ with $-\boldsymbol{k}$ oscillate as $\mathrm{e}^{\mp 2\mathrm{i}\omega_k t}$ and cancel between $\boldsymbol{E}^2$ and $\boldsymbol{B}^2$, because $(\boldsymbol{k} \times \boldsymbol{\epsilon}) \cdot (\boldsymbol{k} \times \boldsymbol{\epsilon}') = k^2\, \boldsymbol{\epsilon} \cdot \boldsymbol{\epsilon}'$ for transverse vectors and $k^2 = \omega_k^2$. The remaining terms sum to $\sum_{\boldsymbol{k}j} 2\mathcal{V}\omega_k^2 |c_{\boldsymbol{k}j}|^2$. The sum matches the energy of a set of harmonic oscillators of frequencies ω_k if $c_{\boldsymbol{k}j} = \mathfrak{a}_{\boldsymbol{k}j}/(2\mathcal{V}\omega_k)^{1/2}$, where the classical amplitude $\mathfrak{a}_{\boldsymbol{k}j}$ of each oscillator is normalized so that its energy is $\omega_k |\mathfrak{a}_{\boldsymbol{k}j}|^2$. Quantization replaces $\mathfrak{a}_{\boldsymbol{k}j}$ by a boson annihilation operator $d_{\boldsymbol{k}j}$ with the commutators of Eq. (2.35):

$$\boldsymbol{A}(\boldsymbol{r}) = \sum_{\boldsymbol{k}j} \frac{1}{(2\mathcal{V}\omega_k)^{1/2}} \big[d_{\boldsymbol{k}j}\, \boldsymbol{\epsilon}_{\boldsymbol{k}j}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} + d_{\boldsymbol{k}j}^\dagger\, \boldsymbol{\epsilon}_{\boldsymbol{k}j}^*\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} \big], \qquad H_{\mathrm{rad}} = \sum_{\boldsymbol{k}j} \omega_k \big(d_{\boldsymbol{k}j}^\dagger d_{\boldsymbol{k}j} + \tfrac{1}{2} \big). \quad \text{(C.133)}$$

The eigenstates of H_{rad} are the photon number states $|\{n_{\boldsymbol{k}j}\}\rangle$, and by Eq. (2.34) the operator $d_{\boldsymbol{k}j}^\dagger$ adds a photon with amplitude $(n_{\boldsymbol{k}j} + 1)^{1/2}$. Emission into a mode that already holds n photons is therefore $n + 1$ times faster than into an empty one; for $\omega \gg T$ the thermal occupation $n_{\mathrm{B}}(\omega)$ is negligible and only spontaneous emission, the term 1 in $n + 1$, remains. Complex polarization vectors describe circular polarization, and the formulas below hold for them.

C.16.2 Minimal coupling

A non-relativistic particle of mass m and charge e_{c}, with $e_{\mathrm{c}} = -e$ for an electron and $e_{\mathrm{c}} = Z_\kappa e$ for a bare nucleus, couples to $\boldsymbol{A}$ by *minimal coupling*, $\boldsymbol{p} \to \boldsymbol{p} - e_{\mathrm{c}}\boldsymbol{A}$. For the electron this is $\boldsymbol{p} + e\boldsymbol{A}$, the non-relativistic limit of Eq. (C.124). Expanding the kinetic energy,

$$\frac{(\boldsymbol{p} - e_{\mathrm{c}}\boldsymbol{A})^2}{2m} = \frac{p^2}{2m} - \frac{e_{\mathrm{c}}}{m}\, \boldsymbol{A} \cdot \boldsymbol{p} + \frac{e_{\mathrm{c}}^2}{2m}\, \boldsymbol{A}^2, \quad \text{(C.134)}$$

where $\boldsymbol{p} \cdot \boldsymbol{A} = \boldsymbol{A} \cdot \boldsymbol{p} - \mathrm{i}\nabla \cdot \boldsymbol{A} = \boldsymbol{A} \cdot \boldsymbol{p}$ in the Coulomb gauge. The term linear in $\boldsymbol{A}$ creates or absorbs one photon; the quadratic term acts at first order only in two-photon processes, such as scattering, and drops out of one-photon emission. For several particles the interaction is the sum over particles, and for a prescribed classical current $\boldsymbol{J}(\boldsymbol{r}, t)$ it is $H' = -\int \mathrm{d}^3 r\, \boldsymbol{J} \cdot \boldsymbol{A}$, the form taken by the linear term when the particle velocity $\boldsymbol{p}/m$ is a given function of time.

C.16.3 The one-photon emission rate

Let the matter make a transition $|i\rangle \to |f\rangle$ with energy release $\omega = E_i - E_f > 0$ and emit one photon into the empty field. Only the $d^\dagger$ part of Eq. (C.133) contributes, and the matrix element of the linear term of Eq. (C.134) is

$$\langle f; 1_{\boldsymbol{k}j}|H'|i;0\rangle = -\frac{e_\mathrm{c}}{m\,(2\mathscr{V}\omega_k)^{1/2}}\,\boldsymbol{\epsilon}^*_{\boldsymbol{k}j}\cdot\boldsymbol{M}_{fi}, \qquad \boldsymbol{M}_{fi} = \langle f|\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\boldsymbol{p}|i\rangle.$$

The golden rule Eq. (2.5), summed over the photon modes with $\sum_{\boldsymbol{k}} \to \mathscr{V}\int \mathrm{d}^3k/(2\pi)^3 = \mathscr{V}\int \omega_k^2 \mathrm{d}\omega_k\,\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}/(2\pi)^3$, gives the rate per solid angle of emission

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}} = \frac{e_\mathrm{c}^2\,\omega}{8\pi^2 m^2}\sum_j |\boldsymbol{\epsilon}^*_{\boldsymbol{k}j}\cdot\boldsymbol{M}_{fi}|^2, \qquad |\boldsymbol{k}| = \omega. \tag{C.135}$$

The prefactor $\omega/(8\pi^2)$ collects 2π from the golden rule, $\omega^2/(2\pi)^3$ from the density of modes after the energy delta function is integrated, and $1/(2\omega)$ from the field normalization.

Dipole limit. When the emitting system is small compared with the photon wavelength, $kr \ll 1$, the factor $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ is replaced by 1. A photon of 10 eV has $k = 5.1\times10^{-3}\,\text{Å}^{-1}$, so across an atom $kr \approx 0.005$. If the unperturbed Hamiltonian is $H_0 = p^2/2m + U(\boldsymbol{r})$, then $[H_0, \boldsymbol{r}] = -\mathrm{i}\boldsymbol{p}/m$, and $\langle f|\boldsymbol{p}|i\rangle = \mathrm{i}m(E_f - E_i)\boldsymbol{r}_{fi} = -\mathrm{i}m\omega\,\boldsymbol{r}_{fi}$, with $\boldsymbol{r}_{fi} = \langle f|\boldsymbol{r}|i\rangle$. The sum over the two polarizations is $\sum_j |\boldsymbol{\epsilon}^*_j\cdot\boldsymbol{d}|^2 = |\boldsymbol{d}|^2 - |\hat{\boldsymbol{k}}\cdot\boldsymbol{d}|^2$ for any vector $\boldsymbol{d}$, by completeness of $\{\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \hat{\boldsymbol{k}}\}$, and the integral of this sum over directions is $4\pi|\boldsymbol{d}|^2(1-\frac{1}{3})$ (Appendix C.3). The total rate is

$$\Gamma_{i\to f} = \frac{e_\mathrm{c}^2\,\omega^3}{3\pi}\,|\boldsymbol{r}_{fi}|^2 = \frac{4}{3}\,\alpha\,\omega^3\,|\boldsymbol{r}_{fi}|^2 \qquad (|e_\mathrm{c}| = e). \tag{C.136}$$

For hydrogen 2p → 1s, $\omega = \frac{3}{4}\times 13.606\,\text{eV}$ and $|\boldsymbol{r}_{fi}|^2 = (2^{15}/3^{10})\,a_0^2$ for each m substate; the lifetime $1/\Gamma$ is 1.60 ns, the measured value. A transition with $\boldsymbol{r}_{fi} = 0$ proceeds through the next terms of the expansion of $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, magnetic-dipole and electric-quadrupole, with rates smaller than the dipole rate by $(kr)^2$.

Emission by a classical current. For a prescribed current, first-order perturbation theory gives the amplitude of one photon in mode $\boldsymbol{k}j$ as $c_{\boldsymbol{k}j} = -\mathrm{i}\int \mathrm{d}t\,\langle 1_{\boldsymbol{k}j}|H'(t)|0\rangle = \mathrm{i}(2\mathscr{V}\omega_k)^{-1/2}\,\boldsymbol{\epsilon}^*_{\boldsymbol{k}j}\cdot\tilde{\boldsymbol{J}}(\boldsymbol{k},\omega_k)$, with $\tilde{\boldsymbol{J}}(\boldsymbol{k},\omega) = \int \mathrm{d}t\,\mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\omega t - \mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\boldsymbol{J}(\boldsymbol{r},t)$ in the conventions of Appendix A.3. Summing $|c_{\boldsymbol{k}j}|^2$ over modes as above gives the number of photons emitted per unit frequency and solid angle,

$$\frac{\mathrm{d}N_\gamma}{\mathrm{d}\omega\,\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}} = \frac{\omega}{16\pi^3}\sum_j |\boldsymbol{\epsilon}^*_{\boldsymbol{k}j}\cdot\tilde{\boldsymbol{J}}(\boldsymbol{k},\omega)|^2. \tag{C.137}$$

A charge at rest that is suddenly set moving with velocity $\boldsymbol{v}$ at $t = 0$ carries $\boldsymbol{J} = e_\mathrm{c}\boldsymbol{v}\,\delta^3(\boldsymbol{r} - \boldsymbol{v}t)\,\Theta(t)$, so $\tilde{\boldsymbol{J}} = e_\mathrm{c}\boldsymbol{v}\int_0^\infty \mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega - \boldsymbol{k}\cdot\boldsymbol{v})t} = \mathrm{i}e_\mathrm{c}\boldsymbol{v}/(\omega - \boldsymbol{k}\cdot\boldsymbol{v} + \mathrm{i}0^+) \simeq \mathrm{i}e_\mathrm{c}\boldsymbol{v}/\omega$ for $v \ll 1$. With the polarization sum and angular integral of the dipole case,

$$\frac{\mathrm{d}N_\gamma}{\mathrm{d}\omega} = \frac{e_\mathrm{c}^2\,v^2}{6\pi^2\,\omega} = \frac{2\alpha}{3\pi}\,\frac{v^2}{\omega} \qquad (|e_\mathrm{c}| = e). \tag{C.138}$$

The energy radiated per unit frequency, $\omega\,\mathrm{d}N_\gamma/\mathrm{d}\omega$, is flat. The kick may be treated as sudden when it lasts less than $1/\omega$. For a nucleus bound in a crystal the restoring forces and the response of the electrons modify Eq. (C.138) at the frequencies of the lattice and of the electrons; Sect. 12.6 includes these modifications through the dynamical Born charge.

C.16.4 A photon mode in a medium

Inside a transparent medium with relative permittivity $\varepsilon_r(\omega) = n_r(\omega)^2$ and no magnetic response, the photon is a mode of the macroscopic field. The dispersion relation becomes $k = n_r(\omega)\,\omega$, and the number of modes per unit frequency grows: $\mathcal{V}\,k^2\,\mathrm{d}k/(2\pi)^3 = \mathcal{V}\,n_r^2\,n_{\mathrm{grp}}\,\omega^2\mathrm{d}\omega/(2\pi)^3$, with $n_{\mathrm{grp}} = \mathrm{d}k/\mathrm{d}\omega = n_r + \omega\,\mathrm{d}n_r/\mathrm{d}\omega$ the group index. The energy stored in a mode of given amplitude grows as well. The time-averaged energy of a monochromatic field in a transparent dispersive medium is $\frac{1}{4}[\mathrm{d}(\omega\varepsilon_r)/\mathrm{d}\omega\,|\boldsymbol{E}_0|^2 + |\boldsymbol{B}_0|^2]$ per unit volume [27]. With $|\boldsymbol{E}_0| = \omega|\boldsymbol{A}_0|$, $|\boldsymbol{B}_0| = k|\boldsymbol{A}_0| = n_r\omega|\boldsymbol{A}_0|$ and $\mathrm{d}(\omega\varepsilon_r)/\mathrm{d}\omega = n_r^2 + 2n_r\omega\,\mathrm{d}n_r/\mathrm{d}\omega$, the bracket is $\omega^2|\boldsymbol{A}_0|^2 \cdot 2n_r n_{\mathrm{grp}}$, larger than in vacuum by the factor $n_r n_{\mathrm{grp}}$. When the quantization is repeated, the normalization $(2\mathcal{V}\omega)^{-1/2}$ of Eq. (C.133) becomes $(2\mathcal{V}\omega\,n_r n_{\mathrm{grp}})^{-1/2}$; for a medium without dispersion, $n_{\mathrm{grp}} = n_r$, the result is the quantization of Ref. [28].

In a rate, the squared normalization contributes $1/(n_r n_{\mathrm{grp}})$ and the density of modes $n_r^2 n_{\mathrm{grp}}$. The group index cancels, and

$$\Gamma^{\mathrm{medium}} = n_r(\omega)\,\Gamma^{\mathrm{vac}}, \qquad \frac{\mathrm{d}N_\gamma^{\mathrm{medium}}}{\mathrm{d}\omega} = n_r(\omega)\,\frac{\mathrm{d}N_\gamma^{\mathrm{vac}}}{\mathrm{d}\omega}, \tag{C.139}$$

for the same matrix element or current. The medium must be transparent at ω, so that ε_r is real and the photon is a propagating mode; near an absorption line the photon is not a well-defined final state. And the emitter must couple to the macroscopic field: the local field at an atom differs from the macroscopic one by a local-field factor, which Eq. (C.139) sets to 1. In the dipole factor $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ the wavevector becomes $n_r\omega$, which leaves the dipole limit intact for n_r of a few.

C.17 Point and space groups

Silicon is mapped onto itself by 48 rotations and reflections, half of them combined with a shift by a quarter of the cube diagonal. The shifted half of the operations makes the (200) X-ray reflection of silicon vanish for any charge distribution in the cell, and forbids some electronic transitions of silicon at the lowest order. Sect. 13.9 uses the shifted operations, glides, to classify the zero-phonon transitions of silicon, and Sect. 14.6 uses point-group characters to decide which crystals allow a response odd in $\boldsymbol{q}$. Both sections start from the crystal structures of Sect. 3.1.

C.17.1 Symmetry operations

A symmetry operation of a crystal is a map $\boldsymbol{r} \to \mathsf{R}\boldsymbol{r} + \boldsymbol{t}$ that maps the crystal onto itself, written $\{\mathsf{R}|\boldsymbol{t}\}$, with R an orthogonal 3×3 matrix and $\boldsymbol{t}$ a translation. Applying $\{\mathsf{R}'|\boldsymbol{t}'\}$ and then $\{\mathsf{R}|\boldsymbol{t}\}$ gives $\boldsymbol{r} \to \mathsf{R}(\mathsf{R}'\boldsymbol{r} + \boldsymbol{t}') + \boldsymbol{t}$, so

$$\{\mathsf{R}|\boldsymbol{t}\}\{\mathsf{R}'|\boldsymbol{t}'\} = \{\mathsf{R}\mathsf{R}' \,|\, \mathsf{R}\boldsymbol{t}' + \boldsymbol{t}\}, \qquad \{\mathsf{R}|\boldsymbol{t}\}^{-1} = \{\mathsf{R}^{-1} \,|\, -\mathsf{R}^{-1}\boldsymbol{t}\}. \tag{C.140}$$

The operations form the *space group*. It contains the lattice translations $\{1|\boldsymbol{R}\}$ of Eq. (3.1), and each R that occurs comes with a translation $\boldsymbol{t}$ fixed up to a lattice vector. The set of the R is the *point group* $\mathcal{P}$, of order $|\mathcal{P}|$. An orthogonal matrix has $\det\mathsf{R} = \pm 1$. With $\det\mathsf{R} = +1$ it is a proper rotation by an angle θ_R about an axis; with $\det\mathsf{R} = -1$ it is improper, the product of the inversion -1 and the proper rotation $-\mathsf{R}$, and we define θ_R as the angle of $-\mathsf{R}$. A mirror reflection, for example, is the inversion times a rotation by π about the normal of the mirror.

If the origin can be chosen so that every R of the point group occurs with $\boldsymbol{t}$ a lattice vector, the space group is *symmorphic*. Otherwise it is *nonsymmorphic*, and some operations carry a

translation that no change of origin removes: a *glide*, a reflection followed by a translation parallel to the mirror, or a *screw*, a rotation followed by a translation along its axis. Applied twice, a glide gives a pure translation, which must be a lattice vector. Crystals have 32 point groups, named in two systems. Silicon and germanium have $m\bar{3}m$ (O_h), the full symmetry of the cube. GaAs has $\bar{4}3m$ (T_d), the full symmetry of the regular tetrahedron. The 24 proper rotations of the cube form 432 (O), and α-quartz and tellurium have 32 (D_3).

C.17.2 Representations and characters

A *representation* of $\mathcal{P}$ assigns to each R a matrix $\mathsf{D}(\mathsf{R})$ with $\mathsf{D}(\mathsf{R})\mathsf{D}(\mathsf{R}') = \mathsf{D}(\mathsf{R}\mathsf{R}')$. Vectors, tensors and functions of direction each carry one: the matrices describe how the components of these objects mix under the operations. A representation is *irreducible* if no basis change splits all its matrices into smaller blocks at once. Its *character* $c(\mathsf{R}) = \operatorname{tr}\mathsf{D}(\mathsf{R})$ is unchanged by basis changes and takes one value on each class of conjugate elements, $\mathsf{S}\mathsf{R}\mathsf{S}^{-1}$ with $\mathsf{S} \in \mathcal{P}$. The irreducible characters $c^{(a)}$ are orthonormal,

$$\frac{1}{|\mathcal{P}|}\sum_{\mathsf{R}\in\mathcal{P}} c^{(a)}(\mathsf{R})^*\, c^{(b)}(\mathsf{R}) = \delta_{ab}, \tag{C.141}$$

so a representation with character c contains the irreducible representation a exactly $n_a = |\mathcal{P}|^{-1}\sum_{\mathsf{R}} c^{(a)}(\mathsf{R})^* c(\mathsf{R})$ times [29, Chap. 3].

The case $a =$ trivial representation, $c^{(a)} = 1$, counts invariants. Let $\mathsf{Q} = |\mathcal{P}|^{-1}\sum_{\mathsf{R}}\mathsf{D}(\mathsf{R})$. Since multiplying all elements of a group by one element only reorders them, $\mathsf{D}(\mathsf{S})\mathsf{Q} = \mathsf{Q}$ for every S; hence $\mathsf{Q}^2 = \mathsf{Q}$, and every vector in the range of Q is invariant. Conversely, an invariant vector $\mathbf{x}$ has $\mathsf{Q}\mathbf{x} = \mathbf{x}$. So Q is the projector onto the invariant subspace. Its trace is the dimension of that subspace:

$$n_{\text{inv}} = \frac{1}{|\mathcal{P}|}\sum_{\mathsf{R}\in\mathcal{P}} c(\mathsf{R}). \tag{C.142}$$

For the point group 32, the six elements fall into three classes: the identity 1, the two rotations C_3 by $\pm 2\pi/3$ about the threefold axis, and the three rotations C_2' by π about the twofold axes perpendicular to it. There are three irreducible representations, labeled A_1, A_2 and E in the standard notation, with characters $(1,1,1)$, $(1,1,-1)$ and $(2,-1,0)$ on the classes $(1, C_3, C_2')$. With the class sizes 1, 2 and 3, Eq. (C.141) checks: for E with itself, $(4 + 2\cdot 1 + 0)/6 = 1$; for E with A_2, $(2 - 2 + 0)/6 = 0$.

C.17.3 Polar and axial tensors

A *polar* vector, such as $\boldsymbol{q}$, a position or a current, transforms with R itself. Its character is $c_{\text{V}}(\mathsf{R}) = \operatorname{tr}\mathsf{R}$. A proper rotation by θ has the eigenvalues 1 and $\mathrm{e}^{\pm\mathrm{i}\theta}$, so $\operatorname{tr}\mathsf{R} = 1 + 2\cos\theta$; an improper one has the negatives. An *axial* vector, such as an angular momentum, a spin or a magnetic field, is a cross product of two polar vectors and does not change sign under the inversion. It transforms with $(\det\mathsf{R})\mathsf{R}$. Hence

$$c_{\text{V}}(\mathsf{R}) = (\det\mathsf{R})\,(1 + 2\cos\theta_{\mathsf{R}}), \qquad c_{\text{A}}(\mathsf{R}) = 1 + 2\cos\theta_{\mathsf{R}}. \tag{C.143}$$

A tensor transforms as the product of its indices, and its character is the product of the characters of its indices. A rank-three polar tensor has c_{V}^3; a second-rank tensor with one polar and one axial index has $c_{\text{V}}c_{\text{A}} = (\det\mathsf{R})(1 + 2\cos\theta_{\mathsf{R}})^2$. For functions of direction, the $2l+1$ spherical harmonics of order l (Appendix C.3) have the character $\sin[(l+\tfrac{1}{2})\theta]/\sin(\theta/2)$ for a proper rotation, multiplied by $(-1)^l$ for an improper one. A crystal property allowed by symmetry is an invariant tensor, and Eq. (C.142) counts how many independent ones exist.

C.17.4 Responses odd in q

Inversion maps $\boldsymbol{q}$ to $-\boldsymbol{q}$, so a centrosymmetric crystal, one whose point group contains -1, has no response odd in $\boldsymbol{q}$ of any kind. In the sum Eq. (C.142) for an odd-rank polar tensor, the elements R and $-\mathsf{R}$ then contribute with opposite signs and cancel. For a scalar response, such as the structure factor $S(\boldsymbol{q},\omega)$, time reversal removes the odd part as well (Appendix A.4): a part odd in $\boldsymbol{q}$ needs a crystal with neither inversion nor time-reversal symmetry, a magnetically ordered one.

A tensor response is different. Write the current response at small q as $\chi_{ij}(\boldsymbol{q},\omega) = \chi^{(0)}_{ij}(\omega) + \mathrm{i}\gamma_{ijl}(\omega)\, q_l + O(q^2)$. The coefficient γ_{ijl} is a rank-three polar tensor. In a crystal with time-reversal symmetry, the argument of Appendix A.4, applied to a product of two currents, gives the reciprocity relation $\chi_{ij}(\boldsymbol{q},\omega) = \chi_{ji}(-\boldsymbol{q},\omega)$. Under time reversal each current takes a sign, the two signs cancel in the product, and the antiunitary conjugation exchanges the order of the two operators. The term linear in $\boldsymbol{q}$ then survives only if γ_{ijl} is antisymmetric in i,j. An antisymmetric pair of indices is equivalent to one axial index, $\gamma_{ijl} = \epsilon_{ijm} g_{ml}$, and the *gyration tensor* g_{ml} has one axial and one polar index. It describes natural optical activity and circular dichroism. By Eqs. (C.142) and (C.143), the number of its independent components in point group $\mathcal{P}$ is

Optical activity allowed by a point group

$$N_{\mathrm{gyr}} = \frac{1}{|\mathcal{P}|} \sum_{\mathsf{R}\in\mathcal{P}} (\det \mathsf{R})\,(1 + 2\cos\theta_{\mathsf{R}})^2. \tag{C.144}$$

Worked example C.1: Gyration invariants of GaAs and of a chiral cubic crystal

The group $\bar{4}3m$ of GaAs has 24 elements in five classes: the identity; eight C_3 ($\theta = 2\pi/3$); three C_2 ($\theta = \pi$); six S_4, improper, with $-S_4$ a rotation by $\pi/2$; and six mirrors σ_d, improper, with $-\sigma_d$ a rotation by π. The values of $(\det \mathsf{R})(1 + 2\cos\theta)^2$ are 9, 0, 1, -1 and -1. With the class sizes, Eq. (C.144) gives $N_{\mathrm{gyr}} = (9 + 0 + 3 - 6 - 6)/24 = 0$. GaAs has no optical activity, although it lacks inversion. It does have one invariant rank-three polar tensor: c^3_{V} takes the values 27, 0, -1, -1 and 1 on the five classes, and $(27 + 0 - 3 - 6 + 6)/24 = 1$. That invariant is the totally symmetric tensor $|\epsilon_{ijl}|$, nonzero for i,j,l all different. The same tensor is behind the piezoelectricity of GaAs and the cubic harmonic xyz at $l = 3$. Reciprocity forbids it in the term linear in $\boldsymbol{q}$ of the current response, because it is symmetric in i,j. The group 432 has the 24 proper rotations of the cube, with values 9, 0, 1 on the identity, the eight C_3 and the three C_2, and 1 on the six C_4 ($\theta = \pi/2$) and the six further C_2'. Then $N_{\mathrm{gyr}} = (9 + 0 + 3 + 6 + 6)/24 = 1$: a cubic crystal without mirrors or inversion has isotropic optical activity, $g_{ml} \propto \delta_{ml}$. A direct enumeration of the matrices of each group reproduces the two gyration counts and the count of rank-three invariants, together with $N_{\mathrm{gyr}} = 0$ for $m\bar{3}m$.

C.17.5 The diamond space group

The diamond structure of silicon and germanium has the space group $Fd\bar{3}m$ (number 227 in the international tables; O_h^7). Its point group is $m\bar{3}m$. With the origin on the atom at $\boldsymbol{\tau}_1 = 0$ of Eq. (3.5), the 24 operations of $\bar{4}3m$ occur with $\boldsymbol{t} = 0$; $\bar{4}3m$ is the site symmetry of each atom. The other 24 occur with $\boldsymbol{t} = \boldsymbol{\tau}_2 = (a/4)(1,1,1)$, and they exchange the two atoms of the basis. The inversion $\{-1|\boldsymbol{\tau}_2\}$, for example, is the inversion through the bond center at

$\boldsymbol{\tau}_2/2$. No choice of origin removes all these translations, and the group is nonsymmorphic. Zincblende GaAs, with different atoms on the two sites, keeps only the first 24 operations; its space group $F\bar{4}3m$ is symmorphic.

Take the mirror m_x, $(x,y,z)\to(-x,y,z)$, which is not in $\bar{4}3m$. It occurs as

$$g=\{\mathsf{m}_x|\boldsymbol{\tau}_2\}:\quad (x,y,z)\to(-x+\tfrac{a}{4},\ y+\tfrac{a}{4},\ z+\tfrac{a}{4}). \tag{C.145}$$

Points with $x=a/8$ keep their x coordinate, so g is a reflection in the plane $x=a/8$ followed by the translation $(0,a/4,a/4)$ within it: a glide, of the kind called a diamond glide. It maps the atom at 0 to $\boldsymbol{\tau}_2$ and the atom at $\boldsymbol{\tau}_2$ to $(0,a/2,a/2)=\boldsymbol{a}_1$, a lattice point. Applied twice, by Eq. (C.140), it gives $\{1|\boldsymbol{w}\}$ with $\boldsymbol{w}=\boldsymbol{\tau}_2+\mathsf{m}_x\boldsymbol{\tau}_2=(a/2)(0,1,1)=\boldsymbol{a}_1$, a lattice translation, as required.

Extinctions. Let $f(\boldsymbol{r})$ be any function invariant under g, $f(\mathsf{m}_x\boldsymbol{r}+\boldsymbol{\tau}_2)=f(\boldsymbol{r})$, and lattice periodic, with components $f_{\boldsymbol{G}}$ (Eq. (A.11)). Substituting $\boldsymbol{r}=\mathsf{m}_x\boldsymbol{r}'+\boldsymbol{\tau}_2$ in $f_{\boldsymbol{G}}=\Omega^{-1}\int_\Omega \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}f(\boldsymbol{r})$ gives $f_{\boldsymbol{G}}=\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_2}f_{\mathsf{m}_x\boldsymbol{G}}$. For $\boldsymbol{G}$ in the mirror plane, $\boldsymbol{G}=(2\pi/a)(0,k,l)$ with k, l even, $\mathsf{m}_x\boldsymbol{G}=\boldsymbol{G}$ and

$$f_{\boldsymbol{G}}=\mathrm{e}^{\mathrm{i}\pi(k+l)/2}f_{\boldsymbol{G}},\qquad \text{so}\qquad f_{\boldsymbol{G}}=0\quad\text{for } k+l\equiv 2\pmod 4. \tag{C.146}$$

The components (020) and (002) vanish, and by the glides with m_y and m_z so does (200). The components (020), (002) and (200) vanish for every function with the symmetry of the crystal: the full crystal potential, the charge density with its bonds, any lattice-periodic response. This symmetry extinction is stronger than the extinction of Worked example 3.2, which came from the sum of identical spherical atoms. The (222) vector lies in no glide plane of the form m_x, m_y, m_z, so the (222) component vanishes only for spherical atoms, and the bond charge makes it nonzero.

Glide parity of Bloch states. Let g act on wavefunctions by $(g\psi)(\boldsymbol{r})=\psi(g^{-1}\boldsymbol{r})$, with $g^{-1}\boldsymbol{r}=\mathsf{m}_x(\boldsymbol{r}-\boldsymbol{\tau}_2)$. If ψ is a Bloch state of crystal momentum $\boldsymbol{k}$, then $(g\psi)(\boldsymbol{r}+\boldsymbol{R})=\mathrm{e}^{\mathrm{i}(\mathsf{m}_x\boldsymbol{k})\cdot\boldsymbol{R}}(g\psi)(\boldsymbol{r})$, so $g\psi$ has crystal momentum $\mathsf{m}_x\boldsymbol{k}$. For $\boldsymbol{k}$ in the mirror plane, $k_x=0$, the operator g maps the Bloch states at $\boldsymbol{k}$ among themselves and commutes with the Hamiltonian, so the states can be chosen as eigenstates of g. Since $g^2=\{1|\boldsymbol{w}\}$ acts as $\psi(\boldsymbol{r}-\boldsymbol{w})=\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{w}}\psi(\boldsymbol{r})$, the eigenvalues are

$$g\,\psi=\eta\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{w}/2}\,\psi,\qquad \eta=\pm1, \tag{C.147}$$

and η is the *glide parity*. It depends on the label: the same state labeled by $\boldsymbol{k}-\boldsymbol{G}$, with $\boldsymbol{G}$ in the plane, has $\eta\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{w}/2}=\eta\,\mathrm{e}^{-\mathrm{i}\pi(k+l)/2}$.

Consider now the vertex $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ of Sect. 3.2 with $\boldsymbol{q}$ in the mirror plane. Conjugation by g gives $g\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}g^{-1}=\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot g^{-1}\boldsymbol{r}}=\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_2}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, using $\mathsf{m}_x\boldsymbol{q}=\boldsymbol{q}$. For eigenstates $|i\rangle$ at $\boldsymbol{k}$ and $|f\rangle$ labeled by the unreduced momentum $\boldsymbol{k}+\boldsymbol{q}$, unitarity of g gives

$$\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle=\langle gf|g\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}g^{-1}|gi\rangle=\eta_f\eta_i\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{w}/2-\boldsymbol{\tau}_2)}\,\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle.$$

The vector $\boldsymbol{w}/2-\boldsymbol{\tau}_2=(a/4)(-1,0,0)$ is normal to the plane, so the phase is 1 for $\boldsymbol{q}$ in the plane, and the matrix element vanishes unless $\eta_f=\eta_i$. For the reduced momentum $\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}$ of the final state, the rule reads

$$\eta_f^{\mathrm{red}}=\eta_i\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{w}/2}, \tag{C.148}$$

so a momentum transfer whose Umklapp vector belongs to the (200) class flips the glide parity, and one with $k+l\equiv0\pmod4$ preserves it. At $\boldsymbol{k}=0$ and $\boldsymbol{q}=\boldsymbol{G}=(2\pi/a)(0,2,0)$, so that the reduced final momentum $\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}$ is again 0, Eq. (C.148) requires a state to have opposite parity to itself, and the diagonal matrix element vanishes: Eq. (C.146) again. A further consequence of the nonsymmorphic operations is that every band of diamond is at least twofold degenerate at the zone-boundary points X, without spin–orbit coupling, a degeneracy that no symmorphic group with the same point group would force [29].

C.18 Continuum elasticity

A longitudinal sound wave in silicon travels at $8.43\,\mathrm{km\,s^{-1}}$ along a cube edge and at $9.35\,\mathrm{km\,s^{-1}}$ along a cube diagonal. Both numbers follow from three elastic constants and the density, and so do the speeds of the two transverse waves in every direction. Sect. 11.1 needs the longitudinal speed averaged over directions and a lower bound on it from the bulk modulus. Sect. 17.5 needs a polarization-weighted average of the inverse speeds, and Sect. 20.3 the acoustic impedances that decide how much of a phonon flux crosses from a crystal into a metal film. All these quantities are long-wavelength properties of the acoustic branches of Sect. 5.2: for Q well inside the Brillouin zone, $\omega_{\boldsymbol{Q}\nu} = c_\nu(\hat{\boldsymbol{Q}})\,Q$, and the crystal responds like an elastic continuum.

C.18.1 Strain, stress and the elastic tensor

At long wavelength all atoms of a cell move together in an acoustic wave, and the displacements $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ of Eq. (A.20) become a smooth field $\boldsymbol{u}(\boldsymbol{r},t)$. A uniform $\boldsymbol{u}$ is a translation and costs no energy. Neither does a rigid rotation, whose gradient $\partial_i u_j$ is antisymmetric. The energy therefore depends on the symmetric part of the gradient, the *strain*

$$u_{ij} = \tfrac{1}{2}(\partial_i u_j + \partial_j u_i), \tag{C.149}$$

which measures changes of length: a segment $\mathrm{d}\boldsymbol{r}$ changes its squared length by $2u_{ij}\,\mathrm{d}r_i\,\mathrm{d}r_j$ to first order. To second order in the strain the energy per unit volume is

$$w_{\mathrm{el}} = \tfrac{1}{2}\,C_{ijkl}\,u_{ij}\,u_{kl}, \tag{C.150}$$

with summation over repeated Cartesian indices. Since u_{ij} is symmetric, only the parts of C_{ijkl} symmetric in ij and in kl contribute, and we take $C_{ijkl} = C_{jikl} = C_{ijlk}$. Since C_{ijkl} is a second derivative of w_{el}, also $C_{ijkl} = C_{klij}$. These symmetries leave 21 independent constants. The *stress* is the derivative $\sigma_{ij} = \partial w_{\mathrm{el}}/\partial u_{ij} = C_{ijkl}u_{kl}$ (Hooke's law); σ_{ij} is the force per unit area, in direction i, exerted across a surface with normal along j by the material on the far side. The net force on a volume element is $\partial_j\sigma_{ij}$ per unit volume, and Newton's law for a medium of mass density ρ_T is

$$\rho_T\,\ddot{u}_i = \partial_j\sigma_{ij} = C_{ijkl}\,\partial_j\partial_l u_k, \tag{C.151}$$

where the second form uses $C_{ijkl}u_{kl} = C_{ijkl}\partial_l u_k$, by the symmetry in kl [30].

Tables and codes store C_{ijkl} as a symmetric 6×6 matrix C_{IJ} in *Voigt notation*. Each symmetric pair of indices becomes one index, $11\to 1$, $22\to 2$, $33\to 3$, $23\to 4$, $13\to 5$, $12\to 6$. The strain components are collected as $\epsilon_I = (u_{11}, u_{22}, u_{33}, 2u_{23}, 2u_{13}, 2u_{12})$. The factor 2 on the shear components is needed because u_{23} and u_{32} both occur in the double sum of Eq. (C.150); with it, $w_{\mathrm{el}} = \frac{1}{2}C_{IJ}\epsilon_I\epsilon_J$ and the stress components σ_I, without factors of 2, are $\sigma_I = C_{IJ}\epsilon_J$. The ordering of the shear rows is a common source of error in practice.

A cubic crystal has three constants: $C_{11} = C_{xxxx}$, $C_{12} = C_{xxyy}$ and $C_{44} = C_{yzyz}$, with all others related to them by the cube's symmetry or zero. The Voigt matrix C_{IJ} then has three distinct eigenvalues: $C_{11} + 2C_{12}$ for a uniform dilation, $C_{11} - C_{12}$ (twice) for the two volume-conserving stretches along the cube axes, and C_{44} (three times) for the shears. A crystal is stable when every strain costs energy, and stability requires

$$C_{11} > |C_{12}|, \qquad C_{11} + 2C_{12} > 0, \qquad C_{44} > 0, \tag{C.152}$$

the Born stability conditions of a cubic crystal. For silicon at room temperature, $C_{11} = 165.6$, $C_{12} = 63.9$ and $C_{44} = 79.5\,\mathrm{GPa}$, with $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$ [31, Table I].

The *bulk modulus* B is the response to a uniform compression. A strain $u_{ij} = \frac{1}{3}\delta\,\delta_{ij}$ changes the volume by the fraction $\delta = \Delta\mathscr{V}/\mathscr{V}$, and in a cubic crystal it produces the stress $\sigma_{ij} = \frac{1}{3}(C_{11} + 2C_{12})\,\delta\,\delta_{ij}$, a pressure $P = -\frac{1}{3}\sigma_{ii}$. Hence

$$B = -\mathscr{V}\frac{\mathrm{d}P}{\mathrm{d}\mathscr{V}} = \tfrac{1}{3}(C_{11} + 2C_{12}), \tag{C.153}$$

which is 97.8 GPa for silicon. In a crystal of lower symmetry a uniform strain and a uniform pressure do not correspond, and the two give different averages, $C_{iijj}/9$ (Voigt) and the inverse of the compliance sum (Reuss); for cubic crystals they coincide with Eq. (C.153).

An isotropic medium, such as a polycrystal with random grain orientations or a glass, has $C_{11} = C_{12} + 2C_{44}$ and only two independent constants. The literature writes them as the Lamé constants, $\lambda = C_{12}$ and the shear modulus $\mu = C_{44}$, and then $B = C_{11} - \frac{4}{3}C_{44}$. Phonon-transport simulations often give silicon such a tensor, with $C_{12} = 52.4$ and $C_{44} = 68.0$ GPa [31, Table I], so that $C_{11} = 188.4$ GPa and the speeds in every direction are 9.0 and 5.4 km s^{-1}. The isotropic tensor is a parameter set of the codes, not a property of silicon: the measured tensor violates the isotropy relation by a factor $2C_{44}/(C_{11} - C_{12}) = 1.56$.

C.18.2 The Christoffel problem

A plane wave $\boldsymbol{u} = \hat{\boldsymbol{e}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{Q}\cdot\boldsymbol{r} - \omega t)}$ solves Eq. (C.151) if

$$\rho_T\, c^2\, \hat{e}_i = \mathsf{M}_{ik}(\hat{\boldsymbol{Q}})\,\hat{e}_k, \qquad \mathsf{M}_{ik}(\hat{\boldsymbol{Q}}) = C_{ijkl}\,\hat{Q}_j\,\hat{Q}_l, \qquad \omega = cQ. \tag{C.154}$$

This is the Christoffel equation, and M is the Christoffel matrix [30]. By the symmetry of C_{ijkl} in kl, $C_{ijkl}\hat{Q}_j\hat{Q}_l\hat{e}_k = C_{ijkl}\hat{Q}_j\hat{Q}_k\hat{e}_l$. Since $C_{ijkl} = C_{klij}$, M is real and symmetric, and since $\hat{e}_i\mathsf{M}_{ik}\hat{e}_k = C_{ijkl}(\hat{e}_i\hat{Q}_j)(\hat{e}_k\hat{Q}_l)$ is the elastic energy of the strain $\frac{1}{2}(\hat{e}_i\hat{Q}_j + \hat{e}_j\hat{Q}_i)$, M is positive definite in a stable crystal. Each direction $\hat{\boldsymbol{Q}}$ therefore has three positive eigenvalues $\rho_T c_\nu(\hat{\boldsymbol{Q}})^2$ with orthonormal polarizations $\hat{\boldsymbol{e}}_\nu(\hat{\boldsymbol{Q}})$. The c_ν are the sound speeds of Sect. 5.2 and the $\hat{\boldsymbol{e}}_\nu$ the unit polarizations of Appendix A.6: Eq. (C.154) is the acoustic block of the dynamical matrix of Eq. (A.22) at small Q, to order Q^2, with $\rho_T = M_{\mathrm{cell}}/\Omega$. In general no polarization is exactly parallel or perpendicular to $\hat{\boldsymbol{Q}}$. The fastest branch, the quasi-longitudinal one, is the branch this book calls LA, with speed $c_{\mathrm{LA}}(\hat{\boldsymbol{Q}})$; the other two branches are quasi-transverse. The group velocity $\partial\omega/\partial\boldsymbol{Q}$ of each branch is not parallel to $\hat{\boldsymbol{Q}}$ either, and the angle between the group velocity and $\hat{\boldsymbol{Q}}$ is the origin of the phonon focusing of Sect. 29.2.

Along the three symmetry directions of a cubic crystal the polarizations are fixed by symmetry and the eigenvalues are

$$\begin{aligned}
&[100]: && \rho_T c_{\mathrm{LA}}^2 = C_{11}, && \rho_T c_{\mathrm{TA}}^2 = C_{44}\ \text{(twice)},\\
&[110]: && \rho_T c_{\mathrm{LA}}^2 = \tfrac{1}{2}(C_{11} + C_{12} + 2C_{44}), && \rho_T c_{\mathrm{TA}}^2 = C_{44}\ (\hat{\boldsymbol{e}} \parallel [001]),\\
& && && \rho_T c_{\mathrm{TA}}^2 = \tfrac{1}{2}(C_{11} - C_{12})\ (\hat{\boldsymbol{e}} \parallel [1\bar{1}0]),\\
&[111]: && \rho_T c_{\mathrm{LA}}^2 = \tfrac{1}{3}(C_{11} + 2C_{12} + 4C_{44}), && \rho_T c_{\mathrm{TA}}^2 = \tfrac{1}{3}(C_{11} - C_{12} + C_{44})\ \text{(twice)}.
\end{aligned} \tag{C.155}$$

Each follows from M_{ik} written out for that $\hat{\boldsymbol{Q}}$. Along [100], for example, $\mathsf{M}_{ik} = C_{i1k1}$, which is $\mathrm{diag}(C_{11}, C_{44}, C_{44})$. For silicon the longitudinal speeds are 8.43, 9.13 and 9.35 km s^{-1} along [100], [110] and [111], and the transverse speeds are 5.84 (twice), 5.84 and 4.67, and 5.09 (twice), in km s^{-1}.

The longitudinal speed in any direction is bounded from below by the bulk modulus. The largest eigenvalue of a symmetric matrix is at least its expectation value in any unit vector, and with the trial vector $\hat{\boldsymbol{Q}}$ itself, $\rho_T c_{\mathrm{LA}}^2 \geq \hat{Q}_i\mathsf{M}_{ik}\hat{Q}_k = C_{ijkl}\hat{Q}_i\hat{Q}_j\hat{Q}_k\hat{Q}_l$, the Rayleigh quotient.

For cubic symmetry the quartic form collects C_{11} from the four equal indices. For each pair of axes it collects $2C_{12}$ from $xxyy$ and $yyxx$, and $4C_{44}$ from $xyxy$, $xyyx$, $yxxy$ and $yxyx$:

$$C_{ijkl}\hat{Q}_i\hat{Q}_j\hat{Q}_k\hat{Q}_l = C_{11}\sum_i \hat{Q}_i^4 + (2C_{12}+4C_{44})\,s_Q = C_{11} - 2(C_{11}-C_{12}-2C_{44})\,s_Q,$$

with $s_Q = \hat{Q}_x^2\hat{Q}_y^2 + \hat{Q}_y^2\hat{Q}_z^2 + \hat{Q}_z^2\hat{Q}_x^2$, where the second form uses $\sum_i \hat{Q}_i^4 = 1 - 2s_Q$. Since s_Q ranges from 0 along [100] to $\frac{1}{3}$ along [111], and the quartic form is linear in s_Q, it lies between its values along [100] and [111]. These two values exceed B by $\frac{2}{3}(C_{11}-C_{12})$ and $\frac{4}{3}C_{44}$, both positive by Eq. (C.152).

Bulk-modulus bound on the longitudinal speed

In a cubic crystal that satisfies Eq. (C.152), in every direction,

$$c_{\mathrm{LA}}(\hat{\boldsymbol{Q}}) \geq \sqrt{B/\rho_T}. \tag{C.156}$$

For silicon the bound is $6.48\,\mathrm{km\,s^{-1}}$, 23 % below the slowest longitudinal speed. Physically, a longitudinal wave compresses the medium as a uniform compression does, plus a shear, and the shear only adds stiffness. A tabulated sound speed that violates Eq. (C.156) for a measured B is wrong.

C.18.3 Sphere averages

Rates that sum over directions of $\boldsymbol{Q}$ contain averages of a function $f(\hat{\boldsymbol{Q}})$ over the unit sphere,

$$\langle f\rangle = \frac{1}{4\pi}\int \mathrm{d}\Omega_{\hat{\boldsymbol{Q}}}\, f(\hat{\boldsymbol{Q}}) = \frac{1}{4\pi}\int_{-1}^{1}\mathrm{d}(\cos\theta)\int_0^{2\pi}\mathrm{d}\phi\, f(\theta,\phi). \tag{C.157}$$

Which function is averaged depends on the rate. Sect. 17.5 uses $\langle\sum_\nu(\hat{\boldsymbol{Q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu\rangle$, in which each branch is weighted by the square of its longitudinal component; Debye-type counts of states use $\langle c_\nu^{-3}\rangle$. For an isotropic medium the averages reduce to single values. For a crystal they differ from one another, and they are computed numerically. At each node of a quadrature grid we diagonalize the 3×3 matrix $\mathsf{M}(\hat{\boldsymbol{Q}})$ and evaluate f. Gauss–Legendre nodes in $\cos\theta$ and equally spaced nodes in ϕ, where the trapezoidal rule is exact for trigonometric polynomials, converge quickly for smooth f. The longitudinal speed is smooth, since the largest eigenvalue never meets the two smaller eigenvalues. The transverse speeds touch along [100] and [111], where they are degenerate, so a function of one transverse branch alone has kinks there; a sum over both transverse branches, as in the polarization-weighted average, is smooth. For a cubic crystal the 48 operations of the point group (Appendix C.17) map the sphere onto itself and leave every eigenvalue unchanged, so the average over $\frac{1}{48}$ of the sphere, bounded by the [100], [110] and [111] directions, equals the full average.

C.18.4 Acoustic impedance and reflection

When a sound wave meets the interface between two media, part of its energy is reflected. Take two media, labeled 1 for $z<0$ and 2 for $z>0$, with densities ρ_1, ρ_2, and a longitudinal wave of frequency ω incident along z, at normal incidence. In medium 1 the displacement is $u_z = A\,[\mathrm{e}^{\mathrm{i}(Q_1 z-\omega t)} + r_{12}\,\mathrm{e}^{\mathrm{i}(-Q_1 z-\omega t)}]$ and in medium 2 it is $u_z = A\,t_{12}^{\mathrm{amp}}\,\mathrm{e}^{\mathrm{i}(Q_2 z-\omega t)}$, with $Q_i = \omega/c_i$ and c_i the longitudinal speed of medium i along z. A welded interface transmits displacement and force, so u_z and $\sigma_{zz} = \rho_i c_i^2\,\partial_z u_z$ are continuous at $z=0$:

$$1 + r_{12} = t_{12}^{\mathrm{amp}}, \qquad \mathrm{i}Q_1\rho_1 c_1^2\,(1-r_{12}) = \mathrm{i}Q_2\rho_2 c_2^2\,t_{12}^{\mathrm{amp}}.$$

Since $Q_i \rho_i c_i^2 = \omega \rho_i c_i$, the second condition contains each medium only through its *acoustic impedance*

$$Z_{\mathrm{ac},i} = \rho_i \, c_i. \tag{C.158}$$

The two conditions give $r_{12} = (Z_{\mathrm{ac},1} - Z_{\mathrm{ac},2})/(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})$ and $t_{12}^{\mathrm{amp}} = 2Z_{\mathrm{ac},1}/(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})$. The energy flux of a wave is the power per unit area $-\sigma_{zz}\dot{u}_z$; for a real wave $u_z = A\cos(Qz - \omega t)$ its time average is $\frac{1}{2} Z_{\mathrm{ac}}\, \omega^2 A^2$. The fraction of the incident flux that crosses the interface is therefore $Z_{\mathrm{ac},2}|t_{12}^{\mathrm{amp}}|^2 / Z_{\mathrm{ac},1}$:

$$t_{12}(0) = \frac{4 Z_{\mathrm{ac},1} Z_{\mathrm{ac},2}}{(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})^2} = 1 - r_{12}^2. \tag{C.159}$$

The argument 0 is the angle of incidence. The result is symmetric in the two media, $t_{12}(0) = t_{21}(0)$, and equals one when the impedances match, whatever the densities and speeds separately are [32].

Silicon and aluminum are almost matched. Along [100], silicon has $Z_{\mathrm{ac}} = 19.6 \times 10^6 \,\mathrm{kg\,m^{-2}\,s^{-1}}$. Aluminum, with $\rho = 2.70 \,\mathrm{g\,cm^{-3}}$ and the elastic constants extrapolated to zero temperature, $C_{11} = 114.3$, $C_{12} = 61.9$ and $C_{44} = 31.6\,\mathrm{GPa}$ [33, Table 2], has $c_{\mathrm{LA}} = 6.51\,\mathrm{km\,s^{-1}}$ along [100] and $Z_{\mathrm{ac}} = 17.6 \times 10^6 \,\mathrm{kg\,m^{-2}\,s^{-1}}$. Aluminum is nearly isotropic, with $2C_{44}/(C_{11} - C_{12}) = 1.21$. The texture of a film therefore changes its longitudinal speed by at most 3 % above the [100] value, and the isotropic average is $6.63\,\mathrm{km\,s^{-1}}$ (Sect. 20.3). Eq. (C.159) then gives $t_{12}(0) = 0.997$: at normal incidence only a third of a percent of a longitudinal flux is reflected.

At oblique incidence the wavevector component along the interface is the same for every wave that the incident one generates, since the boundary conditions must hold at every point of the plane and at every time. For incidence at angle θ_1 on a wave of speed c_1, a reflected or transmitted wave of speed c' leaves at the angle θ' with

$$\frac{\sin\theta_1}{c_1} = \frac{\sin\theta'}{c'}, \tag{C.160}$$

the acoustic Snell law. In general an incident longitudinal wave generates reflected and transmitted waves of both longitudinal and transverse polarization (mode conversion), with amplitudes fixed by the continuity of all three displacement components and of the three stresses σ_{iz}. When $c' > c_1$ the wave c' exists only for $\sin\theta_1 < c_1/c'$; beyond that critical angle it is evanescent and carries no flux. Since all sound speeds of aluminum lie below the longitudinal speed of silicon, the critical angle matters for phonons leaving the film: a longitudinal phonon from aluminum along [100] reaches longitudinal states in silicon only within $\arcsin(6.51/8.43) = 51^\circ$ of the normal. Sect. 20.3 solves the oblique problem and averages the transmission over angles.

C.19 Magnetic neutron scattering

A neutron carries a magnetic moment of -1.913 nuclear magnetons, and in a magnetic crystal it scatters off the unpaired electron spins about as strongly as off the nuclei. The neutron spin couples to the same spin density $\boldsymbol{s}_e(\boldsymbol{q})$ as a dark-matter particle with a spin-dependent coupling to electrons, so Sect. 14.4 can turn measured neutron spectra into dark-matter spin responses. The translation needs the cross-section written with the response matrix S_{ab} of Sect. 4.2, the projection that removes the component of the spin density along the momentum transfer, and the magnetic form factor of an ion. Polarized beams then separate the two components that remain.

C.19.1 The magnetic cross-section

A neutron of mass m_n enters with momentum $\boldsymbol{p}_n$ and leaves with $\boldsymbol{p}'_n$. As for dark matter, $\boldsymbol{q} = \boldsymbol{p}_n - \boldsymbol{p}'_n$ is the momentum given to the target and $\omega = (p_n^2 - p_n'^2)/2m_n$ the energy. The neutron moment is $\boldsymbol{\mu}_n = \gamma_n \mu_N \boldsymbol{\sigma}_n$, with $\gamma_n = -1.913$, the nuclear magneton $\mu_N = e/2m_p$ and the Pauli matrices $\boldsymbol{\sigma}_n$ of the neutron spin [34, Chap. 7]. The interaction of $\boldsymbol{\mu}_n$ with the magnetic field of the electrons is $-\boldsymbol{\mu}_n \cdot \boldsymbol{B}_e(\boldsymbol{r}_n)$.

The spins of the electrons, with $g_e = 2$, form the magnetization $\boldsymbol{M}_e(\boldsymbol{r}) = -2\mu_B\, \boldsymbol{s}_e(\boldsymbol{r})$, where $\boldsymbol{s}_e(\boldsymbol{r})$ is the spin density whose Fourier transform is $\boldsymbol{s}_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\boldsymbol{\sigma}_i/2$ (Sect. 7.1). Without free currents, magnetostatics in Heaviside–Lorentz units reads $\nabla\cdot\boldsymbol{B}_e = 0$ and $\nabla\times\boldsymbol{B}_e = \nabla\times\boldsymbol{M}_e$. In Fourier space, at $\boldsymbol{q} \neq 0$, these are $\boldsymbol{q}\cdot\boldsymbol{B}_e(\boldsymbol{q}) = 0$ and $\boldsymbol{q}\times[\boldsymbol{B}_e(\boldsymbol{q}) - \boldsymbol{M}_e(\boldsymbol{q})] = 0$. The second condition makes $\boldsymbol{B}_e - \boldsymbol{M}_e$ parallel to $\boldsymbol{q}$, and the first condition then fixes $\boldsymbol{B}_e$:

$$B_{e,a}(\boldsymbol{q}) = \sum_b (P_T)_{ab}\, M_{e,b}(\boldsymbol{q}), \qquad (P_T)_{ab} = \delta_{ab} - \hat{q}_a\hat{q}_b. \tag{C.161}$$

Only the component of the magnetization perpendicular to $\boldsymbol{q}$ produces a field at wavevector $\boldsymbol{q}$. A magnetization that is modulated along its own direction, $\boldsymbol{M}_e \parallel \boldsymbol{q}$, is a pattern of magnetic poles whose demagnetizing field cancels the magnetization exactly, so a magnetization parallel to $\boldsymbol{q}$ does not scatter neutrons.

To first order in the interaction (the Born approximation, Sect. 2.2), the amplitude for scattering a neutron is $-(m_n/2\pi)$ times the Fourier transform of its potential, here the operator $-\boldsymbol{\mu}_n\cdot\boldsymbol{B}_e(\boldsymbol{q}) = 2\gamma_n\mu_N\mu_B\,\boldsymbol{\sigma}_n\cdot P_T\boldsymbol{s}_e(\boldsymbol{q})$, with P_T acting on the vector index. With $m_n = m_p$, which holds to 0.14 %, the constants combine as $(m_n/2\pi)\cdot 2\gamma_n(e/2m_p)(e/2m_e) = \gamma_n\, e^2/(4\pi m_e)$. The combination is $\gamma_n r_{\mathrm{cl}}$, with the classical electron radius $r_{\mathrm{cl}} = e^2/(4\pi m_e) = \alpha/m_e = 2.818\,\mathrm{fm}$. The magnetic amplitude operator is therefore

$$\hat{f}_{\mathrm{mag}} = -\gamma_n r_{\mathrm{cl}}\, \boldsymbol{\sigma}_n\cdot\boldsymbol{s}_\perp(\boldsymbol{q}), \qquad \boldsymbol{s}_\perp(\boldsymbol{q}) = P_T\,\boldsymbol{s}_e(\boldsymbol{q}). \tag{C.162}$$

The double-differential cross-section is the flux factor p'_n/p_n times $|\hat{f}_{\mathrm{mag}}|^2$, summed over target final states f with the energy delta function and averaged over the thermal initial states i, with the weights p_i of Eq. (2.62). For an unpolarized beam the average over the neutron spin is $\frac{1}{2}\,\mathrm{Tr}(\sigma_{n,a}\sigma_{n,b}) = \delta_{ab}$. Then $\sum_a s^\dagger_{\perp,a}s_{\perp,a} = \sum_{ab} s^\dagger_{e,a}(P_T)_{ab}s_{e,b}$, since P_T is real, symmetric and equal to its square. Comparing with the definition of S_{ab} in Sect. 4.2, with the operators $O_a = s_{e,a}(\boldsymbol{q})$, we obtain

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\,\mathrm{d}E'} = (\gamma_n r_{\mathrm{cl}})^2\,\frac{p'_n}{p_n}\,\frac{\mathscr{V}}{2\pi}\sum_{a,b}(P_T)_{ab}\,S_{ab}(\boldsymbol{q},\omega), \tag{C.163}$$

for the whole sample of volume $\mathscr{V}$; the factor $\mathscr{V}/2\pi$ undoes the normalization of S_{ab}, which is intensive. The scale is $(\gamma_n r_{\mathrm{cl}})^2 = 0.2906\,\mathrm{b}$, to be compared with squared nuclear scattering lengths of 0.1 to 1 b; in this sense the two kinds of scattering are equally strong. Eq. (C.163) is the standard result of magnetic neutron scattering [34, Chap. 7], [35]. The sum $\sum_{ab}(P_T)_{ab}S_{ab}$ is the combination that Berlin et al. extract from neutron data [36, Eq. (26)]. Their normalization is per formula unit, their spin operator is $\boldsymbol{\sigma}$ in place of $\boldsymbol{\sigma}/2$, and their susceptibility $\chi''_{ab} = 2(1-\mathrm{e}^{-\beta\omega})S_{ab}$ carries $\pi/\mathscr{V}$ where S_{ab} carries $2\pi/\mathscr{V}$ [36, Eq. (B4)]. Since the spin density is localized on the ions, which vibrate, the sum over ions carries the Debye–Waller factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$ of Sect. 5.5; we absorb this factor into S_{ab}.

C.19.2 Ions and the dipole form factor

In the magnetic insulators of Sect. 14.1 the unpaired electrons occupy an open shell of each magnetic ion j at $\boldsymbol{R}_j$. If the shell has a spin $\boldsymbol{S}_j$ and no orbital moment, as in a 3d ion whose

orbital moment is quenched by the crystal field, its spin density is $\boldsymbol{S}_j$ times a normalized density $\varrho_{\rm s}(\boldsymbol{r}-\boldsymbol{R}_j)$ of the unpaired electrons. Then

$$\boldsymbol{s}_e(\boldsymbol{q}) = F^{\rm mag}(q)\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{S}_j, \qquad F^{\rm mag}(q) = \int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\varrho_{\rm s}(\boldsymbol{r}) = \langle j_0(q)\rangle, \tag{C.164}$$

with $F^{\rm mag}(0)=1$. For a spherical shell with radial function $R_{\rm sh}(r)$, the plane-wave expansion Eq. (C.22) keeps only $l=0$, and $\langle j_l(q)\rangle = \int_0^\infty \mathrm{d}r\,r^2 R_{\rm sh}(r)^2 j_l(qr)$ are the radial integrals that tables give for each ion. The form factor falls off when q exceeds the inverse radius of the shell, at a few Å^{-1} for a 3d shell, so magnetic scattering is strongest in the first few Brillouin zones.

If the shell has an orbital moment, as in most rare-earth ions, the orbital currents add to $\boldsymbol{M}_e$. For a J multiplet in the dipole approximation, $qr \lesssim 1$ across the shell, the spin part of the magnetization carries $\langle j_0\rangle$ and the orbital part $\langle j_0\rangle + \langle j_2\rangle$ [34, Chap. 7]. Within the multiplet $\boldsymbol{S} = (g_J-1)\boldsymbol{J}$ and $\boldsymbol{L} = (2-g_J)\boldsymbol{J}$, with g_J the Landé factor of Sect. 14.3 and $\boldsymbol{J}_j$ the total angular momentum of ion j. The moment $2\boldsymbol{S}\langle j_0\rangle + \boldsymbol{L}(\langle j_0\rangle + \langle j_2\rangle)$ then equals $g_J[\langle j_0\rangle + (2/g_J - 1)\langle j_2\rangle]\boldsymbol{J}$, and the neutron couples to

$$\boldsymbol{s}_e(\boldsymbol{q}) \to \frac{g_J}{2}\,F^{\rm mag}(q)\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{J}_j, \qquad F^{\rm mag}(q) = \langle j_0(q)\rangle + \Big(\frac{2}{g_J}-1\Big)\langle j_2(q)\rangle, \tag{C.165}$$

the dipole form factor. For $g_J = 2$ it reduces to Eq. (C.164). The neutron sees the moment $(g_J/2)\boldsymbol{J} = \boldsymbol{S} + \boldsymbol{L}/2$, while a probe that couples to spin alone sees $\boldsymbol{S} = (g_J-1)\boldsymbol{J}$. The ratio of the two moments is the conversion factor for neutron data.

C.19.3 Polarization analysis

A polarized beam resolves $\boldsymbol{s}_\perp$ into components. We add the nuclear amplitude, an operator $\hat{f}_{\rm nuc}$ on the target that does not act on the neutron spin, so that the full amplitude is $\hat{f}_{\rm nuc} - \gamma_n r_{\rm cl}\,\boldsymbol{\sigma}_n\cdot\boldsymbol{s}_\perp$. We take the neutron spin quantized along the polarization direction $\hat{\boldsymbol{P}}$, with the two other axes labeled 1 and 2. The Pauli matrices then give the four spin matrix elements

$$\langle \pm|\hat{f}|\pm\rangle = \hat{f}_{\rm nuc} \mp \gamma_n r_{\rm cl}\,s_{\perp,\hat{P}}, \qquad \langle \mp|\hat{f}|\pm\rangle = -\gamma_n r_{\rm cl}\,(s_{\perp,1} \pm \mathrm{i}\,s_{\perp,2}). \tag{C.166}$$

Scattering without spin flip sees the nuclear amplitude and the component of $\boldsymbol{s}_\perp$ along $\hat{\boldsymbol{P}}$. Scattering with spin flip sees only the magnetic components perpendicular to $\hat{\boldsymbol{P}}$, and the difference between the two spin-flip channels measures the chiral combination $\mathrm{i}(s^\dagger_{\perp,1}s_{\perp,2} - s^\dagger_{\perp,2}s_{\perp,1})$ [37, 38].

The standard frame has x along $\hat{\boldsymbol{q}}$, y perpendicular to it in the scattering plane and z vertical. Since $\boldsymbol{s}_\perp$ has no x component, the spin-flip cross-sections, averaged over the two incident spin states, are proportional to $S_{yy}+S_{zz}$ for $\hat{\boldsymbol{P}} \parallel x$, to S_{zz} for $\hat{\boldsymbol{P}} \parallel y$, and to S_{yy} for $\hat{\boldsymbol{P}} \parallel z$. The first channel contains all the magnetic scattering and no coherent nuclear scattering, and the second and third channels separate the two transverse components [38]. The longitudinal element $S_{xx} = \sum_{ab}\hat{q}_a\hat{q}_b S_{ab}$ appears in none of the channels, polarized or not, because every amplitude in Eq. (C.166) contains $\boldsymbol{s}_e$ only through $\boldsymbol{s}_\perp$. A dark-matter response that needs S_{xx}, such as the contact spin coupling or the absorption of an axion-like particle, therefore cannot be taken from neutron data alone; Sect. 14.4 bounds S_{xx} by positivity and fixes it under spin isotropy.

C.20 Dissipative two-level systems and kinks

A hydrogen atom trapped in a metal, or a group of atoms in a glass, can sit in either of two nearby minima and tunnel between them. In a metal the tunneling is dressed by the conduction electrons, which must rearrange every time the atom moves, and the result is a hopping rate that, for weak coupling ($\alpha_{\rm ohm} < \frac{1}{2}$ in Eq. (C.173)), grows as the temperature falls. A second kind of mobile defect is an extra atom squeezed into a close-packed row of a crystal: it spreads over many neighbors and glides along the row over a barrier of micro-electronvolts. Sect. 16.8 treats the tunneling defect through the spin–boson model, and Sect. 19.5 treats the extra atom through the Frenkel–Kontorova chain.

C.20.1 The spin–boson model

Let $|\mathrm{L}\rangle$ and $|\mathrm{R}\rangle$ be the two localized configurations, with Pauli matrices $\sigma_z = |\mathrm{L}\rangle\langle\mathrm{L}| - |\mathrm{R}\rangle\langle\mathrm{R}|$ and $\sigma_x = |\mathrm{L}\rangle\langle\mathrm{R}| + |\mathrm{R}\rangle\langle\mathrm{L}|$. The energies of the two configurations differ by the asymmetry $\Delta_{\rm as}$, and the overlap of the two localized wavefunctions through the barrier gives the tunneling amplitude Δ_0. The environment, the bath, is a set of harmonic modes d_k of energies ω_k, each displaced in opposite directions by the two configurations:

$$H = \tfrac{1}{2}\Delta_{\rm as}\,\sigma_z - \tfrac{1}{2}\Delta_0\,\sigma_x + \sum_k \omega_k\, d_k^\dagger d_k + \tfrac{1}{2}\,\sigma_z \sum_k \lambda_k (d_k + d_k^\dagger). \tag{C.167}$$

Without the modes, the eigenstates are split by $\Delta_{\rm tl} = (\Delta_{\rm as}^2 + \Delta_0^2)^{1/2}$, the two-level splitting of Sect. 2.5. Any environment whose response to the defect is linear has Gaussian fluctuations, and these are reproduced by a suitable set of oscillators [39]. Everything the bath does then enters through its spectral density

$$J_{\rm s}(\omega) = \pi \sum_k \lambda_k^2\, \delta(\omega - \omega_k). \tag{C.168}$$

Its power law at low ω fixes the long-time decay of the bath correlation, Eq. (C.170) below. For acoustic phonons coupled through the strain, the coupling of a mode grows as $\lambda_k^2 \propto \omega_k$, since the strain amplitude of one phonon is $\propto Q(\rho_T \mathcal{V} \omega_{\boldsymbol{Q}\nu})^{-1/2}$, and the density of modes grows as ω^2; then $J_{\rm s} \propto \omega^3$ (superohmic). For conduction electrons, the bath excitations are electron–hole pairs. If the two configurations scatter electrons differently near the Fermi level, the matrix elements are roughly constant. The number of pairs of energy ω grows as ω, and $J_{\rm s}$ is linear (ohmic),

$$J_{\rm s}(\omega) = 2\pi\, \alpha_{\rm ohm}\, \omega\, \mathrm{e}^{-\omega/\omega_{\rm cut}}, \tag{C.169}$$

with a dimensionless coupling $\alpha_{\rm ohm}$ and a cutoff $\omega_{\rm cut}$ of the order of the Fermi energy. Sect. 16.8 computes $\alpha_{\rm ohm}$ from the scattering phase shifts. Ref. [39] writes α for $\alpha_{\rm ohm}$ and Ref. [40] writes K, and both references use the normalization of Eq. (C.169).

C.20.2 The bath correlation function

In configuration L the term $+\frac{1}{2}\lambda_k(d_k + d_k^\dagger)$ shifts the equilibrium of mode k, and in R the shift is reversed. Completing the square gives $H_{\rm L} = \sum_k \omega_k (d_k^\dagger + \lambda_k/2\omega_k)(d_k + \lambda_k/2\omega_k) + \text{const}$, and $H_{\rm R}$ is the same with $\lambda_k \to -\lambda_k$. The two equilibria differ by $\delta_k = \lambda_k/\omega_k$ in the ladder operators. With $b_k = d_k + \lambda_k/2\omega_k$ and the displacement operator $\hat{D}_k = \hat{D}(\delta_k)$ of Eq. (C.48) in b_k, which shifts b_k by δ_k, the two Hamiltonians are related by $H_{\rm R} = \hat{D} H_{\rm L} \hat{D}^\dagger$, with $\hat{D} = \prod_k \hat{D}_k$, up to constants that cancel.

The quantity a transition needs is the correlation $C(t) = \langle \mathrm{e}^{\mathrm{i}H_{\rm L}t} \mathrm{e}^{-\mathrm{i}H_{\rm R}t} \rangle_{\rm L}$, the overlap at time t between the bath evolving with the defect in L and in R, averaged over the

equilibrium of L. Since $\mathrm{e}^{-\mathrm{i}H_\mathrm{R}t} = \hat{D}\mathrm{e}^{-\mathrm{i}H_\mathrm{L}t}\hat{D}^\dagger$, $C(t) = \langle \hat{D}(t)\hat{D}^\dagger\rangle_\mathrm{L}$, where $\hat{D}(t) = \mathrm{e}^{\mathrm{i}H_\mathrm{L}t}\hat{D}\mathrm{e}^{-\mathrm{i}H_\mathrm{L}t}$ has $b_k \to b_k\mathrm{e}^{-\mathrm{i}\omega_k t}$. For one mode, write $\hat{D}(t) = \mathrm{e}^{\hat{X}}$ and $\hat{D}^\dagger = \mathrm{e}^{\hat{Y}}$ with $\hat{X} = \delta(b^\dagger \mathrm{e}^{\mathrm{i}\omega t} - b\,\mathrm{e}^{-\mathrm{i}\omega t})$ and $\hat{Y} = -\delta(b^\dagger - b)$. Their commutator is the number $[\hat{X},\hat{Y}] = -2\mathrm{i}\delta^2\sin\omega t$, so Eq. (C.47) gives $\mathrm{e}^{\hat{X}}\mathrm{e}^{\hat{Y}} = \mathrm{e}^{\hat{X}+\hat{Y}}\mathrm{e}^{-\mathrm{i}\delta^2\sin\omega t}$. The operator $\hat{X}+\hat{Y} = \delta(b^\dagger u - b\,u^*)$, with $u = \mathrm{e}^{\mathrm{i}\omega t}-1$, is linear, and Bloch's identity Eq. (C.51) gives $\langle \mathrm{e}^{\hat{X}+\hat{Y}}\rangle = \exp[-\frac{1}{2}\delta^2|u|^2(2n_\mathrm{B}(\omega)+1)]$, with $|u|^2 = 2(1-\cos\omega t)$ and $2n_\mathrm{B}+1 = \coth(\omega/2T)$. Multiplying over modes and using Eq. (C.168),

$$C(t) = \mathrm{e}^{-\mathcal{Q}(t)}, \qquad \mathcal{Q}(t) = \frac{1}{\pi}\int_0^\infty \mathrm{d}\omega\,\frac{J_\mathrm{s}(\omega)}{\omega^2}\Big[\coth\frac{\omega}{2T}\,(1-\cos\omega t) + \mathrm{i}\sin\omega t\Big]. \tag{C.170}$$

At $T=0$ and for one mode, $C(t) = \exp[-\delta^2(1-\mathrm{e}^{-\mathrm{i}\omega t})] = \mathrm{e}^{-\delta^2}\sum_n(\delta^2)^n\mathrm{e}^{-\mathrm{i}n\omega t}/n!$: the Poisson distribution of the number of bath quanta emitted when the defect jumps, as in the Franck–Condon problem of Sect. 5.5.

For the ohmic bath at $T \ll \omega_\mathrm{cut}$ and $|t| \gg 1/\omega_\mathrm{cut}$ the integral has a closed form. The imaginary part is $2\alpha_\mathrm{ohm}\int_0^\infty \mathrm{d}\omega\,\mathrm{e}^{-\omega/\omega_\mathrm{cut}}\sin(\omega t)/\omega = 2\alpha_\mathrm{ohm}\arctan(\omega_\mathrm{cut}t) \to \pi\alpha_\mathrm{ohm}\operatorname{sgn} t$. For the real part we write $\coth(\omega/2T) = 1 + 2n_\mathrm{B}(\omega)$ and use one integral twice,

$$\int_0^\infty \frac{\mathrm{d}\omega}{\omega}\,\mathrm{e}^{-\omega/\Omega}\,(1-\cos\omega t) = \tfrac{1}{2}\ln(1+\Omega^2t^2),$$

which follows by differentiating with respect to t, since $\int_0^\infty \mathrm{d}\omega\,\mathrm{e}^{-\omega/\Omega}\sin\omega t = \Omega^2 t/(1+\Omega^2t^2)$. With $\Omega = \omega_\mathrm{cut}$ it gives the zero-temperature part, $\alpha_\mathrm{ohm}\ln(1+\omega_\mathrm{cut}^2t^2) \simeq 2\alpha_\mathrm{ohm}\ln(\omega_\mathrm{cut}|t|)$. The thermal part, where the cutoff can be dropped, has $n_\mathrm{B}(\omega) = \sum_{m\geq1}\mathrm{e}^{-m\omega/T}$, and each term is the same integral with $\Omega = T/m$. Summing, $4\alpha_\mathrm{ohm}\cdot\frac{1}{2}\sum_m\ln(1+T^2t^2/m^2) = 2\alpha_\mathrm{ohm}\ln[\sinh(\pi T|t|)/(\pi T|t|)]$, by the product $\sinh x/x = \prod_{m\geq1}(1+x^2/\pi^2m^2)$. Together,

$$\mathcal{Q}(t) = 2\alpha_\mathrm{ohm}\ln\Big[\frac{\omega_\mathrm{cut}}{\pi T}\sinh(\pi T|t|)\Big] + \mathrm{i}\pi\alpha_\mathrm{ohm}\operatorname{sgn} t. \tag{C.171}$$

The real part grows without bound, linearly in $|t|$ at times beyond $1/T$, so the overlap decays and the bath forgets which configuration it was prepared in. At $T=0$ the overlap decays only as a power, $|C| = (\omega_\mathrm{cut}|t|)^{-2\alpha_\mathrm{ohm}}$. This slow decay is the orthogonality catastrophe of a Fermi sea [39]: the static overlap $\exp[-\frac{1}{2}\sum_k\delta_k^2]$ of the two bath ground states vanishes, since $\sum_k\delta_k^2 = \pi^{-1}\int\mathrm{d}\omega\,J_\mathrm{s}/\omega^2$ diverges logarithmically at small ω.

C.20.3 Incoherent tunneling rate

When the bath relaxes faster than the defect tunnels, the defect hops between L and R at a rate that is second order in Δ_0. The perturbation $O = -\frac{1}{2}\Delta_0|\mathrm{R}\rangle\langle\mathrm{L}|$ moves the defect from L to R, and the thermal golden rule in the correlation form Eq. (2.66), at zero energy given to the external probe, gives $\Gamma_\mathrm{LR} = \int\mathrm{d}t\,\langle O^\dagger(t)O(0)\rangle$. Evolving O under H without the tunneling term gives $\mathrm{e}^{\mathrm{i}(E_\mathrm{L}-E_\mathrm{R})t}$, with $E_\mathrm{L}-E_\mathrm{R} = \Delta_\mathrm{as}$, times $C(t)$:

$$\Gamma_\mathrm{LR} = \frac{\Delta_0^2}{4}\int_{-\infty}^\infty \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\Delta_\mathrm{as}t}\,\mathrm{e}^{-\mathcal{Q}(t)}. \tag{C.172}$$

Expanding $C(t)$ in exponentials, as in the one-mode example, shows that each term is an energy-conserving jump in which the defect gives Δ_as to the bath. Ref. [39] obtains the same rate from the path integral and calls this second-order approximation the noninteracting-blip approximation.

For the ohmic bath, Eq. (C.171) makes the integrand of Eq. (C.172)

$$\mathrm{e}^{\mathrm{i}\Delta_\mathrm{as}t}\,\mathrm{e}^{-\mathcal{Q}(t)} = \mathrm{e}^{\mathrm{i}\Delta_\mathrm{as}t}\Big[\frac{\pi T}{\omega_\mathrm{cut}\,\mathrm{i}\sinh\big(\pi T(t-\mathrm{i}0)\big)}\Big]^{2\alpha_\mathrm{ohm}}.$$

For $t>0$ the factor $\mathrm{i}^{-2\alpha_{\rm ohm}}$ gives the phase $\mathrm{e}^{-\mathrm{i}\pi\alpha_{\rm ohm}}$, and for $t<0$, where $\sinh(\pi T(t-\mathrm{i}0))=|\sinh|\,\mathrm{e}^{-\mathrm{i}\pi}$, the phase is $\mathrm{e}^{+\mathrm{i}\pi\alpha_{\rm ohm}}$. The integrand is analytic in the strip $-1/T<\operatorname{Im}t<0$, and we shift the contour to $t=x-\mathrm{i}/2T$. There $\mathrm{i}\sinh(\pi Tx-\mathrm{i}\pi/2)=\cosh(\pi Tx)$ and $\mathrm{e}^{\mathrm{i}\Delta_{\rm as}t}=\mathrm{e}^{\mathrm{i}\Delta_{\rm as}x}\mathrm{e}^{\Delta_{\rm as}/2T}$. The remaining Fourier transform follows from $y=(1+\mathrm{e}^{-2\xi})^{-1}$, which turns $\operatorname{sech}^{2\alpha}\xi\,\mathrm{e}^{\mathrm{i}\nu\xi}\,\mathrm{d}\xi$ into $2^{2\alpha-1}y^{\alpha+\mathrm{i}\nu/2-1}(1-y)^{\alpha-\mathrm{i}\nu/2-1}\mathrm{d}y$, a Beta-function integral:

$$\int_{-\infty}^{\infty}\mathrm{d}\xi\,\frac{\mathrm{e}^{\mathrm{i}\nu\xi}}{\cosh^{2\alpha}\xi}=2^{2\alpha-1}\,\frac{|\Gamma(\alpha+\mathrm{i}\nu/2)|^2}{\Gamma(2\alpha)},$$

with $\Gamma(z)$ Euler's gamma function. With $\xi=\pi Tx$ and $\nu=\Delta_{\rm as}/\pi T$,

Incoherent rate in an ohmic bath

$$\Gamma_{\rm LR}=\frac{\Delta_0^2}{4\omega_{\rm cut}}\Big(\frac{2\pi T}{\omega_{\rm cut}}\Big)^{2\alpha_{\rm ohm}-1}\mathrm{e}^{\Delta_{\rm as}/2T}\,\frac{|\Gamma(\alpha_{\rm ohm}+\mathrm{i}\Delta_{\rm as}/2\pi T)|^2}{\Gamma(2\alpha_{\rm ohm})},\qquad \Gamma_{\rm RL}=\mathrm{e}^{-\Delta_{\rm as}/T}\,\Gamma_{\rm LR}. \tag{C.173}$$

The reverse rate follows from $\Delta_{\rm as}\to-\Delta_{\rm as}$, and since $|\Gamma|^2$ is even in $\Delta_{\rm as}$ the two rates satisfy detailed balance. For a symmetric defect, by the duplication formula of Γ, the rate is $\Gamma_{\rm LR}=(\Delta_0^2/4\omega_{\rm cut})\,\sqrt{\pi}\,\Gamma(\alpha_{\rm ohm})/\Gamma(\alpha_{\rm ohm}+\frac12)\,(\pi T/\omega_{\rm cut})^{2\alpha_{\rm ohm}-1}$. It scales as $T^{2\alpha_{\rm ohm}-1}$, and for $\alpha_{\rm ohm}<\frac12$ it grows as the temperature falls. As T falls, fewer slow modes are thermally excited, the tunneling amplitude dressed by the bath shrinks as $\Delta_0(\pi T/\omega_{\rm cut})^{\alpha_{\rm ohm}}$, and its square falls as $T^{2\alpha_{\rm ohm}}$. At the same time the bath dephases the two configurations more slowly, over the time $1/T$ of Eq. (C.171), and the rate is roughly the squared amplitude times that time. For $\alpha_{\rm ohm}<\frac12$ the time grows faster than the squared amplitude falls. For $\alpha_{\rm ohm}\ll1$ the relaxation rate $\Gamma_{\rm LR}+\Gamma_{\rm RL}$ of a symmetric defect reduces to $\Delta_0^2/(2\pi\alpha_{\rm ohm}T)$. This relaxation rate is, to leading order in $\alpha_{\rm ohm}$, the overdamped limit $\tilde\Delta_0^2/\gamma_0$ of the Lorentzian of [40, Eq. (10)], with the dephasing rate $\gamma_0=2\pi\alpha_{\rm ohm}T$ and the dressed amplitude $\tilde\Delta_0^2=\Gamma(1-2\alpha_{\rm ohm})\cos(\pi\alpha_{\rm ohm})\,\Delta_0^2(2\pi T/\omega_{\rm cut})^{2\alpha_{\rm ohm}}$ (Sect. 16.8).

The rate depends on Δ_0 and $\omega_{\rm cut}$ only through the combination $\Delta_0^2\,\omega_{\rm cut}^{-2\alpha_{\rm ohm}}$. Defining, for $\alpha_{\rm ohm}<1$, the renormalized tunneling amplitude $\Delta_{\rm r}$ [39] by

$$\Delta_{\rm r}^{2(1-\alpha_{\rm ohm})}=\Gamma(1-2\alpha_{\rm ohm})\cos(\pi\alpha_{\rm ohm})\;\Delta_0^2\,\omega_{\rm cut}^{-2\alpha_{\rm ohm}}, \tag{C.174}$$

the cutoff drops out of every low-energy quantity: for $\Delta_0,T\ll\omega_{\rm cut}$ at fixed $\alpha_{\rm ohm}$, the dynamics depends on $\Delta_{\rm r}$, $\alpha_{\rm ohm}$, $\Delta_{\rm as}$ and T alone. This cutoff independence defines the scaling limit. The form of $\Delta_{\rm r}$ can be understood by an adiabatic argument. Bath modes much faster than the tunneling follow the defect and dress the tunneling amplitude with their overlap, $\exp[-\frac12\sum_{\omega_k>p\Delta_{\rm r}}\delta_k^2]=(p\Delta_{\rm r}/\omega_{\rm cut})^{\alpha_{\rm ohm}}$ for an ohmic bath, with p a number of order one. Requiring the dressed amplitude to be $\Delta_{\rm r}$ itself gives $\Delta_{\rm r}=\Delta_0(p\Delta_{\rm r}/\omega_{\rm cut})^{\alpha_{\rm ohm}}$, and therefore $\Delta_{\rm r}\propto\Delta_0(\Delta_0/\omega_{\rm cut})^{\alpha_{\rm ohm}/(1-\alpha_{\rm ohm})}$, with the prefactor that Eq. (C.174) makes exact. Below $T\sim\Delta_{\rm r}$ the assumption of a fast bath fails, and for $\alpha_{\rm ohm}<\frac12$ a symmetric defect then oscillates coherently, with damping, at a frequency of order $\Delta_{\rm r}$. Eq. (C.173) therefore holds for $\Delta_{\rm r}\ll T\ll\omega_{\rm cut}$, and the growth of the rate on cooling stops near $T\sim\Delta_{\rm r}$. For $\alpha_{\rm ohm}\geq1$ the tunneling amplitude renormalizes to zero and the defect stays localized at $T=0$ [39].

C.20.4 The Frenkel–Kontorova chain

The second model is a row of atoms of mass M_κ, at positions $z_n=na_{\rm ch}+u_n$, joined by springs of constant $K_{\rm ch}$. The row rests in the periodic potential $V_{\rm s}\sin^2(\pi z/a_{\rm ch})$ of the surrounding

crystal, whose period equals the chain spacing $a_{\rm ch}$ [41]:

$$H_{\rm FK} = \sum_n \Big[\frac{p_n^2}{2M_\kappa} + \frac{K_{\rm ch}}{2}\,(u_{n+1}-u_n)^2 + V_{\rm s}\sin^2\frac{\pi u_n}{a_{\rm ch}} \Big]. \tag{C.175}$$

An extra atom in the row, a crowdion, corresponds to u_n going from $a_{\rm ch}$ at $n\to-\infty$ to 0 at $n\to+\infty$: the atoms on the left are each pushed one period, so that the row holds one atom more than the potential has minima [42]. The static equations, $\partial H_{\rm FK}/\partial u_n = 0$, are

$$K_{\rm ch}\,(u_{n+1}-2u_n+u_{n-1}) = \frac{\pi V_{\rm s}}{a_{\rm ch}}\sin\frac{2\pi u_n}{a_{\rm ch}}, \tag{C.176}$$

and the discrete profile is found by minimizing $H_{\rm FK}$ numerically with those boundary values.

When u_n varies slowly from atom to atom, $u_n\to u(z)$ with $u_{n+1}-2u_n+u_{n-1}\to a_{\rm ch}^2\,\partial_z^2 u$, and the phase $\varphi = 2\pi u/a_{\rm ch}$ obeys the sine-Gordon equation

$$\partial_t^2\varphi - c_{\rm ch}^2\,\partial_z^2\varphi + \omega_{\rm s}^2\sin\varphi = 0, \qquad c_{\rm ch} = a_{\rm ch}\sqrt{K_{\rm ch}/M_\kappa}, \qquad \omega_{\rm s}^2 = \frac{2\pi^2 V_{\rm s}}{M_\kappa a_{\rm ch}^2}, \tag{C.177}$$

where $c_{\rm ch}$ is the sound speed of the free chain and $\omega_{\rm s}$ the frequency of an atom oscillating alone in the potential. A static solution has the first integral $\frac12 d_{\rm kink}^2(\partial_z\varphi)^2 = 1-\cos\varphi$, with the kink width $d_{\rm kink} = c_{\rm ch}/\omega_{\rm s} = a_{\rm ch}^2[K_{\rm ch}/(2\pi^2 V_{\rm s})]^{1/2}$. Since $1-\cos\varphi = 2\sin^2(\varphi/2)$, the first integral becomes $d_{\rm kink}\,\partial_z\varphi = \mp 2\sin(\varphi/2)$, and integrating,

$$u(z) = \frac{2a_{\rm ch}}{\pi}\arctan \mathrm{e}^{-(z-z_0)/d_{\rm kink}}, \tag{C.178}$$

which falls from $a_{\rm ch}$ to 0 over a few $d_{\rm kink}$ around the center z_0, as in [42, Eq. (2)]. The first integral states that the spring energy and the substrate energy are equal at every point. Along the kink $\sin^2(\pi u/a_{\rm ch}) = \sin^2(2\arctan \mathrm{e}^{-s}) = \operatorname{sech}^2 s$, with $s = (z-z_0)/d_{\rm kink}$, and with $\sum_n \to \int \mathrm{d}z/a_{\rm ch}$ the rest energy is

$$E_{\rm kink} = \frac{2V_{\rm s}}{a_{\rm ch}}\int \mathrm{d}z\,\operatorname{sech}^2\frac{z-z_0}{d_{\rm kink}} = \frac{4V_{\rm s}d_{\rm kink}}{a_{\rm ch}} = \frac{2a_{\rm ch}}{\pi}\sqrt{2V_{\rm s}K_{\rm ch}}, \tag{C.179}$$

which is [42, Eq. (3)], with $\beta = K_{\rm ch}$ and $V_0 = V_{\rm s}$ there. The sine-Gordon equation is Lorentz invariant with $c_{\rm ch}$ in place of the speed of light. A kink moving at speed v is therefore the static one with $z-z_0 \to (z-z_0-vt)/(1-v^2/c_{\rm ch}^2)^{1/2}$. Its energy is $E_{\rm kink}(1-v^2/c_{\rm ch}^2)^{-1/2} \simeq E_{\rm kink} + \frac12 M_{\rm kink}v^2$, with the kink mass

$$M_{\rm kink} = \frac{E_{\rm kink}}{c_{\rm ch}^2} = \frac{2}{\pi^2}\,\frac{a_{\rm ch}}{d_{\rm kink}}\,M_\kappa. \tag{C.180}$$

A wide kink is light: for $d_{\rm kink} = 3a_{\rm ch}$ it weighs 7 % of an atom, because moving it by one spacing moves each of many atoms by a small fraction of a spacing.

The continuum kink has the same energy wherever its center is. In the discrete chain the energy of a kink depends on where its center is, and a kink must cross the resulting Peierls–Nabarro barrier to move by one spacing. We keep the continuum profile and evaluate the energy on the atoms [42, Eq. (4)], using the equality of the spring and substrate energies to write $E(z_0) \simeq 2V_{\rm s}\sum_n \operatorname{sech}^2[(na_{\rm ch}-z_0)/d_{\rm kink}]$. The Poisson summation formula of Eq. (C.12), here $\sum_n f(na_{\rm ch}) = a_{\rm ch}^{-1}\sum_m \tilde f(2\pi m/a_{\rm ch})$ with $\tilde f(k) = \int \mathrm{d}z\,\mathrm{e}^{-\mathrm{i}kz}f(z)$, and the transform $\int \mathrm{d}z\,\mathrm{e}^{-\mathrm{i}kz}\operatorname{sech}^2(z/d_{\rm kink}) = \pi k d_{\rm kink}^2/\sinh(\pi k d_{\rm kink}/2)$ give

$$E(z_0) = E_{\rm kink} + \frac{8\pi^2 V_{\rm s} d_{\rm kink}^2}{a_{\rm ch}^2}\sum_{m\ge1} m\,\cos\frac{2\pi m z_0}{a_{\rm ch}}\,\operatorname{cosech}\frac{\pi^2 m d_{\rm kink}}{a_{\rm ch}}, \tag{C.181}$$

where the $m = 0$ term reproduces Eq. (C.179). The $m = 1$ term dominates, and the barrier, the difference between the largest and smallest $E(z_0)$, is

$$E_{\mathrm{PN}} \simeq \frac{16\pi^2 V_{\mathrm{s}} d_{\mathrm{kink}}^2}{a_{\mathrm{ch}}^2} \operatorname{cosech} \frac{\pi^2 d_{\mathrm{kink}}}{a_{\mathrm{ch}}} \simeq 32\pi^2 \frac{d_{\mathrm{kink}}^2}{a_{\mathrm{ch}}^2} V_{\mathrm{s}} \, \mathrm{e}^{-\pi^2 d_{\mathrm{kink}}/a_{\mathrm{ch}}}. \tag{C.182}$$

The exponential makes wide kinks nearly free: for $d_{\mathrm{kink}} = 2a_{\mathrm{ch}}$ the barrier is $3 \times 10^{-6} V_{\mathrm{s}}$, a few μeV for a substrate potential of an electronvolt. The energy is lowest with the kink centered between two atoms, $z_0 = a_{\mathrm{ch}}/2$, and highest with an atom at its center. Eq. (C.182) is the leading term for wide kinks. Relaxing the discrete chain of Eq. (C.176) at both centers gives a barrier 17 % lower at $d_{\mathrm{kink}} = 1.23a_{\mathrm{ch}}$, more for narrower kinks (Limit Check 19.7), and the gap between the two barriers closes as the kink widens. Since the continuum profile and the kink mass also assume $d_{\mathrm{kink}} \gg a_{\mathrm{ch}}$, the ratio $d_{\mathrm{kink}}/a_{\mathrm{ch}}$ is the parameter that controls the whole continuum description.

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Appendix D

Long derivations

D.1 Standard-halo integrals

This section evaluates the averages of the standard halo model used in Sects. 1.2 and 1.3: the normalization $N_{\rm esc}$, the laboratory speed distribution $F_\chi(v)$, the mean inverse speed $\langle 1/v\rangle$ and the function $\eta(v_{\rm min})$ for every $v_{\rm min}$. The galactic-frame distribution is $f_{\rm gal}(\boldsymbol{v}_{\rm gal}) = N_{\rm esc}^{-1}\mathrm{e}^{-v_{\rm gal}^2/v_0^2}\Theta(v_{\rm esc}-v_{\rm gal})$, and the laboratory distribution is $f_\chi(\boldsymbol{v}) = f_{\rm gal}(\boldsymbol{v}+\boldsymbol{v}_{\rm E})$ (Eqs. (1.3) and (1.8)). Throughout, $v_{\rm E} = |\boldsymbol{v}_{\rm E}|$ at the time of interest, and we assume $0 < v_{\rm E} < v_{\rm esc}$; the values of Appendix A.8 give $v_{\rm E}/v_{\rm esc} \leq 0.49$. Only one integral is needed:

$$\int_{y_1}^{y_2} \mathrm{e}^{-y^2/v_0^2}\,\mathrm{d}y = \frac{\sqrt{\pi}\,v_0}{2}\left[\mathrm{erf}\left(\frac{y_2}{v_0}\right) - \mathrm{erf}\left(\frac{y_1}{v_0}\right)\right]. \tag{D.1}$$

Eq. (D.1) is the definition of the error function (Appendix C.8).

Normalization

The integrand of $\int \mathrm{d}^3 v_{\rm gal}\,\mathrm{e}^{-v_{\rm gal}^2/v_0^2}\Theta(v_{\rm esc}-v_{\rm gal})$ depends only on $y = v_{\rm gal}$; in spherical coordinates the solid angle contributes 4π and $N_{\rm esc} = 4\pi\int_0^{v_{\rm esc}} y^2\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y$. Integrating by parts with $y\,\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y = \mathrm{d}(-\frac{1}{2}v_0^2\mathrm{e}^{-y^2/v_0^2})$ gives

$$\begin{aligned}\int_0^{v_{\rm esc}} y^2\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y &= \left[-\tfrac{1}{2}v_0^2\,y\,\mathrm{e}^{-y^2/v_0^2}\right]_0^{v_{\rm esc}} + \tfrac{1}{2}v_0^2\int_0^{v_{\rm esc}}\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y \\ &= -\tfrac{1}{2}v_0^2 v_{\rm esc}\mathrm{e}^{-v_{\rm esc}^2/v_0^2} + \frac{\sqrt{\pi}\,v_0^3}{4}\,\mathrm{erf}\left(\frac{v_{\rm esc}}{v_0}\right).\end{aligned}$$

Multiplying by 4π gives Eq. (1.4).

Speed distribution

The cut in Eq. (1.10) removes directions only for $v > v_{\rm esc} - v_{\rm E}$, because the galactic speed $|\boldsymbol{v}+\boldsymbol{v}_{\rm E}|$ ranges over $[\,|v-v_{\rm E}|, v+v_{\rm E}]$ as the direction of $\boldsymbol{v}$ varies. All directions survive if $v + v_{\rm E} < v_{\rm esc}$, none if $|v-v_{\rm E}| > v_{\rm esc}$, that is (since $v_{\rm E} < v_{\rm esc}$) if $v > v_{\rm esc} + v_{\rm E}$; for v between $v_{\rm esc} - v_{\rm E}$ and $v_{\rm esc} + v_{\rm E}$ the surviving directions are $-1 \leq \cos\vartheta < \cos\vartheta_{\rm cut}$ with $v^2 + v_{\rm E}^2 + 2vv_{\rm E}\cos\vartheta_{\rm cut} = v_{\rm esc}^2$. The angular integral is

$$\int_{-1}^{\cos\vartheta_{\rm cut}} \mathrm{e}^{-2vv_{\rm E}\cos\vartheta/v_0^2}\,\mathrm{d}(\cos\vartheta) = \frac{v_0^2}{2vv_{\rm E}}\left[\mathrm{e}^{2vv_{\rm E}/v_0^2} - \mathrm{e}^{-2vv_{\rm E}\cos\vartheta_{\rm cut}/v_0^2}\right].$$

Multiplying by $2\pi v^2 N_{\rm esc}^{-1} \mathrm{e}^{-(v^2+v_{\rm E}^2)/v_0^2}$ and completing the squares gives both branches of Eq. (1.10), with prefactor $\pi v_0^2 v/(v_{\rm E} N_{\rm esc})$. We write the speed distribution as $F_\chi(v) = K_{\rm sp}\, v\,[\mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} - e_{\rm cut}(v)]$, with $K_{\rm sp} = \pi v_0^2/(v_{\rm E} N_{\rm esc})$ and $e_{\rm cut}(v) = \mathrm{e}^{-(v+v_{\rm E})^2/v_0^2}$ below $v_{\rm esc} - v_{\rm E}$, $e_{\rm cut}(v) = \mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ above it.

The function $\eta(v_{\rm min})$

Sect. 1.3 defines $\eta(v_{\rm min}) = \langle \Theta(v - v_{\rm min})/v \rangle = \int_{v_{\rm min}}^{\infty} \mathrm{d}v\, F_\chi(v)/v$. The factor v in F_χ cancels the $1/v$, so for $v_{\rm min} \geq 0$

$$\eta(v_{\rm min}) = K_{\rm sp} \left[\int_{v_{\rm min}}^{v_{\rm esc}+v_{\rm E}} \mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} \mathrm{d}v - \int_{v_{\rm min}}^{v_{\rm esc}+v_{\rm E}} e_{\rm cut}(v)\, \mathrm{d}v \right],$$

where the upper limits are those of F_χ.

(i) $0 \leq v_{\rm min} < v_{\rm esc} - v_{\rm E}$. With $y = v - v_{\rm E}$ the first integral is over $v_{\rm min} - v_{\rm E} < y < v_{\rm esc}$. The second integral has two pieces: $\mathrm{e}^{-(v+v_{\rm E})^2/v_0^2}$ from $v_{\rm min}$ to $v_{\rm esc} - v_{\rm E}$, which with $y = v + v_{\rm E}$ is over $v_{\rm min} + v_{\rm E} < y < v_{\rm esc}$, and the constant $\mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ over an interval of length $2v_{\rm E}$. The Gaussian pieces combine by Eq. (D.1), the $\mathrm{erf}(v_{\rm esc}/v_0)$ terms cancel, and

$$\eta(v_{\rm min}) = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left\{ \frac{\sqrt{\pi}\, v_0}{2} \left[\mathrm{erf}\left(\frac{v_{\rm min} + v_{\rm E}}{v_0} \right) - \mathrm{erf}\left(\frac{v_{\rm min} - v_{\rm E}}{v_0} \right) \right] - 2 v_{\rm E}\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right\}. \tag{D.2}$$

(ii) $v_{\rm esc} - v_{\rm E} \leq v_{\rm min} < v_{\rm esc} + v_{\rm E}$. Only the constant part of $e_{\rm cut}$ remains, over the length $v_{\rm esc} + v_{\rm E} - v_{\rm min}$, and

$$\eta(v_{\rm min}) = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left\{ \frac{\sqrt{\pi}\, v_0}{2} \left[\mathrm{erf}\left(\frac{v_{\rm esc}}{v_0} \right) - \mathrm{erf}\left(\frac{v_{\rm min} - v_{\rm E}}{v_0} \right) \right] - (v_{\rm esc} + v_{\rm E} - v_{\rm min})\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right\}. \tag{D.3}$$

(iii) $v_{\rm min} \geq v_{\rm esc} + v_{\rm E}$. No particle is fast enough: $\eta(v_{\rm min}) = 0$.

Checks. At $v_{\rm min} = v_{\rm esc} - v_{\rm E}$ both Eqs. (D.2) and (D.3) give $(\sqrt{\pi} v_0/2)[\mathrm{erf}(v_{\rm esc}/v_0) - \mathrm{erf}((v_{\rm esc} - 2v_{\rm E})/v_0)] - 2v_{\rm E}\mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ inside the braces, so η is continuous. At $v_{\rm min} = v_{\rm esc} + v_{\rm E}$ Eq. (D.3) vanishes, as branch (iii) requires. At $v_{\rm min} = 0$ the two error functions in Eq. (D.2) are $\mathrm{erf}(v_{\rm E}/v_0)$ and $\mathrm{erf}(-v_{\rm E}/v_0) = -\mathrm{erf}(v_{\rm E}/v_0)$, and $\eta(0)$ is Eq. (1.11). Differentiating Eq. (D.2) or Eq. (D.3) with $\mathrm{d}\,\mathrm{erf}(x)/\mathrm{d}x = (2/\sqrt{\pi})\mathrm{e}^{-x^2}$ gives $\mathrm{d}\eta/\mathrm{d}v_{\rm min} = -F_\chi(v_{\rm min})/v_{\rm min}$, as the definition requires. Without the cut ($v_{\rm esc} \to \infty$, $N_{\rm esc} \to \pi^{3/2} v_0^3$) branch (i) becomes $[\mathrm{erf}((v_{\rm min}+v_{\rm E})/v_0) - \mathrm{erf}((v_{\rm min}-v_{\rm E})/v_0)]/(2v_{\rm E})$, the mean inverse speed of a shifted Maxwellian above $v_{\rm min}$.

D.2 One- and two-body operators

This section proves the second-quantized forms of one- and two-body operators used in Sect. 2.3, Eqs. (2.36) and (2.42), and the rule for changing the single-particle basis, Eq. (2.40). The treatment follows Refs. [1, Chap. 1] and [2, Chap. 1]. We work with fermions first and state the changes for bosons at the end.

States

Let $\{|\gamma\rangle\}$ be an orthonormal single-particle basis, spin included. The antisymmetrized N-particle state with orbitals $\gamma_1, \ldots, \gamma_N$ is

$$|\gamma_1 \ldots \gamma_N\rangle = \frac{1}{\sqrt{N!}} \sum_P (-1)^P\, |\gamma_{P1}\rangle \otimes |\gamma_{P2}\rangle \otimes \cdots \otimes |\gamma_{PN}\rangle, \tag{D.4}$$

where the kth factor of each product is the state of particle k; the wave function of $|\gamma_1 \dots \gamma_N\rangle$ is the Slater determinant of Eq. (2.29). Exchanging two orbital labels changes the sign of the state, and two equal labels give zero. If all labels differ, the state has unit norm, since the $N!$ products are orthonormal. Sect. 2.3 identifies this state with $c^\dagger_{\gamma_1} \cdots c^\dagger_{\gamma_N}|0\rangle$, and the identification defines $c^\dagger_\gamma$ through Eq. (2.30). If a first-quantized operator treats all particles alike, it commutes with every permutation of the particles. The operator then maps antisymmetric states to antisymmetric states, and it is enough to know its action on the states of Eq. (D.4).

Replacing one orbital

The proofs rest on one lemma. Let the labels $\gamma_1, \dots, \gamma_N$ be distinct and let γ be any label. Then

$$c^\dagger_\gamma c_{\gamma_j} \, |\gamma_1 \dots \gamma_j \dots \gamma_N\rangle = |\gamma_1 \dots \gamma \dots \gamma_N\rangle, \tag{D.5}$$

where on the right γ stands in place j and the other labels are unchanged. To prove the lemma, write the state as the string of creation operators $c^\dagger_{\gamma_1} \cdots c^\dagger_{\gamma_N}|0\rangle$. By Eq. (2.32), c_{γ_j} anticommutes with each $c^\dagger_{\gamma_k}$, $k \neq j$, because the labels differ. Moving it to the right past the $j-1$ operators in front of $c^\dagger_{\gamma_j}$ gives a sign $(-1)^{j-1}$. At that position, $c_{\gamma_j} c^\dagger_{\gamma_j} = 1 - c^\dagger_{\gamma_j} c_{\gamma_j}$, and the second term gives zero, because c_{γ_j} can then be moved through the remaining creation operators to the vacuum, which it annihilates. So c_{γ_j} deletes $c^\dagger_{\gamma_j}$ from the string at the cost $(-1)^{j-1}$. Next, $c^\dagger_\gamma$ stands at the front of a string of $N-1$ creation operators; moving it back to place j passes $j-1$ of them, each pass costing a sign by $\{c^\dagger_\gamma, c^\dagger_{\gamma_k}\} = 0$. The two factors $(-1)^{j-1}$ cancel. If γ equals one of the other labels γ_k, $k \neq j$, both sides of Eq. (D.5) vanish: the left because $(c^\dagger_{\gamma_k})^2 = 0$, the right because a label is repeated. If $\gamma = \gamma_j$, both sides return the original state. A bilinear $c^\dagger_\gamma c_{\gamma'}$ with γ' not among $\gamma_1, \dots, \gamma_N$ gives zero, since $c_{\gamma'}$ moves through all creation operators and annihilates the vacuum.

One-body operators

Let $\hat{O}_1 = \sum_{i=1}^N \hat{o}_i$, with $\hat{o}_i$ the one-particle operator $\hat{o}$ acting on the ith factor of each tensor product. Acting on one product in Eq. (D.4), $\hat{o}_i$ changes only the factor $|\gamma_{Pi}\rangle$. After the sums over i and over permutations, each orbital γ_j is replaced in turn by $\hat{o}|\gamma_j\rangle$:

$$\hat{O}_1|\gamma_1 \dots \gamma_N\rangle = \sum_{j=1}^N |\gamma_1 \dots (\hat{o}\gamma_j) \dots \gamma_N\rangle = \sum_{j=1}^N \sum_\gamma \langle\gamma|\hat{o}|\gamma_j\rangle \, |\gamma_1 \dots \gamma \dots \gamma_N\rangle,$$

where $|\gamma_1 \dots (\hat{o}\gamma_j) \dots \gamma_N\rangle$ is Eq. (D.4) with $|\gamma_j\rangle$ replaced by $\hat{o}|\gamma_j\rangle$, and the second step expands $\hat{o}|\gamma_j\rangle$ in the basis, using that Eq. (D.4) is linear in each orbital. By Eq. (D.5) each term on the right is $c^\dagger_\gamma c_{\gamma_j}$ acting on the original state. The bilinear $c^\dagger_\gamma c_{\gamma'}$ gives zero for every γ' not among $\gamma_1, \dots, \gamma_N$. So the sum over j extends to a sum over all labels γ':

$$\hat{O}_1|\gamma_1 \dots \gamma_N\rangle = \sum_{\gamma\gamma'} \langle\gamma|\hat{o}|\gamma'\rangle \, c^\dagger_\gamma c_{\gamma'} \, |\gamma_1 \dots \gamma_N\rangle.$$

$\hat{O}_1$ and the sum of bilinears on the right agree on a basis of every N-particle space, so they are equal on Fock space; the equality is Eq. (2.36). The right-hand side does not refer to N: the same operator acts on states of any particle number.

Two-body operators

Let $\hat{O}_2 = \frac{1}{2}\sum_{i\neq j}\hat{u}_{ij}$, where $\hat{u}_{ij}$ is a two-particle operator acting on the factors i and j, symmetric under the exchange of the two particles. Expand $\hat{u}$ in products of single-particle operators,

$$\hat{u} = \sum_{\gamma_1\gamma_2\gamma_3\gamma_4} \langle\gamma_1\gamma_2|\hat{u}|\gamma_3\gamma_4\rangle \; (|\gamma_1\rangle\langle\gamma_3|) \otimes (|\gamma_2\rangle\langle\gamma_4|),$$

with $\langle\gamma_1\gamma_2|\hat{u}|\gamma_3\gamma_4\rangle = (\langle\gamma_1| \otimes \langle\gamma_2|)\,\hat{u}\,(|\gamma_3\rangle \otimes |\gamma_4\rangle)$ the matrix element between product states, the first factor belonging to the first particle. For a local potential $U_2(\boldsymbol{r}-\boldsymbol{r}')$ the matrix element is the integral given in the caption of Table 2.1. For fixed labels, write $\hat{x}_i = (|\gamma_1\rangle\langle\gamma_3|)_i$ and $\hat{y}_j = (|\gamma_2\rangle\langle\gamma_4|)_j$ for the two single-particle factors of the expansion acting on particles i and j. The sum over pairs with $i \neq j$ is the full double sum minus the diagonal,

$$\sum_{i\neq j}\hat{x}_i\hat{y}_j = \Big(\sum_i \hat{x}_i\Big)\Big(\sum_j \hat{y}_j\Big) - \sum_i (\hat{x}\hat{y})_i,$$

and each term on the right is a one-body operator or a product of two. By the one-body result, $\sum_i \hat{x}_i = c^\dagger_{\gamma_1}c_{\gamma_3}$ and $\sum_j \hat{y}_j = c^\dagger_{\gamma_2}c_{\gamma_4}$, while $\hat{x}\hat{y} = |\gamma_1\rangle\langle\gamma_3|\gamma_2\rangle\langle\gamma_4| = \delta_{\gamma_3\gamma_2}|\gamma_1\rangle\langle\gamma_4|$ gives $\sum_i(\hat{x}\hat{y})_i = \delta_{\gamma_2\gamma_3}c^\dagger_{\gamma_1}c_{\gamma_4}$. With Eq. (2.32),

$$c^\dagger_{\gamma_1}c_{\gamma_3}c^\dagger_{\gamma_2}c_{\gamma_4} - \delta_{\gamma_2\gamma_3}c^\dagger_{\gamma_1}c_{\gamma_4} = -c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}c_{\gamma_3}c_{\gamma_4} = c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}c_{\gamma_4}c_{\gamma_3}.$$

Multiplying by the matrix element, summing over the four labels and including the factor $\frac{1}{2}$ gives

$$\hat{O}_2 = \frac{1}{2}\sum_{\gamma_1\gamma_2\gamma_3\gamma_4} \langle\gamma_1\gamma_2|\hat{u}|\gamma_3\gamma_4\rangle\, c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}c_{\gamma_4}c_{\gamma_3}, \tag{D.6}$$

which is Eq. (2.42). The proof uses no property of $\hat{u}$ other than its expansion, so the result holds for spin-dependent interactions as well. Subtracting the diagonal turns the product of two bilinears into a normal-ordered product; it removes the interaction of each particle with itself, as in the field-operator argument of Sect. 2.3.

Change of basis

Let $\{|\tilde{\gamma}\rangle\}$ be a second orthonormal basis, with the unitary matrix $\mathcal{U}_{\gamma\tilde{\gamma}} = \langle\gamma|\tilde{\gamma}\rangle$. The antisymmetrized state Eq. (D.4) is linear in each orbital, so substituting $|\tilde{\gamma}_k\rangle = \sum_\gamma \mathcal{U}_{\gamma\tilde{\gamma}_k}|\gamma\rangle$ for every k gives

$$|\tilde{\gamma}_1 \ldots \tilde{\gamma}_N\rangle = \sum_{\gamma'_1\cdots\gamma'_N} \mathcal{U}_{\gamma'_1\tilde{\gamma}_1}\cdots\mathcal{U}_{\gamma'_N\tilde{\gamma}_N}\,|\gamma'_1\ldots\gamma'_N\rangle.$$

Written with creation operators, the state $|\tilde{\gamma}_1\ldots\tilde{\gamma}_N\rangle$ is $c^\dagger_{\tilde{\gamma}_1}\cdots c^\dagger_{\tilde{\gamma}_N}|0\rangle$ with $c^\dagger_{\tilde{\gamma}} = \sum_\gamma \mathcal{U}_{\gamma\tilde{\gamma}}c^\dagger_\gamma$, the rule of Eq. (2.40); the adjoint of this rule gives $c_{\tilde{\gamma}}$. Collect the terms in which the labels γ'_k are a permutation of a fixed ordered set $\{\gamma_1 < \cdots < \gamma_N\}$, and reorder each to the standard order at the sign of the permutation. The coefficient of $|\gamma_1\ldots\gamma_N\rangle$ is then the determinant of the $N\times N$ submatrix $\mathcal{U}_{\gamma_j\tilde{\gamma}_k}$. A single determinant in one basis is thus a sum of determinants in the other, with minors of $\mathcal{U}$ as coefficients; the two-orbital bond of Sect. 2.3 is the smallest example.

Sect. 2.3 shows that the anticommutator $\{c_{\tilde{\gamma}}, c^\dagger_{\tilde{\gamma}'}\}$ and the one-body form Eq. (2.36) are preserved. The other two anticommutators vanish, because the new operators are linear combinations of the old annihilation (or creation) operators alone, which anticommute among themselves. Inserting $c^\dagger_\gamma = \sum_{\tilde{\gamma}}\mathcal{U}^*_{\gamma\tilde{\gamma}}c^\dagger_{\tilde{\gamma}}$ and $c_{\gamma'} = \sum_{\tilde{\gamma}'}\mathcal{U}_{\gamma'\tilde{\gamma}'}c_{\tilde{\gamma}'}$ into Eq. (D.6) transforms each of the four labels with its own factor of $\mathcal{U}$ or $\mathcal{U}^*$, and completeness turns the matrix element

into $\langle\tilde\gamma_1\tilde\gamma_2|\hat u|\tilde\gamma_3\tilde\gamma_4\rangle$. The one-body form and the two-body form are therefore the same in every orthonormal basis. For a continuous basis, such as position, the sums become integrals and the Kronecker deltas Dirac deltas; with $\langle \boldsymbol{r}s|\gamma\rangle = \psi_\gamma(\boldsymbol{r},s)$ the rule of Eq. (2.40) gives the field operator of Eq. (2.37) and the field-operator form Eq. (2.41).

Bosons

For bosons, Eq. (D.4) has all signs positive and a repeated label is allowed; the state with occupation numbers n_γ then has norm $\prod_\gamma n_\gamma!$. The normalized states therefore carry the factor $(\prod_\gamma n_\gamma!)^{-1/2}$ and $d^\dagger_\gamma$ acts with $\sqrt{n_\gamma+1}$ (Eq. (2.34)). The lemma Eq. (D.5) holds for the unnormalized states $d^\dagger_{\gamma_1}\cdots d^\dagger_{\gamma_N}|0\rangle$ without any sign. When the label γ_j is repeated, the right-hand side of the lemma is a sum over the occurrences of γ_j, with γ_j replaced in one occurrence in each term; this sum matches the sum over particles in $\hat O_1$. The one-body proof then goes through unchanged with $c\to d$. In the two-body proof the commutator $[d_{\gamma_3}, d^\dagger_{\gamma_2}] = \delta_{\gamma_2\gamma_3}$ replaces the anticommutator, and $d^\dagger_{\gamma_1}d_{\gamma_3}d^\dagger_{\gamma_2}d_{\gamma_4} - \delta_{\gamma_2\gamma_3}d^\dagger_{\gamma_1}d_{\gamma_4} = d^\dagger_{\gamma_1}d^\dagger_{\gamma_2}d_{\gamma_3}d_{\gamma_4}$. Annihilation operators commute, so the order of the two annihilation operators is immaterial and Eq. (D.6) holds with $c\to d$. The change of basis preserves the commutators by the same unitarity argument, and the coefficients of the new states are permanents instead of determinants.

D.3 Wigner–Weisskopf decay

This section derives the exponential decay of a discrete state coupled to a continuum, Eq. (2.55), the Lorentzian distribution of the final states, Eq. (2.56), and the replacement of an infinitesimal by a width in a second-order denominator, Eq. (2.57). The original treatment is that of Weisskopf and Wigner [3]; we follow the resolvent method of Ref. [4, Chap. III], and use the notation of Sects. 2.1 and 2.4.

Exact amplitude equation

A discrete state $|i\rangle$ of energy E_i is coupled by a constant perturbation H' to a set of states $|f\rangle$ of energies E_f; the states $|f\rangle$ become a continuum with density $\mathcal{D}_f(E)$ as the volume grows. We assume $H'_{ii}=0$ and $H'_{ff'}=0$: the perturbation only connects i with the continuum. A nonzero H'_{ii} is a first-order shift and can be put into E_i; couplings within the continuum are left out of this model. With $\xi_f = E_f - E_i$, the amplitude equations Eq. (2.1) are

$$\mathrm{i}\,\dot c_i(t) = \sum_f H'_{if}\,\mathrm{e}^{-\mathrm{i}\xi_f t}\,c_f(t), \qquad \mathrm{i}\,\dot c_f(t) = H'_{fi}\,\mathrm{e}^{\mathrm{i}\xi_f t}\,c_i(t), \tag{D.7}$$

with $c_i(0)=1$ and $c_f(0)=0$. The second equation integrates to $c_f(t) = -\mathrm{i}H'_{fi}\int_0^t \mathrm{d}t'\,\mathrm{e}^{\mathrm{i}\xi_f t'}c_i(t')$. Inserting this solution into the first equation gives a closed equation for c_i,

$$\begin{aligned} \dot c_i(t) &= -\int_0^t \mathrm{d}t'\,K(t-t')\,c_i(t'), \\ K(t-t') &= \sum_f |H'_{fi}|^2\,\mathrm{e}^{-\mathrm{i}\xi_f(t-t')} = \int \mathrm{d}E\,\mathcal{D}_f(E)\,\overline{|H'_{fi}|^2}(E)\,\mathrm{e}^{-\mathrm{i}(E-E_i)(t-t')}. \end{aligned} \tag{D.8}$$

No approximation has been made. The kernel K is the Fourier transform of the product $\mathcal{D}_f\overline{|H'_{fi}|^2}$, which varies on the energy scale E_w of Sect. 2.1. The kernel therefore decays on the time scale $1/E_\mathrm{w}$: the continuum has a memory of that length. For the flat continuum of Worked example 2.1, with $\mathcal{D}_f\overline{|H'_{fi}|^2} = \Gamma_i/2\pi$ on $|E-E_i| < E_\mathrm{w}/2$ and zero outside,

$K(t-t') = (\Gamma_i/\pi)\sin[E_\mathrm{w}(t-t')/2]/(t-t')$, which starts at $\Gamma_i E_\mathrm{w}/2\pi$ and oscillates away within a few $1/E_\mathrm{w}$.

The Markov approximation

At times short compared with $1/E_\mathrm{w}$, c_i has barely changed and $\dot c_i \approx -t\,K(0)$, so $|c_i|^2 \approx 1 - K(0)t^2$: the decay starts quadratically, with the short-time behavior found in Worked example 2.1. At later times, the integral in Eq. (D.8) only receives contributions from $t - t' \lesssim 1/E_\mathrm{w}$. Suppose that c_i changes little over that interval, as it does when its decay rate is small compared with E_w. We can then replace $c_i(t')$ by $c_i(t)$ and extend the lower limit of the integral to $-\infty$:

$$\dot c_i(t) = -c_i(t)\int_{-\infty}^{t} \mathrm{d}t'\, K(t-t').$$

The remaining integral is a sum of terms $\int_{-\infty}^{t}\mathrm{d}t'\,\mathrm{e}^{-\mathrm{i}\xi_f(t-t')}$, which converge with a factor $\mathrm{e}^{-0^+(t-t')}$ and give $-\mathrm{i}/(\xi_f - \mathrm{i}0^+) = \mathrm{i}/(E_i - E_f + \mathrm{i}0^+)$. With the Sokhotski–Plemelj formula Eq. (2.58),

$$\begin{aligned}\int_{-\infty}^{t}\mathrm{d}t'\,K(t-t') &= \mathrm{i}\sum_f \frac{|H'_{fi}|^2}{E_i - E_f + \mathrm{i}0^+}\\ &= \mathrm{i}\ \mathrm{PV}\sum_f \frac{|H'_{fi}|^2}{E_i - E_f} + \pi\sum_f |H'_{fi}|^2\,\delta(E_f - E_i) = \mathrm{i}\Delta E_i + \frac{\Gamma_i}{2},\end{aligned} \tag{D.9}$$

where Γ_i is the golden-rule rate Eq. (2.6) and ΔE_i the level shift of Eq. (2.55). The equation $\dot c_i = -(\Gamma_i/2 + \mathrm{i}\Delta E_i)c_i$ integrates to Eq. (2.55). The Markov approximation requires $\Gamma_i \ll E_\mathrm{w}$, the condition under which the golden-rule window of Eq. (2.7) exists, and holds for $t \gg 1/E_\mathrm{w}$. The memory of the continuum is then short compared with the lifetime, and the probability to decay in the next instant is the same at every time; a constant decay probability per instant defines an exponential law.

The resolvent and the corrections

The size of the corrections to the Markov approximation follows from an exact solution of Eq. (D.8) by Laplace transform. With $\tilde c_i(s) = \int_0^\infty \mathrm{d}t\,\mathrm{e}^{-st}c_i(t)$ for $\mathrm{Re}\,s > 0$, the convolution becomes a product, $s\tilde c_i - 1 = -\tilde K(s)\tilde c_i$, with $\tilde K(s) = \sum_f |H'_{fi}|^2/(s + \mathrm{i}\xi_f)$. Define an energy variable $z = \mathrm{i}s$, measured from E_i; for $\mathrm{Re}\,s > 0$ it lies in the upper half plane. Then

$$\tilde c_i = \frac{\mathrm{i}}{z - R(z)}, \qquad R(z) = \sum_f \frac{|H'_{fi}|^2}{z - \xi_f}, \tag{D.10}$$

and $c_i(t) = (2\pi)^{-1}\int \mathrm{d}z\,\mathrm{e}^{-\mathrm{i}zt}\,\mathrm{i}/(z - R(z))$ along a line just above the real axis [4, Chap. III]. The function $R(z)$ is analytic off the real axis; just above the axis, by Eq. (2.58), $R(x + \mathrm{i}0^+) = \mathrm{PV}\sum_f |H'_{fi}|^2/(x - \xi_f) - \mathrm{i}\pi\sum_f |H'_{fi}|^2\delta(x - \xi_f)$, which at $x = 0$ is $\Delta E_i - \mathrm{i}\Gamma_i/2$. When R is continued below the axis through the continuum, $z - R(z)$ has a zero near $z_\mathrm{p} = \Delta E_i - \mathrm{i}\Gamma_i/2$, as long as R changes little over a range Γ_i around $z = 0$. Closing the contour in the lower half plane picks up the pole of $\tilde c_i$ at this zero and gives

$$c_i(t) = r_\mathrm{p}\,\mathrm{e}^{-\mathrm{i}z_\mathrm{p}t} + c_i^\mathrm{cut}(t), \qquad r_\mathrm{p} = \frac{1}{1 - R'(z_\mathrm{p})},$$

plus the contribution c_i^cut of the branch points at the edges of the continuum. Since R' is of order Γ_i/E_w, the residue r_p of the pole differs from one by a relative amount of that order,

and so does the decay rate, which becomes $\Gamma_i/(1-R')$ when R is expanded to first order about $z=0$. For the flat continuum, $R(z)=(\Gamma_i/2\pi)\ln[(z+E_\mathrm{w}/2)/(z-E_\mathrm{w}/2)]$; just above $z=0$ the function R equals $-\mathrm{i}\Gamma_i/2$, so the level shift vanishes, and $R'(0)=2\Gamma_i/(\pi E_\mathrm{w})$. In Worked example 2.1, $\Gamma_i/E_\mathrm{w}=0.01$ and $R'(0)=0.0064$. The survival probability is then $r_\mathrm{p}^2\mathrm{e}^{-\Gamma_i t/(1-R')}$, which at $t=1/\Gamma_i$ exceeds e^{-1} by 0.0023, or 0.4 % of the depletion. The exact solution for the flat continuum of Worked example 2.1 deviates from the depletion $1-\mathrm{e}^{-\Gamma_i t}$ by this amount once the short-time transient has died out. The branch-point contribution falls off as a power of $1/(E_\mathrm{w}t)$ with a prefactor of order Γ_i/E_w. It overtakes the exponential only after many lifetimes, when the survival probability is already exponentially small, and it affects no rate in this book.

The line shape

With Eq. (2.55), the amplitude of a final state is

$$c_f(t)=-\mathrm{i}H'_{fi}\int_0^t \mathrm{d}t'\,\mathrm{e}^{\mathrm{i}(\xi_f-\Delta E_i)t'-\Gamma_i t'/2}\xrightarrow{t\to\infty}\frac{H'_{fi}}{E_f-E_i-\Delta E_i+\mathrm{i}\Gamma_i/2},$$

whose square is Eq. (2.56). This Lorentzian is the natural line shape of the emitted radiation when the continuum consists of the photon modes [3].

A decaying intermediate state

Finally, let a state $|i\rangle$ be coupled weakly, by a constant H_1, to a state $|m\rangle$ that decays into a continuum through a second coupling H_2, with total rate Γ_w and shift ΔE_m from Eq. (D.9) applied to m. Let H_1 act from the remote past, and keep it to first order, so that $c_i=1$. The amplitude of m then obeys

$$\dot c_m(t)=-\mathrm{i}\,(H_1)_{mi}\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t}-\Big(\frac{\Gamma_\mathrm{w}}{2}+\mathrm{i}\Delta E_m\Big)c_m(t),$$

in which the decay term is the Markov form of the coupling to the continuum, valid under the conditions of the Markov approximation. The solution of this equation with $c_m(-\infty)=0$ is

$$c_m(t)=-\mathrm{i}\,(H_1)_{mi}\int_{-\infty}^t \mathrm{d}t_2\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t_2}\,\mathrm{e}^{-(\Gamma_\mathrm{w}/2+\mathrm{i}\Delta E_m)(t-t_2)}=\frac{(H_1)_{mi}\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t}}{E_i-\tilde E_m+\mathrm{i}\Gamma_\mathrm{w}/2},\qquad(\mathrm{D.11})$$

with $\tilde E_m=E_m+\Delta E_m$. The free propagation $\mathrm{e}^{-\mathrm{i}E_m(t-t_2)}$ of Eq. (2.51) has become a decaying one, and the integral converges without any switching factor. The amplitude of a final state f reached from m by H_2 is, at first order in H_2,

$$c_f(t)=-\mathrm{i}\int_{-\infty}^t \mathrm{d}t_1\,(H_2)_{fm}\,\mathrm{e}^{\mathrm{i}(E_f-E_m)t_1}\,c_m(t_1)=-\mathrm{i}\,\frac{(H_2)_{fm}(H_1)_{mi}}{E_i-\tilde E_m+\mathrm{i}\Gamma_\mathrm{w}/2}\int_{-\infty}^t \mathrm{d}t_1\,\mathrm{e}^{\mathrm{i}(E_f-E_i)t_1},$$

which has the form of the second-order amplitude of Sect. 2.4 with the replacement Eq. (2.57), and leads to Eq. (2.52) for the rate. The derivation needs a weak drive, $|(H_1)_{mi}|\ll|E_i-\tilde E_m+\mathrm{i}\Gamma_\mathrm{w}/2|$, so that the population $|c_m|^2$ stays small and i is not depleted; on resonance the condition reads $|(H_1)_{mi}|\ll\Gamma_\mathrm{w}/2$. The derivation also evaluates Γ_w and ΔE_m at the energy E_m of m, whereas strictly the resolvent gives them at the energy E_i at which m is passed through. The values at E_m and at E_i agree when the decay continuum of m varies little between E_m and E_i, a condition that always holds near resonance, where the width matters.

D.4 The f-sum rule

This section carries out the commutator algebra behind the f-sum rule Eq. (4.30), its species-resolved form Eq. (4.34), and the corrections from nonlocal potentials, Eq. (4.32), and from the mass–velocity term, Eq. (4.33). The results are stated and used in Sect. 4.3; the treatment of the local case follows Refs. [5, Sect. 3.4] and [6, Chap. 2].

One particle and a plane wave

Write $A = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ for one particle, with position $\boldsymbol{r}$ and momentum $\boldsymbol{p}$. Since $[p_\alpha, f(\boldsymbol{r})] = -\mathrm{i}\,\partial_\alpha f$, we have $[p_\alpha, A] = q_\alpha A$, and therefore $A^\dagger p_\alpha A = p_\alpha + q_\alpha$. Conjugation by a unitary operator preserves products and sums, so for any function $F(\boldsymbol{p})$ that is a polynomial or a convergent power series,

$$A^\dagger F(\boldsymbol{p})\, A = F(\boldsymbol{p}+\boldsymbol{q}), \qquad A\, F(\boldsymbol{p})\, A^\dagger = F(\boldsymbol{p}-\boldsymbol{q}). \tag{D.12}$$

The commutator of $F(\boldsymbol{p})$ with A follows by factoring A out on the left, $[F(\boldsymbol{p}), A] = A\,(A^\dagger F A - F) = A\,G(\boldsymbol{p})$ with $G(\boldsymbol{p}) = F(\boldsymbol{p}+\boldsymbol{q}) - F(\boldsymbol{p})$. Commuting once more with $A^\dagger$, and using $A^\dagger A = 1$ and Eq. (D.12), gives

$$[A^\dagger, A\,G(\boldsymbol{p})] = G(\boldsymbol{p}) - A\,G(\boldsymbol{p})\,A^\dagger = G(\boldsymbol{p}) - G(\boldsymbol{p}-\boldsymbol{q}).$$

Inserting G gives the double commutator of any function of the momentum:

$$\big[A^\dagger, [F(\boldsymbol{p}), A]\big] = F(\boldsymbol{p}+\boldsymbol{q}) + F(\boldsymbol{p}-\boldsymbol{q}) - 2F(\boldsymbol{p}). \tag{D.13}$$

The double commutator is a second difference of F, and for the kinetic energy $F = \boldsymbol{p}^2/2m$ it is the number q^2/m. A function of the position alone commutes with A and contributes nothing.

Many particles and several species

The density of each species s, electrons or the nuclei of species κ, is a sum over particles, $n_s(\boldsymbol{q}) = \sum_{i\in s} A_i$, with $A_i = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$ and $\boldsymbol{r}_i$ the position of particle i ($\boldsymbol{X}_I$ for a nucleus). Take a Hamiltonian

$$H = \sum_s \sum_{i\in s} F_s(\boldsymbol{p}_i) + U(\{\boldsymbol{r}_i\}), \tag{D.14}$$

where F_s is a function of the momentum of one particle, $\boldsymbol{p}^2/2m_s$ in the nonrelativistic case, and U is any function of all positions, and of spins, if it does not involve momenta. The Coulomb interactions of Sect. 2.3, external static potentials, the Zeeman energy and the contact hyperfine interaction all belong to U. U commutes with every A_i, and so does $F_s(\boldsymbol{p}_j)$ for $j \neq i$. Hence

$$[H, n_{s'}(\boldsymbol{q})] = \sum_{i\in s'} [F_{s'}(\boldsymbol{p}_i), A_i],$$

an operator built from the positions and momenta of species s' alone. The commutator $[H, n_{s'}(\boldsymbol{q})]$ commutes with $A_j^\dagger$ for every particle j not of species s', and within species s' only the terms with $j = i$ survive in the double commutator. By Eq. (D.13),

$$\big[n_s(\boldsymbol{q})^\dagger, [H, n_{s'}(\boldsymbol{q})]\big] = \delta_{ss'} \sum_{i\in s} \big[F_s(\boldsymbol{p}_i+\boldsymbol{q}) + F_s(\boldsymbol{p}_i-\boldsymbol{q}) - 2F_s(\boldsymbol{p}_i)\big]. \tag{D.15}$$

For $F_s = \boldsymbol{p}^2/2m_s$ each particle contributes q^2/m_s, the double commutator is the number $N_s q^2/m_s$ in every state, and dividing by $\mathcal{V}$ gives Eq. (4.34). The derivation holds at any temperature, with the nuclei moving, and for any interactions in U.

Nonlocal potentials

A nonlocal one-body potential, such as the separable pseudopotential Eq. (24.9) or the Fock exchange operator, is not a function of position: it acts on each electron as an operator $V_{\rm NL}$ with matrix elements $\langle \boldsymbol{r}|V_{\rm NL}|\boldsymbol{r}'\rangle$ between different points. The operator $V_{\rm NL}$ does not commute with A, and since A is unitary,

$$[A^\dagger,[V_{\rm NL},A]] = A^\dagger V_{\rm NL} A + A V_{\rm NL} A^\dagger - 2V_{\rm NL}. \tag{D.16}$$

Summed over electrons and averaged, the double commutator Eq. (D.16) is the extra term of Eq. (4.32); its expectation value in one orbital is twice the $\delta_{\rm NL}$ of Eq. (24.13). In the separable form, $A^\dagger V_{\rm NL} A$ is the sum of Eq. (24.9) with the projector functions multiplied by $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, so the extra term is a sum of projections. At small q we expand $A = 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r} - (\boldsymbol{q}\cdot\boldsymbol{r})^2/2 + \ldots$; the terms of first order cancel between $A^\dagger V_{\rm NL} A$ and $A V_{\rm NL} A^\dagger$, and

$$A^\dagger V_{\rm NL} A + A V_{\rm NL} A^\dagger - 2V_{\rm NL} = -[\boldsymbol{q}\cdot\boldsymbol{r},[\boldsymbol{q}\cdot\boldsymbol{r},V_{\rm NL}]] + O(q^4).$$

The kinetic energy gives $-[\boldsymbol{q}\cdot\boldsymbol{r},[\boldsymbol{q}\cdot\boldsymbol{r},\boldsymbol{p}^2/2m_e]] = q^2/m_e$ by the same expansion, and the sum of the kinetic and nonlocal double commutators is $-\mathrm{i}\,[\boldsymbol{q}\cdot\boldsymbol{r},\boldsymbol{q}\cdot\boldsymbol{v}]$ with the velocity $\boldsymbol{v} = \mathrm{i}[H,\boldsymbol{r}] = \boldsymbol{p}/m_e + \mathrm{i}[V_{\rm NL},\boldsymbol{r}]$. The Thomas–Reiche–Kuhn rule of a nonlocal Hamiltonian is therefore a statement about the commutator of position and velocity, which equals $\mathrm{i}\delta_{\alpha\beta}/m_e$ only when $V_{\rm NL}$ commutes with $\boldsymbol{r}$.

Relativistic corrections

To order $1/m_e^2$, the one-body Hamiltonian of an electron in a potential $U_1(\boldsymbol{r})$ adds three terms to $\boldsymbol{p}^2/2m_e + U_1$:

$$H_{\rm mv} = -\frac{\boldsymbol{p}^4}{8m_e^3}, \qquad H_{\rm Dar} = \frac{\nabla^2 U_1}{8m_e^2}, \qquad H_{\rm so} = \frac{1}{4m_e^2}\,\boldsymbol{\sigma}\cdot(\nabla U_1 \times \boldsymbol{p}).$$

The Darwin term $H_{\rm Dar}$ is a function of position and drops out. The spin–orbit term is linear in the momentum, $H_{\rm so} = \sum_\alpha c_\alpha p_\alpha$, with coefficients c_α that depend on position and spin; the ordering of c_α and p_α does not matter, since $\varepsilon_{\gamma\alpha\beta}\partial_\alpha\partial_\beta U_1 = 0$. The commutator of $H_{\rm so}$ with A is $\sum_\alpha c_\alpha q_\alpha A$, a function of position and spin, which commutes with $A^\dagger$: the double commutator vanishes. The mass–velocity term is of the form Eq. (D.13) with $F = -\boldsymbol{p}^4/8m_e^3$. Writing $\boldsymbol{p}^2 \pm 2\boldsymbol{p}\cdot\boldsymbol{q} + q^2$ for $(\boldsymbol{p}\pm\boldsymbol{q})^2$ and squaring gives

$$(\boldsymbol{p}+\boldsymbol{q})^4 + (\boldsymbol{p}-\boldsymbol{q})^4 - 2\boldsymbol{p}^4 = 2(\boldsymbol{p}^2+q^2)^2 + 8(\boldsymbol{p}\cdot\boldsymbol{q})^2 - 2\boldsymbol{p}^4 = 4p^2q^2 + 8(\boldsymbol{p}\cdot\boldsymbol{q})^2 + 2q^4.$$

The expectation value of $(\boldsymbol{p}\cdot\boldsymbol{q})^2 = q_\alpha q_\beta\, p_\alpha p_\beta$ involves the tensor $\langle p_\alpha p_\beta\rangle$. In an atom, and in a crystal of cubic symmetry, a symmetric second-rank tensor is a multiple of the unit tensor, so $\langle p_\alpha p_\beta\rangle = \delta_{\alpha\beta}\langle p^2\rangle/3$ and $\langle(\boldsymbol{p}\cdot\boldsymbol{q})^2\rangle = q^2\langle p^2\rangle/3$. Per electron, the mass–velocity term then adds

$$-\frac{1}{8m_e^3}\Big[\Big(4+\frac{8}{3}\Big)q^2\langle p^2\rangle + 2q^4\Big] = -\frac{q^2}{m_e}\Big[\frac{5}{3}\,\frac{\langle p^2\rangle/2m_e}{m_e} + \frac{q^2}{4m_e^2}\Big]$$

to the q^2/m_e of the kinetic energy. Averaged over the electrons, $\langle p^2\rangle/2m_e$ is the mean kinetic energy $\bar{K}$, and dividing the sum over electrons by $\mathscr{V}$ gives Eq. (4.33). For a Coulomb Hamiltonian the virial theorem gives $2\langle T\rangle = -\langle U\rangle$ for the total kinetic and potential energies, so $\langle T\rangle = -E$ and $\bar{K} = -E/Z$ per electron of an atom. The relativistic corrections also change the current: the velocity acquires the term $-\boldsymbol{p}^2\boldsymbol{p}/2m_e^3$ from $H_{\rm mv}$ and a spin-dependent term from $H_{\rm so}$, and the continuity equation Eq. (4.29) holds with the corrected current. The two-body Breit terms of the electron–electron interaction contain the momenta of pairs of electrons, contribute to the double commutator at the same order in $1/m_e^2$, and are not evaluated here.

D.5 The dielectric matrix

This section carries out the matrix algebra behind the dielectric matrix Eq. (4.43) and the macroscopic dielectric function Eq. (4.44) of Sect. 4.4. The treatment follows Refs. [7, 8] and [9].

Inversion

Fix $\boldsymbol{q}$ in the first zone and ω, and write every quantity as a matrix in the reciprocal-lattice vectors: the improper response $R = (R_{\boldsymbol{GG'}})$ of Eq. (4.42), a proper response $\tilde{R}$, and the diagonal Coulomb matrix v with elements $v_{\boldsymbol{G}} = v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)$. With $\phi_{\mathrm{tot}} = \phi_{\mathrm{ext}} + v\,\delta\boldsymbol{n}_e$ and $\delta\boldsymbol{n}_e = \tilde{R}\,\phi_{\mathrm{tot}}$ as vectors, eliminating ϕ_{tot} for arbitrary ϕ_{ext} gives

$$R = \tilde{R} + \tilde{R}\,v\,R, \qquad R = (\mathbf{1} - \tilde{R}v)^{-1}\tilde{R}, \tag{D.17}$$

the solution of Eq. (C.57) with $\mathbf{X}_0 = \tilde{R}$ and $\mathbf{K} = v$. The dielectric matrix and its inverse are defined by $\phi_{\mathrm{ext}} = \varepsilon\,\phi_{\mathrm{tot}}$ and $\phi_{\mathrm{tot}} = \varepsilon^{-1}\phi_{\mathrm{ext}}$, so that

$$\varepsilon = \mathbf{1} - v\tilde{R}, \qquad \varepsilon^{-1} = \mathbf{1} + vR. \tag{D.18}$$

The two matrices of Eq. (D.18) are inverses of each other by the push-through identity $v(\mathbf{1} - \tilde{R}v)^{-1} = (\mathbf{1} - v\tilde{R})^{-1}v$, which holds because $(\mathbf{1} - v\tilde{R})\,v = v\,(\mathbf{1} - \tilde{R}v)$. With Eq. (D.17),

$$\mathbf{1} + vR = \mathbf{1} + (\mathbf{1} - v\tilde{R})^{-1}v\tilde{R} = (\mathbf{1} - v\tilde{R})^{-1}\big[(\mathbf{1} - v\tilde{R}) + v\tilde{R}\big] = (\mathbf{1} - v\tilde{R})^{-1}.$$

The order of the factors matters. In element form, $\varepsilon_{\boldsymbol{GG'}} = \delta_{\boldsymbol{GG'}} - v_{\boldsymbol{G}}\tilde{R}_{\boldsymbol{GG'}}$: the Coulomb factor carries the row index, the wavevector at which the potential is produced. This element form is Eq. (4.43).

The long-range term removed

A second form of $\varepsilon_{\mathrm{mac}}$ isolates the one element of v that is singular as $\boldsymbol{q} \to 0$. Split the Coulomb matrix as $v = v_0 + \bar{v}$, where v_0 has the single nonzero element $v_{\boldsymbol{0}} = v_{\mathrm{C}}(q)$ in the head and $\bar{v}$ holds all the other elements. Define $\bar{R}$ by the Dyson equation with $\bar{v}$ alone,

$$\bar{R} = \tilde{R} + \tilde{R}\,\bar{v}\,\bar{R}. \tag{D.19}$$

The matrix $\bar{R}$ is the response to the total potential with the Hartree potential at $\boldsymbol{G} = 0$ left out. Equivalently, it is the response of the electrons with the Coulomb kernel at the single wavevector $\boldsymbol{q}$ removed from their Hamiltonian. The full response follows from $\bar{R}$ by a second Dyson equation with v_0. From $R^{-1} = \tilde{R}^{-1} - v$ and $\bar{R}^{-1} = \tilde{R}^{-1} - \bar{v}$ (Appendix C.7) we have $R^{-1} = \bar{R}^{-1} - v_0$, or

$$R = \bar{R} + \bar{R}\,v_0\,R.$$

Since v_0 connects only the index $\mathbf{0}$ to itself, the head of this equation closes on itself: $(\bar{R}v_0R)_{\boldsymbol{00}} = \bar{R}_{\boldsymbol{00}}\,v_{\mathrm{C}}(q)\,R_{\boldsymbol{00}}$, so $R_{\boldsymbol{00}} = \bar{R}_{\boldsymbol{00}}/[1 - v_{\mathrm{C}}(q)\bar{R}_{\boldsymbol{00}}]$. Then $(\varepsilon^{-1})_{\boldsymbol{00}} = 1 + v_{\mathrm{C}}(q)R_{\boldsymbol{00}} = 1/[1 - v_{\mathrm{C}}(q)\bar{R}_{\boldsymbol{00}}]$, and

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega) = 1 - v_{\mathrm{C}}(q)\,\bar{R}_{\boldsymbol{00}}(\boldsymbol{q},\omega). \tag{D.20}$$

The macroscopic dielectric function Eq. (D.20) has the form of Eq. (4.41) for a homogeneous system, with $\bar{R}$ in place of $\tilde{R}$. The local fields are now inside $\bar{R}$, as the microscopic Hartree potentials $\bar{v}$; the exchange and correlation effects are inside $\tilde{R}$. Since $\bar{R}$ is a causal response to a potential, and the electrons it describes have a local Hamiltonian, the absorptive part $\varepsilon_2 = -v_{\mathrm{C}}(q)\,\mathrm{Im}\,\bar{R}_{\boldsymbol{00}}$ obeys the f-sum rule Eq. (4.47) with the local fields included [9]. In the absorption spectrum computed from Eq. (D.20) the long-range term is absent, and the Coulomb singularity appears only in the loss function.

The limit $\boldsymbol{q} \to 0$

At small q the head $v_{\mathrm{C}}(q) = e^2/q^2$ diverges, while $v_{\boldsymbol{G}}$ stays finite for $\boldsymbol{G} \neq 0$. For an insulator the proper response is analytic at $\boldsymbol{q} = 0$, and the continuity equation fixes how its elements vanish there. Matrix elements of the continuity equation Eq. (4.29) between eigenstates give $(E_f - E_i)\langle f|n_e(\boldsymbol{q})|i\rangle = \langle f|\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})|i\rangle$, so $\langle f|n_e(\boldsymbol{q})|i\rangle = \boldsymbol{q}\cdot\langle f|\boldsymbol{j}_e(\boldsymbol{q})|i\rangle/(E_f - E_i)$, which is $O(q)$ because the gap keeps $E_f - E_i$ away from zero. Each index $\mathbf{0}$ of an element of $\tilde{R}$ brings one factor $\langle f|n_e(\boldsymbol{q})|i\rangle$, and

$$\tilde{R}_{\mathbf{00}} = O(q^2), \qquad \tilde{R}_{\mathbf{0}\boldsymbol{G}},\ \tilde{R}_{\boldsymbol{G}\mathbf{0}} = O(q), \qquad \tilde{R}_{\boldsymbol{G}\boldsymbol{G}'} = O(1).$$

The head of the dielectric matrix, $\varepsilon_{\mathbf{00}} = 1 - (e^2/q^2)\tilde{R}_{\mathbf{00}}$, is then finite. Of the wings, $\varepsilon_{\mathbf{0}\boldsymbol{G}} = -(e^2/q^2)\tilde{R}_{\mathbf{0}\boldsymbol{G}}$ diverges as $1/q$ and $\varepsilon_{\boldsymbol{G}\mathbf{0}} = -v_{\boldsymbol{G}}\tilde{R}_{\boldsymbol{G}\mathbf{0}}$ vanishes as q, and their product in the local-field correction of Sect. 4.4 is finite. So is $\varepsilon_{\mathrm{mac}}$, as Eq. (D.20) shows directly, since $\bar{R}$ contains no singular kernel. The finite limits depend on the direction $\hat{\boldsymbol{q}}$, through the factors $\hat{\boldsymbol{q}}\cdot\langle f|\boldsymbol{j}_e(\mathbf{0})|i\rangle$. The limit of $\varepsilon_{\mathrm{mac}}$ is the longitudinal projection $\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}}$ of the optical tensor of Sect. 4.4, which in a cubic crystal does not depend on $\hat{\boldsymbol{q}}$ (Appendix C.3). A metal differs: its intraband transitions have $E_f - E_i \to 0$, so $\tilde{R}_{\mathbf{00}}$ tends to a constant at $\omega = 0$, and $\varepsilon_{\mathrm{mac}}$ diverges as $1/q^2$, the perfect screening of Sect. 4.4.

Symmetrized and unsymmetrized matrices

Many codes and papers use the symmetrized matrix $\varepsilon^{\mathrm{s}} = \mathbf{1} - v^{1/2}\tilde{R}\,v^{1/2}$, with $v^{1/2}$ the diagonal matrix of $v_{\boldsymbol{G}}^{1/2}$ [10, footnote 3]. The symmetrized matrix is a similarity transform of the matrix ε used here: $\varepsilon^{\mathrm{s}} = v^{-1/2}\,\varepsilon\,v^{1/2}$, since $v^{-1/2}(v\tilde{R})v^{1/2} = v^{1/2}\tilde{R}v^{1/2}$. Inversion gives $(\varepsilon^{\mathrm{s}})^{-1} = v^{-1/2}\varepsilon^{-1}v^{1/2}$, and the elements of this inverse are $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(v_{\boldsymbol{G}'}/v_{\boldsymbol{G}})^{1/2}$. The diagonal elements are the same in both conventions, and they give $\varepsilon_{\mathrm{mac}}$ and the loss function at every $\boldsymbol{G}_q$; the off-diagonal elements differ by the factor $|\boldsymbol{q}+\boldsymbol{G}|/|\boldsymbol{q}+\boldsymbol{G}'|$. When $\tilde{R}$ is Hermitian, so is the symmetrized matrix, and a Hermitian matrix is convenient to invert; the wings of the symmetrized matrix are also finite as $\boldsymbol{q} \to 0$. Mixing off-diagonal elements from the two conventions is an error that no diagonal check reveals.

D.6 The Lindhard function at zero temperature

The density response of free electrons at $T = 0$ is a ratio of logarithms in ω and q, first written down by Lindhard [11]. Sect. 4.5 uses it for the valence electrons of silicon, where it fixes the static screening, the plasmon and the continuum of electron–hole pairs. The derivation follows Ref. [5, Chap. 4] in the notation of Sects. 4.2 and 4.7.

The angular and radial integrals

The independent-electron response is Eq. (4.93) continued to the real axis, $R^0(\boldsymbol{q},\omega) = \mathscr{V}^{-1}\sum_{\boldsymbol{k}}[n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}]/(\omega + \mathrm{i}0^+ + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})$, with the spin in the label. We keep the frequency as a complex number z with $\operatorname{Im} z > 0$, so that every expression below is analytic, and set $z = \omega + \mathrm{i}0^+$ only at the end. At $T = 0$ the occupation is the Fermi sea of Eq. (2.71), $n_{\boldsymbol{k}} = \Theta(k_{\mathrm{F}} - k)$. In the term with $n_{\boldsymbol{k}+\boldsymbol{q}}$, relabel $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$. The occupation becomes $n_{\boldsymbol{k}}$ and the denominator becomes $z + \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}}$, since ξ depends on $|\boldsymbol{k}|$ only. With the two spins and $\sum_{\boldsymbol{k}} \to \mathscr{V}\int \mathrm{d}^3k/(2\pi)^3$,

$$R^0(q,z) = 2\int_{k<k_{\mathrm{F}}} \frac{\mathrm{d}^3k}{(2\pi)^3}\left[\frac{1}{z-\Delta_{\boldsymbol{k}}} - \frac{1}{z+\Delta_{\boldsymbol{k}}}\right], \qquad \Delta_{\boldsymbol{k}} = \frac{(\boldsymbol{k}+\boldsymbol{q})^2 - k^2}{2m_e} = \frac{kqu}{m_e} + \frac{q^2}{2m_e},$$

with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. The resonant term, with $z-\Delta_{\boldsymbol{k}}$, is an electron leaving the sea at $\boldsymbol{k}$ for $\boldsymbol{k}+\boldsymbol{q}$, the transition that absorbs energy; the term with $z+\Delta_{\boldsymbol{k}}$ is its antiresonant partner of Eq. (4.6). Only states inside the sphere appear, and the Pauli principle now enters as a cancellation between the two terms. A transition from $\boldsymbol{k}$ to an occupied $\boldsymbol{k}+\boldsymbol{q}$ appears in the first term at $\boldsymbol{k}$ and in the second term, with the opposite sign, at $-\boldsymbol{k}-\boldsymbol{q}$, which is also inside the sphere; the two contributions cancel.

The variables that make the result simple measure frequency and momentum in units set by the Fermi surface,

$$\nu=\frac{z}{qv_{\mathrm{F}}},\qquad y=\frac{q}{2k_{\mathrm{F}}},\qquad \tilde{k}=\frac{k}{k_{\mathrm{F}}},\qquad \nu_{\pm}=\nu\pm y,$$

with $v_{\mathrm{F}}=k_{\mathrm{F}}/m_e$. Then $z\mp\Delta_{\boldsymbol{k}}=qv_{\mathrm{F}}(\nu\mp y\mp\tilde{k}u)$. The angular integrals are elementary, $\int_{-1}^{1}\mathrm{d}u\,(w-\tilde{k}u)^{-1}=\tilde{k}^{-1}\ln[(w+\tilde{k})/(w-\tilde{k})]$ for complex w off the real segment $[-\tilde{k},\tilde{k}]$, with the principal branch of the logarithm. For $\operatorname{Im}w>0$ the ratio $(w+\tilde{k})/(w-\tilde{k})=1+2\tilde{k}/(w-\tilde{k})$ lies in the lower half plane, away from the cut of the principal branch, so the logarithm is analytic there. With $\mathrm{d}^3k=2\pi k^2\,\mathrm{d}k\,\mathrm{d}u$,

$$R^0(q,z)=\frac{m_ek_{\mathrm{F}}^2}{2\pi^2q}\int_0^1\mathrm{d}\tilde{k}\;\tilde{k}\left[\ln\frac{\nu_-+\tilde{k}}{\nu_--\tilde{k}}-\ln\frac{\nu_++\tilde{k}}{\nu_+-\tilde{k}}\right].$$

The radial integral is the same as in the exchange energy of Sect. 23.1. The antiderivative of the radial integrand, $\frac{1}{2}(\tilde{k}^2-w^2)\ln[(w+\tilde{k})/(w-\tilde{k})]+w\tilde{k}$, holds for complex w, since checking it by differentiation does not use the reality of w. Evaluated between 0 and 1, the antiderivative gives

$$\mathcal{I}(w)=\int_0^1\mathrm{d}\tilde{k}\;\tilde{k}\ln\frac{w+\tilde{k}}{w-\tilde{k}}=w+\frac{1-w^2}{2}\,\ell(w),\qquad \ell(w)=\ln\frac{w+1}{w-1},$$

and $R^0=(m_ek_{\mathrm{F}}^2/2\pi^2q)[\mathcal{I}(\nu_-)-\mathcal{I}(\nu_+)]$. Since $\nu_--\nu_+=-2y=-q/k_{\mathrm{F}}$, the linear terms of $\mathcal{I}$ give a constant. Collecting, with the density of states $\mathcal{N}_{\mathrm{F}}=m_ek_{\mathrm{F}}/2\pi^2$ of one spin at the Fermi level (Eq. (2.71)), we arrive at the Lindhard function,

$$R^0(q,z)=-2\mathcal{N}_{\mathrm{F}}\left\{\frac{1}{2}+\frac{k_{\mathrm{F}}}{4q}\Big[(1-\nu_+^2)\,\ell(\nu_+)-(1-\nu_-^2)\,\ell(\nu_-)\Big]\right\},\qquad \nu_{\pm}=\frac{z}{qv_{\mathrm{F}}}\pm\frac{q}{2k_{\mathrm{F}}}.\tag{D.21}$$

It is analytic in the upper half plane of z, as a retarded function must be (Appendix C.5), and it depends on q and z only through ν and y, apart from the prefactor $2\mathcal{N}_{\mathrm{F}}$.

Real and imaginary parts

On the real axis, $z=\omega+\mathrm{i}0^+$ with $\omega>0$, and both $\nu_{\pm}$ acquire a positive imaginary part. By Eq. (C.73), $\ell(\nu_{\pm}+\mathrm{i}0^+)=\ln|(\nu_{\pm}+1)/(\nu_{\pm}-1)|-\mathrm{i}\pi\,\Theta(1-|\nu_{\pm}|)$. The real part of Eq. (D.21) keeps the form of the equation with every logarithm replaced by the logarithm of the modulus, now with $\nu_{\pm}=\omega/qv_{\mathrm{F}}\pm q/2k_{\mathrm{F}}$ real,

$$\operatorname{Re}R^0(q,\omega)=-2\mathcal{N}_{\mathrm{F}}\left\{\frac{1}{2}+\frac{k_{\mathrm{F}}}{4q}\left[(1-\nu_+^2)\ln\Big|\frac{\nu_++1}{\nu_+-1}\Big|-(1-\nu_-^2)\ln\Big|\frac{\nu_-+1}{\nu_--1}\Big|\right]\right\},\tag{D.22}$$

and the imaginary part collects the step functions $\Theta(1-|\nu_{\pm}|)$,

$$\operatorname{Im}R^0(q,\omega)=-\frac{\pi k_{\mathrm{F}}\mathcal{N}_{\mathrm{F}}}{2q}\Big[(1-\nu_-^2)\,\Theta(1-|\nu_-|)-(1-\nu_+^2)\,\Theta(1-|\nu_+|)\Big].\tag{D.23}$$

For $\omega < 0$ the imaginary part changes sign, since $R^0(q,-\omega) = R^0(q,\omega)^*$ by Eq. (4.5) for a gas with $R^0(-\boldsymbol{q}) = R^0(\boldsymbol{q})$.

The two step functions divide the (q,ω) plane into a region where both are on, a region where only the step of ν_- is on, and a region outside the continuum. Since $\omega > 0$, $\nu_+ > \nu_-$ and $\nu_+ > y > 0$. The condition $|\nu_+| < 1$ is $\omega < qv_\mathrm{F} - q^2/2m_e$, which needs $q < 2k_\mathrm{F}$. The condition $|\nu_-| < 1$ is $|m_e\omega/q - q/2| < k_\mathrm{F}$, which is the band $\omega_- < \omega < \omega_+$ with $\omega_+ = qv_\mathrm{F} + q^2/2m_e$ and ω_- equal to $q^2/2m_e - qv_\mathrm{F}$ for $q > 2k_\mathrm{F}$ and to zero otherwise. Below the line $\omega = qv_\mathrm{F} - q^2/2m_e$ both steps are on. The bracket of Eq. (D.23) is then $\nu_+^2 - \nu_-^2 = 4\nu y = 2m_e\omega/k_\mathrm{F}^2$, and

$$\mathrm{Im}\, R^0 = -\frac{m_e^2\,\omega}{2\pi q} \qquad (\omega < qv_\mathrm{F} - q^2/2m_e).$$

Above that line, up to ω_+, only the step of ν_- is on, and with $k_\mathrm{F}^2(1-\nu_-^2) = k_\mathrm{F}^2 - (m_e\omega/q - q/2)^2$,

$$\mathrm{Im}\, R^0 = -\frac{m_e}{4\pi q}\Big[k_\mathrm{F}^2 - \Big(\frac{m_e\omega}{q} - \frac{q}{2}\Big)^2\Big] \qquad (\text{only } |\nu_-| < 1).$$

Outside the band $\omega_- < \omega < \omega_+$ the imaginary part vanishes: no electron–hole pair has that energy at that momentum. Worked example 4.3 found the same forms by counting the transitions with a delta function; here they appear as the jump of a logarithm across its cut.

Static limit

At $\omega = 0$, $\nu_\pm = \pm y$. The logarithm of the modulus is odd, $\ln|(1-y)/(-1-y)| = -\ln|(1+y)/(1-y)|$, so the two terms in the bracket of Eq. (D.22) add, and $(k_\mathrm{F}/4q)\cdot 2 = 1/4y$. The result is Eq. (4.57).

D.7 The Mermin relaxation-time dielectric function

This section derives the Mermin response of Sect. 4.5 [12], in which collisions at the rate γ_rel relax the electrons toward a local equilibrium that has the instantaneous density of the electrons. The section also compares the Mermin response with the substitution $\omega \to \omega + \mathrm{i}\gamma_\mathrm{rel}$ in the independent-electron response.

The equation of motion

Take independent electrons, $H_0 = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}$ with the spin in the label, in equilibrium with occupations $n_{\boldsymbol{k}} = n_\mathrm{F}(\xi_{\boldsymbol{k}})$, and couple a potential to the density as in Eq. (4.3), $H(t) = H_0 + n_e(\boldsymbol{q})\,\phi_\mathrm{ext}(t)$, with $\boldsymbol{q} \neq 0$. In the random-phase approximation of Sect. 4.5 the potential becomes the total one, external plus induced; here it is simply the potential the electrons feel. The response is carried by the coherences

$$\delta\varrho_{\boldsymbol{k}}(t) = \langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}\rangle(t),$$

the matrix elements of the one-electron density matrix between $\boldsymbol{k}+\boldsymbol{q}$ and $\boldsymbol{k}$, which vanish in equilibrium. Their sum is the induced density, $\delta n(t) \equiv \mathcal{V}^{-1}\sum_{\boldsymbol{k}} \delta\varrho_{\boldsymbol{k}} = \mathcal{V}^{-1}\delta\langle n_e(\boldsymbol{q})^\dagger\rangle$, since $n_e(\boldsymbol{q})^\dagger = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$ (Eq. (2.38)). By Eq. (2.33), $[c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}, H_0] = (\xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}})\, c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$ and $[c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}, n_e(\boldsymbol{q})] = c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} - c_{\boldsymbol{k}+\boldsymbol{q}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$. To first order in the potential the second commutator can be averaged in equilibrium, and $\mathrm{i}\,\partial_t\langle A\rangle = \langle [A,H]\rangle$ gives

$$\mathrm{i}\,\partial_t\,\delta\varrho_{\boldsymbol{k}} = -D_{\boldsymbol{k}}\,\delta\varrho_{\boldsymbol{k}} + (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\,\phi_\mathrm{ext}(t), \qquad D_{\boldsymbol{k}} = \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}.$$

For $\phi_\mathrm{ext} \propto \mathrm{e}^{-\mathrm{i}\omega t}$ the solution is $\delta\varrho_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\,\phi_\mathrm{ext}/(\omega + \mathrm{i}0^+ + D_{\boldsymbol{k}})$, and summing over $\boldsymbol{k}$ gives back $\delta n = R^0(\boldsymbol{q},\omega)\,\phi_\mathrm{ext}$ with the bubble of Eq. (4.93). The equation of motion is the one-electron Liouville equation, and collisions add a relaxation term to it.

Relaxation toward local equilibrium

In the relaxation-time approximation, collisions relax the density matrix toward a reference state at the rate $\gamma_{\rm rel}$:

$$\mathrm{i}\,\partial_t\,\delta\varrho_{\boldsymbol{k}} = -D_{\boldsymbol{k}}\,\delta\varrho_{\boldsymbol{k}} + (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\,\phi_{\rm ext} - \mathrm{i}\gamma_{\rm rel}(\delta\varrho_{\boldsymbol{k}} - \delta\varrho^{\rm loc}_{\boldsymbol{k}}).$$

With the global equilibrium as the reference, $\delta\varrho^{\rm loc}_{\boldsymbol{k}} = 0$, the solution is the collisionless solution with ω replaced by $\omega + \mathrm{i}\gamma_{\rm rel}$, the naive substitution. Mermin's reference is the equilibrium state in which the density already has its instantaneous value, since collisions conserve the number of electrons (Sect. 4.5). A local equilibrium with a local chemical potential $\mu_{\rm ch} + \delta\mu(\boldsymbol{r},t)$ is the equilibrium state in the static potential $-\delta\mu$, and we write this potential as $\phi_{\rm loc}(t)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \text{c.c.}$ The density matrix of the local equilibrium is the time-independent solution of the collisionless equation in a static potential,

$$\delta\varrho^{\rm loc}_{\boldsymbol{k}} = \frac{n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}}{D_{\boldsymbol{k}}}\,\phi_{\rm loc}(t), \qquad \frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\delta\varrho^{\rm loc}_{\boldsymbol{k}} = R^0(\boldsymbol{q},0)\,\phi_{\rm loc}(t),$$

since $(1/\mathcal{V})\sum_{\boldsymbol{k}}(n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})/D_{\boldsymbol{k}}$ is the static bubble. The static response that enters here is the isothermal one, the value of the Matsubara response at $\omega_0^{\rm B} = 0$ in Sect. 4.7, which for free electrons at $\boldsymbol{q} \neq 0$ coincides with $R^0(\boldsymbol{q},\omega\to 0)$. The amplitude $\phi_{\rm loc}$ is fixed by requiring the local-equilibrium state to have the actual density,

$$\frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\delta\varrho^{\rm loc}_{\boldsymbol{k}} = \frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\delta\varrho_{\boldsymbol{k}}, \qquad \text{that is} \qquad \phi_{\rm loc} = \frac{\delta n}{R^0(\boldsymbol{q},0)}. \tag{D.24}$$

For a field at frequency ω, the equation of motion becomes algebraic,

$$(\omega + \mathrm{i}\gamma_{\rm rel} + D_{\boldsymbol{k}})\,\delta\varrho_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\Big[\phi_{\rm ext} + \frac{\mathrm{i}\gamma_{\rm rel}\,\phi_{\rm loc}}{D_{\boldsymbol{k}}}\Big].$$

We divide by the factor on the left and use the partial fractions $1/[D(\omega + \mathrm{i}\gamma_{\rm rel} + D)] = (\omega + \mathrm{i}\gamma_{\rm rel})^{-1}[1/D - 1/(\omega + \mathrm{i}\gamma_{\rm rel} + D)]$; the sum over $\boldsymbol{k}$ then produces two bubbles, one at the complex frequency $\omega + \mathrm{i}\gamma_{\rm rel}$ and one static. With the shorthand $R^0_\gamma = R^0(\boldsymbol{q},\omega + \mathrm{i}\gamma_{\rm rel})$ and $R^0_{\rm st} = R^0(\boldsymbol{q},0)$,

$$\delta n = R^0_\gamma\,\phi_{\rm ext} + \frac{\mathrm{i}\gamma_{\rm rel}}{\omega + \mathrm{i}\gamma_{\rm rel}}(R^0_{\rm st} - R^0_\gamma)\,\phi_{\rm loc}.$$

Inserting Eq. (D.24) and collecting the terms in δn on the left gives

$$\frac{\omega + \mathrm{i}\gamma_{\rm rel}R^0_\gamma/R^0_{\rm st}}{\omega + \mathrm{i}\gamma_{\rm rel}}\,\delta n = R^0_\gamma\,\phi_{\rm ext},$$

and the response $\delta n/\phi_{\rm ext}$ is

$$R^{\rm Mer}(\boldsymbol{q},\omega) = \frac{(1 + \mathrm{i}\gamma_{\rm rel}/\omega)\,R^0(\boldsymbol{q},\omega + \mathrm{i}\gamma_{\rm rel})}{1 + (\mathrm{i}\gamma_{\rm rel}/\omega)\,R^0(\boldsymbol{q},\omega + \mathrm{i}\gamma_{\rm rel})/R^0(\boldsymbol{q},0)}. \tag{D.25}$$

In the random-phase approximation the Mermin dielectric function is $\varepsilon_{\rm Mer} = 1 - v_{\rm C}R^{\rm Mer}$. With $\varepsilon_{\rm Lin} = 1 - v_{\rm C}R^0$ for the Lindhard dielectric function, so that $R^0_\gamma/R^0_{\rm st} = [\varepsilon_{\rm Lin}(\boldsymbol{q},\omega + \mathrm{i}\gamma_{\rm rel}) - 1]/[\varepsilon_{\rm Lin}(\boldsymbol{q},0) - 1]$, the Mermin dielectric function takes the form given by Mermin [12],

$$\varepsilon_{\rm Mer}(\boldsymbol{q},\omega) = 1 + \frac{(1 + \mathrm{i}\gamma_{\rm rel}/\omega)\,\big[\varepsilon_{\rm Lin}(\boldsymbol{q},\omega + \mathrm{i}\gamma_{\rm rel}) - 1\big]}{1 + (\mathrm{i}\gamma_{\rm rel}/\omega)\,\big[\varepsilon_{\rm Lin}(\boldsymbol{q},\omega + \mathrm{i}\gamma_{\rm rel}) - 1\big]/\big[\varepsilon_{\rm Lin}(\boldsymbol{q},0) - 1\big]}. \tag{D.26}$$

Nothing in the derivation used $T = 0$; at finite temperature R^0 and $\varepsilon_{\rm Lin}$ are taken at that temperature.

Causality

The bubble at the shifted frequency, $R^0(\boldsymbol{q}, z + \mathrm{i}\gamma_{\mathrm{rel}})$, is analytic for $\operatorname{Im} z > -\gamma_{\mathrm{rel}}$. The analyticity of the shifted bubble alone does not make R^{Mer} causal, since the denominator of Eq. (D.25), which is proportional to $z + \mathrm{i}\gamma_{\mathrm{rel}} R^0_\gamma / R^0_{\mathrm{st}}$, could vanish in the upper half plane, and a pole there would be a response that grows after the field is gone. The relaxation equation has no growing mode, and the proof uses only its structure.

A zero of the denominator is a free oscillation. With $\phi_{\mathrm{ext}} = 0$ and the time dependence $\mathrm{e}^{-\mathrm{i}zt}$, the equation for δn obtained above by collecting terms reads $(z + \mathrm{i}\gamma_{\mathrm{rel}} R^0_\gamma / R^0_{\mathrm{st}})\, \delta n = 0$, so a nonzero density exists exactly where the denominator vanishes. The nonzero density comes with the nonzero coherences $\delta\varrho_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\, \mathrm{i}\gamma_{\mathrm{rel}} \phi_{\mathrm{loc}} / [D_{\boldsymbol{k}}(z + \mathrm{i}\gamma_{\mathrm{rel}} + D_{\boldsymbol{k}})]$. A zero with $\operatorname{Im} z > 0$ would be a field-free solution of the equation of motion that grows as $\mathrm{e}^{(\operatorname{Im} z)t}$.

No growing field-free solution exists. Write $w_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})/(\xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}})$, which is positive wherever the two occupations differ, because n_{F} falls with energy, and zero where they are equal. Let $\varsigma = \sum_{\boldsymbol{k}} \delta\varrho_{\boldsymbol{k}}$ and $w_{\mathrm{tot}} = \sum_{\boldsymbol{k}} w_{\boldsymbol{k}}$. Since $R^0_{\mathrm{st}} = -w_{\mathrm{tot}}/\mathcal{V}$, the condition Eq. (D.24) makes the reference state $\delta\varrho^{\mathrm{loc}}_{\boldsymbol{k}} = w_{\boldsymbol{k}}\, \varsigma / w_{\mathrm{tot}}$, and the equation of motion without a field is

$$\partial_t\, \delta\varrho_{\boldsymbol{k}} = \mathrm{i} D_{\boldsymbol{k}}\, \delta\varrho_{\boldsymbol{k}} - \gamma_{\mathrm{rel}} \Big(\delta\varrho_{\boldsymbol{k}} - w_{\boldsymbol{k}} \frac{\varsigma}{w_{\mathrm{tot}}} \Big).$$

A coherence with $w_{\boldsymbol{k}} = 0$ obeys $\partial_t\, \delta\varrho_{\boldsymbol{k}} = (\mathrm{i}D_{\boldsymbol{k}} - \gamma_{\mathrm{rel}})\, \delta\varrho_{\boldsymbol{k}}$ and decays at the rate γ_{rel}, so a solution $\propto \mathrm{e}^{-\mathrm{i}zt}$ with $\operatorname{Im} z > 0$ has every coherence with $w_{\boldsymbol{k}} = 0$ equal to zero, and ς is a sum over the coherences with positive $w_{\boldsymbol{k}}$. For these coherences take the weighted norm

$$\Phi_\varrho(t) = \sum_{w_{\boldsymbol{k}} > 0} \frac{|\delta\varrho_{\boldsymbol{k}}|^2}{w_{\boldsymbol{k}}} \geq 0.$$

The term $\mathrm{i}D_{\boldsymbol{k}}$ rotates each phase and does not change Φ_ϱ, and

$$\frac{\mathrm{d}\Phi_\varrho}{\mathrm{d}t} = -2\gamma_{\mathrm{rel}} \Big[\Phi_\varrho - \frac{|\varsigma|^2}{w_{\mathrm{tot}}} \Big] \leq 0,$$

because the Cauchy–Schwarz inequality gives $|\varsigma|^2 = |\sum_{\boldsymbol{k}} (\delta\varrho_{\boldsymbol{k}} / w_{\boldsymbol{k}}^{1/2})\, w_{\boldsymbol{k}}^{1/2}|^2 \leq \Phi_\varrho w_{\mathrm{tot}}$. A growing solution would make $\Phi_\varrho \propto \mathrm{e}^{2(\operatorname{Im} z)t}$ grow, so none exists, and the denominator of Eq. (D.25) has no zero with $\operatorname{Im} z > 0$. At $z = 0$ the numerator and the denominator both contain $\mathrm{i}\gamma_{\mathrm{rel}}/z$, the singularity is removable, and $R^{\mathrm{Mer}} \to R^0(\boldsymbol{q}, 0)$. With the decay at large $|z|$ shown below, R^{Mer} is analytic in the upper half plane and falls off at infinity, a causal response in the sense of Appendix C.5.

Particle conservation

The sum of the equation of motion over $\boldsymbol{k}$ is the continuity equation, and the reference state decides whether it holds. In the term with the potential, $\sum_{\boldsymbol{k}} (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}) = 0$, because the two sums run over the same occupations. In the term with $D_{\boldsymbol{k}}$, the free dispersion gives $-D_{\boldsymbol{k}} = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}} = \boldsymbol{q} \cdot (\boldsymbol{k} + \boldsymbol{q}/2)/m_e$. The factor $(\boldsymbol{k} + \boldsymbol{q}/2)/m_e$ is the plane-wave matrix element $\langle \boldsymbol{k} + \boldsymbol{q} | \frac{1}{2} \{ \boldsymbol{p}/m_e, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \} | \boldsymbol{k} \rangle$ of the paramagnetic current $\boldsymbol{j}_e(\boldsymbol{q}) = \sum_i \frac{1}{2} \{ \boldsymbol{p}_i/m_e, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i} \}$, the velocity of the electrons weighted by the plane wave (Sect. 4.3). By Eq. (2.36), $\boldsymbol{j}_e(\boldsymbol{q})^\dagger = \sum_{\boldsymbol{k}} [(\boldsymbol{k} + \boldsymbol{q}/2)/m_e]\, c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}}$, so that $-\sum_{\boldsymbol{k}} D_{\boldsymbol{k}} \delta\varrho_{\boldsymbol{k}} = \boldsymbol{q} \cdot \delta\langle \boldsymbol{j}_e(\boldsymbol{q})^\dagger \rangle$. At frequency ω the summed equation reads

$$\omega\, \delta\langle n_e(\boldsymbol{q})^\dagger \rangle - \boldsymbol{q} \cdot \delta\langle \boldsymbol{j}_e(\boldsymbol{q})^\dagger \rangle = -\mathrm{i}\gamma_{\mathrm{rel}} \sum_{\boldsymbol{k}} (\delta\varrho_{\boldsymbol{k}} - \delta\varrho^{\mathrm{loc}}_{\boldsymbol{k}}). \tag{D.27}$$

Without collisions the right-hand side is zero, and Eq. (D.27) is the Fourier transform of $\partial_t n_e + \nabla \cdot \boldsymbol{j} = 0$: the density at a crest changes only as fast as the current carries electrons in

or out. The naive substitution, with $\delta\varrho^{\rm loc}=0$, leaves $-\mathrm{i}\gamma_{\rm rel}\,\delta\langle n_e(\boldsymbol{q})^\dagger\rangle$ on the right, a density that decays at the rate $\gamma_{\rm rel}$ whatever the current does. $R^{\rm Mer}$ conserves particle number in the sense that Mermin's condition Eq. (D.24) sets the right-hand side to zero exactly, at every frequency and momentum. Collisions now change only the current, and through the current the density.

Long-wavelength limit

For $q\to 0$ at fixed ω and $\gamma_{\rm rel}$, the frequency $\omega+\mathrm{i}\gamma_{\rm rel}$ lies far outside the continuum, $|\omega+\mathrm{i}\gamma_{\rm rel}|\gg qv_{\rm F}$, and the high-frequency form Eq. (4.58) gives $R^0_\gamma\to \boldsymbol{n}_e q^2/[m_e(\omega+\mathrm{i}\gamma_{\rm rel})^2]$, while $R^0_{\rm st}$ stays finite. The ratio $R^0_\gamma/R^0_{\rm st}$ vanishes as q^2, the denominator of Eq. (D.25) tends to one, and $R^{\rm Mer}\to \boldsymbol{n}_e q^2/[m_e\,\omega(\omega+\mathrm{i}\gamma_{\rm rel})]$, which with $\varepsilon_{\rm Mer}=1-v_{\rm C}R^{\rm Mer}$ is the Drude function of Sect. 4.5. At large ω the same form gives $R^{\rm Mer}\to \boldsymbol{n}_e q^2/m_e\omega^2$, the leading tail of the naive substitution as well, and by Eq. (C.35) the Mermin response and the naive substitution both obey $\int_0^\infty \mathrm{d}\omega\,\omega\ \mathrm{Im}\,R=-\pi\boldsymbol{n}_e q^2/2m_e$.

D.8 Imaginary-time formalism

This section supplies the steps behind the imaginary-time results of Sect. 4.7: the antiperiodicity of the Matsubara Green's function and its Fourier series, the occupation as a frequency sum, and the response of independent electrons written as a sum over a product of two Green's functions and evaluated. The residue formulas for Matsubara sums, Eq. (C.40), and the uniqueness of the analytic continuation are derived in Appendix C.5. The notation is that of Sect. 4.7: operators evolve in imaginary time with $H_\mu=H-\mu_{\rm ch}\hat N_e$, $O(\tau)=\mathrm{e}^{\tau H_\mu}O\,\mathrm{e}^{-\tau H_\mu}$, and averages are taken in $\hat\varrho=\mathrm{e}^{-\beta H_\mu}/\mathcal{Z}$.

Antiperiodicity and the Fourier series

Take $0<\tau<\beta$, so that $-\beta<\tau-\beta<0$. At the negative argument the time ordering in Eq. (4.86) puts $c^\dagger_{\boldsymbol{k}}$ on the left with a minus sign, and with $c_{\boldsymbol{k}}(\tau-\beta)=\mathrm{e}^{(\tau-\beta)H_\mu}c_{\boldsymbol{k}}\,\mathrm{e}^{-(\tau-\beta)H_\mu}$,

$$\mathcal{Z}\,\mathsf{G}(\boldsymbol{k},\tau-\beta)=\mathrm{Tr}\big[\mathrm{e}^{-\beta H_\mu}\,c^\dagger_{\boldsymbol{k}}\,\mathrm{e}^{(\tau-\beta)H_\mu}c_{\boldsymbol{k}}\,\mathrm{e}^{(\beta-\tau)H_\mu}\big]=\mathrm{Tr}\big[\mathrm{e}^{-\tau H_\mu}c^\dagger_{\boldsymbol{k}}\,\mathrm{e}^{\tau H_\mu}\,\mathrm{e}^{-\beta H_\mu}c_{\boldsymbol{k}}\big],$$

where the cyclic property of the trace moved the last factor $\mathrm{e}^{(\beta-\tau)H_\mu}$ to the front, where the last factor combined with $\mathrm{e}^{-\beta H_\mu}$ into $\mathrm{e}^{-\tau H_\mu}$. Cycling $\mathrm{e}^{-\beta H_\mu}c_{\boldsymbol{k}}$ to the front in the same way gives $\mathrm{Tr}[\mathrm{e}^{-\beta H_\mu}c_{\boldsymbol{k}}\,\mathrm{e}^{-\tau H_\mu}c^\dagger_{\boldsymbol{k}}\,\mathrm{e}^{\tau H_\mu}]$, and moving $\mathrm{e}^{\tau H_\mu}$ to the front, where it commutes with $\mathrm{e}^{-\beta H_\mu}$, gives $\mathrm{Tr}[\mathrm{e}^{-\beta H_\mu}c_{\boldsymbol{k}}(\tau)\,c^\dagger_{\boldsymbol{k}}]=-\mathcal{Z}\,\mathsf{G}(\boldsymbol{k},\tau)$. Hence

$$\mathsf{G}(\boldsymbol{k},\tau-\beta)=-\mathsf{G}(\boldsymbol{k},\tau),\qquad 0<\tau<\beta. \tag{D.28}$$

For two bosonic operators the time ordering carries no sign, and the same steps give $R_{ab}(\boldsymbol{q},\tau-\beta)=+R_{ab}(\boldsymbol{q},\tau)$. The traces converge for $-\beta<\tau<\beta$, where $\mathrm{e}^{-\beta H_\mu}\mathrm{e}^{\pm\tau H_\mu}$ still decays at large energies, and this interval is the whole range on which $\mathsf{G}(\boldsymbol{k},\tau)$ and $R_{ab}(\boldsymbol{q},\tau)$ are defined.

If a function on $-\beta<\tau<\beta$ changes sign under $\tau\to\tau-\beta$, it has period 2β, and its Fourier series runs over the frequencies $n\pi/\beta=n\pi T$ with integer n. The coefficient of $\mathrm{e}^{-\mathrm{i}n\pi T\tau}$ is $(2\beta)^{-1}\int_{-\beta}^{\beta}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}n\pi T\tau}\mathsf{G}(\boldsymbol{k},\tau)$. Splitting the integral at $\tau=0$ and shifting the negative half by β with Eq. (D.28) turns the coefficient into $(2\beta)^{-1}[1-\mathrm{e}^{-\mathrm{i}n\pi}]\int_0^\beta\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}n\pi T\tau}\mathsf{G}(\boldsymbol{k},\tau)$. The bracket is 0 for even n and 2 for odd $n=2j+1$, so only the fermionic frequencies $\omega^{\rm F}_j$ remain, and the coefficients give the pair Eq. (4.87). For a periodic function the bracket is $1+\mathrm{e}^{-\mathrm{i}n\pi}$, and only the bosonic frequencies $\omega^{\rm B}_j$ survive.

The occupation as a frequency sum

By Eq. (4.87) at $\tau = -0^+$, $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}\rangle = \mathsf{G}(\boldsymbol{k}, -0^+) = T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}} 0^+}\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}})$, which is Eq. (4.89). The factor $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+}$ cannot be dropped. Since $\mathsf{G} \to 1/(\mathrm{i}\omega_j^{\mathrm{F}})$ at large j, the sum without the factor converges only conditionally, and its symmetric value is $\frac{1}{2}(\mathsf{G}(\boldsymbol{k}, 0^+) + \mathsf{G}(\boldsymbol{k}, 0^-))$, the average of the two sides of the jump at $\tau = 0$; this average equals $\langle c^\dagger c\rangle - \frac{1}{2}$.

To evaluate the sum, let $g(z) = \mathrm{e}^{z0^+}\mathsf{G}(\boldsymbol{k}, z)$ with $\mathsf{G}(\boldsymbol{k}, z) = \int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k}, \omega')/(z - \omega')$. Near each pole $\mathrm{i}\omega_j^{\mathrm{F}}$ of $n_{\mathrm{F}}(z) = 1/(\mathrm{e}^{\beta z} + 1)$ the residue is $-T$ (Appendix C.5), so

$$T\sum_j g(\mathrm{i}\omega_j^{\mathrm{F}}) = -\oint_C \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, n_{\mathrm{F}}(z)\, g(z),$$

with C the small counterclockwise circles of Fig. 4.13a. The function g is analytic off the real axis. In the upper half plane the circles can therefore be merged into the boundary of the half plane, traversed counterclockwise: the line $\omega + \mathrm{i}0^+$ from $-\infty$ to $+\infty$ and an arc at infinity. In the lower half plane they merge into the line $\omega - \mathrm{i}0^+$ from $+\infty$ to $-\infty$ and another arc. On the arcs $|g| \sim 1/|z|$, while $n_{\mathrm{F}}(z)\,\mathrm{e}^{z0^+}$ decays exponentially for $\mathrm{Re}\, z \to +\infty$ through n_{F} and for $\mathrm{Re}\, z \to -\infty$ through the convergence factor, so the arcs contribute nothing. The two lines give

$$T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+}\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = -\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi\mathrm{i}}\, n_{\mathrm{F}}(\omega)\big[\mathsf{G}(\boldsymbol{k}, \omega+\mathrm{i}0^+) - \mathsf{G}(\boldsymbol{k}, \omega-\mathrm{i}0^+)\big] = \int \mathrm{d}\omega\, n_{\mathrm{F}}(\omega)\,\mathsf{A}(\boldsymbol{k}, \omega), \tag{D.29}$$

where the jump $\mathsf{G}(\boldsymbol{k}, \omega+\mathrm{i}0^+) - \mathsf{G}(\boldsymbol{k}, \omega-\mathrm{i}0^+) = -2\pi\mathrm{i}\mathsf{A}(\boldsymbol{k}, \omega)$ follows from the Sokhotski–Plemelj formula Eq. (2.58) applied under the integral that defines $\mathsf{G}(\boldsymbol{k}, z)$. Eq. (D.29) is Eq. (4.78), obtained without the Lehmann form. For a free electron, $\mathsf{A}_0 = \delta(\omega - \xi_{\boldsymbol{k}})$ and the sum is $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$; equivalently, Eq. (C.40) with the single pole of g at $z = \xi_{\boldsymbol{k}}$ gives the residue $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$ directly.

The bubble of independent electrons

Take free electrons, $H_\mu = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}$ with the spin in the label, and $O_a = O_b = n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}}^\dagger c_{\boldsymbol{k}}$ (Eq. (2.38)), with $\boldsymbol{q} \neq 0$. In imaginary time $c_{\boldsymbol{k}}(\tau) = \mathrm{e}^{-\xi_{\boldsymbol{k}}\tau}c_{\boldsymbol{k}}$ and $c_{\boldsymbol{k}}^\dagger(\tau) = \mathrm{e}^{+\xi_{\boldsymbol{k}}\tau}c_{\boldsymbol{k}}^\dagger$, by the steps that gave the real-time form in Sect. 2.3. For $0 < \tau < \beta$, with $n_e(\boldsymbol{q})^\dagger = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$,

$$R^0(\boldsymbol{q}, \tau) = -\frac{1}{\mathscr{V}}\sum_{\boldsymbol{k},\boldsymbol{k}'} \mathrm{e}^{(\xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})\tau}\,\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}\, c_{\boldsymbol{k}'+\boldsymbol{q}}^\dagger c_{\boldsymbol{k}'}\rangle.$$

The Gibbs state of independent electrons is a product over orbitals (Sect. 2.5), and an average of operators is nonzero only if every orbital that loses an electron also gains one. With $\boldsymbol{q} \neq 0$ this condition requires $\boldsymbol{k}' = \boldsymbol{k}$, and then $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}} c_{\boldsymbol{k}+\boldsymbol{q}}^\dagger c_{\boldsymbol{k}}\rangle = n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}})$ with $n_{\boldsymbol{k}} = n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, as in Eq. (2.70). The transform of the τ dependence is

$$\int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{(\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})\tau} = \frac{\mathrm{e}^{\beta(\xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})} - 1}{\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}}.$$

Since $n_{\boldsymbol{k}}\mathrm{e}^{\beta\xi_{\boldsymbol{k}}} = 1 - n_{\boldsymbol{k}}$ and $(1 - n_{\boldsymbol{k}+\boldsymbol{q}})\mathrm{e}^{-\beta\xi_{\boldsymbol{k}+\boldsymbol{q}}} = n_{\boldsymbol{k}+\boldsymbol{q}}$, the numerator times $n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}})$ is $(1 - n_{\boldsymbol{k}})n_{\boldsymbol{k}+\boldsymbol{q}} - n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}}) = n_{\boldsymbol{k}+\boldsymbol{q}} - n_{\boldsymbol{k}}$, and

$$R^0(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}} \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}}, \tag{D.30}$$

the right-hand side of Eq. (4.93). When R^0 is continued to $\omega + \mathrm{i}0^+$, its imaginary part is $-(\pi/\mathcal{V})\sum_{\boldsymbol{k}}[n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}]\,\delta(\omega - \xi_{\boldsymbol{k}+\boldsymbol{q}} + \xi_{\boldsymbol{k}})$, which by Eq. (4.7) is the difference of the independent-electron transition sums of Eq. (2.70).

The same result comes out of a frequency sum. Consider

$$T\sum_{j'} \mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_{j'})\,\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega^{\mathrm{F}}_{j'} + \mathrm{i}\omega^{\mathrm{B}}_{j}) = T\sum_{j'} \frac{1}{(\mathrm{i}\omega^{\mathrm{F}}_{j'} - \xi_{\boldsymbol{k}})(\mathrm{i}\omega^{\mathrm{F}}_{j'} + \mathrm{i}\omega^{\mathrm{B}}_{j} - \xi_{\boldsymbol{k}+\boldsymbol{q}})}.$$

The summand falls off as $1/|z|^2$, so no convergence factor is needed, and the fermionic form of Eq. (C.40) applies with $g(z) = [(z - \xi_{\boldsymbol{k}})(z + \mathrm{i}\omega^{\mathrm{B}}_{j} - \xi_{\boldsymbol{k}+\boldsymbol{q}})]^{-1}$. Its pole at $z = \xi_{\boldsymbol{k}}$ has the residue $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})/(\xi_{\boldsymbol{k}} + \mathrm{i}\omega^{\mathrm{B}}_{j} - \xi_{\boldsymbol{k}+\boldsymbol{q}})$. The other pole of g, at $z = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \mathrm{i}\omega^{\mathrm{B}}_{j}$, has the residue $n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}} - \mathrm{i}\omega^{\mathrm{B}}_{j})/(\xi_{\boldsymbol{k}+\boldsymbol{q}} - \mathrm{i}\omega^{\mathrm{B}}_{j} - \xi_{\boldsymbol{k}})$, and since $\mathrm{e}^{-\mathrm{i}\beta\omega^{\mathrm{B}}_{j}} = 1$ the Fermi function in this residue is $n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})$. The sum of the two residues is

$$T\sum_{j'} \mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_{j'})\,\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega^{\mathrm{F}}_{j'} + \mathrm{i}\omega^{\mathrm{B}}_{j}) = \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\mathrm{i}\omega^{\mathrm{B}}_{j} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}}, \tag{D.31}$$

term by term the summand of Eq. (D.30). The agreement proves Eq. (4.93). Two features recur in every diagram of Sect. 4.8. The bosonic frequency was simplified in the Fermi function before any continuation, $n_{\mathrm{F}}(\xi - \mathrm{i}\omega^{\mathrm{B}}_{j}) = n_{\mathrm{F}}(\xi)$; doing the simplification after the continuation would give a function of ω that is not analytic. And the occupation factors appear as differences, $n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})$, so that a transition between two occupied or two empty orbitals drops out; this cancellation is the Pauli blocking of Sect. 2.5.

D.9 Wick's theorem and the Feynman rules

Every diagram of Sect. 4.8 is one term of an expansion of the Green's function in powers of the interaction. Each term is an average of a long string of free creation and annihilation operators, and Wick's theorem [13] reduces that average to a sum of products of free Green's functions, one product for each way of pairing the operators. This section derives the expansion in imaginary time, where it is exact at any temperature, and at $T = 0$ in real time, where it rests on the adiabatic theorem of Gell-Mann and Low [14]. The conventions for G follow Sect. 4.7: a bare G with a real argument is the time-ordered function, and with an imaginary argument the Matsubara function. The treatment follows Refs. [2, Chaps. 3 and 7] and [1].

The expansion in imaginary time

Split $H_\mu = H_0 + H_{\mathrm{int}}$, with $H_0 = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}$, plus $\sum_{\boldsymbol{Q}} \omega_{\boldsymbol{Q}} d^\dagger_{\boldsymbol{Q}} d_{\boldsymbol{Q}}$ when bosons are present. Operators in the interaction picture evolve with H_0 alone, $O_{\mathrm{I}}(\tau) = \mathrm{e}^{\tau H_0} O\, \mathrm{e}^{-\tau H_0}$, so that $c_{\boldsymbol{k},\mathrm{I}}(\tau) = \mathrm{e}^{-\xi_{\boldsymbol{k}}\tau} c_{\boldsymbol{k}}$ and $c^\dagger_{\boldsymbol{k},\mathrm{I}}(\tau) = \mathrm{e}^{\xi_{\boldsymbol{k}}\tau} c^\dagger_{\boldsymbol{k}}$. Define $U_{\mathrm{I}}(\tau, 0) = \mathrm{e}^{\tau H_0}\mathrm{e}^{-\tau H_\mu}$. Its derivative is $\mathrm{e}^{\tau H_0}(H_0 - H_\mu)\mathrm{e}^{-\tau H_\mu} = -H_{\mathrm{int,I}}(\tau)\, U_{\mathrm{I}}(\tau, 0)$, which has the form of Eq. (2.49) with $\mathrm{i}\,\mathrm{d}t$ replaced by $-\mathrm{d}\tau$. Integrating and iterating as in Eq. (2.50) gives

$$\begin{aligned} U_{\mathrm{I}}(\tau, 0) &= \sum_{n=0}^{\infty} (-1)^n \int_0^\tau \mathrm{d}\tau_1 \int_0^{\tau_1} \mathrm{d}\tau_2 \cdots \int_0^{\tau_{n-1}} \mathrm{d}\tau_n\, H_{\mathrm{int,I}}(\tau_1) \cdots H_{\mathrm{int,I}}(\tau_n) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\tau \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_n\, \mathrm{T}_\tau \big[H_{\mathrm{int,I}}(\tau_1) \cdots H_{\mathrm{int,I}}(\tau_n)\big]. \end{aligned}$$

In the second form each integral runs over the whole interval from 0 to τ. The $n!$ orderings of the times give the same ordered product, since T_τ restores the order and H_{int} contains fermion

operators in pairs, so that no sign arises. The same steps give $U_{\rm I}(\tau,\tau') = \mathrm{e}^{\tau H_0}\mathrm{e}^{-(\tau-\tau')H_\mu}\mathrm{e}^{-\tau' H_0}$ as the ordered exponential over $[\tau',\tau]$.

The Boltzmann factor and the Heisenberg operators of Sect. 4.7 are both expressed through $U_{\rm I}$. The Boltzmann factor is $\mathrm{e}^{-\beta H_\mu} = \mathrm{e}^{-\beta H_0} U_{\rm I}(\beta,0)$, and the full imaginary-time operator is

$$c_{\boldsymbol{k}}(\tau) = \mathrm{e}^{\tau H_\mu} c_{\boldsymbol{k}} \mathrm{e}^{-\tau H_\mu} = U_{\rm I}(\tau,0)^{-1}\, c_{\boldsymbol{k},\rm I}(\tau)\, U_{\rm I}(\tau,0).$$

For $0 < \tau < \beta$, inserting the Boltzmann factor and $c_{\boldsymbol{k}}(\tau)$ into Eq. (4.86) and using $U_{\rm I}(\beta,0)U_{\rm I}(\tau,0)^{-1} = U_{\rm I}(\beta,\tau)$ gives

$$\mathrm{Tr}\big[\mathrm{e}^{-\beta H_\mu} c_{\boldsymbol{k}}(\tau) c_{\boldsymbol{k}}^\dagger\big] = \mathrm{Tr}\big[\mathrm{e}^{-\beta H_0}\, U_{\rm I}(\beta,\tau)\, c_{\boldsymbol{k},\rm I}(\tau)\, U_{\rm I}(\tau,0)\, c_{\boldsymbol{k},\rm I}^\dagger(0)\big].$$

Every factor on the right stands to the left of every factor with an earlier imaginary time, so the product is the time-ordered product of $U_{\rm I}(\beta,0)$ with the two operators. The case $\tau < 0$ works the same way, and the partition function is $\mathcal{Z} = \mathcal{Z}_0 \langle U_{\rm I}(\beta,0)\rangle_0$, where $\langle\cdots\rangle_0 = \mathrm{Tr}[\mathrm{e}^{-\beta H_0}\cdots]/\mathcal{Z}_0$ is the average in the free grand-canonical state. Hence

$$\mathsf{G}(\boldsymbol{k},\tau) = -\frac{\big\langle \mathrm{T}_\tau\, U_{\rm I}(\beta,0)\, c_{\boldsymbol{k},\rm I}(\tau)\, c_{\boldsymbol{k},\rm I}^\dagger(0)\big\rangle_0}{\big\langle U_{\rm I}(\beta,0)\big\rangle_0}. \tag{D.32}$$

Nothing has been assumed beyond the convergence of the series. The density response Eq. (4.90) has the same expansion with $n_e(\boldsymbol{q})^\dagger$ and $n_e(\boldsymbol{q})$ in place of $c_{\boldsymbol{k}}$ and $c_{\boldsymbol{k}}^\dagger$, and a factor $1/\mathcal{V}$.

Inside $H_{\rm int,I}(\tau)$ creation and annihilation operators share one time, and the ordering symbol needs a rule for them. We keep them in the order in which they are written in $H_{\rm int}$, creation operators to the left; equivalently, every creation operator in $H_{\rm int}$ is taken at $\tau + 0^+$.

The expansion at zero temperature

At $T = 0$ the time-ordered function Eq. (4.79) is an average in the interacting ground state $|0\rangle$, which is not known. Switch the interaction on and off slowly, $H_{\rm int}\mathrm{e}^{-\eta_{\rm b}|t|}$ with $\eta_{\rm b} \to 0^+$, and start from the ground state $|\Phi_0\rangle$ of H_0, the Fermi sea. The theorem of Gell-Mann and Low [14] states that $U_{\rm I}(0,-\infty)|\Phi_0\rangle / \langle\Phi_0|U_{\rm I}(0,-\infty)|\Phi_0\rangle$, with $U_{\rm I}$ the real-time operator of Eq. (2.50) with $H' = H_{\rm int}$, tends to an eigenstate of H_μ. We assume that the ground state is nondegenerate and is reached from $|\Phi_0\rangle$ without a level crossing as the interaction is switched on; the eigenstate is then the ground state. Switching off toward $t = +\infty$ leads to the same state, up to a phase, so $\langle\Phi_0|U_{\rm I}(\infty,0)$ is proportional to $\langle 0|$. With $c_{\boldsymbol{k}}(t) = U_{\rm I}(t,0)^{-1} c_{\boldsymbol{k},\rm I}(t)\, U_{\rm I}(t,0)$ the steps that led to Eq. (D.32) go through with real times, and the unknown phases and normalizations cancel in the ratio:

$$\mathsf{G}(\boldsymbol{k},t) = -\mathrm{i}\,\frac{\langle\Phi_0|\mathrm{T}\, U_{\rm I}(\infty,-\infty)\, c_{\boldsymbol{k},\rm I}(t)\, c_{\boldsymbol{k},\rm I}^\dagger(0)|\Phi_0\rangle}{\langle\Phi_0|U_{\rm I}(\infty,-\infty)|\Phi_0\rangle}, \tag{D.33}$$

with $U_{\rm I}(\infty,-\infty) = \sum_n (-\mathrm{i})^n (n!)^{-1} \int \mathrm{d}t_1 \cdots \mathrm{d}t_n\, \mathrm{T}[H_{\rm int,I}(t_1)\cdots H_{\rm int,I}(t_n)]$. At $T > 0$ no expansion of this form exists for real times, since a thermal state is not reached adiabatically from a single free state. A real-time expansion at finite temperature needs a time contour that runs forward and back, and here the imaginary-time form Eq. (D.32) takes the place of the real-time expansion (Sect. 4.7).

Wick's theorem

Let $A_1, \ldots, A_{2n}$ be operators of the free problem, each a creation or an annihilation operator of one mode, multiplied by a number. Each $c_{\boldsymbol{k},\mathrm{I}}(\tau)$ in imaginary time, and each $c_{\boldsymbol{k},\mathrm{I}}(t)$ in real time, is a single-mode operator multiplied by a number. For fermions, the claim is

$$\langle A_1 A_2 \cdots A_{2n} \rangle_0 = \sum_{j=2}^{2n} (-1)^j \, \langle A_1 A_j \rangle_0 \, \langle A_2 \cdots \widehat{A_j} \cdots A_{2n} \rangle_0, \tag{D.34}$$

where the hat removes A_j from the string. Applied repeatedly, Eq. (D.34) writes the average as a sum over all ways of splitting the $2n$ operators into pairs. Each pairing contributes the product of its pair averages, each pair kept in its original order, times the sign of the permutation that brings the members of every pair next to each other. An average of an odd number of operators vanishes, and so does a pair average unless it holds one creation and one annihilation operator of the same mode.

The proof uses only the anticommutators and the cyclic property of the trace [15]. Move A_1 to the right end, one step at a time. Each step past A_j replaces $A_1 A_j$ by $\{A_1, A_j\} - A_j A_1$, and the anticommutator is a number. After $2n-1$ steps,

$$A_1 A_2 \cdots A_{2n} = \sum_{j=2}^{2n} (-1)^j \{A_1, A_j\} \, A_2 \cdots \widehat{A_j} \cdots A_{2n} - A_2 \cdots A_{2n} A_1 .$$

In the average of the last term, the trace moves A_1 back to the front, past the Boltzmann factor: $\mathrm{Tr}[\mathrm{e}^{-\beta H_0} A_2 \cdots A_{2n} A_1] = \mathrm{Tr}[A_1 \mathrm{e}^{-\beta H_0} A_2 \cdots A_{2n}]$. If A_1 annihilates a fermion of energy ξ, then $A_1 \mathrm{e}^{-\beta H_0} = \mathrm{e}^{-\beta\xi} \mathrm{e}^{-\beta H_0} A_1$, because removing the fermion lowers H_0 by ξ; if it creates one, the factor is $\mathrm{e}^{+\beta\xi}$. With $\mathrm{e}^{-\beta\xi_1}$ for either case, where $\xi_1 = \xi$ for an annihilation and $\xi_1 = -\xi$ for a creation operator, the last term averages to $-\mathrm{e}^{-\beta\xi_1} \langle A_1 \cdots A_{2n} \rangle_0$, and

$$\langle A_1 \cdots A_{2n} \rangle_0 = \sum_{j=2}^{2n} (-1)^j \, \frac{\{A_1, A_j\}}{1 + \mathrm{e}^{-\beta\xi_1}} \, \langle A_2 \cdots \widehat{A_j} \cdots A_{2n} \rangle_0 .$$

For $n = 1$ the same equation reads $\langle A_1 A_j \rangle_0 = \{A_1, A_j\}/(1 + \mathrm{e}^{-\beta\xi_1})$; for example, $\langle c_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger \rangle_0 = 1/(1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}}}) = 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$. Inserting this pair average into the general case gives Eq. (D.34). For bosons every anticommutator becomes a commutator, no signs appear, and the denominator is $1 - \mathrm{e}^{-\beta\xi_1}$, so that $\langle d_{\boldsymbol{Q}} d_{\boldsymbol{Q}}^\dagger \rangle_0 = 1 + n_{\mathrm{B}}(\omega_{\boldsymbol{Q}})$. Fermion and boson operators commute, and H_0 is a sum of a fermion part and a boson part, so a mixed average factorizes into a fermion average and a boson average. Linear combinations of single-mode operators, such as $d_{\boldsymbol{Q}} + d_{-\boldsymbol{Q}}^\dagger$, are covered by linearity.

The expansions need two extensions of the theorem: to time-ordered strings, and to averages at $T = 0$. Inside T_τ the operators are first put in time order, at the cost of the sign of that permutation, and Eq. (D.34) is applied to the ordered string. Each pair then appears in time order, as $\langle \mathrm{T}_\tau A_i A_j \rangle_0$, and the sign of the time ordering combines with the sign of the pairing in the ordered string into the sign of the same pairing in the original string. Hence the time-ordered average is the sum over pairings of products of time-ordered pair averages, the contractions, with the sign of the pairing as written. The free contractions are, by Eq. (4.86) and Eq. (4.79),

$$\langle \mathrm{T}_\tau \, c_{\boldsymbol{k},\mathrm{I}}(\tau) \, c_{\boldsymbol{k},\mathrm{I}}^\dagger(\tau') \rangle_0 = -\mathsf{G}_0(\boldsymbol{k}, \tau - \tau'), \qquad \langle \Phi_0 | \mathrm{T} \, c_{\boldsymbol{k},\mathrm{I}}(t) \, c_{\boldsymbol{k},\mathrm{I}}^\dagger(t') | \Phi_0 \rangle = \mathrm{i} \, \mathsf{G}_0(\boldsymbol{k}, t - t'). \tag{D.35}$$

At $T = 0$, the ground state $|\Phi_0\rangle$ of H_0 is the limit $\beta \to \infty$ of the free grand-canonical state, every average in Eq. (D.34) has a limit, and the theorem holds for $\langle \Phi_0 | \cdots | \Phi_0 \rangle$, in real time as in imaginary time.

Disconnected terms cancel

At order n, each pairing in the numerator of Eq. (D.32) splits the n factors $H_{\rm int}$ into two parts. The connected part consists of the m factors that the pairing joins to the two external operators; the vacuum part consists of the remaining $n-m$ factors, which the pairing joins only among themselves. The value of the pairing is the product of the values of the two parts. Since every $H_{\rm int}$ is even in fermion operators, its position under T_τ costs no sign, and the sign of the pairing is the product of the signs of the two parts. There are $n!/[m!\,(n-m)!]$ ways to choose which of the n identical, integrated factors form the connected part, and each gives the same value. With X_m the sum of connected pairings at order m, including the external operators, and Y_l the sum of all pairings of l factors $H_{\rm int}$ alone, the numerator is

$$\sum_n \frac{(-1)^n}{n!} \sum_{m=0}^{n} \frac{n!}{m!\,(n-m)!} X_m\, Y_{n-m} = \Big[\sum_m \frac{(-1)^m}{m!} X_m\Big]\Big[\sum_l \frac{(-1)^l}{l!} Y_l\Big].$$

The second factor is $\langle U_{\rm I}(\beta,0)\rangle_0$, which cancels the denominator. Hence G is -1 times the sum of the connected pairings alone; this result is the linked-cluster theorem. The same argument applies to Eq. (D.33) with $(-\mathrm{i})^n$ in place of $(-1)^n$.

Feynman rules for the Coulomb interaction

Take the electron gas of Eq. (2.47), whose interaction is

$$H_{\rm int} = \frac{1}{2\mathscr{V}} \sum_{\boldsymbol{k}\neq 0} v_{\rm C}(k) \sum_{\boldsymbol{p}\boldsymbol{p}' s s'} c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'} c_{\boldsymbol{p}s}.$$

Under T_τ, with the creation operators at $\tau+0^+$, the order of the four operators is fixed by the ordering symbol, and they may be regrouped with the sign of the permutation. Moving $c_{\boldsymbol{p}s}$ two places to the left costs no sign, and each factor is the product of two bilinears,

$$H_{\rm int,I}(\tau) = \frac{1}{2\mathscr{V}} \sum_{\boldsymbol{k}\neq 0} v_{\rm C}(k) \sum_{\boldsymbol{p}\boldsymbol{p}' s s'} \big[c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c_{\boldsymbol{p}s}\big]\big[c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'}\big](\tau) \qquad (\text{inside } \mathrm{T}_\tau),$$

each bilinear a density component. We draw each factor as a dashed interaction line that carries the momentum $\boldsymbol{k}$ and has the two bilinears as its ends, and each contraction Eq. (D.35) as a directed solid line from a creation operator to the annihilation operator it is paired with. At order n the $2n$ bilinears and the two external operators give $2n+1$ contractions.

The sign. Follow a fermion line from the external $c_{\boldsymbol{k}}(\tau)$ through the bilinears it visits to $c^\dagger_{\boldsymbol{k}}(0)$. Arrange the string so that the bilinears on the line stand in the order of the line, $c(\tau)\,[c_1^\dagger c_1]\,[c_2^\dagger c_2]\cdots[c_m^\dagger c_m]\,c^\dagger(0)$; moving whole bilinears costs no sign. The contractions $c(\tau)c_1^\dagger$, $c_1 c_2^\dagger$, …, $c_m c^\dagger(0)$ are then all between neighbors, the pairing sign is $+1$, and each contraction is $-\mathsf{G}_0$, so the $m+1$ propagators of the open line give $(-1)^{m+1}$. A closed loop through m bilinears, $[c_1^\dagger c_1]\cdots[c_m^\dagger c_m]$, is paired as $c_1 c_2^\dagger$, …, $c_{m-1}c_m^\dagger$ and finally $c_m c_1^\dagger$. All pairs are neighbors except the last. Bringing $c_1^\dagger$ from the front to the end passes $2m-1$ operators and costs a factor -1, in addition to $(-1)^m$ from the m propagators of the loop. With F closed loops, the $2n+1$ propagators therefore carry $(-1)^{2n+1}(-1)^F = -(-1)^F$. The definition Eq. (D.32) contributes -1 and the expansion $(-1)^n$. The total sign of a diagram of order n is $(-1)^n(-1)^F$.

The factors $1/n!$ and $1/2^n$. Two operations map one pairing onto another with the same value: relabeling the n integrated interaction factors among themselves, and exchanging the two ends of any interaction line. The exchange is a relabeling $\boldsymbol{k}\to-\boldsymbol{k}$, $(\boldsymbol{p}s)\leftrightarrow(\boldsymbol{p}'s')$, under

which $v_{\rm C}$ is unchanged. For a connected diagram with two external points the $n!\,2^n$ relabelings give $n!\,2^n$ different pairings, since the external points fix every line of the diagram. Summing over pairings is then the same as summing over topologically distinct diagrams with weight $n!\,2^n$, which cancels the $1/n!$ of the expansion and the $\frac{1}{2}$ in each $H_{\rm int}$.

Frequencies. Each propagator is written as $\mathsf{G}_0(\boldsymbol{p},\tau_a-\tau_b) = T\sum_j \mathrm{e}^{-\mathrm{i}\omega_j^{\rm F}(\tau_a-\tau_b)}\mathsf{G}_0(\boldsymbol{p},\mathrm{i}\omega_j^{\rm F})$ by Eq. (4.87). The four fermion operators of an interaction line share one time τ_v. The integral $\int_0^\beta \mathrm{d}\tau_v$ of the product of their exponentials is β times a Kronecker delta that conserves frequency along the interaction line; the line therefore carries, as a bosonic frequency, the difference of the two fermionic frequencies at either end. The transform $\int_0^\beta \mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega_j^{\rm F}\tau}$ of the external time supplies one more β. Of the $2n+1$ factors T, $n+1$ are cancelled by these factors β, and n remain, one for each independent internal frequency. Momenta are conserved at each end of a line in the same way, and the n factors $1/\mathscr{V}$ go with the n independent internal momentum sums.

When the factors are collected, the Matsubara Green's function $\mathsf{G}(\boldsymbol{k},\mathrm{i}\omega_j^{\rm F})$ at order n is the sum over connected, topologically distinct diagrams with n interaction lines, each diagram evaluated with the rules listed in Sect. 4.8. The interaction line $-v_{\rm C}(k)/\mathscr{V}$ is independent of $\omega^{\rm B}$, and the factor $\mathrm{e}^{\mathrm{i}\omega^{\rm F}0^+}$ of a solid line that begins and ends on the same interaction line comes from the creation operator at $\tau+0^+$. The self-energy $\Sigma(\boldsymbol{k},\mathrm{i}\omega_j^{\rm F})$ is the sum of the diagrams that cannot be cut into two by cutting one solid line, with the two external propagators removed. The Dyson equation of Sect. 4.7 then sums all chains of these diagrams.

At first order the self-energy has a Hartree and a Fock diagram. In the Hartree diagram the interaction line carries $\boldsymbol{k}=0$ into a closed loop. The Hartree diagram is $(-1)(-1)\,[v_{\rm C}(0)/\mathscr{V}]$ $\sum_{\boldsymbol{p}s} T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\rm F}0^+}\mathsf{G}_0(\boldsymbol{p},\mathrm{i}\omega_j^{\rm F}) = v_{\rm C}(0)N_e/\mathscr{V}$ by Eq. (4.89), the electrostatic potential of the uniform density, and it is absent in jellium, where the background removes $\boldsymbol{k}=0$. In the Fock diagram the line carries $\boldsymbol{k}-\boldsymbol{p}$ and there is no loop:

$$\Sigma_{\rm x}(\boldsymbol{k}) = -\frac{1}{\mathscr{V}}\sum_{\boldsymbol{p}} v_{\rm C}(|\boldsymbol{k}-\boldsymbol{p}|)\,T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\rm F}0^+}\mathsf{G}_0(\boldsymbol{p},\mathrm{i}\omega_j^{\rm F}) = -\frac{1}{\mathscr{V}}\sum_{\boldsymbol{p}} v_{\rm C}(|\boldsymbol{k}-\boldsymbol{p}|)\,n_{\rm F}(\xi_{\boldsymbol{p}}), \quad \text{(D.36)}$$

with $\boldsymbol{p}$ of the same spin as $\boldsymbol{k}$. At $T=0$ the Fock self-energy Eq. (D.36) is the exchange self-energy of Eq. (23.5), and the signs of the Hartree and Fock terms agree with Sect. 23.1: the Hartree term repels and exchange attracts. The rules are thus checked at first order against a calculation that used no diagrams.

Exchange of a boson

Let the electrons couple to a bosonic field with modes $\boldsymbol{Q}$ and energies $\omega_{\boldsymbol{Q}} = \omega_{-\boldsymbol{Q}}$,

$$H_{\rm int} = \frac{1}{\sqrt{\mathscr{V}}}\sum_{\boldsymbol{k}\boldsymbol{Q}s} g(\boldsymbol{Q})\,c^\dagger_{\boldsymbol{k}+\boldsymbol{Q},s}c_{\boldsymbol{k}s}\,\varphi_{\boldsymbol{Q}}, \qquad \varphi_{\boldsymbol{Q}} = d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}},$$

with $g(-\boldsymbol{Q}) = g(\boldsymbol{Q})^*$, so that $H_{\rm int}$ is Hermitian, since $\varphi^\dagger_{\boldsymbol{Q}} = \varphi_{-\boldsymbol{Q}}$. For phonons g becomes the vertex $g_{mn\nu}$ of Sect. 6.2 and $d_{\boldsymbol{Q}}$ becomes $b_{\boldsymbol{Q}\nu}$. The boson contraction defines the free boson propagator, $\mathsf{D}_0(\boldsymbol{Q},\tau) = -\langle \mathrm{T}_\tau\,\varphi_{\boldsymbol{Q},\mathrm{I}}(\tau)\,\varphi_{-\boldsymbol{Q},\mathrm{I}}(0)\rangle_0$. For $0<\tau<\beta$, with $d_{\boldsymbol{Q},\mathrm{I}}(\tau) = \mathrm{e}^{-\omega_{\boldsymbol{Q}}\tau}d_{\boldsymbol{Q}}$ and the averages of Eq. (2.69), the propagator is $-[(1+n_{\rm B})\mathrm{e}^{-\omega_{\boldsymbol{Q}}\tau} + n_{\rm B}\mathrm{e}^{\omega_{\boldsymbol{Q}}\tau}]$ with $n_{\rm B} = n_{\rm B}(\omega_{\boldsymbol{Q}})$. Since $\mathrm{e}^{\mathrm{i}\omega_j^{\rm B}\beta} = 1$, the transform is $\int_0^\beta \mathrm{d}\tau\,\mathrm{e}^{(\mathrm{i}\omega_j^{\rm B}\mp\omega_{\boldsymbol{Q}})\tau} = (\mathrm{e}^{\mp\beta\omega_{\boldsymbol{Q}}}-1)/(\mathrm{i}\omega_j^{\rm B}\mp\omega_{\boldsymbol{Q}})$, and with $(1+n_{\rm B})(1-\mathrm{e}^{-\beta\omega_{\boldsymbol{Q}}}) = 1$ and $n_{\rm B}(\mathrm{e}^{\beta\omega_{\boldsymbol{Q}}}-1) = 1$,

$$\mathsf{D}_0(\boldsymbol{Q},\mathrm{i}\omega_j^{\rm B}) = \frac{1}{\mathrm{i}\omega_j^{\rm B}-\omega_{\boldsymbol{Q}}} - \frac{1}{\mathrm{i}\omega_j^{\rm B}+\omega_{\boldsymbol{Q}}} = \frac{2\omega_{\boldsymbol{Q}}}{(\mathrm{i}\omega_j^{\rm B})^2-\omega_{\boldsymbol{Q}}^2}. \quad \text{(D.37)}$$

The temperature has dropped out, as it did for G_0 in Sect. 4.7: thermal factors appear only after a frequency sum.

The interaction is odd in the boson operators, so only even orders $2m$ survive the boson average, and every pairing joins the $2m$ vertices into m boson lines. The counting runs as for the Coulomb case. The expansion gives $(-1)^{2m}/(2m)! = 1/(2m)!$, and the $(2m)!$ relabelings of the vertices, which include the exchange of the two ends of each boson line, cancel the factorial. The fermion signs are the signs for $2m$ bilinears, $-(-1)^F$, and the definition of G adds -1. Each boson line contributes $-\mathsf{D}_0$ from its contraction and $g(\boldsymbol{Q})g(-\boldsymbol{Q})/\mathscr{V} = |g(\boldsymbol{Q})|^2/\mathscr{V}$ from its two vertices. The total is $(-1)^m(-1)^F \prod_{\text{lines}} |g|^2\mathsf{D}_0/\mathscr{V}$, which is the Coulomb result with each line replaced:

$$-\frac{v_{\mathrm{C}}(k)}{\mathscr{V}} \;\longrightarrow\; -\frac{|g(\boldsymbol{Q})|^2\,\mathsf{D}_0(\boldsymbol{Q},\mathrm{i}\omega^{\mathrm{B}})}{\mathscr{V}}. \tag{D.38}$$

The exchange of a boson is an interaction that depends on frequency. At zero frequency, $|g|^2\mathsf{D}_0(\boldsymbol{Q},0) = -2|g|^2/\omega_{\boldsymbol{Q}}$, an attraction. When $|g|^2\mathsf{D}_0$ is continued to real frequencies, it is $2\omega_{\boldsymbol{Q}}|g|^2/(\omega^2-\omega_{\boldsymbol{Q}}^2)$, attractive for $|\omega| < \omega_{\boldsymbol{Q}}$ and repulsive above. An electron polarizes the field, the polarization lags behind the electron, and a second electron that arrives within a time $\sim 1/\omega_{\boldsymbol{Q}}$ is attracted to the polarization. For phonons the same attraction is behind superconductivity (Sect. 7.2). The first diagram of the self-energy is the Fock diagram with a boson line, which Sect. 4.8 evaluates (Eq. (4.98)). Its Hartree partner carries $\boldsymbol{Q} = 0$ and is a static shift of all levels, which vanishes when $g(\boldsymbol{0}) = 0$.

The rules at zero temperature

With Eq. (D.33), the counting is the same with three changes: the expansion gives $(-\mathrm{i})^n$, each contraction is $\mathrm{i}\mathsf{G}_0$ by Eq. (D.35), and the definition of G carries $-\mathrm{i}$. The total factor of a diagram of order n is $-\mathrm{i}\,(-\mathrm{i})^n\,\mathrm{i}^{2n+1}(-1)^F = \mathrm{i}^n(-1)^F$. Transforming with $\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}$ and $\mathsf{G}_0(\boldsymbol{p},t) = \int \mathrm{d}\omega\,(2\pi)^{-1}\mathrm{e}^{-\mathrm{i}\omega t}\mathsf{G}_0(\boldsymbol{p},\omega)$ turns every time integral into $2\pi\delta$ of a frequency, and n integrals $\int \mathrm{d}\omega/2\pi$ remain. The rules for $\mathsf{G}(\boldsymbol{k},\omega)$ are the rules of Sect. 4.8 with the free time-ordered propagator $\mathsf{G}_0(\boldsymbol{p},\omega) = 1/(\omega - \xi_{\boldsymbol{p}} + \mathrm{i}0^+\,\mathrm{sgn}\,\xi_{\boldsymbol{p}})$ of Eq. (4.80), each interaction line $+\mathrm{i}v_{\mathrm{C}}(k)/\mathscr{V}$, and each internal frequency $\int \mathrm{d}\omega/2\pi$. The Fock diagram checks the signs once more: $\mathrm{i}\int \mathrm{d}\omega\,(2\pi)^{-1}\mathrm{e}^{\mathrm{i}\omega 0^+}\mathsf{G}_0(\boldsymbol{p},\omega)$ is evaluated by closing the contour in the upper half plane, where the pole $\xi_{\boldsymbol{p}} + \mathrm{i}0^+$ of an occupied state lies, and it equals $\mathrm{i}\cdot\mathrm{i}\,\Theta(-\xi_{\boldsymbol{p}}) = -\Theta(-\xi_{\boldsymbol{p}})$. With the line $v_{\mathrm{C}}/\mathscr{V}$ the Fock diagram is Eq. (D.36) at $T = 0$. The time-ordered density response, $-(\mathrm{i}/\mathscr{V})\langle \mathrm{T}\, n_e(\boldsymbol{q},t)^\dagger n_e(\boldsymbol{q},0)\rangle$, carries $-\mathrm{i}/\mathscr{V}$ in its definition, and its $2n+2$ contractions give the factor $(-\mathrm{i}/\mathscr{V})(-\mathrm{i})^n\,\mathrm{i}^{2n+2}(-1)^F = (\mathrm{i}/\mathscr{V})\,\mathrm{i}^n(-1)^F$: with the factor i^n in the lines, the response is $+\mathrm{i}/\mathscr{V}$ times its diagrams, where the imaginary-time rule has $-1/\mathscr{V}$. For the bubble the response is $(\mathrm{i}/\mathscr{V})(-1)\sum_{\boldsymbol{k}s}\int \mathrm{d}\omega\,(2\pi)^{-1}\mathsf{G}_0(\boldsymbol{k},\omega)\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q},\omega+\Omega)$, whose frequency integral, done by closing the contour, gives the bubble Eq. (4.93) at $T = 0$ with time-ordered denominators. For a boson the same replacement Eq. (D.38) holds, with the time-ordered propagator $\mathsf{D}_0(\boldsymbol{Q},t) = -\mathrm{i}\langle\Phi_0|\mathrm{T}\,\varphi_{\boldsymbol{Q},\mathrm{I}}(t)\varphi_{-\boldsymbol{Q},\mathrm{I}}(0)|\Phi_0\rangle = -\mathrm{i}\,\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}}|t|}$, whose transform is $\mathsf{D}_0(\boldsymbol{Q},\omega) = 2\omega_{\boldsymbol{Q}}/(\omega^2-\omega_{\boldsymbol{Q}}^2+\mathrm{i}0^+)$. The imaginary-time rules are the ones used in practice at finite temperature, and the zero-temperature rules are their limit after the frequency sums have been done and continued to the real axis (Sect. 4.7).

D.10 The Bethe–Salpeter equation from the ladder

In the bubble of Sects. 4.7 and 4.8 the electron and the hole propagate independently between the two density vertices. The ladder diagrams let them scatter off each other any number of times between the two vertices, and the Bethe–Salpeter equation of Sect. 4.9 is their sum. This section derives the Bethe–Salpeter equation with the rules of Appendix D.9. The

derivation follows the many-body treatments of Hanke and Sham [16] and of Strinati [17]; the first-principles form follows Refs. [9, 18, 19].

Transitions and the pair propagator

Take a basis of orbitals $\psi_m(\boldsymbol{r})$ that diagonalize the one-body part of the problem, with energies ξ_m measured from $\mu_{\rm ch}$ and the spin in the label. In a crystal m stands for a band, a crystal momentum and a spin. The density operator is $n_e(\boldsymbol{q}) = \sum_{mn} \rho_{mn}(\boldsymbol{q})\, c_m^\dagger c_n$ with $\rho_{mn}(\boldsymbol{q}) = \langle m|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle$, by Eq. (2.36). Each term moves one electron from n to m, and we call the pair (mn) a transition. The density response is then a combination of the correlation functions of the pair operators $c_m^\dagger c_n$. Define, in imaginary time,

$$\mathfrak{L}_{mn,m'n'}(\tau) = -\langle \mathrm{T}_\tau\, c_n^\dagger(\tau) c_m(\tau)\; c_{m'}^\dagger(0) c_{n'}(0)\rangle, \tag{D.39}$$

the pair propagator, written in fraktur; it is the two-particle correlation function with both creation and both annihilation times set equal in pairs. Since $n_e(\boldsymbol{q})^\dagger = \sum_{mn} \rho_{mn}(\boldsymbol{q})^* c_n^\dagger c_m$, the response of Eq. (4.90) with $O_a = n_e(\boldsymbol{q}+\boldsymbol{G})$ and $O_b = n_e(\boldsymbol{q}+\boldsymbol{G}')$ is

$$R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \mathrm{i}\omega_j^{\rm B}) = \frac{1}{\mathscr{V}} \sum_{mn,m'n'} \rho_{mn}(\boldsymbol{q}+\boldsymbol{G})^*\, \mathfrak{L}_{mn,m'n'}(\mathrm{i}\omega_j^{\rm B})\, \rho_{m'n'}(\boldsymbol{q}+\boldsymbol{G}'), \tag{D.40}$$

where we take $\boldsymbol{q}$ not equal to a reciprocal-lattice vector, so that $\langle n_e(\boldsymbol{q}+\boldsymbol{G})\rangle = 0$ and only connected contractions contribute. Everything the target does to the pair is in $\mathfrak{L}$.

For independent quasiparticles, Wick's theorem pairs $c_m(\tau)$ with $c_{m'}^\dagger(0)$ and $c_{n'}(0)$ with $c_n^\dagger(\tau)$, and this pairing forces $m' = m$ and $n' = n$. The two contractions form one closed loop through the two pair operators, and by the counting of Appendix D.9 the result is $\mathsf{G}_m(\tau)\mathsf{G}_n(-\tau)$. The transform of this product is the frequency sum Eq. (D.31),

$$\mathfrak{L}^0_{mn,m'n'}(\mathrm{i}\omega_j^{\rm B}) = \delta_{mm'}\delta_{nn'}\, T\sum_{j'} \mathsf{G}_m(\mathrm{i}\omega_{j'}^{\rm F} + \mathrm{i}\omega_j^{\rm B})\, \mathsf{G}_n(\mathrm{i}\omega_{j'}^{\rm F}) = \delta_{mm'}\delta_{nn'}\, \frac{n_{\rm F}(\xi_n) - n_{\rm F}(\xi_m)}{\mathrm{i}\omega_j^{\rm B} - (\xi_m - \xi_n)}, \tag{D.41}$$

diagonal in transitions, with a pole at the transition energy $\xi_m - \xi_n$ and the Pauli factor $n_{\rm F}(\xi_n) - n_{\rm F}(\xi_m)$. With plane waves and the pair propagator of independent quasiparticles, Eq. (D.40) is the bubble Eq. (4.93).

First-order contractions

The interaction is the Coulomb operator of Sect. 2.3 in the orbital basis. Under T_τ, with the creation operators at $\tau + 0^+$, it is a product of two bilinears, as in Appendix D.9,

$$H_{\rm int} = \frac{1}{2}\sum_{abcd} v_{ab,cd}\, [c_a^\dagger c_c]\, [c_b^\dagger c_d], \qquad v_{ab,cd} = \int \mathrm{d}^3r\, \mathrm{d}^3r'\, \psi_a^*(\boldsymbol{r})\psi_c(\boldsymbol{r})\, v_{\rm C}(\boldsymbol{r},\boldsymbol{r}')\, \psi_b^*(\boldsymbol{r}')\psi_d(\boldsymbol{r}'),$$

with $v_{\rm C}(\boldsymbol{r},\boldsymbol{r}') = e^2/(4\pi|\boldsymbol{r}-\boldsymbol{r}'|)$ the transform of Eq. (A.3); spin is conserved at each bilinear. The rules of Appendix D.9 were derived for plane waves, but their proof used only this bilinear structure, and they hold in any orbital basis with $v_{ab,cd}$ in place of $v_{\rm C}(k)/\mathscr{V}$. The expansion Eq. (D.32), applied to Eq. (D.39), gives at first order $+\int_0^\beta \mathrm{d}\tau_1\, \langle \mathrm{T}_\tau\, c_n^\dagger c_m(\tau)\, H_{\rm int}(\tau_1)\, c_{m'}^\dagger c_{n'}(0)\rangle_0$, connected part only. The two bilinears of $H_{\rm int}$ can be contracted with the external ones in three topologically different ways, which we call the exchange, the direct and the self-energy contractions.

In the exchange contraction, each bilinear of $H_{\rm int}$ closes a loop with one external pair operator: $[c_a^\dagger c_c]$ with $c_n^\dagger c_m(\tau)$, which needs $a = m$, $c = n$, and $[c_b^\dagger c_d]$ with $c_{m'}^\dagger c_{n'}(0)$, which

needs $b = n'$, $d = m'$. Bilinears are even in fermion operators and can be regrouped at no cost, and the average factorizes into two free pair averages, each equal to $-\mathfrak{L}^0$. The product is $+\mathfrak{L}^0\mathfrak{L}^0$. The two choices of which bilinear goes with which external pair operator cancel the $\frac{1}{2}$, and the exchange term is $+\mathfrak{L}^0_{mn}(\tau-\tau_1)\,v_{mn',nm'}\,\mathfrak{L}^0_{m'n'}(\tau_1)$, a convolution in τ. In the direct contraction, both bilinears lie on the single loop that runs through the two pair operators. One bilinear is on the electron line, which enters at $c^\dagger_{m'}$ and leaves at c_m; this bilinear requires $a = m$, $c = m'$. The other bilinear is on the hole line and requires $b = n'$, $d = n$. The direct diagram has the same four propagators and the same interaction line as the exchange contraction, but one closed loop instead of two, and by the loop rule of Appendix D.9 it carries the opposite sign: $-\mathfrak{L}^0_{mn}(\tau-\tau_1)\,v_{mn',m'n}\,\mathfrak{L}^0_{m'n'}(\tau_1)$. In the self-energy contractions, both bilinears of H_{int} act on one propagator, or one of them closes a loop by itself; these contractions are the Hartree and Fock self-energies of Appendix D.9 inserted into the electron or the hole line, and they shift the orbital energies.

The exchange and direct contractions define the kernels of the Bethe–Salpeter equation:

$$\begin{aligned} K^{\mathrm{x}}_{\mathrm{full},mn,m'n'} &= \int \mathrm{d}^3r\,\mathrm{d}^3r'\;\psi^*_m(\boldsymbol{r})\psi_n(\boldsymbol{r})\;v_{\mathrm{C}}(\boldsymbol{r},\boldsymbol{r}')\;\psi_{m'}(\boldsymbol{r}')\psi^*_{n'}(\boldsymbol{r}'),\\ K^{\mathrm{d}}_{mn,m'n'} &= \int \mathrm{d}^3r\,\mathrm{d}^3r'\;\psi^*_m(\boldsymbol{r})\psi_{m'}(\boldsymbol{r})\;W^{\mathrm{scr}}(\boldsymbol{r},\boldsymbol{r}')\;\psi^*_{n'}(\boldsymbol{r}')\psi_n(\boldsymbol{r}'). \end{aligned} \tag{D.42}$$

The exchange kernel $K^{\mathrm{x}}_{\mathrm{full}}$ carries the whole Coulomb interaction, including the long-range term that the transition-basis section below splits off. It is the term of the random-phase approximation, and its name refers to the exchange of the electron and the hole between the two transitions. In the direct kernel K^{d} the electron stays at $\boldsymbol{r}$ and scatters from m' to m, while the hole stays at $\boldsymbol{r}'$ and scatters from n' to n. The direct kernel is written above already with the screened interaction that the next step supplies. A diagonal element $K^{\mathrm{d}}_{mn,mn} = \int\!\int |\psi_m(\boldsymbol{r})|^2 W^{\mathrm{scr}}(\boldsymbol{r},\boldsymbol{r}')|\psi_n(\boldsymbol{r}')|^2$ is positive, and it enters with a minus sign.

The statically screened ladder

The Bethe–Salpeter equation keeps all diagrams built by repeating the exchange and direct insertions, with the following choices, which define the approximation. The orbital energies ξ_m are the quasiparticle energies of Sect. 4.7, so that the self-energy contractions are summed into the propagators, with the quasiparticle widths neglected and z_{qp} set to one. A direct interaction line is screened: a chain of bubbles inserted into it turns v_{C} into $v_{\mathrm{C}} + v_{\mathrm{C}}Rv_{\mathrm{C}} = \varepsilon^{-1}v_{\mathrm{C}} = W^{\mathrm{scr}}$, the screened interaction of Sect. 4.8. An exchange line is not screened, since a bubble chain on it is already counted as repeated exchange. These choices are consistent with the GW self-energy of Sect. 4.8: the kernel is the change of the sum of the Hartree potential and the GW self-energy when G changes, with the change of W^{scr} itself neglected [16, 17]. The Hartree term gives $K^{\mathrm{x}}_{\mathrm{full}}$ and the GW term gives $-K^{\mathrm{d}}$.

A screened direct line carries a frequency. In the direct diagram the electron line has the fermionic frequency $\mathrm{i}\omega^{\mathrm{F}}_{j_1} + \mathrm{i}\omega^{\mathrm{B}}_j$ on one side of the interaction line and $\mathrm{i}\omega^{\mathrm{F}}_{j_2} + \mathrm{i}\omega^{\mathrm{B}}_j$ on the other, the hole line correspondingly, and the interaction line is W^{scr} at the bosonic frequency $\mathrm{i}\omega^{\mathrm{F}}_{j_1} - \mathrm{i}\omega^{\mathrm{F}}_{j_2}$. The two internal frequency sums are then coupled through the screened line, and $\mathfrak{L}$ does not close on itself: the pair splits into an electron and a hole at different times. The static approximation replaces $W^{\mathrm{scr}}(\mathrm{i}\omega^{\mathrm{B}})$ by $W^{\mathrm{scr}}(\omega = 0)$ at all frequencies, an interaction that is instantaneous in τ. In the static approximation, the diagram with one direct line is

$$\begin{aligned} &T\sum_{j_1} \mathsf{G}_m(\mathrm{i}\omega^{\mathrm{F}}_{j_1} + \mathrm{i}\omega^{\mathrm{B}}_j)\mathsf{G}_n(\mathrm{i}\omega^{\mathrm{F}}_{j_1})\;(-K^{\mathrm{d}}_{mn,m'n'})\;T\sum_{j_2} \mathsf{G}_{m'}(\mathrm{i}\omega^{\mathrm{F}}_{j_2} + \mathrm{i}\omega^{\mathrm{B}}_j)\mathsf{G}_{n'}(\mathrm{i}\omega^{\mathrm{F}}_{j_2})\\ &\quad = \mathfrak{L}^0_{mn}\,(-K^{\mathrm{d}}_{mn,m'n'})\,\mathfrak{L}^0_{m'n'}, \end{aligned}$$

since each sum separately is Eq. (D.31). The exchange line is instantaneous and carries $\mathrm{i}\omega_j^{\mathrm{B}}$. Every further direct line repeats the step, and the sum of the ladder is the geometric series of Eq. (C.57),

$$\mathfrak{L}(\mathrm{i}\omega_j^{\mathrm{B}}) = \mathfrak{L}^0(\mathrm{i}\omega_j^{\mathrm{B}}) + \mathfrak{L}^0(\mathrm{i}\omega_j^{\mathrm{B}})\,[K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{d}}]\,\mathfrak{L}(\mathrm{i}\omega_j^{\mathrm{B}}), \tag{D.43}$$

a matrix equation in the space of transitions at a single frequency. Eq. (D.43) is the Bethe–Salpeter equation in the static ladder approximation, whose range of validity is stated in Sect. 4.9.

Spin structure of the kernel

For a spin-unpolarized target the orbitals come in pairs $m\uparrow$, $m\downarrow$ with the same spatial part. The exchange kernel couples a transition of either spin to a transition of either spin, because the density at $\boldsymbol{r}'$ sums over both spins. The direct kernel conserves the spin of the electron and of the hole separately, and with $\psi^*_m\psi_n$ requiring equal spins it couples only transitions of the same spin. In the combination $(mn\uparrow + mn\downarrow)/\sqrt{2}$ the kernel is therefore $g_{\mathrm{deg}}K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{d}}$, with $g_{\mathrm{deg}} = 2$ and the kernels now written for spatial orbitals; in $(mn\uparrow - mn\downarrow)/\sqrt{2}$ it is $-K^{\mathrm{d}}$. The two combinations are the singlet and triplet excitons. The density vertex is the same for both spins, so $n_e(\boldsymbol{q})$ creates only singlets, with the weight $\sqrt{g_{\mathrm{deg}}}$ per transition. From here on the labels are spatial, the kernel is the singlet one, and g_{deg} appears in front of the response.

The transition-basis Hamiltonian

At $T = 0$, in an insulator, the Pauli factor of Eq. (D.41) is $+1$ for a transition from an occupied band v to an empty band c. Transitions from v to c are the resonant transitions; the reverse ones, with the factor -1, are antiresonant, and we mark them with a bar. Transitions between two occupied or two empty orbitals have $\mathfrak{L}^0 = 0$ and drop out. In the matrix notation of Appendix C.7, let $\mathbf{F}$ be the diagonal matrix of the Pauli factors, with $\mathbf{F}^2 = \mathbf{1}$, and $\mathbf{D}$ the diagonal matrix of the transition energies; the inverse of Eq. (D.41) is then $\mathbf{F}(\mathrm{i}\omega_j^{\mathrm{B}} - \mathbf{D})$. Eq. (D.43) reads $\mathfrak{L}^{-1} = (\mathfrak{L}^0)^{-1} - \mathbf{K}$ with $\mathbf{K} = g_{\mathrm{deg}}K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{d}}$, and

$$\mathfrak{L} = \left[\mathbf{F}(\mathrm{i}\omega_j^{\mathrm{B}} - \mathbf{D}) - \mathbf{K}\right]^{-1} = \left(\mathrm{i}\omega_j^{\mathrm{B}} - \mathbf{D} - \mathbf{F}\mathbf{K}\right)^{-1}\mathbf{F}. \tag{D.44}$$

The poles of $\mathfrak{L}$ are the eigenvalues of $\mathbf{D} + \mathbf{F}\mathbf{K}$, which in blocks of resonant and antiresonant transitions is

$$\mathbf{D} + \mathbf{F}\mathbf{K} = \begin{pmatrix} H^{\mathrm{exc}}_{\mathrm{full}} & K_{\mathrm{r\bar{r}}} \\ -K^{*}_{\mathrm{r\bar{r}}} & -H^{\mathrm{exc}*}_{\mathrm{full}} \end{pmatrix}, \qquad H^{\mathrm{exc}}_{\mathrm{full},cv,c'v'} = (\xi_c - \xi_v)\,\delta_{cc'}\delta_{vv'} + g_{\mathrm{deg}}K^{\mathrm{x}}_{\mathrm{full},cv,c'v'} - K^{\mathrm{d}}_{cv,c'v'}. \tag{D.45}$$

The lower blocks follow from Eq. (D.42). Exchanging m and n in both transitions conjugates $K^{\mathrm{x}}_{\mathrm{full}}$, since v_{C} is real. It also conjugates K^{d} when $W^{\mathrm{scr}}(\boldsymbol{r},\boldsymbol{r}') = W^{\mathrm{scr}}(\boldsymbol{r}',\boldsymbol{r})$ is real, as the static screened interaction of a target with time-reversal symmetry is. The factor -1 of $\mathbf{F}$ on the antiresonant rows supplies the signs. The two conjugation properties also make $H^{\mathrm{exc}}_{\mathrm{full}}$ Hermitian. In the resonant block the exchange kernel carries the singlet factor g_{deg} [20, Eqs. (18) and (20)].

The coupling block $K_{\mathrm{r\bar{r}}}$ connects the creation of a pair with its annihilation. In a semiconductor the effect of the coupling block on the eigenvalues is of second order in the kernel divided by twice the gap, and the Tamm–Dancoff approximation drops it.

In a crystal the Coulomb line of $K^{\mathrm{x}}_{\mathrm{full}}$ carries $\boldsymbol{q} + \boldsymbol{G}$ for every reciprocal-lattice vector $\boldsymbol{G}$, and the term of $K^{\mathrm{x}}_{\mathrm{full}}$ at $\boldsymbol{G} = 0$, which we call K^{x}_0, is the long-range macroscopic field. The remainder of $K^{\mathrm{x}}_{\mathrm{full}}$ is the exchange kernel proper, $K^{\mathrm{x}} = K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{x}}_0$, the local-field kernel. We write H^{exc} for $H^{\mathrm{exc}}_{\mathrm{full}}$ with K^{x} in place of $K^{\mathrm{x}}_{\mathrm{full}}$. With the band and crystal-momentum labels

of a crystal written out, a transition at fixed $\boldsymbol{q}$ takes an electron from $v\boldsymbol{k}$ to $(c, \boldsymbol{k}+\boldsymbol{q})$, and the resonant problem is Eq. (4.104), a Hermitian eigenvalue problem, whose eigenvalues E_λ and envelopes A^λ depend on $\boldsymbol{q}$. The same problem with $H^{\rm exc}_{\rm full}$ has the eigenpairs $E^{\rm full}_\lambda$ and $A^\lambda_{\rm full}$. In the crystal the kernels Eq. (D.42) become the sums over reciprocal-lattice vectors of Eq. (4.105), with the convention of Appendix A.3, and the full exchange kernel is

$$K^{\rm x}_{{\rm full},cv\boldsymbol{k},c'v'\boldsymbol{k}'} = K^{\rm x}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} + \frac{1}{\mathscr{V}}\, v_{\rm C}(q)\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})\, \rho_{c'v'}(\boldsymbol{k}',\boldsymbol{q})^*. \tag{D.46}$$

The direct kernel follows from Eq. (D.42) with the expansion

$$W^{\rm scr}(\boldsymbol{r},\boldsymbol{r}') = \mathscr{V}^{-1} \sum_{\boldsymbol{p}\boldsymbol{G}\boldsymbol{G}'} \mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\boldsymbol{G})\cdot\boldsymbol{r}}\, W^{\rm scr}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p})\, \mathrm{e}^{-\mathrm{i}(\boldsymbol{p}+\boldsymbol{G}')\cdot\boldsymbol{r}'} :$$

the integral over $\boldsymbol{r}$ gives $\langle c, \boldsymbol{k}+\boldsymbol{q}|\mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\boldsymbol{G})\cdot\boldsymbol{r}}|c', \boldsymbol{k}'+\boldsymbol{q}\rangle$, nonzero for $\boldsymbol{p} = \boldsymbol{k}-\boldsymbol{k}'$, and the integral over $\boldsymbol{r}'$ the complex conjugate of $\langle v\boldsymbol{k}|\mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\boldsymbol{G}')\cdot\boldsymbol{r}}|v'\boldsymbol{k}'\rangle$. No symmetry of the crystal is needed. The exchange kernels carry the momentum $\boldsymbol{q}$ of the pair and the direct kernel the momentum $\boldsymbol{p}$ exchanged between the electron and the hole. All three scale as $1/\mathscr{V} = 1/(N\Omega)$, and the sum over $\boldsymbol{k}'$ in Eq. (4.104) makes $H^{\rm exc}$ act as an integral operator of order one, as the count of Appendix A.3 requires.

Response, structure factor and dielectric function

Continued to $\omega + \mathrm{i}0^+$ (Eq. (4.92)), the resonant block of Eq. (D.44) in the Tamm–Dancoff approximation is the resolvent of the pair Hamiltonian, $(\omega - H^{\rm exc} + \mathrm{i}0^+)^{-1} = \sum_\lambda A^\lambda A^{\lambda\dagger}/(\omega - E_\lambda + \mathrm{i}0^+)$, and likewise for $H^{\rm exc}_{\rm full}$. For the problem Eq. (4.104), Eq. (D.40) with the singlet weight gives

$$\bar{R}_{00}(\boldsymbol{q},\omega) = \frac{g_{\rm deg}}{\mathscr{V}} \sum_\lambda \frac{\left|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\right|^2}{\omega - E_\lambda(\boldsymbol{q}) + \mathrm{i}0^+} + (\text{antiresonant}), \tag{D.47}$$

where the antiresonant part has its poles at $-E_\lambda$. Since $K^{\rm x}$ lacks the term at $\boldsymbol{G}=0$, $\bar{R}_{00}$ is the head of the response that is proper with respect to the long-range Coulomb term. The complete response follows from $\bar{R}_{00}$, because the difference between the kernels with $K^{\rm x}_{\rm full}$ and with $K^{\rm x}$, $g_{\rm deg}K^{\rm x}_0$ with $K^{\rm x}_0 = \mathscr{V}^{-1} v_{\rm C}(q)\rho(\boldsymbol{q})\rho(\boldsymbol{q})^\dagger$ in transition space, has rank one. Let $\bar{\mathfrak{L}}$ solve Eq. (D.43) with $K^{\rm x}$ in place of $K^{\rm x}_{\rm full}$. Then $\mathfrak{L}^{-1} = \bar{\mathfrak{L}}^{-1} - g_{\rm deg}K^{\rm x}_0$, or $\mathfrak{L} = \bar{\mathfrak{L}} + \bar{\mathfrak{L}}\, g_{\rm deg}K^{\rm x}_0\, \mathfrak{L}$, and the head of this equation in Eq. (D.40) closes on itself:

$$R_{00} = \bar{R}_{00} + \bar{R}_{00}\, v_{\rm C}(q)\, R_{00}, \qquad R_{00} = \frac{\bar{R}_{00}}{1 - v_{\rm C}(q)\bar{R}_{00}}.$$

The same R_{00} is Eq. (D.47) with the eigenpairs $E^{\rm full}_\lambda$ and $A^\lambda_{\rm full}$ of the full kernel. It is the complete density response, and its imaginary part enters the dark-matter rate. At $T=0$ and $\omega > 0$ only the resonant part contributes to $\mathrm{Im}\, R_{00}$, and the fluctuation–dissipation theorem Eq. (4.18) gives the exciton contribution to the structure factor,

$$S(\boldsymbol{q},\omega) = \frac{2\pi g_{\rm deg}}{\mathscr{V}} \sum_\lambda \Big|\sum_{cv\boldsymbol{k}} A^\lambda_{{\rm full},cv\boldsymbol{k}}(\boldsymbol{q})\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\Big|^2\, \delta\big(\omega - E^{\rm full}_\lambda(\boldsymbol{q})\big). \tag{D.48}$$

The macroscopic dielectric function needs the proper head. Since $(\varepsilon^{-1})_{00} = 1 + v_{\rm C}R_{00}$ (Sect. 4.4) and $1 + v_{\rm C}R_{00} = 1/(1 - v_{\rm C}\bar{R}_{00})$,

$$\varepsilon_{\rm mac}(\boldsymbol{q},\omega) = \frac{1}{(\varepsilon^{-1})_{00}} = 1 - v_{\rm C}(q)\, \bar{R}_{00}(\boldsymbol{q},\omega). \tag{D.49}$$

The block inversion Eq. (C.55) gives the macroscopic dielectric function in the same way for the dielectric matrix of the random-phase approximation; here the local fields of Sect. 4.6 and the electron–hole attraction are treated in one equation. In the optical limit, with $\rho_{cv} \to \boldsymbol{q}\cdot\boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}})$ by Eq. (3.43), the imaginary part of $\varepsilon_{\mathrm{mac}}$ is the ε_2 of Eq. (4.110).

D.11 The Born–Oppenheimer expansion

A silicon nucleus is 5.12×10^4 times heavier than an electron, with $M = 28.0855\,\mathrm{u} = 26.16\,\mathrm{GeV}$ for the natural isotope mixture, as in Sect. 5.3 and Appendix C.6. Held by force constants of the same order as the electronic force constants, it vibrates $(M/m_e)^{1/2}\approx 226$ times more slowly than the electrons move, and the electrons have time to settle into their ground state for every position of the nuclei. Here we derive the equations that Sects. 5.1 and 6.1 state, count every term in powers of κ_{BO}, and test the counting on a model that can be solved exactly.

The coupled equations

We use the Hamiltonian Eq. (5.2) and the adiabatic states Eq. (5.3) of Sect. 5.1, orthonormal at every u and chosen real, and assume that the levels do not cross in the range of u the nuclei explore. The adiabatic states are the states of the clamped-nuclei problem of Chaps. 3 and 4, and the ground-state energy $E_0(u)$ is the energy surface E^{ad}. Any state of electrons and nuclei can be expanded as $\Psi(r,u)=\sum_i \Psi_i^{\mathrm{nuc}}(u)\,\Psi_i(r;u)$, since the adiabatic states are complete in r at each u. The nuclear Laplacian acts on both factors, $\nabla^2(\Psi_i^{\mathrm{nuc}}\Psi_i) = (\nabla^2\Psi_i^{\mathrm{nuc}})\Psi_i + 2\nabla\Psi_i^{\mathrm{nuc}}\cdot\nabla\Psi_i + \Psi_i^{\mathrm{nuc}}\nabla^2\Psi_i$, while H_{el} acting on Ψ_i yields $E_i(u)\Psi_i$. We insert the expansion into $H\Psi = E\Psi$, multiply by Ψ_j and integrate over the electron coordinates at fixed u. By orthonormality the first term of the Laplacian becomes $\delta_{ji}\nabla^2\Psi_i^{\mathrm{nuc}}$; the other two terms leave brackets that depend on u:

$$\int \mathrm{d}r\;\Psi_j\,\nabla^2_{\boldsymbol{R}\kappa}(\Psi_i^{\mathrm{nuc}}\Psi_i) = \delta_{ji}\,\nabla^2_{\boldsymbol{R}\kappa}\Psi_i^{\mathrm{nuc}} + 2\,\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle\cdot\nabla_{\boldsymbol{R}\kappa}\Psi_i^{\mathrm{nuc}} + \langle\Psi_j|\nabla^2_{\boldsymbol{R}\kappa}\Psi_i\rangle\,\Psi_i^{\mathrm{nuc}}.$$

Multiplying by $-1/2M_\kappa$, summing over the nuclei and over i, and adding $E_j(u)\Psi_j^{\mathrm{nuc}}$ from H_{el} gives the set of coupled equations Eq. (5.4), with the nonadiabatic operators $\mathcal{N}_{ji}$ of Eq. (5.5). The bracket with one gradient is an operator that acts on the nuclear wavefunction through $\nabla_{\boldsymbol{R}\kappa}$; the bracket with the Laplacian is a function of u that multiplies it. The two bracket terms are the only terms that connect one electronic state to another, and setting them all to zero is the Born–Oppenheimer approximation Eq. (5.6).

For real states the derivative coupling $\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle$ has no diagonal element, since $\langle\Psi_0|\nabla\Psi_0\rangle = \frac{1}{2}\nabla\langle\Psi_0|\Psi_0\rangle = 0$. Differentiating this identity once more yields $\langle\Psi_0|\nabla^2\Psi_0\rangle = -\langle\nabla\Psi_0|\nabla\Psi_0\rangle$, so $\mathcal{N}_{00} = \sum_{\boldsymbol{R}\kappa}\langle\nabla_{\boldsymbol{R}\kappa}\Psi_0|\nabla_{\boldsymbol{R}\kappa}\Psi_0\rangle/2M_\kappa$, a small positive potential added to E^{ad}.

The force and the derivative couplings

Both the force on a nucleus and the derivative couplings follow from one identity. Differentiate Eq. (5.3) with respect to $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ and project on Ψ_j:

$$\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}H_{\mathrm{el}}|\Psi_i\rangle + E_j\,\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle = \delta_{ij}\,\nabla_{\boldsymbol{R}\kappa}E_i + E_i\,\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle,$$

where H_{el} acting to the left on Ψ_j gave E_j. For $j=i$ the derivative couplings cancel, and with $i=0$ the remaining terms give the Hellmann–Feynman theorem Eq. (5.10), $-\nabla_{\boldsymbol{R}\kappa}E^{\mathrm{ad}} = -\langle\Psi_0|\nabla_{\boldsymbol{R}\kappa}(V_{eN}+V_{NN})|\Psi_0\rangle$, since only V_{eN} and V_{NN} depend on u. For $j\neq i$ the term δ_{ij} is absent, and V_{NN} drops out because it is a number at fixed u and $\langle\Psi_j|\Psi_i\rangle = 0$. Solving for

the bracket yields Eq. (5.11). The numerator of Eq. (5.11) is the bare electron–phonon vertex, a one-body operator, $\nabla_{\boldsymbol{R}\kappa} V_{eN} = \sum_e \nabla_{\boldsymbol{R}\kappa} v_\kappa(\boldsymbol{r}_e - \boldsymbol{R} - \boldsymbol{\tau}_\kappa - \boldsymbol{u}_{\boldsymbol{R}\kappa})$ with v_κ the electron–nucleus potential.

Power counting

We measure lengths in a_0 and energies in $\alpha^2 m_e$, as in Sect. 5.1. If the displacement is written in units of $\kappa_{\rm BO} a_0$, the nuclear kinetic energy and the harmonic potential are both $\sim \kappa_{\rm BO}^2 \alpha^2 m_e$ by Eq. (5.8). The remaining terms follow by the substitution $u \sim \kappa_{\rm BO} a_0$ and $\nabla \sim 1/(\kappa_{\rm BO} a_0)$ acting on a nuclear wavefunction, with force constants $\Phi^{(n)} \sim \alpha^2 m_e / a_0^n$ and $1/M = \kappa_{\rm BO}^4 / m_e$. The cubic term of the energy surface is $\sim \alpha^2 m_e \, \kappa_{\rm BO}^3$ (Sect. 5.1). The quartic term is $\sim \alpha^2 m_e \, \kappa_{\rm BO}^4$, and so is the cubic term at second order, $(\kappa_{\rm BO}^3)^2/\kappa_{\rm BO}^2$, since its energy denominator is a phonon energy: the anharmonic shifts of the vibrational levels are $\sim \alpha^2 m_e \, \kappa_{\rm BO}^4$. In the nonadiabatic operators Eq. (5.5), the electronic brackets vary on the scale a_0, $\langle \Psi_j | \nabla \Psi_i \rangle \sim 1/a_0$ and $\langle \Psi_j | \nabla^2 \Psi_i \rangle \sim 1/a_0^2$. The first-derivative coupling is then $(1/M)(1/a_0)(1/\kappa_{\rm BO} a_0) = \alpha^2 m_e \, \kappa_{\rm BO}^3$, and the second-derivative coupling $\alpha^2 m_e \, \kappa_{\rm BO}^4$. The diagonal correction $\mathcal{N}_{00}$ is $\sim \kappa_{\rm BO}^4$ as well, but most of it is a constant. Its variation over a zero-point amplitude changes the curvature of the surface by the fraction $\kappa_{\rm BO}^4$, and the vibrational levels by $\alpha^2 m_e \, \kappa_{\rm BO}^6$. The derivative coupling connects the electronic ground state to excited states at an energy $\sim \alpha^2 m_e$ above the ground state, and it therefore shifts the vibrational levels only at second order, by $(\kappa_{\rm BO}^3)^2 \alpha^2 m_e = \alpha^2 m_e \, \kappa_{\rm BO}^6$. Every nonadiabatic correction to the vibrational energies is therefore of relative order $\kappa_{\rm BO}^4 = m_e/M$, two powers of $\kappa_{\rm BO}$ beyond the anharmonic ones.

The counting assumed that the electronic excitation energies ΔE are of order $\alpha^2 m_e$. When they are smaller, the derivative coupling Eq. (5.11) grows as $1/\Delta E$. Its matrix element between vibronic states is $(1/M)\langle \Psi_j | \nabla \Psi_0 \rangle \cdot \nabla \sim (1/M)(F_{j0}/\Delta E)(1/\ell_{\rm zp})$, with F_{j0} the matrix element of the force in the numerator of Eq. (5.11); with $1/M = \omega_{\rm ph} \ell_{\rm zp}^2$ for $\ell_{\rm zp}^2 = 1/(M\omega_{\rm ph})$ the matrix element is $\omega_{\rm ph} \, \ell_{\rm zp} F_{j0}/\Delta E$. Divided by the energy ΔE that separates the two vibronic states, the matrix element gives the amplitude Eq. (5.9) with which an excited electronic state is mixed into the ground state.

A solvable harmonic model

A solvable model tests the counting without approximation: one electron bound harmonically to one nucleus, which is itself bound to a fixed point. In one dimension,

$$H = \frac{p^2}{2m_e} + \frac{P^2}{2M} + \frac{k_{\rm e}}{2}(x - X)^2 + \frac{K_{\rm N}}{2} X^2,$$

with x and p the electron's coordinate and momentum, X and P the nucleus's, and two spring constants of the same electronic order. The adiabatic states are oscillator states centered on the nucleus, with the excitation energy $\omega_{\rm e} = (k_{\rm e}/m_e)^{1/2}$ and the surface $E_0(X) = \omega_{\rm e}/2 + K_{\rm N} X^2/2$, so the Born–Oppenheimer frequency is $\omega_{\rm BO} = (K_{\rm N}/M)^{1/2}$. The diagonal correction $\mathcal{N}_{00} = m_e \omega_{\rm e}/(4M)$ is a constant, since the ground state only translates with X, and it leaves the frequency unchanged. The exact frequencies are the normal-mode frequencies of two coupled oscillators. With the force constants as a matrix, the equation of motion requires $\det \left[\begin{smallmatrix} k_{\rm e} - m_e\omega^2 & -k_{\rm e} \\ -k_{\rm e} & k_{\rm e} + K_{\rm N} - M\omega^2 \end{smallmatrix} \right] = 0$, that is $m_e M \omega^4 - [m_e(k_{\rm e} + K_{\rm N}) + M k_{\rm e}] \, \omega^2 + k_{\rm e} K_{\rm N} = 0$. The lower root for ω^2, expanded in m_e/M, is

$$\omega^2 = \frac{K_{\rm N}}{M} \Big[1 - \frac{m_e}{M} + \Big(1 - \frac{K_{\rm N}}{k_{\rm e}} \Big) \frac{m_e^2}{M^2} + O\Big(\frac{m_e^3}{M^3} \Big) \Big]. \tag{D.50}$$

At first order the nucleus vibrates as if its mass were $M + m_e$: the electron is dragged along and adds its mass. The lag of the electron behind the nucleus depends on the ratio of the two springs and enters only at the next order. The leading relative correction is $m_e/M = \kappa_{\mathrm{BO}}^4$ for ω^2, and $m_e/2M$ for the frequency itself, and it comes entirely from the off-diagonal coupling at second order, as the counting predicts. For $K_{\mathrm{N}} = k_{\mathrm{e}}/2$ and $m_e/M = 10^{-2}$, 10^{-3} and 10^{-4}, the exact ω^2 differs from $(K_{\mathrm{N}}/M)(1 - m_e/M)$ by 5.1×10^{-5}, 5.0×10^{-7} and 5.0×10^{-9} of its value, the m_e^2/M^2 term of Eq. (D.50) plus, at the largest mass ratio, a visible m_e^3/M^3 term.

D.12 The Fröhlich vertex

In a polar crystal the field of a longitudinal optical wave is long-ranged (Sect. 5.4). This field couples the longitudinal phonon to an electron through the Fröhlich vertex [21], whose strength diverges as $1/Q$. This section derives the vertex from the energy function Eq. (5.38) of the displacements and the field, in the form Sect. 6.2 uses, following Refs. [22], [23, Sect. II.D.2] and [24].

Take one mode $(\boldsymbol{Q}, \nu)$ at small $\boldsymbol{Q}$ in the expansion Eq. (A.20),

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = (2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} B_{\boldsymbol{Q}\nu},$$

and replace the cell position $\boldsymbol{R}$ by a continuous $\boldsymbol{r}$, which is allowed for $|\boldsymbol{Q}|$ small compared with the reciprocal-lattice vectors. The condition $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$ of Sect. 5.4 gives the field of the wave,

$$\boldsymbol{\mathcal{E}}(\boldsymbol{r}) = -\hat{\boldsymbol{Q}}\, \frac{e}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z_\kappa^* \cdot \hat{\boldsymbol{Q}}}{(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\, \hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}.$$

An electron, with charge $-e$, feels the force $-e\boldsymbol{\mathcal{E}}$, and its potential energy $V^{\mathrm{L}}(\boldsymbol{r})$ obeys $\nabla V^{\mathrm{L}} = e\boldsymbol{\mathcal{E}}$. For a wave $\propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$ the gradient is $\mathrm{i}\boldsymbol{Q}$. The potential energy is then $V^{\mathrm{L}} = e\, \hat{\boldsymbol{Q}} \cdot \boldsymbol{\mathcal{E}}/(\mathrm{i}Q)$, or

$$V^{\mathrm{L}}_{\boldsymbol{Q}\nu}(\boldsymbol{r}) = \mathrm{i}\, \frac{e^2}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z_\kappa^* \cdot \boldsymbol{Q}}{(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\, \boldsymbol{Q} \cdot \varepsilon_\infty \cdot \boldsymbol{Q}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}.$$

Summed over modes and written in second quantization over Bloch states with Eq. (2.36), the potential energy V^{L} has the form Eq. (A.24), with the matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$ the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{Q})$ of Eq. (A.16). Reading off the factor of $N^{-1/2} c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}} B_{\boldsymbol{Q}\nu}$ gives the long-range vertex

$$g^{\mathrm{L}}_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \mathrm{i}\, \frac{e^2}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z_\kappa^* \cdot \boldsymbol{Q}}{(2M_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\, \boldsymbol{Q} \cdot \varepsilon_\infty \cdot \boldsymbol{Q}}\, \rho_{mn}(\boldsymbol{k}, \boldsymbol{Q}). \tag{D.51}$$

It is the $\boldsymbol{G} = 0$ term of Ref. [25, Eq. (4)], with $e_{\mathrm{SI}}^2/\varepsilon_0^{\mathrm{SI}} \to e^2$, the factor $N^{-1/2}$ moved into Eq. (A.24) and Z^* transposed to our index order. The terms with $\boldsymbol{G} \neq 0$ of that equation come from the periodic arrangement of the dipoles and are regular at $\boldsymbol{Q} \to 0$. Eq. (D.51) obeys the Hermiticity condition Eq. (A.26). Under $\boldsymbol{Q} \to -\boldsymbol{Q}$ and $(m, \boldsymbol{k}) \leftrightarrow (n, \boldsymbol{k} + \boldsymbol{Q})$, the polarization vector becomes $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ by Eq. (A.21), the factor $\boldsymbol{Q}$ in the numerator changes sign, and $\rho_{nm}(\boldsymbol{k} + \boldsymbol{Q}, -\boldsymbol{Q}) = \rho_{mn}(\boldsymbol{k}, \boldsymbol{Q})^*$; the product is the complex conjugate of Eq. (D.51), since $\mathrm{i} \to -\mathrm{i}$ is supplied by the sign of $\boldsymbol{Q}$. Only longitudinal modes couple, through $Z_\kappa^* \cdot \boldsymbol{Q}$, and the vertex grows as $1/Q$ at small Q.

For the cubic two-atom crystal the longitudinal optical mode at small $\boldsymbol{Q}$ has $\boldsymbol{e}_1 = (M_2/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{Q}}$ and $\boldsymbol{e}_2 = -(M_1/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{Q}}$, which satisfy Eq. (A.21). The sum in Eq. (D.51) is then $Z^*Q\,[(M_2/M_{\mathrm{cell}}M_1)^{1/2} + (M_1/M_{\mathrm{cell}}M_2)^{1/2}] = Z^*Q/\sqrt{\mu_{\mathrm{red}}}$, and

$$g^{\mathrm{L}} = \mathrm{i}\, \frac{e^2 Z^*}{\Omega\, \varepsilon_\infty\, Q\, (2\omega_{\mathrm{L}}\mu_{\mathrm{red}})^{1/2}}\, \rho_{mn}(\boldsymbol{k}, \boldsymbol{Q}).$$

By Eqs. (5.43) and (5.45), $e^2 Z^{*2}/(\Omega\varepsilon_\infty\mu_{\rm red}) = \omega_{\rm L}^2 - \omega_{\rm T}^2 = \omega_{\rm L}^2(1-\varepsilon_\infty/\varepsilon_0)$, so the Born charge and the masses drop out, and $g^{\rm L}$ is the Fröhlich vertex Eq. (6.15) for $Z^* > 0$, with an overall minus sign otherwise.

D.13 The Allen–Heine–Cardona self-energy

This section derives the Fan–Migdal and Allen–Heine self-energies of Sect. 6.3 with the rules of Appendix D.9 and continues them to real frequency. The field-theoretic forms follow Ref. [26, Sects. V.B and IX.A], in the conventions of Appendix A.6.

Second-order coupling

Let $V(\boldsymbol{r};\{\boldsymbol{u}\})$ be the one-electron potential as a function of the displacements $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ of all nuclei from equilibrium. The expansion of V in Sect. 6.3 has the derivatives $\partial V/\partial u_a$ and $\partial^2 V/\partial u_a \partial u_b$, with $a = (\boldsymbol{R}\kappa\alpha)$; they are total derivatives at equilibrium, which include the self-consistent response of the electrons. In the vertex $g^{\rm AH}$ of Eq. (6.18) the two factors $1/\sqrt{N}$ of Eq. (A.20) combine into the single $1/N$, and the matrix element is of order one in N for the same reason as in Eq. (A.25). Ref. [26, Eqs. (39) and (40)] uses the same normalization and calls the vertex $g^{\rm DW}$. The tadpole is the term of $H_{\rm ep}$ with $\boldsymbol{Q} = 0$ and one boson line attached to a closed electron loop. The tadpole would describe a static shift of the equilibrium positions, and it vanishes because the expansion is made about the minimum of the Born–Oppenheimer surface, where $\langle B_{\boldsymbol{0}\nu}\rangle = 0$.

The Fan–Migdal self-energy

The second-order self-energy from $H_{\rm ep}$ is the boson-exchange diagram of Appendix D.9, with two changes. The vertex of Eq. (A.24) carries $1/\sqrt{N}$ instead of $1/\sqrt{\mathcal{V}}$, so the replacement Eq. (D.38) reads $-|g|^2 \mathsf{D}_0/N$. And the vertex carries band indices: an electron in $(n,\boldsymbol{k})$ goes to $(m,\boldsymbol{k}+\boldsymbol{Q})$ with $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\,B_{\boldsymbol{Q}\nu}$ and returns with $g_{nm\nu}(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q})\,B_{-\boldsymbol{Q}\nu}$; by Eq. (A.26), the return vertex $g_{nm\nu}(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q})$ equals $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^*$. We neglect the off-diagonal elements in n, an approximation that is exact for a nondegenerate level and, once summed over the zone, for a level whose partners form one irreducible representation (Sect. 6.3). Then

$$\Sigma^{\rm FM}_{n\boldsymbol{k}}(\mathrm{i}\omega_j^{\rm F}) = -\frac{1}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, T\sum_{j'} \mathsf{D}_0(\boldsymbol{Q}\nu,\mathrm{i}\omega_{j'}^{\rm B})\,\mathsf{G}_0(m,\boldsymbol{k}+\boldsymbol{Q},\mathrm{i}\omega_j^{\rm F}-\mathrm{i}\omega_{j'}^{\rm B}), \qquad \text{(D.52)}$$

with $\mathsf{D}_0(\boldsymbol{Q}\nu,z) = 1/(z-\omega_{\boldsymbol{Q}\nu}) - 1/(z+\omega_{\boldsymbol{Q}\nu})$ from Eq. (D.37) and $\mathsf{G}_0(m,\boldsymbol{p},z) = 1/(z-\xi_{m\boldsymbol{p}})$, where $\xi_{m\boldsymbol{p}} = \varepsilon_{m\boldsymbol{p}} - \mu_{\rm ch}$.

The frequency sum is done with the bosonic form of Eq. (C.40). Write $\omega_0 = \omega_{\boldsymbol{Q}\nu}$, $\xi = \xi_{m,\boldsymbol{k}+\boldsymbol{Q}}$ and $z_0 = \mathrm{i}\omega_j^{\rm F} - \xi$. The function $g(z) = \mathsf{D}_0(z)\,\mathsf{G}_0(\mathrm{i}\omega_j^{\rm F} - z)$ falls off as $1/|z|^3$ and has three simple poles. At $z=\omega_0$ the residue of $n_{\rm B}(z)g(z)$ is $n_{\rm B}(\omega_0)/(z_0-\omega_0)$. At $z=-\omega_0$ it is $-n_{\rm B}(-\omega_0)/(z_0+\omega_0) = [1+n_{\rm B}(\omega_0)]/(z_0+\omega_0)$. At $z=z_0$, where $\mathsf{G}_0(\mathrm{i}\omega_j^{\rm F}-z) = -1/(z-z_0)$, it is $-n_{\rm B}(z_0)\,\mathsf{D}_0(z_0)$, and since $\mathrm{e}^{\mathrm{i}\beta\omega_j^{\rm F}} = -1$, $n_{\rm B}(\mathrm{i}\omega_j^{\rm F}-\xi) = -1/(\mathrm{e}^{-\beta\xi}+1) = -[1-n_{\rm F}(\xi)]$. The three residues add up to $[n_{\rm B}+1-n_{\rm F}]/(z_0-\omega_0) + [n_{\rm B}+n_{\rm F}]/(z_0+\omega_0)$, and the frequency sum is minus this total. This minus sign cancels the minus sign in front of Eq. (D.52). Continuing $\mathrm{i}\omega_j^{\rm F} \to \omega + \mathrm{i}0^+$ by the argument of Appendix D.8, and measuring ω from $\mu_{\rm ch}$ as in Sect. 4.7, we obtain the Fan–Migdal self-energy Eq. (6.19), with $f_{m\boldsymbol{p}} = n_{\rm F}(\xi_{m\boldsymbol{p}})$ and $n_{\boldsymbol{Q}\nu} = n_{\rm B}(\omega_{\boldsymbol{Q}\nu})$. At $T=0$ it has poles only at $\omega = \xi_m + \omega_{\boldsymbol{Q}\nu}$ for empty m and at $\xi_m - \omega_{\boldsymbol{Q}\nu}$ for occupied m.

The imaginary part, Eq. (6.27), follows from Eq. (2.58). Both weights of the imaginary part are nonnegative, since $n+1-f \geq 0$ and $n+f \geq 0$, so $\mathrm{Im}\,\Sigma^{\rm FM} \leq 0$, as Eq. (4.82) requires.

The Allen–Heine self-energy

The second-order vertex enters at first order. To lowest order in the coupling, the phonons in $H_{\mathrm{ep}}^{(2)}$ may be replaced by their thermal average in the free lattice. The operators $B_{\boldsymbol{Q}\nu}$ and $B_{-\boldsymbol{Q}\nu} = B_{\boldsymbol{Q}\nu}^{\dagger}$ commute, so their product needs no ordering rule, and by Eq. (2.69)

$$\langle B_{\boldsymbol{Q}\nu} B_{\boldsymbol{Q}'\nu'} \rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\, \delta_{\nu\nu'}\, \langle b_{\boldsymbol{Q}\nu} b_{\boldsymbol{Q}\nu}^{\dagger} + b_{-\boldsymbol{Q}\nu}^{\dagger} b_{-\boldsymbol{Q}\nu} \rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\, \delta_{\nu\nu'}\, (2n_{\boldsymbol{Q}\nu} + 1).$$

The average of $H_{\mathrm{ep}}^{(2)}$ is then a one-electron operator that conserves $\boldsymbol{k}$. A static one-body term enters the Dyson equation directly as a self-energy, by the first-order rule of Appendix D.9 for a potential that carries no interaction line. The diagonal element of the averaged $H_{\mathrm{ep}}^{(2)}$ in band n is the Allen–Heine self-energy Eq. (6.20).

The adiabatic limit

The adiabatic formula Eq. (6.22) also follows without diagrams, by the route of Allen and Heine [27]. Freeze the nuclei at displacements u_a. To second order in u, nondegenerate perturbation theory gives

$$\begin{aligned} \varepsilon_{n\boldsymbol{k}}(\{u\}) - \varepsilon_{n\boldsymbol{k}} = \sum_a \langle n\boldsymbol{k}|\partial_a V|n\boldsymbol{k}\rangle\, u_a + \frac{1}{2}\sum_{ab} \Big[\langle n\boldsymbol{k}|\partial_a\partial_b V|n\boldsymbol{k}\rangle \\ + 2\,\mathrm{Re}\sum_{m\boldsymbol{k}'}{}' \frac{\langle n\boldsymbol{k}|\partial_a V|m\boldsymbol{k}'\rangle\langle m\boldsymbol{k}'|\partial_b V|n\boldsymbol{k}\rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}'}} \Big] u_a u_b, \end{aligned} \tag{D.53}$$

where $\partial_a = \partial/\partial u_a$ and the prime removes the states degenerate with $(n, \boldsymbol{k})$. Averaging over the harmonic lattice removes the linear term and replaces $u_a u_b$ by $\langle u_a u_b \rangle$. With Eq. (A.20), the average couples $\boldsymbol{Q}$ only to $-\boldsymbol{Q}$ with the weight $(2n_{\boldsymbol{Q}\nu}+1)/(2N\sqrt{M_\kappa M_{\kappa'}}\,\omega_{\boldsymbol{Q}\nu})$ times the polarization vectors, and the two terms in the square bracket become, term by term, the Fan sum and g^{AH} of Eq. (6.22). The diagrammatic form adds what the frozen picture misses: the phonon energies in the denominators, the separate weights of emission and absorption, and the imaginary part.

Translation invariance and the rigid-ion approximation

The second derivatives of V in g^{AH} require second-order density-functional perturbation theory. Translation invariance expresses them through products of first derivatives, which a phonon calculation already supplies. Displace the whole crystal rigidly by $\boldsymbol{t}$, on top of an arbitrary configuration $\{u\}$. The potential follows, $V(\boldsymbol{r};\{\boldsymbol{u}+\boldsymbol{t}\}) = V(\boldsymbol{r}-\boldsymbol{t};\{\boldsymbol{u}\})$, and the eigenvalues do not change. Hence $\sum_{\boldsymbol{R}\kappa} \partial\varepsilon_{n\boldsymbol{k}}/\partial u_{\boldsymbol{R}\kappa}^{\alpha} = 0$ for every configuration. At equilibrium the identity is the first sum rule,

$$\sum_{\boldsymbol{R}\kappa} \langle n\boldsymbol{k}|\partial_{\boldsymbol{R}\kappa\alpha} V|n\boldsymbol{k}\rangle = \sum_{\kappa} \langle n\boldsymbol{k}|\partial_{\kappa\alpha,\boldsymbol{0}} V|n\boldsymbol{k}\rangle = 0, \tag{D.54}$$

by the Hellmann–Feynman theorem, with $\partial_{\kappa\alpha,\boldsymbol{Q}} V$ of Appendix A.6; it states that the diagonal vertex of a uniform translation vanishes, the acoustic sum rule of Sect. 6.2. Differentiating the identity once more with respect to $u_{\boldsymbol{R}'\kappa'}^{\beta}$ and using the second-order coefficient of Eq. (D.53) gives the second sum rule,

$$\sum_{\boldsymbol{R}\kappa} \langle n\boldsymbol{k}|\partial_{\boldsymbol{R}\kappa\alpha}\partial_{\boldsymbol{R}'\kappa'\beta} V|n\boldsymbol{k}\rangle = -2\,\mathrm{Re}\sum_{\kappa}\sum_{m}{}' \frac{\langle n\boldsymbol{k}|\partial_{\kappa\alpha,\boldsymbol{0}} V|m\boldsymbol{k}\rangle\, \langle m\boldsymbol{k}|\partial_{\boldsymbol{R}'\kappa'\beta} V|n\boldsymbol{k}\rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}, \tag{D.55}$$

where $\partial_{\kappa\alpha,\mathbf{0}} V = \sum_{\boldsymbol{R}} \partial_{\boldsymbol{R}\kappa\alpha} V$ conserves $\boldsymbol{k}$, so that only $\boldsymbol{k}' = \boldsymbol{k}$ survives. Both rules are exact; they are Ref. [26, Eqs. (193) and (194)].

The left side of Eq. (D.55) is a sum over all sites, and it does not yet give the on-site element that g^{AH} needs. The rigid-ion approximation supplies it by writing the potential as a sum of atom-centered terms that move with their nuclei, $V(\boldsymbol{r}) = \sum_{\boldsymbol{R}\kappa} v_\kappa(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\kappa - \boldsymbol{u}_{\boldsymbol{R}\kappa})$, with $\boldsymbol{\tau}_\kappa$ the position of species κ in the cell. Then $\partial^2 V / \partial u_{\boldsymbol{R}\kappa\alpha} \partial u_{\boldsymbol{R}'\kappa'\beta}$ vanishes unless $(\boldsymbol{R}\kappa) = (\boldsymbol{R}'\kappa')$, and the left side of Eq. (D.55) is the on-site element $\langle n\boldsymbol{k}|\partial_\alpha \partial_\beta v_{\kappa'}(\boldsymbol{r} - \boldsymbol{R}' - \boldsymbol{\tau}_{\kappa'})|n\boldsymbol{k}\rangle$. On the right, the matrix element of the single-site operator $\partial_{\boldsymbol{R}'\kappa'\beta} V$ between two states of the same $\boldsymbol{k}$ is the same for every cell $\boldsymbol{R}'$, by the translation symmetry of Bloch states, and equals $N^{-1}\langle m\boldsymbol{k}|\partial_{\kappa'\beta,\mathbf{0}} V|n\boldsymbol{k}\rangle$. Hence

$$\langle n\boldsymbol{k}|\partial_\alpha \partial_\beta v_{\kappa'}|n\boldsymbol{k}\rangle = -\frac{2}{N} \operatorname{Re} \sum_\kappa \sum_m{}' \frac{\langle n\boldsymbol{k}|\partial_{\kappa\alpha,\mathbf{0}} V|m\boldsymbol{k}\rangle \, \langle m\boldsymbol{k}|\partial_{\kappa'\beta,\mathbf{0}} V|n\boldsymbol{k}\rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}, \tag{D.56}$$

for every site of species κ'. The second derivative of the potential has been expressed through first derivatives at $\boldsymbol{Q} = 0$.

In the same approximation the Allen–Heine term is the on-site element averaged with the displacement tensor $B_\kappa = \langle \boldsymbol{u}_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \rangle$ of Eq. (A.28). Only the diagonal terms $(\boldsymbol{R}\kappa) = (\boldsymbol{R}'\kappa')$ of the quadratic potential survive, and by Eqs. (A.20) and (A.21),

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = \frac{1}{2} \sum_{\boldsymbol{R}\kappa'} \sum_{\alpha\beta} B^{\alpha\beta}_{\kappa'} \, \langle n\boldsymbol{k}|\partial_\alpha \partial_\beta v_{\kappa'}|n\boldsymbol{k}\rangle, \qquad B^{\alpha\beta}_\kappa = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} \frac{2n_{\boldsymbol{Q}\nu} + 1}{2 M_\kappa \omega_{\boldsymbol{Q}\nu}} \, e^\alpha_{\kappa\nu}(\boldsymbol{Q}) \, e^\beta_{\kappa\nu}(\boldsymbol{Q})^*.$$

The sum over cells gives a factor N. We insert Eq. (D.56) and symmetrize between $(\kappa\alpha)$ and $(\kappa'\beta)$; the symmetrization leaves the real part of the product unchanged. The result is Eq. (6.24), with V the Kohn–Sham potential. Writing B_κ as its sum over modes gives Ref. [26, Eqs. (195)–(198)], where the $\boldsymbol{Q} = 0$ matrix elements are rebuilt from the vertices $g_{mn\nu}(\boldsymbol{k}, \mathbf{0})$ of the optical modes.

D.14 BCS gap and coherence factors

The mean field

In the reduced Hamiltonian Eq. (7.23), with $\xi_{\boldsymbol{k}} = \xi_{-\boldsymbol{k}}$, write each pair operator as its average plus a fluctuation, $c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} = b_{\boldsymbol{k}} + (c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} - b_{\boldsymbol{k}})$, and drop the product of two fluctuations, which vanishes relative to the kept terms as $\mathcal{V} \to \infty$ (Sect. 7.2). The mean-field Hamiltonian is

$$H_{\mathrm{MF}} = \sum_{\boldsymbol{k}}{}' \Big[\xi_{\boldsymbol{k}} \big(c^\dagger_{\boldsymbol{k}\uparrow} c_{\boldsymbol{k}\uparrow} + c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}\downarrow} \big) - \Delta_{\mathrm{BCS}} \big(c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow} + c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} \big) \Big] + \frac{\mathcal{V} \Delta^2_{\mathrm{BCS}}}{V_{\mathrm{BCS}}},$$

plus the unpaired states outside the shell. For each $\boldsymbol{k}$ the summand of H_{MF} is Eq. (C.61) with $c_1 = c_{\boldsymbol{k}\uparrow}$, $c_2 = c_{-\boldsymbol{k}\downarrow}$, $\xi_0 = \xi_{\boldsymbol{k}}$ and $g_0 = \Delta_{\mathrm{BCS}}$. By Appendix C.7, the operators Eq. (7.25) with $u^{\mathrm{B}}_{\boldsymbol{k}} = \cos\theta_{\boldsymbol{k}}$, $v^{\mathrm{B}}_{\boldsymbol{k}} = \sin\theta_{\boldsymbol{k}}$, are fermions, and the pair terms drop out of H_{MF} when $\tan 2\theta_{\boldsymbol{k}} = \Delta_{\mathrm{BCS}} / \xi_{\boldsymbol{k}}$. Then $\cos 2\theta_{\boldsymbol{k}} = \xi_{\boldsymbol{k}} / E_{\boldsymbol{k}}$ and $\sin 2\theta_{\boldsymbol{k}} = \Delta_{\mathrm{BCS}} / E_{\boldsymbol{k}}$, and these two relations give Eq. (7.26). The quasiparticle $\gamma^\dagger_{\boldsymbol{k}1}$ carries momentum $\boldsymbol{k}$ and spin up, $\gamma^\dagger_{\boldsymbol{k}2}$ momentum $-\boldsymbol{k}$ and spin down.

The ground state, annihilated by all γ, is the BCS state Eq. (7.27). To check that every γ annihilates the BCS state, note that $\gamma_{\boldsymbol{k}1}$ commutes with the factors of the product at other wavevectors, which contain fermion operators in pairs, so it suffices to act on the factor at $\boldsymbol{k}$. With $u = u^{\mathrm{B}}_{\boldsymbol{k}}$, $v = v^{\mathrm{B}}_{\boldsymbol{k}}$, $c_{\boldsymbol{k}\uparrow}|0\rangle = 0$, $c_{\boldsymbol{k}\uparrow} c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle = c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle$ and $(c^\dagger_{-\boldsymbol{k}\downarrow})^2 = 0$,

$$(u c_{\boldsymbol{k}\uparrow} - v c^\dagger_{-\boldsymbol{k}\downarrow})(u + v c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow})|0\rangle = uv \, c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle - vu \, c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle = 0.$$

In the same way $\gamma_{\boldsymbol{k}2}$ yields $-uv\,c^\dagger_{\boldsymbol{k}\uparrow}|0\rangle + vu\,c^\dagger_{\boldsymbol{k}\uparrow}|0\rangle = 0$, where the minus sign comes from moving $c_{-\boldsymbol{k}\downarrow}$ past $c^\dagger_{\boldsymbol{k}\uparrow}$.

The gap equation

Inverting Eq. (7.25) gives $c_{\boldsymbol{k}\uparrow} = u\gamma_{\boldsymbol{k}1} + v\gamma^\dagger_{\boldsymbol{k}2}$ and $c_{-\boldsymbol{k}\downarrow} = u\gamma_{\boldsymbol{k}2} - v\gamma^\dagger_{\boldsymbol{k}1}$, with the subscripts on u and v omitted. In thermal equilibrium the quasiparticles are independent fermions of energy $E_{\boldsymbol{k}}$, so $\langle\gamma^\dagger\gamma\rangle = n_{\rm F}(E_{\boldsymbol{k}})$, and

$$b_{\boldsymbol{k}} = \langle c_{-\boldsymbol{k}\downarrow}c_{\boldsymbol{k}\uparrow}\rangle = uv\big[\langle\gamma_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}2}\rangle - \langle\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1}\rangle\big] = uv\,[1-2n_{\rm F}(E_{\boldsymbol{k}})] = \frac{\Delta_{\rm BCS}}{2E_{\boldsymbol{k}}}\tanh\frac{E_{\boldsymbol{k}}}{2T}.$$

Inserted into Eq. (7.24), $b_{\boldsymbol{k}}$ gives the gap equation of Eq. (7.29), whose right side decreases with both $\Delta_{\rm BCS}$ and T. At $T=0$ the integral is $\operatorname{arsinh}(\omega_{\rm D}/\Delta_{\rm BCS})$, and

$$\Delta_{\rm BCS}(0) = \frac{\omega_{\rm D}}{\sinh[1/(\mathcal{N}_{\rm F}V_{\rm BCS})]} \simeq 2\omega_{\rm D}\,{\rm e}^{-1/(\mathcal{N}_{\rm F}V_{\rm BCS})},$$

the exponential form being valid for weak coupling, $\mathcal{N}_{\rm F}V_{\rm BCS} \ll 1$.

The critical temperature

At T_c the gap closes, and the gap equation with $\Delta_{\rm BCS}=0$ fixes T_c:

$$\frac{1}{\mathcal{N}_{\rm F}V_{\rm BCS}} = \int_0^{\omega_{\rm D}} {\rm d}\xi\,\frac{\tanh(\xi/2T_c)}{\xi}.$$

The integral is evaluated through the Matsubara expansion of the hyperbolic tangent. We use the fermionic form of Eq. (C.40) with $g(z) = 1/(x^2-z^2)$. The poles of g at $z=\pm x$ have residues $\mp 1/2x$, so $T\sum_j 1/(x^2+(\omega^{\rm F}_j)^2) = [n_{\rm F}(-x)-n_{\rm F}(x)]/2x = \tanh(x/2T)/2x$. Then

$$\int_0^{\omega_{\rm D}} {\rm d}\xi\,\frac{\tanh(\xi/2T)}{\xi} = 2T\sum_j\int_0^{\omega_{\rm D}}\frac{{\rm d}\xi}{\xi^2+(\omega^{\rm F}_j)^2} = \sum_{j=0}^{\infty}\frac{2}{\pi}\,\frac{\arctan[X/(j+\frac12)]}{j+\frac12}, \qquad X=\frac{\omega_{\rm D}}{2\pi T},$$

where the terms j and $-j-1$ were combined and $\omega^{\rm F}_j = 2\pi T(j+\frac12)$. Split $(2/\pi)\arctan(X/y) = \Theta(X-y)+r(y/X)$, with $r(t) = (2/\pi)\arctan(1/t) - \Theta(1-t)$. The step function cuts the sum at $j+\frac12 < X$. The sum over the remainder terms r, for $X \gg 1$, becomes the integral $\int_0^\infty {\rm d}t\,r(t)/t$, which vanishes. On $t<1$, $r(t) = -(2/\pi)\arctan t$; on $t>1$, $r(t) = (2/\pi)\arctan(1/t)$; and the substitution $t\to 1/t$ maps one piece onto minus the other. The truncated sum is a sum over odd integers, $\sum_{j=0}^{J}(j+\frac12)^{-1} = 2(H_{2J+2} - \frac12 H_{J+1})$ with the harmonic numbers $H_n = \sum_{m=1}^n 1/m = \ln n + \gamma_{\rm E} + O(1/n)$, and for $J \simeq X \gg 1$ the truncated sum is $\ln J + \gamma_{\rm E} + 2\ln 2$. Altogether the integral is $\ln(4{\rm e}^{\gamma_{\rm E}}X) = \ln(2{\rm e}^{\gamma_{\rm E}}\omega_{\rm D}/\pi T)$, with corrections of order $(T/\omega_{\rm D})^2$, and

$$T_c = \frac{2{\rm e}^{\gamma_{\rm E}}}{\pi}\,\omega_{\rm D}\,{\rm e}^{-1/(\mathcal{N}_{\rm F}V_{\rm BCS})} = 1.134\,\omega_{\rm D}\,{\rm e}^{-1/(\mathcal{N}_{\rm F}V_{\rm BCS})}.$$

In $\Delta_{\rm BCS}(0)/T_c$ the unknown $\omega_{\rm D}$ and the exponential cancel, and the result is the ratio of Eq. (7.29).

Coherence factors

For the density $n_e(\boldsymbol{q})$ and the spin density $s^z_e(\boldsymbol{q})$ of Sect. 7.2, write $u = u^{\rm B}_{\boldsymbol{k}}$, $v = v^{\rm B}_{\boldsymbol{k}}$ and primes for $\boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}$. With the inverted Eq. (7.25),

$$c^\dagger_{\boldsymbol{k}'\uparrow}c_{\boldsymbol{k}\uparrow} = u'u\,\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1} + u'v\,\gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2} + v'u\,\gamma_{\boldsymbol{k}'2}\gamma_{\boldsymbol{k}1} + v'v\,\gamma_{\boldsymbol{k}'2}\gamma^\dagger_{\boldsymbol{k}2}.$$

For the spin-down part, relabel $\boldsymbol{k} \to -\boldsymbol{k}-\boldsymbol{q}$, so that $\sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow} c_{\boldsymbol{k}\downarrow} = \sum_{\boldsymbol{k}} c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}'\downarrow}$, and expand:

$$c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}'\downarrow} = uu'\,\gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}'2} - uv'\,\gamma^\dagger_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}'1} - vu'\,\gamma_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}'2} + vv'\,\gamma_{\boldsymbol{k}1}\gamma^\dagger_{\boldsymbol{k}'1}.$$

Collect the terms by process. The term that creates two quasiparticles, $\gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2}$, comes with $u'v$ from spin up and, after $-\gamma^\dagger_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}'1} = \gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2}$, with uv' from spin down. The term that scatters a quasiparticle from $\boldsymbol{k}$ to $\boldsymbol{k}'$, $\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1}$, comes with uu' from spin up and, after $\gamma_{\boldsymbol{k}1}\gamma^\dagger_{\boldsymbol{k}'1} = -\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1}$ for $\boldsymbol{q} \neq 0$, with $-vv'$ from spin down. The density adds the contributions of the two spins and the spin density subtracts them; the resulting factors are Eq. (7.31). The scattering of the second branch, $\gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}'2}$, has the same factors. Expressed through the energies, with Eq. (7.26), the squares of the coherence factors are Eq. (7.32).

D.15 Conserving BCS density response

Nambu form

The matrices of Eq. (7.35), together with the unit matrix, obey the Pauli algebra, $\mathsf{s}_1\mathsf{s}_2 = \mathrm{i}\boldsymbol{\eta}$ and cyclically. Writing out the components gives $O_{\boldsymbol{\eta}}(\boldsymbol{q}) = \sum_{\boldsymbol{k}}[c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow}c_{\boldsymbol{k}\uparrow} - c_{-\boldsymbol{k}-\boldsymbol{q}\downarrow}c^\dagger_{-\boldsymbol{k}\downarrow}] = n_e(\boldsymbol{q})$ for $\boldsymbol{q} \neq 0$, after anticommuting and relabeling $-\boldsymbol{k}-\boldsymbol{q} \to \boldsymbol{k}$ in the spin-down term. The pair operators are $O_{\mathsf{s}_1}(\boldsymbol{q}) = B^\dagger_{\boldsymbol{q}} + B_{-\boldsymbol{q}}$ and $O_{\mathsf{s}_2}(\boldsymbol{q}) = -\mathrm{i}(B^\dagger_{\boldsymbol{q}} - B_{-\boldsymbol{q}})$, with $B^\dagger_{\boldsymbol{P}} = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{P}\uparrow}c^\dagger_{-\boldsymbol{k}\downarrow}$ the creation operator of a pair of total momentum $\boldsymbol{P}$. The matrix $h_{\boldsymbol{k}} = \xi_{\boldsymbol{k}}\boldsymbol{\eta} - \Delta_{\mathrm{BCS}}\mathsf{s}_1$ has the eigenvalues $\pm E_{\boldsymbol{k}}$, with projectors $P^\pm_{\boldsymbol{k}} = \frac{1}{2}(1 \pm h_{\boldsymbol{k}}/E_{\boldsymbol{k}})$, and $\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}) = (\mathrm{i}\omega^{\mathrm{F}} - h_{\boldsymbol{k}})^{-1} = \sum_\pm P^\pm_{\boldsymbol{k}}/(\mathrm{i}\omega^{\mathrm{F}} \mp E_{\boldsymbol{k}})$.

The interaction in the pairing channel

To let the pair field respond at wavevector $\boldsymbol{q}$, the interaction must scatter pairs of every total momentum, not only $\boldsymbol{P} = 0$ as in the reduced Hamiltonian Eq. (7.23). Take the contact attraction

$$H_{\mathrm{int}} = -\frac{V_{\mathrm{BCS}}}{\mathscr{V}}\sum_{\boldsymbol{P}} B^\dagger_{\boldsymbol{P}} B_{\boldsymbol{P}} = -V_{\mathrm{BCS}}\int \mathrm{d}^3 r\, \psi^\dagger_\uparrow\psi^\dagger_\downarrow\psi_\downarrow\psi_\uparrow(\boldsymbol{r}),$$

with the pair states restricted to the shell $|\xi| < \omega_{\mathrm{D}}$. Without the restriction, H_{int} is a function of the local densities $\psi^\dagger_\sigma\psi_\sigma$ alone and commutes with the density. On a lattice it is the on-site term $-V_{\mathrm{BCS}}\sum_i n_{i\uparrow}n_{i\downarrow}$, which needs no shell; in the continuum the shell makes it slightly nonlocal, and its commutator with $n_e(\boldsymbol{q})$ is of relative order $v_{\mathrm{F}}q/\omega_{\mathrm{D}}$, negligible at the wavevectors of interest. The Hamiltonian $H_0 + H_{\mathrm{int}}$ therefore obeys the continuity equation Eq. (4.29), $[H, n_e(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})$. With $B^\dagger_{\boldsymbol{P}} = \frac{1}{2}(O_{\mathsf{s}_1} + \mathrm{i}O_{\mathsf{s}_2})(\boldsymbol{P})$ and $B_{\boldsymbol{P}} = \frac{1}{2}(O_{\mathsf{s}_1} - \mathrm{i}O_{\mathsf{s}_2})(-\boldsymbol{P})$,

$$H_{\mathrm{int}} = -\frac{V_{\mathrm{BCS}}}{4\mathscr{V}}\sum_{\boldsymbol{P}}\sum_{a=1,2} O_{\mathsf{s}_a}(\boldsymbol{P})\,O_{\mathsf{s}_a}(-\boldsymbol{P}) + (\text{one-body terms}),$$

where the one-body terms come from reordering and shift the chemical potential. Replacing one factor of each product by its average, in the two possible ways, gives the mean-field term $-(V_{\mathrm{BCS}}/2\mathscr{V})\sum_a\langle O_{\mathsf{s}_a}(\boldsymbol{0})\rangle O_{\mathsf{s}_a}(\boldsymbol{0})$, and with $\langle O_{\mathsf{s}_1}(\boldsymbol{0})\rangle = 2\sum_{\boldsymbol{k}} b_{\boldsymbol{k}}$ the mean-field term is $-\Delta_{\mathrm{BCS}}O_{\mathsf{s}_1}(\boldsymbol{0})$, the pair term of H_{MF}.

Ward-identity violation of the bare bubble

The quasiparticle bubble with vertices A and B follows from Wick's theorem for the Nambu field as in Appendix D.8. It carries a trace over the Nambu index:

$$R^0_{AB}(\boldsymbol{q}, \mathrm{i}\omega^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}} T\sum_j \mathrm{Tr}[A\,\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega^{\mathrm{F}}_j + \mathrm{i}\omega^{\mathrm{B}})\,B\,\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_j)].$$

With the spectral form of G_0, the frequency sum is Eq. (D.31) with $\xi_{\boldsymbol{k}} \to s'E_{\boldsymbol{k}}$ and $\xi_{\boldsymbol{k}+\boldsymbol{q}} \to sE_{\boldsymbol{k}+\boldsymbol{q}}$. At $T=0$, $n_{\mathrm{F}}(-E)=1$ and $n_{\mathrm{F}}(E)=0$, and after continuation

$$R^0_{AB}(\boldsymbol{q},\omega) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}}\left[\frac{\mathrm{Tr}[AP^+_{\boldsymbol{k}+\boldsymbol{q}}BP^-_{\boldsymbol{k}}]}{\omega - E_{\boldsymbol{k}} - E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+} - \frac{\mathrm{Tr}[AP^-_{\boldsymbol{k}+\boldsymbol{q}}BP^+_{\boldsymbol{k}}]}{\omega + E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+}\right]. \tag{D.57}$$

For $A=B=\boldsymbol{\eta}$, $\boldsymbol{\eta}h_{\boldsymbol{k}'}\boldsymbol{\eta} = \xi'\boldsymbol{\eta} + \Delta_{\mathrm{BCS}}\mathsf{s}_1$, and the trace is $\frac{1}{2}[1-(\xi\xi'-\Delta^2_{\mathrm{BCS}})/EE']$, the case-I creation factor of Eq. (7.32).

At $\boldsymbol{q}=0$ the traces follow from the Pauli algebra of the Nambu matrices. Write $\mathsf{s}_3 \equiv \boldsymbol{\eta}$, so that $h_{\boldsymbol{k}}/E_{\boldsymbol{k}} = \sum_a n_a \mathsf{s}_a$ with the unit vector $\boldsymbol{n} = (-\Delta_{\mathrm{BCS}}/E, 0, \xi/E)$. Multiplying out with $\mathsf{s}_a\mathsf{s}_b = \delta_{ab} + \mathrm{i}\epsilon_{abc}\mathsf{s}_c$ gives

$$\mathrm{Tr}[\mathsf{s}_aP^+\mathsf{s}_bP^-] = \delta_{ab} - n_an_b - \mathrm{i}\epsilon_{abc}n_c, \qquad \mathrm{Tr}[\mathsf{s}_aP^-\mathsf{s}_bP^+] = \delta_{ab} - n_an_b + \mathrm{i}\epsilon_{abc}n_c.$$

With $F(\omega)$ of Sect. 7.2, the symmetric parts combine into $4E/(\omega^2-4E^2)$ and the antisymmetric parts into $2\omega/(\omega^2-4E^2)$, and the two combinations give the bubbles Eq. (7.38). In $R^0_{\mathsf{s}_2\mathsf{s}_2}$ we wrote $4E/(\omega^2-4E^2) = [\omega^2-(\omega^2-4E^2)]/[E(\omega^2-4E^2)]$ and used the gap equation at $T=0$, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}} 1/E_{\boldsymbol{k}} = 2/V_{\mathrm{BCS}}$ (Eq. (7.29)).

In the commutator Eq. (7.36) the kinetic term gives $\sum_{\boldsymbol{k}}(\xi_{\boldsymbol{k}+\boldsymbol{q}}-\xi_{\boldsymbol{k}})(c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow}c_{\boldsymbol{k}\uparrow} + c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow}c_{\boldsymbol{k}\downarrow}) = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})$ and the pair term gives the remaining terms of Eq. (7.36).

The conserving response

Let an external potential ϕ_{ext} couple to $n_e(\boldsymbol{q})$. The quasiparticles then see the external field and the change of the pair field that the external field induces, $\delta H = -(V_{\mathrm{BCS}}/2\mathscr{V})\sum_a \delta\langle O_{\mathsf{s}_a}(-\boldsymbol{q})\rangle O_{\mathsf{s}_a}(\boldsymbol{q})$, which is the mean-field term of H_{int} at wavevector $\boldsymbol{q}$. The quasiparticles respond to the external field and to δH with the bare bubbles. In the channel basis $(\boldsymbol{\eta}, \mathsf{s}_1, \mathsf{s}_2)$ this response is Eq. (7.37), with all bubbles from Eq. (D.57).

The response obeys the continuity equation because of a property of mean-field dynamics. Under $H_{\mathrm{MF}} + \delta H$ the state stays a quasi-free state, one for which Wick's theorem holds, at every time. For any one-body operator A, the average $\langle[H_0 + H_{\mathrm{int}}, A]\rangle$ in a quasi-free state, evaluated with Wick's theorem, equals $\langle[H_{\mathrm{MF}} + \delta H, A]\rangle$, provided the mean fields are the self-consistent ones: each pair of contractions of H_{int} is a mean field acting on the remaining two operators. The Hartree mean fields of H_{int}, which are density–density terms, commute with n_e and may be kept or dropped without changing the argument; we drop them, since their static part only shifts μ_{ch}. Hence $\mathrm{d}\langle n_e(\boldsymbol{q})\rangle/\mathrm{d}t = \mathrm{i}\langle[H_0 + H_{\mathrm{int}}, n_e(\boldsymbol{q})]\rangle = \mathrm{i}\boldsymbol{q}\cdot\langle\boldsymbol{j}_e(\boldsymbol{q})\rangle$, since H_{int} commutes with the density. The term $2\mathrm{i}\Delta_{\mathrm{BCS}}\delta\langle O_{\mathsf{s}_2}\rangle$ that the bare dynamics produces is cancelled, to linear order, by the commutator of n_e with δH, which contains $\langle O_{\mathsf{s}_1}(\boldsymbol{0})\rangle\delta\langle O_{\mathsf{s}_2}\rangle$ with the opposite sign. A response that obeys continuity at all times satisfies the f-sum rule, by the double-commutator argument of Sect. 4.3, with the averages taken in the mean-field ground state, and it vanishes at $\boldsymbol{q}=0$, where n_e is the conserved particle number.

D.16 Polarization of Dirac electrons

The Dirac form

Start from the block Hamiltonian Eq. (7.41), with $M(\delta\boldsymbol{k}) = \sum_i \delta k_i M_i$ the 2×2 matrix of the velocity between the conduction and the valence doublet. As Sect. 7.3 shows, the Kramers degeneracy of both bands at every $\delta\boldsymbol{k}$ requires $MM^\dagger \propto \mathbb{1}$ for every $\delta\boldsymbol{k}$. For $\delta\boldsymbol{k}$ along the axis

i the requirement reads $M_i M_i^\dagger \propto \mathbb{1}$. For $\delta\boldsymbol{k}$ along the diagonal of the axes i and j it reads $(M_i + M_j)(M_i + M_j)^\dagger \propto \mathbb{1}$, which with the first condition requires

$$M_i M_i^\dagger \propto \mathbb{1}, \qquad M_i M_j^\dagger + M_j M_i^\dagger \propto \mathbb{1}, \tag{D.58}$$

with real coefficients of proportionality. We write the Pauli matrices that act within a doublet as $\sigma_1^{\mathrm{d}}, \sigma_2^{\mathrm{d}}, \sigma_3^{\mathrm{d}}$, and $\boldsymbol{c}\cdot\boldsymbol{\sigma}^{\mathrm{d}} = \sum_l c_l \sigma_l^{\mathrm{d}}$.

Step 1: reduce to one direction. Assume $M_1 \neq 0$. By Eq. (D.58), $M_1 M_1^\dagger = m^2 \mathbb{1}$ with $m > 0$, so $W = M_1/m$ is unitary. Define $N_i = M_i M_1^{-1}$, so that $M_i = N_i M_1$ and $N_1 = \mathbb{1}$. Then $M_i M_1^\dagger = N_i M_1 M_1^\dagger = m^2 N_i$ and $M_i M_i^\dagger = N_i M_1 M_1^\dagger N_i^\dagger = m^2 N_i N_i^\dagger$, and Eq. (D.58) with $j = 1$ becomes

$$N_i + N_i^\dagger \propto \mathbb{1}, \qquad N_i N_i^\dagger \propto \mathbb{1}.$$

Step 2: the form of each N_i. Every 2×2 matrix is the sum of a Hermitian part $\frac{1}{2}(N + N^\dagger)$ and an anti-Hermitian part, which is i times a Hermitian matrix, and every Hermitian matrix is a real combination of $\mathbb{1}$ and the σ_l^{d}. The first condition makes the Hermitian part a real multiple $x_i\mathbb{1}$, and the anti-Hermitian part is $\mathrm{i}(y_i\mathbb{1} + \boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}})$ with real y_i and $\boldsymbol{c}_i$. Hence $N_i = z_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}}$ with $z_i = x_i + \mathrm{i}y_i$. With $(\boldsymbol{c}\cdot\boldsymbol{\sigma}^{\mathrm{d}})^2 = c^2\mathbb{1}$,

$$N_i N_i^\dagger = (z_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}})(z_i^*\mathbb{1} - \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}}) = (|z_i|^2 + c_i^2)\mathbb{1} + 2y_i\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}},$$

and the second condition requires $y_i = 0$ unless $\boldsymbol{c}_i = 0$. A direction with $\boldsymbol{c}_i = 0$ and $y_i \neq 0$ is excluded by Eq. (D.58) for a pair $i, j \neq 1$: if $N_i = z_i\mathbb{1}$ and $N_j = x_j\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_j\cdot\boldsymbol{\sigma}^{\mathrm{d}}$ with $\boldsymbol{c}_j \neq 0$, then $N_i N_j^\dagger + N_j N_i^\dagger = 2x_j x_i\mathbb{1} + 2y_i\,\boldsymbol{c}_j\cdot\boldsymbol{\sigma}^{\mathrm{d}}$, which requires $y_i = 0$. The remaining case is that all $\boldsymbol{c}_i$ vanish. Then $M = (\delta k_1 + z_2\delta k_2 + z_3\delta k_3)M_1$ is one complex number times M_1. The bands $\pm(\Delta_{\mathrm{D}}^2 + m^2|\delta k_1 + z_2\delta k_2 + z_3\delta k_3|^2)^{1/2}$ then do not disperse along the line in $\delta\boldsymbol{k}$ on which this complex number vanishes, and we exclude this nongeneric case. Every N_i is therefore a real combination $x_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}}$, described by the real four-vector $(x_i, \boldsymbol{c}_i)$, and N_1 is the four-vector $(1,0,0,0)$.

Step 3: remove M_1. A change of basis within the valence doublet, $M \to M U_{\mathrm{v}}^\dagger$ with the unitary $U_{\mathrm{v}} = W$, turns $M_i = N_i M_1 = m N_i W$ into $m N_i$. Then $M = m\sum_i \delta k_i\,(x_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\mathrm{d}})$, and m can be absorbed into the velocities.

Step 4: rotate the four-vectors. A matrix $Q = x\mathbb{1} + \mathrm{i}\,\boldsymbol{c}\cdot\boldsymbol{\sigma}^{\mathrm{d}}$ with real $(x, \boldsymbol{c})$ has $\det Q = x^2 + c^2$, the squared length of its four-vector. A change of basis in both doublets, $Q \to U_{\mathrm{c}} Q U_{\mathrm{v}}^\dagger$ with U_{c} and U_{v} unitary of unit determinant, maps matrices of the form of Q linearly onto matrices of the same form and preserves the determinant. It therefore acts on the four-vector as an orthogonal transformation, and continuity from $U_{\mathrm{c}} = U_{\mathrm{v}} = \mathbb{1}$ makes it a rotation. Every rotation of four-dimensional space arises in this way. Choosing $U_{\mathrm{c}} = U_{\mathrm{v}}$ gives the rotations of the three components $\boldsymbol{c}$ that leave x fixed. Choosing $U_{\mathrm{c}} = \mathrm{e}^{\mathrm{i}\phi\sigma_3^{\mathrm{d}}}$ and $U_{\mathrm{v}} = \mathrm{e}^{-\mathrm{i}\phi\sigma_3^{\mathrm{d}}}$ gives $Q \to \mathrm{e}^{\mathrm{i}\phi\sigma_3^{\mathrm{d}}} Q\, \mathrm{e}^{\mathrm{i}\phi\sigma_3^{\mathrm{d}}}$, which on $Q = \mathbb{1}$ yields $\cos 2\phi\,\mathbb{1} + \mathrm{i}\sin 2\phi\,\sigma_3^{\mathrm{d}}$, a rotation in the plane of x and c_3. Rotations of $\boldsymbol{c}$ together with rotations in one plane that contains the x axis generate all rotations of four-dimensional space. The three four-vectors $(x_i, \boldsymbol{c}_i)$ span a subspace of at most three dimensions, and a rotation can bring this subspace into the subspace orthogonal to $(1,0,0,0)$; the rotation removes every component along $\mathbb{1}$. A phase $-\mathrm{i}$ in the conduction doublet then removes the factor i, and

$$M = \sum_{i,l} \delta k_i\, A_{il}\, \sigma_l^{\mathrm{d}}$$

with a real 3×3 matrix A.

Step 5: principal axes. The singular-value decomposition writes $A = R\,\mathrm{diag}(v_1, v_2, v_3)\,R'^{\mathrm{T}}$ with two rotations R and R' and real v_i; when both rotations are chosen proper, one of the v_i

may be negative. The rotation R is a choice of crystal axes for $\delta\boldsymbol{k}$. The rotation R' is undone by the same unitary U in both doublets, since $U\sigma_l^{\mathrm{d}}U^\dagger = \sum_{l'} R'_{ll'}\sigma_{l'}^{\mathrm{d}}$ for a suitable U, by the correspondence between unitary 2×2 matrices of unit determinant and rotations. In the resulting basis $M = \sum_i v_i\,\delta k_i\,\sigma_i^{\mathrm{d}}$, which is the block of the Dirac Hamiltonian Eq. (7.42).

The bubble

Take one isotropic Dirac cone, $H_{\boldsymbol{k}} = v_{\mathrm{D}}\boldsymbol{\alpha}_{\mathrm{D}}\cdot\boldsymbol{k} + \Delta_{\mathrm{D}}\beta_{\mathrm{D}}$, with $\boldsymbol{k}$ measured from the node and $\boldsymbol{\alpha}_{\mathrm{D}}$ and β_{D} the 4×4 Dirac matrices, which anticommute pairwise and square to one. Then $H_{\boldsymbol{k}}^2 = E_{\boldsymbol{k}}^2$, and each band $\pm E_{\boldsymbol{k}}$ is doubly degenerate, with the projector $P_{\boldsymbol{k}}^{\pm} = \frac{1}{2}(1 \pm H_{\boldsymbol{k}}/E_{\boldsymbol{k}})$ of rank two. The density matrix element between Bloch states near the cone is the overlap of the band spinors: for q small compared with the reciprocal lattice, the factor $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ acts on the slowly varying envelope and conserves the spinor index [28, App. A.3]. Summed over the two degenerate states of each band, the squared overlap is $\mathrm{Tr}[P_{\boldsymbol{k}'}^{+}P_{\boldsymbol{k}}^{-}]$, and with $\mathrm{Tr}(\alpha_{\mathrm{D}}^i\alpha_{\mathrm{D}}^j) = 4\delta_{ij}$, $\mathrm{Tr}\,\beta_{\mathrm{D}}^2 = 4$ and $\mathrm{Tr}(\alpha_{\mathrm{D}}^i\beta_{\mathrm{D}}) = 0$,

$$\mathcal{F}(\boldsymbol{k},\boldsymbol{k}') = \frac{1}{2}\,\mathrm{Tr}\big[P_{\boldsymbol{k}'}^{+}P_{\boldsymbol{k}}^{-}\big] = \frac{1}{2}\Big(1 - \frac{v_{\mathrm{D}}^2\,\boldsymbol{k}\cdot\boldsymbol{k}' + \Delta_{\mathrm{D}}^2}{E_{\boldsymbol{k}}E_{\boldsymbol{k}'}}\Big) \tag{D.59}$$

per state, in agreement with Ref. [28, Eq. (A.7)]. At $T = 0$, with the chemical potential in the gap, the occupations are 1 in the lower band and 0 in the upper. Of the bubble Eq. (4.93), continued to real frequency, the interband terms remain, with the degeneracy g_{deg} of Sect. 7.3:

$$R_{\mathrm{D}}^0(\boldsymbol{q},\omega) = \frac{g_{\mathrm{deg}}}{\mathscr{V}}\sum_{|\boldsymbol{k}|<\Lambda_{\mathrm{D}}} \mathcal{F}(\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q})\left[\frac{1}{\omega - E_{\boldsymbol{k}} - E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+} - \frac{1}{\omega + E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+}\right].$$

Reduction to one integral

The bracket depends on $\boldsymbol{k}$ only through the pair energy $S = E + E'$, with $E = E_{\boldsymbol{k}}$ and $E' = E_{\boldsymbol{k}+\boldsymbol{q}}$. Use $k = |\boldsymbol{k}|$ and $k' = |\boldsymbol{k}+\boldsymbol{q}|$ as coordinates; with the azimuth about $\boldsymbol{q}$ integrated out, $\mathrm{d}^3k = 2\pi k k'\,\mathrm{d}k\,\mathrm{d}k'/q$ over the region $|k-k'| \le q \le k+k'$. Since $k\,\mathrm{d}k = E\,\mathrm{d}E/v_{\mathrm{D}}^2$, the measure becomes $\mathrm{d}^3k = 2\pi EE'\,\mathrm{d}E\,\mathrm{d}E'/(v_{\mathrm{D}}^4 q)$. In the overlap, $2\boldsymbol{k}\cdot\boldsymbol{k}' = k^2 + k'^2 - q^2$ and $v_{\mathrm{D}}^2k^2 = E^2 - \Delta_{\mathrm{D}}^2$, so

$$\mathcal{F} = \frac{v_{\mathrm{D}}^2q^2 - (E-E')^2}{4EE'}.$$

The factors EE' cancel. With $S = E + E'$ and $D = E - E'$, $\mathrm{d}E\,\mathrm{d}E' = \frac{1}{2}\mathrm{d}S\,\mathrm{d}D$, and the boundary of the region follows from $(k^2 + k'^2 - q^2)^2 \le 4k^2k'^2$. Rewritten with the energies, the boundary condition is $D^2(S^2 - v_{\mathrm{D}}^2q^2) \le v_{\mathrm{D}}^2q^2(S^2 - v_{\mathrm{D}}^2q^2 - 4\Delta_{\mathrm{D}}^2)$. Write $u = S^2 - v_{\mathrm{D}}^2q^2$, the squared energy of the pair in its rest frame, and $D_{\mathrm{m}} = v_{\mathrm{D}}q(1 - 4\Delta_{\mathrm{D}}^2/u)^{1/2}$. Then $|D| \le D_{\mathrm{m}}$, pairs exist only for $u \ge 4\Delta_{\mathrm{D}}^2$, and

$$\int_{-D_{\mathrm{m}}}^{D_{\mathrm{m}}}\mathrm{d}D\,(v_{\mathrm{D}}^2q^2 - D^2) = 2D_{\mathrm{m}}\Big(v_{\mathrm{D}}^2q^2 - \frac{D_{\mathrm{m}}^2}{3}\Big) = \frac{4}{3}v_{\mathrm{D}}^3q^3\rho(u), \quad \rho(u) = \Big(1 - \frac{4\Delta_{\mathrm{D}}^2}{u}\Big)^{1/2}\Big(1 + \frac{2\Delta_{\mathrm{D}}^2}{u}\Big).$$

Collecting the factors gives $\mathscr{V}^{-1}\sum_{\boldsymbol{k}}\mathcal{F}\,h(S) = (q^2/24\pi^2v_{\mathrm{D}})\int\mathrm{d}S\,\rho(S^2 - v_{\mathrm{D}}^2q^2)\,h(S)$ for any function h. The bracket in R_{D}^0 is $2S/[(\omega+\mathrm{i}0^+)^2 - S^2]$, and with $\mathrm{d}u = 2S\,\mathrm{d}S$,

$$R_{\mathrm{D}}^0(\boldsymbol{q},\omega) = -\frac{g_{\mathrm{deg}}\,q^2}{24\pi^2v_{\mathrm{D}}}\int_{4\Delta_{\mathrm{D}}^2}^{4v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2}\mathrm{d}u\,\frac{\rho(u)}{u - s - \mathrm{i}0^+\,\mathrm{sgn}\,\omega}, \qquad s = \omega^2 - v_{\mathrm{D}}^2q^2. \tag{D.60}$$

The cutoff $|\boldsymbol{k}| < \Lambda_{\mathrm{D}}$ has become $u < 4v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2$, which differs from the exact boundary by terms of relative order q/Λ_{D} and $\Delta_{\mathrm{D}}^2/v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2$. The cutoff breaks the Lorentz invariance of the response (its dependence on $\boldsymbol{q}$ and ω through q^2 and s alone) only in these neglected terms.

The dielectric function

In Eq. (7.46) the factor q^2 of Eq. (D.60) cancels the Coulomb $1/q^2$, and with $e^2 = 4\pi\alpha$ and $\tilde\alpha = \alpha/(\varepsilon_{\rm bg} v_{\rm D})$,

$$\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_{\rm bg}} = 1 + \frac{g_{\rm deg}\tilde\alpha}{6\pi}\int_{4\Delta_{\rm D}^2}^{U} \mathrm{d}u\, \frac{\rho(u)}{u - s - \mathrm{i}0^+\,\mathrm{sgn}\,\omega}, \qquad U = 4v_{\rm D}^2\Lambda_{\rm D}^2.$$

By Eq. (2.58), for $\omega > 0$ the imaginary part is $\mathrm{Im}\,\varepsilon/\varepsilon_{\rm bg} = (g_{\rm deg}\tilde\alpha/6)\,\rho(s)\,\Theta(s - 4\Delta_{\rm D}^2)$, which is Eq. (7.49).

For the real part, split $\rho(u)/(u-s) = \rho(u)/u + s\rho(u)/[u(u-s)]$. The integral of ρ/u is elementary with $u = 4\Delta_{\rm D}^2/(1-b^2)$, in which $\rho\,\mathrm{d}u/u = b^2(3-b^2)(1-b^2)^{-1}\mathrm{d}b = [b^2 - 2 + 2/(1-b^2)]\mathrm{d}b$; integrated to $b_U = (1 - 4\Delta_{\rm D}^2/U)^{1/2}$ it gives $\frac{1}{3} - 2 + \ln[(1+b_U)/(1-b_U)] = \ln(U/\Delta_{\rm D}^2) - \frac{5}{3}$ for $U \gg \Delta_{\rm D}^2$. The integral of the second term converges, and its cutoff may be sent to infinity. Write $\ln(1 - s/a) = -s\int_a^\infty \mathrm{d}u/[u(u-s)]$ with $a = \Delta_{\rm D}^2/x(1-x)$ and integrate over x with the weight $6x(1-x)$. After the integrals are exchanged, a given u contributes for $x(1-x) \geq \Delta_{\rm D}^2/u$, between $x_\pm = \frac{1}{2}(1 \pm b)$ with $b = (1 - 4\Delta_{\rm D}^2/u)^{1/2}$, and $\int_{x_-}^{x_+} 6x(1-x)\,\mathrm{d}x = b(3-b^2)/2 = \rho(u)$. Hence $s\int \rho\,\mathrm{d}u/[u(u-s)] = -6\int_0^1 \mathrm{d}x\, x(1-x)\ln[1 - x(1-x)s/\Delta_{\rm D}^2]$, and Eq. (7.48) follows, with the branch of the logarithm that gives $-\mathrm{i}\pi$ where its argument is negative; this branch reproduces Eq. (7.49).

D.17 Effective-mass theory of shallow donors

The envelope equation

The donor adds to the crystal Hamiltonian the potential $U(\boldsymbol{r})$ of its excess charge. Far from the core, at distances large compared with the lattice constant, the potential is the Coulomb potential of a unit charge, $-e^2/(4\pi\varepsilon\, r)$ in Heaviside–Lorentz units ($e^2 = 4\pi\alpha$). The host screens it by its static dielectric constant ε:

$$U(\boldsymbol{r}) = -\frac{e^2}{4\pi\varepsilon\, r} = -\frac{\alpha}{\varepsilon\, r}.$$

The dielectric constant is a pure number. For silicon, which is nonpolar and has no lattice contribution, it is $\varepsilon = \varepsilon_0 = \varepsilon_\infty = 11.7$ (Sect. 5.4), and from here on we write it ε_0, the static constant, as in Sect. 7.4. Expand the donor state in the Bloch states of Sect. 3.2, $\psi = \sum_{n\boldsymbol{k}} A_n(\boldsymbol{k})\,\psi_{n\boldsymbol{k}}$. The Schrödinger equation becomes

$$(\varepsilon_{n\boldsymbol{k}} - E)A_n(\boldsymbol{k}) + \sum_{n'\boldsymbol{k}'} \langle\psi_{n\boldsymbol{k}}|U|\psi_{n'\boldsymbol{k}'}\rangle\, A_{n'}(\boldsymbol{k}') = 0.$$

A slowly varying U simplifies the matrix element. Its Fourier components are concentrated at $|\boldsymbol{k} - \boldsymbol{k}'| \sim 1/a^*$, small compared with the zone, so for the cell-periodic parts $u_{n\boldsymbol{k}}$ of Eq. (3.20), $\langle\psi_{n\boldsymbol{k}}|U|\psi_{n'\boldsymbol{k}'}\rangle \simeq \tilde U(\boldsymbol{k} - \boldsymbol{k}')\,\langle u_{n\boldsymbol{k}}|u_{n'\boldsymbol{k}'}\rangle_{\rm cell}$, where $\tilde U(\boldsymbol{p}) = \mathscr{V}^{-1}\int \mathrm{d}^3r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}U(\boldsymbol{r})$ and the cell average is normalized to $\delta_{nn'}$ at $\boldsymbol{k} = \boldsymbol{k}'$. For $\boldsymbol{k}'$ close to $\boldsymbol{k}$ the overlap of different bands is of order $|\boldsymbol{k} - \boldsymbol{k}'|$ times a ratio of a velocity matrix element to an energy gap, by Eq. (3.42), and it couples bands only at the next order in $1/(a^* k_{\rm zone})$. To leading order the bands decouple, and the conduction-band amplitude obeys

$$(\varepsilon_{c\boldsymbol{k}} - E)A(\boldsymbol{k}) + \sum_{\boldsymbol{k}'} \tilde U(\boldsymbol{k} - \boldsymbol{k}')\, A(\boldsymbol{k}') = 0.$$

Near one conduction minimum $\boldsymbol{k}_j$, write $\boldsymbol{k} = \boldsymbol{k}_j + \delta\boldsymbol{k}$ and define the envelope $F_j(\boldsymbol{r}) = \mathscr{V}^{-1/2}\sum_{\delta\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\delta\boldsymbol{k}\cdot\boldsymbol{r}} A(\boldsymbol{k}_j + \delta\boldsymbol{k})$. Multiplication by $\delta\boldsymbol{k}$ becomes $-\mathrm{i}\nabla$ acting on F_j, and the

convolution with $\tilde{U}$ becomes multiplication by $U(\boldsymbol{r})$. With the band near the minimum from Sect. 3.3, $\varepsilon_{c,\boldsymbol{k}_j+\delta\boldsymbol{k}} = E_\mathrm{g} + \delta k_\parallel^2/2m_\parallel^* + \delta k_\perp^2/2m_\perp^*$ (energies from the valence maximum), and z along the valley axis, the equation for the amplitude becomes the envelope equation Eq. (7.53), with $E = E_\mathrm{g} - E_\mathrm{b}^\mathrm{D}$. The derivation assumed three things: that the envelope extends over many cells, $a^* \gg a$; that the binding is small compared with the energy over which the band departs from its quadratic form; and that the potential is the screened Coulomb potential everywhere that matters. The third assumption fails within a cell or two of the core, in the central-cell region treated below.

For $m_\parallel^* = m_\perp^* = m^*$, Eq. (7.53) is the hydrogen atom with $m_e \to m^*$ and $\alpha \to \alpha/\varepsilon_0$, and Eq. (7.54) follows from the hydrogenic energies and length scale by the same substitution, with $a_0 = 1/(\alpha m_e) = 0.529\,\text{Å}$ and $\mathrm{Ry} = 13.606\,\mathrm{eV}$ [29].

Anisotropic valleys

With unequal masses, Eq. (7.53) does not separate, and Kohn and Luttinger solved it variationally with an envelope squeezed along the heavy axis [30],

$$F(\boldsymbol{r}) = (\pi r_\perp^2 r_\parallel)^{-1/2} \exp\Big[-\Big(\frac{x^2+y^2}{r_\perp^2} + \frac{z^2}{r_\parallel^2}\Big)^{1/2}\Big],$$

with the radii $r_\perp$ across the valley axis and $r_\parallel$ along it; the envelope is normalized to one. In the scaled coordinates $(x/r_\perp, y/r_\perp, z/r_\parallel)$ the envelope is the hydrogen ground state $\mathrm{e}^{-r}/\sqrt{\pi}$, for which each Cartesian component of $\langle|\nabla F|^2\rangle$ is $\frac{1}{3}$. The kinetic energy is therefore $(2/3)(2m_\perp^* r_\perp^2)^{-1} + (1/3)(2m_\parallel^* r_\parallel^2)^{-1}$. For the potential, write $r = R\,(r_\perp^2 \sin^2\theta + r_\parallel^2 \cos^2\theta)^{1/2}$ in the scaled spherical coordinates (R, θ). The radial integral $\int_0^\infty R\,\mathrm{e}^{-2R}\,\mathrm{d}R = \frac{1}{4}$ separates, and for $r_\parallel < r_\perp$

$$\Big\langle \frac{1}{r} \Big\rangle = \int_0^1 \frac{\mathrm{d}\mu}{\big[r_\perp^2 - (r_\perp^2 - r_\parallel^2)\mu^2\big]^{1/2}} = \frac{\arcsin e_\mathrm{ecc}}{r_\perp e_\mathrm{ecc}}, \qquad e_\mathrm{ecc} = \Big(1 - \frac{r_\parallel^2}{r_\perp^2}\Big)^{1/2}.$$

With lengths measured in units of $a_\perp^* = \varepsilon_0/(m_\perp^*\alpha)$ and energies in units of $\mathrm{Ry}_\perp^* = m_\perp^*\alpha^2/2\varepsilon_0^2$, and with e_ecc the eccentricity of the spheroid, the energy is

$$\frac{E(r_\perp, r_\parallel)}{\mathrm{Ry}_\perp^*} = \frac{2}{3r_\perp^2} + \frac{\gamma_m}{3r_\parallel^2} - \frac{2\arcsin e_\mathrm{ecc}}{r_\perp e_\mathrm{ecc}}, \qquad \gamma_m = \frac{m_\perp^*}{m_\parallel^*}. \tag{D.61}$$

For $\gamma_m = 1$ the minimum is at $r_\perp = r_\parallel = 1$ with $E = -1$, the hydrogenic result. For silicon, $\gamma_m = 0.191/0.916 = 0.209$, $\mathrm{Ry}_\perp^* = 18.98\,\mathrm{meV}$ and $a_\perp^* = 32.4\,\text{Å}$. Minimizing Eq. (D.61) numerically gives $E = -1.564\,\mathrm{Ry}_\perp^*$ at $r_\perp = 24.3\,\text{Å}$ and $r_\parallel = 14.0\,\text{Å}$:

$$E_\mathrm{b}^\mathrm{D}(\mathrm{Si}) = 29.7\,\mathrm{meV} \qquad \text{(effective-mass theory, variational)}. \tag{D.62}$$

The radius $r_\perp$ is four and a half lattice constants, $r_\parallel$ two and a half. The variational energy is an upper bound on the exact effective-mass energy.

Valleys and the central cell

At the order of Eq. (7.53) the six valley states $F_j\psi_{c\boldsymbol{k}_j}$ of silicon are degenerate (Sect. 7.4). The valley states couple through the part δU of the donor potential that the envelope equation leaves out. The donor potential differs from $-\alpha/\varepsilon_0 r$ only within about a lattice constant of the core, where the screening is incomplete and the ion has its own chemistry. Since δU is

short-ranged and each envelope is nearly constant over the range of δU, the matrix elements between valley states are $\langle j|\delta U|j'\rangle \simeq F_j(0)F_{j'}(0)\int \mathrm{d}^3r\,\psi^*_{c\boldsymbol{k}_j}\delta U\,\psi_{c\boldsymbol{k}_{j'}}$.

The tetrahedral symmetry of the site leaves three distinct elements: Δ_0 on the diagonal, Δ_g between opposite valleys and Δ_f between perpendicular ones. The 6×6 matrix of δU in the valley states has the eigenvectors

$$\begin{array}{lll} A_1: & (1,1,1,1,1,1)/\sqrt{6}, & E=\Delta_0+\Delta_\mathrm{g}+4\Delta_\mathrm{f},\\ E: & (1,1,-1,-1,0,0)/2 \text{ and } (1,1,1,1,-2,-2)/\sqrt{12}, & E=\Delta_0+\Delta_\mathrm{g}-2\Delta_\mathrm{f},\\ T_2: & (1,-1,0,0,0,0)/\sqrt{2} \text{ and its two partners}, & E=\Delta_0-\Delta_\mathrm{g}, \end{array} \tag{D.63}$$

with the valleys ordered as $(+x,-x,+y,-y,+z,-z)$. In the T_2 states each pair of opposite valleys enters with opposite signs, and the four perpendicular couplings cancel. The A_1 state has the largest density at the core, $|\psi(0)|^2=6|F(0)|^2|u_c(0)|^2$; the E and T_2 states have nodes there.

D.18 The operator basis

The reduction of Sect. 8.1 takes an interaction written for a dark-matter particle and one free fermion and turns it into operators of the whole target. Natural units are used, and $\boldsymbol{q}$ is the momentum the dark matter loses (Sect. 1.3).

D.18.1 Galilean invariants

The dark matter goes from $\boldsymbol{p}$ to $\boldsymbol{p}'=\boldsymbol{p}-\boldsymbol{q}$ and the target fermion, of mass m, from $\boldsymbol{k}$ to $\boldsymbol{k}'=\boldsymbol{k}+\boldsymbol{q}$. A Galilean boost by a velocity $\boldsymbol{u}$ shifts $\boldsymbol{p}$ and $\boldsymbol{p}'$ by $m_\chi\boldsymbol{u}$, and $\boldsymbol{k}$ and $\boldsymbol{k}'$ by $m\boldsymbol{u}$. The momentum transfer $\boldsymbol{q}$ is invariant, and so is the difference between the mean velocities of the dark matter and the fermion,

$$\boldsymbol{v}^\perp=\frac{\boldsymbol{p}+\boldsymbol{p}'}{2m_\chi}-\frac{\boldsymbol{k}+\boldsymbol{k}'}{2m}=\bar{\boldsymbol{v}}_\chi-\boldsymbol{v}_\mathrm{t}, \tag{D.64}$$

with $\boldsymbol{v}_\mathrm{t}\equiv(\boldsymbol{k}+\boldsymbol{k}')/2m$ (used only in this section). Of the four momenta, conservation leaves three and invariance two, $\boldsymbol{q}$ and $\boldsymbol{v}^\perp$.

For free particles, with the dark matter changing its mass by Δ_χ when the scattering is inelastic, energy conservation reads $p^2/2m_\chi+k^2/2m=p'^2/2m_\chi+k'^2/2m+\Delta_\chi$. Since $p^2-p'^2=\boldsymbol{q}\cdot(\boldsymbol{p}+\boldsymbol{p}')$ and $k'^2-k^2=\boldsymbol{q}\cdot(\boldsymbol{k}+\boldsymbol{k}')$, energy conservation becomes

$$\boldsymbol{q}\cdot\boldsymbol{v}^\perp=\Delta_\chi, \tag{D.65}$$

and $\boldsymbol{v}^\perp$ is transverse to $\boldsymbol{q}$ for elastic scattering on any free target, whatever its initial momentum. In the rest frame of the target, $\boldsymbol{k}=0$, the elastic condition is $\boldsymbol{q}\cdot\boldsymbol{v}=q^2/2\mu_{\chi T}$ with $\mu_{\chi T}$ the reduced mass, as in Sect. 1.3.

D.18.2 Bilinears between plane waves

With the spinors Eq. (C.125), $u=(E+m)^{1/2}(\xi,\boldsymbol{\sigma}\cdot\boldsymbol{k}\,\xi/(E+m))$, and the states normalized to one particle per volume, each bilinear is divided by $(2E\cdot 2E')^{1/2}$. Write $\boldsymbol{S}=\boldsymbol{\sigma}/2$ and $\boldsymbol{K}=\boldsymbol{k}+\boldsymbol{k}'$. To second order in k/m, $[(E+m)/2E]^{1/2}=1-k^2/8m^2$ and $1/(E+m)=1/2m$. With $(\boldsymbol{\sigma}\cdot\boldsymbol{k}')(\boldsymbol{\sigma}\cdot\boldsymbol{k})=\boldsymbol{k}'\cdot\boldsymbol{k}+\mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{k}'\times\boldsymbol{k})$ and $\boldsymbol{k}'\times\boldsymbol{k}=\boldsymbol{q}\times\boldsymbol{K}/2$, the scalar and the time

component of the vector are

$$\begin{aligned}\frac{\bar u' u}{(4EE')^{1/2}} &= 1 - \frac{k^2+k'^2}{8m^2} - \frac{\boldsymbol{k}'\cdot\boldsymbol{k}}{4m^2} - \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2} = 1 - \frac{K^2}{8m^2} - \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2},\\ \frac{\bar u'\gamma^0 u}{(4EE')^{1/2}} &= 1 - \frac{k^2+k'^2}{8m^2} + \frac{\boldsymbol{k}'\cdot\boldsymbol{k}}{4m^2} + \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2} = 1 - \frac{q^2}{8m^2} + \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2}.\end{aligned} \tag{D.66}$$

The second line is the vertex Eq. (C.132) found there by the Foldy–Wouthuysen transformation. The scalar and the vector time component differ at second order: the scalar loses $K^2/8m^2 = v_\mathrm{t}^2/2$, the vector time component only $q^2/8m^2$. The other bilinears are needed to first order. With the blocks of Eq. (C.123) and $\sigma^i\sigma^j = \delta^{ij} + \mathrm{i}\epsilon^{ijl}\sigma^l$,

$$\frac{\bar u'\gamma^i u}{2m} = \frac{K^i}{2m} - \frac{\mathrm{i}(\boldsymbol{q}\times\boldsymbol{S})^i}{m}, \qquad \frac{\bar u'\gamma^0\gamma^5 u}{2m} = \frac{2\boldsymbol{S}\cdot\boldsymbol{K}}{2m}, \qquad \frac{\bar u'\gamma^i\gamma^5 u}{2m} = 2S^i, \qquad \frac{\bar u'\mathrm{i}\gamma^5 u}{2m} = -\frac{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{S}}{m}, \tag{D.67}$$

with the two-component spinors understood on either side. For the spatial vector, $\bar u'\gamma^i u = u'^\dagger\alpha^i_\mathrm{D} u = \xi'^\dagger[\sigma^i(\boldsymbol{\sigma}\cdot\boldsymbol{k}) + (\boldsymbol{\sigma}\cdot\boldsymbol{k}')\sigma^i]\xi$ to this order, and the Pauli algebra gives $K^i - \mathrm{i}\epsilon^{ijl}q^j\sigma^l$. The bilinears of Eq. (D.67) agree with Ref. [31, Table I], whose $\boldsymbol{q}$ has the same sign. For the dark matter, whose momentum changes by $-\boldsymbol{q}$, replace $\boldsymbol{q}\to-\boldsymbol{q}$ and $\boldsymbol{K}/2m \to \bar{\boldsymbol{v}}_\chi$.

The exchange of a massive vector mediator between two currents adds the spatial components of the currents to the potential of Sect. 1.4. The Proca field obeys $(\partial^2 + m_V^2)A'^\mu = y_\chi J^\mu_\chi$ for a source $-y_\chi A'_\mu J^\mu_\chi$, each component like the time component in Eq. (C.78). For $\omega \ll q$ its transform is $y_\chi J^\mu_\chi/(q^2+m_V^2)$, and a second current with coupling y has the energy $y(A'^0 J^0 - \boldsymbol{A}'\cdot\boldsymbol{J})$. The potential is

$$\tilde V = \frac{y_\chi y}{q^2+m_V^2}\,(J^0_\chi J^0 - \boldsymbol{J}_\chi\cdot\boldsymbol{J}), \tag{D.68}$$

with the currents the bilinears of Eqs. (D.66) and (D.67). The leading term of $\tilde V$ is $\mathcal{O}_1$, and the spatial product is the source of the corrections estimated in Worked example 8.1.

D.18.3 From one particle to many

Each operator of Table 8.1 has, between plane waves, a matrix element $f(\boldsymbol{k},\boldsymbol{k}')\,\delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$ with f a polynomial of degree at most two in $\boldsymbol{K}$, times at most one spin. The one-body operator with these matrix elements is unique, and for the three monomials it is

$$1 \to \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}, \qquad \frac{\boldsymbol{K}}{2m} \to \frac{1}{2}\Big\{\frac{\boldsymbol{p}}{m}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\Big\}, \qquad \frac{K^2}{4m^2} \to \frac{1}{4m^2}\{\boldsymbol{p},\{\boldsymbol{p},\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}\}. \tag{D.69}$$

The proof is one line each: $\langle\boldsymbol{k}'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\boldsymbol{k}\rangle = \delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$, and a factor $\boldsymbol{p}$ on the left of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ contributes $\boldsymbol{k}'$, while a factor $\boldsymbol{p}$ on the right contributes $\boldsymbol{k}$. The double anticommutator is $p^2X + 2\boldsymbol{p}\cdot X\boldsymbol{p} + Xp^2$ with $X = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and its matrix element is $k'^2 + 2\boldsymbol{k}'\cdot\boldsymbol{k} + k^2 = K^2$. Writing $K^2 = q^2 + 4\boldsymbol{k}\cdot\boldsymbol{k}'$ gives the second form of $\mathcal{T}_s$ in Eq. (8.3). The spin $\boldsymbol{S}$ commutes with $\boldsymbol{r}$ and $\boldsymbol{p}$ and simply multiplies. For a target of many particles of species s, the operator is the sum over particles, as for any one-body operator (Eq. (2.36)), and in second quantization it is $\sum_{\boldsymbol{k}} f(\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q})\,c^\dagger_{\boldsymbol{k}+\boldsymbol{q}}c_{\boldsymbol{k}}$ with the spin indices of $\boldsymbol{S}$ between the operators. The resulting operators are the rows of Eq. (8.3). In position space, the density row is the transform $n_e(\boldsymbol{q}) = \int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\hat n_e(\boldsymbol{r})$ of the density operator of Sect. 4.2, and the other rows are transforms of the corresponding densities.

D.18.4 Reduction of the operators

With Eq. (D.64), each $\mathcal{O}_i$ is a polynomial in $\boldsymbol{v}_\mathrm{t}$ and $\boldsymbol{S}_e$ with coefficients built from $\bar{\boldsymbol{v}}_\chi$, $\boldsymbol{S}_\chi$ and $\boldsymbol{q}$. Expanding and replacing $1\to n$, $S_e^\beta \to s^\beta$, $v_\mathrm{t}^\alpha \to j^\alpha$, $v_\mathrm{t}^\alpha S_e^\beta \to \mathcal{J}^{\alpha\beta}$ and $v_\mathrm{t}^2 \to (2/m)\mathcal{T}$

gives Table 8.1. The cases $\mathcal{O}_3$, $\mathcal{O}_{12}$ and $\mathcal{O}_{15}$ show every pattern. For $\mathcal{O}_3$,

$$\mathrm{i}\boldsymbol{S}_e\cdot\Big(\frac{\boldsymbol{q}}{m}\times\boldsymbol{v}^\perp\Big)=\frac{\mathrm{i}}{m}\Big[(\boldsymbol{q}\times\bar{\boldsymbol{v}}_\chi)\cdot\boldsymbol{S}_e-\epsilon_{\alpha\beta\gamma}S_e^\alpha q^\beta v_{\mathrm{t}}^\gamma\Big]\;\to\;\frac{\mathrm{i}}{m}\Big[(\boldsymbol{q}\times\bar{\boldsymbol{v}}_\chi)\cdot\boldsymbol{s}-\epsilon_{\alpha\beta\gamma}q^\beta\mathcal{J}^{\gamma\alpha}\Big],$$

where the velocity index of $\mathcal{J}$ comes first. For $\mathcal{O}_{12}$, $(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)^\alpha=\epsilon_{\alpha\beta\gamma}S_e^\beta(\bar v_\chi^\gamma-v_{\mathrm{t}}^\gamma)$, so $\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)\to\boldsymbol{S}_\chi\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}_\chi)-\epsilon_{\alpha\beta\gamma}S_\chi^\alpha\mathcal{J}^{\gamma\beta}$. For $\mathcal{O}_{15}$ the same cross product is dotted into $\boldsymbol{q}$, $(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)\cdot\boldsymbol{q}\to\boldsymbol{q}\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}_\chi)-\epsilon_{\alpha\beta\gamma}q^\alpha\mathcal{J}^{\gamma\beta}$, and multiplied by $-\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m^2$. The operators without $\boldsymbol{S}_e$ need only n and $\boldsymbol{j}$, those linear in $\boldsymbol{S}_e$ without $\boldsymbol{v}^\perp$ only $\boldsymbol{s}$, and those with both need $\boldsymbol{s}$ and $\mathcal{J}$. Operator $\mathcal{O}_2$ is the one case of degree two, Eq. (8.9).

D.18.5 Longitudinal identities

Let the target Hamiltonian consist of the kinetic energies of its particles and of potentials that depend on their positions alone, as H_{C} of Eq. (2.48) does. Then $[H,n_s(\boldsymbol{q})]=\boldsymbol{q}\cdot\boldsymbol{j}_s(\boldsymbol{q})$ for each species, by the commutator of Eq. (4.29) with $m_e\to m_s$, and between eigenstates with $E_f-E_i=\omega$,

$$\langle f|\boldsymbol{q}\cdot\boldsymbol{j}_s(\boldsymbol{q})|i\rangle=\omega\,\langle f|n_s(\boldsymbol{q})|i\rangle. \tag{D.70}$$

If, in addition, H commutes with every spin $\boldsymbol{S}_j$ of the species, the same commutator multiplied by S_j^β gives $[H,s_s^\beta(\boldsymbol{q})]=q^\alpha\mathcal{J}_s^{\alpha\beta}(\boldsymbol{q})$, and $\langle f|q^\alpha\mathcal{J}_s^{\alpha\beta}|i\rangle=\omega\langle f|s_s^\beta|i\rangle$. Spin–orbit coupling, the hyperfine interaction and a Zeeman term break this second identity and leave the first intact. On the dark-matter side, Eq. (1.21) with $\bar{\boldsymbol{v}}_\chi=\boldsymbol{v}-\boldsymbol{q}/2m_\chi$ gives $\boldsymbol{q}\cdot\bar{\boldsymbol{v}}_\chi=\omega+\Delta_\chi$ on shell. Every occurrence of $\boldsymbol{q}\cdot\boldsymbol{v}^\perp$ in an operator therefore becomes, inside a rate,

$$\boldsymbol{q}\cdot\bar{\boldsymbol{v}}_\chi\,n_s-\boldsymbol{q}\cdot\boldsymbol{j}_s\;\to\;(\omega+\Delta_\chi-\omega)\,n_s=\Delta_\chi\,n_s, \tag{D.71}$$

which is Eq. (8.5), the many-body form of Eq. (D.65). For $\mathcal{O}_2$ the identity applies only to the cross term. The square of the longitudinal electron velocity is not fixed by ω in a bound target, because the potential energy changes along with the kinetic energy. The three rows of Eq. (8.9) are then needed as they stand, and the cancellation of the longitudinal parts used in Worked example 8.2 is exact only for free electrons.

D.18.6 Nuclei

Write the nucleons of nucleus I at $\boldsymbol{X}_I+\boldsymbol{y}_j$, with momenta $(m_N/M_\kappa)\boldsymbol{P}_I+\boldsymbol{\pi}_j$, where $\boldsymbol{P}_I$ is the momentum of the nucleus and $\sum_j\boldsymbol{\pi}_j=0$. A nucleon operator $\sum_j c^{(N_j)}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{X}_I+\boldsymbol{y}_j)}$, with N_j the type of nucleon j, has, between nuclear ground states, the element $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\langle0|\sum_j c^{(N_j)}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}_j}|0\rangle$. For equal proton and neutron distributions the element is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}[Z_\kappa c^{(p)}+(A_\kappa-Z_\kappa)c^{(n)}]F_N(q)$ with F_N of Eq. (2.26), the first form of Eq. (8.8). The nucleon velocity $(\boldsymbol{k}+\boldsymbol{k'})/2m_N$ splits into $(\boldsymbol{P}_I+\boldsymbol{P}_I')/2M_\kappa$ and an internal part built from $\boldsymbol{\pi}_j$. The internal part has zero ground-state expectation value, and its contribution at first order in $\boldsymbol{q}\cdot\boldsymbol{y}_j$ involves the orbital and spin–orbit structure of the nucleus [32, Eqs. (26) and (27)]; dropping it leaves the rows $\boldsymbol{j}_\kappa$ and $\mathcal{T}_\kappa$ with the nuclear mass.

For the spin, $\boldsymbol{W}=\sum_j c^{(N_j)}\boldsymbol{S}_j$ is a vector operator. Within the multiplet of nuclear spin J_{nuc} the Wigner–Eckart theorem makes every vector operator proportional to $\boldsymbol{J}^{\mathrm{nuc}}$, $\langle M'|\boldsymbol{W}|M\rangle=w\,\langle M'|\boldsymbol{J}^{\mathrm{nuc}}|M\rangle$. Taking $M'=M=J_{\mathrm{nuc}}$ and the z components gives $w\,J_{\mathrm{nuc}}=c^{(p)}\langle S_p\rangle+c^{(n)}\langle S_n\rangle$, where $\langle S_p\rangle=\langle J_{\mathrm{nuc}},J_{\mathrm{nuc}}|\sum_{j\in p}S_j^z|J_{\mathrm{nuc}},J_{\mathrm{nuc}}\rangle$ and similarly for neutrons. With this w, the Wigner–Eckart form is the second form of Eq. (8.8). Averaged over M and summed over M' and the components, $\sum_\alpha|\langle M'|J_{\mathrm{nuc}}^\alpha|M\rangle|^2$ gives $\mathrm{Tr}(\boldsymbol{J}_{\mathrm{nuc}}^2)/(2J_{\mathrm{nuc}}+1)=J_{\mathrm{nuc}}(J_{\mathrm{nuc}}+1)$, the factor used for ^{29}Si in Sect. 8.1.

D.18.7 Spin traces

The tensor Eq. (8.7) is Hermitian, since exchanging a and b conjugates the trace. It is positive semidefinite: for any complex numbers x_a, $\sum_{ab} x_a^* \mathfrak{D}_{ab} x_b = (2|\boldsymbol{S}_\chi|+1)^{-1}\,\mathrm{Tr}_\chi(Y^\dagger Y) \geq 0$ with $Y = \sum_b x_b \tilde{V}_b$. For a dark-matter spin S, $\mathrm{Tr}_\chi\, \boldsymbol{S}_\chi = 0$ and $\mathrm{Tr}_\chi(S_\chi^\alpha S_\chi^\beta) = \delta_{\alpha\beta}(2S+1)S(S+1)/3$, so the average of $S_\chi^\alpha S_\chi^\beta$ is $\delta_{\alpha\beta}/4$ for $S = \frac{1}{2}$. Products of three spin components have a trace proportional to $\epsilon_{\alpha\beta\gamma}$ for spin $\frac{1}{2}$, and they arise only when a coefficient quadratic in $\boldsymbol{S}_\chi$ meets a linear one, and none of the operators of Table 8.1 produces this combination.

D.18.8 Absorption and the energy deposited

An absorbed boson enters through $\int \mathrm{d}^3x\, \phi(x)\, \mathcal{B}(\boldsymbol{x})$, with $\mathcal{B}$ a bilinear of the target fields and $\phi \propto \mathrm{e}^{-\mathrm{i}m_\phi t + \mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$. The spatial integral is the Fourier transform at $\boldsymbol{q} = \boldsymbol{p}_\phi$, so the plane-wave rules above apply with $\boldsymbol{q} = \boldsymbol{p}_\phi$. For the axion-like particle, $\hat{\psi}^\dagger \boldsymbol{\sigma}\cdot \overleftrightarrow{\nabla}\hat{\psi} = \hat{\psi}^\dagger\boldsymbol{\sigma}\cdot\nabla\hat{\psi} - (\nabla\hat{\psi}^\dagger)\cdot\boldsymbol{\sigma}\hat{\psi}$ has the element $\boldsymbol{\sigma}\cdot(\mathrm{i}\boldsymbol{k} + \mathrm{i}\boldsymbol{k}') = 4\mathrm{i}m_e\, \boldsymbol{S}\cdot\boldsymbol{K}/2m_e$, since ∇ on the annihilated state gives $\mathrm{i}\boldsymbol{k}$ and on the created state $-\mathrm{i}\boldsymbol{k}'$. The second term of Eq. (1.39) is then $(\mathrm{i}g_{aee}/4m_e^2)(\partial_t\phi)\, 4\mathrm{i}m_e\, \mathcal{J}_e^{\alpha\alpha} = -(g_{aee}/m_e)(\partial_t\phi)\, \mathcal{J}_e^{\alpha\alpha}$, and the first is $-(g_{aee}/m_e)(\nabla\phi)\cdot\boldsymbol{s}_e$; the interaction energy, minus the Lagrangian, is Eq. (8.10). For the scalar, $\hat{\psi}^\dagger \overleftrightarrow{\nabla}^2 \hat{\psi}$ has the element $-k^2 - k'^2 - 2\boldsymbol{k}'\cdot\boldsymbol{k} = -K^2$, and $(1/8m_e^2)(-K^2) = -(1/m_e)(K^2/8m_e)$ is $-1/m_e$ times the element $K^2/8m_e$ of $\mathcal{T}_e$. The element $(1/8m_e^2)(-K^2)$ is the order-v^2 term of the scalar bilinear in Eq. (D.66), and with $H = -\mathcal{L}$ it gives Eq. (8.11).

The energy deposited is Eq. (8.12). For a non-relativistic outgoing particle of mass $m_{\mathrm{out}} = m_\chi + \Delta_\chi$,

$$\omega = \Big(m_\chi + \frac{p^2}{2m_\chi}\Big) - \Big(m_\chi + \Delta_\chi + \frac{|\boldsymbol{p}-\boldsymbol{q}|^2}{2m_{\mathrm{out}}}\Big) = \boldsymbol{q}\cdot\boldsymbol{v} - \frac{q^2}{2m_\chi} - \Delta_\chi + \frac{|\boldsymbol{p}-\boldsymbol{q}|^2}{2}\Big(\frac{1}{m_\chi} - \frac{1}{m_{\mathrm{out}}}\Big),$$

where $p^2 - |\boldsymbol{p}-\boldsymbol{q}|^2 = 2\boldsymbol{q}\cdot\boldsymbol{p} - q^2$ was used. The first three terms are $\omega_{\boldsymbol{q}}$, and with $1/m_\chi - 1/m_{\mathrm{out}} = \Delta_\chi/(m_\chi m_{\mathrm{out}})$ the result is Eq. (8.13). The same formula holds for fermionic absorption into a massive $\tilde{\nu}$ with $m_{\tilde{\nu}} = m_\chi + \Delta_\chi$. For a massless $\tilde{\nu}$, $E_{\mathrm{out}} = |\boldsymbol{p}-\boldsymbol{q}|$, which for $v \to 0$ is q, and $\omega = m_\chi - q$, Eq. (8.14). With a free nucleus of mass M as the target, $\omega = q^2/2M$ and $q = m_\chi - q^2/2M$, so $q = m_\chi(1 - m_\chi/2M + \dots)$ and the recoil energy is $m_\chi^2/2M$ to first order in m_χ/M.

D.19 Master rate formulas at finite temperature

The rates of Sect. 8.2 follow from the golden rule with the interaction of Eq. (8.6) and a target in thermal equilibrium. Natural units are used; $\boldsymbol{q}$ is the momentum the dark matter loses and ω the energy the target gains.

D.19.1 Scattering

The dark matter is a plane wave of momentum $\boldsymbol{p} = m_\chi\boldsymbol{v}$ in the box of volume $\mathcal{V}$ (Appendix A.3), in the spin state $|s\rangle$, one of $2|\boldsymbol{S}_\chi|+1$. The target is in the eigenstate $|i\rangle$ with the Gibbs weight p_i. By Eq. (8.6), the matrix element to the dark-matter state $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$, $|s'\rangle$ and the target state $|f\rangle$ is

$$\mathcal{A}_{fi} = \frac{1}{\mathcal{V}}\sum_b \langle s'|\tilde{V}_b(\boldsymbol{q}, \bar{\boldsymbol{v}}_\chi)|s\rangle\, \langle f|O_b(\boldsymbol{q})|i\rangle.$$

Energy conservation between the initial energy $E_i + E(\boldsymbol{p})$ and the final energy $E_f + E(\boldsymbol{p}')$ is $\delta(\omega_{\boldsymbol{q}} - E_f + E_i)$, with $\omega_{\boldsymbol{q}} = E(\boldsymbol{p}) - E(\boldsymbol{p}')$ the energy of Eq. (1.21), mass splitting included.

The golden rule Eq. (2.5), averaged over the initial states as in Eq. (2.64), and averaged over s and summed over s', gives

$$\Gamma = \sum_{\boldsymbol{p}'} \frac{2\pi}{2|\boldsymbol{S}_\chi|+1} \sum_{s,s'} \sum_{i,f} p_i \, |\mathcal{A}_{fi}|^2 \, \delta(\omega_{\boldsymbol{q}} - E_f + E_i).$$

Squaring the sum over b produces the pairs (a,b), with the conjugate of the term a:

$$|\mathcal{A}_{fi}|^2 = \frac{1}{\mathscr{V}^2} \sum_{a,b} \langle s'|\tilde{V}_a|s\rangle^* \langle s'|\tilde{V}_b|s\rangle \; \langle i|O_a^\dagger|f\rangle \langle f|O_b|i\rangle.$$

Since $\langle s'|\tilde{V}_a|s\rangle^* = \langle s|\tilde{V}_a^\dagger|s'\rangle$, the sum over s' is a completeness relation and the sum over s a trace:

$$\frac{1}{2|\boldsymbol{S}_\chi|+1} \sum_{s,s'} \langle s|\tilde{V}_a^\dagger|s'\rangle\langle s'|\tilde{V}_b|s\rangle = \frac{\mathrm{Tr}_\chi[\tilde{V}_a^\dagger \tilde{V}_b]}{2|\boldsymbol{S}_\chi|+1} = \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v}),$$

the tensor Eq. (8.7). It depends on $\boldsymbol{v}$ and $\boldsymbol{q}$ through $\bar{\boldsymbol{v}}_\chi = \boldsymbol{v} - \boldsymbol{q}/2m_\chi$. The final momenta are $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$, and the sum over them is $\sum_{\boldsymbol{p}'} = \mathscr{V} \int \mathrm{d}^3q/(2\pi)^3$ (Eq. (2.10)). Collecting the factors gives

$$\Gamma(\boldsymbol{v}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v}) \; \frac{2\pi}{\mathscr{V}} \sum_{i,f} p_i \, \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle \, \delta(\omega_{\boldsymbol{q}} - E_f + E_i),$$

and the last factor is $S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}})$ of Eq. (4.13). The resulting rate is Eq. (8.15). One power of $1/\mathscr{V}$ from the squared amplitude cancels against the sum over final momenta, the other is absorbed in S_{ab}, and the rate of one particle in the box is independent of $\mathscr{V}$.

The box holds $\rho_\chi \mathscr{V}/m_\chi$ dark-matter particles with velocities distributed as $f_\chi(\boldsymbol{v})$, and a target mass $\rho_T \mathscr{V}$. The rate per unit target mass is then $(\rho_\chi/m_\chi\rho_T) \int \mathrm{d}^3v \, f_\chi(\boldsymbol{v}) \, \Gamma(\boldsymbol{v})$. Inserting $1 = \int \mathrm{d}\omega \, \delta(\omega - \omega_{\boldsymbol{q}})$ under the $\boldsymbol{q}$ integral and replacing $\omega_{\boldsymbol{q}}$ by ω in S_{ab} gives the spectrum Eq. (8.16).

D.19.2 Index order of the contraction

The squared amplitude pairs $\tilde{V}_a^\dagger$ with $O_a^\dagger$, so the contraction is $\sum_{ab} \mathfrak{D}_{ab} S_{ab} = \mathrm{Tr}[\mathfrak{D} S^{\mathrm{T}}]$. Both matrices are Hermitian. Write each as a real symmetric part plus i times a real antisymmetric part, $\mathfrak{D} = \mathfrak{D}_{\mathrm{R}} + \mathrm{i}\mathfrak{D}_{\mathrm{I}}$ and $S = S_{\mathrm{R}} + \mathrm{i}S_{\mathrm{I}}$. Then $S^{\mathrm{T}} = S_{\mathrm{R}} - \mathrm{i}S_{\mathrm{I}}$, and since the trace of a symmetric matrix times an antisymmetric one vanishes,

$$\mathrm{Tr}[\mathfrak{D} S^{\mathrm{T}}] = \mathrm{Tr}[\mathfrak{D}_{\mathrm{R}} S_{\mathrm{R}}] + \mathrm{Tr}[\mathfrak{D}_{\mathrm{I}} S_{\mathrm{I}}], \qquad \mathrm{Tr}[\mathfrak{D} S] = \mathrm{Tr}[\mathfrak{D}_{\mathrm{R}} S_{\mathrm{R}}] - \mathrm{Tr}[\mathfrak{D}_{\mathrm{I}} S_{\mathrm{I}}].$$

Both traces are real. Only $\mathrm{Tr}[\mathfrak{D} S^{\mathrm{T}}]$ is the rate, and only it is guaranteed nonnegative: $S^{\mathrm{T}} = S^*$ is positive semidefinite when S is, and the trace of a product of two positive semidefinite matrices is nonnegative.

D.19.3 Absorption from a quantized field

Quantize a real scalar field in the box,

$$\phi(\boldsymbol{x}) = \sum_{\boldsymbol{k}} \frac{1}{\sqrt{2\omega_{\boldsymbol{k}} \mathscr{V}}} (a_{\boldsymbol{k}} \, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} + a_{\boldsymbol{k}}^\dagger \, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}), \qquad \omega_{\boldsymbol{k}} = (m_\phi^2 + k^2)^{1/2},$$

with $[a_{\boldsymbol{k}}, a_{\boldsymbol{k}'}^\dagger] = \delta_{\boldsymbol{k}\boldsymbol{k}'}$; a vector field carries in addition a polarization vector for each mode. The coupling integrates the field against target bilinears, and the plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}$ turns each

bilinear into the row at $\boldsymbol{q}=\boldsymbol{k}$ (Sect. 8.1.5); for the density, $\int \mathrm{d}^3x\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}\hat{n}_e(\boldsymbol{x}) = n_e(\boldsymbol{k})$. With g_a^ϕ the coefficient of row a for the mode $\boldsymbol{k}$, derivatives of the field included,

$$H_{\rm int} = \sum_{\boldsymbol{k}} \frac{1}{\sqrt{2\omega_{\boldsymbol{k}}\mathcal{V}}}\Big[a_{\boldsymbol{k}} \sum_a g_a^\phi\, O_a(\boldsymbol{k}) + a_{\boldsymbol{k}}^\dagger \sum_a g_a^{\phi\,*}\, O_a(\boldsymbol{k})^\dagger\Big].$$

From the state with N_ϕ quanta in mode $\boldsymbol{k}$ and the target in $|i\rangle$, absorption leads to $N_\phi-1$ quanta and $|f\rangle$, with the matrix element $\sqrt{N_\phi/2\omega_{\boldsymbol{k}}\mathcal{V}}\,\sum_b g_b^\phi\langle f|O_b(\boldsymbol{k})|i\rangle$ and energy conservation $\delta(\omega_{\boldsymbol{k}} - E_f + E_i)$. The golden rule, thermally averaged and summed over f, gives

$$\Gamma_{N_\phi\to N_\phi-1} = \frac{N_\phi}{2\omega_{\boldsymbol{k}}}\sum_{a,b} g_a^{\phi\,*} g_b^\phi\, \frac{2\pi}{\mathcal{V}} \sum_{i,f} p_i\, \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\, \delta(\omega_{\boldsymbol{k}} - E_f + E_i) = N_\phi\, \Gamma_{\rm abs},$$

with $\Gamma_{\rm abs}$ of Eq. (8.19) at $\boldsymbol{p}_\phi = \boldsymbol{k}$. The rate is N_ϕ times the rate per quantum.

Emission leads to $N_\phi+1$ quanta, with the matrix element $\sqrt{(N_\phi+1)/2\omega_{\boldsymbol{k}}\mathcal{V}}\,\sum_b g_b^{\phi\,*}\langle f|O_b^\dagger|i\rangle$ and $\delta(\omega_{\boldsymbol{k}} + E_f - E_i)$. When the names of i and f are exchanged, as in the proof of Eq. (2.68), the weight becomes $p_f = p_i\,\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}$ and the squared element becomes $|\sum_b g_b^\phi\langle f|O_b|i\rangle|^2$, so that $\Gamma_{N_\phi\to N_\phi+1} = (N_\phi+1)\,\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}\Gamma_{\rm abs}$. The field loses quanta at the net rate

$$\Gamma_{N_\phi\to N_\phi-1} - \Gamma_{N_\phi\to N_\phi+1} = N_\phi\,\Gamma_{\rm abs}\Big[1 - \Big(1+\frac{1}{N_\phi}\Big)\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}\Big].$$

The term $1/N_\phi$ is spontaneous emission. For a halo field it is negligible, and per quantum the net rate is $\Gamma_{\rm net} = (1-\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}})\Gamma_{\rm abs}$. By the fluctuation–dissipation theorem Eq. (4.18), with $x_a = g_a^\phi$,

$$\begin{aligned}(1-\mathrm{e}^{-\beta\omega})\sum_{a,b} x_a^* x_b S_{ab} &= \mathrm{i}\sum_{a,b} x_a^* x_b (R_{ab} - R_{ba}^*) \\ &= \mathrm{i}\Big[\sum_{a,b} x_a^* x_b R_{ab} - \Big(\sum_{a,b} x_a^* x_b R_{ab}\Big)^*\Big] = -2\ \mathrm{Im}\sum_{a,b} x_a^* x_b R_{ab},\end{aligned}$$

where the middle step relabels $a\leftrightarrow b$ in the term with R_{ba}^*. Dividing by $2\omega_{\boldsymbol{k}}$ gives the second form of Eq. (8.19). The modes are occupied with N_ϕ quanta. The number density of quanta, $\sum_{\boldsymbol{k}} N_\phi/\mathcal{V}$, is ρ_χ/m_ϕ up to corrections of order v^2, and the quanta are distributed in $\boldsymbol{k} = m_\phi\boldsymbol{v}$ as f_χ. The event rate per unit target mass is $(\rho_\chi/m_\phi\rho_T)\langle\Gamma_{\rm abs}\rangle$.

D.19.4 Absorption from a classical field

For $N_\phi \gg 1$ the field can be treated as classical. Write it as a sum of modes with complex amplitudes,

$$\phi(\boldsymbol{x},t) = \sum_{\boldsymbol{k}}\big[\phi_{\boldsymbol{k}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega_{\boldsymbol{k}}t)} + \phi_{\boldsymbol{k}}^*\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega_{\boldsymbol{k}}t)}\big].$$

A virialized halo is a superposition of many modes with independent, uniformly distributed phases. By the central limit theorem the field at any point is then a Gaussian random variable, and all its correlations follow from $\langle\phi_{\boldsymbol{k}}\phi_{\boldsymbol{k}'}^*\rangle = \delta_{\boldsymbol{k}\boldsymbol{k}'}\langle|\phi_{\boldsymbol{k}}|^2\rangle$, with $\langle\phi_{\boldsymbol{k}}\phi_{\boldsymbol{k}'}\rangle = 0$. The energy density, $\frac12\dot\phi^2 + \frac12(\nabla\phi)^2 + \frac12 m_\phi^2\phi^2$, has the average $\sum_{\boldsymbol{k}}(\omega_{\boldsymbol{k}}^2 + k^2 + m_\phi^2)\langle|\phi_{\boldsymbol{k}}|^2\rangle = \sum_{\boldsymbol{k}} 2\omega_{\boldsymbol{k}}^2\langle|\phi_{\boldsymbol{k}}|^2\rangle$. The number density of quanta in mode $\boldsymbol{k}$ is its energy density divided by $\omega_{\boldsymbol{k}}$,

$$n_{\boldsymbol{k}} = 2\omega_{\boldsymbol{k}}\,\langle|\phi_{\boldsymbol{k}}|^2\rangle, \qquad \sum_{\boldsymbol{k}} n_{\boldsymbol{k}} \simeq \frac{\rho_\chi}{m_\phi}.$$

For a single mode, $\phi = \phi_0\cos(m_\phi t)$ has $|\phi_{\boldsymbol{0}}| = \phi_0/2$ and $\rho_\chi = \frac12 m_\phi^2\phi_0^2$, as in Sect. 2.1.

The interaction is now a sum of harmonic perturbations,

$$H'(t) = \sum_{\boldsymbol{k}} [\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{k}} t}\, \phi_{\boldsymbol{k}}\, \hat{O}_{\boldsymbol{k}} + \mathrm{e}^{\mathrm{i}\omega_{\boldsymbol{k}} t}\, \phi^*_{\boldsymbol{k}}\, \hat{O}^\dagger_{\boldsymbol{k}}], \qquad \hat{O}_{\boldsymbol{k}} = \sum_a g^\phi_a\, O_a(\boldsymbol{k}).$$

The first-order amplitude of Sect. 2.1 is a sum over modes, and its square contains products $\phi_{\boldsymbol{k}}\phi^*_{\boldsymbol{k}'}$. The Gaussian field average sets the products with $\boldsymbol{k} \neq \boldsymbol{k}'$ to zero. In a single realization, the time average sets them to zero once the observation time exceeds the coherence time t_{coh}, the inverse spread of the frequencies $\omega_{\boldsymbol{k}}$. The rate is then the sum over modes of the harmonic rate Eq. (2.8), with $\hat{O} = \phi_{\boldsymbol{k}}\hat{O}_{\boldsymbol{k}}$. Summed over f and thermally averaged, the absorption term gives the rate in the box

$$\begin{aligned}\sum_{\boldsymbol{k}} 2\pi\, \langle |\phi_{\boldsymbol{k}}|^2 \rangle \sum_{i,f} p_i\, |\langle f|\hat{O}_{\boldsymbol{k}}|i\rangle|^2\, \delta(\omega_{\boldsymbol{k}} - E_f + E_i) &= \mathscr{V} \sum_{\boldsymbol{k}} \frac{n_{\boldsymbol{k}}}{2\omega_{\boldsymbol{k}}} \sum_{a,b} g^{\phi\,*}_a g^\phi_b\, S_{ab}(\boldsymbol{k}, \omega_{\boldsymbol{k}}) \\ &= \mathscr{V} \sum_{\boldsymbol{k}} n_{\boldsymbol{k}}\, \Gamma_{\mathrm{abs}}(\boldsymbol{k}).\end{aligned}$$

Each quantum per unit volume is absorbed at the rate Γ_{abs}, as in the quantized calculation. The emission term gives $\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}$ times the absorption rate, with no spontaneous part. The quantized and classical descriptions agree for $N_\phi \gg 1$.

D.19.5 In-medium mixing of the dark photon

A dark photon A' couples to the charges of the target with ζe in place of e, so the charges respond to the vector potential $\zeta \boldsymbol{A}' + \boldsymbol{A}$, where $\boldsymbol{A}$ is the ordinary electromagnetic field that the target itself produces. Take a transverse mode of frequency ω and wavevector $\boldsymbol{k}$ with $\omega^2 - k^2 = m_\phi^2$. The induced current is fixed by the photon self-energy of Sect. 4.4, $\boldsymbol{J} = -\Pi_{\mathrm{em}}(\zeta \boldsymbol{A}' + \boldsymbol{A})$. The homogeneous case of this relation is the medium's wave equation $(\omega^2 - k^2 - \Pi_{\mathrm{em}})\boldsymbol{A} = 0$. The current sources the ordinary field through $(k^2 - \omega^2)\boldsymbol{A} = \boldsymbol{J}$, so that $\boldsymbol{A} = -\boldsymbol{J}/m_\phi^2$ at this frequency and wavevector. Eliminating $\boldsymbol{A}$ gives

$$\boldsymbol{J} = -\zeta\, \frac{m_\phi^2\, \Pi_{\mathrm{em}}}{m_\phi^2 - \Pi_{\mathrm{em}}}\, \boldsymbol{A}'.$$

The dark photon therefore sees the self-energy $\zeta^2 m_\phi^2 \Pi_{\mathrm{em}}/(m_\phi^2 - \Pi_{\mathrm{em}})$ instead of $\zeta^2 \Pi_{\mathrm{em}}$, and its absorption rate is $-1/\omega$ times the imaginary part of this self-energy. Since $\mathrm{Im}[\Pi_{\mathrm{em}}(m_\phi^2 - \Pi^*_{\mathrm{em}})] = m_\phi^2\, \mathrm{Im}\, \Pi_{\mathrm{em}}$,

$$\Gamma_{\mathrm{net}} = -\frac{\zeta^2}{\omega}\, \frac{m_\phi^4\, \mathrm{Im}\, \Pi_{\mathrm{em}}}{|m_\phi^2 - \Pi_{\mathrm{em}}|^2} = \zeta^2_{\mathrm{eff}}\, \sigma_1(\omega),$$

with $\mathrm{Im}\, \Pi_{\mathrm{em}} = -\omega\sigma_1$ from $\Pi_{\mathrm{em}} = -\mathrm{i}\omega\sigma_{\mathrm{cond}}$ and ζ_{eff} of Sect. 1.4 [33, Eqs. (5) and (6), 34, Eqs. (20)–(22)]. To first order in $\Pi_{\mathrm{em}}/m_\phi^2$, the in-medium factor in Γ_{net} is one and the rate is $\zeta^2\sigma_1$, the result of the proper response alone. The photon-reducible part of the response matters when m_ϕ is comparable to the plasma frequency, 16.6 eV in silicon (Sect. 4.3).

D.19.6 Detailed balance for the dark matter

Let the dark matter go from $\boldsymbol{p}$ to one final momentum $\boldsymbol{p}'$, depositing $\omega = E(\boldsymbol{p}) - E(\boldsymbol{p}')$, where the two states may belong to different mass eigenstates χ_1 and χ_2 of equal spin. By the golden rule and the steps that led to Eq. (8.15), the rate into $\boldsymbol{p}'$ is

$$\Gamma(\boldsymbol{p} \to \boldsymbol{p}') = \frac{2\pi}{\mathscr{V}^2} \sum_{a,b} \mathfrak{D}_{ab} \sum_{i,f} p_i\, \langle i|O^\dagger_a|f\rangle \langle f|O_b|i\rangle\, \delta(\omega - E_f + E_i).$$

The reverse transition, from $\boldsymbol{p}'$ and the target state $|f\rangle$ to $\boldsymbol{p}$ and $|i\rangle$, has the matrix element of the Hermitian interaction between the exchanged states, the complex conjugate of the forward one:

$$\langle \boldsymbol{p}, s; i|H_{\rm int}|\boldsymbol{p}', s'; f\rangle = \frac{1}{\mathscr{V}}\sum_b \langle s|\tilde{V}_b^\dagger|s'\rangle\,\langle i|O_b^\dagger|f\rangle.$$

Squaring, averaging over s' and summing over s gives

$$\sum_{a,b}\frac{\mathrm{Tr}_\chi[\tilde{V}_a\tilde{V}_b^\dagger]}{2|\boldsymbol{S}_\chi|+1}\,\langle f|O_a|i\rangle\langle i|O_b^\dagger|f\rangle,$$

and the cyclic trace is $\mathrm{Tr}_\chi[\tilde{V}_b^\dagger\tilde{V}_a] = (2|\boldsymbol{S}_\chi|+1)\,\mathfrak{D}_{ba}$. The initial target state is now $|f\rangle$, with weight p_f, and energy conservation is $\delta(-\omega - E_i + E_f)$, the same delta function. Relabeling $a \leftrightarrow b$ gives

$$\mathit{\Gamma}(\boldsymbol{p}' \to \boldsymbol{p}) = \frac{2\pi}{\mathscr{V}^2}\sum_{a,b}\mathfrak{D}_{ab}\sum_{i,f} p_f\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\,\delta(\omega - E_f + E_i),$$

and on the support of the delta function $p_f = p_i\,\mathrm{e}^{-\beta\omega}$. The two rates therefore obey Eq. (8.20). Only the Hermiticity of the interaction and the Gibbs weights entered, not the target's symmetries or the form of the coupling. If the dark-matter states had different spins, the ratio of the two rates would carry in addition the ratio of their multiplicities from the spin averages.

D.20 Moment matrices and Ward identities

This section derives the relations of Sect. 8.3 beyond the first moment. The species-resolved f-sum rule, Eq. (4.34), and the algebra behind it are derived in Appendix D.4, and we use them without repeating the derivation. The Hamiltonian of the electrons and nuclei is $H = \sum_i \boldsymbol{p}_i^2/2m_i + U$, with U a function of the positions and spins of all particles; species are labeled s, particles i, j, and nuclei of species κ also I, with positions $\boldsymbol{X}_I$.

D.20.1 Moments as commutator brackets

Sect. 8.3.2 derives the moment formula Eq. (8.22). The same expansion as in Eq. (4.11), carried to higher order, makes the coefficient of $z^{-(n+1)}$ in $R_{ab}(\boldsymbol{q}, z)$ equal to $\mathscr{V}^{-1}\langle[O_a^\dagger, [H, O_b]_n]\rangle$ for every n; for odd n the coefficient is $\mathscr{M}_{ab}^n$.

In a stationary state the average of $[H, Y]$ vanishes for any Y, so by the Jacobi identity $\langle[X,[H,Y]]\rangle = \langle[[X,H],Y]\rangle$. For the third moment the identity moves one commutator to the left,

$$\mathscr{M}_{ab}^3 = \frac{1}{\mathscr{V}}\big\langle\,[[O_a^\dagger, H],\,[H,[H,O_b]]]\,\big\rangle. \tag{D.72}$$

D.20.2 Commutators of a density

For one particle of mass m, write $A = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$. The shift $\boldsymbol{p}A = A(\boldsymbol{p}+\boldsymbol{q})$ (Appendix D.4) gives, for any function $h(\boldsymbol{p})$,

$$\Big[\frac{\boldsymbol{p}^2}{2m},\,A\,h(\boldsymbol{p})\Big] = A\,f(\boldsymbol{p})\,h(\boldsymbol{p}), \qquad f(\boldsymbol{p}) = \frac{(\boldsymbol{p}+\boldsymbol{q})^2 - \boldsymbol{p}^2}{2m} = \frac{\boldsymbol{q}\cdot\boldsymbol{p} + q^2/2}{m}, \tag{D.73}$$

where f is the energy a free particle of momentum $\boldsymbol{p}$ gains from the kick $\boldsymbol{q}$. With $h = 1$, Eq. (D.73) is the continuity equation, $[H, A] = Af = \frac{1}{2}\{\boldsymbol{q}\cdot\boldsymbol{p}/m, A\}$. Applying the commutator

with H once more, now with the potential energy, which commutes with A but not with f, gives

$$[H, Af] = Af^2 + \frac{\mathrm{i}}{m}\, A\, \boldsymbol{q}\cdot\nabla U, \tag{D.74}$$

since $[U, \boldsymbol{q}\cdot\boldsymbol{p}] = \mathrm{i}\boldsymbol{q}\cdot\nabla U$. Summed over the particles of species s, $[H, n_s] = \boldsymbol{q}\cdot\boldsymbol{j}_s = \sum_i A_i f_i$, and $[H,[H,n_s]] = \sum_i [A_i f_i^2 + (\mathrm{i}/m_s)A_i \boldsymbol{q}\cdot\nabla_i U]$. The adjoint is $(Af)^\dagger = A^\dagger g(\boldsymbol{p})$, with $g(\boldsymbol{p}) = (\boldsymbol{q}\cdot\boldsymbol{p} - q^2/2)/m$, and $[n_s^\dagger, H] = \sum_i A_i^\dagger g_i = \boldsymbol{q}\cdot\boldsymbol{j}_s^\dagger$.

D.20.3 Third moments

By Eq. (D.72), $\mathscr{M}^3_{ss'} = \mathscr{V}^{-1}\langle[\sum_{i\in s} A_i^\dagger g_i,\, \sum_{j\in s'}(A_j f_j^2 + (\mathrm{i}/m_{s'})A_j \boldsymbol{q}\cdot\nabla_j U)]\rangle$. Three kinds of terms survive.

Same particle, kinetic. Using $A^\dagger g(\boldsymbol{p})A = g(\boldsymbol{p}+\boldsymbol{q}) = f(\boldsymbol{p})$ and $Af(\boldsymbol{p})A^\dagger = f(\boldsymbol{p}-\boldsymbol{q}) = g(\boldsymbol{p})$ gives

$$[A^\dagger g, Af^2] = f^3 - g^3 = (f-g)(f^2+fg+g^2) = \frac{q^2}{m}\Big[\frac{3(\boldsymbol{q}\cdot\boldsymbol{p})^2}{m^2} + \Big(\frac{q^2}{2m}\Big)^2\Big].$$

Same particle, potential. $[A^\dagger g, (\mathrm{i}/m)A\,\boldsymbol{q}\cdot\nabla U] = (\mathrm{i}/m)\{f\,\boldsymbol{q}\cdot\nabla U - \boldsymbol{q}\cdot\nabla U\, g\} = (\mathrm{i}/m)\{[f, \boldsymbol{q}\cdot\nabla U] + \boldsymbol{q}\cdot\nabla U(f-g)\}$. The commutator in the braces is $-(\mathrm{i}/m)(\boldsymbol{q}\cdot\nabla)^2 U$, and the last term, $(\mathrm{i}q^2/m^2)\boldsymbol{q}\cdot\nabla U$, has zero average, because $\nabla_i U = -\dot{\boldsymbol{p}}_i$ and $\langle\dot{\boldsymbol{p}}_i\rangle = 0$ in a stationary state. The same-particle potential term is therefore $m^{-2}(\boldsymbol{q}\cdot\nabla_i)^2 U$.

Different particles. Kinetic terms of different particles commute. In the potential term, g_i fails to commute only with the dependence of $\nabla_j U$ on $\boldsymbol{r}_i$: $[A_i^\dagger g_i, (\mathrm{i}/m_{s'})A_j\boldsymbol{q}\cdot\nabla_j U] = (m_s m_{s'})^{-1} A_i^\dagger A_j (\boldsymbol{q}\cdot\nabla_i)(\boldsymbol{q}\cdot\nabla_j)U$.

Collecting the three kinds of terms, with $A_i^\dagger A_j = \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_i - \boldsymbol{r}_j)}$ and the same-particle potential term included as $i = j$, gives

$$\mathscr{M}^3_{ss'} = \delta_{ss'}\,\frac{n_s q^2}{m_s}\Big[\Big(\frac{q^2}{2m_s}\Big)^2 + \frac{3q^2\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2\rangle_s}{m_s^2}\Big] + \frac{1}{m_s m_{s'}\mathscr{V}}\Big\langle\sum_{i\in s}\sum_{j\in s'}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)}(\boldsymbol{q}\cdot\nabla_i)(\boldsymbol{q}\cdot\nabla_j)U\Big\rangle, \tag{D.75}$$

which is Eq. (8.27) for $s = s'$. For $s = e$ and $s' = \kappa$ only the part of U that depends on both an electron and a nucleus survives, $\sum_{j,I} w_\kappa(\boldsymbol{r}_j - \boldsymbol{X}_I)$ with $w_\kappa(\boldsymbol{y}) = -Z_\kappa\alpha/y$, for which $(\boldsymbol{q}\cdot\nabla_j)(\boldsymbol{q}\cdot\nabla_I)w_\kappa = -(\boldsymbol{q}\cdot\nabla)^2 w_\kappa$. With the conditional density defined by $N_\kappa n_e^{(\kappa)}(\boldsymbol{y}) = \langle\sum_j\sum_{I\in\kappa}\delta^3(\boldsymbol{r}_j - \boldsymbol{X}_I - \boldsymbol{y})\rangle$, where N_κ is the number of nuclei of species κ, the average becomes an integral over $\boldsymbol{y}$, and the mixed moment becomes Eq. (8.24). A spring $U = \frac{1}{2}K(\boldsymbol{r}-\boldsymbol{X})^2$ between one electron and one nucleus has $(\boldsymbol{q}\cdot\nabla_r)(\boldsymbol{q}\cdot\nabla_X)U = -Kq^2$, which gives $\mathscr{M}^3_{e\kappa} = -Kq^2\langle\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{X})}\rangle/(m_e M_\kappa\mathscr{V}) \to -Kq^2/(m_e M_\kappa\mathscr{V})$ at small q. The same number follows from the spectrum. In the relative mode, at $\omega_\mathrm{r}^2 = K/\mu$ with $\mu = m_e M_\kappa/(m_e + M_\kappa)$, the electron and nucleus displacements are $M_\kappa/(m_e+M_\kappa)$ and $-m_e/(m_e+M_\kappa)$ times the relative displacement, whose squared transition amplitude is $1/2\mu\omega_\mathrm{r}$. The cross weight $-q^2[m_e M_\kappa/(m_e+M_\kappa)^2]/2\mu\omega_\mathrm{r}$, times $2\omega_\mathrm{r}^3/\mathscr{V}$ for the two terms of Eq. (8.22), is $-Kq^2/(m_e M_\kappa\mathscr{V})$; the center-of-mass mode, in which the pair recoils as a whole with $\omega = q^2/2(m_e + M_\kappa)$, contributes at order q^6.

D.20.4 The mixed moment in a crystal

In Fourier space, $(\boldsymbol{q}\cdot\nabla)^2 w_\kappa$ has the transform $-(\boldsymbol{q}\cdot\boldsymbol{k})^2\tilde{w}_\kappa(\boldsymbol{k})$, with $\tilde{w}_\kappa(\boldsymbol{k}) = -Z_\kappa e^2/k^2$. For a static crystal, $n_e^{(\kappa)}(\boldsymbol{y}) = \sum_{\boldsymbol{G}} n_{\boldsymbol{G}}^{(\kappa)}\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{y}}$ with $n_{\boldsymbol{0}}^{(\kappa)} = n_e$ and $\sum_{\boldsymbol{G}} n_{\boldsymbol{G}}^{(\kappa)} = n_e^{(\kappa)}$, and the integral in Eq. (8.24) picks $\boldsymbol{k} = \boldsymbol{q} - \boldsymbol{G}$:

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q}) = -\frac{Z_\kappa e^2\, n_\kappa}{m_e M_\kappa}\sum_{\boldsymbol{G}} n_{\boldsymbol{G}}^{(\kappa)}\,\frac{[\boldsymbol{q}\cdot(\boldsymbol{q}-\boldsymbol{G})]^2}{|\boldsymbol{q}-\boldsymbol{G}|^2}. \tag{D.76}$$

The term $\boldsymbol{G}=0$ is exactly $n_e q^2$ at every q; it does not vanish faster than q^2 as $q\to 0$, because the Coulomb transform $1/k^2$ is evaluated at $k=q$. For $\boldsymbol{G}\neq 0$ and $q\ll G$, the weight is $(\boldsymbol{q}\cdot\hat{\boldsymbol{G}})^2[1+O(q/G)]$. On a site of cubic symmetry the coefficients $n_{\boldsymbol{G}}^{(\kappa)}$ are equal over each star of $\boldsymbol{G}$, and the average of $(\boldsymbol{q}\cdot\hat{\boldsymbol{G}})^2$ over a cubic star is $q^2/3$. The sum is then $q^2[n_e+(n_e^{(\kappa)}-n_e)/3]$, which is Eq. (8.25). The correction of first order in q/G is odd in $\boldsymbol{G}$; it cancels between $\boldsymbol{G}$ and $-\boldsymbol{G}$ when the site has inversion symmetry, $n_{-\boldsymbol{G}}^{(\kappa)}=n_{\boldsymbol{G}}^{(\kappa)}$, and survives otherwise. Since the density is real, $n_{-\boldsymbol{G}}^{(\kappa)}=(n_{\boldsymbol{G}}^{(\kappa)})^*$, and the surviving odd part is $\mathrm{i}\,\mathrm{Im}\,n_{\boldsymbol{G}}^{(\kappa)}$: the first-order correction is purely imaginary and enters $\mathrm{Im}\,\mathscr{M}^3_{e\kappa}$ alone, while the real part is corrected at order $(q/G)^2$ at every cubic site. In the middle regime of Sect. 8.3.3, q exceeds G for some stars, and the weight of these stars tends to q^2 instead of $q^2/3$.

For a density with spherical symmetry about the nucleus the angular integral can be done in real space. We write $(\boldsymbol{q}\cdot\nabla)^2(1/y)=-(4\pi/3)q^2\delta^3(\boldsymbol{y})+2q^2P_2(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{y}})/y^3$, with P_2 the Legendre polynomial; the angular integral of the plane wave against P_2 is $\int \mathrm{d}\Omega_y\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}P_2(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{y}})=-4\pi j_2(qy)$. With $w_\kappa=-Z_\kappa e^2/(4\pi y)$, Eq. (8.24) becomes

$$\mathscr{M}^3_{e\kappa}=-\frac{n_\kappa}{m_e M_\kappa}\Big(-\frac{Z_\kappa e^2}{4\pi}\Big)\Big[-\frac{4\pi}{3}q^2\bar{n}_e^{(\kappa)}(0)-8\pi q^2\int_0^\infty \mathrm{d}y\,\frac{\bar{n}_e^{(\kappa)}(y)\,j_2(qy)}{y}\Big],$$

which is Eq. (8.26). For a uniform density the integral is $n_e\int_0^\infty \mathrm{d}x\,j_2(x)/x=n_e/3$, and the mixed moment reproduces the $\boldsymbol{G}=0$ term of Eq. (D.76), $n_e q^2$ in total. The same check runs in momentum space: the spherical form equals $\int\frac{\mathrm{d}^3p}{(2\pi)^3}A(p/q)\,\tilde{n}(p)$ with $A(x)=\frac{1}{2}\int_{-1}^1\mathrm{d}u\,(1-xu)^2/(1+x^2-2xu)$, which tends to 1 for structure much smoother than $1/q$ and to $1/3$ for structure much sharper.

D.20.5 Existence of the fourth moment

The even moment $\mathscr{M}^4_{ss}$ is $\mathscr{V}^{-1}\langle X^\dagger X+XX^\dagger\rangle$ with $X=[H,[H,n_s]]$, a sum of nonnegative terms. By Eq. (D.74), X contains $(\mathrm{i}/m_s)\sum_i A_i\boldsymbol{q}\cdot\nabla_i U$, whose square contains the mean square of $\boldsymbol{q}\cdot\nabla_i U$. For the Coulomb attraction of a nucleus, $|\nabla w_\kappa|^2=(Z_\kappa\alpha)^2/y^4$, and against an electron density that is finite at $y=0$ its average $\int 4\pi y^2\mathrm{d}y\,n(y)\,(Z_\kappa\alpha)^2/y^4$ diverges at small y. The kinetic part of X, $\sum_i A_i f_i^2$, has a finite mean square, set by the average of p^4, which is finite even with the cusp of the wavefunction at the nucleus. By the triangle inequality for the norm $\langle X^\dagger X\rangle^{1/2}$ the kinetic part cannot cancel the divergence, and $\mathscr{M}^4$ is infinite for both electrons and nuclei. The third moment involves $\nabla^2 w_\kappa$ only, which is a delta function at the nucleus, and stays finite. In the Born–Oppenheimer approximation U is replaced, for the nuclei, by a smooth surface, and the force on a nucleus is bounded.

D.20.6 Momentum flux

Let $\boldsymbol{g}(\boldsymbol{q})=\sum_i\frac{1}{2}\{\boldsymbol{p}_i,A_i\}=\sum_i A_i(\boldsymbol{p}_i+\boldsymbol{q}/2)$. The kinetic part of its commutator with H follows from Eq. (D.73) with $h=p^\alpha+q^\alpha/2$:

$$\Big[\sum_j\frac{\boldsymbol{p}_j^2}{2m_j},\,g^\alpha\Big]=q_\beta\sum_i\frac{A_i}{m_i}\Big(p_i+\frac{q}{2}\Big)^\alpha\Big(p_i+\frac{q}{2}\Big)^\beta.$$

For a pair potential $U=\sum_{i<j}w_{ij}(\boldsymbol{r}_i-\boldsymbol{r}_j)$ the potential part is

$$\mathrm{i}\sum_i\partial_i^\alpha U\,A_i=\mathrm{i}\sum_{i<j}\partial^\alpha w_{ij}(\boldsymbol{r}_{ij})\,(A_i-A_j),$$

since the force of j on i is minus the force of i on j. With $\boldsymbol{r}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j$,

$$A_i - A_j = \mathrm{i} q_\beta r_{ij}^\beta \int_0^1 \mathrm{d}\lambda \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_j + \lambda \boldsymbol{r}_{ij})},$$

which contains the plane wave averaged along the segment between the two particles. The kinetic and potential parts both carry a factor q_β, and

$$\mathsf{T}^{\alpha\beta}(\boldsymbol{q}) = \sum_i \frac{A_i}{m_i} \Big(p_i + \frac{q}{2}\Big)^\alpha \Big(p_i + \frac{q}{2}\Big)^\beta - \sum_{i<j} \partial^\alpha w_{ij}(\boldsymbol{r}_{ij})\, r_{ij}^\beta \int_0^1 \mathrm{d}\lambda \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_j + \lambda \boldsymbol{r}_{ij})} \tag{D.77}$$

is the momentum flux of Eq. (8.30): the momentum carried by the particles plus the stress transmitted along each pair. There must be no external potential, which would add a force $\sum_i \nabla_i V_{\mathrm{ext}}$ that no factor of q cancels. The interactions must be pairwise, or more generally translation invariant with forces that sum to zero; spin-dependent interactions that are functions of positions and spins satisfy this condition.

The soft acoustic limit follows in two steps. Between eigenstates, Eq. (8.28) gives $\omega_{fi}\langle f|n_{\mathrm{M}}|i\rangle = q_\alpha \langle f|g^\alpha|i\rangle$ and Eq. (8.30) gives $\omega_{fi}\langle f|g^\alpha|i\rangle = q_\beta \langle f|\mathsf{T}^{\alpha\beta}|i\rangle$, so for $\omega_{fi} \neq 0$

$$\langle f|n_{\mathrm{M}}(\boldsymbol{q})|i\rangle = \frac{q_\alpha q_\beta}{\omega_{fi}^2}\, \langle f|\mathsf{T}^{\alpha\beta}(\boldsymbol{q})|i\rangle.$$

Inserted into Eq. (4.13) in the second slot, or in both, the relation gives Eq. (8.31). The matrix elements of $\mathsf{T}^{\alpha\beta}(\boldsymbol{q})$ between the ground state and a state of finite energy are finite as $q \to 0$, where they become the matrix elements of the total stress, so the mass-density amplitude to a state of finite energy is of order q^2.

D.20.7 The boost from momentum conservation

Eq. (8.32) also follows from the superadiabatic amplitude Eq. (8.53) summed over all nuclei at small q, where $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \to 1$ inside the electronic factor. The adiabatic states move rigidly with a rigid translation of all nuclei, so $\sum_I \nabla_I \Psi_j = -\sum_i \nabla_i \Psi_j$, and the summed derivative coupling is $\sum_I \boldsymbol{d}_{j0}^I = -\langle \Psi_j | \sum_i \nabla_i |\Psi_0\rangle = -\mathrm{i}\langle \Psi_j|\boldsymbol{P}_e|\Psi_0\rangle$, with $\boldsymbol{P}_e$ the total electron momentum. Since $\boldsymbol{P}_e = \mathrm{i} m_e [H_e, \sum_i \boldsymbol{r}_i]$, $\langle \Psi_j|\boldsymbol{P}_e|\Psi_0\rangle = \mathrm{i} m_e (E_j - E_0)\langle \Psi_j | \sum_i \boldsymbol{r}_i |\Psi_0\rangle$. The summed amplitude Eq. (8.53) is then $-\mathrm{i}\boldsymbol{q}\cdot\sum_I \boldsymbol{d}_{j0}^I/[M_\kappa(E_j - E_0)] = -(m_e/M_\kappa)\langle \Psi_j|\mathrm{i}\boldsymbol{q}\cdot\sum_i \boldsymbol{r}_i|\Psi_0\rangle$, which is $-(m_e/M_\kappa)\langle \Psi_j|n_e(\boldsymbol{q})|\Psi_0\rangle$ to first order in q. The two routes agree.

D.20.8 The electron third moment

Multiply Eq. (8.29), with $O_B = n_e$, by ω^3, add the term with $-\boldsymbol{q}$, and integrate. By Eq. (8.30) the right side is $q_\alpha q_\beta$ times the first moment of n_e and $\mathsf{T}^{\alpha\beta}$, $\mathscr{V}^{-1}\langle [n_e^\dagger, [H, \mathsf{T}^{\alpha\beta}]]\rangle$, and the outer commutator with $n_e^\dagger$ brings one more factor of q. Hence $\sum_{s'} m_{s'} \mathscr{M}^3_{es'} = O(q^3)$, and the terms of order q^2 of $m_e \mathscr{M}^3_{ee}$ and $\sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa}$ must cancel.

We check the cancellation directly. In Eq. (D.75) with $s = s' = e$, the kinetic bracket is of order q^4. In the potential term, the electron–nucleus energy contributes only at $i = j$, $\sum_{i,I} (\boldsymbol{q}\cdot\nabla)^2 w_\kappa(\boldsymbol{r}_i - \boldsymbol{X}_I)$, whose average over a cubic crystal keeps the trace, $q^2\nabla^2 w_\kappa/3 = Z_\kappa e^2 q^2 \delta^3/3$, and gives $(e^2q^2/3m_e^2)\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa \boldsymbol{n}_e^{(\kappa)}$. The electron–electron energy $\sum_{i<j} e^2/(4\pi r_{ij})$ contributes at $i = j$ and at $i \neq j$, and the two contributions combine into $\sum_{i\neq j} (\boldsymbol{q}\cdot\nabla)^2 w_{ee}(\boldsymbol{r}_{ij})[1 - \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_{ij}}]$. The bracket vanishes at $\boldsymbol{r}_{ij} = 0$, so the contact term and the short-range exchange–correlation hole contribute only at higher order in q. Against the uniform density $\boldsymbol{n}_e$, the term 1 gives $\boldsymbol{n}_e \int \mathrm{d}^3 r\, (\boldsymbol{q}\cdot\nabla)^2 w_{ee} = -e^2 q^2 \boldsymbol{n}_e/3$, since the quadrupolar part averages out over angles and the contact part remains. The phase term gives $-\boldsymbol{n}_e \int \mathrm{d}^3 r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} (\boldsymbol{q}\cdot\nabla)^2 w_{ee} = \boldsymbol{n}_e q^4 \tilde{w}_{ee}(q) =$

$+e^2q^2\boldsymbol{n}_e$, with $\tilde{w}_{ee} = e^2/q^2$. Together the electron pairs contribute $\frac{2}{3}e^2q^2\boldsymbol{n}_e$ per electron. The periodic part of the density cancels between the two terms on cubic sites, star by star as in Eq. (D.76). Altogether

$$m_e\mathscr{M}^3_{ee} = \frac{e^2q^2}{3m_e}\Big[\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa \boldsymbol{n}_e^{(\kappa)} + 2\boldsymbol{n}_e^2\Big] + O(q^3),$$

which is Eq. (8.33). By Eq. (8.25) and neutrality, $\sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa} = -(e^2q^2/3m_e)[\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa \boldsymbol{n}_e^{(\kappa)} + 2\boldsymbol{n}_e^2]$, and the terms of order q^2 of $m_e\mathscr{M}^3_{ee}$ and $\sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa}$ cancel, as they must. In jellium the nuclei are a rigid uniform background with $\nabla^2 U_{\rm bg} = e^2\boldsymbol{n}_e$ at every electron, the contact sum becomes $\boldsymbol{n}_e^2$, and $\mathscr{M}^3_{ee} = (\boldsymbol{n}_e q^2/m_e)(e^2\boldsymbol{n}_e/m_e) = \boldsymbol{n}_e q^2\omega_p^2/m_e$ at small q, the third moment of the electron gas.

D.21 Screening for every external coupling

This section derives the screened response Eq. (8.35) for any pair of target rows, its total-charge and electron-density forms, the exactness theorem Eq. (8.38) in full, and the classification of Coulomb lines of Sect. 8.4.4. The notation follows Sects. 4.4 and 8.4: $\boldsymbol{q}$ lies in the first zone, the reciprocal-lattice vectors $\boldsymbol{G}$ label the channel $\{\boldsymbol{q}+\boldsymbol{G}\}$, $v_{\boldsymbol{G}} \equiv v_{\rm C}(|\boldsymbol{q}+\boldsymbol{G}|)$, and the dark-matter vertex is at the row $\boldsymbol{G}_q$ of the unreduced momentum (Eq. (A.12)).

D.21.1 The screened response

The Coulomb Hamiltonian Eq. (2.48) couples the species only through $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$, by the term $(2\mathscr{V})^{-1}\sum_{\boldsymbol{k}\neq 0} v_{\rm C}(k)\, n_Z(\boldsymbol{k})^\dagger n_Z(\boldsymbol{k})$ (the self-energies of the point charges aside). In the presence of a weak external field, let $\delta\boldsymbol{n}_{Z,\boldsymbol{G}} = \delta\langle n_Z(\boldsymbol{q}+\boldsymbol{G})^\dagger\rangle/\mathscr{V}$ be the induced charge-number density at $\boldsymbol{q}+\boldsymbol{G}$. The induced density adds to the Hamiltonian the mean-field term $v_{\boldsymbol{G}}\,\delta\boldsymbol{n}_{Z,\boldsymbol{G}}\, n_Z(\boldsymbol{q}+\boldsymbol{G})$ plus its conjugate, a field of amplitude $v_{\boldsymbol{G}}\delta\boldsymbol{n}_{Z,\boldsymbol{G}}$ on the row $n_Z(\boldsymbol{q}+\boldsymbol{G})$. The proper response of the channel is defined by Eq. (8.34) as the response to the external fields ϕ_b and to these Hartree fields together. The proper response $\tilde{R}$ includes everything else that the Coulomb interaction does, including the Coulomb lines at momenta outside the channel and the fluctuations of the channel components beyond their mean.

Write the induced charge as a column vector $\delta\boldsymbol{n}_Z$ over $\boldsymbol{G}$, the proper charge–charge block as the matrix $P_{\boldsymbol{G}\boldsymbol{G}'} = \tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}$, and v as the diagonal matrix of $v_{\boldsymbol{G}}$. For a field on row b alone, Eq. (8.34) with $a = Z_{\boldsymbol{G}}$ reads

$$\delta\boldsymbol{n}_Z = \tilde{R}_{Zb}\,\phi_b + P\,v\,\delta\boldsymbol{n}_Z, \qquad \delta\boldsymbol{n}_Z = (1-Pv)^{-1}\tilde{R}_{Zb}\,\phi_b, \tag{D.78}$$

with $\tilde{R}_{Zb}$ the column of $\tilde{R}_{Z_{\boldsymbol{G}}b}$. Inserting this solution into Eq. (8.34) for a general row a gives

$$\delta O_a = [\tilde{R}_{ab} + \tilde{R}_{aZ}\,v\,(1-Pv)^{-1}\tilde{R}_{Zb}]\phi_b,$$

with $\tilde{R}_{aZ}$ the row vector of $\tilde{R}_{aZ_{\boldsymbol{G}}}$. The push-through identity $v(1-Pv)^{-1} = (1-vP)^{-1}v$ follows from $(1-vP)v = v(1-Pv)$ by multiplying with the two inverses, and it turns the bracket into

$$R_{ab} = \tilde{R}_{ab} + \tilde{R}_{aZ}\,\varepsilon^{-1}v\,\tilde{R}_{Zb}, \qquad \varepsilon = 1 - vP, \tag{D.79}$$

which is Eq. (8.35) with $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\boldsymbol{G}}\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}$. The same steps give the Dyson form $R_{ab} = \tilde{R}_{ab} + \tilde{R}_{aZ}\,v\,R_{Zb}$, since $R_{Zb} = (1-Pv)^{-1}\tilde{R}_{Zb}$ by Eq. (D.78). For the electron density alone, $a = b = e$ and $n_Z = -n_e$, the two forms reduce to Eq. (4.40) and Eq. (4.43), and the one-species algebra of Appendix D.5 is the special case. The matrix ε is invertible at

every real ω except at the self-sustained longitudinal oscillations of the target, plasmons and longitudinal optical phonons, where R itself has its poles.

With a single $\boldsymbol{G}$, P and v are numbers, and Eq. (D.79) is Eq. (8.36). Written as a matrix over the rows a, b of the species, the screening term $v\tilde{R}_{aZ}\tilde{R}_{Zb}/\varepsilon$ is a rank-one correction of $\tilde{R}$, and Eq. (8.36) is the Sherman–Morrison form of the inverse in species space.

Total charge. Take $a = b = Z$, so that $\tilde{R}_{aZ} = \tilde{R}_{Zb} = P$. Since $vP = 1 - \varepsilon$ and $1 - \varepsilon$ commutes with ε^{-1},

$$vR_{ZZ} = vP + vP\,\varepsilon^{-1}\,vP = (1-\varepsilon)\big[1 + \varepsilon^{-1}(1-\varepsilon)\big] = (1-\varepsilon)\,\varepsilon^{-1} = \varepsilon^{-1} - 1. \tag{D.80}$$

The element of Eq. (D.80) at $\boldsymbol{G} = \boldsymbol{G}' = \boldsymbol{G}_q$ is $v_{\mathrm{C}}(q)R_{ZZ}(\boldsymbol{q},\omega) = (\varepsilon^{-1})_{00}(\boldsymbol{q},\omega) - 1$. With the fluctuation–dissipation theorem Eq. (4.18), $S_{ZZ} = -2\,\mathrm{Im}\,R_{ZZ}/(1 - \mathrm{e}^{-\beta\omega}) = (2q^2/e^2)\,L/(1 - \mathrm{e}^{-\beta\omega})$, with $L = \mathrm{Im}[-(\varepsilon^{-1})_{00}]$ of the total dielectric matrix. No approximation enters: the relation holds for moving nuclei, any temperature and any ω.

Electron density. The electron density is $n_e = -n_Z + \sum_\kappa Z_\kappa n_\kappa$, and the response of n_e is bilinear in its two rows, $-n_Z$ and $\sum_\kappa Z_\kappa n_\kappa$. Hence

$$R_{ee} = R_{ZZ} - \sum_\kappa Z_\kappa\big(R_{Z\kappa} + R_{\kappa Z}\big) + \sum_{\kappa\kappa'} Z_\kappa Z_{\kappa'} R_{\kappa\kappa'}. \tag{D.81}$$

With the nuclei clamped, each n_κ is a c-number Bragg sum, its commutators vanish, and every term with a nuclear row drops out: $R_{ee} = R_{ZZ}$, and the electron density is screened by the loss function of the total dielectric matrix, which is then the electronic one. With moving nuclei the nuclear terms describe the charge that the nuclei carry and the electrons they drag, and they make the electron-channel function $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}}R_{ee}$ of Appendix A.7 differ from ε^{-1}.

D.21.2 Exactness of the loss-function form

Let the coupling be $O_c = \sum_s c_s n_s(\boldsymbol{q})$, with $c_s = c_1^{(s)}$ of Sect. 8.1, real, independent of $\boldsymbol{q}$ (nuclear form factors set to one), and S_{cc} its structure factor. By Eq. (D.80), Eq. (8.38) is the statement $S_{cc} = \lambda_Z^2 S_{ZZ}$ at every $\boldsymbol{q}$ and every $\omega \neq 0$; at $\omega = 0$ the static Bragg terms of a clamped lattice differ, and the statement holds there only for the charge itself, with $D = 0$ below. The target is a crystal: the small-momentum step below uses its long-wavelength longitudinal acoustic phonon.

Sufficiency. Suppose $O_c - \lambda_Z n_Z = D$ has no matrix elements between states of different energy. In the Lehmann form Eq. (4.13), $\langle f|O_c|i\rangle = \lambda_Z\langle f|n_Z|i\rangle$ for every pair with $E_f \neq E_i$, and $S_{cc} = \lambda_Z^2 S_{ZZ}$ at every $\omega \neq 0$. For $c_e = -\lambda_Z$, $c_\kappa = \lambda_Z Z_\kappa$, $D = 0$. With clamped nuclei, $D = (c_e + \lambda_Z)n_e + \sum_\kappa (c_\kappa - \lambda_Z Z_\kappa)n_\kappa$ is static exactly when $c_e = -\lambda_Z$, for any c_κ; the sign of λ_Z is free.

Necessity, large momentum. Let the nuclei move. For q large compared with the inverse size of every electron orbital and with $q^2/2M_\kappa$ large compared with the phonon energies, the impulse regime of Sect. 8.5, every particle recoils as if free during the time $1/\omega$ that the response probes. The self term of each particle in $\langle O_c(\boldsymbol{q},t)^\dagger O_c(\boldsymbol{q},0)\rangle$ then gives a recoil peak centered at $q^2/2m_s$. Its width is of order $q\,p_s/m_s$, with p_s the root-mean-square momentum of the particle in the target, and its weight is $\int \mathrm{d}\omega\, S_{cc}/2\pi = c_s^2 n_s$ per species, by the zeroth moment of the self term. The terms between two different particles involve $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i(t)}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j(0)}\rangle$, the Fourier transform of a pair correlation that is smooth on the scale $1/q$, and vanish as $q \to \infty$; in particular the electrons no longer follow the nuclei, and the electron–nucleus interference disappears. The distance between the peaks of two masses grows as q^2 and their widths as q, so at large enough q each mass has a separate peak. Group the nuclear species by mass. Since nuclei of different atomic number have different masses, the nuclei of one group

share an atomic number Z_g; let n_g be their total density. Equality of S_{cc} and $\lambda_Z^2 S_{ZZ}$ peak by peak requires

$$c_e^2 = \lambda_Z^2, \qquad \sum_{\kappa \in g} c_\kappa^2 n_\kappa = \lambda_Z^2 Z_g^2 \, n_g \quad \text{for every mass group } g. \tag{D.82}$$

Necessity, small momentum. At $\boldsymbol{q} \to 0$ and the frequency of a longitudinal acoustic phonon, the target is displaced locally as a whole: a long-wavelength acoustic phonon is a slow, local translation of the crystal, and the electrons go with it. A displacement field $\boldsymbol{U}(\boldsymbol{r})$ that varies on the scale $1/q$ moves each particle from $\boldsymbol{r}_j$ to $\boldsymbol{r}_j + \boldsymbol{U}(\boldsymbol{r}_j)$, and $n_s(\boldsymbol{q}) \to \sum_{j \in s} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}[1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{U}(\boldsymbol{r}_j)]$. For the mode of wavevector $\boldsymbol{q}$, $\boldsymbol{U}(\boldsymbol{r}) \propto \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and the amplitude to create the phonon is proportional to $q \sum_s c_s N_s$, with N_s the number of particles of species s, up to corrections of relative order q. The weight of the acoustic peak in S_{cc} is therefore $q^2 (\sum_s c_s n_s)^2$ times a factor common to all couplings. For n_Z the sum is the net charge-number density, which vanishes by neutrality, $n_e = \sum_g Z_g n_g$, and the acoustic weight of S_{ZZ} starts at order q^4, from the charge that a long-wavelength strain induces. Equality as $q \to 0$ requires

$$c_e n_e + \sum_\kappa c_\kappa n_\kappa = 0. \tag{D.83}$$

Conclusion. Choose the sign of λ_Z so that $c_e = -\lambda_Z$ with $\lambda_Z > 0$; Eq. (D.83) becomes $\sum_\kappa c_\kappa n_\kappa = \lambda_Z n_e = \lambda_Z \sum_g Z_g n_g$. By the Cauchy–Schwarz inequality with the weights n_κ, and by Eq. (D.82),

$$\sum_{\kappa \in g} c_\kappa n_\kappa \le \Big(\sum_{\kappa \in g} c_\kappa^2 n_\kappa \Big)^{1/2} n_g^{1/2} = \lambda_Z Z_g \, n_g,$$

with equality only if $c_\kappa = \lambda_Z Z_g$ for every species of the group. Summed over the groups, the left-hand side reaches $\lambda_Z \sum_g Z_g n_g$ only if equality holds in every group. Hence $c_e = -\lambda_Z$ and $c_\kappa = \lambda_Z Z_\kappa$ for every κ: $O_c = \lambda_Z n_Z$. With the nuclei clamped there are no nuclear recoil peaks and no acoustic phonon, and Eq. (D.82) leaves only $c_e^2 = \lambda_Z^2$, which sufficiency already covers. Sufficiency and necessity together prove Eq. (8.38).

The hypothesis on the masses excludes isobars of different atomic number in one target, which no crystal of this book contains. Without it, two species of equal mass and different Z would share a recoil peak, and Eq. (D.82) would constrain only a combination of their couplings.

D.21.3 Coulomb-line classes

The classification of Sect. 8.4.4 is a statement about the diagrams of the response at the random-phase level, also called time-dependent Hartree, at order g^2 in the rate with one phonon line. In this approximation every induced density acts on the electrons only through its Hartree field. A diagram is then built from fermion loops, each a Kohn–Sham response of some order. A loop takes its input fields, external, Hartree or ionic, and returns one induced density. That density reaches the rest of the diagram through one Coulomb line, or it is the output of the response. Each loop therefore has exactly one line toward the output, and the loops and Coulomb lines form a tree. Lines inside a loop, the exchange and ladder diagrams of Sect. 4.8, are absent at this level.

Now add the phonon line, which joins two ion vertices on two loops of the diagram.

The diagram without the phonon line is connected. Then the diagram is a single tree, and adding one line to a tree closes exactly one cycle. Every Coulomb line off the cycle is a bridge: cutting it separates a piece that is attached to the rest only through that line. Momentum conservation fixes the momentum of the line to the total momentum entering that piece. The piece contains no part of the cycle and at most the two external vertices, so the momentum is

$\boldsymbol{q}$ up to a reciprocal-lattice vector and the frequency is ω. The Coulomb lines off the cycle are the external lines, and their chains are resummed by Eq. (D.79). On the cycle, let A and B be the loops at which the paths from the two external vertices meet it; a loop that carries an external vertex and lies on the cycle is its own meeting point. The cycle consists of the phonon line and a path from one ion vertex to the other that passes through A and B. Momentum is conserved at every loop, and the only momenta that enter the cycle from outside are $\boldsymbol{q}$ at A and $-\boldsymbol{q}$ at B (or the reverse). Follow the cycle from the ion vertex at one end, with $\boldsymbol{Q}$ the momentum of the phonon line. The momentum is $\boldsymbol{Q}+\boldsymbol{G}$ up to the first of A and B, it is $\boldsymbol{Q}+\boldsymbol{q}+\boldsymbol{G}$ between A and B, and it is $\boldsymbol{Q}+\boldsymbol{G}$ again from the second of them to the other ion vertex. The frequencies follow in the same way: the frequency ω' of the phonon line on the segments next to the ion vertices, $\omega-\omega'$ or $\omega+\omega'$ between A and B, and $\omega'=\pm\omega_{\boldsymbol{Q}\nu}$ when the phonon is on shell in the final state. The segments next to the ion vertices are the vertex lines, and the segment between A and B is the pair segment; when $A=B$, it is empty. Every Coulomb line is thus external, vertex or pair, and only one of these.

The chains on the vertex segments, summed over all diagrams, dress the bare potential of the displaced ion with every Coulomb chain at $(\boldsymbol{Q}+\boldsymbol{G},\omega')$ before it reaches the first loop that carries an external vertex. The dressed potential is $\varepsilon^{-1}(\boldsymbol{Q},\omega')$ acting on the bare ionic potential, Eq. (6.10) at the frequency ω'. The chains on the pair segment between two loops sum to $v+vRv=\varepsilon^{-1}v$ at $(\boldsymbol{q}+\boldsymbol{Q},\omega\mp\omega_{\boldsymbol{Q}\nu})$, the screened interaction W^{scr} of Sect. 4.8.

The diagram without the phonon line falls into two trees. Then the phonon line is a bridge, each tree contains one external vertex, and the phonon line carries $(\boldsymbol{q}+\boldsymbol{G},\omega)$. This phonon line is the ionic polarization inserted into the external chain, and summing such insertions turns the electronic dielectric matrix into the total one. This case does not describe a pair with a phonon in the final state: its phonon line is virtual and screens the dark-matter vertex. In the amplitude of the phonon-assisted process the virtual phonon line would be a second phonon line, at $(\boldsymbol{q}+\boldsymbol{G},\omega)$, beside the real phonon at $\boldsymbol{Q}$, and it lies outside the one-phonon diagrams of the first case. Using the total ε of Eq. (D.79) in the external and pair chains includes the virtual phonon line once.

The assignment uses only the position of a line relative to the cycle and the external vertices, not the values of the momenta. It remains unique when the momenta coincide modulo $\boldsymbol{G}$, at $\boldsymbol{q}\to 0$, where the pair and vertex segments both carry $\boldsymbol{Q}$, or at $\boldsymbol{Q}\to 0$, where the pair segment and the external lines both carry $\boldsymbol{q}$.

The static vertex. Density-functional perturbation theory evaluates the vertex chains at $\omega'=0$. For densities the reality of the response in real space gives $R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q},\omega')^*=R_{-\boldsymbol{G},-\boldsymbol{G}'}(-\boldsymbol{Q},-\omega')$, and time reversal of the equilibrium state gives $R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q},\omega')=R_{-\boldsymbol{G}',-\boldsymbol{G}}(-\boldsymbol{Q},\omega')$. Together, $R(\boldsymbol{Q},-\omega')=R(\boldsymbol{Q},\omega')^\dagger$. Below the gap the response has no absorptive part, its anti-Hermitian part vanishes, and $R(\boldsymbol{Q},-\omega')=R(\boldsymbol{Q},\omega')$; the same holds for the proper response and for ε. The response is analytic in ω' for $|\omega'|<E_{\mathrm{g}}$, since its poles lie at the excitation energies, and an even analytic function differs from its value at zero by a term of order ω'^2. With the scale E_{g} set by the nearest poles, the static vertex errs by a relative amount of order $(\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}})^2$. Without time reversal, in a magnetic insulator, only the reality relation remains, which ties ω' to $-\omega'$ at $-\boldsymbol{Q}$; the linear term survives, and the error is of order $\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}}$.

D.22 The decomposition theorem

A response summed over sectors of final states equals the full response only if the sector projectors commute with the Hamiltonian, and none of the projectors of Sect. 8.6 does so exactly. The notation follows Sect. 8.6, and the adiabatic states, surfaces and couplings follow

Sects. 5.1 and 6.1.

D.22.1 Sectors and the bin bound

Let $\mathcal{P}_A$ be orthogonal projectors with $\sum_A \mathcal{P}_A = 1$, write $H = H_\mathrm{d} + H_\mathrm{off}$ with $H_\mathrm{d} = \sum_A \mathcal{P}_A H \mathcal{P}_A$, and fix one initial state $|i\rangle$ with $\psi_b = O_b|i\rangle$. A measured spectrum is never resolved better than some bin, and we describe the bin by a shape $w_\mathrm{b}(x) \geq 0$ with $\int \mathrm{d}x\, w_\mathrm{b} = 1$, its width $\Delta\omega_\mathrm{bin}$, the kernel $f_\Delta(x) = w_\mathrm{b}(x/\Delta\omega_\mathrm{bin})/\Delta\omega_\mathrm{bin}$ and the transform $\hat{w}_\mathrm{b}(s) = \int \mathrm{d}x\, w_\mathrm{b}(x)\, \mathrm{e}^{-\mathrm{i}sx}$. Without the factor $2\pi/\mathcal{V}$, the binned response of one initial state is $S^\mathrm{bin}_{ab}(\omega) = \langle i|O_a^\dagger f_\Delta(\omega + E_i - H) O_b|i\rangle$. Since $f_\Delta(\omega + E_i - H_\mathrm{d})$ commutes with every $\mathcal{P}_A$, it has no blocks between different sectors, and

$$S^\mathrm{bin}_{ab} = \sum_A \langle i|O_a^\dagger\, \mathcal{P}_A\, f_\Delta(\omega + E_i - H_\mathrm{d})\, \mathcal{P}_A\, O_b|i\rangle + C^\mathrm{cross}_{ab},$$
$$C^\mathrm{cross}_{ab} = \langle O_a i|\big[f_\Delta(\omega + E_i - H) - f_\Delta(\omega + E_i - H_\mathrm{d})\big]|\psi_b\rangle.$$

Each sector term is a restriction by orthogonal projectors, positive for $a = b$ and positive semidefinite as a matrix. To bound C^cross_{ab}, write the kernel as a time integral, $f_\Delta(\omega + E_i - H) = \int (\mathrm{d}t/2\pi)\, \hat{w}_\mathrm{b}(\Delta\omega_\mathrm{bin} t)\, \mathrm{e}^{\mathrm{i}(\omega + E_i)t} \mathrm{e}^{-\mathrm{i}Ht}$, and compare the evolutions under H and H_d with the Duhamel formula, which follows by differentiating $\mathrm{e}^{\mathrm{i}Ht}\mathrm{e}^{-\mathrm{i}H_\mathrm{d}t}$:

$$(\mathrm{e}^{-\mathrm{i}Ht} - \mathrm{e}^{-\mathrm{i}H_\mathrm{d}t})\psi_b = -\mathrm{i}\int_0^t \mathrm{d}t'\, \mathrm{e}^{-\mathrm{i}H(t-t')}\, H_\mathrm{off}\, \mathrm{e}^{-\mathrm{i}H_\mathrm{d}t'}\, \psi_b.$$

With $\mathfrak{h}_\mathrm{off} = \sup_{t'} \|H_\mathrm{off} \mathrm{e}^{-\mathrm{i}H_\mathrm{d}t'}\psi_b\|/\|\psi_b\|$ the norm of the right side is at most $|t|\, \mathfrak{h}_\mathrm{off} \|\psi_b\|$. We insert this bound in the time integral and substitute $y = \Delta\omega_\mathrm{bin} t$; the Cauchy–Schwarz inequality then gives

$$|C^\mathrm{cross}_{ab}| \leq \mathfrak{c}_\mathrm{b}\, \frac{\mathfrak{h}_\mathrm{off}}{\Delta\omega^2_\mathrm{bin}}\, \|O_a|i\rangle\|\, \|O_b|i\rangle\|, \qquad \mathfrak{c}_\mathrm{b} = \frac{1}{2\pi}\int \mathrm{d}y\, |y|\, |\hat{w}_\mathrm{b}(y)|. \tag{D.84}$$

For a Gaussian bin of standard deviation $\Delta\omega_\mathrm{bin}$, $w_\mathrm{b}(x) = \mathrm{e}^{-x^2/2}/\sqrt{2\pi}$, the transform is $\mathrm{e}^{-y^2/2}$, the integral $\int \mathrm{d}y\, |y|\, \mathrm{e}^{-y^2/2} = 2$, and $\mathfrak{c}_\mathrm{b} = 1/\pi$, the constant of Eq. (8.63). A rectangular bin has a transform that falls only as $1/y$, and $\mathfrak{c}_\mathrm{b}$ diverges: the bound requires bins with smooth edges. Only the state ψ_b that the probe creates enters $\mathfrak{h}_\mathrm{off}$, not the norm of H_off over the whole space, which is proportional to the volume for a local coupling.

D.22.2 The electron-following projector

The hypotheses of Sect. 8.6 are the conditions under which the space-adiabatic theorem applies. The electronic Hamiltonian $H_\mathrm{el}(u)$ must be a smooth function of the nuclear positions; with nuclei of finite size, rigid charge distributions $Z_\kappa F^\mathrm{ch}_\kappa$ with F^ch_κ the nuclear charge form factor, the electron–nucleus potential is smooth and Spohn and Teufel's regularity condition holds, whereas for point nuclei it fails [35, Sect. 2]. The electron–electron repulsion does not depend on u and does not enter. The ground-state energy $E_0(u)$ must be nondegenerate and separated from the rest of the spectrum of $H_\mathrm{el}(u)$ by a gap, here at least E_g, on the configurations that the vibrational state visits, up to exponentially small tails. The lattice must be harmonic with perturbative anharmonicity, and $T \ll E_\mathrm{g}$.

In units where the electronic scales are one and the nuclear momenta are measured in units of $(M/m_e)^{1/2}$, the nuclear kinetic energy is $-\kappa^4_\mathrm{BO}\nabla^2/2$, the small parameter of the semiclassical expansion is $(m_e/M)^{1/2} = \kappa^2_\mathrm{BO}$, and the full Hamiltonian is the quantization of the operator-valued symbol $\frac{1}{2}p^2 + H_\mathrm{el}(u)$ [36, Eq. (38)]. Theorem 2.1 of Panati, Spohn

and Teufel then provides an orthogonal projector Π whose symbol is a series $\sum_j \kappa_{\mathrm{BO}}^{2j}\pi_j$ with leading term the spectral projector of the electronic ground state, $\pi_0(u) = |\Psi_0(u)\rangle\langle\Psi_0(u)|$, and with $[H,\Pi]$ smaller than any power of κ_{BO} [36, Thm. 2.1]. The Born–Oppenheimer Hamiltonian grows quadratically in the nuclear momentum while the gap does not, and the theorem is applied below an arbitrary fixed energy through an auxiliary symbol that equals the true symbol on the region of phase space where the classical energy $\frac{1}{2}p^2 + E_0(u)$ lies below the fixed energy [36, Sect. 4.5]. Truncating the series after $j = n$ defines $\mathcal{P}_{\mathrm{ad}}$, and since $(1-\mathcal{P}_{\mathrm{ad}})H\mathcal{P}_{\mathrm{ad}} = (1-\mathcal{P}_{\mathrm{ad}})[H,\mathcal{P}_{\mathrm{ad}}]\mathcal{P}_{\mathrm{ad}}$, the leak is of the order of the first omitted term, $\kappa_{\mathrm{BO}}^{2(n+1)}$, in operator norm on the states below that energy. This bound is Eq. (8.47). Low-lying vibrational states lie below any fixed energy above the zero-point energy, so the bound applies to them with the same order. The constants depend on the number of slow coordinates, and for this reason the proof covers a finite crystal.

The first correction to the state follows more simply from the coupled equations Eq. (5.4). Let a state on the ground surface have the nuclear wavefunction Ψ_0^{nuc} and the energy E, and let Ψ_j^{nuc} be its component on the excited adiabatic state Ψ_j, as in Sect. 5.1. To first order in the coupling, $[T_N + E_j(u) - E]\Psi_j^{\mathrm{nuc}} = -\mathcal{N}_{j0}\Psi_0^{\mathrm{nuc}}$, with T_N the kinetic energy of the nuclei. The derivative coupling dominates $\mathcal{N}_{j0}\Psi_0^{\mathrm{nuc}}$, which is $-M_\kappa^{-1}\boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\mathrm{nuc}}$ by Eqs. (5.5) and (6.1), and the operator in brackets on the left is $E_j(u) - E_0(u) \equiv \Delta E_j$ up to vibrational energies. Hence

$$\Psi_j^{\mathrm{nuc}} = \frac{\boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\mathrm{nuc}}}{M_\kappa\,\Delta E_j} = \frac{\mathrm{i}\,\boldsymbol{d}_{j0}\cdot\boldsymbol{p}\,\Psi_0^{\mathrm{nuc}}}{M_\kappa\,\Delta E_j}, \tag{D.85}$$

with $\boldsymbol{p} = -\mathrm{i}\nabla$ the nuclear momentum, a sum over nuclei being understood. Eq. (D.85) is the first-order term of the unitary of Panati, Spohn and Teufel, written in physical units [36, below Eq. (39)].

The same order of the expansion changes the mass. The second-order effective Hamiltonian on the ground surface is [36, Eq. (40)], for a real nondegenerate state with vanishing Berry connection,

$$h_{\mathrm{BO}} = \frac{p^2}{2M} + E_0(u) + \frac{1}{2M}\sum_{j\neq 0}|\boldsymbol{d}_{j0}|^2 - \sum_{j\neq 0}\frac{(\boldsymbol{p}\cdot\boldsymbol{d}_{0j})(\boldsymbol{d}_{j0}\cdot\boldsymbol{p})}{M^2\,\Delta E_j},$$

in physical units for one nucleus. The third term is the diagonal correction of Born and Huang (Appendix D.11). The last term is quadratic in the momentum. With $\boldsymbol{d}_{j0} = \boldsymbol{F}_{j0}/(E_0 - E_j)$ from Eq. (5.11) it equals $-\boldsymbol{p}\cdot\delta M\cdot\boldsymbol{p}/2M^2$ with

$$\delta M = 2\sum_{j\neq 0}\frac{\mathrm{Re}[\boldsymbol{F}_{0j}\boldsymbol{F}_{j0}]}{\Delta E_j^3}, \tag{D.86}$$

a positive tensor, and together with $p^2/2M$ the last term gives the kinetic energy $\frac{1}{2}\boldsymbol{p}\cdot(M+\delta M)^{-1}\cdot\boldsymbol{p}$ to first order. For an atom the adiabatic states depend on the position $\boldsymbol{X}$ of the nucleus only through $\boldsymbol{r}_e - \boldsymbol{X}$, and Sect. 6.1 showed that $\boldsymbol{d}_{j0} = m_e\Delta E_j\langle\Psi_j|\sum_e \boldsymbol{r}_e|\Psi_0\rangle$. Then $\boldsymbol{F}_{j0} = -m_e\Delta E_j^2\langle\Psi_j|\sum_e \boldsymbol{r}_e|\Psi_0\rangle$, and a diagonal element of Eq. (D.86) is $2m_e^2\sum_j \Delta E_j\,|\langle\Psi_j|\sum_e x_e|\Psi_0\rangle|^2 = m_e\sum_j f_{j0}^{\mathrm{osc}}$ with the oscillator strengths of Eq. (4.36). By the Thomas–Reiche–Kuhn rule Eq. (4.37), $\delta M = N_e m_e$ for an atom with N_e electrons: the whole cloud is dragged, and the atom moves with its full mass, as in the solvable model of Appendix D.11.

D.22.3 The kick and the electronic sectors

Consider one nucleus at $\boldsymbol{X}$ and the kick $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$, which is diagonal in the electronic label at each $\boldsymbol{X}$. The tilted initial state is $\sum_k \Psi_k^{\mathrm{nuc}}\Psi_k$, with Ψ_k^{nuc} from Eq. (D.85). The tilted final state on surface j is $\sum_k \Psi_k^{\mathrm{nuc}\prime}\Psi_k$, dominated by its vibrational state $\Psi_j^{\mathrm{nuc}\prime}$. By the same argument

with j and 0 exchanged, the component of the tilted final state on the ground surface is $\Psi_0^{\mathrm{nuc}\prime} = \boldsymbol{d}_{0j}\cdot\nabla\Psi_j^{\mathrm{nuc}\prime}/[M(E_0-E_j)] = \boldsymbol{d}_{j0}^*\cdot\nabla\Psi_j^{\mathrm{nuc}\prime}/(M\Delta E_j)$, where we used $\boldsymbol{d}_{0j}=-\boldsymbol{d}_{j0}^*$, the anti-Hermiticity of the derivative coupling. The complex conjugate of $\Psi_0^{\mathrm{nuc}\prime}$ that enters the overlap below then carries $\boldsymbol{d}_{j0}$ itself. The leading term $\langle\Psi_j^{\mathrm{nuc}\prime}\Psi_j|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\Psi_0\rangle$ vanishes by orthogonality of the adiabatic states at each $\boldsymbol{X}$, and to first order the amplitude is

$$\langle\Psi_j^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_j^{\mathrm{nuc}}\rangle + \langle\Psi_0^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\rangle = \frac{1}{M\Delta E_j}\int \mathrm{d}\boldsymbol{X}\left[\Psi_j^{\mathrm{nuc}\prime *}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\,\boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\mathrm{nuc}} + (\boldsymbol{d}_{j0}\cdot\nabla\Psi_j^{\mathrm{nuc}\prime *})\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\Psi_0^{\mathrm{nuc}}\right].$$

In the second term we integrate by parts. We treat $\boldsymbol{d}_{j0}$ and ΔE_j as constant over the extent of the nuclear wavefunctions, neglecting corrections of relative order $\ell_{\mathrm{zp}}|\nabla\ln\boldsymbol{d}_{j0}| \sim \kappa_{\mathrm{BO}}$; the gradient then falls on $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\Psi_0^{\mathrm{nuc}}$ and gives $-\Psi_j^{\mathrm{nuc}\prime *}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}(\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{d}_{j0}\Psi_0^{\mathrm{nuc}} + \boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\mathrm{nuc}})$. The gradients of Ψ_0^{nuc} cancel between the two terms, and

$$\langle\widetilde{\Psi^{\mathrm{nuc}\prime}\Psi}_j|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\widetilde{\Psi^{\mathrm{nuc}}\Psi}_0\rangle = -\mathrm{i}\,\frac{\boldsymbol{q}\cdot\boldsymbol{d}_{j0}}{M\Delta E_j}\,\langle\Psi_j^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\rangle, \tag{D.87}$$

which is Eq. (8.53). The cancellation makes the result independent of how the nucleus moved before the kick; only the momentum transferred appears. Summing the squared modulus over the complete set of vibrational states $\Psi_j^{\mathrm{nuc}\prime}$ of surface j removes the nuclear factor, since $\sum|\langle\Psi_j^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\rangle|^2 = 1$, and gives P_{Mig} of Eq. (8.55).

D.22.4 Partial sum rules

At $T \ll E_{\mathrm{g}}$ the initial states lie in the electron-following sector. For a projector that commutes with H, the response of sector ad is the full response of the restricted operators $\bar{O} = \mathcal{P}_{\mathrm{ad}}O\mathcal{P}_{\mathrm{ad}}$ under $H_{\mathrm{ad}} = \mathcal{P}_{\mathrm{ad}}H\mathcal{P}_{\mathrm{ad}}$, and the first-moment identity Eq. (4.28) applies to it inside the sector: $\mathscr{M}^1_{ab}|_{\mathrm{ad}} = \mathscr{V}^{-1}\langle[\bar{O}_a^\dagger,[H_{\mathrm{ad}},\bar{O}_b]]\rangle$. Under the rotation by $U_n^\dagger$, the sector becomes functions of u times $\Psi_0(u)$, H_{ad} becomes h_{BO}, and the electron density becomes $\bar{\rho}_e(\boldsymbol{q};u)$ of Eq. (8.51) up to the tilt, of relative order κ_{BO}^2. A function $F(u)$ commutes with $E_0(u)$, and the double commutator with the kinetic energy of the nuclei is $[F^\dagger,[p^2/2M,F]] = |\nabla F|^2/M$, from $[p^2,F] = -\nabla^2F - 2\mathrm{i}(\nabla F)\cdot\boldsymbol{p}$. Summed over nuclei,

$$\mathscr{M}^1_{ee}\big|_{\mathrm{ad}} = \frac{1}{\mathscr{V}}\Big\langle\sum_{\boldsymbol{R}\kappa}\frac{|\nabla_{\boldsymbol{R}\kappa}\bar{\rho}_e|^2}{M_\kappa}\Big\rangle, \qquad \mathscr{M}^1_{e\kappa}\big|_{\mathrm{ad}} = \frac{1}{\mathscr{V}}\Big\langle\sum_{I\in\kappa}\frac{\nabla_I\bar{\rho}_e^\dagger\cdot\nabla_I\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}}{M_\kappa}\Big\rangle,$$

with $\nabla_I\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} = \mathrm{i}\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$. The mass correction changes these two moments by the relative amount $\delta M/M \sim \kappa_{\mathrm{BO}}^4$. For rigid clouds, $\bar{\rho}_e = \sum_I f_{\kappa(I)}(\boldsymbol{q})\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, the gradient with respect to $\boldsymbol{X}_I$ is $\mathrm{i}\boldsymbol{q}f_\kappa\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, and the averages give $q^2\sum_\kappa \boldsymbol{n}_\kappa|f_\kappa|^2/M_\kappa$ and $q^2\boldsymbol{n}_\kappa f_\kappa^*/M_\kappa$, the forms quoted in Eqs. (8.54) and (8.57). The total of $\mathscr{M}^1_{e\kappa}$ over all sectors is zero by Eq. (4.34), and the electronic sectors carry the negative of the electron-following part.

For the nuclear density the share of the first moment carried by the electronic sectors follows from Eq. (D.87). Each final state lies above the initial one by ΔE_j plus the vibrational energy of $\Psi_j^{\mathrm{nuc}\prime}$ relative to Ψ_0^{nuc}. Weighted with the squared amplitude Eq. (D.87) and summed over the vibrational states, the electronic energies give $\sum_j|\boldsymbol{q}\cdot\boldsymbol{d}_{j0}|^2/(M^2\Delta E_j) = \sum_j|\boldsymbol{q}\cdot\boldsymbol{F}_{j0}|^2/(M^2\Delta E_j^3) = \boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/2M^2$, and the vibrational energies give the first moment of the kick on the excited surface, E_R by the argument of Eq. (5.55), times P_{Mig}. The first moment summed over $\pm\boldsymbol{q}$ is twice the one-sided one, and per nucleus the electronic sectors carry $\mathscr{M}^1|_{0+} = \boldsymbol{n}_\kappa[P_{\mathrm{Mig}}q^2/M + \boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/M^2]$. The total is $\boldsymbol{n}_\kappa q^2/M$, and the electron-following sector carries the difference between the total and the electronic share,

$\boldsymbol{n}_\kappa[(1-P_{\mathrm{Mig}})q^2/M - \boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/M^2]$. To first order in the two small quantities the electron-following share is $\boldsymbol{n}_\kappa(1-P_{\mathrm{Mig}})\,\boldsymbol{q}\cdot(M+\delta M)^{-1}\cdot\boldsymbol{q}$, the same result that follows from h_{BO} with the depleted weight $1-P_{\mathrm{Mig}}$ of $\bar{n}_\kappa$, and it is Eq. (8.56).

D.22.5 Labels inside the off-shell window

Let H_0 contain the electronic excitations at clamped, renormalized bands and the harmonic phonons, and let H_{ep} couple them; the eigenstates of H_0 carry a definite phonon number. Take ω inside the off-shell window, so that every zero-phonon state $|m\rangle$ at the same total momentum has $E_m \geq \omega + \delta_{\mathrm{w}}$. An exact final state at ω grows out of a one-phonon state $|f\rangle$ of H_0. The Lippmann–Schwinger equation $|\tilde{f}\rangle = |f\rangle + (E_f - H_0 + \mathrm{i}0^+)^{-1}H_{\mathrm{ep}}|\tilde{f}\rangle$, projected on the zero-phonon states, has denominators $E_f - E_m \leq -\delta_{\mathrm{w}}$ that never vanish, and its iteration converges for $|g| \ll \delta_{\mathrm{w}}$. The zero-phonon component of $|\tilde{f}\rangle$ is $\sum_m |m\rangle g_{mf}/(E_f-E_m) + O(g^3/\delta_{\mathrm{w}}^3)$, with squared norm $\sum_m |g_{mf}|^2/(E_f-E_m)^2 \leq \theta_{\mathrm{ep}}$ when δ_{off} in Eq. (8.40) is taken as the smallest denominator, δ_{w}. Consider any two definitions of the label that agree at $g=0$ and are continuous in g, such as the phonon number of the leading component, a Schrieffer–Wolff number, or the number of phonons in the outgoing state; both assign $|\tilde{f}\rangle$ to $\mathcal{P}_+$ and differ by at most this weight. The response at ω is $(2\pi/\mathcal{V})\sum_f |\langle\tilde{f}|O_b|i\rangle|^2\delta(\omega-E_f)$, all of it in $\mathcal{P}_+$ to relative order θ_{ep}. To leading order $\langle\tilde{f}|O_b|i\rangle = g_{fm}[\mathcal{A}_f^{\mathrm{off}} + \rho_m/(\omega-E_m)]$, summed over m, which is Eq. (8.59) with no delta term, since $\omega - \tilde{E}_m$ never vanishes on the support. The bare projector $\mathcal{P}_0^{(0)}$, by contrast, gives $\langle\tilde{f}|\mathcal{P}_0^{(0)}O_b|i\rangle = \sum_m g_{mf}^*\rho_m/(E_f-E_m)$, the resonant part of the same amplitude, and it would assign to $\mathcal{P}_0$ a weight of the size of the rate.

D.22.6 The resonant amplitude

Outside the window, keep one zero-phonon pair $|m\rangle$ at energy E_m and the one-phonon states $|f\rangle$ it couples to, with $H = E_m|m\rangle\langle m| + \sum_f E_f|f\rangle\langle f| + \sum_f(g_{fm}|f\rangle\langle m| + g_{fm}^*|m\rangle\langle f|)$; nonresonant pairs and the paths through occupied states only enter the amplitudes $\mathcal{A}_f^{\mathrm{off}}$, and degenerate pairs need the matrix form of the same algebra. The partitioning Eq. (C.56) with $P = |m\rangle\langle m|$ gives the level's Green's function $\mathsf{G}_m(z) = [z - E_m - \Sigma_m(z)]^{-1}$ with $\Sigma_m(z) = \sum_f |g_{fm}|^2/(z-E_f)$, as in Worked example 4.12. The scattering states $|f^-\rangle$ of the continuum satisfy $\langle f^-|m\rangle = g_{fm}\mathsf{G}_m(E_f+\mathrm{i}0^+)$ and $\langle f^-|f'\rangle = \delta_{ff'} + g_{fm}\mathsf{G}_m(E_f+\mathrm{i}0^+)\,g_{f'm}^*/(E_f - E_{f'} + \mathrm{i}0^+)$, the Lippmann–Schwinger solution of the model [37]. If no bound state splits off, they are complete, and the response of the source $O_b|i\rangle = \rho_m|m\rangle + \sum_f g_{fm}\mathcal{A}_f^{\mathrm{off}}|f\rangle$ is $(2\pi/\mathcal{V})\sum_f |\langle f^-|O_b|i\rangle|^2\delta(\omega-E_f)$. The resulting amplitude is Eq. (8.58).

Let $\mathcal{A}_f^{\mathrm{off}}$ take the value $\mathcal{A}_\nu^{\mathrm{off}}$ on the energy shell of branch ν, and use $\sum_{f\in\nu}|g_{fm}|^2\delta(\omega-E_f) = \Gamma_m^{(\nu)}(\omega)/2\pi$. The response in branch ν is $(2\pi/\mathcal{V})(\Gamma_m^{(\nu)}/2\pi)|\mathcal{A}_\nu^{\mathrm{off}} + \mathsf{G}_m(\rho_m + \Upsilon_m)|^2$, and we expand its three terms near $\tilde{E}_m$, with $x = \omega - \tilde{E}_m$, against a test function smooth on δ_{F}. Near the pole $\mathsf{G}_m \simeq z_{\mathrm{qp}}/(x + \mathrm{i}z_{\mathrm{qp}}\Gamma_m/2)$, and Eq. (2.60) with the width $z_{\mathrm{qp}}\Gamma_m$ gives $|\mathsf{G}_m|^2 = (2\pi z_{\mathrm{qp}}/\Gamma_m)\delta(x) + \mathrm{fp}(1/x^2) + O(g^2)$. In the cross term $2\,\mathrm{Re}[\mathcal{A}_\nu^{\mathrm{off}*}\mathsf{G}_m\rho_m]$ the propagator becomes $\mathrm{PV}(1/x) - \mathrm{i}\pi\delta(x)$ by Eq. (C.26), which yields $2\,\mathrm{Re}(\mathcal{A}_\nu^{\mathrm{off}*}\rho_m)\,\mathrm{PV}(1/x) + 2\pi\,\mathrm{Im}(\mathcal{A}_\nu^{\mathrm{off}*}\rho_m)\delta(x)$. In $|\rho_m + \Upsilon_m|^2 = |\rho_m|^2 + 2\,\mathrm{Re}(\rho_m^*\Upsilon_m) + O(g^4)$, the vertex correction at the pole is $\Upsilon_m = \mathrm{PV}\,\Upsilon_m - (\mathrm{i}/2)\sum_\mu \Gamma_m^{(\mu)}\mathcal{A}_\mu^{\mathrm{off}}$, and $2\,\mathrm{Re}(\rho_m^*\Upsilon_m) = 2\,\mathrm{Re}(\rho_m^*\,\mathrm{PV}\,\Upsilon_m) + \sum_\mu \Gamma_m^{(\mu)}\,\mathrm{Im}(\rho_m^*\mathcal{A}_\mu^{\mathrm{off}})$. With $c_\mu = \mathrm{Im}(\rho_m^*\mathcal{A}_\mu^{\mathrm{off}})$ and $\bar{c} = \sum_\mu \Gamma_m^{(\mu)}c_\mu/\Gamma_m$, the coefficient of $(2\pi/\mathcal{V})\delta(x)$ in branch ν collects to

$$\frac{\Gamma_m^{(\nu)}}{\Gamma_m}\Big[z_{\mathrm{qp}}|\tilde{\rho}_m|^2 + \Gamma_m(\bar{c} - c_\nu)\Big] = z_{\mathrm{qp}}|\tilde{\rho}_m|^2\,\frac{\Gamma_m^{(\nu)}}{\Gamma_m}(1 + \bar{c}^{\mathrm{F}} - c_\nu^{\mathrm{F}}) + O(g^4), \tag{D.88}$$

with $c_\nu^{\mathrm{F}} = \Gamma_m c_\nu/|\rho_m|^2$. Summed over ν with the weights $\Gamma_m^{(\nu)}/\Gamma_m$, which add up to one, the Fano terms cancel, and the delta part is S^0 of Eq. (8.59). The remainder of the response in

branch ν, $(\Gamma_m^{(\nu)}/2\pi)[|\rho_m|^2\,\mathrm{fp}(1/x^2) + 2\,\mathrm{Re}(\mathcal{A}_\nu^{\mathrm{off}*}\rho_m)\,\mathrm{PV}(1/x) + |\mathcal{A}_\nu^{\mathrm{off}}|^2]$, summed over ν, is S^+. The neglected terms are of three kinds: order g^4, from $z_{\mathrm{qp}}^2 \to 1$ in the finite part and from $|\Upsilon_m|^2$; the term $-\pi\Gamma_m\delta''(x)/4$ of Eq. (2.60), which against a test function smooth on δ_{F} is of relative order $\Gamma_m^2/\delta_{\mathrm{F}}^2$; and the shift of the finite part from E_m to $\tilde{E}_m$, of order g^4. The sum of S^0 and S^+ therefore agrees with the exact response to $O(g^4, \Gamma_m^2/\delta_{\mathrm{F}}^2)$. If the pair also decays by channels other than phonon emission, impact ionization for instance, the same algebra holds with ν and μ running over every channel. Each channel has its own off-shell amplitude $\mathcal{A}_\nu^{\mathrm{off}}$, the self-energy and the vertex correction Υ_m sum over all of them, the Lorentzian carries the total width $\Gamma_m^{\mathrm{tot}} = \sum_\nu \Gamma_m^{(\nu)}$, and Eq. (D.88) becomes Eq. (8.62). The deviations $c_\nu^{\mathrm{F}} - \bar{c}^{\mathrm{F}}$ then average to zero over all channels; restricted to the phonon branches but normalized by Γ_m^{tot}, they would not, and the number of zero-phonon events would change. As before, the delta function itself needs no width, and only the cascade needs Γ_m^{tot}. A cascade that lets the zero-phonon pair emit its first phonon with these branching ratios, Fano terms included, reproduces the exact distribution of the first emission at order g^2.

The residue identity Eq. (8.60) follows from the Kramers–Kronig form $\mathrm{Re}\,\Sigma_m(\omega) = (2\pi)^{-1}\,\mathrm{PV}\int \mathrm{d}\omega'\,\Gamma_m(\omega')/(\omega - \omega')$ and Eq. (C.30). The ambiguity estimate needs a bound on $1 - z_{\mathrm{qp}}^{-1} = \partial_\omega\,\mathrm{Re}\,\Sigma_m$. Split the finite-part integral at $|\omega' - \tilde{E}_m| = \delta_{\mathrm{F}}$. Outside, $|\int \mathrm{d}\omega'\,\Gamma_m/(\omega' - \tilde{E}_m)^2| \le 2\hat{\Gamma}_m/\delta_{\mathrm{F}}$, with $\hat{\Gamma}_m$ the largest width. Inside, $\Gamma_m(\tilde{E}_m)$ times $\mathrm{fp}\int_{-\delta_{\mathrm{F}}}^{\delta_{\mathrm{F}}} \mathrm{d}y/y^2 = -2/\delta_{\mathrm{F}}$ contributes $2\Gamma_m/\delta_{\mathrm{F}}$ in modulus, the linear term of Γ_m drops out by symmetry, and the curvature contributes at most $\delta_{\mathrm{F}}\|\Gamma_m''\|$, which is $\hat{\Gamma}_m/\delta_{\mathrm{F}}$ if Γ_m varies on the scale δ_{F}. Hence $|1 - z_{\mathrm{qp}}^{-1}| \le (5/2\pi)\hat{\Gamma}_m/\delta_{\mathrm{F}}$. Raising the residue to a power between 1 and 2 changes the zero-phonon weight by at most this amount relative to w_0; this bound is the first term of Eq. (8.61). Replacing the delta function by a normalized kernel of width Γ_m moves the tail weight $2\int_{x_{\mathrm{w}}}^\infty \mathrm{d}x\,(\Gamma_m/2\pi)/x^2 = \Gamma_m/\pi x_{\mathrm{w}}$ outside a window of half-width x_{w}; this tail weight is the second term. Assigning to the line all weight within $|x| < x_{\mathrm{w}}$ moves into it the finite-part, principal-value and $|\mathcal{A}^{\mathrm{off}}|^2$ terms of S^+ there. Their pole parts reproduce the second term; their smooth parts are bounded by $\hat{\Gamma}_m x_{\mathrm{w}}/\delta_{\mathrm{F}}^2$ times $|\rho_m|^2$ for the finite part, $2|\rho_m||\mathcal{A}^{\mathrm{off}}|\delta_{\mathrm{F}}$ for the principal value and $|\mathcal{A}^{\mathrm{off}}|^2\delta_{\mathrm{F}}^2$ for the last term. Divided by $w_0 \simeq |\rho_m|^2$ they add up to the third term. Each step is first order in $\Gamma_m/\delta_{\mathrm{F}}$, and the bound holds to that order.

D.22.7 Real amplitudes under inversion and time reversal

Let PT be the product of inversion and time reversal. It is antiunitary and commutes with H if both symmetries hold. On states of N electrons $\mathsf{T}^2 = (-1)^N$, $\mathsf{P}^2 = 1$ and the two commute, so $(\mathsf{PT})^2 = (-1)^N$; on phonons $(\mathsf{PT})^2 = 1$. For even N, then, $(\mathsf{PT})^2 = 1$ on the whole space. An antiunitary involution that commutes with a Hermitian operator leaves each eigenspace invariant, and every eigenspace has an orthonormal basis of invariant vectors: for any v in it, $v + \mathsf{PT}v$ and $\mathrm{i}(v - \mathsf{PT}v)$ are invariant, and v is a combination of them. Choose an invariant orthonormal basis for the eigenstates of H_0 within each sector; the choice is possible because PT commutes with the phonon number. For the one-phonon states the choice fixes the phase of each polarization vector $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$, since inversion and time reversal each reverse $\boldsymbol{Q}$ and PT maps the mode $(\boldsymbol{Q}, \nu)$ to itself. For two invariant states a, b and an invariant operator O, antiunitarity gives $\langle a|O|b\rangle = \langle \mathsf{PT}a|\mathsf{PT}Ob\rangle^* = \langle a|Ob\rangle^*$, a real number. The density is invariant: inversion maps $n_e(\boldsymbol{q})$ to $n_e(-\boldsymbol{q})$, and time reversal, which conjugates $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, maps it back. So is H_{ep}, as part of H. Every ρ_m and g_{fm} is real, and so is every off-shell amplitude $\mathcal{A}_f^{\mathrm{off}}$, a sum of products of real matrix elements and real denominators. Hence $c_\nu = 0$ and the Fano coefficients vanish, provided the invariant basis of a degenerate zero-phonon manifold also diagonalizes its level-shift matrix, a condition that holds for one-body self-energies and can fail for level shifts from the electron–hole vertex (Sect. 8.6). With spin–orbit coupling the single-electron states form Kramers doublets with $(\mathsf{PT})^2 = -1$, but the argument concerns

many-electron states with even N and needs no real single-particle gauge.

D.22.8 Gapless targets

For a state ψ_b that decays by the golden rule into a continuum of final states f with a flat density $\mathcal{D}_f$ over a width ΔE_c, $\Gamma = 2\pi|\langle f|H_\mathrm{off}|\psi_b\rangle|^2\mathcal{D}_f/\|\psi_b\|^2$ and $\|H_\mathrm{off}\psi_b\|^2 = \sum_f |\langle f|H_\mathrm{off}|\psi_b\rangle|^2 = \Gamma\,\Delta E_\mathrm{c}\|\psi_b\|^2/2\pi$, which is the estimate of Sect. 8.6. Without any dressing, $U_\mathrm{qp} = 1$, the estimate does not become a bound. The diagonal part H_d conserves the bare numbers, but the probe's quasiparticle is created on top of the filled Fermi sea, and H_off also acts on the sea, creating a phonon and a pair away from the probe's excitations. The squared norm of the resulting disconnected term grows with the number of cells, and so does $\mathfrak{h}_\mathrm{off}$. The estimate describes the connected part, in which H_off acts on the quasiparticle $|n\boldsymbol{k}\rangle$ itself. For the electron–phonon part, the state $H_\mathrm{ep}|n\boldsymbol{k}\rangle$ contains one phonon emitted or absorbed, and its squared norm is $N^{-1}\sum_{m\nu\boldsymbol{Q}} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2$ times $n_{\boldsymbol{Q}\nu}+1-f_{m,\boldsymbol{k}+\boldsymbol{Q}}$ for emission and $n_{\boldsymbol{Q}\nu}+f_{m,\boldsymbol{k}+\boldsymbol{Q}}$ for absorption, at most $2n_{\boldsymbol{Q}\nu}+1$ in total. This squared norm is $-\pi^{-1}\int \mathrm{d}\omega\ \mathrm{Im}\,\Sigma^\mathrm{FM}_{n\boldsymbol{k}}(\omega)$ from Eq. (6.27), the zeroth moment of the Fan–Migdal spectral function. The electron–electron part adds the analogous sum over final states with two particles and a hole and the phonon number unchanged; the two sets of final states differ in phonon number, so they are orthogonal and the sums add. For a pair source, the electron and hole contributions add. Taken over the bands, these moments form a matrix whose largest eigenvalue bounds the square of the connected part. In aluminum the electron–phonon moments alone give an eigenvalue of about $0.03\,\mathrm{eV}^2$ (Sect. 8.6), so the electron–phonon contribution to the connected part is at most $0.17\,\mathrm{eV}$; the electron–electron moments cannot lower the eigenvalue and are not estimated. Separating the connected part from the disconnected part needs the dressed vacuum U_qp and a commutator bound uniform in N, a problem that is open (Sect. 8.6).

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